Fundamentals of Statistics

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Preface

This is a book about the mathematics underneath the statistical methods you already use. It covers what a graduate introduction to statistical inference covers: the limit theorems that justify large-sample reasoning, statistical models, estimators and how to judge them, confidence intervals, hypothesis tests, maximum likelihood, the chi-squared family, Bayesian inference, linear regression, generalized linear models and principal component analysis. It assumes you have met most of these before as recipes, and it asks the question an introductory course has no room for: why is any of this allowed?

Who it is for. A reader with a solid undergraduate statistics course behind them. You know the sample mean, the normal distribution, confidence intervals and t-tests, and you have had a semester of calculus. Every step of every derivation is written out; nothing is left as an exercise that the text then relies on.

The mathematics is all here: the ϵ-N definition of a limit, the proofs, the inequalities. At the same time, the limit theorems of Part II are used constantly in the rest of the book, and nobody carries their exact statements around in their head. So every later use of the Law of Large Numbers, the Central Limit Theorem, Slutsky's theorem, the continuous mapping theorem or the delta method restates in one line what that theorem gives you, at the point where it is used, marked *Recall*. You can read any chapter after Chapter 6 without turning back to Part II.

How it is organized. One chapter per idea, ten to nineteen pages, in parts ordered by what has to be understood before what. Chapter 1 introduces the example that runs through the whole book and the three questions statistics asks about it. Chapter 2 defines the object every later chapter lives inside. Part II is the probability: where an average settles, how it fluctuates on the way, and what you may do to a limit and still know where you are. Part III answers the three questions with guarantees, and the rest of the book is methods.

Every chapter opens with a box, 'In this chapter', that says what is coming and how it connects to the chapter before, and closes with a paragraph on what comes next (the last chapter, on where the book ends), a summary and exercises. Five kinds of box run through the text:

- *Intuition*: the picture or the argument in words behind a result, usually before the result is proved.
- *From undergraduate statistics*: the version of the idea you already met, and what the general version adds. There are many of these: the statistics you already know is the natural place to start from.
- *Pitfall*: a mistake that is easy to make.
- *Worked example*: the computation carried through with numbers.
- *Does this make sense?*: how to tell that an answer is absurd without redoing the work. Extreme cases, bounds, a second route to the same number.

Every definition is followed by a short block that says the same thing in plain words and then says what it means for the running example. Every proof is preceded by one line of strategy, marked either *core* (follow it: the proof carries an idea the rest of the book uses) or *for completeness* (the statement is what you need, and the strategy line carries the idea; read the proof when you want to). Solutions to all exercises are at the back of the book, and each exercise says on which page.

The mathematics you need. A solid introductory statistics course and one semester of calculus, provided you are willing to read a definition slowly. Where a proof leans on a mathematical idea that an introductory course leaves implicit (the ϵ-N definition of a limit, the mean value theorem, Markov's inequality, what 'injective' means), the text marks the step with a gray pointer such as '(toolkit A.9)' to the Mathematical Toolkit, Appendix A, which explains that idea in a page, usually with a picture, and says where else the book uses it. The toolkit is where to go when a step feels unjustified. It is not a prerequisite to read first.

The minimal path through Part II. Chapters 3 to 5 are about limits of random variables, and they exist to deliver three facts: averages settle on their expectations, the error of an average is Gaussian on the $1/\sqrt{n}$ scale, and there are rules for transforming limits and plugging estimates into them. Read those three chapters for those three facts and take Chapter 3's opening box as the working translation of the word 'asymptotic' wherever it appears. The remaining modes of convergence are in Appendix B, for orientation only.

Two conventions that bite. This book writes the Gaussian distribution as $\text{Gauss}(\mu, \sigma^2)$ with the *variance* as the second parameter, so $\text{Gauss}(85, 4)$ has standard deviation 2. Much of the software you will meet takes the standard deviation instead. And it writes q_α for the quantile with probability α *above* it, so $q_{0.05} = 1.645$ and $q_{0.025} = 1.96$, while most tables and software index a quantile by the probability below. Both conventions are standard in mathematical statistics and both are traps when you move between sources. Chapter 6 explains them with pictures; every formula in this book follows them.

Figures. Every figure is generated by a short script that ships with the book's source, so any of them can be redrawn with other numbers. Nothing in a figure is schematic: when a curve is drawn, it is the real curve, and when a probability is shaded, the shaded area is that probability.

Where the material comes from. The syllabus is that of MIT's 18.6501x, *Fundamentals of Statistics*, which I worked through and took notes on, and this book grew out of those notes. You do not need the course to read this book, and after this paragraph it is not mentioned again.

Notation

The symbols that recur, in the order a reader meets them. These are the standard conventions of mathematical statistics; the two that differ from most software are marked.

Symbol	Meaning
$X_1, \dots, X_n$	the sample: n random variables, i.i.d. (independent, identically distributed) unless said otherwise
$x_1, \dots, x_n$	the realized values, numbers; likewise $\bar{X}_n$ is a random variable and $\bar{x}_n$ a number (Chapter 3)
$X_1, \dots, X_n \overset{iid}{\sim} \text{Bern}(p)$	the X_i are i.i.d. with the named distribution; Bern, Bin, Pois, Expo, Unif, Gauss
$\text{Gauss}(\mu, \sigma^2)$	Gaussian with mean μ and *variance* σ^2 (most software takes the standard deviation)
φ, Φ	density and CDF of $\text{Gauss}(0,1)$
q_α	the quantile with probability α *above* it: $\mathbb{P}(Z > q_\alpha) = \alpha$, so $q_{0.05} = 1.645$, $q_{0.025} = 1.96$ (software indexes by the probability below)
$\mathbb{P}$, $\mathbb{E}$, $\mathbb{V}\text{ar}$, $\mathbb{C}\text{ov}$	probability, expectation, variance, covariance; $\mathbb{P}_\theta$, $\mathbb{E}_\theta$ when they are computed under the distribution with parameter θ
$\mathbb{1}(A)$	the indicator: 1 if A holds, 0 otherwise; $\mathbb{E}[\mathbb{1}(A)] = \mathbb{P}(A)$
$\bar{R}_n$, $\hat{p}$	the kiss experiment: $R_i = 1$ if couple i turns right, $\hat{p} = \bar{R}_n$; $n = 124$, 80 right turns, $\bar{R}_n = 0.645$
$\bar{T}_n$, $\hat{\lambda}$	the subway inter-arrival times, $T_i \overset{iid}{\sim} \text{Expo}(\lambda)$, $\hat{\lambda} = 1/\bar{T}_n$; $n = 64$, $\bar{T}_n = 6.23$
$X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$	convergence in probability (Definition 3.1)
$X_n \xrightarrow[n\to\infty]{\text{(d)}} X$	convergence in distribution (Definition 4.1); the limit may be written as a distribution, $X_n \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1)$
$X_n \xrightarrow[n\to\infty]{\text{a.s.}} X$, $X_n \xrightarrow[n\to\infty]{\text{a.s./}\mathbb{P}} X$	convergence almost surely (Appendix B); both arrows at once, which is how the two laws of large numbers are written together
Ω, ω	the sample space, one point per complete outcome of the study, and a point of it; a random variable is a function on Ω (Section 2.3)
$(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$	a statistical model: observation space E, where one observation lands, and a family of candidate distributions indexed by $\theta \in \Theta$; $E^n = \Omega$ for fixed n (Definition 2.1)
θ, θ^*, $\hat{\theta}_n$	a parameter, the true parameter, an estimator of it from n observations
Bias, $R(\hat{\theta}_n)$	bias $\mathbb{E}_\theta[\hat{\theta}_n] - \theta$ and quadratic risk $\mathbb{E}_\theta[(\hat{\theta}_n - \theta)^2]$ (Chapter 7)
$\mathcal{I} \ni \theta$	the random interval $\mathcal{I}$ contains θ; same as $\theta \in \mathcal{I}$, written to put the random object first (Chapter 8)
α	a small probability: a test's level, or $1-$ (an interval's level)

Symbol	Meaning
$a^\top$, $A^\top$	the transpose of a vector or matrix; vectors are columns, so $a^\top b$ is a number
∇h, $\nabla^2 h$	gradient and Hessian of a function of several variables (toolkit A.12); for $g : \mathbb{R}^d \to \mathbb{R}^k$, ∇g is the $d \times k$ matrix of partial derivatives
$\mathrm{TV}(\mathbb{P}, \mathbb{Q})$, $\mathrm{KL}(\mathbb{P}, \mathbb{Q})$	total variation distance and Kullback–Leibler divergence between two distributions (Chapter 10)
$L_n(\mathbf{x}; \theta)$, $\ell_n(\theta)$	the likelihood of the sample at θ and its logarithm; $\hat{\theta}_n$ or $\hat{\theta}_n^{\mathrm{MLE}}$ its maximizer (Chapter 11)
$I(\theta)$	the Fisher information of one observation (Chapter 12); a $d \times d$ matrix when $\theta \in \mathbb{R}^d$
$\mathrm{Gauss}_d(m, \Sigma)$	the Gaussian vector in $\mathbb{R}^d$ with mean m and positive definite covariance matrix Σ (toolkit A.14)
χ^2_k, t_n	the chi-squared distribution with k degrees of freedom and Student's t with n (Chapter 14)
T_n^{W}, T_n^{LR}, T_n^{S}	Wald's, the likelihood ratio and the score statistic (Chapter 15)
$o_{\mathbb{P}}(1)$	a sequence of random quantities that converges in probability to 0 (Chapter 15)
N_j, $\mathrm{Mult}(n, \mathbf{p})$	the count in category j; the multinomial distribution of $(N_1, \ldots, N_K)$ (Chapter 16)
F_n, $X_{(i)}$	the empirical CDF; the ith smallest observation (Chapter 17)
$\pi(\theta)$, $\pi(\theta \mid \mathbf{x})$	prior and posterior density (or PMF) of θ, with $\mathbf{x} = (x_1, \ldots, x_n)$ (Chapter 18)
$\mathrm{Beta}(a, b)$, $\mathrm{Gamma}(\alpha, \beta)$	the Beta distribution on $(0, 1)$; the Gamma distribution with shape α and rate β (Chapter 18)
$\propto$	'is proportional to, as a function of θ': $\pi(\theta \mid \mathbf{x}) \propto L_n(\mathbf{x}; \theta)\pi(\theta)$ (Chapter 18)
$\mathbb{E}[Y \mid X = x]$, $f(x)$	conditional expectation; the regression function of Y on X (toolkit A.15) (Chapter 20)
S_{xx}, S_{xy}, S_{yy}, r, R^2	$\sum_i (X_i - \bar{X})^2$, $\sum_i (X_i - \bar{X})(Y_i - \bar{Y})$, $\sum_i (Y_i - \bar{Y})^2$, the sample correlation and the coefficient of determination; $\hat{a}$, $\hat{b}$ the least squares line, e_i its residuals (Chapter 20)
X, P, γ_j	the $n \times p$ design matrix, the projection $X(X^\top X)^{-1}X^\top$ onto its column space, the jth diagonal entry of $(X^\top X)^{-1}$ (Chapter 21)
$F_{k,l}$	the F distribution with (k, l) degrees of freedom, the ratio of two independent chi-squared variables over their degrees of freedom (Chapters 14 and 21)
$\eta(\theta)$, $T(y)$, $B(\theta)$, $h(y)$	natural parameters, sufficient statistics, normalizer and base weight of an exponential family $h(y)\exp(\eta(\theta)^\top T(y) - B(\theta))$ (Chapter 22)
$b(\theta)$, ϕ, $V(\mu)$, g_c	in the canonical form $\exp((y\theta - b(\theta))/\phi + c(y, \phi))$: the function with $b' = \mathbb{E}[Y]$ and $\phi b'' = \mathbb{V}\mathrm{ar}(Y)$, the dispersion, the variance function $\mathbb{V}\mathrm{ar}(Y) = \phi V(\mu)$, and the canonical link $(b')^{-1}$ (Chapter 22)
g, $x_i^\top \beta$, μ_i, W	the link function, the linear predictor, the mean of Y_i and the diagonal weight matrix $b''(x_i^\top \beta)$ of a generalized linear model (Chapter 23)
Σ, S, λ_j, v_j	the covariance matrix of a random vector and the sample covariance matrix (divisor n); the eigenvalues of S in decreasing order and their orthonormal eigenvectors, the principal directions (Chapter 24)
r.v., CDF, PDF, PMF	random variable; cumulative distribution function $F(x) = \mathbb{P}(X \le x)$; density; probability mass function
CLT, LLN, CMT	Central Limit Theorem, Law of Large Numbers, continuous mapping theorem

(toolkit A.k)	a pointer to entry k of the Mathematical Toolkit, Appendix A, which explains the piece of mathematics being used

Part I

The Kiss Experiment and the Statistical Model

The book opens with its running example and with the object every later chapter lives in: the kiss experiment (Chapter 1) and the statistical model (Chapter 2), which turns the three assumptions of that experiment into a family of candidate distributions with a parameter to learn. The model comes first so that the limit theorems of Part II arrive with a job to do.

Chapter 1

The Kiss Experiment and What Statistics Is For

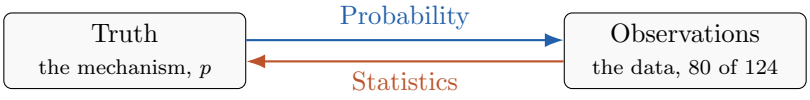
In this chapter

- Probability and statistics as the same picture read in opposite directions: from a known truth to the data it will produce, and from observed data back to the truths that could have produced it.
- The book's running example, the kiss experiment, and the three modeling assumptions that turn 80 out of 124 into a mathematical problem.
- Why the sample proportion is a random variable, what population it is a sample from, and the three questions about it that the next chapters answer.
- The trinity of statistical inference: estimation, confidence intervals and hypothesis testing, each stated for the kiss experiment and each pointed to the chapter that delivers it.

1.1 Probability and statistics are the same picture read in opposite directions

Probability studies randomness. Sometimes the physical process is completely known: dice, cards, roulette, fair coins. Roll one die. Alice gets \$1 if the number of dots is at most 3; Bob gets \$2 if the number of dots is at most 2. Who would you rather be? You don't need any data to answer: Alice expects $1 \times \frac{3}{6} = \$0.50$ per roll, Bob expects $2 \times \frac{2}{6} \approx \$0.67$, so Bob. Roll two dice and win \$100 if you called the sum correctly: you call 7 because six of the thirty-six equally likely outcomes sum to 7 and no other total has as many. In both cases the mechanism is known, and probability deduces what the data will look like.

Statistics is the reverse problem. The mechanism is unknown, you observe data it produced, and you want to say something about the mechanism. The two directions fit into a single diagram, the central dogma of probability and statistics.



A drug trial shows both directions at work. *Probability:* previous studies showed the drug to be 80% effective. Then in a study of 100 patients you can anticipate that on average 80 will be cured, and that at least 65 will be cured with probability 99.99%. *Statistics:* you observe that 78 of 100 patients were cured. You will be able to conclude, by the end of Chapter 8, that you are 95% confident the drug is effective on between 69.9% and 86.1% of patients. (That sentence is the standard shorthand for a confidence interval; Section 8.8 says exactly what it does and does not claim.) Figure 1.1 draws both halves. On the left, the effectiveness $p = 0.80$ is known, so the number cured out of 100 has a known distribution, the Binomial, and the darker bars from 65 upward carry the computable 0.9999. On the right, 78 of 100 were cured and p is unknown; the curve shows how plausible each value of p makes the data (it is the likelihood of Chapter 11, scaled to peak at 1), and the shaded band beneath it is the 95% confidence interval $[0.70, 0.86]$ of Chapter 8.

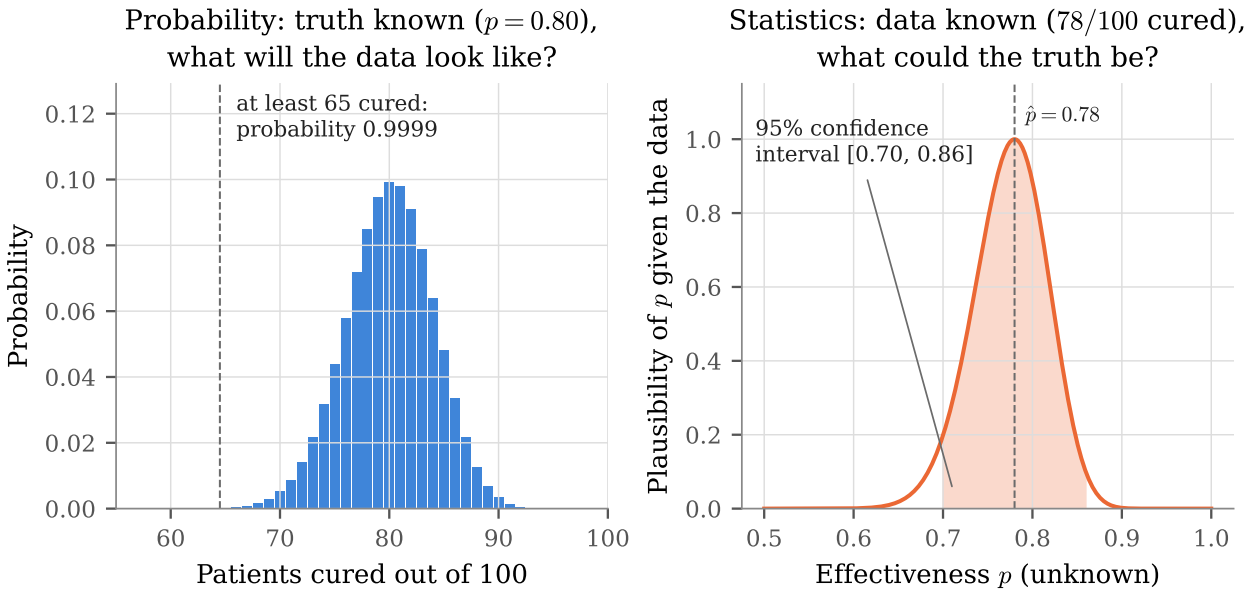


Figure 1.1: The drug trial. Left: the Binomial distribution of the number cured out of 100 when $p = 0.80$; the bars from 65 up are darker. Right: the scaled likelihood of p after 78 cures; the shaded band is the 95% confidence interval.

Notice what the right of the figure is *not*. It isn't a probability distribution for p: in this book p is a fixed unknown number, not a random variable, and the curve is a plausibility ranking of candidate truths, not a density. The one place where p does get a distribution is Part VII, starting with Chapter 18, and the difference is the whole point of that chapter.

From undergraduate statistics: descriptive and inferential statistics

An introductory course divides statistics into descriptive (summarize the data you have) and inferential (say something about the population you don't have). This book is entirely about the second, and it adds the step the introductory course leaves implicit: a mathematical model of how the data were produced, precise enough that the inferential statements can be proved rather than asserted. The model is a list of candidates for the thing on the left of the diagram, and every method in the book is a way of traveling the orange arrow with a guarantee attached.

1.2 The kiss experiment

Newborn babies, lying on their backs, turn their heads to the right more often than to the left. A 2003 paper in *Nature* asked whether that preference survives into adulthood, and looked for it in an unexpected place: which way couples turn their heads when they kiss.[1] The author watched kissing couples in public places (airports, train stations, beaches and parks) in the United States, Germany and Turkey, and recorded for each couple which way their heads turned.

Let p denote the proportion of couples who turn their heads to the right when kissing. The statistical experiment: observe n kissing couples and record each outcome, 1 for right and 0 for left. Estimate p with the proportion $\hat{p}$ of couples who turned right. Of the $n = 124$ couples observed, 80 turned to the right, so

$$\hat{p} = \frac{80}{124} = 64.5\%.$$

Two things about the number 124 because the whole book turns on them. It is small, and it is small for a reason: each observation needs an observer in a public place at the right moment, close enough to see which way the heads turn, and a kiss is over in seconds. And it is the only sample there is: the study is published, and the question is what these 124 observations can be made to say. Everything that follows is about how much a sample of this size can be made to say.

Now some random intuition. 64.5% is much larger than 50%, so there seems to be a preference for turning right. But suppose the data had been right, right, left, that is $n = 3$: that's 66.7% to the right, an even bigger number. Better evidence? Obviously not; intuitively you need a large enough sample size to make a call. How large? Another way to put the problem: for $n = 124$, what is the minimum number of couples turning right that you would need to see to be convinced that $p > 50\%$? 63? 72? 75? 80? Mathematical modeling is needed to understand the accuracy of this procedure, and the number turns out to be 72 (Section 1.5).

From undergraduate statistics: experiments, observational studies and random sampling

An introductory course reserves 'experiment' for a study that imposes a treatment, which makes the kiss experiment an observational study: nobody told the couples which way to turn. This book's 'statistical experiment' is any procedure whose outcome is random, as in probability theory. An introductory course also teaches what randomness in the design buys: random assignment licenses a conclusion about cause and effect, and random sampling licenses a generalization from the sample to a population. The aspirin trial of Section 15.7 assigned 22,071 physicians at random to aspirin or placebo, so its gap of 17 against 9 heart attacks per 1,000 physicians can be credited to aspirin. The kiss experiment has neither kind of randomness: the 124 couples were whoever kissed where the observer was watching. The model of Section 1.3 takes as given what a random sample would have supplied: random outcomes, a common p and independence. The common p is the proportion of all kissing couples who turn right if the way the 124 were found has nothing to do with the way they turned.

[1] O. Güntürkün, 'Adult persistence of head-turning asymmetry', *Nature* 421 (2003), 711.

1.3 Modeling assumptions

Formally, the procedure consists of the following. For $i = 1, \ldots, n$, define $R_i = 1$ if the i-th couple turns to the right and $R_i = 0$ otherwise. The estimator of p is the sample average

$$\hat{p} = \bar{R}_n = \frac{1}{n} \sum_{i=1}^{n} R_i.$$

What is the accuracy of this estimator? To answer, you propose a statistical model that describes, or approximates well, the experiment. You think of the R_i's as random variables so that $\hat{p}$ is also a random variable, and then you study its fluctuations.

Coming up with a model consists of making assumptions on the observations R_i, $i = 1, \ldots, n$, in order to draw statistical conclusions. Here are the three assumptions this model makes:

1. Each R_i is a random variable.
2. Each of the r.v.s R_i is Bernoulli with parameter p.
3. $R_1, \ldots, R_n$ are mutually independent.

Let's discuss these assumptions, since each one is a decision rather than a fact.

1. Randomness is a way of modeling lack of information. With perfect information about the conditions of the kiss (the couple's heights, which side each was standing on, which way each habitually turns, what is going on in their minds), physics and psychology might let you predict the outcome. You don't have that information, so you treat the outcome as random.
2. Given (1), the R_i's are necessarily Bernoulli r.v.s since $R_i \in \{0, 1\}$; the only question is their parameters. They could have a different parameter for each couple, $R_i \sim \text{Bern}(p_i)$, but you don't have enough information in the data to estimate 124 separate p_i's from 124 observations. So you simply assume the observations come from the same process: $p_i = p$ for all i.
3. Independence is reasonable because the couples were observed in many different places, in three countries, at different times. It would fail if, say, several of the 124 observations were the same couple seen more than once, or if couples watched each other and copied.

A collection of r.v.s $X_1, \ldots, X_n$ satisfying (2) and (3), each with the same distribution and all mutually independent, is called **independent and identically distributed**, abbreviated i.i.d., and the book writes $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bern}(p)$. Nearly every model in this book starts from an i.i.d. sample.

Intuition: complicated process = simple process + random noise

That equation is the one-line philosophy of statistical modeling. Some processes are random in fact (a biased coin, measurement error, which customer walks in next). Others are deterministic but far too complex to describe (which way a particular couple turns). Statistical modeling treats the second kind like the first: pick a plausible simple process (every couple turns right with the same probability p) and a noise distribution (Bernoulli), and let the noise absorb everything you chose not to model. Good modeling consists in choosing a simple process and a noise distribution that are plausible for the data at hand, and the trade is always the same: more assumptions buy fewer parameters to estimate, at the price of being wrong if the assumptions are. Chapter 2 makes this precise with the definition of a statistical model.

1.4 Population versus sample

In the model, what is $\hat{p} = \bar{R}_n$ a sample *from*? Here is a thought experiment that answers it. Assume there is a total population of 5,000 couples who kiss in airports, and assume for the sake of argument that $p = 35\%$, or that $p = 65\%$. What do samples of size 124 look like in each case? Figure 1.2 answers by drawing the exact distribution of $\bar{R}_n$ under the i.i.d. model for each assumed p, together with the distribution for samples of size 3. (For a population of 5,000 the draws are without replacement, which changes the standard deviation of the picture by about one percent; the bridge box below says why the book works with the i.i.d. model instead.) With $n = 3$, on the left of the figure, the two distributions overlap heavily: two or more right turns out of 3 has probability 0.28 under $p = 0.35$ and 0.72 under $p = 0.65$, so three couples settle nothing. With $n = 124$, on the right, the distributions have separated, and the observed 0.645 is typical under $p = 0.65$ and essentially impossible under $p = 0.35$: a proportion that large has probability about 2×10^{-11}.

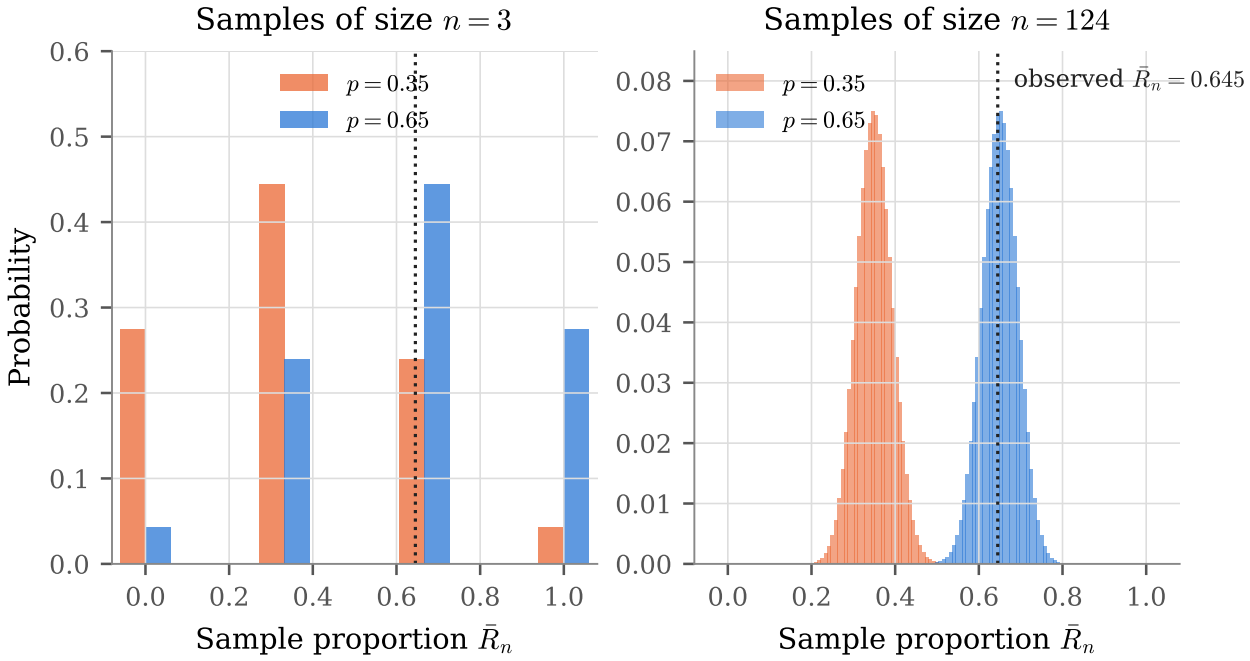


Figure 1.2: The exact Binomial distribution of the sample proportion $\bar{R}_n$ when $p = 0.35$ (orange) and when $p = 0.65$ (blue), for samples of size $n = 3$ (left) and $n = 124$ (right). The dotted line is the observed 0.645.

The figure is the whole subject in miniature. Under each candidate truth, probability tells you what the data would look like. Statistics runs that backwards: the observed data are typical under some candidate truths and absurd under others, and the first group is your answer. The rest of the book is about doing this with guarantees, and about doing it when the candidate truths form a continuum rather than two values.

From undergraduate statistics: population, sample, parameter, statistic

An introductory course describes a population (all kissing couples), a parameter of it (p, the proportion who turn right), a sample (124 couples) and a statistic computed from the sample ($\hat{p} = 0.645$) that estimates the parameter. This book keeps all four words and changes what sits

behind the first one. The population is no longer a finite list of 5,000 couples; it is a probability distribution, Bern(p), and a sample is n independent draws from it. The change is what makes the mathematics go: a finite population would make the draws slightly dependent (drawing without replacement) and would make the answer depend on the population size, while the distribution model gives clean results that hold whatever the size of the population and whether or not you can even list it.

Why does the model need probability? Because you need to understand probabilistic aspects of the distribution of the random variable

$$\hat{p} = \bar{R}_n = \frac{1}{n}\sum_{i=1}^{n} R_i.$$

Specifically, you need to be able to answer questions such as:

- Is the expected value of $\hat{p}$ close to the unknown p?
- Does $\hat{p}$ take values close to p with high probability?
- Is the variance of $\hat{p}$ large? That is, does $\hat{p}$ fluctuate a lot?

Most of the probability tools needed are about averages of independent random variables, which is why Part II is about exactly that, once Chapter 2 has said what a model is. For the kiss experiment the three answers are: $\mathbb{E}[\hat{p}] = p$ exactly; $\hat{p}$ is close to p with high probability once n is large (the Law of Large Numbers, Chapter 3); and $\mathbb{V}\mathrm{ar}(\hat{p}) = p(1-p)/n$, which at $n = 124$ (and evaluated at $p = 0.645$, since p itself is unknown) makes the typical fluctuation of $\hat{p}$ about 0.043 (the Central Limit Theorem, Chapter 4, says the fluctuations are Gaussian).

Pitfall: is 64.5% an estimator?

No: 64.5% is an estimate, and an estimate is not an estimator. An **estimator** is a rule for turning data into a guess, $\hat{p} = \bar{R}_n$, and because the data are random the estimator is a random variable; it has an expectation, a variance and a distribution, and those are what Figure 1.2 draws. An **estimate** is the number the rule produced on the one sample you have, $\bar{r}_n = 0.645$. It is one draw from the estimator's distribution: before the study, $\bar{R}_n$ could have come out anywhere from 0 to 1 with probabilities set by p, and the study produced one of those values. Once observed it is a fixed number; the distribution belongs to the rule that produced it, not to the number. Every statement about accuracy in this book is a statement about the estimator; the estimate is what you report. Chapter 3 returns to this distinction under the name 'theoretical sample mean', and Section 2.3 walks the whole chain from the sample space to the number. Outside that discussion and this box, the book follows the usual practice and writes the upper-case symbol with its realized value, $\bar{R}_n = 0.645$, trusting the context to say which of the two is meant.

1.5 The trinity of statistical inference

Everything statistical inference does falls under three headings, together called the trinity of statistical inference. Here is each one as a sentence about the kiss experiment, with the chapter that makes the sentence precise.

1. **Estimation.** '$\hat{p} = \bar{R}_n$ is an estimator for the proportion p of couples who turn their heads to the right.' What makes it a good one? It is unbiased, its variance shrinks like $1/n$, and it is consistent. Chapter 7 defines these properties and compares estimators by them; Chapter 11 shows that $\bar{R}_n$ is also the maximum likelihood estimator.

2. **Confidence intervals.** '$[0.56, 0.73]$ is a 95% confidence interval for p.' A range of truths under which the observed data are not surprising, with a guarantee about how often the procedure captures the truth. Chapter 8 builds this interval three ways and explains what the 95% does and does not mean.

3. **Hypothesis testing.** 'We find statistical evidence that more couples turn their heads to the right when kissing.' A yes/no decision about p with a controlled chance of being wrong. Chapter 9 sets up the decision and Chapter 15 supplies the general tests. Here is the answer to the question of Section 1.2: if the truth were $p = 1/2$, that is, if couples had no preference for either side, then the number turning right among 124 couples would reach 72 or more with probability 0.044 and 75 or more with probability 0.012. So 63 right turns would be unremarkable (probability 0.46 of at least that many), 72 would be mild evidence, and 80, which has probability 0.0008 of being reached under $p = 1/2$, is strong evidence. So 72 is the threshold at the conventional 5% level for this one-sided question; the two-sided version of Chapter 9, which also guards against $p < 1/2$, asks for 74. (Both numbers come from the exact Binomial distribution of the count; Chapter 9 also derives the Gaussian-approximation versions, 72 and 73, and says why they can differ.)

All three are statements about the same random variable $\bar{R}_n$, built from the same three modeling assumptions, and Figure 1.3 draws them on one axis. Change an assumption, for instance drop independence, and all three change. In the figure, every statement is about where p sits on the line. The estimate is a point, $\hat{p} = 0.645$. The 95% confidence interval is a bracket around it, $[0.56, 0.73]$: the truths under which the data are not surprising. The test compares the data to one particular truth, $p = \frac{1}{2}$: if the observed proportion lands at or beyond the threshold $72/124 = 0.581$, in the shaded region, the data count as evidence against $p = \frac{1}{2}$ and for $p > \frac{1}{2}$, and the threshold is set so that this happens only 5% of the time when $p = \frac{1}{2}$ really is the truth. Chapter 8 and Chapter 9 build the bracket and the threshold; the point is where both start.

From undergraduate statistics: the conditions for a one-proportion interval

An introductory course computes the one-proportion z-interval $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n} = 0.645 \pm 0.084$, the $[0.56, 0.73]$ above, after checking three conditions: a random sample, a sample of at most 10% of the population, and at least 10 successes and 10 failures (some texts ask for 15). The first two are assumptions (2) and (3) of Section 1.3 in checklist form. Drawing without replacement from N couples makes the R_i slightly dependent and puts a second factor in the standard deviation of $\hat{p}$,

$$\sqrt{\mathbb{V}\mathrm{ar}(\hat{p})} = \sqrt{\frac{p(1-p)}{n}}\sqrt{\frac{N-n}{N-1}},$$

and the 10% condition keeps that factor above $\sqrt{0.9} \approx 0.95$; for the 5,000 couples of Section 1.4 it is 0.988. The third condition is not about how the data were produced but about whether the Gaussian approximation behind the interval is accurate at this n: the kiss data pass with 80 right turns and 44 left, and the interval's actual coverage at $p = 0.645$ is 0.939 (Section 8.5).

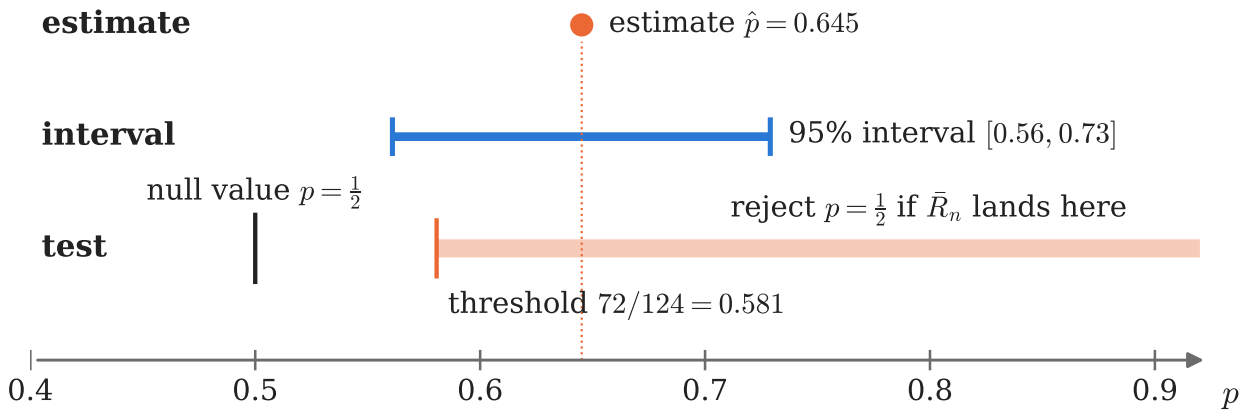


Figure 1.3: The trinity on the p axis for the kiss data: the estimate $\hat{p} = 0.645$ (orange dot), the 95% confidence interval $[0.56, 0.73]$ (blue bracket), and the test of $p = \frac{1}{2}$ with its threshold 0.581 and rejection region (orange shading).

1.6 What this book is and is not

This book has two aims: to introduce the mathematical theory behind statistical methods, and to supply theoretical guarantees for the methods you already use. By the end you should be able to take a real-life situation like the kiss experiment, state it as a statistical problem in mathematical terms, choose methods appropriate to it, and say what those methods do and do not license you to conclude. Four things recur throughout: the derivation of a standard method rather than its recipe, the justification of a quantitative statement from a stated set of modeling assumptions, the mathematics that arises in statistics and is worth seeing in its own right, and a toolbox of arguments that transfers to models the book never mentions.

Three things are outside its scope. Applied modeling, meaning which data to collect and how to clean what you collected, is a separate subject, and the kiss experiment illustrates the point: Section 1.3 takes the 124 observations as given and says nothing about where and when to watch, or how to decide which way a pair of heads turned. Implementation is not treated at all, and no formula here is tied to a particular piece of software. And the aim is not a list of named methods: the general machinery of Chapter 15 delivers whole families of tests at once, including most of the named ones, which is the reason for building it. Where the undergraduate version of an idea makes the general version easier to see, that version appears in a box labeled 'From undergraduate statistics'.

1.7 What comes next

The three questions of Section 1.4 are all about the random variable $\hat{p} = \bar{R}_n$: its expectation, where it concentrates, how much it fluctuates. The first has a one-line answer, $\mathbb{E}[\hat{p}] = p$. Before any of them is answered, Chapter 2 turns the three assumptions of Section 1.3 into an object, the statistical model, so that everything later can be stated for any model and not only this one. The other two questions are about what happens as n grows, and 'as n grows' is a limit, so Part II is about limits of random variables: Chapter 3 says where $\bar{R}_n$ ends up, Chapter 4 says what its fluctuations look like

on the way, and Chapter 5 says what you may do to those limits (transform them, plug estimates into them) and still know where you are. Chapter 6 then collects what you need about the Gaussian distribution, which is where all of the limits land. If you are comfortable with limits of random variables already, Chapter 3 opens with a box that says which parts of Part II are the minimal path.

1.8 Summary

- Probability goes from a known mechanism to the data it produces; statistics goes from observed data back to the mechanisms that could have produced them. Same diagram, opposite arrows.
- The kiss experiment: $n = 124$ couples, 80 turned right, $\hat{p} = \bar{R}_n = 0.645$. It is the running example of the book.
- Three modeling assumptions turn the data into mathematics: each R_i is random, each is $\text{Bern}(p)$ with the same p, and they are independent. Together: $R_1, \ldots, R_n \overset{iid}{\sim} \text{Bern}(p)$. Each assumption is a choice with a justification.
- The population is a probability distribution, not a list; a sample is n independent draws. The estimator $\hat{p}$ is a random variable; the estimate 0.645 is one realization of it.
- The trinity of inference: estimation (is $\hat{p}$ a good guess?), confidence intervals (which truths are compatible with the data?), hypothesis testing (is $p > 1/2$?). Every inferential method in this book is one of the three.

1.9 Exercises

Exercise 1.1. Two dice are rolled and you win \$100 if you called the sum. Which total should you call, and what is your expected winning? Why is this a probability question rather than a statistics question? *(Solution on page 414.)*

Exercise 1.2. In the drug example, verify the probability statement: if the drug cures each patient independently with probability 0.8, show that the number cured out of 100 has mean 80 and standard deviation 4, and explain why 'at least 65 cured' has probability very close to 1. *(Solution on page 414.)*

Exercise 1.3. Suppose the kiss data had been $n = 3$ couples, two turning right. Under the hypothetical truths $p = 0.35$ and $p = 0.65$, compute the probability of observing at least two right turns. What does the comparison tell you? *(Solution on page 414.)*

Exercise 1.4. Which of the three modeling assumptions would be violated in each case, and in which direction would it push $\hat{p}$? (a) The observer recorded each couple twice by mistake. (b) Half the couples were observed at a single wedding reception where the photographer told everyone to turn right. (c) The observer only recorded couples whose faces were visible from where the observer stood. *(Solution on page 414.)*

Chapter 2

Statistical Models and Identifiability

In this chapter

- What a statistical model is: an observation space and a family of candidate distributions indexed by a parameter. The whole of the rest of the book happens inside this object.
- The chain from the abstract sample space to the numbers in the data file, walked once for the kiss experiment: outcomes, random variables as functions on them, and observations as those functions evaluated at the outcome that happened.
- The trade that modeling always makes: assumptions buy fewer parameters to learn, at the price of being wrong if the assumptions are. The siblings example, drawn both ways.
- Well-specified models, and parametric versus nonparametric ones.
- Identifiability: when the distribution of the data pins down the parameter, and when it doesn't, with the standard example drawn in the parameter space so you can see what fails.
- Gaussian mixture models, as a generative story and as an example of a model whose parameter is identifiable only up to relabeling.

Chapter 1 made three assumptions about the kiss experiment and called the result a model. This chapter says what a model is in general, so that everything from Part III on can be stated for any model and not only the Bernoulli one.

2.1 The rationale behind statistical modeling

Chapter 1 was about one estimator, $\hat{p} = \bar{R}_n$, under one set of assumptions. To say what estimators, intervals and tests are in general, you first have to fix the object they all live in: a model, which is the three assumptions of Section 1.3 written as a mathematical object. The rationale behind statistical modeling can be outlined as follows:

- Let $X_1, \ldots, X_n$ be n independent copies of X. The goal of statistics is to learn the distribution of X.

  NB: It's crucial to understand this notation, this way of specifying problems, and, most importantly, the relationship between the n i.i.d. r.v.s $X_1, \ldots, X_n$ and the r.v. X. Each of the n i.i.d. r.v.s $X_1, \ldots, X_n$ has the same distribution as the r.v. X. The relationship between X and $X_1, \ldots, X_n$ can be understood as follows:

1. X is the underlying r.v. of interest, with an unknown distribution that you want to learn.
2. $X_1, \ldots, X_n$ are n r.v.s that form your sample of observations drawn from the same distribution as X. This is all-important: When you see something like 'Suppose $X_1, \ldots, X_n$ are n independent copies of X' you should *automatically interpret these n variables as your sample data.*

By observing the values of $X_1, \ldots, X_n$, you can make inferences about the distribution of X.

- If $X \in \{0, 1\}$, then it's Bernoulli and you only have to learn the parameter p. That was the kiss experiment.
- It can be more complicated. Take a (partial) dataset of the number of siblings, including self, collected from college students:

$$2, 3, 2, 4, 1, 3, 1, 1, 1, 1, 1, 2, 2, 3, 2, 2, 2, 3, 2, 1, 3, 1, 2, 3, \ldots$$

  If $X \in \{1, 2, 3, 4, 5, 6, \geq 7\}$, each with a different probability $\mathbb{P}(X = x)$, then you could make no assumption and try to learn the whole PMF: that's 7 parameters to learn (six, really, since they sum to one).
- Or you could assume that $X - 1 \sim \text{Pois}(\lambda)$. That's only 1 parameter, λ, to learn. Figure 2.1 shows both on the siblings data. The empirical PMF is seven numbers, six of them free since they sum to one; the model is one number, $\hat{\lambda}$, set to the sample mean minus one. The model fits well here; where such a model didn't fit, the fault would be the assumption's, not the data's.
- **NB:** It's important to keep in mind that parameters define distributions. What is meant is that, suppose you're considering the family of Gaussian distributions. Once you specify μ and σ^2, you've fixed the distribution. Learning the parameter and learning the distribution are the same task. (The Gaussian family appears in several examples of this chapter, written $\text{Gauss}(\mu, \sigma^2)$ with the variance second, and with Φ for the CDF of $\text{Gauss}(0, 1)$; Chapter 6 is its full treatment, and the notation page has what is needed here.)

Intuition: assumptions are currency

Every assumption you add to a model buys you something and costs you something. '$X - 1$ is Poisson' reduces seven unknowns to one, which means 24 observations go a long way instead of being spread over seven cells, and it lets you say things about families of 8 siblings that you never observed. What it costs is the possibility of being wrong: if the real PMF isn't Poisson-shaped, no amount of data will make the model right, and the estimate of λ will be a precise estimate of the wrong thing. The phrase for this is that a model should be plausible, and the mathematics that follows assumes it is. Checking the assumption is the job of the goodness-of-fit tests in Chapter 16.

2.2 Statistical models

Definition 2.1 (Statistical model)**.** Let the observed outcome of a statistical experiment be a sample $X_1, \ldots, X_n$ of n i.i.d. r.v.s in some observation space E (usually $E \subseteq \mathbb{R}$; the requirement

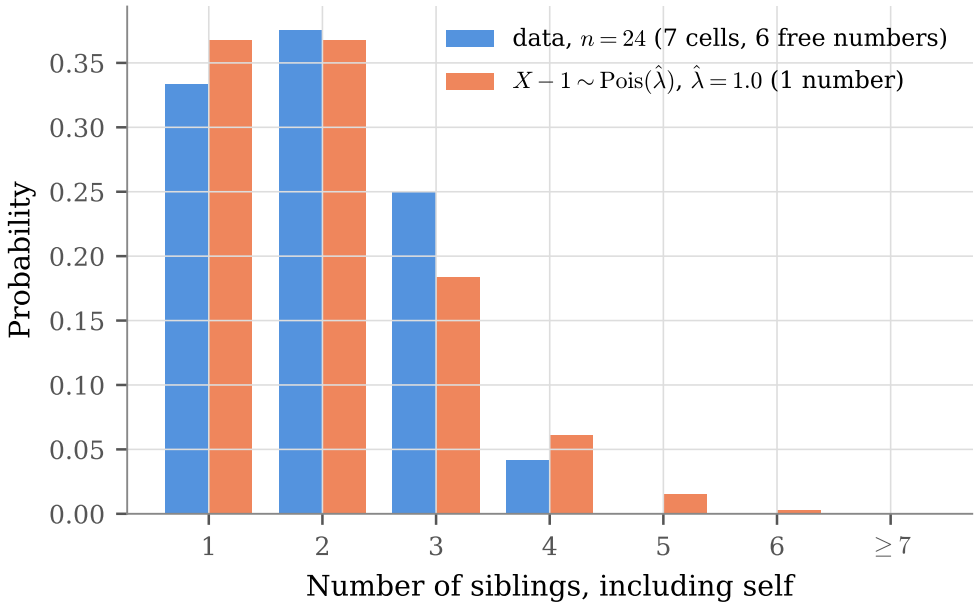


Figure 2.1: The siblings data ($n = 24$ values) as an empirical PMF (blue), next to the fitted one-parameter model $X - 1 \sim \text{Pois}(\hat{\lambda})$ (orange), with $\hat{\lambda}$ the sample mean minus one.

that E be measurable is a technicality, see (toolkit A.5)) and denote by $\mathbb{P}$ their common distribution. A **statistical model** associated to that statistical experiment is a pair:

$$(E, (\mathbb{P}_\theta)_{\theta \in \Theta}),$$

where:

- E is the **observation space** for X; that is, the set containing all possible values of X. It is therefore also where your observations, $X_1, \ldots, X_n$, live. (The word *outcome* is reserved for a point of the sample space Ω of Section 2.3, which records a whole study rather than one observation.)
- $(\mathbb{P}_\theta)_{\theta \in \Theta}$ is a family of probability distributions on E, the candidate distributions for X.
- Θ is the **parameter space**; that is, a set consisting of the possible values of θ.

> **In words.** A model is a list of candidate explanations for the data: where an observation can land (E), and every distribution you are willing to consider as the one that produced it, with a label θ on each.
>
> **For the kiss data.** $E = \{0, 1\}$, the candidates are the Bern(p) distributions, one for each $p \in (0, 1)$, and the label is p itself. 'Learn the model' means 'find out which label is right'.

Note: A **probability measure** or **probability distribution** takes an event $A \subseteq E$ as input and returns $\mathbb{P}_\theta(A)$ (toolkit A.5). It is assumed to satisfy the usual axioms of probability. Most importantly, a probability measure is not a PMF or PDF. A probability measure assigns probabilities to events in the observation space, while a PMF or PDF assigns probabilities (or densities) to specific values of a discrete or continuous r.v. The two determine each other, so in practice you write down the family by writing down its PMFs or PDFs, as in the examples below, but the object in the definition is the distribution.

Figure 2.2 is the picture to keep in mind: a model is a family of distributions, and the parameter is the label that picks one out. In the top row, two one-parameter families; sliding the parameter moves you through the family. In the bottom row, the Gaussian family has a two-dimensional parameter space, drawn on the left, and each point of it is one distribution, drawn on the right. The arrow $\theta \mapsto \mathbb{P}_\theta$ from the left of the figure to the right is what the definition is about, and Section 2.6 asks whether it ever sends two points to the same distribution.

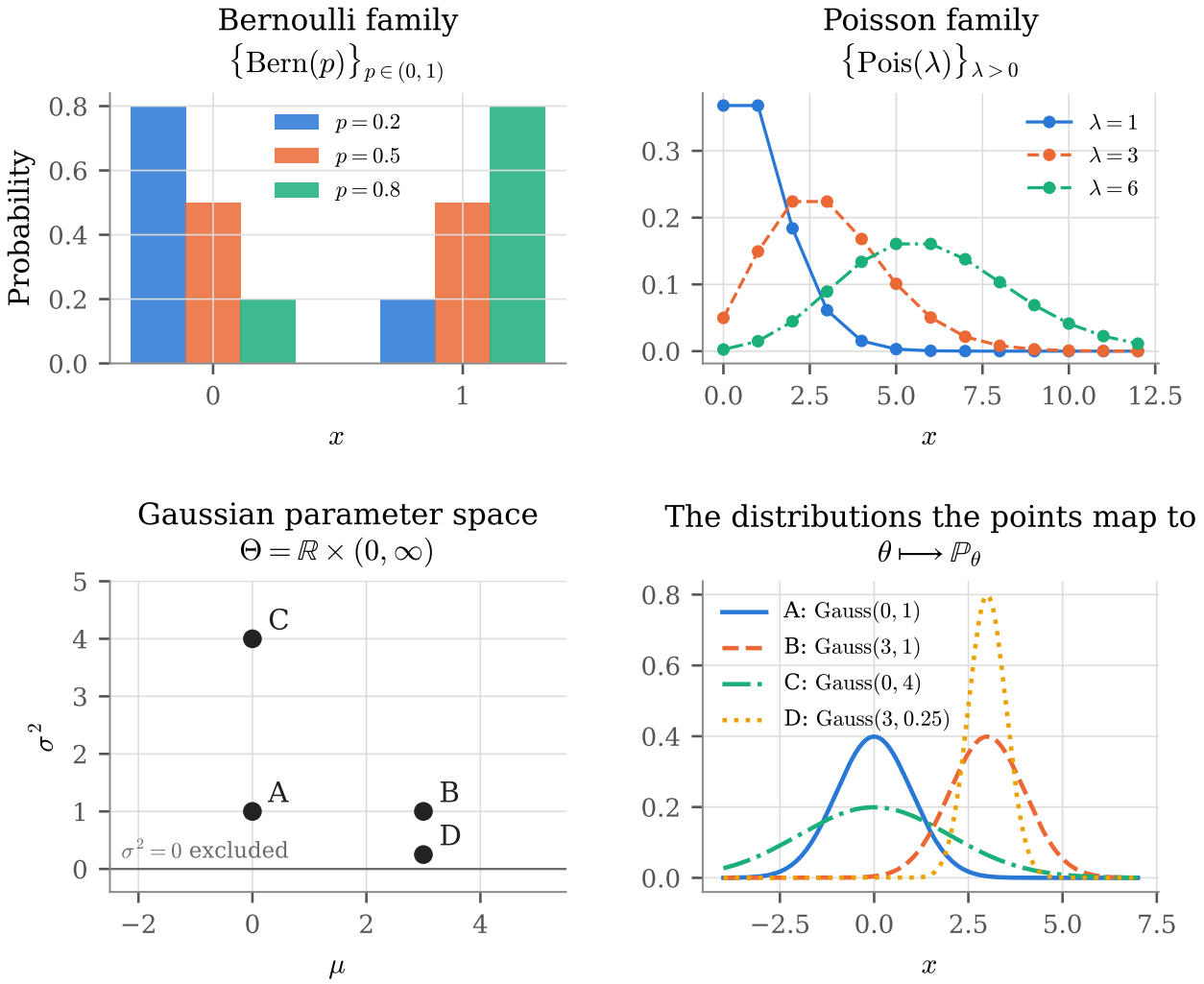


Figure 2.2: Top: the Bernoulli family at three values of p, and the Poisson family at three values of λ. Bottom: four points A to D in the Gaussian parameter space (μ, σ^2), left, and the densities they map to, right.

2.2.1 Examples of parametric models

For parametric models (defined in Section 2.5: the parameter is a list of finitely many numbers), the parameters live in some space that's a subset of $\mathbb{R}^d$ and the parameters can be represented by a vector. Here are some canonical examples of parametric models:

1. For n Bernoulli trials: $\big(\{0,1\}, (\mathrm{Bern}(p))_{p\in(0,1)}\big)$. The vectors of parameters thus consist of

singletons (p).

2. If $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$ for some unknown $\lambda > 0$: $(\mathbb{N}, (\text{Pois}(\lambda))_{\lambda>0})$. The vectors of parameters thus consist of singletons (λ).

3. If $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$ for some unknown $\mu \in \mathbb{R}$ and $\sigma^2 > 0$:

$$\left(\mathbb{R}, (\text{Gauss}(\mu, \sigma^2))_{(\mu,\sigma^2)\in\mathbb{R}\times(0,\infty)}\right).$$

   The vector of parameters has length 2 and is given by (μ, σ^2).

4. If $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}_d(\mu, \mathbb{I}_d)$ for some unknown $\mu \in \mathbb{R}^d$: $\left(\mathbb{R}^d, (\text{Gauss}_d(\mu, \mathbb{I}_d))_{\mu\in\mathbb{R}^d}\right)$. The vector of parameters is μ itself, of length d.

5. If $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([0, a])$ for some unknown $a > 0$: $(\mathbb{R}_+, (\text{Unif}([0, a]))_{a>0})$. Not $([0, a], \ldots)$: see the pitfall below.

Sometimes there's no simple notation like $\text{Bern}(p)$ for the family and you have to be more explicit. The linear regression model: $(X_1, Y_1), \ldots, (X_n, Y_n) \in \mathbb{R}^d \times \mathbb{R}$ are i.i.d. with $Y_i = \beta^\top X_i + \epsilon_i$, $\epsilon_i \overset{iid}{\sim} \text{Gauss}(0, 1)$ independent of $X_i \sim \text{Gauss}_d(0, \mathbb{I}_d)$, for an unknown $\beta \in \mathbb{R}^d$. Here $E = \mathbb{R}^d \times \mathbb{R}$, $\Theta = \mathbb{R}^d$, and $\mathbb{P}_\beta$ is the joint distribution of the pair (X, Y) that this recipe produces. Chapter 21 is about this model in a slightly different form: the X_i treated as fixed numbers rather than Gaussian draws, and the error variance σ^2 unknown, which makes $\Theta = \mathbb{R}^d \times (0, \infty)$ and makes the Y_i independent but no longer identically distributed, since each has its own mean $\beta^\top x_i$. Treating the predictors as fixed is what lets the regression chapter condition on them and still have a model in the sense of Definition 2.1.

Pitfall: the observation space must not depend on the parameter

Which of these is a statistical model for n Bernoulli trials: $\left(\{1\}, (\text{Bern}(p))_{p\in(0,1)}\right)$ or $\left(\{0, 1\}, (\text{Bern}(p))_{p\in(0.2,0.4)}\right)$? Only the second. The first has an observation space that leaves out a value the distributions put mass on; a Bernoulli r.v. takes the value 0, so $\{1\}$ isn't where the observations live. The second is a model, just one that assumes p is between 0.2 and 0.4 (which may or may not be well-specified, the next section's question). Similarly, for $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([0, a])$ with a unknown, the model is $(\mathbb{R}_+, (\text{Unif}([0, a]))_{a>0})$ and not $([0, a], \ldots)$: the observation space is fixed before you know a, so it has to be a set that contains every observation for every candidate a. Any such set will do ($\mathbb{R}$ works too); the convention in this book is the smallest natural one. An observation space that mentions the unknown parameter is not an observation space.

None of this is new to a reader who has taken an introductory course; only the bookkeeping is.

From undergraduate statistics: the model was always there

In an introductory course the model is a sentence: 'assume the data are normally distributed', 'assume each trial succeeds with the same probability'. This book makes the sentence into an object with three parts, and the point of doing so is that every later statement can name exactly what it depends on. 'The sample mean is unbiased' is a statement about $\mathbb{E}_\theta[\bar{X}_n]$ for every $\theta \in \Theta$; 'the t-test has level 5%' is a statement that holds for every member of the Gaussian family; 'the estimator is consistent' means $\hat{\theta}_n \to \theta$ under $\mathbb{P}_\theta$ for every θ. When a method fails in practice, it

is often because the data came from outside the family, and you can only say that if the family is written down.

2.3 From the sample space to the data and back

The definition says the observations live in an observation space E, while a random variable is, by definition, a function on a sample space Ω (toolkit A.5); Chapter 3 returns to that definition when it needs sequences of random variables. These are two different sets, and many books call both of them the sample space, which is why this book keeps the two names apart: E is where one observation lands, and Ω is where one whole study lands. The gap between them is where the single most common confusion in this subject lives: what is the relationship between the abstract space on which probability is defined and the concrete numbers in the data file? This section walks the chain once, slowly, for the kiss experiment, with Figure 2.3 as the map. On the left of the figure is the sample space Ω, one point per complete possible outcome of the study, with the one that happened marked. In the middle, the data are a function on Ω, one row of 0s and 1s per outcome, and the marked row is the data set you have. On the right, the statistic $\bar{R}_n$ is another function on Ω; over all outcomes weighted by their probabilities it has a distribution, the bars, and the marked outcome gives the one value you observe, $\bar{r}_n = 8/12$. Nothing travels from right to left: the dashed arrow is inference, which holds the one value up against the distribution of the rule.

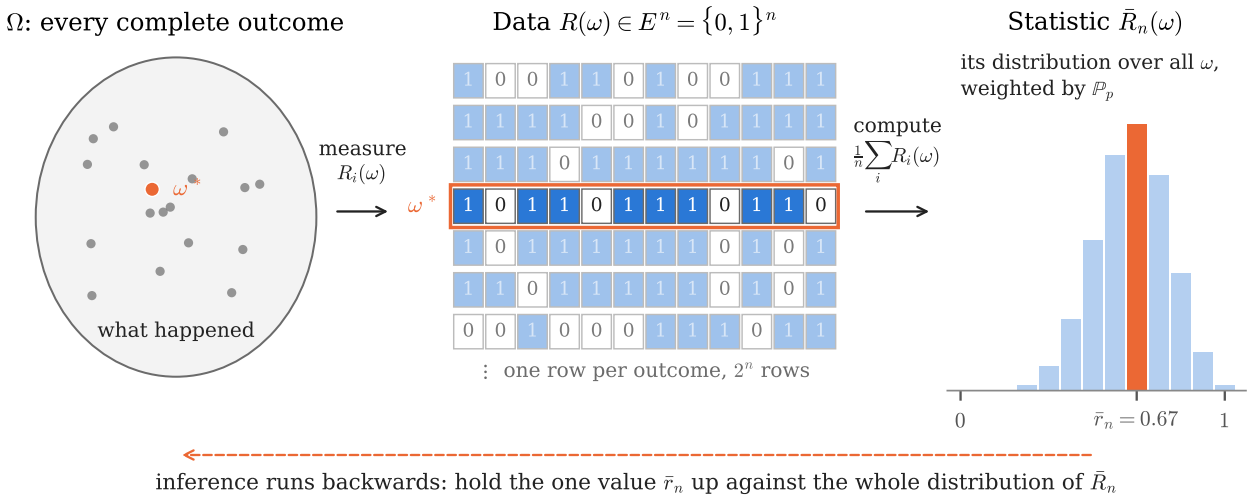


Figure 2.3: The sample space Ω (left), the data as one row of 0s and 1s per outcome (middle), and the statistic $\bar{R}_n$ with its distribution (right), for a kiss experiment with $n = 12$. The outcome that happened is marked in orange.

Layer one: the sample space. An outcome ω of the study is everything that could happen, in full detail: which 124 couples were observed, where and in which order, and which way each one turned. The set of all such outcomes is the sample space, written Ω. It is as abstract as the subject gets, and you never see an element of it. What you see is what your measuring instruments report about the one ω that occurred. For the $n = 12$ experiment of Figure 2.3, one ω is a complete record of all twelve couples at once, say

$$\omega = (R, L, R, R, L, R, R, R, L, R, R, L),$$

one entry per couple, and it is one row of the middle panel of the figure. So Ω holds not twelve things but $2^{12} = 4{,}096$ of them, one for every way the twelve couples could have come out. What n counts is the random variables defined on Ω, which is the width of a row; 2^n counts the points of Ω, which is the height of the grid. For the experiment as it was run, $n = 124$ and Ω has 2^{124} points, a number with 38 digits.

Layer two: random variables are the measuring instruments. A random variable is a function on Ω (toolkit A.5). The instrument for couple i is R_i, a function $\Omega \to \{0, 1\}$ defined by

$$R_i(\omega) = \begin{cases} 1 & \text{if the } i\text{-th entry of } \omega \text{ is } R, \\ 0 & \text{otherwise.} \end{cases}$$

Reading the tuple above, $R_1(\omega) = 1$, $R_2(\omega) = 0$ and $R_{12}(\omega) = 0$. The data vector $(R_1, \ldots, R_n)$ is a function on Ω too, with values in $E^n = \{0, 1\}^n$; one row of Figure 2.3 per outcome. And $\bar{R}_n = \frac{1}{n}\sum_i R_i$ is yet another function on Ω. That is all 'the estimator is a random variable' means: a rule that could be applied to any outcome, written down before you know which outcome you will get.

Pitfall: counting the sample space

Three counts in Figure 2.3 are easy to run together, and they answer three different questions. There are 4,096 outcomes, the points of Ω. There are 12 random variables defined on them, $R_1, \ldots, R_{12}$, one per couple. And there are 13 values that $\bar{R}_n$ can take, namely $0/12, 1/12, \ldots, 12/12$. The right-hand panel of the figure therefore has thirteen bars and not 4,096: many outcomes give the same average, and the height of a bar is the total probability of all the rows that produce that average. The mistake to avoid is reading the sample space as having n elements, one per observation. It has 2^n, one per data set you could have gotten, and n is how many numbers each of those data sets contains.

Layer three: the observations are those functions evaluated at the outcome that happened. Nature picked one ω^*. Your data are $r_1 = R_1(\omega^*), \ldots, r_n = R_n(\omega^*)$, and your estimate is $\bar{r}_n = \bar{R}_n(\omega^*) = 0.645$. Going from the sample space to the observations is 'plug in the ω that occurred'. A realization is one evaluation of a function; a distribution is how often each value would come up across all of Ω, weighted by probability. The realized number 0.645 is one draw from the distribution of $\bar{R}_n$; after it is observed it is a fixed number, and the distribution belongs to the rule, not to the number (the pitfall in Section 1.2).

Going back is not an inverse. You never recover ω^* from the data, and you do not need to. What inference does instead is ask, for each candidate θ: across all the outcomes that could have occurred under $\mathbb{P}_\theta$, how often would $\bar{R}_n$ have landed near the value it took? The one number is held up against the whole distribution of the rule, and the candidates under which it would have been typical are the plausible ones. That is the dashed arrow in Figure 2.3, and it is what every confidence interval and test in this book does.

Intuition: the trick that makes Ω concrete

Since Ω only ever enters through the random variables defined on it, you may as well take Ω to be the set of data sets you could have gotten. For the kiss experiment that is $\Omega = \{0, 1\}^{124}$, a finite set with 2^{124} elements, and under the model $\mathbb{P}_p(\omega) = p^{\#\text{ones}}(1-p)^{\#\text{zeros}}$. Every row of Figure 2.3 is a point of this Ω, and $R_i(\omega)$ is just 'the i-th entry of the row'. This is why Definition 2.1 mentions E and never Ω: for a fixed n the set E^n of possible data sets *is* the

sample space, and a model is a family of probabilities on it. The abstract Ω of probability theory is needed only when one space has to carry infinitely many random variables at once, $R_1, R_2, R_3, \ldots$ without end, which is exactly what the limit theorems of Part II need in order to say 'as $n \to \infty$'. For any fixed n, think $\Omega = E^n$ and nothing is lost.

The same shift from an abstract Ω to a definition on E^n is what turns a simulated idea from a first course into a defined object.

From undergraduate statistics: the sampling distribution, defined rather than simulated

An introductory course introduces the sampling distribution of the sample mean by rerunning the study: draw many samples, compute $\bar{x}$ for each, draw the histogram. That histogram is the right of Figure 2.3, approximated by simulation; this chapter defines it instead, without rerunning anything, by putting a probability on Ω and asking how $\bar{R}_n(\omega)$ is distributed under it. The two pictures agree: each simulated sample is one ω, an introductory course's 'many samples' are many points of Ω, and the limit of the histogram as the number of samples grows is the distribution of the random variable. Figure 3.5 in Chapter 3 draws the same picture as rows and columns: a row is an ω, the row average is $\bar{R}_n(\omega)$, and the column of row averages is the sampling distribution. Your data set is the one row nature handed you.

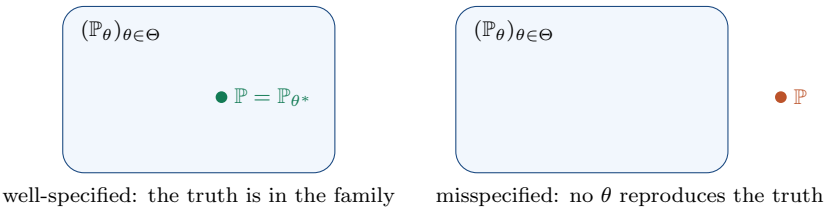
2.4 Well-specified models

It is almost always assumed that a statistical model is well-specified.

Definition 2.2 (Well-specified)**.** A model $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ is **well-specified** if there exists some $\theta \in \Theta$ such that $\mathbb{P} = \mathbb{P}_\theta$, where $\mathbb{P}$ is the true distribution that generates the observed data $X_1, \ldots, X_n$. In other words, $\mathbb{P} \in (\mathbb{P}_\theta)_{\theta\in\Theta}$. If $\mathbb{P} \notin (\mathbb{P}_\theta)_{\theta\in\Theta}$, then the model is **misspecified**. That particular θ such that $\mathbb{P} = \mathbb{P}_\theta$ is the **true parameter**, often written θ^*, and is unknown.

In words. The true distribution is on your list of candidates. If it isn't, no label is right, and 'the true parameter' has no meaning.

For the kiss data. If couples really do turn right with a single probability p^*, independently, the model is well-specified and $p^* \in (0, 1)$ is the true parameter. If, say, the probability differed by country, no single Bern(p) reproduces the data-generating process and the model is misspecified.



well-specified: the truth is in the family misspecified: no θ reproduces the truth

Here's an example to illustrate the meaning of well-specified models:

- Suppose you're modeling the outcome of a coin toss using a Bernoulli distribution with parameter p, where p represents the probability of getting heads. Suppose further that the *true* probability of getting heads is $p^* = 0.6$.
- Now, if your model considers all possible values of $p \in (0, 1)$, then it's a well-specified model because it includes the true probability distribution, the Bern(0.6) distribution, as one of the possible distributions in the family.
- On the other hand, if your model only considers values of p between, say, 0 and 0.5, then it's not well-specified because it doesn't include the true probability distribution, Bern(0.6), in the family of distributions.

The above example addresses misspecification when the true parameter value is not contained in your model's parameter space. But a model misspecification could also be due to applying the 'wrong' distribution: for example, by modeling the toss of a coin using a Gaussian distribution, or the siblings data of Figure 2.1 with a shifted Poisson when the true PMF has a second bump at 5.

One aim of a statistical experiment is to *estimate* the true parameter θ^*. Another is to check θ^*'s properties when they have a specific meaning determined by the context of the study. As an example of this second aim you may want to know if $\theta^* > 2$ or if $\theta^* \neq 1/2$, which is the question of Chapter 9.

2.5 Parametric, nonparametric and semiparametric models

It will often be assumed that the parameter space Θ is a subset of $\mathbb{R}^d$ for some $d \geq 1$. In this case, when $\Theta \subseteq \mathbb{R}^d$ for $d \geq 1$, the model is called **parametric**. Hence, Θ contains vectors of parameters with length d; a finite-dimensional space.

On the other hand, when Θ is infinite-dimensional, the model is called **nonparametric**. The elements of Θ are then functions (a density, a CDF) rather than vectors of numbers.

Thus, the distinction between parametric and nonparametric models is a distinction between the *length* of the vectors in the parameter space Θ. It is not a distinction of the *number* of vectors living in the space. For example, there is an uncountably infinite number of vectors living in the parameter space of a Bernoulli model where $\Theta = (0, 1)$, but all the vectors are singletons: any number between 0 and 1. What makes a model nonparametric is that no finite list of numbers can specify a member of the family; you'd need the value of a function at infinitely many points.

Nonparametric models typically include assumptions on the functions that describe the r.v. For a continuous r.v. there are two: the PDF and the CDF. Here are two examples of nonparametric models:

1. If $X_1, \ldots, X_n \in \mathbb{R}$ are i.i.d. r.v.s with unknown unimodal[1] PDF f:

$$\left(\mathbb{R}, (\mathbb{P}_f)_{\{f : f \text{ is a unimodal PDF}\}}\right).$$

2. If $X_1, \ldots, X_n \in [0, 1]$ are i.i.d. r.v.s with unknown invertible CDF F:

$$\left([0, 1], (\mathbb{P}_F)_{\{F : F \text{ is an invertible CDF}\}}\right).$$

[1] A unimodal PDF increases on $(-\infty, a)$ and decreases on (a, ∞) for some a.

Figure 2.4 draws the difference. On the left, one number, λ, moves you along a single curve through the space of densities; the whole parametric family is this one-dimensional path. On the right are four members of the first nonparametric family, all unimodal densities; no finite set of numbers indexes them, the family is the whole space of such shapes, and these are four points in it.

From undergraduate statistics: nonparametric and distribution-free tests

Some introductory courses teach nonparametric, or distribution-free, tests, such as the sign test and the Wilcoxon tests, for data that do not look normal. The word 'nonparametric' has this section's meaning: the test's guarantee holds for every member of a nonparametric model. The sign test of whether a median equals m_0 assumes only that $X_1, \ldots, X_n$ are i.i.d. from some continuous distribution, a family indexed by a whole CDF rather than a list of numbers. It counts the observations above m_0, and when m_0 is the median the count is $\mathrm{Bin}(n, \frac{1}{2})$ whatever the CDF is, so one threshold gives the same level for every continuous distribution with median m_0: that is what 'distribution-free' means. Keeping only the signs turns the problem into the kiss experiment's: with right turns as the plus signs, the one-sided 5% threshold of 72 out of 124 in Section 1.5 is the sign test's threshold. The Kolmogorov–Smirnov test of Chapter 17 is distribution-free in the same sense.

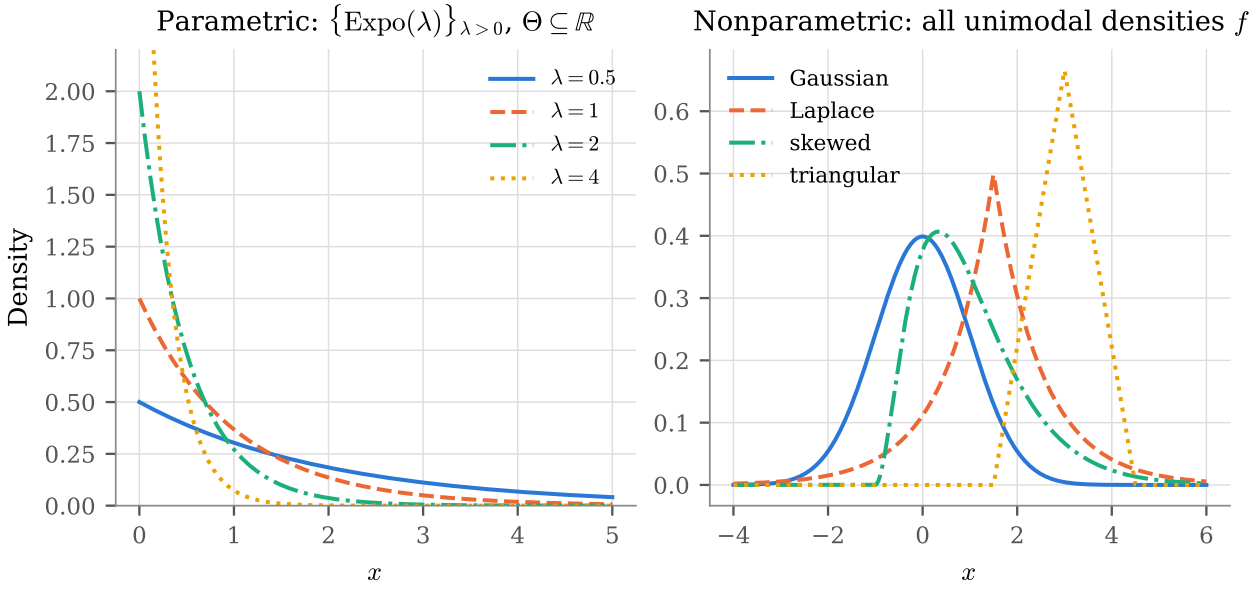


Figure 2.4: Left: four members of the parametric family $\{\mathrm{Expo}(\lambda)\}_{\lambda>0}$, at $\lambda = 0.5, 1, 2, 4$. Right: four members of the nonparametric family of all unimodal densities: Gaussian, Laplace, a skewed density and a triangular one.

A third kind: if $\Theta = \Theta_1 \times \Theta_2$ where Θ_1 is finite-dimensional and Θ_2 is infinite-dimensional, the model is **semiparametric**. In these models you only care to estimate the finite-dimensional parameter, and the infinite-dimensional one is called a **nuisance parameter**. The standard example is the Cox proportional hazards model from survival analysis, where the conditional CDF of a survival time Y given covariates $X = x$ has the form

$$F(t) = 1 - \exp\left(-\int_0^t h(u)e^{\beta^\top x}\,\mathrm{d}u\right),$$

with h an unknown non-negative nuisance function and $\beta \in \mathbb{R}^d$ the parameter of interest. This book

does not develop such models; the example is here so that you have seen what a semiparametric parameter space looks like, with $\Theta_1 = \mathbb{R}^d$ for β and Θ_2 the space of non-negative functions h.

From undergraduate statistics: the regression conditions as a model

An introductory course checks four conditions, sometimes remembered as LINE, before any inference about a regression slope: the mean of y is a linear function of x, and the errors are independent, normal and of equal variance at every x. Together they are one parametric model, with the x_i treated as fixed: L is the line in $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, and I, N and E together say $\epsilon_1, \ldots, \epsilon_n \overset{iid}{\sim} \text{Gauss}(0, \sigma^2)$, so the parameter is $(\beta_0, \beta_1, \sigma^2) \in \mathbb{R}^2 \times (0, \infty)$. Replace N by 'the errors are i.i.d. from some distribution with mean 0 and finite variance' and the model is semiparametric: (β_0, β_1) is the parameter of interest and the error distribution is an infinite-dimensional nuisance parameter. Least squares still estimates the line without bias in that model; what N adds is the exact t distribution of the slope's t statistic, as it does for the t-test of a mean (Chapter 14).

2.6 Identifiability

The first thing to ask of a model is whether it's well-specified. You can't verify it, since the true $\mathbb{P}$ is exactly what you don't know, but you can ask whether it's plausible, and the mathematics that follows assumes it. If it isn't, θ^* doesn't exist and 'estimating the true parameter' has no meaning; what the estimators of Part IV estimate in that case is the subject of Chapter 10. Presuming your model is well-specified, you also need to ascertain whether the parameter of the model is *identifiable*. Identifiability is a property of the parameterization: of the map from Θ to the family, not of the family itself, and not of any estimator or estimation procedure.

Definition 2.3 (Identifiability)**.** The parameter $\theta \in \Theta$ is **identifiable** if the map $\theta \mapsto \mathbb{P}_\theta$ is injective (one-to-one, (toolkit A.3)). That is:

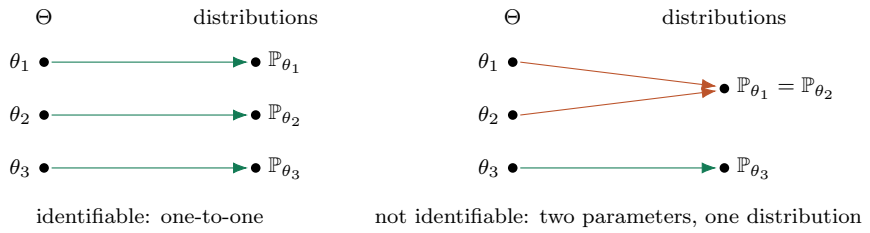
$$\theta_1 \neq \theta_2 \implies \mathbb{P}_{\theta_1} \neq \mathbb{P}_{\theta_2},$$

or equivalently, taking the contrapositive:

$$\mathbb{P}_{\theta_1} = \mathbb{P}_{\theta_2} \implies \theta_1 = \theta_2.$$

In words. Different labels, different distributions. Then knowing the distribution (which is what infinitely many data would tell you) pins down the label.

For the kiss data. $p_1 \neq p_2$ gives $\text{Bern}(p_1) \neq \text{Bern}(p_2)$, since the two put different probabilities on 1; p is identifiable. The example below shows a model where two different labels give the same distribution.



This definition means that distinct values of θ should correspond to distinct probability distributions. Intuitively, if the parameter θ is identifiable you can go from $\mathbb{P}_\theta$ to θ. But if θ is *not* identifiable you can have multiple θ's that give the same $\mathbb{P}$. This means that even if you had an infinite number of observations, so that you knew $\mathbb{P}$ exactly, you'd have no way of deciding which of those θ's produced it.

Alternatively, think of it this way: if you are handed $\mathbb{P}_\theta$, can you actually recover θ? If not, if θ is not identifiable, then there are multiple θ that yield the distribution you were handed. So if you're going to use the parameter to learn about a distribution, then you must make sure the parameter is identifiable. Data can only ever tell you about $\mathbb{P}$; identifiability is what lets you translate that into a statement about θ.

The formal way of convincing someone that the parameter is *not* identifiable is to exhibit two different parameter values that give the same distribution, as the next example shows.

2.6.1 Example: a non-identifiable parameter

Example 2.4. Let $X_i = \mathbb{1}(Y_i \geq 0)$ for i.i.d. r.v.s $Y_1, \ldots, Y_n \sim \text{Gauss}(\mu, \sigma^2)$, where μ and σ^2 are unknown and the Y_i are *not observed*; you only see whether each one is non-negative. Then μ and σ^2 are *not* identifiable from the common distribution of the X_i's.

To see why, first note that, by definition, the X_i are Bernoulli r.v.s since the indicator function $\mathbb{1}(Y_i \geq 0)$ simply tells you if Y_i is non-negative or not and assigns 1 or 0 accordingly. So you want to determine whether you can find μ and σ^2 just from observing the Bernoulli r.v.s X_i.

Now, as the X_i's are Bernoulli, what is the relevant parameter? It's the probability that $Y \geq 0$, so that $X_i \sim \text{Bern}(\mathbb{P}(Y \geq 0))$, where Y represents any one of the n i.i.d. r.v.s Y_i. Because Y is a non-standard Gaussian r.v., standardize it to get:

$$\mathbb{P}(Y \geq 0) = \mathbb{P}\left(\frac{Y-\mu}{\sigma} \geq -\frac{\mu}{\sigma}\right) = 1 - \Phi\left(-\frac{\mu}{\sigma}\right) = \Phi\left(\frac{\mu}{\sigma}\right).$$

So the parameter of this Bernoulli is $p = \Phi(\mu/\sigma)$. As Φ is invertible, you can solve for μ/σ to get:

$$\frac{\mu}{\sigma} = \Phi^{-1}(p).$$

So you can learn p (it's a Bernoulli proportion), apply Φ^{-1}, and this gives you μ/σ. Hence the ratio μ/σ is identifiable.

However, μ and σ^2 by themselves are *not* identifiable. Why? One way to see it is to count dimensions. The parameter of a Bernoulli r.v. is one number between 0 and 1. The parameter of the Gaussian is two numbers, μ and σ^2. A continuous map from a two-dimensional space onto a one-dimensional one can't be one-to-one: it has to send whole curves of (μ, σ^2) pairs to the same p.

But the formal way mentioned above requires a counterexample. So, let $(\mu_1, \sigma_1^2) = (1, 4)$ and $(\mu_2, \sigma_2^2) = (2, 16)$. Then:

$$\frac{\mu_1}{\sqrt{\sigma_1^2}} = \frac{1}{\sqrt{4}} = \frac{1}{2}, \quad \text{and} \quad \frac{\mu_2}{\sqrt{\sigma_2^2}} = \frac{2}{\sqrt{16}} = \frac{1}{2}.$$

Hence different parameter vectors, $(1, 4)$ and $(2, 16)$, both yield the same Bernoulli parameter $p = \Phi(1/2) = 0.691$, so $\mathbb{P}_{(1,4)} = \mathbb{P}_{(2,16)}$ while $(1, 4) \neq (2, 16)$. Figure 2.5 shows the two Gaussians,

which look nothing alike, putting the same mass above zero, and shows in the parameter space what the counterexample is a sample of: a whole ray of parameters producing one distribution. On the left, Gauss$(1, 4)$ and Gauss$(2, 16)$ are different distributions, but the only thing you observe is whether $Y \geq 0$, and both put probability $\Phi(1/2) = 0.691$ there. On the right, in the (μ, σ) plane, every point on the orange ray $\mu/\sigma = 1/2$ produces that same Bernoulli, and every other ray produces some other one. The observable distribution identifies the ray, that is, μ/σ, and nothing more; the counterexample above is two points on the orange ray.

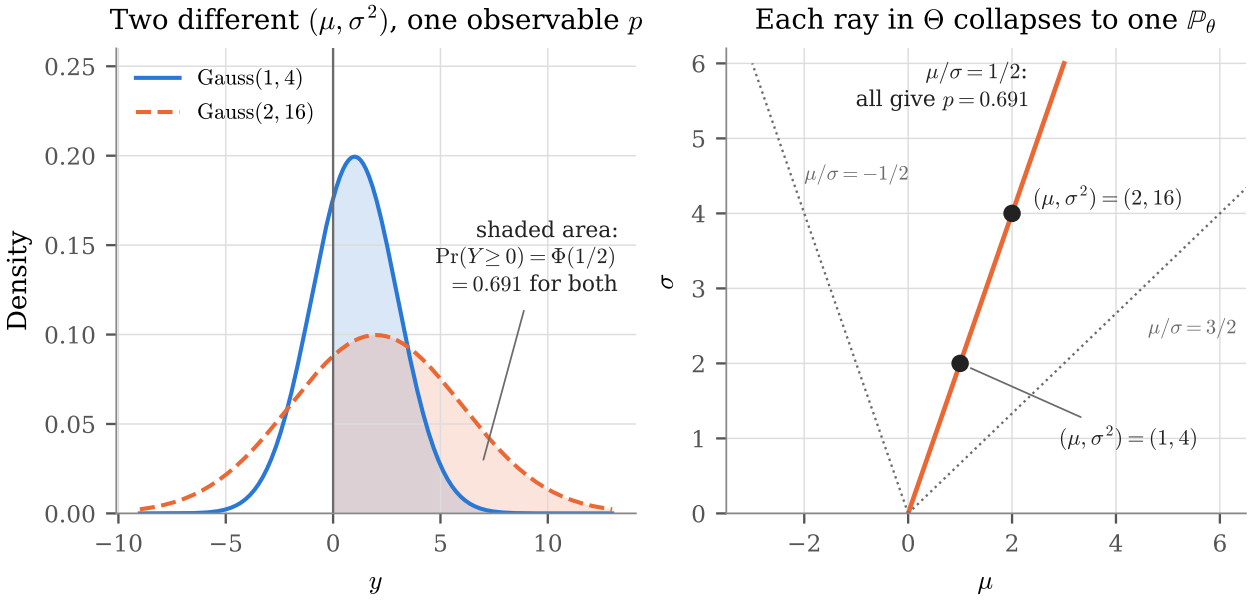


Figure 2.5: Example 2.4. Left: the densities of Gauss$(1, 4)$ and Gauss$(2, 16)$, each with its mass above zero shaded. Right: the (μ, σ) plane, with the ray $\mu/\sigma = 1/2$ in orange, the counterexample's two points marked, and two other rays dotted.

2.6.2 More examples

Three variations are worth doing in your head because the pattern is always the same: write down the distribution of what you actually observe as a function of θ, and ask whether that function is one-to-one.

Example 2.5 (Censored Exponential)**.** Let $X \sim \text{Expo}(\lambda)$ with $\lambda > 0$ unknown, and suppose you observe only $Y = \mathbb{1}(X > 5)$, a censored version of X: you can't see X, only whether it exceeded 5. Then $Y \sim \text{Bern}(f(\lambda))$ with $f(\lambda) = \mathbb{P}(X > 5) = \int_5^\infty \lambda e^{-\lambda x}\, dx = e^{-5\lambda}$. The model for Y is $\big(\{0, 1\}, (\text{Bern}(e^{-5\lambda}))_{\lambda > 0}\big)$, and λ *is* identifiable because $\lambda \mapsto e^{-5\lambda}$ is strictly decreasing, hence injective: from $p = \mathbb{P}(Y = 1)$ you recover $\lambda = -\ln(p)/5$. Censoring loses information (you'll estimate λ less precisely than from the X's), but it doesn't lose identifiability. Compare Example 2.4, where the trouble was not the censoring but the two-dimensional parameter behind a one-dimensional observable.

Example 2.6 (Squaring a uniform)**.** Let $X_i = Y_i^2$ where $Y_1, \ldots, Y_n \overset{iid}{\sim} \text{Unif}([0, a])$ are unobserved. Is a identifiable? Yes: X_i takes values in $[0, a^2]$, and $\mathbb{P}(X_i \leq x) = \mathbb{P}(Y_i \leq \sqrt{x}) = \sqrt{x}/a$ for $0 \leq x \leq a^2$. Different a give different CDFs (for instance, different supports), so the map $a \mapsto \mathbb{P}_a$ is injective.

Example 2.7 (Thresholding a uniform at half its range)**.** Let $X_i = \mathbb{1}(Y_i \geq a/2)$ where $Y_1, \ldots, Y_n \overset{iid}{\sim} \text{Unif}([0, a])$ are unobserved. Is a identifiable? No: $\mathbb{P}(Y_i \geq a/2) = 1/2$ for *every* $a > 0$, so $X_i \sim \text{Bern}(1/2)$ whatever a is. The map $a \mapsto \mathbb{P}_a$ is constant, the worst possible failure of injectivity: the data carry no information about a at all.

Figure 2.6 draws the three maps side by side; identifiability is the shape of the curve. In each example the observation is a Bernoulli whose parameter p is a function of θ. On the left, the censored Exponential has $p = e^{-5\lambda}$, strictly decreasing, so each p comes from exactly one λ. In the middle, the uniform thresholded at half its range has $p = \frac{1}{2}$ for every a, so the observation says nothing about a. On the right, the Gaussian sign example has $p = \Phi(\mu/\sigma)$, drawn against μ for three values of σ: one value of p is reached by a whole ray of (μ, σ) pairs, the ray of Figure 2.5. Identifiable means the curve never takes the same value twice.

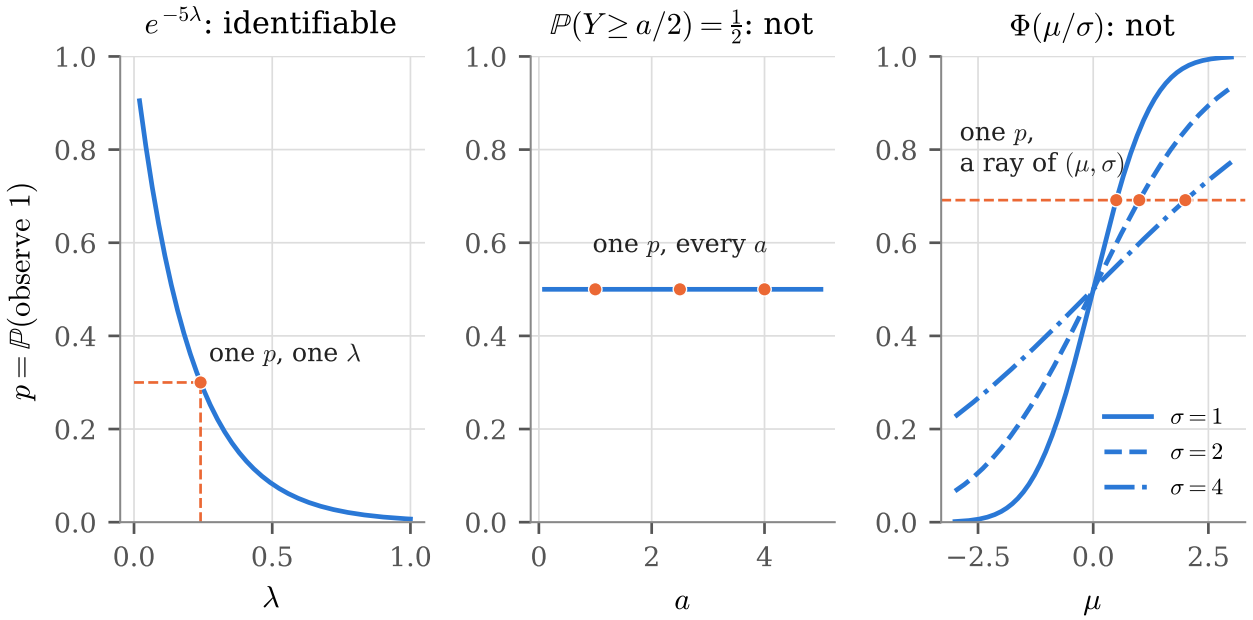


Figure 2.6: The observable Bernoulli parameter p as a function of θ. Left: the censored Exponential, $p = e^{-5\lambda}$. Middle: the uniform thresholded at half its range, $p = \frac{1}{2}$. Right: the Gaussian sign example, $p = \Phi(\mu/\sigma)$ against μ, one line style per σ.

From undergraduate statistics: collinearity, a non-identifiability you have already met

In a regression course, if two predictors are perfectly collinear (say $x_2 = 2x_1$), the software drops one of them or leaves its coefficient blank. That is non-identifiability. The model $Y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$ with $x_2 = 2x_1$ produces exactly the same distribution of Y for every (β_1, β_2) with the same value of $\beta_1 + 2\beta_2$: a whole line of parameter vectors, one distribution, exactly like the ray in Figure 2.5. The data can tell you $\beta_1 + 2\beta_2$ and nothing more, and no sample size fixes it. Dropping a predictor is the software's way of choosing one point on the line. Chapter 21 returns to this when it needs the design matrix to have full rank.

2.7 Gaussian mixture models

A mixture of Gaussians is best described in terms of the generative model. You first generate a discrete variable that determines which of the component Gaussians to use, and then generate an observation from that chosen Gaussian density.

More formally, suppose you have two Gaussian r.v.s, $X^{(1)}$ and $X^{(2)}$, such that $X^{(1)} \sim \text{Gauss}(\mu_1, \sigma_1^2)$ and $X^{(2)} \sim \text{Gauss}(\mu_2, \sigma_2^2)$. Their PDFs are:

$$f_1(x) = \frac{1}{\sigma_1\sqrt{2\pi}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right),$$
$$f_2(x) = \frac{1}{\sigma_2\sqrt{2\pi}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma_2^2}\right).$$

Let $\pi \in (0,1)$ be the *mixture weight*. A **mixture** of the two Gaussians has the PDF:

$$\begin{aligned} f(x) &= \pi f_1(x) + (1-\pi) f_2(x) \\ &= \pi \left(\frac{1}{\sigma_1\sqrt{2\pi}}\right) \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right) + (1-\pi)\left(\frac{1}{\sigma_2\sqrt{2\pi}}\right) \exp\left(-\frac{(x-\mu_2)^2}{2\sigma_2^2}\right). \end{aligned}$$

(The symbol π does double duty here, as the weight and as the constant 3.14159... inside the normalizing constants; both uses are standard and the context always tells them apart.)

To better understand a mixture of Gaussian distributions, think of it as a latent variable model.[2] Proceed in 3 steps:

1. **Latent component selection:** Sample $Z \sim \text{Bern}(\pi)$, so $Z = 1$ with probability π and $Z = 0$ with probability $1 - \pi$.

2. **Independent sampling:** Sample $X^{(1)} \sim \text{Gauss}(\mu_1, \sigma_1^2)$ and $X^{(2)} \sim \text{Gauss}(\mu_2, \sigma_2^2)$, independent of Z and of each other. Both values are conceptually sampled regardless of Z's value.

3. **Mixture realization:** Define the r.v. X by:

   $$X = ZX^{(1)} + (1-Z)X^{(2)}.$$

   Only one of the two is used: $X = X^{(1)}$ if $Z = 1$ and $X = X^{(2)}$ if $Z = 0$.

Question: How can this be a *mixture* if each sample comes from exactly *one* Gaussian?

Answer: The answer lies in distinguishing between:

- **Sample generation**: Each individual observation is generated from only one component, determined by Z.
- **Distribution properties**: The overall probability distribution of X is a weighted mixture of both Gaussians because you don't get to see Z.

[2] Recall that a latent variable is an unobserved variable that can only be inferred indirectly from observable variables.

Mathematically, for any possible value x, the law of total probability over Z gives the density

$$f(x) = \mathbb{P}(Z = 1)f_1(x) + \mathbb{P}(Z = 0)f_2(x) = \pi f_1(x) + (1 - \pi)f_2(x),$$

so both components contribute to the overall distribution, weighted by their probabilities. From the population perspective, if you generate n samples from this model, approximately $n\pi$ of them come from Gauss(μ_1, σ_1^2) and approximately $n(1 - \pi)$ from Gauss(μ_2, σ_2^2). Figure 2.7 shows both perspectives. On the left, the mixture density is the sum of the two weighted components. On the right, each of the 400 draws came from exactly one component, and its color records which; the black mixture density over the histogram is what you see when you cannot tell.

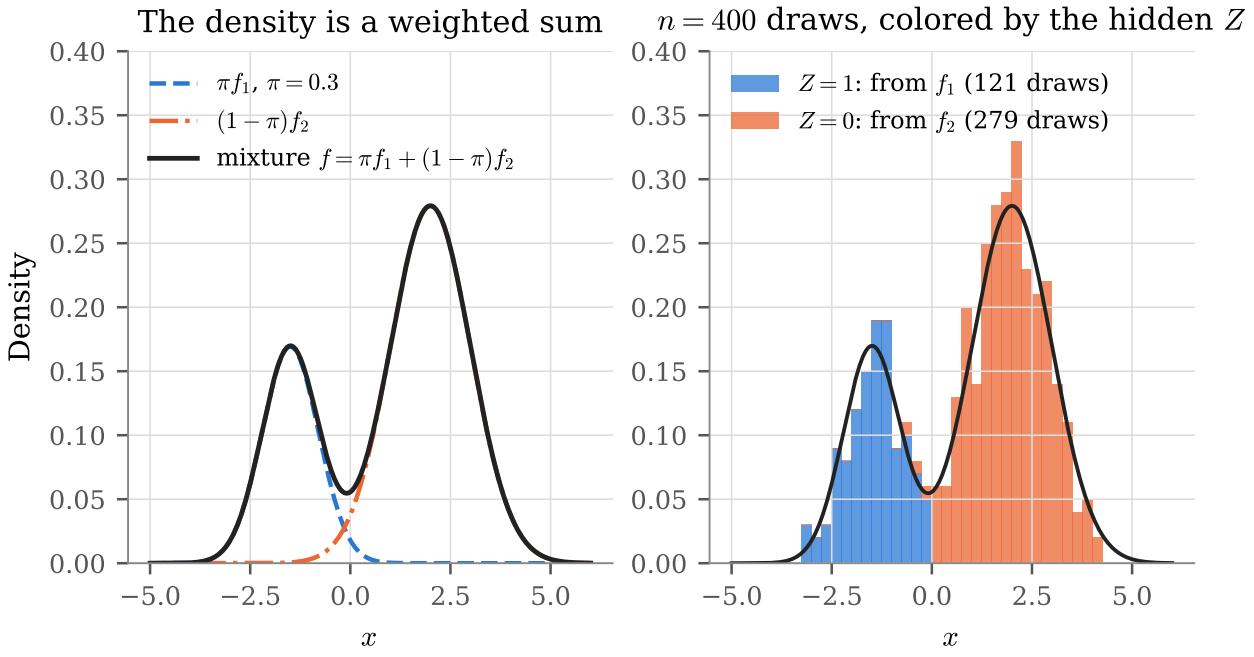


Figure 2.7: A two-component Gaussian mixture with $\pi = 0.3$, Gauss$(-1.5, 0.5)$ (blue) and Gauss$(2, 1)$ (orange). Left: the two weighted components and their sum, the mixture density (black). Right: 400 draws, histogrammed and colored by the latent Z, with the mixture density over them.

An alternative specification of the same r.v. is $X = (X^{(1)})^Z (X^{(2)})^{1-Z}$, which picks the same component as before. The latent variable perspective is especially useful for inference algorithms like EM (expectation-maximization), where you need to estimate both the component parameters $(\mu_1, \sigma_1^2, \mu_2, \sigma_2^2)$ and the mixture weight π without ever observing which component produced which observation.

From undergraduate statistics: a histogram with two peaks

An introductory course reads a histogram with two peaks as a sign that the data may mix two groups, and suggests separating them if some variable records the group. The Gaussian mixture is that reading as a model, for when no variable does: Z is the unrecorded group, each component is the distribution within one group, and π is the share of the first group. With the groups recorded, π would be estimated by the proportion of observations in the first group, a sample proportion like $\hat{p}$, and each component from its own group's mean and variance. Without them, Section 11.8 estimates all five parameters at once; for the waiting times between eruptions of the Old Faithful geyser, with peaks near 55 and 80 minutes, it puts about a third of the waits

in the short group.

2.7.1 Possible Gaussian mixture models

There are several scenarios for a model with mixtures of Gaussians, depending on the constraints imposed on its 5 parameters. Here are 3 possible examples. It should be noted that, for clarity but with an abuse of notation, the PDFs of the two Gaussian r.v.s are not explicitly written below; each distribution is written generically as $\text{Gauss}(\mu_1, \sigma_1^2)$.

1. Model where all 5 parameters are unknown:

$$\left(\mathbb{R}, (\pi \cdot \text{Gauss}(\mu_1, \sigma_1^2) + (1-\pi) \cdot \text{Gauss}(\mu_2, \sigma_2^2))_{\pi \in (0,1),\, \mu_1, \mu_2 \in \mathbb{R},\, \sigma_1^2, \sigma_2^2 \in (0,\infty)}\right).$$

2. Model with known variances, and for simplicity suppose $\sigma_1^2 = \sigma_2^2 = 1$:

$$\left(\mathbb{R}, (\pi \cdot \text{Gauss}(\mu_1, 1) + (1-\pi) \cdot \text{Gauss}(\mu_2, 1))_{\pi \in (0,1),\, \mu_1, \mu_2 \in \mathbb{R}}\right).$$

3. Model with only unknown means, and for simplicity suppose $\sigma_1^2 = \sigma_2^2 = 1$ and $\pi = 1/2$:

$$\left(\mathbb{R}, \left(\tfrac{1}{2} \cdot \text{Gauss}(\mu_1, 1) + \tfrac{1}{2} \cdot \text{Gauss}(\mu_2, 1)\right)_{\mu_1, \mu_2 \in \mathbb{R}}\right).$$

Pitfall: mixtures are identifiable only up to relabeling

Take the third model and the two parameter vectors $(\mu_1, \mu_2) = (-1, 2)$ and $(\mu_1, \mu_2) = (2, -1)$. They're different points of $\Theta = \mathbb{R}^2$, and they give exactly the same density: $\frac{1}{2}f_{-1} + \frac{1}{2}f_2 = \frac{1}{2}f_2 + \frac{1}{2}f_{-1}$. So by Definition 2.3 the parameter is not identifiable; the same happens in the first two models with $(\pi, \mu_1, \sigma_1^2, \mu_2, \sigma_2^2) \leftrightarrow (1-\pi, \mu_2, \sigma_2^2, \mu_1, \sigma_1^2)$. This failure is called *label switching*, and it's harmless in a way Example 2.4 is not: the two parameter vectors describe the same two components with their names swapped, so the data do pin down the components, just not which one is 'first'. The fix is a convention such as $\mu_1 < \mu_2$, which restricts Θ to half the plane and removes the swap; that nothing else collides, so that the map is then injective, is a theorem about finite Gaussian mixtures that is not proved here. The lesson is that non-identifiability is sometimes a defect of the labeling rather than of the information in the data, and the first thing to ask when you find it is which of the two you're looking at.

2.8 What comes next

A model fixes the family of candidate distributions. Three things are done inside it, and they are Part III: guess θ (Chapter 7), attach a range to the guess (Chapter 8), and decide a yes/no question about θ (Chapter 9). All three are stated for a general model $(E, (\mathbb{P}_\theta)_{\theta \in \Theta})$ and computed, as always, for the kiss model. All three also rest on what an average of n draws does as n grows, and that is Part II, which comes first: where the average settles (Chapter 3), how it fluctuates on the way (Chapter 4), what may be done to those limits (Chapter 5), and the Gaussian distribution the limits land on (Chapter 6).

2.9 Summary

- A statistical model is $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$: an observation space and a family of candidate distributions for X, indexed by a parameter. Parameters define distributions; learning one is learning the other. The observation space must not depend on θ.
- Assumptions buy fewer parameters: the siblings PMF has six, the shifted Poisson one. The price is misspecification if the assumption is wrong.
- Three layers: the sample space Ω (every complete outcome), random variables (functions on Ω, the measuring instruments and the rules applied to their readings), and the observations (those functions evaluated at the one outcome that happened). An estimate is one draw from the estimator's distribution. Inference does not invert the chain; it holds the one value up against the distribution of the rule. For fixed n, take $\Omega = E^n$.
- Well-specified: the true $\mathbb{P}$ is some $\mathbb{P}_{\theta^*}$ in the family. Everything from Part III on assumes this; Chapter 10 says what estimators estimate when it fails.
- Parametric: $\Theta \subseteq \mathbb{R}^d$. Nonparametric: Θ is a space of functions. Semiparametric: a finite-dimensional parameter of interest plus an infinite-dimensional nuisance parameter.
- Identifiable: $\theta \mapsto \mathbb{P}_\theta$ is injective, so knowing the distribution of the data would tell you θ. To show a parameter is not identifiable, exhibit two parameter values with the same distribution. Observing only a function of the data can destroy identifiability (two numbers behind one observable proportion) or leave it intact (one number behind one proportion, injectively); what matters is whether the map from the parameter to the observable distribution is one-to-one.
- A Gaussian mixture is a latent-variable model: pick a component with $Z \sim \text{Bern}(\pi)$, then draw from it. Its density is $\pi f_1 + (1-\pi) f_2$. Its parameter is identifiable only up to swapping the component labels.

2.10 Exercises

Exercise 2.1. Which of the following is a statistical model? (1) $\big(\{1\}, (\text{Bern}(p))_{p\in(0,1)}\big)$; (2) $\big(\{0,1\}, (\text{Bern}(p))_{p\in(0.2,0.4)}\big)$; (3) both; (4) neither. *(Solution on page 414.)*

Exercise 2.2. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([0,a])$ for some unknown $a > 0$. Which is the associated statistical model: (1) $([0,a], (\text{Unif}([0,a]))_{a>0})$; (2) $(\mathbb{R}_+, (\text{Unif}([0,a]))_{a>0})$; (3) $(\mathbb{R}, (\text{Unif}([0,a]))_{a>0})$; (4) none? *(Solution on page 414.)*

Exercise 2.3. Which of these models are parametric? (A) $E = \{0,1,2,\ldots\}$ and $\{\mathbb{P}_\theta\}$ is the set of all distributions on E. (B) $E = \{0,1\}$, $(\text{Bern}(\theta))_{\theta\in[0,1]}$. (C) $E = \mathbb{R}$, $(\text{Gauss}(0,\sigma^2))_{\sigma^2>0}$. (D) $E = \{1,2,3,4\}$ and $\mathbb{P}_{(p_1,\ldots,p_4)}$ puts probability p_i on i, with $p_i \geq 0$ and $\sum p_i = 1$. (E) $E = \mathbb{R}$, $(\text{Gauss}(\mu,\sigma^2))_{\mu\in\mathbb{R},\sigma^2>0}$. (F) $E = [0,\infty)$, $(\text{Unif}([0,\theta]))_{\theta>0}$. (G) $E = [0,1]$ and $\{\mathbb{P}_f\}$ is the set of all distributions with a continuous density f on $[0,1]$. *(Solution on page 415.)*

Exercise 2.4. You observe $X_i = \mathbb{1}(Y_i \geq c)$ for $Y_i \overset{iid}{\sim} \text{Gauss}(\mu, 1)$ with $\sigma^2 = 1$ known and c a known constant. Is μ identifiable? What if instead σ^2 is unknown and $\mu = 0$ is known? *(Solution on page 415.)*

Exercise 2.5. In the two-component mixture with all five parameters unknown, is the parameter $(\pi, \mu_1, \sigma_1^2, \mu_2, \sigma_2^2)$ identifiable on the restricted parameter space where $\mu_1 < \mu_2$? What about the point $\pi = 1$, if it were allowed? *(Solution on page 415.)*

Part II

Probability for Statistics

The three questions of Chapter 1 are about the random variable $\bar{R}_n$ as n grows, and 'as n grows' is a limit. This part has the three limit facts every later chapter uses, the Law of Large Numbers (Chapter 3), the Central Limit Theorem (Chapter 4) and the rules for working with limits (Chapter 5), and then the Gaussian distribution and the quantile notation that the rest of the book is written in (Chapter 6). If limits are not your difficulty, Chapter 3's opening box says what can be skipped.

Chapter 3

The Law of Large Numbers

In this chapter

- Why limits of random variables need different tools than limits of numbers.
- The sample mean on two scales: collapsing on μ (this chapter) and, magnified by $\sqrt{n}$, settling into a Gaussian (the next).
- Convergence in probability, the mode of convergence that says an estimator gets close to what it estimates.
- The Weak Law of Large Numbers, with Chebyshev's inequality as the proof.

The running example is the kiss experiment of Chapter 1: $n = 124$ couples, 80 of whom turned right, so $\hat{p} = \bar{R}_n = 0.645$.

Chapter 2 fixed the object inference happens in; this part supplies the three facts about averages that Part III stands on.

The next three chapters. The rest of the book rests on three limit facts about averages: the Law of Large Numbers (averages settle on their expectations, this chapter), the Central Limit Theorem (the error of an average is Gaussian on the $1/\sqrt{n}$ scale, Chapter 4), and the rules for transforming limits and plugging in estimates (continuous mapping, Slutsky, the delta method, Chapter 5). Those three chapters are the minimal path. Two more modes of convergence, almost sure and in mean, complete the standard hierarchy; that material is in Appendix B because nothing later is proved with it; where a later chapter mentions almost sure convergence, the in-probability version is the one in use. Read it once for orientation and don't spend time mastering it.

Intuition: what 'asymptotic' means

An asymptotic result is a license to replace an unknown sampling distribution by its limiting distribution, usually a Gaussian, with an error the theorem does not tell you. That is all 'asymptotic' ever means in this book. The definitions below are about infinite sequences, while every application is about one fixed sample size, usually $n = 124$; the theorems say $n \to \infty$ and they get applied at $n = 64$ without further comment. The gap is real, and Section 4.4 in the next chapter says what the alternatives are and why statistical practice accepts the trade. When a later chapter says 'by the CLT' or 'asymptotically normal', read it as: for the n at hand, treat this as Gaussian and carry on.

3.1 Why limits of random variables need new tools

Chapter 1 asked whether $\hat{p}$ is close to p once n is large. That is a question about a limit, and limits of random variables need their own definition. In calculus you say that a sequence of real numbers x_n converges to a limit x if, for any $\epsilon > 0$, $|x_n - x| < \epsilon$ for all large n (toolkit A.1). Convergence is more subtle in probability. Returning to calculus, suppose that $x_n = x$ for all n. Then, trivially, $\lim_{n\to\infty} x_n = x$.

Consider a probabilistic version of this example. Suppose that $X_1, X_2, \ldots$ is a sequence of i.i.d. r.v.s such that $X_i \overset{iid}{\sim} \text{Gauss}(0,1)$. Since these all have the same distribution, you might think X_n 'converges' to $X \sim \text{Gauss}(0,1)$. But this can't be right: $X_n - X \sim \text{Gauss}(0,2)$ for every n, so $\mathbb{P}(|X_n - X| < \epsilon)$ is the same small number for every n and never improves. (In fact $\mathbb{P}(X_n = X) = 0$ for all n, since two independent continuous r.v.s are equal with probability zero.[1]) This suggests limiting processes involving probabilities require different tools than those limiting processes from calculus.

The central ideas in this chapter and the next are the two most famous results in all of probability theory: the Law(s) of Large Numbers and the Central Limit Theorem. An understanding of why these results are valid requires an understanding of limiting processes that involve r.v.s. These limits are conceptually more general than the limits of undergraduate calculus since, as limits of r.v.s, these limits are not limits of numbers but limits of *functions*.

So what is meant by 'convergence of random variables'? And what is a sequence of r.v.s? Recall that, by definition, an r.v. is a mapping from a sample space, Ω, to the real number line (toolkit A.5). Suppose, to fix ideas, that the outcomes can be listed (a continuous r.v. needs a richer sample space, but the picture is the same), so you can write:

$$\Omega = \{\omega_1, \ldots, \omega_k\}.$$

This means an r.v. X defined on Ω can be written as:

$$X(\omega_i) = x_i, \text{ for } i = 1, \ldots, k.$$

When you have a sequence of r.v.s it's useful to keep in mind that there's always an underlying sample space. In particular, each X_n is a function from Ω to $\mathbb{R}$:

$$X_n(\omega_i) = x_{n,i}, \text{ for } i = 1, \ldots, k.$$

So a sequence of r.v.s is a sequence of functions $X_n : \Omega \to \mathbb{R}$. Pick an outcome ω_i and you get an ordinary sequence of numbers $x_{1,i}, x_{2,i}, x_{3,i}, \ldots$, one such sequence for every outcome. A *realization* of the sequence is what you see when the experiment is run once and the outcome happens to be ω_i.

This chapter and the next examine the two modes of convergence this book works with; Appendix B has the full set. Why are there different modes of convergence? Remember that to say a sequence

[1] Why is this true? Let Y_1 and Y_2 be continuous r.v.s with joint density f_{Y_1,Y_2}. Then

$$\mathbb{P}(Y_1 = Y_2) = \iint_D f_{Y_1,Y_2}(y_1, y_2)\, \mathrm{d}y_1\, \mathrm{d}y_2, \qquad D = \{(y_1, y_2) \in \mathbb{R}^2 : y_1 = y_2\}.$$

The set D is a line living in two dimensions and hence has no area: you're integrating a bounded function over a region of zero area, so the integral is 0. This is the two-dimensional version of the fact that the 'area' under a density at a single point is zero.

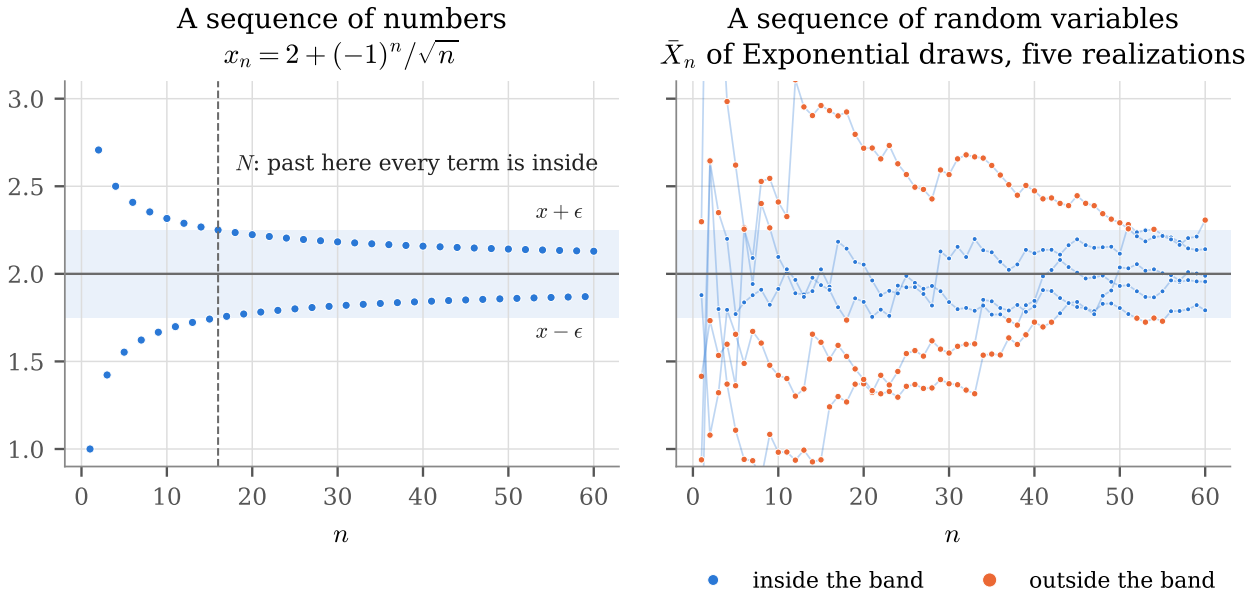


Figure 3.1: Left: the sequence $x_n = 2 + (-1)^n/\sqrt{n}$, with the band of half-width ϵ around 2 and the index N dashed. Right: five realizations of the running mean $\bar{X}_n$ of Exponential draws with mean 2; blue points are inside the band, orange points outside.

$(X_i)_{i=1}^{\infty}$ converges to X means that the X_i's are getting 'closer and closer' to X. Different modes of convergence relate to different ways of defining what 'closer' means. Figure 3.1 shows the difficulty: for a sequence of numbers there is a single index N past which every term is within ϵ of the limit, while for a sequence of r.v.s the index that works depends on the outcome of the experiment and may not exist for some outcomes. On the right of the figure, each path is one outcome of the experiment, and each eventually settles inside the band, but at a different index; no N works for every path, since some realization will leave the band after any N you name. The modes of convergence are different ways of saying 'the orange points become rare.'

Two of the modes, convergence almost surely and convergence in probability, compare the *values* of X_n and X outcome by outcome, so X_n and X must live on the same sample space. The third, convergence in distribution, compares the CDFs of X_n and X and never looks at values at all; X_n and X can be defined on entirely different probability spaces. But as r.v.s have CDFs, the difference between sequences of r.v.s and sequences of CDFs isn't as significant as you might expect.

So it's important to emphasize here that the 'things' converging are r.v.s, or their CDFs, and that they converge to an r.v., or a CDF.[2]

3.2 A first look: the sample mean on two scales

To motivate the notion of convergence of r.v.s, consider the following example. Suppose that $X_1, \ldots, X_n$ are i.i.d. r.v.s with mean μ and variance σ^2 and define the (theoretical, non-realized)

[2]Just because a sequence of r.v.s converges to an r.v. doesn't mean the limiting r.v. has an interesting distribution or what you normally think of as a 'distribution'. In fact, one mode of convergence explained below (convergence in probability), is almost entirely concerned with the convergence of sequences of r.v.s to a constant or 'degenerate' r.v.

sample mean:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

First, consider the behavior of the distribution of $\bar{X}_n$ when n is large. It seems reasonable that $\bar{X}_n$ will be close to μ if n is sufficiently large; that is, the r.v. $\bar{X}_n - \mu$ should have a distribution that, for large n, is concentrated around 0 or, more precisely:

$$\mathbb{P}(|\bar{X}_n - \mu| \leq \epsilon) \approx 1,$$

for any small positive number ϵ. This type of limiting process, and more specifically, that as n increases without limit the probability equals 1, is an example of *convergence in probability*, and also illustrates a convergence process to a constant (here, μ), as discussed in the previous footnote.

Moreover because $\mathbb{V}\text{ar}(\bar{X}_n) = \sigma^2/n$,[3] it follows that $\mathbb{V}\text{ar}(\bar{X}_n) \to 0$ as $n \to \infty$. Because the variance of $\bar{X}_n$ eventually vanishes as n increases, the limit tells you nothing about the *shape* of the distribution of $\bar{X}_n$: every density on the left of Figure 3.2 is on its way to becoming a spike at μ.

It's possible though to look at the difference between $\bar{X}_n$ and μ on a 'magnified' scale. To that end, multiply the difference $\bar{X}_n - \mu$ by $\sqrt{n}$, which makes the mean and variance of the resulting r.v. constant. Specifically, define the r.v. Y_n by:

$$Y_n = \sqrt{n}(\bar{X}_n - \mu),$$

and note that:

$$\mathbb{E}[Y_n] = \mathbb{E}[\sqrt{n}(\bar{X}_n - \mu)] = \sqrt{n}(\mathbb{E}[\bar{X}_n] - \mu) = \sqrt{n}(\mu - \mu) = 0,$$

and

$$\mathbb{V}\text{ar}(Y_n) = \mathbb{V}\text{ar}(\sqrt{n}(\bar{X}_n - \mu)) = n\,\mathbb{V}\text{ar}(\bar{X}_n - \mu) = n\,\mathbb{V}\text{ar}(\bar{X}_n) = \sigma^2.$$

Intuition: what's special about $\sqrt{n}$?

Why does multiplying by $\sqrt{n}$, and not by n or by $\log n$, yield a constant mean and variance?

- **The mean:** Multiplying $\bar{X}_n - \mu$ by anything doesn't impact its expected value because $\bar{X}_n$ is an unbiased estimator of μ, so the expected value of the difference is 0 to begin with.
- **The variance:** As n increases, $\mathbb{V}\text{ar}(\bar{X}_n) = \sigma^2/n$ decreases. When you multiply an r.v. by $\sqrt{n}$ the variance is scaled by a factor of n, which exactly cancels the division by n. Multiplying by n instead would scale the variance by n^2 and the result would blow up; multiplying by $\log n$ would not be enough and the result would still collapse to a spike.

The standard deviation of $\bar{X}_n$ is $\sigma/\sqrt{n}$, so $\sqrt{n}$ is the magnification that holds the spread fixed.

Figure 3.2 shows the same sample mean of Expo(1) draws on both scales. On the left, the density of $\bar{X}_n$ collapses onto $\mu = 1$ as n grows; this is the Law of Large Numbers. On the right, after the $\sqrt{n}$ magnification, the density of $\sqrt{n}(\bar{X}_n - \mu)$ keeps a fixed spread and its shape settles down to a Gaussian; this is the Central Limit Theorem. At $n = 1$ the magnified density is the skewed exponential itself, shifted; by $n = 64$ it is hard to tell from Gauss$(0, \sigma^2)$.

[3] That $\mathbb{V}\text{ar}(\bar{X}_n) = \sigma^2/n$ follows from the properties of the variance and the independence of the X_i. Specifically:

$$\mathbb{V}\text{ar}(\bar{X}_n) = \frac{1}{n^2}\,\mathbb{V}\text{ar}(X_1 + \cdots + X_n) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

From undergraduate statistics: the gambler's fallacy

An introductory course states the Law of Large Numbers for coin tosses (the proportion of heads settles near $\frac{1}{2}$) and warns against the gambler's fallacy: the belief that after a run of heads the coin is due for tails. The law and the warning are both right, since the proportion settles by swamping, not by compensation. The tosses are independent, so after 10 heads in a row the expected excess of heads over $n/2$ stays at 5 for every $n \geq 10$, and the expected proportion, $\frac{1}{2} + 5/n$, tends to $\frac{1}{2}$ only because the excess is divided by n. In general the excess is the proportion's error multiplied by n rather than $\sqrt{n}$, and its standard deviation $\sqrt{n}/2$ grows (5 at $n = 100$, 50 at $n = 10{,}000$) while the proportion's, $1/(2\sqrt{n})$, shrinks (0.05, then 0.005). The running sums of Figure 3.6, later in this chapter, are heads minus tails, twice this excess.

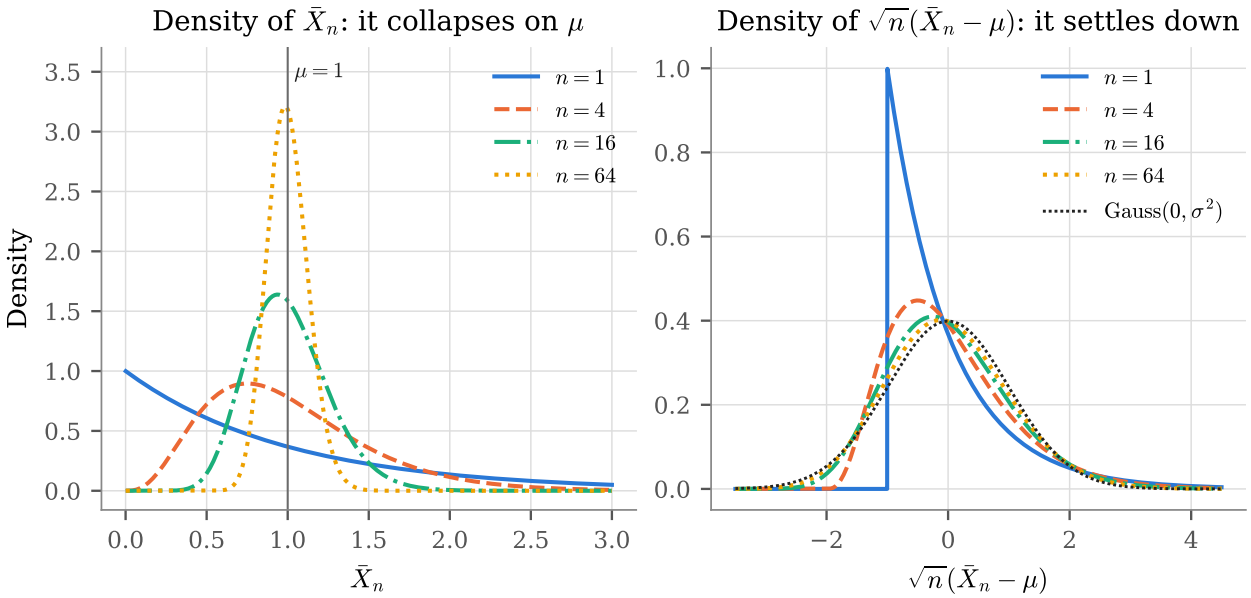


Figure 3.2: The sample mean of Expo(1) draws at $n = 1, 4, 16, 64$. Left: the density of $\bar{X}_n$, with $\mu = 1$ marked. Right: the density of $\sqrt{n}(\bar{X}_n - \mu)$, with the Gauss$(0, \sigma^2)$ density dotted in black.

You can now consider the behavior of the CDF of Y_n as n increases. If this sequence of CDFs has a limit, then you can use the limiting CDF to approximate the CDF of Y_n (and hence of $\bar{X}_n$).

For example, suppose you have that:

$$\mathbb{P}(Y_n \leq y) = \mathbb{P}(\sqrt{n}(\bar{X}_n - \mu) \leq y) \approx F_0(y).$$

Then, to say something about the CDF of $\bar{X}_n$, subtract μ from $\bar{X}_n$ and multiply this difference by $\sqrt{n}$, to get:

$$\mathbb{P}(\bar{X}_n \leq x) = \mathbb{P}(\sqrt{n}(\bar{X}_n - \mu) \leq \sqrt{n}(x - \mu)) \approx F_0(\sqrt{n}(x - \mu)),$$

provided n is sufficiently large to make the approximation valid. This type of limiting process is an example of *convergence in distribution*, which is the subject of Chapter 4.

What if instead of transforming $\bar{X}_n$ into Y_n, you 'standardized' it by dividing Y_n by σ. That is, define the r.v. Z_n by:

$$Z_n = \sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right).$$

Now, calculate $\mathbb{E}[Z_n]$ and $\mathbb{V}\text{ar}(Z_n)$:

$$\mathbb{E}[Z_n] = \mathbb{E}\left[\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right)\right] = \frac{\sqrt{n}}{\sigma}(\mathbb{E}[\bar{X}_n] - \mu) = \frac{\sqrt{n}}{\sigma}(\mu - \mu) = 0.$$

Thus, $\mathbb{E}[Y_n] = \mathbb{E}[Z_n] = 0$. But observe that $\mathbb{V}\text{ar}(Z_n) \neq \mathbb{V}\text{ar}(Y_n) = \sigma^2$:

$$\begin{aligned}
\mathbb{V}\text{ar}(Z_n) &= \mathbb{V}\text{ar}\left(\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right)\right) \\
&= \frac{n}{\sigma^2}\,\mathbb{V}\text{ar}\left(\bar{X}_n\right) \\
&= \frac{n}{\sigma^2}\,\mathbb{V}\text{ar}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) \\
&= \frac{1}{n\sigma^2}\,\mathbb{V}\text{ar}\left(\sum_{i=1}^{n} X_i\right) \\
&= \frac{n\sigma^2}{n\sigma^2} \\
&= 1.
\end{aligned}$$

Thus, despite the extra scaling factor, $\sqrt{n}$, the r.v. Z_n has a 'standard' look to it: Its mean is 0 and its variance is 1. It may not come as a surprise that Z_n converges in distribution to a standard Gaussian distribution.

From undergraduate statistics: the standard error

An introductory course states, usually without proof, that the sampling distribution of the sample mean has mean μ, has standard deviation $\sigma/\sqrt{n}$ (the *standard error* of the mean), and is approximately normal when n is large. The two halves of Figure 3.2 are the two halves of that statement. The standard error shrinking to zero is what drives the Law of Large Numbers. The shape becoming normal is the Central Limit Theorem. Dividing $\bar{X}_n - \mu$ by the standard error $\sigma/\sqrt{n}$ is exactly the standardization that produces Z_n:

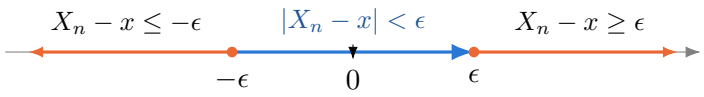
$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right).$$

This chapter proves the first half; Chapter 4 makes 'approximately normal when n is large' into a theorem and says what 'large' buys you.

3.3 Convergence in probability

Convergence in probability considers the distance between a sequence of r.v.s X_n and some number x being at least as large as some arbitrary $\epsilon > 0$. That is, in formal terms you're concerned with whether the probability of $|X_n - x| \geq \epsilon$ converges to 0.

A first step in such problems, and limit problems generally, is to grasp the meaning of $|X_n - x| \geq \epsilon$. An absolute value can be thought of as the distance from 0. So $|X_n - x|$ is the distance of the number $X_n - x$ from 0, and the event $|X_n - x| \geq \epsilon$ says that this distance is at least ϵ, which happens either when $X_n - x \geq \epsilon$ or when $X_n - x \leq -\epsilon$. A visualization along the real line is instructive.



If a sequence of r.v.s converges in probability to a number, say, 0, then you say the sequence converges to the zero *random variable* (not the number 0). Now for the definition.

Definition 3.1 (Convergence in probability)**.** A sequence of r.v.s, $(X_n)_{n\ge1}$, **converges in probability** to an r.v. X if, for every $\epsilon > 0$:

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| \ge \epsilon) = 0, \text{ or equivalently } \lim_{n\to\infty} \mathbb{P}(|X_n - X| < \epsilon) = 1.$$

If $(X_n)_{n\ge1}$ converges in probability to some r.v. X, then this is denoted by $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$.

> **In words.** Read the definition from the inside out. Fix a tolerance ϵ. For each n, $\mathbb{P}(|X_n - X| \ge \epsilon)$ is a number, the probability mass that X_n puts outside the ϵ-band around X. The definition says this sequence of *numbers* goes to 0 in the ordinary calculus sense (toolkit A.1), and that it must do so for every tolerance, however small. Nothing is said about any single outcome; everything is said about probabilities.
>
> **For the kiss data.** $\bar{R}_n \xrightarrow[n\to\infty]{\mathbb{P}} p$ means: pick any tolerance, say 0.01; the probability that the sample proportion turning right misses p by more than 0.01 can be made as small as you like by observing enough couples. It does not say how many couples; Section 3.4 and Chapter 4 do.

Figure 3.3 shows the picture for the sample mean, where X is the constant μ. The orange mass outside the band is $\mathbb{P}(|\bar{X}_n - \mu| \ge \epsilon)$, and it goes from about one half at $n = 5$ to essentially zero at $n = 100$. Shrink ϵ and the same thing happens, only it takes a larger n.

Though convergence in probability is defined as a sequence of r.v.s converging to an r.v., in statistical applications the limit is almost always a constant: the sample mean converges to μ, an estimator converges to the parameter it estimates. There's a reason the interesting limits are constants. If the r.v.s in the sequence are independent and the sequence converges in probability, then the limiting r.v. must be a constant.[4] The sample mean $\bar{X}_n$ isn't a sequence of independent r.v.s (each $\bar{X}_n$ shares X_1 with all the others), so this fact doesn't apply to it directly, but the conclusion is the same: the randomness averages out.

3.3.1 Example: the minimum of uniforms

Example 3.2. Let $X_1, X_2, \ldots \overset{iid}{\sim} \text{Unif}([0,1])$ and define the minimum of the first n by $X_{(1)} = \min_{i=1,\ldots,n} X_i$. Then:

$$X_{(1)} \xrightarrow[n\to\infty]{\mathbb{P}} 0.$$

[4] A sketch, not a proof, and not needed later: the limit of a sequence can't depend on any finite number of its terms (throw away the first million terms and the limit is unchanged). If the terms are independent, the limit is therefore independent of every one of them, hence independent of everything that's random in the experiment, hence independent of itself. An r.v. independent of itself satisfies $\mathbb{V}\text{ar}(X) = \mathbb{C}\text{ov}(X, X) = 0$, so it's a constant. The precise statement is Kolmogorov's zero-one law.

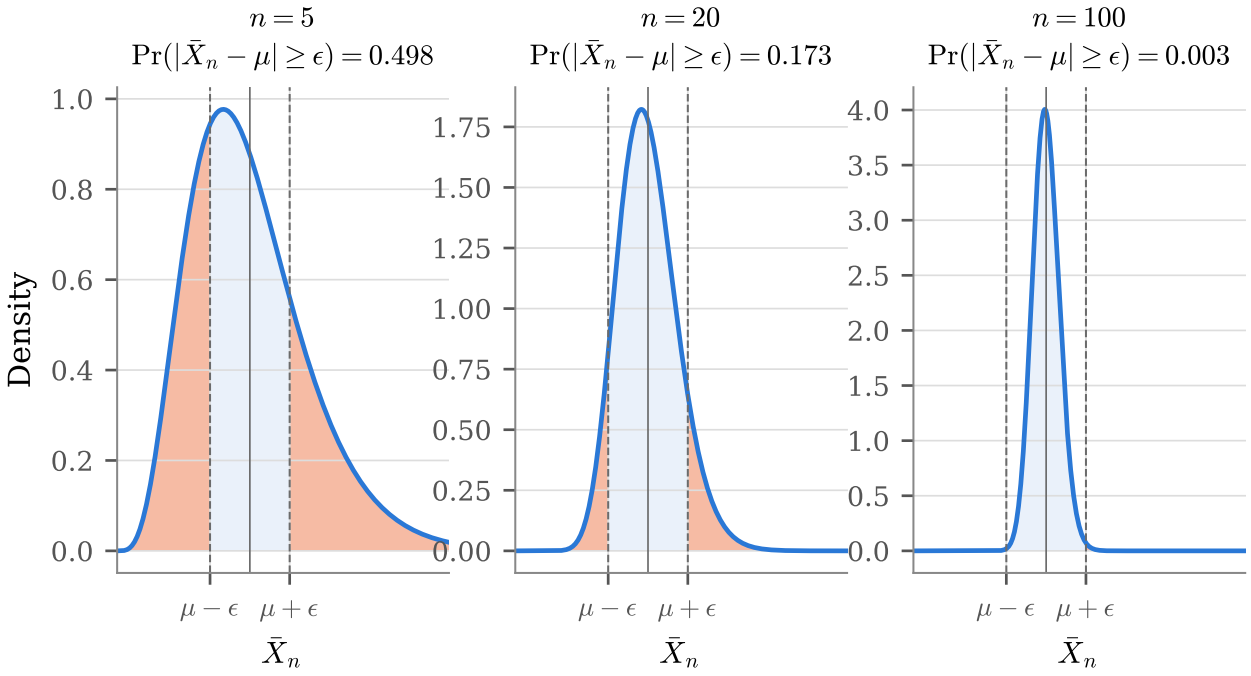


Figure 3.3: The density of $\bar{X}_n$ for Expo(1) draws at $n = 5$, 20 and 100, with the band $[\mu - \epsilon, \mu + \epsilon]$ for $\epsilon = 0.3$ between the dashed lines. The orange shading is the mass outside the band.

Proof strategy (core). Write the event $|X_{(1)} - 0| \geq \epsilon$ as an event about the whole sample, use independence to turn its probability into a product of n equal factors, and let $n \to \infty$. This is the template for every direct convergence proof that follows.

Proof.

- Let $\epsilon > 0$. Write the claim using the definition of convergence in probability: you need $\mathbb{P}(|X_{(1)} - 0| \geq \epsilon) \to 0$.
- Now, you'll simplify the expression in such a way as to eventually take the limit of the expression as $n \to \infty$.
- To that end, observe that:

$$\begin{aligned}
\mathbb{P}(|X_{(1)} - 0| \geq \epsilon) &= \mathbb{P}(|X_{(1)}| \geq \epsilon) \\
&= \mathbb{P}(X_{(1)} \geq \epsilon) \\
&= \mathbb{P}(X_1 \geq \epsilon, X_2 \geq \epsilon, \ldots, X_n \geq \epsilon) \\
&= \mathbb{P}(X_1 \geq \epsilon)\mathbb{P}(X_2 \geq \epsilon) \cdots \mathbb{P}(X_n \geq \epsilon) \\
&= \mathbb{P}(X_1 \geq \epsilon)^n.
\end{aligned}$$

  The movement from the second to the third equality holds because, as $X_{(1)}$ is the minimum of the X_i's, the minimum is at least ϵ exactly when every X_i is at least ϵ: the single event is equivalent to the joint event. The fourth equality results since the X_i's are independent, and the fifth since they're identically distributed.

- You know that $X_1 \sim \text{Unif}([0, 1])$. Therefore, $f_{X_1}(x) = 1$ for all values in X_1's support.

- This implies that $\mathbb{P}(X_1 \geq \epsilon) = \int_\epsilon^1 1 \, dx = 1 - \epsilon$, and further that:

$$\mathbb{P}(X_1 \geq \epsilon)^n = (1 - \epsilon)^n.$$

  Also, note that $1 - \epsilon$ is some number between 0 and 1 (for $\epsilon \geq 1$ the probability is already 0, so there's nothing to prove).

- Now you're ready to take the limit of this probability and confirm the statement:

$$\lim_{n \to \infty} \mathbb{P}(|X_{(1)} - 0| \geq \epsilon) = \lim_{n \to \infty} (1 - \epsilon)^n = 0,$$

  and hence $X_{(1)} \xrightarrow[n \to \infty]{\mathbb{P}} 0$, as desired.

□

Figure 3.4 is the proof as a picture. The density of $X_{(1)}$ is $n(1-x)^{n-1}$ on $[0, 1]$; as n grows the mass piles up against 0, and the mass beyond ϵ, which is the $(1 - \epsilon)^n$ of the proof, goes to zero. The proof computes the shaded tail; the picture shows why it has to shrink. (The peaks of the $n = 10$ and $n = 30$ curves are at 10 and 30, above the top of the plot.)

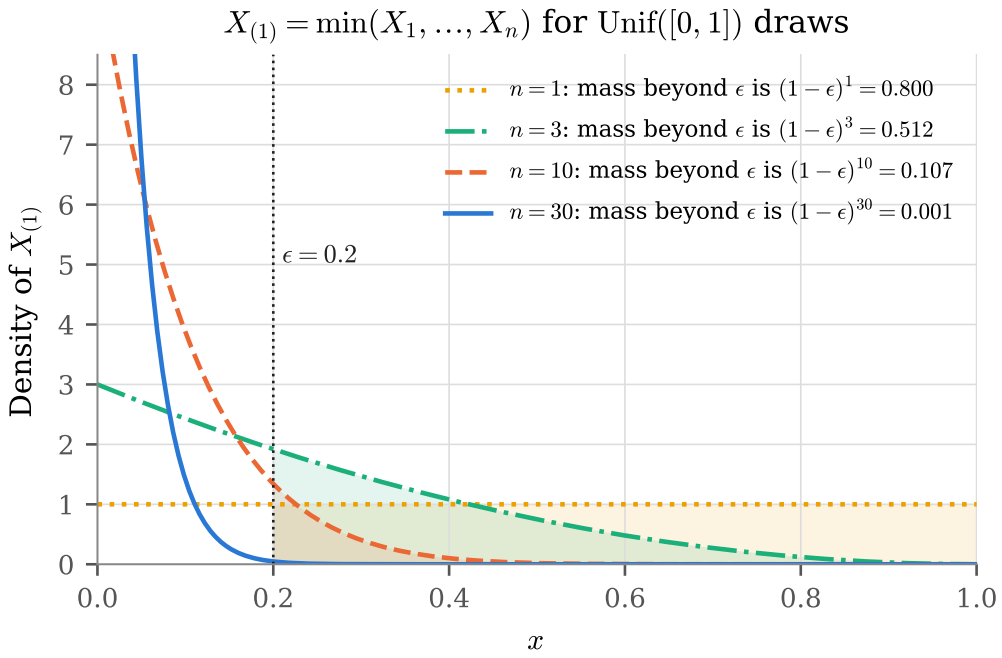


Figure 3.4: The density $n(1 - x)^{n-1}$ of $X_{(1)} = \min(X_1, \ldots, X_n)$ for Unif([0, 1]) draws at $n = 1, 3, 10, 30$, with the mass beyond $\epsilon = 0.2$ shaded for each. The $n = 10$ and $n = 30$ curves are cut off at the top.

The proof is the template for every direct proof of convergence in probability: write the event $|X_n - X| \geq \epsilon$, rewrite it as an event about the original sample, use independence to turn a joint probability into a product, and take the limit.

3.3.2 A workhorse: shrinking variance

Most of the time you won't compute $\mathbb{P}(|X_n - x| \geq \epsilon)$ directly. You'll use the following criterion, which only needs a mean and a variance.

Proposition 3.3 (Shrinking-variance criterion)**.** *Let $(T_n)_{n\geq 1}$ be a sequence of r.v.s with $\mathbb{E}[T_n] \to \mu$ and $\mathbb{V}ar(T_n) \to 0$. Then $T_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$.*

Proof strategy (core). Chebyshev's inequality bounds the probability of a large deviation by a variance; a variance going to 0 drags the probability with it. The triangle inequality handles the mean not being exactly μ (toolkit A.7).

Proof. The argument is Chebyshev's inequality. Recall its two ingredients. *Markov's inequality:* if $Y \geq 0$ then $\mathbb{P}(Y \geq a) \leq \mathbb{E}[Y]/a$ for every $a > 0$. *Chebyshev's inequality:* applying Markov's inequality to $Y = (T - \mathbb{E}[T])^2$ with $a = \epsilon^2$ gives, for any r.v. T with finite variance,

$$\mathbb{P}(|T - \mathbb{E}[T]| \geq \epsilon) \leq \frac{\mathbb{V}\text{ar}(T)}{\epsilon^2}.$$

Now fix $\epsilon > 0$. For n large enough that $|\mathbb{E}[T_n] - \mu| < \epsilon/2$, the event $|T_n - \mu| \geq \epsilon$ forces $|T_n - \mathbb{E}[T_n]| \geq \epsilon/2$ (triangle inequality), so

$$\mathbb{P}(|T_n - \mu| \geq \epsilon) \leq \mathbb{P}(|T_n - \mathbb{E}[T_n]| \geq \epsilon/2) \leq \frac{4\,\mathbb{V}\text{ar}(T_n)}{\epsilon^2} \to 0. \qquad \square$$

Example 3.4. Let $X_n \sim \text{Bern}(1/n)$, so X_n is 1 with probability $1/n$ and 0 otherwise. For $0 < \epsilon < 1$, $\mathbb{P}(|X_n - 0| \geq \epsilon) = \mathbb{P}(X_n = 1) = 1/n \to 0$, so $X_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$. Alternatively, $\mathbb{E}[X_n] = 1/n \to 0$ and $\mathbb{V}\text{ar}(X_n) = \frac{1}{n}(1 - \frac{1}{n}) \to 0$, and Proposition 3.3 gives the same conclusion.

3.4 The Weak Law of Large Numbers (WLLN)

Frequently, statisticians are concerned with situations in which the limiting r.v. is a constant and the r.v.s in the sequence are sample means. The most famous result along these lines is the Weak Law of Large Numbers (WLLN), which is stated here but explained and proved below.

Theorem 3.5 (Weak Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be i.i.d. r.v.s with $\mathbb{E}[X_i] = \mu$ and $\mathbb{V}ar(X_i) = \sigma^2 < \infty$, and let $\bar{X}_n$ be the sample mean. Then, for every $\epsilon > 0$:*

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| < \epsilon) = 1.$$

That is, $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$.

From undergraduate statistics: probability as long-run relative frequency

An introductory course defines the probability of an event as the proportion of times it happens in a very long series of repetitions. Theorem 3.5 makes that sentence a theorem. Repeat the experiment independently and let $X_i = \mathbb{1}(A \text{ happens on repetition } i)$ for the event A: then $\mu = \mathbb{P}(A)$, the variance $\mathbb{P}(A)(1 - \mathbb{P}(A)) \leq \frac{1}{4}$ is always finite, $\bar{X}_n$ is the relative frequency of A,

and the theorem says $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{P}(A)$. This case, proved by Jacob Bernoulli and published in 1713, was the first law of large numbers. For the kiss data, A is 'the couple turns right' and $\bar{R}_n = 80/124 = 0.645$ is its relative frequency in 124 repetitions. The theorem cannot serve as the definition, since its conclusion is itself a statement about probabilities: the axioms of probability come first, and the frequency meaning follows from them. The version about one endless run, which is what 'long run' describes, is the Strong Law (Theorem B.3), and Figure B.1 draws six such runs of a fair coin.

Let's motivate this all-important result. Suppose you have some probability distribution with a certain mean, μ, and variance, σ^2, which is assumed finite. Then you draw n independent r.v.s, $X_1, \ldots, X_n$, from the distribution so that the X_i's are independent and identically distributed (i.i.d. r.v.s). At this point, make sure you understand that each X_i is an exact copy of the parent r.v. you are 'drawing' these r.v.s from. Also, make sure to keep in mind that you're 'drawing' r.v.s and not values of r.v.s (i.e., numbers). You can think of this as carrying out a long experiment during which these r.v.s are drawn. Once you've drawn all of these r.v.s, you calculate their average that yields the **(theoretical) sample mean**, $\bar{X}_n$:

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}.$$

Question: Why 'theoretical' sample mean?

Answer: In order to avoid confusion with the descriptive statistic that also goes by 'sample mean', $\bar{X}_n$ is the theoretical sample mean because *it's a random variable*, not a number. The sample here, $X_1, \ldots, X_n$, is a sample of n r.v.s, each of which has the same distribution. If the values of the X_i's are realized, then the descriptive sample mean, the realized value, is easily calculated as:

$$\bar{x}_n = \frac{x_1 + \cdots + x_n}{n},$$

where the x_i's are the realized values of the X_i's. This explanation is given in the hopes that when you see $\bar{X}_n$ your mind doesn't go to the number $\bar{x}_n$. In the kiss experiment, $\bar{R}_n$ is the r.v. 'proportion turning right in a sample of n couples' and $\bar{r}_n = 80/124 = 0.645$ is the number the one sample happened to produce.

Notice that $\bar{X}_n$ is itself an r.v. because it's a function of r.v.s. The sample mean isn't the true mean, μ, of the parent distribution; that is, $\bar{X}_n \neq \mu$, although as demonstrated below $\mathbb{E}[\bar{X}_n] = \mu$. But, $\mu = \mathbb{E}[X_i]$ for all i and is a number (not an r.v.). This is how it should be since the distribution of every X_i is an exact copy of X's distribution. As such, the mean of X must equal the mean of the X_i's. The sample mean is the simplest and most natural way for trying to estimate the true mean, and the Weak Law of Large Numbers (WLLN) offers support to this notion.

Before explaining in more detail the WLLN consider some properties of the sample mean, $\bar{X}_n$. By linearity, its expectation is:

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{X_1 + \cdots + X_n}{n}\right] = \frac{n\mu}{n} = \mu.$$

The expectation $\mathbb{E}[\bar{X}_n]$ involves two different kinds of averaging, and it's worth separating them. The sample mean, $\bar{X}_n$, averages over the n draws of one experiment. The expectation, $\mathbb{E}[\bar{X}_n]$,

averages over *all* possible outcomes of that experiment, weighted by their probabilities, which is a theoretical average because you never observe all possible outcomes. The statement $\mathbb{E}[\bar{X}_n] = \mu$ says that the second kind of average of the first kind of average lands on the true mean. Figure 3.5 draws the two directions. Each row is one run of the experiment; averaging along the row gives that run's sample mean, a different number each time. The expectation averages down the column, over every possible run weighted by its probability; only eight runs are shown, and the expectation uses all 2^{10} of them. The first average is a random variable; the second is the number p.

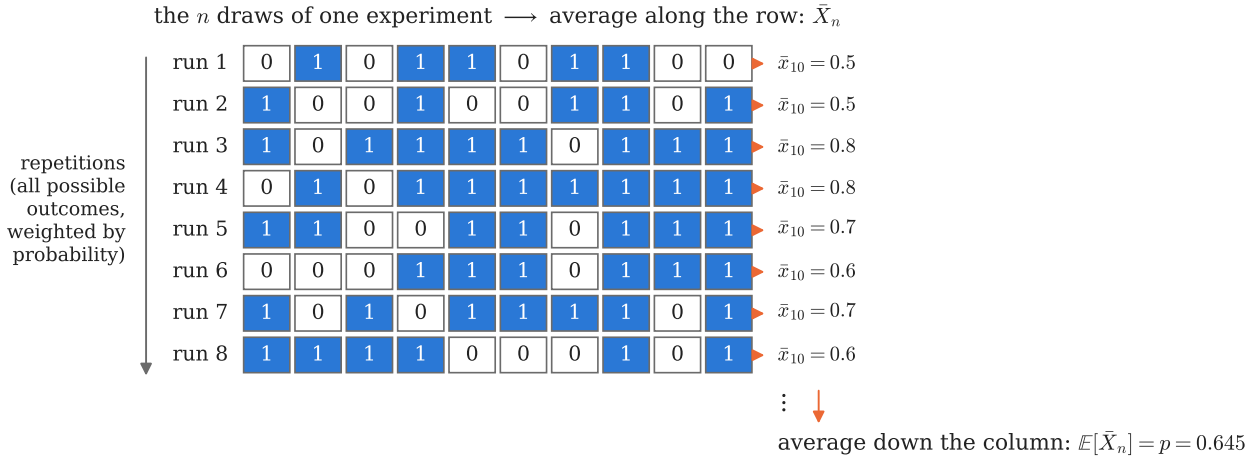


Figure 3.5: Eight runs of $n = 10$ Bernoulli draws with $p = 0.645$, one run per row, with each run's sample mean $\bar{x}_{10}$ at the right end of its row. The orange arrows mark the two averages: along a row and down the column.

Now, consider the variance of $\bar{X}_n$:

$$\mathbb{V}\mathrm{ar}(\bar{X}_n) = \frac{\mathbb{V}\mathrm{ar}(X_1 + \cdots + X_n)}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

This follows since the X_i's are independent so that the variance of the sum is the sum of the individual variances.

Intuition: why the variance of the sample mean is scaled by $1/n$

Averaging n independent draws cancels noise. Think of each X_i as $\mu + U_i$ where the U_i are independent, mean-zero errors. Then $\bar{X}_n = \mu + \bar{U}_n$, and the errors in $\bar{U}_n$ have no reason to line up: some are positive, some negative, and summing n of them produces a total whose typical size grows like $\sqrt{n}$, not like n (variances add, so standard deviations add in quadrature). Dividing by n then leaves a typical error of size $\sigma\sqrt{n}/n = \sigma/\sqrt{n}$. That's why $\mathbb{V}\mathrm{ar}(\bar{X}_n) = \sigma^2/n$ and why doubling the precision costs four times the data. Figure 3.6 is that claim drawn: the sum of n cancelling errors reaches about $\sqrt{n}$, not n. On the left, if the steps never cancelled, the sum after n steps would be n, which leaves the top of the figure by step 70; because they cancel, the sum stays inside a band that widens like $\sqrt{n}$. On the right, the simulated points sit on $\sqrt{n}$, which is exact here: $\mathbb{E}[S_n^2] = n$, since each step has variance 1 and the cross terms have expectation zero.

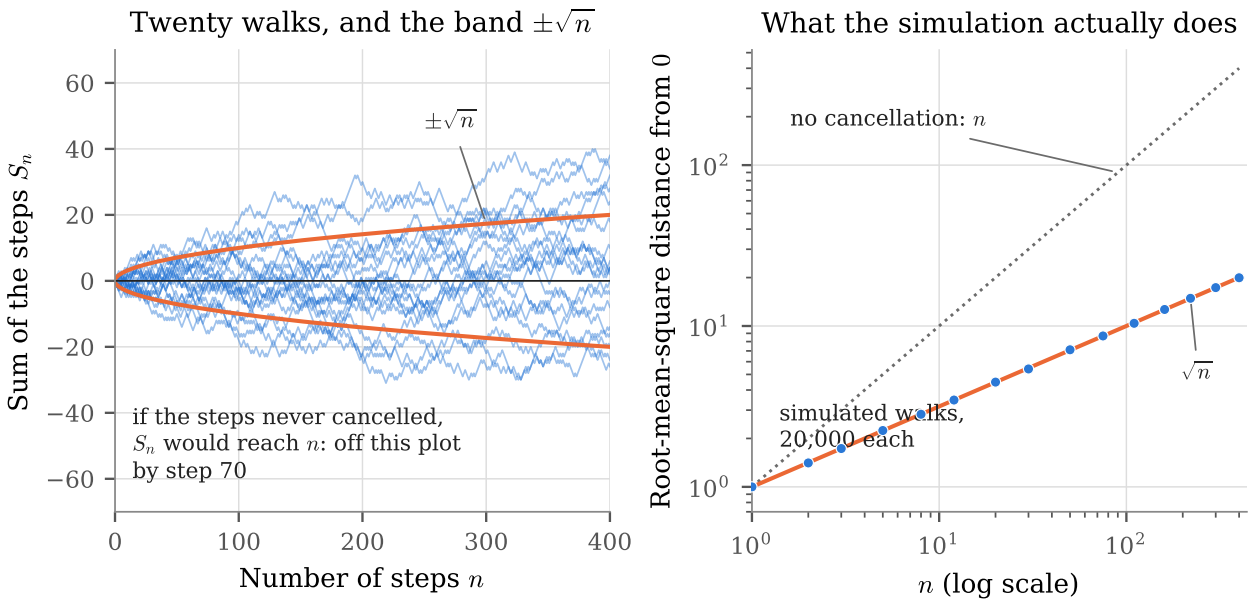


Figure 3.6: Left: twenty running sums S_n of equally likely ± 1 steps, with the band $\pm\sqrt{n}$ in orange. Right: the root-mean-square distance of S_n from zero over 20,000 simulated runs (blue dots), with $\sqrt{n}$ (orange) and n (dashed gray), on logarithmic axes.

Does this make sense? The cost of precision

Any time you produce a standard error, ask what it would take to halve it. The answer is always four times the data, and it follows from the $\sqrt{n}$: to replace $\sigma/\sqrt{n}$ by $\sigma/(2\sqrt{n})$ you need $\sqrt{n}$ to double, so n must quadruple. If a calculation of yours implies that twice the data halves the error, you have divided by n somewhere that should have been $\sqrt{n}$. For the kiss experiment, the standard error at $n = 124$ is 0.043; getting it to 0.02 takes not 250 couples but about 570.

Now, Chebyshev's inequality says that the probability that the distance of some r.v. from its mean is larger than some number, say ϵ, is bounded above by the variance of the r.v. divided by ϵ^2. In the present context, the variance is σ^2/n, and so the upper bound is $\dfrac{\sigma^2}{n\epsilon^2}$. Formally, this is expressed as:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}.$$

The WLLN states that if ϵ is a fixed number and you take the limit of this probability as $n \to \infty$, then the limit converges to 0. That is, the WLLN states that for any $\epsilon > 0$, as $n \to \infty$ you have:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) = \mathbb{P}\left(\left|\frac{X_1 + \cdots + X_n}{n} - \mu\right| \geq \epsilon\right) \to 0.$$

So the probability of falling far from the mean diminishes to zero as you draw more and more samples.

From undergraduate statistics: Chebyshev's rule

Some introductory courses teach Chebyshev's rule beside the empirical rule: whatever the shape of the data, at least 75% of the values lie within two standard deviations of the mean and at least 88.9% within three, against 95% and 99.7% for a normal shape. The rule is Chebyshev's inequality with $\epsilon = k\sigma$: at least $1 - 1/k^2$ of any distribution with finite variance lies within k standard deviations of its mean. The proof below is the rule applied to $\bar{X}_n$, whose standard deviation is the standard error $\sigma/\sqrt{n}$: a fixed distance ϵ from μ is $k = \epsilon\sqrt{n}/\sigma$ standard errors, k grows like $\sqrt{n}$, and the guaranteed share $1 - 1/k^2 = 1 - \sigma^2/(n\epsilon^2)$ climbs to 1. For the kiss data, $\epsilon = 0.084$ is 1.95 standard errors at $n = 124$, so at least 73 of every 100 samples of 124 couples land that close (the worked example below); four times as many couples double k to 3.9 and raise the guarantee to 93 in 100.

To make the interpretation of the WLLN more concrete, suppose again that you're conducting one long experiment, and during that experiment you draw n i.i.d. r.v.s, $X_1, \ldots, X_n$, from the same parent distribution. One way of thinking about the X_i is that each one equals the true mean, μ, plus some measurement noise, U_i, where $\mathbb{E}[U_i] = 0$ for all i and the U_i are independent. You can then express these noisy measurements as $X_i = \mu + U_i$ for $i = 1, \ldots, n$.

So you've got a collection of noisy measurements and then you compute their average. What the Weak Law of Large Numbers says is that the sample mean, $\bar{X}_n$, is unlikely to be far off from the true mean, μ, of the parent distribution. And by far off is meant at least ϵ distance away. The sample mean is, thus, a good estimator of the true mean.

Here's the proof of the WLLN, which is easily demonstrated with Chebyshev's inequality.

Proof strategy (core). Two lines: Chebyshev's inequality with $\mathbb{V}\text{ar}(\bar{X}_n) = \sigma^2/n$, then $n \to \infty$. This is the proof to know by heart.

Proof.

- Let ϵ be some positive real number. Then by Chebyshev's inequality:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\mathbb{V}\text{ar}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}.$$

- As $n \to \infty$, the right-hand side goes to 0, which implies $\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon)$ goes to 0, too.

□

Worked example: the kiss experiment and Chebyshev

In the kiss experiment $R_i \overset{iid}{\sim} \text{Bern}(p)$, so $\mu = p$ and $\sigma^2 = p(1-p)$. How likely is it that the sample proportion misses p by more than 0.084 (about 10 couples out of 124)? Chebyshev's inequality with $n = 124$, $\epsilon = 0.084$ and the worst case $p(1-p) \leq 1/4$ gives

$$\mathbb{P}(|\bar{R}_n - p| \geq 0.084) \leq \frac{p(1-p)}{n\epsilon^2} \leq \frac{0.25}{124 \times 0.084^2} \approx 0.29,$$

and with $p(1-p)$ evaluated at 0.645 the bound is 0.26. The bound is true but loose: the CLT (Chapter 4) says the probability is really about 0.05. Chebyshev's inequality is the right tool for proving that something converges, and the wrong tool for saying how fast.

Figure 4.5 in the next chapter shows all the bounds on the same axes.

From undergraduate statistics: consistency

In an introductory course you were told 'with a bigger sample the estimate gets closer to the truth'; the WLLN is that sentence with every word defined. An estimator that converges in probability to the parameter it estimates is called *consistent* (Definition 7.4). The WLLN says the sample mean is a consistent estimator of the population mean, and the sample proportion is a consistent estimator of the population proportion. Note what it does not say: it doesn't say $\bar{X}_n$ equals μ for any finite n, and it doesn't say how big n must be for a given accuracy. The CLT and Hoeffding's inequality, both in Chapter 4, handle the second question.

3.5 What comes next

The Law of Large Numbers says where $\bar{X}_n$ ends up: at μ, with the variance σ^2/n shrinking to nothing. That is the left of Figure 3.2, and it leaves the obvious question open. If every density on that side is on its way to becoming a spike, what does $\bar{X}_n$ look like *along the way*? The right of the figure already gave the answer in a picture: magnify the error by $\sqrt{n}$ and its shape settles down to a Gaussian, whatever the parent distribution. Making that picture into a theorem needs the second mode of convergence, and it's the subject of the next chapter.

3.6 Summary

- A sequence of r.v.s is a sequence of functions on the sample space; one outcome fixes one ordinary sequence of numbers. No single index N works for every outcome, so convergence has to be defined through probabilities.
- 'Asymptotic' means: for the n at hand, replace the unknown sampling distribution by its limit, usually a Gaussian, and carry on; the theorem does not say how large n must be.
- $\bar{X}_n$ is a random variable, the theoretical sample mean; $\bar{x}_n$ is the number one sample produced. Every theorem is about the first. Its expectation is a second average, over all possible samples, of the first.
- **Convergence in probability:** $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$ if $\mathbb{P}(|X_n - X| \geq \epsilon) \to 0$ for every $\epsilon > 0$. The mass outside any band around the limit disappears.
- **Shrinking-variance criterion:** $\mathbb{E}[T_n] \to \mu$ and $\mathbb{V}\mathrm{ar}(T_n) \to 0$ imply $T_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$. This is Chebyshev's inequality, and it's how consistency is usually proved.
- **Weak Law of Large Numbers:** $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$ whenever the variance is finite because $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathbb{V}\mathrm{ar}(\bar{X}_n) = \sigma^2/n$. Estimators built from averages are consistent.
- Chebyshev's bound $\sigma^2/(n\epsilon^2)$ is a guarantee at every n but a loose one: 0.26 in the kiss experiment where the truth is 0.05.

3.7 Exercises

Exercise 3.1. Let $X_n \sim \mathrm{Bern}(1/n)$. (a) For $0 < \epsilon < 1$, what is $\mathbb{P}(|X_n| > \epsilon)$? (b) Does (X_n) converge in probability, and to what? (c) Does it converge in distribution? (d) What is $\lim_n \mathbb{E}[\cos(X_n)]$?

(Solution on page 415.)

Exercise 3.2. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Expo}(\lambda)$, so $\mathbb{E}[X_1] = 1/\lambda$ and $\mathbb{V}\text{ar}(X_1) = 1/\lambda^2$. Use the shrinking-variance criterion to show that $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} 1/\lambda$, and find the smallest n for which Chebyshev's inequality guarantees $\mathbb{P}(|\bar{X}_n - 1/\lambda| \geq 0.1/\lambda) \leq 0.05$. *(Solution on page 415.)*

Exercise 3.3. In the kiss model with $p = 0.645$, how many couples would Chebyshev's inequality require before $\mathbb{P}(|\bar{R}_n - p| \geq 0.02) \leq 0.05$? Is the answer sensitive to p? *(Solution on page 415.)*

Chapter 4

The Central Limit Theorem

In this chapter

- Convergence in distribution, the mode of convergence that compares CDFs rather than values, and why its definition only looks at continuity points.
- The Central Limit Theorem in its three equivalent forms, and what it says at the sample sizes you actually have.
- Hoeffding's inequality: a finite-sample guarantee for bounded data when n is too small to trust the CLT.
- What the limit theorems buy you, and which of their conclusions could be reached without limits.

Chapter 3 ended with $\bar{X}_n$ collapsing onto μ and the question of what it looks like on the way there. This chapter answers it.

4.1 Convergence in distribution

Convergence in distribution is weaker than convergence in probability (and, in the full hierarchy of Appendix B, the weakest mode there is). All it says is that the CDF of X_n converges to the CDF of X as n goes to infinity, *but doesn't require any dependence between the X_n's and X*. That is, the r.v.s X_n and X can be defined on entirely different probability spaces. The CDF of these r.v.s is used instead of a PDF or PMF because all r.v.s, whether they be discrete, continuous or mixed, have CDFs.

Question: Why does not requiring any dependence between the X_n's and X make convergence in distribution weak? You might think that, by not introducing a dependence assumption, you're actually making the statement stronger.

Answer: By not imposing any dependence structure between X_n and X, convergence in distribution allows for a wider range of scenarios where the CDFs of X_n converge to the CDF of X. It doesn't guarantee that the actual values of X_n get closer to X as n increases, which is a requirement for stronger forms of convergence like convergence in probability or convergence almost surely. Convergence in distribution only concerns the limiting behavior of the distributions, not the actual values of the r.v.s. A statement that holds in more scenarios is a weaker statement because it tells you less about any one of them.

Definition 4.1 (Convergence in distribution)**.** A sequence $(X_n)_{n\geq 1}$ of r.v.s **converges in distribution** to an r.v. X, denoted $X_n \xrightarrow[n\to\infty]{(\mathrm{d})} X$, if:

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x),$$

for all x at which F_X is continuous.

> **In words.** For each fixed x, $F_{X_n}(x) = \mathbb{P}(X_n \leq x)$ is a number, and the definition asks that this sequence of numbers converge to $\mathbb{P}(X \leq x)$, at every x where the limit's CDF has no jump (toolkit A.6). It compares two distributions, not two random variables: nothing is said about X_n and X being close on any outcome, and they need not even live on the same sample space.
>
> **For the kiss data.** The CLT will say that the standardized error $\sqrt{n}(\bar{R}_n - p)/\sqrt{p(1-p)}$ converges in distribution to Gauss$(0,1)$: for every x, the probability that the standardized error is below x approaches $\Phi(x)$. With $n = 124$ that is the license to compute $\mathbb{P}(|\bar{R}_n - p| \leq 0.084) \approx \Phi(1.95) - \Phi(-1.95) = 0.95$.

4.1.1 Why only the continuity points?

The qualifier 'at which F_X is continuous' looks like a technicality. It isn't; without it the most basic examples fail.

Example 4.2. Let $T_n \sim \mathrm{Unif}\left(\left[5 - \frac{1}{2n}, 5 + \frac{1}{2n}\right]\right)$, a uniform on an interval of width $1/n$ centered at 5. Every sensible notion of convergence should say $T_n \to 5$, and indeed $T_n \xrightarrow[n\to\infty]{\mathbb{P}} 5$ since $\mathbb{P}(|T_n - 5| > \delta) = 0$ as soon as $\frac{1}{2n} \leq \delta$. The CDF F_n of T_n is 0 to the left of the interval, rises linearly to 1 across it, and in particular $F_n(5) = \frac{1}{2}$ for *every* n. The CDF of the constant 5 is the step function $F(t) = \mathbb{1}(t \geq 5)$, with $F(5) = 1$. So $F_n(t) \to F(t)$ for every $t \neq 5$ but not at $t = 5$, the one point where F jumps. With the continuity-point qualifier, $T_n \xrightarrow[n\to\infty]{(\mathrm{d})} 5$; without it, T_n wouldn't converge in distribution to anything. Figure 4.1 shows the picture: the F_n close in on the step function everywhere except at the jump, where all of them pass through $\frac{1}{2}$, and the definition of convergence in distribution excludes exactly that point.

The general principle: a jump in F_X at x means X has positive probability of equaling x exactly, and a sequence approaching from the left or the right can straddle that jump forever. Away from the jumps there's nothing to straddle.

4.1.2 Two equivalent definitions

Alternatively, convergence in distribution can be defined using the expectation of a function, provided the function is continuous and bounded. This is the form that generalizes: on spaces where there is no CDF to write down, it becomes the definition of convergence in distribution.

Proposition 4.3 (Equivalent characterizations)**.** *For real-valued r.v.s the following are equivalent:*

1. $F_{X_n}(x) \to F_X(x)$ *for all* x *at which* F_X *is continuous;*
2. $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ *for every continuous and bounded function* $f : \mathbb{R} \to \mathbb{R}$.

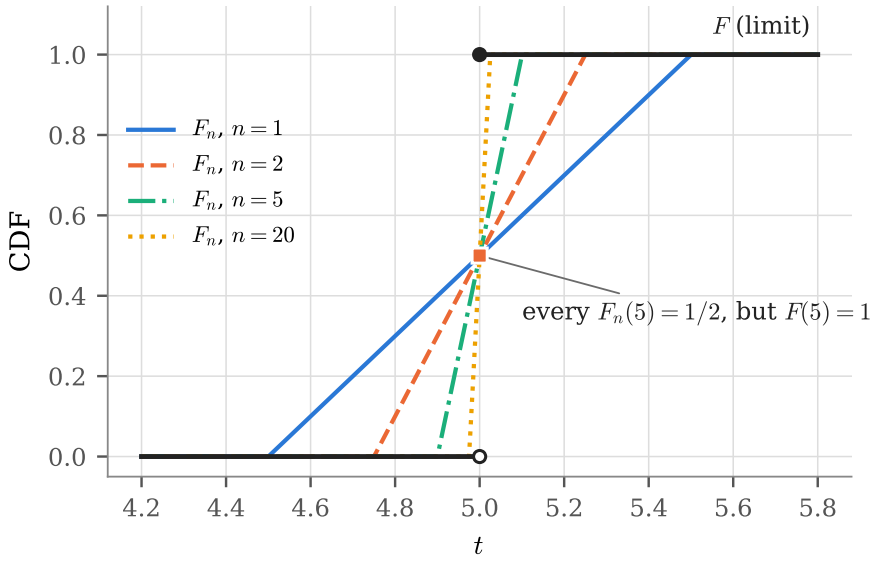


Figure 4.1: The CDFs F_n of $T_n \sim \text{Unif}([5-\frac{1}{2n}, 5+\frac{1}{2n}])$ for $n = 1, 2, 5, 20$, and the CDF F of the constant 5 in black. The orange square marks the point $(5, \frac{1}{2})$ that every F_n passes through.

If moreover X has a density (for instance X is Gaussian), then $X_n \xrightarrow[n\to\infty]{(d)} X$ implies convergence of interval probabilities:

$$\mathbb{P}(a \le X_n \le b) \to \mathbb{P}(a \le X \le b), \qquad \textit{for all } a < b.$$

Note that in (2) the things converging are numbers, $\mathbb{E}[f(X_n)]$, so the convergence is ordinary convergence of numbers. The second form is the useful one in proofs because it doesn't single out any particular x; the first form is the useful one in computations because CDFs are what you can write down. The last statement is why the CLT is enough to compute probabilities: $\mathbb{P}(a \le Z_n \le b) \approx \Phi(b) - \Phi(a)$, which is what every asymptotic confidence interval and test rests on.

One implication between the two modes is used on nearly every page from here on, so here it is as a statement with a proof.

Proposition 4.4 (In probability implies in distribution)**.** *If $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$, then $X_n \xrightarrow[n\to\infty]{(d)} X$.*

Proof strategy (for completeness; skip on a first reading). Sandwich $F_{X_n}(x)$ between $F_X(x-\epsilon)$ and $F_X(x+\epsilon)$, up to the probability that X_n and X differ by more than ϵ, which goes to 0; then let $\epsilon \to 0$ and use continuity of F_X at x. The statement is what matters; the proof is a standard ϵ argument.

Proof. Fix a continuity point x of F_X and an $\epsilon > 0$. If $X_n \le x$ then either $X \le x+\epsilon$ or $|X_n - X| > \epsilon$, so

$$F_{X_n}(x) \le F_X(x+\epsilon) + \mathbb{P}(|X_n - X| > \epsilon).$$

In the same way, if $X \le x - \epsilon$ then either $X_n \le x$ or $|X_n - X| > \epsilon$, so

$$F_X(x-\epsilon) - \mathbb{P}(|X_n - X| > \epsilon) \le F_{X_n}(x).$$

Let $n \to \infty$: the probability terms vanish by convergence in probability, and $F_{X_n}(x)$ is squeezed, in the limit, between $F_X(x-\epsilon)$ and $F_X(x+\epsilon)$. Now let $\epsilon \to 0$: both bounds tend to $F_X(x)$ because F_X is continuous at x. □

The continuity-point qualifier is what the last line uses: at a jump of F_X the two bounds would not meet, and Example 4.2 shows the conclusion can indeed fail there.

Example 4.5. In Example 3.4, $X_n \sim \text{Bern}(1/n)$ converges in probability, hence in distribution, to 0. The function $f(x) = \cos x$ is continuous and bounded, so $\mathbb{E}[\cos(X_n)] \to \cos(0) = 1$. You can check this directly: $\mathbb{E}[\cos X_n] = (1 - \frac{1}{n})\cos 0 + \frac{1}{n}\cos 1 \to 1$.

4.1.3 Example 1: i.i.d. random variables

If $(X_n)_{n\geq 1}$ is a sequence of i.i.d. r.v.s, then convergence in distribution isn't very interesting because they converge to the CDF of any one r.v. in the sequence. That is, if $X_1, X_2, \ldots$ is a sequence of i.i.d. r.v.s with CDF $F_X(x)$, then $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$. This is because:

$$F_{X_n}(x) = F_X(x), \text{ for all } x \text{ implies that } \lim_{n\to\infty} F_{X_n}(x) = F_X(x), \text{ for all } x.$$

This is the example from Section 3.1: the sequence of standard Gaussians converges in distribution to a standard Gaussian even though $\mathbb{P}(X_n = X) = 0$. Convergence in distribution is the mode that makes sense of that example.

4.1.4 Example 2: the maximum of uniforms

Example 4.6. Let $X_1, X_2, \ldots \overset{iid}{\sim} \text{Unif}([0,1])$ and define the maximum of the first n by $X_{(n)} = \max_{i=1,\ldots,n} X_i$. Then:

$$n(1 - X_{(n)}) \xrightarrow[n\to\infty]{\text{(d)}} \text{Expo}(1).$$

Before the proof, a word on why the maximum is being magnified by n rather than $\sqrt{n}$. The maximum of n uniforms piles up just below 1; by the same argument as in Example 3.2 it converges in probability to 1, so the unmagnified sequence has a degenerate limit. The gap $1 - X_{(n)}$ is of order $1/n$ (with n points spread over $[0,1]$ the largest one sits about $1/n$ from the top), so n is the magnification that holds its size fixed. The CLT's $\sqrt{n}$ is for sums; a maximum is not a sum and gets a different rate and a different (non-Gaussian) limit.

Proof strategy (core). Write the CDF of $n(1-X_{(n)})$ as a probability about the whole sample, use independence to get a power $(1 - x/n)^n$, and recognize the limit e^{-x} (toolkit A.10). Same template as Example 3.2, with a CDF in place of a tail probability.

Proof.

- Define the CDF of the r.v. $n(1 - X_{(n)})$ at a point $x > 0$:

$$F_{n(1-X_{(n)})}(x) = \mathbb{P}(n(1 - X_{(n)}) \leq x).$$

- Now, simplify the CDF in such a way as to eventually take the limit as $n \to \infty$.

- To that end, observe that:

$$\begin{aligned}
\mathbb{P}(n(1-X_{(n)}) \leq x) &= \mathbb{P}(n - nX_{(n)} \leq x) \\
&= \mathbb{P}\left(X_{(n)} \geq 1 - \frac{x}{n}\right) \\
&= 1 - \mathbb{P}\left(X_{(n)} < 1 - \frac{x}{n}\right) \\
&= 1 - \mathbb{P}\left(X_1 < 1 - \frac{x}{n}, X_2 < 1 - \frac{x}{n}, \ldots, X_n < 1 - \frac{x}{n}\right) \\
&= 1 - \mathbb{P}\left(X_1 < 1 - \frac{x}{n}\right)\mathbb{P}\left(X_2 < 1 - \frac{x}{n}\right)\cdots\mathbb{P}\left(X_n < 1 - \frac{x}{n}\right) \\
&= 1 - \mathbb{P}\left(X_1 < 1 - \frac{x}{n}\right)^n.
\end{aligned}$$

  The movement from the third to the fourth equality holds because, as $X_{(n)}$ is the maximum of the X_i's, the maximum is below a level exactly when every X_i is below it: the single event is equivalent to the joint event. The fifth equality results since the X_i's are independent.

- Now, recall that the X_i's, and thus X_1, are uniformly distributed on $(0, 1)$. This means its CDF is linear, emanating from the origin rising to 1 with a slope of 1; in other words, plug any value of the support into the CDF and you get the same value back. Hence, for n large enough that $1 - x/n \in (0, 1)$:

$$\mathbb{P}\left(X_1 < 1 - \frac{x}{n}\right) = 1 - \frac{x}{n}.$$

- Returning to the last right-hand side from above, you have that:

$$1 - \mathbb{P}\left(X_1 < 1 - \frac{x}{n}\right)^n = 1 - \left(1 - \frac{x}{n}\right)^n,$$

  which by taking the limit as $n \to \infty$ gives:

$$\lim_{n\to\infty}\left[1 - \left(1 - \frac{x}{n}\right)^n\right] = 1 - e^{-x},$$

  using the calculus fact $(1 - x/n)^n \to e^{-x}$.

- To summarize, then, you have:

$$\lim_{n\to\infty} \mathbb{P}(n(1 - X_{(n)}) \leq x) = 1 - e^{-x} \quad \text{for every } x > 0,$$

  and $1 - e^{-x}$ is the CDF of the Exponential with $\lambda = 1$ (for $x \leq 0$ both sides are 0). This proves the result.

□

Figure 4.2 shows the result by simulation. On the left, the maximum of $n = 50$ uniforms crowds the top of the interval. On the right, the magnified gap $n(1 - X_{(n)})$ is already very close to Expo(1) at $n = 50$.

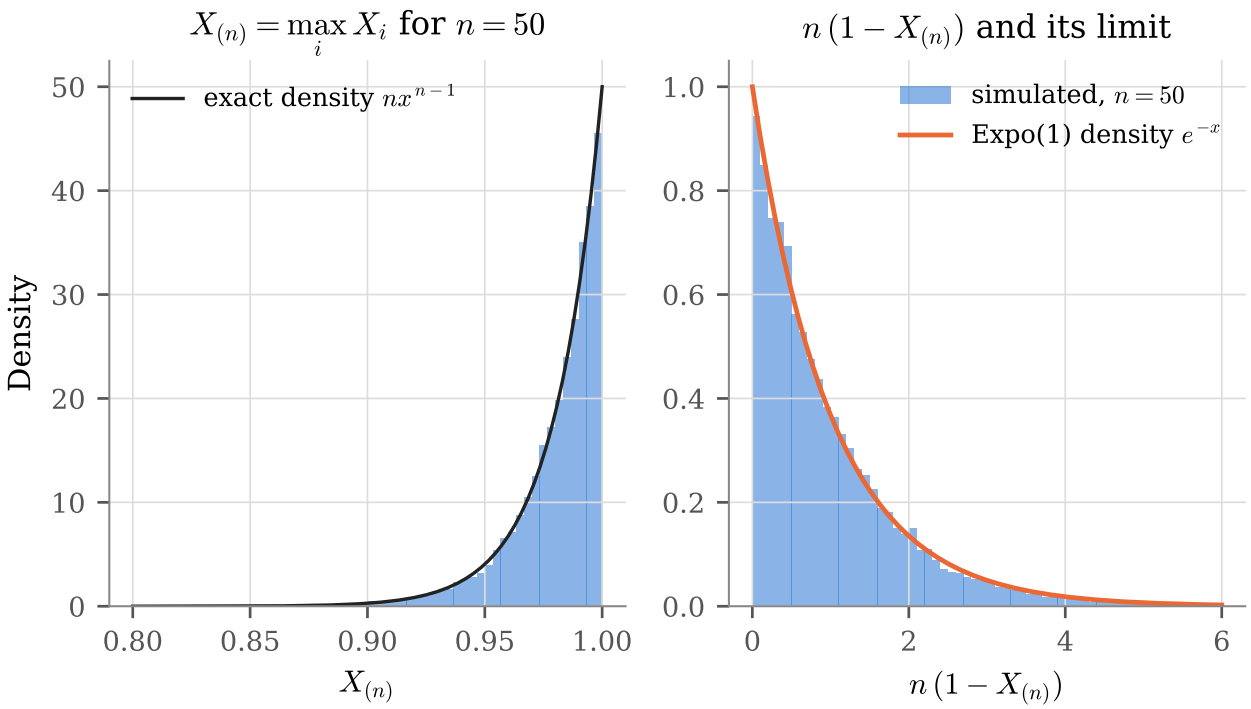


Figure 4.2: Example 4.6 by simulation, 20,000 replications at $n = 50$. Left: histogram of the maximum $X_{(n)}$ of n uniforms, with its exact density nx^{n-1} in black. Right: histogram of $n(1-X_{(n)})$, with the Expo(1) density in orange.

Pitfall: convergence in distribution says nothing about closeness

Let $Z \sim \text{Gauss}(0,1)$ and set $X_n = -Z$ for every n. By the symmetry of the Gaussian, $X_n \sim \text{Gauss}(0,1)$ for every n, so $X_n \xrightarrow[n\to\infty]{(\text{d})} Z$. But $|X_n - Z| = 2|Z|$, which is not small and never becomes small. So X_n does not converge in probability to Z, even though it converges in distribution to it. Convergence in distribution is a statement about two CDFs, not about two columns of numbers getting closer. The same example shows why you can't add limits in distribution: $X_n + Z = 0$ for every n, while the sum of two independent Gauss$(0,1)$ r.v.s is Gauss$(0,2)$. Knowing the limiting distributions of X_n and Y_n separately doesn't determine the limiting distribution of $X_n + Y_n$; you need to know how they're related. Chapter 5 says what you can do instead.

4.1.5 Limits that are constants

One case of convergence in distribution is used constantly later, so here it is as a statement with a proof.

Proposition 4.7. *If $X_n \xrightarrow[n\to\infty]{(d)} c$ for a constant c, then $X_n \xrightarrow[n\to\infty]{\mathbb{P}} c$.*

Proof strategy (for completeness; skip on a first reading). The CDF of a constant is a step; convergence of the CDFs at two continuity points, one on each side of c, squeezes the mass outside the ϵ-band to 0.

Proof. The CDF of the constant c is the step $F(x) = \mathbb{1}(x \geq c)$, continuous everywhere except at c. Fix $\epsilon > 0$. The points $c - \epsilon$ and $c + \epsilon/2$ are continuity points, so $F_{X_n}(c - \epsilon) \to F(c - \epsilon) = 0$ and $F_{X_n}(c + \epsilon/2) \to F(c + \epsilon/2) = 1$. Therefore

$$\mathbb{P}(|X_n - c| \geq \epsilon) = \mathbb{P}(X_n \leq c - \epsilon) + \mathbb{P}(X_n \geq c + \epsilon) \leq F_{X_n}(c - \epsilon) + (1 - F_{X_n}(c + \epsilon/2)) \to 0 + 0. \quad \square$$

For a constant limit, then, 'in probability' and 'in distribution' are the same thing, which is why the Law of Large Numbers could equally be written with the arrow for convergence in distribution, and why Slutsky's theorem in Chapter 5 needs only convergence in probability for the sequence with the constant limit.

4.2 The Central Limit Theorem

As in the previous section, let $X_1, X_2, \ldots$ be i.i.d. r.v.s with mean μ and variance σ^2. The Law of Large Numbers says that as $n \to \infty$, $\bar{X}_n$ converges to the constant μ. But what is $\bar{X}_n$'s distribution along the way to becoming the constant μ? This is addressed by the Central Limit Theorem (CLT), an all-important theorem in statistics.

The CLT states that for large n, the distribution of $\bar{X}_n$ *after standardization* approaches a standard Gaussian distribution. Recall that if you standardize an r.v., the resulting r.v. is measured in units of standard deviation. In the present case, the convergence process examines the typical deviations of $\bar{X}_n$ around μ. Now the statement of the CLT.

Theorem 4.8 (Central Limit Theorem)**.** *Let $X_1, \ldots, X_n$ be i.i.d. r.v.s such that $\mathbb{E}[X_i] = \mu$ and $\mathbb{V}ar(X_i) = \sigma^2 < \infty$ for $i = 1, \ldots, n$. Then:*

$$\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \xrightarrow[n\to\infty]{(d)} \mathit{Gauss}(0, 1).$$

In words, the CDF of the left-hand side converges to Φ, the CDF of the standard Gaussian distribution.

An equivalent statement of the CLT in which $\bar{X}_n$ remains mean-centered by μ but omits the standardization by σ is given as follows:

$$\sqrt{n}\left(\bar{X}_n - \mu\right) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, \sigma^2).$$

Question 1: Why does this second statement hold?

Answer 1: It holds because of the way the Gaussian family behaves under scaling. Let $X \sim \mathrm{Gauss}(0, 1)$, so its PDF is:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right).$$

Now let $Y = \sigma X$ for some $\sigma > 0$. Then $\mathbb{E}[Y] = \mathbb{E}[\sigma X] = \sigma\,\mathbb{E}[X] = 0$. Hence the mean remains 0. But note what happens to the variance of X after multiplying it by σ:

$$\begin{aligned}\mathbb{V}\mathrm{ar}(Y) = \mathbb{E}[(Y - \mathbb{E}[Y])^2] &= \mathbb{E}[(\sigma X - 0)^2] \\ &= \sigma^2\,\mathbb{E}[X^2] \\ &= \sigma^2\,\mathbb{V}\mathrm{ar}(X) \\ &= \sigma^2,\end{aligned}$$

and Y is still Gaussian (a linear function of a Gaussian is Gaussian), so $\sigma X \sim \mathrm{Gauss}(0, \sigma^2)$. Multiplying both sides of the first statement by the constant σ therefore gives the second; the step 'multiply both sides of a limit in distribution by a constant' is justified by the continuous mapping theorem of Chapter 5.

Question 2: In the statement of the CLT the r.v. $\sqrt{n}\left(\dfrac{\bar{X}_n - \mu}{\sigma}\right)$ converges to the standard Gaussian distribution. The term in parentheses standardizes $\bar{X}_n$; that is, the mean, μ, is subtracted from $\bar{X}_n$, and the result divided by the standard deviation, σ. But why multiply this standardized term by $\sqrt{n}$?

Answer 2: Because σ is the standard deviation of a single X_i, not of $\bar{X}_n$. The standard deviation of the sample mean equals $\sigma/\sqrt{n}$.[1] Standardizing $\bar{X}_n$ properly means dividing by *its* standard deviation, and division by a fraction is multiplication by its reciprocal:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right).$$

So the $\sqrt{n}$ isn't an extra factor; it's the part of the standardization that's easy to lose track of.

A third and equivalent statement of the CLT uses a standardized sum. Define the standardized sum of $X_1 + \cdots + X_n$ by:

$$Z_n = \frac{n\bar{X}_n - n\mu}{\sqrt{n}\sigma},$$

where $n\bar{X}_n$ is just $X_1 + \cdots + X_n$. As Z_n is standardized it follows that $\mathbb{E}(Z_n) = 0$ and $\mathbb{V}\mathrm{ar}(Z_n) = 1$. Further, let Z be a standard Gaussian r.v. Then the CLT states that:

$$\lim_{n\to\infty} \mathbb{P}(Z_n \le z) = \mathbb{P}(Z \le z), \text{ for every } z,$$

where $\mathbb{P}(Z \le z)$ is the standard Gaussian CDF $\Phi(z)$. The reason why the CLT states that the CDF of Z_n (and not its PDF) converges to the standard Gaussian CDF is because the result applies to *all* r.v.s with finite variance, and all r.v.s have CDFs. The X_i can be discrete, as in the kiss experiment, in which case Z_n has no PDF at all.

Question 3: Why do you standardize the sums of the X_i's for the approximation?

[1] To see why, let $X_1, \ldots, X_n$ be i.i.d. r.v.s. By independence, you have that:

$$\mathbb{V}\mathrm{ar}(\bar{X}_n) = \mathbb{V}\mathrm{ar}\left(\frac{X_1}{n} + \cdots + \frac{X_n}{n}\right) = \frac{\sigma^2}{n^2} + \cdots + \frac{\sigma^2}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Answer 3: Suppose the $X_1 + \cdots + X_n$ aren't standardized. Then $\mathbb{E}(X_1 + \cdots + X_n) = n\mu$ and $\mathbb{V}\mathrm{ar}(X_1 + \cdots + X_n) = n\sigma^2$. Both are finite for every n, but neither stays bounded as $n \to \infty$: the unstandardized sum drifts off (its mean is $n\mu$) and spreads out (its standard deviation is $\sigma\sqrt{n}$), so no fixed distribution can describe it and its CDF has no limit that is a CDF. Standardizing pins the sum down, with $\mathbb{E}(Z_n) = 0$ and $\mathbb{V}\mathrm{ar}(Z_n) = 1$ for all n, so that a limiting distribution has a chance to exist.

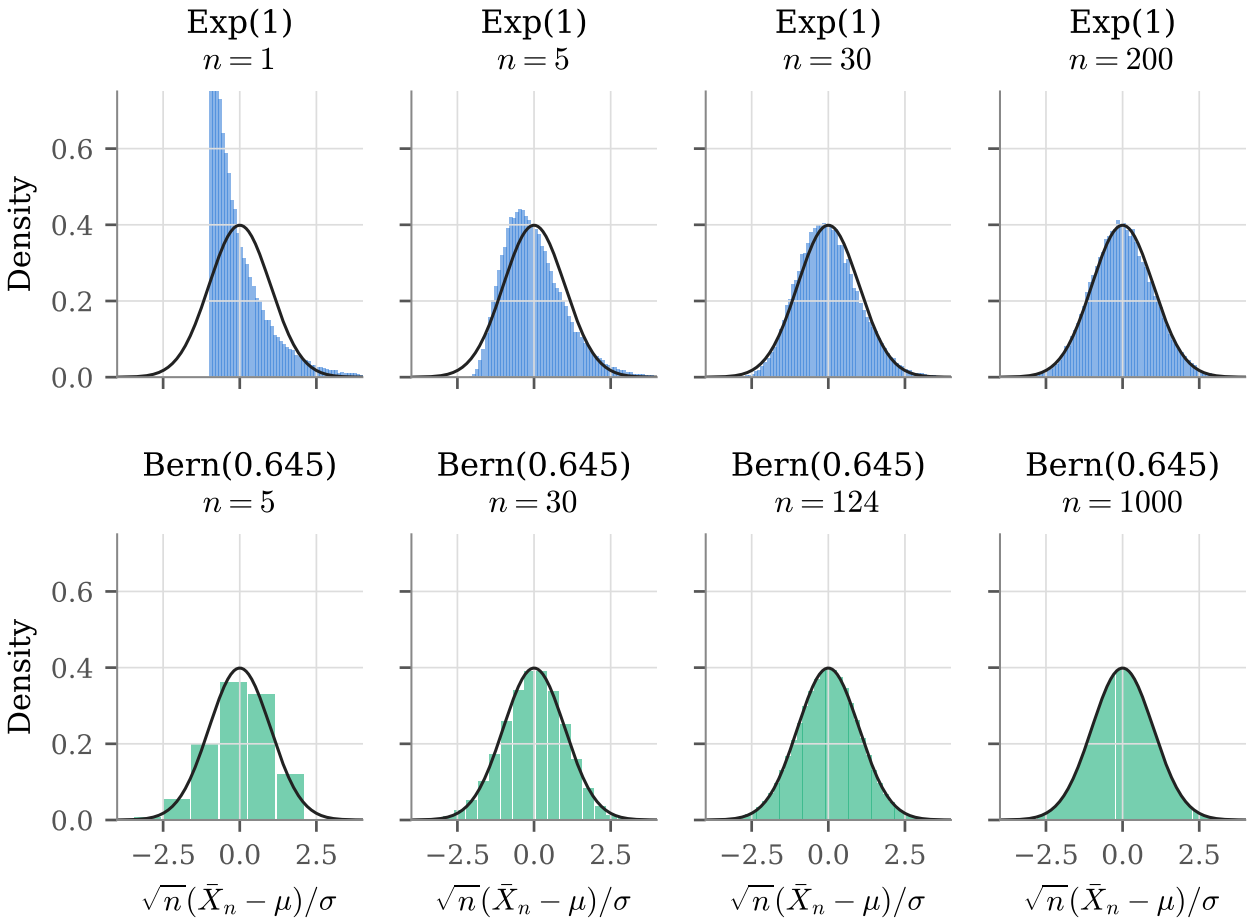


Figure 4.3: The standardized sample mean $\sqrt{n}(\bar{X}_n - \mu)/\sigma$, with the Gauss$(0, 1)$ density in black over it. Top row: Expo(1) draws at $n = 1, 5, 30, 200$, histograms from 50,000 simulated samples. Bottom row: Bern(0.645) draws at $n = 5, 30, 124, 1000$, the exact Binomial distribution as bars.

From undergraduate statistics: the '$n \geq 30$' rule

Introductory courses offer a rule of thumb: the normal approximation to the sample mean is fine once $n \geq 30$. The CLT contains no such number. It's a limit statement, and how fast the limit is reached depends on the parent distribution: above all on how asymmetric and heavy-tailed it is. In the top row of Figure 4.3 the parent is the strongly skewed Expo(1), and the skew is still visible at $n = 30$. In the bottom row the parent is the kiss experiment's Bern(0.645); the standardized mean lives on a lattice, visible as bars, and by $n = 124$ the Gaussian is already an excellent description. For a symmetric parent, $n = 5$ can be enough; for a heavily skewed one, $n = 200$ may not be. What is a theorem is the *rate*: the Berry–Esseen bound says the gap between the CDF of Z_n and Φ is at most a universal constant times $\mathbb{E}|X - \mu|^3/(\sigma^3\sqrt{n})$, so the

error shrinks like $1/\sqrt{n}$, slowly when the ratio $\mathbb{E}|X-\mu|^3/\sigma^3$ is large, as it is for skewed and heavy-tailed parents. A finer expansion (the Edgeworth expansion) shows that, for a continuous parent, the leading term of the error is proportional to the parent's skewness, which is why symmetric parents reach the Gaussian faster in practice.

4.2.1 The essence of the CLT: a 'stochastic' bound

Due to the so-called 'empirical rule', values between -3 and 3 in the support of a standard Gaussian account for nearly 100% of its probability mass (99.7% of it). By the CLT, this implies that, for n large enough for the approximation to apply, the following inequality holds with probability about 99.7%:

$$|\bar{X}_n - \mu| \leq \frac{3\sigma}{\sqrt{n}},$$

since, for every n, the two events are the same:

$$\begin{aligned}
\sqrt{n}\left(\frac{\bar{X}_n-\mu}{\sigma}\right) \in [-3,3] &\Leftrightarrow -3 \leq \sqrt{n}\left(\frac{\bar{X}_n-\mu}{\sigma}\right) \leq 3 \\
&\Leftrightarrow \left|\sqrt{n}\left(\frac{\bar{X}_n-\mu}{\sigma}\right)\right| \leq 3 \\
&\Leftrightarrow \frac{\sqrt{n}}{\sigma}|\bar{X}_n-\mu| \leq 3 \\
&\Leftrightarrow |\bar{X}_n-\mu| \leq \frac{3\sigma}{\sqrt{n}}.
\end{aligned}$$

The essence of the CLT is that, despite $\bar{X}_n - \mu$ being an r.v. with an unbounded support (when the X_i are unbounded), it's *stochastically* bounded in the sense that, with high probability, it's between $-3\sigma/\sqrt{n}$ and $3\sigma/\sqrt{n}$. Replace 3 by 1.96 and 'with high probability' by 95% and you have the recipe for every confidence interval in Chapter 8. Figure 4.4 draws the bound as a funnel around the running average, and it is also the chapter's two theorems in one picture. The funnel narrows like $1/\sqrt{n}$: that is the Law of Large Numbers. At any fixed n past the first few dozen, about 5% of the paths are outside the inner funnel and almost none outside the outer one (in this simulation, 6% of the points with $n \geq 30$ are outside the inner funnel): that is the stochastic bound. The inner funnel at $n = 124$ is the width of the confidence interval of Chapter 8.

From undergraduate statistics: the '$np \geq 10$ and $n(1-p) \geq 10$' condition

An introductory course says the sample proportion is approximately normal, with mean p and standard deviation $\sqrt{p(1-p)/n}$, once $np \geq 10$ and $n(1-p) \geq 10$ (some books say 5, some 15). That is Theorem 4.8 for Bern(p) data, with $\mu = p$ and $\sigma^2 = p(1-p)$; the inner funnel of Figure 4.4 is p plus or minus 1.96 of those standard deviations. The condition is stated in counts rather than in n because the Bernoulli parent's skewness, $(1-2p)/\sqrt{p(1-p)}$, grows without bound as p nears 0 or 1, and the skewness of the count is that divided by $\sqrt{n}$. With both counts at least 10 the count's skewness is at most $1/\sqrt{10} = 0.32$ in size, whatever p is; at $p = 0.01$ the condition asks for $n = 1{,}000$, where the '$n \geq 30$' rule would accept $n = 30$ and a skewness of 1.8. For the kiss data the counts, with $\hat{p}$ in place of p, are the observed 80 and 44, and the count's skewness is -0.05. Section 9.6 takes up the continuity correction, which repairs

the count's discreteness rather than its skewness.

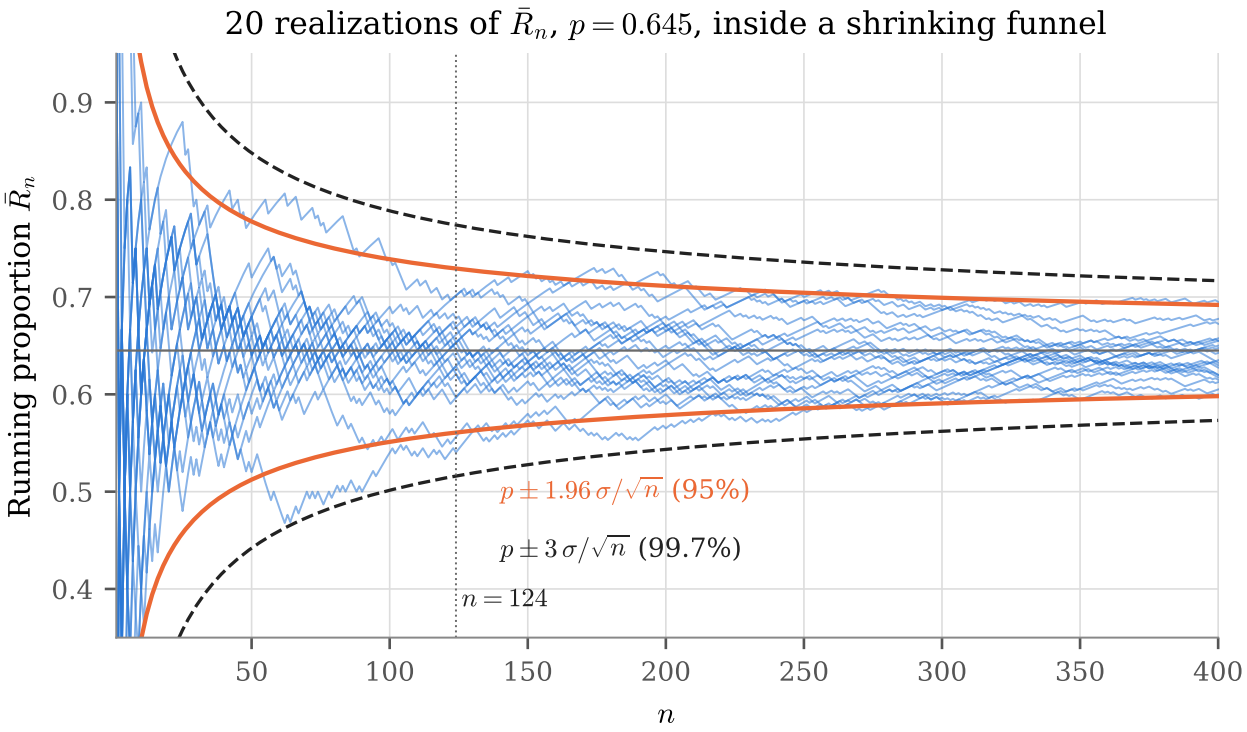


Figure 4.4: Twenty realizations of the running proportion $\bar{R}_n$ in the kiss model, $p = 0.645$, with the curves $p \pm 1.96\,\sigma/\sqrt{n}$ (orange) and $p \pm 3\,\sigma/\sqrt{n}$ (dashed black), where $\sigma = \sqrt{p(1-p)}$. The dotted line marks $n = 124$.

4.3 Hoeffding's inequality: help for small sample sizes

What if your sample size, n, is not large enough to apply the CLT? Hoeffding's inequality is a helpful tool when n is small that effectively delivers the same essential message as the CLT: $\bar{X}_n$ is close to μ within some stochastic bound. The statement isn't as strong as the CLT and requires additional restrictions on the r.v.s. Namely, the r.v.s must be bounded almost surely, that is, with probability one (toolkit A.5), which rules out any r.v. with an arbitrarily large or small support (e.g., Gaussian and Exponential r.v.s). But despite the stronger restrictions, its power lies in providing a bound on the distance between $\bar{X}_n$ and μ that is a theorem rather than an approximation, for every n, however small. Here's the statement.

Theorem 4.9 (Hoeffding's inequality, 1963)**.** *Let n be some positive integer and $X, X_1, \ldots, X_n$ be i.i.d. r.v.s such that $\mu = \mathbb{E}[X]$ and X is almost surely bounded within $[a, b]$ where $a < b$. Then:*

$$\mathbb{P}\left(|\bar{X}_n - \mu| \geq \epsilon\right) \leq 2\exp\left(-\frac{2n\epsilon^2}{(b-a)^2}\right), \quad \textit{for all } \epsilon > 0.$$

Compare this with Chebyshev's bound $\sigma^2/(n\epsilon^2)$. Both go to zero as n grows, but Hoeffding's bound goes to zero *exponentially* fast in n while Chebyshev's goes like $1/n$. The price is the boundedness

assumption, which Chebyshev doesn't need. Note also what Hoeffding's inequality is not: it's a bound, not an approximation. The CLT says $\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \approx 2(1 - \Phi(\epsilon\sqrt{n}/\sigma))$ for large n without saying how large; Hoeffding says $\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq 2e^{-2n\epsilon^2/(b-a)^2}$ for every $n \geq 1$, full stop.

Worked example: the kiss experiment, three ways

The R_i are Bernoulli, so they're bounded in $[0, 1]$ and $(b-a)^2 = 1$. With $n = 124$ and $\epsilon = 0.084$:

- Chebyshev (with $p(1-p)$ evaluated at 0.645): $\mathbb{P}(|\bar{R}_n - p| \geq 0.084) \leq 0.26$; with the worst case $p(1-p) \leq 1/4$ instead, 0.29.
- Hoeffding: $\mathbb{P}(|\bar{R}_n - p| \geq 0.084) \leq 2\exp(-2 \times 124 \times 0.084^2) = 2e^{-1.75} \approx 0.35$.
- CLT: $\mathbb{P}(|\bar{R}_n - p| \geq 0.084) \approx 2\left(1 - \Phi\left(\frac{0.084\sqrt{124}}{\sqrt{0.645 \times 0.355}}\right)\right) = 2(1 - \Phi(1.95)) \approx 0.05$.

At this n the exact Binomial probability is 0.05, so the CLT approximation is right on and both bounds are loose, with Chebyshev slightly tighter than Hoeffding. Figure 4.5 shows that Hoeffding overtakes Chebyshev at about $n = 165$ and is far ahead by $n = 400$, as the exponential must eventually beat $1/n$. In the figure the CLT approximation sits on top of the exact probability for n beyond a few dozen. The jaggedness of the exact curve is real: it comes from the lattice of possible values of $\bar{R}_n$ moving relative to the fixed ϵ. The two bounds are guarantees; the CLT number is an approximation that happens, here, to be excellent.

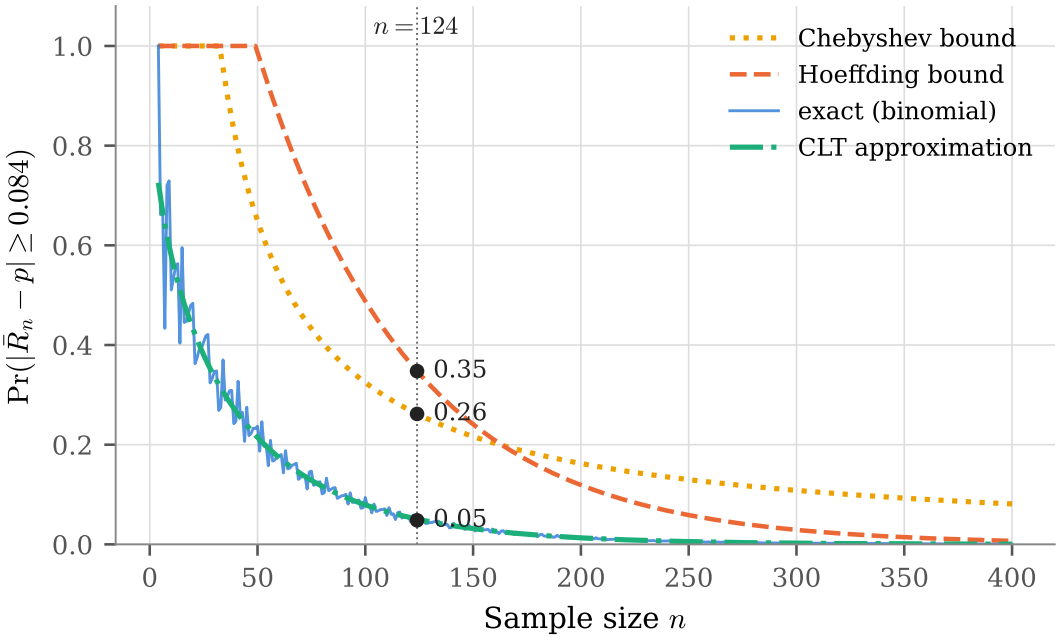


Figure 4.5: $\mathbb{P}(|\bar{R}_n - p| \geq 0.084)$ in the kiss model with $p = 0.645$, as a function of n: the exact Binomial probability, the CLT approximation, and the Chebyshev and Hoeffding upper bounds. The dots mark $n = 124$.

4.4 What the limit theorems buy you

A fair question at this point: couldn't the results of this chapter and the last be proved without limits? More of them than you would guess, and it's worth seeing which because it shows what the limit theorems are actually for.

Results that have a finite-sample version.

- The Law of Large Numbers is Chebyshev's inequality with n sent to infinity. The finite-n content is the inequality itself, $\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \sigma^2/(n\epsilon^2)$; the limit only tidies the statement. Hoeffding's inequality (Section 4.3) is the same idea with a better bound; the 'conservative' confidence interval of Chapter 8 uses the same trick of replacing the unknown variance by its worst case, $p(1-p) \leq 1/4$. Both are guarantees that hold at $n = 2$ as well as at $n = 10^6$.
- When the parent distribution is known, an exact sampling distribution replaces the CLT. For Bernoulli data the sum $n\bar{R}_n$ is Binomial, so the kiss experiment's proportion has exact tests and an exact interval (the Clopper–Pearson interval, which for these data is $[0.55, 0.73]$). For Exponential data the sum is Gamma, which gives an exact interval for the subway rate λ (an exercise in Chapter 8). For Gaussian data the standardized mean has a t distribution and the ratio of variances an F distribution, which is why the t-test of Chapter 14 and Gaussian linear regression in Chapter 21 need no limit theorem at all.
- The CLT itself has a finite-sample version: the Berry–Esseen theorem mentioned in Section 4.2 bounds the gap between the CDF of the standardized mean and Φ at every n. The limit statement is that theorem with the error term dropped.

What has no finite-sample substitute. What you can't do without a limit theorem is get a statement that is both universal and sharp. Chebyshev and Hoeffding hold for every parent too, but as bounds that may sit far from the truth; an approximation that is accurate for every parent at once is the whole content of the CLT. It's the only reason the same recipe, estimate $\pm 1.96\times$ standard error, works for proportions, means, rates, regression coefficients and maximum likelihood estimators of every model in this book. Without it you must either know the parent exactly, and then compute its exact sampling distribution, which is possible for the three families above and intractable for most others, or accept an inequality, which is honest but often far from the truth: Hoeffding says 0.35 where the exact answer is 0.05 in the kiss experiment (Figure 4.5). The asymptotic normality of a general maximum likelihood estimator (Chapter 12) and Wilks' theorem for the likelihood ratio test (Chapter 15) have no finite-sample versions; their exact distributions depend on the model and are usually unobtainable.

So the asymptotic route is a trade: one formula for every model, in exchange for not knowing how large n must be. The price is paid in the places where the approximation is poor, which the Berry–Esseen bound says are the heavy-tailed and asymmetric parents, and small n.

4.5 What comes next

You now have two limits for the sample mean: it collapses onto μ (Chapter 3) and, magnified by $\sqrt{n}$, its error is Gaussian (this chapter). Almost nothing in statistics is literally a sample mean. The estimator of the subway rate is $1/\bar{T}_n$, the standard error you can compute has $\hat{p}$ where the CLT has p, and the odds are a ratio of two things. The next chapter is the toolkit for carrying the two limits through such operations: Slutsky's theorem for plugging in, the continuous mapping theorem for transforming, and the delta method for the CLT of a function of an average.

4.6 Summary

- **Convergence in distribution:** $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$ if $F_{X_n}(x) \to F_X(x)$ at every continuity point of F_X; equivalently $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ for every bounded continuous f. It compares distributions, not values: $X_n = -Z$ converges in distribution to Z without ever being close to it.
- Convergence in probability implies convergence in distribution (Proposition 4.4). The converse can fail, as the $X_n = -Z$ example shows, unless the limit is a constant, in which case it holds (Proposition 4.7).
- **Central Limit Theorem:** $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1)$, whatever the parent distribution with finite variance. With overwhelming probability $|\bar{X}_n - \mu| \le 3\sigma/\sqrt{n}$: a funnel that narrows like $1/\sqrt{n}$. How fast the Gaussian arrives depends on the parent, above all on its asymmetry; the '$n \ge 30$' rule is a convention.
- Averages are not the only things with limits, and $\sqrt{n}$ and the Gaussian are not the only rate and limit: the maximum of n uniforms has an error of order $1/n$ and an Exponential limit (Example 4.6).
- **Hoeffding's inequality:** for data in $[a,b]$, $\mathbb{P}(|\bar{X}_n - \mu| \ge \epsilon) \le 2e^{-2n\epsilon^2/(b-a)^2}$ for every n. A guarantee, not an approximation, and exponentially better than Chebyshev for large n.
- The limit theorems' content is universality: one formula for every parent. The price is not knowing how large n must be. When the parent is known, exact distributions (Binomial, Gamma, t, F) need no limits at all.

4.7 Exercises

Exercise 4.1. Let $T_n \sim \text{Unif}([5 - \frac{1}{2n}, 5 + \frac{1}{2n}])$. (a) For a fixed $0 < \delta < 1$, find N such that $\mathbb{P}(|T_n - 5| > \delta) = 0$ whenever $n \ge N$. (b) What is the limit of (T_n)? (c) Let F_n be the CDF of T_n and F the CDF of the limit. For which t does $F_n(t) \to F(t)$? *(Solution on page 415.)*

Exercise 4.2. Let $Y_1, Y_2, \ldots \overset{iid}{\sim} \text{Unif}([0,1])$ and $M_n = \max(Y_1, \ldots, Y_n)$. (a) For $0 < \delta < 1$, find $\mathbb{P}(|M_n - 1| > \delta)$. (b) Does M_n converge in probability to a constant? (c) Find the CDF of M_n on $[0,1]$. *(Solution on page 415.)*

Exercise 4.3. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([0,b])$, so $\sigma^2 = b^2/12$, with n too small to trust the CLT. Find the Chebyshev and Hoeffding upper bounds on $\mathbb{P}(|\bar{X}_n - \mathbb{E}[X]| \ge c\,\sigma/\sqrt{n})$ for $c = 2$ and $c = 6$. *(Solution on page 416.)*

Exercise 4.4. In the kiss experiment, use the CLT to find the smallest n for which $\mathbb{P}(|\bar{R}_n - p| \ge 0.02) \approx 0.05$ when $p = 0.645$, and compare with the Chebyshev answer of Section 3.7. *(Solution on page 416.)*

Chapter 5

Working with Limits: Slutsky, the Continuous Mapping Theorem and the Delta Method

In this chapter

- Which arithmetic survives a limit: sums, products and quotients for convergence in probability, and the one case (Slutsky's theorem) where it works for convergence in distribution.
- The continuous mapping theorem: a continuous function of a convergent sequence converges.
- The delta method: the Central Limit Theorem for a function of an average, with the derivative as the scale factor.
- A map from every asymptotic result in the rest of the book back to these tools.

You now have two limits in hand: $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$ from Chapter 3 and $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow[n\to\infty]{\text{(d)}}$ Gauss$(0, \sigma^2)$ from Chapter 4. Almost nothing in statistics is literally an average, so the question is what you may do to these limits and still know where you end up.

5.1 Arithmetic and convergence

For convergence in probability and convergence almost surely, the arithmetic properties of limits of real-valued sequences have analogous arithmetic properties. (Almost sure convergence is defined in Appendix B; on the minimal path, read the rules below for convergence in probability, the only one of the two the rest of the book uses.)

Let $(X_n)_{n\geq 1}$ and $(Y_n)_{n\geq 1}$ be sequences of r.v.s and suppose that:

$$X_n \xrightarrow[n\to\infty]{\text{a.s./}\mathbb{P}} X \text{ and } Y_n \xrightarrow[n\to\infty]{\text{a.s./}\mathbb{P}} Y.$$

Then:

1. **Additive property:** $X_n + Y_n \xrightarrow[n\to\infty]{\text{a.s./}\mathbb{P}} X + Y$.

2. **Multiplicative property:** $X_n Y_n \xrightarrow[n\to\infty]{\text{a.s./}\mathbb{P}} XY$.

3. **Division property:** If $Y \neq 0$ almost surely, then:

$$\frac{X_n}{Y_n} \xrightarrow[n\to\infty]{\text{a.s./}\mathbb{P}} \frac{X}{Y}.$$

For almost sure convergence these are just the calculus rules applied outcome by outcome: on the probability-one set where both sequences of numbers converge, their sums, products and quotients converge to the sums, products and quotients of the limits. For convergence in probability the rules hold too, though the proof is a little longer.

NB: These rules **don't** apply to convergence in distribution. The pitfall box in Section 4.1 gave the counterexample: $X_n = -Z$ and $Y_n = Z$ both converge in distribution to Gauss$(0, 1)$, but $X_n + Y_n = 0$, while two independent sequences with the same marginal limits would have sum converging to Gauss$(0, 2)$. The marginal limits don't determine the limit of the sum because they say nothing about how X_n and Y_n are related. This is where Slutsky's theorem comes in: it's the special case where one of the two limits is a constant, so there's no relationship to worry about.

From undergraduate statistics: variances add, standard deviations do not

An introductory course combines independent random variables with the rule 'variances add': $\mathbb{V}\text{ar}(X \pm Y) = \mathbb{V}\text{ar}(X) + \mathbb{V}\text{ar}(Y)$, for a difference as well as a sum. Without independence there is a third term, $\mathbb{V}\text{ar}(X \pm Y) = \mathbb{V}\text{ar}(X) + \mathbb{V}\text{ar}(Y) \pm 2\,\mathbb{C}\text{ov}(X, Y)$, and the covariance is one part of the relation between X_n and Y_n that the marginal limits leave out. In the counterexample above both variables have variance 1: $X_n = -Y_n$ gives $\mathbb{V}\text{ar}(X_n + Y_n) = 0$, independence gives 2, and $X_n = Y_n$ gives 4, the case in which the standard deviations add, $1 + 1 = 2$. Independence is the hypothesis behind the two-sample standard error $\sqrt{s_1^2/n_1 + s_2^2/n_2}$: two independent kiss studies of 124 couples give two sample proportions with standard error 0.043 each, and their difference has standard error $\sqrt{0.043^2 + 0.043^2} = 0.061$, not 0.086. Before-and-after measurements on the same subjects are usually positively correlated, so the two-sample formula overstates the variance of their difference, and the paired analysis of an introductory course works with the differences instead.

5.2 Slutsky's theorem

For convergence in distribution, Slutsky's theorem is the main tool, and the kiss experiment shows why you need one. The CLT says

$$\sqrt{n}\,\frac{\bar{R}_n - p}{\sqrt{p(1-p)}} \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, 1),$$

but the denominator involves the unknown p, so you cannot compute the left side from data. You'd like to replace $\sqrt{p(1-p)}$ by $\sqrt{\hat{p}(1-\hat{p})}$ with $\hat{p} = \bar{R}_n$. Write the thing you want as a product:

$$\sqrt{n}\,\frac{\bar{R}_n - p}{\sqrt{\hat{p}(1-\hat{p})}} = \underbrace{\sqrt{n}\,\frac{\bar{R}_n - p}{\sqrt{p(1-p)}}}_{\xrightarrow[n\to\infty]{\text{(d)}}\text{Gauss(0,1) by the CLT}} \cdot \underbrace{\sqrt{\frac{p(1-p)}{\hat{p}(1-\hat{p})}}}_{\xrightarrow[n\to\infty]{\mathbb{P}}\text{1 by the LLN and CMT}}.$$

The first factor converges in distribution and the second in probability, to the constant 1. What is the limit of the product? The pitfall of Section 4.1 says that in general the two marginal limits don't determine it. Slutsky's theorem says that when one of the limits is a constant, they do.

Theorem 5.1 (Slutsky)**.** *Let $(X_n)_{n\geq 1}$ and $(Y_n)_{n\geq 1}$ be sequences of r.v.s such that:*

$$X_n \xrightarrow[n\to\infty]{(d)} X \text{ and } Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y,$$

where X is an r.v. and y is a given real number; that is, the limit of Y_n is deterministic. Then:

1. ***Additive property:*** $X_n + Y_n \xrightarrow[n\to\infty]{(d)} X + y$.
2. ***Multiplicative property:*** $X_n Y_n \xrightarrow[n\to\infty]{(d)} yX$.
3. ***Division property:*** *If $y \neq 0$, then:*

$$\frac{X_n}{Y_n} \xrightarrow[n\to\infty]{(d)} \frac{X}{y}.$$

Slutsky's theorem is useful precisely because it allows you to combine multiple r.v.s with known asymptotics, and retain this knowledge; that is, you know what the resultant distribution will converge to as $n \to \infty$. Its typical use: the CLT hands you a sequence converging in distribution to a Gaussian, the Law of Large Numbers hands you a sequence converging in probability to a constant, and Slutsky lets you multiply or divide the one by the other. Note that, by Proposition 4.7, the hypothesis $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$ could equally be written $Y_n \xrightarrow[n\to\infty]{(\mathrm{d})} y$; the theorem needs the limit to be a constant, not a particular mode.

Worked example: the plug-in standardization in the kiss experiment, finished

Back to the product above. The second factor converges in probability to 1 because $\hat{p} \xrightarrow[n\to\infty]{\mathbb{P}} p$ (Law of Large Numbers) and $x \mapsto \sqrt{p(1-p)/(x(1-x))}$ is continuous at p (continuous mapping theorem, next). Slutsky's multiplicative property then gives

$$\sqrt{n}\,\frac{\bar{R}_n - p}{\sqrt{\hat{p}(1-\hat{p})}} \xrightarrow[n\to\infty]{(\mathrm{d})} 1 \cdot \mathrm{Gauss}(0,1) = \mathrm{Gauss}(0,1).$$

This is the plug-in route to the confidence interval for p, and it's the justification for every $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ you've ever computed.

The same two-line argument settles a substitution every introductory course makes without comment.

From undergraduate statistics: why you're allowed to use s in place of σ

Every introductory course replaces the unknown σ by the sample standard deviation s in the standard error, and for a mean switches from z to t. Slutsky's theorem is the reason this works for large n: $s \xrightarrow[n\to\infty]{\mathbb{P}} \sigma$ by the Law of Large Numbers applied to the data and to their squares (for the squares in its finite-mean form, Theorem B.3) and the continuous mapping theorem of

Section 5.3, so

$$\frac{\bar{X}_n - \mu}{s/\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \cdot \frac{\sigma}{s} \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1) \cdot 1.$$

The t distribution is the exact finite-sample answer when the data are Gaussian (Chapter 14); Slutsky is the large-sample answer for any parent with finite variance.

5.3 Continuous mapping theorem (CMT)

The Continuous Mapping Theorem (CMT) states that if there's some random variable X for which $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$, then $f(X_n) \xrightarrow[n\to\infty]{\text{(d)}} f(X)$, too, provided f is a continuous function (a function whose graph has no jumps; a small change in the input gives a small change in the output). In short, CMT states that a continuous function preserves the asymptotic limits of an r.v.

But the CMT is actually more general than explained above. Specifically, it applies to all three modes of convergence, almost surely (Appendix B) and in probability as well as in distribution. Here's its more general statement.

Theorem 5.2 (Continuous mapping theorem)**.** *If f is a continuous function and $(X_n)_{n\geq 1}$ is a sequence of r.v.s, then:*

$$X_n \xrightarrow[n\to\infty]{a.s./\mathbb{P}/(d)} X \quad \Rightarrow \quad f(X_n) \xrightarrow[n\to\infty]{a.s./\mathbb{P}/(d)} f(X).$$

For almost sure convergence the theorem is the calculus fact that a continuous function of a convergent sequence of numbers converges, applied outcome by outcome. For convergence in distribution it follows from the $\mathbb{E}[g(X_n)]$ characterization of Proposition 4.3, since $g \circ f$ is continuous and bounded whenever g is. Only continuity at the points where the limit lives is needed: if $X_n \xrightarrow[n\to\infty]{\mathbb{P}} p$ with $p > 0$, then $1/X_n \xrightarrow[n\to\infty]{\mathbb{P}} 1/p$ even though $f(x) = 1/x$ isn't continuous at 0.

Example 5.3. By the Law of Large Numbers, $\bar{R}_n \xrightarrow[n\to\infty]{\mathbb{P}} p$. Therefore $f(\bar{R}_n) \xrightarrow[n\to\infty]{\mathbb{P}} f(p)$ for any continuous f: the sample odds $\bar{R}_n/(1-\bar{R}_n)$ converge to the true odds $p/(1-p)$, the sample log-odds converge to the true log-odds, and so on. Consistency survives continuous transformations.

Pitfall: what the CMT does not give you

By the CLT, $\sqrt{n}(\bar{R}_n - p) \xrightarrow[n\to\infty]{\text{(d)}} Y$ with $Y \sim \text{Gauss}(0, p(1-p))$, so by the CMT

$$f\left(\sqrt{n}(\bar{R}_n - p)\right) \xrightarrow[n\to\infty]{\text{(d)}} f(Y).$$

That is a true statement, and almost never the one you want. What you want is the limit of $\sqrt{n}\left(f(\bar{R}_n) - f(p)\right)$: the magnified error of the transformed *estimator*, not a transformation of the magnified error. The CMT is silent on this. The delta method is the tool, and Figure 5.1 shows the two objects side by side, for $f(x) = x^2$ in the kiss model with $p = 0.645$ and $n = 124$; p^2 is the probability that two independent couples both turn right. On the left is f applied to the magnified error, $(\sqrt{n}(\bar{R}_n - p))^2$, which by the CMT converges to the square of a Gaussian,

$\sigma^2\chi_1^2$ with $\sigma^2 = p(1-p)$, a chi-squared shape. True, and useless for saying how accurate $\bar{R}_n^2$ is. On the right is the magnified error of the transformed estimator, $\sqrt{n}(\bar{R}_n^2 - p^2)$, which the delta method says is Gauss$(0, (2p)^2 p(1-p))$, standard deviation 0.62.

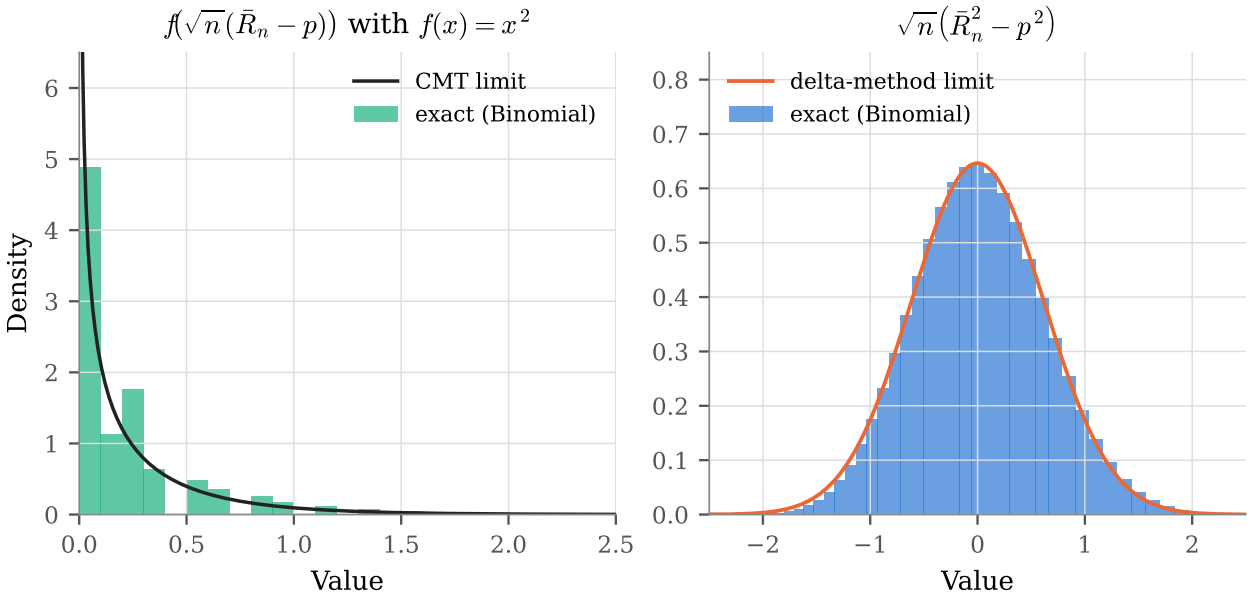


Figure 5.1: Two different random variables for $f(x) = x^2$ in the kiss model, $p = 0.645$, $n = 124$. Left: $(\sqrt{n}(\bar{R}_n - p))^2$ with its CMT limit (black curve). Right: $\sqrt{n}(\bar{R}_n^2 - p^2)$ with its delta-method limit (orange curve). The bars are the exact Binomial distribution of $n\bar{R}_n$, binned.

5.4 The delta method

The delta method is what you need when you're taking the limit of something like $\sqrt{n}\left(g(\bar{X}_n) - g(\mu)\right)$: you know the CLT for $\bar{X}_n$, and you want the CLT for a function of it.

Theorem 5.4 (Delta method)**.** *Let $(Z_n)_{n\geq 1}$ be a sequence of r.v.s such that:*

$$\sqrt{n}(Z_n - \theta) \xrightarrow[n\to\infty]{(d)} Gauss(0, \sigma^2),$$

for some $\theta \in \mathbb{R}$ and $\sigma^2 > 0$. A sequence like $(Z_n)_{n\geq 1}$ that satisfies this condition is said to be asymptotically normal around θ, and σ^2 is its asymptotic variance. Let $g : \mathbb{R} \to \mathbb{R}$ be continuously differentiable at the point θ. Then:

1. $g(Z_n) \xrightarrow[n\to\infty]{\mathbb{P}} g(\theta)$;
2. $(g(Z_n))_{n\geq 1}$ *is also asymptotically normal, around* $g(\theta)$;
3. *more precisely,*

$$\sqrt{n}\left(g(Z_n) - g(\theta)\right) \xrightarrow[n\to\infty]{(d)} Gauss\left(0,\, g'(\theta)^2\sigma^2\right).$$

Intuition: a differentiable function is locally linear

Figure 5.2 is the whole idea. It is drawn for $g(x) = 1/x$ with the numbers of the subway example below, $\theta = 6.23$ and $\sigma/\sqrt{n} = 0.78$; the dotted lines map $\theta \pm 2\sigma/\sqrt{n}$ across. For large n, Z_n sits in a narrow Gaussian blob of width $\sigma/\sqrt{n}$ around θ. On a blob that narrow, the curve g is indistinguishable from its tangent line at θ, which has slope $g'(\theta)$. A linear function maps a Gaussian to a Gaussian, shifting the center to $g(\theta)$ and multiplying the width by the slope. So $g(Z_n)$ sits in a Gaussian blob of width $|g'(\theta)|\,\sigma/\sqrt{n}$ around $g(\theta)$, which is statement (3). Where g is steep, errors in Z_n are amplified; where it's flat, they're damped.

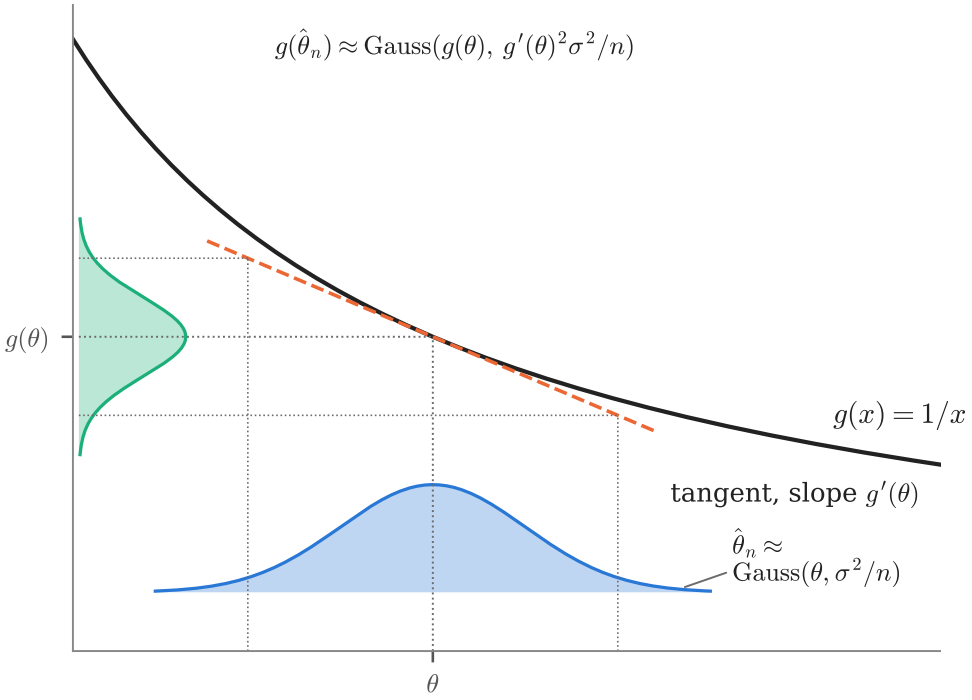


Figure 5.2: The delta method for $g(x) = 1/x$, $\theta = 6.23$, $\sigma/\sqrt{n} = 0.78$: the input Gaussian in blue on the horizontal axis, its image in green on the vertical axis, the tangent at θ dashed in orange, and dotted guides at $\theta \pm 2\sigma/\sqrt{n}$.

Proof strategy (core). The mean value theorem (toolkit A.9) writes $g(Z_n) - g(\theta)$ as a slope times $Z_n - \theta$. The slope converges in probability to $g'(\theta)$ (CMT), and $\sqrt{n}(Z_n - \theta)$ converges in distribution (the assumption); Slutsky multiplies the two.

Proof. By the mean value theorem there's a point $\tilde{\theta}_n$ between Z_n and θ such that

$$g(Z_n) - g(\theta) = g'(\tilde{\theta}_n)(Z_n - \theta).$$

Multiply through by $\sqrt{n}$:

$$\sqrt{n}\,(g(Z_n) - g(\theta)) = g'(\tilde{\theta}_n) \cdot \sqrt{n}(Z_n - \theta).$$

Now take the two factors in turn. First, $Z_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta$: by Slutsky, $Z_n - \theta = \frac{1}{\sqrt{n}} \cdot \sqrt{n}(Z_n - \theta) \xrightarrow[n\to\infty]{\text{(d)}} 0 \cdot \text{Gauss}(0, \sigma^2) = 0$, and convergence in distribution to a constant is convergence in probability (Proposition 4.7). This proves (1) via the CMT, and since $\tilde{\theta}_n$ is squeezed between Z_n and θ,

also $\tilde{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta$, hence $g'(\tilde{\theta}_n) \xrightarrow[n\to\infty]{\mathbb{P}} g'(\theta)$ by the CMT and the continuity of g' at θ. Second, $\sqrt{n}(Z_n - \theta) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, \sigma^2)$ by assumption. Slutsky's multiplicative property with $y = g'(\theta)$ gives

$$g'(\tilde{\theta}_n) \cdot \sqrt{n}(Z_n - \theta) \xrightarrow[n\to\infty]{(\mathrm{d})} g'(\theta) \cdot \mathrm{Gauss}(0, \sigma^2) = \mathrm{Gauss}(0, g'(\theta)^2\sigma^2),$$

which is (3), and (2) is (3) read as a sentence. □

The proof is three lines because Slutsky and the CMT did the work in advance. Note that nothing requires g to be invertible or monotone; $g(x) = \sqrt{2}\sin x$ is a perfectly good choice. Note also that the asymptotic variance $g'(\theta)^2\sigma^2$ generally depends on the unknown θ, even when σ^2 doesn't; the plug-in trick of the previous section handles that.

Worked example: inter-arrival times of the T

You observe the times between arrivals of the subway at Kendall, $T_1, \ldots, T_n$, and model them as i.i.d. Expo(λ), so $\mathbb{E}[T_1] = 1/\lambda$ and $\mathbb{V}\mathrm{ar}(T_1) = 1/\lambda^2$. The natural estimator of λ is $\hat{\lambda} = 1/\bar{T}_n$.

- *Consistency.* $\bar{T}_n \xrightarrow[n\to\infty]{\mathbb{P}} 1/\lambda$ by the Law of Large Numbers, so $\hat{\lambda} = 1/\bar{T}_n \xrightarrow[n\to\infty]{\mathbb{P}} \lambda$ by the CMT ($f(x) = 1/x$ is continuous at $1/\lambda > 0$).
- *The CLT for* $\bar{T}_n$. $\sqrt{n}\left(\bar{T}_n - \frac{1}{\lambda}\right) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, \lambda^{-2})$.
- *The delta method for* $\hat{\lambda}$. Take $g(x) = 1/x$, so $g'(x) = -1/x^2$ and $g'(1/\lambda)^2 = \lambda^4$. Then

$$\sqrt{n}\left(\hat{\lambda} - \lambda\right) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}\left(0,\, \lambda^4 \cdot \lambda^{-2}\right) = \mathrm{Gauss}(0, \lambda^2).$$

With $n = 64$ and $\bar{T}_n = 6.23$ minutes: $\hat{\lambda} = 0.1605$ per minute, and the standard error of $\hat{\lambda}$ (the standard deviation $\lambda/\sqrt{n}$ of its approximating Gaussian, with $\hat{\lambda}$ plugged in for λ) is about 0.020. So λ is pinned down to roughly 0.16 ± 0.04, which becomes the plug-in confidence interval $[0.12, 0.20]$ of Chapter 8.

Worked example: odds in the kiss experiment

The odds of turning right are $\omega = p/(1-p)$, estimated by $\hat{\omega} = \bar{R}_n/(1-\bar{R}_n)$. With $g(p) = p/(1-p)$ you get $g'(p) = 1/(1-p)^2$, and the CLT gives asymptotic variance $p(1-p)$ for $\bar{R}_n$, so

$$\sqrt{n}\,(\hat{\omega} - \omega) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}\left(0,\, \frac{p(1-p)}{(1-p)^4}\right) = \mathrm{Gauss}\left(0,\, \frac{p}{(1-p)^3}\right).$$

At $p = 0.645$ and $n = 124$: $\hat{\omega} = 1.82$ and the standard error is $\sqrt{0.645/0.355^3/124} \approx 0.34$. The derivative $1/(1-p)^2 \approx 7.9$ is large, so the odds are estimated far less precisely, in absolute terms, than the proportion they're computed from: a steep g amplifies the error. Figure 5.3 puts this next to a flat g, where the same input error nearly disappears. The input on both sides is $\bar{R}_n \approx \mathrm{Gauss}(p, p(1-p)/n)$ with standard deviation 0.043. On the left, the odds have slope 7.9 at $p = 0.645$, and the output standard deviation is $7.9 \times 0.043 = 0.34$. On the right, the variance $g(p) = p(1-p)$ has slope -0.29 there, and the output standard deviation is $0.29 \times 0.043 = 0.012$. Same data and the same precision in p; the delta method's $|g'(\theta)|$ is the whole difference in the precision of $g(p)$.

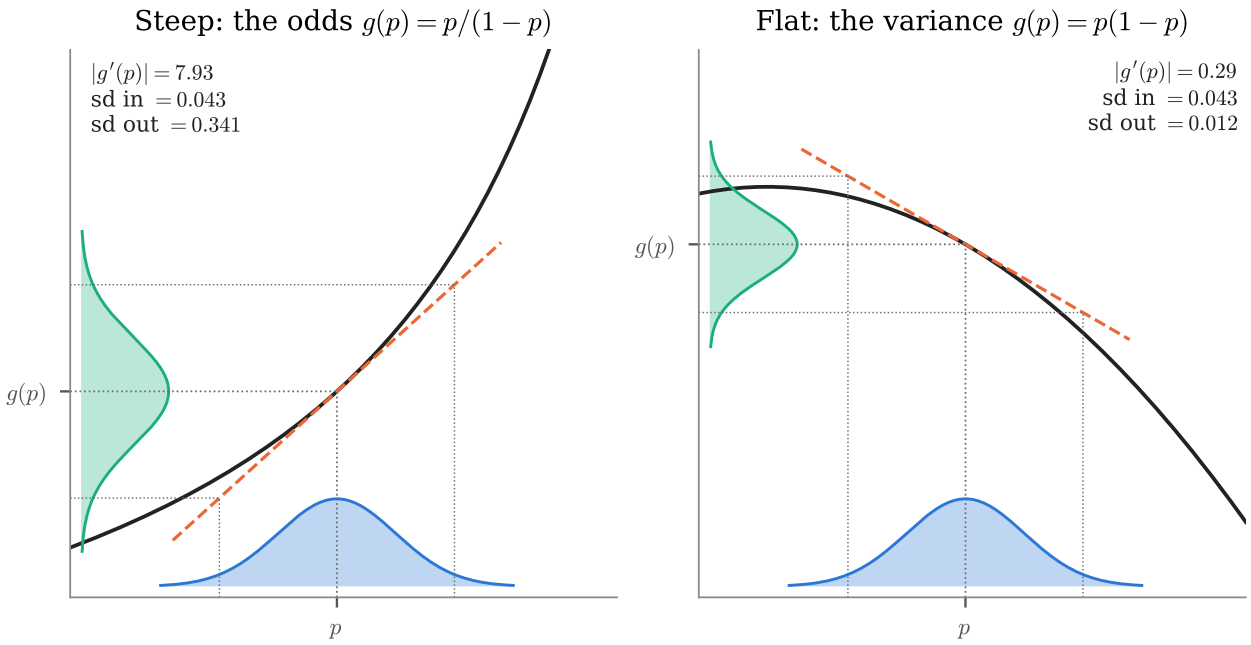


Figure 5.3: The input $\bar{R}_n \approx \text{Gauss}(p, p(1-p)/n)$, in blue on the horizontal axis, pushed through two functions, with the output in green on the vertical axis and the tangent at p dashed in orange. Left: the odds $g(p) = p/(1-p)$. Right: the variance $g(p) = p(1-p)$.

From undergraduate statistics: propagation of error

If you took a laboratory course you met the rule 'if $y = g(x)$ and x has uncertainty δx, then y has uncertainty $\delta y \approx |g'(x)|\,\delta x$'. The delta method is that rule for standard errors, with a theorem attached:

$$\text{se}\left(g(\hat{\theta}_n)\right) \approx |g'(\theta)|\ \text{se}(\hat{\theta}_n), \qquad \text{se}(\hat{\theta}_n) = \frac{\sigma}{\sqrt{n}}.$$

It also says what the lab rule leaves out: the transformed estimate isn't just 'uncertain by δy', it's approximately Gaussian, so you can attach a probability to the error bar.

Remark. When $g'(\theta) = 0$ the theorem still holds but says only that $\sqrt{n}(g(Z_n) - g(\theta)) \xrightarrow[n\to\infty]{\text{(d)}} 0$, that is, the error of $g(Z_n)$ is of smaller order than $1/\sqrt{n}$. To see the actual limit you magnify by n instead of $\sqrt{n}$ and get a second-order expansion: $n(g(Z_n) - g(\theta)) \xrightarrow[n\to\infty]{\text{(d)}} \frac{\sigma^2 g''(\theta)}{2}\chi^2_1$. No result later in this book needs this case.

Remark (Vector versions). Everything in this chapter has a version for vectors, with a covariance matrix in place of σ^2 and a matrix of partial derivatives in place of $g'(\theta)$. If $\sqrt{n}(Z_n - \theta) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0, \Sigma)$ and $g : \mathbb{R}^d \to \mathbb{R}^k$ is continuously differentiable at θ, with $\nabla g(\theta)$ the $d \times k$ matrix of partial derivatives, then

$$\sqrt{n}\,(g(Z_n) - g(\theta)) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_k\left(0,\ \nabla g(\theta)^\top \Sigma\, \nabla g(\theta)\right),$$

and the multivariate CLT says that the vector of sample means of i.i.d. vectors with covariance Σ satisfies the hypothesis with Σ itself. The proofs are the ones above with matrices in place of

numbers. The method of moments (Chapter 13) and the chi-squared goodness-of-fit test (Chapter 16) use these, and the book states them again there in the form each needs.

From undergraduate statistics: the confidence interval for an odds ratio

Courses in biostatistics and epidemiology teach a confidence interval for an odds ratio: the log of the estimated odds ratio plus or minus $1.96\sqrt{1/a + 1/b + 1/c + 1/d}$, with a, b, c, d the four counts of the two-by-two table, exponentiated at both ends. The square root is the delta method. For one group with S successes in n trials, the log odds $g(p) = \log\big(p/(1-p)\big)$ has $g'(p) = 1/\big(p(1-p)\big)$, so Theorem 5.4 gives it approximate variance $g'(p)^2\, p(1-p)/n = \big(1/p + 1/(1-p)\big)/n$, which with $\hat{p} = S/n$ plugged in is $1/S + 1/(n-S)$. For the kiss data the log odds of turning right is $\log(80/44) = 0.598$ with standard error $\sqrt{1/80 + 1/44} = 0.188$, and exponentiating the ends of $0.598 \pm 1.96 \times 0.188$, which keeps the level (Section 8.6), gives $[1.26, 2.63]$ for the odds. An odds ratio compares two independent groups, so its log is the difference of two independent log odds, and the variance of that difference adds all four reciprocals.

5.5 Where the limit theorems are used in the rest of the book

Every asymptotic statement in the later chapters is one of the facts of Chapters 3 and 4 or this chapter in disguise, except for the results the table marks 'stated, not proved': Donsker's theorem about the whole empirical CDF, behind the Kolmogorov–Smirnov test; the Bernstein–von Mises theorem of Chapter 18; the coverage rates of Bayesian bounds and regions in Chapter 19; the consistency of the robust standard error in Chapters 20 and 21; and the Central Limit Theorem for eigenvalues in Chapter 24. The table says which. 'LLN' means the Law of Large Numbers (Theorem 3.5), 'CLT' the Central Limit Theorem (Theorem 4.8), 'CMT' the continuous mapping theorem (Theorem 5.2), 'Slutsky' Theorem 5.1, and 'delta' the delta method (Theorem 5.4).

Result	Chapter	Rests on
Consistency of $\bar{X}_n$, $\hat{p}$, and any estimator built from an average	Chapter 7	LLN, then CMT for functions of averages
Asymptotic confidence intervals; the plug-in interval	Chapter 8	CLT; Slutsky for replacing p by $\hat{p}$ in the standard error
Confidence interval for $1/\bar{T}_n$ and other transformed estimators	Chapter 8	delta
Level and p-value of a test built on a standardized average	Chapter 9	CLT, Slutsky
Consistency of the maximum likelihood estimator	Chapter 12	LLN applied to the average log-likelihood, plus the KL divergence argument
Asymptotic normality of the MLE, with variance $1/I(\theta)$	Chapter 12	CLT applied to the average score, then a Taylor expansion exactly as in the delta method proof, with Fisher information as the slope
Method of moments: consistency and asymptotic variance	Chapter 13	LLN, then the (multivariate) delta method
Wald test	Chapter 15	asymptotic normality of the MLE, Slutsky for the estimated information, CMT (squaring gives a χ^2)

Result	Chapter	Rests on
Likelihood ratio test (Wilks' theorem)	Chapter 15	a second-order Taylor expansion, the same device as the delta method, plus Slutsky
Score test	Chapter 15	multivariate CLT applied to the score, then CMT
t-test	Chapter 14	nothing here: exact for Gaussian data (Cochran's theorem)
Chi-squared goodness-of-fit test	Chapter 16	multivariate CLT for the rotated standardized deviations of the counts, then CMT; with estimated parameters, the nuisance-parameter theorem of Chapter 15
Kolmogorov–Smirnov and Lilliefors tests	Chapter 17	Glivenko–Cantelli (a uniform LLN), proved there from the LLN at finitely many points; Donsker's theorem (a CLT for the whole empirical CDF), stated without proof
Bayesian posteriors concentrate on the truth (proved for Beta priors); Bernstein–von Mises	Chapter 18	LLN and Chebyshev for the concentration; Bernstein–von Mises stated, not proved
Bayesian confidence regions have asymptotic coverage $1-\alpha$; Jeffreys' prior is flat on the variance-stabilizing scale	Chapter 19	asymptotic normality of the MLE, Bernstein–von Mises and Slutsky; the delta method; coverage rates (Welch and Peers) stated, not proved
Inference for least squares with non-Gaussian errors	Chapters 20 and 21	LLN and CMT for the spread of the x's, CLT for the average of $(X_i-\mu)\epsilon_i$, Slutsky for the ratio; consistency of the robust standard error stated, not proved (exact t and F when errors are Gaussian)
Maximum likelihood in an exponential family	Chapter 22	CLT for the sample mean; the MLE result, its regularity conditions verified there for every canonical family
Asymptotic normality of GLM estimates	Chapter 23	the MLE result again
Sample covariance, eigenvalues and principal directions converge	Chapter 24	LLN entrywise (with fourth moments), then CMT for the eigenvalues and, where an eigenvalue is isolated, the eigenvectors; the CLT for eigenvalues stated, not proved

5.6 What comes next

Every limit in these three chapters landed on the same distribution. Chapter 6 collects what the rest of the book needs about it: the density, the three properties that let you standardize anything Gaussian, how to turn a Gaussian probability into a table lookup, and this book's convention for quantiles, which is the notation every confidence interval and test from Chapter 8 on is written in. If you know the Gaussian well, the quantile section is the one part not to skip.

5.7 Summary

- For convergence in probability (and almost surely), limits add, multiply and divide like limits of numbers. For convergence in distribution they do not: the marginal limits of X_n and Y_n don't determine the limit of $X_n + Y_n$.
- **Slutsky's theorem:** if $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$ and $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$ with y a constant, then $X_n + Y_n \xrightarrow[n\to\infty]{\text{(d)}} X + y$, $X_n Y_n \xrightarrow[n\to\infty]{\text{(d)}} yX$ and $X_n / Y_n \xrightarrow[n\to\infty]{\text{(d)}} X/y$. This justifies every plug-in: replacing σ by s, or p by $\hat{p}$, in a standard error.
- **Continuous mapping theorem:** $X_n \to X$ in any of the three modes implies $f(X_n) \to f(X)$ in the same mode, for f continuous at the limit. Consistency survives transformations. What it does not give you is the limit of $\sqrt{n}(f(\bar{X}_n) - f(\mu))$; that is the delta method's job.
- **Delta method:** if $\sqrt{n}(Z_n - \theta) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, \sigma^2)$ and g is continuously differentiable at θ, then $\sqrt{n}(g(Z_n) - g(\theta)) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, g'(\theta)^2\sigma^2)$. A narrow Gaussian blob pushed through a smooth curve comes out Gaussian, with its width multiplied by the slope. This is propagation of error with a theorem attached.

5.8 Exercises

Exercise 5.1. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bern}(p)$. Use the delta method to find the asymptotic distribution of $\sqrt{n}\left(\bar{X}_n(1 - \bar{X}_n) - p(1-p)\right)$. For which p does the delta method fail to give a Gaussian limit, and why? *(Solution on page 416.)*

Exercise 5.2. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Expo}(\lambda)$. Show that $\sqrt{n}(\log \bar{X}_n + \log \lambda) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, 1)$, whatever λ is. Why might a statistician prefer to build a confidence interval for $\log(1/\lambda)$ rather than for $1/\lambda$? *(Solution on page 416.)*

Exercise 5.3. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bern}(p)$ with $0 < p < 1$. Show that

$$\sqrt{n}\,\frac{\bar{X}_n - p}{\sqrt{\bar{X}_n(1 - \bar{X}_n)}} \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, 1),$$

naming the theorem used at each step. *(Solution on page 416.)*

Chapter 6

The Gaussian Distribution and Quantiles

In this chapter

- The Gaussian density: what each piece of the formula does, why the CDF has no closed form, and why the tails are so thin that the distribution lives in a finite interval for practical purposes.
- The three properties used constantly from here on: closure under affine transformations, standardization, and symmetry. Plus the one that usually goes unstated: sums of independent Gaussians are Gaussian.
- How to turn a Gaussian probability into a table lookup, forwards and backwards.
- Quantiles, in this book's convention q_α (the number with mass α to its *right*), the four values worth memorizing, and the one-sided versus two-sided distinction that every confidence interval and test depends on.

This chapter is a review. If you know it, read the quantile section for the notation and move on.

6.1 Why the Gaussian is everywhere

Chapters 3 to 5 kept producing Gaussian limits; this chapter collects what you need about the Gaussian to use them, and fixes the quantile notation of the rest of the book. Because of the Central Limit Theorem (Chapter 4), the Gaussian (also called the normal) distribution is ubiquitous in statistics: every average of enough independent, identically distributed draws with finite variance is approximately Gaussian, whatever the draws come from, so every estimator built from an average inherits a Gaussian error distribution for large n. That is where the Gaussian enters statistical inference, and it stays: confidence intervals, test statistics, the asymptotic distribution of the maximum likelihood estimator, the error term of linear regression. The distribution is named after Carl Friedrich Gauss (1777–1855), who arrived at it in the context of the method of least squares, which is the subject of Chapter 20.

6.2 The density and its CDF

Definition 6.1 (Gaussian distribution)**.** A continuous r.v. X has the **Gaussian distribution** with parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ if its PDF is given by:

$$f_{\mu,\sigma^2}(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad \text{for } x \in (-\infty, \infty).$$

You write this as $X \sim \text{Gauss}(\mu, \sigma^2)$, and then $\mathbb{E}[X] = \mu$ and $\mathbb{V}\text{ar}(X) = \sigma^2$. The case $\mu = 0$, $\sigma^2 = 1$ is the **standard Gaussian**, whose density and CDF get their own letters:

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad \Phi(z) = \int_{-\infty}^{z} \varphi(t)\, \mathrm{d}t.$$

> **In words.** A Gaussian variable is one whose probabilities are areas under the bell-shaped curve above (toolkit A.6); μ says where the bell is centered and σ how wide it is. Two numbers fix the whole distribution.
>
> **For the kiss data.** The CLT says $\bar{R}_n$ behaves, for large n, like a Gaussian with $\mu = p$ and $\sigma^2 = p(1-p)/n$: centered at the truth, with width 0.043 at $n = 124$ and $p = 0.645$.

Take the formula apart. The constant $\dfrac{1}{\sigma\sqrt{2\pi}}$ is a *normalizing constant*: it normalizes the total area under the PDF to 1, and plays no other role. Everything about the shape is in the exponent. Consider the quadratic there:

$$-\frac{(x-\mu)^2}{2\sigma^2} = -\left(\frac{x^2 - 2\mu x + \mu^2}{2\sigma^2}\right).$$

It is zero at $x = \mu$ and negative everywhere else, so the density peaks at μ and falls off symmetrically on both sides. If σ is small, that is, if the standard deviation is small, then the quadratic is more negative at the same distance from μ, so the exponential falls off faster and the density is narrower. A small standard deviation and a narrow density are the same thing seen twice. Figure 6.1 shows the effect of each parameter. On the left, Gauss$(0, 4)$ is Gauss$(0, 1)$ stretched to twice the width, and its peak drops to half the height to keep the area at 1; Gauss$(10, 4)$ is Gauss$(0, 4)$ moved ten units to the right with no change of shape. So μ moves the density without changing its shape, and σ stretches it without moving its center. On the right, the three shaded regions of the standard Gaussian have probabilities 0.6827, 0.9545 and 0.9973: the '68–95–99.7 rule' of an introductory course, stated exactly.

From undergraduate statistics: the empirical rule is a Gaussian fact

An introductory course applies the 68–95–99.7 rule to data with a bell-shaped histogram, and the restriction matters: the three numbers are the Gaussian's. For the Exponential waits of the subway, the shares within one, two and three standard deviations of the mean are 86.5%, 95.0% and 98.2%, so 1.8% of the waits lie beyond three standard deviations, nearly seven times the Gaussian's 0.27%. The guarantee that holds for every shape with a finite variance is Chebyshev's rule, from the bridge in Section 3.4: at least 75% within two standard deviations and 88.9% within three.

Two consequences of the exponent are worth stating on their own.

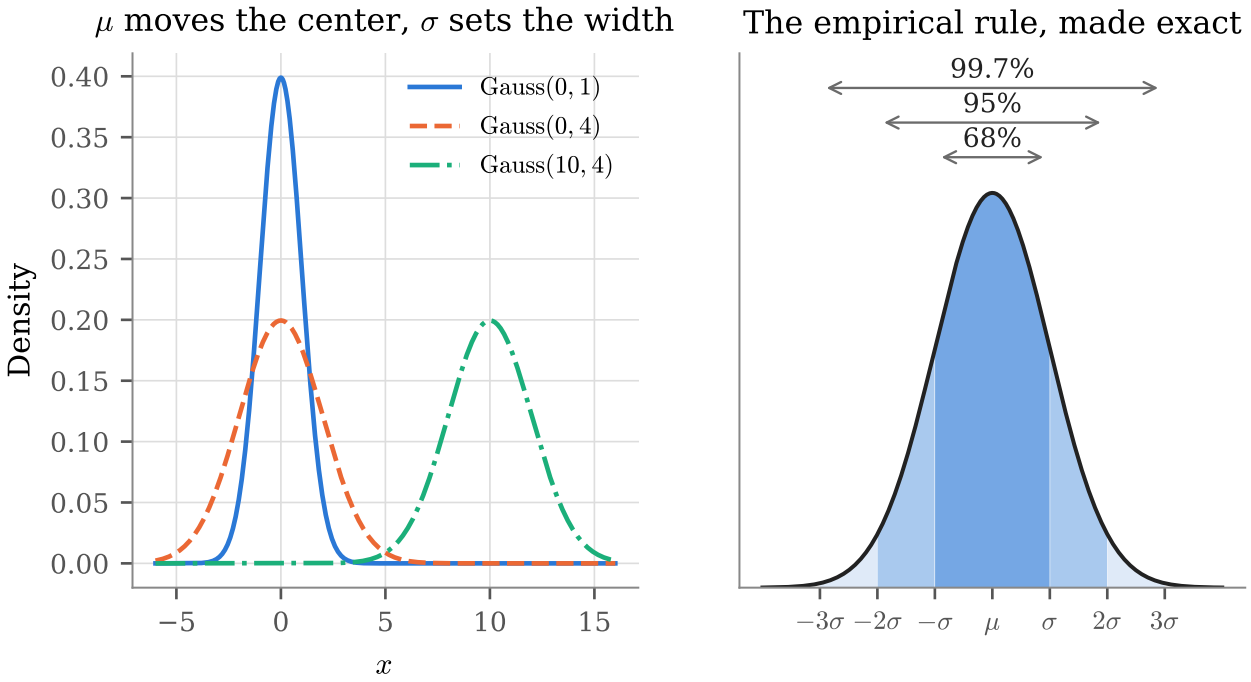


Figure 6.1: Left: the densities of Gauss$(0, 1)$ (solid), Gauss$(0, 4)$ (dashed) and Gauss$(10, 4)$ (dash-dot). Right: the standard Gaussian density, shaded within one, two and three standard deviations of the mean.

- **The tails are thin.** The density decays like $e^{-x^2/(2\sigma^2)}$, faster than any exponential $e^{-c|x|}$, so for practical purposes a Gaussian lives in a finite interval: 99.7% of the mass is within 3σ of μ, and the probability of being more than 5σ away is about one in two million. This is what licenses a statement like 'with overwhelming probability $|\bar{X}_n - \mu| \leq 3\sigma/\sqrt{n}$' (Section 4.2).
- **The CDF has no closed form.**

$$\mathbb{P}(X \leq x) = F_{\mu,\sigma^2}(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) \mathrm{d}t$$

must be left as an integral: it's mathematically impossible to write the antiderivative of $e^{-t^2/2}$ in terms of elementary functions. Still, F_{μ,σ^2} is a perfectly well-defined function: input some value x and it returns the accumulated area under the PDF from $-\infty$ up to x. You get its values from tables or from software, and Section 6.4 says how.

6.3 Properties of Gaussian distributions

1. **Affine transformation.** Perhaps the most useful property of the Gaussian family is that it is invariant under affine transformations. Given the r.v. $X \sim \text{Gauss}(\mu, \sigma^2)$, then

$$aX + b \sim \text{Gauss}(a\mu + b, a^2\sigma^2) \qquad \text{for all } a \neq 0,\ b \in \mathbb{R}.$$

Shifting and rescaling a Gaussian produces another Gaussian; the mean and variance change in the way means and variances always do under $aX + b$, and the shape stays Gaussian, which is the content of the statement.

Proof strategy (for completeness; skip on a first reading). Write the CDF of $aX + b$ as a CDF of X at a transformed point, differentiate (chain rule) to get the density, and recognize the Gaussian formula with new parameters (toolkit A.6).

Proof. Take $a > 0$ (the case $a < 0$ is the same with the inequality flipped). The CDF of $Y = aX + b$ is

$$F_Y(y) = \mathbb{P}(aX + b \le y) = \mathbb{P}\left(X \le \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right).$$

Differentiate with respect to y (chain rule) to get the density:

$$\begin{aligned} f_Y(y) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right) &= \frac{1}{a\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right) \\ &= \frac{1}{(a\sigma)\sqrt{2\pi}} \exp\left(-\frac{(y - (a\mu + b))^2}{2(a\sigma)^2}\right), \end{aligned}$$

which is the Gauss$(a\mu + b, a^2\sigma^2)$ density. □

2. **Standardization.** If $X \sim \text{Gauss}(\mu, \sigma^2)$, then:

$$Z = \frac{X - \mu}{\sigma} \sim \text{Gauss}(0, 1).$$

This is the affine property with $a = 1/\sigma$ and $b = -\mu/\sigma$, and it is the single most used fact about the Gaussian because it reduces every Gaussian probability to a probability about the one fixed distribution that is tabulated. That is because of standardization, you have that:

$$\mathbb{P}(u \le X \le v) = \mathbb{P}\left(\frac{u-\mu}{\sigma} \le Z \le \frac{v-\mu}{\sigma}\right) = \Phi\left(\frac{v-\mu}{\sigma}\right) - \Phi\left(\frac{u-\mu}{\sigma}\right),$$

where Z on the right-hand side is as above, $Z = \frac{X-\mu}{\sigma}$.

Question: Why does this last result follow?

Answer: Consider $\mathbb{P}(u \le X \le v)$ where $X \sim \text{Gauss}(\mu, \sigma^2)$. First, mean-center the variables by subtracting the mean of X in the inequalities and then divide each term by the standard deviation.

$$\begin{aligned} \mathbb{P}(u \le X \le v) &= \mathbb{P}(u - \mu \le X - \mu \le v - \mu) \\ &= \mathbb{P}\left(\frac{u-\mu}{\sigma} \le \frac{X-\mu}{\sigma} \le \frac{v-\mu}{\sigma}\right) \\ &= \mathbb{P}\left(\frac{u-\mu}{\sigma} \le Z \le \frac{v-\mu}{\sigma}\right). \end{aligned}$$

Observe that the inequalities are preserved because adding a number and multiplying by a positive number doesn't change an inequality's direction.

3. **Symmetry.** If $X \sim \text{Gauss}(0, \sigma^2)$, then $-X \sim \text{Gauss}(0, \sigma^2)$ (the affine property with $a = -1$, $b = 0$). Consequently, for $x > 0$:

$$\begin{aligned}\mathbb{P}(|X| > x) &= \mathbb{P}(X > x \text{ or } X < -x) \\ &= \mathbb{P}(X > x) + \mathbb{P}(X < -x) \\ &= \mathbb{P}(X > x) + \mathbb{P}(-X > x) \\ &= 2\mathbb{P}(X > x).\end{aligned}$$

   Intuitively, when you negate a Gaussian r.v. you mirror its distribution about the vertical axis. This mirroring doesn't change the shape or the spread of the distribution; it only changes the sign of the values. As a result, the negated Gaussian random variable $-X$ maintains the same mean (here, 0) and the same variance σ^2 as the original r.v. X. For the standard Gaussian the symmetry reads $\Phi(-z) = 1 - \Phi(z)$, which is why some tables print only positive z.

4. **Sums of independent Gaussians are Gaussian.** If $X_1 \sim \text{Gauss}(\mu_1, \sigma_1^2)$ and $X_2 \sim \text{Gauss}(\mu_2, \sigma_2^2)$ are independent, then

$$X_1 + X_2 \sim \text{Gauss}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

   The mean and variance are what they would be for any independent pair; the point is again that the shape stays Gaussian. (The proof is omitted here; it follows from the convolution formula, or in one line from moment generating functions.) Combined with the affine property, it gives the fact behind every exact small-sample result in this book: if $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$ then

$$\bar{X}_n \sim \text{Gauss}\left(\mu, \frac{\sigma^2}{n}\right) \qquad \text{exactly, for every } n,$$

   with no appeal to the Central Limit Theorem. For Gaussian data the CLT's conclusion holds at $n = 2$.

From undergraduate statistics: z-scores

Standardization is the z-score of an introductory course: 'how many standard deviations above the mean is this observation?' An introductory course states that the z-score of a normal observation has the standard normal distribution, which is why a single table serves every μ and σ. Here the same move is applied not to an observation but to an estimator: $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ is the z-score of $\bar{X}_n$, and the CLT says it is approximately standard Gaussian. Every 'z-interval' and 'z-test' you learned is that sentence plus a quantile from Section 6.5.

6.4 Computing probabilities, forwards and backwards

Since the CDF has no formula, every Gaussian probability is computed by standardizing and then looking up Φ, in a table or in software. There are two directions. *Forwards:* given a value, find a probability; standardize to a z-score and read $\Phi(z)$. *Backwards:* given a probability, find the value; read the table in reverse to get $z = \Phi^{-1}(\text{probability})$ and unstandardize, $x = \mu + \sigma z$. Figure 6.2 shows both on the graph of Φ, with the numbers of the exam example that follows. Forwards, a z-score goes in on the horizontal axis and a probability comes out on the vertical axis: $\Phi(2.5) = 0.9938$. Backwards, a probability goes in on the vertical axis and a z-score comes out on

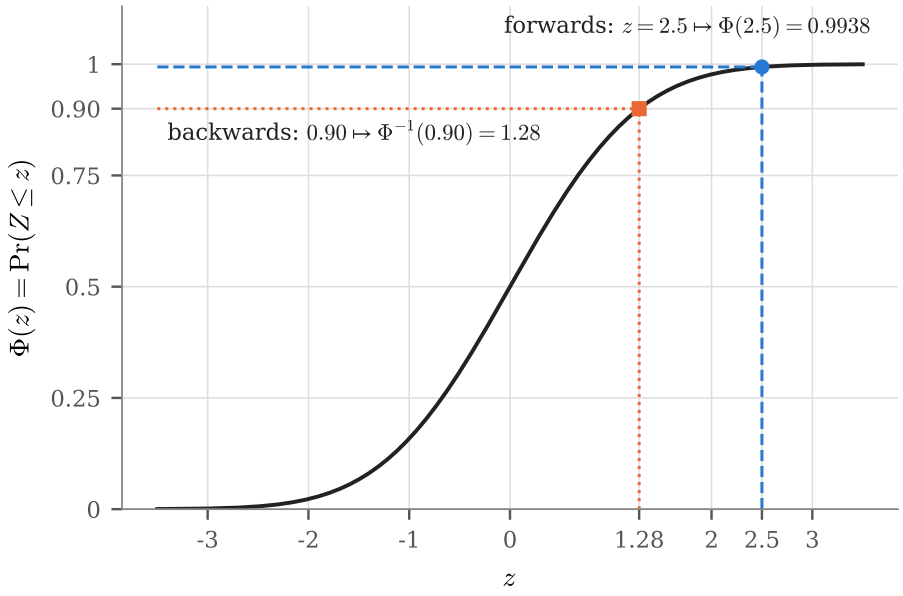


Figure 6.2: The standard Gaussian CDF Φ with the two readings of the exam example below: forwards from $z = 2.5$ to $\Phi(2.5) = 0.9938$ (blue dashes), backwards from 0.90 to $\Phi^{-1}(0.90) = 1.28$ (orange dots).

the horizontal axis: $\Phi^{-1}(0.90) = 1.28$. Tables print the forward direction; for the backward direction you scan the table for the probability, or ask software for the quantile directly.

Worked example: exam scores

Assume that the score distribution for a final exam is approximately $X \sim \text{Gauss}(85, 4)$, that is, mean 85 and standard deviation $\sigma = 2$.

- $\mathbb{P}(X > 90)$. Standardize: $z = (90 - 85)/2 = 2.5$. Then $\mathbb{P}(X > 90) = \mathbb{P}(Z > 2.5) = 1 - \Phi(2.5) = 1 - 0.9938 = 0.0062$. About six students in a thousand score above 90.
- $\mathbb{P}(80 < X < 90)$. The endpoints standardize to -2.5 and 2.5, so $\mathbb{P}(80 < X < 90) = \Phi(2.5) - \Phi(-2.5) = 2\Phi(2.5) - 1 = 0.9876$, using the symmetry $\Phi(-z) = 1 - \Phi(z)$.
- The score x such that $\mathbb{P}(X < x) = 0.90$ (the 90th percentile). For this you read the table backwards: the z with $\Phi(z) = 0.90$ is $z = 1.2816$, so $x = 85 + 2 \times 1.2816 = 87.6$.

Pitfall: variance or standard deviation?

This book writes $\text{Gauss}(\mu, \sigma^2)$: the second parameter is the *variance*. So $\text{Gauss}(85, 4)$ has standard deviation 2, and $\text{Gauss}(10, 4)$ in Figure 6.1 is twice as wide as $\text{Gauss}(0, 1)$, not four times. Most software and many textbooks take the standard deviation instead, and the mix-up fails silently: for the exam scores, $\mathbb{P}(X \leq 90)$ is 0.994 with standard deviation 2, but a 4 entered where the 2 was meant answers a different question and gives 0.894, with no warning that anything went wrong. Check which convention is in force every time, and when you write $\text{Gauss}(\mu, \sigma^2)$ yourself, write the square.

6.5 Quantiles

To grasp the idea of a quantile it's instructive to consider an example involving a closely related idea, a percentile. Suppose you score a 72 on an exam. Then your percentile is the fraction of students who score *below* a 72. In terms of probability, percentiles are computed using the CDF of exam scores: Letting X denote the score of a random student on the exam, and to make things easy supposing X is a continuous r.v., then your percentile is $\mathbb{P}(X \leq 72) = F_X(72)$.

In general, a student with score x has percentile $F_X(x)$. Going the other way, if you start with a percentile, say 0.95, then $F^{-1}(0.95)$ is the score on the exam that has that percentile. A percentile is also called a quantile, which is why F^{-1} is called the **quantile function**.[1] The function F converts scores to quantiles, and the function F^{-1} converts quantiles to scores: the forward and backward readings of Figure 6.2. And finally, note that a quantile (of a specific order) is a number in the support of an r.v., not a probability. Now for the formal definition, in the convention this book uses throughout.

Definition 6.2 (Quantile of order $1-\alpha$)**.** Let $\alpha \in (0,1)$. The **quantile of order** $1-\alpha$ of an r.v. X is the number q_α such that

$$\mathbb{P}(X \leq q_\alpha) = 1 - \alpha.$$

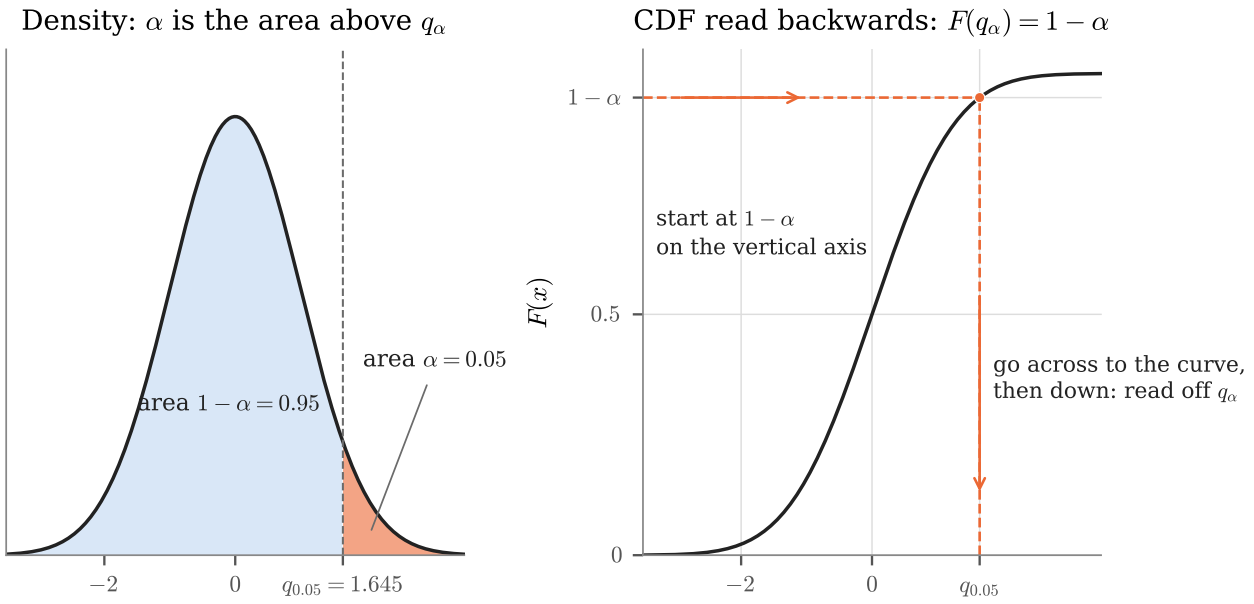


Figure 6.3: The definition of q_α for the standard Gaussian, $\alpha = 0.05$. Left: the density, with area $1-\alpha$ below q_α (blue) and area α above it (orange). Right: the CDF, with the backward reading from $1-\alpha$ to q_α (orange dashes).

In words. q_α is the cutoff with probability α above it and $1-\alpha$ below: read the CDF backwards at $1-\alpha$, as in Figure 6.3 (toolkit A.6). The subscript names the tail it cuts off, so read $q_{0.05}$ as 'the cutoff with 5% above it'.

For the kiss data. The 1.96 in $\bar{R}_n \pm 1.96 \times 0.043$ is $q_{0.025}$: it leaves 2.5% in each tail, 5% in both, so the interval catches the standardized error 95% of the time.

[1] More precisely, a percentile is a quantile expressed as a percentage.

How to read the notation. The symbol q_α carries two numbers and it is easy to attach the subscript to the wrong one. The subscript α is a *probability*, the area of the upper tail in Figure 6.3; the value q_α is a *number on the horizontal axis*, a cutoff in the same units as X. The left of the figure shows the first as a shaded area and the second as the cutoff that leaves area α to its right and $1-\alpha$ to its left. The right shows how to find the cutoff: read the CDF backwards, starting at height $1-\alpha$ on the vertical axis, going across to the curve, then dropping down to the horizontal axis; the number you land on is q_α, and for the standard Gaussian and $\alpha = 0.05$ it is 1.645. So $q_{0.05} = 1.645$ says: the probability that a standard Gaussian exceeds 1.645 is 0.05. The phrase 'quantile of order $1-\alpha$' in the definition refers to the area *below* the cutoff, 0.95 for $q_{0.05}$, which is how the inverse CDF is indexed; the subscript refers to the area *above*. Both describe the same point.

Remark: It might seem weird to define quantiles in terms of their complements; that is, the quantile is of order $1-\alpha$ and not of order α. The reason is what the quantile is for. In the formulas that follow, α is usually a small tail probability (a 5% level, a 2.5% tail), and q_α is the cutoff that leaves exactly that much probability in the upper tail. Indexing the cutoff by the tail it cuts off, rather than by the mass below it, is what makes $q_{0.05}$ mean 'the 5% critical value' in every formula that follows. In an introductory course the same number arrived as $z^* = 1.645$ or $z_{0.05}$ from the back of the book; $q_{0.05}$ is that z^* with the tail it cuts off written into the name.

Properties of quantiles. In the following let F be the CDF of some r.v. X, assumed continuous and strictly increasing so that F^{-1} exists.

- By the definitions of CDFs and quantiles, it follows that $F(q_\alpha) = 1-\alpha$.
- How do you find q_α, the quantile of order $1-\alpha$? Since $F(q_\alpha) = 1-\alpha$, you just take the inverse of F, meaning $F^{-1}(1-\alpha) = q_\alpha$.
- Note also that $\mathbb{P}(X > q_\alpha) = \alpha$. This follows from the definition of quantiles by considering the complement of $\mathbb{P}(X \le q_\alpha)$, and it's the form you'll use most: q_α is the number exceeded with probability α.
- Let $X = Z \sim \text{Gauss}(0,1)$. Then α equals the probability that $|Z|$ is greater than $q_{\alpha/2}$. That is, $\mathbb{P}(|Z| > q_{\alpha/2}) = \alpha$. This is the symmetry property: $\mathbb{P}(|Z| > q_{\alpha/2}) = 2\mathbb{P}(Z > q_{\alpha/2}) = 2 \cdot \alpha/2$.

The last property is the two-sided version, and the difference between it and the one-sided definition is drawn in Figure 6.4. For a one-sided statement ('Z exceeds the cutoff') the cutoff is q_α; for a two-sided statement ('Z is beyond the cutoff in either direction') the cutoff is $q_{\alpha/2}$ because the probability α has to be split between two tails. With $\alpha = 0.05$: on the left of the figure, $q_{0.05} = 1.645$ leaves 5% of the mass in the upper tail; on the right, $q_{0.025} = 1.96$ leaves 2.5% in each tail, so $|Z| > 1.96$ has probability 5%. Same α, different cutoff.

Some important quantiles of the standard Gaussian, which recur so often that it is worth knowing them by heart:

α	10%	5%	2.5%	0.5%
q_α	1.28	1.645	1.96	2.576

Read across: $\mathbb{P}(Z > 1.28) = 10\%$, $\mathbb{P}(Z > 1.645) = 5\%$, $\mathbb{P}(Z > 1.96) = 2.5\%$, $\mathbb{P}(Z > 2.576) = 0.5\%$. And read two-sided: $\mathbb{P}(|Z| > 1.645) = 10\%$, $\mathbb{P}(|Z| > 1.96) = 5\%$, $\mathbb{P}(|Z| > 2.576) = 1\%$. The 1.96 of every 95% confidence interval is $q_{0.025}$, not $q_{0.05}$.

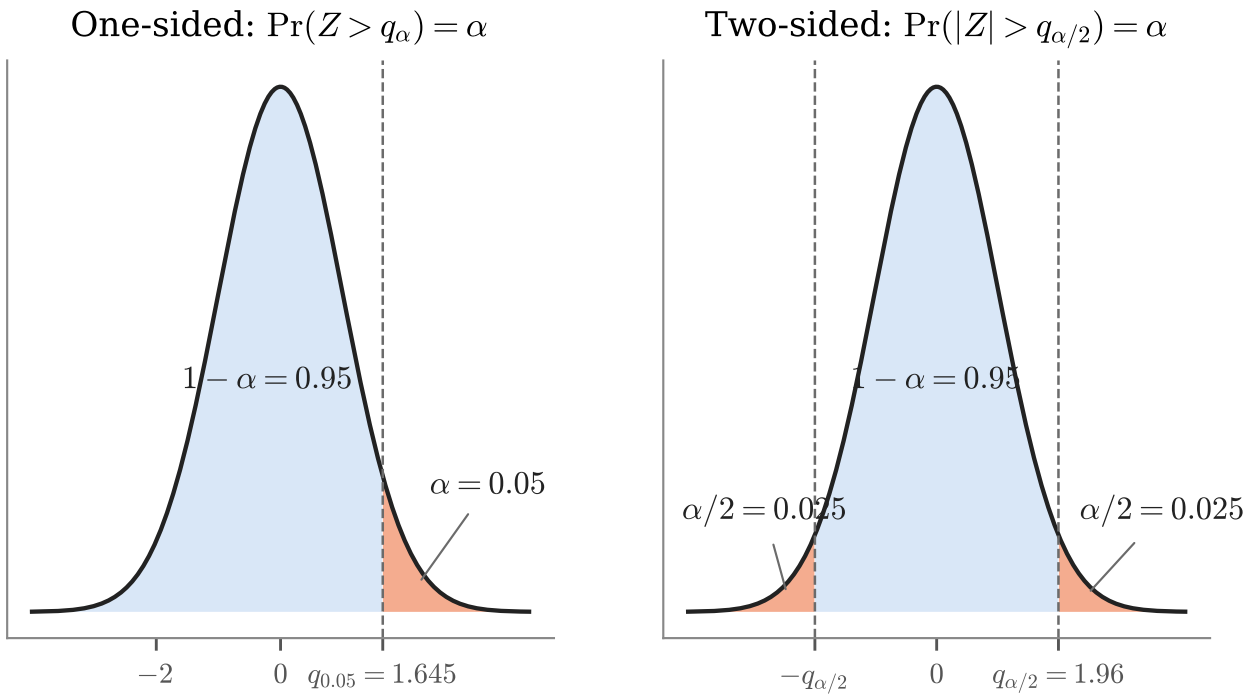


Figure 6.4: The standard Gaussian density with $\alpha = 0.05$. Left: the one-sided cutoff $q_{0.05} = 1.645$, with the tail of area 0.05 in orange. Right: the two-sided cutoffs $\pm q_{0.025} = \pm 1.96$, with the two tails of area 0.025 each in orange.

Pitfall: which α is that?

Two conventions collide here and both are in common use. This book's q_α has mass α *above* it: $q_{0.05} = 1.645$. Software and many textbooks index a quantile by the mass *below* it: on that convention the 0.95-quantile is 1.645 and the 0.05-quantile is -1.645. So this book's q_α is the other convention's $(1-\alpha)$-quantile, and its $q_{\alpha/2}$ for a 95% interval is the other convention's 0.975-quantile. When a formula in this book says $q_{\alpha/2}$ and you reach for a table or software that counts from below, look up $1-\alpha/2$. Figure 6.3 is the picture to keep in mind: the subscript is the area in the upper tail.

Example 6.3 (Quantiles of the Exponential)**.** Let $X \sim \text{Expo}(\lambda)$, with CDF $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. To find the number x below which the mass is p, solve $F_X(x) = p$:

$$\begin{aligned} 1 - e^{-\lambda x} &= p \\ e^{-\lambda x} &= 1 - p \\ -\lambda x &= \ln(1-p) \\ x &= -\frac{\ln(1-p)}{\lambda}. \end{aligned}$$

So, for example, the 50th percentile, or median, of an Exponential r.v. with parameter λ is $x = -\ln(1/2)/\lambda = \ln 2/\lambda \approx 0.69/\lambda$, which is less than the mean $1/\lambda$ because the distribution is skewed to the right. In this book's notation, $q_\alpha = -\ln(\alpha)/\lambda$: the subway inter-arrival time of Chapter 5, with $\hat{\lambda} = 0.16$ per minute, exceeds $q_{0.05} = -\ln(0.05)/0.16 \approx 19$ minutes about one time in twenty. Figure 6.5 shows the three numbers on the density. The median, $\ln 2/\lambda = 4.3$ minutes,

splits the mass in half and sits left of the mean of $1/\lambda = 6.2$ minutes because the long right tail pulls the mean up; the quantile $q_{0.05} = 18.7$ minutes leaves 5% of the mass above it. For the Gaussian the median and the mean coincide; for a skewed distribution they don't.

From undergraduate statistics: quartiles and the $1.5 \times$ IQR rule

The quartiles of an introductory course are quantiles, and in this book's notation the subscript still names the mass above: the first quartile is $q_{0.75}$, the median $q_{0.5}$ and the third quartile $q_{0.25}$. The rule behind the box plot flags as a possible outlier any value beyond the fences, 1.5 interquartile ranges below the first quartile and above the third. For a Gaussian the quartiles are $\mu \pm 0.674\sigma$, so the interquartile range is 1.35σ, the fences sit at $\mu \pm 2.70\sigma$, and 0.70% of the mass lies beyond them: about 7 flagged values in every 1,000 Gaussian observations. For the subway's waits, with $\lambda = 0.16$ per minute, the quartiles are 1.8 and 8.7 minutes, the lower fence is below zero and the upper fence is 19.0 minutes, just above $q_{0.05} = 18.7$: about 48 waits in every 1,000 lie beyond it.

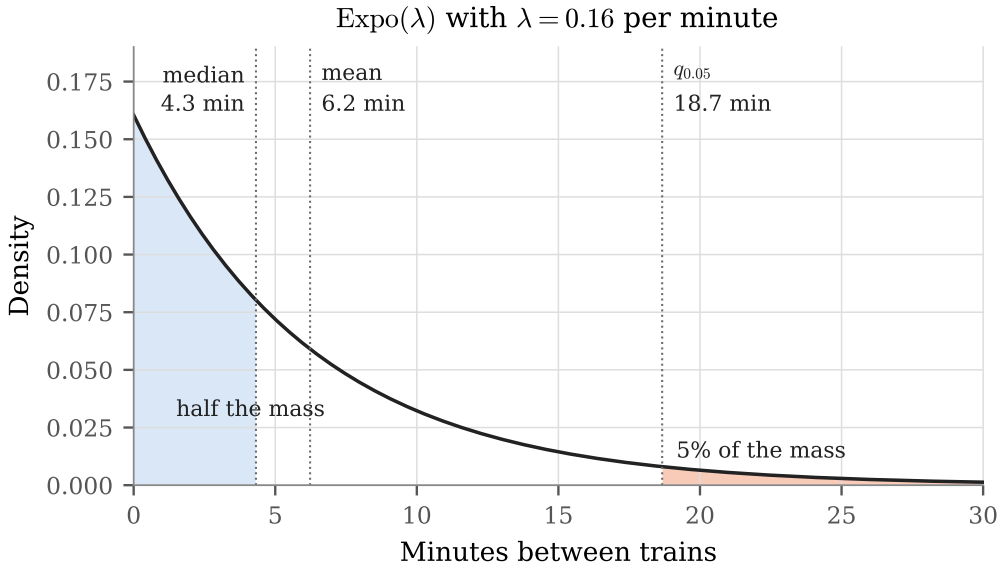


Figure 6.5: The Expo(λ) density of the subway inter-arrival times, $\lambda = 0.16$ per minute, with the median, the mean and $q_{0.05}$ marked by dotted lines. The half of the mass below the median is blue; the 5% above $q_{0.05}$ is orange.

6.6 What comes next

This closes the probability part of the book. Part III turns the three questions of Chapter 1 into definitions with guarantees, stated for the general model of Chapter 2: Chapter 7 says what an estimator is and how to judge one; Chapter 8 and Chapter 9 attach the two kinds of guarantee, an interval and a decision. The Gaussian quantiles of this chapter appear in every formula of the last two.

6.7 Summary

- $X \sim \text{Gauss}(\mu, \sigma^2)$ has density $\frac{1}{\sigma\sqrt{2\pi}} \exp(-(x-\mu)^2/(2\sigma^2))$. The second parameter is the variance. The CDF has no closed form; the tails are thin enough that 3σ contains 99.7% of the mass.
- Affine transformations, in particular standardization $Z = (X-\mu)/\sigma \sim \text{Gauss}(0,1)$, preserve Gaussianity; so do sums of independent Gaussians. The sample mean of Gaussian data is exactly $\text{Gauss}(\mu, \sigma^2/n)$.
- Every Gaussian probability is a table lookup after standardization: forwards for probabilities, backwards for quantiles. $\Phi(-z) = 1 - \Phi(z)$.
- The quantile q_α of order $1-\alpha$ satisfies $\mathbb{P}(X > q_\alpha) = \alpha$. For the standard Gaussian, $\mathbb{P}(|Z| > q_{\alpha/2}) = \alpha$. Memorize $q_{0.05} = 1.645$ and $q_{0.025} = 1.96$, and keep straight which one a question needs.

6.8 Exercises

Exercise 6.1. Let $Z \sim \text{Gauss}(0,1)$. Compute $\mathbb{P}(Z \le 1)$, $\mathbb{P}(Z \ge -1)$ and $\mathbb{P}(|Z| > 1)$. *(Solution on page 416.)*

Exercise 6.2. Let $X \sim \text{Gauss}(\mu, \sigma^2)$ and $Y = 2X$. Write down the PDF of Y, first from the affine property and then by the change-of-variables formula $f_Y(y) = f_X(g^{-1}(y))/|g'(g^{-1}(y))|$ for $Y = g(X)$. *(Solution on page 416.)*

Exercise 6.3. Let $X \sim \text{Gauss}(2,2)$ (mean 2, variance 2) and let $x > 0$. Write $\mathbb{P}(X \ge -x)$ as $\Phi(g(x))$ for a function g that is positive for $x > 0$. *(Solution on page 416.)*

Exercise 6.4. Verify the three numbers of the empirical rule from a table, and find $\mathbb{P}(|X-\mu| > 2\sigma)$ for any Gaussian X. Why doesn't the answer depend on μ or σ? *(Solution on page 417.)*

Exercise 6.5. For $\alpha = 0.10$, what is the cutoff c such that $\mathbb{P}(Z > c) = 0.10$, and what is the cutoff c' such that $\mathbb{P}(|Z| > c') = 0.10$? Which of the two belongs in a 90% confidence interval of the form $\bar{X}_n \pm c\,\sigma/\sqrt{n}$? *(Solution on page 417.)*

Part III

Foundations of Inference

The three questions of Chapter 1 become definitions with guarantees, stated for the general model of Chapter 2 and computed for the kiss experiment. An estimator (Chapter 7) is a guess inside the model, judged by bias, variance and risk; a confidence interval (Chapter 8) and a test (Chapter 9) are the two ways of attaching a guarantee to the guess.

Chapter 7

Estimators: Bias, Variance and Risk

In this chapter

- What a statistic is, what an estimator is, and the sense in which a statistic throws information away (and when, by the sufficiency principle, that's fine).
- The two large-sample properties an estimator should have: consistency (it gets to the truth) and asymptotic normality (it gets there at rate $1/\sqrt{n}$ with a Gaussian error), and why the asymptotic variance is not the limit of the variance.
- The two finite-sample measures: bias and variance, with Jensen's inequality as the reason transformed estimators pick up bias.
- Quadratic risk = variance + bias^2, and the trade-off it makes visible: a biased estimator can beat an unbiased one, and no estimator wins for every value of the parameter.

The running example is again the kiss experiment, $R_1, \dots, R_n \overset{iid}{\sim} \text{Bern}(p)$ with $n = 124$, now with several competing estimators of p. Chapter 2 fixed the object inside which inference happens, and Part II supplied the three limit facts about averages; the first task inside the model is to guess θ from the data, and this chapter says what a guess is and how to judge one.

7.1 Statistics and estimators

Definition 7.1 (Statistic, estimator)**.** A **statistic** is any measurable function of the sample: $T = T(X_1, \dots, X_n)$. For example, $\bar{X}_n$, $\max_i X_i$, $X_1 + \log(1 + |X_n|)$, the sample variance, and so on. As a statistic is a function of your data, it's an r.v.

An **estimator** of θ is a statistic $\hat{\theta}_n = \hat{\theta}_n(X_1, \dots, X_n)$ whose expression doesn't depend on θ.

> **In words.** A statistic is anything you can compute from the data. An estimator is a statistic you intend as a guess for θ, and its formula may use the data and known constants only, never θ.
>
> **For the kiss data.** $\bar{R}_n$ is a statistic and an estimator of p. $\bar{R}_n - p$ is a random variable but not a statistic, since you cannot compute it. $\max_i R_i$ is a statistic (it is 1 unless every couple turned left) and a useless estimator of p.

The word 'measurable' is a technicality from probability theory (toolkit A.5); the working rule of thumb is that if you can compute it exactly once you're given the data, it's measurable. Things

that are only implicitly defined, such as a sup or inf over an infinite family (toolkit A.2), can fail the condition, but every statistic this book writes down is a formula in the observations and passes it.

The clause 'doesn't depend on θ' is the whole content of the second definition. $\bar{R}_n$ is an estimator of p in the kiss model; $\bar{R}_n - p$ is neither a statistic nor an estimator because you can't compute it from the data alone: its formula involves the thing you're trying to estimate. (It is a perfectly good random variable, and it has a name, the **estimation error**, which is the quantity the rest of this chapter measures.) The constant 0.5 is an estimator of p (a bad one, as Section 7.8 will show), and so is X_1: the definition says what an estimator is, not what a good one is. The rest of the chapter is about telling them apart.

Recall from Chapter 1 the distinction between an estimator and an estimate: the estimator $\hat{\theta}_n$ is a random variable, with a distribution, an expectation and a variance; the *estimate* $\hat{\theta}_n(x_1, \ldots, x_n)$ is the number it produces on your one sample. Every property defined in this chapter is a property of the random variable.

7.2 A statistic reduces the data

Let $\mathbf{X} = (X_1, \ldots, X_n)$ be the vector of observations and $\mathbf{x} = (x_1, \ldots, x_n)$ a realized sample. Any statistic $T(\mathbf{X})$ defines a form of data reduction. Anyone who uses only the observed value of the statistic, $T(\mathbf{x})$, rather than the entire observed sample $\mathbf{x}$, will treat as equal two samples, $\mathbf{x}$ and $\mathbf{y}$, that satisfy $T(\mathbf{x}) = T(\mathbf{y})$ even though the actual sample values may differ.

Data reduction in terms of a particular statistic can be thought of as a partition of the sample space E^n. For illustration, suppose a sample of size n is drawn and $\bar{x} = 8$. There are many points in E^n which have the same mean of 8, and you can think of them as belonging to the set $\{(x_1, \ldots, x_n) : \bar{x} = 8\}$. In fact, all points on the hyperplane

$$x_1 + \cdots + x_n = 8n$$

yield the mean $\bar{x} = 8$, and thus the hyperplane is the set. But there are of course many values that $\bar{X}_n$ can take, and thus there are many such sets. In this sense, the sample mean, or any statistic, partitions the sample space into a collection of sets (equivalence classes). Figure 7.1 draws this for $n = 2$, where the sample space is the plane and the hyperplanes are lines, next to the quite different partition induced by the maximum. On the left, the samples $(1.5, 4.5)$ and $(4.5, 1.5)$ lie on the same line and are the same sample to anyone who only sees $\bar{x} = 3$. On the right, the level sets of the maximum are L-shapes, and $(1.5, 4.5)$ and $(4.5, 4.5)$ are the same sample to anyone who only sees $\max = 4.5$. Each statistic keeps one number and discards the rest, and which rest is discarded depends on the statistic.

Let $\mathcal{T} = \{t = T(\mathbf{x}) \text{ for some } \mathbf{x} \in E^n\}$ be the image of E^n under T; that is, $\mathcal{T}$ is the set of all possible values of the statistic. Then T partitions the sample space into the sets $A_t = \{\mathbf{x} : T(\mathbf{x}) = t\}$ for $t \in \mathcal{T}$. Samples $\mathbf{x}_1$ and $\mathbf{x}_2$ belong to the same set if and only if $T(\mathbf{x}_1) = T(\mathbf{x}_2)$. The statistic summarizes the data in that, rather than reporting the entire sample $\mathbf{x}$, it reports only that $T(\mathbf{x}) = t$ or, equivalently, $\mathbf{x} \in A_t$.

7.2.1 The sufficiency principle

A **sufficient statistic** for a parameter θ is a statistic that, in a certain sense, captures all the information about θ contained in the sample. Any additional information in the sample, besides the

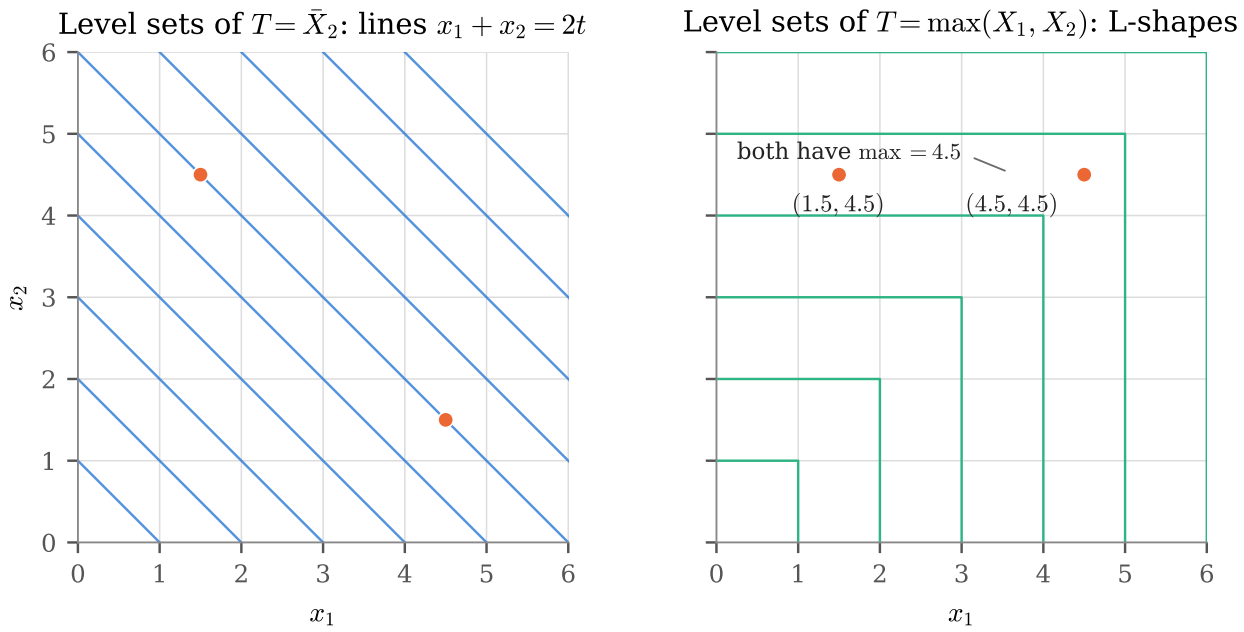


Figure 7.1: Level sets of the sample mean (left, the lines $x_1 + x_2 = 2t$) and of the maximum (right, L-shapes) for a sample of size $n = 2$, a point in the plane. Orange dots mark two samples on one level set.

value of the sufficient statistic, does not contain any more information about θ. This leads to the data reduction technique called the sufficiency principle.

Sufficiency principle: If $T(\mathbf{X})$ is a sufficient statistic for θ, then any inference about θ should depend on the sample $\mathbf{X}$ only through the value $T(\mathbf{X})$. That is, if $\mathbf{x}$ and $\mathbf{y}$ are two sample points such that $T(\mathbf{x}) = T(\mathbf{y})$, then the inference about θ should be the same whether $\mathbf{X} = \mathbf{x}$ or $\mathbf{X} = \mathbf{y}$ is observed.

Definition 7.2 (Sufficient statistic)**.** A statistic $T(\mathbf{X})$ is a **sufficient statistic for** θ if the conditional distribution of the sample $\mathbf{X}$ given the value of $T(\mathbf{X})$ does not depend on θ.

> **In words.** Once you know the value of T, the rest of the sample looks the same whatever θ is, so the rest carries no further information about θ.
>
> **For the kiss data.** Given that 80 of 124 couples turned right, every ordering of the 80 ones and 44 zeros is equally likely, whatever p is; the count is sufficient and the order tells you nothing more about p. The example below does the computation.

Example 7.3. In the kiss model, the number of couples turning right $S = \sum_i R_i$ is sufficient for p. Given $S = s$, every arrangement of s ones and $n - s$ zeros is equally likely:

$$\mathbb{P}_p(\mathbf{X} = \mathbf{x} \mid S = s) = \frac{p^s(1-p)^{n-s}}{\binom{n}{s} p^s(1-p)^{n-s}} = \frac{1}{\binom{n}{s}},$$

which doesn't involve p. Once you know 80 of 124 couples turned right, the order in which they did so tells you nothing further about p. This is why every method in this book for this model is a function of $\bar{R}_n$ alone: by the sufficiency principle, nothing else in the sample tells you more about p, if the model is right. (The order of the R_i is exactly what you would look at to check whether it is,

as with the wedding reception of exercise 4(b) in Chapter 1, whose couples would show up as a run of 1s.)

Sufficiency has a large theory of its own, which this book does not take up; it appears once more, in Chapter 22, where the exponential families make the sufficient statistic visible. It is here because it says precisely what the data reduction in Figure 7.1 is allowed to throw away: everything except T, whenever the conditional distribution of the sample given T is free of θ.

7.3 Consistency

The best estimator would be the one that equals the parameter of interest. Given the probabilistic nature of statistics, ascertaining rules to derive such an estimator is hopeless. But there are desirable properties you want estimators to have.

For example, as you collect more and more data you should get a better and better idea of which distribution generated the data. In the limit, as your sample size increases to infinity you should know exactly which distribution it is. Specifically, you want an estimator that converges to the true parameter θ as your sample size goes to infinity. An estimator satisfying this property is called *consistent*, and it comes in two forms, depending on whether the convergence is in probability or almost surely.

Definition 7.4 (Consistency)**.** An estimator $\hat{\theta}_n$ of θ is **weakly consistent** if

$$\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta \quad \text{(with respect to } \mathbb{P}_\theta\text{)},$$

that is, $\mathbb{P}_\theta\left(|\hat{\theta}_n - \theta| \geq \epsilon\right) \to 0$ for every $\epsilon > 0$, and **strongly consistent** if $\hat{\theta}_n \xrightarrow[n\to\infty]{\text{a.s.}} \theta$ (almost sure convergence is defined in Appendix B; the unqualified word 'consistent' is used for either, in this book and in common usage). Here $\mathbb{P}_\theta$ is the distribution that generated the data, and the statement is required to hold for every $\theta \in \Theta$.

> **In words.** With enough data the estimator lands as close to the truth as you like, with probability as high as you like, and this must be so whichever value the truth has.
>
> **For the kiss data.** $\bar{R}_n \xrightarrow[n\to\infty]{\mathbb{P}} p$ is the Law of Large Numbers, so $\bar{R}_n$ is consistent. The constant estimator 0.5 is not: it never moves, so it only gets close to p if p happens to be 0.5, and consistency has to hold for every p.

The Law of Large Numbers is the consistency of $\bar{X}_n$ for μ; its weak form (Theorem 3.5) gives weak consistency, its strong form (Theorem B.3 in Appendix B) strong consistency.

Recall. The Law of Large Numbers says that an average of independent draws settles on the number it is an average of: for any width you name, the probability that $\bar{X}_n$ sits outside a band of that width around μ goes to zero. It says nothing about how fast, and nothing about the shape of what is left; those are the Central Limit Theorem's business.

The continuous mapping theorem (Chapter 5) then makes any continuous function of a consistent estimator consistent for the corresponding function of the parameter.

Recall. The continuous mapping theorem says that applying a continuous function to a converging sequence gives a converging sequence, with the function applied to the limit: if $\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta$ and g is continuous at θ,

then $g(\hat{\theta}_n) \xrightarrow[n\to\infty]{\mathbb{P}} g(\theta)$. It is the rule that lets you take square roots and reciprocals of estimators without starting over.

Consistency only takes a stand on where the distribution of $\hat{\theta}_n$ *concentrates*, not on the whole distribution. This is evident from its definition; consistency is defined in terms of convergence in probability or almost surely, and both are statements about the limit being a point. But consistency usually isn't enough to do inference because you want to know not just where the estimator concentrates but also the precision with which it concentrates, and how fast.

Example 7.5 (How fast is consistent?)**.** Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bern}(p)$. What is the smallest c such that $n^c(\bar{X}_n - p)$ does *not* converge to 0 in probability? By the CLT, $\sqrt{n}(\bar{X}_n - p)/\sigma \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1)$ with $\sigma = \sqrt{p(1-p)}$, so for $c < 1/2$,

$$n^c(\bar{X}_n - p) = \frac{\sigma}{n^{1/2-c}} \cdot \frac{\sqrt{n}}{\sigma}(\bar{X}_n - p) \xrightarrow[n\to\infty]{\text{(d)}} 0 \cdot \text{Gauss}(0,1) = 0$$

by Slutsky, which is convergence in probability to 0 since the limit is a constant. At $c = 1/2$ the limit is $\text{Gauss}(0, \sigma^2)$, not 0. So $c = 1/2$: the estimation error $\bar{X}_n - p$ is of exact order $1/\sqrt{n}$, and such an estimator is called $\sqrt{n}$*-consistent*. That is more than consistency says, and it's the second property.

7.4 Asymptotic normality

This is where the property of asymptotic normality enters, which unlike consistency is *defined in terms of convergence in distribution*. Asymptotic normality is a property of an estimator that, if present, gives it CLT-like behavior: a Gaussian limiting distribution. Unlike the CLT, it's not a theorem; it's a property that a given estimator may or may not have, and specifically a property of estimators, hence concerned with inference. For reasons addressed below, it involves some scaling in order to circumvent a problem with the variance of the limiting distribution.

Definition 7.6 (Asymptotic normality)**.** An estimator $\hat{\theta}_n$ of θ is **asymptotically normal** if

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, \sigma^2).$$

The variance of this limiting distribution, σ^2, is called the **asymptotic variance** of $\hat{\theta}_n$.

> **In words.** Magnify the estimation error by $\sqrt{n}$ and it settles into a bell curve centered at 0. Unmagnified, the error is approximately $\text{Gauss}(0, \sigma^2/n)$: shrinking like $1/\sqrt{n}$, with a known shape.
>
> **For the kiss data.** $\sqrt{n}(\bar{R}_n - p) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, p(1-p))$ by the CLT, so $\bar{R}_n$ is asymptotically normal with asymptotic variance $p(1-p) = 0.229$ at $p = 0.645$, and its error at $n = 124$ is approximately $\text{Gauss}(0, 0.043^2)$.

NB$_1$: You might imagine the 'natural' form of the asymptotic variance would simply be the limit of $\mathbb{V}\text{ar}(\hat{\theta}_n)$ as n goes to infinity. **This is not what it is.** Why not? Think about the estimator $\bar{X}_n$. Its variance is $\mathbb{V}\text{ar}(\bar{X}_n) = \sigma^2/n$, and hence, as $n \to \infty$, it converges to 0. It'd be silly to let

'asymptotic variance' refer to something that's always 0. Hence, the asymptotic variance σ^2 is *not* the limit of the variance of $\hat{\theta}_n$. Figure 7.2 shows the two quantities side by side for the kiss estimator. On the left, $\mathbb{V}\mathrm{ar}(\hat{p}) = p(1-p)/n$ goes to zero, and its limit carries no information about precision. On the right, $n \cdot \mathbb{V}\mathrm{ar}(\hat{p}) = p(1-p) = 0.229$ is constant, and that constant is the asymptotic variance, the variance of the limiting Gaussian of $\sqrt{n}(\hat{p} - p)$.

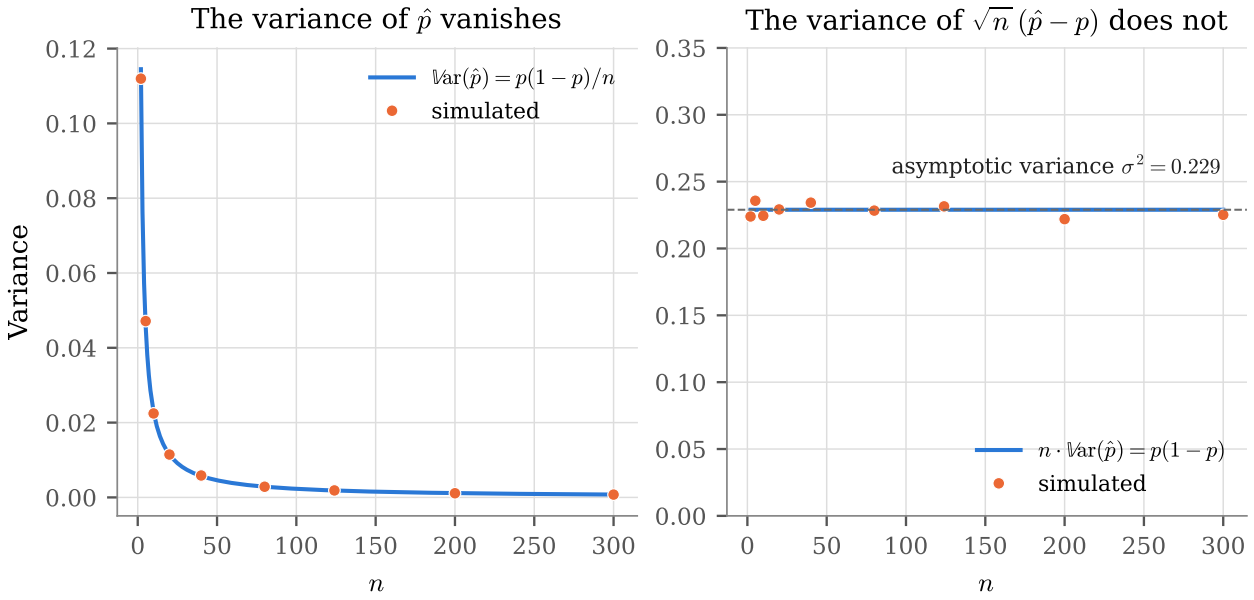


Figure 7.2: For $\hat{p} = \bar{R}_n$ with $p = 0.645$, against n. Left: $\mathbb{V}\mathrm{ar}(\hat{p}) = p(1-p)/n$. Right: $n \cdot \mathbb{V}\mathrm{ar}(\hat{p})$, with the asymptotic variance as a gray dashed line. Blue curves are exact; orange dots are variances estimated from 4,000 simulated samples at each n.

NB$_2$: This brings us to the rescaling in the definition of asymptotic normality. The rescaling by $\sqrt{n}$ is crucial because it holds the *spread* of the error fixed as the sample size increases, which is what allows a non-degenerate Gaussian limit to exist at all. To understand why, again think about $\bar{X}_n$: without the rescaling, $\bar{X}_n - \mu$ converges to the degenerate distribution at 0, which is consistency restated and says nothing about precision. With it, $\sqrt{n}(\bar{X}_n - \mu)$ has variance σ^2 for every n and a Gaussian limit. For the sample mean the asymptotic variance therefore coincides with $n\,\mathbb{V}\mathrm{ar}(\bar{X}_n)$; in general it is defined as the variance of the limiting Gaussian, which is the only definition that always makes sense, since convergence in distribution does not by itself say anything about variances. This is the two-scales picture of Section 3.2, and Example 7.5 says why $\sqrt{n}$ and not some other power.

NB$_3$: Note also that you subtract the unknown parameter θ from $\hat{\theta}_n$ in the definition. This focuses attention on the deviation of the estimator from its true value, $\hat{\theta}_n - \theta$, which is the estimation error, the quantity you actually care about.

Intuition: what asymptotic normality buys

For large n the error of an asymptotically normal estimator is approximately Gauss$(0, \sigma^2/n)$. That one sentence gives you a standard error, $\sigma/\sqrt{n}$; an error bar, $\hat{\theta}_n \pm 1.96\sigma/\sqrt{n}$ with probability about 95%; and a test of any hypothesis about θ. Consistency says the error vanishes; asymptotic normality says how big it is at the n you have and what shape it has. Chapter 8 is entirely about using the sentence. When the asymptotic variance depends on θ (as $p(1-p)$ does), you'll replace it by a consistent estimate and lean on Slutsky, as in Chapter 5.

7.5 Bias

Consistency and asymptotic normality are properties of the limit. The next two are properties of $\hat{\theta}_n$ at the sample size you actually have.

Definition 7.7 (Bias)**.** The **bias** of an estimator $\hat{\theta}_n$ of θ is

$$\text{Bias}(\hat{\theta}_n) = \mathop{\mathbb{E}}_{\theta}[\hat{\theta}_n] - \theta.$$

If $\text{Bias}(\hat{\theta}_n) = 0$ for every $\theta \in \Theta$, the estimator is called **unbiased**.

> **In words.** Average the estimator over all the samples you could have drawn; the bias is how far that average sits from the truth.
>
> **For the kiss data.** $\mathbb{E}[\bar{R}_n] = p$, so $\bar{R}_n$ is unbiased; the constant 0.5 has bias $0.5 - p$, which is -0.145 if $p = 0.645$.

An unbiased estimator is right on average: if you could repeat the experiment many times, the estimates would be scattered around θ with no systematic lean. It says nothing about how widely they're scattered.

Worked example: four estimators of p: bias

Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bern}(p)$ and consider four candidate estimators of p, the first three of which look respectable and the fourth of which does not.

1. $\hat{p} = \bar{X}_n$: $\mathbb{E}[\bar{X}_n] = p$, so the bias is 0.
2. $\hat{p} = X_1$: $\mathbb{E}[X_1] = p$, bias 0. Unbiased, and it ignores $n - 1$ observations.
3. $\hat{p} = (X_1 + X_2)/2$: $\mathbb{E} = p$, bias 0.
4. $\hat{p} = \sqrt{\mathbb{1}(X_1 = 1, X_2 = 1)}$: the indicator is 1 with probability p^2, and the square root of 0 or 1 is itself, so $\mathbb{E}[\hat{p}] = p^2$ and the bias is $p^2 - p = -p(1-p) < 0$. This estimator is systematically too low.

The fourth one is a trap for anyone who moves a square root through an expectation: $\sqrt{\mathbb{E}[\cdot]}$ is not $\mathbb{E}[\sqrt{\cdot}]$.

Worked example: a biased sample mean

Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([a, a+1])$ with a unknown. Then $\mathbb{E}[\bar{X}_n] = \mathbb{E}[X_1] = \int_a^{a+1} x \, \mathrm{d}x = a + \frac{1}{2}$, so $\bar{X}_n$ has bias $\frac{1}{2}$ as an estimator of a, for every n: more data doesn't help because the lean is built in. The fix is obvious once the bias is known: $\bar{X}_n - \frac{1}{2}$ is unbiased for a. A known bias that doesn't depend on the unknown parameter is as good as no bias at all; the dangerous biases are the ones you don't know, and the awkward ones are those that depend on the parameter, like the subway bias below.

7.5.1 Jensen's inequality

The fourth estimator above is a case of a general fact: applying a convex (or concave) function to an unbiased estimator produces a biased one, and the direction of the bias is predictable.

A function $g : \mathbb{R} \to \mathbb{R}$ is **convex** if for all $x_1 < x_2$ and all $0 \le t \le 1$,

$$g(tx_1 + (1-t)x_2) \le tg(x_1) + (1-t)g(x_2) :$$

the graph of g lies below every chord. Now read the right-hand side probabilistically. Let X take the value x_1 with probability t and x_2 with probability $1-t$. Then $tx_1 + (1-t)x_2 = \mathbb{E}[X]$ and $tg(x_1) + (1-t)g(x_2) = \mathbb{E}[g(X)]$, so convexity says $g(\mathbb{E}[X]) \le \mathbb{E}[g(X)]$ for every two-point X. It holds in general.

Theorem 7.8 (Jensen's inequality)**.** *For any r.v. X with finite mean and any convex function g,*

$$g(\mathbb{E}[X]) \le \mathbb{E}[g(X)].$$

For a concave g the inequality reverses.

The two-point argument above is the whole idea; the general case replaces the chord by the tangent line at $\mathbb{E}[X]$, which for a convex g lies below the curve everywhere (toolkit A.8); evaluating that line at X and taking expectations kills its linear term and leaves exactly $g(\mathbb{E}[X]) \le \mathbb{E}[g(X)]$.

Figure 7.3 draws the two-point case for $g(x) = 1/x$ and an X equal to 1 or 4 with probability $\frac{1}{2}$ each. The point $(\mathbb{E}[X], \mathbb{E}[g(X)])$ is the midpoint of the chord, and the chord lies above the curve, so $\mathbb{E}[g(X)] = 0.625$ exceeds $g(\mathbb{E}[X]) = 0.4$. For any convex g, averaging first and transforming second gives less than transforming first and averaging second.

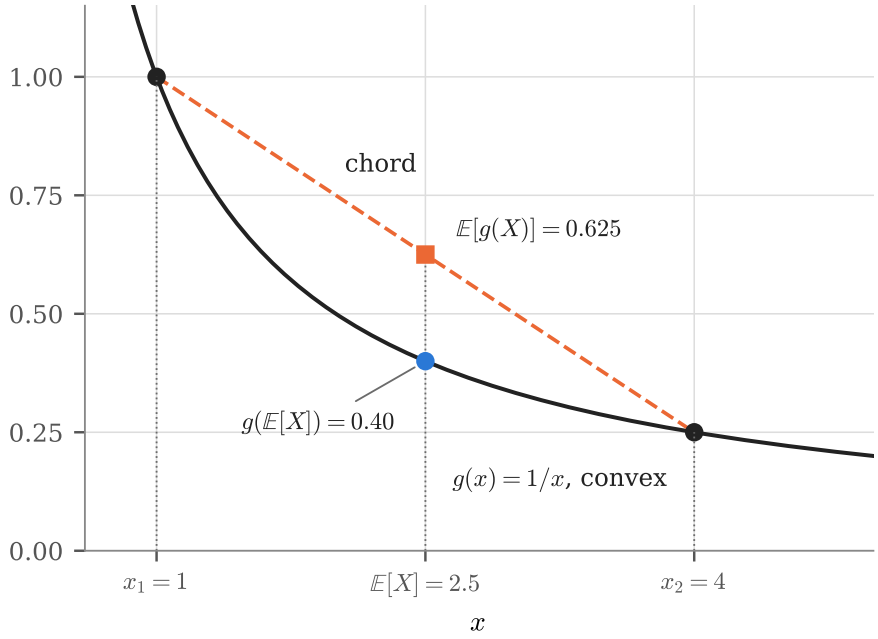


Figure 7.3: Jensen's inequality for $g(x) = 1/x$ (solid curve) and an X equal to 1 or 4 with probability $\frac{1}{2}$ each. The orange dashed line is the chord; the blue dot is $(\mathbb{E}[X], g(\mathbb{E}[X]))$ and the orange square is $(\mathbb{E}[X], \mathbb{E}[g(X)])$.

Example 7.9 (The subway rate again)**.** In Chapter 5 the rate of the T was estimated by $\hat{\lambda} = 1/\bar{T}_n$, with $\mathbb{E}[\bar{T}_n] = 1/\lambda$. Since $g(x) = 1/x$ is convex on $(0, \infty)$, Jensen gives $\mathbb{E}[1/\bar{T}_n] \ge 1/\,\mathbb{E}[\bar{T}_n] = \lambda$: the estimator is biased upward, for every n. The bias does vanish as n grows, but not because the

estimator is consistent and asymptotically normal (neither property says anything about the bias); it vanishes because an exact computation gives $\mathbb{E}[1/\bar{T}_n] = \frac{n}{n-1}\lambda$, a bias of $\lambda/(n-1)$, which is of order $1/n$ and so is swamped by the $1/\sqrt{n}$ standard error. The exercises take this up. Biased and consistent is a common combination, and nothing is wrong with it.

From undergraduate statistics: why the sample variance divides by $n-1$

An introductory course divides the sum of squared deviations by $n-1$, and the reason is a bias calculation. For $X_1, \ldots, X_n$ i.i.d. with mean μ and variance σ^2, the natural estimator $\frac{1}{n}\sum(X_i - \bar{X}_n)^2$ has expectation

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2\right] - \mathbb{E}\left[(\bar{X}_n - \mu)^2\right] = \sigma^2 - \frac{\sigma^2}{n} = \frac{n-1}{n}\sigma^2,$$

using $\sum(X_i - \bar{X}_n)^2 = \sum(X_i - \mu)^2 - n(\bar{X}_n - \mu)^2$. Its bias is $-\sigma^2/n$: the deviations from $\bar{X}_n$ are systematically smaller than the deviations from μ because $\bar{X}_n$ was chosen to be in the middle of this sample. Multiplying by $n/(n-1)$ removes the bias exactly, which is the $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X}_n)^2$ you were taught. Both versions are consistent; the correction matters at $n = 5$ and not at $n = 500$.

7.6 Variance

An estimator is a random variable, so it has a variance, $\mathbb{V}\mathrm{ar}(\hat{\theta}_n) = \mathbb{E}[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2]$, which measures how widely it scatters around its own mean (not around θ). Small variance means the estimate you got is close to the estimate you would have got from a different sample; it says nothing about whether either is close to θ.

Worked example: four estimators of p: variance

For the same four estimators, with $\mathbb{V}\mathrm{ar}(X_i) = p(1-p)$:

1. $\bar{X}_n$: $\mathbb{V}\mathrm{ar} = p(1-p)/n$, since the variance of a sum of independent terms is the sum of the variances.
2. X_1: $\mathbb{V}\mathrm{ar} = p(1-p)$. Unbiased, but as variable as a single coin flip, however large n is.
3. $(X_1 + X_2)/2$: $\mathbb{V}\mathrm{ar} = p(1-p)/2$.
4. $\sqrt{\mathbb{1}(X_1 = 1, X_2 = 1)}$: a Bernoulli with parameter p^2, so $\mathbb{V}\mathrm{ar} = p^2(1-p^2)$.

The first three are unbiased, so among them the comparison is purely on variance, and $\bar{X}_n$ wins by a factor of n over X_1. This is the sense in which using all the data matters: an unbiased estimator that ignores observations is paying in variance for no return in bias. Figure 7.4 draws all four sampling distributions on one axis. $\bar{X}_n$ is a narrow spike at p. X_1 is two bars, at 0 and 1: unbiased, since its mean is p, and never anywhere near p. $(X_1 + X_2)/2$ is three bars. The square-root indicator is two bars with its mean at $p^2 = 0.42$, off the truth. Bias is where the triangle sits relative to the orange line, variance is how spread out the bars are, and only the first estimator does well on both.

For the uniform example, $\mathbb{V}\mathrm{ar}(\bar{X}_n) = \mathbb{V}\mathrm{ar}(X_1)/n = \frac{1}{12n}$, and the bias-corrected $\bar{X}_n - \frac{1}{2}$ has the same variance, since subtracting a constant changes nothing.

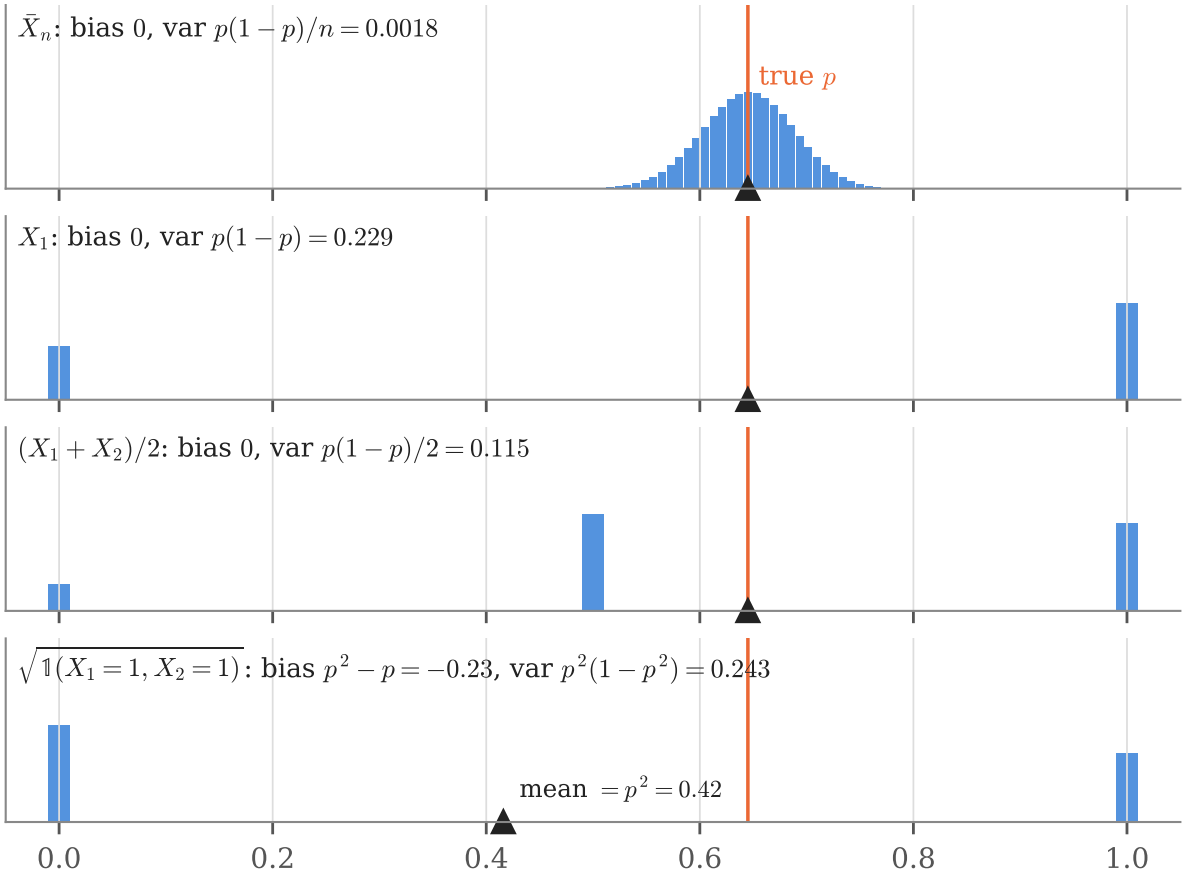


Figure 7.4: The exact sampling distributions (blue bars) of the four estimators of p at $n = 124$, $p = 0.645$. The orange line is the true p; the black triangle under each axis is the estimator's mean.

7.7 Quadratic risk

It may not be possible to find an estimator that is both unbiased and has the smallest variance. Is there a compromise that can be made between a biased estimator and a high-variance one? That is, can you find an estimator that simply has a small bias and a small variance? That's where the quadratic risk comes in: a single number that charges for both.

Definition 7.10 (Quadratic risk)**.** The **quadratic risk** (or **risk**) of an estimator $\hat{\theta}_n \in \mathbb{R}$ is

$$R(\hat{\theta}_n) = \mathop{\mathbb{E}}_{\theta}\left[|\hat{\theta}_n - \theta|^2\right],$$

the expected squared distance from the estimator to the truth. It is more commonly called the **mean squared error**, and the two names mean the same thing.

> **In words.** Square the miss, average over all possible samples. One number that punishes both a systematic lean and a wide scatter.
>
> **For the kiss data.** $R(\bar{R}_n) = p(1-p)/n = 0.00185$ at $p = 0.645$, $n = 124$; a typical miss is its square root, 0.043.

Theorem 7.11 (Bias-variance decomposition)**.**

$$R(\hat{\theta}_n) = \mathbb{V}ar(\hat{\theta}_n) + Bias(\hat{\theta}_n)^2.$$

Proof strategy (core). Add and subtract $\mathbb{E}[\hat{\theta}_n]$ inside the square, expand, and watch the cross term vanish because $\mathbb{E}[\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n]] = 0$.

Proof. First note that $|\hat{\theta}_n - \theta|^2 = (\hat{\theta}_n - \theta)^2$. Add and subtract $\mathbb{E}[\hat{\theta}_n]$ inside the square and expand:

$$\begin{aligned}
R(\hat{\theta}_n) &= \mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n] + \mathbb{E}[\hat{\theta}_n] - \theta)^2\right] \\
&= \mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2\right] + \mathbb{E}\left[(\mathbb{E}[\hat{\theta}_n] - \theta)^2\right] + 2\,\mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])(\mathbb{E}[\hat{\theta}_n] - \theta)\right] \\
&= \mathbb{V}\mathrm{ar}(\hat{\theta}_n) + (\mathbb{E}[\hat{\theta}_n] - \theta)^2 + 2(\mathbb{E}[\hat{\theta}_n] - \theta)\ \mathbb{E}\left[\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n]\right] \\
&= \mathbb{V}\mathrm{ar}(\hat{\theta}_n) + \mathrm{Bias}(\hat{\theta}_n)^2 + 2(\mathbb{E}[\hat{\theta}_n] - \theta) \cdot 0.
\end{aligned}$$

In the second term, $\mathbb{E}[\hat{\theta}_n] - \theta$ is a number, so its expectation is itself; in the third, the same number comes out of the expectation, and what remains, $\mathbb{E}[\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n]]$, is zero. □

From undergraduate statistics: the MSE in an ANOVA table

An introductory course's ANOVA table reports MSE, the mean square for error: the squared deviations of the observations from their group means, summed and divided by $N - k$ for N observations in k groups (simple regression output divides the sum of squared residuals by $N - 2$ instead). It shares a name with the quadratic risk, but the squares it averages are residuals, not an estimator's misses. MSE is a statistic, computed from the data, and an unbiased estimator of σ^2, the common variance of the model's errors. The quadratic risk is an expectation under $\mathbb{P}_\theta$, a function of the unknown parameter, and no data set produces it. The two meet in the standard error: with one group the MSE is the sample variance S^2, and S^2/n, the square of the estimated standard error $S/\sqrt{n}$, is an unbiased estimator of σ^2/n, which is the risk of $\bar{X}_n$ since $\bar{X}_n$ is unbiased.

The decomposition is the reason a biased estimator can be the better one: if accepting a little bias buys a large reduction in variance, the risk goes down. It also makes a point that is easy to miss when you only ever compute bias and variance separately: the risk is a function of θ, through both terms, and an estimator that is best for one value of θ need not be best for another. Figure 7.5 shows both facts in the kiss model. On the left, at $p = 0.645$ and $n = 124$, $\bar{R}_n$ is unbiased with standard deviation 0.043. The shrinkage estimator $(n\bar{R}_n + 10)/(n + 20)$ pulls the estimate toward $\frac{1}{2}$, as if 20 extra coin flips had come up half heads; it is biased by -0.020 but has standard deviation 0.037, and its risk, 0.00177, is smaller than $\bar{R}_n$'s 0.00185. On the right, the risk of four estimators is plotted against the true p. Shrinkage beats $\bar{R}_n$ for p between 0.34 and 0.66 and loses outside; the constant 0.5 is unbeatable at $p = 0.5$ and terrible elsewhere; X_1 is unbiased and hopeless everywhere. No curve lies below all the others, so 'best estimator' is not a well-posed question without saying for which p.

Intuition: bias and variance as a target

Think of the truth θ as the bull's-eye and each estimate as a shot. Variance is how tight the group is; bias is how far the center of the group is from the bull's-eye. The risk is the average squared distance of a shot from the bull's-eye, and the decomposition says it's the

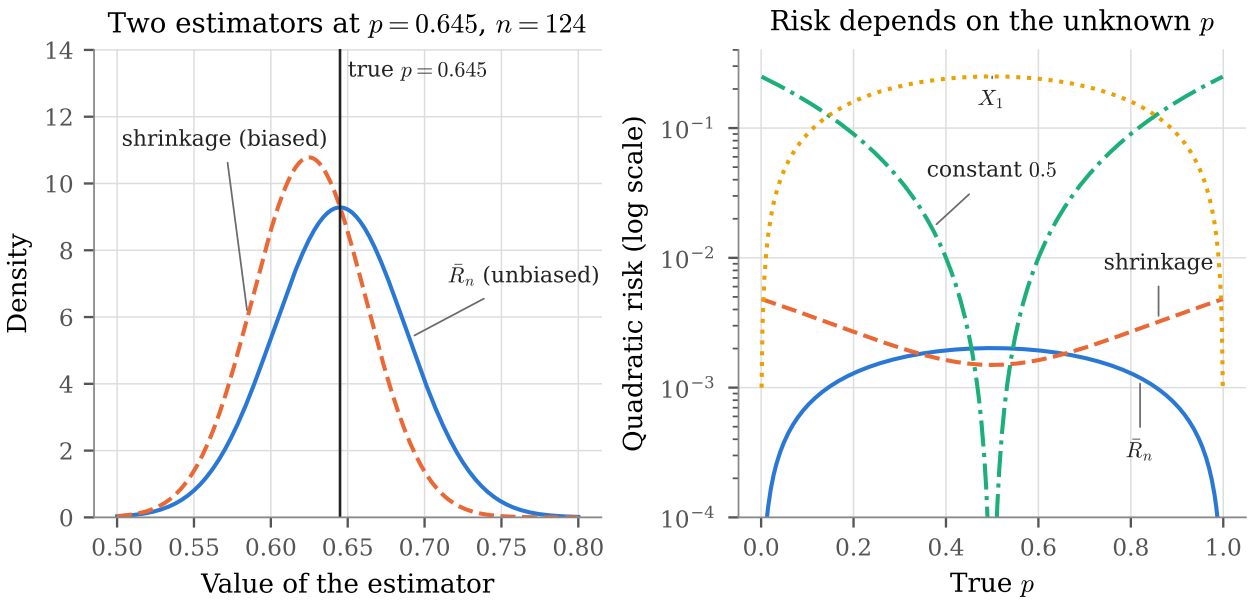


Figure 7.5: The kiss model, $n = 124$. Left: Gaussian approximations to the sampling distributions of $\bar{R}_n$ (solid) and the shrinkage estimator (dashed) at $p = 0.645$. Right: risk against the true p, log scale: $\bar{R}_n$ (solid), shrinkage (dashed), the constant 0.5 (dash-dot), X_1 (dotted).

squared distance of the group's center plus the group's spread: a tight group off to one side can be closer on average than a loose group centered on the target. The shrinkage estimator in Figure 7.5 is the tight group off to one side. What the target picture doesn't show is that moving the bull's-eye (changing p) changes which group is closer, and that's the right of Figure 7.5. Figure 7.6 draws the four combinations. The cross in each is $\mathbb{E}[\hat{\theta}_n]$, so bias is the distance from the cross to the bull's-eye, variance is the spread of the dots around the cross, and the risk is the average squared distance from a dot to the bull's-eye, which the decomposition splits into the two. Lower left is the shrinkage estimator of Figure 7.5: a tight group slightly off center, closer on average than the loose group centered on the target at upper right.

The target picture also settles a question the earlier bridge on $n - 1$ left open: Unbiased is not the same as closest.

From undergraduate statistics: the $n - 1$ story, finished

Among estimators of the form $c \sum_i (X_i - \bar{X}_n)^2$, the unbiased S^2 is not the one with the smallest quadratic risk. For Gaussian data, dividing by $n+1$ instead gives a biased estimator with strictly smaller quadratic risk than both the n and $n - 1$ versions. It is rarely used, since unbiasedness is a convention that makes results comparable across studies. At $n = 12$ the risks are $0.182\sigma^4$, $0.160\sigma^4$ and $0.154\sigma^4$ for the divisors $n - 1$, n and $n + 1$, and Exercise 14.3 proves the general case.

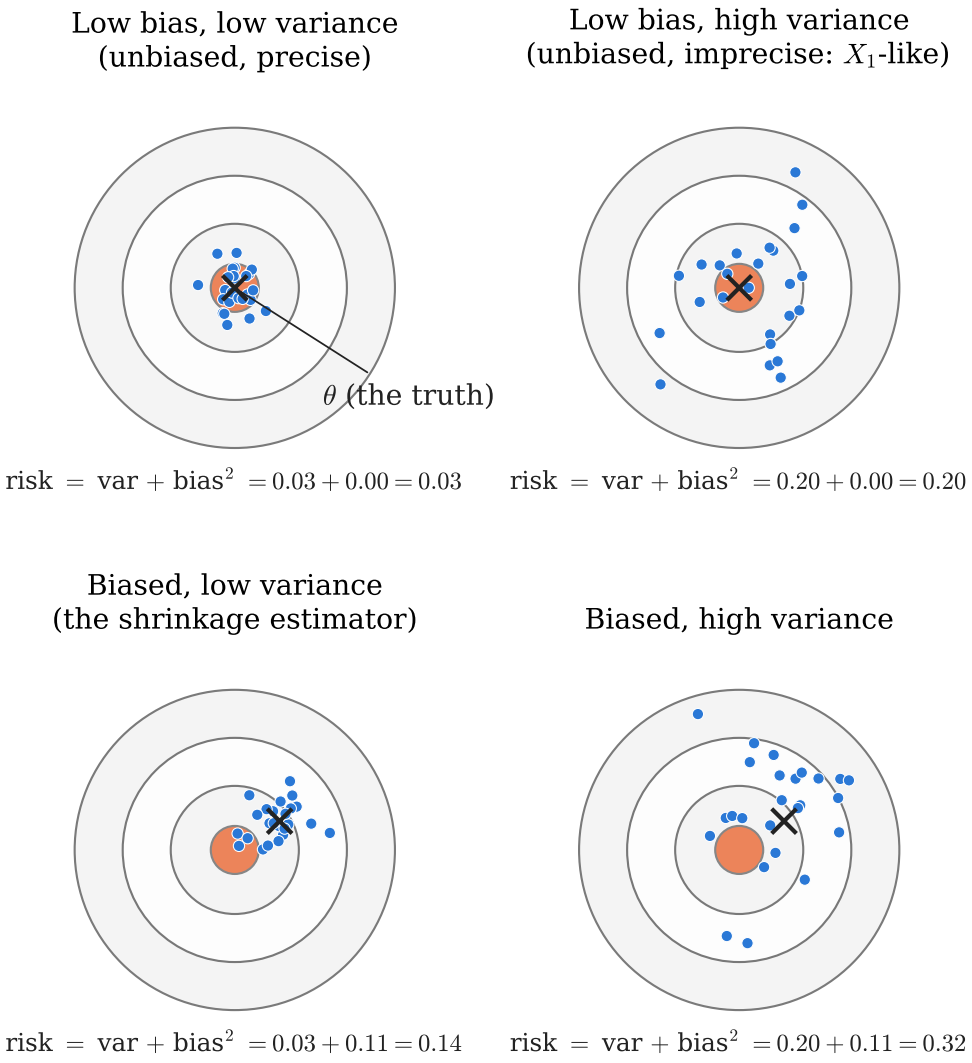


Figure 7.6: Bias and variance as a target, in the four combinations. The bull's-eye is the truth θ; each blue dot is the estimate from one sample; the black cross is the center of the group, $\mathbb{E}[\hat{\theta}_n]$.

7.8 Strengths and weaknesses of estimators

Everything in this chapter can be applied at once to three estimators of p for $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bern}(p)$, n large: $\bar{X}_n$, the constant 0.5, and X_1. Four criteria: bias, consistency, whether the estimator is efficiently computable (in this section, whether there is an explicit formula for it; the maximum likelihood estimators of Chapter 23 have no closed form and are found by an iterative algorithm instead), and whether the quadratic risk tends to 0 as $n \to \infty$.

Estimator	Bias	Consistent?	Explicit formula?	Risk $\to 0$?
$\bar{X}_n$	0	yes (LLN)	yes	yes: $p(1-p)/n$
0.5	$0.5 - p$, nonzero unless $p = \frac{1}{2}$	no, unless $p = \frac{1}{2}$	yes	no: $(0.5-p)^2$ for every n
X_1	0	no: it's 0 or 1, never near p	yes	no: $p(1-p)$ for every n

The constant 0.5 uses none of the data, so nothing can be learned from it; its only virtue is that it's exactly right when $p = \frac{1}{2}$. X_1 is unbiased, which sounds good, and uses one observation out of n, which is the actual problem: its risk doesn't shrink, so its accuracy is limited no matter how many samples are collected. $\bar{X}_n$ has every property on the list. Unbiasedness is desirable and may be the first thing to check, but the comparison with X_1 shows why it isn't enough: there are estimators that are unbiased and have a lot of variability, and risk is the number that catches them.

From undergraduate statistics: random sampling for bias, a larger sample for variability

An introductory course often draws the four targets of Figure 7.6 with two rules: random sampling reduces bias, and a larger sample reduces variability. The two rules act on the two terms of Theorem 7.11. Random sampling makes each R_i a draw from Bern(p) with p the population's proportion, which is all that $\mathbb{E}[\bar{R}_n] = p$ needs. A design that over-represents part of the population draws from Bern(p') instead, with $p' \neq p$, so $\bar{R}_n$ is unbiased and consistent for p', and as an estimator of p it has

$$R(\bar{R}_n) = \frac{p'(1-p')}{n} + (p' - p)^2,$$

of which a larger sample shrinks only the first term. The 1936 *Literary Digest* poll had about 2.4 million returned ballots and predicted that Landon would beat Roosevelt 57% to 43%; of the votes cast for the two, Roosevelt won 62% (Freedman, Pisani and Purves, *Statistics*). With $p' = 0.43$ and $p = 0.62$ the first term is 1.0×10^{-7}, a standard deviation of 0.0003, and the second is 0.036: the risk was almost all bias, a miss of 19 points.

7.9 What comes next

Asymptotic normality hands you a sentence, 'the error of $\hat{\theta}_n$ is approximately Gauss$(0, \sigma^2/n)$', and the next two chapters are the two ways of using it. Chapter 8 turns it into a range of plausible values of θ with a guaranteed capture rate; Chapter 9 turns it into a yes/no decision about θ with a guaranteed error rate. Both need the quantiles of Chapter 6 and the plug-in step of Chapter 5, since σ^2 usually depends on θ.

7.10 Summary

- A statistic is a function of the data; an estimator of θ is a statistic whose formula doesn't involve θ. Both are random variables. A statistic partitions the sample space and keeps one number per cell; a sufficient statistic keeps everything the sample says about θ.

- Consistent: $\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta$ under $\mathbb{P}_\theta$, for every θ. The sample mean is $\sqrt{n}$-consistent: its error is of exact order $1/\sqrt{n}$ (Example 7.5). Asymptotically normal: $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, \sigma^2)$; σ^2 is the asymptotic variance, the variance of the limiting Gaussian, which for the sample mean equals $n\,\mathbb{V}\text{ar}(\bar{X}_n)$ and is never the limit of $\mathbb{V}\text{ar}(\hat{\theta}_n)$, which for any estimator whose variance shrinks is 0 and carries no information.
- Bias: $\mathbb{E}[\hat{\theta}_n] - \theta$. Unbiased means right on average. Convex or concave functions of unbiased estimators are biased, in the direction Jensen's inequality says: $g(\mathbb{E}[X]) \le \mathbb{E}[g(X)]$ for convex g.
- Variance: spread of $\hat{\theta}_n$ around its own mean. Using more of the data lowers it.
- Quadratic risk: $\mathbb{E}[(\hat{\theta}_n - \theta)^2] = \mathbb{V}\text{ar} + \text{Bias}^2$. It depends on θ; a biased estimator can have lower risk than an unbiased one, and no estimator has the lowest risk for every θ.
- Of $\bar{X}_n$, the constant $\frac{1}{2}$ and X_1, only $\bar{X}_n$ is unbiased, consistent and has risk tending to 0. Unbiasedness alone, as X_1 shows, is not enough.

7.11 Exercises

Exercise 7.1. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([a, a+1])$. Find $\mathbb{E}[\bar{X}_n]$, the bias of $\bar{X}_n$ for a, $\mathbb{V}\text{ar}(\bar{X}_n)$, and the quadratic risk of $\bar{X}_n - \frac{1}{2}$. *(Solution on page 417.)*

Exercise 7.2. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bern}(p)$ and $\hat{p} = \sqrt{\mathbb{1}(X_1 = 1, X_2 = 1)}$. Find its bias, variance and quadratic risk, and say whether it is consistent. *(Solution on page 417.)*

Exercise 7.3. In the kiss model with $n = 124$, compute the bias, variance and risk of the shrinkage estimator $\tilde{p}_n = (n\bar{R}_n + 10)/(n + 20)$ at $p = 0.645$ and at $p = 0.9$, and compare each with $\bar{R}_n$. *(Solution on page 417.)*

Exercise 7.4. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Expo}(\lambda)$ and $\hat{\lambda} = 1/\bar{X}_n$. Use Jensen's inequality to determine the sign of the bias, and use the fact (which you may take as given) that $\bar{X}_n$ has the Gamma distribution with $\mathbb{E}[1/\bar{X}_n] = \frac{n}{n-1}\lambda$ for $n \ge 2$ to find the bias exactly. Is there an unbiased estimator of λ of the form $c/\bar{X}_n$? *(Solution on page 417.)*

Exercise 7.5. Show that an estimator whose bias and variance both tend to 0 is consistent. Is the converse true? *(Solution on page 417.)*

Chapter 8

Confidence Intervals

In this chapter

- What a confidence interval is: a *random* interval, computable from the data alone, that traps the fixed parameter with a guaranteed probability. The two words that carry the definition are 'random' and 'computable'.
- The recipe that produces every interval in this book: find a pivot (a quantity whose distribution you know), trap it between two quantiles, solve for the parameter. An exact pivot gives an exact interval; the CLT gives an asymptotic one.
- Three ways around a pivot whose scale involves the unknown parameter: conservative, solve, plug-in. One picture explains all three and shows how they nest.
- What drives the width (the level, through a quantile; the sample size, through $1/\sqrt{n}$), and what the 95% does and does not mean.

The running examples are the kiss experiment ($n = 124$, $\bar{R}_n = 0.645$), for which Chapter 1 promised the interval $[0.56, 0.73]$, and the subway inter-arrival times of Chapter 5 ($n = 64$, $\bar{T}_n = 6.23$).

8.1 What a confidence interval is

Chapter 7 produced a number, $\hat{p} = 0.645$, and a sampling distribution around it. A confidence interval converts the two into a statement with a guarantee: a range of values of p, and a probability that the range is right.

Definition 8.1 (Confidence interval)**.** Let $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ be a statistical model based on observations $X_1, \dots, X_n$, with $\Theta \subseteq \mathbb{R}$, and let $\alpha \in (0, 1)$.

A **confidence interval of level** $1-\alpha$ for θ is a random interval $\mathcal{I} = [L(X_1, \dots, X_n),\, U(X_1, \dots, X_n)]$ whose endpoints are statistics (their formulas don't involve θ), such that

$$\mathbb{P}_\theta\left(\mathcal{I} \ni \theta\right) \geq 1 - \alpha \quad \text{for all } \theta \in \Theta.$$

A confidence interval of **asymptotic level** $1 - \alpha$ is a random interval $\mathcal{I} = \mathcal{I}_n$ of the same kind such that

$$\lim_{n\to\infty} \mathbb{P}_\theta\left(\mathcal{I}_n \ni \theta\right) \geq 1 - \alpha \quad \text{for all } \theta \in \Theta.$$

In words. A rule that turns a sample into an interval, computable from the sample alone, such that the interval it produces contains the true parameter at least $100(1-\alpha)\%$ of the time, whatever the true parameter is. 'Asymptotic' means the guarantee holds in the limit of large samples.

For the kiss data. The rule '$\bar{R}_n \pm 1.96\sqrt{\bar{R}_n(1-\bar{R}_n)/n}$' is a confidence interval of asymptotic level 95%; on this sample it produced $[0.56, 0.73]$. The rule '$\bar{R}_n \pm 1.96\sqrt{p(1-p)/n}$' is not one, since p is in its formula.

The notation $\mathcal{I} \ni \theta$ says '$\mathcal{I}$ contains θ'. It means the same as $\theta \in \mathcal{I}$, and this book writes it backwards on purpose: the random object, the thing the probability is about, is the interval on the left, not the parameter on the right. Both appearances of θ in $\mathbb{P}_\theta(\mathcal{I} \ni \theta)$ are the same fixed number doing two jobs: as the subscript it fixes which distribution the sample is drawn from, and inside the parentheses it is the target that the interval either contains or misses. Nothing in the expression is random except $\mathcal{I}$.

Three things in the definition deserve a second look.

1. *The interval is random; the parameter is not.* L and U are functions of the sample, so they're random variables, and $\mathcal{I}$ moves from sample to sample. θ is a fixed number that doesn't move. The probability is the probability that the moving interval lands on the fixed point.
2. *The endpoints must be computable.* An interval whose formula contains θ is not a confidence interval, however nice its coverage probability: you can't compute it. This clause is what the three 'solutions' of Section 8.4 are solutions to.
3. *The guarantee holds for every θ.* You don't know θ, so a guarantee that holds for some values of it and not others is no guarantee. This clause is what makes one of the intervals in Exercise 8.2 fail.

The inequality $\geq 1-\alpha$ has a consequence worth stating once: a confidence interval of level 95% is also a confidence interval of level 90%, of level 50%, of any level below 95%. It is not, in general, one of level 98%. 'Level' is a floor on the coverage, not its exact value.

The word 'level' is used for tests too (Chapter 9), where it denotes α, the probability of rejecting a true hypothesis; for an interval the level is $1-\alpha$, the probability of a correct capture. Same α, complementary conventions, and the context always says which is meant.

Intuition: a procedure, not a number

The confidence interval is the *rule* that turns a sample into an interval, $(x_1, \ldots, x_n) \mapsto [L, U]$, and the level is a property of the rule: the fraction of samples on which the rule's output contains θ. Once you apply the rule to your one sample you get two numbers, $[0.56, 0.73]$. Those numbers are not random and the parameter is not random, so the pair has no probability attached: $[0.56, 0.73]$ either contains p or it doesn't, and you will never know which. What you know is that the rule that produced it is right 95% of the time. Figure 8.1 is this sentence as a picture. The kiss experiment was simulated 100 times with $n = 124$ and $p = 0.645$, and the plug-in interval of Section 8.4 was computed each time. The parameter is the fixed vertical line; the intervals move, and each is centered at its own $\hat{p}$. Sorted by $\hat{p}$, the six that miss are the samples whose proportion came out unusually low or high. Ninety-four of the hundred contain 0.645 and six don't, and the level 95% is the long-run rate of the first kind, not a statement about any one of the hundred intervals.

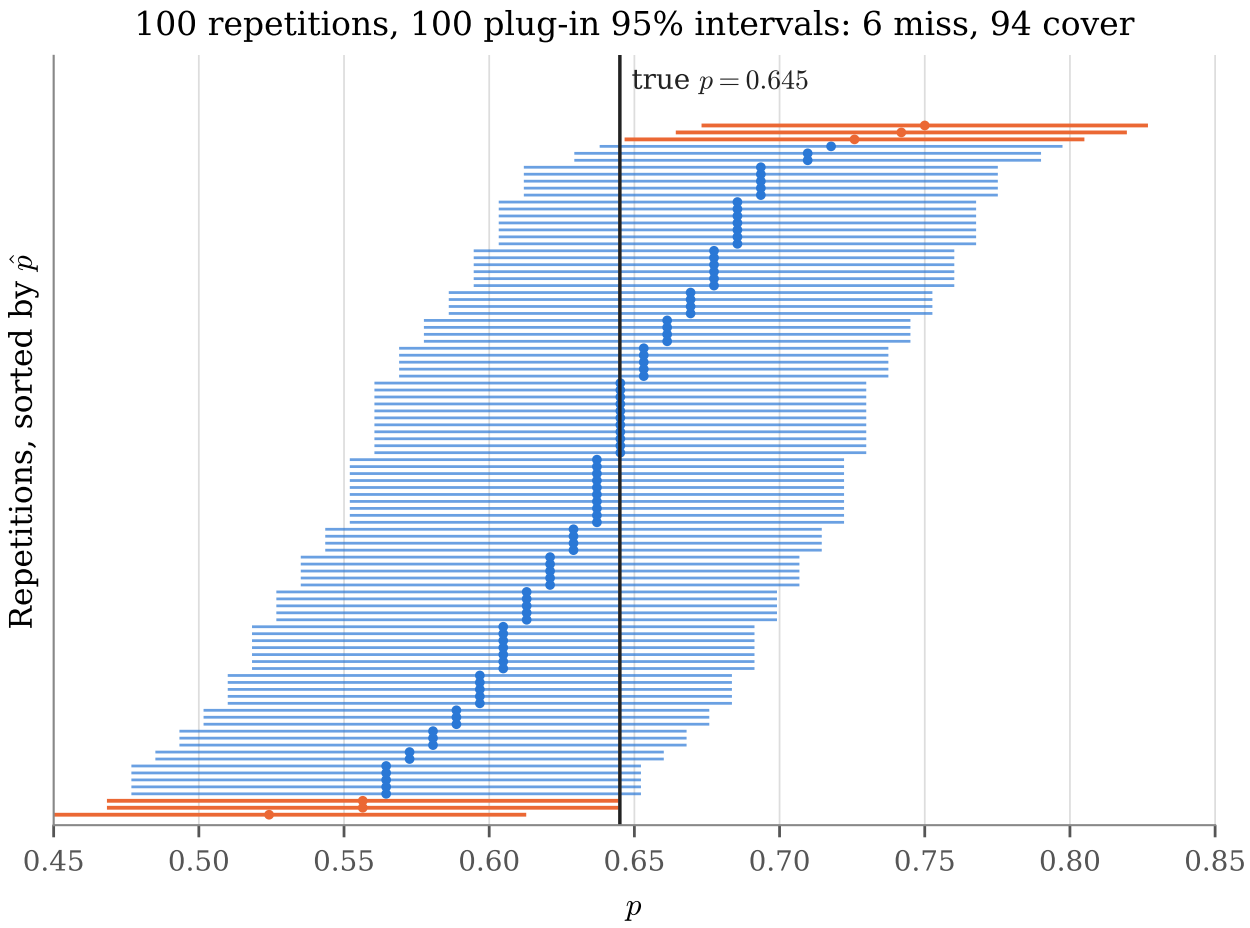


Figure 8.1: One hundred simulated kiss experiments with $n = 124$ and $p = 0.645$, sorted by $\hat{p}$ (dot), each with its 95% plug-in interval. The black vertical line is p; intervals that contain it are blue, the six that miss are orange.

8.2 The recipe, on an exact case

Every interval in this book is built the same way, and the cleanest place to see the construction is the one case where nothing is asymptotic.

Worked example: Gaussian data, known variance

Let $X_1, \dots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$ with σ^2 known and μ unknown. By Chapter 6, $\bar{X}_n \sim \text{Gauss}(\mu, \sigma^2/n)$ exactly, for every n, so

$$Z = \sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma} \sim \text{Gauss}(0,1) \quad \text{for every } n \text{ and every } \mu.$$

A quantity like Z, a function of the data and the parameter whose distribution doesn't depend on the parameter, is called a **pivot**. The recipe has three steps.

1. *Trap the pivot.* With $q_{\alpha/2}$ the quantile of Section 6.5, $\mathbb{P}(-q_{\alpha/2} \le Z \le q_{\alpha/2}) = 1 - \alpha$.
2. *Rewrite the event.* $-q_{\alpha/2} \le \sqrt{n}(\bar{X}_n - \mu)/\sigma \le q_{\alpha/2}$ is the same event as $|\bar{X}_n - \mu| \le q_{\alpha/2}\,\sigma/\sqrt{n}$.

3. *Solve for the parameter.* That event is the same as

$$\mu \in \left[\bar{X}_n - q_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{X}_n + q_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right] =: \mathcal{I}.$$

The event in step 3 is the event in step 1, written differently, so $\mathbb{P}_\mu(\mathcal{I} \ni \mu) = 1-\alpha$ for every μ and every n. The endpoints involve $\bar{X}_n$, σ (known), n and a quantile, none of which is μ, so $\mathcal{I}$ is a confidence interval of level $1-\alpha$, with no limit anywhere. For $\alpha = 0.05$ it is $\bar{X}_n \pm 1.96\,\sigma/\sqrt{n}$.

The three steps are the construction behind the formula an introductory course hands out.

From undergraduate statistics: estimate ± margin of error

The interval you learned is 'point estimate ± (critical value) × (standard error)'. The recipe says where each piece comes from: the point estimate is the center because the pivot is centered at the parameter; the critical value is the quantile that traps the pivot; the standard error is the pivot's scale, $\sigma/\sqrt{n}$. When σ is unknown, an introductory course replaces it by s and switches to a t critical value: that is a different pivot, $\sqrt{n}(\bar{X}_n - \mu)/s$, with an exact Student distribution for Gaussian data, and Chapter 14 derives it. For large n and any parent distribution with finite variance, the replacement is justified by Slutsky instead (Section 5.2), which is the route this chapter takes.

One-sided intervals. Nothing forces step 1 to trap the pivot symmetrically. $\mathbb{P}(Z \le q_\alpha) = 1-\alpha$ as well, and solving $\sqrt{n}(\bar{X}_n - \mu)/\sigma \le q_\alpha$ for μ gives

$$\mu \in \left[\bar{X}_n - q_\alpha\frac{\sigma}{\sqrt{n}},\ \infty\right),$$

a **lower confidence bound**: 'μ is at least this much', with level $1-\alpha$. Trapping $Z \ge -q_\alpha$ gives the upper bound $(-\infty, \bar{X}_n + q_\alpha\sigma/\sqrt{n}]$. Note the quantile: q_α (1.645 at 95%), not $q_{\alpha/2}$ (1.96) because all of the 5% of misses is spent on one side. One-sided intervals are the confidence-interval face of one-sided tests, and Exercise 8.6 makes the correspondence exact.

8.3 The kiss experiment: an interval that isn't one

Now $R_1, \dots, R_n \overset{iid}{\sim} \text{Bern}(p)$, $\hat{p} = \bar{R}_n$, and there is no exact pivot in sight: the distribution of $\bar{R}_n$ is Binomial, scaled, and it depends on p in a way that no simple transformation removes. What there is instead is the CLT (Theorem 4.8).

Recall. The Central Limit Theorem says that the error of an average, magnified by $\sqrt{n}$ and divided by the standard deviation of one observation, has a distribution that settles down to Gauss$(0,1)$, whatever the observations were drawn from. You do not need to know that distribution to use the theorem; you need only its mean and variance.

$$\sqrt{n}\,\frac{\bar{R}_n - p}{\sqrt{p(1-p)}} \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1).$$

The left side is an *asymptotic* pivot: its distribution depends on p for every finite n, but its limit doesn't. Convergence in distribution means convergence of the CDF at every continuity point of

the limit (Section 4.1), and the Gaussian CDF is continuous everywhere, so

$$\lim_{n\to\infty} \mathbb{P}_p\left(-q_{\alpha/2} \le \sqrt{n}\,\frac{\bar{R}_n - p}{\sqrt{p(1-p)}} \le q_{\alpha/2}\right) = \mathbb{P}(-q_{\alpha/2} \le Z \le q_{\alpha/2}) = 1-\alpha.$$

Steps 2 and 3 of the recipe then give

$$\lim_{n\to\infty} \mathbb{P}_p\left(\left[\bar{R}_n - q_{\alpha/2}\sqrt{\frac{p(1-p)}{n}},\ \bar{R}_n + q_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right] \ni p\right) = 1-\alpha. \tag{8.1}$$

This looks like the Gaussian case with σ replaced by $\sqrt{p(1-p)}$, and it has the right asymptotic coverage, *but it is not a confidence interval*: the endpoints contain p. You can't compute $\bar{R}_n \pm 1.96\sqrt{p(1-p)/124}$ without knowing the thing you're estimating. The pivot's scale depends on the parameter, and the recipe's step 3 is stuck.

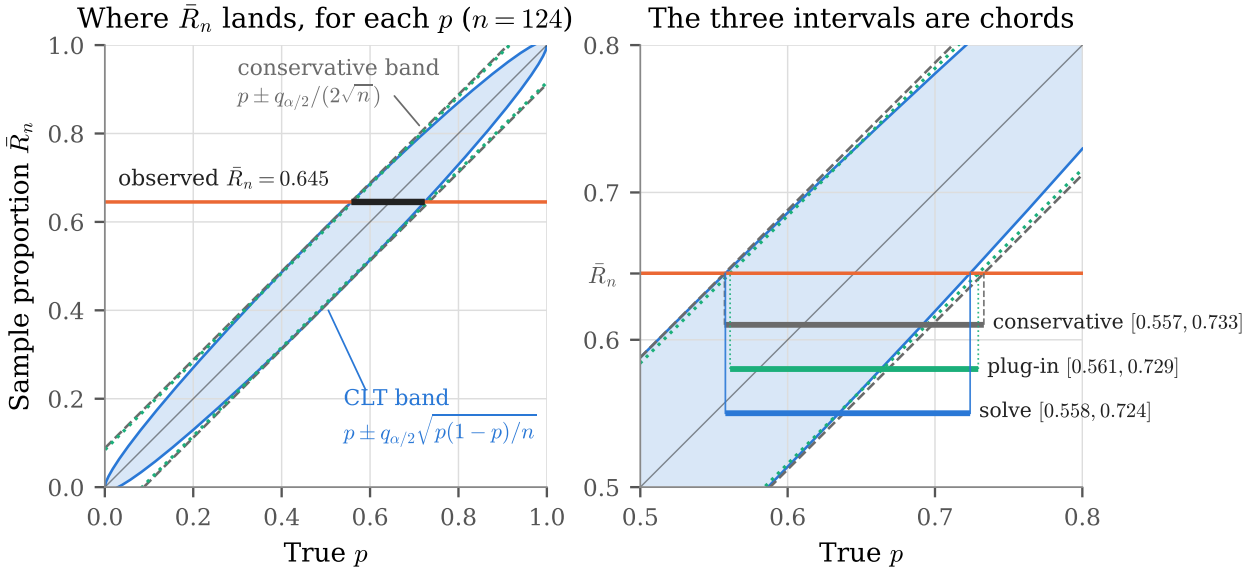


Figure 8.2: Sample proportion against the true p, $n = 124$. Left: the CLT band $p \pm q_{\alpha/2}\sqrt{p(1-p)/n}$ (blue), the widest band (gray dashes), a band of constant width (aqua dots) and the observed $\bar{R}_n = 0.645$ (orange line). Right: the three chords, to scale.

8.4 Three ways to get unstuck

Figure 8.2 is the picture to hold on to. For each candidate value of p on the horizontal axis, the CLT band is the set of sample proportions that value would plausibly produce: by Equation (8.1), $\bar{R}_n$ lands in $p \pm q_{\alpha/2}\sqrt{p(1-p)/n}$ about 95% of the time, and the band is a lens, widest at $p = \frac{1}{2}$ and pinched at the ends. Reading the picture downwards, from a p to its band, is probability. Reading it sideways, from the observed $\bar{R}_n = 0.645$ (the horizontal line) to the set of p whose bands reach it, is statistics, and that set is the confidence interval. With the lens itself, it is the chord the line cuts out of the lens, and that chord is the solve interval. The three solutions differ in which band they read. The conservative interval replaces the lens by the widest band it fits in, and gets a longer chord. The plug-in interval replaces the lens by a band of constant width $q_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$, the

width the lens has at $p = \hat{p}$, and gets a chord of nearly the same length, centered at $\hat{p}$. The right of the figure shows the three chords to scale, with $n = 124$.

Solution 1: conservative

Whatever p is, $p(1-p) \le \frac{1}{4}$, with equality at $p = \frac{1}{2}$. So the event in Equation (8.1) is contained in a larger event with a computable boundary:

$$|\bar{R}_n - p| \le q_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \quad \Longrightarrow \quad |\bar{R}_n - p| \le \frac{q_{\alpha/2}}{2\sqrt{n}}.$$

Define

$$\mathcal{I}_{\text{conserv}} = \left[\bar{R}_n - \frac{q_{\alpha/2}}{2\sqrt{n}},\ \bar{R}_n + \frac{q_{\alpha/2}}{2\sqrt{n}}\right].$$

Its coverage is at least that of the CLT event, so $\lim_{n\to\infty} \mathbb{P}_p(\mathcal{I}_{\text{conserv}} \ni p) \ge 1-\alpha$ for every p: an asymptotic confidence interval, with the inequality of the definition used in earnest. The coverage is exactly $1-\alpha$ in the limit only at $p = \frac{1}{2}$; elsewhere the interval is wider than it needs to be, which is what 'conservative' means. In the figure it reads the widest band instead of the lens.

With $n = 124$: $q_{0.025}/(2\sqrt{124}) = 1.96/22.27 = 0.088$, so $\mathcal{I}_{\text{conserv}} = [0.557, 0.733]$.

Solution 2: solve

Keep the lens and do step 3 properly. The event $|\bar{R}_n - p| \le q_{\alpha/2}\sqrt{p(1-p)/n}$ is, after squaring,

$$(\bar{R}_n - p)^2 \le \frac{q_{\alpha/2}^2}{n}\,p(1-p), \qquad \text{that is,} \qquad \left(1 + \frac{q_{\alpha/2}^2}{n}\right)p^2 - \left(2\bar{R}_n + \frac{q_{\alpha/2}^2}{n}\right)p + \bar{R}_n^2 \le 0.$$

The left side is a convex quadratic in p, so the inequality holds exactly between its two roots $p_1 < p_2$, and

$$\mathcal{I}_{\text{solve}} = [p_1, p_2]$$

is the *same event* as the one in Equation (8.1), now with computable endpoints: $\lim_{n\to\infty} \mathbb{P}_p(\mathcal{I}_{\text{solve}} \ni p) = 1-\alpha$, with equality. The quadratic formula gives the roots in closed form,

$$p_{1,2} = \frac{1}{1 + q_{\alpha/2}^2/n}\left(\bar{R}_n + \frac{q_{\alpha/2}^2}{2n} \ \mp\ q_{\alpha/2}\sqrt{\frac{\bar{R}_n(1-\bar{R}_n)}{n} + \frac{q_{\alpha/2}^2}{4n^2}}\right), \tag{8.2}$$

an interval known as the Wilson score interval. The word 'score' is explained in Section 9.8: this interval is what you get by inverting a standard test of $H_0 : p = p_0$, the score test, which measures distance from p_0 in units of the standard error computed at p_0. Note that it is *not* centered at $\bar{R}_n$: its center is pulled toward $\frac{1}{2}$, by an amount of order $1/n$ because the lens is wider there.

With $n = 124$, $\bar{R}_n = 80/124$ and $q_{0.025} = 1.96$ the quadratic is $1.0310\,p^2 - 1.3213\,p + 0.4162 \le 0$, with roots $p_1 = 0.558$ and $p_2 = 0.724$: $\mathcal{I}_{\text{solve}} = [0.558, 0.724]$.

Pitfall: rounding the quadratic before solving it

Solve for p_1 and p_2 from the coefficients as computed, not from rounded versions of them. A quadratic's roots are sensitive to its coefficients, and this quadratic is a bad case. Its constant term is 0.4162; carrying that to two decimals as 0.41 moves the roots three points each way, to $[0.53, 0.75]$, and rounding it up to 0.42 instead gives $[0.59, 0.69]$, wrong in the other direction. Two decimals are not enough here either way, and the error is large enough to change the conclusion: the $[0.53, 0.75]$ that the truncated coefficients produce is wider than the conservative interval $[0.557, 0.733]$, while the true solve interval $[0.558, 0.724]$ is the narrowest of the three. The solve interval can never be the widest, since Proposition 8.2 shows it always sits inside the conservative interval, so an interval that comes out wider than the conservative one is an arithmetic error and nothing else.

Solution 3: plug-in

Replace the unknown scale $\sqrt{p(1-p)}$ by its estimate $\sqrt{\hat{p}(1-\hat{p})}$, and let Slutsky's theorem pay for it.

Recall. Slutsky's theorem is the rule that lets you substitute an estimate for an unknown constant inside a limit: if one quantity settles into a distribution and a second converges in probability to a constant, their product and their sum behave as if the second simply *were* that constant. It is what makes every 'estimate ± standard error' formula legitimate.

This is the worked example of Section 5.2: since $\hat{p} \xrightarrow[n\to\infty]{\mathbb{P}} p$ (LLN) and $x \mapsto \sqrt{p(1-p)/(x(1-x))}$ is continuous at p (CMT),

$$\sqrt{n}\,\frac{\bar{R}_n - p}{\sqrt{\hat{p}(1-\hat{p})}} = \underbrace{\sqrt{n}\,\frac{\bar{R}_n - p}{\sqrt{p(1-p)}}}_{\xrightarrow[n\to\infty]{(\mathrm{d})}\,\mathrm{Gauss}(0,1)} \cdot \underbrace{\sqrt{\frac{p(1-p)}{\hat{p}(1-\hat{p})}}}_{\xrightarrow[n\to\infty]{\mathbb{P}}\,1} \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0,1).$$

The left side is a new asymptotic pivot whose scale no longer involves p, so the recipe runs to the end:

$$\mathcal{I}_{\text{plug-in}} = \left[\bar{R}_n - q_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}},\ \bar{R}_n + q_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right], \qquad \lim_{n\to\infty} \mathbb{P}_p(\mathcal{I}_{\text{plug-in}} \ni p) = 1-\alpha.$$

In the figure it reads a band of constant width, the lens's width at $\hat{p}$. The price is a second asymptotic approximation on top of the CLT: Slutsky's theorem is a statement about $n \to \infty$, and at finite n the estimated scale is itself noisy.

With $n = 124$: $1.96\sqrt{0.645 \times 0.355/124} = 0.084$, so $\mathcal{I}_{\text{plug-in}} = [0.561, 0.729]$.

The three side by side

Interval	95%, $n = 124$	Reads	Asymptotic coverage
conservative	$[0.557, 0.733]$	the widest band	$\geq 1-\alpha$; $= 1-\alpha$ only at $p = \frac{1}{2}$
solve	$[0.558, 0.724]$	the lens itself	$= 1-\alpha$; one approximation (CLT)
plug-in	$[0.561, 0.729]$	a band of half-width $q_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$	$= 1-\alpha$; two approximations (CLT, Slutsky)

The conservative and plug-in intervals round to the $[0.56, 0.73]$ that Chapter 1 promised, the solve interval to $[0.56, 0.72]$, and Chapter 1's drug example is the same computation: 78 of 100 cured gives $\hat{p} = 0.78$ and $\mathcal{I}_{\text{plug-in}} = 0.78 \pm 1.96\sqrt{0.78 \times 0.22/100} = [0.699, 0.861]$, the $[0.70, 0.86]$ quoted there. There are more intervals still; the exact interval built from the Binomial distribution of $n\bar{R}_n$ rather than from the CLT (the Clopper–Pearson interval mentioned in Section 4.4) gives $[0.554, 0.729]$ for the kiss data, and Newcombe (1998) compares seven of them on coverage and width.[1] That they all agree to within a point is the CLT doing its job at $n = 124$.

Proposition 8.2 (How the three nest)**.** *For every sample and every n: $\mathcal{I}_{solve} \subseteq \mathcal{I}_{conserv}$ and $\mathcal{I}_{plug\text{-}in} \subseteq \mathcal{I}_{conserv}$, and both $\mathcal{I}_{solve}$ and $\mathcal{I}_{plug\text{-}in}$ contain $\bar{R}_n$.*

Proof strategy (core). Everything follows from $p(1-p) \leq \frac{1}{4}$: a p that passes the solve test passes the conservative one, and the plug-in half-width is at most the conservative half-width.

Proof. If $p \in \mathcal{I}_{\text{solve}}$ then $|\bar{R}_n - p| \leq q_{\alpha/2}\sqrt{p(1-p)/n} \leq q_{\alpha/2}/(2\sqrt{n})$, so $p \in \mathcal{I}_{\text{conserv}}$. The plug-in half-width is $q_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n} \leq q_{\alpha/2}/(2\sqrt{n})$, the conservative one. Finally $p = \bar{R}_n$ satisfies the solve inequality trivially ($0 \leq$ something non-negative), and the plug-in interval is centered there. □

In the picture: the lens lies inside the widest band, so any chord of the lens lies inside the corresponding chord of the band. Solve and plug-in need not nest with each other; here the solve interval is shifted toward $\frac{1}{2}$ and slightly shorter.

Intuition: which one to use

Use the solve interval for a proportion, the plug-in interval for everything else. The defense: the solve interval has the exact asymptotic level with one approximation, has a closed form (Equation (8.2)) and, as Figure 8.3 shows, holds up at finite n far better than the plug-in interval near $p = 0$ or 1, where the latter breaks; the only reason it isn't universal is that 'solve for the parameter' is a quadratic here and may be intractable elsewhere. The plug-in interval is the one every formula sheet gives, is what the next chapters use, and is fine whenever $n\hat{p}(1-\hat{p})$ is comfortably large; Chapter 15 identifies it as an inverted Wald test and, for models other than the proportion, recommends the likelihood ratio interval in its place. The conservative interval is for two jobs: a guaranteed level, and planning sample sizes before any data exist (Section 8.7), since it is the only one of the three whose width you know in advance.

[1] R. Newcombe, 'Two-sided confidence intervals for the single proportion: comparison of seven methods', *Statistics in Medicine* 17 (1998).

8.5 What 'asymptotic level' buys at a finite n

Asymptotic level $1-\alpha$ is a promise about $n \to \infty$. For a proportion you can check what it delivers at the n you have because the distribution of $n\bar{R}_n$ is Binomial and the coverage probability is a finite sum:

$$\mathbb{P}_p(\mathcal{I} \ni p) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} \, \mathbb{1}\big(\mathcal{I}(k) \ni p\big),$$

where $\mathcal{I}(k)$ is the interval the rule produces when k of the n couples turn right. Figure 8.3 plots this against p for the three intervals, at $n = 124$ and at $n = 20$. The conservative interval over-covers away from $p = \frac{1}{2}$: its coverage is 0.99995 at $p = 0.05$. The plug-in interval under-covers, mildly at $n = 124$ (0.939 at $p = 0.645$) and badly at $n = 20$ near the edges. The solve interval stays near 0.95 at $n = 124$ (0.952 at $p = 0.645$); at $n = 20$ it stays above 0.92 for p between 0.1 and 0.9 and above 0.89 for p between 0.01 and 0.99, while on the first of those ranges the plug-in interval falls to 0.81, and at $p = 0.01$ to 0.18, below the bottom of the plot. The saw teeth are real, not noise: $\bar{R}_n$ takes only $n+1$ values, so there are only $n+1$ possible intervals, and the coverage jumps whenever p crosses an endpoint of one of them.

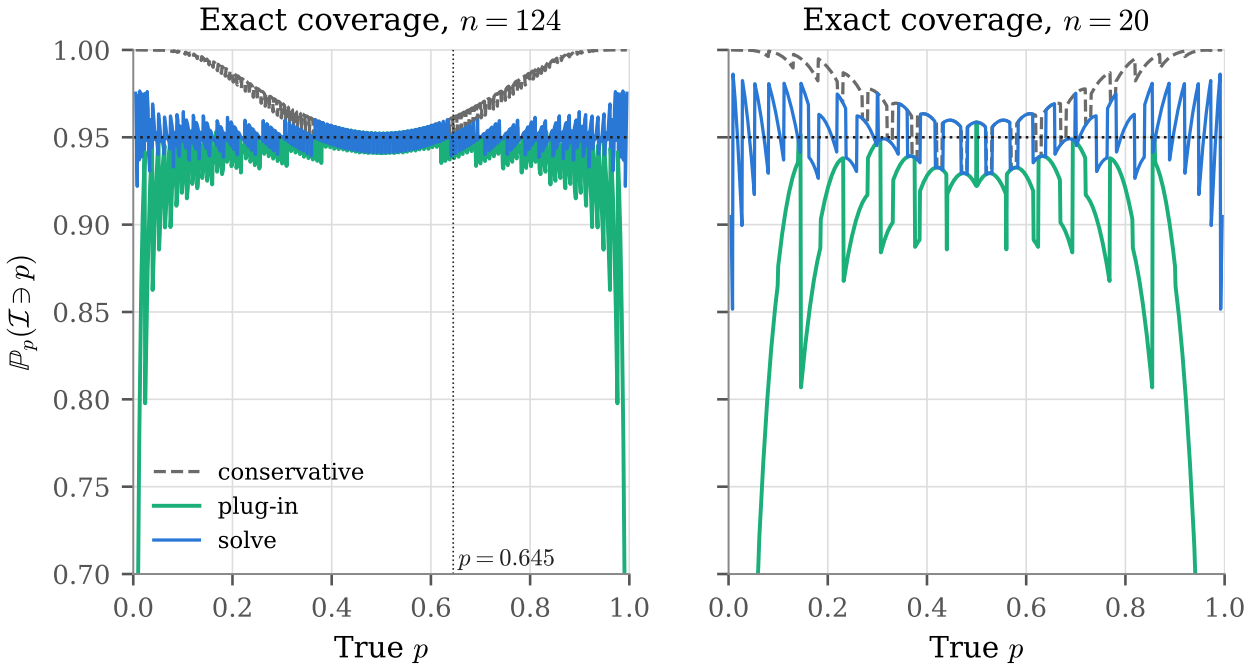


Figure 8.3: Exact coverage probability $\mathbb{P}_p(\mathcal{I} \ni p)$ of the three 95% intervals against the true p: conservative (gray dashes), plug-in (aqua) and solve (blue), with the nominal 0.95 as a dotted line. Left, $n = 124$; right, $n = 20$.

Three lessons from the figure. The asymptotic promise is kept at $n = 124$ to within a point or two for p in the middle of $(0, 1)$, which is where the kiss data put it. The plug-in interval is the one that fails first as n shrinks or p approaches the edges because its width is $\sqrt{\hat{p}(1-\hat{p})/n}$ and $\hat{p}$ near 0 or 1 makes it think it is very precise (at $\hat{p} = 0$ its width is zero). And the conservative interval is not free: its extra width buys coverage you didn't ask for, and that over-coverage is wasted precision.

This is the chapter's instance of the book's stance on asymptotics (the intuition box that opens Chapter 3, and Section 4.4): the theorem licenses the Gaussian replacement and is silent about the

error at your n. For a proportion the error can be computed, as here. For most models it can't, and the plug-in interval is used on trust and on the experience that n in the dozens with a parameter away from the boundary is usually enough.

From undergraduate statistics: the plus-four interval

Some introductory courses teach a plus-four interval for a proportion: add two successes and two failures, so that $\tilde{p} = (S+2)/(n+4)$ with $S = n\bar{R}_n$, and report $\tilde{p} \pm 1.96\sqrt{\tilde{p}(1-\tilde{p})/(n+4)}$ (Agresti and Coull, *The American Statistician*, 1998). It is a close approximation to the solve interval that needs no quadratic formula. By Equation (8.2) the solve interval is centered at

$$\frac{\bar{R}_n + q_{\alpha/2}^2/(2n)}{1 + q_{\alpha/2}^2/n} = \frac{S + q_{\alpha/2}^2/2}{n + q_{\alpha/2}^2},$$

and at 95%, $q_{0.025}^2 = 3.84$, so $(S+2)/(n+4)$ is that center with 3.84 rounded to 4. For the kiss data $\tilde{p} = 82/128 = 0.641$ and the solve center is $81.92/127.84 = 0.641$; the plus-four interval is $[0.558, 0.724]$, the solve interval to three decimals, where the plug-in interval is $[0.561, 0.729]$. At $n = 20$ its exact coverage stays above 0.92 for every p between 0.01 and 0.99. Chapter 18 reads $\tilde{p}$ as a posterior mean.

8.6 The subway: when the scale depends on the parameter in a new way

The three solutions were introduced for a proportion, where the scale $\sqrt{p(1-p)}$ is bounded. The inter-arrival-time example of Section 5.4 shows what changes when it isn't. $T_1, \dots, T_n \overset{iid}{\sim} \text{Expo}(\lambda)$, $\hat{\lambda} = 1/\bar{T}_n$, and the delta method gave

Recall. The delta method says what happens to a transformed estimator: if $\hat{\theta}_n$ is approximately Gaussian around θ with standard deviation $\sigma/\sqrt{n}$, then $g(\hat{\theta}_n)$ is approximately Gaussian around $g(\theta)$ with standard deviation $|g'(\theta)|\sigma/\sqrt{n}$. The slope of g at θ stretches or compresses the error.

$$\sqrt{n}\,(\hat{\lambda} - \lambda) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, \lambda^2), \qquad \text{so} \qquad \lim_{n\to\infty} \mathbb{P}_\lambda\left(|\hat{\lambda} - \lambda| \le q_{\alpha/2}\frac{\lambda}{\sqrt{n}}\right) = 1 - \alpha.$$

Again the half-width involves the parameter, now as a factor λ rather than $\sqrt{p(1-p)}$, and Figure 8.4 shows the consequence: the band is a wedge, not a lens. For each true λ, the delta method says $\hat{\lambda}$ lands in $\lambda \pm q_{\alpha/2}\lambda/\sqrt{n}$ about 95% of the time, and that band keeps opening as λ grows, with no widest point. The observed $\hat{\lambda} = 1/6.23 = 0.1605$ is the horizontal line, and the solve interval is the chord it cuts from the wedge, which is not centered at $\hat{\lambda}$ because the wedge is wider to the right. The plug-in interval uses a band of constant width $q_{\alpha/2}\hat{\lambda}/\sqrt{n}$ and is centered. No band of constant width contains the wedge, so there is nothing for a conservative interval to read.

1. *Conservative:* there is nothing to bound. λ ranges over $(0, \infty)$, so $\sup_\lambda q_{\alpha/2}\lambda/\sqrt{n} = \infty$ and the 'conservative interval' is $(0, \infty)$, which has level 100% and says nothing.
2. *Solve:* the inequality is linear in λ this time. $|\hat{\lambda} - \lambda| \le q_{\alpha/2}\lambda/\sqrt{n}$ is the same as $\lambda(1 - q_{\alpha/2}/\sqrt{n}) \le$

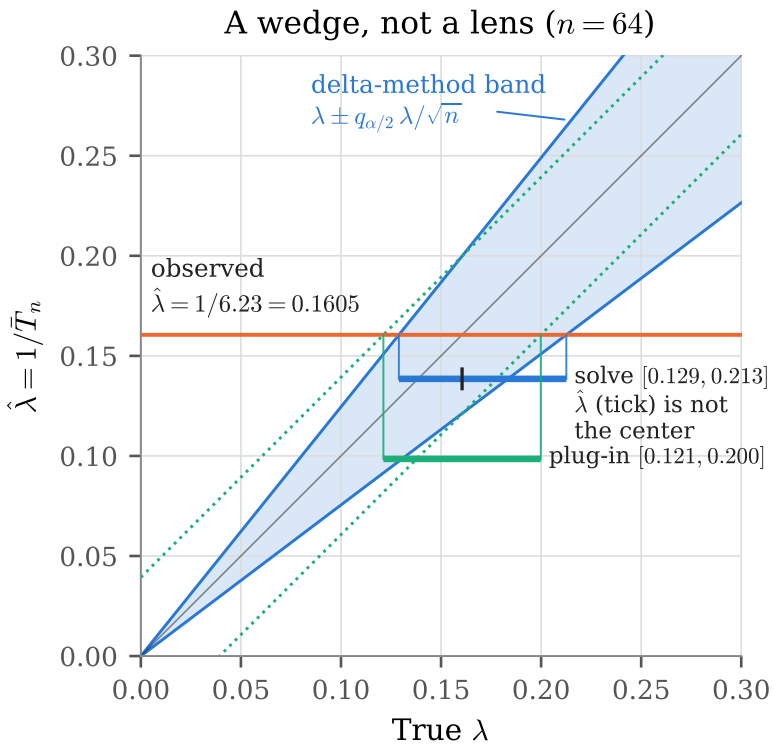


Figure 8.4: The subway rate, $n = 64$: $\hat\lambda$ against the true λ. The delta-method band $\lambda \pm q_{\alpha/2}\lambda/\sqrt{n}$ (blue), the plug-in band of constant width (aqua dots), the observed $\hat\lambda = 0.1605$ (orange line) and the two chords below it.

$\hat\lambda \le \lambda(1 + q_{\alpha/2}/\sqrt{n})$, and if $n > q_{\alpha/2}^2$ (so that $1 - q_{\alpha/2}/\sqrt{n} > 0$) this solves to

$$\mathcal{I}_{\text{solve}} = \left[\frac{\hat\lambda}{1 + q_{\alpha/2}/\sqrt{n}},\ \frac{\hat\lambda}{1 - q_{\alpha/2}/\sqrt{n}} \right].$$

3. *Plug-in:* replace λ in the half-width by $\hat\lambda$ (Slutsky, since $\hat\lambda \xrightarrow[n\to\infty]{\mathbb{P}} \lambda$):

$$\mathcal{I}_{\text{plug-in}} = \left[\hat\lambda \left(1 - \frac{q_{\alpha/2}}{\sqrt{n}}\right),\ \hat\lambda \left(1 + \frac{q_{\alpha/2}}{\sqrt{n}}\right) \right].$$

With $n = 64$ and $\bar T_n = 6.23$ minutes: $\hat\lambda = 0.1605$ per minute and $q_{0.025}/\sqrt{64} = 0.245$, so

$$\mathcal{I}_{\text{solve}} = [0.129, 0.213] \approx [0.13, 0.21], \qquad \mathcal{I}_{\text{plug-in}} = [0.121, 0.200] \approx [0.12, 0.20].$$

The two differ more than the kiss intervals did because $n = 64$ is smaller and $q/\sqrt{n} = 0.245$ is not small: $1/(1 - 0.245) = 1.32$ against $1 + 0.245 = 1.245$. The solve interval reaches further to the right, where the wedge is wider: large values of λ produce more variable $\hat\lambda$, so more of them are compatible with the observed 0.16.

Remark. There is a second way to see the solve interval that explains its asymmetry. Build the plug-in interval for the *mean* $1/\lambda$ first: the CLT gives $\sqrt{n}(\bar T_n - 1/\lambda) \xrightarrow[n\to\infty]{(\text{d})} \text{Gauss}(0, 1/\lambda^2)$, and plugging in gives $\bar T_n(1 \pm q_{\alpha/2}/\sqrt{n}) = [4.70, 7.76]$ minutes. Now map it through $x \mapsto 1/x$, which

is decreasing: $\lambda \in [1/7.76, 1/4.70] = [0.129, 0.213]$, which is $\mathcal{I}_{\text{solve}}$ exactly. A confidence interval for θ becomes a confidence interval for $g(\theta)$, with the same level, by applying any continuous and strictly monotone g to its endpoints, in the right order ($[g(L), g(U)]$ for increasing g, $[g(U), g(L)]$ for decreasing g) because $\theta \in [L, U]$ and $g(\theta) \in g([L, U])$ are the same event. The delta method is needed when g is not monotone (the odds of Chapter 5 are, $p(1-p)$ is not), or when you want the interval for $g(\theta)$ to be symmetric about $g(\hat{\theta}_n)$, and symmetry is not a virtue in itself.

8.7 What makes an interval wide

The half-width of the conservative and plug-in intervals is exactly

$$\text{half-width} = q_{\alpha/2} \times \frac{\widehat{\sigma}}{\sqrt{n}},$$

a quantile times an estimated standard error, and the solve intervals have the same size up to terms of smaller order. The two factors are the two things you control.

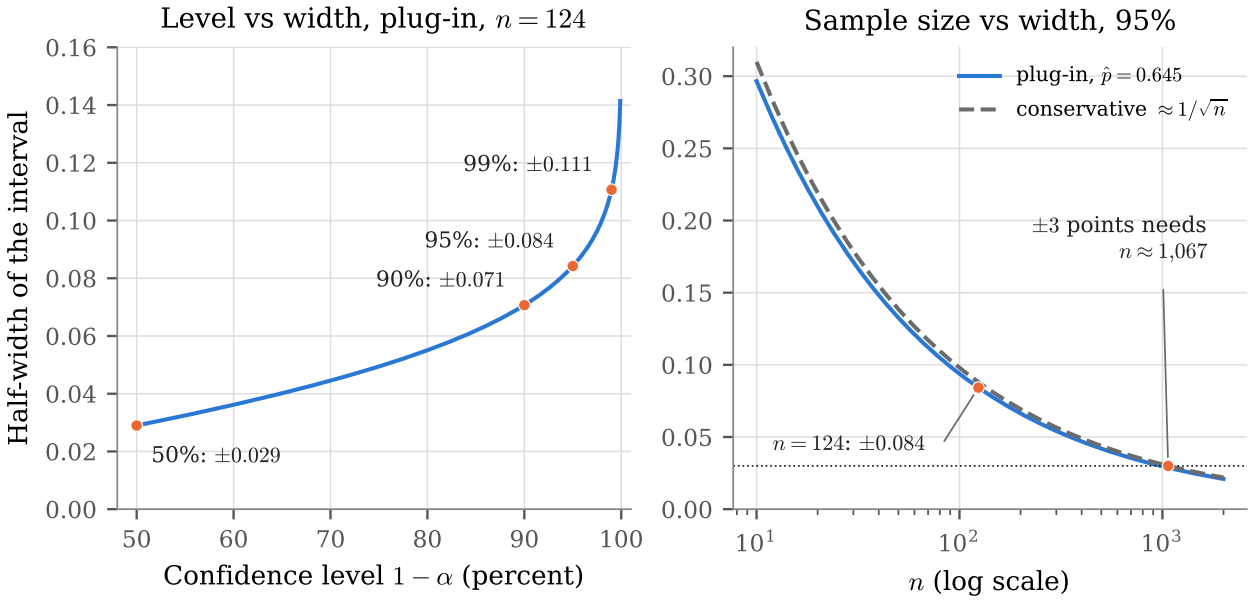


Figure 8.5: Left: the half-width of the plug-in interval at $\hat{p} = 0.645$, $n = 124$, against the confidence level, with orange dots at the usual levels. Right: the half-width at 95% against n, plug-in at $\hat{p} = 0.645$ (blue) and conservative (gray dashes).

The level. Raising the confidence level widens the interval, through the quantile, and the cost accelerates (Figure 8.5, left). The quantile $q_{\alpha/2}$ does the work: 0.67 at 50%, 1.645 at 90%, 1.96 at 95%, 2.58 at 99%, 3.29 at 99.9%, and infinite at 100%. Going from 90% to 95% costs 19% more width; from 95% to 99%, 31% more; and 100% confidence costs the whole parameter space, since $q_{\alpha/2} \to \infty$ as $\alpha \to 0$. This is why 95% is a convention and not a law: it is a point on a curve where the interval is still informative and the miss rate is still small. A 98% interval of the same type always contains the 95% one because $q_{0.01} > q_{0.025}$; but a 98% interval need not contain a 95% interval built differently. $[0, \bar{R}_n + q_{0.02}\,\widehat{\sigma}/\sqrt{n}]$ has asymptotic level 98% and does not contain the 95% lower-bound interval $[\bar{R}_n - q_{0.05}\,\widehat{\sigma}/\sqrt{n},\ 1]$, nor the other way around: the 98% interval reaches

down to 0 but stops short of 1, the 95% one reaches up to 1 but stops short of 0, so each contains points the other does not. Nesting holds within one family of intervals indexed by α, not between two intervals of different shapes whose levels happen to differ.

The sample size. The width shrinks like $1/\sqrt{n}$ (Figure 8.5, right), which is the rate of the CLT and the rate of every interval in this book built from a sample mean. Both curves on the right fall at that rate, the plug-in half-width at $\hat{p} = 0.645$ and the conservative half-width $q_{0.025}/(2\sqrt{n}) = 0.98/\sqrt{n}$, which is the pollster's 'about $1/\sqrt{n}$'; the marked point is the $n \approx 1{,}067$ that a margin of ± 3 points needs. The practical form of $1/\sqrt{n}$: to halve the width you need four times the data; to get one more decimal place you need a hundred times the data. Intervals that shrink faster exist when the model has special structure (Exercise 8.5 has one that shrinks like $1/n$), but they are built on the model's exact distribution rather than on the CLT, and they are fragile when the model is wrong.

From undergraduate statistics: the pollster's ± 3 points

National polls report a 'margin of error of ± 3 percentage points' with about 1,000 respondents, whatever the question. That is the conservative interval: its half-width $q_{0.025}/(2\sqrt{n}) = 0.98/\sqrt{n} \approx 1/\sqrt{n}$ doesn't depend on p, so a pollster can state it before seeing a single answer. Setting $0.98/\sqrt{n} = 0.03$ gives $n = (1.96/0.06)^2 = 1{,}067$. The same arithmetic plans any proportion study: the sample size for a margin m at 95% is $n = (0.98/m)^2$, and the conservative bound is the right one to use because at the planning stage there is no $\hat{p}$ to plug in.

8.8 What the 95% means, and three things it doesn't

Take $\mathcal{I}_{\text{plug-in}} = [0.12, 0.20]$ for the subway rate. The sentence '$[0.12, 0.20]$ is a confidence interval of asymptotic level 95% for λ' means: *if you were to repeat the experiment*, collecting 64 new inter-arrival times and computing the interval each time, the parameter λ would be inside the resulting interval about 95% of the time, with the approximation improving as n grows. This is the frequentist interpretation, and Figure 8.1 is a hundred such repetitions. Three readings that sound the same and are not:

Pitfall: three wrong readings

1. *'The probability that $\lambda \in [0.12, 0.20]$ is 0.95.'* No. λ is a fixed number and $[0.12, 0.20]$ is a fixed interval; $\mathbb{P}(\lambda \in [0.12, 0.20])$ is 1 or 0, and the theory doesn't tell you which. The 0.95 belongs to the procedure. (The sentence 'you are 95% confident that λ is in $[0.12, 0.20]$', as in the drug example of Chapter 1, is the accepted shorthand for the procedure statement, and is fine as long as it's read that way.)
2. *'95% of the inter-arrival times fall in $[0.12, 0.20]$.'* No. The interval is for the parameter λ, not for the data; it shrinks to a point as $n \to \infty$ while the data's spread stays what it is.
3. *'95% of the future estimates $\hat{\lambda}$ will fall in $[0.12, 0.20]$.'* No. A future $\hat{\lambda}$ is centered at λ, not at this sample's 0.16, and lands in this particular interval about 95% of the time only if this sample's 0.16 happens to be close to λ, and less the further it is off.

The first reading is the one everyone wants, a probability statement about the parameter given the data. It is available, but at a price: you must treat λ as random, with a distribution before the data (a prior), and compute its distribution after the data (the posterior). That is the Bayesian approach of Chapter 18, and the interval it produces, which Chapter 19 calls a Bayesian confidence region, does carry the probability 0.95 in the sense of reading 1. For large n the two kinds of interval nearly

coincide, a consequence of the Bernstein–von Mises theorem (Theorem 18.12); what differs is what the number 0.95 is a probability *of*.

From undergraduate statistics: what the margin of error leaves out

An introductory course warns that a margin of error covers only random sampling error, not undercoverage, nonresponse or the wording of a question. The definition says why: the 95% is a statement about $\mathbb{P}_\theta(\mathcal{I} \ni \theta)$, a probability computed under the model, and for a poll the model says each R_i is a draw from $\text{Bern}(p)$ with p the proportion in the whole population. If the design makes the draws come from $\text{Bern}(p')$ with $p' \neq p$, the interval covers p' about 95% of the time and covers p with a probability that goes to 0 as n grows, since it shrinks around p'. The *Literary Digest* poll of Section 7.8 is the extreme case: from about 2.4 million ballots its plug-in interval for Roosevelt's share is 0.43 ± 0.0006, 19 points below the 0.62 he won.

8.9 What comes next

A confidence interval answers 'which values of θ are compatible with the data?'. A test answers a narrower question, 'is this particular value compatible?', and Exercise 8.6 already showed that the two are the same construction: the value $\frac{1}{2}$ is rejected exactly when it falls outside a one-sided interval. Chapter 9 starts from the decision rather than the interval, which brings in two new things the interval didn't need: two kinds of error, with a level for one and a power for the other, and the p-value, which reports a test without fixing α in advance. The thresholds 72 and 74 that Chapter 1 quoted are derived there.

8.10 Summary

- A confidence interval of level $1 - \alpha$ is a random interval with computable endpoints such that $\mathbb{P}_\theta(\mathcal{I} \ni \theta) \geq 1 - \alpha$ for every θ; asymptotic level replaces the probability by its limit. Level is a floor: a 95% interval is also a 90% one.
- The recipe: find a pivot, trap it between quantiles ($q_{\alpha/2}$ two-sided, q_α one-sided), solve for the parameter. Gaussian data with known variance give an exact pivot and an exact interval; the CLT gives an asymptotic pivot and an asymptotic interval.
- When the pivot's scale involves the parameter, the interval it suggests isn't computable. Three fixes: bound the scale (conservative, level $\geq 1 - \alpha$, widest), solve the inequality for the parameter (solve, exact asymptotic level, one approximation), or estimate the scale and invoke Slutsky (plug-in, exact asymptotic level, two approximations). In the inversion picture they read the widest band, the lens, and a band of constant width.
- Kiss experiment, 95%: conservative $[0.557, 0.733]$, solve $[0.558, 0.724]$, plug-in $[0.561, 0.729]$; they agree to within a point, and here the solve interval is the narrowest, not the widest. Subway: solve $[0.13, 0.21]$, plug-in $[0.12, 0.20]$, no conservative interval.
- At finite n the three behave differently: for a proportion the exact coverage can be computed from the Binomial, and at $n = 124$ the plug-in interval covers 0.939 and the solve interval 0.952 at $p = 0.645$, while at $n = 20$ the plug-in interval fails near the edges and the solve interval holds. Use solve for a proportion, plug-in otherwise.
- A continuous, strictly monotone map carries a confidence interval for θ to one for $g(\theta)$ with the same level; the subway solve interval is the plug-in interval for $1/\lambda$ mapped through $x \mapsto 1/x$.

- Half-width $= q_{\alpha/2}\,\widehat{\sigma}/\sqrt{n}$: the level acts through the quantile (and 100% costs everything), the sample size through $1/\sqrt{n}$ (four times the data, half the width; ± 3 points needs about 1,000).
- The level is the long-run hit rate of the procedure. A realized interval has no probability attached; the statement '$\mathbb{P}(p \in [0.56, 0.73]) = 0.95$' is Bayesian, and Chapter 18 says what it costs.

8.11 Exercises

Exercise 8.1. For the kiss data, find a 98% asymptotic confidence interval for p of the conservative and of the plug-in type. If $[0.34, 0.57]$ is a 95% confidence interval for some proportion, is it also a 98% one? A 90% one? *(Solution on page 417.)*

Exercise 8.2. A second study observes 150 couples, 75 turning each way, so $\bar{R}_n = 0.5$. Which of the following are realizations of 50% asymptotic confidence intervals for p: (1) $[0, 0.5]$; (2) $[0.5, 1]$; (3) $[0.466, 0.533]$; (4) $[0.48, 0.52]$? *(Solution on page 418.)*

Exercise 8.3. Prove that $\mathcal{I}_{\text{plug-in}} \subseteq \mathcal{I}_{\text{conserv}}$ always, and find the sample for which the two coincide. Then show that $\mathcal{I}_{\text{solve}} = \mathcal{I}_{\text{plug-in}}$ never happens unless $\bar{R}_n = \frac{1}{2}$, by comparing centers. *(Solution on page 418.)*

Exercise 8.4. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(\theta, \theta)$ for an unknown $\theta > 0$: the mean and the variance are the same unknown number. (a) Find an exact pivot based on $\bar{X}_n$. (b) Write the interval $\mathcal{I}_\theta$, depending on θ, with exact coverage 90%. (c) Find the plug-in interval of asymptotic level 90%. (d) Find the solve interval, and say whether its level is exact or asymptotic. *(Solution on page 418.)*

Exercise 8.5. Let $X_1, \ldots, X_n$ be i.i.d. with density $f(x) = e^{-(x-a)}\mathbb{1}(x \geq a)$, a unit Exponential shifted by an unknown $a \in \mathbb{R}$. (a) Show that $\hat{a}_1 = \bar{X}_n - 1$ is consistent and find its asymptotic distribution and the two-sided plug-in interval of asymptotic level $1 - \alpha$. (b) Let $\hat{a}_2 = \min_i X_i$. Show that $n(\hat{a}_2 - a) \sim \text{Expo}(1)$ exactly, and build a one-sided interval of exact level $1 - \alpha$ from it. (c) Compare the two intervals' widths and their robustness. *(Solution on page 418.)*

Exercise 8.6. For the kiss data, build a 95% lower confidence bound for p of the plug-in type and of the conservative type. Then show that '$\frac{1}{2}$ is below the conservative lower bound' is the same event as '$n\bar{R}_n \geq 72$', the one-sided threshold that Chapter 1 quoted and Chapter 9 derives. *(Solution on page 418.)*

Exercise 8.7. The subway data have an exact interval that uses no limit theorem. If $T_1, \ldots, T_n \overset{iid}{\sim} \text{Expo}(\lambda)$ then $\lambda \sum_i T_i = \lambda n \bar{T}_n$ has the Gamma distribution with shape n and rate 1, whatever λ is (you may take this as given). Use it as a pivot to build a confidence interval for λ of exact level 95%, and compute it for $n = 64$, $\bar{T}_n = 6.23$, given that the 2.5% and 97.5% points of that Gamma distribution are 49.29 and 80.60. *(Solution on page 419.)*

Chapter 9

Hypothesis Tests: Errors, Level, Power and p-Values

In this chapter

- What a test is: a yes/no rule applied to the data, written as an indicator of a rejection region. The null hypothesis is the claim the data are allowed to overturn; the alternative is what you conclude when they do.
- The two ways a test can be wrong, why they are not treated alike, and the two numbers that describe a test: its level, a cap on one error, and its power, one minus the other.
- The recipe behind every test in this book: a test statistic whose distribution under the null hypothesis you know, and a quantile of that distribution as the threshold. One-sided and two-sided versions, and why the null hypothesis makes the unknown parameter known.
- The test for the kiss data, done exactly and done by the CLT: thresholds 72 and 74 one way, 72 and 73 the other, and why they differ.
- The p-value: the smallest level at which the data reject, and the one number a test reports.
- Tests and confidence intervals are the same construction seen from two sides.

The running example is the kiss experiment: $n = 124$ couples, 80 turning right, and the question Chapter 1 opened with, whether 80 is more than chance would produce if couples had no preference for either side.

9.1 A yes/no question about a parameter

A confidence interval reports a range. Often what is wanted is a decision. The examples below are the kind a manager or a scientist actually faces, and each is a yes/no question about a parameter of a model.

- **Boarding a plane.** Two airlines board differently: one from the rear of the plane to the front ('R2F'), the other by window seats first, then middle, then aisle ('WilMA'). Which is faster? The data are $n = 72$ R2F flights averaging 24.2 minutes with a standard deviation of 2.1, and $m = 56$ WilMA flights averaging 15.9 minutes with a standard deviation of 1.3. With boarding times modeled as $X \sim \text{Gauss}(\mu_1, \sigma_1^2)$ and $Y \sim \text{Gauss}(\mu_2, \sigma_2^2)$ and the two

samples independent, the question is: is $\mu_1 = \mu_2$, or is $\mu_1 > \mu_2$? Two samples, one question: a **two-sample test**.

- **Waiting in the emergency room.** The national average wait is 30 minutes. Patients at one hospital say its wait is longer. With $X_1, \ldots, X_n$ the waits of n random patients there and $\mu = \mathbb{E}[X_1]$, the question is: is $\mu > 30$? One sample: a **one-sample test**.
- **The kiss experiment.** If couples have no preference for either side, $p = \frac{1}{2}$. Of 124 couples, 80 turned right. Is $p = \frac{1}{2}$, or not?

The obvious first answer to the boarding question is '$\bar{X}_n = 24.2 > 15.9 = \bar{Y}_m$, so $\mu_1 > \mu_2$'. What is wrong with it is this: by bad luck the 72 R2F flights might all have been full and the 56 WilMA flights half empty, and sample means move from sample to sample. A better rule puts a buffer around each mean and asks whether the buffers still separate:

$$\text{conclude } \mu_1 > \mu_2 \text{ if } \bar{X}_n - \text{buffer} > \bar{Y}_m + \text{buffer}.$$

For the one-sample version, conclude $\mu > 30$ if $\bar{X}_n > 30 + \text{buffer}$. What should the buffer be? Something that measures the size of the random fluctuations of $\bar{X}_n$, and Chapter 4 says what that is: a few multiples of $\sigma/\sqrt{n}$. Figure 9.1 draws the heuristic for the boarding data and for two one-sample examples, with a buffer of two standard errors around each estimate. With $n = 72$ and $m = 56$ the buffers around 24.2 and 15.9 minutes reach about half a minute (R2F) and a third of a minute (WilMA) on each side, nowhere near each other, and the conclusion is not in doubt. For a coin tossed 30 times with 13 heads, the buffer 0.43 ± 0.18 reaches well past $\frac{1}{2}$, and the heuristic says nothing. For a survey of 4,971 Americans in which 32% get news on YouTube, the buffer 0.32 ± 0.013 just reaches the 33% being tested, and again the heuristic is silent. For a proportion the buffer is $2\sqrt{\hat{p}(1-\hat{p})/n}$, and it shrinks like $1/\sqrt{n}$.

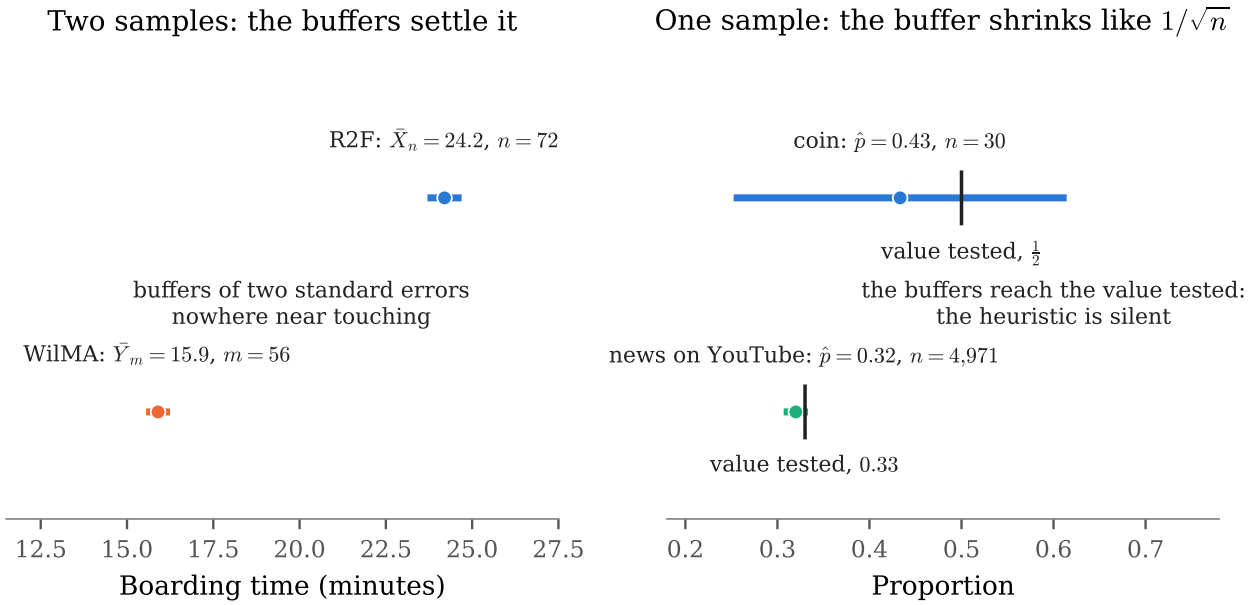


Figure 9.1: Buffers of two standard errors around each estimate (dot). Left: the boarding data, R2F (blue) and WilMA (orange). Right: two one-sample proportions, the coin (blue) and the YouTube survey (aqua), each with the value being tested as a black tick.

The rest of the chapter makes the heuristic precise: what counts as a buffer, what guarantee it comes with, and what to report when the data land near the edge.

From undergraduate statistics: statistically significant

Every one of these questions was answered in an introductory course with a z-test or a t-test and the phrase 'the difference is statistically significant at the 5% level'. This chapter is that procedure, written so that each piece has a name and a guarantee: the hypotheses, the rejection region, the two errors, the level, the power and the p-value. 'Significant at 5%' will turn out to mean 'the test of level 5% rejected', and '$p < 0.05$' will be a theorem rather than a rule of thumb.

9.2 Null and alternative hypotheses

Let $X_1, \dots, X_n$ be a sample of i.i.d. r.v.s and $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ a statistical model (Chapter 2). Let Θ_0 and Θ_1 be disjoint subsets of Θ. The two hypotheses are

$$H_0 : \theta \in \Theta_0 \qquad \text{and} \qquad H_1 : \theta \in \Theta_1.$$

H_0 is the **null hypothesis**, H_1 the **alternative hypothesis**. If you believe the true θ is in one of the two sets, you test H_0 against H_1: you look in the data for evidence against H_0, and you decide whether to **reject** H_0.[1]

The two hypotheses do not play symmetric roles, and this asymmetry is the first thing to internalize. The data are used only to try to disprove H_0. If they fail to, that is not evidence that H_0 is true; it is a lack of evidence against it. The standard phrase is 'innocent until proven guilty': the null hypothesis is the defendant, the data are the evidence, and a verdict of not guilty is not a finding of innocence. So the conclusions of a test are 'reject H_0' and 'fail to reject H_0', never 'accept H_0'.

Which claim goes in the null? The one that gets the benefit of the doubt: the status quo, the claim of no effect, the value you would act on if the data said nothing. For the kiss experiment the null is $p = \frac{1}{2}$, no preference; for the emergency room it is 'the wait is not longer than the national average'; for the boarding methods it is 'no difference'. The alternative is the claim that needs evidence to be adopted, and it is the one the study was run to look for.

Simple and composite. A hypothesis is **simple** if its set is a single point, $\Theta_k = \{\theta_k\}$, and **composite** otherwise, typically a half-line $\{\theta > \theta_k\}$ or $\{\theta < \theta_k\}$ or everything but a point, $\{\theta \neq \theta_k\}$. When $\Theta \subseteq \mathbb{R}$ and H_0 is simple, $H_0 : \theta = \theta_0$, the terminology is:

- **two-sided test**: $H_1 : \theta \neq \theta_0$, departures in either direction count as evidence;
- **one-sided test**: $H_1 : \theta > \theta_0$ or $H_1 : \theta < \theta_0$, only one direction counts.

One-sided tests are also written with a composite null, $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$; Section 9.4 explains why the two forms of the null lead to the same test. For the examples above: the boarding question is one-sided ($\mu_1 > \mu_2$); the emergency room is one-sided ($\mu > 30$); the fair coin is two-sided ($p \neq \frac{1}{2}$); and the test for the kiss data can be put either way, two-sided as $H_0 : p = \frac{1}{2}$ against $H_1 : p \neq \frac{1}{2}$, or one-sided as Chapter 1 asked it, $H_1 : p > \frac{1}{2}$. Both are done below.

[1] Some treatments require Θ_0 and Θ_1 to partition Θ, and others, this book among them, ask only that the two sets be disjoint. The difference matters when values of θ in neither set are possible, in which case the test is silent about them. In every example in this book the two sets partition Θ.

Pitfall: one-sided is a modeling decision, not a reading of the data

Writing $H_1 : \mu_1 > \mu_2$ for the boarding methods says that $\mu_1 < \mu_2$ is not a possibility you entertain. The modeling assumption reduces the number of ways the hypothesis $\mu_1 = \mu_2$ can be rejected: $\mu_1 < \mu_2$ is not allowed, so a sample pointing that way counts as no evidence against the null at all. That decision has to be made before looking at the data. Choosing the side after seeing which way $\bar{X}_n - \bar{Y}_m$ came out is a way of rejecting more often than the level promises: a nominal 5% becomes an actual 10%, since a two-sided question is then being answered with a one-sided threshold. Section 9.6 shows the thresholds differ (72 against 74 right turns for the kiss data), which is exactly the room this mistake exploits.

9.3 What a test is

Definition 9.1 (Test, rejection region)**.** A **test** is a statistic $\psi : E^n \to \{0, 1\}$: a function of the sample $X_1, \ldots, X_n$ taking the values 0 and 1, such that

- if $\psi(X_1, \ldots, X_n) = 0$, H_0 is not rejected;
- if $\psi(X_1, \ldots, X_n) = 1$, H_0 is rejected.

The **rejection region** of ψ is the set of samples on which it rejects,

$$R_\psi = \{(x_1, \ldots, x_n) \in E^n : \psi(x_1, \ldots, x_n) = 1\},$$

so that $\psi = \mathbb{1}(R_\psi)$ (toolkit A.4).

> **In words.** A test is a rule, fixed before the data arrive, that sorts every possible sample into two bins: reject or don't. The rejection region is the 'reject' bin, and the test is its indicator. Because the rule is applied to a random sample, ψ is a random variable, a Bernoulli one, and asking how often it equals 1 is a probability question, like asking how often an interval covers.
>
> **For the kiss data.** 'Reject $p = \frac{1}{2}$ if 72 or more of the 124 couples turn right' is the test $\psi = \mathbb{1}(n\bar{R}_n \geq 72)$. Its rejection region is every sequence of 124 zeros and ones with at least 72 ones. On the sample we have, $n\bar{R}_n = 80$ and $\psi = 1$.

The definition contains no probability: a test is a partition of the sample space E^n of Section 2.3 into two pieces. Probability enters when you ask how the test behaves under each candidate $\mathbb{P}_\theta$, which is the next section.

In practice every test in this book has the form

$$\psi = \mathbb{1}(T_n > c)$$

for a statistic $T_n = T_n(X_1, \ldots, X_n)$, called the **test statistic**, and a number c, the **threshold**; the rejection region is then $\{T_n > c\}$ (for a count, '≥ 72' is '> 71', so the strict inequality is only a matter of where c sits). The test statistic compresses the sample into one number that measures how far the data are from what H_0 predicts, and the threshold says how far is too far. Everything a test does is in those two choices, and the sections that follow are about making them.

From undergraduate statistics: the test statistic you already know

In an introductory course the test statistic was $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$, or t with s in place of σ, and the threshold was a number from a table, 1.96 or 1.645 or a t critical value. That is $\psi = \mathbb{1}(|T_n| > c)$ with T_n the standardized mean. What this chapter adds is the function ψ itself as an object, so that 'the test rejects with probability α when H_0 is true' is a statement about a random variable with a definition, and so that tests with other statistics (likelihood ratios in Chapter 15, chi-squared distances in Chapter 16) are the same object with a different T_n.

9.4 Two kinds of error, level and power

A test can be wrong in two ways, and the standard table lays them out with the truth in the rows and the decision in the columns:

	not rejected ($\psi = 0$)	rejected ($\psi = 1$)
H_0 true ($\theta \in \Theta_0$)	correct	**type 1 error**
H_1 true ($\theta \in \Theta_1$)	**type 2 error**	correct

A **type 1 error** is rejecting H_0 when it is true: convicting the innocent. A **type 2 error** is failing to reject H_0 when H_1 is true: acquitting the guilty. Since the truth is one fixed θ and the decision is random, the natural measure of each error is a probability, and since the probability depends on which θ is true, each error is a *function* of θ on its own hypothesis.

Definition 9.2 (Type 1 error, type 2 error, power)**.** For a test ψ of $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$:

- the **type 1 error** of ψ is the function

$$\alpha_\psi : \Theta_0 \to [0,1], \qquad \theta \mapsto \mathbb{P}_\theta(\psi = 1);$$

- the **type 2 error** of ψ is the function

$$\beta_\psi : \Theta_1 \to [0,1], \qquad \theta \mapsto \mathbb{P}_\theta(\psi = 0);$$

- the **power** of ψ is the number (toolkit A.2)

$$\pi_\psi = \inf_{\theta \in \Theta_1} \left(1 - \beta_\psi(\theta)\right).$$

In words. Fix a truth θ and imagine rerunning the study many times. If θ is in the null set, the fraction of reruns on which the test rejects is the type 1 error at that θ; if θ is in the alternative, the fraction on which it fails to reject is the type 2 error at that θ, and the fraction on which it rejects is the power at that θ. The power of the test is the worst case of that last fraction over the whole alternative: the least the test can be relied on to detect a departure from H_0.

For the kiss data. For $\psi = \mathbb{1}(n\bar{R}_n \geq 72)$ and the one-sided hypotheses $H_0 : p = \frac{1}{2}$, $H_1 : p > \frac{1}{2}$: the type 1 error is the single number $\mathbb{P}_{1/2}(n\bar{R}_n \geq 72) = 0.044$. The type 2 error at $p = 0.645$ is $\mathbb{P}_{0.645}(n\bar{R}_n \leq 71) = 0.057$, so the test detects a true p of 0.645 with probability 0.943. The power in the definition's sense is $\inf_{p>1/2}$ of that detection probability, which is 0.044, approached as $p \downarrow \frac{1}{2}$: no test can reliably detect a p just above $\frac{1}{2}$ with 124 couples, and the infimum records that.

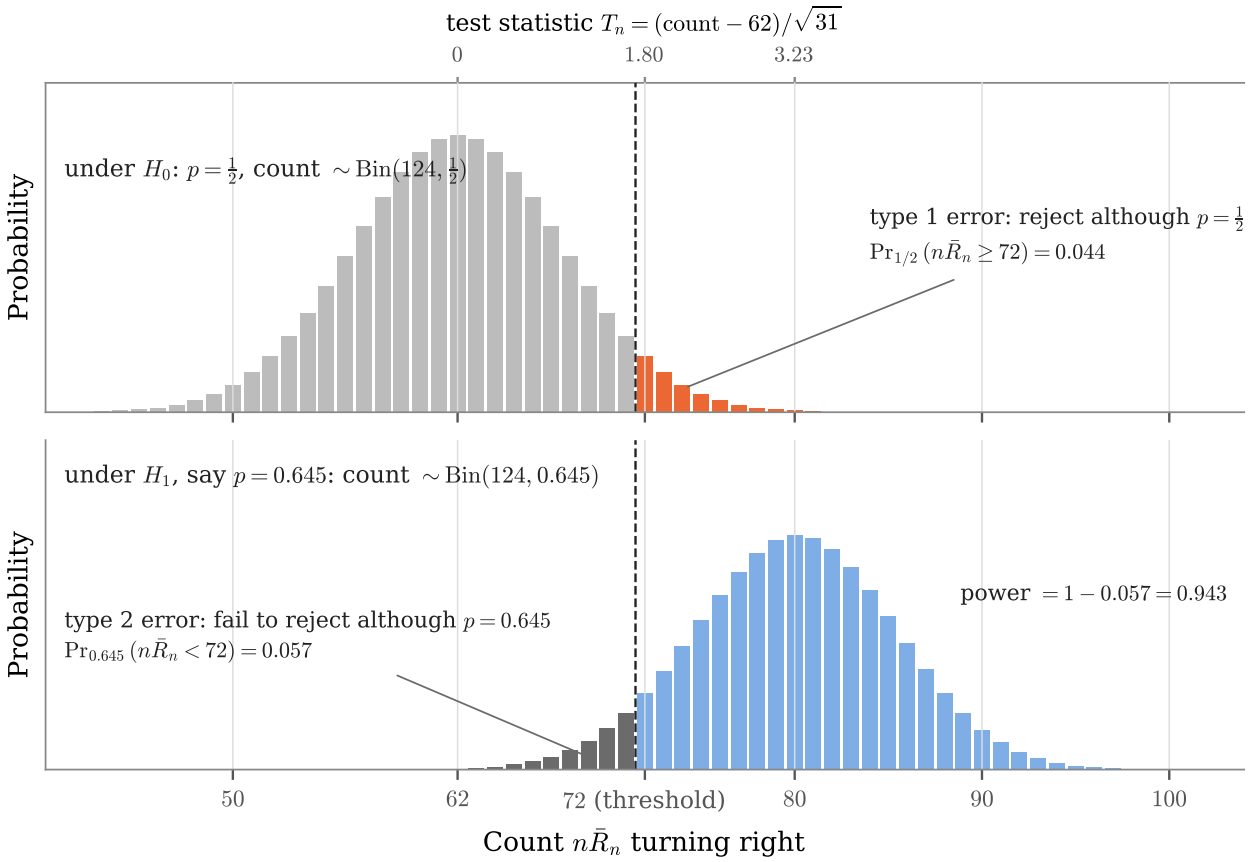


Figure 9.2: The test 'reject $p = \frac{1}{2}$ if the count is 72 or more', threshold dashed. Top: the count under $p = \frac{1}{2}$, mass at 72 and above in orange. Bottom: the count under $p = 0.645$, mass below 72 in gray, the rest in blue.

Figure 9.2 is the picture to hold on to. The same threshold cuts two different distributions of the count: the one H_0 predicts and one of the many H_1 allows. At the top, if $p = \frac{1}{2}$ the count is $\text{Bin}(124, \frac{1}{2})$, centered at 62, and the orange mass at 72 and above is the type 1 error, 0.044. At the bottom, if instead $p = 0.645$ the count is $\text{Bin}(124, 0.645)$, centered at 80; the gray mass below 72 is the type 2 error at that p, 0.057, and the blue mass is the power there, 0.943. What falls on the wrong side of the cut under each is the corresponding error. The scale along the top of each is the test statistic $T_n = (\text{count} - 62)/\sqrt{31}$ of Section 9.5: the threshold 72 is $T_n = 1.80$, and the observed 80 is 3.23. And the picture shows the trade: sliding the threshold to the right makes the type 1 error smaller and the type 2 error larger (the orange shrinks and the gray grows), sliding it left does the reverse, and at a fixed n you cannot shrink both. The two distributions pull apart as n grows (their standard deviations are $\sqrt{p(1-p)/n}$ on the proportion scale), and that is the only lever that lowers both errors at once.

Rather than two functions on two sets, it is often clearer to look at one function on all of Θ, the probability of rejecting,

$$\theta \mapsto \mathbb{P}_\theta(\psi = 1),$$

which is α_ψ on Θ_0 and $1 - \beta_\psi$ on Θ_1.[2] Figure 9.3 draws it for the kiss data, computed exactly from the Binomial distribution of the count. On the left, at $n = 124$, the one-sided test rejects for a

[2]Two conventions for the letter β are in common use. This book writes β_ψ for the type 2 error, as in Definition 9.2.

count of 72 or more and the two-sided test for 74 or more or 50 or fewer. At $p = \frac{1}{2}$ both curves are at or below 0.05 (they are 0.044 and 0.038); to the right both climb to 1, the one-sided test faster since it spends its whole 5% on that side; to the left the two-sided test climbs and the one-sided test, which does not watch that side, stays at 0. On the right, the one-sided test at $n = 30$ (threshold 20) sits beside the same test at $n = 124$: more data makes the curve steeper, so both errors fall at once. A good test has a rejection curve that is low on Θ_0 and climbs quickly to 1 on Θ_1; the level, defined next, caps the low part, and the power is the lowest point of the high part.

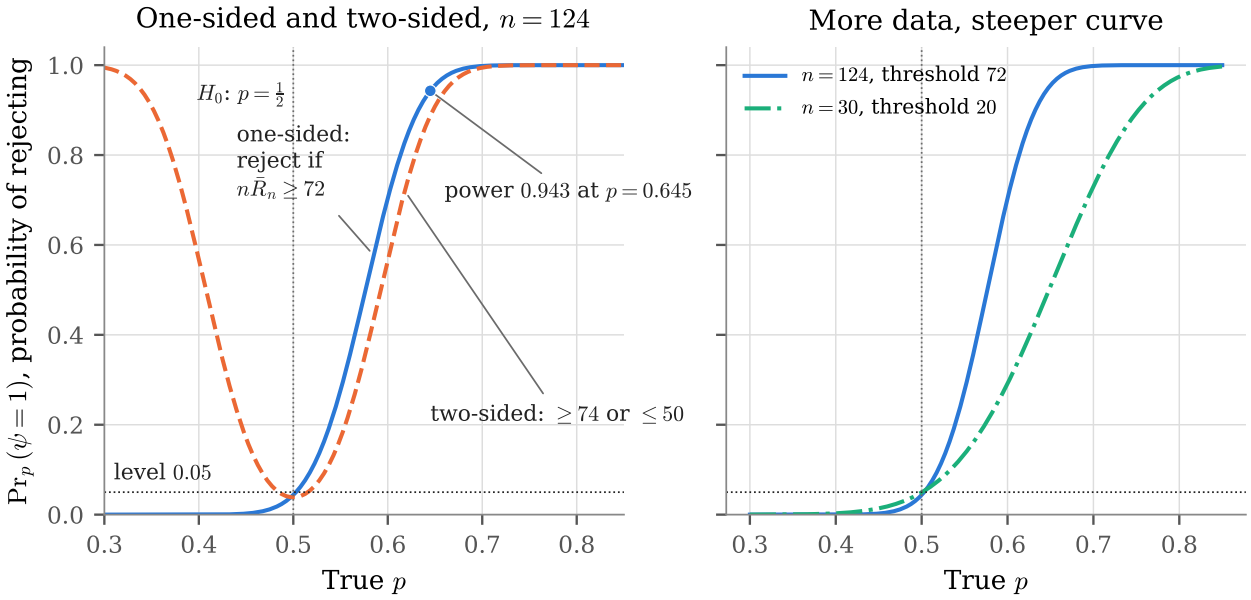


Figure 9.3: The probability of rejecting against the true p, with the level 0.05 dotted. Left, $n = 124$: one-sided, count ≥ 72 (solid), and two-sided, ≥ 74 or ≤ 50 (dashed). Right: one-sided at $n = 124$ (solid) and $n = 30$ (dash-dot).

From undergraduate statistics: α, β and the power calculation

An introductory course calls the type 1 error probability α and the type 2 error probability β, says 'power $= 1 - \beta$', and may have made you compute the sample size needed for 80% power against a specific alternative. All of that is here. The only addition is that α and β are functions of θ, not single numbers: a type 2 error of 0.057 is a type 2 error *at* $p = 0.645$, and at $p = 0.55$ the same test has a type 2 error of 0.72, which is why failing to reject is not accepting: this test misses a true p of 0.55 nearly three times in four. The power calculation of an introductory course picks one alternative and reads one point off Figure 9.3's curve; the power π_ψ of Definition 9.2 takes the infimum over all of them, which for a composite alternative that touches Θ_0 is the rejection probability at the edge of Θ_0, 0.044 here: at most the level, and useless as a summary of what the test can detect. The one-point version is the useful one in planning a study, and the infimum is the one that describes the test over the whole alternative.

Definition 9.3 (Level, asymptotic level)**.** Let $\alpha \in (0, 1)$. A test ψ has **level** α if

$$\alpha_\psi(\theta) \leq \alpha \quad \text{for all } \theta \in \Theta_0,$$

Elsewhere $\beta(\theta)$ names this rejection probability itself, the power function, which is $1 - \beta_\psi(\theta)$ on Θ_1 and $\alpha_\psi(\theta)$ on Θ_0. Check which one a source means before carrying a formula across.

and **asymptotic level** α if

$$\lim_{n\to\infty} \alpha_{\psi_n}(\theta) \le \alpha \quad \text{for all } \theta \in \Theta_0,$$

where ψ_n is the test built on n observations (toolkit A.1).

> **In words.** A test of level 5% is one that convicts the innocent at most 5% of the time, whichever innocent θ is the truth. The level is a cap on the worst case of the type 1 error over the whole null set, not the type 1 error at one point. 'Asymptotic' means the cap is guaranteed only in the limit of large samples, which is what a CLT-based threshold delivers.
>
> **For the kiss data.** $\psi = \mathbb{1}(n\bar{R}_n \ge 72)$ has level 5% for $H_0 : p = \frac{1}{2}$ exactly, since $0.044 \le 0.05$; it also has level 5% for the composite null $H_0 : p \le \frac{1}{2}$, since $\mathbb{P}_p(n\bar{R}_n \ge 72)$ only decreases as p falls below $\frac{1}{2}$. The threshold 71 would not do: $\mathbb{P}_{1/2}(n\bar{R}_n \ge 71) = 0.063$.

Two remarks on the definition, both consequences of the asymmetry of Section 9.2. First, the level is a guarantee about the type 1 error only. A test of level 5% promises nothing about its type 2 error; $\psi \equiv 0$, the test that never rejects, has level 5% (its type 1 error is 0) and is useless. The standard way of choosing a test is to fix the level first and then, among tests of that level, prefer the one with the most power. Second, the level is a cap, so a test of level 5% is also a test of level 10%: 'level 5%' is a one-directional bound, as 'level 95%' was for an interval in Chapter 8, the worst the procedure can do rather than what it does.

Pitfall: level α for a test, level $1-\alpha$ for an interval

The same word, the same α, complementary conventions. For a confidence interval the level is $1-\alpha$, the probability of a correct capture. For a test the level is α, the probability of an incorrect rejection. Both conventions describe the procedure by the guarantee it gives, and the guarantee for a test is about the error it controls. Section 9.8 shows the two procedures are one object, which is why the same α appears in both.

Intuition: why the worst case over Θ_0

For a one-sided question such as 'can we conclude that at most a third of Americans get news on YouTube?' you would write $H_0 : p \ge 0.33$ against $H_1 : p < 0.33$, a composite null. Which p in Θ_0 should the threshold be computed from? The definition answers: the one that makes rejection most likely, since the level must hold for every p in Θ_0. The test rejects when $\hat{p}$ is small, and a small $\hat{p}$ is likeliest when p is as small as Θ_0 allows, that is, at the boundary $p = 0.33$. So the composite null $p \ge 0.33$ is tested with the threshold of the simple null $p = 0.33$, and no separate computation is needed. The same reasoning gives the threshold 72 of the kiss data for $H_0 : p \le \frac{1}{2}$. This is why one-sided tests are freely written either way.

9.5 Building a test from the Central Limit Theorem

The general-purpose test is the standardized mean, and, as in the plug-in interval of Chapter 8, all three limit tools of Part II work together in its one argument.

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2, both finite, and let μ_0 be a fixed number. To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \ne \mu_0$, take

$$T_n = \sqrt{n}\,\frac{\bar{X}_n - \mu_0}{\sigma}.$$

If H_0 is true then $\mu = \mu_0$ and the CLT (Theorem 4.8) says $T_n \xrightarrow[n\to\infty]{(d)} \text{Gauss}(0,1)$. So under H_0 the probability that $|T_n|$ exceeds $q_{\alpha/2}$ tends to $\mathbb{P}(|Z| > q_{\alpha/2}) = \alpha$, and the test

$$\psi_\alpha = \mathbb{1}(|T_n| > q_{\alpha/2})$$

has asymptotic level α. The one-sided versions use one tail: for $H_1 : \mu > \mu_0$, $\psi_\alpha = \mathbb{1}(T_n > q_\alpha)$; for $H_1 : \mu < \mu_0$, $\psi_\alpha = \mathbb{1}(T_n < -q_\alpha)$. Figure 6.4 in Chapter 6 is the picture: $q_{\alpha/2}$ leaves $\alpha/2$ in each tail, q_α leaves α in one.

Proposition 9.4 (The standardized-mean test)**.** *Under the assumptions above, with σ known or replaced by any consistent estimator $\hat{\sigma}_n$:*

1. *$\psi_\alpha = \mathbb{1}(|T_n| > q_{\alpha/2})$ has asymptotic level α for $H_0 : \mu = \mu_0$, and $\mathbb{1}(T_n > q_\alpha)$ has asymptotic level α for $H_0 : \mu \le \mu_0$.*
2. *Under any $\mu \neq \mu_0$ in the alternative, $\mathbb{P}_\mu(\psi_\alpha = 1) \to 1$: the type 2 error at every fixed alternative tends to 0.*

Proof strategy (core). Under H_0 the CLT gives the limit of the rejection probability, and it is a limit of numbers, computed from Φ. With $\hat{\sigma}_n$ in place of σ, Slutsky supplies the same limit. Under H_1 the LLN sends $\bar{X}_n - \mu_0$ to a nonzero constant, so T_n runs off to $\pm\infty$.

Proof. (1) Under $\mathbb{P}_{\mu_0}$, $T_n \xrightarrow[n\to\infty]{(d)} Z \sim \text{Gauss}(0,1)$ by the CLT, and the set $\{|z| > q_{\alpha/2}\}$ has boundary points $\pm q_{\alpha/2}$ at which Z has no mass, so convergence in distribution gives convergence of the probabilities (Proposition 4.3):

$$\alpha_{\psi_n}(\mu_0) = \mathbb{P}_{\mu_0}(|T_n| > q_{\alpha/2}) \to \mathbb{P}(|Z| > q_{\alpha/2}) = \alpha.$$

This is a limit of a sequence of numbers, one per n, and not a mode of convergence of random variables. The random variable T_n is the thing that converges in distribution; the probability it assigns to the fixed set $\{|z| > q_{\alpha/2}\}$ is a number for each n, and those numbers converge in the ordinary calculus sense (toolkit A.1). If σ is replaced by $\hat{\sigma}_n \xrightarrow[n\to\infty]{\mathbb{P}} \sigma$, then $\sqrt{n}(\bar{X}_n - \mu_0)/\hat{\sigma}_n = \big(\sqrt{n}(\bar{X}_n - \mu_0)/\sigma\big) \cdot (\sigma/\hat{\sigma}_n)$ is a product of something converging in distribution and something converging in probability to the constant 1, and Slutsky's theorem (Theorem 5.1) gives the same limit. For the one-sided test with composite null $\mu \le \mu_0$: at $\mu = \mu_0$ the same argument gives $\mathbb{P}_{\mu_0}(T_n > q_\alpha) \to \mathbb{P}(Z > q_\alpha) = \alpha$; at $\mu < \mu_0$, write the event $\{T_n > q_\alpha\}$ as $\{(\bar{X}_n - \mu_0)/\sigma > q_\alpha/\sqrt{n}\}$: the left side converges in probability to the negative number $(\mu - \mu_0)/\sigma$ by the LLN, while the right side tends to 0, so the event's probability tends to $0 \le \alpha$ (once $q_\alpha/\sqrt{n}$ is within ϵ of 0, the event is contained in $\{|(\bar{X}_n - \mu_0)/\sigma - (\mu - \mu_0)/\sigma| > |\mu - \mu_0|/\sigma - \epsilon\}$, whose probability tends to 0).

(2) Under $\mu \neq \mu_0$, write $\{|T_n| > q_{\alpha/2}\}$ as $\{|(\bar{X}_n - \mu_0)/\sigma| > q_{\alpha/2}/\sqrt{n}\}$: the left side converges in probability to the positive number $|\mu - \mu_0|/\sigma$ while the right side tends to 0, so the probability tends to 1 by the same containment argument. For the one-sided test, $\mu > \mu_0$ sends $(\bar{X}_n - \mu_0)/\sigma$ to a positive number and $\mathbb{P}_\mu(T_n > q_\alpha) \to 1$ the same way. □

Part (2) is the reason for liking this test: against any fixed departure from H_0, enough data will detect it. A test with this property is called **consistent**, by analogy with a consistent estimator. Part (1) is the promise Section 5.5 made: the level of a test built on a standardized average rests on the CLT, and on Slutsky as soon as the scale is estimated.

Intuition: under the null the parameter becomes known

The reason tests are easier than intervals is that under the null hypothesis the value of the unknown parameter becomes known, so there is nothing to plug in. A confidence interval for p had to deal with the unknown p in the standard error $\sqrt{p(1-p)/n}$, which is what the three solutions of Section 8.4 were for. A test of $H_0 : p = \frac{1}{2}$ computes everything *as if* $p = \frac{1}{2}$ because the level is a statement about what happens when H_0 is true. So the standard error is $\sqrt{\frac{1}{2} \cdot \frac{1}{2}/n} = 1/(2\sqrt{n})$, a number, and the test statistic is

$$T_n = \sqrt{n}\,\frac{\bar{R}_n - \frac{1}{2}}{\sqrt{\frac{1}{2} \cdot \frac{1}{2}}} = 2\sqrt{n}\left(\bar{R}_n - \tfrac{1}{2}\right),$$

with nothing unknown in it. The alternative only enters through the power, which is computed separately and does not affect the threshold.

Worked example: three tests of a proportion

Each is the two-sided or one-sided test above with σ the null standard deviation $\sqrt{p_0(1-p_0)}$.

- **The kiss experiment**, $H_0 : p = \frac{1}{2}$ against $H_1 : p \neq \frac{1}{2}$. $T_n = \sqrt{124}\,(0.645 - 0.5)/0.5 = 3.23$, and $3.23 > 1.96 = q_{0.025}$: reject at asymptotic level 5%. The one-sided version compares with $q_{0.05} = 1.645$ and rejects too.
- **The fair coin**, 13 heads in 30 tosses, $H_0 : p = \frac{1}{2}$ against $H_1 : p \neq \frac{1}{2}$. $T_n = \sqrt{30}\,(13/30 - 0.5)/0.5 = -0.73$, and $0.73 < 1.96$: do not reject. A value of -0.73 is a perfectly plausible draw from a standard Gaussian: about 47% of draws fall further from zero than that.
- **News on YouTube**, $\hat{p} = 0.32$ of $n = 4{,}971$, $H_0 : p \geq 0.33$ against $H_1 : p < 0.33$. A one-sided test with the threshold computed at the boundary $p = 0.33$: $T_n = \sqrt{4971}\,(0.32 - 0.33)/\sqrt{0.33 \times 0.67} = -1.50$, and $-1.50 > -1.645 = -q_{0.05}$: do not reject at level 5%. The data lean the alternative's way but not far enough, which is the same thing Figure 9.1 showed with the buffer just reaching 0.33.

Note that the three conclusions came from the same kind of comparison, T_n against a Gaussian quantile; what differed was the quantile, $q_{0.025}$ for a two-sided question and $q_{0.05}$ for a one-sided one, and, for the one-sided test, that only one sign counts. It is easier to reject one-sided than two-sided because a one-sided test puts its whole α in the tail it watches.

Example 9.5 (A Poisson rate)**.** Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$, so $\mathbb{E}[X_i] = \mathbb{V}\text{ar}(X_i) = \lambda$, and let $\hat{\lambda} = \bar{X}_n$. The CLT gives $\sqrt{n}(\hat{\lambda} - \lambda)/\sqrt{\lambda} \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1)$. To test $H_0 : \lambda = 2$ against $H_1 : \lambda \neq 2$ at asymptotic level 5%, use $T_n = \sqrt{n}(\hat{\lambda} - 2)/\sqrt{2}$, with the null value 2 in the variance as well as the mean, and reject when $|T_n| > 1.96$. To test $H_0 : \lambda \geq 2$ against $H_1 : \lambda < 2$, reject when $T_n < -1.645$; at any $\lambda > 2$ the LLN sends T_n to $+\infty$ and the rejection probability to 0, so the level is set at the boundary $\lambda = 2$ as in the intuition box above. A fourth pair of hypotheses, $H_0 : |\lambda - 2| \leq 1$ against $H_1 : |\lambda - 2| > 1$, has a null set that is an interval, $[1, 3]$; the worst case over it is at the two endpoints, 1 and 3, the test has to be sized at both, and Exercise 9.5 works it out.

9.6 The kiss experiment, exactly and by the CLT

Chapter 1 quoted two thresholds, 72 for the one-sided question and 74 for the two-sided one, and said both come from the exact Binomial distribution of the count. The CLT test of the previous section gives 72 and 73. Here are both computations and the reason for the gap.

Exactly. Under $H_0 : p = \frac{1}{2}$ the count $n\bar{R}_n$ is exactly Bin$(124, \frac{1}{2})$, with no approximation. A one-sided test of level 5% rejects for counts $\geq k$ with k the smallest integer such that $\mathbb{P}_{1/2}(n\bar{R}_n \geq k) \leq 0.05$. The tail probabilities are 0.063 at $k = 71$, 0.044 at $k = 72$, so $k = 72$. A two-sided test of level 5% rejects for counts $\geq k$ or $\leq 124 - k$, and by symmetry needs $\mathbb{P}_{1/2}(n\bar{R}_n \geq k) \leq 0.025$: the tail is 0.029 at $k = 73$ and 0.019 at $k = 74$, so $k = 74$, with the mirror threshold 50. These are the **exact Binomial tests** that Section 4.4 promised: the finite-sample alternatives to the CLT that exist whenever the parent distribution is known. Their type 1 errors are 0.044 and 0.038, not 0.05: with a discrete statistic the level can only be met from below, since the tail probability jumps at each integer and 0.05 is not one of the values it takes.

By the CLT. The one-sided test rejects when $2\sqrt{n}(\bar{R}_n - \frac{1}{2}) > 1.645$, that is, when $n\bar{R}_n > 62 + 1.645 \times \sqrt{31} = 71.2$, that is, for counts ≥ 72. The two-sided test rejects when $|n\bar{R}_n - 62| > 1.96 \times \sqrt{31} = 10.9$, that is, for counts ≥ 73 or ≤ 51. So the one-sided thresholds agree and the two-sided ones differ by one count.

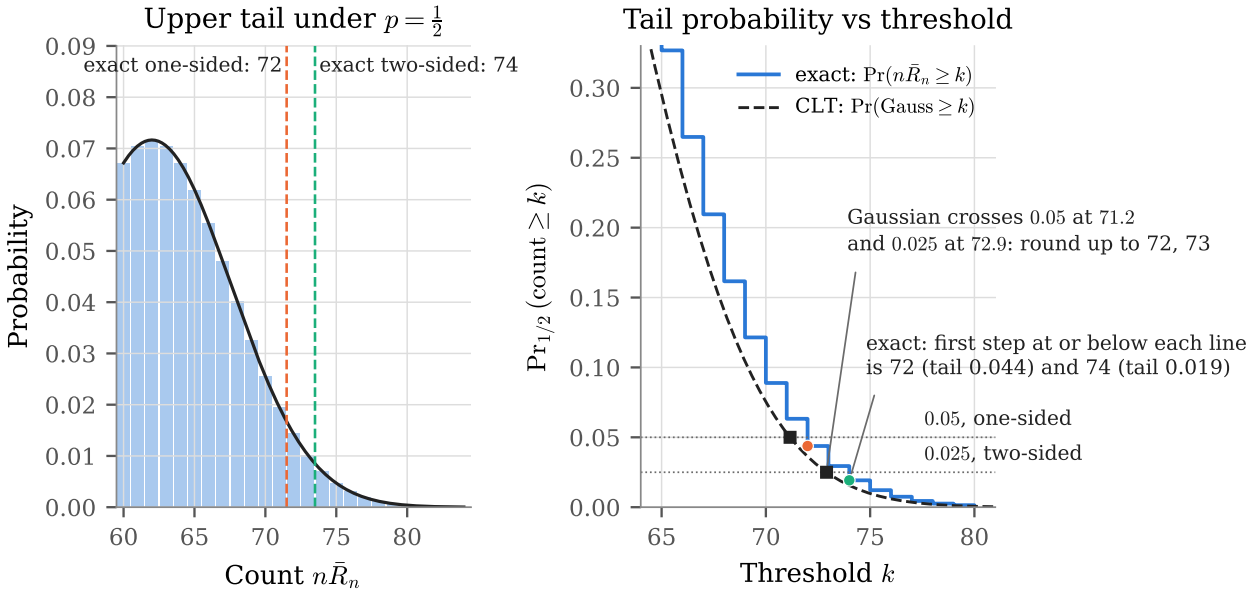


Figure 9.4: Left: the upper tail of Bin$(124, \frac{1}{2})$ (bars) and the Gauss$(62, 31)$ density (curve), with the exact thresholds 72 (orange) and 74 (aqua) dashed. Right: $\mathbb{P}_{1/2}(\text{count} \geq k)$ against k, exact (staircase) and Gaussian (dashed), with the levels 0.05 and 0.025 dotted.

Why they differ. Figure 9.4 shows the two tail probabilities side by side. On the left is the Gauss$(62, 31)$ density the CLT puts in place of the Bin$(124, \frac{1}{2})$ bars; on the right is the tail probability $\mathbb{P}_{1/2}(\text{count} \geq k)$ as a function of the threshold k, a staircase for the exact distribution and a smooth curve for the Gaussian. The exact threshold is the first step at or below the level line, 72 for 0.05 and 74 for 0.025; the CLT threshold is where the curve crosses the line, 71.2 and 72.9, rounded up to 72 and 73. The Gaussian curve is a good approximation of the staircase, off by about 0.005 at $k = 73$, but 0.005 is enough to put the exact tail (0.029) above the line 0.025 and the Gaussian

tail (0.024) below it, and that is the whole disagreement. The gap is mostly discreteness, not the CLT's inaccuracy: the Gaussian approximates the probability of the count being ≥ 73 better if the threshold is placed at 72.5 (the **continuity correction** of an introductory course), and with it the two-sided threshold becomes $62 + 0.5 + 10.9 = 73.4$, rounding up to 74, in agreement with the exact test. This book does not use the correction. For a proportion, whenever the Gaussian approximation lands close enough to the line that the correction would change the decision, the exact Binomial test is available and settles the question outright. An asymptotic level 5% is not an exact level 5%, and at a threshold where the exact tail is near the line the two can round in different directions. For 80 right turns, far from either threshold, the decision is the same every way: reject.

From undergraduate statistics: the binomial test and the z-test for a proportion

An introductory course presents these as two tests: the 'exact binomial test' when n is small, the 'z-test for a proportion' when np_0 and $n(1-p_0)$ are both at least 10. They test the same hypothesis with the same statistic, the count, and differ only in whether the count's distribution under H_0 is used as it is or replaced by its Gaussian limit. The rule of thumb keeps the count's skewness small (the '$np \geq 10$' bridge in Section 4.2); it does nothing about discreteness, which is why the two-sided thresholds above still differ by one count with $np_0 = 62$.

9.7 p-values

A test as defined reports one bit, reject or not, at one level fixed in advance. That loses information: a T_n of 1.97 and a T_n of 3.23 both reject at 5%, and the second is much stronger evidence. It also forces the reader to accept the writer's α. The p-value fixes both.

Definition 9.6 (p-value)**.** Let $(\psi_\alpha)_{\alpha\in(0,1)}$ be a family of tests of H_0 indexed by their level, nested so that rejecting at level α implies rejecting at every level above α. The **(asymptotic) p-value** of the family on a given sample is the smallest (asymptotic) level α at which ψ_α rejects H_0 on that sample.[3] It is random: it depends on the sample.

In words. Run the test at every level at once. At large α it rejects easily, at small α it is hard to satisfy; the p-value is the level where it switches. Equivalently, for the standardized-mean tests of this chapter, it is the probability under H_0 of a test statistic at least as extreme as the one observed: the tail area beyond the observed value.

For the kiss data. $T_n = 3.23$. Two-sided, the p-value is $\mathbb{P}(|Z| > 3.23) = 2(1 - \Phi(3.23)) = 0.0012$; one-sided, $\mathbb{P}(Z > 3.23) = 0.0006$. The exact Binomial tails give 0.0016 and 0.0008. Under $p = \frac{1}{2}$, 80 or more right turns out of 124 is a one-in-a-thousand event.

Figure 9.5 shows the p-value two ways. On the left, the density of T_n under H_0 is the standard Gaussian by the CLT, the 5% rejection region is $|T_n| > 1.96$, and the p-value is the area beyond ± 3.23, too small to see at this scale: the probability under H_0 of something at least this extreme, 0.0012. On the right, the threshold $q_{\alpha/2}$ is drawn as a function of the level, with the observed value as a horizontal line; the test rejects at every level whose threshold lies below the line, and the smallest such level, where the curve crosses, is the p-value. The two readings are the same number because of how the thresholds are built. For the two-sided test, ψ_α rejects when $|T_n| > q_{\alpha/2}$, and

[3] Strictly an infimum: at the boundary level the threshold equals the observed statistic and the strict inequality in ψ_α fails. The boundary case has probability zero, and the golden rule's $\leq$ settles it by convention.

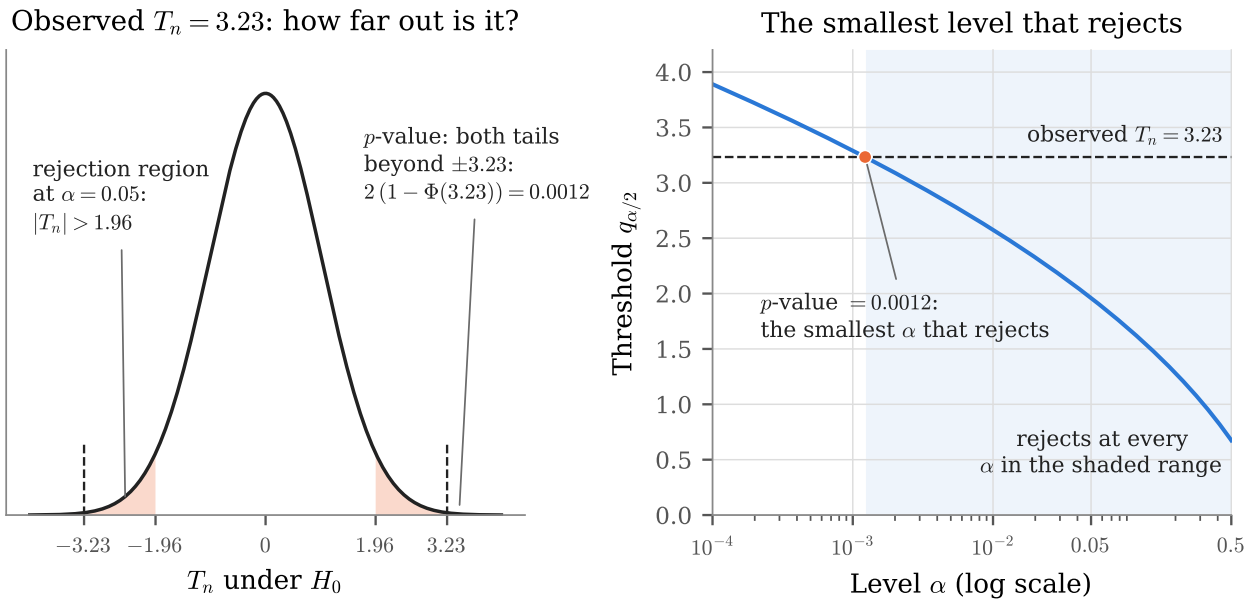


Figure 9.5: The p-value for $T_n = 3.23$. Left: the density of T_n under H_0, the rejection region at 5% (light orange), the area beyond ± 3.23 (dark orange), the observed value (dashed). Right: the threshold $q_{\alpha/2}$ against the level α, with the observed value dashed.

$q_{\alpha/2}$ decreases as α grows, so the levels at which the test rejects are exactly those with $q_{\alpha/2} < |t|$ for the observed t, that is, those with $\alpha/2 > \mathbb{P}(Z > |t|)$, that is, $\alpha > \mathbb{P}(|Z| > |t|)$. The smallest such α is $\mathbb{P}(|Z| > |t|)$, the tail area. For a one-sided test with $H_1 : \mu > \mu_0$ the same argument gives $\mathbb{P}(Z > t)$. Since $\mathbb{P}(|Z| > |t|) = 2\mathbb{P}(Z > |t|)$, a two-sided p-value is twice the one-sided one when the data lean the alternative's way; when they lean the other way the one-sided p-value is above $\frac{1}{2}$ and the one-sided test cannot reject. That is the one-sided test's advantage of Section 9.5 seen from the other end: it spends its whole α on one side and is blind to the other.

The golden rule. The consequence of the definition is the rule worth memorizing:

$$p\text{-value} \le \alpha \quad \Longleftrightarrow \quad H_0 \text{ is rejected by } \psi_\alpha \text{ at (asymptotic) level } \alpha.$$

So one number replaces the whole family of tests. A reader who wants level 5% compares the p-value with 0.05; one who wants 1% compares with 0.01; and the smaller the p-value, the more levels reject, the more confidently H_0 is rejected.

From undergraduate statistics: $p < 0.05$

'Reject if $p < 0.05$' was the rule of an introductory course, and the golden rule is its justification: the p-value is the smallest level that rejects, so a p-value below 0.05 means the level-5% test rejects, no more and no less. Two habits from that course transfer intact. The p-value is computed from the null distribution of the statistic, which is why it needed μ_0 (or p_0) and not any guess at the alternative. And a one-sided p-value is half the two-sided one for a symmetric statistic when the data lean the alternative's way, so the choice of sides made in Section 9.2 before seeing the data is the choice that fixes the p-value.

Pitfall: what a p-value is not

A p-value of 0.0012 is the probability, computed as if H_0 were true, of data at least as extreme as these. It is not the probability that H_0 is true; in this part of the book H_0 is true or false and has no probability (Chapter 18 is where hypotheses get probabilities, and the number it produces is a different one). It is not the probability that the result 'was due to chance'. And it is not a measure of the size of the effect: with n large enough, a true p of 0.51 produces a tiny p-value (at $n = 100{,}000$, $T_n = 2\sqrt{n} \times 0.01 = 6.3$), since by Proposition 9.4 the test detects any fixed departure eventually. The p-value answers one question, how surprising the data would be under H_0, and the confidence interval of Chapter 8 answers the other, how big the departure is. Report both.

Intuition: the p-value is a statistic, and under H_0 it is uniform

The p-value is a function of the sample, so it is a random variable, and it has a distribution under each θ. Under a simple H_0, for a test statistic with a continuous distribution, that distribution is uniform on $[0, 1]$: a p-value below 0.05 happens 5% of the time, below 0.01 1% of the time, which is just the level restated (Exercise 9.6 proves it). Under H_1 the distribution piles up near 0, and the more so the larger n. So a small p-value is not proof; it is an event that is rare under H_0 and common under H_1.

9.8 Tests and confidence intervals are one construction

Exercise 8.6 in Chapter 8 found that '$\frac{1}{2}$ is below the conservative lower confidence bound' is the same event as '$n\bar{R}_n \geq 72$', the one-sided test's rejection. That was not a coincidence of the kiss data.

Proposition 9.7 (Duality of tests and confidence intervals)**.** *Let $\Theta \subseteq \mathbb{R}$.*

1. *Suppose that for every $\theta_0 \in \Theta$ there is a test $\psi(\theta_0)$ of $H_0 : \theta = \theta_0$ with level α. Then the set of values not rejected,*
$$\mathcal{I} = \{\theta_0 \in \Theta : \psi(\theta_0) = 0\},$$
is a confidence set of level $1 - \alpha$ for θ.
2. *Conversely, if $\mathcal{I}$ is a confidence interval of level $1 - \alpha$ for θ, then for every θ_0 the test $\psi(\theta_0) = \mathbb{1}(\theta_0 \notin \mathcal{I})$ has level α for $H_0 : \theta = \theta_0$.*

The same holds with 'asymptotic' in front of both levels.

Proof strategy (core). Both directions are one line: the event 'the true θ is not in $\mathcal{I}$' is the event 'the test of the true θ rejects', whose probability is the type 1 error.

Proof. (1) For any true θ, $\mathbb{P}_\theta(\mathcal{I} \not\ni \theta) = \mathbb{P}_\theta(\psi(\theta) = 1) = \alpha_{\psi(\theta)}(\theta) \leq \alpha$, since $\psi(\theta)$ has level α for the null $\theta = \theta$, which is true. So $\mathbb{P}_\theta(\mathcal{I} \ni \theta) \geq 1 - \alpha$ for every θ: the coverage condition of Definition 8.1, applied to a set rather than an interval. (2) Under $H_0 : \theta = \theta_0$ the truth is θ_0, and $\mathbb{P}_{\theta_0}(\psi(\theta_0) = 1) = \mathbb{P}_{\theta_0}(\mathcal{I} \not\ni \theta_0) \leq \alpha$ by the coverage guarantee. The asymptotic version replaces each probability by its limit. □

Figure 9.6 draws part (1) for the kiss data, running the test for every p_0 at once. Each vertical segment is the set of sample proportions the level-5% test of $H_0 : p = p_0$ does not reject; a candidate

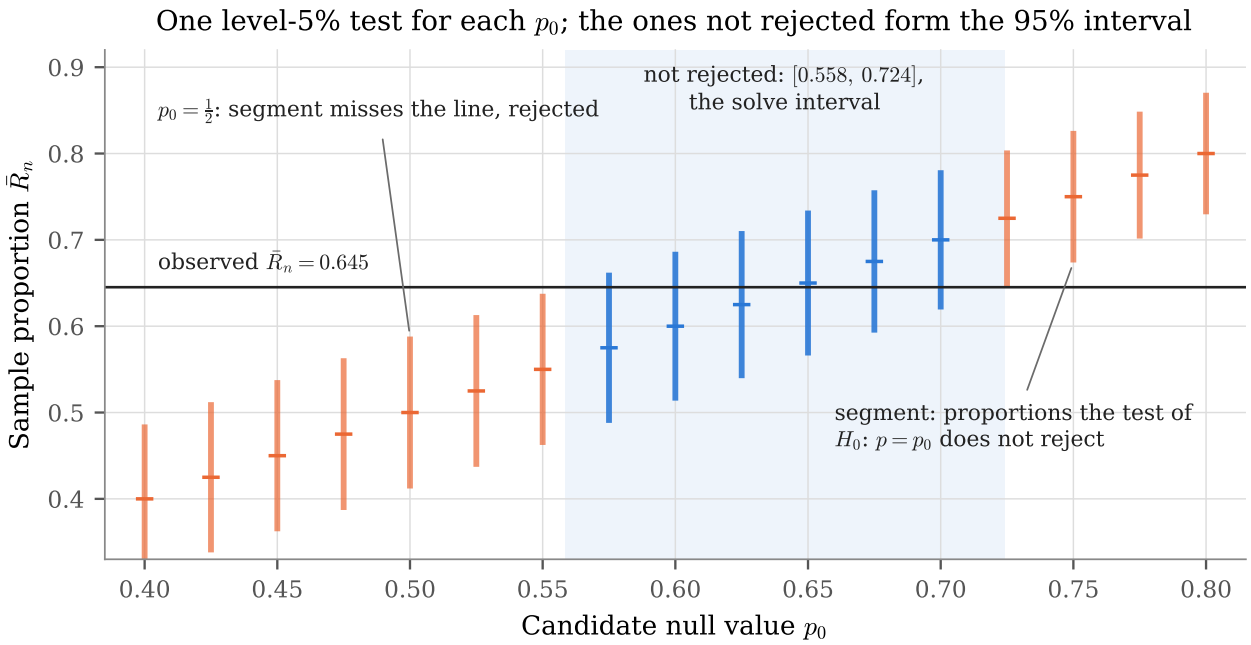


Figure 9.6: The duality for the kiss data. At each candidate p_0 (horizontal axis), a vertical segment $p_0 \pm 1.96\sqrt{p_0(1-p_0)/124}$; the horizontal line is the observed $\bar{R}_n = 0.645$. Segments crossing the line are blue, segments missing it are orange, and the shaded strip is $[0.558, 0.724]$.

whose segment crosses the line of the observed $\bar{R}_n = 0.645$ is not rejected, one whose segment misses it is rejected, and $\frac{1}{2}$ misses. The shaded strip of candidates not rejected is $[0.558, 0.724]$, the solve interval of Chapter 8 exactly. The test being inverted is

$$\psi(p_0) = \mathbb{1}\left\{|\bar{R}_n - p_0| > q_{\alpha/2}\sqrt{p_0(1-p_0)/n}\right\},$$

which measures the distance from p_0 in units of the standard error computed *at* p_0, the value the null hypothesis asserts. For a proportion, a test built that way, from the standard error under the null rather than from an estimate, is the **score test**, which Chapter 15 defines for any model. The candidates that survive are those with $|\bar{R}_n - p_0| \le q_{\alpha/2}\sqrt{p_0(1-p_0)/n}$, which is the inequality that Section 8.4 solved to get the solve interval $[0.558, 0.724]$. So the solve interval is the set of null values the score test does not reject, and the score test of $p = \frac{1}{2}$ rejects exactly because $\frac{1}{2}$ is outside the interval. This is why Section 8.4 called the result the Wilson score interval: it is the score test, inverted. Inverting the test built from the *estimated* standard error $\sqrt{\bar{R}_n(1-\bar{R}_n)/n}$ instead gives the plug-in interval $[0.561, 0.729]$. That one is **Wald's test**, which Chapter 15 treats in general. Score and Wald are the same idea evaluated at two different places, the null value and the estimate, and they are the two tests you will meet most often. And the one-sided test of Exercise 8.6 inverts to a one-sided interval, a lower confidence bound. Every interval in Chapter 8 is a test turned inside out, and every test here is an interval asked about one point.

From undergraduate statistics: if the confidence interval excludes the null value, reject

An introductory course states this as a shortcut: 'the 95% confidence interval is $[0.56, 0.73]$, which does not contain 0.5, so the difference is significant at 5%'. Proposition 9.7 is why the shortcut is legitimate, and it also says when it is exact and when it is approximate. The shortcut

reproduces the test's decision exactly when the test and the interval are built from the same standard error, the solve interval for the score test and the plug-in interval for Wald's, and only approximately otherwise. Using the plug-in interval to decide a score test can disagree with it on a borderline sample, since the two standard errors differ: $\sqrt{\bar{R}_n(1-\bar{R}_n)/n} = 0.043$ against $\sqrt{\frac{1}{2}\cdot\frac{1}{2}/n} = 1/(2\sqrt{n}) = 0.045$ here. Far from the border, as with 80 picks, every version agrees.

9.9 What comes next

This chapter built one test, the standardized mean against a Gaussian quantile, and the ideas that describe any test. The remaining tests in this book are the same object with a different statistic and a different null distribution. Chapter 14 replaces the Gaussian quantile by a Student quantile when n is small and the data are Gaussian, which needs the chi-squared distribution and Cochran's theorem for the sample variance; it also finishes the boarding example as a two-sample test. Chapter 15 gives three general recipes for a test statistic in any parametric model, Wald's test (the standardized maximum likelihood estimator), the likelihood ratio test and the score test, with chi-squared limits in place of Gaussian ones. Chapter 16 and Chapter 17 test the model itself rather than a parameter in it, with the same vocabulary: a statistic, a null distribution, a threshold, a p-value. In all of them the level is the cap on the type 1 error, the power is what the test can detect, and the p-value is what gets reported.

9.10 Summary

- Hypotheses: $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$, disjoint. The data are used only to try to reject H_0; failing to reject is not accepting. Simple ($\{\theta_0\}$) or composite; one-sided or two-sided, and the side is chosen before the data are seen.
- A test is $\psi : E^n \to \{0, 1\}$, the indicator of its rejection region; in practice $\psi = \mathbb{1}(T_n > c)$ for a test statistic T_n and a threshold c.
- Type 1 error $\alpha_\psi(\theta) = \mathbb{P}_\theta(\psi = 1)$ on Θ_0; type 2 error $\beta_\psi(\theta) = \mathbb{P}_\theta(\psi = 0)$ on Θ_1; power $\pi_\psi = \inf_{\Theta_1}(1 - \beta_\psi)$. At fixed n the two errors trade off; only n lowers both.
- Level α: $\alpha_\psi \leq \alpha$ on all of Θ_0, a cap on the worst case; asymptotic level: the same in the limit. For a composite null the worst case is at the boundary, so $H_0 : \theta \leq \theta_0$ and $H_0 : \theta = \theta_0$ get the same test. For a test the level is α; for an interval it is $1 - \alpha$.
- The standardized-mean test: $T_n = \sqrt{n}(\bar{X}_n - \mu_0)/\sigma$, reject if $|T_n| > q_{\alpha/2}$ (two-sided) or $T_n > q_\alpha$ (one-sided). Asymptotic level α by the CLT, plus Slutsky if σ is estimated; consistent (power $\to 1$ at every fixed alternative) by the LLN. Under H_0 the parameter is known, so no plug-in is needed.
- Kiss experiment: $T_n = 3.23$, reject at 5% either way. Exact Binomial thresholds 72 (one-sided) and 74 (two-sided); CLT thresholds 72 and 73. They differ because the exact tail is a staircase and 0.025 falls between two of its steps.
- p-value: the smallest level at which the data reject; for these tests, the tail area beyond the observed statistic under H_0. Golden rule: p-value $\leq \alpha$ if and only if ψ_α rejects. Kiss experiment: 0.0012 two-sided, 0.0006 one-sided. Not the probability that H_0 is true.
- Duality: the null values a level-α test does not reject form a level-$(1 - \alpha)$ confidence set, and a level-$(1 - \alpha)$ interval rejects every null value outside it at level α. The solve interval inverts

the score test, which uses the standard error at the null value; the plug-in interval inverts Wald's test, which uses the estimated standard error.

9.11 Exercises

Exercise 9.1. For 80 of 124 couples turning right, formulate the two-sided hypothesis problem, compute the p-value by the CLT and exactly, and conclude. Then do the one-sided version. *(Solution on page 419.)*

Exercise 9.2. (From OpenIntro Statistics.) Students count the chocolate chips in 32 cookies and find a mean of 14.77 with a standard deviation of 4.37. The packaging claims at least 20 chips per cookie. One student says the claim is unreasonable; another says the difference might be chance. Formulate the hypotheses, compute a p-value and decide, saying what approximation you used. *(Solution on page 419.)*

Exercise 9.3. A machine learning procedure claims to reduce the rate of surgery on benign tumors below the state of the art, 95%, while keeping the true positive rate. On 297 benign tumors it recommended surgery for 206. Let p be the proportion of benign tumors on which it prescribes surgery. Formulate the hypotheses, compute the p-value and conclude. *(Solution on page 419.)*

Exercise 9.4. For the fair coin (13 heads in 30 tosses), compute the two-sided p-value by the CLT and exactly, and comment on the gap. *(Solution on page 419.)*

Exercise 9.5. For $X_1, \dots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$, build a test of asymptotic level 5% for $H_0 : |\lambda - 2| \leq 1$ against $H_1 : |\lambda - 2| > 1$. *(Solution on page 419.)*

Exercise 9.6. Let T be a test statistic with a continuous, strictly increasing CDF F_0 under H_0, and let the test reject for large T, so the p-value is $P = 1 - F_0(T)$. Show that under H_0, P is uniform on $[0, 1]$, and deduce that $\mathbb{P}_{H_0}(P \leq \alpha) = \alpha$ for every α: the golden rule gives exact level α. *(Solution on page 420.)*

Exercise 9.7. Show that the set of $p_0 \in (0, 1)$ not rejected by the test $\psi(p_0) = \mathbb{1}(|\bar{R}_n - p_0| > q_{\alpha/2}\sqrt{p_0(1-p_0)/n})$ is exactly the solve interval of Section 8.4, and that inverting the test with the plug-in standard error gives the plug-in interval. *(Solution on page 420.)*

Part IV

Methods of Estimation

Part III judged estimators; this part produces them. The distances between distributions of Chapter 10 say what 'close to the truth' means; maximum likelihood (Chapters 11 and 12) is the main method, with its consistency and asymptotic normality through the Fisher information; the method of moments and M-estimation (Chapter 13) are the two generalizations.

Chapter 10

Distances Between Distributions: Total Variation and Kullback–Leibler

In this chapter

- Why 'the estimate is close to the truth' should be measured between distributions and not between parameters, and an example where a parameter difference of 0.01 means almost nothing in one place and a great deal in another.
- Total variation: the largest disagreement between two distributions over all events, equal to the area between their densities. What it says about tests: the best possible test between two distributions has $\alpha + \beta = 1 - \mathrm{TV}$.
- Why total variation, for all its good properties, can't be estimated by an average of the data.
- The Kullback–Leibler divergence: the average evidence one observation supplies against the wrong distribution. Its properties, its asymmetry, why it adds up over independent observations, and why near the truth it is a parabola.
- Estimating the KL divergence by the Law of Large Numbers, and the discovery that minimizing the estimate is maximizing the likelihood.
- What an estimator estimates when the model is wrong: the member of the family closest to the truth in KL. For the Gaussian family that is the member with the right mean and variance.

Part III judged estimators that were handed to you. This chapter is the first step toward producing them: it says what 'close' means, and the answer leads straight to maximum likelihood in Chapter 11.

10.1 What 'close to the truth' should mean

Go back to the setup of Chapter 2. You have a statistical model $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$, data $X_1, \dots, X_n \overset{iid}{\sim} \mathbb{P}_{\theta^*}$ from a true parameter θ^* that you don't know, and an estimator $\hat{\theta}_n$. What would it mean for $\hat{\theta}_n$ to be a good estimator?

The answer of Chapter 7 was that $\hat{\theta}_n$ should be close to θ^* as numbers: small bias, small variance, small quadratic risk $\mathbb{E}[(\hat{\theta}_n - \theta^*)^2]$. That is a fine answer when the parameter is the thing you care

about, like the share of couples who turn right. But a parameter is a label for a distribution. What you actually use a fitted model for is its distribution: the probability it assigns to an interval, a tail, a range of outcomes. So the question you'd really like answered is whether the fitted distribution $\mathbb{P}_{\hat{\theta}_n}$ is close to the true one $\mathbb{P}_{\theta^*}$, in the sense that for every event A,

$$\mathbb{P}_{\hat{\theta}_n}(A) \approx \mathbb{P}_{\theta^*}(A).$$

If that holds, every probability you compute from the fitted model is about right, and that is the whole point of fitting it.

Intuition: a parameter difference has no fixed meaning

Is a difference of 0.01 between two parameters small? It depends on where in the parameter space it happens. Two Gaussians with standard deviation 1 whose means differ by 0.01 are, for every practical purpose, the same distribution: no event has probabilities under the two that differ by more than 0.004. Two Gaussians with mean 0 whose standard deviations are 0.01 and 0.02 also have parameters 0.01 apart, and they are obviously different distributions: one is twice as wide as the other, and there is an event whose probability is 0.32 higher under one than under the other. Figure 10.1 draws both. On the left the two curves coincide to the width of a line, and the total variation distance of Section 10.2 between them is 0.004. On the right they share only the gray area, 0.677 of their mass, so the distance is 0.323.

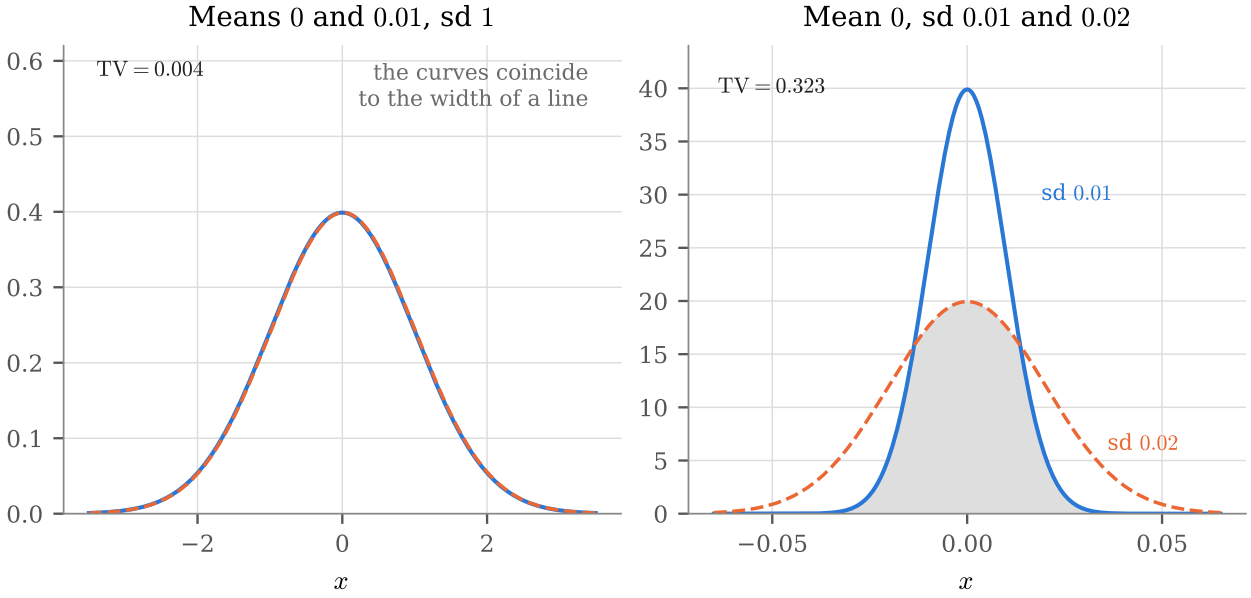


Figure 10.1: Two pairs of Gaussian densities whose parameters differ by 0.01. Left: means 0 and 0.01, standard deviation 1. Right: mean 0, standard deviations 0.01 and 0.02; the gray area is the mass they share.

There is a second reason to measure closeness between distributions, and it matters as much as the first. Section 2.4 asked whether the model is well-specified, whether the true distribution is in the family at all. If it isn't, there is no θ^*, and 'the distance from $\hat{\theta}_n$ to θ^*' has no meaning. A distance between distributions still has one: you can ask which member of the family is closest to the truth, even when none of them is the truth. Section 10.7 answers that question.

The chapter builds two such measures. The first, total variation, is the natural one, and it is a true distance. The second, the Kullback–Leibler divergence, is not a true distance, and it is the one statistics uses, for a reason that takes Section 10.4 to explain.

10.2 Total variation

The requirement '$\mathbb{P}(A) \approx \mathbb{Q}(A)$ for every event A' already contains a definition. Look at all the events, find the one on which the two distributions disagree most, and call that disagreement the distance.

Definition 10.1 (Total variation distance)**.** Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability distributions on the same observation space E. The **total variation distance** between them is

$$\mathrm{TV}(\mathbb{P}, \mathbb{Q}) = \max_{A \subset E} |\mathbb{P}(A) - \mathbb{Q}(A)|,$$

the largest difference between the probabilities they assign to one event.

> **In words.** Run through every question you could ask about one draw (is it positive? is it between 2 and 3? is it even?) and record how differently the two distributions answer. The total variation distance is the worst case. If it is 0.01, no question you could ever ask gets answers more than one percentage point apart. That is exactly the guarantee you want from a fitted model: every probability it reports is within TV of the true one.
>
> **For the kiss data.** One couple either turns right or doesn't, so there are four events: nothing, 'right', 'left' and everything. Compare Bern(0.645), the fitted distribution, with Bern($\frac{1}{2}$), 'no preference'. The empty event and the whole space get probabilities 0 and 1 from both. 'Right' gets 0.645 and 0.5; 'left' gets 0.355 and 0.5. The largest difference is 0.145, so TV = 0.145. As a natural frequency: of 1,000 couples, the fitted model predicts 645 right turns and 'no preference' predicts 500; the two descriptions disagree about 145 couples in a thousand.

For an observation space with four events the maximum is easy. For an observation space like $\mathbb{R}$ there are uncountably many events, and searching over them is not a calculation anyone wants to do. The next result turns the maximum into a sum or an integral.

Proposition 10.2 (Total variation as an area)**.** *If $\mathbb{P}$ and $\mathbb{Q}$ have PMFs p and q on a discrete E,*

$$\mathrm{TV}(\mathbb{P}, \mathbb{Q}) = \frac{1}{2} \sum_{x \in E} |p(x) - q(x)|;$$

if they have densities p and q,

$$\mathrm{TV}(\mathbb{P}, \mathbb{Q}) = \frac{1}{2} \int_E |p(x) - q(x)| \, \mathrm{d}x.$$

The maximum in Definition 10.1 is attained at the event $A^\star = \{x : p(x) > q(x)\}$, where $\mathbb{P}$ puts more mass than $\mathbb{Q}$.

Proof strategy (core). The event that favors $\mathbb{P}$ most collects every point where p beats q and no other point. Its advantage is the area of the region where p is above q; the region where q is above p has the same area, since both densities have total mass 1. Half the total area between the curves is therefore the area of either piece.

Proof. The discrete case; the continuous one replaces sums by integrals. Split E into $A^\star = \{p > q\}$ and its complement $\{p \le q\}$. For any event A,

$$\mathbb{P}(A) - \mathbb{Q}(A) = \sum_{x \in A} \big(p(x) - q(x)\big) \le \sum_{x \in A \cap A^\star} \big(p(x) - q(x)\big) \le \sum_{x \in A^\star} \big(p(x) - q(x)\big) = \mathbb{P}(A^\star) - \mathbb{Q}(A^\star).$$

The first inequality drops the points of A where $p - q \le 0$, which can only increase the sum; the second adds the points of $A^\star$ that A missed, where $p - q > 0$, which can only increase it again. The same argument with $\mathbb{P}$ and $\mathbb{Q}$ exchanged gives $\mathbb{Q}(A) - \mathbb{P}(A) \le \mathbb{Q}(A^{\star c}) - \mathbb{P}(A^{\star c})$. Both p and q sum to 1, so

$$\sum_{x \in A^\star} \big(p(x) - q(x)\big) + \sum_{x \notin A^\star} \big(p(x) - q(x)\big) = 1 - 1 = 0,$$

which says that $\mathbb{P}(A^\star) - \mathbb{Q}(A^\star) = \mathbb{Q}(A^{\star c}) - \mathbb{P}(A^{\star c})$: the area where p is above q equals the area where q is above p. Call this common value D. Every event has $|\mathbb{P}(A) - \mathbb{Q}(A)| \le D$, and $A^\star$ attains it, so $\mathrm{TV}(\mathbb{P}, \mathbb{Q}) = D$. Finally $\sum_x |p(x) - q(x)|$ is the sum of the two pieces, $D + D = 2D$, and so $D = \frac{1}{2}\sum_x |p(x) - q(x)|$. □

Figure 10.2 is the proposition as a picture. Lay the two densities on top of each other. Where they overlap, the gray region, they agree. Where they don't, the area splits into two pieces of equal size, one where p is above and one where q is above, and the total variation distance is the area of either piece. Equivalently, $1 - \mathrm{TV}$ is the area of the overlap, the probability mass the two distributions share. On the left, for Gauss$(0, 1)$ and Gauss$(1, 1)$, the blue piece and the orange piece each have area 0.383, which is the total variation distance, and the gray overlap has area $1 - 0.383 = 0.617$. On the right, the two distributions of the number of right turns among 124 couples share only 0.100 of their mass, so TV = 0.900; the two PMFs cross between 71 and 72.

Worked example: four total variation distances

- **Two Bernoullis.** For Bern(p) and Bern(q) the PMFs differ by $|p - q|$ at 1 and by $|(1 - p) - (1 - q)| = |p - q|$ at 0, so

  $$\mathrm{TV}(\mathrm{Bern}(p), \mathrm{Bern}(q)) = \tfrac{1}{2}\big(|p - q| + |p - q|\big) = |p - q|.$$

  For the kiss data, $|0.645 - 0.5| = 0.145$, as the in-words box found by listing events. For Bern$(\frac{1}{2})$ against Bern$(\frac{1}{10})$, 0.4.

- **Two Gaussians with the same variance.** The densities of Gauss$(0, 1)$ and Gauss$(\mu, 1)$, $\mu > 0$, cross at the midpoint $\mu/2$, and $A^\star = \{x < \mu/2\}$ is where the first is above. So $\mathrm{TV} = \mathbb{P}(Z < \mu/2) - \mathbb{P}(Z + \mu < \mu/2) = \Phi(\mu/2) - \Phi(-\mu/2) = 2\Phi(\mu/2) - 1$ for $Z \sim$ Gauss$(0, 1)$. With $\mu = 1$ this is 0.383, the left of Figure 10.2; with $\mu = 0.01$ it is 0.004, the left of Figure 10.1.

- **An exponential and a uniform.** Expo(1) has density e^{-x} on $[0, \infty)$, and Unif$([0, 1])$ has density 1 on $[0, 1]$. The two need a common observation space, and $[0, \infty)$ contains

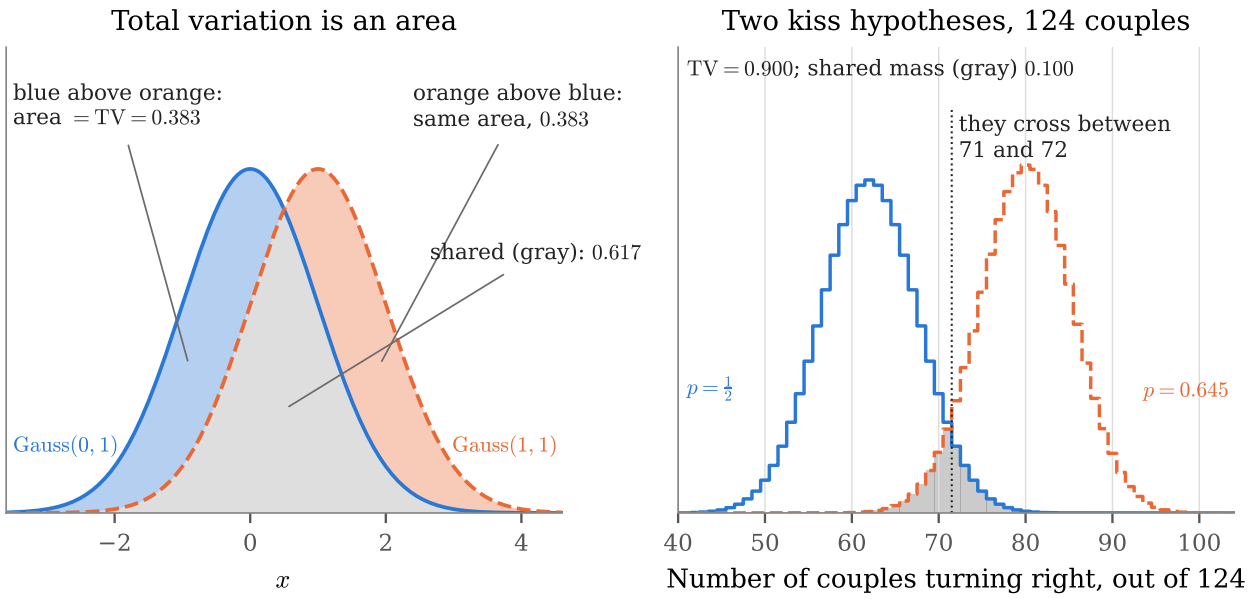


Figure 10.2: Total variation as an area. Left: Gauss$(0,1)$ and Gauss$(1,1)$: blue where the first density is higher, orange where the second is, gray where they overlap. Right: PMFs of the right-turn count among 124 couples under $p = \frac{1}{2}$ and 0.645, shared mass in gray.

both. On $[0,1]$, $1 \geq e^{-x}$; beyond 1, only the exponential has mass. So

$$\mathrm{TV} = \frac{1}{2}\left(\int_0^1 (1 - e^{-x})\,\mathrm{d}x + \int_1^\infty e^{-x}\,\mathrm{d}x\right) = \frac{1}{2}\left((1 - 1 + e^{-1}) + e^{-1}\right) = e^{-1} = 0.368.$$

- **Number of right turns.** $\mathrm{Bin}(124, \frac{1}{2})$ against $\mathrm{Bin}(124, 0.645)$: half the sum of 125 absolute differences, computed by machine, is 0.900. One couple's distributions are 0.145 apart; 124 couples' are 0.900 apart. Section 10.3 says what that growth means.

Proposition 10.3 (Total variation is a distance)**.** *For distributions $\mathbb{P}$, $\mathbb{Q}$, $\mathbb{V}$ on the same E:*

1. ***symmetry:*** $\mathrm{TV}(\mathbb{P},\mathbb{Q}) = \mathrm{TV}(\mathbb{Q},\mathbb{P})$*;*
2. ***bounds:*** $0 \leq \mathrm{TV}(\mathbb{P},\mathbb{Q}) \leq 1$*, with* $\mathrm{TV} = 1$ *exactly when there is an event that has probability* 1 *under one and* 0 *under the other;*
3. ***definiteness:*** $\mathrm{TV}(\mathbb{P},\mathbb{Q}) = 0$ *only if* $\mathbb{P} = \mathbb{Q}$*;*
4. ***triangle inequality:*** $\mathrm{TV}(\mathbb{P},\mathbb{Q}) \leq \mathrm{TV}(\mathbb{P},\mathbb{V}) + \mathrm{TV}(\mathbb{V},\mathbb{Q})$.

Proof strategy (for completeness; skip on a first reading). Each property holds for every single event, and taking the maximum over events keeps it.

Proof. (1) $|\mathbb{P}(A) - \mathbb{Q}(A)| = |\mathbb{Q}(A) - \mathbb{P}(A)|$. (2) Every $|\mathbb{P}(A) - \mathbb{Q}(A)|$ is a difference of two numbers in $[0,1]$, so it lies in $[0,1]$. It equals 1 only if one probability is 1 and the other is 0. (3) If the maximum over events is 0, every event has the same probability under both, and that is what $\mathbb{P} = \mathbb{Q}$ means. (4) For each event, $|\mathbb{P}(A) - \mathbb{Q}(A)| \leq |\mathbb{P}(A) - \mathbb{V}(A)| + |\mathbb{V}(A) - \mathbb{Q}(A)|$ by the triangle inequality for numbers (toolkit A.7), and each term on the right is at most the corresponding total variation. Take the maximum over A on the left. □

A total variation of 1 means the two distributions live on separate parts of the observation space: one draw tells you, with certainty, which one it came from. A total variation of 0 means they are the same. Everything else is in between, and Section 10.3 makes 'in between' precise.

From undergraduate statistics: effect sizes and overlap

An introductory course measured how far apart two groups are with Cohen's d, the difference in means in units of the standard deviation, and called 0.2 small, 0.5 medium and 0.8 large. For two Gaussians with a common standard deviation, d is a parameter distance, and the second item of the worked example converts it to a distance between distributions: $\mathrm{TV} = 2\Phi(d/2) - 1$. Figure 10.3 draws the three conventional sizes. The total variation distances are 0.080, 0.197 and 0.311: a 'large' effect still leaves the two distributions sharing 69% of their mass, a 'medium' one leaves them sharing 80%, and a 'small' one 92%. The shared area $1 - \mathrm{TV}$ is what some texts call the overlapping coefficient, and TV is the share of each group that has no counterpart in the other. With equal numbers in the two groups, the best rule for guessing which group one person came from is right with probability $(1 + \mathrm{TV})/2$, which at $d = 0.8$ is 0.66.

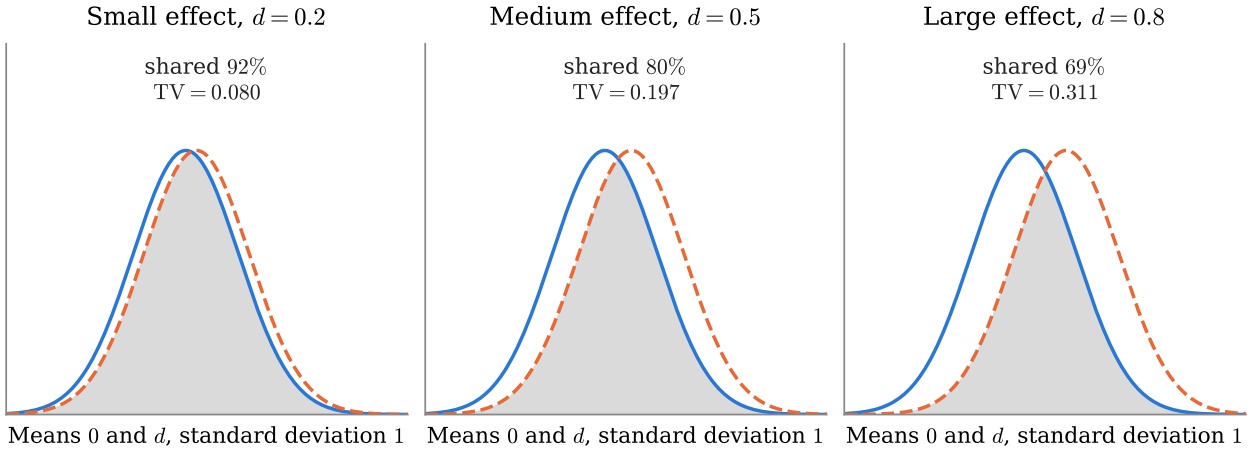


Figure 10.3: Cohen's three conventional effect sizes, $d = 0.2$, 0.5 and 0.8, as total variation distances between Gauss$(0, 1)$ (blue, solid) and Gauss$(d, 1)$ (orange, dashed). The shared mass is gray.

Pitfall: total variation does not know which outcomes are near each other

Let $X \sim \mathrm{Bern}(\frac{1}{2})$ and $a = 0.001$. The random variable $X + a$ is X shifted by a thousandth, and any reasonable notion of 'close' should call the two distributions close. Total variation calls them as far apart as two distributions can be: X lives on $\{0, 1\}$, $X + a$ on $\{0.001, 1.001\}$, the event $\{0, 1\}$ has probability 1 under the first and 0 under the second, and $\mathrm{TV} = 1$. Total variation compares probabilities point by point; it has no idea that 0 and 0.001 are neighbors. The same thing happens with the CLT.

Recall. The Central Limit Theorem (Theorem 4.8): for iid data with mean μ and variance σ^2, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow[n\to\infty]{\text{(d)}} \mathrm{Gauss}(0, 1)$.

For $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Bern}(\frac{1}{2})$, $2\sqrt{n}(\bar{X}_n - \frac{1}{2}) \xrightarrow[n\to\infty]{\text{(d)}} \mathrm{Gauss}(0, 1)$, but $2\sqrt{n}(\bar{X}_n - \frac{1}{2})$ takes only the $n+1$ values $2\sqrt{n}(k/n - \frac{1}{2})$, a set that has probability 1 under its own distribution and probability 0 under any continuous one. So the total variation distance between the standardized mean and its Gaussian limit is 1 for every n. Convergence in distribution (Definition 4.1) is not convergence

in total variation, and it can't be: a discrete distribution is never close to a continuous one in total variation, however finely its mass is spread.

10.3 What total variation says about tests

Total variation has an operational meaning that makes 'in between' concrete: it measures how well data from one distribution can be told apart from data from the other.

Suppose you have to decide, from one draw X, whether it came from $\mathbb{P}$ or from $\mathbb{Q}$. This is the test of Chapter 9 with $H_0 : X \sim \mathbb{P}$ and $H_1 : X \sim \mathbb{Q}$, both simple. A test is an event R, the rejection region, and its two errors are $\alpha = \mathbb{P}(R)$, rejecting when $\mathbb{P}$ is true, and $\beta = \mathbb{Q}(R^c)$, not rejecting when $\mathbb{Q}$ is true. How small can the two errors be together?

Proposition 10.4 (The best test and total variation)**.** *For every test R of $\mathbb{P}$ against $\mathbb{Q}$,*

$$\alpha + \beta = \mathbb{P}(R) + \mathbb{Q}(R^c) \geq 1 - \mathrm{TV}(\mathbb{P}, \mathbb{Q}),$$

with equality for the test that rejects where $\mathbb{Q}$'s density is larger, $R = \{x : q(x) > p(x)\}$.

Proof strategy (core). Rewrite $\alpha + \beta$ as 1 minus the advantage $\mathbb{Q}(R) - \mathbb{P}(R)$ the rejection region gives the alternative. That advantage can never exceed TV, and the region where q beats p achieves it.

Proof. $\mathbb{Q}(R^c) = 1 - \mathbb{Q}(R)$, so $\alpha + \beta = 1 - \big(\mathbb{Q}(R) - \mathbb{P}(R)\big)$. By Definition 10.1, $\mathbb{Q}(R) - \mathbb{P}(R) \leq \mathrm{TV}(\mathbb{P}, \mathbb{Q})$ for every event R, which gives the inequality. By Proposition 10.2 with the roles of $\mathbb{P}$ and $\mathbb{Q}$ exchanged, the event $\{q > p\}$ attains the maximum, which gives the equality. □

Intuition: total variation is distinguishability

Read Proposition 10.4 at its two ends. If $\mathrm{TV} = 0$, the two distributions are the same, and every test has $\alpha + \beta = 1$ exactly, no better than a test that ignores the data. If $\mathrm{TV} = 1$, the two live on separate parts of E, and the test 'reject if X lands in $\mathbb{Q}$'s part' makes no errors at all. In between, $1 - \mathrm{TV}$ is the irreducible error: the total of the two error rates that no test, however clever, can get below. In Figure 10.2 it is the gray overlap. A draw that lands in the overlap could have come from either distribution, and that is where every test makes its mistakes.

Worked example: how distinguishable is 'no preference' from the kiss data?

For one couple, $\mathrm{TV}(\mathrm{Bern}(\frac{1}{2}), \mathrm{Bern}(0.645)) = 0.145$, so the best test based on a single couple has $\alpha + \beta \geq 0.855$: almost no better than a coin flip.
For 124 couples, the relevant distributions are those of the count of right turns, $\mathrm{Bin}(124, \frac{1}{2})$ and $\mathrm{Bin}(124, 0.645)$; the order in which the couples turned carries no information about p, so the count is all that matters. Their total variation is 0.900, so the best test has $\alpha + \beta = 0.100$. The right of Figure 10.2 shows where the best test rejects: where the 0.645 PMF is above the $\frac{1}{2}$ PMF, which is from 72 right turns up. Its two errors are

$$\alpha = \mathbb{P}_{1/2}(\text{count} \geq 72) = 0.0438, \qquad \beta = \mathbb{P}_{0.645}(\text{count} \leq 71) = 0.0567,$$

and they add up to 0.100. In natural frequencies: of 1,000 studies of 124 couples in a world with no preference, the best test cries 'preference' in about 44; of 1,000 studies in a world where

$p = 0.645$, it misses the preference in about 57. No test of 124 couples does better on both counts at once.
That threshold, 72, is the one Section 9.6 found for the one-sided exact test at level 5%. Both tests reject for large counts, which is no accident; that they pick the same threshold is a coincidence of these particular numbers: the level-5% test fixes α near 0.05 and accepts whatever β results, while the best test of Proposition 10.4 minimizes the sum and accepts whatever split results.

Does this make sense? More data, more distinguishable

Going from one couple to 124 took the total variation from 0.145 to 0.900. That is the direction it has to go: more data can only make two hypotheses easier to tell apart, since you could always ignore the extra data. But it doesn't go up by a factor of 124, and it couldn't: total variation is capped at 1. Total variation doesn't add over independent observations, and it has no simple formula for n observations in terms of one. The divergence of Section 10.5 does, which is one of several reasons it is the one that statistics uses.

10.4 Why total variation can't be estimated by an average

Here is the plan the chapter started with. Total variation measures how close $\mathbb{P}_\theta$ is to the truth $\mathbb{P}_{\theta^*}$. So estimate the function

$$\theta \mapsto \mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta^*})$$

from the data, and take as $\hat{\theta}_n$ the θ that makes the estimate smallest. If the estimate is good, its minimizer should be near the true minimizer, which is θ^* (the only θ at distance 0, if the model is identifiable).

The obstacle is the word 'estimate'. The one general-purpose estimation tool in this book is the Law of Large Numbers.

Recall. The Law of Large Numbers (Theorem 3.5): if $X_1, \ldots, X_n \overset{iid}{\sim} \mathbb{P}_{\theta^*}$ and h is a function with $\mathbb{V}\mathrm{ar}_{\theta^*}(h(X)) < \infty$, then $\frac{1}{n}\sum_{i=1}^n h(X_i) \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}_{\theta^*}[h(X)]$. An expectation under the truth is estimated by an average over the data.

So the question is whether $\mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta^*})$ is an expectation under $\mathbb{P}_{\theta^*}$ of something you can compute. It can be written as one. Factor $p_{\theta^*}(x)$ out of the integrand of Proposition 10.2:

$$\mathrm{TV}(\mathbb{P}_\theta, \mathbb{P}_{\theta^*}) = \frac{1}{2}\int \left|\frac{p_\theta(x)}{p_{\theta^*}(x)} - 1\right| p_{\theta^*}(x)\,\mathrm{d}x = \mathop{\mathbb{E}}_{\theta^*}\left[\frac{1}{2}\left|\frac{p_\theta(X)}{p_{\theta^*}(X)} - 1\right|\right].$$

But the function inside the expectation contains p_{θ^*}, the density you don't know, and it sits inside the absolute value tangled up with p_θ. To average it over the data you would need to evaluate it at each X_i, and to evaluate it you would need θ^*, so no average of the data estimates it. For an observation space with a few values there is a way around this: plug in the observed frequencies. For the kiss data, $\mathrm{TV}(\mathrm{Bern}(q), \mathrm{Bern}(p^*)) = |q - p^*|$ is estimated by $|q - \hat{p}|$, whose minimizer is $\hat{p}$. For continuous data the plug-in fails outright: the observed frequencies form a discrete distribution, and by the pitfall of Section 10.2 a discrete distribution sits at total variation 1 from every continuous candidate.

What would rescue the plan is a measure of distance in which the unknown p_{θ^*} separates from p_θ, so that the part involving θ^* is a constant that doesn't depend on θ. A constant can be left unestimated: it doesn't move the minimizer. The logarithm turns a ratio into a difference, $\log(p_{\theta^*}/p_\theta) = \log p_{\theta^*} - \log p_\theta$, and that is how the next divergence is built.

10.5 The Kullback–Leibler divergence

Here is a second way to compare two distributions, built from a question about evidence. Suppose the truth is $\mathbb{P}$, with PMF or density p, and you are weighing it against an alternative $\mathbb{Q}$, with q. One observation x arrives. How much does it favor $\mathbb{P}$ over $\mathbb{Q}$? A natural measure is the ratio $p(x)/q(x)$, how many times more likely x is under $\mathbb{P}$ than under $\mathbb{Q}$, and its logarithm $\log\big(p(x)/q(x)\big)$ turns the product of the ratios for independent observations into a sum. This log-ratio is positive when x favors $\mathbb{P}$, negative when it favors $\mathbb{Q}$ and 0 when it favors neither. A single observation can point the wrong way. On average, over observations drawn from the truth $\mathbb{P}$, it should point the right way. That average is the divergence.

Definition 10.5 (Kullback–Leibler divergence)**.** Let $\mathbb{P}$ and $\mathbb{Q}$ be distributions on E with PMFs or densities p and q. The **Kullback–Leibler (KL) divergence** of $\mathbb{Q}$ from $\mathbb{P}$ is

$$\mathrm{KL}(\mathbb{P},\mathbb{Q}) = \mathop{\mathbb{E}}_{\mathbb{P}}\left[\log\frac{p(X)}{q(X)}\right] = \begin{cases} \displaystyle\sum_{x\in E} p(x)\log\frac{p(x)}{q(x)} & \text{(discrete)},\\[2ex] \displaystyle\int_E p(x)\log\frac{p(x)}{q(x)}\,\mathrm{d}x & \text{(continuous)}, \end{cases}$$

where the sum or integral runs over the points with $p(x) > 0$, and $\mathrm{KL}(\mathbb{P},\mathbb{Q}) = +\infty$ if $q(x) = 0$ at some point where $p(x) > 0$. It is also written $\mathrm{KL}(\mathbb{P}\,\|\,\mathbb{Q})$, and called the **relative entropy**.

In words. Draw X from the truth $\mathbb{P}$ and compute how much more likely X is under $\mathbb{P}$ than under the alternative $\mathbb{Q}$, on the log scale. Average that over many draws. The result is the average evidence one observation supplies for the truth against $\mathbb{Q}$. The divergence is large when $\mathbb{Q}$ makes the observations that actually happen look improbable, and it is infinite when $\mathbb{Q}$ calls something impossible that $\mathbb{P}$ produces: one such observation refutes $\mathbb{Q}$ outright. The logarithm is natural (base e), and the unit is the *nat*; with base-2 logarithms the unit is the *bit*, and 1 nat is 1.44 bits.

For the kiss data. Take the truth to be the fitted $\mathrm{Bern}(\hat{p})$, $\hat{p} = 80/124 = 0.6452$, and the alternative $\mathrm{Bern}(\frac{1}{2})$. A couple turning right favors $\hat{p}$ by $\log(0.6452/0.5) = 0.2549$; a couple turning left favors $\frac{1}{2}$ by $\log(0.5/0.3548) = 0.3430$. Right turns happen 64.5% of the time, so on average

$$\mathrm{KL}(\mathrm{Bern}(\hat{p}),\mathrm{Bern}(\tfrac{1}{2})) = 0.6452\times 0.2549 - 0.3548\times 0.3430 = 0.0428 \text{ nats}.$$

Each couple, on average, supplies 0.043 nats of evidence that $p = 0.645$ rather than $\frac{1}{2}$. Left turns are individually stronger evidence the other way; right turns are more frequent, and frequency wins.

Two conventions in the definition need a word. Points where $p(x) = 0$ never occur when $X \sim \mathbb{P}$, so they contribute nothing to an expectation under $\mathbb{P}$; this is also the convention $0\log(0/q) = 0$, which agrees with the limit $t\log t \to 0$ as $t \to 0^+$. Points where $q(x) = 0$ but $p(x) > 0$ are the opposite: they occur, and when they do, the log-ratio is $+\infty$. An expectation that puts positive probability on $+\infty$ is $+\infty$.

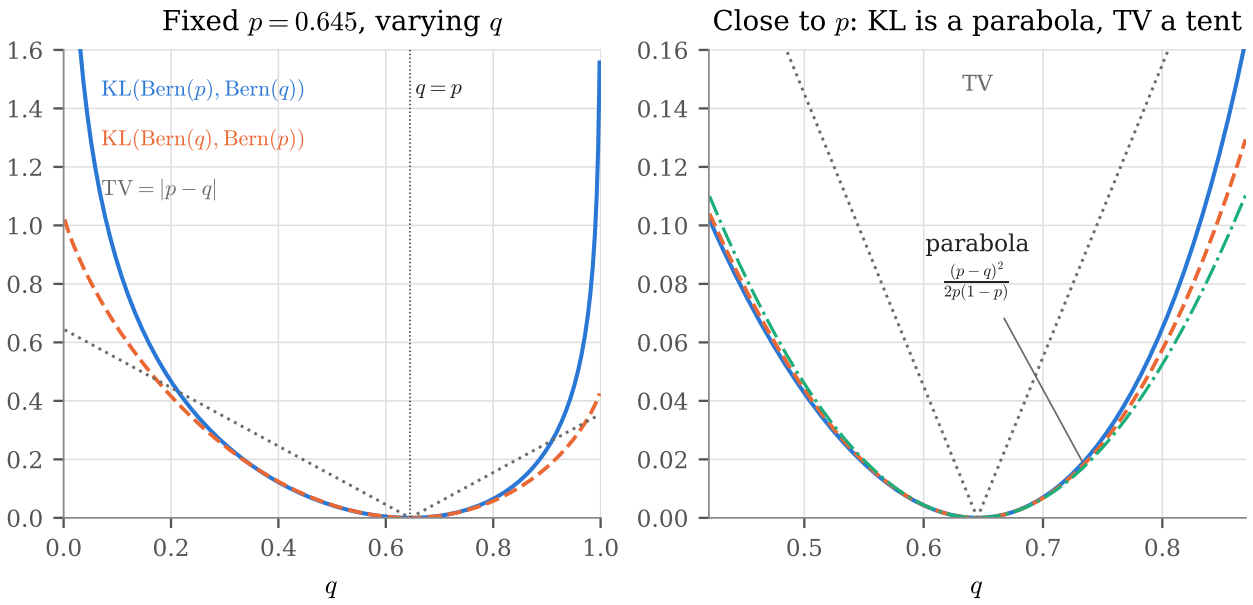


Figure 10.4: KL divergence in both orders and total variation between Bern(p) and Bern(q) for $p = 0.645$, as functions of q; KL(Bern(p), Bern(q)) is blue. Left: the whole range. Right: near $q = p$, with the parabola $(p - q)^2/(2p(1 - p))$.

Worked example: KL divergences you can compute by hand

- **Two Bernoullis.** With two outcomes the sum has two terms:

$$\mathrm{KL}(\mathrm{Bern}(p), \mathrm{Bern}(q)) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$

  Figure 10.4 plots it with $p = 0.645$ fixed, in both orders, beside the total variation $|p - q|$. On the left, over the whole range of q, total variation is a tent with straight sides; the two KL divergences are smooth at $q = p$ and differ from each other away from it, and KL(Bern(p), Bern(q)) goes to infinity as q approaches 0 or 1, where Bern(q) calls one of the two outcomes impossible. At $q = \frac{1}{2}$ the total variation is 0.145 and the two KL divergences are 0.0428 and 0.0440. Two values to keep: $\mathrm{KL}(\mathrm{Bern}(\frac{1}{2}), \mathrm{Bern}(\frac{1}{10})) = 0.511$ and $\mathrm{KL}(\mathrm{Bern}(\frac{1}{10}), \mathrm{Bern}(\frac{1}{2})) = 0.368$.

- **Two Gaussians with the same variance.** For $\mathbb{P} = \mathrm{Gauss}(a, \sigma^2)$ and $\mathbb{Q} = \mathrm{Gauss}(b, \sigma^2)$ the normalizing constants cancel in the ratio, and

$$\log \frac{p(x)}{q(x)} = \frac{(x - b)^2 - (x - a)^2}{2\sigma^2} = \frac{(a - b)(2x - a - b)}{2\sigma^2}.$$

  This is linear in x, so its expectation under $X \sim \mathrm{Gauss}(a, \sigma^2)$ is the same expression at $x = a$:

$$\mathrm{KL}(\mathrm{Gauss}(a, \sigma^2), \mathrm{Gauss}(b, \sigma^2)) = \frac{(a - b)(2a - a - b)}{2\sigma^2} = \frac{(a - b)^2}{2\sigma^2}.$$

  Half the squared distance between the means in standard-deviation units. For the left pair of Figure 10.1, means 0 and 0.01 with $\sigma = 1$ give 0.00005. The right pair, standard deviations 0.01 and 0.02, does not have a common variance and needs the general formula of Exercise 10.6: $\mathrm{KL}(\mathrm{Gauss}(0, 0.01^2), \mathrm{Gauss}(0, 0.02^2)) = 0.318$, and 0.807 the other way.

- **Two exponentials.** For $\text{Expo}(\lambda)$ against $\text{Expo}(\mu)$, $\log(p(x)/q(x)) = \log(\lambda/\mu) - (\lambda-\mu)x$, again linear in x, with $\mathbb{E}[X] = 1/\lambda$:

$$\text{KL}(\text{Expo}(\lambda), \text{Expo}(\mu)) = \log\frac{\lambda}{\mu} - \frac{\lambda-\mu}{\lambda} = \log\frac{\lambda}{\mu} + \frac{\mu}{\lambda} - 1.$$

  $\text{KL}(\text{Expo}(1), \text{Expo}(2)) = 0.307$ and $\text{KL}(\text{Expo}(2), \text{Expo}(1)) = 0.193$.

Does this make sense? The Gaussian formula

Check $(a-b)^2/(2\sigma^2)$ three ways. At $a = b$ it is 0, which a divergence between a distribution and itself must be. It is symmetric in a and b, which is right here (swapping the two Gaussians is the same as reflecting the line, and reflecting changes nothing), though KL is not symmetric in general. And it is unchanged if a, b and σ are all converted to a new unit, say minutes instead of hours: a divergence between distributions can't depend on the units of the measurement, and $(a-b)/\sigma$ doesn't.

Proposition 10.6 (Properties of the KL divergence)**.** *For distributions $\mathbb{P}$ and $\mathbb{Q}$ on E:*

1. ***nonnegativity:*** $\text{KL}(\mathbb{P}, \mathbb{Q}) \geq 0$*;*
2. ***definiteness:*** $\text{KL}(\mathbb{P}, \mathbb{Q}) = 0$ *only if* $\mathbb{P} = \mathbb{Q}$*;*
3. ***no symmetry:*** *in general* $\text{KL}(\mathbb{P}, \mathbb{Q}) \neq \text{KL}(\mathbb{Q}, \mathbb{P})$*;*
4. ***no triangle inequality:*** *in general* $\text{KL}(\mathbb{P}, \mathbb{Q}) \not\leq \text{KL}(\mathbb{P}, \mathbb{V}) + \text{KL}(\mathbb{V}, \mathbb{Q})$.

Properties 1 and 2 make KL behave like a distance in the way that matters most: it is zero exactly when the distributions agree. Properties 3 and 4 mean it is not a distance, which is why it is called a divergence.

Proof strategy (core). The log-ratio $\log(p/q)$ is $-\log$ of the ratio q/p. The curve $-\log y$ lies above its tangent line $1-y$ at $y = 1$, touching it only there. The average of the ratio $q(X)/p(X)$ under $\mathbb{P}$ is at most 1, since it is the total mass of q on the points where p is positive, so the average of the tangent line is at least 0.

Proof. (1) If $\text{KL}(\mathbb{P}, \mathbb{Q}) = +\infty$ there is nothing to prove, so assume $q > 0$ wherever $p > 0$. Let $X \sim \mathbb{P}$ and $Y = q(X)/p(X)$, a positive random variable. Then $\log(p(X)/q(X)) = -\log Y$. The function $g(y) = -\log y$ is convex ($g''(y) = 1/y^2 > 0$), so it lies above its tangent line at $y = 1$, which is $1-y$, and touches it only at $y = 1$ (toolkit A.8):

$$-\log y \geq 1 - y \quad \text{for all } y > 0, \text{ with equality only at } y = 1.$$

Taking expectations,

$$\text{KL}(\mathbb{P}, \mathbb{Q}) = \mathbb{E}[-\log Y] \geq 1 - \mathbb{E}[Y].$$

In the discrete case, $\mathbb{E}[Y] = \sum_{x:p(x)>0} p(x) \cdot q(x)/p(x) = \sum_{x:p(x)>0} q(x) \leq 1$; in the continuous case the same with an integral. So $1 - \mathbb{E}[Y] \geq 0$. (This is the same idea as Jensen's inequality, Theorem 7.8: the tangent-line argument is its proof.)

(2) If $\text{KL}(\mathbb{P}, \mathbb{Q}) = 0$, both inequalities in (1) are equalities. The first, $\mathbb{E}[-\log Y] = 1 - \mathbb{E}[Y]$, is an average of the nonnegative gap $-\log Y - (1-Y)$ that comes out 0, so the gap is 0 with probability 1, which forces $Y = 1$: $q(x) = p(x)$ at every point where $p(x) > 0$. Then q already has total mass 1 on those points, so it has none anywhere else, and $q = p$.

(3) $\text{KL}(\text{Bern}(\frac{1}{2}), \text{Bern}(\frac{1}{10})) = 0.511$ and $\text{KL}(\text{Bern}(\frac{1}{10}), \text{Bern}(\frac{1}{2})) = 0.368$.

(4) With the Gaussian formula,

$$\mathrm{KL}(\mathrm{Gauss}(0,1),\mathrm{Gauss}(2,1)) = 2, \quad \text{while}$$
$$\mathrm{KL}(\mathrm{Gauss}(0,1),\mathrm{Gauss}(1,1)) + \mathrm{KL}(\mathrm{Gauss}(1,1),\mathrm{Gauss}(2,1)) = 0.5 + 0.5 = 1.$$

The direct route is twice as long as the detour: KL grows like a squared distance, and squares of distances don't satisfy the triangle inequality. □

Intuition: why the order matters: who is surprised by what

Think of $\log(1/q(x))$ as how surprised you are to see x if you believe $\mathbb{Q}$. Then $\mathrm{KL}(\mathbb{P},\mathbb{Q})$ is how much more surprised, on average, a believer in $\mathbb{Q}$ is than a believer in $\mathbb{P}$, when the data really come from $\mathbb{P}$. Take $\mathbb{P} = \mathrm{Bern}(\frac{1}{2})$, a fair coin, and $\mathbb{Q} = \mathrm{Bern}(\frac{1}{10})$, a coin that rarely shows heads. With the fair coin as the truth, each head is 5 times likelier under the truth than under $\mathbb{Q}$, worth $\log 5 = 1.609$ in favor of the truth, and heads come half the time; each tail favors $\mathbb{Q}$ by $\log(0.9/0.5) = 0.588$. On average, $0.5 \times 1.609 - 0.5 \times 0.588 = 0.511$. Reverse the roles, so the $\frac{1}{10}$ coin is the truth. Now each tail favors the truth by 0.588 and comes 90% of the time, while the rare heads favor the fair-coin believer by 1.609: on average $0.9 \times 0.588 - 0.1 \times 1.609 = 0.368$. The decisive outcome, heads, is common in the first case and rare in the second. The divergence is largest when the alternative calls rare what the truth makes common. At the extreme, $\mathrm{KL}(\mathrm{Bern}(\frac{1}{2}),\mathrm{Bern}(0)) = +\infty$, since the first head refutes 'heads never happen'; but $\mathrm{KL}(\mathrm{Bern}(0),\mathrm{Bern}(\frac{1}{2})) = \log 2 = 0.693$, finite, since a stream of tails is perfectly possible for a fair coin, just less likely each time.

10.5.1 Independent observations add

The single most useful property of the KL divergence is the one total variation lacks.

Proposition 10.7 (Additivity over independent observations)**.** *If $\mathbb{P}^{\otimes n}$ and $\mathbb{Q}^{\otimes n}$ denote the distributions of n independent draws from $\mathbb{P}$ and from $\mathbb{Q}$, then*

$$\mathrm{KL}(\mathbb{P}^{\otimes n},\mathbb{Q}^{\otimes n}) = n\,\mathrm{KL}(\mathbb{P},\mathbb{Q}).$$

Proof strategy (for completeness; skip on a first reading). The log of a product is a sum, and an expectation of a sum is a sum of expectations.

Proof. The joint PMF or density of n independent draws is the product, so the log-ratio of the joint distributions is a sum:

$$\log \frac{\prod_{i=1}^n p(x_i)}{\prod_{i=1}^n q(x_i)} = \sum_{i=1}^n \log \frac{p(x_i)}{q(x_i)}.$$

Under $\mathbb{P}^{\otimes n}$ each $X_i \sim \mathbb{P}$, and the expectation of a sum is the sum of the expectations, each equal to $\mathrm{KL}(\mathbb{P},\mathbb{Q})$. □

Evidence from independent observations accumulates, and on the log scale it accumulates by addition. For the kiss data, 124 couples supply on average $124 \times 0.0428 = 5.30$ nats of evidence for $p = 0.645$ over $p = \frac{1}{2}$, a likelihood ratio of about $e^{5.30} \approx 200$. The count of right turns is enough to compute that evidence, so the same additivity gives $\mathrm{KL}(\mathrm{Bin}(n,p),\mathrm{Bin}(n,q)) = n\,\mathrm{KL}(\mathrm{Bern}(p),\mathrm{Bern}(q))$, which Exercise 10.5 asks you to check directly. Compare total variation, which went from 0.145 for one couple to 0.900 for 124, by no rule you could state, and could never pass 1.

10.5.2 Near the truth, KL is a parabola

Figure 10.4 shows something about the shape of KL that the formulas don't make obvious. Total variation between Bernoullis is $|p-q|$, a tent with a corner at $q=p$. KL is smooth there: flat at the bottom, curving up on both sides, the shape of a parabola. It must be flat at $q=p$, since KL has a minimum of 0 there and is differentiable. A second-order Taylor expansion (toolkit A.9) in q around p gives the curvature. With $f(q) = p\log(p/q) + (1-p)\log((1-p)/(1-q))$, $f(p)=0$; $f'(q) = -p/q + (1-p)/(1-q)$, so $f'(p)=0$; and $f''(q) = p/q^2 + (1-p)/(1-q)^2$, so $f''(p) = 1/p + 1/(1-p) = 1/(p(1-p))$. Hence

$$\mathrm{KL}(\mathrm{Bern}(p), \mathrm{Bern}(q)) \approx \frac{1}{2} f''(p)(q-p)^2 = \frac{(p-q)^2}{2p(1-p)} \quad \text{for } q \text{ near } p.$$

The right of Figure 10.4 zooms in on q near p, and both KL curves hug this parabola there. For the kiss data, at $q=\frac{1}{2}$, the parabola gives 0.0460 against the exact 0.0428.

Look at what the curvature is: $1/(p(1-p))$, one over the variance of a single Bernoulli observation. The Gaussian formula has the same shape, $(a-b)^2/(2\sigma^2)$, with curvature $1/\sigma^2$. In both, KL near the truth is half the squared parameter distance measured in units of the noise in one observation. Where an observation is noisy, nearby parameters are hard to tell apart and KL rises slowly; where it is precise, KL rises fast. Chapter 12 gives the curvature its name, the Fisher information, and shows that it sets the variance of the best estimators.

From undergraduate statistics: the G-test and Pearson's chi-squared

Some introductory courses teach a test of a proportion, or of a table of counts, called the G-test or likelihood-ratio chi-squared test: $G^2 = 2\sum O\log(O/E)$, summed over cells, with O the observed and E the expected counts. That statistic is a KL divergence. For the kiss data, the cells are right and left, $O=(80,44)$ and, under $p=\frac{1}{2}$, $E=(62,62)$:

$$G^2 = 2\left(80\log\tfrac{80}{62} + 44\log\tfrac{44}{62}\right) = 2\times 124\times \mathrm{KL}\big(\mathrm{Bern}(\hat{p}), \mathrm{Bern}(\tfrac{1}{2})\big) = 10.60,$$

twice the 5.30 nats of evidence above. Pearson's chi-squared statistic for the same table is $\sum(O-E)^2/E = 10.45$, which is also $z^2 = 3.233^2$ from the test of Section 9.6. Pearson's statistic is a parabola approximation to G^2, with the variance taken at the null value. The expansion above was in q around p; expanding instead in p around q gives the same parabola with the variance at q, $(p-q)^2/(2q(1-q))$. With $q=\frac{1}{2}$ that is $2(\hat{p}-\frac{1}{2})^2 = 0.0421$, and $248\times 0.0421 = 10.45$, Pearson's statistic. With the variance at $\hat{p}$ instead, $248\times 0.0460 = 11.41$, the square of the Wald statistic $\sqrt{124}(\hat{p}-\frac{1}{2})/\sqrt{\hat{p}(1-\hat{p})} = 3.38$ of Chapter 15. Both are compared with a chi-squared distribution with one degree of freedom, whose 95% quantile is $3.84 = 1.96^2$, and both reject. Chapter 16 develops the chi-squared test in general.

10.5.3 KL controls total variation

KL is not a distance, but small KL forces small total variation.

Theorem 10.8 (Pinsker's inequality)**.** *For any distributions* $\mathbb{P}$ *and* $\mathbb{Q}$*,*

$$\mathrm{TV}(\mathbb{P},\mathbb{Q}) \le \sqrt{\tfrac{1}{2}\mathrm{KL}(\mathbb{P},\mathbb{Q})}.$$

Nothing later in the book depends on Pinsker's inequality, so its proof is in Appendix C.1 of Appendix C. It is a page of calculus that reuses the tangent-line bound from the proof of Proposition 10.6. For the kiss data, $\sqrt{0.0428/2} = 0.146$, against TV $= 0.145$: nearly tight. The inequality is what lets you work with KL, which is convenient, and still conclude something about total variation, which is what you wanted in Section 10.1: if the fitted distribution is close to the truth in KL, then every probability it reports is close to the true probability. The converse fails: total variation can be small while KL is infinite. $\mathbb{P} = \text{Pois}(1/n)$ and $\mathbb{Q}$, the distribution that puts all its mass at 0, have TV $= 1 - e^{-1/n}$, which goes to 0, while $\text{KL}(\mathbb{P}, \mathbb{Q}) = +\infty$ for every n, since $\mathbb{P}$ gives positive probability to 1 and $\mathbb{Q}$ calls it impossible (Exercise 10.8).

10.6 Estimating the KL divergence: the road to maximum likelihood

Return to the plan of Section 10.4 with KL in place of total variation. The quantity to make small is $\theta \mapsto \text{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$, the divergence of the candidate from the truth. The truth goes first: the average is taken under the distribution the data come from, which is the only distribution you can average under. Split the logarithm of the ratio into a difference:

$$\text{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = \mathop{\mathbb{E}}_{\theta^*}\big[\log p_{\theta^*}(X)\big] - \mathop{\mathbb{E}}_{\theta^*}\big[\log p_\theta(X)\big].$$

The first term involves the unknown θ^*, but it does not involve θ: as a function of θ it is a constant. The second term is an expectation under the truth of a function, $\log p_\theta$, that you can compute for any θ you like.

Recall. The Law of Large Numbers (Theorem 3.5): an average of $h(X_1), \dots, h(X_n)$ over iid data converges in probability to $\mathbb{E}_{\theta^*}[h(X)]$, provided $h(X)$ has finite variance.

Apply it to $h = \log p_\theta$, one θ at a time:

$$\widehat{\text{KL}}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = \text{constant} - \frac{1}{n}\sum_{i=1}^{n} \log p_\theta(X_i).$$

You don't know the constant, and you don't need to. The θ that minimizes $\widehat{\text{KL}}$ is the same whatever the constant is, and it is the θ that maximizes $\frac{1}{n}\sum_i \log p_\theta(X_i)$. Since log turns a product into a sum and is increasing (toolkit A.10),

$$\operatorname*{argmin}_\theta \widehat{\text{KL}}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = \operatorname*{argmax}_\theta \sum_{i=1}^{n} \log p_\theta(X_i) = \operatorname*{argmax}_\theta \prod_{i=1}^{n} p_\theta(X_i).$$

The product $\prod_i p_\theta(X_i)$ is the probability, or density, of the data you observed, computed as if θ were the truth. The θ that makes your data most probable is the **maximum likelihood estimator**, and minimizing the estimated KL divergence is the same thing as finding it. Chapter 11 takes this as its starting point.

Worked example: the kiss data

For Bernoulli data, $\log p_q(x) = x \log q + (1 - x)\log(1 - q)$, and the average over 124 couples of

whom 80 turned right is

$$\frac{1}{124}\sum_{i=1}^{124}\log p_q(R_i) = \hat{p}\log q + (1-\hat{p})\log(1-q), \qquad \hat{p} = \tfrac{80}{124} = 0.645.$$

Its derivative in q is $\hat{p}/q - (1-\hat{p})/(1-q)$, which is zero at $q = \hat{p}$, positive before and negative after. The estimated KL is smallest at the sample proportion, $\hat{p} = 0.645$, the estimator you would have used anyway.

Figure 10.5 shows the estimate at work. To draw it, the unknown constant has to be filled in, so the figure does what only a simulation can: it fixes the truth at $p^* = 0.645$ and draws samples from it. The black curve is the true $\mathrm{KL}(\mathrm{Bern}(p^*), \mathrm{Bern}(q))$. Each colored curve is the estimate from one simulated sample, $\frac{1}{n}\sum_i \log(p^*(R_i)/p_q(R_i))$, the same curve as the true one with the expectation replaced by an average; there are five samples at each of $n = 20$, 124 and 1,000. Each estimated curve has its minimum, marked by a dot, at its own sample proportion, which is the maximum likelihood estimate. Every estimated curve passes through 0 at $q = p^*$ and dips below zero near its minimum. As n grows the estimated curves close in on the true one, and their minimizers close in on p^*, which is the true curve's minimizer. That is the argument Chapter 12 turns into a proof that the maximum likelihood estimator is consistent.

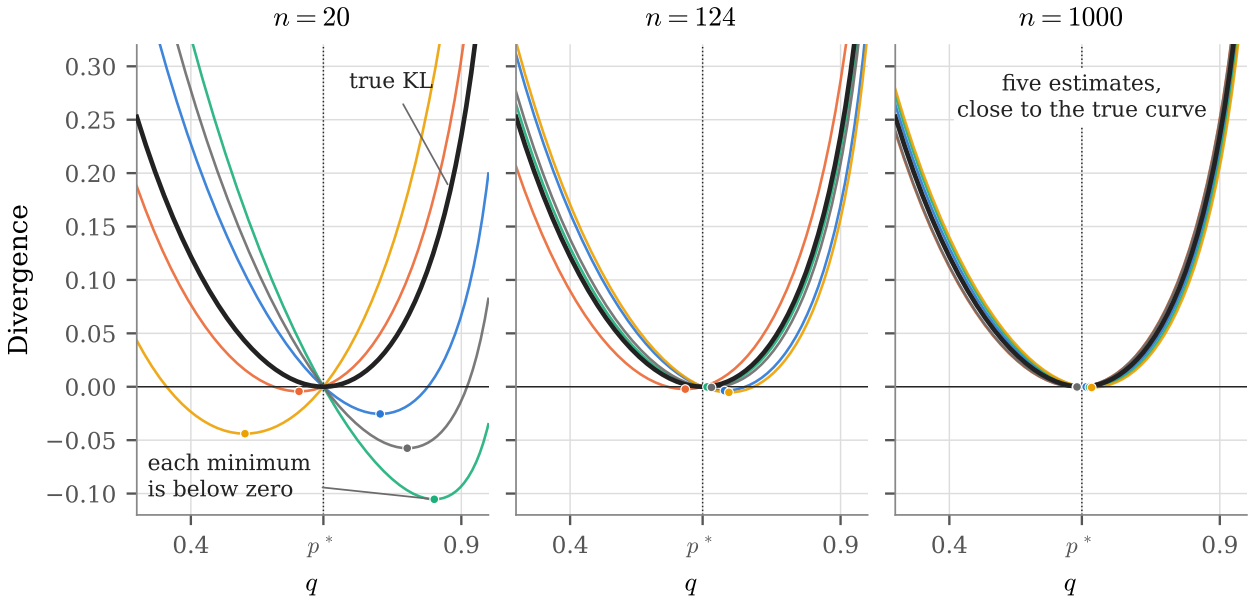


Figure 10.5: The true $\mathrm{KL}(\mathrm{Bern}(p^*), \mathrm{Bern}(q))$ for $p^* = 0.645$ (black) and five estimates of it (colors), each from one simulated sample of n couples, for $n = 20$, 124 and 1,000. The dots mark the minimizer of each estimate.

Does this make sense? An estimated divergence below zero

Every estimated curve in Figure 10.5 dips below zero, and a KL divergence is never negative. Is the estimate broken? No. At $q = p^*$ the estimate is exactly 0 (each term is $\log 1$), and at its own minimizer $\hat{p}$ it must be lower still. The estimate is an average, and an average of quantities whose expectation is positive can come out negative in a finite sample; the LLN promises only

that it ends up near the expectation. What the dip measures is how much better the sample proportion fits this sample than the truth does, which is always at least a little, since $\hat{p}$ was chosen by looking at the sample. For $n = 20$ the dips are visible; for $n = 1{,}000$ they are invisible at this scale.

Pitfall: the constant you dropped was the distance

Minimizing $\widehat{\mathrm{KL}}$ finds the best θ, but the minimum value of $-\frac{1}{n}\sum_i \log p_\theta(X_i)$ is not an estimate of the distance from the truth. The dropped constant $\mathbb{E}_{\theta^*}[\log p_{\theta^*}(X)]$ is unknown, so the number you computed is the divergence plus an unknown shift. The maximized log-likelihood of the kiss data, $80 \log 0.645 + 44 \log 0.355 = -80.6$, says nothing on its own about how well the Bernoulli model fits. Only differences of log-likelihoods, between two values of θ or two models fitted to the same data, cancel the constant and mean something. That is why the likelihood ratio tests of Chapter 15 compare two maximized log-likelihoods and never interpret one alone.

10.7 When the model is wrong: the closest member of the family

Everything so far assumed the truth is $\mathbb{P}_{\theta^*}$ for some θ^* in the model. Chapter 2 warned that the assumption can fail, and promised to say what estimators estimate when it does. Here is the answer.

Let the truth be any distribution $\mathbb{P}$, in the family or not.

Recall. The Law of Large Numbers (Theorem 3.5): for iid data from $\mathbb{P}$, $\frac{1}{n}\sum_i h(X_i) \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}_{\mathbb{P}}[h(X)]$ when $h(X)$ has finite variance. Nothing in it requires $\mathbb{P}$ to belong to a model.

The estimated divergence is built the same way: $\mathrm{KL}(\mathbb{P}, \mathbb{P}_\theta) = \mathbb{E}_{\mathbb{P}}[\log p(X)] - \mathbb{E}_{\mathbb{P}}[\log p_\theta(X)]$, the first term is still a constant, and the LLN still estimates the second by $\frac{1}{n}\sum_i \log p_\theta(X_i)$, since the data still come from $\mathbb{P}$. So maximum likelihood still minimizes an estimate of $\theta \mapsto \mathrm{KL}(\mathbb{P}, \mathbb{P}_\theta)$. What has changed is the minimum: no θ makes the divergence 0, so the minimizer is the member of the family that is closest to the truth, not the truth.

Definition 10.9 (KL projection)**.** Let $\mathbb{P}$ be a distribution and $(\mathbb{P}_\theta)_{\theta\in\Theta}$ a family on the same observation space. A **KL projection** of $\mathbb{P}$ onto the family is a parameter

$$\theta^\circ = \operatorname*{argmin}_{\theta\in\Theta} \mathrm{KL}(\mathbb{P}, \mathbb{P}_\theta),$$

when the minimum exists and is attained at a single point. If $\mathbb{P} = \mathbb{P}_{\theta^*}$ is in the family and the model is identifiable, $\theta^\circ = \theta^*$.

In words. Look over every distribution the model allows and pick the one the truth diverges from least. If the truth is in the model, that is the truth itself, at divergence 0. If not, it is the model's best impersonation of the truth. Maximum likelihood, applied to data from a truth outside the model, estimates θ°: under conditions that Chapter 12 states, $\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta^\circ$. So a wrong model fitted by maximum likelihood is not meaningless. It converges to something definite, the closest member of the family, and the question is only whether the closest member is close enough for your purpose.

For the kiss data. The Bernoulli family contains every distribution on $\{0, 1\}$, so for the kiss

experiment the model cannot be wrong in this sense: whatever the truth is, one couple's turn is some Bern(p). What can fail is the iid assumption of Chapter 1: couples influencing each other, or one observer's couples differing from another's. A Bernoulli model with dependent data is misspecified in a way that no projection onto a family of iid models repairs.

Proposition 10.10 (KL projection onto the Gaussian and Poisson families)**.** *Let $\mathbb{P}$ be the distribution of a random variable X.*

1. *If X has a density p with $\mathbb{E}|\log p(X)| < \infty$ and finite variance, the KL projection onto $(Gauss(\mu, \sigma^2))_{\mu\in\mathbb{R},\sigma^2>0}$ is $\mu^\circ = \mathbb{E}[X]$, $(\sigma^2)^\circ = \mathbb{V}ar(X)$: the Gaussian with the same mean and variance.*
2. *If X takes values in $\{0, 1, 2, \ldots\}$, with $\mathbb{E}[X] > 0$, $\mathbb{E}[\log X!] < \infty$ and $\mathbb{E}|\log p(X)| < \infty$ for its PMF p, the KL projection onto $(Pois(\lambda))_{\lambda>0}$ is $\lambda^\circ = \mathbb{E}[X]$: the Poisson with the same mean.*

The conditions on $\log p(X)$ keep the constant $\mathbb{E}[\log p(X)]$ finite. Without them the divergence can be $+\infty$ for every member of the family (a discrete X against any Gaussian, say), and what maximum likelihood targets is then the minimizer of $-\mathbb{E}[\log p_\theta(X)]$ alone, which the same calculation finds.

Proof strategy (for completeness; skip on a first reading). Only the term $-\mathbb{E}[\log p_\theta(X)]$ depends on the parameter. For both families it involves X only through $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ (plus a term free of the parameter), so it is an explicit function of the parameters. Minimize it by calculus.

Proof. (1) For the Gaussian density, $-\log p_{\mu,\sigma^2}(x) = \frac{1}{2}\log(2\pi\sigma^2) + (x-\mu)^2/(2\sigma^2)$, and

$$-\mathbb{E}[\log p_{\mu,\sigma^2}(X)] = \tfrac{1}{2}\log(2\pi\sigma^2) + \frac{\mathbb{E}[(X-\mu)^2]}{2\sigma^2} = \tfrac{1}{2}\log(2\pi\sigma^2) + \frac{\mathbb{V}\mathrm{ar}(X) + (\mathbb{E}[X]-\mu)^2}{2\sigma^2},$$

using the bias-variance split $\mathbb{E}[(X-\mu)^2] = \mathbb{V}\mathrm{ar}(X) + (\mathbb{E}[X]-\mu)^2$ from Chapter 7. For any σ^2, the μ that minimizes this is $\mu = \mathbb{E}[X]$, which kills the squared term. With that μ, the derivative in σ^2 of $\frac{1}{2}\log(2\pi\sigma^2) + \mathbb{V}\mathrm{ar}(X)/(2\sigma^2)$ is $\frac{1}{2\sigma^2} - \frac{\mathbb{V}\mathrm{ar}(X)}{2\sigma^4}$, zero at $\sigma^2 = \mathbb{V}\mathrm{ar}(X)$, negative before and positive after, so this is the minimum.

(2) For the Poisson PMF, $-\log p_\lambda(x) = \lambda - x\log\lambda + \log(x!)$, and $-\mathbb{E}[\log p_\lambda(X)] = \lambda - \mathbb{E}[X]\log\lambda + \mathbb{E}[\log X!]$. The last term doesn't depend on λ. The derivative of the rest is $1 - \mathbb{E}[X]/\lambda$, zero at $\lambda = \mathbb{E}[X]$, negative before and positive after. □

Worked example: fitting a Gaussian to exponential data

Let the truth be Expo(1), which has mean 1 and variance 1, and fit the Gaussian family. The KL projection is Gauss(1, 1). From the proof, with $\mathbb{V}\mathrm{ar}(X) = 1$ and $\mathbb{E}[X] = 1$, and with the constant term $\mathbb{E}[\log e^{-X}] = -\mathbb{E}[X] = -1$,

$$\mathrm{KL}(\mathrm{Expo}(1), \mathrm{Gauss}(\mu, \sigma^2)) = \tfrac{1}{2}\log(2\pi\sigma^2) + \frac{1 + (1-\mu)^2}{2\sigma^2} - 1.$$

At the projection the divergence is $\frac{1}{2}\log(2\pi) + \frac{1}{2} - 1 = 0.419$. The minimum is not 0, since no Gaussian is an exponential. Figure 10.6 shows the projection against two other candidates, Gauss(0.7, 0.6), which is taller and sits closer to the exponential's peak at 0, and Gauss(1.5, 1.5), which covers its long right tail better. Both diverge more, 0.572 and 0.538. The right of the figure draws the divergence over a grid of means and variances: its minimum, 0.419, is at mean 1 and variance 1, the exponential's own mean and variance, and the three dots are the three Gaussians on the left. The divergence weights each point by how often the truth produces it,

and it penalizes a candidate that calls likely outcomes unlikely; matching the mean and the variance is the best a Gaussian can do on that criterion.

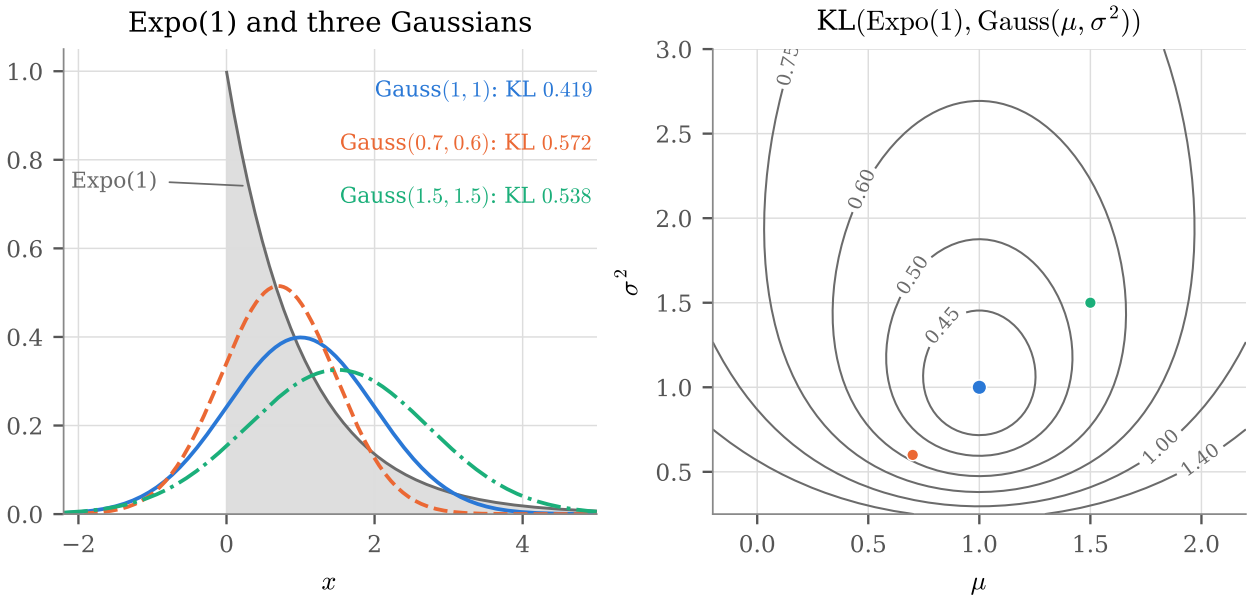


Figure 10.6: Projecting Expo(1) onto the Gaussian family. Left: the exponential density (gray) and three Gaussians, each labeled with its KL divergence from it. Right: $\mathrm{KL}(\mathrm{Expo}(1), \mathrm{Gauss}(\mu, \sigma^2))$ over a grid of means and variances, with dots at the three Gaussians on the left.

The Poisson half of Proposition 10.10 pays off the debt of Chapter 2. The siblings data of Figure 2.1 were fitted with $X - 1 \sim \mathrm{Pois}(\hat{\lambda})$ and $\hat{\lambda}$ the sample mean minus one. If the true distribution of the number of siblings is not a shifted Poisson, say with an extra bump at 5, the fit is still estimating something definite: the shifted Poisson whose mean matches the true mean, $\lambda^\circ = \mathbb{E}[X] - 1$, which is the KL projection. The sample mean minus one estimates it by the LLN, which needs no assumption that the shifted Poisson is right. Whether that projection is a good description of the truth is a separate question, and it is the question the goodness-of-fit tests of Chapter 16 ask.

From undergraduate statistics: 'fit a normal with the sample mean and standard deviation'

An introductory course fits a normal distribution to data by plugging in $\bar{x}$ and s, and justifies it, if at all, by saying that the normal is determined by its mean and standard deviation. Proposition 10.10 is a better justification, and it holds whether or not the data are normal: $\mathrm{Gauss}(\mathbb{E}[X], \mathbb{V}\mathrm{ar}(X))$ is the Gaussian closest to the truth in KL, and $\bar{x}$ and s^2 estimate its two parameters. What the procedure can't tell you is whether the closest Gaussian is close. For exponential data it is 0.419 nats away, and probabilities computed from it can be badly wrong: under $\mathrm{Gauss}(1, 1)$ the probability of a negative value is $\Phi(-1) = 0.16$, under Expo(1) it is 0. A QQ plot (Chapter 17) is how you see that in data.

Pitfall: the order of the arguments is not a choice

$\mathrm{KL}(\mathbb{P}, \mathbb{P}_\theta)$ and $\mathrm{KL}(\mathbb{P}_\theta, \mathbb{P})$ are different functions of θ with different minimizers in general. Statistics minimizes the first, with the truth first, and not because it is better in some abstract

sense: it is the only one that can be estimated. The first is an expectation under the truth $\mathbb{P}$, which the data average. The second is an expectation under the candidate $\mathbb{P}_\theta$, and it has the unknown p inside it in a way that does not split off as a constant. Because KL is not symmetric, the two orders are different problems, and only the truth-first order is an average under the distribution the data come from.

10.8 What comes next

This chapter ended at maximum likelihood without defining a likelihood. Chapter 11 defines it, for discrete and continuous models, and computes the maximum likelihood estimator for the families of this book: Bernoulli, Poisson, Gaussian, exponential, uniform. Chapter 12 proves what Figure 10.5 shows, that the estimator converges to the minimizer of the KL divergence, and then uses the curvature of KL at its minimum, the $1/(p(1-p))$ of Figure 10.4, to find its asymptotic variance: that curvature is the Fisher information. Distances between distributions return in Chapter 17, where the largest gap between an empirical CDF and a candidate CDF is the test statistic.

10.9 Summary

- Closeness should be measured between distributions, not parameters: a parameter difference of 0.01 is TV $= 0.004$ for two means with $\sigma = 1$ and TV $= 0.323$ for standard deviations 0.01 and 0.02. A distance between distributions also makes sense when the model is wrong.
- Total variation: $\mathrm{TV}(\mathbb{P}, \mathbb{Q}) = \max_A |\mathbb{P}(A) - \mathbb{Q}(A)| = \frac{1}{2}\sum |p - q|$ (or $\frac{1}{2}\int$), attained at $\{p > q\}$; the area where one density is above the other; $1 - \mathrm{TV}$ is the shared mass. A true distance with values in $[0, 1]$. $\mathrm{TV}(\mathrm{Bern}(p), \mathrm{Bern}(q)) = |p - q|$; kiss data 0.145 for one couple, 0.900 for 124.
- The best test of $\mathbb{P}$ against $\mathbb{Q}$ has $\alpha + \beta = 1 - \mathrm{TV}$. For 124 couples it rejects $p = \frac{1}{2}$ from 72 right turns, with $\alpha = 0.0438$ and $\beta = 0.0567$.
- Total variation ignores the geometry of the observation space: $\mathrm{TV}(X, X + 0.001) = 1$, and a discrete distribution is at distance 1 from every continuous one, so the CLT is not convergence in total variation.
- Total variation can't be estimated by an average: written as an expectation under the truth, it has the unknown density inside an absolute value. Plugging in frequencies works for a few discrete outcomes and fails for continuous data.
- KL divergence: $\mathrm{KL}(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{\mathbb{P}}[\log(p(X)/q(X))]$, the average evidence per observation for $\mathbb{P}$ over $\mathbb{Q}$; $+\infty$ if $\mathbb{Q}$ calls impossible what $\mathbb{P}$ produces. Nonnegative and zero only for $\mathbb{P} = \mathbb{Q}$ (Jensen), but not symmetric and no triangle inequality. Kiss data: 0.0428 and, reversed, 0.0440.
- Formulas: Bernoulli $p\log(p/q) + (1-p)\log((1-p)/(1-q))$; Gaussians with a common variance $(a-b)^2/(2\sigma^2)$; exponentials $\log(\lambda/\mu) + \mu/\lambda - 1$.
- KL adds over independent observations: $\mathrm{KL}(\mathbb{P}^{\otimes n}, \mathbb{Q}^{\otimes n}) = n\mathrm{KL}(\mathbb{P}, \mathbb{Q})$; $124 \times 0.0428 = 5.30$ nats for the kiss data. Near the truth it is a parabola with curvature one over the variance of one observation, $(p-q)^2/(2p(1-p))$ for Bernoulli. $2n\mathrm{KL}$ is the G^2 statistic, and its parabola approximation is Pearson's chi-squared.
- Pinsker: $\mathrm{TV} \le \sqrt{\mathrm{KL}/2}$; the converse fails.

- $\mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$ is a constant minus $\mathbb{E}_{\theta^*}[\log p_\theta(X)]$, and the LLN estimates the second term by $\frac{1}{n}\sum \log p_\theta(X_i)$. Minimizing the estimate is maximizing the likelihood $\prod p_\theta(X_i)$. Kiss data: the minimizer is $\hat{p} = 0.645$.
- When the model is wrong, maximum likelihood estimates the KL projection, the member of the family closest to the truth. For the Gaussian family it has the true mean and variance; for the Poisson family, the true mean. Expo(1) projects to Gauss(1, 1) at divergence 0.419.

10.10 Exercises

Exercise 10.1. Compute $\mathrm{TV}(\mathrm{Bern}(\frac{1}{2}), \mathrm{Bern}(\frac{1}{2}))$, $\mathrm{TV}(\mathrm{Bern}(\frac{1}{2}), \mathrm{Bern}(\frac{1}{3}))$ and $\mathrm{TV}(\mathrm{Bern}(\frac{1}{2}), \mathrm{Bern}(\frac{1}{10}))$ from Definition 10.1, by listing the events, and check each against the formula $|p - q|$. *(Solution on page 420.)*

Exercise 10.2. Show that $\mathrm{TV}(\mathrm{Expo}(1), \mathrm{Unif}([0, 1])) = 1/e$, and identify the event $A^\star$ of Proposition 10.2. *(Solution on page 420.)*

Exercise 10.3. Let $X \sim \mathrm{Bern}(\frac{1}{2})$ and $a \in (0, 1)$. Show that $\mathrm{TV}(X, X + a) = 1$, meaning the total variation distance between the distributions of X and $X+a$. Then show that $\mathrm{TV}(2\sqrt{n}(\bar{X}_n - \frac{1}{2}), Z) = 1$ for every n, where $X_1, \dots, X_n \overset{iid}{\sim} \mathrm{Bern}(\frac{1}{2})$ and $Z \sim \mathrm{Gauss}(0, 1)$, and explain why this does not contradict the CLT. *(Solution on page 420.)*

Exercise 10.4. For one observation from $\mathrm{Bern}(p_0)$ or $\mathrm{Bern}(p_1)$ with $p_0 < p_1$, list all four tests (rejection regions $\emptyset$, $\{0\}$, $\{1\}$, $\{0, 1\}$), compute $\alpha + \beta$ for each, and confirm that the smallest is $1 - |p_1 - p_0|$, as Proposition 10.4 says. Evaluate for $p_0 = \frac{1}{2}$, $p_1 = 0.645$. *(Solution on page 420.)*

Exercise 10.5. Compute $\mathrm{KL}(\mathrm{Bin}(n, p), \mathrm{Bin}(n, q))$ directly from the Binomial PMF and show that it equals $n\,\mathrm{KL}(\mathrm{Bern}(p), \mathrm{Bern}(q))$. Evaluate for $n = 124$, $p = 80/124$, $q = \frac{1}{2}$. *(Solution on page 420.)*

Exercise 10.6. Show that

$$\mathrm{KL}(\mathrm{Gauss}(a, \sigma^2), \mathrm{Gauss}(b, \tau^2)) = \log\frac{\tau}{\sigma} + \frac{\sigma^2 + (a-b)^2}{2\tau^2} - \frac{1}{2}.$$

Check that it reduces to $(a - b)^2/(2\sigma^2)$ when $\tau = \sigma$, and evaluate it for $\mathrm{Gauss}(0, 0.01^2)$ against $\mathrm{Gauss}(0, 0.02^2)$ and the other way round. *(Solution on page 421.)*

Exercise 10.7. Show that $\mathrm{KL}(\mathrm{Expo}(\lambda), \mathrm{Expo}(\mu)) = \log(\lambda/\mu) + \mu/\lambda - 1$, check that it is nonnegative by studying $t \mapsto t - 1 - \log t$ with $t = \mu/\lambda$, and check that it is 0 only when $\lambda = \mu$. *(Solution on page 421.)*

Exercise 10.8. Let $\mathbb{P}_n = \mathrm{Pois}(1/n)$ and let δ_0 be the distribution that puts all its mass at 0. Compute $\mathrm{TV}(\mathbb{P}_n, \delta_0)$, $\mathrm{KL}(\mathbb{P}_n, \delta_0)$ and $\mathrm{KL}(\delta_0, \mathbb{P}_n)$, and say what happens as $n \to \infty$. *(Solution on page 421.)*

Exercise 10.9. Let $X_1, \dots, X_n \overset{iid}{\sim} \mathrm{Pois}(\lambda^*)$. Write $\widehat{\mathrm{KL}}(\mathrm{Pois}(\lambda^*), \mathrm{Pois}(\lambda))$ up to its constant, and find the λ that minimizes it. *(Solution on page 421.)*

Exercise 10.10. Data $X_i \in \{0, 1, 2\}$ come from a distribution with $\mathbb{P}(X = 0) = 0.5$, $\mathbb{P}(X = 1) = 0.2$, $\mathbb{P}(X = 2) = 0.3$, and are fitted with the model $X \sim \mathrm{Bin}(2, \theta)$. Find the KL projection θ° and the divergence at it. Is the model well-specified? *(Solution on page 421.)*

Chapter 11

The Likelihood and Maximum Likelihood

In this chapter

- The likelihood: the probability (or density) of the data you actually observed, read as a function of the parameter. One formula, two readings, and why the switch between them is the whole idea.
- Why a density can stand in for a probability when the data are continuous.
- The likelihoods of the standard families: Bernoulli, Poisson, Gaussian, exponential, uniform, and a mixture of Gaussians.
- The maximum likelihood estimator, and how to find a maximum: concave functions, the first-order condition, and the second-derivative test in one and in several variables.
- The maximum likelihood estimators of the standard families, computed for the kiss data, the siblings data, the subway data and the boarding data. The uniform family, where calculus fails and the answer is the largest observation.
- When the estimator doesn't exist or isn't unique, and what to do when there is no formula: the EM algorithm for a mixture, run on the waiting times of Old Faithful.

Chapter 10 ended with a principle: minimizing an estimate of the KL divergence from the truth is the same as maximizing $\prod_i p_\theta(X_i)$. This chapter names that product and learns to maximize it.

11.1 From the KL divergence to the likelihood

Recall. Chapter 10: for data $X_1, \ldots, X_n \overset{iid}{\sim} \mathbb{P}_{\theta^*}$, $\mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) = \text{constant} - \mathbb{E}_{\theta^*}[\log p_\theta(X)]$; the Law of Large Numbers estimates the expectation by $\frac{1}{n}\sum_i \log p_\theta(X_i)$ (when $\log p_\theta(X)$ has finite variance); and the θ that minimizes the estimated divergence is the θ that maximizes $\prod_{i=1}^n p_\theta(X_i)$.

That product is the subject of this chapter. It is the probability, or the density, of the data you have, computed as if θ were the truth. Chapter 10 reached it from a question about distances. There is a second road to it, more direct and older. Ask of each candidate θ: if this were the truth, how probable would the data I actually saw be? A candidate under which your data would be a fluke is a poor candidate. A candidate under which your data are typical is a good one. Ranking candidates

by the probability they give your data, and picking the top one, is maximum likelihood. The two roads arrive at the same estimator, and Chapter 12 uses the first one to prove that it works.

11.2 The likelihood of a discrete model

Start with a model whose observations take values in a countable set E: the kiss data in $\{0,1\}$, the siblings counts in $\{1,2,3,\ldots\}$. Each $\mathbb{P}_\theta$ has a PMF $p_\theta(x) = \mathbb{P}_\theta(X = x)$.

Definition 11.1 (Likelihood, discrete model)**.** Let $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ be a statistical model for $X_1, \ldots, X_n$ i.i.d., with E countable and $\mathbb{P}_\theta$ having PMF p_θ. The **likelihood** of the model is the function

$$L_n : E^n \times \Theta \to \mathbb{R}$$

$$(x_1, \ldots, x_n; \theta) \mapsto \mathbb{P}_\theta(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} p_\theta(x_i).$$

The product form is independence: the joint PMF of independent observations is the product of the individual PMFs. We write L for L_n when n is clear, and $L(\theta)$ when the data are fixed.

In words. Take the data you observed and a candidate θ, and compute the probability that the model with parameter θ would produce exactly those data. That number, for every candidate at once, is the likelihood. The data are held fixed; the parameter varies.

For the kiss data. Each couple contributes p if it turned right and $1-p$ if it turned left, so for the observed 80 right turns and 44 left turns

$$L(p) = p^{80}(1-p)^{44}.$$

At $p = 0.645$ this is 9.4×10^{-36}; at $p = \frac{1}{2}$ it is 4.7×10^{-38}. Both are tiny, and they should be: there are 2^{124} possible sequences of turns, and any one sequence with 80 right turns is improbable under every p, its probability never more than 9.4×10^{-36}. What carries information is the ratio. The observed sequence is about 200 times more probable under $p = 0.645$ than under $p = \frac{1}{2}$, and $\log 200 = 5.30$, the evidence in nats that Chapter 10 found.

Intuition: one formula, two readings

You have written $\mathbb{P}_\theta(X = x)$ many times, always as a function of x with θ held fixed: the PMF. The same expression is a function of both x and θ, and the likelihood is what you get by switching which one you hold fixed. Figure 11.1 makes the switch visible for the kiss model. The left of the figure shades $\mathbb{P}_p(\text{count} = k)$ over a grid of counts k and parameters p. Read it along a horizontal line, p fixed: you get the distribution of the count, the probability question of Chapter 1, which sums to 1 over k; the middle of the figure is this slice along the dashed aqua line, $p = 0.645$. Read it along a vertical line, the count fixed at the 80 you observed: you get the likelihood, the statistics question, a curve over p that peaks where the data are most probable; the right of the figure is this slice along the orange line, and it peaks at $80/124 = 0.645$. The likelihood of the full sequence of 124 turns is this curve divided by the constant $\binom{124}{80}$, which doesn't change where it peaks. Probability fixes the parameter and asks about the data; statistics fixes the data and asks about the parameter.

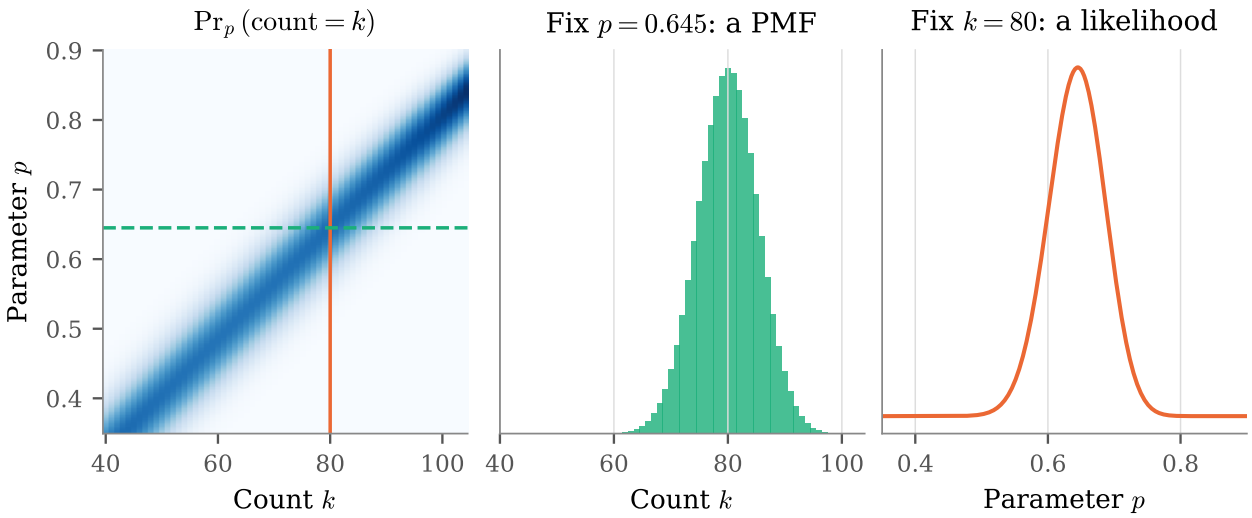


Figure 11.1: $\mathbb{P}_p(\text{count} = k)$ for 124 couples. Left: over counts k (horizontal) and parameters p (vertical), darker more probable. Middle: the slice along the dashed aqua line, $p = 0.645$. Right: the slice along the orange line, $k = 80$.

Pitfall: the likelihood is not a distribution for θ

The likelihood curve in Figure 11.1 looks like a density for p, and it isn't one. The curve drawn there (the version for the count) has area 1/125, not 1, and the sequence likelihood $p^{80}(1-p)^{44}$ has area 1.0×10^{-36}; nothing in the construction makes the area mean anything: rescale it by any constant and every comparison it supports is unchanged. Until Chapter 18, p is a fixed unknown number, not a random variable, so 'the probability that p is between 0.6 and 0.7' has no meaning here. The likelihood is a plausibility ranking of candidate truths by how well they predicted your data. This is why the right of Figure 1.1 in Chapter 1, which is this kind of curve for the drug data scaled to peak at 1, has no vertical scale that could be read as a probability. That chapter is where the parameter gets a distribution, and the likelihood is one of its two ingredients.

Worked example: the likelihoods of the two discrete families

- **Bernoulli.** For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bern}(p)$, $E = \{0, 1\}$ and $\Theta = (0, 1)$. The PMF can be written in one line as $p_p(x) = p^x(1-p)^{1-x}$, which is p at $x = 1$ and $1 - p$ at $x = 0$. Then

$$L(x_1, \ldots, x_n; p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^{\sum_i x_i}(1-p)^{n-\sum_i x_i}.$$

  The data enter only through the number of ones, $\sum_i x_i$.

- **Poisson.** For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$, $E = \{0, 1, 2, \ldots\}$ and $\Theta = (0, \infty)$:

$$L(x_1, \ldots, x_n; \lambda) = \prod_{i=1}^{n} e^{-\lambda}\frac{\lambda^{x_i}}{x_i!} = e^{-n\lambda}\frac{\lambda^{\sum_i x_i}}{x_1! \cdots x_n!}.$$

  For the siblings data of Chapter 2, modeled as $X_i - 1 \sim \text{Pois}(\lambda)$, the 24 values of $x_i - 1$ add up to 24, and the likelihood is $e^{-24\lambda}\lambda^{24}$ divided by a constant that doesn't involve λ.

Three properties of a likelihood follow straight from the product.

- **Order doesn't matter.** A product can be taken in any order, so the likelihood of $(1, 2, 3)$ is the likelihood of $(2, 3, 1)$. For the Bernoulli and Poisson families it depends on the data only through their sum, up to a factor that does not involve the parameter; the sum is the sufficient statistic of Chapter 7: the likelihood never looks at anything else in the data.
- **It updates one observation at a time.** If L_n is the likelihood after n observations, then $L_{n+1}(\theta) = L_n(\theta) \cdot p_\theta(x_{n+1})$. For Poisson data $1, 2$ the likelihood is $e^{-2\lambda}\lambda^3/2$; a third observation 3 multiplies it by $e^{-\lambda}\lambda^3/6$, giving $e^{-3\lambda}\lambda^6/12$ (Exercise 11.1).
- **It is as smooth in θ as the PMF is.** If each $p_\theta(x)$ is a continuous function of θ, so is the product. A discrete model can have a perfectly smooth likelihood: the data are discrete, the parameter is not.

11.3 The likelihood of a continuous model

For continuous data the probability of observing exactly $x_1, \ldots, x_n$ is 0 under every candidate, which would make every likelihood 0 and rank nothing. The fix is to use densities.

Definition 11.2 (Likelihood, continuous model)**.** Let $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ be a statistical model for $X_1, \ldots, X_n$ i.i.d., where each $\mathbb{P}_\theta$ has a density f_θ (toolkit A.6). The **likelihood** of the model is

$$L_n : E^n \times \Theta \to \mathbb{R}$$
$$(x_1, \ldots, x_n; \theta) \mapsto \prod_{i=1}^{n} f_\theta(x_i).$$

In words. The same recipe as for a discrete model, with each observation contributing the height of its density instead of its probability. A candidate whose density is high where your data fell scores well; one whose density is low there scores badly.

For the kiss data. The kiss data are discrete, so the discrete definition applies and this one doesn't. The two definitions are one idea, and from here on the chapter writes p_θ for either a PMF or a density.

Intuition: why the density is enough

A recorded measurement is never exact. A subway gap written down as 5 minutes was somewhere near 5, say within $h/2$ of it for a small h. The probability of that is the area of a thin strip under the density, about $f_\lambda(5) \cdot h$. Compare two candidate rates, and the h cancels:

$$\frac{\mathbb{P}_{\lambda_1}(\text{gap within } h/2 \text{ of } 5)}{\mathbb{P}_{\lambda_2}(\text{gap within } h/2 \text{ of } 5)} \approx \frac{f_{\lambda_1}(5)\,h}{f_{\lambda_2}(5)\,h} = \frac{f_{\lambda_1}(5)}{f_{\lambda_2}(5)}.$$

Figure 11.2 draws this for the rates 0.1 and 0.2 and a gap recorded as 5 minutes. On the left, the taller Expo(0.2) strip contains the Expo(0.1) strip, and its orange part is the extra probability the faster rate gives the same gap. On the right, as the width h shrinks, the ratio of the two strip probabilities settles at the ratio of the densities, $f_{0.1}(5)/f_{0.2}(5) = 0.824$. Every use of a likelihood is a comparison between parameters, so the density is all it needs.

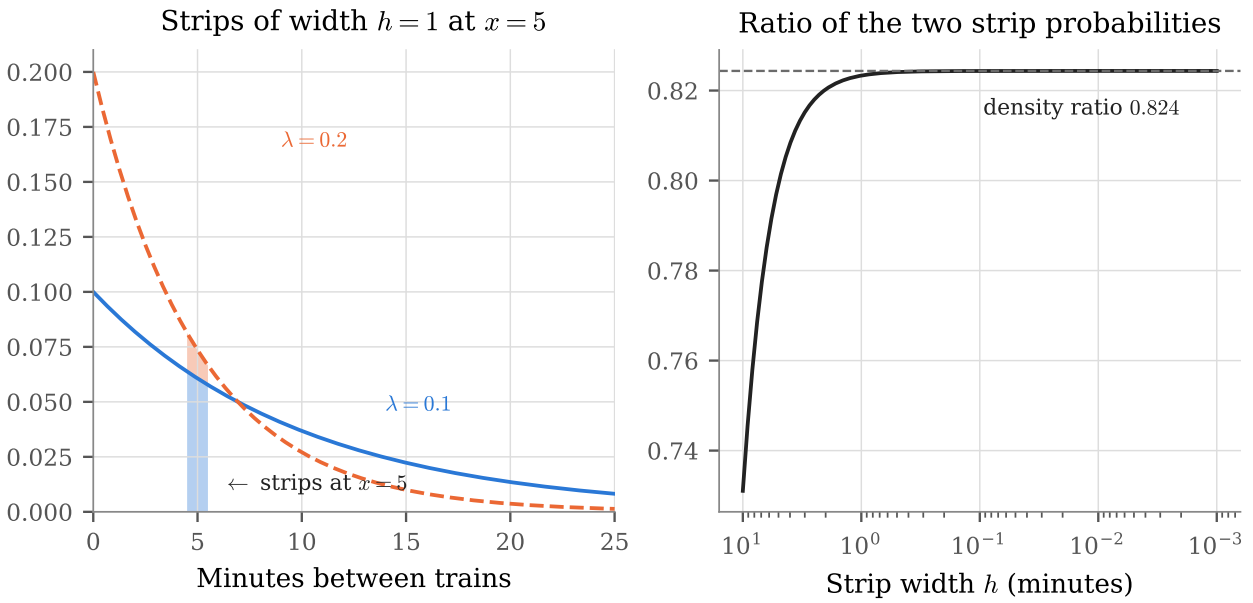


Figure 11.2: Left: the Expo(0.1) and Expo(0.2) densities, with the probability of a gap within $\frac{1}{2}$ minute of 5 as a strip under each; the Expo(0.1) strip is blue, the Expo(0.2) strip's extra height orange. Right: the ratio of the strip probabilities against the width h.

Worked example: the likelihoods of the continuous families

- **Gaussian.** For $X_1, \dots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$, $E = \mathbb{R}$ and $\Theta = \mathbb{R} \times (0, \infty)$:

$$L(x_1, \dots, x_n; \mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$
$$= \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2\right).$$

- **Exponential.** For $X_1, \dots, X_n \overset{iid}{\sim} \text{Expo}(\lambda)$, $E = (0, \infty)$ and $\Theta = (0, \infty)$. The density is $f_\lambda(x) = \lambda e^{-\lambda x} \mathbb{1}(x > 0)$, so

$$L(x_1, \dots, x_n; \lambda) = \lambda^n e^{-\lambda \sum_i x_i} \prod_{i=1}^{n} \mathbb{1}(x_i > 0) = \lambda^n e^{-\lambda \sum_i x_i} \, \mathbb{1}\big(\min_i x_i > 0\big).$$

  The product of indicators is 1 exactly when every x_i is positive, which is when the smallest one is (toolkit A.4). The indicator doesn't involve λ, and data from the model are always positive, so it is usually dropped: $L(\lambda) = \lambda^n e^{-\lambda \sum_i x_i}$.

- **Uniform.** For $X_1, \dots, X_n \overset{iid}{\sim} \text{Unif}([0, \theta])$, $E = [0, \infty)$ and $\Theta = (0, \infty)$. The density is $f_\theta(x) = \frac{1}{\theta} \mathbb{1}(0 \le x \le \theta)$, so

$$L(x_1, \dots, x_n; \theta) = \frac{1}{\theta^n} \prod_{i=1}^{n} \mathbb{1}(0 \le x_i \le \theta) = \frac{1}{\theta^n} \, \mathbb{1}\big(\min_i x_i \ge 0\big) \, \mathbb{1}\big(\max_i x_i \le \theta\big).$$

Every $x_i \le \theta$ holds exactly when the largest does. Here the indicator involving the maximum *does* depend on θ, and it cannot be dropped: it is what makes this likelihood unusual (Section 11.6). The other indicator doesn't involve θ and is dropped as before: $L(\theta) = \theta^{-n}\mathbb{1}(\theta \ge \max_i x_i)$.

- **Mixture of two Gaussians.** For the model of Section 2.7, with weight π and components $\text{Gauss}(\mu_1, \sigma_1^2)$ and $\text{Gauss}(\mu_2, \sigma_2^2)$, the density of one observation is a sum, and

$$L = \prod_{i=1}^{n} \left(\frac{\pi}{\sigma_1\sqrt{2\pi}} e^{-(x_i-\mu_1)^2/(2\sigma_1^2)} + \frac{1-\pi}{\sigma_2\sqrt{2\pi}} e^{-(x_i-\mu_2)^2/(2\sigma_2^2)} \right).$$

A product of sums does not simplify, and that is the source of the trouble in Section 11.8.

Pitfall: an indicator you drop must not involve the parameter

Dropping a factor from a likelihood is safe exactly when the factor doesn't depend on θ: it multiplies every candidate's score by the same number and changes no comparison. The exponential's $\mathbb{1}(\min_i x_i > 0)$ is such a factor. The uniform's $\mathbb{1}(\max_i x_i \le \theta)$ is not, and dropping it leaves θ^{-n}, which grows without bound as $\theta \to 0$: the best candidate would be one under which your data were impossible.

11.4 The maximum likelihood estimator

Definition 11.3 (Maximum likelihood estimator)**.** Let $X_1, \dots, X_n$ be an i.i.d. sample from a statistical model $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ with likelihood L. The **maximum likelihood estimator** (MLE) of θ is

$$\hat{\theta}_n^{\text{MLE}} = \operatorname*{argmax}_{\theta\in\Theta} L(X_1, \dots, X_n; \theta),$$

provided the maximum exists and is attained at a single point.

In words. Among all the candidate truths the model allows, pick the one under which the data you actually got were most probable. Before the data arrive, $\hat{\theta}_n^{\text{MLE}}$ is a random variable, a function of $X_1, \dots, X_n$ like every estimator in Chapter 7; once they arrive it is a number, the maximum likelihood estimate.

For the kiss data. $L(p) = p^{80}(1-p)^{44}$ is largest at $p = 80/124 = 0.645$ (Section 11.6 does the calculus), so the maximum likelihood estimate is the sample proportion. In general $\hat{p}^{\text{MLE}} = \bar{R}_n$, the estimator Chapter 1 started from because it was the obvious one. In models where no estimator is obvious, the same principle still produces one.

In practice you maximize the logarithm of the likelihood, the **log-likelihood**

$$\ell_n(\theta) = \log L(X_1, \dots, X_n; \theta) = \sum_{i=1}^{n} \log p_\theta(X_i).$$

Because log is strictly increasing, L and ℓ_n have their maximum at the same θ (toolkit A.10). There are three reasons to prefer ℓ_n. It turns a product of n factors into a sum of n terms, and sums are easier to differentiate. It keeps the numbers in a sane range: $L(0.645) = 9.4 \times 10^{-36}$ for the

kiss data, and for a few thousand observations a likelihood is smaller than any number a computer stores, while $\ell_n(0.645) = -80.65$ is an ordinary number. And $\frac{1}{n}\ell_n(\theta)$ is an average, which is what the Law of Large Numbers and the Central Limit Theorem know how to handle; Chapter 12 uses that.

Figure 11.3 shows both curves for the kiss data. The likelihood is a narrow peak at 0.645 on a scale of 10^{-36}: its height there is 9.4×10^{-36}, against 0.047×10^{-36} at $p = \frac{1}{2}$. The log-likelihood is a concave hill whose top is at the same place, where the tangent is flat: its maximum is -80.65, its value at $p = \frac{1}{2}$ is -85.95, and the gap, 5.30, is 124 times the KL divergence 0.0428 of Chapter 10.

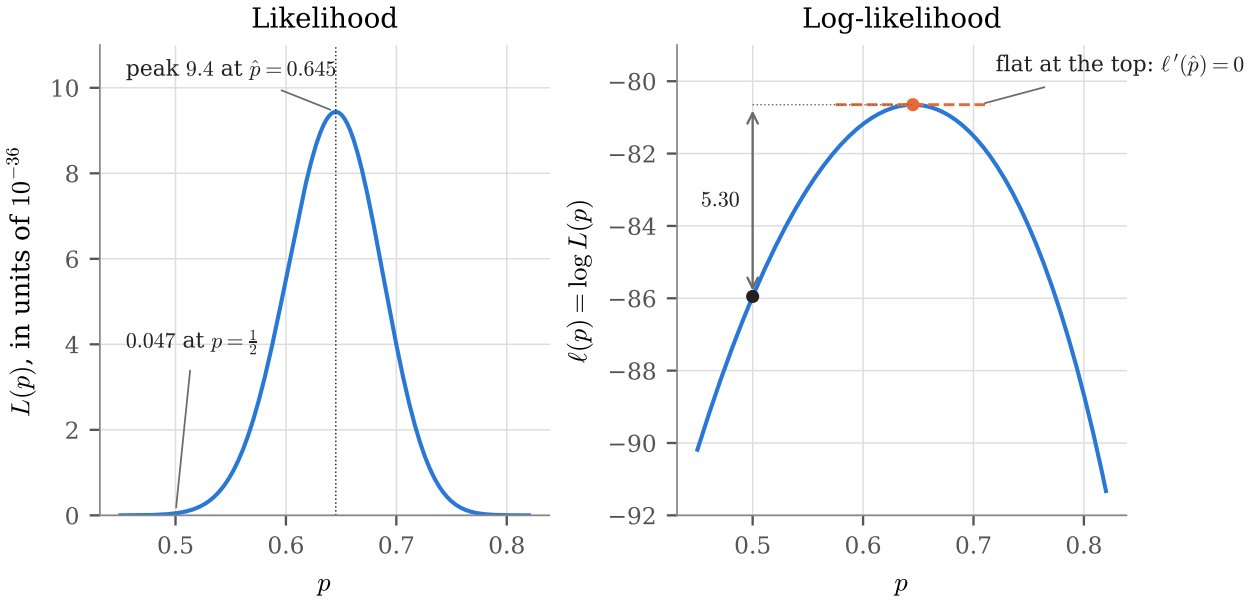


Figure 11.3: The kiss data. Left: the likelihood $L(p) = p^{80}(1-p)^{44}$, in units of 10^{-36}. Right: the log-likelihood $\ell(p) = 80 \log p + 44 \log(1-p)$, with its tangent at the maximum.

Does this make sense? The sharpness of the peak

How sharply the log-likelihood curves down at its top should say something about how precisely the data pin down p: a flat top means many values of p fit about equally well. Check it. For the kiss data $\ell(p) = 80 \log p + 44 \log(1-p)$, so $\ell''(p) = -80/p^2 - 44/(1-p)^2$, and at $\hat{p}$ this is $-n/(\hat{p}(1-\hat{p})) = -541.7$. Then $1/\sqrt{541.7} = 0.0430$, which is exactly the standard error $\sqrt{\hat{p}(1-\hat{p})/n}$ of Chapter 8, since $\ell''(\hat{p}) = -n/(\hat{p}(1-\hat{p}))$. For the subway data the same computation gives $\hat{\lambda}/\sqrt{n} = 0.020$, the standard error of Chapter 5. It is not a coincidence, and Chapter 12 proves it in general: the curvature of the log-likelihood at its peak estimates n times the Fisher information, and one over its square root is the standard error of the MLE.

11.5 Finding a maximum

The definition says 'argmax', and an argmax has to be found. Maximizing an arbitrary function is hard: it can have many hills, and a point where the slope is zero can be the top of a small hill, the bottom of a valley or neither. The right of Figure 11.4 is $\theta \mapsto \prod_i (\theta - x_i)$ for five numbers x_i: four points with zero slope, marked by dots, and on the whole line no maximum at all. Setting the

derivative to 0 finds all four without saying which, if any, is a maximum. The left of the figure is the Poisson log-likelihood of the siblings data, $-24\lambda + 24\log\lambda$ up to a constant: strictly concave, one hill, with its single maximum at $\hat{\lambda} = 1$, where the slope is 0. The functions that are easy to maximize are the concave ones.

Definition 11.4 (Concave and convex functions)**.** A twice-differentiable function $h : \Theta \to \mathbb{R}$ on an interval $\Theta \subset \mathbb{R}$ is **concave** if $h''(\theta) \leq 0$ for every $\theta \in \Theta$, and **strictly concave** if $h''(\theta) < 0$ for every θ. (The 'strictly' version here is stronger than the usual chord definition, which also allows $h'' = 0$ at isolated points, as for $-\theta^4$; it is the version this book uses.) It is (strictly) **convex** if $-h$ is (strictly) concave, that is, $h'' \geq 0$ ($h'' > 0$).

> **In words.** A concave function's slope never increases: it is a hill, or a straight line, never a valley. Every chord lies below the graph (toolkit A.8). On their natural domains, $-\theta^2$, $\sqrt{\theta}$, $\log\theta$ and $\sin\theta$ on $[0, \pi]$ are concave; $2\theta - 3$ is both concave and convex; θ^2 and e^θ are convex.
>
> **For the kiss data.** $\ell''(p) = -80/p^2 - 44/(1-p)^2$ is negative for every $p \in (0, 1)$, so the kiss log-likelihood is strictly concave: one hill, no valleys, as the right of Figure 11.3 shows.

Proposition 11.5 (Maximizing a strictly concave function)**.** *Let h be strictly concave and differentiable on an open interval Θ. If $h'(\hat{\theta}) = 0$ at some $\hat{\theta} \in \Theta$, then $\hat{\theta}$ is the unique maximizer of h on Θ. If h' never vanishes, h has no maximum on Θ.*

Proof strategy (for completeness; skip on a first reading). Strict concavity makes the slope strictly decreasing, so it can cross zero at most once, and on each side of the crossing the function moves away from the top.

Proof. Since $h'' < 0$, h' is strictly decreasing. If $h'(\hat{\theta}) = 0$, then $h' > 0$ to the left of $\hat{\theta}$ and $h' < 0$ to the right: h increases up to $\hat{\theta}$ and decreases after, so $h(\theta) < h(\hat{\theta})$ for every $\theta \neq \hat{\theta}$. At an interior maximizer the derivative must be 0, since otherwise moving a little in the direction of the slope would increase h; so if h' never vanishes there is no interior maximizer, and on an open interval there is no other kind. □

So the recipe for a one-parameter model is: write ℓ_n, check that it is strictly concave, solve $\ell_n'(\theta) = 0$. The Bernoulli, Poisson and exponential log-likelihoods of Section 11.6 pass the check; the Gaussian needs a two-step argument instead; the uniform fails it entirely.

For a parameter with several coordinates, $\theta = (\theta_1, \ldots, \theta_d)$, the derivative becomes the **gradient** $\nabla\ell_n(\theta)$, the vector of partial derivatives, and the second derivative becomes the **Hessian** $\nabla^2\ell_n(\theta)$, the $d \times d$ matrix of second partial derivatives (toolkit A.12). The first-order condition is $\nabla\ell_n(\hat{\theta}) = 0$, a system of d equations. A sufficient condition for strict concavity (on a Θ that contains the segment between any two of its points) is a negative definite Hessian at every θ, which for $d = 2$ means its top-left entry is negative and its determinant positive. Under it, the solution of $\nabla\ell_n = 0$, if there is one, is the unique maximizer, as in one dimension. When there is no formula for the solution, numerical methods find it; for a concave ℓ_n they are reliable, and the subject of convex optimization is how to do it fast.

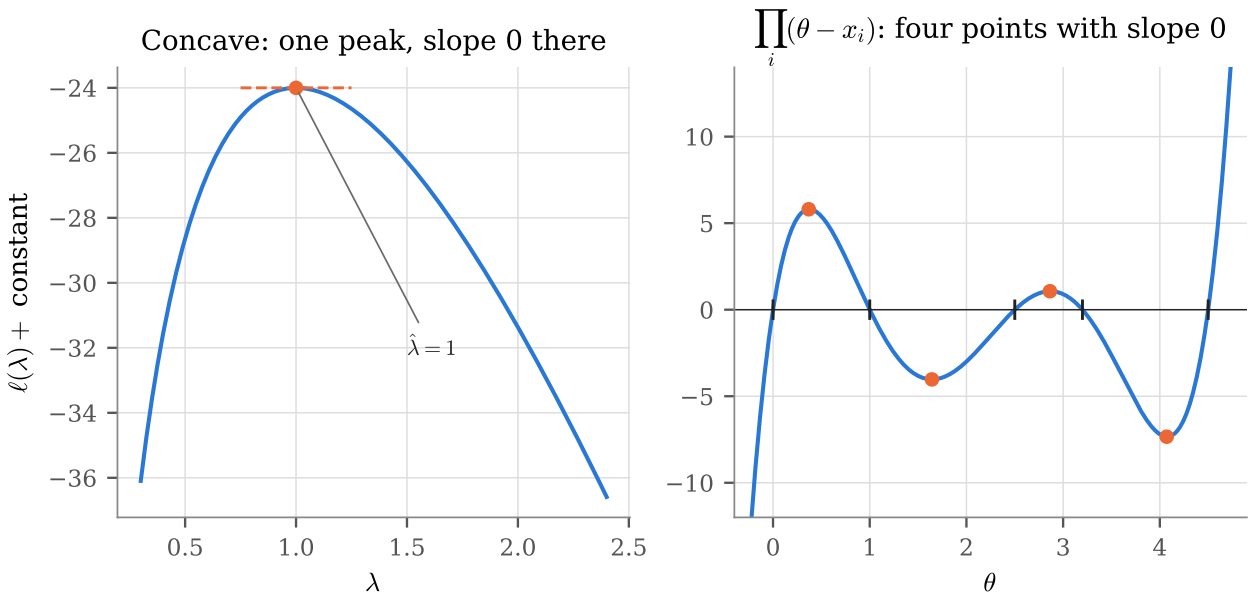


Figure 11.4: Easy and hard functions to maximize. Left: the Poisson log-likelihood of the siblings data, $-24\lambda + 24\log\lambda$ up to a constant. Right: $\prod_i(\theta - x_i)$ for the five values x_i marked on the axis, with dots where the slope is 0.

11.6 Maximum likelihood estimators of the standard families

Worked example: Bernoulli: the kiss data

For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Bern}(p)$ with $S = \sum_i X_i$ ones,

$$\ell_n(p) = S\log p + (n-S)\log(1-p), \quad \ell_n'(p) = \frac{S}{p} - \frac{n-S}{1-p}, \quad \ell_n''(p) = -\frac{S}{p^2} - \frac{n-S}{(1-p)^2} < 0.$$

Setting $\ell_n'(p) = 0$ gives $S(1-p) = (n-S)p$, so $p = S/n$:

$$\hat{p}^{\text{MLE}} = \bar{X}_n.$$

(If $S = 0$ or $S = n$ there is no solution in $(0,1)$; Section 11.7 deals with that.) For the kiss data, $\hat{p} = 80/124 = 0.645$, the estimate of every earlier chapter.

Worked example: Poisson: the siblings data

For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$,

$$\ell_n(\lambda) = -n\lambda + \Big(\sum_i X_i\Big)\log\lambda - \sum_i \log(X_i!), \quad \ell_n'(\lambda) = -n + \frac{\sum_i X_i}{\lambda}, \quad \ell_n''(\lambda) = -\frac{\sum_i X_i}{\lambda^2} < 0$$

(when at least one X_i is positive). The first-order condition gives

$$\hat{\lambda}^{\text{MLE}} = \bar{X}_n.$$

For the siblings data, with $X_i - 1 \sim \text{Pois}(\lambda)$, the sample mean of the $x_i - 1$ is $24/24 = 1$, so $\hat{\lambda} = 1.0$: the fit drawn in Figure 2.1, and the left of Figure 11.4.

Worked example: exponential: the subway data

For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Expo}(\lambda)$,

$$\ell_n(\lambda) = n \log \lambda - \lambda \sum_i X_i, \quad \ell_n'(\lambda) = \frac{n}{\lambda} - \sum_i X_i, \quad \ell_n''(\lambda) = -\frac{n}{\lambda^2} < 0,$$

and

$$\hat{\lambda}^{\text{MLE}} = \frac{n}{\sum_i X_i} = \frac{1}{\bar{X}_n}.$$

For the 64 subway gaps averaging 6.23 minutes, $\hat{\lambda} = 0.1605$ per minute, the estimator of Chapter 5. Chapter 7 showed it is biased upward, $\mathbb{E}[1/\bar{T}_n] = \frac{n}{n-1}\lambda$: a maximum likelihood estimator need not be unbiased.

Worked example: Gaussian: the boarding data

For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$ the parameter has two coordinates, and

$$\ell_n(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (X_i - \mu)^2.$$

Maximize in μ first, for a fixed σ^2. The only term involving μ is $-\frac{1}{2\sigma^2} \sum_i (X_i - \mu)^2$, and the split

$$\sum_{i=1}^{n} (X_i - \mu)^2 = \sum_{i=1}^{n} (X_i - \bar{X}_n)^2 + n(\bar{X}_n - \mu)^2$$

(write $X_i - \mu = (X_i - \bar{X}_n) + (\bar{X}_n - \mu)$, square, and note the cross terms add to 0) shows it is largest at $\mu = \bar{X}_n$, whatever σ^2 is. Now put $\mu = \bar{X}_n$ and maximize in σ^2. With $D = \sum_i (X_i - \bar{X}_n)^2$, the function $-\frac{n}{2} \log \sigma^2 - D/(2\sigma^2)$ has derivative $-\frac{n}{2\sigma^2} + \frac{D}{2\sigma^4}$, positive for $\sigma^2 < D/n$ and negative after. Chaining the two steps, for every (μ, σ^2),

$$\ell_n(\mu, \sigma^2) \le \ell_n(\bar{X}_n, \sigma^2) \le \ell_n(\bar{X}_n, D/n),$$

with equality only at $(\bar{X}_n, D/n)$, so

$$\hat{\mu}^{\text{MLE}} = \bar{X}_n, \qquad \widehat{\sigma^2}^{\text{MLE}} = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2.$$

At the maximum the gradient is 0 and the Hessian is the diagonal matrix with entries $-n/\widehat{\sigma^2}$ and $-n/(2\widehat{\sigma^2}^2)$, negative definite. (The log-likelihood is not concave in σ^2 everywhere, which is why the argument went one coordinate at a time instead of checking the Hessian at every point. It also needs $D > 0$: if every observation is the same number, the likelihood grows without bound as $\sigma^2 \to 0$.)

For the 72 flights boarded rear to front, with mean 24.2 minutes and standard deviation $s = 2.1$ (taken to be computed with $n - 1$, the usual convention), $\hat{\mu} = 24.2$ and $\widehat{\sigma^2} = \frac{71}{72} \times 2.1^2 = 4.35$. Figure 11.5 draws the log-likelihood surface, measured down from its maximum. On the left, each contour is labeled by how far it lies below the maximum at $(24.2, 4.35)$. The middle slice, at $\sigma^2 = 4.35$, is symmetric in μ. The right slice, at $\mu = 24.2$, is lopsided: variances above the maximum are penalized less than variances below it.

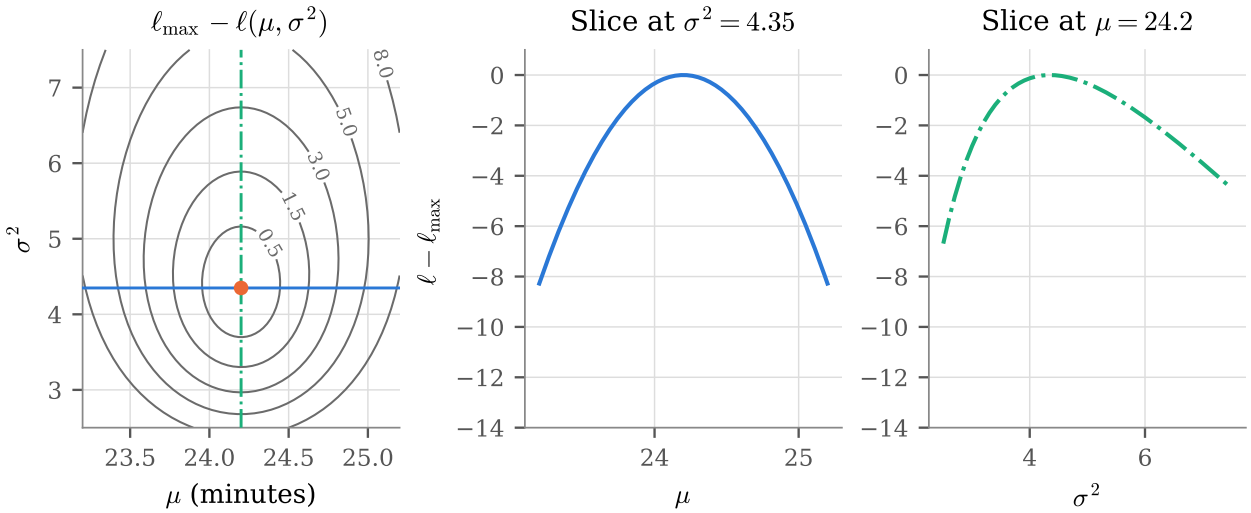


Figure 11.5: The Gaussian log-likelihood of the boarding data, measured down from its maximum. Left: contours over (μ, σ^2), with the two slices marked as lines (blue solid, aqua dash-dot). Middle: the slice at $\sigma^2 = 4.35$. Right: the slice at $\mu = 24.2$.

From undergraduate statistics: n or $n-1$?

The maximum likelihood estimator of the variance divides by n. The sample variance you were taught divides by $n-1$. They differ by the factor $(n-1)/n$, which is 0.986 for the boarding data: 4.35 against 4.41. Chapter 7 showed that dividing by n is biased low by exactly that factor and that dividing by $n-1$ removes the bias; it also showed that removing the bias is not the same as minimizing the risk. Maximum likelihood doesn't aim at unbiasedness, so it lands on n. For large n the difference is negligible, and every large-sample result in Chapter 12 holds for either.

From undergraduate statistics: least squares is maximum likelihood

The Gaussian log-likelihood contains $-\frac{1}{2\sigma^2}\sum_i (X_i - \mu)^2$, so maximizing it over μ is minimizing a sum of squared deviations. The same is true when μ is replaced by a regression line $a + bx_i$: if the errors around the line are Gaussian with a common variance, the maximum likelihood estimates of a and b are the least squares estimates of an introductory regression course. Chapter 20 takes that observation up with a second variable in the picture.

Worked example: uniform: where calculus fails

For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([0, \theta])$ the likelihood is $L(\theta) = \theta^{-n} \mathbb{1}(\theta \geq \max_i X_i)$. Below the largest observation the likelihood is 0 (a θ smaller than an observation makes that observation impossible). From the largest observation on it is θ^{-n}, which decreases, with derivative $-n\theta^{-n-1}$, never 0. The maximum is at the jump:

$$\hat{\theta}^{\text{MLE}} = \max_i X_i.$$

Figure 11.6 draws it for five observations whose largest is 7.6: the likelihood is zero until $\theta = 7.6$, then θ^{-5}, decreasing, and the maximum is at the jump, where no derivative is zero. Setting the derivative to 0 would have found only the flat stretch below $\max_i X_i$, where the likelihood is

0; the recipe of Section 11.5 assumes a smooth function with its maximum inside Θ, and this likelihood is neither smooth at its maximum nor concave. The estimator is also biased: the largest of n draws always falls short of θ, and $\mathbb{E}[\max_i X_i] = \frac{n}{n+1}\theta$ (Exercise 11.4). Multiplying by $\frac{n+1}{n}$ removes the bias. The right of the figure draws the density of the maximum of five draws when $\theta = 10$: it never exceeds 10, and its mean is 8.33, which is $\frac{5}{6} \times 10$.

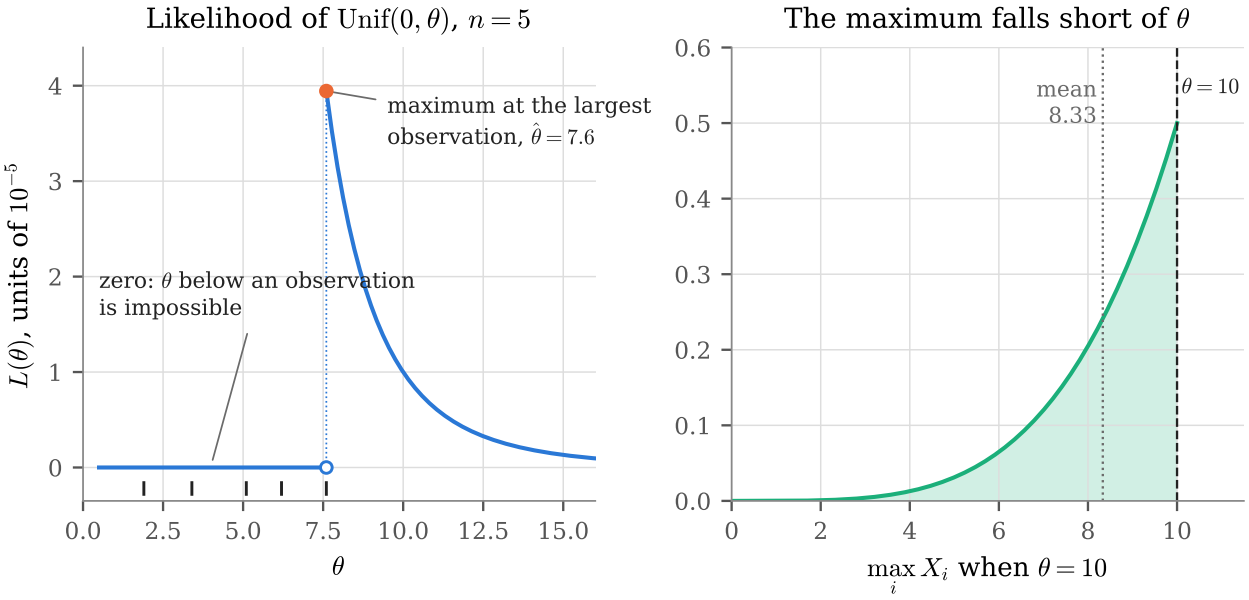


Figure 11.6: Left: the likelihood of Unif$([0, \theta])$ for five observations, marked by ticks below the axis. Right: the density $5m^4/10^5$ of the maximum m of five draws from Unif$([0, 10])$.

Intuition: the German tank problem

The uniform model once estimated an enemy's tank production. During the Second World War, Allied statisticians estimated German tank production from the serial numbers on captured tanks. If serial numbers run $1, 2, \ldots, N$ and the captured ones are a random sample, N plays the role of θ, the largest serial number seen is the maximum likelihood estimate, and correcting it for the shortfall of the maximum gives a better one. Their serial-number estimates, which refined this idea and used markings on several components (chassis, gearboxes, wheels), for three months, beside the figures from conventional intelligence and the German records found after the war:[a]

Month	Statistical estimate	Intelligence estimate	German records
June 1940	169	1,000	122
June 1941	244	1,550	271
August 1942	327	1,550	342

The model is crude (serial numbers are not quite a random sample), and its estimates were within 50 tanks of the records while the intelligence estimates were off by a factor of about four to eight.

[a] R. Ruggles and H. Brodie, 'An empirical approach to economic intelligence in World War II', *Journal of the American Statistical Association* 42 (1947), 72–91.

One more property makes maximum likelihood convenient: it doesn't care how the parameter is written.

Proposition 11.6 (Invariance)**.** *Let $g : \Theta \to g(\Theta)$ be one-to-one, and reparameterize the model by $\eta = g(\theta)$. If $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of η.*

Proof strategy (for completeness; skip on a first reading). Reparameterizing relabels the candidates without changing any candidate's likelihood, so the winner is the same distribution under its new label.

Proof. The candidate labeled η in the new parameterization is the distribution $\mathbb{P}_{g^{-1}(\eta)}$, so its likelihood is $L(g^{-1}(\eta))$. This is largest when $g^{-1}(\eta) = \hat{\theta}$, that is, at $\eta = g(\hat{\theta})$. □

For the subway data, the mean gap $\beta = 1/\lambda$ has MLE $1/\hat{\lambda} = \bar{X}_n = 6.23$ minutes. For the boarding data, the standard deviation has MLE $\sqrt{4.35} = 2.09$ minutes. Invariance does not carry unbiasedness along: $\bar{X}_n$ is unbiased for β and $1/\bar{X}_n$ is biased for λ, though both are maximum likelihood estimators.

11.7 When the maximum likelihood estimator fails to exist or isn't unique

Definition 11.3 said 'provided the maximum exists and is attained at a single point'. Both can fail, in ordinary models.

Pitfall: three ways the definition can fail

- **No maximum: the data sit on the boundary.** If every observation in a Bernoulli sample is 0 ($S = 0$), then $L(p) = (1-p)^n$ increases as p decreases and has no maximum on $\Theta = (0, 1)$: the supremum is at $p = 0$, which the model excludes. For the kiss data this can't happen in practice (under $p = 0.645$ its probability is 0.355^{124}), but for $n = 5$ observations of an outcome with $p = 0.1$ it happens 59% of the time. The usual repair is to allow $p \in [0, 1]$, and then $\hat{p} = 0$.

- **More than one maximum.** For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([\theta, \theta + 1])$, $L(\theta) = \mathbb{1}(\theta \leq \min_i X_i)\,\mathbb{1}(\theta + 1 \geq \max_i X_i)$, which equals 1 on the whole interval $[\max_i X_i - 1, \min_i X_i]$ and 0 elsewhere. Every point of that interval is a maximizer (Exercise 11.5).

- **An unbounded likelihood.** In the mixture of two Gaussians with unknown variances, put $\mu_1 = x_1$ and let $\sigma_1 \to 0$. The first factor of the likelihood contains $\pi/(\sigma_1\sqrt{2\pi})$, which goes to infinity, while every other factor stays above $(1 - \pi)$ times the second component's density at its point, which is positive. The likelihood is unbounded: a component of zero width sitting on one data point 'explains' that point infinitely well. The maximum likelihood estimator of a mixture with free variances does not exist, and what is computed in practice is the best local maximum away from these spikes.

11.8 No formula: mixtures and the EM algorithm

The left of Figure 11.7 is a histogram of the 272 waiting times, in minutes, between eruptions of the Old Faithful geyser in Yellowstone National Park.[1] It has two humps, one near 55 minutes and one near 80, and a single Gaussian (orange, dashed) fits it badly. A mixture of two Gaussians is the natural model: each wait comes from a short-wait regime or a long-wait regime, and you don't get to see which.

The log-likelihood of the mixture,

$$\ell_n = \sum_{i=1}^{n} \log\Big(\pi\, f_{\mu_1,\sigma_1^2}(X_i) + (1-\pi)\, f_{\mu_2,\sigma_2^2}(X_i)\Big),$$

has the log of a sum in every term, and setting its gradient to 0 gives five equations with no closed-form solution. The **EM algorithm** (expectation-maximization, given its general form and its name in 1977 and still in everyday use) gets around this with one idea: pretend you know which regime each observation came from.

The complete data

Recall the generative story of Section 2.7: draw $Z_i \sim \text{Bern}(\pi)$; if $Z_i = 1$, draw X_i from component 1, otherwise from component 2. If you observed the labels Z_i along with the X_i, the log-likelihood of the *complete data* (X_i, Z_i) would split:

$$\ell_n^{\text{comp}} = \sum_{i=1}^{n} \Big[Z_i\big(\log\pi + \log f_{\mu_1,\sigma_1^2}(X_i)\big) + (1-Z_i)\big(\log(1-\pi) + \log f_{\mu_2,\sigma_2^2}(X_i)\big)\Big].$$

There is no log of a sum: each observation contributes to exactly one component. Maximizing it is two separate Gaussian fits and one Bernoulli fit, each done in Section 11.6: $\hat\pi$ is the share of observations labeled 1, and $\hat\mu_1$, $\hat\sigma_1^2$ are the mean and divide-by-n variance of the observations labeled 1 (and likewise for component 2).

The E step: guess the labels

You don't observe the Z_i. The best guess for Z_i given X_i, under the current parameter values, is its conditional expectation, which is the probability that X_i came from component 1. By Bayes' formula,

$$w_i = \mathbb{E}[Z_i \mid X_i] = \mathbb{P}(Z_i = 1 \mid X_i) = \frac{\pi\, f_{\mu_1,\sigma_1^2}(X_i)}{\pi\, f_{\mu_1,\sigma_1^2}(X_i) + (1-\pi)\, f_{\mu_2,\sigma_2^2}(X_i)} \in (0,1).$$

A wait of 50 minutes gets w_i close to 1; a wait of 85 gets w_i close to 0; a wait of 67, between the humps, gets something in the middle. Guessing $Z_i \approx \mathbb{E}[Z_i] = \pi$, the same for every observation, would ignore the data entirely; conditioning on X_i is what makes the guess useful.

[1] The version of the data distributed with W. Härdle, *Smoothing Techniques* (Springer, 1991); A. Azzalini and A. W. Bowman, 'A look at some data on the Old Faithful geyser', *Applied Statistics* 39 (1990), 357–365, analyze a longer record.

The M step: fit as if the guesses were labels

Replace each Z_i in ℓ_n^{comp} by w_i and maximize. This is legitimate because ℓ_n^{comp} is linear in each Z_i, so its expectation given the data, under the current parameter values, is ℓ_n^{comp} with each Z_i replaced by w_i. The same calculus as before, with weights, gives

$$\hat{\pi} = \frac{1}{n}\sum_i w_i, \quad \hat{\mu}_1 = \frac{\sum_i w_i X_i}{\sum_i w_i}, \quad \hat{\sigma}_1^2 = \frac{\sum_i w_i (X_i - \hat{\mu}_1)^2}{\sum_i w_i},$$

and the same for component 2 with weights $1 - w_i$ (Exercise 11.9). Each observation is split between the components in proportion to how likely it is to belong to each.

Worked example: the EM algorithm

1. Start from some values of $\pi, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2$.
2. **E step:** compute w_i for every observation from the current values.
3. **M step:** update all five parameters by the weighted formulas.
4. Repeat steps 2 and 3 until the parameters stop changing.

For Old Faithful, starting from means 50 and 70, both standard deviations 10 and $\pi = \frac{1}{2}$, the algorithm settles on weight 0.36 for Gauss$(54.6, 5.87^2)$ and 0.64 for Gauss$(80.1, 5.87^2)$: about a third of the waits are short, averaging 55 minutes, and the rest are long, averaging 80. The fitted mixture is the blue curve in Figure 11.7. Its log-likelihood is -1034.0, against -1095.3 for the best single Gaussian.

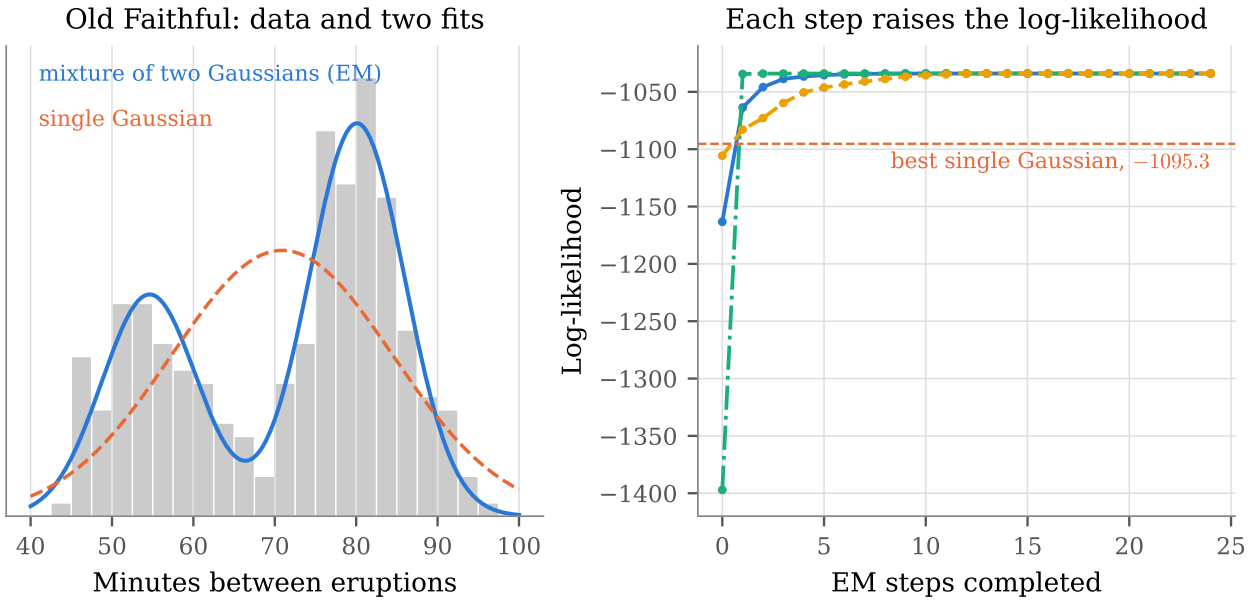


Figure 11.7: Old Faithful waiting times. Left: histogram, best single Gaussian (orange, dashed), EM mixture fit (blue). Right: the log-likelihood after each EM step from starting means $50, 70$ (blue, solid), $40, 95$ (aqua, dash-dot) and $65, 75$ (yellow, dashed); orange line, the single Gaussian.

The right of Figure 11.7 tracks the log-likelihood at the start and after each EM step, from three starting points: the means 50 and 70 of the worked example, and also 40 and 95, and 65 and 75. It shows the property that makes EM trustworthy as far as it goes: every step raises the log-likelihood or leaves it unchanged, and all three runs climb to the same -1034.0, well above the

single Gaussian's −1095.3. That fact is proved in Appendix C.2 of Appendix C, with an argument that uses the nonnegativity of the KL divergence; nothing later in the book depends on it. What it does not guarantee is that EM reaches the highest point. It climbs whatever hill it starts near, which may be a local maximum or one of the spikes of Section 11.7. In practice EM is run from several starting points and the run with the largest log-likelihood is kept. EM is guaranteed to climb, not to reach the top: the results that guarantee it reaches the maximum likelihood estimate assume a starting point close enough to it, or well-separated components.

Pitfall: which component is which

EM's labels 'component 1' and 'component 2' mean nothing. Start it with the means the other way round and it returns the same fit with the labels swapped: weight 0.64 on Gauss$(80.1, 5.87^2)$ and 0.36 on Gauss$(54.6, 5.87^2)$. Both have the same likelihood, since they are the same distribution. This is the label switching of Section 2.7: the mixture's parameter is identifiable only up to relabeling, so every peak of its likelihood has a twin with the labels swapped.

11.9 What comes next

This chapter defined the maximum likelihood estimator and computed it. Chapter 12 asks how good it is, and answers with three results that hold for every well-behaved model at once. It is consistent, by the argument Figure 10.5 drew: the average log-likelihood converges to a function maximized at the truth. It is asymptotically normal, by the Central Limit Theorem applied to the derivative of the log-likelihood. And its asymptotic variance is one over the Fisher information, the curvature that the sanity check of Section 11.4 found matching the standard errors of earlier chapters. Chapter 13 generalizes in two directions: the method of moments, an older and simpler recipe that often gives the same answer, and M-estimation, which maximizes or minimizes averages other than the log-likelihood.

11.10 Summary

- The likelihood $L(\theta) = \prod_i p_\theta(x_i)$ (PMFs) or $\prod_i f_\theta(x_i)$ (densities) is the probability or density of the observed data under each candidate θ: the joint PMF read with the data fixed and the parameter varying. It is not a distribution for θ; only ratios of likelihoods mean something. Kiss data: $L(p) = p^{80}(1-p)^{44}$, 200 times larger at 0.645 than at $\frac{1}{2}$.
- A density works in place of a probability because a recorded value stands for a small interval, whose width cancels in every comparison between parameters.
- The likelihood ignores the order of the data, updates one observation at a time, and is as smooth in θ as the PMF or density. Factors that don't involve θ (the exponential's indicator) can be dropped; factors that do (the uniform's $\mathbb{1}(\theta \geq \max_i x_i)$) cannot.
- MLE: $\hat{\theta}^{\text{MLE}} = \operatorname{argmax}_\theta L = \operatorname{argmax}_\theta \ell_n$, with $\ell_n = \sum_i \log p_\theta(X_i)$. If ℓ_n is strictly concave, the solution of $\ell_n' = 0$ (or $\nabla \ell_n = 0$) is the unique maximizer; in several dimensions, a negative definite Hessian everywhere is enough for strict concavity.
- Bernoulli $\bar{X}_n$ (0.645 for the kiss data); Poisson $\bar{X}_n$ (1.0 for the siblings); exponential $1/\bar{X}_n$ (0.1605 per minute for the subway); Gaussian $\bar{X}_n$ and $\frac{1}{n}\sum(X_i - \bar{X}_n)^2$ (24.2 and 4.35 for boarding); uniform $\max_i X_i$, found at a jump rather than where a derivative vanishes, and biased low by the factor $n/(n+1)$.

- The curvature of the log-likelihood at its peak gives the standard error: $1/\sqrt{541.7} = 0.043$ for the kiss data.
- Invariance: the MLE of $g(\theta)$ is $g(\hat{\theta})$ for one-to-one g. Unbiasedness is not invariant.
- The MLE can fail to exist (all-zero Bernoulli data on $(0,1)$; mixtures with free variances, whose likelihood is unbounded) or be non-unique ($\text{Unif}([\theta, \theta+1])$).
- EM for mixtures: E step $w_i = \mathbb{P}(Z_i = 1 \mid X_i)$ by Bayes' formula; M step, weighted proportions, means and variances. No step lowers the log-likelihood; it can stop at a local maximum, so run it from several starts. Old Faithful: weight 0.36 on mean 54.6, 0.64 on mean 80.1, log-likelihood -1034.0 against -1095.3 for one Gaussian.

11.11 Exercises

Exercise 11.1. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$. You observe $X_1 = 1$, $X_2 = 2$. Compute $L_2(1, 2; \lambda)$. A third observation $X_3 = 3$ arrives: compute $L_3(1, 2, 3; \lambda)$ in two ways, from the definition and by updating L_2. Then compute $L_3(2, 3, 1; \lambda)$ and explain why it is the same. *(Solution on page 421.)*

Exercise 11.2. Let $(\mathbb{Z}, (\mathbb{P}_\theta)_{\theta \in \mathbb{R}})$ be a discrete model whose PMF $p_\theta(x)$ is a continuous function of θ for each fixed x. True or false: for fixed data $x_1, \ldots, x_n$, the likelihood $\theta \mapsto L_n(x_1, \ldots, x_n; \theta)$ is continuous. Give a discrete model with a smooth likelihood. *(Solution on page 421.)*

Exercise 11.3. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(0, \sigma^2)$ with $\sigma^2 > 0$ unknown. Find the MLE of σ^2, check that the log-likelihood has a unique maximum, and use invariance to find the MLE of σ. *(Solution on page 422.)*

Exercise 11.4. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([0, \theta])$ and $M = \max_i X_i$. Show that $\mathbb{P}(M \le m) = (m/\theta)^n$ for $0 \le m \le \theta$, find the density of M, and show that $\mathbb{E}[M] = \frac{n}{n+1}\theta$. Which multiple of M is unbiased? *(Solution on page 422.)*

Exercise 11.5. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([\theta, \theta+1])$ with $\theta \in \mathbb{R}$. Write the likelihood and show that every θ in $[\max_i X_i - 1, \min_i X_i]$ maximizes it. Why is that interval never empty? *(Solution on page 422.)*

Exercise 11.6. A basketball player takes shots until the first one goes in; the number of shots is $X \sim \text{Geom}(p)$, with $\mathbb{P}(X = k) = (1-p)^{k-1}p$ for $k = 1, 2, \ldots$. Over n independent rounds the numbers of shots are $X_1, \ldots, X_n$. Find the MLE of p, and evaluate it for rounds of 3, 1, 2, 5 and 1 shots. *(Solution on page 422.)*

Exercise 11.7. Let $X_1, \ldots, X_n$ be i.i.d. with density $f_a(x) = e^{-(x-a)}\mathbb{1}(x \ge a)$, an exponential shifted to start at an unknown $a \in \mathbb{R}$. Find the MLE of a. *(Solution on page 422.)*

Exercise 11.8. Let $X_1, \ldots, X_n$ be i.i.d. with the Laplace density $f_m(x) = \frac{1}{2}e^{-|x-m|}$, $m \in \mathbb{R}$, and n odd. Show that the MLE of m is the sample median. *(Solution on page 422.)*

Exercise 11.9. Derive the M-step formulas for π and μ_1: maximize $\sum_i [w_i(\log \pi + \log f_{\mu_1, \sigma_1^2}(X_i)) + (1 - w_i)(\log(1-\pi) + \log f_{\mu_2, \sigma_2^2}(X_i))]$ over π and over μ_1. *(Solution on page 422.)*

Exercise 11.10. In the Bernoulli model with $\Theta = (0, 1)$, show that the MLE fails to exist when all n observations are 0. If $p = 0.1$, how likely is that for $n = 5$ and for $n = 50$? *(Solution on page 423.)*

Chapter 12

Properties of the MLE: Consistency, Fisher Information and Asymptotic Normality

In this chapter

- Three facts that hold for the maximum likelihood estimator of every well-behaved model at once: it is consistent, it is asymptotically normal, and its asymptotic variance is one over a number you can compute from the model alone, the Fisher information.
- Why it is consistent: the average log-likelihood is an average, the Law of Large Numbers sends it to a fixed curve, and that curve peaks at the truth. The picture Chapter 10 drew, made into an argument.
- The score: the slope of one observation's log-likelihood. It averages to zero at the truth, and its variance is the Fisher information, which is also the curvature of the log-likelihood at its peak. Both definitions, and why they agree.
- The Fisher information of the Bernoulli, Poisson, exponential and Gaussian models, and what each says about how precisely its parameter can be learned.
- Why the MLE is asymptotically normal: the Central Limit Theorem applied to the score, then one step of the delta method's reasoning. The result is the interval $\hat{\theta} \pm 1.96/\sqrt{nI(\hat{\theta})}$, for any model.
- The Cramér–Rao bound: no unbiased estimator has smaller variance than $1/(nI(\theta))$, so the MLE's precision is the best available.
- What the theorem needs, and the models it does not cover: the uniform model, whose error shrinks like $1/n$, not $1/\sqrt{n}$.

Chapter 11 computed the maximum likelihood estimator. This chapter says how good it is, and makes good on two promises: the consistency argument sketched under Figure 10.5, and the curvature-equals-standard-error coincidence of the sanity check in Section 11.4.

12.1 Three questions about an estimator

Chapter 7 set out what to ask of an estimator: does it get to the truth as the sample grows (consistency), how fast and with what shape of error (asymptotic normality and the asymptotic variance), and how does it compare with the alternatives. For the sample proportion $\bar{R}_n$ those questions were answered by the Law of Large Numbers and the Central Limit Theorem, applied directly to an average. The maximum likelihood estimator of an arbitrary model is not an average. It is the argmax of a random function, and no theorem in Part II says anything about argmaxes.

This chapter's whole trick is to turn the argmax back into an average. It happens twice. For consistency, the thing averaged is the log-likelihood of one observation: $\frac{1}{n}\ell_n(\theta)$ is an average, and the LLN handles it. For asymptotic normality, the thing averaged is the slope of the log-likelihood of one observation, the score: $\frac{1}{n}\ell_n'(\theta)$ is an average, and the CLT handles it. Everything else is bookkeeping, with one Taylor expansion.

12.2 Consistency

Recall. The Law of Large Numbers (Theorem 3.5): if $X_1, \dots, X_n$ are i.i.d. and $h(X)$ has finite variance, then $\frac{1}{n}\sum_{i=1}^{n} h(X_i) \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[h(X)]$.

Fix a candidate θ and look at the average log-likelihood,

$$\frac{1}{n}\ell_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log p_\theta(X_i).$$

With $h = \log p_\theta$, the LLN says this converges in probability, under the truth $\mathbb{P}_{\theta^*}$, to

$$M(\theta) = \mathop{\mathbb{E}}_{\theta^*}\big[\log p_\theta(X)\big] = \mathop{\mathbb{E}}_{\theta^*}\big[\log p_{\theta^*}(X)\big] - \mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta),$$

the split Chapter 10 made. The first term does not depend on θ. The second is a KL divergence, which is 0 at $\theta = \theta^*$ and positive everywhere else if the model is identifiable (Proposition 10.6). So M is a fixed curve, not random, and it has a unique peak at the truth.

Figure 12.1 draws what happens, for one simulated sample of couples at each of $n = 10$, 124 and 1,000. The random curve $\frac{1}{n}\ell_n$, dashed, settles onto the fixed curve $M(p) = p^* \log p + (1 - p^*)\log(1 - p)$, black, as n grows, and its peak, the MLE, slides toward the peak of M, the truth. At $n = 10$ the sample curve peaks at 0.80, far from $p^* = 0.645$; at $n = 1{,}000$ the two curves are indistinguishable and the peaks nearly coincide. The bottom row shows the same thing across 20,000 simulated studies: the sampling distribution of $\hat{p}$ tightens around p^*, with standard deviation $\sqrt{p^*(1-p^*)/n}$ equal to 0.151, 0.043 and 0.015.

Intuition: the peaks follow the curves

The LLN works one θ at a time: for each fixed candidate, the average log-likelihood at that candidate converges. The theorem below needs more, that the whole random curve stays close to the whole fixed curve at once, so that the location of its peak converges too. Here is why that is plausible, and where it can fail. If two curves are within ϵ of each other everywhere, the peak of one is within ϵ of the peak of the other in height. If the fixed curve M drops off decisively away from its peak, as it does in Figure 12.1, a point far from θ^* has M well below $M(\theta^*)$, and

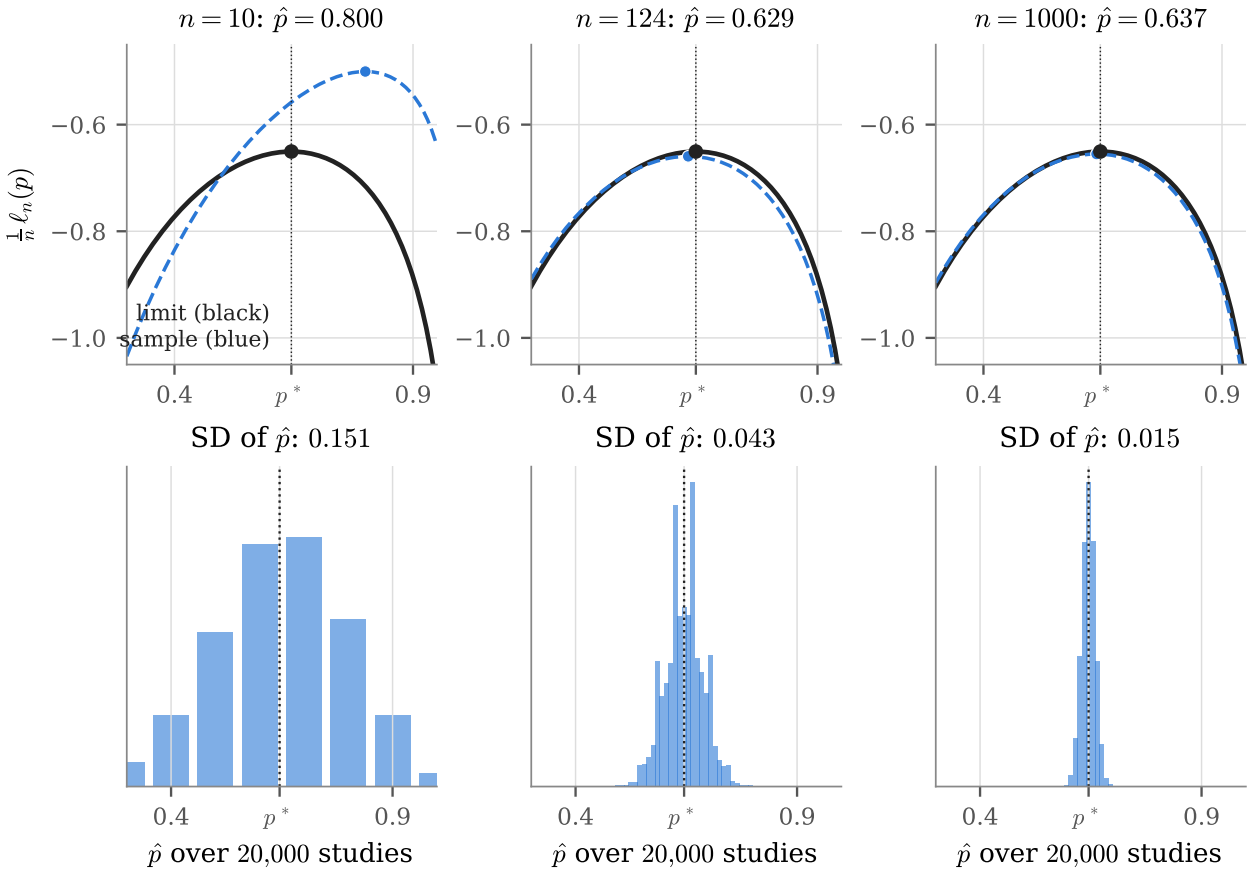


Figure 12.1: The kiss model, $p^* = 0.645$, at $n = 10$, 124 and 1,000. Top: the average log-likelihood of one simulated sample (blue, dashed; peak, blue dot) and its limit $M(p)$ (black; peak, black dot). Bottom: $\hat{p}$ over 20,000 simulated studies.

the random curve there is also well below its value near θ^*, so the random peak cannot be far away. What can go wrong is a parameter space where M never drops off decisively (an infinite Θ with M flattening toward its edge), or a model where the random curve is close to M at every single θ but not uniformly, with thin spikes that move around. The 'technical conditions' in the theorem rule those out, and once the parameter space is restricted to a bounded closed set around the truth, as the paragraph after the theorem explains, they hold for every model in this book except the ones Section 12.6 names.

Theorem 12.1 (Consistency of the MLE)**.** *Let* $X_1, \dots, X_n \overset{iid}{\sim} \mathbb{P}_{\theta^*}$ *in an identifiable model* $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ *with* $\Theta \subset \mathbb{R}^d$*, and let* $\hat{\theta}_n$ *be the maximum likelihood estimator. Suppose that* Θ *is bounded and closed, that* $\log p_\theta(x)$ *is continuous in* θ *for every* x*, and that* $\mathbb{E}_{\theta^*}\left[\left(\sup_{\theta\in\Theta}|\log p_\theta(X)|\right)^2\right] < \infty$*. Then*

$$\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta^* \quad \text{under } \mathbb{P}_{\theta^*}.$$

If the model is misspecified, so that the data come from a $\mathbb{P}$ *outside the family, the same conclusion holds with* θ^* *replaced by the KL projection* θ° *of Definition 10.9, provided it is unique.*

Proof strategy (core). Under the hypotheses, a uniform law of large numbers (proved in Appendix C.3 of Appendix C) upgrades the pointwise LLN to $\sup_\theta|\frac{1}{n}\ell_n(\theta) - M(\theta)| \xrightarrow[n\to\infty]{\mathbb{P}} 0$. Then the argument of the intuition box applies: outside any neighborhood of θ^*, M is at least some $\delta > 0$ below $M(\theta^*)$, and once the random curve is uniformly within $\delta/2$ of M, its maximizer must be inside the neighborhood.

Proof. Write $A_n = \sup_{\theta\in\Theta}|\frac{1}{n}\ell_n(\theta) - M(\theta)|$. The uniform law of large numbers (Theorem C.3) says $A_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$ under the stated hypotheses; the one-θ-at-a-time version is the LLN with $h = \log p_\theta$, and the uniform version is where the continuity, boundedness and the square-integrable envelope $\sup_\theta|\log p_\theta(X)|$ are used. It is the one step of this chapter proved outside it.

Fix $\epsilon > 0$ and let $N = \{\theta : \|\theta - \theta^*\| < \epsilon\}$. Its complement in Θ, $\Theta \setminus N$, is bounded and closed, and M is continuous on it (continuity of $\log p_\theta$ in θ plus the envelope condition), so M attains a maximum on $\Theta \setminus N$ at some θ_1. By identifiability and Proposition 10.6, $M(\theta_1) < M(\theta^*)$; write $\delta = M(\theta^*) - M(\theta_1) > 0$.

Now suppose $A_n < \delta/2$. For every $\theta \in \Theta \setminus N$,

$$\tfrac{1}{n}\ell_n(\theta) \le M(\theta) + A_n < M(\theta_1) + \delta/2 = M(\theta^*) - \delta/2 < M(\theta^*) - A_n \le \tfrac{1}{n}\ell_n(\theta^*).$$

So every θ outside N has a strictly smaller average log-likelihood than θ^* does, and the maximizer $\hat\theta_n$ must lie in N. Hence $\mathbb{P}_{\theta^*}(\|\hat\theta_n - \theta^*\| \ge \epsilon) \le \mathbb{P}_{\theta^*}(A_n \ge \delta/2) \to 0$.

For the misspecified case replace θ^* by θ° and $\mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$ by $\mathrm{KL}(\mathbb{P}, \mathbb{P}_\theta)$ throughout; uniqueness of θ° is what gives $\delta > 0$. □

None of the families in this book has a bounded closed parameter space: the Bernoulli's is $(0, 1)$, the Poisson's $(0, \infty)$, and on the closed interval $[0, 1]$ the Bernoulli envelope $\sup_p|\log p_p(X)|$ is infinite. The theorem is applied to them by restricting Θ to a bounded closed set Θ' that contains θ^* in its interior, for instance $[a, 1-a]$ for the Bernoulli with $0 < a < \min(p^*, 1-p^*)$. The conclusion is unchanged as long as the MLE over all of Θ lands in Θ' with probability tending to 1, and for every family in this chapter it does, since ℓ_n falls to $-\infty$ at the edge of Θ: for the kiss data, $80\log p + 44\log(1-p) \to -\infty$ at both ends of $(0, 1)$, so $\bar R_n \in [a, 1-a]$ once $\bar R_n$ is within $\min(p^*, 1-p^*) - a$ of p^*, which happens with probability tending to 1.

> **In words.** The average log-likelihood is an average, so it converges to a fixed curve. That curve is a constant minus the KL divergence from the truth, so it peaks at the truth. If the random curve hugs the fixed one closely enough everywhere, its peak is near the truth. The same argument, with the truth outside the model, lands on the member of the family closest to the truth.
>
> **For the kiss data.** $M(p) = 0.645\log p + 0.355\log(1-p)$, the black curve in Figure 12.1, peaks at 0.645. The theorem says $\hat p = \bar R_n \xrightarrow[n\to\infty]{\mathbb{P}} p^*$, which Chapter 7 already knew from the LLN directly. The theorem's value is that it needs no formula for the estimator: for the mixture of Section 11.8, once the labels are fixed by requiring $\mu_1 < \mu_2$ so that the model is identifiable, it says the same thing, though the MLE has no formula.

12.3 The score and the Fisher information

Consistency says the MLE gets there. The next question is how fast, and the answer is in the shape of the log-likelihood near its peak. Chapter 11 ended with a sanity check: for the kiss data,

the curvature $-\ell_n''(\hat{p}) = 541.7$ and $1/\sqrt{541.7} = 0.043$, the standard error. This section defines the quantity behind that.

Everything is built from one observation. Write

$$\ell(\theta) = \log p_\theta(X)$$

for the log-likelihood of a single $X \sim \mathbb{P}_\theta$, a random function of θ, and assume it is twice differentiable in θ. Its derivative $\ell'(\theta)$ is called the **score**. For n observations, $\ell_n(\theta) = \sum_i \log p_\theta(X_i)$ is a sum of n independent copies of ℓ, so ℓ_n' is a sum of n independent scores and ℓ_n'' a sum of n independent second derivatives.

Intuition: what the score is

Fix the truth θ^* and draw one observation. The score $\ell'(\theta^*)$ is the slope, at the truth, of that observation's log-likelihood: how much the observation pushes the estimate of θ up or down from θ^*. For the kiss model (Figure 12.4, left, which draws the tangent at p^* on each of the two log-likelihoods) a couple turning right has log-likelihood $\log p$, with slope $1/p^* = 1.55$ at $p^* = 0.645$: it pushes the estimate up. A couple turning left has log-likelihood $\log(1-p)$, with slope $-1/(1-p^*) = -2.82$: it pushes down, harder, since left turns are the rarer event. Weighted by how often each happens, $0.645 \times 1.55 - 0.355 \times 2.82 = 0$, to rounding. At the truth the pushes cancel on average, which is the first fact below. How much they vary around zero, their variance, measures how much one observation can move the estimate: that is the Fisher information.

Proposition 12.2 (The score has mean zero)**.** *Under regularity conditions that allow differentiating under the integral sign (stated in Section 12.6), for every $\theta \in \Theta$,*

$$\mathop{\mathbb{E}}_{\theta}[\ell'(\theta)] = 0.$$

Proof strategy (for completeness; skip on a first reading). The density integrates to 1 for every θ, so the derivative of that integral in θ is 0. Moving the derivative inside turns it into the expectation of the score.

Proof. Take the continuous case; the discrete case replaces the integral by a sum. For every θ, $\int p_\theta(x)\,\mathrm{d}x = 1$. Differentiate both sides in θ, moving the derivative inside the integral:

$$0 = \frac{\mathrm{d}}{\mathrm{d}\theta}\int p_\theta(x)\,\mathrm{d}x = \int \frac{\partial p_\theta(x)}{\partial \theta}\,\mathrm{d}x. \qquad (*)$$

Now $\ell'(\theta) = \partial_\theta \log p_\theta(X) = \partial_\theta p_\theta(X)/p_\theta(X)$, so

$$\mathop{\mathbb{E}}_{\theta}[\ell'(\theta)] = \int \frac{\partial_\theta p_\theta(x)}{p_\theta(x)}\, p_\theta(x)\,\mathrm{d}x = \int \partial_\theta p_\theta(x)\,\mathrm{d}x = 0$$

by $(*)$. □

Definition 12.3 (Fisher information)**.** For a one-parameter model satisfying the regularity conditions of Section 12.6, the **Fisher information** at θ is

$$I(\theta) = \mathop{\mathbb{V}\mathrm{ar}}_{\theta}\left(\ell'(\theta)\right) = \mathop{\mathbb{E}}_{\theta}\left[\ell'(\theta)^2\right],$$

the variance of the score of one observation. For a parameter $\theta \in \mathbb{R}^d$ the score is the gradient $\nabla\ell(\theta)$ and the **Fisher information matrix** is its $d \times d$ covariance matrix, $I(\theta) = \mathbb{C}\mathrm{ov}_\theta(\nabla\ell(\theta))$.

In words. Draw one observation from $\mathbb{P}_\theta$ and ask how steeply its log-likelihood rises or falls at θ. That slope is zero on average. Its spread, the Fisher information, measures how much one observation typically says about θ: a large spread means observations push hard in one direction or the other, and each one carries a lot of information. n independent observations carry n times as much, since the variance of a sum of independent scores is the sum of their variances.

For the kiss data. $\ell'(p) = R/p - (1-R)/(1-p)$. The score takes the value $1/p$ with probability p and $-1/(1-p)$ with probability $1-p$, and its variance is $I(p) = 1/(p(1-p))$ (Section 12.3.1). At $p = 0.645$, $I = 4.37$, and 124 couples carry $nI(\hat{p}) = 541.7$ units of information, the curvature the sanity check of Chapter 11 found.

There is a second way to compute the same number, and it is the one that explains the sanity check.

Theorem 12.4 (Fisher information as curvature)**.** *Under the regularity conditions of Section 12.6,*

$$I(\theta) = -\mathop{\mathbb{E}}_{\theta}\left[\ell''(\theta)\right].$$

In several dimensions, $I(\theta) = -\,\mathbb{E}_\theta[\nabla^2 \ell(\theta)]$, the negative of the expected Hessian.

Proof strategy (core). Differentiate the identity $(*)$ once more, to get $\int \partial_\theta^2 p_\theta = 0$. Write ℓ'' by the quotient rule as $\partial_\theta^2 p_\theta / p_\theta$ minus the squared score. The expectation of the first piece is 0 by the new identity, and the expectation of the second is $I(\theta)$.

Proof. Differentiating $(*)$ in θ once more, again moving the derivative inside,

$$\int \frac{\partial^2 p_\theta(x)}{\partial \theta^2}\,\mathrm{d}x = 0. \tag{$**$}$$

By the quotient rule applied to $\ell'(\theta) = \partial_\theta p_\theta(X)/p_\theta(X)$,

$$\ell''(\theta) = \frac{\partial_\theta^2 p_\theta(X)\cdot p_\theta(X) - \big(\partial_\theta p_\theta(X)\big)^2}{p_\theta(X)^2} = \frac{\partial_\theta^2 p_\theta(X)}{p_\theta(X)} - \left(\frac{\partial_\theta p_\theta(X)}{p_\theta(X)}\right)^2 = \frac{\partial_\theta^2 p_\theta(X)}{p_\theta(X)} - \ell'(\theta)^2.$$

Take expectations under $\mathbb{P}_\theta$. The first term gives $\int \partial_\theta^2 p_\theta(x)\,\mathrm{d}x = 0$ by $(**)$. The second gives $-\,\mathbb{E}_\theta[\ell'(\theta)^2] = -I(\theta)$ by Definition 12.3. So $\mathbb{E}_\theta[\ell''(\theta)] = -I(\theta)$. The matrix version is the same computation entry by entry. □

Intuition: information is curvature

$\ell''(\theta)$ is how sharply one observation's log-likelihood curves at θ, and the theorem says the Fisher information is the average of that curvature, with the sign flipped so that it comes out positive. Figure 12.2 draws it for the kiss model. The two possible log-likelihoods, $\log p$ and $\log(1-p)$, are each curved; their average under p^*, the black curve, peaks at p^* and curves down there with second derivative $-I(p^*) = -4.37$. The aqua parabola through the peak has exactly that curvature. The right of the figure draws $I(p) = 1/(p(1-p))$ itself: smallest at $p = \frac{1}{2}$ and growing without bound at either end. For n observations the log-likelihood is a sum, its curvature near θ^* is about $nI(\theta^*)$, and a sharply curved peak is one whose location is well determined: move a little away from $\hat{\theta}$ and the log-likelihood drops a lot, so few candidates fit nearly as well.

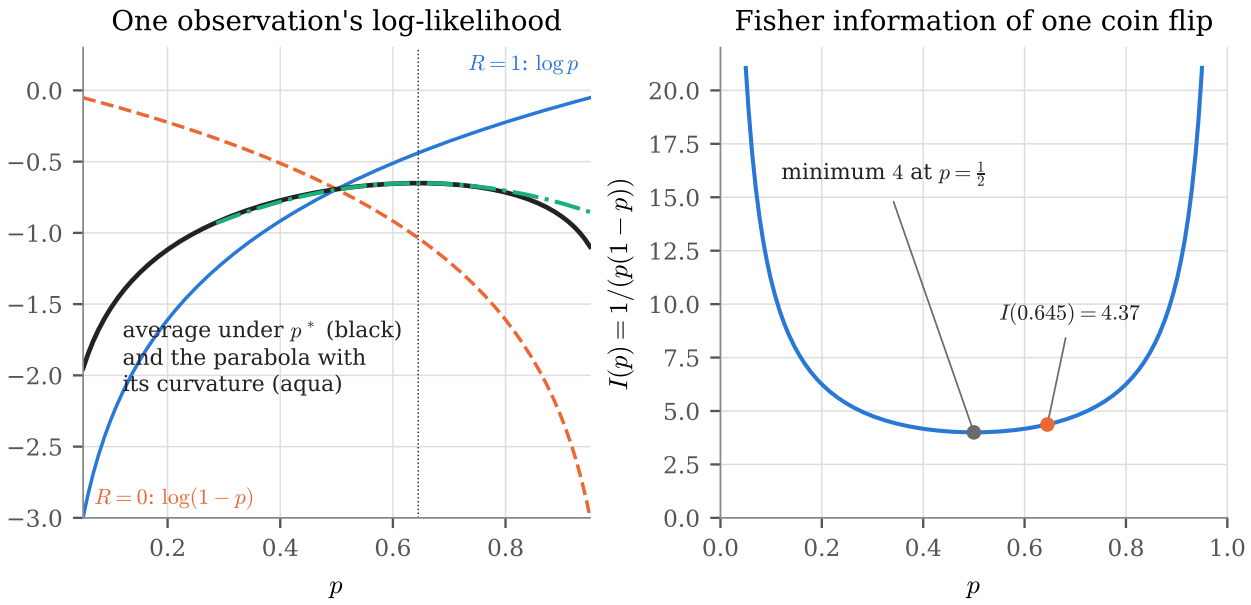


Figure 12.2: Left: one couple's log-likelihood, $\log p$ (right turn, blue) and $\log(1-p)$ (left turn, orange, dashed), their average at $p^* = 0.645$ (black), and a parabola through its peak (aqua). Right: $I(p) = 1/(p(1-p))$ for the Bernoulli model.

Does this make sense? Which coin is most informative?

The right of Figure 12.2 says a single flip of a fair coin carries the least information about p, $I(\frac{1}{2}) = 4$, and a flip of a heavily biased coin carries the most, $I(0.9) = 11.1$. Is that backwards? A fair coin's outcome is the most unpredictable, so it feels the most informative. But the question is not how surprising the flip is; it is how much the flip changes your estimate of p. The variance of $\hat{p}$ is $p(1-p)/n$, which is largest at $p = \frac{1}{2}$: a fair coin's p is the hardest to pin down in absolute terms, since its flips vary the most. For $p = 0.9$ nearly every flip is heads, the sample proportion barely moves from sample to sample, and p is learned to within a narrow margin. The Fisher information is the reciprocal of that variance per observation, $1/(p(1-p))$, so it says exactly this.

12.3.1 Fisher information of the standard families

Worked example: four Fisher informations

In each case the curvature formula of Theorem 12.4 is the quicker route; the variance of the score is computed once as a check.

- **Bernoulli.** $\ell(p) = R\log p + (1-R)\log(1-p)$, so $\ell'(p) = R/p - (1-R)/(1-p)$ and $\ell''(p) = -R/p^2 - (1-R)/(1-p)^2$. With $\mathbb{E}_p[R] = p$,

$$I(p) = -\mathop{\mathbb{E}}_p[\ell''(p)] = \frac{p}{p^2} + \frac{1-p}{(1-p)^2} = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}.$$

  As a check, $\ell'(p) = R\big(\frac{1}{p} + \frac{1}{1-p}\big) - \frac{1}{1-p}$ is a linear function of R, so $\mathbb{V}\mathrm{ar}_p(\ell'(p)) = \big(\frac{1}{p(1-p)}\big)^2 \mathbb{V}\mathrm{ar}(R) = \frac{p(1-p)}{p^2(1-p)^2} = \frac{1}{p(1-p)}$. The same.

- **Poisson.** $\ell(\lambda) = -\lambda + X\log\lambda - \log(X!)$, $\ell''(\lambda) = -X/\lambda^2$, and $\mathbb{E}_\lambda[X] = \lambda$ gives $I(\lambda) = 1/\lambda$.
- **Exponential.** $\ell(\lambda) = \log\lambda - \lambda X$, $\ell''(\lambda) = -1/\lambda^2$ with no X in it, so $I(\lambda) = 1/\lambda^2$. For the subway rate $\hat\lambda = 0.1605$ per minute, $I = 38.8$.
- **Gaussian, both parameters unknown.** With $\theta = (\mu, \sigma^2)$ and $v = \sigma^2$, $\ell = -\frac{1}{2}\log(2\pi v) - (X-\mu)^2/(2v)$. The second partial derivatives are

$$\frac{\partial^2\ell}{\partial\mu^2} = -\frac{1}{v}, \qquad \frac{\partial^2\ell}{\partial\mu\,\partial v} = -\frac{X-\mu}{v^2}, \qquad \frac{\partial^2\ell}{\partial v^2} = \frac{1}{2v^2} - \frac{(X-\mu)^2}{v^3}.$$

  Under $\mathbb{P}_\theta$, $\mathbb{E}[X-\mu] = 0$ and $\mathbb{E}[(X-\mu)^2] = v$, so

$$I(\mu, \sigma^2) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 1/(2\sigma^4) \end{pmatrix}.$$

  The off-diagonal zero says that, for a Gaussian, learning about μ and learning about σ^2 don't interfere with each other.

Figure 12.3 draws two of these as functions of the parameter: $1/\lambda^2$ for the exponential, where the subway rate 0.16 has $I = 38.8$, and $1/\lambda$ for the Poisson, where the siblings fit $\hat\lambda = 1$ has $I = 1$. The right of the figure shows what the information buys: the width $2 \times 1.96/\sqrt{nI(p)}$ of the 95% interval for a proportion at $n = 100$, which is widest at $p = \frac{1}{2}$ and is 0.19 at $p = 0.645$.

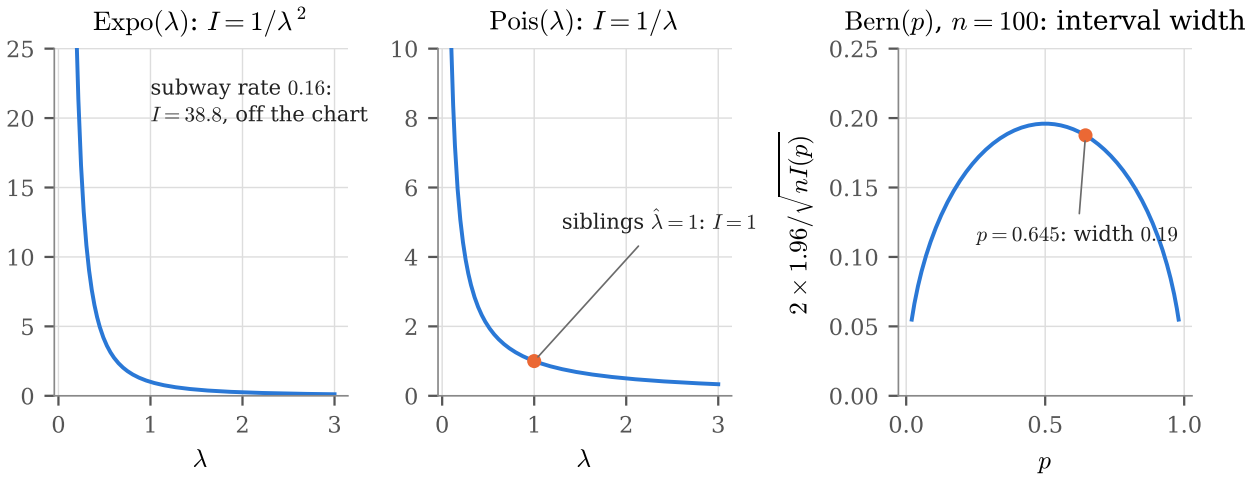


Figure 12.3: Fisher information in three one-parameter families. Left: $I(\lambda) = 1/\lambda^2$ for Expo(λ). Middle: $I(\lambda) = 1/\lambda$ for Pois(λ). Right: the width $2 \times 1.96/\sqrt{nI(p)}$ of the 95% interval for a proportion at $n = 100$, as a function of p.

Pitfall: Fisher information depends on how the parameter is written

The exponential model has $I(\lambda) = 1/\lambda^2$ in the rate λ. Written in the mean gap $\beta = 1/\lambda$ it has $I(\beta) = 1/\beta^2$, a different function (Exercise 12.2). The information is a property of the model *and* the parameterization: it measures information per unit of θ, and a unit of λ is not a unit of β. The general rule is $I(\beta) = I(\lambda)\,(\mathrm{d}\lambda/\mathrm{d}\beta)^2$, which is the delta method's $g'(\theta)^2$ seen from the other side. What does not depend on the parameterization is the conclusion drawn from it: the

95% interval for β obtained from $I(\beta)$ is the image under $\beta = 1/\lambda$ of the interval for λ, to first order.

12.4 Asymptotic normality

Recall. The Central Limit Theorem (Theorem 4.8): if $Y_1, Y_2, \ldots$ are i.i.d. with mean m and variance $s^2 < \infty$, then $\sqrt{n}(\bar{Y}_n - m) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, s^2)$.

Apply it to the scores $Y_i = \ell_i'(\theta^*)$, where $\ell_i(\theta) = \log p_\theta(X_i)$. They are i.i.d., with mean 0 (Proposition 12.2) and variance $I(\theta^*)$ (Definition 12.3), so

$$\sqrt{n} \cdot \frac{1}{n}\ell_n'(\theta^*) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \ell_i'(\theta^*) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}\big(0, I(\theta^*)\big).$$

The middle of Figure 12.4 shows it for the kiss model at $n = 124$: the average score at p^* over 20,000 simulated studies, with the Gauss$(0, I(p^*)/n)$ the CLT predicts, standard deviation 0.188.

That is a statement about the slope of the log-likelihood at the truth. The MLE is where the slope is zero. To connect them, expand the slope around the truth.

Recall. The mean value theorem (toolkit A.9): for a differentiable g, $g(b) - g(a) = g'(c)(b - a)$ for some c between a and b. Slutsky's theorem (Theorem 5.1): if $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$ and $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$, a constant, then $X_n Y_n \xrightarrow[n\to\infty]{\text{(d)}} yX$ and $X_n + Y_n \xrightarrow[n\to\infty]{\text{(d)}} X + y$. The continuous mapping theorem (Theorem 5.2): if $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$ and f is continuous at y, then $f(Y_n) \xrightarrow[n\to\infty]{\mathbb{P}} f(y)$.

Theorem 12.5 (Asymptotic normality of the MLE)**.** *Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathbb{P}_{\theta^*}$ in a model satisfying the regularity conditions of Section 12.6, with θ^* in the interior of Θ and $I(\theta^*) > 0$, and let $\hat{\theta}_n$ be the MLE. Then*

$$\sqrt{n}\,(\hat{\theta}_n - \theta^*) \xrightarrow[n\to\infty]{(d)} \textit{Gauss}\Big(0, \frac{1}{I(\theta^*)}\Big) \quad \textit{under } \mathbb{P}_{\theta^*}.$$

For $\theta \in \mathbb{R}^d$, the limit is the d-dimensional Gaussian with mean 0 and covariance matrix $I(\theta^*)^{-1}$.

Proof strategy (core). The slope of ℓ_n is 0 at $\hat{\theta}_n$ and, by the CLT, approximately Gauss$(0, nI(\theta^*))$ at θ^*. The mean value theorem writes the difference between the two slopes as (curvature somewhere in between) $\times$ $(\hat{\theta}_n - \theta^*)$. The curvature in between is an average of second derivatives, which the LLN sends to $-I(\theta^*)$ once consistency pulls the in-between point to θ^*. Dividing the Gaussian by $-I(\theta^*)$ gives the result, with Slutsky doing the division.

Proof. Since θ^* is interior and $\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta^*$ (Theorem 12.1), with probability tending to 1 the MLE is interior too, and then the first-order condition holds: $\ell_n'(\hat{\theta}_n) = 0$. Apply the mean value theorem to ℓ_n' between θ^* and $\hat{\theta}_n$: there is a $\tilde{\theta}_n$ between them with

$$0 = \ell_n'(\hat{\theta}_n) = \ell_n'(\theta^*) + \ell_n''(\tilde{\theta}_n)\,(\hat{\theta}_n - \theta^*).$$

Rearrange and scale by $\sqrt{n}$:

$$\sqrt{n}\,(\hat{\theta}_n - \theta^*) = \frac{\frac{1}{\sqrt{n}}\ell_n'(\theta^*)}{-\frac{1}{n}\ell_n''(\tilde{\theta}_n)}.$$

The numerator converges in distribution to $\text{Gauss}(0, I(\theta^*))$ by the CLT applied to the scores, as above. For the denominator, $-\frac{1}{n}\ell_n''(\theta^*) = -\frac{1}{n}\sum_i \ell_i''(\theta^*) \xrightarrow[n\to\infty]{\mathbb{P}} -\mathbb{E}_{\theta^*}[\ell''(\theta^*)] = I(\theta^*)$ by the LLN and Theorem 12.4; and since $\tilde{\theta}_n$ is squeezed between θ^* and $\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta^*$, the bound $|\ell_i'''(\theta)| \le K(X_i)$ near θ^* from the regularity conditions gives, by the mean value theorem once more,

$$\left|\tfrac{1}{n}\ell_n''(\tilde{\theta}_n) - \tfrac{1}{n}\ell_n''(\theta^*)\right| \le \left(\tfrac{1}{n}\sum_{i=1}^{n} K(X_i)\right)|\hat{\theta}_n - \theta^*| \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}_{\theta^*}[K(X)]\cdot 0 = 0,$$

by the LLN for the average of the $K(X_i)$ and Slutsky's product rule; so the denominator converges in probability to $I(\theta^*)$ as well. Since $I(\theta^*) > 0$, the function $x \mapsto 1/x$ is continuous there, and the continuous mapping theorem gives $1/(-\frac{1}{n}\ell_n''(\tilde{\theta}_n)) \xrightarrow[n\to\infty]{\mathbb{P}} 1/I(\theta^*)$. Slutsky's theorem multiplies the two:

$$\sqrt{n}\,(\hat{\theta}_n - \theta^*) \xrightarrow[n\to\infty]{\text{(d)}} \frac{1}{I(\theta^*)}\cdot \text{Gauss}(0, I(\theta^*)) = \text{Gauss}\Big(0, \frac{I(\theta^*)}{I(\theta^*)^2}\Big) = \text{Gauss}\Big(0, \frac{1}{I(\theta^*)}\Big).$$

In d dimensions (toolkit A.12) each coordinate of the gradient gets its own in-between point $\tilde{\theta}_{n,j}$, all squeezed to θ^*; the denominator becomes the matrix whose row j is $-\frac{1}{n}$ times row j of the Hessian evaluated at $\tilde{\theta}_{n,j}$, which converges in probability to $I(\theta^*)$ entry by entry; the numerator converges to a Gaussian vector with covariance $I(\theta^*)$ by the multivariate CLT (Section 5.5); and the inverse matrix multiplies it, giving covariance $I^{-1}II^{-1} = I(\theta^*)^{-1}$. □

The right of Figure 12.4 is the theorem for the kiss model: $\hat{p}$ over the same 20,000 studies, with the $\text{Gauss}(p^*, 1/(nI(p^*)))$ it predicts, standard deviation 0.043. Beside the middle histogram it is the same picture, since to first order $\hat{p} - p^*$ is the average score divided by $I(p^*)$, which is what the rearranged mean value step in the proof says.

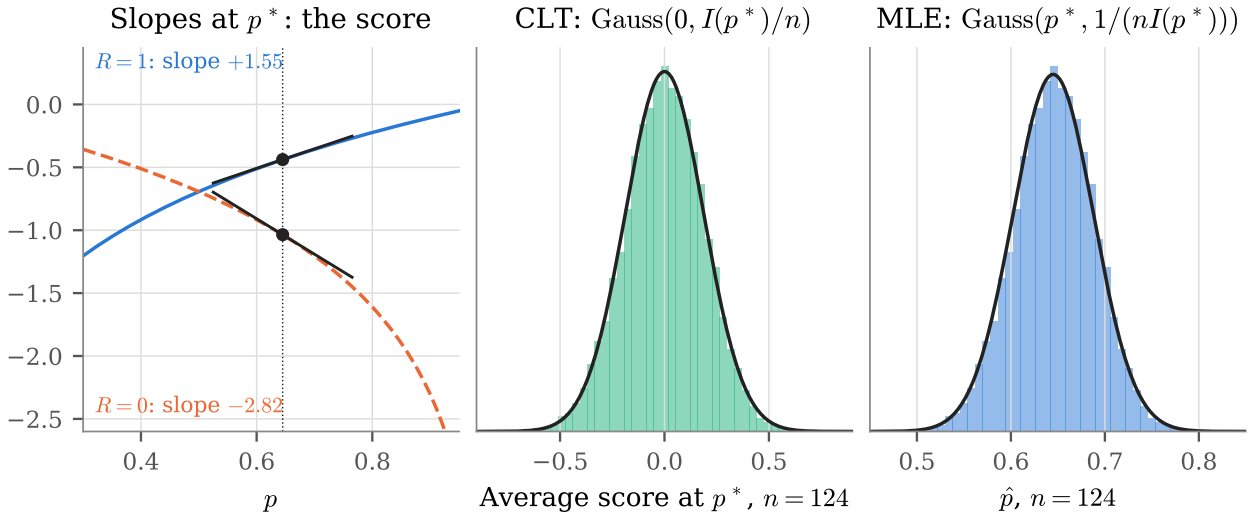


Figure 12.4: Kiss model, $p^* = 0.645$, $n = 124$. Left: $\log p$ (blue) and $\log(1-p)$ (orange, dashed), with tangents at p^*. Middle: the average score at p^* over 20,000 studies, with the CLT's Gaussian. Right: $\hat{p}$ over those studies, with the theorem's Gaussian.

Intuition: why the variance is one over the information

Two forces set the size of the MLE's error. The slope of the log-likelihood at the truth is random, with standard deviation $\sqrt{nI}$ (the CLT): that is how far the data push. The curvature near the peak is about nI (the LLN): that is how hard the log-likelihood resists. Displacement is push divided by resistance, $\sqrt{nI}/(nI) = 1/\sqrt{nI}$, which is the standard deviation of $\hat{\theta}_n$. More information means both a bigger push and a bigger resistance, and the resistance wins, since it grows like n while the push grows like $\sqrt{n}$.

In words. For a large sample, the maximum likelihood estimator is approximately Gaussian, centered at the truth, with standard deviation $1/\sqrt{nI(\theta^*)}$. Everything about its precision is in one number computed from the model, and the recipe estimate $\pm 1.96\,/\sqrt{nI(\hat{\theta}_n)}$ gives a 95% confidence interval for any regular model, with no need to work out the estimator's distribution by hand.

For the kiss data. $I(p) = 1/(p(1-p))$, so $1/(nI(p^*)) = p^*(1-p^*)/n$: the theorem reproduces the CLT for $\bar{R}_n$ exactly, with asymptotic variance 0.229 and standard error 0.043 at $n = 124$. The interval $0.645 \pm 1.96 \times 0.043 = [0.561, 0.729]$ is the plug-in interval of Chapter 8. In natural frequencies: of 1,000 studies of 124 couples with $p^* = 0.645$, about 683 land within 0.043 of the truth (the exact Binomial count is 698).

From undergraduate statistics: one recipe for every standard error

In an introductory course each estimator came with its own standard error formula: $\sqrt{\hat{p}(1-\hat{p})/n}$ for a proportion, $s/\sqrt{n}$ for a mean, something from software for a regression coefficient. Theorem 12.5 is where those formulas come from, and it is one formula: the standard error of a maximum likelihood estimator is $1/\sqrt{nI(\hat{\theta})}$, or in practice $1/\sqrt{-\ell_n''(\hat{\theta})}$, the reciprocal square root of the curvature of the log-likelihood at its peak. For a proportion that is $\sqrt{\hat{p}(1-\hat{p})/n}$, exactly. For a Gaussian mean it is $\sqrt{\widehat{\sigma^2}/n}$, which is $s/\sqrt{n}$ up to the factor $\sqrt{(n-1)/n}$. For a regression coefficient, a logistic regression coefficient, a Poisson rate, the same recipe, which is what software computes when it reports a standard error next to a maximum likelihood estimate.

Worked example: standard errors from the Fisher information

- **Subway rate.** $I(\lambda) = 1/\lambda^2$, so the standard error of $\hat{\lambda}$ is $\hat{\lambda}/\sqrt{n} = 0.1605/8 = 0.020$ per minute, the number Chapter 5 obtained by the delta method. A 95% interval: 0.1605 ± 0.039, or $[0.121, 0.200]$.
- **Siblings.** $I(\lambda) = 1/\lambda$, so the standard error of $\hat{\lambda} = 1.0$ from 24 students is $\sqrt{1.0/24} = 0.204$: with so few observations, the Poisson rate is known only to about ± 0.4.
- **Boarding times.** The inverse information matrix is $\mathrm{diag}(\sigma^2, 2\sigma^4)$, so $\hat{\mu}$ has standard error $\sqrt{4.35/72} = 0.25$ minutes and $\widehat{\sigma^2}$ has standard error $\sqrt{2 \times 4.35^2/72} = 0.73$. The mean boarding time 24.2 is known to a quarter of a minute; the variance 4.35 only to within about 1.4.

12.5 The Cramér–Rao bound

Is $1/(nI(\theta))$ a good variance, or merely the MLE's variance? It is the best possible, in the following sense.

Theorem 12.6 (Cramér–Rao bound)**.** *Under the regularity conditions of Section 12.6, let $T = T(X_1, \dots, X_n)$ be any unbiased estimator of θ with finite variance. Then for every θ,*

$$\operatorname*{\mathbb{V}ar}_{\theta}(T) \geq \frac{1}{nI(\theta)}.$$

Proof strategy (for completeness; skip on a first reading). Unbiasedness, differentiated in θ, says the covariance of T with the total score $\ell_n'(\theta)$ is exactly 1. The Cauchy–Schwarz inequality bounds a covariance by the product of standard deviations, and the total score has variance $nI(\theta)$.

Proof. Unbiasedness says $\int T(x)\, p_\theta^{(n)}(x)\, \mathrm{d}x = \theta$ for every θ, where $p_\theta^{(n)} = \prod_i p_\theta(x_i)$ is the joint density. Differentiate in θ, moving the derivative inside (which the finite variance of T and the regularity conditions permit), and use $\partial_\theta p_\theta^{(n)} = p_\theta^{(n)}\, \ell_n'(\theta)$:

$$1 = \int T(x)\, \ell_n'(\theta)\, p_\theta^{(n)}(x)\, \mathrm{d}x = \operatorname*{\mathbb{E}}_{\theta}\left[T\, \ell_n'(\theta)\right] = \operatorname*{\mathbb{C}ov}_{\theta}\left(T, \ell_n'(\theta)\right),$$

the last step because $\mathbb{E}_\theta[\ell_n'(\theta)] = 0$. By the Cauchy–Schwarz inequality for covariances, $\mathbb{C}\mathrm{ov}(T,S)^2 \leq \mathbb{V}\mathrm{ar}(T)\,\mathbb{V}\mathrm{ar}(S)$ (take $c = \mathbb{C}\mathrm{ov}(T,S)/\mathbb{V}\mathrm{ar}(S)$ in $0 \leq \mathbb{V}\mathrm{ar}(T - cS) = \mathbb{V}\mathrm{ar}(T) - 2c\,\mathbb{C}\mathrm{ov}(T,S) + c^2\,\mathbb{V}\mathrm{ar}(S)$), with $S = \ell_n'(\theta)$ and $\mathbb{V}\mathrm{ar}_\theta(S) = nI(\theta)$ (a sum of n independent scores):

$$1 \leq \operatorname*{\mathbb{V}ar}_{\theta}(T) \cdot nI(\theta).$$

□

An unbiased estimator whose variance equals the bound is called **efficient**. The sample proportion is efficient for p: $\mathbb{V}\mathrm{ar}(\bar{R}_n) = p(1-p)/n = 1/(nI(p))$ exactly, at every n. Figure 12.5 draws the bound for the kiss model at $n = 124$, with the variance of $\bar{R}_n$ lying exactly on it, and beside them the variance and the mean squared error of the shrinkage estimator $(n\bar{R}_n + 10)/(n + 20)$ of Chapter 7; the mean squared error rises above the bound once p is more than about 0.16 from $\frac{1}{2}$. The MLE in general is not unbiased at finite n (the subway $1/\bar{T}_n$ is not), so the bound does not apply to it directly; what Theorem 12.5 says is that its asymptotic variance equals the bound, so that in large samples nothing unbiased can do better.

Pitfall: a bound for unbiased estimators only

The shrinkage estimator in Figure 12.5 has variance below the Cramér–Rao bound for every p. That does not contradict the theorem: the estimator is biased, and the theorem is about unbiased ones. Bias can buy variance, as Chapter 7 showed, and the honest comparison for a biased estimator is its mean squared error, which for the shrinkage estimator is below $\bar{R}_n$'s only near $p = \frac{1}{2}$. A general version of the bound, with the bias's derivative in the numerator, covers biased estimators too; it is not needed here.

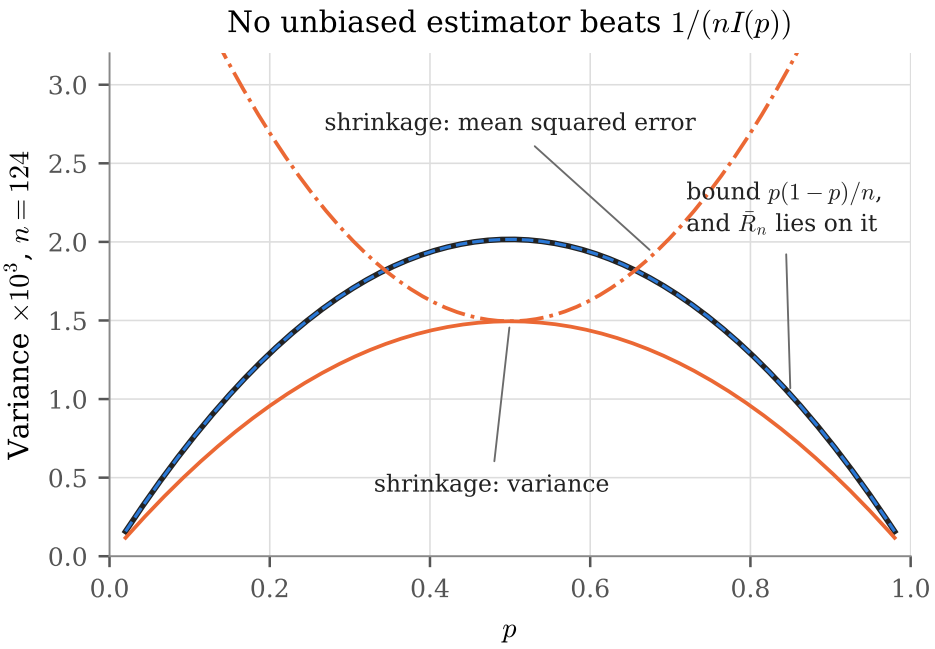


Figure 12.5: The kiss model at $n = 124$: the Cramér–Rao bound $p(1-p)/n$ (black), the variance of $\bar{R}_n$ (blue, dashed), and the variance (orange, solid) and mean squared error (orange, dash-dot) of the shrinkage estimator $(n\bar{R}_n + 10)/(n + 20)$ of Chapter 7.

12.6 What the theorems need, and the models they do not cover

The results above hold for **regular** models, which means:

1. the parameter is identifiable (Section 2.6);
2. the support of $\mathbb{P}_\theta$, the set where $p_\theta > 0$, does not depend on θ;
3. the true θ^* is not on the boundary of Θ;
4. $p_\theta(x)$ is three times differentiable in θ, the derivatives can be moved inside the integral (which gave $(*)$ and $(**)$, and the corresponding step in the Cramér–Rao proof), $\ell''(\theta^*)$ has finite variance, and $|\ell'''(\theta)| \le K(X)$ near θ^* for a K with finite variance;
5. $I(\theta) > 0$ near θ^*, and in d dimensions $I(\theta)$ is invertible.

Every family in this book except the uniform satisfies all five. Here is what each one guards against.

- **Identifiability** is what makes the limit curve M peak at one point. In the Gaussian mixture of Section 2.7, two parameter values give the same distribution, so M has two peaks and the MLE converges to one of them, with the labels decided by the starting point.

- **Fixed support** is what allows differentiating under the integral sign. For $\mathrm{Unif}([0,\theta])$ the support is $[0,\theta]$, which moves with θ; the identity $(*)$ fails, the score is not defined at the observations' maximum, and the theorems do not apply. And they are false there: the MLE $\max_i X_i$ has error of exact order $1/n$, not $1/\sqrt{n}$, as Exercise 11.4 found. The limit of $n(\theta - \max_i X_i)$ is exponential, not Gaussian (Exercise 12.5). The uniform MLE is better than regular, not worse: its error shrinks faster than any regular estimator's.

- **Interior truth** is what allows $\ell_n'(\hat{\theta}_n) = 0$. If θ^* sits on the boundary, say $\Theta = [0,\infty)$ for a

Gaussian mean with $\mu^* = 0$, the MLE is $\max(\bar{X}_n, 0)$: equal to 0 half the time and to the positive half of a Gaussian otherwise, not Gaussian.

- **Smoothness** is used in the mean value step and the LLN for ℓ''. The Laplace model of Exercise 11.8, whose log-likelihood has a corner at each observation, needs a separate argument (and its MLE, the median, is asymptotically normal all the same, with variance 1).
- **Positive information** is what makes the division in the proof legitimate. $I(\theta^*) = 0$ means the log-likelihood is flat at the truth to second order: the data say nothing about θ to first order, and the error shrinks more slowly than $1/\sqrt{n}$.

Does this make sense? Faster than $1/\sqrt{n}$

For Unif$([0, \theta])$ with $\theta = 10$, Exercise 11.4 gave $\mathbb{E}[\max_i X_i] = \frac{n}{n+1}\theta$, so the bias of the MLE is $-\theta/(n+1)$: about -0.1 at $n = 100$ and -0.01 at $n = 1{,}000$, shrinking tenfold when n grows tenfold. A regular estimator's error shrinks by $\sqrt{10} = 3.2$ when n grows tenfold. So the uniform MLE is converging faster than any regular model allows, and the Cramér–Rao bound, which would put a floor of order $1/n$ under its variance, does not hold for it either. No contradiction: the bound needs $(*)$, which needs a fixed support.

12.7 What comes next

This chapter finished the theory of maximum likelihood: consistent, asymptotically normal, with asymptotic variance $1/I(\theta^*)$ and the Cramér–Rao bound saying that is as good as unbiased estimation gets. Chapter 13 puts two other recipes beside it. The method of moments matches sample moments to population moments, and the delta method gives its asymptotic variance, which is at least $1/I(\theta^*)$ and usually larger. M-estimation replaces $\log p_\theta$ by any function ρ whose expectation peaks at the quantity of interest; the argument of this chapter, an average, a limit curve, a score and a curvature, goes through word for word, with $I(\theta)$ replaced by a sandwich of two matrices. Chapter 15 builds tests from this chapter's theorem: Wald's test standardizes $\hat{\theta}_n - \theta_0$ by $1/\sqrt{nI(\hat{\theta}_n)}$, and the likelihood ratio test compares $\ell_n(\hat{\theta}_n)$ with $\ell_n(\theta_0)$, whose difference is approximately $\frac{1}{2}nI(\theta^*)(\hat{\theta}_n - \theta_0)^2$ by the same Taylor expansion, and so approximately half a chi-squared.

12.8 Summary

- Consistency: $\frac{1}{n}\ell_n(\theta) \xrightarrow[n\to\infty]{\mathbb{P}} M(\theta) = \text{constant} - \mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta)$ by the LLN; M peaks at θ^* (or at the KL projection θ° if the model is wrong); under a uniform version of the LLN the peak of $\frac{1}{n}\ell_n$ follows the peak of M, so $\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta^*$.
- The score $\ell'(\theta)$ is the slope of one observation's log-likelihood. $\mathbb{E}_\theta[\ell'(\theta)] = 0$, from differentiating $\int p_\theta = 1$.
- Fisher information: $I(\theta) = \mathbb{V}\mathrm{ar}_\theta(\ell'(\theta)) = -\mathbb{E}_\theta[\ell''(\theta)]$, the variance of the score and the expected curvature of the log-likelihood. n observations carry $nI(\theta)$. Bernoulli $1/(p(1-p))$ (kiss: 4.37, 124 couples: 541.7); Poisson $1/\lambda$; exponential $1/\lambda^2$; Gaussian $\mathrm{diag}(1/\sigma^2, 1/(2\sigma^4))$. It changes with the parameterization: $I(\beta) = I(\lambda)(\mathrm{d}\lambda/\mathrm{d}\beta)^2$.

- Asymptotic normality: $\sqrt{n}(\hat{\theta}_n - \theta^*) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, 1/I(\theta^*))$, by the CLT on the scores, the mean value theorem on ℓ_n', the LLN on ℓ_n'' and Slutsky. Push $\sqrt{nI}$ over resistance nI. The 95% interval is $\hat{\theta}_n \pm 1.96/\sqrt{nI(\hat{\theta}_n)}$ for any regular model: kiss $[0.561, 0.729]$, subway $[0.121, 0.200]$.
- Cramér–Rao: every unbiased estimator has $\mathbb{V}\text{ar} \geq 1/(nI(\theta))$. $\bar{R}_n$ attains it. Biased estimators can go below it (shrinkage), and are judged by mean squared error.
- Regular models: identifiable, fixed support, interior truth, three derivatives that pass under the integral, positive information. The uniform model fails fixed support, and its MLE converges at rate $1/n$ with a non-Gaussian limit; mixtures fail identifiability.

12.9 Exercises

Exercise 12.1. For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$, compute $I(\lambda)$ both as $\mathbb{V}\text{ar}(\ell'(\lambda))$ and as $-\mathbb{E}[\ell''(\lambda)]$, state the asymptotic distribution of $\hat{\lambda} = \bar{X}_n$, and check it against the CLT applied directly to $\bar{X}_n$. *(Solution on page 423.)*

Exercise 12.2. For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Expo}(\lambda)$, let $\beta = 1/\lambda$ be the mean gap. Write the log-likelihood in β, compute $I(\beta)$, and verify $I(\beta) = I(\lambda)(\mathrm{d}\lambda/\mathrm{d}\beta)^2$. State the asymptotic distribution of $\hat{\beta} = \bar{X}_n$ and check it against the CLT. *(Solution on page 423.)*

Exercise 12.3. For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$ with σ^2 known, compute $I(\mu)$ and show that $\bar{X}_n$ is efficient: its variance equals the Cramér–Rao bound at every n. *(Solution on page 423.)*

Exercise 12.4. For the kiss data, compute the total score $\ell_n'(p)$ at $p = \frac{1}{2}$, 0.6, 0.645 and 0.7, and the standardized score $\ell_n'(p_0)/\sqrt{nI(p_0)}$ at $p_0 = \frac{1}{2}$. Compare the latter with the test statistic of Section 9.6. *(Solution on page 423.)*

Exercise 12.5. For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Unif}([0, \theta])$ and $M = \max_i X_i$, show that $n(\theta - M)/\theta \xrightarrow[n\to\infty]{\text{(d)}} \text{Expo}(1)$. Conclude that the MLE's error is of order $1/n$ and its limit is not Gaussian. *(Solution on page 423.)*

Exercise 12.6. For the geometric model of Exercise 11.6, $\mathbb{P}(X = k) = (1-p)^{k-1}p$, compute $I(p)$ and the standard error of $\hat{p} = 1/\bar{X}_n$ for the five rounds of $3, 1, 2, 5, 1$ shots. *(Solution on page 423.)*

Exercise 12.7. Show directly that the Fisher information of n i.i.d. observations, defined as the variance of the total score $\ell_n'(\theta)$, is $nI(\theta)$, and that it is also $-\mathbb{E}_\theta[\ell_n''(\theta)]$. *(Solution on page 423.)*

Exercise 12.8. Data come from Expo(1) but are fitted with the Gaussian family. Proposition 10.10 says the KL projection is Gauss(1, 1). What does Theorem 12.1 say the Gaussian MLEs $(\bar{X}_n, \frac{1}{n}\sum(X_i - \bar{X}_n)^2)$ converge to, and does that agree with the LLN applied directly? *(Solution on page 424.)*

Chapter 13

Method of Moments and M-Estimation

In this chapter

- The method of moments: match the sample mean, mean square and so on to their population versions, and solve for the parameter. Older than maximum likelihood, often the same answer, always a formula.
- Its asymptotic variance, by the delta method, and why it is never smaller than the maximum likelihood estimator's.
- M-estimation: the general recipe behind both. Pick a loss whose expected value is smallest at the quantity you want, average the loss over the data, minimize. The sample mean, the median, every quantile and the MLE are all instances.
- The asymptotic variance of an M-estimator, $J^{-1}KJ^{-1}$, by the argument of Chapter 12 with two matrices in place of one. The Laplace median has asymptotic variance 1, as Chapter 12 promised.
- Robustness: when the data have heavy tails or a few wild values, the median and Huber's estimator keep working and the mean does not. The Cauchy distribution, for which the sample mean never settles down.

Chapter 12 finished maximum likelihood. This chapter puts the two other estimation recipes beside it and shows all three are one idea.

13.1 Three recipes, one idea

Every estimator in Part III was pulled out of the air: the sample proportion for p, the sample mean for μ, $1/\bar{T}_n$ for the subway rate. Each looked obvious, and Chapter 11 showed that maximum likelihood produces each of them from a principle. This chapter gives two more principles. The method of moments is the older and simpler one: write the parameter as a function of the population mean (and mean square, and so on), then replace the population moments by sample moments. M-estimation is the more general: choose a loss function, average it over the data, minimize. Maximum likelihood is M-estimation with the negative log-likelihood as the loss, and the method of moments is M-estimation with a squared-distance loss, so this chapter also explains what the three have in common and why the limit theorems of Chapter 12 apply to all of them.

13.2 The method of moments

Recall. The Law of Large Numbers (Theorem 3.5): for i.i.d. $X_1, \dots, X_n$ and a function h with $\mathbb{V}\mathrm{ar}(h(X)) < \infty$, $\frac{1}{n}\sum_{i=1}^n h(X_i) \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[h(X)]$.

The kth **moment** of a distribution $\mathbb{P}_\theta$ is $m_k(\theta) = \mathbb{E}_\theta[X^k]$: the mean for $k = 1$, the mean square for $k = 2$. Its sample version is $\hat{m}_k = \frac{1}{n}\sum_{i=1}^n X_i^k$, and by the LLN (with $h(x) = x^k$, which needs $\mathbb{E}[X^{2k}] < \infty$), $\hat{m}_k \xrightarrow[n\to\infty]{\mathbb{P}} m_k(\theta^*)$. So the sample moments are consistent estimators of the population moments. The method of moments turns that into an estimator of θ by inverting the map from parameters to moments.

Definition 13.1 (Method of moments estimator)**.** Let $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ be a model with $E \subset \mathbb{R}$ and $\Theta \subset \mathbb{R}^d$, and let $M(\theta) = (m_1(\theta), \dots, m_d(\theta))$ be its **moment map**, assumed one-to-one. The **method of moments estimator** is

$$\hat{\theta}_n^{\mathrm{MM}} = M^{-1}(\hat{m}_1, \dots, \hat{m}_d),$$

the parameter whose first d population moments equal the first d sample moments, provided one exists.

> **In words.** A model with d unknowns needs d equations. The method of moments writes down d of them: the population mean equals the sample mean, the population mean square equals the sample mean square, and so on up to d. Solve for the d unknowns. The hope is that d moments pin down d parameters, which is what 'one-to-one' asks of the moment map.
>
> **For the kiss data.** One parameter, one equation: $m_1(p) = \mathbb{E}_p[R] = p$, so $M(p) = p$, M^{-1} is the identity, and $\hat{p}^{\mathrm{MM}} = \hat{m}_1 = \bar{R}_n = 0.645$. For a proportion the method of moments is the sample proportion, as every method so far has been.

Figure 13.1 draws the definition for two families. On the right, the exponential family has one parameter and the moment map $m_1(\lambda) = 1/\lambda$ is a curve. Reading it backwards from the subway's sample mean of 6.23 minutes gives $\hat{\lambda} = 1/6.23 = 0.1605$, the same estimate as maximum likelihood. On the left, the Gaussian family has two parameters and the moment map sends (μ, σ^2) to $(m_1, m_2) = (\mu, \sigma^2 + \mu^2)$. The parameters with the sample's $\hat{m}_1 = 70.9$ lie on a vertical line, those with its $\hat{m}_2 = 5{,}210$ on a curve, and the two cross once, at the estimate: $\hat{\mu} = 70.9$ minutes and $\widehat{\sigma^2} = 184.1$ for the 272 Old Faithful waits of Chapter 11.

Worked example: method of moments for the standard families

- **Bernoulli.** $m_1(p) = p$, so $\hat{p} = \bar{X}_n$.
- **Poisson.** $m_1(\lambda) = \lambda$, so $\hat{\lambda} = \bar{X}_n$. For the siblings data with $X - 1 \sim \mathrm{Pois}(\lambda)$, $m_1 = \lambda + 1$ and $\hat{\lambda} = \bar{X}_n - 1 = 1.0$, which is how Chapter 2 fitted it in the first place.
- **Exponential.** $m_1(\lambda) = 1/\lambda$, so $\hat{\lambda} = 1/\bar{X}_n$.
- **Gaussian.** $m_1 = \mu$ and $m_2 = \sigma^2 + \mu^2$. Solving, $\hat{\mu} = \hat{m}_1 = \bar{X}_n$ and $\widehat{\sigma^2} = \hat{m}_2 - \hat{m}_1^2 = \frac{1}{n}\sum_i X_i^2 - \bar{X}_n^2 = \frac{1}{n}\sum_i (X_i - \bar{X}_n)^2$: the divide-by-$n$ variance, which is the MLE as well.
- **Uniform.** For $\mathrm{Unif}([0, \theta])$, $m_1 = \theta/2$, so $\hat{\theta}^{\mathrm{MM}} = 2\bar{X}_n$. This one differs from the MLE $\max_i X_i$, and it can be absurd: for the five observations of Figure 11.6, with mean 4.84, it

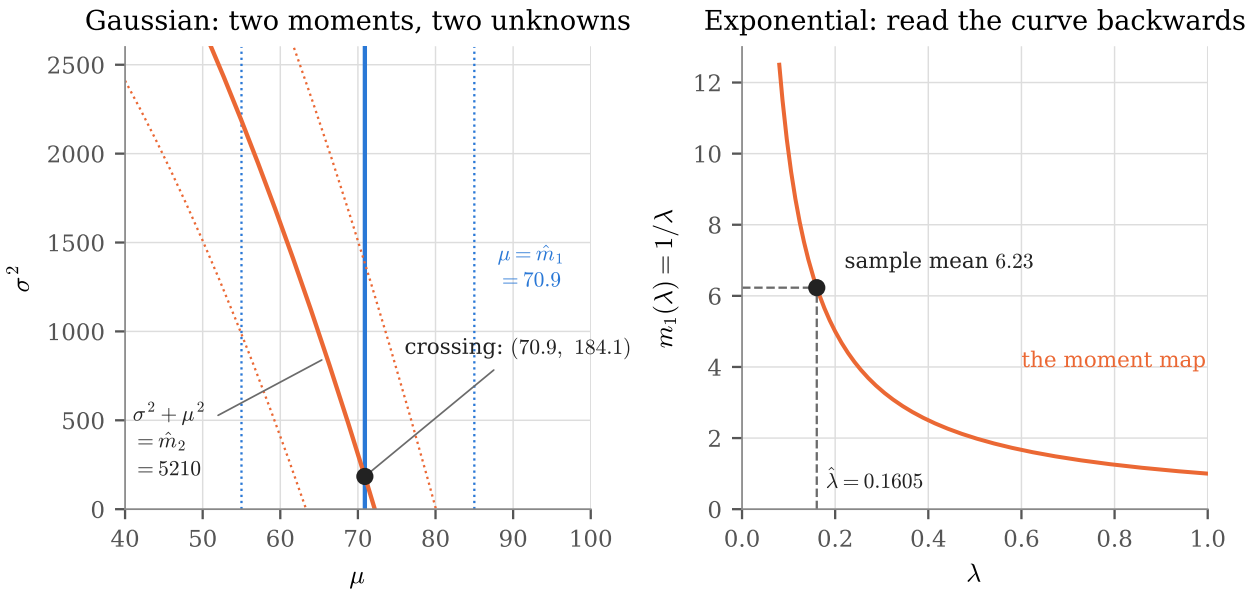


Figure 13.1: Left: the Gaussian moment map, with the parameters having the Old Faithful sample mean (blue) and sample mean square (orange) crossing at the estimate; dotted: other values of m_1 and m_2. Right: the exponential moment map $m_1 = 1/\lambda$, read backwards from the subway sample mean.

gives 9.68, which is fine, but a sample of $1, 1, 1, 1, 7.6$ gives 4.64, below an observation the model then calls impossible.

For the Bernoulli, Poisson, exponential and Gaussian families the method of moments is the maximum likelihood estimator; for the uniform it is not, and for the Laplace model of Exercise 11.8 the method of moments gives the sample mean while maximum likelihood gives the median. The agreement for the first four is not a coincidence, and Chapter 22 explains it: in an exponential family the likelihood equations match the moments of the sufficient statistics, which for these four families are the moments of X itself.

From undergraduate statistics: the method of moments is how your first statistics course estimated everything

An introductory course estimates a population mean by the sample mean, a variance by the sample variance, a proportion by the sample proportion, a correlation by the sample correlation. Each of those is the method of moments: a population quantity written in terms of moments, with sample moments plugged in. Karl Pearson named the method in 1894, before maximum likelihood existed, and it is still the first thing to try when a model's likelihood is awkward, since solving d polynomial equations is usually easier than maximizing a function. What the introductory course did not say is how precise the plug-in is, which is the next section.

13.2.1 Asymptotic variance of the method of moments

Recall. The delta method (Theorem 5.4): if $\sqrt{n}(Z_n-\theta) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,\sigma^2)$ and g is continuously differentiable at θ, then $\sqrt{n}(g(Z_n)-g(\theta)) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,g'(\theta)^2\sigma^2)$. The Central Limit Theorem (Theorem 4.8): for i.i.d. Y_i with mean m and variance $s^2<\infty$, $\sqrt{n}(\bar{Y}_n-m) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,s^2)$. Slutsky's theorem (Theorem 5.1): if $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$ and $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$, then $X_nY_n \xrightarrow[n\to\infty]{\text{(d)}} yX$ and, for $y\neq 0$, $X_n/Y_n \xrightarrow[n\to\infty]{\text{(d)}} X/y$. Vector versions (the vector-versions remark of Chapter 5): for i.i.d. vectors V_i with mean m and finite covariance matrix Σ, $\sqrt{n}(\bar{V}_n-m) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0,\Sigma)$, a Gaussian vector, meaning that every linear combination $a^\top Z$ of its coordinates is Gaussian, with variance $a^\top\Sigma a$; and for $g:\mathbb{R}^d\to\mathbb{R}^d$ continuously differentiable at m with Jacobian matrix G (row j is the gradient of the jth coordinate of g), $\sqrt{n}(g(\bar{V}_n)-g(m)) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0,G\Sigma G^\top)$.

A method of moments estimator is a function, M^{-1}, of sample moments, and sample moments are averages. So the CLT gives the limit of the moments and the delta method carries it through M^{-1}.

Theorem 13.2 (Asymptotic normality of the method of moments)**.** *Suppose $X_1,\dots,X_n \overset{iid}{\sim} \mathbb{P}_{\theta^*}$, the moment map M is one-to-one, its image $M(\Theta)$ contains a neighborhood of $M(\theta^*)$ (so that the estimator exists for sample moments near the population ones), M^{-1} is continuously differentiable at $M(\theta^*)$, and $\mathbb{E}_{\theta^*}[X^{2d}]<\infty$. Let $\Sigma(\theta^*)$ be the $d\times d$ covariance matrix of the vector $(X,X^2,\dots,X^d)$ under $\mathbb{P}_{\theta^*}$, and ∇M^{-1} the Jacobian matrix of M^{-1} at $M(\theta^*)$: the $d\times d$ matrix whose row j is the gradient of the jth coordinate of M^{-1} (the transpose of the matrix in the vector-versions remark of Chapter 5, which puts the coordinates in columns). Then*

$$\sqrt{n}\left(\hat{\theta}_n^{MM}-\theta^*\right) \xrightarrow[n\to\infty]{(d)} \mathit{Gauss}_d\big(0,\Gamma(\theta^*)\big), \qquad \Gamma(\theta^*)=\nabla M^{-1}\,\Sigma(\theta^*)\,(\nabla M^{-1})^\top.$$

For $d=1$: $\Gamma=\big((M^{-1})'(m_1)\big)^2\,\mathbb{V}ar_{\theta^*}(X)$.

Proof strategy (for completeness; skip on a first reading). The vector of sample moments is an average of i.i.d. vectors, so the multivariate CLT gives $\sqrt{n}(\hat{m}-M(\theta^*)) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0,\Sigma)$. The estimator is $M^{-1}(\hat{m})$, and the multivariate delta method pushes the Gaussian through M^{-1}: the covariance gets multiplied by the derivative matrix on both sides.

Proof. The ith observation contributes the vector $(X_i,X_i^2,\dots,X_i^d)$; these vectors are i.i.d. with mean $M(\theta^*)$ and covariance $\Sigma(\theta^*)$, finite by the moment condition, and $\hat{m}=(\hat{m}_1,\dots,\hat{m}_d)$ is their average. The multivariate CLT of the recall gives $\sqrt{n}(\hat{m}-M(\theta^*)) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0,\Sigma(\theta^*))$. For $d=1$ the delta method applied to $g=M^{-1}$ finishes the proof. For general d, the multivariate delta method is the same argument coordinate by coordinate: the mean value theorem writes $M^{-1}(\hat{m})-M^{-1}(M(\theta^*))$ as a matrix of derivatives, evaluated at in-between points that converge to $M(\theta^*)$, times $\hat{m}-M(\theta^*)$; the derivative matrix converges in probability to ∇M^{-1} by continuity, and Slutsky's theorem (Theorem 5.1) multiplies it into the Gaussian limit. A matrix A times a Gaussian vector Z with covariance Σ is Gaussian with covariance $A\Sigma A^\top$: each linear combination $a^\top AZ=(A^\top a)^\top Z$ is Gaussian with variance $a^\top A\Sigma A^\top a$, and when A is invertible, the linear-map rule of the Toolkit (toolkit A.14) shows that AZ is $\text{Gauss}_d(0,A\Sigma A^\top)$ in the sense defined there. □

Worked example: the exponential rate, both ways

For Expo(λ), $M(\lambda) = 1/\lambda$, $M^{-1}(m) = 1/m$ and $(M^{-1})'(m) = -1/m^2$, which at $m = 1/\lambda$ is $-\lambda^2$. With $\mathbb{V}\mathrm{ar}(X) = 1/\lambda^2$, the theorem gives asymptotic variance $\lambda^4 \cdot \lambda^{-2} = \lambda^2$: the same as the maximum likelihood estimator's $1/I(\lambda) = \lambda^2$ from Chapter 12, as it must be, since the two estimators are the same function of the data. For the subway, standard error $\hat{\lambda}/\sqrt{n} = 0.020$ once more.

Does this make sense? Could a different moment do better?

The Poisson model has $m_1 = \lambda$, and also $m_2 = \lambda + \lambda^2$. Nothing in the definition says you must use the first moment; you could match the second instead, solving $\hat{m}_2 = \lambda + \lambda^2$ for $\lambda = \big(-1+\sqrt{1+4\hat{m}_2}\big)/2$. Is that any better? The theorem says no. With $g(m) = (-1+\sqrt{1+4m})/2$, $g'(m_2) = 1/(2\lambda+1)$, and $\mathbb{V}\mathrm{ar}(X^2) = \lambda + 6\lambda^2 + 4\lambda^3$ for a Poisson, so the asymptotic variance is $(\lambda + 6\lambda^2 + 4\lambda^3)/(2\lambda+1)^2$, which at $\lambda = 1$ is $11/9 = 1.22$ against $\lambda = 1$ for the sample mean. The second-moment estimator is 22% noisier. It throws away information by squaring, and an estimator built on higher moments is usually worse than one built on lower ones, since higher sample moments have larger variance (usually, not always: the derivative of M^{-1} matters as much as the variance of the moment).

Proposition 13.3 (Method of moments is never better than maximum likelihood)**.** *In a regular model (Section 12.6), under the hypotheses of Theorem 13.2, the asymptotic variance of the method of moments estimator is at least that of the maximum likelihood estimator:* $\Gamma(\theta^*) \geq 1/I(\theta^*)$ *for* $d = 1$*, and in general* $\Gamma - I^{-1}$ *is positive semidefinite, meaning* $a^\top(\Gamma - I^{-1})a \geq 0$ *for every vector* a*: no linear combination of the parameters is estimated more precisely by moments than by likelihood.*

Proof strategy (for completeness; skip on a first reading). The delta step of Theorem 13.2 writes the method of moments estimator as an average plus a negligible term, and the score is an average too. The multivariate CLT makes the pair jointly Gaussian in the limit, their covariance works out to the identity by the same differentiate-under-the-integral step as in Cramér–Rao, and a variance is never negative.

Proof. Take $d = 1$ first. Write $T_n = \hat{\theta}_n^{\mathrm{MM}}$ and $c = (M^{-1})'(m_1(\theta^*))$. The mean value theorem step in the proof of Theorem 13.2 gives

$$\sqrt{n}(T_n - \theta^*) = c\,\sqrt{n}(\hat{m}_1 - m_1(\theta^*)) + R_n, \qquad R_n \xrightarrow[n\to\infty]{\mathbb{P}} 0,$$

since the derivative at the in-between point differs from c by a quantity converging to zero in probability, multiplied by a sequence converging in distribution (Slutsky). So $\sqrt{n}(T_n - \theta^*)$ is, up to R_n, the standardized average of the i.i.d. variables $U_i = c(X_i - m_1(\theta^*))$. The score is the average of the i.i.d. variables $W_i = \ell'(\theta^*; X_i)$, with mean zero (Proposition 12.2) and variance $I(\theta^*)$ (Definition 12.3). Apply the multivariate CLT to the pairs (U_i, W_i): the pair $\big(\sqrt{n}\bar{U}_n, \sqrt{n}\bar{W}_n\big)$ converges to a Gaussian vector (U, W) with $\mathbb{V}\mathrm{ar}(U) = \Gamma$, $\mathbb{V}\mathrm{ar}(W) = I$ and

$$\begin{aligned}\mathbb{C}\mathrm{ov}(U, W) = \mathbb{C}\mathrm{ov}(U_1, W_1) &= c\,\mathop{\mathbb{E}}_{\theta^*}\big[X\,\ell'(\theta^*; X)\big] = c\int x\,\frac{\partial}{\partial\theta}p_\theta(x)\Big|_{\theta^*}\,\mathrm{d}x \\ &= c\,\frac{\mathrm{d}}{\mathrm{d}\theta}m_1(\theta)\Big|_{\theta^*} = c\,M'(\theta^*) = 1,\end{aligned}$$

where the mean-zero score let us drop $m_1(\theta^*)$, the exchange of derivative and integral is the regularity condition used in Theorem 12.6, and the last step is the inverse function rule $(M^{-1})'(M(\theta^*)) =$

$1/M'(\theta^*)$. Adding R_n does not change the limit (Slutsky). Now $\mathbb{V}\mathrm{ar}(U - W/I) = \Gamma - 2 \cdot \frac{1}{I} + \frac{I}{I^2} = \Gamma - \frac{1}{I} \geq 0$, which is the claim.

For general d the same steps give a Gaussian vector (U, W) in $\mathbb{R}^{2d}$, with U having covariance Γ, W the score vector with covariance I, and $\mathbb{C}\mathrm{ov}(U, W) = \nabla M^{-1}\, \nabla M =$ the identity matrix, since the matrix of $\partial m_k / \partial \theta_j$ is the Jacobian ∇M and the inverse function rule inverts it. For any vector a, $\mathbb{V}\mathrm{ar}\left(a^\top U - a^\top I^{-1} W\right) = a^\top \Gamma a - 2a^\top I^{-1} a + a^\top I^{-1} I I^{-1} a = a^\top (\Gamma - I^{-1}) a \geq 0$. □

So when maximum likelihood is available, it is at least as precise. The method of moments is the tool elsewhere: when the likelihood has no tractable maximum (the mixture of Section 11.8 is a classic case, and Pearson's 1894 paper fitted a mixture of two Gaussians by matching five moments), or when it is the same estimator anyway and the moment version is easier to compute.

13.3 M-estimation

Both recipes so far minimized something. Maximum likelihood minimized $-\frac{1}{n}\sum_i \log p_\theta(X_i)$. The method of moments, for $d = 1$, solves $\hat{m}_1 = m_1(\theta)$, which is where $\frac{1}{n}\sum_i (X_i - m_1(\theta))^2$ has zero derivative with respect to m_1. Here is the general pattern.

Definition 13.4 (M-estimator)**.** Let $X_1, \ldots, X_n$ be i.i.d. from a distribution $\mathbb{P}$ on E, and let $\mu^* \in \mathcal{M} \subset \mathbb{R}^d$ be a quantity attached to $\mathbb{P}$ that you want to estimate. A **loss function** for μ^* is a function $\rho : E \times \mathcal{M} \to \mathbb{R}$ with $\mathbb{E}|\rho(X, \mu)| < \infty$ for every μ, such that the **expected loss**

$$Q(\mu) = \mathbb{E}\left[\rho(X, \mu)\right]$$

is smallest at $\mu = \mu^*$, and only there. The **M-estimator** of μ^* is the minimizer of the **empirical loss**,

$$\hat{\mu}_n = \operatorname*{argmin}_{\mu \in \mathcal{M}} Q_n(\mu), \qquad Q_n(\mu) = \frac{1}{n}\sum_{i=1}^{n} \rho(X_i, \mu).$$

In words. You want a number attached to the distribution: its mean, its median, a parameter. Find a loss that charges each observation according to how far it is from a candidate μ, chosen so that the average charge over the whole distribution is smallest at the number you want. You cannot compute that average, but the LLN says the average over the data is close to it, so minimize the data's average instead. No statistical model is needed: the recipe asks only that Q have its minimum at μ^*.

For the kiss data. With $\rho(r, \mu) = (r - \mu)^2$, $Q(\mu) = \mathbb{E}[(R - \mu)^2] = \mathbb{V}\mathrm{ar}(R) + (p - \mu)^2$ is smallest at $\mu = p$, and $Q_n(\mu) = \frac{1}{n}\sum_i (R_i - \mu)^2$ is smallest at $\bar{R}_n = 0.645$. The sample proportion is an M-estimator for the squared loss, which is its fourth derivation in this book.

Worked example: losses and what they estimate

- **Squared loss,** $\rho(x, \mu) = (x - \mu)^2$**.** $Q(\mu) = \mathbb{V}\mathrm{ar}(X) + (\mathbb{E}[X] - \mu)^2$, minimized at $\mu^* = \mathbb{E}[X]$ (when $\mathbb{V}\mathrm{ar}(X) < \infty$). The M-estimator is the sample mean: for the Old Faithful waits, Q_n is minimized at 70.9 minutes, the blue dot in Figure 13.2. In $\mathbb{R}^d$, with $\rho = \|x - \mu\|^2$, it is the vector of sample means.

- **Absolute loss,** $\rho(x, \mu) = |x - \mu|$**.** $Q(\mu) = \mathbb{E}|X - \mu|$ is minimized at a median of X

(Proposition 13.5; this needs $\mathbb{E}|X| < \infty$, and Section 13.4 says what to do when that fails). The M-estimator is the sample median.

- **Check loss, $\rho(x, \mu) = C_\alpha(x - \mu)$ with $C_\alpha(u) = \alpha u$ for $u \geq 0$ and $-(1-\alpha)u$ for $u < 0$.** A tilted absolute value, charging α per unit above μ and $1 - \alpha$ per unit below. Q is minimized at the point with probability α below it, the α-quantile (Exercise 13.4). The M-estimator is the sample quantile. With $\alpha = \frac{1}{2}$ it is the median again.
- **Negative log-likelihood, $\rho(x, \theta) = -\log p_\theta(x)$,** in a model with $\mathbb{P} = \mathbb{P}_{\theta^*}$. $Q(\theta) = -\mathbb{E}_{\theta^*}[\log p_\theta(X)] = \mathrm{KL}(\mathbb{P}_{\theta^*}, \mathbb{P}_\theta) + \text{constant}$, minimized at θ^* when the model is identifiable (Chapter 10). The M-estimator is the MLE.
- **Huber's loss, $\rho(x, \mu) = h_\delta(x - \mu)$** with $h_\delta(u) = u^2/2$ for $|u| \leq \delta$ and $\delta(|u| - \delta/2)$ beyond. Quadratic near zero, linear far away. For a symmetric distribution Q is minimized at the center. Section 13.4 says what it is for.

Figure 13.2 draws three of the losses and their empirical versions on the Old Faithful waits. The squared loss is a parabola in the residual (drawn as $\frac{1}{2}(x-\mu)^2$, so that it and Huber's loss coincide on $[-\delta, \delta]$; halving a loss changes neither its minimizer nor, as Section 13.3.1 shows, the variance of its M-estimator); the absolute loss is a V; Huber's is the parabola inside $[-\delta, \delta]$ and the V's straight sides outside, where the blue and aqua curves separate. On the right, each empirical loss is minimized at a different center of the same data: the squared loss at the mean, 70.9 minutes; the absolute loss at the median, 76; Huber's loss with $\delta = 5$ minutes at 75.0, between the two. The data are not symmetric (the two humps of Figure 11.7 have unequal weights), so the three 'centers' are three different numbers, each the right answer to its own loss.

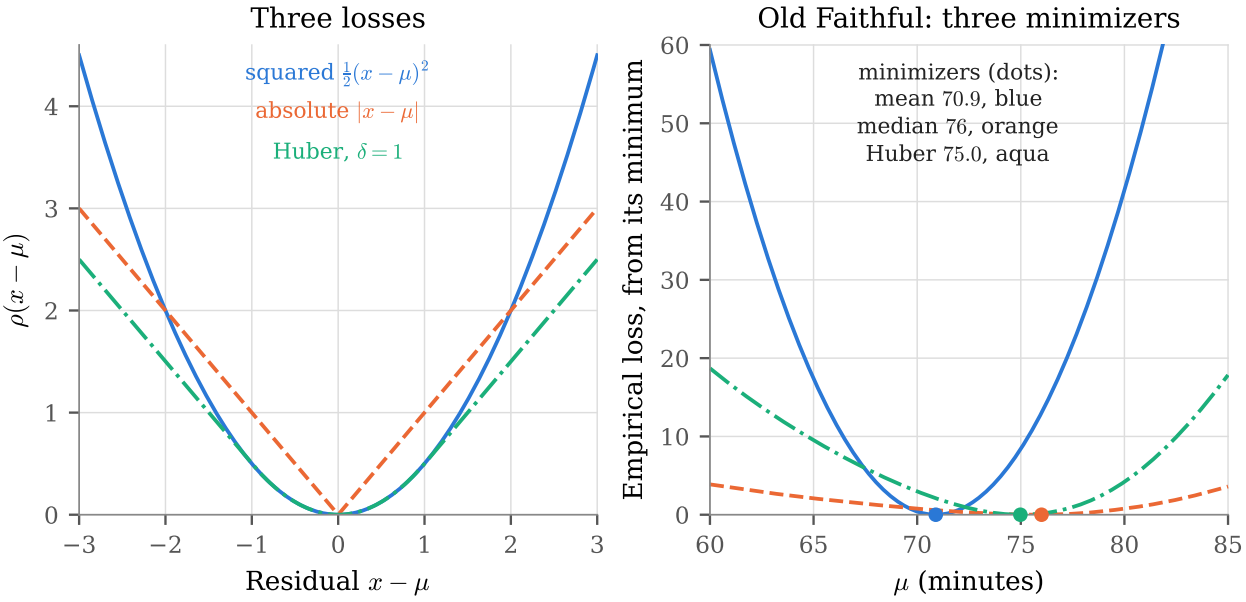


Figure 13.2: Left: the squared (blue), absolute (orange, dashed) and Huber (aqua, dash-dot, $\delta = 1$) losses as functions of the residual. Right: the three empirical losses for the Old Faithful waits, each measured from its minimum, with the minimizers marked.

Proposition 13.5 (The median minimizes expected absolute deviation)**.** *Let X have $\mathbb{E}|X| < \infty$ and*

a continuous density f, and let m be a median, $\mathbb{P}(X < m) = \mathbb{P}(X > m) = \frac{1}{2}$. Then $Q(\mu) = \mathbb{E}|X - \mu|$ is minimized at $\mu = m$, and at no other point if $f(m) > 0$.

Proof strategy (core). The derivative of Q at μ is the probability below μ minus the probability above it: each unit you move μ to the right, the observations below it each cost one more and those above each cost one less. That derivative is negative below the median, zero at it and positive above.

Proof. Split the expectation at μ:

$$Q(\mu) = \int_{\mu}^{\infty} (x - \mu) f(x)\,\mathrm{d}x + \int_{-\infty}^{\mu} (\mu - x) f(x)\,\mathrm{d}x.$$

Differentiate in μ. The first integral has derivative $-\int_{\mu}^{\infty} f(x)\,\mathrm{d}x$ (the integrand's derivative in μ is $-f$, and the boundary term $(\mu - \mu)f(\mu)$ is 0); the second has derivative $\int_{-\infty}^{\mu} f(x)\,\mathrm{d}x$ by the same reasoning (toolkit A.6). So

$$Q'(\mu) = \mathbb{P}(X < \mu) - \mathbb{P}(X > \mu) = 2F(\mu) - 1,$$

where F is the CDF. This is at most 0 for $\mu < m$, zero at m and at least 0 for $\mu > m$, so Q decreases until m and increases after: m is a minimizer. If $f(m) > 0$, then F is strictly increasing near m, so Q' is strictly negative just below m and strictly positive just above, and m is the only minimizer (the Laplace and Gaussian densities are positive everywhere, so this applies to them). □

Does this make sense? Is the kiss proportion a median too?

The median of Bern(0.645) is 1: more than half the mass sits there. So the absolute loss, which estimates the median, would estimate '1' for the kiss experiment, and indeed the sample median of 80 ones and 44 zeros is 1. That is correct and useless. A loss chooses what it estimates (an M-estimator estimates whatever its Q is minimized at), and for a proportion the squared loss, which estimates the mean p, is the one that answers the question.

13.3.1 Asymptotic normality of M-estimators

The argument of Chapter 12 used only that the log-likelihood is a sum of i.i.d. terms. Every empirical loss is one, so the argument goes through, with one change: the two roles the Fisher information played, as the variance of the score and as the curvature, are now played by two different numbers.

Definition 13.6 (The J and K of a loss)**.** For a loss ρ twice differentiable in μ, with $\mu \in \mathbb{R}^d$, define

$$J(\mu) = \mathbb{E}\left[\nabla_{\mu}^2 \rho(X, \mu)\right], \qquad K(\mu) = \mathbb{C}\mathrm{ov}\left(\nabla_{\mu} \rho(X, \mu)\right),$$

the expected Hessian of the loss (the curvature of Q at μ) and the covariance matrix of its gradient (as the Fisher information matrix of Definition 12.3 is the covariance of the score). For $d = 1$ they are $J(\mu) = \mathbb{E}[\partial_{\mu}^2 \rho(X, \mu)]$ and $K(\mu) = \mathbb{V}\mathrm{ar}(\partial_{\mu} \rho(X, \mu))$.

> **In words.** K is how much one observation's push on the estimate varies: the analogue of the variance of the score. J is how hard the expected loss resists being pushed: the analogue of the curvature. For the negative log-likelihood the two coincide and both equal the Fisher information, by Theorem 12.4; for any other loss they are different numbers.

For the kiss data. With the squared loss, $\partial_\mu \rho = -2(R-\mu)$ and $\partial_\mu^2 \rho = 2$, so $J = 2$ and $K = 4\,\mathbb{V}\mathrm{ar}(R) = 4p(1-p)$: at $p = 0.645$, $K = 0.916$.

Recall. Theorem 12.5 and its proof: for an estimator defined by $\sum_i g_i'(\hat{\mu}) = 0$, the mean value theorem writes $0 = \sum_i g_i'(\mu^*) + \sum_i g_i''(\tilde{\mu})(\hat{\mu} - \mu^*)$; the CLT handles the first sum, the LLN the second, Slutsky the ratio.

Theorem 13.7 (Asymptotic normality of M-estimators)**.** *Let $X_1, \dots, X_n$ be i.i.d. from $\mathbb{P}$ and let μ^* be the unique minimizer of Q, in the interior of $\mathcal{M}$ (the analogue of condition 3 of Section 12.6). Suppose the hypotheses of Theorem 12.1 hold with ρ in place of $-\log p_\theta$ (a bounded closed $\mathcal{M}$, or the restriction argument given there; ρ continuous in μ; $\mathbb{E}[\sup_\mu |\rho(X,\mu)|] < \infty$), so that $\hat{\mu}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu^*$. Suppose also that ρ is three times differentiable in μ near μ^*, that derivatives in μ can be moved inside $\mathbb{E}$ at μ^*, that $K(\mu^*)$ and $\mathbb{E}[\partial_\mu^2 \rho(X,\mu^*)]$ are finite with $J(\mu^*)$ invertible (the analogue of condition 5), and that $|\partial_\mu^3 \rho(X,\mu)| \le B(X)$ near μ^* for some B with $\mathbb{E}[B(X)] < \infty$ (the analogue of condition 4). Then*

$$\sqrt{n}\,(\hat{\mu}_n - \mu^*) \xrightarrow[n\to\infty]{(d)} \mathit{Gauss}_d\Big(0,\ J(\mu^*)^{-1} K(\mu^*) J(\mu^*)^{-1}\Big).$$

For $d = 1$ the asymptotic variance is $K(\mu^*)/J(\mu^*)^2$.

Proof strategy (for completeness; skip on a first reading). Word for word the proof of Theorem 12.5, with ρ in place of $-\log p_\theta$. The gradient of the empirical loss is zero at $\hat{\mu}_n$ and, by the CLT, approximately Gauss$(0, K/n)$ at μ^* (its mean is zero there because μ^* minimizes Q). The curvature between is about J by the LLN. Displacement is push over resistance: $(\text{sd}\ \sqrt{K/n})/J$.

Proof. Consistency is Theorem 12.1 with $-\log p_\theta$ replaced by ρ: the proof there used the uniform LLN and the unique minimizer of the limit, and nothing about the likelihood. For normality, take $d = 1$. Since μ^* minimizes Q and is interior, $Q'(\mu^*) = \mathbb{E}[\partial_\mu \rho(X,\mu^*)] = 0$ (differentiating under the expectation, which the regularity conditions allow). So the terms $\partial_\mu \rho(X_i,\mu^*)$ are i.i.d. with mean 0 and variance $K(\mu^*)$, and the CLT gives $\frac{1}{\sqrt{n}} \sum_i \partial_\mu \rho(X_i,\mu^*) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, K(\mu^*))$. Since $\hat{\mu}_n$ is interior with probability tending to 1, $Q_n'(\hat{\mu}_n) = 0$, and the mean value theorem gives a $\tilde{\mu}_n$ between $\hat{\mu}_n$ and μ^* with

$$0 = Q_n'(\hat{\mu}_n) = Q_n'(\mu^*) + Q_n''(\tilde{\mu}_n)(\hat{\mu}_n - \mu^*), \qquad \text{so} \qquad \sqrt{n}(\hat{\mu}_n - \mu^*) = -\,\frac{\sqrt{n}\,Q_n'(\mu^*)}{Q_n''(\tilde{\mu}_n)}.$$

The numerator is $\frac{1}{\sqrt{n}} \sum_i \partial_\mu \rho(X_i,\mu^*)$, which converges in distribution to Gauss$(0,K)$. The denominator is $Q_n''(\tilde{\mu}_n) = \frac{1}{n} \sum_i \partial_\mu^2 \rho(X_i,\tilde{\mu}_n)$. It converges in probability to $J(\mu^*)$ by the LLN at μ^* and the bound B on the third derivative, used exactly as K was in the proof of Theorem 12.5: the difference between the average second derivative at $\tilde{\mu}_n$ and at μ^* is at most $\bar{B}_n |\hat{\mu}_n - \mu^*|$, an average converging to $\mathbb{E}[B(X)]$ times a quantity converging to 0. Slutsky's theorem gives the limit $-\mathrm{Gauss}(0,K)/J = \mathrm{Gauss}(0, K/J^2)$ (the sign is absorbed, since $-Z$ has the same distribution as Z). In d dimensions the denominator is the matrix J, inverted, applied to a Gaussian vector of covariance K, giving $J^{-1} K J^{-1}$. □

Intuition: why the variance has three factors

For maximum likelihood the push and the resistance were the same number, I, and push over resistance gave $\sqrt{I}/I = 1/\sqrt{I}$. For a general loss they come apart. The push has standard deviation $\sqrt{K}$; the resistance is J; the displacement has standard deviation $\sqrt{K}/J$, and its variance is K/J^2. In matrix form that is $J^{-1}KJ^{-1}$, the curvature inverted on both sides of the push's covariance, which is why it is called the **sandwich**. A loss with a stiff Q (large J) and a quiet gradient (small K) gives a precise estimator.

Worked example: the sandwich for the sample mean

With the squared loss, $J = 2$ and $K = 4\,\mathbb{V}\mathrm{ar}(X)$, so the asymptotic variance is $4\,\mathbb{V}\mathrm{ar}(X)/4 = \mathbb{V}\mathrm{ar}(X)$: the CLT for the sample mean, recovered. For the kiss data, $p(1-p) = 0.229$.

Worked example: the Laplace median has asymptotic variance 1

Let $X_1, \ldots, X_n$ have the Laplace density $f(x) = \frac{1}{2}e^{-|x-m|}$ of Exercise 11.8, and estimate the center m by the sample median, the M-estimator for $\rho = |x-\mu|$. The absolute value has a corner, so Definition 13.6 does not apply directly; the way around is to compute J as the curvature of Q itself. From the proof of Proposition 13.5, $Q'(\mu) = 2F(\mu) - 1$, so $J = Q''(m) = 2f(m) = 2 \times \frac{1}{2} = 1$. For K, $\partial_\mu |X-\mu| = -\mathrm{sign}(X-\mu)$ away from the corner, which is ± 1 with probability $\frac{1}{2}$ each at $\mu = m$: variance 1. So the asymptotic variance is $K/J^2 = 1$. The sample mean of Laplace data has asymptotic variance $\mathbb{V}\mathrm{ar}(X) = 2$. The median is twice as precise, and its standard error is $1/\sqrt{n}$ against the mean's $\sqrt{2/n}$. (That the theorem's conclusion holds despite the corner is a separate argument, which this book does not reproduce; the formula $1/(4f(m)^2)$ for the median's asymptotic variance is standard.)

13.4 Robustness

Figure 13.3 runs the sample mean and the sample median side by side on three symmetric distributions, 20,000 samples of $n = 100$ each. For Gaussian data the mean is the better estimator of the center: its standard deviation is 0.100 against the median's 0.124, matching the asymptotic variances 1 and $\pi/2$ (the median's $1/(4f(0)^2)$ with $f(0) = 1/\sqrt{2\pi}$). In about 61% of samples the mean lands closer to the truth. For Laplace data the order reverses, 0.141 against 0.108, and the median is closer in 62% of samples. The mean's 0.141 is exactly the $\sqrt{2/100}$ of the worked example; the median's 0.108 is still 8% above its asymptotic $1/\sqrt{100} = 0.100$ because the Laplace density has a corner at the center and the median's distribution approaches its Gaussian limit slowly there. For Cauchy data, with density $f(x) = 1/(\pi(1 + (x-m)^2))$, the mean does not settle at all: its histogram is spread across the whole axis and beyond, with a standard deviation of 72 over the 20,000 samples, while the median's is 0.158, close to its asymptotic $(\pi/2)/10 = 0.157$.

Pitfall: the Cauchy mean

The Cauchy density has tails so heavy that $\mathbb{E}|X| = \int |x| f(x)\,dx$ is infinite: the integrand behaves like $1/(\pi|x|)$ for large $|x|$, and $\int_1^A dx/x = \log A \to \infty$. The mean does not exist, the LLN's hypothesis fails, and the sample mean does not converge to anything. Worse, the average of n Cauchy observations has exactly the same Cauchy distribution as a single observation: averaging

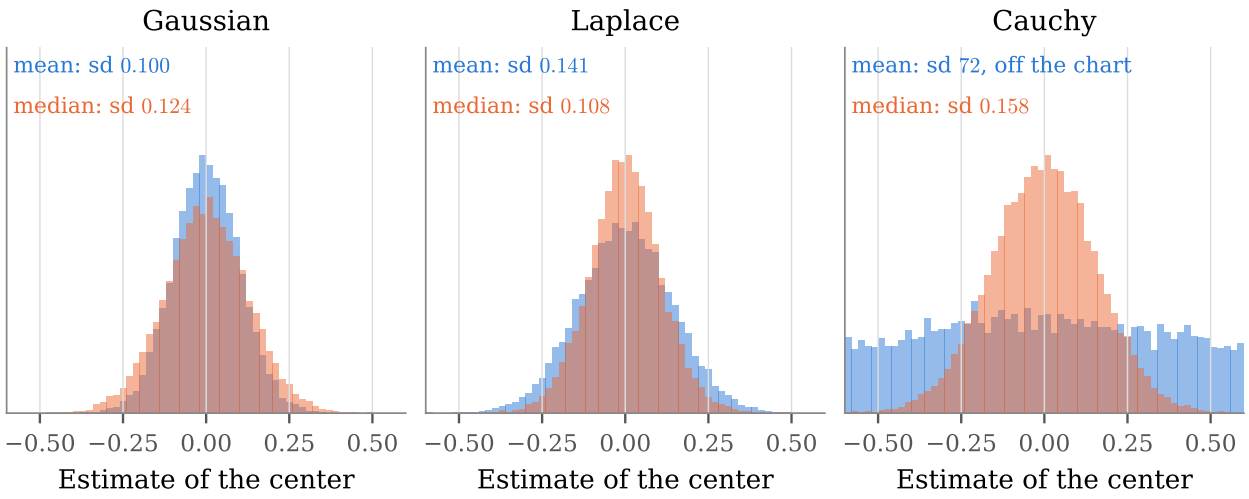


Figure 13.3: The sample mean (blue) and sample median (orange) of $n = 100$ observations, over 20,000 samples, for Gaussian, Laplace and Cauchy data centered at 0.

100 of them buys nothing. The method of moments is unavailable (no moments), the sample mean is useless, and the median, the MLE and Huber's estimator all work, with the Cauchy MLE of the center attaining the Cramér–Rao variance 2, against the median's $\pi^2/4 = 2.47$. (Since $\mathbb{E}|X| = \infty$, the absolute loss has infinite expectation here and Definition 13.4 does not literally apply; the fix is the loss $|x - \mu| - |x|$, which is bounded by $|\mu|$, has finite expectation for every distribution, and has the same minimizer and the same M-estimator, since subtracting $|x|$ shifts Q_n by a constant.)

An estimator that keeps working when the data have heavy tails, or when a few observations are wrong (a decimal point in the wrong place, a sensor fault), is called **robust**. The sample median is the simplest robust estimator of a center: move the largest observation to 10^{10} and the median does not change, while the mean goes to infinity. Its price is efficiency on clean Gaussian data, where it is $\pi/2 = 1.57$ times noisier than the mean, so that it needs 57% more observations for the same precision. Huber's loss is the compromise.

Intuition: Huber's loss: a mean in the middle, a median in the tails

$h_\delta(u)$ is $u^2/2$ for $|u| \le \delta$ and $\delta(|u| - \delta/2)$ beyond, joined so that the function and its slope are continuous (Figure 13.2, left). Its derivative is the **clip** function: u inside $[-\delta, \delta]$ and $\pm\delta$ outside. So an observation within δ of the current center pushes it the way the squared loss would, in proportion to its distance, and an observation beyond δ pushes with the fixed force δ, the way the absolute loss would, however far away it is. A wild observation can push at most δ: that is the robustness. Observations in the bulk are treated as the mean treats them: that is the efficiency. The constant δ sets where 'the bulk' ends, and a common choice on data with scale 1 is $\delta = 1.345$, which costs only 5% in variance on Gaussian data (1.05 against the mean's 1) while capping the influence of any observation.

Worked example: Huber on Laplace data

Take the Laplace center m and $\rho(x,\mu) = h_\delta(x-\mu)$. Huber's loss is twice differentiable except at $\pm\delta$, so Definition 13.6 applies (as with the median, the kink means Theorem 13.7 does not literally cover it, and the standard argument that extends the theorem to this case is not reproduced here): $\partial_\mu^2 \rho = \mathbb{1}(|X-\mu| \le \delta)$, and

$$J(m) = \mathbb{P}(|X-m| \le \delta) = \int_{-\delta}^{\delta} \tfrac{1}{2} e^{-|u|}\,\mathrm{d}u = 1 - e^{-\delta}.$$

The gradient $-\mathrm{clip}_\delta(X-\mu)$ has mean zero at m by symmetry, and

$$K(m) = \mathbb{E}\left[\mathrm{clip}_\delta(X-m)^2\right] = \int_{-\delta}^{\delta} u^2\, \tfrac{1}{2} e^{-|u|}\,\mathrm{d}u + \delta^2 \mathbb{P}(|X-m| > \delta) = 2 - e^{-\delta}(2\delta + 2),$$

the integral by two integrations by parts. With $\delta = 1$: $J = 0.632$, $K = 0.528$ and the asymptotic variance $K/J^2 = 1.32$, between the median's 1 and the mean's 2. As $\delta \to 0$ the variance tends to 1 (Huber becomes the median) and as $\delta \to \infty$ it tends to 2 (Huber becomes the mean); at $\delta = 0.5$ it is 1.17, at $\delta = 2$ it is 1.59.

From undergraduate statistics: trimmed means and 'robust standard errors'

An introductory course may have mentioned the trimmed mean, the average after discarding the top and bottom 10%, as a compromise between mean and median. Huber's estimator is the principled version: instead of discarding the extremes it caps their influence, and it comes with a standard error, $\sqrt{K/(nJ^2)}$ with J and K estimated from the data. The sandwich formula also underlies the 'robust' or 'heteroskedasticity-consistent' standard errors of regression software: least squares is an M-estimator with the squared loss, and when the errors are not identically distributed K is no longer a multiple of J, so the usual standard error (σ^2 times the inverse of $\mathbb{E}[XX^\top]$) is wrong and the sandwich $J^{-1}KJ^{-1}$ is right. Chapter 20 gives the version for one predictor, and Chapter 21 returns to the general one.

13.5 What comes next

Part IV is complete: three recipes, each consistent and asymptotically normal, each with a variance you can compute and plug in to get estimate $\pm$ 1.96 standard error. Maximum likelihood has the smallest variance when the model is right; the method of moments is the same estimator for the common families and a fallback when the likelihood is awkward; M-estimation with a robust loss gives up a little precision for protection against heavy tails and bad data. Chapter 14 opens Part V by handling the one situation none of this covers, a small Gaussian sample where the limit theorems have not taken hold, with exact distributions in place of asymptotic ones. Chapter 15 turns the asymptotic normality of Chapter 12 into tests. Chapter 22 explains why the method of moments and maximum likelihood kept agreeing.

13.6 Summary

- Method of moments: $\hat{\theta}^{\mathrm{MM}} = M^{-1}(\hat{m}_1, \ldots, \hat{m}_d)$, the parameter whose first d moments match the sample's. Bernoulli, Poisson, exponential and Gaussian all give the MLE; Unif$([0,\theta])$ gives $2\bar{X}_n$, not the MLE.
- Its asymptotic variance, by the multivariate CLT and the delta method, is $\nabla M^{-1}\,\Sigma\,(\nabla M^{-1})^{\top}$; for $d=1$, $((M^{-1})')^2\,\mathbb{V}\mathrm{ar}(X)$. It is never below $1/I$. Higher moments make worse estimators (11/9 against 1 for the Poisson rate).
- M-estimator: $\hat{\mu}_n = \operatorname{argmin} \frac{1}{n}\sum_i \rho(X_i, \mu)$ for a loss whose expectation Q is minimized at μ^*. Squared loss gives the mean, absolute loss the median ($Q' = 2F - 1$), check loss the quantile, negative log-likelihood the MLE, Huber's loss a robust center.
- Asymptotic variance $J^{-1}KJ^{-1}$ with $J = \mathbb{E}[\nabla^2\rho]$ (resistance) and $K = \mathbb{C}\mathrm{ov}(\nabla\rho)$ (push); for the MLE $J = K = I$. Laplace median 1 against mean 2; Gaussian median $\pi/2$ against mean 1; Huber with $\delta = 1$ on Laplace data 1.32.
- Cauchy data have no mean; the sample mean of 100 of them is as noisy as one. The median, Huber and the MLE still work.

13.7 Exercises

Exercise 13.1. For the four observations $0.5, 1.8, -2.3, 0.9$ from a Gauss(μ, σ^2) model, compute the method of moments estimates of μ and σ. *(Solution on page 424.)*

Exercise 13.2. For $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Unif}([0,\theta])$, find the method of moments estimator and its asymptotic variance, and compare its standard deviation at $n = 100$, $\theta = 10$ with the MLE's, whose error is about θ/n in size (Exercise 12.5). *(Solution on page 424.)*

Exercise 13.3. The Gamma family with shape $a > 0$ and rate $b > 0$ has $m_1 = a/b$ and $m_2 = a(a+1)/b^2$. Find the method of moments estimators of a and b. (The MLE of this family has no closed form, which is why the method of moments is used for it.) *(Solution on page 424.)*

Exercise 13.4. Show that $Q(\mu) = \mathbb{E}[C_\alpha(X - \mu)]$, with the check loss $C_\alpha(u) = \alpha u$ for $u \geq 0$ and $-(1-\alpha)u$ for $u < 0$, is minimized at the α-quantile of a continuous X with $\mathbb{E}|X| < \infty$, the point q with $F(q) = \alpha$. *(Solution on page 424.)*

Exercise 13.5. For Gaussian data with variance σ^2, compute the asymptotic variance of the sample median by the $1/(4f(m)^2)$ formula and compare it with the sample mean's. How many more observations does the median need for the same precision? *(Solution on page 424.)*

Exercise 13.6. For Gaussian data with variance 1 and Huber's loss with $\delta = 1.345$, $J = \mathbb{P}(|Z| \leq \delta) = 2\Phi(\delta) - 1$ and $K = \mathbb{E}[\min(Z^2, \delta^2)]$. Given $J = 0.821$ and $K = 0.710$, compute the asymptotic variance and the efficiency relative to the sample mean. *(Solution on page 424.)*

Exercise 13.7. Accept that the average of n i.i.d. Cauchy observations has the same Cauchy distribution as one of them. Explain why this does not contradict the Law of Large Numbers or the Central Limit Theorem, and what it says about $\bar{X}_{1000}$ as an estimate of the center. *(Solution on page 424.)*

Exercise 13.8. Show that for the loss $\rho(x,\theta) = -\log p_\theta(x)$ in a regular model, $J(\theta^*) = K(\theta^*) = I(\theta^*)$, so that Theorem 13.7 reduces to Theorem 12.5. *(Solution on page 425.)*

Part V

Parametric Hypothesis Tests

The tests of Chapter 9 were built on one standardized average. This part supplies the distributions the remaining parametric tests use, chi-squared and Student's t, with Cochran's theorem behind them (Chapter 14), and the three general-purpose tests built on the likelihood, Wald's, the likelihood ratio and the score test (Chapter 15).

Chapter 14

Chi-Squared, Student's t, Cochran's Theorem and the t-Test

In this chapter

- Why a small Gaussian sample needs its own theory: the standardized mean with an estimated standard deviation is not Gaussian, and at $n = 5$ the Gaussian 5% rule rejects a true null 12% of the time.
- The chi-squared distribution: a sum of k squared standard Gaussians, with mean k and variance $2k$. It is the distribution of the sample variance and, in Chapter 15, of every squared test statistic.
- Cochran's theorem: for Gaussian data the sample mean and the sample variance are independent, and $(n-1)S_n^2/\sigma^2$ is chi-squared with $n-1$ degrees of freedom. The proof rotates the data, and the picture is a round cloud of points.
- Student's t distribution, $Z/\sqrt{V/k}$: heavier tails than the Gaussian, converging to it as k grows, and the Cauchy distribution of Chapter 13 when $k = 1$.
- The t-test and the t-interval, for one sample and for two (pooled when the variances agree, Welch's approximation when they do not), with exact level for Gaussian data and asymptotic level for any data.
- Many tests at once: the family-wise error rate and Bonferroni's correction, the false discovery rate and the Benjamini–Hochberg procedure.

Chapter 9 built one test from the Central Limit Theorem and Chapter 13 closed the estimation recipes. This chapter removes the limit theorems from the test and asks what is exactly true at every n.

14.1 When the sample is small

A home is a candidate for a grid-connected wind turbine if its mean wind speed is at least 10 miles per hour. Wind was measured once a month for a year: $n = 12$ observations, mean 14.3, standard deviation 4.7. Is the mean wind speed above 10? The test of Chapter 9 would compute $T_n = \sqrt{12}\,(14.3 - 10)/4.7 = 3.17$, compare it with $q_{0.05} = 1.645$, and reject. That test's guarantee was an *asymptotic* level: by the CLT the standardized mean is approximately Gaussian, and by

Slutsky's theorem replacing σ by an estimate does not change the limit. Both are statements about $n \to \infty$, and here $n = 12$. With twelve numbers there is no limit to appeal to.

What is exactly true at $n = 12$? Suppose the twelve measurements are modeled as $X_1, \dots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$. Then by Chapter 6, $\bar{X}_n \sim \text{Gauss}(\mu, \sigma^2/n)$ exactly, so

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma} \sim \text{Gauss}(0, 1) \qquad \text{for every } n,$$

with no CLT. If σ were known, the test of Chapter 9 would have exact level α at $n = 12$, or at $n = 2$. It is not known, and replacing it by the sample standard deviation

$$S_n = \sqrt{S_n^2}, \qquad S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2,$$

is where the exactness is lost: $\sqrt{n}(\bar{X}_n - \mu)/S_n$ is not Gaussian. Figure 14.1 shows what it is instead. On the left, 50,000 samples of $n = 5$ Gaussian observations were drawn, and for each the ratio $T = \sqrt{n}(\bar{X}_n - \mu)/S_n$ was computed. The histogram is symmetric and bell-shaped, but it is not the standard Gaussian drawn over it in black: the peak is lower and the tails are fatter. A ratio whose denominator is itself random, and sometimes small, takes large values more often than a ratio with a fixed denominator. The dashed orange curve, which fits the histogram, is the distribution this chapter derives. On the right, the price of ignoring the difference: the Gaussian rule 'reject when $|T| > 1.96$', meant to reject a true null 5% of the time, rejects it 12% of the time at $n = 5$ and 8% at $n = 10$, and is still 6% at $n = 30$. Of 1,000 studies with five Gaussian observations each and a true null, the Gaussian rule rejects about 121, not 50.

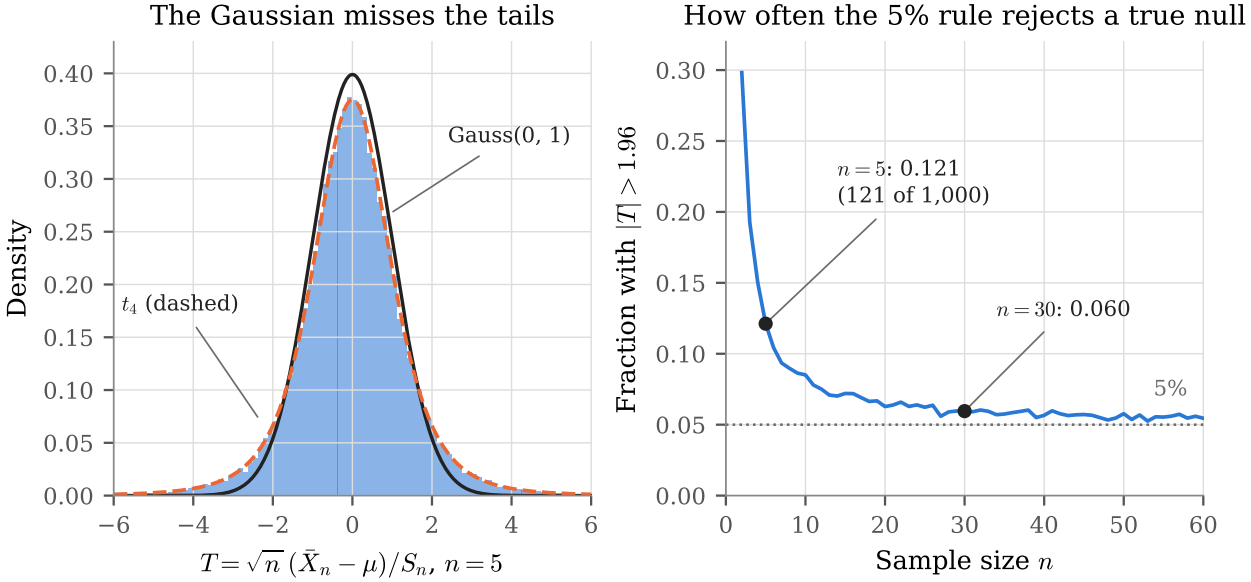


Figure 14.1: Left: $T = \sqrt{n}(\bar{X}_n - \mu)/S_n$ over 50,000 Gaussian samples of $n = 5$ (histogram), with the Gauss$(0, 1)$ density (black) and the t_4 density (orange, dashed). Right: the fraction of samples with $|T| > 1.96$ against n; dotted: 5%.

Intuition: dividing by an estimate adds noise

The numerator $\sqrt{n}(\bar{X}_n - \mu)$ is Gaussian with standard deviation σ. Dividing by the constant σ standardizes it. Dividing by S_n divides by a random variable that is near σ on average but wanders, and when the sample happens to be unusually tight, S_n is small and the ratio is inflated. At $n = 5$, S_n is below 0.6σ in about one sample in six, and each such sample stretches its T by a factor above 1.6. That is where the fat tails come from, and it is why the effect fades as n grows: $S_n \xrightarrow[n\to\infty]{\mathbb{P}} \sigma$ by the Law of Large Numbers, so for large n dividing by S_n is almost dividing by σ.

The plan of the chapter is to find the exact distribution of $\sqrt{n}(\bar{X}_n - \mu)/S_n$. Write it as

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{S_n} = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{\sqrt{S_n^2/\sigma^2}} = \frac{Z}{\sqrt{S_n^2/\sigma^2}}, \qquad Z \sim \text{Gauss}(0,1). \tag{14.1}$$

Dividing numerator and denominator by σ changes nothing, and it makes both pieces free of μ and σ: the numerator is a standard Gaussian, and the denominator is S_n^2/σ^2, whose distribution does not depend on μ or σ either, since shifting the data leaves S_n^2 unchanged and scaling them by a constant scales S_n^2 by its square. Two questions remain: what is the distribution of S_n^2/σ^2, and how does it relate to the numerator? The chi-squared distribution answers the first (Section 14.2), Cochran's theorem the second (Section 14.3), and Student's t distribution is the name of the ratio (Section 14.4). Because the ratio's distribution depends on neither μ nor σ, it is a pivot in the sense of Chapter 8, and the test and interval built on it (Section 14.5) are exact at every n.

14.2 The chi-squared distribution

S_n^2/σ^2 is, up to the factor $1/(n-1)$, a sum of squared deviations divided by σ^2: a sum of squares of things that are nearly standard Gaussians. So the distribution of a sum of squared standard Gaussians is the one to study.

Definition 14.1 (Chi-squared distribution)**.** For a positive integer k, the **chi-squared distribution with k degrees of freedom**, written χ^2_k, is the distribution of

$$V = Z_1^2 + Z_2^2 + \cdots + Z_k^2, \qquad Z_1, \ldots, Z_k \overset{iid}{\sim} \text{Gauss}(0,1).$$

Its quantile with probability α above it is written $q_\alpha(\chi^2_k)$.

In words. Take k independent standard Gaussians, square each, add. The result is positive, since squares are, and it is the squared length of the vector $(Z_1, \ldots, Z_k)$: the squared distance from the origin of a random point whose coordinates are independent standard Gaussians. 'Degrees of freedom' counts the independent Gaussians that went in.

For the kiss data. Chapter 9 tested $p = \frac{1}{2}$ with the standardized statistic $T_n = \sqrt{124}\,(0.645 - 0.5)/0.5 = 3.23$, approximately Gauss$(0,1)$ under the null, and rejected two-sidedly when $|T_n| > 1.96$. The same test can be written with one squared Gaussian: $T_n^2 = 10.45$ is approximately χ^2_1 under the null, and $|T_n| > 1.96$ is the same event as $T_n^2 > 1.96^2 = 3.84 = q_{0.05}(\chi^2_1)$. The two-sided p-value $\mathbb{P}(|Z| > 3.23) = \mathbb{P}(\chi^2_1 > 10.45) = 0.0012$ comes out the same either way. Squaring turns a two-sided Gaussian test into a one-sided chi-squared test, which is how Chapter 15 tests several

parameters at once.

Proposition 14.2 (Properties of the chi-squared distribution)**.** *Let* $V \sim \chi^2_k$.

1. $\mathbb{E}[V] = k$ *and* $\mathbb{V}ar(V) = 2k$.
2. *If* $V \sim \chi^2_k$ *and* $W \sim \chi^2_l$ *are independent, then* $V + W \sim \chi^2_{k+l}$.
3. V *has density* $f_k(v) = \dfrac{1}{2^{k/2}\,\Gamma(k/2)}\, v^{k/2-1} e^{-v/2}$ *for* $v > 0$, *where* $\Gamma(x) = \int_0^\infty s^{x-1} e^{-s}\,\mathrm{d}s$ *is the Gamma function, which satisfies* $\Gamma(x+1) = x\Gamma(x)$, $\Gamma(n) = (n-1)!$ *for integers and* $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. *For* $k = 1$, $f_1(v) = e^{-v/2}/\sqrt{2\pi v}$; *for* $k = 2$, $f_2(v) = \frac{1}{2}e^{-v/2}$, *the Expo*$(\frac{1}{2})$ *density.*
4. *As* $k \to \infty$, $(V - k)/\sqrt{2k} \xrightarrow[n\to\infty]{(d)}$ *Gauss*$(0, 1)$.

Recall. The Central Limit Theorem (Theorem 4.8): for i.i.d. Y_i with mean m and variance $s^2 < \infty$, $\sqrt{k}(\bar{Y}_k - m) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, s^2)$.

Proof strategy (for completeness; skip on a first reading). Everything follows from the definition as a sum of independent squares: means and variances add, two sums of squares concatenate, and a sum of k i.i.d. terms obeys the CLT. The density for $k = 1$ is a change of variables; the general density is a convolution, done in Appendix C.4.

Proof. (1) $\mathbb{E}[Z^2] = \mathbb{V}\mathrm{ar}(Z) = 1$, so $\mathbb{E}[V] = k$. For the variance, $\mathbb{V}\mathrm{ar}(Z^2) = \mathbb{E}[Z^4] - (\mathbb{E}[Z^2])^2 = \mathbb{E}[Z^4] - 1$, and integrating by parts with $u = z^3$ and $\mathrm{d}v = z\varphi(z)\,\mathrm{d}z = -\mathrm{d}\varphi(z)$, where φ is the standard Gaussian density,

$$\mathbb{E}[Z^4] = \int_{-\infty}^{\infty} z^3 \cdot z\varphi(z)\,\mathrm{d}z = \Big[-z^3\varphi(z)\Big]_{-\infty}^{\infty} + 3\int_{-\infty}^{\infty} z^2\varphi(z)\,\mathrm{d}z = 0 + 3\,\mathbb{E}[Z^2] = 3.$$

So $\mathbb{V}\mathrm{ar}(Z^2) = 2$, and the variance of a sum of k independent terms is $2k$.

(2) The distribution of $V + W$ depends only on the joint distribution of (V, W), which under independence is the same however the two are constructed, so take V to be a sum of k squared independent standard Gaussians and W a sum of l others, independent of the first k. Then $V + W$ is a sum of $k + l$ squared i.i.d. standard Gaussians.

(3) For $k = 1$ and $v > 0$, $\mathbb{P}(Z^2 \le v) = \mathbb{P}(-\sqrt{v} \le Z \le \sqrt{v}) = 2\Phi(\sqrt{v}) - 1$, and differentiating in v (toolkit A.6) gives the density $2\varphi(\sqrt{v})/(2\sqrt{v}) = e^{-v/2}/\sqrt{2\pi v}$, which is the formula with $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. The general formula follows by adding one squared Gaussian at a time, a convolution computed in Appendix C.4; nothing later depends on the formula itself, only on the definition and on (1), (2) and (4).

(4) $V = \sum_{i=1}^k Z_i^2$ is a sum of k i.i.d. terms with mean 1 and variance 2, so by the CLT, $\sqrt{k}(V/k - 1) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, 2)$, which is the claim after dividing by $\sqrt{2}$. □

Figure 14.2 draws the densities. For $k = 1$ the density is infinite at 0 (a standard Gaussian is near 0 often, so its square is near 0 very often) and falls away; for $k = 2$ it is the exponential; from $k = 3$ on it starts at 0, rises to a peak and has a long right tail. The peak sits at $k - 2$ and the mean at k, so the distribution is skewed to the right, with the mean pulled past the peak by the tail. As k grows the shape becomes symmetric: on the right of the figure, χ^2_{30} is close to Gauss$(30, 60)$, the Gaussian with its mean and variance, as statement (4) says it should be, with a remaining lean to the right.

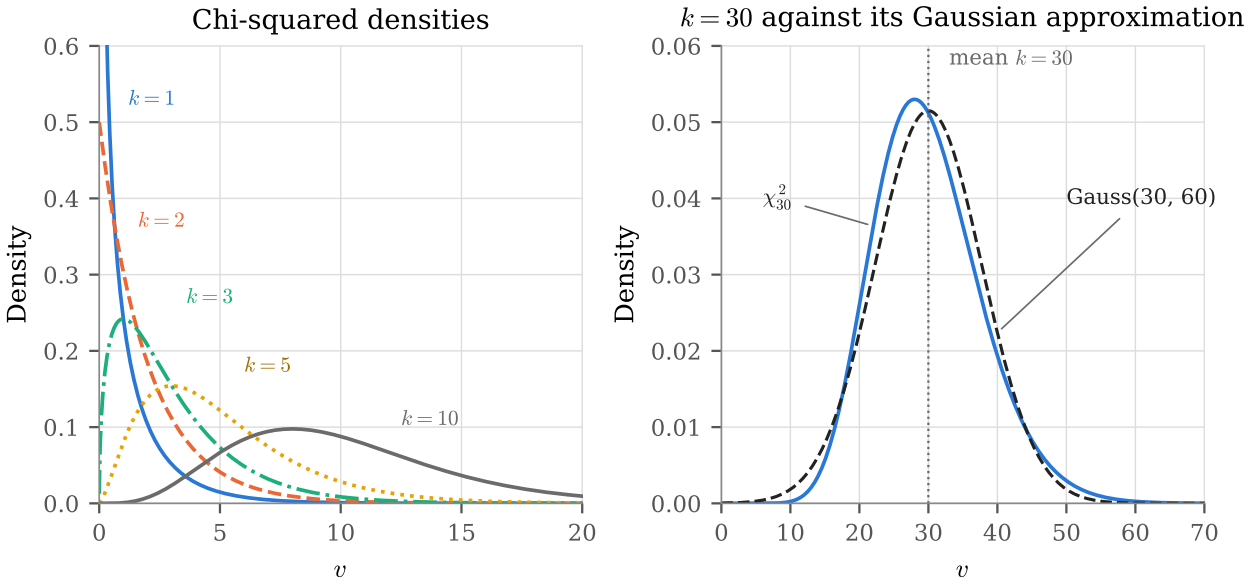


Figure 14.2: Left: the χ^2_k densities for $k = 1, 2, 3, 5, 10$. Right: χ^2_{30} (blue) with its mean marked (dotted) and the Gaussian with the same mean and variance, Gauss$(30, 60)$ (black, dashed).

Does this make sense? Where should the peak be?

A χ^2_k variable has mean k and standard deviation $\sqrt{2k}$: for $k = 10$, mean 10 and standard deviation 4.5; for $k = 100$, mean 100 and standard deviation 14. The spread grows like $\sqrt{k}$ while the mean grows like k, so relative to its mean the distribution tightens: V/k has standard deviation $\sqrt{2/k}$, which is the Law of Large Numbers for the average of k squared Gaussians. That is why the sample variance, an average of squared deviations, is a reliable estimate of σ^2 for large n and an unreliable one for small n: at $n = 5$ its relative standard deviation is $\sqrt{2/4} = 0.71$, so a sample variance half or double the truth is routine.

Worked example: two sums of squares

A known mean. If $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(0, \sigma^2)$, then $X_i/\sigma \overset{iid}{\sim} \text{Gauss}(0, 1)$ and

$$\frac{1}{\sigma^2}\sum_{i=1}^{n} X_i^2 = \sum_{i=1}^{n}\left(\frac{X_i}{\sigma}\right)^2 \sim \chi^2_n.$$

So the sample second moment $V_n = \frac{1}{n}\sum_i X_i^2$ satisfies $nV_n/\sigma^2 \sim \chi^2_n$, with mean n: $\mathbb{E}[V_n] = \sigma^2$, and $\mathbb{V}\text{ar}(V_n) = 2\sigma^4/n$. For $n = 12$ measurements with a known mean of zero, the variance estimate has relative standard deviation $\sqrt{2/12} = 0.41$.

Darts. A dart lands at (Z_1, Z_2) with independent standard Gaussian coordinates. The probability of landing inside the unit disk is $\mathbb{P}(Z_1^2 + Z_2^2 \le 1) = \mathbb{P}(\chi^2_2 \le 1) = 1 - e^{-1/2} = 0.393$, since χ^2_2 is Expo$(\frac{1}{2})$: about 39 of every 100 throws.

14.3 Cochran's theorem

Now the denominator of (14.1). Writing $Z_i = (X_i - \mu)/\sigma$, which are i.i.d. Gauss$(0,1)$,

$$\frac{(n-1)S_n^2}{\sigma^2} = \sum_{i=1}^{n} \left(\frac{X_i - \bar{X}_n}{\sigma}\right)^2 = \sum_{i=1}^{n} (Z_i - \bar{Z}_n)^2.$$

If $\bar{Z}_n$ were replaced by the true mean 0, this would be $\sum_i Z_i^2 \sim \chi^2_n$ by the worked example. It is not: the deviations are measured from the sample mean, which was itself computed from the same Z_i, and the n terms $Z_i - \bar{Z}_n$ are neither independent nor standard Gaussian (they sum to zero, so any $n-1$ of them determine the last). The answer is that one degree of freedom is spent on the mean: the sum is χ^2_{n-1}, and it is independent of $\bar{Z}_n$.

Intuition: a Gaussian cloud is round, so rotate it

Draw $n = 2$ independent standard Gaussians as a point (Z_1, Z_2) in the plane and repeat 1,500 times (Figure 14.3, left). The cloud is round: its density $\frac{1}{2\pi}e^{-(z_1^2+z_2^2)/2}$ depends only on the distance from the origin, so the cloud looks the same from every direction. Now describe each point not by its coordinates along the usual axes but along two rotated axes, u_1 in the direction $(1,1)/\sqrt{2}$ and u_2 in the direction $(1,-1)/\sqrt{2}$. The coordinate along u_1 is $Y_1 = (Z_1 + Z_2)/\sqrt{2} = \sqrt{2}\,\bar{Z}_2$: it carries the sample mean. The coordinate along u_2 is $Y_2 = (Z_1 - Z_2)/\sqrt{2}$, and $Y_2^2 = (Z_1 - Z_2)^2/2 = (Z_1 - \bar{Z}_2)^2 + (Z_2 - \bar{Z}_2)^2$: it carries the sum of squared deviations. Because the cloud is round, the picture in the rotated coordinates (right) is the same round cloud: Y_1 and Y_2 are again independent standard Gaussians. So the sample mean and the sum of squared deviations are built from two independent standard Gaussians, one each, and the sum of squares is $Y_2^2 \sim \chi^2_1$: one degree of freedom, not two because the other one went to the mean. For general n the same rotation works in n dimensions: one axis for the mean, $n-1$ for the deviations.

The fact that makes the picture a proof is in the Toolkit (toolkit A.13): if $Z = (Z_1, \ldots, Z_n)$ has i.i.d. Gauss$(0,1)$ coordinates and Q is an orthogonal matrix (a rotation, possibly with reflections: $Q^\top Q = I$), then $Y = QZ$ again has i.i.d. Gauss$(0,1)$ coordinates, and $\|Y\| = \|Z\|$.

Theorem 14.3 (Cochran's theorem for the sample mean and variance)**.** *Let* $X_1, \ldots, X_n \overset{iid}{\sim}$ *Gauss*(μ, σ^2) *with* $n \geq 2$*. Then*

1. $\bar{X}_n$ *and* S_n^2 *are independent;*
2. $\dfrac{(n-1)S_n^2}{\sigma^2} \sim \chi^2_{n-1}$.

Proof strategy (core). Standardize to $Z_i = (X_i - \mu)/\sigma$ and rotate: pick an orthogonal Q whose first row is $(1, \ldots, 1)/\sqrt{n}$. The first rotated coordinate is $\sqrt{n}\bar{Z}_n$; the sum of squared deviations is the squared length of the other $n-1$ coordinates, since rotation preserves length. The rotated coordinates are i.i.d. standard Gaussians, so the two pieces are independent and the second is a sum of $n-1$ squares.

Proof. Let $Z_i = (X_i - \mu)/\sigma$, so $Z = (Z_1, \ldots, Z_n)$ has i.i.d. Gauss$(0,1)$ coordinates, $\bar{X}_n = \mu + \sigma\bar{Z}_n$ and $(n-1)S_n^2/\sigma^2 = \sum_i (Z_i - \bar{Z}_n)^2$. It is enough to prove the two statements for $\bar{Z}_n$ and $\sum_i (Z_i - \bar{Z}_n)^2$.

Let Q be an $n \times n$ orthogonal matrix whose first row is $(1/\sqrt{n}, \ldots, 1/\sqrt{n})$; one exists because any unit vector can be extended to an orthonormal basis (toolkit A.13). Set $Y = QZ$. By the Toolkit,

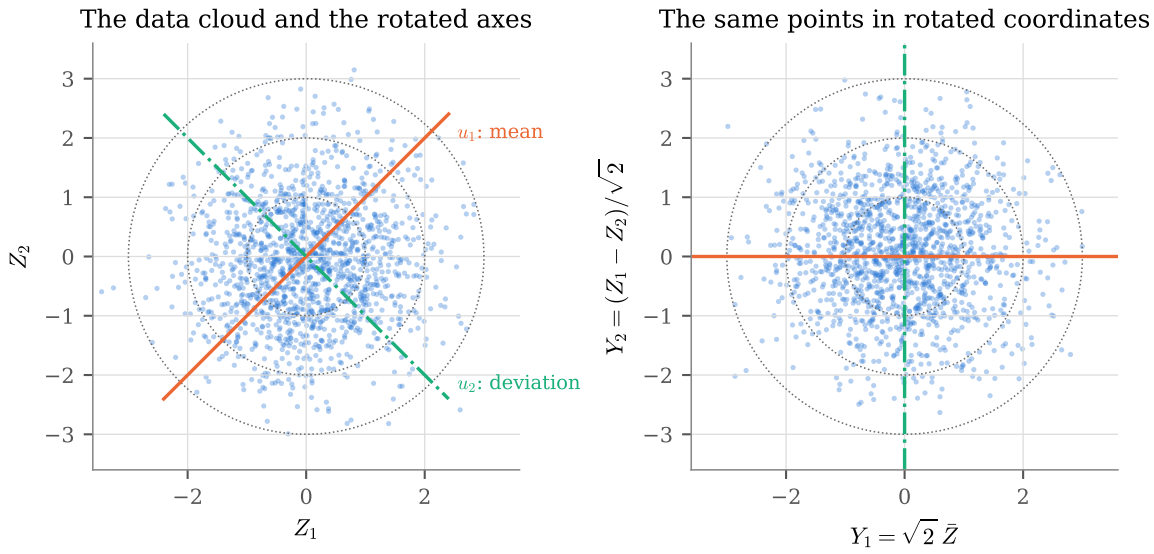


Figure 14.3: Left: 1,500 draws of (Z_1, Z_2) with $Z_1, Z_2 \overset{iid}{\sim} \text{Gauss}(0,1)$, the axis $u_1 = (1,1)/\sqrt{2}$ (orange) and $u_2 = (1,-1)/\sqrt{2}$ (aqua, dash-dot). Right: the same points in the rotated coordinates $Y_1 = \sqrt{2}\bar{Z}$ and $Y_2 = (Z_1 - Z_2)/\sqrt{2}$. Gray circles: contours of the density.

$Y_1, \ldots, Y_n$ are i.i.d. Gauss$(0,1)$, and $\|Y\|^2 = \|Z\|^2$. The first coordinate is

$$Y_1 = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Z_i = \sqrt{n}\,\bar{Z}_n.$$

Expanding the squared deviations,

$$\sum_{i=1}^{n} (Z_i - \bar{Z}_n)^2 = \sum_{i=1}^{n} Z_i^2 - n\bar{Z}_n^2 = \|Z\|^2 - Y_1^2 = \|Y\|^2 - Y_1^2 = Y_2^2 + \cdots + Y_n^2,$$

where the first equality expands the square (the cross term is $-2\bar{Z}_n \sum_i Z_i = -2n\bar{Z}_n^2$) and the third is $\|QZ\| = \|Z\|$. So $\bar{Z}_n = Y_1/\sqrt{n}$ is a function of Y_1 alone and $\sum_i (Z_i - \bar{Z}_n)^2$ a function of $(Y_2, \ldots, Y_n)$ alone; since the joint density of Y factors into a function of y_1 times a function of $(y_2, \ldots, y_n)$ (toolkit A.13), any function of the first is independent of any function of the rest, which is (1). And $Y_2^2 + \cdots + Y_n^2$ is a sum of $n-1$ squared i.i.d. standard Gaussians, which is (2). □

Corollary 14.4 (Mean and variance of the sample variance)**.** *For* $X_1, \ldots, X_n \overset{iid}{\sim}$ *Gauss*(μ, σ^2), $\mathbb{E}[S_n^2] = \sigma^2$ *and* $\mathbb{V}ar(S_n^2) = \dfrac{2\sigma^4}{n-1}$.

Proof strategy (for completeness; skip on a first reading). S_n^2 is a chi-squared variable with $n-1$ degrees of freedom, scaled by $\sigma^2/(n-1)$; scale its mean and variance.

Proof. $S_n^2 = \sigma^2 V/(n-1)$ with $V \sim \chi^2_{n-1}$, so $\mathbb{E}[S_n^2] = \sigma^2(n-1)/(n-1)$ and $\mathbb{V}\text{ar}(S_n^2) = \sigma^4 \cdot 2(n-1)/(n-1)^2$. □

The unbiasedness was proved for any distribution in Chapter 7; the variance formula is new, and Gaussian. For the wind data, $n = 12$: the sample variance has standard deviation $\sigma^2\sqrt{2/11} = 0.43\sigma^2$, so the observed $4.7^2 = 22.1$ is a rough estimate of σ^2, which is exactly why the chapter cannot pretend S_n is σ.

Does this make sense? Cochran at $n = 2$, by hand

With two observations, $\bar{X}_2 = (X_1 + X_2)/2$ and $S_2^2 = (X_1 - X_2)^2/2$ (both deviations from the mean equal $\pm(X_1 - X_2)/2$). The theorem says $S_2^2/\sigma^2 = \big((X_1 - X_2)/(\sqrt{2}\sigma)\big)^2 \sim \chi_1^2$, which checks: $X_1 - X_2 \sim \text{Gauss}(0, 2\sigma^2)$ by the sum rule of Chapter 6, so $(X_1 - X_2)/(\sqrt{2}\sigma)$ is a standard Gaussian and its square is χ_1^2. It also says the sum $X_1 + X_2$ and the difference $X_1 - X_2$ are independent, which is the 45° rotation of Figure 14.3. Independence here is not obvious: both are functions of the same two numbers. It holds because the cloud is round, and Exercise 14.2 verifies it directly from the joint density.

Pitfall: the independence is a Gaussian specialty

Figure 14.4 plots the sample mean against the sample variance for 4,000 samples of $n = 5$. For Gaussian data (left) the cloud has no structure: knowing that a sample's mean is high says nothing about its variance. For exponential data with the same mean and variance (right), samples with a high mean have a high variance, with a correlation of 0.70. The reason is the shape of the exponential density: a large sample mean usually means the sample contains one of the distribution's occasional large values, which also inflates the variance. Cochran's theorem is false for non-Gaussian data, and so is everything built on it, including the exactness of the t-test. Section 14.5 says what survives.

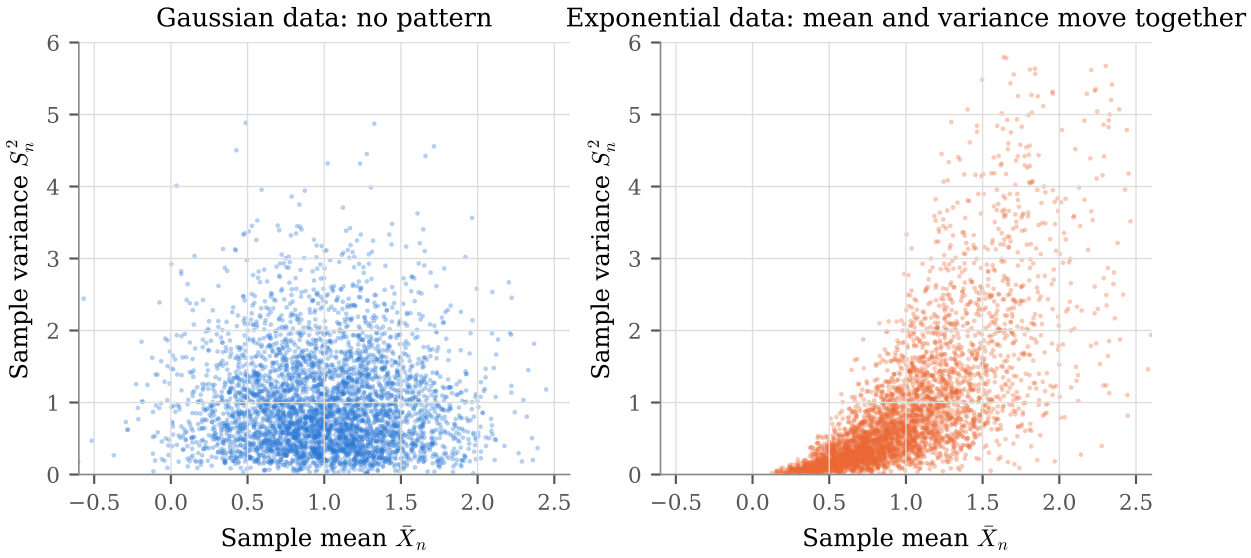


Figure 14.4: The sample mean and the sample variance of 4,000 samples of $n = 5$. Left: Gauss(1, 1) data (blue). Right: Expo(1) data (orange), which also has mean 1 and variance 1.

From undergraduate statistics: degrees of freedom, and the $n - 1$ for the third time

'Degrees of freedom' in an introductory course is a number looked up to find the right row of a table, and $n - 1$ is explained, if at all, as 'one degree of freedom is lost estimating the mean'. Cochran's theorem is that sentence made exact. The n deviations $X_i - \bar{X}_n$ live in an $(n - 1)$-dimensional space, since they sum to zero, and in the rotated coordinates they are exactly $n - 1$ independent Gaussians: $n - 1$ is the count of independent squared Gaussians in the sample variance, and it is the degrees of freedom of its chi-squared distribution. Chapter 7

gave two earlier readings of the same $n-1$: it removes the bias of the divide-by-n variance, and it is not the choice that minimizes risk (Exercise 14.3 finishes that story with the variance formula of Corollary 14.4).

Remark (The general theorem). What is called Cochran's theorem in full is a statement about splitting $\|Z\|^2$ into several sums of squares, one per subspace in an orthogonal decomposition of $\mathbb{R}^n$: each piece is chi-squared with the dimension of its subspace as degrees of freedom, and the pieces are independent. Theorem 14.3 is the case of two subspaces, the line through $(1,\dots,1)$ and everything perpendicular to it. Chapter 21 uses the general case, with the subspace spanned by the regressors in place of the line, to get the exact distribution of the residual variance in Gaussian linear regression; the proof is the one above with a different orthogonal basis.

14.4 Student's t distribution

Both pieces of (14.1) are now known and independent: the numerator is $Z \sim \text{Gauss}(0,1)$ and the denominator is $\sqrt{V/(n-1)}$ with $V \sim \chi^2_{n-1}$. The ratio has a name.

Definition 14.5 (Student's t distribution)**.** For a positive integer k, **Student's t distribution with k degrees of freedom**, written t_k, is the distribution of

$$T = \frac{Z}{\sqrt{V/k}}, \qquad Z \sim \text{Gauss}(0,1),\ V \sim \chi^2_k,\ Z \text{ and } V \text{ independent.}$$

Its quantile with probability α above it is written $q_\alpha(t_k)$.

In words. A standard Gaussian divided by an independent estimate of its own scale, where the estimate is the root of an average of k squared standard Gaussians; the average has mean 1 and wanders less as k grows. The ratio is symmetric about 0 like Z, and wider than Z because the denominator is sometimes small.

For the kiss data. The t statistic is $\sqrt{n}(\bar{R}_n - \frac{1}{2})/S_n$ with $S_n^2 = \frac{n}{n-1}\hat{p}(1-\hat{p}) = 0.231$: $T = \sqrt{124}\,(0.645-0.5)/0.480 = 3.36$, against the two-sided threshold $q_{0.025}(t_{123}) = 1.979$ in place of 1.960. The threshold moved by 0.02 because at $n = 124$ the denominator hardly wanders; the data are 0s and 1s, so the exactness of Theorem 14.3 does not apply to them, and Section 14.5 says why the test is sound anyway.

Proposition 14.6 (Properties of Student's t)**.** *Let $T \sim t_k$.*

1. *T is symmetric about 0: $-T \sim t_k$, so $q_\alpha(t_k) = -q_{1-\alpha}(t_k)$.*
2. *T has density $g_k(t) = \dfrac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi}\,\Gamma\left(\frac{k}{2}\right)}\left(1+\dfrac{t^2}{k}\right)^{-(k+1)/2}$. For $k=1$ this is the Cauchy density $1/(\pi(1+t^2))$ of Chapter 13.*
3. *$\mathbb{E}[T] = 0$ for $k \geq 2$ and $\mathbb{V}ar(T) = \dfrac{k}{k-2}$ for $k \geq 3$; for $k = 1$ the mean does not exist, and for $k = 2$ the variance is infinite.*
4. *As $k \to \infty$, $t_k \xrightarrow[n\to\infty]{(d)}$ Gauss$(0,1)$: for every t, $\mathbb{P}(T_k \leq t) \to \Phi(t)$, and $q_\alpha(t_k) \to q_\alpha$.*

Recall. The Law of Large Numbers (Theorem 3.5): for i.i.d. Y_i with $\mathbb{V}\mathrm{ar}(Y_i) < \infty$, $\bar{Y}_k \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[Y_1]$. The continuous mapping theorem (Theorem 5.2): if $W_k \xrightarrow[n\to\infty]{\mathbb{P}} w$ and g is continuous, then $g(W_k) \xrightarrow[n\to\infty]{\mathbb{P}} g(w)$. Slutsky's theorem (Theorem 5.1): if $X_k \xrightarrow[n\to\infty]{(\mathrm{d})} X$ and $W_k \xrightarrow[n\to\infty]{\mathbb{P}} w \neq 0$, then $X_k/W_k \xrightarrow[n\to\infty]{(\mathrm{d})} X/w$.

Proof strategy (for completeness; skip on a first reading). Symmetry is the symmetry of Z. The limit is the Law of Large Numbers for the denominator plus Slutsky: V/k is an average of k squared Gaussians, so it tends to 1. The density is a calculation done in Appendix C.4 and the moments are an exercise; nothing later depends on them.

Proof. (1) $-T = (-Z)/\sqrt{V/k}$ and $-Z \sim \mathrm{Gauss}(0,1)$ is independent of V, so $-T$ satisfies the definition. For the quantiles, $\mathbb{P}(T > q) = \alpha$ gives $\mathbb{P}(-T < -q) = \alpha$, that is, $\mathbb{P}(T < -q) = \alpha$, so $-q$ has probability $1-\alpha$ above it.

(4) Write $V = Z_1^2 + \cdots + Z_k^2$, independent of Z. Then V/k is the average of k i.i.d. variables with mean 1 and variance 2, so $V/k \xrightarrow[n\to\infty]{\mathbb{P}} 1$ by the Law of Large Numbers, and $\sqrt{V/k} \xrightarrow[n\to\infty]{\mathbb{P}} 1$ by the continuous mapping theorem. The numerator Z does not depend on k at all (a constant sequence converges in distribution to itself), so by Slutsky's theorem $Z/\sqrt{V/k} \xrightarrow[n\to\infty]{(\mathrm{d})} Z/1 = Z$. Convergence in distribution to a continuous limit is convergence of the CDFs F_k at every point, and it carries the quantiles along. Fix $\epsilon > 0$. Since Φ is strictly increasing, $\Phi(q_\alpha - \epsilon) < 1-\alpha < \Phi(q_\alpha + \epsilon)$, and since $F_k(q_\alpha \pm \epsilon) \to \Phi(q_\alpha \pm \epsilon)$, for all large k we have $F_k(q_\alpha - \epsilon) < 1-\alpha < F_k(q_\alpha + \epsilon)$; the solution $q_\alpha(t_k)$ of $F_k(q) = 1-\alpha$ then lies in $(q_\alpha - \epsilon, q_\alpha + \epsilon)$.

(2) is computed in Appendix C.4 by conditioning on V and integrating. (3) is Exercise 14.4, from $\mathbb{E}[1/V] = 1/(k-2)$ for $V \sim \chi^2_k$. □

Figure 14.5 draws the densities on the left. Every t_k is a lower, wider bell than the Gaussian, and the tails are where the difference lives: at $t = 3$ the Gaussian density is 0.0044 while t_5's is 0.017, four times larger, and t_1's is 0.032. The t_1 curve is the Cauchy density, the distribution whose mean does not exist and whose sample mean never settles (Chapter 13); here it appears as the most extreme member of the family, the ratio of a Gaussian to the absolute value of a single other Gaussian. By $k = 30$ the curve is nearly on top of the Gaussian. On the right, the exact version of Figure 14.1's simulation: the probability that $|t_{n-1}|$ exceeds 1.96, which is the true type 1 error of the Gaussian 5% rule on Gaussian samples of size n. It is 0.30 at $n = 2$, 0.12 at $n = 5$, 0.076 at $n = 12$ and 0.060 at $n = 30$, and it is still 0.055 at $n = 60$. The table of quantiles tells the same story from the other side.

k	$q_{0.10}(t_k)$	$q_{0.05}(t_k)$	$q_{0.025}(t_k)$	$q_{0.01}(t_k)$	$q_{0.005}(t_k)$
1	3.08	6.31	12.71	31.82	63.66
2	1.89	2.92	4.30	6.96	9.92
5	1.48	2.02	2.57	3.36	4.03
11	1.36	1.80	2.20	2.72	3.11
30	1.31	1.70	2.04	2.46	2.75
100	1.29	1.66	1.98	2.36	2.63
∞ (Gaussian)	1.28	1.645	1.96	2.33	2.58

With one degree of freedom the 2.5% threshold is 12.7 instead of 1.96: a single sample variance is so unreliable that the test demands a sample mean more than twelve estimated standard errors from the null before it will reject at 5%. With 30 degrees of freedom the threshold is 2.04, and the

introductory rule of thumb that '$n \geq 30$ is large' is the observation that the row for 30 is within 4% of the Gaussian row at the levels an introductory course uses, 5% and 2.5%.

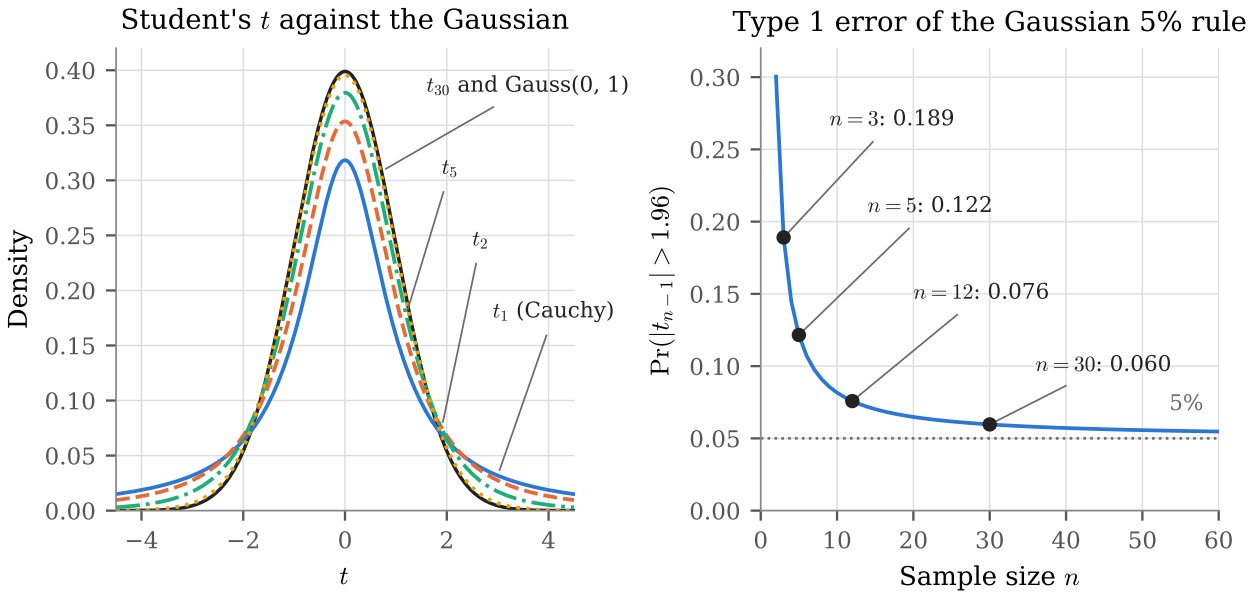


Figure 14.5: Left: the densities of t_1 (blue), t_2 (orange, dashed), t_5 (aqua, dash-dot) and t_{30} (yellow, dotted), with Gauss$(0, 1)$ in black. Right: $\mathbb{P}(|t_{n-1}| > 1.96)$ against n; dotted: 5%.

Theorem 14.7 (The t pivot)**.** *If* $X_1, \ldots, X_n \overset{iid}{\sim} Gauss(\mu, \sigma^2)$ *with* $n \geq 2$*, then*

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{S_n} \sim t_{n-1} \qquad \textit{exactly, for every } n, \mu \textit{ and } \sigma^2.$$

Proof strategy (core). Decompose as in (14.1); the numerator is a standard Gaussian, the denominator is $\sqrt{V/(n-1)}$ with $V \sim \chi^2_{n-1}$ by Cochran, and the two are independent by Cochran.

Proof. By (14.1),

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{S_n} = \frac{Z}{\sqrt{V/(n-1)}}, \qquad Z = \sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma}, \quad V = \frac{(n-1)S_n^2}{\sigma^2}.$$

$Z \sim \text{Gauss}(0, 1)$ because $\bar{X}_n \sim \text{Gauss}(\mu, \sigma^2/n)$ (Chapter 6). $V \sim \chi^2_{n-1}$ by Theorem 14.3(2). Z is a function of $\bar{X}_n$ and V a function of S_n^2, which are independent by Theorem 14.3(1). So the ratio satisfies Definition 14.5 with $k = n - 1$. □

The distribution was found in 1908 by William Sealy Gosset, a chemist at the Guinness brewery in Dublin who needed to compare small batches of barley; the brewery forbade its staff from publishing, so the paper appeared under the pseudonym 'Student'.

From undergraduate statistics: the t critical value you were taught

An introductory course says: when σ is unknown, use s and look up the t table with $n - 1$ degrees of freedom instead of the z table. Theorem 14.7 is the reason, and it also says what the rule assumes: the t table is exact when the data are Gaussian and only then. What that course

called 'the t distribution gets closer to the normal as n grows' is Proposition 14.6(4), and 'use z when $n \geq 30$' is a judgment that 2.04 is close enough to 1.96.

14.5 The t-test and the t-interval

With an exact pivot, the test and the interval follow the recipe of Chapters 8 and 9 with the Student quantile in place of the Gaussian one.

Definition 14.8 (The one-sample t-test)**.** Let $X_1, \dots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$ with μ and σ^2 unknown, let μ_0 be a fixed number, and let

$$T_n = \sqrt{n}\,\frac{\bar{X}_n - \mu_0}{S_n}.$$

Student's t-test at level α is

$$\begin{aligned}
&H_0 : \mu = \mu_0,\ H_1 : \mu \neq \mu_0 : \quad \psi_\alpha = \mathbb{1}(|T_n| > q_{\alpha/2}(t_{n-1})), \quad p\text{-value } \mathbb{P}(|t_{n-1}| > |T_n^{\text{obs}}|);\\
&H_0 : \mu \leq \mu_0,\ H_1 : \mu > \mu_0 : \quad \psi_\alpha = \mathbb{1}(T_n > q_\alpha(t_{n-1})), \quad\ p\text{-value } \mathbb{P}(t_{n-1} > T_n^{\text{obs}});\\
&H_0 : \mu \geq \mu_0,\ H_1 : \mu < \mu_0 : \quad \psi_\alpha = \mathbb{1}(T_n < -q_\alpha(t_{n-1})), \quad\ p\text{-value } \mathbb{P}(t_{n-1} < T_n^{\text{obs}}).
\end{aligned}$$

where T_n^{obs} is the value of T_n computed from the data. The **t-interval** is

$$\mathcal{I} = \big[\bar{X}_n - q_{\alpha/2}(t_{n-1})\,S_n/\sqrt{n},\ \bar{X}_n + q_{\alpha/2}(t_{n-1})\,S_n/\sqrt{n}\big].$$

In words. Standardize the sample mean's distance from the null value by its estimated standard error $S_n/\sqrt{n}$, and compare with a Student threshold instead of a Gaussian one. The threshold is larger, by a lot for small n and by a little for large n, to pay for the noise in S_n. The interval is the usual 'estimate ± threshold × standard error' with the Student threshold.

For the kiss data. $T = 3.36$ against $q_{0.025}(t_{123}) = 1.979$: reject $p = \frac{1}{2}$ at 5%, with p-value $\mathbb{P}(|t_{123}| > 3.36) = 0.0010$. The t-interval is $0.645 \pm 1.979 \times 0.480/\sqrt{124} = [0.560, 0.730]$, against the plug-in interval $[0.561, 0.729]$ of Chapter 8: at $n = 124$ the two recipes agree to within 0.001.

Proposition 14.9 (Exact level, exact coverage, consistency)**.** *For $X_1, \dots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$:*

1. *Each test in Definition 14.8 has level exactly α at every $n \geq 2$: the type 1 error is α when $\mu = \mu_0$, and at most α everywhere in a composite null $\mu \leq \mu_0$ or $\mu \geq \mu_0$.*
2. *The t-interval has coverage exactly $1 - \alpha$ at every $n \geq 2$.*
3. *Under any fixed alternative $\mu \neq \mu_0$, $\mathbb{P}_\mu(\psi_\alpha = 1) \to 1$ as $n \to \infty$.*

Proof strategy (for completeness; skip on a first reading). At $\mu = \mu_0$ the statistic is the pivot of Theorem 14.7, whose quantile was chosen to leave probability α outside. For a composite null, moving μ into the null shifts T_n the wrong way for rejecting, so the type 1 error can only fall. The interval is the pivot trapped between its quantiles and solved for μ. Consistency is the pivot staying put while $\sqrt{n}(\mu - \mu_0)/S_n$ runs off to infinity.

Proof. (1) Write $T_n = T_n^* + \sqrt{n}(\mu - \mu_0)/S_n$ with $T_n^* = \sqrt{n}(\bar{X}_n - \mu)/S_n \sim t_{n-1}$ by Theorem 14.7, whatever μ is. When $\mu = \mu_0$, $T_n = T_n^*$ and $\mathbb{P}_{\mu_0}(|T_n| > q_{\alpha/2}(t_{n-1})) = \alpha$ by the definition of the quantile and the symmetry of t_{n-1}; likewise $\mathbb{P}_{\mu_0}(T_n > q_\alpha(t_{n-1})) = \alpha$. For the composite null $\mu \leq \mu_0$ of the second test, $\sqrt{n}(\mu - \mu_0)/S_n \leq 0$ since $S_n > 0$, so $T_n \leq T_n^*$ and

$$\mathbb{P}_\mu\big(T_n > q_\alpha(t_{n-1})\big) \leq \mathbb{P}\big(T_n^* > q_\alpha(t_{n-1})\big) = \alpha,$$

with equality at $\mu = \mu_0$. The third test is the mirror image.

(2) $\mathbb{P}_{\mu,\sigma^2}(\mu \in \mathcal{I}) = \mathbb{P}\big(|\sqrt{n}(\bar{X}_n - \mu)/S_n| \le q_{\alpha/2}(t_{n-1})\big) = \mathbb{P}(|t_{n-1}| \le q_{\alpha/2}(t_{n-1})) = 1 - \alpha$.

(3) Take the two-sided test and $\mu > \mu_0$; the one-sided cases are identical. The threshold $q_{\alpha/2}(t_{n-1})$ depends on n, but it is at most $c = q_{\alpha/2}(t_1)$, the largest quantile in the family (12.7 at $\alpha = 0.05$), so it is enough to show $\mathbb{P}_\mu(T_n > c) \to 1$. Fix $M > 0$. Since $T_n^* \sim t_{n-1} \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0,1)$ by Proposition 14.6(4), $\mathbb{P}(T_n^* > -M) \to \Phi(M)$. Since $S_n \xrightarrow[n\to\infty]{\mathbb{P}} \sigma$ (the Law of Large Numbers applied to the X_i and to the X_i^2, which have finite variance for Gaussian data, then the continuous mapping theorem for the square root, as in Chapter 7), $\sqrt{n}(\mu - \mu_0)/S_n \xrightarrow[n\to\infty]{\mathbb{P}} +\infty$: $\mathbb{P}(\sqrt{n}(\mu - \mu_0)/S_n > c + M) \to 1$. On the intersection of the two events, $T_n = T_n^* + \sqrt{n}(\mu - \mu_0)/S_n > -M + c + M = c$, so

$$\mathbb{P}_\mu(T_n > c) \ge \mathbb{P}(T_n^* > -M) - \mathbb{P}\big(\sqrt{n}(\mu - \mu_0)/S_n \le c + M\big) \to \Phi(M) - 0.$$

M was arbitrary and $\Phi(M) \to 1$ as $M \to \infty$, so $\mathbb{P}_\mu(T_n > c) \to 1$. □

Worked example: the wind turbine, properly

$n = 12$, $\bar{X}_n = 14.3$, $S_n = 4.7$, $\mu_0 = 10$, $H_0 : \mu \le 10$ against $H_1 : \mu > 10$. The statistic is $T = \sqrt{12}\,(14.3 - 10)/4.7 = 3.17$, the threshold is $q_{0.05}(t_{11}) = 1.80$ (not 1.645), and the test rejects: the home has enough wind. The p-value is $\mathbb{P}(t_{11} > 3.17) = 0.0045$: of 1,000 homes whose true mean wind is exactly 10, about 4 would produce a year of measurements this far above it. The Gaussian rule would have reported $\mathbb{P}(Z > 3.17) = 0.0008$, overstating the evidence by a factor of six because it trusts a standard deviation estimated from twelve numbers as if it were exact. The 95% t-interval for the mean wind speed is $14.3 \pm 2.20 \times 4.7/\sqrt{12} = [11.3, 17.3]$ miles per hour; the Gaussian version $14.3 \pm 1.96 \times 4.7/\sqrt{12} = [11.6, 17.0]$ is narrower, and its true coverage at $n = 12$ is $\mathbb{P}(|t_{11}| \le 1.96) = 92.4\%$, not 95%.

Worked example: the cookies, with the t version promised in Chapter 9

Exercise 9.2 had 32 cookies with a mean of 14.77 chips and standard deviation 4.37 against a claim of at least 20: $T = \sqrt{32}\,(14.77 - 20)/4.37 = -6.77$. Chapter 9 compared it with the Gaussian and got a p-value of 6×10^{-12}. With 31 degrees of freedom the p-value is $\mathbb{P}(t_{31} < -6.77) = 7 \times 10^{-8}$: ten thousand times larger, and still decisive. This is the typical pattern for n around 30: the t correction changes p-values that are small by orders of magnitude, and changes decisions only when the Gaussian p-value was already near the level.

Pitfall: z in place of t at small n

Using 1.96 with five observations is not a rounding error. Figure 14.5 (right) gives the true type 1 error of the Gaussian rule on Gaussian data: 12% at $n = 5$, 8% at $n = 10$. Of 1,000 five-observation studies with a true null, about 122 reject at the nominal 5%; the same 1,000 studies analyzed with $q_{0.025}(t_4) = 2.78$ reject 50. In the other direction, a reported two-sided '$p = 0.03$' from a Gaussian table at $n = 5$ corresponds to $|T| = 2.17$, whose t_4 p-value is 0.10.

Pitfall: which variance goes in the denominator

Theorem 14.7 uses S_n^2 with $n-1$. If the divide-by-n variance $\hat{\sigma}_n^2 = \frac{1}{n}\sum_i (X_i - \bar{X}_n)^2$ of Chapter 11

is used instead, the pivot changes: since $S_n^2 = \frac{n}{n-1}\hat{\sigma}_n^2$,

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{S_n} = \sqrt{n-1}\,\frac{\bar{X}_n - \mu}{\hat{\sigma}_n} \sim t_{n-1}.$$

So $\sqrt{n}(\bar{X}_n - \mu)/\hat{\sigma}_n$ is $\sqrt{n/(n-1)}$ times a t_{n-1} variable, and comparing it with t_{n-1} quantiles rejects too often; at $n = 5$ the factor is 1.12. Exercise 14.5 works through the version with the maximum likelihood estimators.

Recall. The Central Limit Theorem (Theorem 4.8): for i.i.d. X_i with mean μ and variance $\sigma^2 < \infty$, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1)$. Slutsky's theorem (Theorem 5.1): if $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$ and $W_n \xrightarrow[n\to\infty]{\mathbb{P}} w \neq 0$, then $X_n/W_n \xrightarrow[n\to\infty]{\text{(d)}} X/w$.

From undergraduate statistics: when the t-test is safe without Gaussian data

The exactness in Proposition 14.9 needs Gaussian data; the kiss data are not Gaussian, nor are most data. What survives is the asymptotic guarantee of Chapter 9: for any parent distribution with finite variance, under H_0 write $T_n = \big(\sqrt{n}(\bar{X}_n - \mu_0)/\sigma\big)/(S_n/\sigma)$; the numerator converges in distribution to $\text{Gauss}(0,1)$ by the CLT, the denominator converges in probability to 1 since $S_n \xrightarrow[n\to\infty]{\mathbb{P}} \sigma$, so $T_n \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1)$ by Slutsky's theorem, and $q_\alpha(t_{n-1}) \to q_\alpha$ by Proposition 14.6(4). So the t-test has asymptotic level α for every parent. The t-test is therefore never worse than the Gaussian test for large n and exactly right for Gaussian data at every n, which is why it is the default. For small n and non-Gaussian data neither guarantee applies. Mild departures from the Gaussian shape (a little skew, slightly heavy tails) move the true level only slightly because the sample mean is already close to Gaussian at modest n; heavy tails and outliers move it a lot, for the reasons of Chapter 13, and a robust center is then the better estimator. Whether a sample looks Gaussian can itself be tested, which is what the Lilliefors test and the QQ plot of Chapter 17 do.

14.6 Two samples

The boarding question of Chapter 9 compared two airlines, and a clinical trial compares a treated group with a control group. The model: $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(\mu_1, \sigma_1^2)$ and $Y_1, \ldots, Y_m \overset{iid}{\sim} \text{Gauss}(\mu_2, \sigma_2^2)$, the two samples independent, and the hypothesis $H_0 : \mu_1 = \mu_2$ against a one- or two-sided alternative. The natural statistic is the standardized difference of means. By the sum rule of Chapter 6, $\bar{X}_n - \bar{Y}_m \sim \text{Gauss}\big(\mu_1 - \mu_2, \sigma_1^2/n + \sigma_2^2/m\big)$ exactly, so with known variances

$$\frac{\bar{X}_n - \bar{Y}_m - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}} \sim \text{Gauss}(0,1).$$

With estimated variances the denominator becomes $\sqrt{S_1^2/n + S_2^2/m}$, and whether the result is a t variable depends on whether the two variances are equal.

Theorem 14.10 (The pooled two-sample t pivot)**.** *If in addition $\sigma_1^2 = \sigma_2^2 = \sigma^2$, let*

$$S_p^2 = \frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}$$

be the ***pooled variance****, the average of the two sample variances weighted by their degrees of freedom. Then*

$$\frac{\bar{X}_n - \bar{Y}_m - (\mu_1 - \mu_2)}{S_p\sqrt{1/n + 1/m}} \sim t_{n+m-2} \qquad \textit{exactly.}$$

Proof strategy (for completeness; skip on a first reading). Cochran once per sample gives two independent chi-squareds; their sum is chi-squared with the degrees of freedom added; it is independent of both sample means; the ratio is a t by definition.

Proof. By Theorem 14.3 applied to each sample, $(n-1)S_1^2/\sigma^2 \sim \chi^2_{n-1}$ and $(m-1)S_2^2/\sigma^2 \sim \chi^2_{m-1}$; they are independent because the samples are, so their sum $V = (n+m-2)S_p^2/\sigma^2 \sim \chi^2_{n+m-2}$ by Proposition 14.2(2). The four quantities $\bar{X}_n, S_1^2, \bar{Y}_m, S_2^2$ are mutually independent (the first two are independent by Cochran, the last two likewise, and the pairs are independent of each other), so V is independent of $\bar{X}_n - \bar{Y}_m$. With $\sigma_1^2 = \sigma_2^2$ the standardized difference is $Z = (\bar{X}_n - \bar{Y}_m - (\mu_1 - \mu_2))/(\sigma\sqrt{1/n + 1/m}) \sim \text{Gauss}(0,1)$, and

$$\frac{\bar{X}_n - \bar{Y}_m - (\mu_1 - \mu_2)}{S_p\sqrt{1/n + 1/m}} = \frac{Z}{\sqrt{S_p^2/\sigma^2}} = \frac{Z}{\sqrt{V/(n+m-2)}},$$

which is t_{n+m-2} by Definition 14.5. □

The tests and intervals follow as in Definition 14.8, with $\mu_1 - \mu_2$ in place of μ and t_{n+m-2} in place of t_{n-1}; for a composite null such as $\mu_1 - \mu_2 \le 0$ the argument of Proposition 14.9(1) applies unchanged, since the shift term it adds to the pivot is then nonpositive.

When the variances differ, the statistic

$$T = \frac{\bar{X}_n - \bar{Y}_m - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n + S_2^2/m}}$$

is still a standard Gaussian divided by an independent estimate of its scale, but the estimate is no longer a scaled chi-squared: $S_1^2/n + S_2^2/m$ is a weighted sum of two chi-squareds with different weights and different degrees of freedom, and its distribution depends on the unknown ratio σ_1^2/σ_2^2. There is no exact pivot. The standard remedy, **Welch's t-test**, treats T as approximately t_ν with the degrees of freedom chosen so that a scaled χ^2_ν matches the denominator's mean and variance.

Proposition 14.11 (The Welch–Satterthwaite degrees of freedom)**.** *Let $D = S_1^2/n + S_2^2/m$ and $\delta = \sigma_1^2/n + \sigma_2^2/m = \mathbb{E}[D]$. The scaled chi-squared $\delta\, W/\nu$ with $W \sim \chi^2_\nu$ that has the same mean and variance as D has*

$$\nu = \frac{(\sigma_1^2/n + \sigma_2^2/m)^2}{\dfrac{(\sigma_1^2/n)^2}{n-1} + \dfrac{(\sigma_2^2/m)^2}{m-1}},$$

and the estimate $\hat{\nu}$ with S_1^2, S_2^2 in place of σ_1^2, σ_2^2 satisfies $\min(n-1, m-1) \le \hat{\nu} \le n+m-2$. Welch's test compares T with the quantiles of $t_{\hat{\nu}}$, rounding $\hat{\nu}$ down.

Proof strategy (for completeness; skip on a first reading). Match two moments: the mean matches automatically, and equating the variances, $2\delta^2/\nu$ against the sum of the two sample variances' variances from Corollary 14.4, gives ν. The bounds are a one-line inequality each.

Proof. $\mathbb{E}[\delta W/\nu] = \delta = \mathbb{E}[D]$ for every ν. $\mathbb{V}\text{ar}(\delta W/\nu) = 2\delta^2/\nu$, and by Corollary 14.4 and independence, $\mathbb{V}\text{ar}(D) = \frac{2\sigma_1^4}{n^2(n-1)} + \frac{2\sigma_2^4}{m^2(m-1)}$. Setting the two equal and solving for ν gives the formula.

For the bounds, write $a = S_1^2/n$ and $b = S_2^2/m$, so $\hat{\nu} = (a+b)^2/\big(a^2/(n-1) + b^2/(m-1)\big)$. Since $a^2/(n-1) + b^2/(m-1) \le (a^2+b^2)/\min(n-1, m-1) \le (a+b)^2/\min(n-1, m-1)$, $\hat{\nu} \ge \min(n-1, m-1)$. By the Cauchy–Schwarz inequality, $(a+b)^2 = \big(\sqrt{n-1}\cdot\frac{a}{\sqrt{n-1}} + \sqrt{m-1}\cdot\frac{b}{\sqrt{m-1}}\big)^2 \le (n+m-2)\big(\frac{a^2}{n-1} + \frac{b^2}{m-1}\big)$, so $\hat{\nu} \le n+m-2$. □

The bounds say what the approximation does: when one sample dominates the variance of the difference, the degrees of freedom are about that sample's $n-1$; when the two contribute equally with equal sizes, $\hat{\nu}$ is near $n+m-2$, the pooled count. As $m \to \infty$ with n fixed, $b \to 0$ and $\hat{\nu} \to n-1$: the one-sample case. Rounding $\hat{\nu}$ down is the conservative direction, since a smaller ν means a larger quantile.

Worked example: a clinical trial with small groups

A drug is meant to lower LDL cholesterol. The treated group, $n = 12$, shows a mean decrease of 156.4 with $S_1 = 22.5$; the control group, $m = 22$, a mean decrease of 132.7 with $S_2 = 8.7$. $H_0 : \mu_1 - \mu_2 \le 0$ against $H_1 : \mu_1 - \mu_2 > 0$. The variances differ by a factor of $(22.5/8.7)^2 = 6.7$, so pooling is not appropriate and Welch's test is the one to use:

$$T = \frac{156.4 - 132.7}{\sqrt{22.5^2/12 + 8.7^2/22}} = 3.51, \qquad \hat{\nu} = \frac{(22.5^2/12 + 8.7^2/22)^2}{\frac{(22.5^2/12)^2}{11} + \frac{(8.7^2/22)^2}{21}} = 12.8,$$

rounded down to 12, between the bounds 11 and 32 and close to the lower one because the treated group's variance dominates. The threshold at 5% is $q_{0.05}(t_{12}) = 1.78$ and the p-value is $\mathbb{P}(t_{12} > 3.51) = 0.0022$: strong evidence that the drug lowers LDL more than the control. Of 1,000 trials of this size in which the drug does nothing beyond the control, about 2 would produce a T this large. The Gaussian p-value would have been 0.0002, ten times smaller.

Worked example: the boarding question, finished

Chapter 9 opened with $n = 72$ R2F flights averaging 24.2 minutes (standard deviation 2.1) and $m = 56$ WilMA flights averaging 15.9 (standard deviation 1.3), and settled the question with a heuristic buffer. Welch's test: $T = (24.2 - 15.9)/\sqrt{2.1^2/72 + 1.3^2/56} = 8.3/0.302 = 27.4$ with $\hat{\nu} = 120$, against a threshold of $q_{0.05}(t_{120}) = 1.66$. The p-value is below 10^{-15}; the pooled version, $T = 25.9$ with 126 degrees of freedom, says the same. The buffer heuristic was right, and now it comes with a level: the difference of 8.3 minutes is 27 estimated standard errors, and nothing about estimating two variances from 72 and 56 flights can change that.

Remark (The F distribution). Whether the two variances are equal can itself be tested. For independent $V \sim \chi^2_k$ and $W \sim \chi^2_l$, the ratio $(V/k)/(W/l)$ has the F **distribution with** (k, l) **degrees of freedom**, $F_{k,l}$, and by Theorem 14.3, $(S_1^2/\sigma_1^2)/(S_2^2/\sigma_2^2) \sim F_{n-1,m-1}$ exactly for two Gaussian samples (Exercise 14.8). Under $H_0 : \sigma_1^2 = \sigma_2^2$ the ratio S_1^2/S_2^2 is an $F_{n-1,m-1}$ pivot, and large or small values reject. The test is sensitive to non-Gaussian tails, more so than the t-test because fourth moments enter the variance of a sample variance. The F distribution's main role in this book is in Chapter 21, where a ratio of two independent sums of squares from the general Cochran theorem tests several regression coefficients at once.

14.7 Many tests at once

In 2009 a team of neuroscientists placed a dead Atlantic salmon in an fMRI scanner, showed it a series of photographs of people in social situations, and asked it to judge the emotion in each. The scanner divides the fish into $N = 8{,}064$ small volumes, and in each one a test at level $\alpha = 0.001$ asked whether the measured signal responded to the photographs. A small cluster of volumes inside the brain cavity came out 'significant'. The fish was dead; every null hypothesis was true; and a handful of rejections is exactly what the level promises. A test at level 0.001 rejects a true null once in 1,000 tries, so 8,064 tests produce about 8 false rejections, and the probability that at least one of 8,064 independent tests rejects is $1 - 0.999^{8064} = 0.9997$. (The study's own test used the scanner's time series; the simplest model with the same arithmetic is $n = 15$ photograph responses per volume, i.i.d. Gaussian with mean μ, and a t-test of $\mu = 0$, which is what the simulation below runs.) Figure 14.6 (left) repeats the experiment with simulated noise: 8,064 t-tests on samples of 15 standard Gaussians, every null true. The p-values are uniformly spread between 0 and 1 (Exercise 9.6), about 161 per bar of width 0.02, and the inset shows the p-values below 0.02 at a finer scale: the bar below 0.001 holds this run's false discoveries, 5 of them, against the 8 expected.

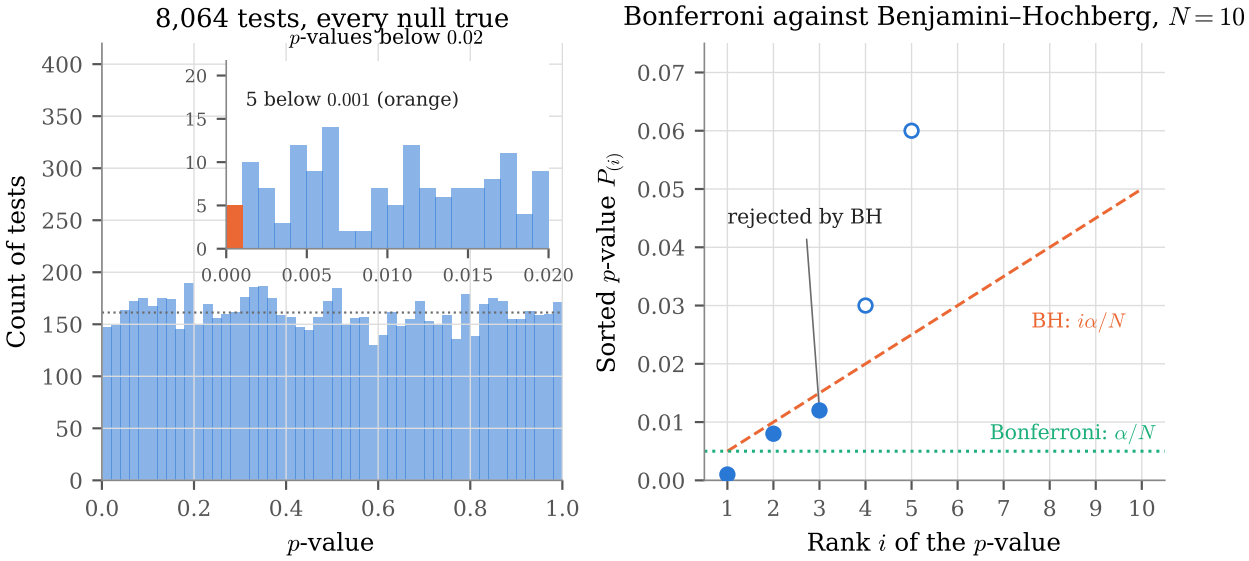


Figure 14.6: Left: p-values of 8,064 two-sided t-tests at $n = 15$; dotted: expected count per bar; inset: p-values below 0.02, those below 0.001 in orange. Right: sorted p-values, five above the scale, with Bonferroni's α/N (aqua, dotted) and Benjamini–Hochberg's $i\alpha/N$ (orange, dashed); filled dots: rejected by the latter.

The level controls each test's error separately; it says nothing about the batch. Two quantities describe the batch. Let N tests be run, with N_0 of the nulls true, and let R be the number of rejections and V the number of false rejections (rejections of true nulls) among them; V is a new use of the letter, unrelated to the chi-squared variables above.

Definition 14.12 (Family-wise error rate and false discovery rate)**.** The **family-wise error rate** is $\mathrm{FWER} = \mathbb{P}(V \geq 1)$, the probability of at least one false rejection. The **false discovery rate** is $\mathrm{FDR} = \mathbb{E}\left[V/\max(R, 1)\right]$, the expected fraction of the rejections that are false (taken as 0 when nothing is rejected).

In words. The FWER asks for no false alarm at all in the whole batch; controlling it at 5% means that in 95 of 100 batches, every rejection is a true discovery. The FDR tolerates false alarms as long as they are a small fraction of what is reported; controlling it at 5% means that, on average, 5% of the findings you announce are wrong. The first is the stricter demand: since $V/\max(R,1) \le \mathbb{1}(V \ge 1)$, FDR $\le$ FWER, and anything that controls the FWER controls the FDR, not conversely.

For the kiss data. One test, $N = 1$: if the null is true both rates are the level, α, and if it is false both are 0. The salmon's 8,064 tests at $\alpha = 0.001$ have FWER $= 0.9997$, and since every null is true, every rejection is false and FDR $= \mathbb{P}(R \ge 1) = 0.9997$ as well.

Both procedures below work with the p-values $P_1, \dots, P_N$ of the N tests, and both use one fact from Chapter 9: under a true simple null with a continuous test statistic, the p-value is uniform on $[0,1]$, so $\mathbb{P}(P_i \le u) = u$ (Exercise 9.6); under a composite null the monotonicity argument of Proposition 14.9(1) gives $\mathbb{P}(P_i \le u) \le u$, which is all Bonferroni needs.

Proposition 14.13 (Bonferroni's correction)**.** *Suppose $\mathbb{P}(P_i \le u) \le u$ for every true null i and every $u \in [0,1]$, and reject test i when $P_i \le \alpha/N$. Then* FWER $\le \alpha$*, whatever the dependence between the tests.*

Proof strategy (core). The probability of a union is at most the sum of the probabilities, and each true null is rejected with probability α/N.

Proof. The event $\{V \ge 1\}$ is the union over the true nulls i of $\{P_i \le \alpha/N\}$. The indicator of a union is at most the sum of the indicators, since if the union occurs at least one indicator equals 1 (toolkit A.4), so taking expectations,

$$\text{FWER} = \mathbb{P}\Big(\bigcup_{i \text{ true null}} \{P_i \le \alpha/N\} \Big) \le \sum_{i \text{ true null}} \mathbb{P}(P_i \le \alpha/N) \le N_0 \cdot \frac{\alpha}{N} \le \alpha.$$

No independence was used. □

For the salmon, Bonferroni's level per test is $0.001/8064 = 1.2 \times 10^{-7}$, and the two-sided t_{14} threshold rises from 4.14 to 9.77: no volume survives, which is the right answer for a dead fish. The cost is that for a live subject with a few truly active volumes, a threshold of 9.77 estimated standard errors will miss most of them. Bonferroni buys 'no false alarm' at the price of power, and the price grows with N.

The Benjamini–Hochberg procedure gives up on 'no false alarm' and controls the FDR instead. Sort the p-values, $P_{(1)} \le P_{(2)} \le \cdots \le P_{(N)}$, and compare the ith smallest with the rising line $i\alpha/N$ rather than with the flat α/N.

Theorem 14.14 (Benjamini–Hochberg)**.** *Let $i_{\max} = \max\{i : P_{(i)} \le i\alpha/N\}$ (and 0 if no i qualifies), and reject the $i_{\max}$ tests with the smallest p-values. If the p-values of the true nulls are uniform on $[0,1]$ and independent of each other and of the other p-values, then*

$$\text{FDR} = \frac{N_0}{N}\,\alpha \le \alpha.$$

The proof is in Appendix C.5; nothing later in the book depends on it. The picture on the right of Figure 14.6 is the procedure: ten sorted p-values, $0.001, 0.008, 0.012, 0.03, 0.06, \dots$, against the

Bonferroni line at 0.005 and the Benjamini–Hochberg line $0.005\,i$. Bonferroni rejects only the first. Benjamini–Hochberg finds the last point under its line, the third ($0.012 \le 0.015$; the fourth, 0.03, is above 0.02), and rejects the first three, including the second, whose p-value 0.008 is above 0.005 on its own. The rising line is the point: a batch with several small p-values is evidence that some nulls are false, and the procedure lets the small ones vouch for each other. The theorem's N_0/N says the control is conservative when many nulls are false; when every null is true, $V = R$, so FDR = FWER and the procedure holds the probability of any false rejection at exactly α, where Bonferroni holds it at most α.

Pitfall: FWER and FDR are different promises

'At most 5% of my 100 discoveries are false on average' and 'with probability 95% none of my discoveries is false' are regularly conflated. A procedure with FDR ≤ 0.05 that reports 100 discoveries is expected to contain about 5 false ones; a procedure with FWER ≤ 0.05 contains none in 19 batches of 20. Which one is wanted depends on what a false discovery costs: a follow-up experiment, or a published claim.

14.8 What comes next

This chapter is the one place in the book where exact small-sample distributions replace limit theorems, and the price was a Gaussian model for the data. Chapter 15 returns to large samples and arbitrary parametric models: Wald's test standardizes the maximum likelihood estimator of Chapter 12 and compares its square with a chi-squared quantile, which is the squared-Gaussian reading of Definition 14.1 in d dimensions, and the likelihood ratio test does the same with the log-likelihood. Both have chi-squared limits with degrees of freedom counted as here, one per constraint tested. Chapter 16 uses the same chi-squared limit to test whether data fit a discrete distribution, and Chapter 17 tests whether a sample is Gaussian in the first place, which is the question this chapter's exactness depends on. Chapter 21 uses Cochran's theorem in its general form for the residual variance of a regression, and the t and F distributions for the tests on its coefficients.

14.9 Summary

- For Gaussian data, $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ is exactly standard Gaussian at every n; replacing σ by S_n breaks this, and the Gaussian 5% rule has type 1 error 12% at $n = 5$.
- χ^2_k is the sum of k squared independent standard Gaussians: mean k, variance $2k$, additive, approximately Gauss$(k, 2k)$ for large k; χ^2_2 is Expo$(\frac{1}{2})$.
- Cochran: for Gaussian data $\bar{X}_n$ and S_n^2 are independent and $(n-1)S_n^2/\sigma^2 \sim \chi^2_{n-1}$, proved by rotating the standardized data so that one coordinate carries the mean and $n-1$ carry the deviations. $\mathbb{V}\mathrm{ar}(S_n^2) = 2\sigma^4/(n-1)$. The independence fails for non-Gaussian data.
- t_k is $Z/\sqrt{V/k}$ with Z and $V \sim \chi^2_k$ independent: symmetric, heavier tails than the Gaussian (variance $k/(k-2)$), converging to it as $k \to \infty$ by the LLN and Slutsky; t_1 is the Cauchy.
- The t pivot: $\sqrt{n}(\bar{X}_n - \mu)/S_n \sim t_{n-1}$ exactly for Gaussian data. The t-test and t-interval have exact level and coverage at every n for Gaussian data, and asymptotic level for any parent with finite variance.

- Two samples: with equal variances the pooled statistic is exactly t_{n+m-2}; with unequal variances Welch's statistic is approximately $t_{\hat{\nu}}$, with $\hat{\nu}$ from matching the denominator's mean and variance, between $\min(n,m)-1$ and $n+m-2$.
- Many tests: Bonferroni's α/N controls the probability of any false rejection, with no assumption on dependence; Benjamini–Hochberg's rising threshold $i\alpha/N$ controls the expected fraction of false rejections, for independent p-values, and rejects more.

14.10 Exercises

Exercise 14.1. Let $X_1,\dots,X_n \overset{iid}{\sim} \text{Gauss}(3,\sigma^2)$ with σ^2 unknown and the mean known to be 3. (a) Find a constant a and an integer k such that $a\sum_i (X_i-3)^2 \sim \chi^2_k$. (b) Compute the mean and variance of the estimator $\tilde{\sigma}^2 = \frac{1}{n}\sum_i (X_i-3)^2$ and compare its variance with that of S_n^2. (c) A dart lands at (Z_1,Z_2) with i.i.d. standard Gaussian coordinates. What is the probability that it lands within distance 2 of the origin? *(Solution on page 425.)*

Exercise 14.2. Let $X_1, X_2 \overset{iid}{\sim} \text{Gauss}(0,1)$ and set $U=(X_1+X_2)/\sqrt{2}$, $W=(X_1-X_2)/\sqrt{2}$. Starting from the joint density of (X_1,X_2), show that (U,W) has the same joint density, and conclude that U and W are independent standard Gaussians. Deduce Theorem 14.3 for $n=2$. *(Solution on page 425.)*

Exercise 14.3. For $X_1,\dots,X_n \overset{iid}{\sim} \text{Gauss}(\mu,\sigma^2)$ consider the estimators $\hat{\sigma}^2_c = c\sum_i (X_i-\bar{X}_n)^2$ of σ^2. Using Theorem 14.3, find the bias, variance and quadratic risk of $\hat{\sigma}^2_c$ as functions of c, and show that the risk is minimized at $c=1/(n+1)$, not at $1/(n-1)$ (unbiased) or $1/n$ (maximum likelihood). Compute the three risks for $n=12$ and $\sigma=1$. *(Solution on page 425.)*

Exercise 14.4. Let $V\sim\chi^2_k$ with density f_k from Proposition 14.2. (a) Show that $\mathbb{E}[1/V]=1/(k-2)$ for $k\geq 3$, using $\Gamma(x+1)=x\Gamma(x)$. (b) Deduce that $\mathbb{V}\text{ar}(T)=k/(k-2)$ for $T\sim t_k$, $k\geq 3$, and explain why $\mathbb{E}[T]=0$ needs $k\geq 2$. (c) What are the variances of t_3, t_5 and t_{30}, and what do they say about the table of quantiles in Section 14.4? *(Solution on page 425.)*

Exercise 14.5. Let $X_1,\dots,X_n \overset{iid}{\sim} \text{Gauss}(\mu,\sigma^2)$ with maximum likelihood estimators $\hat{\mu}=\bar{X}_n$ and $\hat{\sigma}^2=\frac{1}{n}\sum_i (X_i-\bar{X}_n)^2$ (Chapter 11). (a) Show that $\sqrt{n-1}\,(\hat{\mu}-\mu)/\hat{\sigma}\sim t_{n-1}$. (b) Propose a test of $H_0:\mu\geq 0$ against $H_1:\mu<0$ with exact level α and prove its level, including for $\mu>0$. *(Solution on page 425.)*

Exercise 14.6. A 95% Gaussian interval $\bar{X}_n \pm 1.96\,S_n/\sqrt{n}$ is computed from n Gaussian observations. (a) Write its exact coverage in terms of the t_{n-1} distribution and evaluate it for $n=3,5,12,30$ using Figure 14.5 (the value at $n=3$ is 0.189). (b) How large must n be for the coverage to reach 94%? *(Solution on page 425.)*

Exercise 14.7. A larger version of the clinical trial has $n=70$ treated and $m=50$ control patients, with mean decreases 156.4 and 132.7 and sample variances $S_1^2=5198.4$ and $S_2^2=3867.0$ (standard deviations 72.1 and 62.2, far larger than in the small trial). Test $H_0:\mu_1-\mu_2\leq 0$ against $H_1:\mu_1-\mu_2>0$ at 5% with Welch's test, report the one-sided p-value, and say what changes if the two-sided p-value is reported by mistake. *(Solution on page 426.)*

Exercise 14.8. Let $X_1,\dots,X_n \overset{iid}{\sim} \text{Gauss}(\mu_1,\sigma_1^2)$ and $Y_1,\dots,Y_m \overset{iid}{\sim} \text{Gauss}(\mu_2,\sigma_2^2)$ be independent samples. (a) Show that $(S_1^2/\sigma_1^2)/(S_2^2/\sigma_2^2)\sim F_{n-1,m-1}$. (b) For the boarding data ($S_1=2.1$, $n=72$; $S_2=1.3$, $m=56$), compute the ratio S_1^2/S_2^2 and explain why, with $\mathbb{P}(F_{71,55}>2.61)<0.001$, the pooled test was not the right choice even though it gave the same answer. *(Solution on page 426.)*

Exercise 14.9. (a) For the salmon's 8,064 tests at level 0.001, compute the expected number of false rejections, the FWER assuming independence, and Bonferroni's per-test level. (b) The scanner is used on a live subject with $N = 8{,}064$ volumes of which $N_0 = 8{,}000$ have no true activity, and each test has level 0.001. Without correction, how many false rejections are expected, and how many does Bonferroni at $\alpha = 0.05$ expect? (c) If the uncorrected tests reject 50 volumes in all, what fraction of them are expected to be false? *(Solution on page 426.)*

Exercise 14.10. Ten tests give p-values $0.001, 0.008, 0.012, 0.03, 0.06, 0.2, 0.35, 0.5, 0.7, 0.9$. (a) Which tests does Bonferroni reject at $\alpha = 0.05$? (b) Which does Benjamini–Hochberg reject? (c) Suppose the fourth and fifth p-values were 0.022 and 0.024 instead of 0.03 and 0.06. What changes, and why does the fifth p-value affect whether the fourth test is rejected? *(Solution on page 426.)*

Chapter 15

Wald, Likelihood Ratio and Score Tests

In this chapter

- Three general-purpose tests for any regular parametric model, all read off the log-likelihood: Wald's test measures how far the maximum likelihood estimate is from the null value, the likelihood ratio test how far the log-likelihood drops from its top to the null, the score test how steeply the log-likelihood climbs at the null.
- Gaussian vectors in several dimensions, and the one fact the tests need about them: the squared length of a Gaussian vector, measured in units of its own noise, is chi-squared.
- Each statistic is approximately chi-squared when the null is true, with one degree of freedom per constraint the null imposes. Wilks' theorem for the likelihood ratio is proved by a second-order Taylor expansion.
- The Neyman–Pearson lemma: for one null value against one alternative, the likelihood ratio test is the most powerful test of its level. The one-sided test of the kiss data in Chapter 9 is the best test of its level against every $p > \frac{1}{2}$ at once.
- The three tests agree in large samples and can disagree near the threshold in small ones. The likelihood ratio test does not depend on how the parameter is written and usually keeps its level best, and it is the one to use when both fits are available.
- Nuisance parameters and implicit hypotheses: the constrained maximum, the profile likelihood and two-sample comparisons. The t-test, the two-proportion test and the chi-squared tests for a table of counts are all instances.

Chapter 14 built exact tests for Gaussian data. This chapter builds approximate tests for every regular model, from the asymptotic normality of the maximum likelihood estimator proved in Chapter 12.

15.1 Three ways to measure the distance to a null

Every test in Chapter 9 asked one question: is the estimate far from the null value, measured in standard errors? For the kiss data the estimate is $\hat{p} = 0.645$, the null value is $p_0 = \frac{1}{2}$, and the answer was yes. But 'far, in standard errors' can be measured in more than one way, and Chapter 10 already met three numbers for the same question, 11.41, 10.60 and 10.45, each compared with the

same threshold 3.84. This chapter says where the three come from, why each is approximately chi-squared, and which to use.

All three are read off the log-likelihood $\ell_n(p) = 80 \log p + 44 \log(1-p)$ of Chapter 11, which peaks at $\hat{p} = 0.645$. Figure 15.1 draws it measured from its top. On the left, the blue curve falls from its top to the null value by 5.30. That drop, doubled, is the **likelihood ratio** statistic, 10.60: it compares how well the best value under the null and the best value overall explain the data, on the log scale. The orange parabola is the one that matches the log-likelihood at its top, with the same peak and the same curvature $nI(\hat{p}) = 541.7$ (Chapter 12). Its drop to the null, doubled, is **Wald's** statistic, $nI(\hat{p})(\hat{p} - p_0)^2 = 11.41$: the squared distance from estimate to null value in units of the standard error $1/\sqrt{nI(\hat{p})} = 0.043$. On the right, the **score** statistic never visits the top. It stands at the null value and measures the slope there, $\ell_n'(\frac{1}{2}) = 72$ (Exercise 12.4), against the curvature $nI(\frac{1}{2}) = 496$ that the model predicts there: $72^2/496 = 10.45$. The aqua parabola has the log-likelihood's slope at p_0 and that curvature, and the score statistic is twice its rise from p_0 to its own top. A steep climb at the null says the top is far away. For a proportion that top is exactly at $\hat{p}$, since $72/496 = 0.145 = \hat{p} - p_0$, so Wald's parabola and the score parabola differ only in their curvature, and $11.41/10.45 = 541.7/496 = 1.09$.

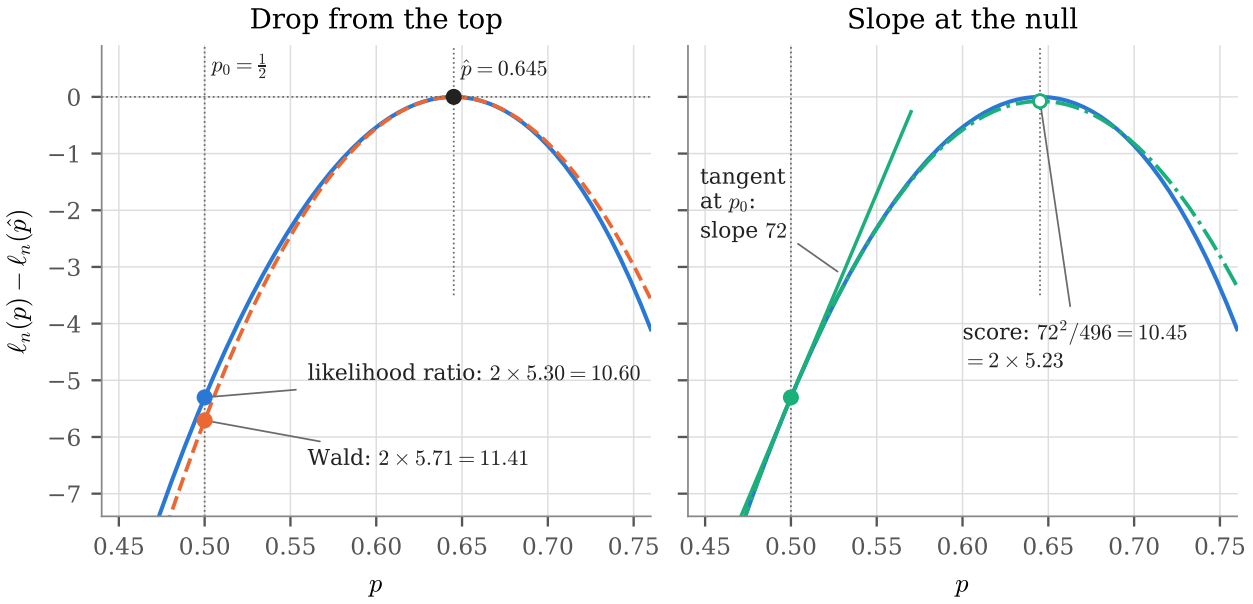


Figure 15.1: The kiss data's log-likelihood $\ell_n(p) - \ell_n(\hat{p})$ (blue). Left: Wald's parabola, fitted at the top (orange, dashed). Right: the tangent at $p_0 = \frac{1}{2}$ (aqua) and the score parabola (aqua, dash-dot), fitted at the null.

If the log-likelihood were exactly a parabola, the three numbers would be one number: a parabola drops from its top by half its curvature times the squared distance, and its slope at a point is its curvature times the distance to the top. They differ only as much as the log-likelihood departs from a parabola between p_0 and $\hat{p}$, which for the kiss data is little.

Intuition: near its top, every log-likelihood is a Gaussian one

Chapter 12 showed that the curvature of the log-likelihood near its top is about $nI(\theta^*)$, and that $\hat{\theta}_n$ is approximately Gauss$(\theta^*, 1/(nI(\theta^*)))$. A parabola with that curvature is exactly the log-likelihood of one observation $\hat{\theta}$ from a Gaussian with known variance $1/(nI)$, and for that

model the three ways of measuring distance give the same number, the squared standardized distance $nI(\hat{\theta} - \theta_0)^2$, which is a squared standard Gaussian when θ_0 is the truth. So for large n every regular model, seen from near its top, is that Gaussian model, and all three statistics are approximately χ^2_1. The rest of the chapter makes this precise, and repeats it in d dimensions, where the squared standardized distance becomes χ^2_d.

15.2 Gaussian vectors and the chi-squared distribution

With one parameter, a squared standardized distance is a squared standard Gaussian, which is χ^2_1 (Chapter 14). With d parameters the estimate is a vector, the errors in its coordinates have different sizes and are correlated, and 'standardized distance' needs a definition. The limits in Chapters 12 and 13 are Gaussian vectors $\text{Gauss}_d(m, \Sigma)$, and the Toolkit (toolkit A.14) gives the definition this book uses: Z has the distribution of $m + AW$, where W has d independent standard Gaussian coordinates and A is an invertible matrix with $AA^\top = \Sigma$, a positive definite matrix. The distribution depends on A only through Σ, every nonzero linear combination of the coordinates is Gaussian, and Σ is the covariance matrix of Z. One fact about such vectors carries every test in this chapter.

Lemma 15.1 (The quadratic form of a Gaussian vector)**.** *If $Z \sim \text{Gauss}_d(m, \Sigma)$ with Σ positive definite, then $(Z - m)^\top \Sigma^{-1} (Z - m) \sim \chi^2_d$.*

Proof strategy (core). Represent $Z - m$ as AW. The matrix Σ^{-1} undoes A on both sides, and what is left is the squared length of W.

Proof. By definition $Z - m$ has the distribution of AW, with $AA^\top = \Sigma$, so the quadratic form has the distribution of $(AW)^\top \Sigma^{-1} AW$. Since $\Sigma^{-1} = (AA^\top)^{-1} = (A^\top)^{-1} A^{-1}$,

$$(AW)^\top \Sigma^{-1} AW = W^\top A^\top (A^\top)^{-1} A^{-1} A\, W = W^\top W = W_1^2 + \cdots + W_d^2,$$

a sum of d squared independent standard Gaussians, which is χ^2_d by Definition 14.1. □

Figure 15.2 shows what the quadratic form measures. On the left are 2,000 draws of a two-dimensional Gaussian vector whose coordinates have standard deviations 1 and 1.5 and correlation 0.8: an elongated, tilted cloud. The ellipse is the set where $z^\top \Sigma^{-1} z = 5.99$, the 95% quantile of χ^2_2, and 102 of the 2,000 points fall outside it, close to the 100 the lemma predicts. On the right are the same points mapped back by A^{-1}: the round cloud of W, with the ellipse mapped to the circle of radius $\sqrt{5.99} = 2.45$. A point is outside the ellipse exactly when its image is outside the circle. So $(z - m)^\top \Sigma^{-1} (z - m)$ is the squared distance from the center after the correlation and the unequal spreads have been undone: distance in units of the noise, direction by direction. Two points at the same ordinary distance from the center can be at very different statistical distances: $(1, 1)$ lies along the cloud and is well inside the ellipse, while $(1, -1)$ lies across it and is outside (Exercise 15.1).

15.3 Wald's test

Recall. Asymptotic normality of the MLE (Theorem 12.5): in a regular model with θ^* interior and $I(\theta^*)$ invertible, $\sqrt{n}(\hat{\theta}_n - \theta^*) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0, I(\theta^*)^{-1})$. The continuous mapping theorem for vectors (Theorem 5.2

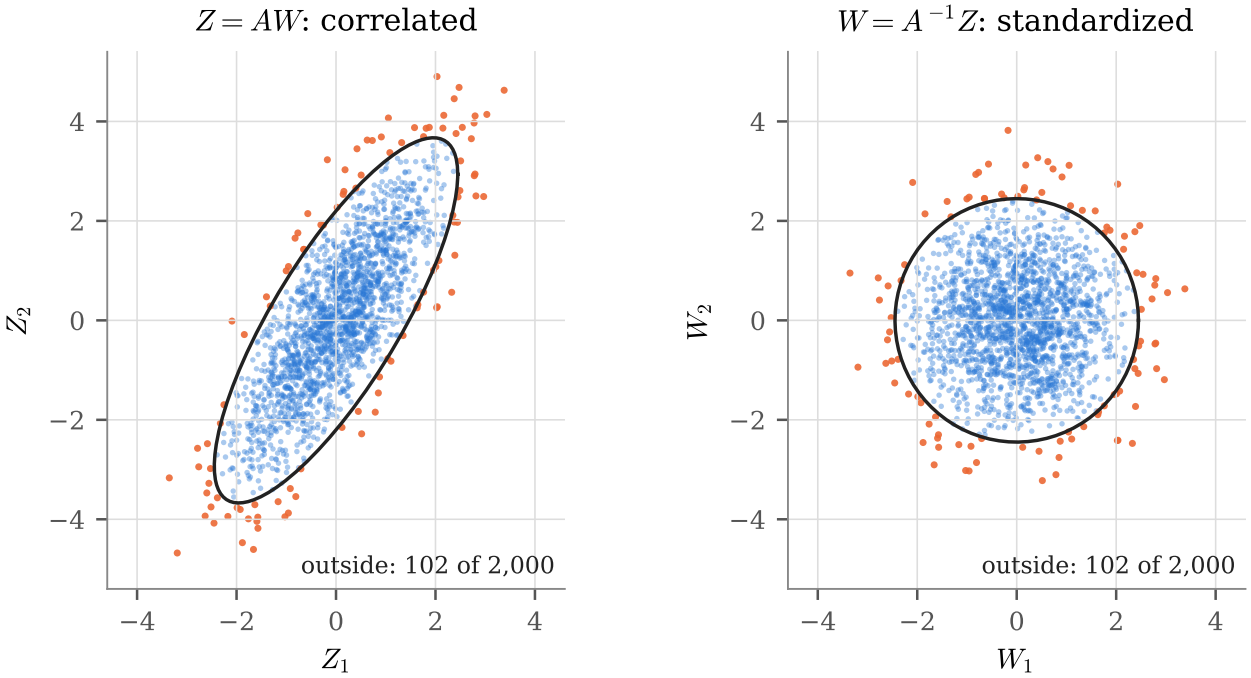


Figure 15.2: Left: 2,000 draws of $Z = AW$, with $\Sigma = AA^\top$ having standard deviations 1 and 1.5 and correlation 0.8, and the ellipse $z^\top \Sigma^{-1} z = 5.99$. Right: the same points mapped by A^{-1}, and the circle of radius 2.45. Orange: points outside.

and the vector-versions remark of Chapter 5): if $V_n \xrightarrow[n\to\infty]{\text{(d)}} V$ in $\mathbb{R}^d$ and $g : \mathbb{R}^d \to \mathbb{R}^k$ is continuous, then $g(V_n) \xrightarrow[n\to\infty]{\text{(d)}} g(V)$; if $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$ and g is continuous at y, then $g(Y_n) \xrightarrow[n\to\infty]{\mathbb{P}} g(y)$. A matrix counts as the vector of its entries, so this covers sums, products and inverses of random matrices that converge entry by entry. Slutsky's theorem (Theorem 5.1): if $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$ and $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$, a constant, then $X_n + Y_n \xrightarrow[n\to\infty]{\text{(d)}} X + y$ and $X_n Y_n \xrightarrow[n\to\infty]{\text{(d)}} yX$. Convergence in distribution to a constant is convergence in probability to it (Proposition 4.7).

Definition 15.2 (Wald's test)**.** Let $(E, (\mathbb{P}_\theta)_{\theta\in\Theta})$ with $\Theta \subset \mathbb{R}^d$ be a regular model (Section 12.6) with Fisher information $I(\theta)$, let $\hat\theta_n$ be the maximum likelihood estimator, and let θ_0 be a fixed interior point of Θ. To test $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$, **Wald's statistic** is

$$T_n^{\mathrm{W}} = n\,(\hat\theta_n - \theta_0)^\top I(\hat\theta_n)\,(\hat\theta_n - \theta_0),$$

and **Wald's test** of asymptotic level α rejects when $T_n^{\mathrm{W}} > q_\alpha(\chi^2_d)$. For $d = 1$, $T_n^{\mathrm{W}} = (Z_n^{\mathrm{W}})^2$ with $Z_n^{\mathrm{W}} = \sqrt{nI(\hat\theta_n)}\,(\hat\theta_n - \theta_0)$, and the one-sided tests reject $H_0 : \theta \le \theta_0$ when $Z_n^{\mathrm{W}} > q_\alpha$ and $H_0 : \theta \ge \theta_0$ when $Z_n^{\mathrm{W}} < -q_\alpha$.

> **In words.** Estimate θ by maximum likelihood, measure how far the estimate is from the null value in units of its own estimated noise (one standard error in one dimension, the estimated noise ellipse in several), square, and compare with the chi-squared quantile with as many degrees of freedom as the parameter has coordinates.

For the kiss data. $I(p) = 1/(p(1-p))$, so $Z^{\mathrm{W}} = \sqrt{124}\,(0.645 - 0.5)/\sqrt{0.645 \times 0.355} = 3.38$ and $T^{\mathrm{W}} = 11.41 > 3.84$: reject $p = \frac{1}{2}$, with p-value $\mathbb{P}(\chi^2_1 > 11.41) = 0.0007$. The subscript n is dropped for a value computed from data: T_n^{W} is a random variable, a function of the sample, and $T^{\mathrm{W}} = 11.41$ is the number it takes for the 124 couples observed. This is the test of Chapter 9 with the standard error estimated at $\hat{p}$ instead of computed at p_0, and Section 9.8 showed that it inverts to the plug-in interval $[0.561, 0.729]$.

Wald's statistic and the likelihood ratio statistic below both have the form $V_n^\top M_n V_n$, a scaled estimation error measured by a matrix that converges to the Fisher information. One lemma handles both.

Lemma 15.3 (Quadratic forms in the limit)**.** *Let V_n be random vectors in $\mathbb{R}^d$ with $V_n \xrightarrow[n\to\infty]{(d)} \mathrm{Gauss}_d(0, I^{-1})$ for a positive definite matrix I, and let M_n be random $d \times d$ matrices with $M_n \xrightarrow[n\to\infty]{\mathbb{P}} I$ entry by entry. Then $V_n^\top M_n V_n \xrightarrow[n\to\infty]{(d)} \chi^2_d$, and $V_n^\top M_n V_n - V_n^\top I V_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$.*

Proof strategy (for completeness; skip on a first reading). $V_n^\top I V_n$ converges to the quadratic form of a Gaussian vector, which is χ^2_d. The difference $V_n^\top (M_n - I) V_n$ is at most the squared length of V_n times the size of $M_n - I$, a sequence that converges in distribution times one that goes to zero.

Proof. The map $v \mapsto v^\top I v$ is continuous, so $V_n^\top I V_n \xrightarrow[n\to\infty]{(\mathrm{d})} Z^\top I Z$ with $Z \sim \mathrm{Gauss}_d(0, I^{-1})$, and $Z^\top I Z \sim \chi^2_d$ by Lemma 15.1 with $\Sigma = I^{-1}$, which is positive definite since I is (toolkit A.14). For the difference, write $B_n = M_n - I$. For any vectors u, v and matrix B,

$$|u^\top B v| = \Big|\sum_{j,k} u_j B_{jk} v_k\Big| \le \sum_{j,k} |u_j||v_k||B_{jk}| \le \|u\|\,\|v\| \sum_{j,k} |B_{jk}|, \tag{15.1}$$

since $|u_j| \le \|u\|$ and $|v_k| \le \|v\|$. Take $u = v = V_n$ and $B = B_n$. The squared length $\|V_n\|^2$ converges in distribution to $\|Z\|^2$ by the continuous mapping theorem, and $\sum_{j,k} |B_{n,jk}| \xrightarrow[n\to\infty]{\mathbb{P}} 0$ since each entry does. By Slutsky's product rule the product $\|V_n\|^2 \sum_{j,k} |B_{n,jk}|$ converges in distribution to $0 \cdot \|Z\|^2 = 0$, which is convergence in probability to 0. Since $|V_n^\top B_n V_n|$ is at most that product, $\mathbb{P}(|V_n^\top B_n V_n| > \epsilon)$ is at most the probability that the product exceeds ϵ, which tends to 0 for every $\epsilon > 0$. So $V_n^\top M_n V_n - V_n^\top I V_n = V_n^\top B_n V_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$, and Slutsky's additive rule gives $V_n^\top M_n V_n \xrightarrow[n\to\infty]{(\mathrm{d})} \chi^2_d$. □

Theorem 15.4 (Wald's test)**.** *In the setting of Definition 15.2, suppose also that $I(\theta)$ is continuous at θ_0 and invertible there.*

1. *If H_0 is true, $T_n^{\mathrm{W}} \xrightarrow[n\to\infty]{(d)} \chi^2_d$, so $\mathbb{P}_{\theta_0}(T_n^{\mathrm{W}} > q_\alpha(\chi^2_d)) \to \alpha$: Wald's test has asymptotic level α. For $d = 1$, $Z_n^{\mathrm{W}} \xrightarrow[n\to\infty]{(d)} \mathrm{Gauss}(0,1)$ and the one-sided tests have asymptotic level α.*
2. *If $\theta^* \neq \theta_0$ is an interior point at which I is continuous and invertible, $\mathbb{P}_{\theta^*}(T_n^{\mathrm{W}} > q_\alpha(\chi^2_d)) \to 1$.*

Proof strategy (for completeness; skip on a first reading). Under the null the scaled estimation error is approximately $\text{Gauss}_d(0, I(\theta_0)^{-1})$, and $I(\hat\theta_n)$ estimates $I(\theta_0)$; apply Lemma 15.3. Under an alternative, T_n^{W}/n settles at a positive number, so T_n^{W} grows like n.

Proof. (1) Under H_0, $V_n = \sqrt{n}(\hat\theta_n - \theta_0) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0, I(\theta_0)^{-1})$ by Theorem 12.5. Since $\hat\theta_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta_0$ (Theorem 12.1) and I is continuous at θ_0, the continuous mapping theorem gives $M_n = I(\hat\theta_n) \xrightarrow[n\to\infty]{\mathbb{P}} I(\theta_0)$ entry by entry. So $T_n^{\mathrm{W}} = V_n^\top M_n V_n \xrightarrow[n\to\infty]{\text{(d)}} \chi^2_d$ by Lemma 15.3, and since the χ^2_d CDF is continuous, $\mathbb{P}_{\theta_0}(T_n^{\mathrm{W}} > q) \to \mathbb{P}(\chi^2_d > q)$ for every q. For $d = 1$, $Z_n^{\mathrm{W}} = \sqrt{I(\hat\theta_n)} \cdot \sqrt{n}(\hat\theta_n - \theta_0)$, where the first factor converges in probability to $\sqrt{I(\theta_0)}$ and the second in distribution to $\text{Gauss}(0, 1/I(\theta_0))$, so by Slutsky $Z_n^{\mathrm{W}} \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, 1)$. For the one-sided null $\theta \le \theta_0$ and a true $\theta < \theta_0$: the factor $\sqrt{nI(\hat\theta_n)}$ is never negative and $q_\alpha > 0$ (as $\alpha < \frac{1}{2}$), so $Z_n^{\mathrm{W}} > q_\alpha$ forces $\hat\theta_n > \theta_0$, and $\mathbb{P}_\theta(\hat\theta_n > \theta_0) \le \mathbb{P}_\theta(|\hat\theta_n - \theta| \ge \theta_0 - \theta) \to 0$ by consistency (Theorem 12.1); at θ_0 itself the probability tends to α. The other one-sided test is the mirror image.

(2) Under θ^*, $\hat\theta_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta^*$, so by the continuous mapping theorem $T_n^{\mathrm{W}}/n = (\hat\theta_n - \theta_0)^\top I(\hat\theta_n)(\hat\theta_n - \theta_0) \xrightarrow[n\to\infty]{\mathbb{P}} c = (\theta^* - \theta_0)^\top I(\theta^*)(\theta^* - \theta_0)$, and $c > 0$ since $I(\theta^*)$ is positive definite (toolkit A.14). Then $\mathbb{P}(T_n^{\mathrm{W}} > q) \ge \mathbb{P}(T_n^{\mathrm{W}}/n > c/2) \to 1$ as soon as $nc/2 > q$. □

Does this make sense? Why does the threshold grow with the number of parameters?

The 5% thresholds are $q_{0.05}(\chi^2_1) = 3.84 = 1.96^2$, $q_{0.05}(\chi^2_2) = 5.99$, $q_{0.05}(\chi^2_3) = 7.81$ and $q_{0.05}(\chi^2_5) = 11.07$. Each coordinate contributes its own squared standardized error, with mean 1, so a statistic that adds up d of them is about d by chance alone, and the threshold must sit above that: for large d it is roughly $d + 1.645\sqrt{2d}$, the mean plus 1.645 standard deviations of χ^2_d. Testing two parameters jointly is not the same as testing each at 5%: with independent estimates, rejecting when either of two 5% tests rejects has type 1 error $1 - 0.95^2 = 9.75\%$, the multiple-testing effect of Chapter 14, and the joint threshold 5.99 is what keeps the chance at 5%.

15.4 The likelihood ratio test

Wald's test standardizes a distance, and the standardization is a choice: the curvature is taken at $\hat\theta_n$, and the distance is measured in whatever coordinates the parameter happens to be written in. A test can be built from the likelihood itself, by asking how much better the data are explained by the best value overall than by the best value under the null. When both hypotheses are single points, that comparison is not only natural but optimal.

Proposition 15.5 (Neyman–Pearson lemma)**.** *Let $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$ be simple hypotheses, let $L_n(\theta)$ be the likelihood, and for a constant $c > 0$ let*

$$\psi^* = \mathbb{1}\{L_n(\theta_1) > c\, L_n(\theta_0)\}.$$

Every test ψ with $\mathbb{P}_{\theta_0}(\psi = 1) \le \mathbb{P}_{\theta_0}(\psi^* = 1)$ has $\mathbb{P}_{\theta_1}(\psi = 1) \le \mathbb{P}_{\theta_1}(\psi^* = 1)$: no test with at most the type 1 error of ψ^* has more power.

Proof strategy (core). ψ^* rejects exactly on the data sets where the alternative explains the data more than c times better than the null. So $\psi^* - \psi$ always has the sign of $L_n(\theta_1) - cL_n(\theta_0)$, and summing over data sets turns that into a comparison of the two powers.

Proof. For a possible data set $x = (x_1, \ldots, x_n)$ write $L_1(x)$ and $L_0(x)$ for its likelihood (the joint PMF or density) under θ_1 and θ_0. At every x,

$$\big(\psi^*(x) - \psi(x)\big)\big(L_1(x) - cL_0(x)\big) \geq 0 :$$

if $\psi^*(x) = 1$, then $L_1(x) - cL_0(x) > 0$ and $\psi^*(x) - \psi(x) = 1 - \psi(x) \geq 0$; if $\psi^*(x) = 0$, then $L_1(x) - cL_0(x) \leq 0$ and $\psi^*(x) - \psi(x) = -\psi(x) \leq 0$. Sum over all x (integrate, for densities):

$$\sum_x \big(\psi^*(x) - \psi(x)\big)L_1(x) \geq c \sum_x \big(\psi^*(x) - \psi(x)\big)L_0(x).$$

The left side is $\mathbb{P}_{\theta_1}(\psi^* = 1) - \mathbb{P}_{\theta_1}(\psi = 1)$, and the right side is c times $\mathbb{P}_{\theta_0}(\psi^* = 1) - \mathbb{P}_{\theta_0}(\psi = 1) \geq 0$. □

For the kiss data with $\theta_0 = \frac{1}{2}$ and any $p_1 > \frac{1}{2}$, the likelihood ratio is $L_n(p_1)/L_n(\frac{1}{2}) = (2p_1)^S\big(2(1-p_1)\big)^{n-S}$, where S is the number of right turns. Since $2p_1 > 1 > 2(1-p_1)$, the ratio increases with S, so 'the ratio exceeds c' is the same event as 'S is at least some threshold'. The test of Chapter 9 that rejects when $S \geq 72$, whose exact type 1 error is 0.0438, is therefore ψ^* for every $p_1 > \frac{1}{2}$ at once, with c depending on p_1 and the rejection region not: for $p_1 = 0.645$, every c with $0.93 \leq c < 1.69$ gives it. So against every alternative $p > \frac{1}{2}$ it is the most powerful test whose type 1 error is at most its own: its power is 0.943 at $p = 0.645$ and 0.704 at $p = 0.6$, and no test of 124 couples with a type 1 error no larger than this test's does better at either. The best test of Proposition 10.4, which minimized the sum of the two error probabilities by rejecting where the alternative's PMF exceeds the null's, is ψ^* with $c = 1$.

With composite hypotheses there is no single alternative to compare with, and the natural substitute compares the best fit under each.

Definition 15.6 (Likelihood ratio test)**.** Let $\Theta_0 \subset \Theta$ be the null set, $\hat{\theta}_n$ the maximum likelihood estimator over Θ, and $\hat{\theta}_n^{\mathrm{c}}$ the **constrained maximum likelihood estimator**, the maximizer of the likelihood over Θ_0. The **likelihood ratio statistic** is

$$T_n^{\mathrm{LR}} = 2 \log \frac{L_n(\hat{\theta}_n)}{L_n(\hat{\theta}_n^{\mathrm{c}})} = 2\big(\ell_n(\hat{\theta}_n) - \ell_n(\hat{\theta}_n^{\mathrm{c}})\big) \geq 0,$$

and the **likelihood ratio test** of asymptotic level α rejects when $T_n^{\mathrm{LR}} > q_\alpha(\chi^2_k)$, where k is the number of constraints the null imposes (Theorems 15.7 and 15.12). For a simple null $\Theta_0 = \{\theta_0\}$, $\hat{\theta}_n^{\mathrm{c}} = \theta_0$ and $k = d$.

In words. Fit the model twice, once freely and once with the null imposed, and compare the two maximized log-likelihoods. The statistic is never negative, since maximizing over a larger set can only do better, and it is twice the log of how many times better the free fit explains the data. Only this difference means anything: a maximized log-likelihood on its own carries an unknown constant (Chapter 10).

For the kiss data. The free fit is $\hat{p} = 0.645$ with $\ell_n = -80.65$, the null fit is $p = \frac{1}{2}$ with $\ell_n = -85.95$, and $T^{\mathrm{LR}} = 2 \times 5.30 = 10.60$, the G^2 statistic of Chapter 10. The free fit explains

the data $e^{5.30} \approx 200$ times better, and the p-value is $\mathbb{P}(\chi^2_1 > 10.60) = 0.0011$.

Recall. The Law of Large Numbers (Theorem 3.5): for i.i.d. random variables with mean μ and finite variance, $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$. Applied entry by entry, it holds for averages of i.i.d. random vectors and matrices.

Theorem 15.7 (Wilks' theorem, simple null)**.** *In a regular model with $H_0 : \theta = \theta_0$, θ_0 interior, $I(\theta_0)$ invertible and I continuous at θ_0: if H_0 is true, $T_n^{\mathrm{LR}} \xrightarrow[n\to\infty]{(d)} \chi^2_d$, and $T_n^{\mathrm{LR}} - T_n^{\mathrm{W}} \xrightarrow[n\to\infty]{\mathbb{P}} 0$.*

Proof strategy (core). Expand the log-likelihood to second order around its top. The first-order term vanishes there, since the slope at the top is zero, and what is left is the drop of a parabola: the squared distance $\hat{\theta}_n - \theta_0$ weighted by the curvature at an in-between point, which converges to the Fisher information. That is Wald's statistic with the curvature measured somewhere else.

Proof. With probability tending to 1, $\hat{\theta}_n$ is interior and $\nabla \ell_n(\hat{\theta}_n) = 0$. Apply Taylor's theorem with the Lagrange remainder (toolkit A.9) to $h(t) = \ell_n\big(\hat{\theta}_n + t(\theta_0 - \hat{\theta}_n)\big)$ on $[0,1]$: $h(1) = h(0) + h'(0) + \frac{1}{2}h''(\tau)$ for some τ between 0 and 1. Here $h'(0) = \nabla \ell_n(\hat{\theta}_n)^\top(\theta_0 - \hat{\theta}_n) = 0$ and $h''(\tau) = (\theta_0 - \hat{\theta}_n)^\top \nabla^2 \ell_n(\tilde{\theta}_n)(\theta_0 - \hat{\theta}_n)$, with $\tilde{\theta}_n = \hat{\theta}_n + \tau(\theta_0 - \hat{\theta}_n)$ on the segment between $\hat{\theta}_n$ and θ_0 (toolkit A.12). So

$$T_n^{\mathrm{LR}} = 2\big(h(0) - h(1)\big) = V_n^\top H_n V_n, \qquad V_n = \sqrt{n}(\hat{\theta}_n - \theta_0), \quad H_n = -\tfrac{1}{n}\nabla^2 \ell_n(\tilde{\theta}_n).$$

Since $\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta_0$, with probability tending to 1 the whole segment from $\hat{\theta}_n$ to θ_0, and $\tilde{\theta}_n$ with it, lies in the neighborhood of θ_0 where the bound $K(X)$ on third derivatives of condition 4 holds. Then $H_n \xrightarrow[n\to\infty]{\mathbb{P}} I(\theta_0)$ entry by entry, by the step in the proof of Theorem 12.5: $-\frac{1}{n}\nabla^2 \ell_n(\theta_0) \xrightarrow[n\to\infty]{\mathbb{P}} -\mathbb{E}_{\theta_0}[\nabla^2 \ell(\theta_0)] = I(\theta_0)$ by the Law of Large Numbers and Theorem 12.4, and the bound on third derivatives controls the move from θ_0 to $\tilde{\theta}_n$. And $V_n \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}_d(0, I(\theta_0)^{-1})$ by Theorem 12.5. Lemma 15.3 gives $T_n^{\mathrm{LR}} \xrightarrow[n\to\infty]{(\mathrm{d})} \chi^2_d$. Finally, with $I = I(\theta_0)$, $T_n^{\mathrm{LR}} - T_n^{\mathrm{W}} = V_n^\top(H_n - I)V_n - V_n^\top(I(\hat{\theta}_n) - I)V_n$, and both terms converge in probability to 0 by the second part of Lemma 15.3. □

In one dimension the proof reads $T_n^{\mathrm{LR}} = \big(-\frac{1}{n}\ell_n''(\tilde{\theta}_n)\big) \cdot n(\hat{\theta}_n - \theta_0)^2$: the drop of the log-likelihood is the squared distance times the curvature somewhere between estimate and null value. At finite n the exact distribution of T_n^{LR} depends on the model and has a closed form only in special cases (Exercise 15.5 and Exercise 15.6); Wilks' theorem has no finite-sample version, and the chi-squared limit is the general answer. Section 15.6 measures how good it is.

Pitfall: a null on the boundary

Wilks' theorem needs the null value in the interior of Θ, condition 3 of Section 12.6. Testing whether a variance is zero, or whether a mixture needs its second component, puts the null on the edge of the parameter set, and the limit changes. For Gaussian data with known variance, $\Theta = [0, \infty)$ and $H_0 : \mu = 0$, the constrained and free fits coincide whenever $\bar{X}_n \le 0$, so T_n^{LR} is exactly 0 half the time, and it is $n\bar{X}_n^2/\sigma^2$ the other half: a 50:50 mixture of 0 and χ^2_1. The threshold 3.84 then gives level 2.5%, not 5% (Exercise 15.9).

15.5 The score test

Definition 15.8 (Score test)**.** In a regular model with $H_0 : \theta = \theta_0$, let $U_n(\theta_0) = \frac{1}{\sqrt{n}}\nabla\ell_n(\theta_0)$ be the standardized score at the null value. The **score statistic** is

$$T_n^{\mathrm{S}} = U_n(\theta_0)^\top I(\theta_0)^{-1} U_n(\theta_0),$$

and the **score test** of asymptotic level α rejects when $T_n^{\mathrm{S}} > q_\alpha(\chi^2_d)$. For $d = 1$, $T_n^{\mathrm{S}} = (Z_n^{\mathrm{S}})^2$ with $Z_n^{\mathrm{S}} = \ell_n'(\theta_0)/\sqrt{nI(\theta_0)}$.

> **In words.** Stand at the null value and ask how steeply the log-likelihood climbs there, in units of how steeply it climbs by chance when the null is true. Nothing is estimated: the slope and its yardstick are both computed at the null.
>
> **For the kiss data.** $\ell_n'(\frac{1}{2}) = 80/0.5 - 44/0.5 = 72$ and $nI(\frac{1}{2}) = 124 \times 4 = 496$ (Exercise 12.4), so $Z^{\mathrm{S}} = 72/\sqrt{496} = 3.23$ and $T^{\mathrm{S}} = 10.45$. This is the statistic of Chapter 9 and Pearson's chi-squared statistic for the right and left counts (Chapter 10), and it inverts to the solve interval $[0.558, 0.724]$ (Section 9.8).

Recall. The Central Limit Theorem for vectors (the vector-versions remark of Chapter 5): for i.i.d. random vectors Y_i with mean m and positive definite covariance matrix Σ, $\sqrt{n}(\bar{Y}_n - m) \xrightarrow[n\to\infty]{\text{(d)}} \mathrm{Gauss}_d(0, \Sigma)$.

Theorem 15.9 (The score test)**.** *In a regular model with $I(\theta_0)$ invertible, if $H_0 : \theta = \theta_0$ is true, then $U_n(\theta_0) \xrightarrow[n\to\infty]{(d)} Gauss_d(0, I(\theta_0))$ and $T_n^{\mathrm{S}} \xrightarrow[n\to\infty]{(d)} \chi^2_d$.*

Proof strategy (core). When the null is true, the score is a sum of i.i.d. vectors with mean zero and covariance $I(\theta_0)$. The CLT makes it Gaussian, and the quadratic form lemma makes the statistic χ^2_d.

Proof. Under H_0 the vectors $\nabla \log p_{\theta_0}(X_i)$ are i.i.d. with mean 0 (Proposition 12.2, coordinate by coordinate) and covariance matrix $I(\theta_0)$ (Definition 12.3), and $U_n(\theta_0)$ is $\sqrt{n}$ times their average. The multivariate CLT gives $U_n(\theta_0) \xrightarrow[n\to\infty]{\text{(d)}} \mathrm{Gauss}_d(0, I(\theta_0))$, the continuous mapping theorem applied to $u \mapsto u^\top I(\theta_0)^{-1} u$ gives $T_n^{\mathrm{S}} \xrightarrow[n\to\infty]{\text{(d)}} Z^\top I(\theta_0)^{-1} Z$ with $Z \sim \mathrm{Gauss}_d(0, I(\theta_0))$, and that is χ^2_d by Lemma 15.1. □

The proof uses no estimator, only the Central Limit Theorem. That is the score test's practical attraction: when the model with the alternative is hard to fit and the null model is easy, the score test needs only the easy fit.

Chapter 9 described the score test as the standardized estimate with the standard error computed at the null value. For a proportion the two descriptions agree: $\ell_n'(p_0)/\sqrt{nI(p_0)} = \sqrt{n}(\hat{p} - p_0)/\sqrt{p_0(1-p_0)}$. In general they do not, and Definition 15.8 is the general one. For the subway model of the next section, $\ell_n'(\lambda_0) = n/\lambda_0 - n\bar{T}_n$, so $Z_n^{\mathrm{S}} = \sqrt{n}(1 - \lambda_0\bar{T}_n) = \sqrt{n}(\hat{\lambda} - \lambda_0)/\hat{\lambda}$, which is Wald's statistic in λ, with the standard error at $\hat{\lambda}$.

15.6 Three tests, one limit

To compare the statistics write $o_{\mathbb{P}}(1)$ for any sequence of random quantities that converges in probability to 0; for vectors and matrices, entry by entry. By Slutsky's theorem, adding $o_{\mathbb{P}}(1)$ to a

sequence does not change its limit in distribution, and $o_{\mathbb{P}}(1)$ times a sequence that converges in distribution is again $o_{\mathbb{P}}(1)$.

Theorem 15.10 (Three tests, one limit)**.** *In a regular model with $H_0 : \theta = \theta_0$, θ_0 interior, $I(\theta_0)$ invertible and I continuous at θ_0: if H_0 is true,*

$$T_n^{\mathrm{W}} = T_n^{\mathrm{S}} + o_{\mathbb{P}}(1), \qquad T_n^{\mathrm{LR}} = T_n^{\mathrm{S}} + o_{\mathbb{P}}(1),$$

and all three converge in distribution to χ^2_d.

Proof strategy (for completeness; skip on a first reading). All three equal the score statistic plus something that vanishes. The proof of Theorem 12.5 writes the estimation error as the score divided by a curvature that converges to $I(\theta_0)$; substitute it into the likelihood ratio statistic and use Lemma 15.3 for the curvature's remaining error.

Proof. The proof of Theorem 12.5 applies the mean value theorem to each coordinate of the gradient, between θ_0 and $\hat\theta_n$. With probability tending to 1 this gives $0 = \frac{1}{\sqrt{n}}\nabla\ell_n(\hat\theta_n) = U_n - H_n'V_n$, where $U_n = U_n(\theta_0)$, $V_n = \sqrt{n}(\hat\theta_n - \theta_0)$, and row j of H_n' is row j of $-\frac{1}{n}\nabla^2\ell_n$ evaluated at its own in-between point. Those points converge in probability to θ_0, so $H_n' \xrightarrow[n\to\infty]{\mathbb{P}} I = I(\theta_0)$ entry by entry. The matrix H_n of the proof of Theorem 15.7 differs: it comes from one Taylor expansion of ℓ_n itself, so all its entries are evaluated at one point and it is symmetric, while H_n', with its rows taken at different points, need not be. Since $\det H_n' \xrightarrow[n\to\infty]{\mathbb{P}} \det I \neq 0$, H_n' is invertible with probability tending to 1, and then $V_n = (H_n')^{-1}U_n$. Writing $A^{-\top}$ for $(A^{-1})^\top$,

$$T_n^{\mathrm{LR}} = V_n^\top H_n V_n = U_n^\top M_n U_n, \qquad M_n = (H_n')^{-\top} H_n (H_n')^{-1}.$$

Matrix products and inverses are continuous where the inverses exist, so $M_n \xrightarrow[n\to\infty]{\mathbb{P}} I^{-1}II^{-1} = I^{-1}$ entry by entry. By the bound (15.1) with $u = v = U_n$, $|U_n^\top(M_n - I^{-1})U_n| \le \|U_n\|^2 \sum_{j,k}|(M_n - I^{-1})_{jk}|$, where $\|U_n\|^2$ converges in distribution (Theorem 15.9) and the sum converges in probability to 0; as in the proof of Lemma 15.3, $T_n^{\mathrm{LR}} - T_n^{\mathrm{S}} = U_n^\top(M_n - I^{-1})U_n = o_{\mathbb{P}}(1)$. Theorem 15.7 gives $T_n^{\mathrm{W}} - T_n^{\mathrm{LR}} = o_{\mathbb{P}}(1)$, and the limits are Theorems 15.4, 15.7 and 15.9. □

At finite n the three statistics differ, and near the threshold the difference can decide the test. Suppose the subway line of Chapter 5 advertises a train every 5 minutes on average. The 64 observed gaps of Chapter 5 average $\bar T_n = 6.23$ minutes; with gaps modeled as Expo(λ), the null is $\lambda_0 = 0.2$ per minute and $\hat\lambda = 1/6.23 = 0.1605$. The log-likelihood is $\ell_n(\lambda) = n\log\lambda - \lambda n\bar T_n$ and $I(\lambda) = 1/\lambda^2$, and the three statistics are

$$T^{\mathrm{W}} = \frac{n(\hat\lambda - \lambda_0)^2}{\hat\lambda^2} = 3.87, \qquad T^{\mathrm{LR}} = 2n\Big(\log\frac{\hat\lambda}{\lambda_0} - 1 + \frac{\lambda_0}{\hat\lambda}\Big) = 3.34, \qquad T^{\mathrm{S}} = n(1 - \lambda_0\bar T_n)^2 = 3.87,$$

the score statistic equal to Wald's for this model, as the last section found. At level 5% the threshold is 3.84: Wald and score reject, with p-value 0.049, and the likelihood ratio test does not, with p-value 0.068. Which is right? This model has a referee: $\lambda n\bar T_n$ is exactly Gamma$(n, 1)$ (Exercise 8.7), which gives an exact test. Its two-sided p-value, twice the smaller tail, is 0.062, siding with the likelihood ratio.

Figure 15.3 draws each statistic as a function of the null value. Each curve is zero at the estimate and grows away from it; the null values where a curve stays below 3.84 form that test's 95% confidence

interval (Proposition 9.7), drawn as a bar underneath. For the kiss data, on the left, the three curves nearly coincide and the three intervals agree to within 0.006: Wald's $[0.561, 0.729]$ (plug-in), the likelihood ratio's $[0.559, 0.726]$, and the score's $[0.558, 0.724]$ (solve). For the subway, on the right, they come apart near $\lambda_0 = 0.2$, which sits just outside Wald's interval $[0.1212, 0.1998]$ and just inside the likelihood ratio's $[0.1243, 0.2031]$. The likelihood ratio interval is within 0.001 of the exact Gamma interval $[0.1236, 0.2022]$ at both ends; Wald's is off by about 0.0024 at each.

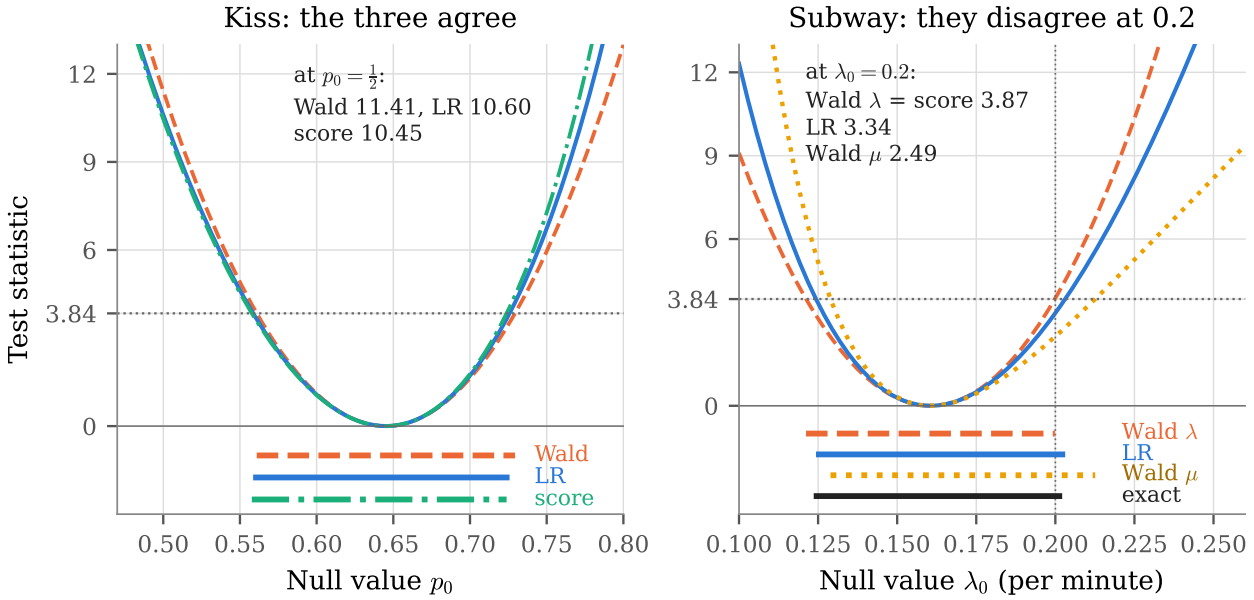


Figure 15.3: The statistics against the null value, with the threshold 3.84 (dotted) and the 95% intervals (bars). Left: kiss data; Wald (orange), likelihood ratio (LR, blue), score (aqua). Right: subway; Wald in λ (equal to score), LR, Wald in $\mu = 1/\lambda$ (yellow), exact (black).

The yellow curve on the right is the most instructive. It is Wald's statistic for the same hypothesis written in the mean wait $\mu = 1/\lambda$ instead of the rate: $\mu_0 = 5$ minutes, $\hat{\mu} = 6.23$, $I(\mu) = 1/\mu^2$, and $T^{\mathrm{W}} = n(\hat{\mu} - \mu_0)^2/\hat{\mu}^2 = 2.49$, with p-value 0.114. The hypothesis 'a train every 5 minutes' is one statement about one distribution, and Wald's test returns 3.87 or 2.49 depending on which letter is used to label it.

Pitfall: Wald's test depends on how the parameter is written

Figure 15.4 shows why. It draws the subway log-likelihood twice, against the rate λ on the left and against the mean wait μ on the right, with Wald's parabola in each coordinate. The log-likelihood has the same value at the same distribution whatever the coordinate (Proposition 11.6), so the blue dots at the null sit at the same height, -1.67, in both, and the likelihood ratio statistic is 3.34 either way. But a curve that is close to a parabola in λ is a different shape in $1/\lambda$, and Wald replaces the log-likelihood by a parabola in whatever coordinate it is handed. In λ its parabola falls below the curve at the null, at -1.94, and Wald overstates the evidence; in μ it stays above, at -1.25, and Wald understates it. The score statistic, being the slope over the curvature at the null, is also unchanged by relabeling (Exercise 15.10).

How close each test's type 1 error is to its nominal level can be computed exactly for this model, since $\lambda_0 n \bar{T}_n$ has a known Gamma distribution under the null. Each statistic is a function of $\lambda_0 \bar{T}_n$ alone, so the type 1 errors do not depend on λ_0, and Figure 15.5 takes $\lambda_0 = 1$. On its left, the

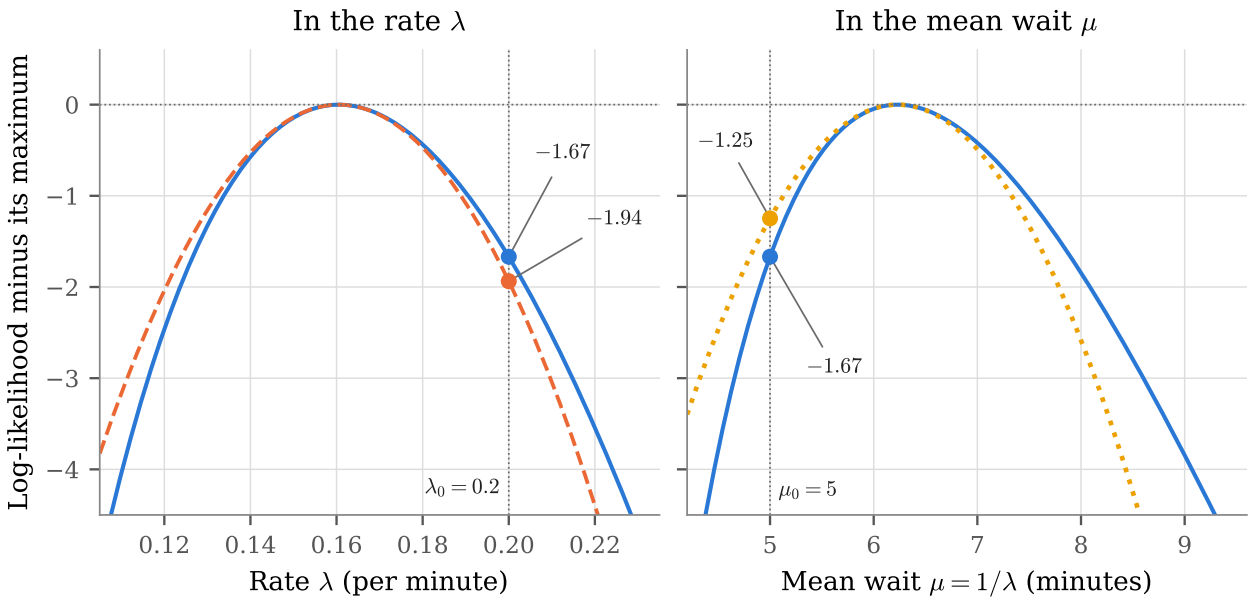


Figure 15.4: The subway log-likelihood minus its maximum (blue) against the rate λ (left) and the mean wait $\mu = 1/\lambda$ (right), with Wald's parabola in each coordinate (orange, dashed; yellow, dotted) and the null $\lambda_0 = 0.2$, $\mu_0 = 5$ marked.

likelihood ratio statistic of 50,000 samples of only $n = 10$ exponential observations already follows the χ^2_1 density closely: 5.2% of them exceed 3.84. On its right is the exact type 1 error at nominal level 5% of Wald's test in λ (which is also the score test), the likelihood ratio test and Wald's test in μ. The likelihood ratio test is at 5.4% for $n = 5$, 5.2% for $n = 10$ and 5.0% for $n = 64$. Wald's test in λ is a little conservative, at 4.4% for $n = 5$. Wald's test in μ rejects a true null 13% of the time at $n = 5$ and 7.4% at $n = 20$, and is still at 5.8% at $n = 64$. Of 1,000 subway studies of 20 gaps on a line that keeps its schedule, about 74 would accuse it under Wald's test in μ and about 51 under the likelihood ratio test.

Which test to use. When the model can be fitted both with and without the null, use the likelihood ratio test. The evidence is in this section's figures: it does not depend on how the parameter is written (Figure 15.4); its type 1 error is usually the closest of the three to the nominal level in small samples (Figure 15.5); and its intervals follow the shape of the likelihood, which is why for the subway they nearly coincide with the exact interval (Figure 15.3). Wald's statistic is what software prints beside every estimate, as estimate over standard error, and it is the natural screen when a model has many parameters; rerun any borderline case as a likelihood ratio test. The score test is the tool when only the null model is easy to fit, for instance when asking whether a term should be added to a model that has already been fitted without it. For intervals this replaces the default of Chapter 8, where the plug-in interval, which is the inverted Wald test, served every model but the proportion. For a single proportion the solve interval of that chapter, the inverted score test, remains the choice: it has a closed form and holds its level well at small n (Figure 8.3).

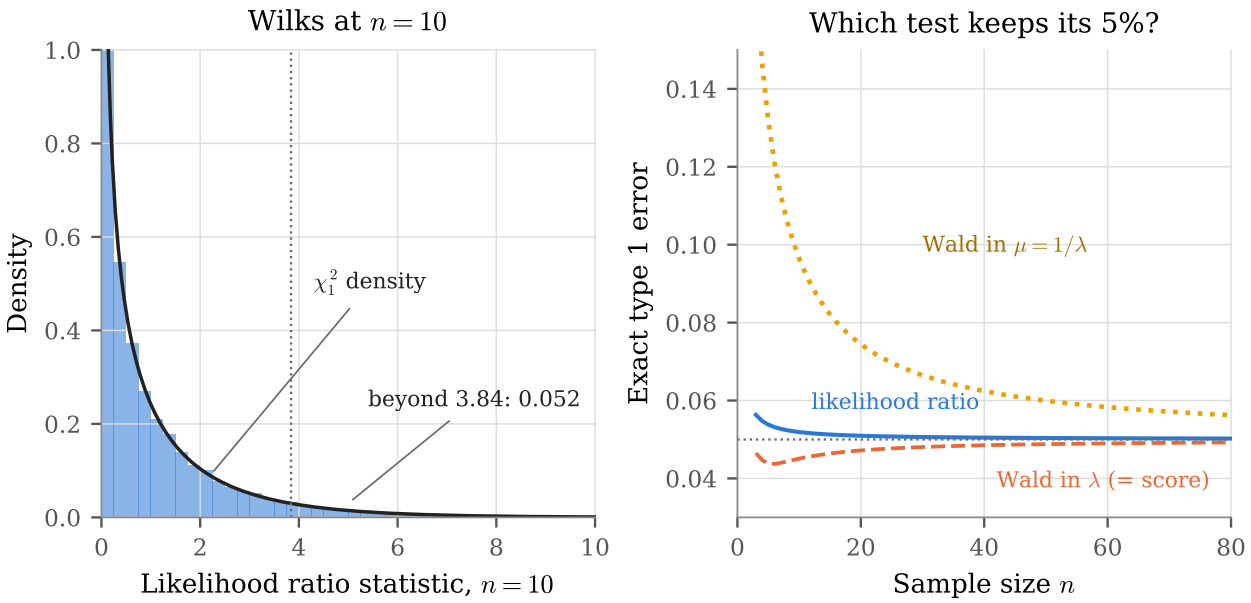


Figure 15.5: Exponential data, $\lambda_0 = 1$. Left: the likelihood ratio statistic over 50,000 null samples of $n = 10$, with the χ^2_1 density (black). Right: exact type 1 error at nominal 5% against n: Wald in λ (equal to score), likelihood ratio, Wald in μ.

15.7 Nuisance parameters and implicit hypotheses

Most hypotheses fix only part of the parameter. 'Is the mean wind speed 10?' says nothing about the variance, which must still be estimated; 'is the heart-attack rate the same with aspirin as with placebo?' says nothing about what the common rate is. Write the parameter as $\theta = (\nu, \beta)$, with $\nu \in \mathbb{R}^r$ the coordinates the null leaves free, called **nuisance parameters** (ν for nuisance), and $\beta \in \mathbb{R}^k$ the coordinates the null fixes, $H_0 : \beta = \beta_0$, so that $d = r + k$. The constrained estimator is $\hat{\theta}^{\mathrm{c}}_n = (\hat{\nu}^{\mathrm{c}}_n, \beta_0)$, where $\hat{\nu}^{\mathrm{c}}_n$ maximizes $\ell_n(\nu, \beta_0)$ over ν. Write the Fisher information in blocks, $I_{\nu\nu}$ ($r \times r$), $I_{\nu\beta}$, $I_{\beta\nu}$ and $I_{\beta\beta}$ ($k \times k$), and $(I^{-1})_{\beta\beta}$ for the $\beta\beta$ block of the inverse matrix, which by Theorem 12.5 is the asymptotic covariance matrix of $\sqrt{n}(\hat{\beta}_n - \beta^*)$. Its inverse is $I_{\beta|\nu} = I_{\beta\beta} - I_{\beta\nu} I_{\nu\nu}^{-1} I_{\nu\beta}$, as the proof of Theorem 15.12 shows. That matrix is the covariance of the part of the score in the β coordinates that the score in the ν coordinates cannot predict linearly: the information about β that survives not knowing ν. When $I_{\beta\nu} = 0$, not knowing ν costs nothing in the limit, and $I_{\beta|\nu} = I_{\beta\beta}$.

Definition 15.11 (The three tests with nuisance parameters)**.** For $H_0 : \beta = \beta_0$ with ν free,

$$
\begin{aligned}
T^{\mathrm{W}}_n &= n\,(\hat{\beta}_n - \beta_0)^\top \big[(I(\hat{\theta}_n)^{-1})_{\beta\beta}\big]^{-1} (\hat{\beta}_n - \beta_0), \\
T^{\mathrm{LR}}_n &= 2\big(\ell_n(\hat{\theta}_n) - \ell_n(\hat{\theta}^{\mathrm{c}}_n)\big), \\
T^{\mathrm{S}}_n &= \tfrac{1}{n} \nabla \ell_n(\hat{\theta}^{\mathrm{c}}_n)^\top I(\hat{\theta}^{\mathrm{c}}_n)^{-1} \nabla \ell_n(\hat{\theta}^{\mathrm{c}}_n),
\end{aligned}
$$

each compared with $q_\alpha(\chi^2_k)$.

> **In words.** Wald measures the fixed coordinates' estimation error with their own asymptotic covariance, which already accounts for having to estimate the free ones. The likelihood ratio lets the free coordinates do their best under each hypothesis. The score test fits the null model, free coordinates and all, and asks how steeply the log-likelihood climbs from there. Each is compared

with a chi-squared with one degree of freedom per fixed coordinate: what counts is the number of constraints, not the number of parameters.

For the kiss data. The Bernoulli model has one parameter and nothing left to free, so these reduce to the tests above. A nuisance parameter appears as soon as the question is comparative: 'do couples in the United States and in Germany turn right equally often?' has $d = 2$ parameters and one constraint, and the common proportion is the nuisance parameter.

Theorem 15.12 (The three tests with nuisance parameters)**.** *In a regular model with $\theta^* = (\nu^*, \beta_0)$ interior, I invertible at θ^* and continuous there: if $H_0 : \beta = \beta_0$ is true, T_n^{W}, T_n^{LR} and T_n^{S} all converge in distribution to χ^2_k, and their differences are $o_{\mathbb{P}}(1)$.*

Proof strategy (for completeness; skip on a first reading). Expand the free and the constrained fits around θ^*, as in Theorem 15.7. Each statistic becomes, up to $o_{\mathbb{P}}(1)$, the same quadratic form in one Gaussian vector of dimension k: the part of the score in the fixed directions that the free directions cannot explain. A block factorization of I identifies its covariance, and Lemma 15.1 finishes.

Proof. Write $I = I(\theta^*)$ and $U_n = \frac{1}{\sqrt{n}}\nabla \ell_n(\theta^*) = (U_{n,\nu}, U_{n,\beta})$. The block $I_{\nu\nu}$ is positive definite, hence invertible (toolkit A.14): for $a_\nu \neq 0$, take $a = (a_\nu, 0)$ in $a^\top I a > 0$. Let $C = I_{\beta\nu} I_{\nu\nu}^{-1}$ and $I_{\beta|\nu} = I_{\beta\beta} - I_{\beta\nu} I_{\nu\nu}^{-1} I_{\nu\beta}$.

A block identity. With $L = \begin{pmatrix} \mathrm{Id}_r & 0 \\ C & \mathrm{Id}_k \end{pmatrix}$ and $D = \begin{pmatrix} I_{\nu\nu} & 0 \\ 0 & I_{\beta|\nu} \end{pmatrix}$, multiplying out gives $I = LDL^\top$, using $I_{\nu\beta} = I_{\beta\nu}^\top$. Both diagonal blocks of $D = L^{-1} I L^{-\top}$ are positive definite, since I is. So $I^{-1} = L^{-\top} D^{-1} L^{-1}$, and since $L^{-1}u = (u_\nu, u_\beta - C u_\nu)$ for any $u = (u_\nu, u_\beta)$,

$$u^\top I^{-1} u = u_\nu^\top I_{\nu\nu}^{-1} u_\nu + v^\top I_{\beta|\nu}^{-1} v, \qquad v = u_\beta - C u_\nu. \tag{$\dagger$}$$

Multiplying out $L^{-\top} D^{-1} L^{-1}$ also shows that the β rows of $I^{-1}u$ are $I_{\beta|\nu}^{-1} v$, and that $(I^{-1})_{\beta\beta} = I_{\beta|\nu}^{-1}$.

The two fits. As in the proofs of Theorems 12.5 and 15.10, $\sqrt{n}(\hat\theta_n - \theta^*) = I^{-1} U_n + o_{\mathbb{P}}(1)$ and $2(\ell_n(\hat\theta_n) - \ell_n(\theta^*)) = U_n^\top I^{-1} U_n + o_{\mathbb{P}}(1)$. The null model, with parameter ν alone, is itself regular with score $U_{n,\nu}$ and information $I_{\nu\nu}$ at ν^*, so the same two statements hold for it: $\sqrt{n}(\hat\nu_n^{\mathrm{c}} - \nu^*) = I_{\nu\nu}^{-1} U_{n,\nu} + o_{\mathbb{P}}(1)$ and $2(\ell_n(\hat\theta_n^{\mathrm{c}}) - \ell_n(\theta^*)) = U_{n,\nu}^\top I_{\nu\nu}^{-1} U_{n,\nu} + o_{\mathbb{P}}(1)$.

The three statistics. Let $v_n = U_{n,\beta} - C U_{n,\nu}$. Subtracting the two expansions and using ($\dagger$), $T_n^{\mathrm{LR}} = v_n^\top I_{\beta|\nu}^{-1} v_n + o_{\mathbb{P}}(1)$. For Wald, the β rows of $I^{-1}U_n$ give $\sqrt{n}(\hat\beta_n - \beta_0) = I_{\beta|\nu}^{-1} v_n + o_{\mathbb{P}}(1)$, and $[(I(\hat\theta_n)^{-1})_{\beta\beta}]^{-1} \xrightarrow[n\to\infty]{\mathbb{P}} I_{\beta|\nu}$, so $T_n^{\mathrm{W}} = v_n^\top I_{\beta|\nu}^{-1} I_{\beta|\nu} I_{\beta|\nu}^{-1} v_n + o_{\mathbb{P}}(1) = v_n^\top I_{\beta|\nu}^{-1} v_n + o_{\mathbb{P}}(1)$. For the score, expand the gradient at $\hat\theta_n^{\mathrm{c}}$ around θ^* as in Theorem 12.5: $\frac{1}{\sqrt{n}}\nabla\ell_n(\hat\theta_n^{\mathrm{c}}) = U_n - I\sqrt{n}(\hat\theta_n^{\mathrm{c}} - \theta^*) + o_{\mathbb{P}}(1)$. Its ν part is exactly 0, the first-order condition of the constrained fit, and its β part is $U_{n,\beta} - I_{\beta\nu} I_{\nu\nu}^{-1} U_{n,\nu} + o_{\mathbb{P}}(1) = v_n + o_{\mathbb{P}}(1)$. With $I(\hat\theta_n^{\mathrm{c}})^{-1} \xrightarrow[n\to\infty]{\mathbb{P}} I^{-1}$, whose $\beta\beta$ block is $I_{\beta|\nu}^{-1}$, $T_n^{\mathrm{S}} = v_n^\top I_{\beta|\nu}^{-1} v_n + o_{\mathbb{P}}(1)$. (Each $o_{\mathbb{P}}(1)$ step is the bound (15.1): with $u = v$ for a quadratic form, and with $u \neq v$ for the cross terms that an $o_{\mathbb{P}}(1)$ vector creates inside one.)

The limit. $v_n = B U_n$ with $B = (-C \;\; \mathrm{Id}_k)$, a $k \times d$ matrix of rank k. By the multivariate CLT, $U_n \xrightarrow[n\to\infty]{(\mathrm{d})} Z \sim \mathrm{Gauss}_d(0, I)$, so $v_n \xrightarrow[n\to\infty]{(\mathrm{d})} BZ$, which is $\mathrm{Gauss}_k(0, BIB^\top)$ (toolkit A.14), and multiplying out, $BIB^\top = I_{\beta|\nu}$. By the continuous mapping theorem and Lemma 15.1, $v_n^\top I_{\beta|\nu}^{-1} v_n \xrightarrow[n\to\infty]{(\mathrm{d})} \chi^2_k$, and Slutsky adds back the $o_{\mathbb{P}}(1)$ terms. □

Worked example: the wind turbine, three ways

The twelve monthly wind speeds of Chapter 14 have mean 14.3 and standard deviation $S = 4.7$; the question is $H_0 : \mu = 10$ with σ^2 unknown. Here $\beta = \mu$, $\nu = \sigma^2$, $k = 1$. The free fit is $\hat{\mu} = 14.3$, $\hat{\sigma}^2 = \frac{11}{12}S^2 = 20.25$; the constrained fit keeps $\mu = 10$ and has $\hat{\sigma}^2_{\mathrm{c}} = \hat{\sigma}^2 + (14.3 - 10)^2 = 38.74$, the variance measured around 10 instead of around 14.3. With $I(\mu, \sigma^2) = \mathrm{diag}(1/\sigma^2, 1/(2\sigma^4))$ (Chapter 12) the three statistics are

$$T^{\mathrm{W}} = \frac{n(\hat{\mu} - 10)^2}{\hat{\sigma}^2} = 10.96, \qquad T^{\mathrm{LR}} = n \log \frac{\hat{\sigma}^2_{\mathrm{c}}}{\hat{\sigma}^2} = 7.78, \qquad T^{\mathrm{S}} = \frac{n(\hat{\mu} - 10)^2}{\hat{\sigma}^2_{\mathrm{c}}} = 5.73,$$

with χ^2_1 p-values 0.0009, 0.0053 and 0.017. Here $I_{\mu\sigma^2} = 0$, so not knowing σ^2 costs nothing in the limit, and at $n = 12$ the statistics differ only through how they estimate it: Wald's and the score statistic are the same squared distance $n(\hat{\mu} - 10)^2/\sigma^2$, with σ^2 estimated at 20.25 and at 38.74. Each is an increasing function of the t statistic $T = \sqrt{12}(14.3 - 10)/4.7 = 3.17$: $T^{\mathrm{W}}_n = \frac{n}{n-1}T^2$, $T^{\mathrm{LR}}_n = n \log\left(1 + \frac{T^2}{n-1}\right)$ (Exercise 15.6) and $T^{\mathrm{S}}_n = \frac{nT^2}{n-1+T^2}$. With $x = T^2/(n-1)$ these are nx, $n\log(1+x)$ and $nx/(1+x)$, and since $x/(1+x) \le \log(1+x) \le x$, the likelihood ratio statistic always lies between the other two. So all three reject exactly when $|T|$ is large, as the t-test does, and differ only in where the chi-squared approximation puts the threshold. The exact two-sided t-test gives $p = 0.0089$, twice the one-sided 0.0045 of Chapter 14, and once more the likelihood ratio is the closest.

Figure 15.6 shows what the constrained fit does. On the left are contours of the log-likelihood over (μ, σ^2), with the peak at the free fit. The dotted line is the null, $\mu = 10$, and the orange dot is the highest point on it, $(10, 38.7)$, where the line just touches a contour; that contour, dashed, is 3.89 below the peak, half of $T^{\mathrm{LR}} = 7.78$. On the right is the **profile log-likelihood**, $\ell^{\mathrm{p}}_n(\mu) = \max_{\sigma^2} \ell_n(\mu, \sigma^2)$, the height of the highest point on each vertical line: a one-parameter log-likelihood for μ with the nuisance parameter maximized out. The likelihood ratio statistic is twice its drop from the top to $\mu_0 = 10$. Cutting the profile at $-3.84/2 = -1.92$ gives the 95% likelihood ratio interval for the mean wind speed, $[11.5, 17.1]$, a little narrower than the exact t interval $[11.3, 17.3]$: with 12 observations the chi-squared approximation is still a little optimistic.

Implicit hypotheses. Sometimes the null is a relation among the parameters rather than a fixed value of some of them: two proportions are equal, a mean equals a standard deviation. Write it as $H_0 : g(\theta) = 0$ for a continuously differentiable $g : \mathbb{R}^d \to \mathbb{R}^k$ whose $d \times k$ matrix of partial derivatives $\nabla g(\theta)$ (the convention of Chapter 5) has rank k, meaning that its k columns are linearly independent: no constraint is a combination of the others.

Recall. The delta method for vectors (Theorem 5.4 and the vector-versions remark of Chapter 5): if $\sqrt{n}(Z_n - \theta) \xrightarrow[n\to\infty]{\text{(d)}} \mathrm{Gauss}_d(0, \Sigma)$ and g is continuously differentiable at θ, then $\sqrt{n}(g(Z_n) - g(\theta)) \xrightarrow[n\to\infty]{\text{(d)}} \mathrm{Gauss}_k(0, \nabla g(\theta)^\top \Sigma \nabla g(\theta))$.

Proposition 15.13 (Wald's test for an implicit hypothesis)**.** *Let $g : \mathbb{R}^d \to \mathbb{R}^k$ be continuously differentiable, with $\nabla g(\theta^*)$ of rank k, and let $\Gamma(\theta) = \nabla g(\theta)^\top I(\theta)^{-1} \nabla g(\theta)$ and $T^{\mathrm{W}}_n = n\, g(\hat{\theta}_n)^\top \Gamma(\hat{\theta}_n)^{-1} g(\hat{\theta}_n)$. In a regular model with θ^* interior, $I(\theta^*)$ invertible and I continuous at θ^*, if $g(\theta^*) = 0$, then $T^{\mathrm{W}}_n \xrightarrow[n\to\infty]{(d)} \chi^2_k$.*

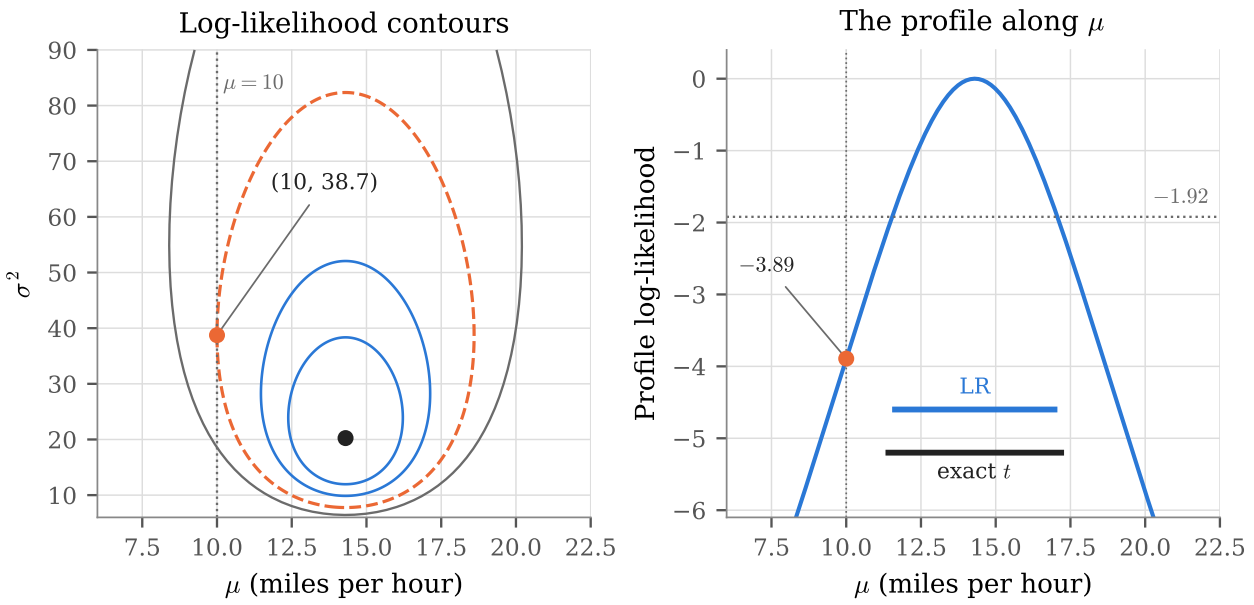


Figure 15.6: Wind speeds, $n = 12$, $H_0 : \mu = 10$. Left: contours of $\ell_n(\mu, \sigma^2)$ minus its maximum at -1, -2, -3.89 (orange, dashed) and -6, the peak (black) and the best point on $\mu = 10$ (orange). Right: the profile log-likelihood, with the likelihood ratio (LR) and exact t intervals.

Proof strategy (for completeness; skip on a first reading). The delta method makes $\sqrt{n}\, g(\hat{\theta}_n)$ approximately $\text{Gauss}_k(0, \Gamma(\theta^*))$; Wald's statistic is its quadratic form with Γ estimated.

Proof. By Theorem 12.5 and the delta method, $V_n = \sqrt{n}\, g(\hat{\theta}_n) = \sqrt{n}(g(\hat{\theta}_n) - g(\theta^*)) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_k(0, \Gamma(\theta^*))$. $\Gamma(\theta^*)$ is invertible: for $a \neq 0$, $\nabla g(\theta^*)a \neq 0$ since ∇g has rank k, so $a^\top \Gamma a = (\nabla g\, a)^\top I^{-1} (\nabla g\, a) > 0$. By continuity $\Gamma(\hat{\theta}_n)^{-1} \xrightarrow[n\to\infty]{\mathbb{P}} \Gamma(\theta^*)^{-1}$, and Lemma 15.3 with I replaced by $\Gamma(\theta^*)^{-1}$ gives $T_n^{\text{W}} = V_n^\top \Gamma(\hat{\theta}_n)^{-1} V_n \xrightarrow[n\to\infty]{\text{(d)}} \chi^2_k$. □

The likelihood ratio and score tests need no new theory. Near θ^*, k independent smooth constraints can be turned into 'fix k coordinates' by a smooth one-to-one change of parameter $\theta = h(\eta)$ (the implicit function theorem of calculus, which this book takes as given), and Theorem 15.12 applies in η. The likelihood ratio and score statistics are unchanged by such a change. The likelihood ratio statistic compares two maximized values of the same function, whatever its coordinates. For the score statistic, let J be the $d \times d$ matrix with entries $\partial\theta_i/\partial\eta_j$, invertible since the change is one-to-one and smooth both ways; the chain rule makes the score in η equal to $J^\top \nabla \ell_n$ and the information $J^\top I J$, so $(J^\top \nabla \ell_n)^\top (J^\top I J)^{-1} J^\top \nabla \ell_n = \nabla \ell_n^\top I^{-1} \nabla \ell_n$ (Exercise 15.10 does one dimension). The change of parameter is defined only near θ^*, but both maximizers lie there with probability tending to 1, so both statistics converge to χ^2_k.

Several independent samples. The theorems of this chapter were proved for one i.i.d. sample, and they hold for several independent samples whose sizes grow in fixed proportions, each contributing its own information: $nI(\theta)$ becomes $n_1 I_1(\theta) + n_2 I_2(\theta)$ for two samples, and the proofs apply the CLT and the LLN to each sample.

Worked example: aspirin and heart attacks

The preliminary report of the Physicians' Health Study (Steering Committee of the Physicians' Health Study Research Group, *New England Journal of Medicine* 318, 1988, 262–264) randomized 22,071 male physicians to aspirin or placebo. Among 11,034 on placebo, 189 had a heart attack ($\hat{p}_1 = 0.0171$); among 11,037 on aspirin, 104 did ($\hat{p}_2 = 0.0094$). In natural frequencies: about 17 heart attacks per 1,000 physicians on placebo, 9 per 1,000 on aspirin. With $\theta = (p_1, p_2)$ and $g(\theta) = p_1 - p_2$, $\nabla g = (1, -1)^\top$, and since the samples are independent the delta method's variance for $\hat{p}_1 - \hat{p}_2$ is the sum of the two estimated variances, $\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2$, so

$$Z^{\mathrm{W}} = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}} = \frac{0.00771}{0.00154} = 5.00, \qquad T^{\mathrm{W}} = 25.04.$$

The score test fits the null, a common rate $\hat{p} = 293/22{,}071 = 0.0133$. There the score in p_j is $n_j(\hat{p}_j - \hat{p})/(\hat{p}(1-\hat{p}))$ and the information is diagonal with entries $n_j/(\hat{p}(1-\hat{p}))$, so

$$T^{\mathrm{S}} = \frac{n_1(\hat{p}_1 - \hat{p})^2 + n_2(\hat{p}_2 - \hat{p})^2}{\hat{p}(1-\hat{p})} = \frac{(\hat{p}_1 - \hat{p}_2)^2}{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)} = 25.01,$$

the square of the pooled two-proportion statistic, and Pearson's chi-squared for the 2×2 table (Chapter 16). The likelihood ratio statistic compares the fits with two rates and with one: $T^{\mathrm{LR}} = 25.37$, the G^2 for the table. With $k = 1$ constraint, all three have p-values below 10^{-6}. Wald's 95% interval for the difference is $0.0077 \pm 1.96 \times 0.00154 = [0.0047, 0.0107]$: with 95% confidence, aspirin prevented between about 5 and 11 heart attacks per 1,000 physicians over the study. With samples this large the three statistics agree to within 1.5%.

From undergraduate statistics: the named tests are three recipes

Most of the tests in an introductory course are one of the three, applied to a particular model. The z-test for a mean with known variance is all three at once, since its log-likelihood is an exact parabola (Exercise 15.5). The one-sample t-test is the likelihood ratio test for the Gaussian mean with unknown variance, with the exact t distribution in place of the chi-squared limit (the worked example above). The one-proportion z-test with the null standard error is the score test, and with the estimated standard error it is Wald's. The two-proportion z-test with the pooled proportion is the score test, and Pearson's chi-squared test for a table of counts is its many-cell version; the G-test is the likelihood ratio test for the same table (Chapter 16). The F-test for a group of regression coefficients is the likelihood ratio test for a Gaussian linear model (Chapter 21), and the deviance tests of generalized linear models are likelihood ratio tests (Chapter 23). That is what Chapter 1 meant by machinery that delivers whole families of tests at once.

15.8 What comes next

Part V is complete: exact tests for small Gaussian samples and three general recipes for large samples from any regular model. Every test so far assumed the model. Chapter 16 tests the model itself for discrete data, and its two statistics are this chapter's score and likelihood ratio statistics for the table of observed counts, with $K - 1$ degrees of freedom for K categories, one fewer for each parameter estimated from the counts. Chapter 17 does the same for continuous distributions, with

a statistic that is not built from a likelihood. Chapter 21 returns to the likelihood ratio test with Gaussian errors, where the general Cochran theorem makes it exact and it becomes the F-test.

15.9 Summary

- Three statistics for $H_0 : \theta = \theta_0$, read off the log-likelihood: Wald's $n(\hat\theta_n - \theta_0)^\top I(\hat\theta_n)(\hat\theta_n - \theta_0)$ (distance in units of the estimated noise); the likelihood ratio $2(\ell_n(\hat\theta_n) - \ell_n(\theta_0))$ (twice the drop from the top); the score $\frac{1}{n}\nabla\ell_n(\theta_0)^\top I(\theta_0)^{-1}\nabla\ell_n(\theta_0)$ (slope at the null over its yardstick). Kiss data: 11.41, 10.60, 10.45.
- A Gaussian vector $Z \sim \text{Gauss}_d(m, \Sigma)$ has the distribution of $m + AW$ with $AA^\top = \Sigma$; $(Z-m)^\top\Sigma^{-1}(Z-m) \sim \chi^2_d$, the squared distance in units of the noise.
- Under the null all three converge to χ^2_d, and their differences converge in probability to 0. Wilks' theorem for the likelihood ratio is a second-order Taylor expansion plus Theorem 12.5; the score test needs only the CLT.
- Neyman–Pearson: for simple hypotheses, rejecting when $L_n(\theta_1)/L_n(\theta_0)$ is large is most powerful. The kiss test $S \geq 72$, with type 1 error 0.0438, is the most powerful test against every $p > \frac{1}{2}$ among tests whose type 1 error is no larger.
- In small samples the three can disagree near the threshold (subway, $\lambda_0 = 0.2$: 3.87, 3.34, 3.87). Wald depends on the parameterization (3.87 in λ, 2.49 in $1/\lambda$); likelihood ratio and score do not. The likelihood ratio test is the default when both fits are available.
- With nuisance parameters, the degrees of freedom are the number of constraints; the likelihood ratio uses the constrained fit, equivalently the profile likelihood. Wind data: 10.96, 7.78, 5.73, all increasing in $|T|$. Implicit hypotheses $g(\theta) = 0$ use the delta method. Aspirin: 25.04, 25.37, 25.01.
- Wilks' theorem needs an interior null; on the boundary the limit is a mixture.

15.10 Exercises

Exercise 15.1. (a) Show that $\mathbb{P}(\chi^2_2 > q) = e^{-q/2}$, and deduce $q_{0.05}(\chi^2_2) = -2\log 0.05 = 5.99$. (b) Show that $q_\alpha(\chi^2_1) = q^2_{\alpha/2}$. (c) For $Z \sim \text{Gauss}_2(0, \Sigma)$ with $\Sigma = \begin{pmatrix} 1 & 1.2 \\ 1.2 & 2.25 \end{pmatrix}$, as in Figure 15.2, compute $z^\top\Sigma^{-1}z$ for $z = (1, 1)$ and $z = (1, -1)$, and say which is inside the 95% ellipse. *(Solution on page 426.)*

Exercise 15.2. The drug of Chapter 1 was believed to be 80% effective, and a new trial found it effective in 78 of 100 patients. Compute Wald's, the likelihood ratio and the score statistics for $H_0 : p = 0.8$ against $H_1 : p \neq 0.8$, and the p-values. What do you conclude, and does the choice of test matter here? *(Solution on page 426.)*

Exercise 15.3. For $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$ and $H_0 : \lambda = \lambda_0$, show that $T_n^{\text{W}} = n(\hat\lambda - \lambda_0)^2/\hat\lambda$, $T_n^{\text{S}} = n(\hat\lambda - \lambda_0)^2/\lambda_0$ and $T_n^{\text{LR}} = 2n\big(\hat\lambda\log(\hat\lambda/\lambda_0) - (\hat\lambda - \lambda_0)\big)$, with $\hat\lambda = \bar X_n$. The siblings data of Chapter 2 are modeled as $X_i - 1 \sim \text{Pois}(\lambda)$: evaluate the three statistics on the counts minus one ($n = 24$, $\hat\lambda = 1.0$) for $\lambda_0 = 1.5$. *(Solution on page 426.)*

Exercise 15.4. A coin lands heads with probability 0.25 or 0.75, and you test $H_0 : p = 0.25$ against $H_1 : p = 0.75$ with the likelihood ratio test $\psi^* = \mathbb{1}(L_n(0.75) > L_n(0.25))$ ($c = 1$). (a) Show that for

n flips with S heads the test rejects exactly when $S > n/2$. (b) For $n = 16$, compute its level and its power, and decide for 6 heads. (c) What does Proposition 15.5 say about any other test with level at most this one's? *(Solution on page 427.)*

Exercise 15.5. For $X_1, \dots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$ with σ^2 known and $H_0 : \mu = \mu_0$, show that Wald's, the likelihood ratio and the score statistics are all equal to $n(\bar{X}_n - \mu_0)^2/\sigma^2$, and that under H_0 this is exactly χ^2_1 at every n. *(Solution on page 427.)*

Exercise 15.6. For $X_1, \dots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, \sigma^2)$ with both unknown and $H_0 : \mu = \mu_0$: (a) show that $T_n^{\text{LR}} = n \log\big(1 + T^2/(n-1)\big)$, where $T = \sqrt{n}(\bar{X}_n - \mu_0)/S_n$ is the t statistic of Chapter 14. (b) Conclude that the likelihood ratio test rejects exactly when $|T|$ is large. (c) For $n = 12$, which threshold on $|T|$ does the χ^2_1 threshold 3.84 correspond to, and what is its exact level? *(Solution on page 427.)*

Exercise 15.7. Let $X_1, \dots, X_{100}$ be i.i.d. with density $\theta x^{-\theta-1}$ for $x > 1$ (a Pareto distribution), and $H_0 : \theta = 2$. (a) Find the maximum likelihood estimator and $I(\theta)$. (b) With $\hat{\theta} = 2.45$, compute T^{LR} and T^{W} and decide at 5%. (c) Show that $\log X_i \sim \text{Expo}(\theta)$, and explain why the numbers resemble the subway's. *(Solution on page 427.)*

Exercise 15.8. Five observations $0.2, -0.1, -1.9, -0.4, -1.8$ are modeled as $\text{Gauss}(\mu, \sigma^2)$. Test $H_0 : (\mu, \sigma^2) = (0, 1)$, a simple null with $d = 2$, by Wald's test (with I at the estimate, and also with I at the null value), the likelihood ratio test and the score test, using $I(\mu, \sigma^2) = \text{diag}(1/\sigma^2, 1/(2\sigma^4))$. *(Solution on page 427.)*

Exercise 15.9. Let $X_1, \dots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, 1)$ with $\Theta = [0, \infty)$, and test $H_0 : \mu = 0$. (a) Show that the maximum likelihood estimator is $\max(\bar{X}_n, 0)$ and that $T_n^{\text{LR}} = n\bar{X}_n^2 \mathbb{1}(\bar{X}_n > 0)$. (b) Show that under H_0, $\mathbb{P}(T_n^{\text{LR}} > c) = \frac{1}{2}\mathbb{P}(\chi^2_1 > c)$ for $c > 0$, so that the threshold 3.84 gives level 2.5%. (c) Which threshold gives level 5%? *(Solution on page 427.)*

Exercise 15.10. Let $\theta = h(\eta)$ with h one-to-one and differentiable with $h' \neq 0$, in a one-parameter model. (a) Show that the likelihood ratio statistic is the same whether computed in θ or in η. (b) Show that the score statistic is the same in both, using $\frac{\mathrm{d}}{\mathrm{d}\eta}\ell = \ell'(\theta)h'(\eta)$ and $I_\eta(\eta) = I_\theta(h(\eta))h'(\eta)^2$, the rule in the pitfall of Section 12.3.1, which Exercise 12.2 checks for the exponential. (c) Check (b) on the subway: compute the score statistic in the mean wait μ for $\mu_0 = 5$. *(Solution on page 427.)*

Part VI

Nonparametric Hypothesis Tests

Every method so far assumed the model was well-specified. This part tests that assumption: goodness of fit for discrete distributions by the chi-squared test (Chapter 16), and for continuous ones by comparing the empirical CDF with the candidate, the Kolmogorov–Smirnov and Lilliefors tests and the QQ plot (Chapter 17).

Chapter 16

Goodness of Fit for Discrete Distributions

In this chapter

- Testing a whole distribution instead of one parameter: the null names every probability, and the alternative is every other distribution. When the data fall into K categories, that alternative is a model with $K-1$ free probabilities, so the tests of Chapter 15 apply to it.
- Pearson's chi-squared statistic, observed minus expected, squared, over expected, and why its limit has $K-1$ degrees of freedom: the counts add up to n, so one direction of the standardized deviations is dead, and in the limit the other $K-1$ are independent standard Gaussians.
- Pearson's statistic is the score statistic of the model with one category dropped, G^2 is the likelihood ratio statistic, and Neyman's statistic is Wald's statistic in the coordinates $p_1, \ldots, p_{K-1}$. For counts, Pearson's keeps its level best.
- Testing a family, such as the Poisson, whose parameters are estimated from the counts: one degree of freedom is lost for each parameter. Deaths by horse kick in the Prussian army, and the siblings data of Chapter 2.
- Tables of counts: tests of independence and of homogeneity, with $(r-1)(c-1)$ degrees of freedom, and the aspirin trial of Chapter 15.
- What the test cannot do: it needs large expected counts, failing to reject is not confirming, and a fit can be too good (Mendel's peas).

Chapter 15 built three tests for any regular parametric model. This chapter applies them to the model that assumes nothing about a discrete distribution except the list of values it can take.

16.1 Testing a whole distribution

Every test so far has been about a parameter of a model taken on trust: whether $p = \frac{1}{2}$ in the Bernoulli model of the kiss data, whether the mean wind speed is 10 in a Gaussian model, whether the subway's rate is 0.2 in an exponential one. The model itself was never in question. This chapter questions it, for data that fall into finitely many categories.

The classic case is Gregor Mendel's. His laws of inheritance predict that peas from plants hybrid for two traits, seed shape (round dominant over wrinkled) and seed color (yellow dominant over green),

come in four kinds in the ratio $9:3:3:1$: round and yellow with probability $\frac{9}{16}$, round and green $\frac{3}{16}$, wrinkled and yellow $\frac{3}{16}$, and wrinkled and green $\frac{1}{16}$. In one experiment (Mendel, *Versuche über Pflanzen-Hybriden*, 1866) he counted 556 peas:

	round yellow	round green	wrinkled yellow	wrinkled green	total
observed	315	108	101	32	556
expected under $9:3:3:1$	312.75	104.25	104.25	34.75	556
observed minus expected	2.25	3.75	−3.25	−2.75	0

The expected counts are $556 \times \frac{9}{16}$ and so on. The question is whether deviations of a few peas, in counts of 32 to 315, are the size chance produces when the theory is right. Notice the last column: the deviations add up to 0, since both rows add up to 556. That fact will cost a degree of freedom.

This is a test of a whole distribution. The null hypothesis names all four probabilities, and the alternative is every other distribution on the four kinds of pea, not the other values of one parameter. A test of this kind is a **goodness-of-fit test**: it asks whether the data fit a stated distribution. For a finite set of outcomes the alternative 'any other distribution' is not as large as it sounds. A distribution on K outcomes is a list of K probabilities that add up to 1, so it has $K-1$ free numbers, and the model of all such distributions is a parametric model with $K-1$ parameters. Chapter 17 meets the continuous case, where 'any other distribution' cannot be described by finitely many numbers.

Definition 16.1 (The categorical model)**.** Let $E = \{a_1, \ldots, a_K\}$ be a set of K categories and

$$\Delta_K = \left\{\mathbf{p} = (p_1, \ldots, p_K) : p_j \geq 0 \text{ for every } j,\ p_1 + \cdots + p_K = 1\right\}$$

the **probability simplex**. For $\mathbf{p} \in \Delta_K$, $\mathbb{P}_\mathbf{p}$ is the distribution on E with $\mathbb{P}_\mathbf{p}(X = a_j) = p_j$, and the **categorical model** is $(E, (\mathbb{P}_\mathbf{p})_{\mathbf{p}\in\Delta_K})$. For $X_1, \ldots, X_n \overset{iid}{\sim} \mathbb{P}_\mathbf{p}$ the **counts** are $N_j = \#\{i : X_i = a_j\}$, with $N_1 + \cdots + N_K = n$, and the likelihood is

$$L_n(\mathbf{p}) = \prod_{i=1}^{n} \mathbb{P}_\mathbf{p}(X = X_i) = p_1^{N_1} p_2^{N_2} \cdots p_K^{N_K}.$$

The vector of counts has the **multinomial distribution** $\text{Mult}(n, \mathbf{p})$:

$$\mathbb{P}(N_1 = n_1, \ldots, N_K = n_K) = \frac{n!}{n_1! \cdots n_K!}\, p_1^{n_1} \cdots p_K^{n_K} \qquad \text{for } n_1 + \cdots + n_K = n,$$

where the multinomial coefficient counts the orders in which n_1 observations can land in the first category, n_2 in the second, and so on.

> **In words.** Each observation lands in one of K boxes, the model allows any probabilities for the boxes, and the data reduce to how many observations landed in each. The order in which they arrived carries no information about $\mathbf{p}$, since the likelihood depends on the data only through the counts.
>
> **For the kiss data.** $K = 2$, with the boxes 'right' and 'left', $\mathbf{p} = (p, 1-p)$ and counts $(N_1, N_2) = (80, 44)$. $\text{Mult}(124, (p, 1-p))$ is the Binomial distribution of the number of right turns, and the categorical model with two categories is the Bernoulli model.

The goodness-of-fit question is $H_0 : \mathbf{p} = \mathbf{p}^0$ against $H_1 : \mathbf{p} \neq \mathbf{p}^0$, for a stated $\mathbf{p}^0 \in \Delta_K$ whose entries are all positive; for Mendel, $\mathbf{p}^0 = (\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16})$. The natural estimate of $\mathbf{p}$ is the vector of proportions, and it is the maximum likelihood estimator. The proof needs no Lagrange multiplier for the constraint $p_1 + \cdots + p_K = 1$: the KL divergence of Chapter 10 takes care of it.

Proposition 16.2 (The maximum likelihood estimator is the vector of proportions)**.** *In the categorical model the maximum likelihood estimator is* $\hat{\mathbf{p}} = (N_1/n, \ldots, N_K/n)$*, and for every* $\mathbf{p} \in \Delta_K$*,*

$$\ell_n(\hat{\mathbf{p}}) - \ell_n(\mathbf{p}) = n\,\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}}).$$

Proof strategy (core). The log-likelihood is n times the average of $\log p_j$ over the empirical distribution of the data. Its drop from $\hat{\mathbf{p}}$ to any other $\mathbf{p}$ is n times a KL divergence, which is never negative and is zero only at $\hat{\mathbf{p}}$.

Proof. Write $\hat{p}_j = N_j/n$. If $p_j = 0$ for some j with $N_j > 0$, then $\ell_n(\mathbf{p}) = -\infty$ and the KL divergence is $+\infty$ (Definition 10.5), so the identity holds and $\mathbf{p}$ is not a maximizer. Otherwise, with the sums over the j for which $N_j > 0$,

$$\ell_n(\mathbf{p}) = \sum_j N_j \log p_j = n \sum_j \hat{p}_j \log p_j, \qquad \text{so} \qquad \ell_n(\hat{\mathbf{p}}) - \ell_n(\mathbf{p}) = n \sum_j \hat{p}_j \log \frac{\hat{p}_j}{p_j} = n\,\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}}).$$

By Proposition 10.6 the right side is at least 0, and it is 0 only if $\mathbb{P}_{\mathbf{p}} = \mathbb{P}_{\hat{\mathbf{p}}}$, that is, $\mathbf{p} = \hat{\mathbf{p}}$. So $\hat{\mathbf{p}}$ is the unique maximizer. □

The counts are not independent of one another: they share the n observations, so a surplus in one category is a deficit somewhere else.

Proposition 16.3 (Mean and covariance of the counts)**.** *If* $(N_1, \ldots, N_K) \sim \mathrm{Mult}(n, \mathbf{p})$*, then* $\mathbb{E}[N_j] = np_j$*,* $\mathbb{V}ar(N_j) = np_j(1 - p_j)$*, and* $\mathbb{C}ov(N_j, N_k) = -np_jp_k$ *for* $j \neq k$*. The covariance matrix* $n\big(\mathrm{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top\big)$ *is not invertible: the vector* $(1, \ldots, 1)$ *is in its null space.*

Proof strategy (for completeness; skip on a first reading). Each count is a sum of n independent indicators, one per observation. Two different indicators of one observation are never both 1, which makes their covariance $-p_jp_k$.

Proof. $N_j = \sum_{i=1}^n \mathbb{1}(X_i = a_j)$. For one observation, $\mathbb{E}[\mathbb{1}(X = a_j)] = p_j$ and $\mathbb{V}\mathrm{ar}(\mathbb{1}(X = a_j)) = p_j(1 - p_j)$, and for $j \neq k$ the product $\mathbb{1}(X = a_j)\mathbb{1}(X = a_k)$ is 0, so the covariance of the two indicators is $0 - p_jp_k$. Indicators of different observations are independent, so means, variances and covariances add over the n observations. Finally, the jth entry of the covariance matrix times $(1, \ldots, 1)$ is $\sum_k \mathbb{C}\mathrm{ov}(N_j, N_k) = \mathbb{C}\mathrm{ov}(N_j, N_1 + \cdots + N_K) = \mathbb{C}\mathrm{ov}(N_j, n) = 0$. □

For Mendel's null, $\mathbb{C}\mathrm{ov}(N_1, N_2) = -556 \times \frac{9}{16} \times \frac{3}{16} = -58.6$, a correlation of -0.54 between the round-yellow and the round-green counts. A covariance matrix that is not invertible is not positive definite, so the Gaussian vectors of the Toolkit (toolkit A.14) do not cover the limit of the full vector of counts. The next section works around this by rotating the deviations, and Section 16.3 by dropping a category.

16.2 Pearson's chi-squared test

Each count has its own natural scale. Under the null, N_j has mean np_j^0, and its deviation $N_j - np_j^0$ should be measured against something of the size of $\sqrt{np_j^0}$: a deviation of 3 is large for a category that expects 4 and small for one that expects 400. Squaring the scaled deviations and adding them up gives the statistic Karl Pearson proposed in 1900.

Definition 16.4 (Pearson's chi-squared test)**.** For $H_0 : \mathbf{p} = \mathbf{p}^0$ against $H_1 : \mathbf{p} \neq \mathbf{p}^0$, with every $p_j^0 > 0$, **Pearson's chi-squared statistic** is

$$T_n^{\mathrm{S}} = \sum_{j=1}^{K} \frac{(N_j - np_j^0)^2}{np_j^0} = n \sum_{j=1}^{K} \frac{(\hat{p}_j - p_j^0)^2}{p_j^0},$$

and **Pearson's chi-squared test** of asymptotic level α rejects when $T_n^{\mathrm{S}} > q_\alpha(\chi^2_{K-1})$. Its asymptotic p-value is $\mathbb{P}(\chi^2_{K-1} > T^{\mathrm{S}})$, with T^{S} the observed value. The superscript is the one of Chapter 15: Section 16.3 shows that T_n^{S} is a score statistic.

> **In words.** For each category, take the observed count minus the count the null expects, square it, divide by the expected count, and add over the categories. Compare the total with the chi-squared distribution with $K - 1$ degrees of freedom, one fewer than the number of categories.
>
> **For the kiss data.** Under $\mathbf{p}^0 = (\frac{1}{2}, \frac{1}{2})$ the expected counts are 62 and 62, and $T^{\mathrm{S}} = (80 - 62)^2/62 + (44 - 62)^2/62 = 10.45 > 3.84 = q_{0.05}(\chi^2_1)$: reject, with p-value 0.0012. This is the square of the statistic 3.23 of Chapter 9, and Pearson's statistic of Chapter 10.

For Mendel's peas the four terms are $2.25^2/312.75 = 0.016$, $3.75^2/104.25 = 0.135$, $3.25^2/104.25 = 0.101$ and $2.75^2/34.75 = 0.218$, so $T^{\mathrm{S}} = 0.47$. The smallest category contributes the most, although its deviation is not the largest, since its deviation is measured against an expected count of only 35. With $K - 1 = 3$ degrees of freedom the 5% threshold is 7.81, and the p-value is $\mathbb{P}(\chi^2_3 > 0.47) = 0.93$. Figure 16.1 draws both steps: the observed counts against the expected ones, and the observed statistic against its null distribution. If Mendel's theory is exactly right, about 925 of 1,000 repetitions of his experiment would land farther from 9 : 3 : 3 : 1, in this measure, than his did.

Recall. The Central Limit Theorem for vectors (the vector-versions remark of Chapter 5): for i.i.d. random vectors Y_i with mean m and positive definite covariance matrix Σ, $\sqrt{n}(\bar{Y}_n - m) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0, \Sigma)$. The continuous mapping theorem for vectors (Theorem 5.2 and the vector-versions remark of Chapter 5): if $V_n \xrightarrow[n\to\infty]{\text{(d)}} V$ in $\mathbb{R}^d$ and $g : \mathbb{R}^d \to \mathbb{R}^k$ is continuous, then $g(V_n) \xrightarrow[n\to\infty]{\text{(d)}} g(V)$; if $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$ and g is continuous at y, then $g(Y_n) \xrightarrow[n\to\infty]{\mathbb{P}} g(y)$. The Law of Large Numbers (Theorem 3.5): the average of i.i.d. random variables with finite variance converges in probability to their mean; applied entry by entry, it holds for averages of i.i.d. random vectors.

Theorem 16.5 (Pearson's chi-squared test)**.** *Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathbb{P}_{\mathbf{p}}$ in the categorical model, and let $\mathbf{p}^0 \in \Delta_K$ have every $p_j^0 > 0$.*

1. *If $H_0 : \mathbf{p} = \mathbf{p}^0$ is true, then $T_n^{\mathrm{S}} \xrightarrow[n\to\infty]{(d)} \chi^2_{K-1}$, so Pearson's test has asymptotic level α.*
2. *If $\mathbf{p} \neq \mathbf{p}^0$, then $T_n^{\mathrm{S}}/n \xrightarrow[n\to\infty]{\mathbb{P}} \sum_j (p_j - p_j^0)^2/p_j^0 > 0$, and the probability that the test rejects tends to 1.*

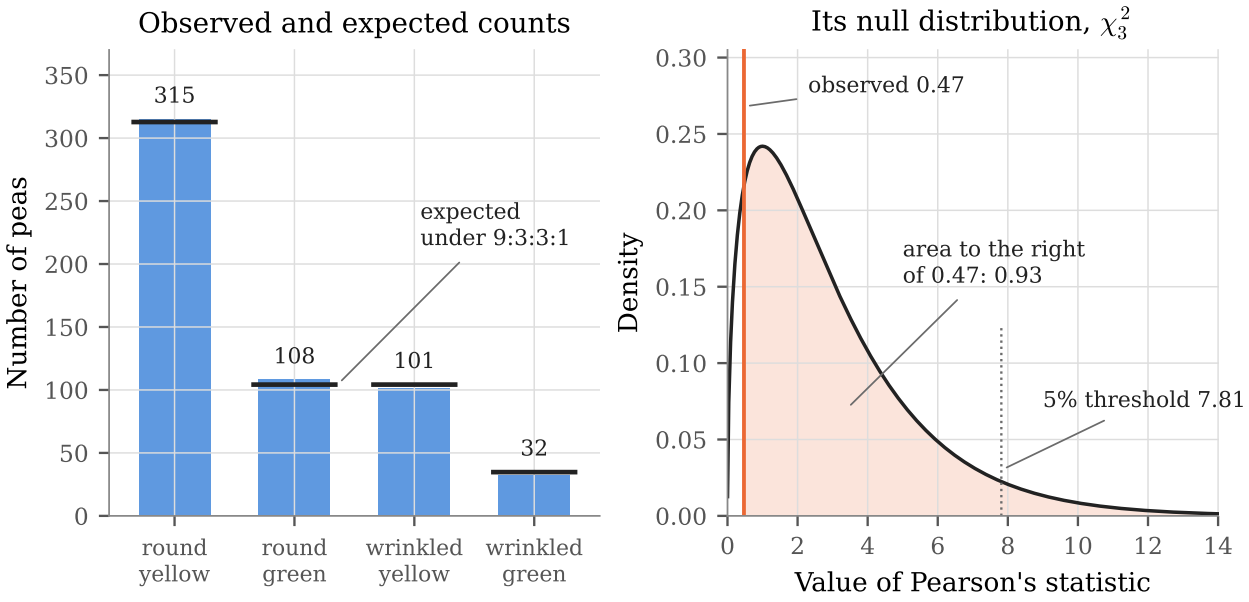


Figure 16.1: Mendel's 556 peas. Left: observed counts (bars) and counts expected under $9:3:3:1$ (black ticks). Right: the χ^2_3 density, the observed statistic 0.47 (orange), the area to its right (shaded), the 5% threshold (dotted).

Proof strategy (core). Write T_n^{S} as the squared length of the vector of standardized deviations $R_j = \sqrt{n}(\hat{p}_j - p_j^0)/\sqrt{p_j^0}$. Since the counts add up to n, that vector is perpendicular to the unit vector $u = (\sqrt{p_1^0}, \dots, \sqrt{p_K^0})$. Rotate so that u becomes the first axis, as in the proof of Cochran's theorem (Theorem 14.3): the first rotated coordinate is exactly 0, and by the CLT the other $K-1$ are approximately independent standard Gaussians, whose squares add up to a χ^2_{K-1}.

Proof. (1) Let $Y_i = (\mathbb{1}(X_i = a_1), \dots, \mathbb{1}(X_i = a_K))$, so that $\hat{\mathbf{p}} = \bar{Y}_n$, and let $D = \mathrm{diag}(\mathbf{p}^0)$. The vector $R = \sqrt{n}\, D^{-1/2}(\hat{\mathbf{p}} - \mathbf{p}^0)$ has coordinates $R_j = \sqrt{n}(\hat{p}_j - p_j^0)/\sqrt{p_j^0}$, so $T_n^{\mathrm{S}} = \|R\|^2$. Let $u = (\sqrt{p_1^0}, \dots, \sqrt{p_K^0}) = D^{-1/2}\mathbf{p}^0$, a unit vector since $\|u\|^2 = \sum_j p_j^0 = 1$. Then

$$u^\top R = \sqrt{n} \sum_{j=1}^{K} (\hat{p}_j - p_j^0) = 0$$

because both $\hat{\mathbf{p}}$ and $\mathbf{p}^0$ add up to 1. Let Q be a $K \times K$ orthogonal matrix whose first row is $u^\top$ (toolkit A.13), and Q_2 the $(K-1) \times K$ matrix of its other rows. Since Q preserves length,

$$T_n^{\mathrm{S}} = \|QR\|^2 = (u^\top R)^2 + \|Q_2 R\|^2 = \|Q_2 R\|^2.$$

Now $Q_2 R = \sqrt{n}\,\bar{W}_n$, where the vectors $W_i = Q_2 D^{-1/2}(Y_i - \mathbf{p}^0)$ are i.i.d. with mean 0. By Proposition 16.3 with $n = 1$, Y_i has covariance matrix $D - \mathbf{p}^0(\mathbf{p}^0)^\top$, so W_i has covariance matrix

$$Q_2 D^{-1/2}\big(D - \mathbf{p}^0(\mathbf{p}^0)^\top\big) D^{-1/2} Q_2^\top = Q_2\big(\mathrm{Id}_K - uu^\top\big) Q_2^\top = Q_2 Q_2^\top - (Q_2 u)(Q_2 u)^\top = \mathrm{Id}_{K-1},$$

using $D^{-1/2}\mathbf{p}^0 = u$, and then $Q_2 Q_2^\top = \mathrm{Id}_{K-1}$ and $Q_2 u = 0$, since the rows of Q are orthonormal. The identity matrix is positive definite, so the multivariate CLT gives $Q_2 R \xrightarrow[n\to\infty]{\text{(d)}} Z$, a vector of

$K-1$ independent standard Gaussians, and the continuous mapping theorem applied to $v \mapsto \|v\|^2$ gives $T_n^{\mathrm{S}} \xrightarrow[n\to\infty]{\text{(d)}} \|Z\|^2$, which is χ^2_{K-1} by Definition 14.1. Since the χ^2_{K-1} CDF is continuous, $\mathbb{P}(T_n^{\mathrm{S}} > q_\alpha(\chi^2_{K-1})) \to \alpha$.

(2) By the Law of Large Numbers, $\hat{p}_j \xrightarrow[n\to\infty]{\mathbb{P}} p_j$ for each j, and by the continuous mapping theorem $T_n^{\mathrm{S}}/n = \sum_j (\hat{p}_j - p_j^0)^2/p_j^0 \xrightarrow[n\to\infty]{\mathbb{P}} c = \sum_j (p_j - p_j^0)^2/p_j^0$, which is positive since $\mathbf{p} \neq \mathbf{p}^0$. Then $\mathbb{P}(T_n^{\mathrm{S}} > q) \geq \mathbb{P}(T_n^{\mathrm{S}}/n > c/2) \to 1$ as soon as $nc/2 > q$. □

Intuition: three counts, two directions of freedom

Figure 16.2 draws the proof for $K = 3$, with $\mathbf{p}^0 = (0.5, 0.3, 0.2)$ and 2,000 samples of $n = 400$. On the left are the first two estimated proportions of each sample; the third is whatever they leave. The cloud is tilted and stretched, since the two proportions are negatively correlated (-0.66 here): a sample with more of the first category has less room for the others. The curve is where $T_n^{\mathrm{S}} = 5.99$, the 95% quantile of χ^2_2. On the right are the standardized deviations R of the same samples, drawn in the plane perpendicular to u, where they lie exactly. There the cloud is round, the curve becomes the circle of radius $\sqrt{5.99} = 2.45$, and 100 of the 2,000 samples fall outside it, the 5% the theorem predicts. It is the picture of Figure 15.2, one dimension down from the three counts: three counts give chance only two directions in which to push.

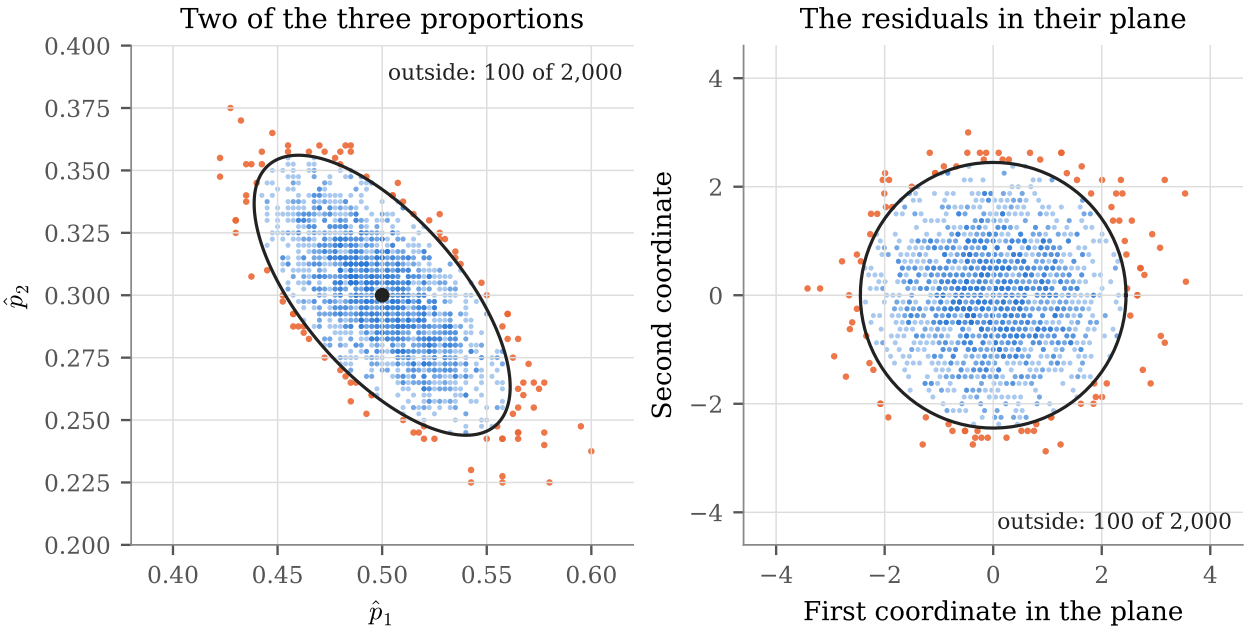


Figure 16.2: Three categories, $\mathbf{p}^0 = (0.5, 0.3, 0.2)$, 2,000 samples of $n = 400$. Left: the first two proportions and the curve $T_n^{\mathrm{S}} = 5.99$. Right: the standardized deviations in their plane, and the circle of radius 2.45. Orange: samples outside.

Does this make sense? Why divide by the expected count and not by the binomial variance?

Under the null each N_j alone is $\mathrm{Bin}(n, p_j^0)$, with variance $np_j^0(1-p_j^0)$, so a standardized count would divide by that. Pearson's statistic divides by np_j^0 instead, which makes each term $1-p_j^0$ times a squared standardized binomial count. The factors $1-p_j^0$ add up to $K-1$, which is exactly the mean of χ^2_{K-1}: under the null, $\mathbb{E}[T_n^{\mathrm{S}}] = \sum_j (1-p_j^0) = K-1$ at every n (Exercise 16.1). For $K=2$ the bookkeeping is exact: since $N_2 - np_2^0 = -(N_1 - np_1^0)$ and $p_1^0 + p_2^0 = 1$,

$$\frac{(N_1 - np_1^0)^2}{np_1^0} + \frac{(N_2 - np_2^0)^2}{np_2^0} = (N_1 - np_1^0)^2\Big(\frac{1}{np_1^0} + \frac{1}{np_2^0}\Big) = \frac{(N_1 - np_1^0)^2}{np_1^0(1-p_1^0)},$$

the single squared standardized binomial count, which is χ^2_1 in the limit.

Pitfall: the counts are not independent

Adding up the K squared deviations $(N_j - np_j^0)^2$, each divided by its own binomial variance $np_j^0(1-p_j^0)$, and comparing the sum with χ^2_K treats the counts as independent. They are not (Proposition 16.3). For $K=2$ the two terms are equal, since $N_2 - np_2^0 = -(N_1 - np_1^0)$ and $p_2^0(1-p_2^0) = p_1^0(1-p_1^0)$, so in the limit the sum is twice a single χ^2_1, and it exceeds the χ^2_2 threshold 5.99 with probability $\mathbb{P}(\chi^2_1 > 3.00) = 8.3\%$, not 5% (Exercise 16.11). Pearson's denominators and the $K-1$ degrees of freedom come as a pair, and changing only one of them is just as wrong: for $K=2$, Pearson's statistic against the χ^2_2 threshold rejects a true null 1.4% of the time, and the sum above against the χ^2_1 threshold 16.6%.

16.3 Pearson's test is a score test

The categorical model is a parametric model, so the three tests of Chapter 15 apply to it after one repair. The coordinates $p_1, \dots, p_K$ are tied together by $p_1 + \cdots + p_K = 1$: none of them can move while the others stay fixed, so the parameter set has no interior in $\mathbb{R}^K$, and the covariance matrix of the counts is singular (Proposition 16.3). The repair, announced at the end of Section 16.1, is to drop one category, which leaves a positive definite information (toolkit A.14). Parameterize by the first $K-1$ probabilities, $\mathbf{p}_{-K} = (p_1, \dots, p_{K-1})$, which is $\mathbf{p}$ without its last entry and ranges over the open set of vectors with positive entries that add up to less than 1; then $p_K = 1 - p_1 - \cdots - p_{K-1}$. This model is regular in the sense of Section 12.6: different parameters give different distributions, every distribution in it gives each category positive probability, the parameter set is open, the log-likelihood is smooth with bounded derivatives near any point of it, and its information is invertible (part 1 below).

Proposition 16.6 (Three statistics for counts)**.** *In the categorical model parameterized by* $\mathbf{p}_{-K} = (p_1, \dots, p_{K-1})$*:*

1. *the Fisher information is* $I(\mathbf{p}_{-K}) = \mathrm{diag}(1/p_1, \dots, 1/p_{K-1}) + \frac{1}{p_K}\mathbf{1}\mathbf{1}^\top$*, with* $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^{K-1}$*, and its inverse is* $I(\mathbf{p}_{-K})^{-1} = \mathrm{diag}(p_1, \dots, p_{K-1}) - \mathbf{p}_{-K}\mathbf{p}_{-K}^\top$*;*
2. *for* $H_0 : \mathbf{p} = \mathbf{p}^0$*, the score statistic of Definition 15.8 is Pearson's statistic* T_n^{S}*; the likelihood*

ratio statistic of Definition 15.6 is

$$T_n^{\mathrm{LR}} = 2\sum_{j=1}^{K} N_j \log \frac{N_j}{np_j^0} = 2n\,\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}^0}),$$

the G^2 statistic, with the terms where $N_j = 0$ counted as 0; and, when every $N_j > 0$, Wald's statistic of Definition 15.2 is ***Neyman's statistic*** $T_n^{\mathrm{W}} = \sum_j (N_j - np_j^0)^2/N_j$*;*

3. *none of the three depends on which category was dropped.*

Proof strategy (for completeness; skip on a first reading). Differentiate the log-likelihood in the $K-1$ free coordinates. The information is a diagonal matrix plus a constant one, and its inverse is the covariance matrix of the first $K-1$ indicators, which multiplying out confirms. The score at $\mathbf{p}^0$ turns out to be exactly n times the information times the estimation error, so the score statistic is a quadratic form in that error, which adds up to Pearson's statistic; Wald's statistic is the same form with the information at the estimate.

Proof. (1) For one observation, $\log p(X) = \sum_{j<K} \mathbb{1}(X = a_j)\log p_j + \mathbb{1}(X = a_K)\log(1 - p_1 - \cdots - p_{K-1})$, so for $j, k < K$

$$\frac{\partial}{\partial p_j}\log p(X) = \frac{\mathbb{1}(X = a_j)}{p_j} - \frac{\mathbb{1}(X = a_K)}{p_K},$$
$$\frac{\partial^2}{\partial p_j\,\partial p_k}\log p(X) = -\frac{\mathbb{1}(X = a_j)}{p_j^2}\mathbb{1}(j = k) - \frac{\mathbb{1}(X = a_K)}{p_K^2}.$$

Taking expectations and changing the sign (Theorem 12.4) gives $I_{jk} = \mathbb{1}(j = k)/p_j + 1/p_K$. Write $\mathbf{q} = \mathbf{p}_{-K}$ for short and $\mathrm{diag}(1/\mathbf{q})$ for $\mathrm{diag}(1/p_1, \ldots, 1/p_{K-1})$, so that $\mathrm{diag}(1/\mathbf{q})\,\mathbf{q} = \mathbf{1}$, and note $\mathbf{1}^\top\mathbf{q} = 1 - p_K$. Multiplying out,

$$\Big(\mathrm{diag}(1/\mathbf{q}) + \tfrac{1}{p_K}\mathbf{1}\mathbf{1}^\top\Big)\big(\mathrm{diag}(\mathbf{q}) - \mathbf{q}\mathbf{q}^\top\big) = \mathrm{Id} - \mathbf{1}\mathbf{q}^\top + \tfrac{1}{p_K}\mathbf{1}\mathbf{q}^\top - \tfrac{1-p_K}{p_K}\mathbf{1}\mathbf{q}^\top = \mathrm{Id}.$$

(2) Let $\mathbf{h} = \hat{\mathbf{p}}_{-K} - \mathbf{p}^0_{-K}$ be the estimation error; since the proportions and the null probabilities both add up to 1, $\sum_{k<K} h_k = -(\hat{p}_K - p_K^0)$. Summing the score of one observation over the sample, $\partial\ell_n/\partial p_j$ at $\mathbf{p}^0$ is $N_j/p_j^0 - N_K/p_K^0 = n(\hat{p}_j/p_j^0 - \hat{p}_K/p_K^0)$ for $j < K$. By part (1), the jth entry of $I(\mathbf{p}^0_{-K})\mathbf{h}$ is the same, divided by n:

$$\frac{\hat{p}_j - p_j^0}{p_j^0} + \frac{1}{p_K^0}\sum_{k<K} h_k = \frac{\hat{p}_j}{p_j^0} - 1 - \frac{\hat{p}_K - p_K^0}{p_K^0} = \frac{\hat{p}_j}{p_j^0} - \frac{\hat{p}_K}{p_K^0}.$$

So the score is exactly linear in the estimation error, $U_n = \frac{1}{\sqrt{n}}\nabla\ell_n(\mathbf{p}^0_{-K}) = \sqrt{n}\,I(\mathbf{p}^0_{-K})\mathbf{h}$, and the score statistic is

$$U_n^\top I(\mathbf{p}^0_{-K})^{-1}U_n = n\,\mathbf{h}^\top I(\mathbf{p}^0_{-K})\mathbf{h} = n\sum_{j<K}\frac{h_j^2}{p_j^0} + \frac{n}{p_K^0}\Big(\sum_{j<K} h_j\Big)^2$$
$$= n\sum_{j=1}^{K}\frac{(\hat{p}_j - p_j^0)^2}{p_j^0} = T_n^{\mathrm{S}}.$$

Wald's statistic $n\mathbf{h}^\top I(\hat{\mathbf{p}}_{-K})\mathbf{h}$ is the same quadratic form with the information at the estimate, which puts $\hat{p}_j$ in place of p_j^0 in the denominators: it is $n\sum_j(\hat{p}_j - p_j^0)^2/\hat{p}_j = T_n^{\mathrm{W}}$. The likelihood ratio statistic is $2(\ell_n(\hat{\mathbf{p}}) - \ell_n(\mathbf{p}^0)) = 2n\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}^0})$ by Proposition 16.2, and $2n\sum_j \hat{p}_j\log(\hat{p}_j/p_j^0) = 2\sum_j N_j\log(N_j/(np_j^0))$.

(3) The three final formulas treat the K categories alike. □

Corollary 16.7 (Three tests for counts)**.** *Under $H_0 : \mathbf{p} = \mathbf{p}^0$, with every $p_j^0 > 0$, Pearson's statistic, G^2 and Neyman's statistic all converge in distribution to χ^2_{K-1}, and their differences converge in probability to* 0*. (Neyman's statistic is defined on the event that every $N_j > 0$, whose probability tends to* 1*.)*

Proof strategy (for completeness; skip on a first reading). The model with one category dropped is regular, with $K-1$ parameters and an information that is invertible and continuous, so Theorem 15.10 applies.

Proof. By Proposition 16.6 the three statistics are the score, likelihood ratio and Wald statistics of a regular model with $K-1$ parameters, $\mathbf{p}^0_{-K}$ interior, and I invertible and continuous at $\mathbf{p}^0_{-K}$, and Theorem 15.10 gives the common χ^2_{K-1} limit and differences that converge in probability to 0. The event that some $N_j = 0$ has probability at most $\sum_j (1-p_j^0)^n$, which tends to 0. □

The corollary proves part (1) of Theorem 16.5 again, from Theorem 15.9: the rotation of Section 16.2 and the dropped category are two coordinate systems for the same $(K-1)$-dimensional plane of deviations. The proof of the proposition also shows that the score is exactly n times the information times the estimation error, so Pearson's statistic is Wald's statistic with the information at the null value instead of at the estimate; texts that define Wald's test that way call Pearson's test a Wald test.

For Mendel's peas $G^2 = 0.475$ and Neyman's statistic is 0.487, against Pearson's 0.470, with p-values from 0.92 to 0.93: with 556 observations the three agree, as the corollary says they should. In smaller samples they do not, and since the null distribution of the counts is known exactly, how far each strays from its nominal level can be computed exactly, by adding up the multinomial probabilities of all the tables of counts that the statistic rejects. Figure 16.3 does this for Mendel's null at every n from 8 to 150. Pearson's test stays between 3.6% and 6.1% throughout, even at $n = 8$, where the wrinkled green category expects half a pea. The likelihood ratio test runs above 5% for almost every n, reaching 7.9% at $n = 14$ and still at 5.2% at $n = 150$. Neyman's test, which divides by the observed counts and so is infinite whenever a category is empty, rejects a true null more than 16% of the time below $n = 38$, off the top of the figure (77% at $n = 8$), 15% of the time at $n = 40$ and still 8% at $n = 150$.

Which test to use. For counts, use Pearson's statistic. This reverses the advice of Chapter 15, where the likelihood ratio test kept its level best, and it is not a quirk of Mendel's null: exact computations for many nulls and sample sizes find the same (Larntz, *Journal of the American Statistical Association*, 1978). Part of the reason is in the sanity check above: Pearson's statistic has exactly the mean of its limit, $K-1$, at every n. G^2 remains the statistic to report when a likelihood ratio is wanted, for instance to compare nested models, and with large expected counts the two agree.

From undergraduate statistics: observed minus expected, squared, over expected

The chi-squared goodness-of-fit test of an introductory course is this chapter's test: $\sum (O-E)^2/E$ over the categories, with O the observed and E the expected counts, compared with the chi-squared distribution with 'number of categories minus one' degrees of freedom, provided every expected count is at least 5. What this chapter adds: the minus one is a lost direction, since the counts add up to n (Theorem 16.5); the statistic is the score statistic of a model with $K-1$ parameters, so the G-test is its likelihood ratio twin (Proposition 16.6); and the rule of 5 is a cautious guideline, which Pearson's statistic often beats (Figure 16.3).

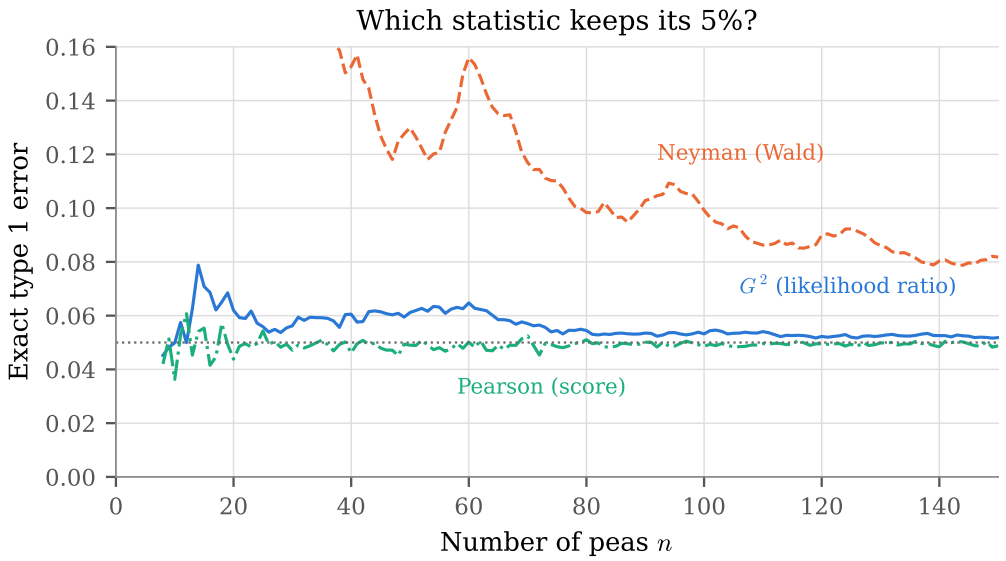


Figure 16.3: Mendel's null 9 : 3 : 3 : 1: exact type 1 error at the χ^2_3 threshold 7.81 against the sample size, for Pearson's (score; aqua, dash-dot), G^2 (likelihood ratio; blue) and Neyman's (Wald; orange, dashed) statistics. Dotted: 5%.

16.4 Testing a family of distributions

A model is more often a family than a single distribution. Ladislaus von Bortkiewicz counted the soldiers killed by the kick of a horse in 14 corps of the Prussian army over the 20 years from 1875 to 1894 (*Das Gesetz der kleinen Zahlen*, 1898). Leaving out four corps whose composition differed from the rest, as he did, gives 200 corps-years: 109 had no such death, 65 had one, 22 had two, 3 had three and 1 had four. A death by horse kick is a rare event with many independent chances to happen, the situation in which counts are close to Poisson, so the natural model is that the deaths in a corps-year are Pois(λ), with λ unknown. The null hypothesis is now a family: H_0: $\mathbf{p} = \mathbf{p}(\lambda)$ for some $\lambda > 0$, where $p_j(\lambda)$ is the Poisson probability of category j. Nobody states λ in advance, so it has to be estimated, and the fitted distribution is then tested against the counts it was fitted to.

Definition 16.8 (The chi-squared test for a family)**.** Let $\{\mathbf{p}(\theta) : \theta \in \Theta\}$, with $\Theta \subset \mathbb{R}^d$ open, be a family of distributions on $E = \{a_1, \dots, a_K\}$ with every $p_j(\theta) > 0$ and $d < K - 1$. To test H_0: $\mathbf{p} = \mathbf{p}(\theta)$ for some $\theta \in \Theta$, against H_1: $\mathbf{p}$ is not in the family, let $\hat{\theta}$ maximize $\sum_j N_j \log p_j(\theta)$, the maximum likelihood estimator of θ from the counts. **Pearson's statistic for the family** is

$$T_n^{\mathrm{S}} = \sum_{j=1}^{K} \frac{\big(N_j - n p_j(\hat{\theta})\big)^2}{n p_j(\hat{\theta})},$$

and the test of asymptotic level α rejects when $T_n^{\mathrm{S}} > q_\alpha(\chi^2_{K-1-d})$. The likelihood ratio version is $G^2 = 2\sum_j N_j \log\big(N_j/(n p_j(\hat{\theta}))\big)$, with the terms where $N_j = 0$ counted as 0, compared with the same threshold.

> **In words.** Fit the family to the counts by maximum likelihood, compute the expected counts from the fitted distribution, and compute Pearson's statistic as before; then lose one degree of freedom for each parameter fitted.

For the kiss data. With $K = 2$ categories and the Bernoulli family, $d = 1$ leaves $K - 1 - d = 0$ degrees of freedom: the fitted distribution reproduces the two counts exactly, and there is nothing left to test. A goodness-of-fit test of a family needs more free probabilities than the family has parameters, $K - 1 > d$, so at least $d + 2$ categories.

Recall. The Law of Large Numbers and the continuous mapping theorem, as before Theorem 16.5: the proportions are averages of i.i.d. indicators, so $\hat{\mathbf{p}} \xrightarrow[n\to\infty]{\mathbb{P}} \mathbf{p}$, and $g(\hat{\mathbf{p}}) \xrightarrow[n\to\infty]{\mathbb{P}} g(\mathbf{p})$ for any g continuous at $\mathbf{p}$.

Theorem 16.9 (The chi-squared test for a family)**.** *In Definition 16.8, suppose that $\theta \mapsto \mathbf{p}(\theta)$ is three times continuously differentiable; that its $d \times K$ matrix of partial derivatives has rank d at θ^*, meaning that its d rows are linearly independent, so that no change of θ leaves $\mathbf{p}$ unchanged to first order; and that $\mathbf{p}(\theta)$ is close to $\mathbf{p}(\theta^*)$ only when θ is close to θ^*: for every $\varepsilon > 0$ there is a $\delta > 0$ with $\|\mathbf{p}(\theta) - \mathbf{p}(\theta^*)\| \geq \delta$ whenever $\|\theta - \theta^*\| \geq \varepsilon$. If H_0 holds with $\mathbf{p} = \mathbf{p}(\theta^*)$, then with probability tending to 1 the estimator $\hat{\theta}$ exists, $\hat{\theta} \xrightarrow[n\to\infty]{\mathbb{P}} \theta^*$, and Pearson's statistic for the family and G^2 both converge in distribution to χ^2_{K-1-d}, with a difference that converges in probability to 0.*

Proof strategy (for completeness; skip on a first reading). First, the fit settles near the truth: the KL divergence from the proportions to members of the family far from $\mathbf{p}(\theta^*)$ stays away from 0, while to $\mathbf{p}(\theta^*)$ it vanishes. Then, in the model of Section 16.3 with $K - 1$ free probabilities, the family near $\mathbf{p}(\theta^*)$ is a smooth surface of dimension d, so the null imposes $K - 1 - d$ constraints. The constrained maximum likelihood estimate is $\mathbf{p}(\hat{\theta})$; the score statistic there is Pearson's statistic for the family, by the identity of Proposition 16.6, and the likelihood ratio statistic is G^2. Theorem 15.12 does the rest.

Proof. The fit settles near the truth. Write $\mathbf{p}^* = \mathbf{p}(\theta^*)$. Since $\sum_j N_j \log p_j(\theta) = n \sum_j \hat{p}_j \log \hat{p}_j - n\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}(\theta)})$, the estimator $\hat{\theta}$ minimizes the KL divergence from the proportions to the family. For any two distributions $\mathbf{p}$ and $\mathbf{q}$ on the K categories, with every $q_j > 0$, Jensen's inequality (Theorem 7.8) for the convex function $-\log$, with the sums over the j for which $p_j > 0$, gives

$$\mathrm{KL}(\mathbb{P}_{\mathbf{p}}, \mathbb{P}_{\mathbf{q}}) = -2 \sum_j p_j \log \sqrt{q_j / p_j} \geq -2 \log \sum_j \sqrt{p_j q_j},$$

and $1 - \sum_j \sqrt{p_j q_j} = \frac{1}{2} \sum_j (\sqrt{p_j} - \sqrt{q_j})^2 \geq \frac{1}{8} \|\mathbf{p} - \mathbf{q}\|^2$, since $(\sqrt{p_j} - \sqrt{q_j})^2 = (p_j - q_j)^2 / (\sqrt{p_j} + \sqrt{q_j})^2$ and the denominator is at most 4. Fix $\varepsilon > 0$ small enough that the closed ball B of radius ε around θ^* lies in Θ, and take $\delta \leq 1$ from the hypothesis. For $\theta \notin B$, with $\mathbf{q} = \mathbf{p}(\theta)$,

$$\sum_j \sqrt{\hat{p}_j q_j} \leq \sum_j \sqrt{p^*_j q_j} + A_n \leq 1 - \frac{\delta^2}{8} + A_n, \qquad A_n = \sum_j \left| \sqrt{\hat{p}_j} - \sqrt{p^*_j} \right|,$$

using $q_j \leq 1$, and $A_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$ and $\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}^*}) \xrightarrow[n\to\infty]{\mathbb{P}} 0$ by the Law of Large Numbers and the continuous mapping theorem, every p^*_j being positive. So with probability tending to 1, $A_n \leq \delta^2/16$ and $\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}^*}) < \kappa = -2\log(1 - \delta^2/16)$, and then every $\theta \notin B$ has $\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}(\theta)}) \geq \kappa$: the log-likelihood is lower at θ than at θ^*. On B, which is closed and bounded, the continuous log-likelihood attains a maximum (toolkit A.2), which is then a maximum over all of Θ. So with probability tending to 1, $\hat{\theta}$ exists and lies within ε of θ^*, for every ε: $\hat{\theta} \xrightarrow[n\to\infty]{\mathbb{P}} \theta^*$.

The limit. Work in the categorical model with one category dropped, which is regular (Section 16.3). Near $\mathbf{p}^*$ the members of the family are, by the last hypothesis, the $\mathbf{p}(\theta)$ with θ near θ^*; they form the image of a smooth map whose derivative has rank d (dropping the last column of the $d \times K$ matrix

keeps the rank, since the columns add up to 0), a d-dimensional surface in the $(K-1)$-dimensional parameter set, and by the implicit function theorem (taken as given, as in Section 15.7) it is the set where $K-1-d$ independent smooth constraints hold. With probability tending to 1 both $\hat{\mathbf{p}}$ and $\mathbf{p}(\hat{\theta})$ lie in this neighborhood, and $\mathbf{p}(\hat{\theta})$ is the maximizer of the likelihood over the family, by the definition of $\hat{\theta}$. The identity in part (2) of Proposition 16.6 holds at every point $\mathbf{p}^0$ with positive entries, so at $\mathbf{p}^0 = \mathbf{p}(\hat{\theta})$ it says that the score statistic of Definition 15.11 is Pearson's statistic for the family; and the likelihood ratio statistic is $2\big(\ell_n(\hat{\mathbf{p}}) - \ell_n(\mathbf{p}(\hat{\theta}))\big) = 2n\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}(\hat{\theta})}) = G^2$ by Proposition 16.2. By Theorem 15.12 and the paragraph on implicit hypotheses in Section 15.7, both converge to χ^2_{K-1-d}, with $K-1-d$ the number of constraints, and their difference converges in probability to 0, since both statistics are unchanged by the change of parameter. □

Intuition: each fitted parameter uses up a direction

With $\mathbf{p}^0$ stated, the standardized deviations had one dead direction, u. Fitting θ moves the expected counts as close to the observed ones as the family allows, along the d directions in which the family can move, so what is left of the deviations is, up to terms that vanish, perpendicular to those directions too: $K-1-d$ free directions remain. Figure 16.4 shows the loss for Pois(1) data grouped into the three categories {0}, {1} and {2 or more}, in 20,000 samples of $n = 1{,}000$. With λ known, Pearson's statistic follows χ^2_2. With λ estimated from the three counts, it follows χ^2_1, and its average drops from 2.0 to 1.0: the fit absorbs a whole degree of freedom. Comparing the fitted statistic with χ^2_2, as if λ had been known, would reject a true Poisson model 1.4% of the time instead of 5%.

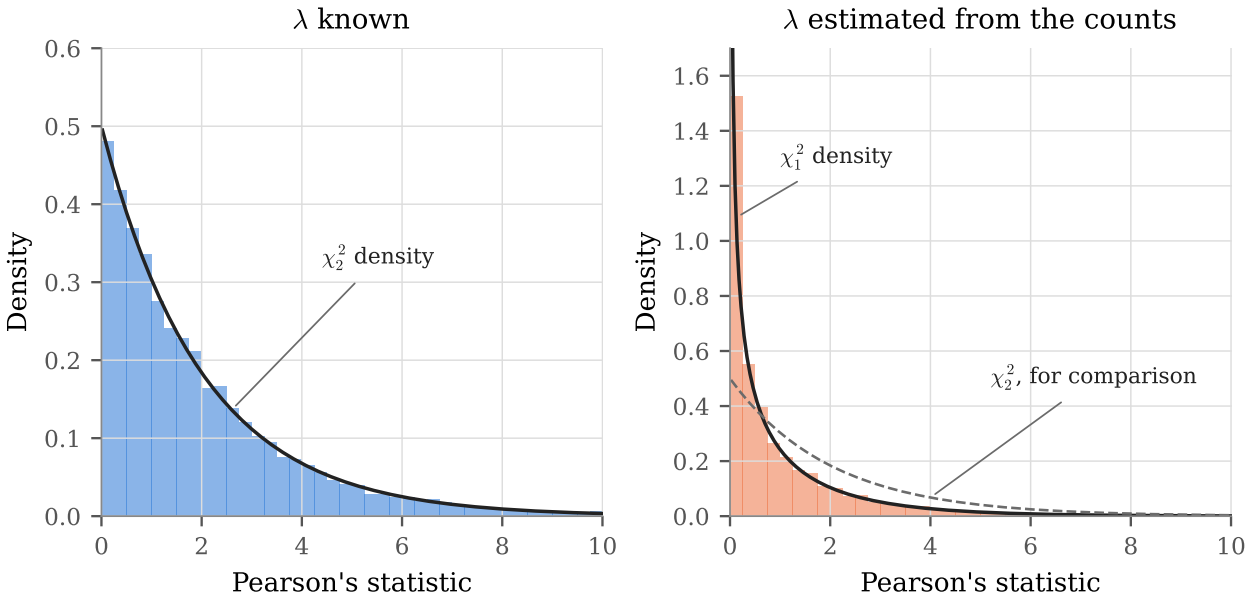


Figure 16.4: Pearson's statistic in 20,000 samples of $n = 1{,}000$ from Pois(1), cells {0}, {1}, {2 or more} (histograms). Left: λ known; χ^2_2 density (black). Right: λ estimated from the counts; χ^2_1 density (black), χ^2_2 (gray, dashed).

Pearson made that last mistake himself. His 1900 paper used $K-1$ degrees of freedom whether or not parameters had been estimated; Fisher showed in 1922, for tables of counts, and in 1924 in general, that each estimated parameter costs one, and Pearson disputed it in print (Stigler, *Statistical Science*, 2008).

Worked example: deaths by horse kick

The 200 corps-years had 122 deaths in all, so $\hat{\lambda} = 122/200 = 0.61$ deaths per corps-year. Under Pois(0.61) the expected numbers of corps-years with 0, 1, 2, 3 and '4 or more' deaths are 108.67, 66.29, 20.22, 4.11 and 0.71. Two expected counts below 5, one of them below 1, strain the chi-squared approximation (Section 16.6), so pool the last two categories into '3 or more', with 4 corps-years observed and 4.82 expected. That leaves $K = 4$ categories and $d = 1$ parameter, so $K - 1 - d = 2$ degrees of freedom, and

$$T^{\mathrm{S}} = \frac{0.33^2}{108.67} + \frac{1.29^2}{66.29} + \frac{1.78^2}{20.22} + \frac{0.82^2}{4.82} = 0.001 + 0.025 + 0.157 + 0.140 = 0.32,$$

with p-value $\mathbb{P}(\chi^2_2 > 0.32) = 0.85$; $G^2 = 0.33$. Strictly, the theorem wants λ estimated from the four counts, which gives 0.608 rather than 0.61, since the pooled category hides whether its corps-years had three deaths or four; the statistic is 0.32 either way. One expected count in four is still just below 5, a little more than the guideline of Section 16.6 allows; pooling '2 or more' as well, with λ estimated again from the three counts (0.614), gives $T^{\mathrm{S}} = 0.056$ on 1 degree of freedom, $p = 0.81$, the same verdict. Nothing here argues against the Poisson model, and the left of Figure 16.5 shows why: the observed and expected counts agree almost category by category. That was Bortkiewicz's point: rare events with many chances to happen follow the Poisson distribution, the 'law of small numbers' of his title.

Worked example: the siblings data, checked

Chapter 2 modeled the number of siblings, self included, of 24 students by $X - 1 \sim \mathrm{Pois}(\lambda)$, and left the check of that assumption to this chapter. The 24 values of $X - 1$ are 0 eight times, 1 nine times, 2 six times and 3 once, with mean $\hat{\lambda} = 1.0$. Under Pois(1.0) the expected counts for 0, 1, 2 and '3 or more' are 8.83, 8.83, 4.41 and 1.93. Two of these four are below 5, more than the guideline of Section 16.6 allows, and pooling the last two leaves $K = 3$ categories, with observed counts $(8, 9, 7)$ and expected $(8.83, 8.83, 6.34)$, and $K - 1 - d = 1$ degree of freedom. Estimated from the three counts, as the theorem asks, λ is 1.08, the expected counts become $(8.12, 8.80, 7.08)$, and $T^{\mathrm{S}} = 0.007$, with p-value 0.93. With the sample mean 1.0 instead, $T^{\mathrm{S}} = 0.15$; the pitfall below says to compare it with both χ^2_1 and χ^2_2, which give p-values 0.70 and 0.93. Either way the data give no reason to doubt the Poisson model (Figure 16.5, right). They give little reason to trust it either: if the 24 values came from a geometric distribution with the same mean, which puts half its probability on 0, the test with λ estimated from the three counts would reject only about one time in five.

Pitfall: estimate from the counts you test

Theorem 16.9 needs $\hat{\theta}$ to be the maximum likelihood estimator from the counts. In practice the parameter is often estimated from the raw data, the sample mean for a Poisson, which carries more information than counts with a pooled category. Then the limit is no longer χ^2_{K-1-d}: Chernoff and Lehmann (*Annals of Mathematical Statistics*, 1954) showed that it is χ^2_{K-1-d} plus d further independent χ^2_1 variables with weights between 0 and 1, so its tail probabilities lie between those of χ^2_{K-1-d} and χ^2_{K-1}, and the test with $K - 1 - d$ degrees of freedom rejects too often. In the setting of Figure 16.4 the excess is small, a type 1 error of 5.4% at the χ^2_1 threshold against 5.0% for the estimate from the counts; for Gaussian data grouped into four cells that are equally likely under the true distribution, with the mean and variance estimated

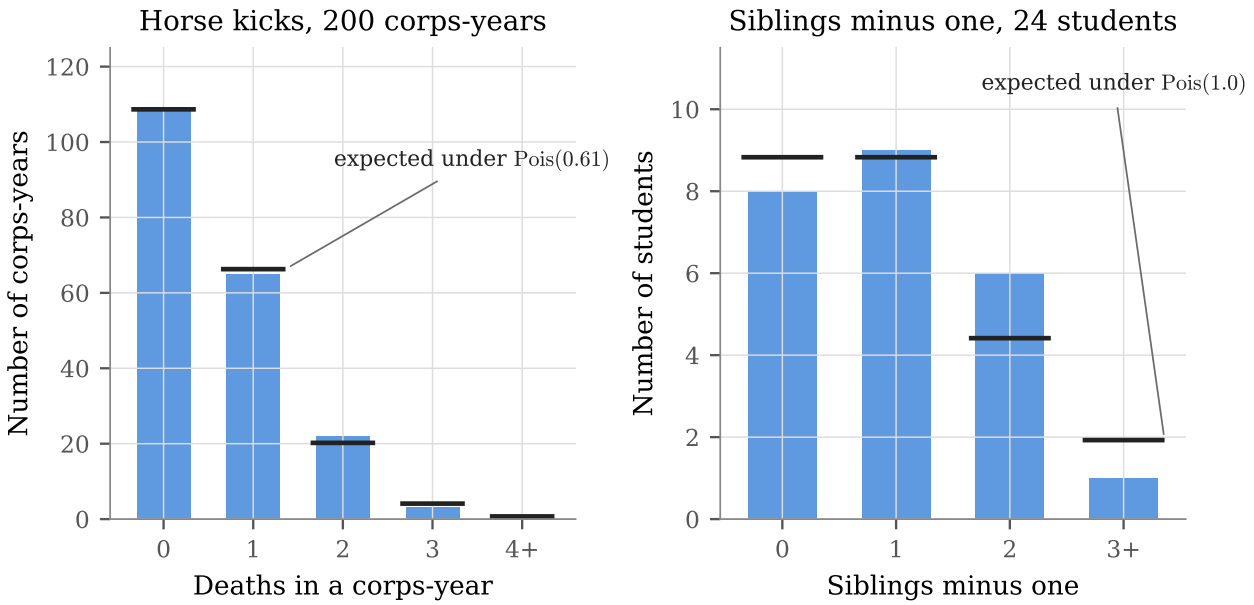


Figure 16.5: Observed counts (bars) and the counts expected under a Poisson model with λ estimated (black ticks). Left: deaths by horse kick in 200 corps-years, Pois(0.61). Right: the siblings data, 24 students, Pois(1.0).

from the raw data, it is 10.5% in the limit. The correct p-value lies between the χ^2_{K-1-d} and the χ^2_{K-1} p-values: report both, as for the siblings data, and if they fall on opposite sides of α, estimate from the counts instead. When the categories are the possible values themselves, as for the binomial family of Exercise 16.8, the raw data are the counts and the issue disappears.

16.5 Tables of counts

The aspirin trial of Chapter 15 is a table of counts, with two rows, placebo and aspirin, and two columns, heart attack or not:

	heart attack	no heart attack	total
placebo	189	10,845	11,034
aspirin	104	10,933	11,037
total	293	21,778	22,071

Chapter 15 asked whether the two heart-attack rates are equal. The same question can be asked of any table: is the row variable unrelated to the column variable? Tables arise from two designs. In the first, n units are each classified by two variables, the row variable with r categories and the column variable with c, and the rc cells form one categorical model, with p_{jk} the probability of cell (j, k); the variables are **independent** when $p_{jk} = a_j b_k$ for every j and k, with (a_j) the distribution of the row variable and (b_k) that of the column variable. In the second, the row totals are fixed by design, as in the aspirin trial, where 11,034 physicians were assigned to placebo and 11,037 to aspirin; each row is its own categorical sample, and the null of **homogeneity** is that all the rows have the same distribution over the columns. Both designs lead to the same test.

Definition 16.10 (The chi-squared test for a table)**.** Let N_{jk} be the counts of an $r \times c$ table, with row totals $N_{j\cdot}$, column totals $N_{\cdot k}$ and grand total n. The **expected counts** are $E_{jk} = N_{j\cdot}N_{\cdot k}/n$, and the **chi-squared test for the table** compares

$$T_n^{\mathrm{S}} = \sum_{j=1}^{r}\sum_{k=1}^{c}\frac{(N_{jk}-E_{jk})^2}{E_{jk}} \qquad \text{or} \qquad G^2 = 2\sum_{j=1}^{r}\sum_{k=1}^{c}N_{jk}\log\frac{N_{jk}}{E_{jk}}$$

with $q_\alpha\big(\chi^2_{(r-1)(c-1)}\big)$, the terms of G^2 where $N_{jk}=0$ counted as 0.

In words. If the row and column variables have nothing to do with each other, the expected count in a cell is the row total times the column's share of all the observations, $N_{j\cdot} \times N_{\cdot k}/n$. The degrees of freedom count the cells left free once the row and column totals are known: fill in $(r-1)(c-1)$ cells, and the totals determine the rest.

For the kiss data. The 124 couples were observed in public places in the United States, Germany and Turkey. Their counts by country would form a 3×2 table of country by direction of the head turn, and this test would ask whether couples in the three countries turn right equally often, with $(3-1)(2-1) = 2$ degrees of freedom.

Corollary 16.11 (Degrees of freedom for a table)**.** *Suppose every cell probability is positive, and either the rc cells are one categorical sample of size n and the null is independence, or the r rows are independent categorical samples whose sizes grow in fixed proportions and the null is homogeneity. Under the null, both statistics of Definition 16.10 converge in distribution to $\chi^2_{(r-1)(c-1)}$. Under independence the maximum likelihood estimates are $\hat{a}_j = N_{j\cdot}/n$ and $\hat{b}_k = N_{\cdot k}/n$; under homogeneity the fitted common distribution is $(N_{\cdot k}/n)_k$; in both cases the fitted expected counts are the E_{jk}.*

Proof strategy (for completeness; skip on a first reading). Count parameters. Independence is a family with $(r-1)+(c-1)$ parameters inside a model with $rc-1$, which leaves $(r-1)(c-1)$ constraints; homogeneity is one distribution with $c-1$ parameters inside r separate ones with $c-1$ each, which leaves the same number. The log-likelihood under either null splits into pieces that are each maximized by proportions.

Proof. Independence: the table is a categorical model with $K = rc$ categories, and the family $p_{jk} = a_j b_k$ has $d = (r-1)+(c-1)$ free parameters, since each of (a_j) and (b_k) adds up to 1. The map from (a,b) to the cell probabilities is a polynomial, and it is undone by the linear map $a_j = \sum_k p_{jk}$, $b_k = \sum_j p_{jk}$, so it has rank d, and cell probabilities close to the true ones come only from (a,b) close to the true ones: the hypotheses of Theorem 16.9 hold. The family's log-likelihood $\sum_{j,k} N_{jk}(\log a_j + \log b_k) = \sum_j N_{j\cdot}\log a_j + \sum_k N_{\cdot k}\log b_k$ is the sum of two categorical log-likelihoods, maximized by $\hat{a}_j = N_{j\cdot}/n$ and $\hat{b}_k = N_{\cdot k}/n$ (Proposition 16.2), so $n\hat{a}_j\hat{b}_k = E_{jk}$, and Theorem 16.9 gives the limit χ^2 with $rc-1-(r-1)-(c-1) = (r-1)(c-1)$ degrees of freedom.

Homogeneity: the rows are independent categorical samples of sizes $N_{j\cdot}$, so the free model has $r(c-1)$ parameters and the null, one common distribution (b_k), has $c-1$, which imposes $r(c-1)-(c-1) = (r-1)(c-1)$ constraints. Under the null the log-likelihood is $\sum_k N_{\cdot k}\log b_k$, maximized at $\hat{b}_k = N_{\cdot k}/n$, so the fitted expected count in cell (j,k) is $N_{j\cdot}\hat{b}_k = E_{jk}$. In the free model the log-likelihood is a sum of row log-likelihoods in separate parameters, so the score is the row scores stacked and the information is block diagonal, one block per row; the score statistic at the fitted null is therefore the sum over the rows of the score statistic of each row at $\hat{b}$, which is Pearson's statistic for that row (Proposition 16.6), and the sum is T_n^{S}. The likelihood ratio statistic is G^2 by Proposition 16.2 applied row by row, and Theorem 15.12, with the paragraphs on implicit hypotheses and on several independent samples in Section 15.7, gives the limit. □

Worked example: aspirin and heart attacks, as a table

The expected number of heart attacks on placebo is $11{,}034 \times 293/22{,}071 = 146.48$, and on aspirin 146.52; the observed numbers are 189 and 104. The expected counts without a heart attack are 10,887.52 and 10,890.48. All four deviations are ± 42.52, which is the one degree of freedom made visible: in a 2×2 table with fixed totals, one cell determines the other three. So

$$T^{\mathrm{S}} = \frac{42.52^2}{146.48} + \frac{42.52^2}{146.52} + \frac{42.52^2}{10{,}887.52} + \frac{42.52^2}{10{,}890.48} = 12.34 + 12.34 + 0.17 + 0.17 = 25.01,$$

with $(2-1)(2-1) = 1$ degree of freedom, and $G^2 = 25.37$. These are the score and likelihood ratio statistics of Chapter 15, which also showed that T^{S} is the square of the pooled two-proportion statistic. The p-value is below one in a million: if aspirin made no difference, a statistic this large would appear in fewer than one of a million trials like this one.

16.6 What a goodness-of-fit test can and cannot tell you

Small expected counts. The chi-squared limit needs every expected count to be large. A widely used guideline, usually credited to Cochran (*Biometrics*, 1954), asks that no expected count be below 1 and at most a fifth of them below 5. Figure 16.3 shows that the guideline is cautious for Pearson's statistic, which held its level for Mendel's null at $n = 8$ with an expected count of one half, less cautious for G^2, and not enough for Neyman's statistic. When the guideline fails there are two remedies. Pool neighboring categories, as for the horse kicks, at the price of a coarser question. Or, for a stated null, compute the p-value by simulation: the null distribution of the counts is $\mathrm{Mult}(n, \mathbf{p}^0)$, known exactly, so draw many tables from it, compute the statistic for each, and report the fraction at least as large as the observed one (Exercise 16.4).

Failing to reject is not confirming. Mendel's p-value of 0.93 says that his counts are consistent with $9:3:3:1$; it does not say that the ratio is $9:3:3:1$. In a simulation of 100,000 repetitions of his experiment, in a world where wrinkled green peas are about a quarter rarer than the theory says (the ratio $9:3:3:0.75$), Pearson's test rejects the theory in only 18% of them, and if wrinkled green peas were about half as common as the theory says ($9:3:3:0.5$), in 75%. With 2,000 peas the two figures become 64% and essentially 100%. Exercise 16.12 computes how the statistic grows with n under the first of these alternatives.

Large samples find every departure. By part (2) of Theorem 16.5, T_n^{S} grows like $n \sum_j (p_j - p_j^0)^2/p_j^0$ for any $\mathbf{p} \neq \mathbf{p}^0$, however close. Real data are never exactly Poisson or exactly $9:3:3:1$, so with enough of them the test rejects, and whether the departure it finds matters is a separate question, answered by the size of the deviations, as in Figures 16.1 and 16.5, rather than by the p-value.

Does this make sense? Can a fit be too good?

Fisher (*Annals of Science*, 1936) went through all of Mendel's experiments and added up their chi-squared statistics: 41.6 on 84 degrees of freedom. Independent chi-squared variables add (Proposition 14.2), so if Mendel's theories were right and the experiments independent, the total would be approximately χ^2_{84}, with mean 84 and standard deviation $\sqrt{168} = 13.0$. A total of 41.6 is 3.3 standard deviations below the mean, and the probability of a fit at least this good is

about 3 in 100,000 (Fisher, with the tables of his day, put it at 7 in 100,000). The upper tail of Pearson's statistic measures misfit; across many experiments, its lower tail measures a fit too close for chance. Fisher concluded that the data had been adjusted toward the theory, perhaps by an assistant who knew what Mendel expected, and the case has been argued many times since (for example, Pires and Branco, *Statistical Science*, 2010).

16.7 What comes next

Chapter 17 turns to continuous data. Grouping a continuous sample into categories and running Pearson's test works, but it discards where in its category each observation fell, and the answer depends on the categories chosen. The empirical CDF uses every observation as it is, and its largest distance from a stated CDF is the Kolmogorov–Smirnov statistic. Estimating parameters changes that statistic's null distribution, as estimating from the raw data did in this chapter, and the Lilliefors test is the repair; QQ plots make the same comparison as a picture.

16.8 Summary

- A goodness-of-fit test tests a whole distribution. For K categories, 'any other distribution' is the categorical model, parametric with $K-1$ free probabilities; the maximum likelihood estimator is the vector of proportions, and $\ell_n(\hat{\mathbf{p}}) - \ell_n(\mathbf{p}) = n\mathrm{KL}(\mathbb{P}_{\hat{\mathbf{p}}}, \mathbb{P}_{\mathbf{p}})$.
- The counts are multinomial and negatively correlated, $\mathbb{C}\mathrm{ov}(N_j, N_k) = -np_jp_k$, and their covariance matrix is not invertible.
- Pearson's $T_n^{\mathrm{S}} = \sum_j (N_j - np_j^0)^2/(np_j^0) \xrightarrow[n\to\infty]{\text{(d)}} \chi^2_{K-1}$: the standardized deviations are perpendicular to $(\sqrt{p_1^0}, \ldots, \sqrt{p_K^0})$, and in the other $K-1$ directions they are independent standard Gaussians in the limit. Mendel: 0.47, $p = 0.93$. Kiss data: 10.45.
- Pearson's statistic is the score statistic of the model with one category dropped, and also Wald's statistic with the information at the null; $G^2 = 2\sum_j N_j \log(N_j/(np_j^0))$ is the likelihood ratio statistic, and Neyman's $\sum_j (N_j - np_j^0)^2/N_j$ is Wald's statistic in the coordinates $p_1, \ldots, p_{K-1}$. For counts, Pearson's keeps its level best.
- A family with d parameters estimated from the counts: χ^2_{K-1-d}. Horse kicks: 0.32 on 2 degrees of freedom; siblings: no evidence against the Poisson model. Estimating from the raw data puts the limit between χ^2_{K-1-d} and χ^2_{K-1} (10.5% instead of 5% for a Gaussian in four cells).
- Tables: expected counts (row total)(column total)/n and $(r-1)(c-1)$ degrees of freedom, for independence and for homogeneity. Aspirin: 25.01 and 25.37 on 1 degree of freedom.
- Expected counts below 1, or many below 5: pool, or simulate the p-value. Not rejecting is not confirming, large samples reject every departure, and a fit can be too good (Fisher on Mendel: 41.6 on 84 degrees of freedom).

16.9 Exercises

Exercise 16.1. (a) Show that under $H_0 : \mathbf{p} = \mathbf{p}^0$, $\mathbb{E}[T_n^{\mathrm{S}}] = K - 1$ exactly, for every n. (b) Show that, by contrast, the mean of Neyman's statistic is infinite at every n if the statistic is given the value $+\infty$ when a count is 0, and explain why that does not contradict Corollary 16.7. *(Solution on page 428.)*

Exercise 16.2. For the kiss data and $\mathbf{p}^0 = (\frac{1}{2}, \frac{1}{2})$: (a) compute Pearson's statistic, G^2 and Neyman's statistic, and compare them with the three statistics of Chapter 15. (b) Show that for $K = 2$ Neyman's statistic equals $n(\hat{p} - p_0)^2/(\hat{p}(1 - \hat{p}))$, the square of Wald's statistic for a proportion. *(Solution on page 428.)*

Exercise 16.3. In his first experiment Mendel classified 7,324 seeds from hybrid plants by shape: 5,474 round and 1,850 wrinkled. His theory predicts the ratio 3 : 1. Compute the expected counts, Pearson's statistic and its p-value, and G^2. *(Solution on page 428.)*

Exercise 16.4. A die is rolled 15 times, and the faces 1 to 6 come up 3, 1, 1, 2, 2 and 6 times. (a) Compute Pearson's statistic for H_0: the die is fair, and the chi-squared p-value. (b) Suppose 60 rolls had given four times these counts. Recompute. (c) Why is the chi-squared approximation doubtful in (a), and how would you compute the p-value by simulation? (The exact p-value, obtained by adding up the null probabilities of all the tables of 15 rolls whose statistic is at least 7.0, is 0.256.) *(Solution on page 428.)*

Exercise 16.5. *Fortune* magazine recorded the zodiac signs of 256 heads of the 400 largest companies (De Veaux, Velleman and Bock, *Intro Stats*). From Aries to Pisces the counts were 23, 20, 18, 23, 20, 19, 18, 21, 19, 22, 24 and 29. (a) Test whether the twelve signs are equally likely: give Pearson's statistic, its degrees of freedom and its p-value. (b) Pisces has 29, more than a third above its expected share. Taken on its own, as if Pisces had been named before the data were seen, how surprising is that? (c) Why is (b) the wrong question once Pisces has been picked out because its count is the largest? (When all twelve signs are equally likely, the largest of the twelve counts is 29 or more in 56% of samples of 256.) *(Solution on page 428.)*

Exercise 16.6. A random sample of 275 jurors from a small county had 205 White, 26 Black, 25 Hispanic and 19 jurors of other groups. Among the county's registered voters, the people eligible for jury duty, the four groups make up 72%, 7%, 12% and 9% (Diez, Barr and Çetinkaya-Rundel, *OpenIntro Statistics*). (a) Compute the expected counts, Pearson's statistic and its p-value. Is there evidence that the jurors are not representative of the voters? (b) If the jurors' proportions stayed the same, from about how many jurors on would the test reject at the 5% level? *(Solution on page 429.)*

Exercise 16.7. (a) Expand $f(x) = x \log(x/m)$ around $x = m$, for a constant $m > 0$, and deduce that $G^2 = T_n^{\mathrm{S}} +$ (higher-order terms), using that the observed and expected counts have the same total. (b) In the kiss data $G^2 = 10.60$ and $T^{\mathrm{S}} = 10.45$. Which terms of the expansion account for the difference? *(Solution on page 429.)*

Exercise 16.8. A sample of 1,000 observations on $\{0, 1, 2, 3\}$ has counts 339, 455, 180 and 26. Test whether it comes from the family $\mathrm{Bin}(3, \theta)$: find the maximum likelihood estimate of θ from the counts, the expected counts, Pearson's statistic, its degrees of freedom and its p-value. *(Solution on page 429.)*

Exercise 16.9. The British Board of Trade's inquiry into the sinking of the *Titanic* in 1912 reported, by class (first, second, third, crew), these numbers of survivors and deaths: 203 and 122; 118 and 167; 178 and 528; 212 and 673 (Dawson, *Journal of Statistics Education*, 1995). (a) Compute the expected counts under independence of class and survival. (b) Compute Pearson's statistic and its degrees of freedom. (c) Which cell contributes the most, and what does it say? *(Solution on page 429.)*

Exercise 16.10. For an $r \times c$ table with expected counts $E_{jk} = N_{j\cdot}N_{\cdot k}/n$: (a) show that the expected counts have the same row and column totals as the observed ones; (b) deduce that the deviations $N_{jk} - E_{jk}$ add up to 0 along every row and every column, so that the $(r-1)(c-1)$ deviations in the first $r-1$ rows and $c-1$ columns determine all of them; (c) for $r = c = 2$, show that the four deviations have the same absolute value. *(Solution on page 429.)*

Exercise 16.11. For $K = 2$, let $V_n = \sum_{j=1}^{2}(N_j - np_j^0)^2/(np_j^0(1-p_j^0))$, the sum of the two squared standardized binomial counts. (a) Show that $V_n = 2T_n^{\mathrm{S}}$. (b) Find the limit of V_n under the null, and the probability that it exceeds $q_{0.05}(\chi^2_2) = 5.99$ in the limit. *(Solution on page 429.)*

Exercise 16.12. Take Mendel's null $\mathbf{p}^0 = (9,3,3,1)/16$ and the alternative $\mathbf{p} = (9,3,3,0.75)/15.75$, in which wrinkled green peas are about a quarter rarer. (a) Compute $c = \sum_j (p_j - p_j^0)^2/p_j^0$. (b) Show that under $\mathbf{p}$, $\mathbb{E}[T_n^{\mathrm{S}}] = nc + \sum_j p_j(1-p_j)/p_j^0$. (c) Evaluate this mean at $n = 556$ and at $n = 2{,}000$, and compare it with the threshold 7.81 and with the simulated rejection rates of Section 16.6. *(Solution on page 430.)*

Chapter 17

Kolmogorov–Smirnov, Lilliefors and QQ Plots

In this chapter

- The empirical CDF, which estimates a whole distribution at once. At each point it is a sample proportion, so the Law of Large Numbers and the Central Limit Theorem apply point by point; the Glivenko–Cantelli theorem makes the first hold at all points at once, and Donsker's theorem does the same for the second.
- The Kolmogorov–Smirnov test of a stated continuous distribution: the largest vertical gap between the empirical CDF and the candidate. Its null distribution is the same for every continuous candidate, so one table, or one simulation, serves them all.
- Other measures of the gap, Cramér–von Mises and Anderson–Darling, and what binning the data for a chi-squared test gives up.
- Testing a family: estimating its parameters changes the null distribution, and the Kolmogorov–Smirnov table then gives a test that almost never rejects. The Lilliefors test for normality is the repair.
- QQ plots: the same comparison drawn as quantiles against quantiles, and how to read heavy tails, light tails and skew from one. Newcomb's measurements of the speed of light and the eruptions of Old Faithful.

Chapter 16 tested distributions on finitely many categories by counting. This chapter tests continuous distributions without grouping the data at all.

17.1 The empirical CDF

A continuous distribution has no natural categories. Chapter 16 could still test one by cutting the line into cells and counting, but the answer depends on where the cuts go, and each observation is reduced to the name of its cell. The object that describes a distribution on the line without any cutting is its CDF, $F(t) = \mathbb{P}(X \leq t)$, and it has an obvious estimate: the proportion of the sample at or below t.

Definition 17.1 (The empirical CDF)**.** For a sample $X_1, \ldots, X_n$, the **empirical CDF** is the

random function

$$F_n(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(X_i \le t) = \frac{\#\{i : X_i \le t\}}{n}, \qquad t \in \mathbb{R}.$$

With the observations sorted, $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$ (the **order statistics**), F_n is a staircase that is 0 below $X_{(1)}$, climbs $1/n$ at each observation, and is 1 from $X_{(n)}$ on.

> **In words.** At each t, count the observations at or below t and divide by n (toolkit A.4). The result is the CDF of the distribution that puts probability $1/n$ on each data point.
>
> **For the kiss data.** With $R_i = 1$ for a right turn, $F_n(t)$ is 0 for $t < 0$, $44/124 = 0.355$ for $0 \le t < 1$, and 1 from $t = 1$ on. Its step at 1 has height $\hat{p} = 0.645$: for two-valued data the empirical CDF carries exactly the information in $\hat{p}$.

The left of Figure 17.1 shows two empirical CDFs of simulated waits from the exponential distribution with mean 6.23 minutes, the model of the subway data, against the true CDF. With 10 observations the staircase is coarse and strays as far as 0.25 from the curve; with 100 it stays within about 0.08.

Recall. The Law of Large Numbers (Theorem 3.5): for i.i.d. random variables with mean μ and finite variance, $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$. The Central Limit Theorem (Theorem 4.8): if the variance σ^2 is also positive, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, \sigma^2)$. Slutsky's theorem (Theorem 5.1): if $X_n \xrightarrow[n\to\infty]{(\mathrm{d})} X$ and $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$, a constant, then $X_n Y_n \xrightarrow[n\to\infty]{(\mathrm{d})} yX$; it lets an estimated standard error stand in for the true one.

Proposition 17.2 (The empirical CDF at fixed points)**.** *Let $X_1, \ldots, X_n$ be i.i.d. with CDF F. For each t, $nF_n(t) \sim \mathrm{Bin}(n, F(t))$, so $\mathbb{E}[F_n(t)] = F(t)$ and $\mathbb{V}ar(F_n(t)) = F(t)(1-F(t))/n$; for $s \le t$, $\mathbb{C}ov(F_n(s), F_n(t)) = F(s)(1-F(t))/n$. As $n \to \infty$, $F_n(t) \xrightarrow[n\to\infty]{\mathbb{P}} F(t)$, and if $0 < F(t) < 1$, $\sqrt{n}(F_n(t) - F(t)) \xrightarrow[n\to\infty]{(d)} Gauss\big(0, F(t)(1-F(t))\big)$.*

Proof strategy (for completeness; skip on a first reading). $F_n(t)$ is the average of the i.i.d. indicators $\mathbb{1}(X_i \le t)$, each Bernoulli with mean $F(t)$; the Law of Large Numbers and the CLT do the rest.

Proof. The indicators $\mathbb{1}(X_i \le t)$ are independent and each is $\mathrm{Bern}(F(t))$, so their sum $nF_n(t)$ is $\mathrm{Bin}(n, F(t))$, with mean $nF(t)$ and variance $nF(t)(1-F(t))$. For $s \le t$, $\mathbb{1}(X \le s)\mathbb{1}(X \le t) = \mathbb{1}(X \le s)$, so the two indicators of one observation have covariance $F(s) - F(s)F(t) = F(s)(1-F(t))$; indicators of different observations are independent, and dividing the sum of the n covariances by n^2 gives the formula. The two limits are the Law of Large Numbers and the CLT for the average of the indicators, whose variance $F(t)(1-F(t))$ is finite, and positive when $0 < F(t) < 1$. □

So at any one point the empirical CDF is an ordinary proportion, and Chapter 8 already gives an interval for it. For the waiting times between eruptions of Old Faithful (Chapter 11), 83 of the 272 waits were 60 minutes or less, so $F_{272}(60) = 0.305$, and the plug-in 95% interval for $F(60)$ is $0.305 \pm 1.96\sqrt{0.305 \times 0.695/272} = [0.250, 0.360]$.

From undergraduate statistics: the cumulative relative frequency plot

An introductory course groups data into classes and plots the cumulative relative frequency at the end of each class, a plot sometimes called an ogive. The empirical CDF is that plot with every observation its own class: no class width to choose, and nothing lost to grouping. Read at

one t it is the sample proportion of Chapter 8, with the standard error $\sqrt{F(t)(1-F(t))/n}$ of any proportion.

17.2 All points at once: the Glivenko–Cantelli theorem

A goodness-of-fit test compares the whole curve F_n with a candidate, so it needs F_n to be close to F everywhere at the same time. Proposition 17.2 does not say that. It says that at each fixed t a large gap is unlikely; but there are infinitely many t, and small chances at infinitely many places need not add up to a small chance of a large gap somewhere. Functions can converge at every point without converging uniformly: x^n tends to 0 at every x in $[0,1)$, yet $\sup_{0\le x<1} x^n = 1$ for every n (toolkit A.2). For CDFs uniform convergence does hold, and the reason is that CDFs never go down.

Recall. The Law of Large Numbers, as above: the average of i.i.d. indicators converges in probability to their common mean.

Theorem 17.3 (Glivenko–Cantelli)**.** *Let $X_1, \dots, X_n$ be i.i.d. with CDF F, any CDF. Then*

$$\sup_{t\in\mathbb{R}} |F_n(t) - F(t)| \xrightarrow[n\to\infty]{\mathbb{P}} 0.$$

Proof strategy (core). Cut the line at the $k-1$ points where F crosses $\frac{1}{k}, \frac{2}{k}, \dots$. Between two cuts both F and F_n only go up, and F rises by at most $\frac{1}{k}$, so the gap anywhere is at most the gap at the cuts plus $\frac{1}{k}$. The Law of Large Numbers controls the gaps at finitely many cuts; then let k grow.

Proof. Write $F(t-) = \mathbb{P}(X < t)$, which equals $F(t)$ unless F jumps at t, and likewise $F_n(t-) = \frac{1}{n}\#\{i : X_i < t\}$. F is continuous from the right and has a limit from the left at every t: as m grows, the events $\{X \le t + \frac{1}{m}\}$ shrink to $\{X \le t\}$, so $F(t+\frac{1}{m})$ decreases to $F(t)$, and the events $\{X \le t - \frac{1}{m}\}$ grow to $\{X < t\}$, so $F(t-\frac{1}{m})$ increases to $F(t-)$. Fix an integer $k \ge 2$. For $j = 1, \dots, k-1$ let $t_j = \inf\{t : F(t) \ge j/k\}$, a finite number since F rises from 0 to 1, and set $t_0 = -\infty$ and $t_k = +\infty$, with $F(t_0) = F_n(t_0) = 0$ and $F(t_k-) = F_n(t_k-) = 1$. Since $F(t) < j/k$ for every $t < t_j$, and F is continuous from the right, $F(t_j-) \le j/k \le F(t_j)$; therefore

$$F(t_j-) - F(t_{j-1}) \le \frac{j}{k} - \frac{j-1}{k} = \frac{1}{k}, \qquad j = 1, \dots, k.$$

Take any t, and the j with $t_{j-1} \le t < t_j$. (If F jumps across several of the levels j/k at once, the corresponding t_j coincide and the stretches between them are empty; every t still lies in exactly one stretch.) Both F and F_n are nondecreasing, so

$$F_n(t) - F(t) \le F_n(t_j-) - F(t_{j-1}) \le F_n(t_j-) - F(t_j-) + \frac{1}{k}$$

and

$$F_n(t) - F(t) \ge F_n(t_{j-1}) - F(t_j-) \ge F_n(t_{j-1}) - F(t_{j-1}) - \frac{1}{k}.$$

Hence $\sup_t |F_n(t) - F(t)| \le M_n + 1/k$, where M_n is the largest of the $2(k-1)$ numbers $|F_n(t_j) - F(t_j)|$ and $|F_n(t_j-) - F(t_j-)|$, $j = 1, \dots, k-1$. Each of them converges to 0 in probability by the Law of Large Numbers: $F_n(t_j)$ is the average of the indicators $\mathbb{1}(X_i \le t_j)$, with mean $F(t_j)$, and $F_n(t_j-)$ the average of $\mathbb{1}(X_i < t_j)$, with mean $F(t_j-)$. So $\mathbb{P}(M_n > \varepsilon)$, which is at most the sum of the

$2(k-1)$ probabilities that one of them exceeds ε (the probability of a union is at most the sum of the probabilities, as in the proof of Proposition 14.13), tends to 0 for every $\varepsilon > 0$. Given $\varepsilon > 0$, choose $k > 2/\varepsilon$: then $\mathbb{P}\big(\sup_t |F_n(t) - F(t)| > \varepsilon\big) \le \mathbb{P}(M_n > \varepsilon/2) \to 0$. □

With the Strong Law of Large Numbers (Theorem B.3) in place of the weak one, the same argument, run for every $k = 2, 3, \ldots$ at once, shows that the largest gap tends to 0 with probability 1, which is how the theorem is usually stated (Glivenko and Cantelli, both in 1933); nothing later needs that form. Appendix C proves a uniform Law of Large Numbers (Theorem C.3) for functions that are continuous in a parameter ranging over a closed, bounded set; the indicators $\mathbb{1}(x \le t)$ are neither continuous in t nor indexed by a bounded set, and monotonicity does the work instead.

Intuition: a staircase in a corridor

The right of Figure 17.1 draws the proof with $k = 4$, for a continuous F. The quartiles t_1, t_2, t_3 of the true CDF cut the line into four stretches, and on each stretch the true CDF climbs through one gray box, from its lower-left corner to its upper-right corner, a rise of $\frac{1}{4}$. The empirical CDF is a staircase that also only goes up. If at the three cuts it is within δ of the curve, then inside a stretch it can be no lower than its value at the left cut and no higher than its value at the right cut, so it stays within $\delta + \frac{1}{4}$ of the curve everywhere. Three proportions control a whole function; with k stretches, $k-1$ proportions control it to within $\frac{1}{k}$ plus their own errors.

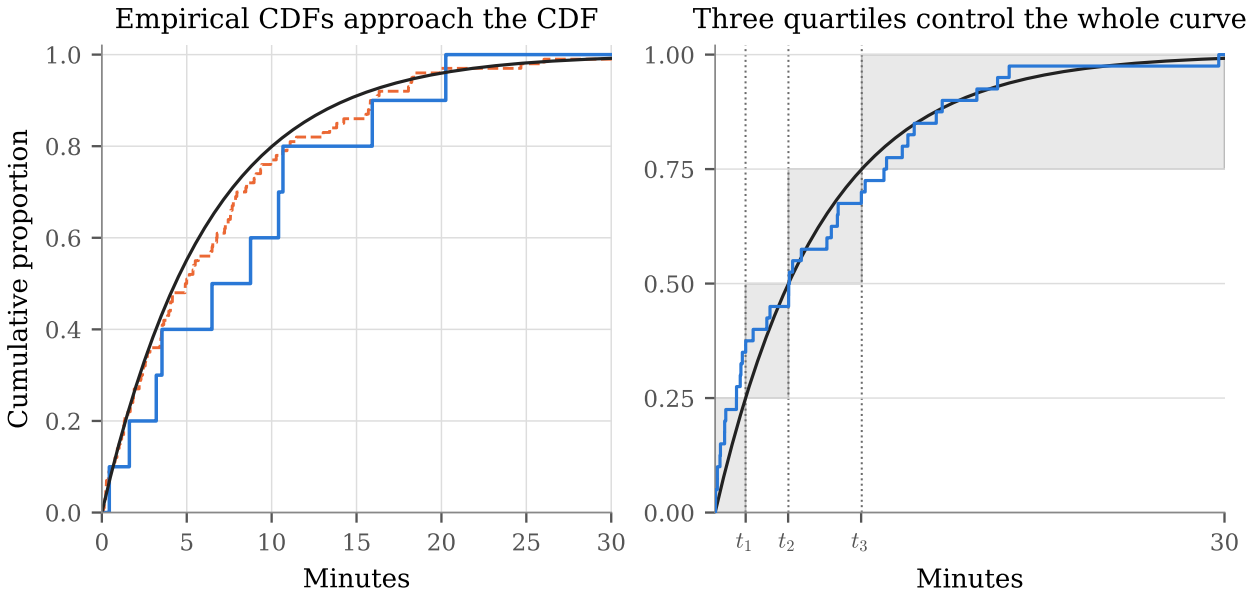


Figure 17.1: Simulated waits, mean 6.23; the exponential CDF (black) in both halves. Left: empirical CDFs for $n = 10$ (blue) and $n = 100$ (orange, dashed). Right: quartiles t_1, t_2, t_3 (dotted), four gray boxes, and an empirical CDF for $n = 40$ (blue).

The largest gap is itself a distance between distributions, the **Kolmogorov distance** $\sup_t |F(t) - G(t)|$ between the distributions with CDFs F and G. It compares the probabilities of the half-lines $(-\infty, t]$ only, where total variation compares those of all events, so it is never larger than the total variation distance (Exercise 17.6). That restriction is what makes it estimable. Section 10.4 showed that the empirical distribution of continuous data sits at total variation 1 from the truth at every n; at

Kolmogorov distance it converges to the truth, by the Glivenko–Cantelli theorem. For the exponential Expo(1) and Gauss(1, 1), its closest Gaussian in KL divergence (Chapter 10), the Kolmogorov distance is 0.159, attained at 0, below which the Gaussian puts probability $\Phi(-1) = 0.159$ and the exponential none; their total variation distance is 0.319.

How fast does the largest gap shrink? In 4,000 simulated samples its median is 0.247 at $n = 10$, 0.081 at $n = 100$ and 0.026 at $n = 1{,}000$, and $\sqrt{n}$ times the median is close to 0.8 at all three. The gap shrinks like $1/\sqrt{n}$, as the error of an average does, and the next section finds the distribution of $\sqrt{n}$ times it.

17.3 The size of the gap: Donsker's theorem

Multiplying by $\sqrt{n}$, as the CLT does for an average, turns the gap into a random curve, $t \mapsto \sqrt{n}(F_n(t) - F(t))$. By Proposition 17.2, at each t it is approximately Gauss$(0, F(t)(1 - F(t)))$, and its values at two points $s \le t$ have covariance $F(s)(1 - F(t))$. The variance is largest where $F = \frac{1}{2}$ and falls to 0 at both ends, where F_n and F are both pinned near 0 or near 1. For uniform data, $F(u) = u$ on $[0, 1]$, the curve is $u \mapsto \sqrt{n}(G_n(u) - u)$, with G_n the empirical CDF of n independent Unif$(0, 1)$ draws $U_1, \dots, U_n$, and the covariance at $u \le v$ is $u(1 - v) = \min(u, v) - uv$.

Recall. The Central Limit Theorem for vectors (the vector-versions remark of Chapter 5): for i.i.d. random vectors Y_i with mean m and positive definite covariance matrix Σ, $\sqrt{n}(\bar{Y}_n - m) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0, \Sigma)$.

At finitely many points $u_1 < \cdots < u_k$ in $(0, 1)$, the values of this curve are $\sqrt{n}$ times the centered average of the i.i.d. vectors $(\mathbb{1}(U_i \le u_1), \dots, \mathbb{1}(U_i \le u_k))$, and the CLT for vectors makes them jointly Gaussian in the limit (the covariance matrix is positive definite; Exercise 17.3 checks two points). The limit of the whole curve is a random function with those Gaussian values.

Theorem 17.4 (Kolmogorov; Donsker)**.** *Let $X_1, \dots, X_n$ be i.i.d. with a continuous CDF F. Then*

$$\sqrt{n} \sup_{t \in \mathbb{R}} |F_n(t) - F(t)| \xrightarrow[n\to\infty]{(d)} K,$$

*where K is the largest absolute value of a **Brownian bridge**: a random continuous function B on $[0, 1]$ with $B(0) = B(1) = 0$ whose values at any points $0 < u_1 < \cdots < u_k < 1$ form a Gaussian vector with mean 0 and $\mathbb{C}ov(B(u), B(v)) = \min(u, v) - uv$. The distribution of K, **Kolmogorov's distribution**, is*

$$\mathbb{P}(K > x) = 2 \sum_{j=1}^{\infty} (-1)^{j-1} e^{-2j^2 x^2}, \qquad x > 0,$$

and its 95% quantile is $q_{0.05}(K) = 1.358$ (1.224 for 90%, 1.628 for 99%).

This theorem is stated without proof. The part about finitely many points follows from the CLT for vectors, as above; the step from finitely many points to the largest gap over all of them needs a theory of convergence for random functions that this book does not develop. Donsker's theorem proper says that the whole curve $u \mapsto \sqrt{n}(G_n(u) - u)$ converges in distribution to B; this book uses only its consequence for the largest gap. Kolmogorov (*Giornale dell'Istituto Italiano degli Attuari*, 1933) found the limit distribution, and Smirnov (*Annals of Mathematical Statistics*, 1948) tabulated it; Doob (*Annals of Mathematical Statistics*, 1949) saw, and Donsker (*Annals of Mathematical Statistics*, 1952) proved, that it is the distribution of the largest absolute value of the limiting curve,

the Brownian bridge. Notice what the limit does not depend on: F. Every continuous distribution leads to the same K, which the next section explains.

The left of Figure 17.2 shows four curves $\sqrt{n}(G_n(u) - u)$ for $n = 1{,}000$ uniform draws. They already look like the limit: pinned to 0 at both ends, wandering most in the middle, and, like about 19 in 20 such curves, staying inside ± 1.358. The right compares the distribution of $\sqrt{n}$ times the largest gap at $n = 100$ with Kolmogorov's density; at $n = 100$ the exact 95% point is 1.340, against the limit's 1.358.

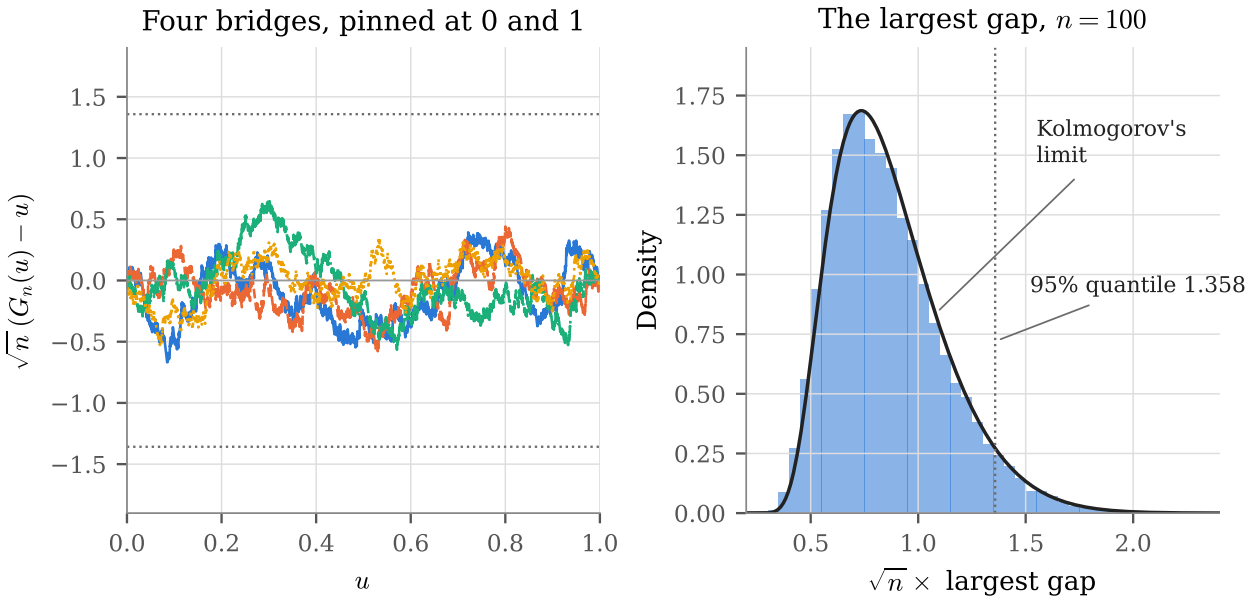


Figure 17.2: Uniform data. Left: four curves $\sqrt{n}(G_n(u) - u)$ for $n = 1{,}000$, and ± 1.358 (dotted). Right: $\sqrt{n}$ times the largest gap in 20,000 samples of $n = 100$ (histogram), Kolmogorov's density (black) and its 95% quantile (dotted).

Does this make sense? Why is the threshold 1.36 and not 0.98?

At the single point $u = \frac{1}{2}$ the bridge has variance $\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$, so $|B(\frac{1}{2})|$ exceeds $1.96 \times \frac{1}{2} = 0.98$ with probability 5%. K is the largest $|B(u)|$ over the whole curve, so it exceeds 0.98 more often, and its 95% point must be higher. It is 1.358, which is the 95% point of the largest of about eight independent values distributed like $|B(\frac{1}{2})|$: values of the bridge at nearby points move together, so infinitely many points behave like a handful. The same contrast set the thresholds of the multiple-testing corrections of Section 14.7: the largest of many statistics needs a larger threshold than any one of them.

A confidence band. For continuous F, since $\mathbb{P}(\sqrt{n}\sup_t |F_n(t) - F(t)| \le 1.358) \to 0.95$, the band $F_n(t) \pm 1.358/\sqrt{n}$ contains the entire curve F with probability tending to 95%. For Old Faithful, $n = 272$ and the half-width is $1.358/\sqrt{272} = 0.082$, so at 60 minutes the band is $[0.223, 0.387]$, wider than the pointwise interval $[0.250, 0.360]$ of Section 17.1, since the band must cover every t at once. The band does not even need large n. Massart (*Annals of Probability*, 1990) proved the Dvoretzky–Kiefer–Wolfowitz inequality with its best constant: for every n, every $\varepsilon > 0$ and every F,

$$\mathbb{P}\Big(\sup_t |F_n(t) - F(t)| > \varepsilon\Big) \le 2e^{-2n\varepsilon^2}.$$

Setting the right side to 0.05 gives $\varepsilon = \sqrt{\log 40/(2n)} = 1.3581/\sqrt{n}$, the asymptotic half-width to four digits, so the band with this half-width covers F with probability at least 95% at every n, whatever F is. (The inequality is stated without proof; nothing later depends on it.) A band tests every candidate at once: a stated CDF is rejected at the 5% level when it leaves the band somewhere, and a family is rejected when every member does. The Gaussian CDF closest to the Old Faithful staircase in Kolmogorov distance has mean 72.5 and standard deviation 15.1 and is 0.102 away, outside the band, so no Gaussian fits the waits.

17.4 The Kolmogorov–Smirnov test

To test whether data come from a stated continuous distribution, measure the largest gap between its CDF and the empirical CDF. The alternative is every other distribution, which no finite list of parameters describes; over all densities the likelihood has no maximum (a density can be made as tall as we like at the observed points), so the tests of Chapter 15 do not apply, and the statistic is built from the empirical CDF alone.

Definition 17.5 (The Kolmogorov–Smirnov test)**.** Let F^0 be a continuous CDF. For $H_0 : F = F^0$ against $H_1 : F \neq F^0$, the **Kolmogorov–Smirnov statistic** is

$$D_n = \sup_{t \in \mathbb{R}} |F_n(t) - F^0(t)|,$$

and the **Kolmogorov–Smirnov test** of level α rejects when $D_n > d_\alpha(n)$, the $1-\alpha$ quantile of D_n under H_0. The test of asymptotic level α rejects when $\sqrt{n} D_n > q_\alpha(K)$, which is 1.358 for $\alpha = 0.05$.

> **In words.** Draw the candidate CDF and the staircase of the data on the same axes, find the largest vertical distance between them, and reject when it is larger than chance alone would make it. The threshold shrinks like $1/\sqrt{n}$: about $1.36/\sqrt{n}$ at the 5% level.
>
> **For the kiss data.** The kiss data are 0s and 1s, and the candidate $p = \frac{1}{2}$ has a two-step CDF, so $D_n = |F_n(0) - F^0(0)| = |0.355 - 0.5| = 0.145$, the gap in the proportion of left turns, and $\sqrt{124}\, D_n = 1.62$, half the statistic 3.23 of Chapter 9. But F^0 is not continuous, and the theory below does not apply: with this discrete null the threshold 1.358 would reject a true null about 5 times in 1,000 at $n = 124$ (0.66% in the limit), not 50 times (Exercise 17.4). Discrete data belong to the tests of Chapters 9 and 16.

The supremum runs over infinitely many t, but only $2n$ numbers need checking.

Proposition 17.6 (Computing the statistic)**.** *If F^0 is continuous,*

$$D_n = \max_{1 \le i \le n} \max\Big(\frac{i}{n} - F^0(X_{(i)}),\ F^0(X_{(i)}) - \frac{i-1}{n} \Big).$$

Proof strategy (for completeness; skip on a first reading). Between two observations the staircase is flat and F^0 only rises, so the largest gaps sit at the steps.

Proof. On $[X_{(i)}, X_{(i+1)})$, $F_n = i/n$ and F^0 is nondecreasing, so $F_n - F^0$ is largest at the left end, where it is $i/n - F^0(X_{(i)})$, and $F^0 - F_n$ approaches its supremum $F^0(X_{(i+1)}) - i/n$ at the right end, by the continuity of F^0. On $(-\infty, X_{(1)})$, $F_n = 0$ and the gap $F^0(t)$ approaches $F^0(X_{(1)}) - 0$; on $[X_{(n)}, \infty)$, $F_n = 1$ and the gap $1 - F^0(t)$ is largest at $X_{(n)}$, where it is $n/n - F^0(X_{(n)})$. These

are the $2n$ terms of the formula. If several observations are tied, $X_{(i)} = \cdots = X_{(j)}$, the staircase jumps from $(i-1)/n$ to j/n there; the terms $j/n - F^0(X_{(j)})$ and $F^0(X_{(i)}) - (i-1)/n$ are the gaps at that value and just below it, and the other terms of the block are no larger. □

Worked example: ten random numbers

A random number generator should produce independent Unif$(0,1)$ values. Ten of its outputs, sorted and rounded to two decimals, are in the second row below. Here $F^0(u) = u$, so the two terms of Proposition 17.6 are the last two rows:

i	1	2	3	4	5	6	7	8	9	10
$u_{(i)}$	0.08	0.17	0.38	0.51	0.54	0.61	0.80	0.87	0.94	0.98
$i/10 - u_{(i)}$	0.02	0.03	−0.08	−0.11	−0.04	−0.01	−0.10	−0.07	−0.04	0.02
$u_{(i)} - (i-1)/10$	0.08	0.07	0.18	0.21	0.14	0.11	0.20	0.17	0.14	0.08

The largest entry is $0.51 - 0.30 = 0.21$: just below 0.51 the staircase is still at 0.30, while the uniform CDF has reached 0.51. So $D_{10} = 0.21$. The exact 5% threshold for $n = 10$ is $d_{0.05}(10) = 0.409$, and the exact p-value is $\mathbb{P}(D_{10} \geq 0.21) = 0.70$: nothing here suggests that the generator is broken. The asymptotic version agrees: $\sqrt{10} \times 0.21 = 0.664 < 1.358$, with p-value $\mathbb{P}(K > 0.664) = 0.77$. The left of Figure 17.3 draws the staircase, the uniform CDF and this largest gap, in orange.

The worked example used the uniform candidate, and nothing in it was special to the uniform.

Lemma 17.7 (The probability integral transform)**.** *If X has a continuous CDF F, then $F(X) \sim$ Unif$(0,1)$.*

Proof strategy (core). The event $F(X) \leq u$ is the event that X is at or below the last point where F equals u, and that has probability u.

Proof. Fix $u \in (0,1)$. Since F is continuous and runs from 0 to 1, it takes the value u (the intermediate value theorem), and since it is nondecreasing and continuous, the set where $F(t) = u$ is a closed, bounded interval $[a,b]$, possibly a single point. If $X \leq b$ then $F(X) \leq F(b) = u$; if $X > b$ then $F(X) > u$, since b is the last point where F equals u. So $\mathbb{P}(F(X) \leq u) = \mathbb{P}(X \leq b) = F(b) = u$, which is the CDF of Unif$(0,1)$. □

Theorem 17.8 (The Kolmogorov–Smirnov test)**.** *Let $X_1, \ldots, X_n$ be i.i.d. with CDF F, and let F^0 be continuous.*

1. *If $F = F^0$, then D_n has the distribution of $\sup_{0 \leq u \leq 1} |G_n(u) - u|$, where G_n is the empirical CDF of n i.i.d.* Unif$(0,1)$ *variables. It is the same for every continuous F^0, so the test of Definition 17.5 has level α, and by Theorem 17.4 the test with threshold $q_\alpha(K)/\sqrt{n}$ has asymptotic level α.*
2. *If $F \neq F^0$, then $D_n \xrightarrow[n\to\infty]{\mathbb{P}} \sup_t |F(t) - F^0(t)| > 0$, and the probability that either test rejects tends to* 1.

Proof strategy (core). Push every observation through F^0. Under H_0 the transformed values are uniform, and the vertical gaps between the staircase and the candidate do not change, since the transformation only moves points along the horizontal axis.

Proof. (1) Suppose first that F^0 is strictly increasing on the interval where $0 < F^0 < 1$. Let $U_i = F^0(X_i)$; under H_0 these are i.i.d. Unif$(0,1)$ by Lemma 17.7, and with probability 1 every

X_i lies in that interval. For such an X_i and every t, $X_i \le t$ exactly when $U_i \le F^0(t)$, so $F_n(t) = G_n(F^0(t))$, where G_n is the empirical CDF of $U_1, \dots, U_n$. Hence

$$D_n = \sup_t |G_n(F^0(t)) - F^0(t)| = \sup_{0<u<1} |G_n(u) - u|,$$

since $F^0(t)$ takes every value in $(0, 1)$ as t runs over the line; at $u = 0$ and $u = 1$ the gap is 0 with probability 1. If F^0 is flat on a stretch, extend the stretch as far as F^0 stays flat: the result is a closed stretch of probability 0. There are at most countably many of these longest flat stretches (they do not overlap and each has positive length, so each contains a rational number of its own), so with probability 1 no observation falls in one, endpoints included. For every other value of X_i the equivalence above still holds: if $X_i > t$ and $F^0(X_i) = F^0(t)$, then F^0 is flat on $[t, X_i]$, a stretch containing X_i. The threshold $d_\alpha(n)$ is a quantile of this one distribution, which gives the level; and the uniform distribution is continuous, so Theorem 17.4 applied to $U_1, \dots, U_n$ gives the asymptotic level.

(2) By the triangle inequality, at every t, $|F_n(t) - F^0(t)|$ differs from $|F(t) - F^0(t)|$ by at most $|F_n(t) - F(t)|$, so D_n differs from $c = \sup_t |F(t) - F^0(t)|$ by at most $\sup_t |F_n(t) - F(t)|$, which converges to 0 in probability by Theorem 17.3. Hence $D_n \xrightarrow[n\to\infty]{\mathbb{P}} c$, and $c > 0$ since $F \ne F^0$. Both thresholds tend to 0: $q_\alpha(K)/\sqrt{n}$ plainly, and $d_\alpha(n)$ since, under H_0, the Glivenko–Cantelli theorem gives $\mathbb{P}(D_n > \varepsilon) < \alpha$ for large n and any $\varepsilon > 0$, so that $d_\alpha(n) \le \varepsilon$. Once both thresholds are below $c/2$, the probability of rejecting is at least $\mathbb{P}(D_n > c/2)$, which tends to 1. □

Intuition: one table for every continuous distribution

The right of Figure 17.3 takes the ten generator outputs of the worked example and sends each through the exponential quantile function, $x = -5\log(1 - u)$, which turns uniform numbers into exponential waits with mean 5 minutes (Exercise 17.1). Against the exponential CDF the new staircase has the same steps at the same heights, and every gap is the same, including $D_{10} = 0.21$. That is the theorem: an increasing transformation stretches the horizontal axis and leaves heights alone, and the Kolmogorov–Smirnov statistic measures only heights. So the null distribution of D_n can be computed once, for uniform data, and used for every continuous F^0. Massey (*Journal of the American Statistical Association*, 1951) published such a table, for n up to 35, and a simulation of uniform samples reproduces any entry of it.

The exact thresholds approach $1.358/\sqrt{n}$ from below: for $n = 5$, 10, 20, 50 and 100 they are 0.563, 0.409, 0.294, 0.188 and 0.134, against 0.607, 0.429, 0.304, 0.192 and 0.136. So the asymptotic test is slightly conservative, and from $n = 50$ on the two thresholds agree to within about 2%.

17.5 Other measures of the gap

The Kolmogorov–Smirnov statistic keeps only the largest gap. Two other classical statistics add up the squared gaps over the whole line, weighting each t by the probability the candidate puts near it:

$$W_n^2 = n \int_{-\infty}^{\infty} \big(F_n(t) - F^0(t)\big)^2 f^0(t)\,\mathrm{d}t, \qquad A_n^2 = n \int_{-\infty}^{\infty} \frac{\big(F_n(t) - F^0(t)\big)^2}{F^0(t)\big(1 - F^0(t)\big)} f^0(t)\,\mathrm{d}t,$$

the **Cramér–von Mises** and **Anderson–Darling** statistics, with f^0 the density of F^0 (Anderson and Darling, *Annals of Mathematical Statistics*, 1952, introduced the second). The substitution

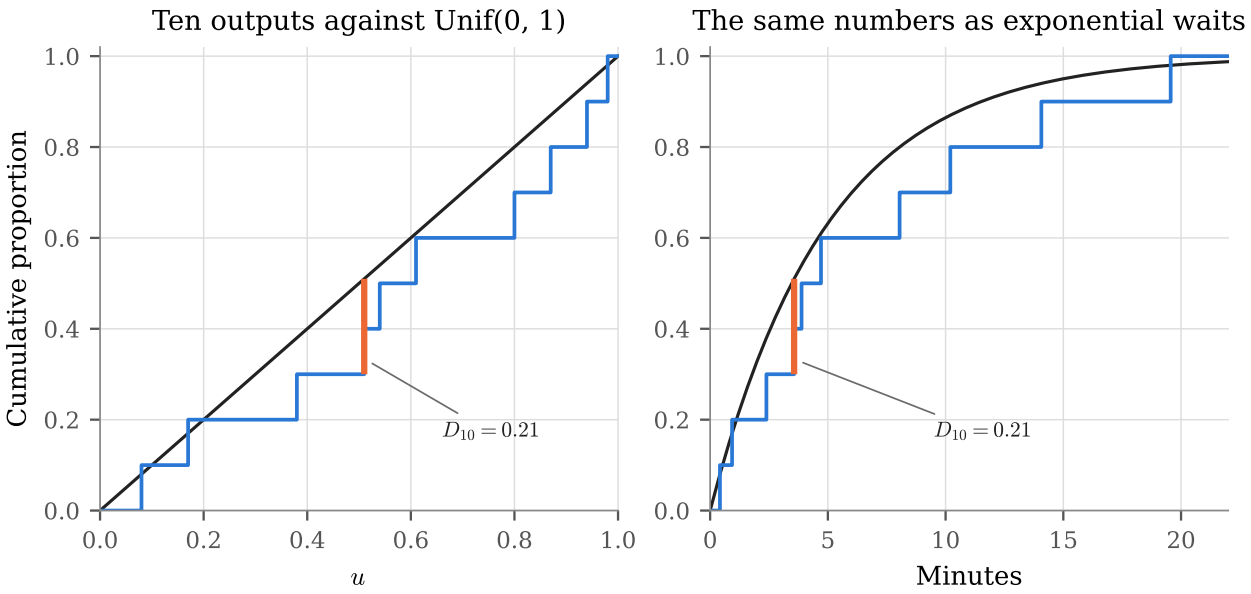


Figure 17.3: Left: the empirical CDF (blue) of ten generator outputs against the Unif$(0, 1)$ CDF (black), largest gap in orange. Right: the same values as exponential waits, $x = -5\log(1 - u)$, against the exponential CDF with mean 5.

$u = F^0(t)$, with $\mathrm{d}u = f^0(t)\,\mathrm{d}t$, turns both into integrals over $[0, 1]$ of the uniform staircase G_n of Theorem 17.8, so both are distribution-free in the same way. With $U_{(i)} = F^0(X_{(i)})$ they are computed as

$$W_n^2 = \frac{1}{12n} + \sum_{i=1}^{n} \left(U_{(i)} - \frac{2i-1}{2n} \right)^2, \qquad A_n^2 = -n - \frac{1}{n} \sum_{i=1}^{n} (2i-1) \Big(\log U_{(i)} + \log \left(1 - U_{(n+1-i)} \right) \Big)$$

(Exercise 17.7 derives the first), and their 5% thresholds are about 0.46 and 2.49 for large n.

The weight in A_n^2 has a reason. The gap at t has variance $F(t)(1 - F(t))/n$ (Proposition 17.2), which falls to 0 in the tails. The Kolmogorov–Smirnov statistic and W_n^2 measure raw gaps, which are naturally small in the tails, so they hardly look there. A_n^2 divides each squared gap by its variance, as Pearson's statistic divided each squared deviation by its expected count in Chapter 16, and a gap in the tails then counts as much as one in the middle.

Figure 17.4 puts the three ideas side by side for a Gaussian null with $n = 50$: the Kolmogorov–Smirnov test, the Anderson–Darling test, and Pearson's test on the counts in 4 or in 10 cells equally likely under the null, each with level 5% or just below, its threshold simulated under the null. When the mean is off by 0.4 standard deviations they reject 69%, 78%, 57% and 42% of the time. Binning gives something up: each observation becomes the name of its cell, and with more cells each count is noisier. When the spread is 1.5 times too large the order changes: 36%, 81%, 40% and 60%. A change of spread moves the tails most, where the Kolmogorov–Smirnov statistic hardly looks, and when the spread is 0.7 times what it should be, the two binned tests (47% and 42%) beat both of the others (17% and 26%). No test wins everywhere; the Anderson–Darling test loses least across these alternatives.

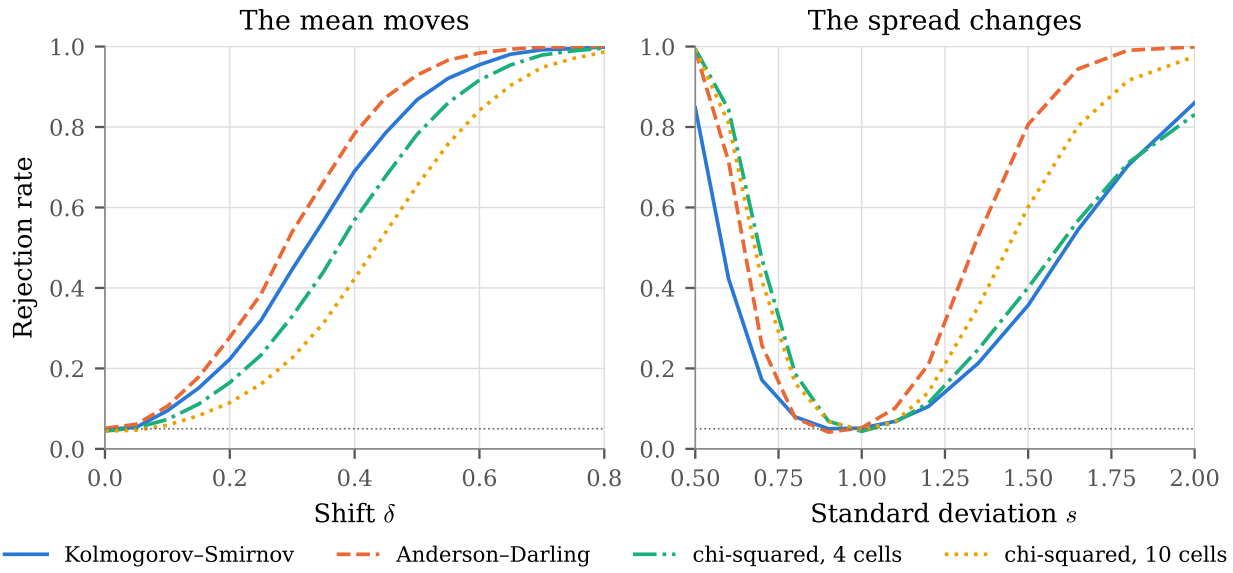


Figure 17.4: Rejection rates of four 5% tests of Gauss$(0, 1)$ with $n = 50$: Kolmogorov–Smirnov, Anderson–Darling, and Pearson's test on 4 or 10 equally likely cells. Left: data from Gauss$(\delta, 1)$. Right: data from Gauss$(0, s^2)$. Gray dotted line: 5%.

17.6 Testing a family: the Lilliefors test

The question asked of data is rarely 'is this Gauss$(0, 1)$?' and usually 'is this Gaussian?', with the mean and variance unknown; before the t-test of Chapter 14, for instance. The obvious move is to plug in estimates,

$$\hat{D}_n = \sup_t |F_n(t) - \Phi\Big(\frac{t - \bar{X}_n}{S_n}\Big)|,$$

with $\bar{X}_n$ and S_n the sample mean and standard deviation (with divisor $n - 1$, as in Chapter 14; the thresholds below use it), and to compare $\hat{D}_n$ with the Kolmogorov–Smirnov threshold.

Pitfall: the Kolmogorov–Smirnov table with estimated parameters

The fitted Gaussian is chosen to match the data, so it sits closer to the empirical CDF than the true Gaussian does, and $\hat{D}_n$ is systematically smaller than D_n. The left of Figure 17.5 shows the two null distributions for $n = 50$: 95% of the time $\sqrt{n}\hat{D}_n$ is below 0.88, while the Kolmogorov–Smirnov threshold is 1.33. Using that threshold for $\hat{D}_n$ gives a test that rejects a true Gaussian model fewer than 2 times in 10,000 samples, at $n = 50$ and at $n = 100$ alike, where it should reject 500 times. Chapter 16 met the same effect for the chi-squared test: a parameter fitted to the data makes the fit look better than it is, which there cost a degree of freedom for each parameter fitted from the counts, and left no exact correction when the parameters came from the raw data (a Gaussian in four cells, compared with χ^2_1 instead of χ^2_3, rejected 10.5% of the time in the limit instead of 5%: the correction overshot). For the Kolmogorov–Smirnov statistic the effect is larger, and no change of degrees of freedom repairs it.

The repair is to find the null distribution of $\hat{D}_n$ itself, and it can be found once and for all.

Proposition 17.9 (The fitted statistic is pivotal)**.** *Let* $X_1, \dots, X_n \overset{iid}{\sim}$ *Gauss*(μ, σ^2), $n \geq 2$. *The*

distribution of $\hat{D}_n$ does not depend on μ or σ^2: it is the distribution of the same statistic computed from n i.i.d. standard Gaussians.

Proof strategy (core). Standardize the data. Shifting and rescaling a sample shifts and rescales its empirical CDF, its mean and its standard deviation together, so every gap stays where it was.

Proof. Write $X_i = \mu + \sigma Z_i$ with $Z_1, \dots, Z_n \overset{iid}{\sim} \text{Gauss}(0,1)$, and let $\bar{Z}_n$, S_n^Z and F_n^Z be the mean, standard deviation and empirical CDF of the Z_i. Then $\bar{X}_n = \mu + \sigma\bar{Z}_n$ and $S_n = \sigma S_n^Z$. For every t, put $s = (t-\mu)/\sigma$: $X_i \le t$ exactly when $Z_i \le s$, so $F_n(t) = F_n^Z(s)$, and $(t - \bar{X}_n)/S_n = (s - \bar{Z}_n)/S_n^Z$. As t runs over the line so does s, so

$$\hat{D}_n = \sup_s |F_n^Z(s) - \Phi\Big(\frac{s - \bar{Z}_n}{S_n^Z}\Big)|,$$

a function of $Z_1, \dots, Z_n$ alone. □

Definition 17.10 (The Lilliefors test for normality)**.** For H_0: the X_i are i.i.d. Gauss(μ, σ^2) for some μ and some $\sigma^2 > 0$, against H_1: they are not, the **Lilliefors test** of level α rejects when $\hat{D}_n > \ell_\alpha(n)$, the $1-\alpha$ quantile of $\hat{D}_n$ under H_0, computed by simulating samples of n standard Gaussians (Proposition 17.9).

In words. Fit the Gaussian with the sample mean and standard deviation, measure the largest gap as before, and compare it with a threshold made for fitted Gaussians, about two thirds of the Kolmogorov–Smirnov one.

For the kiss data. The Gaussian with the kiss data's mean 0.645 and standard deviation 0.480 gives $\hat{D}_{124} = 0.415$, attained just below 1, where the staircase is at 0.355 and the fitted Gaussian at 0.770; the 5% threshold for $n = 124$ is 0.080. Two-valued data are far from Gaussian, which is why the t-test of Chapter 14 is only asymptotically valid for them.

n	5	10	20	50	100
Kolmogorov–Smirnov, $d_{0.05}(n)$	0.563	0.409	0.294	0.188	0.134
Lilliefors, $\ell_{0.05}(n)$	0.343	0.262	0.192	0.125	0.089

The table gives 5% thresholds for both tests; the Lilliefors values come from 200,000 simulated samples for each n, and for large n they are about $0.9/\sqrt{n}$, against $1.36/\sqrt{n}$. Lilliefors (*Journal of the American Statistical Association*, 1967) made the first such table by simulation. The proof of Proposition 17.9 works for any family obtained by shifting and rescaling one distribution, with estimates that shift and rescale along with the data, and also for the exponential scale family (Exercise 17.8). For families without this invariance the null distribution depends on the true parameter, and simulating samples from the fitted distribution is the usual approximation.

Worked example: Newcomb's measurements of the speed of light

In 1882 Simon Newcomb measured, 66 times, the time light takes to travel to a distant mirror and back, a known distance of about 7.4 kilometers; the values are in nanoseconds above 24,800 (Stigler, *Annals of Statistics*, 1977). Sixty-four of them lie between 16 and 40; the other two are −44 and −2. All 66 have mean 26.2 and standard deviation 10.7, and the fitted Gaussian is $\hat{D}_{66} = 0.231$ from the staircase, against the 5% threshold $\ell_{0.05}(66) = 0.109$: none of 40,000

simulated Gaussian samples came this far, and the test rejects. Without the two low values, the 64 that remain have mean 27.75 and standard deviation 5.08, and $\hat{D}_{64} = 0.090$ is below the threshold $\ell_{0.05}(64) = 0.111$: no evidence against a Gaussian shape. The two values that sink the test stand out at once in the QQ plot of Section 17.7. The modern value of the speed of light corresponds to 33.0 in these units, so the 64 good measurements also share a systematic error of about 5, which no test of shape can see.

The right of Figure 17.5 is the Old Faithful staircase against its fitted Gaussian, with mean 70.9 and standard deviation 13.6. The fitted Gaussian puts probability 0.62 below 75 minutes, while only 46% of the observed waits are that short; the gap, $\hat{D}_{272} = 0.155$, is almost three times the threshold $\ell_{0.05}(272) = 0.054$, and none of 20,000 simulated samples came this far. The waits are a mixture of two kinds of eruption (Figure 11.7), and a single Gaussian is the wrong shape.

Rounded data. Newcomb's values are whole numbers and the waits are whole minutes, so both samples contain ties. Proposition 17.6 still computes the statistic, but rounding alone makes the staircase take bigger steps, which moves the null distribution of $\hat{D}_n$ up; the thresholds above are for unrounded data. Gaussian samples with the fitted mean and standard deviation, rounded the same way, give 5% thresholds of 0.116 for Newcomb's 66 values, 0.131 for his 64 good ones, where the p-value of $\hat{D}_{64} = 0.090$ rises from 0.21 to 0.56, and 0.063 for the 272 waits. The two low values still sink Newcomb's test, the other 64 still pass, and Old Faithful's 0.155 is still far above its threshold.

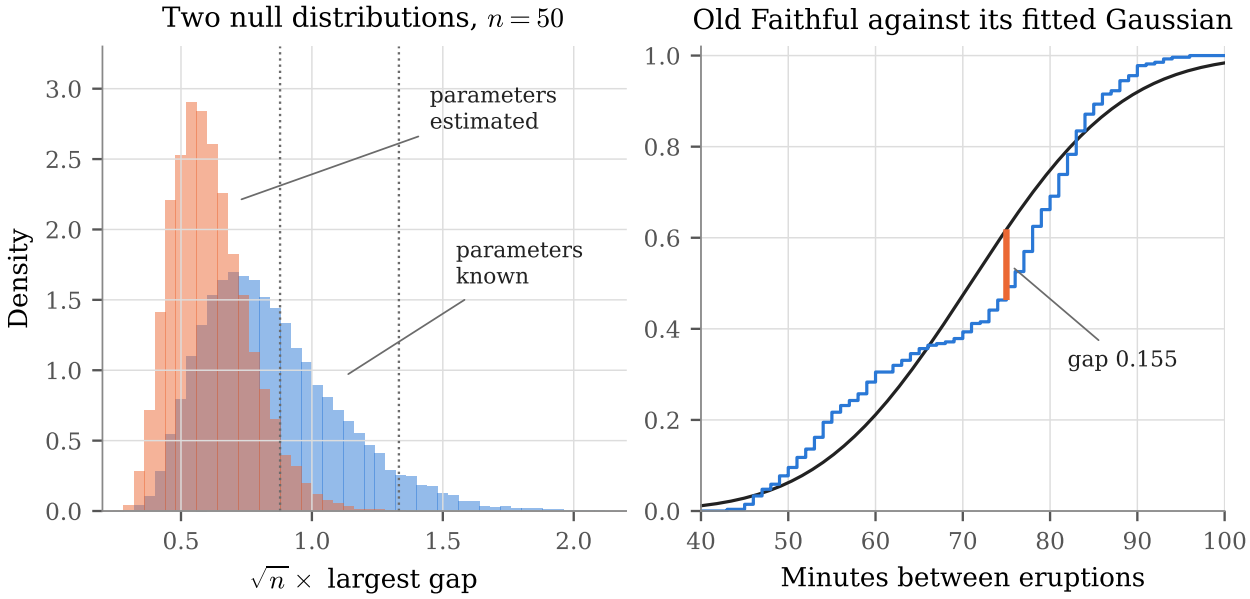


Figure 17.5: Left: for $n = 50$, $\sqrt{n}$ times the largest gap to the true Gaussian (blue) and the fitted one (orange); 95% points 1.33 and 0.88 dotted. Right: Old Faithful's waits (blue), fitted Gaussian (black), largest gap (orange).

Not rejecting is weak evidence when the sample is small. With 12 observations, the size of the wind-turbine sample of Chapter 14, the Lilliefors test detects exponential data only 36% of the time and data from Student's t with 3 degrees of freedom 18% of the time; with 20 observations, 58% and 26%.

From undergraduate statistics: checking normality before a t-test

An introductory course asks for a look at a histogram, or a normal probability plot, before a t-test on a small sample, and software often reports a formal test of normality, commonly the Shapiro–Wilk test, which is more powerful than Lilliefors' against many alternatives. The Lilliefors test is the formal version of the histogram check, and the QQ plot of the next section is the normal probability plot. Formal tests are limited the way Chapter 16 found: a small sample shows little, as the 36% above says, and a large sample finds every departure, including ones that do not matter for a t-test, whose level the CLT protects at large n (Chapter 14).

17.7 QQ plots

A test answers yes or no. A QQ plot shows where the data depart from a candidate, and how. It compares quantiles instead of CDF heights: the ith smallest observation $X_{(i)}$ is a sample quantile of order about i/n, and if the data come from F it should be close to the population quantile of the same order.

Definition 17.11 (QQ plot)**.** Let F be a continuous CDF, strictly increasing where $0 < F < 1$, with quantile function F^{-1} (Chapter 6). The **QQ plot** of a sample against F is the set of points

$$\left(F^{-1}\left(\frac{i-1/2}{n}\right),\ X_{(i)}\right), \qquad i = 1, \ldots, n.$$

Against $F = \Phi$, the standard Gaussian CDF, it is called a **normal QQ plot**.

In words. Sort the data, and pair the ith smallest value with the point below which the candidate puts a fraction $(i - \frac{1}{2})/n$ of its probability. Some authors use i/n and drop the last point, whose quantile $F^{-1}(1)$ is infinite; the picture is the same.

For the kiss data. Against Gaussian quantiles the kiss data plot as two flat rows, 44 points at height 0 and 80 at height 1. A two-valued sample gives a staircase, never a line.

If $X \sim \text{Gauss}(\mu, \sigma^2)$, its quantile of order p is $\mu + \sigma\Phi^{-1}(p)$, so the normal QQ plot of Gaussian data lies near a straight line with slope σ and height μ at 0, whatever μ and σ are. That is why one normal QQ plot checks the whole Gaussian family: the question is whether the points lie on some line, not on a particular one. The figures below draw the line through the points at the first and third quartiles, which outliers cannot pull.

Intuition: reading the ends of a QQ plot

For data with a single peak, the middle of a QQ plot is usually close to straight, and the ends carry the information. Figure 17.6 shows four samples of 200 against Gaussian quantiles. Heavy tails, as for Student's t with 3 degrees of freedom: the smallest values are smaller and the largest larger than a Gaussian with the same quartiles would give, so the left end falls below the line and the right end rises above it. Light tails, as for the uniform: the opposite, with both ends bending back toward the horizontal. Skewed right, as for the exponential: the lower tail is short and the upper tail long, so the points bend upward, above the line at both ends and below it in the middle. Skewed left: the mirror image, bending down. The skewed-right plot is the picture Chapter 10 promised for exponential data and their closest Gaussian.

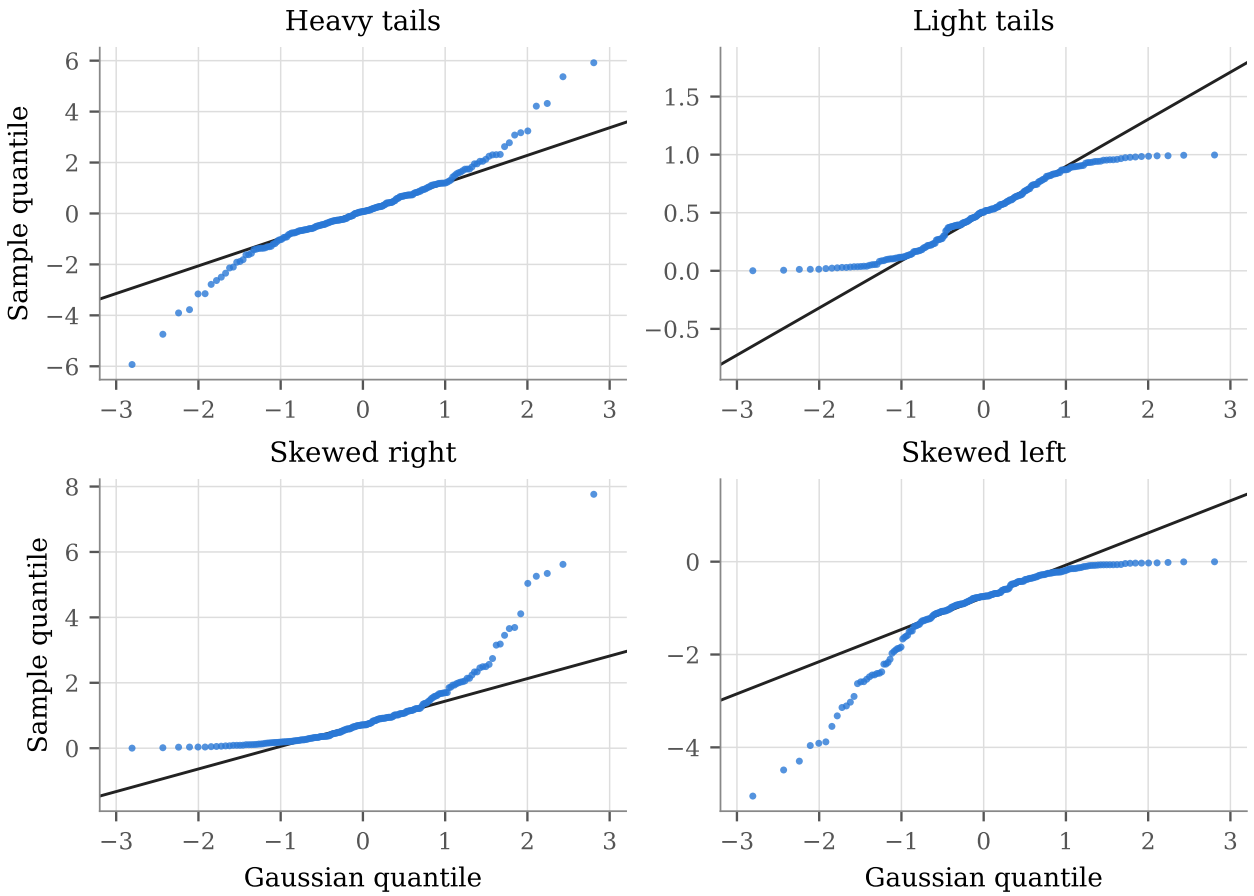


Figure 17.6: Normal QQ plots of samples of $n = 200$ (blue), each with the line through its quartiles (black). Heavy tails: Student's t, 3 degrees of freedom. Light tails: Unif$(0, 1)$. Skewed right: Expo(1). Skewed left: minus an Expo(1).

Figure 17.7 reads two real samples beside a simulated one. On the left are 66 standard Gaussians, for calibration: the points wander around the line, most at the ends, where order statistics are noisiest, and a pattern in a small sample should be judged against plots like this one. In the middle, Newcomb's 64 good measurements line up, and -44 and -2 sit far below the line; no test statistic points to the two culprits as directly. On the right, Old Faithful's waits rise gently, then steeply, then gently again: two clumps, around 55 and 80 minutes, with few waits in between.

The QQ plot and the Kolmogorov–Smirnov statistic compare the same two curves: the QQ plot measures the distance between them horizontally, at a fixed probability, and the statistic vertically, at a fixed t. In the tails, where a CDF is nearly flat, two CDFs can be far apart horizontally and still close vertically: the extreme quantiles differ a lot while the probabilities differ little. So the QQ plot magnifies the tails, which is where the Kolmogorov–Smirnov statistic hardly looks.

17.8 What comes next

Every method in the book so far has treated the parameter as a fixed unknown and the data as the only random thing. Chapter 18 begins Part VII, where the parameter gets a distribution of its

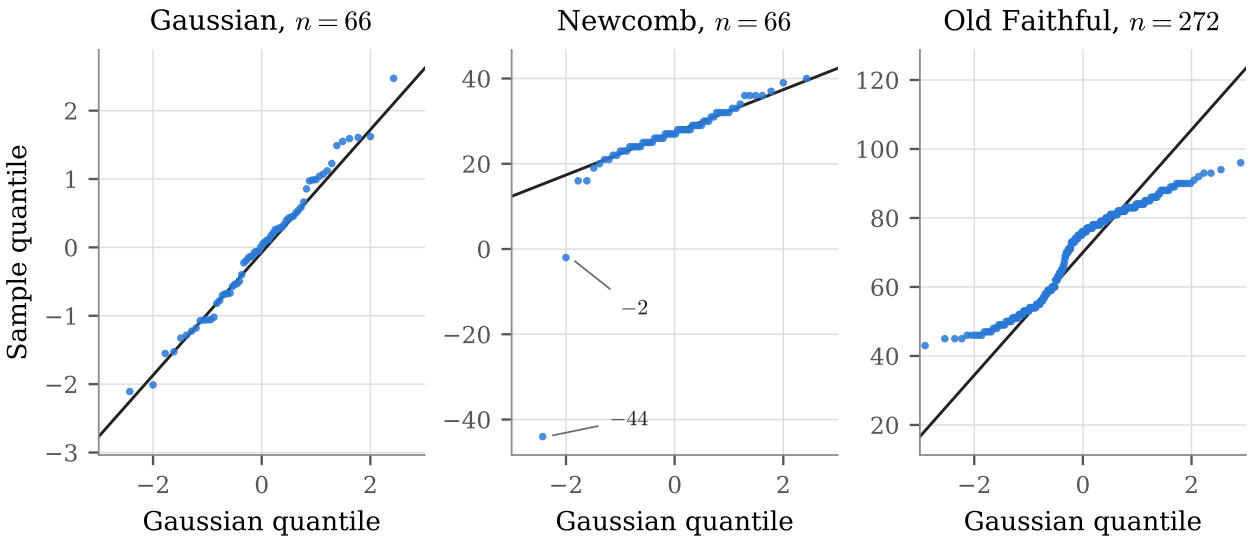


Figure 17.7: Normal QQ plots (blue) with the line through the quartiles (black). Left: 66 simulated standard Gaussians. Middle: Newcomb's 66 measurements, nanoseconds above 24,800. Right: Old Faithful's 272 waits, in minutes.

own: a prior, updated by the data into a posterior, with the kiss data's p the first example. The likelihood of Chapter 11 is one of the two ingredients, and the answer is a distribution for p rather than an interval or a test.

17.9 Summary

- The empirical CDF $F_n(t)$ is the proportion of the sample at or below t: at each t a binomial proportion, with variance $F(t)(1-F(t))/n$. Old Faithful: $F_{272}(60) = 0.305$.
- Glivenko–Cantelli: $\sup_t |F_n(t) - F(t)| \xrightarrow[n\to\infty]{\mathbb{P}} 0$ for every F, proved by cutting the line at finitely many quantiles and using monotonicity. The Kolmogorov distance can be estimated where total variation cannot.
- Kolmogorov and Donsker: for continuous F, $\sqrt{n}\sup_t |F_n(t) - F(t)| \xrightarrow[n\to\infty]{(\mathrm{d})} K$, the largest absolute value of a Brownian bridge; $q_{0.05}(K) = 1.358$. The band $F_n \pm 1.358/\sqrt{n}$ covers F with probability close to 95% for large n when F is continuous, and with 1.3581 in place of 1.358 it covers F with probability at least 95% at every n, whatever F is (Dvoretzky–Kiefer–Wolfowitz).
- The Kolmogorov–Smirnov test of a stated continuous F^0 uses D_n, the largest of the $2n$ numbers $i/n - F^0(X_{(i)})$ and $F^0(X_{(i)}) - (i-1)/n$. By the probability integral transform its null distribution is the same for every continuous F^0; the test is consistent against every alternative.
- Cramér–von Mises and Anderson–Darling add up squared gaps; Anderson–Darling divides each by its variance and so sees the tails. Binning for a chi-squared test loses power against shifts.
- With parameters estimated, the Kolmogorov–Smirnov table gives a test that almost never rejects. For the Gaussian family the fitted statistic is pivotal, and the Lilliefors test simulates its null distribution: thresholds about $0.9/\sqrt{n}$ against $1.36/\sqrt{n}$. Newcomb: rejected with the two low values, not without them. Old Faithful: 0.155 against 0.054 (0.063 allowing for

rounding).

- A QQ plot pairs $X_{(i)}$ with $F^{-1}((i-\frac{1}{2})/n)$. A line means a good fit up to location and scale; heavy tails put the left end below the line and the right end above it, light tails bend both ends toward the horizontal, and skew bends the whole plot one way.

17.10 Exercises

Exercise 17.1. Let F be a continuous CDF, strictly increasing where $0 < F < 1$, and let $U \sim \text{Unif}(0,1)$. (a) Show that $F^{-1}(U)$ has CDF F. (b) For the exponential distribution with mean 5, find F^{-1}, and turn the uniform values 0.08 and 0.51 into exponential waits. (c) Show that Lemma 17.7 needs continuity: for $X \sim \text{Bern}(\frac{1}{2})$, find the distribution of $F(X)$. *(Solution on page 430.)*

Exercise 17.2. Five values are 0.8, 0.7, 0.4, 0.7 and 0.2. Compute the Kolmogorov–Smirnov statistic for H_0: the data are $\text{Unif}(0,1)$, directly from the empirical CDF, and compare it with the exact 5% threshold $d_{0.05}(5) = 0.563$ and with $1.358/\sqrt{5}$. What is odd about the data under a continuous null? *(Solution on page 430.)*

Exercise 17.3. Let G_n be the empirical CDF of n i.i.d. $\text{Unif}(0,1)$ variables and $0 < u < v < 1$. (a) Show that $\sqrt{n}(G_n(u) - u)$ and $\sqrt{n}(G_n(v) - v)$ have covariance $u(1-v)$. (b) Show that the pair converges in distribution to a Gaussian vector with this covariance, checking that the covariance matrix is positive definite. (c) Where is the limiting variance largest, and what is the correlation of the limit at $u = \frac{1}{4}$ and $v = \frac{3}{4}$? *(Solution on page 430.)*

Exercise 17.4. For the kiss data and F^0 the CDF of $\text{Bern}(\frac{1}{2})$: (a) show that $D_n = |\hat{p} - \frac{1}{2}|$ and that $\sqrt{n}D_n = |T_n|/2$, where $T_n = \sqrt{n}(\hat{p} - \frac{1}{2})/\frac{1}{2}$ is the statistic of Chapter 9; (b) under H_0, find the limit of $\sqrt{n}D_n$ and the limit of the probability that it exceeds 1.358; (c) say which step of the proof of Theorem 17.8 fails. *(Solution on page 430.)*

Exercise 17.5. Of the 272 Old Faithful waits, 83 are at most 60 minutes. (a) Compute the plug-in 95% interval for $F(60)$ and the 95% band of Section 17.3 at 60. (b) Why is the band wider? (c) The Gaussian CDF closest to the empirical CDF in Kolmogorov distance is 0.102 away. What does the band say about the Gaussian family, and at what level? *(Solution on page 430.)*

Exercise 17.6. (a) Show that the Kolmogorov distance between two distributions is at most their total variation distance. (b) For Expo(1) and Gauss(1, 1) the Kolmogorov distance is 0.159 and the total variation distance is 0.319; explain the value 0.159. (c) For a sample from a continuous F, the empirical distribution is at total variation distance 1 from F at every n (Section 10.4). Why does its Kolmogorov distance from F tend to 0 all the same? *(Solution on page 431.)*

Exercise 17.7. Let $U_{(1)} < \cdots < U_{(n)}$ be the order statistics of $U_i = F^0(X_i)$, with $U_{(0)} = 0$ and $U_{(n+1)} = 1$. (a) Show that $W_n^2 = n\int_0^1 (G_n(u) - u)^2 \, \mathrm{d}u$, with G_n the empirical CDF of the U_i. (b) Using that $G_n = i/n$ on $[U_{(i)}, U_{(i+1)})$, show that $W_n^2 = \frac{1}{12n} + \sum_i \left(U_{(i)} - \frac{2i-1}{2n}\right)^2$. (c) Compute W_{10}^2 for the ten generator outputs of the worked example in Section 17.4, and compare it with the 5% threshold 0.46. *(Solution on page 431.)*

Exercise 17.8. For the exponential family, let $\hat{D}_n = \sup_{t \geq 0} |F_n(t) - (1 - e^{-t/\bar{X}_n})|$, the largest gap to the exponential distribution with the sample mean as its mean. Show that if the X_i are i.i.d. Expo(λ), the distribution of $\hat{D}_n$ does not depend on λ, so that its thresholds can be found by simulating Expo(1) samples. *(Solution on page 431.)*

Exercise 17.9. For $X \sim \text{Expo}(1)$, the normal QQ plot of a large sample lies near the curve $h(x) = F^{-1}(\Phi(x)) = -\log(1 - \Phi(x))$. (a) Compute $h(-2)$, $h(0)$ and $h(2)$, using $\Phi(-2) = 0.023$. (b) Compare the slopes of the chords from -2 to 0 and from 0 to 2, and explain the upward bend of the skewed-right plot in Figure 17.6. (c) What is the curve for $\text{Unif}(0, 1)$ data, and how does it explain the light-tailed plot? *(Solution on page 431.)*

Part VII

Bayesian Statistics

The one part of the book where the parameter gets a distribution. Priors, posteriors and conjugacy (Chapter 18) set up the machinery; Jeffreys priors and Bayesian confidence regions (Chapter 19) finish it, and say how the result relates to the frequentist intervals of Chapter 8.

Chapter 18

Priors, Posteriors and Conjugacy

In this chapter

- The Bayesian model: a prior distribution for the parameter, the statistical model read as the distribution of the data given the parameter, and the posterior that Bayes' formula makes from the two. Computed first on a grid of nine values of the kiss proportion.
- Conjugate priors, which keep the posterior in the prior's family: Beta for a proportion, Gamma for a rate, Gaussian for a mean. The kiss data, the subway waits and the wind turbine, and the prior read as extra observations.
- Updating one observation at a time, and why it gives the same posterior as all the data at once.
- Summaries of the posterior: the posterior mean, which is best under squared error, the median, the mode, and the probability that the next couple turns right.
- Probabilities of intervals and of hypotheses: with a flat prior they come close to confidence levels and one-sided p-values while answering a different question, and a point null gets a very different number. The Bernstein–von Mises theorem: with enough data the prior stops mattering, and the posterior is the Gaussian of Chapter 12.

Chapter 17 ended the book's tests, all of which treated the parameter as a fixed unknown number. This chapter gives it a distribution.

18.1 A distribution for the parameter

Every chapter so far has treated p, the proportion of couples who turn their heads to the right, as a fixed unknown number. The data are random and p is not, so every conclusion has been a statement about a procedure: an interval that covers p in 95% of repetitions, a test that rejects a true null 5% of the time. Chapter 8 named the statement most people want instead, 'the probability that p is between 0.56 and 0.73 is 0.95', and noted that it needs p to have a distribution.

The Bayesian approach gives it one. Before the data, what is known about p is described by a probability distribution, the **prior**; after the data, by the conditional distribution of p given the data, the **posterior**. The model for the data is the one used all along, now read as the distribution of the data given p. Nothing here claims that p is random in the world: the couples have whatever preference they have. The distribution describes knowledge about p, and the data change the knowledge. That is the difference Chapter 1 promised, and the rest of the chapter computes with it.

A prior can be read in three ways, and the mathematics is the same for all of them. As a degree of belief: someone who is 90% sure that p lies between 0.4 and 0.6 can write that down as a distribution, and the rules of probability are what keep such beliefs consistent (if someone's betting odds break those rules, a bookmaker can choose bets against them that win whatever happens: the Dutch book argument). As a state of information: what was known before the study, from earlier studies or from constraints such as $0 \le p \le 1$. As a device: a way to build estimators, which are then judged by the criteria of Chapter 7 like any others.

From undergraduate statistics: Bayes' rule

An introductory course states Bayes' rule for two events, often with a medical test: the probability of the disease given a positive test is the probability of a positive test given the disease, times the prevalence, divided by the overall probability of a positive test. The formula of this chapter is the same rule with 'the parameter equals θ' in place of the disease and the observed data in place of the positive test. The prevalence becomes the prior, the test's accuracy becomes the likelihood, and the answer is the posterior. The denominator, the overall chance of what was observed, adds the numerator up over every value the parameter could take.

18.2 Bayes' formula for a parameter

Start with a parameter that can take only finitely many values, where the rule is the one for events. Suppose that before the kiss study p was thought equally likely to be any of $0.1, 0.2, \dots, 0.9$: prior probability $\frac{1}{9}$ each. A sequence of 124 outcomes with 80 right turns has probability $p^{80}(1-p)^{44}$ under each candidate (the probability of exactly 80 right turns in any order adds a binomial coefficient, the same for every candidate, which cancels). Bayes' rule multiplies each prior probability by this likelihood and divides by the sum over the nine candidates:

$$\mathbb{P}(p = 0.6 \mid \text{data}) = \frac{\frac{1}{9} \cdot 0.6^{80}\, 0.4^{44}}{\sum_{j=1}^{9} \frac{1}{9}(j/10)^{80}(1 - j/10)^{44}} = 0.58.$$

The prior probabilities cancel, since they are equal, and what remains is the likelihood at 0.6 as a share of its sum over the grid. In natural frequencies: picture 900,000 repetitions of the study, 100,000 for each candidate. Of the 100,000 with $p = 0.6$, about 4,378 see exactly 80 right turns; of those with $p = 0.7$, about 3,160; with $p = 0.5$, about 37; with $p = 0.8$, about 2; with the other five values, essentially none. Among the roughly 7,580 repetitions that saw what was seen, 58% had $p = 0.6$, 42% had $p = 0.7$, and half a percent had $p = \frac{1}{2}$. That is the posterior, drawn in Figure 18.1: the likelihood decides which candidates survive, and the prior, flat here, decides nothing.

For a parameter that varies continuously, sums become integrals.

Definition 18.1 (Prior and posterior)**.** Let $(E, (\mathbb{P}_\theta)_{\theta \in \Theta})$ be a statistical model for $X_1, \dots, X_n$ with likelihood $L_n(x_1, \dots, x_n; \theta)$ (Definitions 11.1 and 11.2), and write $\mathbf{x} = (x_1, \dots, x_n)$. A **prior** is a probability distribution on Θ with density π, or with PMF π when Θ is finite or countable (this π is unrelated to the mixture weight of Chapters 2 and 11). The **Bayesian model** takes θ random with distribution π and, given θ, the observations i.i.d. $\mathbb{P}_\theta$. The **posterior** is the conditional distribution of θ given $X_1 = x_1, \dots, X_n = x_n$, with density (or PMF)

$$\pi(\theta \mid \mathbf{x}) = \frac{L_n(\mathbf{x}; \theta)\, \pi(\theta)}{m(\mathbf{x})}, \qquad m(\mathbf{x}) = \int_\Theta L_n(\mathbf{x}; t)\, \pi(t)\, \mathrm{d}t,$$

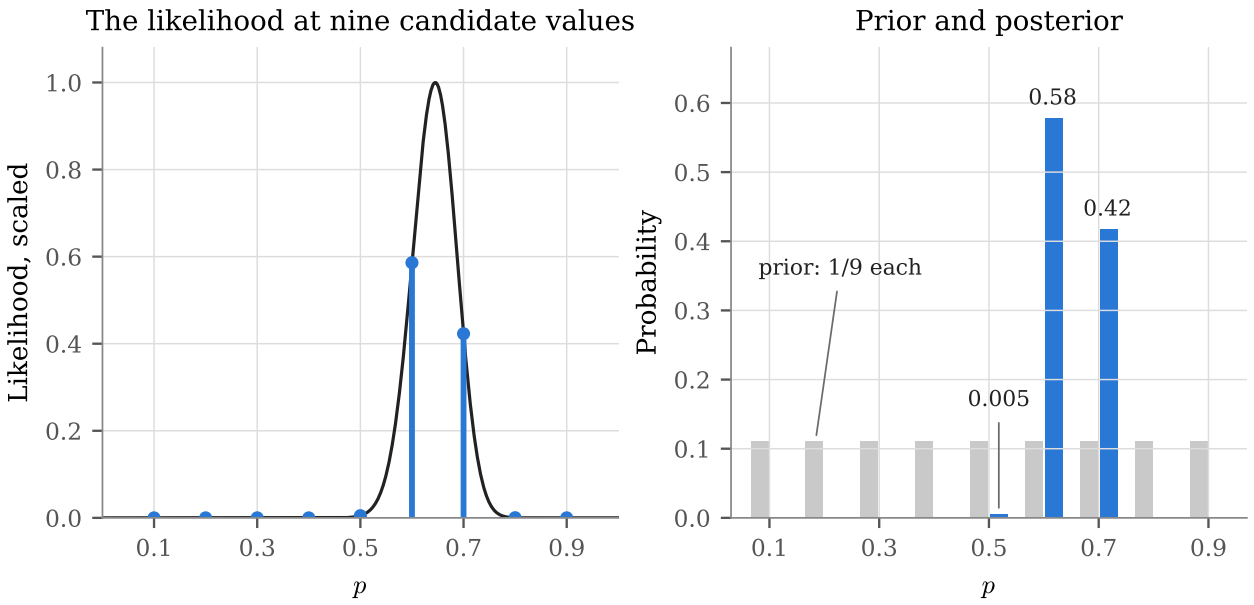


Figure 18.1: The kiss data and a prior of $\frac{1}{9}$ on each of $p = 0.1, 0.2, \ldots, 0.9$. Left: the likelihood $p^{80}(1-p)^{44}$ scaled to maximum 1 (black) and its values at the nine candidates (blue). Right: prior (gray) and posterior (blue).

defined whenever $0 < m(\mathbf{x}) < \infty$, with a sum in place of the integral when Θ is finite or countable. The denominator $m(\mathbf{x})$ is the **marginal likelihood** of the data.

> **In words.** Posterior is likelihood times prior, rescaled to total probability 1. Both are read as functions of θ with the data held fixed, and the marginal likelihood is the rescaling constant: the probability, or density, that the model and the prior together gave the data before they were seen.
>
> **For the kiss data.** The likelihood is $L_n = p^{80}(1-p)^{44}$, and with the flat prior $\pi(p) = 1$ on $(0, 1)$ the posterior density is $p^{80}(1-p)^{44}/m$, where $m = \int_0^1 t^{80}(1-t)^{44}\,\mathrm{d}t$. It is the black curve on the left of Figure 18.1, rescaled to be a density.

The condition $0 < m(\mathbf{x}) < \infty$ rules out two things: data that every allowed θ calls impossible, where $m(\mathbf{x}) = 0$, and a likelihood so large over some values of θ that the integral diverges, which cannot happen for discrete data, since then $L_n \le 1$ and so $m(\mathbf{x}) \le 1$. The condition holds in every example of this chapter. The formula is the definition of a conditional density. The joint density of the parameter and the data is $\pi(\theta)L_n(\mathbf{x};\theta)$, the density of θ times the density of the data given θ; the density of the data alone is this joint density integrated over θ, which is $m(\mathbf{x})$; and a conditional density is a joint density divided by the density of what is conditioned on, the continuous version of $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$. The same formula covers four cases: a parameter that is discrete or continuous, and data that are discrete or continuous. For discrete data L_n is a product of PMFs and for continuous data a product of densities; for a discrete parameter m is a sum over Θ and for a continuous one an integral. (With discrete data and a continuous parameter the joint distribution is a PMF in the data and a density in θ.)

Since $m(\mathbf{x})$ does not depend on θ, it never needs computing to find the shape of the posterior. A nonnegative function with a finite, nonzero integral is proportional to exactly one density, so it is enough to compute $L_n(\mathbf{x};\theta)\pi(\theta)$, drop every factor that does not involve θ, and recognize what is

left. The symbol $\propto$, 'is proportional to as a function of θ', records this:

$$\pi(\theta \mid \mathbf{x}) \propto L_n(\mathbf{x};\theta)\,\pi(\theta).$$

Pitfall: a prior of zero cannot be undone

Where $\pi(\theta) = 0$, the product $L_n\pi$ is 0, so the posterior is 0 there whatever the data say. In the grid of Figure 18.1, the value 0.645 had prior probability 0, and the posterior puts nearly all its weight on its two neighbors, 0.6 and 0.7. A prior uniform on $[0.3, 0.8]$ would rule out every p above 0.8 for good, even if every couple turned right. A prior should give some probability to every value that is not impossible.

18.3 The Beta prior for a proportion

A grid works, but a family of densities on $(0,1)$ that matches the shape of the likelihood does better.

Definition 18.2 (The Beta distribution)**.** For $a, b > 0$, the **Beta distribution** Beta(a,b) has density

$$f(p) = \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}, \qquad 0 < p < 1, \qquad B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}\,\mathrm{d}t = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

where $\Gamma(a) = \int_0^\infty t^{a-1}e^{-t}\,\mathrm{d}t$, so that $\Gamma(k) = (k-1)!$ for $k = 1, 2, 3, \ldots$. Its mean is $a/(a+b)$, its variance $ab/\big((a+b)^2(a+b+1)\big)$, and, when $a, b > 1$, its mode is $(a-1)/(a+b-2)$ (Exercise 18.1). Beta$(1,1)$ is Unif$(0,1)$.

> **In words.** A flexible family on $(0,1)$: a larger a pulls the mass toward 1, a larger b toward 0, and $a+b$ sets how concentrated it is.
>
> **For the kiss data.** Beta$(1,1)$ is the flat prior. Beta$(33,33)$ has mean $\frac{1}{2}$ and puts 90% of its probability between 0.4 and 0.6: the prior of a skeptic who doubts a strong preference either way.

Theorem 18.3 (Beta prior, Bernoulli data)**.** *Let $p \sim$ Beta(a,b) and, given p, let $X_1, \ldots, X_n$ be i.i.d. Bern(p), with $S = \sum_i X_i$. The posterior is* Beta$(a+S, b+n-S)$.

Proof strategy (core). Multiply the prior by the likelihood and keep only what depends on p: what is left has the shape of a Beta density, with each exponent raised by a count.

Proof. By Definition 18.1,

$$\pi(p \mid \mathbf{x}) \propto p^S(1-p)^{n-S} \cdot \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)} \propto p^{a+S-1}(1-p)^{b+n-S-1}, \qquad 0 < p < 1.$$

Up to a constant this is the Beta$(a+S, b+n-S)$ density, and since both integrate to 1 the constant is 1. □

Intuition: the prior as imaginary couples

The parameters tell the story: updating adds the S right turns to a and the $n - S$ left turns to b, so in the posterior a and b play the part of right and left turns seen before the study. The posterior mean is

$$\frac{a+S}{a+b+n} = w\,\frac{a}{a+b} + (1-w)\,\frac{S}{n}, \qquad w = \frac{a+b}{a+b+n},$$

a weighted average of the prior mean and the sample proportion, with the prior weighted by its $a+b$ imaginary couples against the n real ones. For the kiss data the flat prior counts as 2 imaginary couples in the posterior mean (its density, $p^0(1-p)^0$, adds nothing to the likelihood), weight $w = 2/126 = 0.016$, and the posterior mean is $81/126 = 0.643$; Beta$(10, 10)$ is 20 couples, $w = 0.139$, mean $90/144 = 0.625$; the skeptic's Beta$(33, 33)$ is 66 couples, $w = 0.347$, mean $113/190 = 0.595$. As n grows with the prior fixed, w falls like $1/n$.

Figure 18.2 draws the three priors and their posteriors. The flat prior's posterior, Beta$(81, 45)$, is the likelihood made into a density, with standard deviation 0.043. The skeptic's posterior, Beta$(113, 77)$, is pulled a third of the way back toward $\frac{1}{2}$, yet even the skeptic now puts probability 0.996 on $p > \frac{1}{2}$. The 124 couples persuade someone who started 90% sure that any preference was weak that couples do prefer the right, though not that the preference is strong: the skeptic still puts 0.56 on $p < 0.6$.

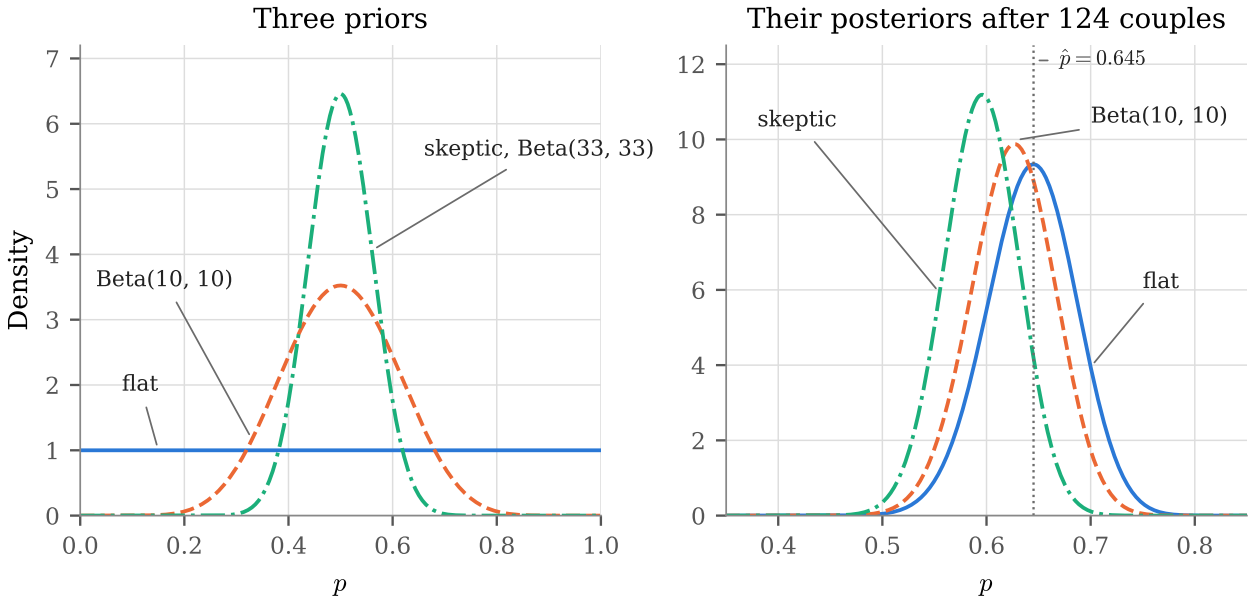


Figure 18.2: Left: the priors Beta$(1, 1)$ (blue), Beta$(10, 10)$ (orange, dashed) and the skeptic's Beta$(33, 33)$ (aqua, dash-dot). Right: their posteriors after 80 right turns among 124 couples, and $\hat{p} = 0.645$ (dotted).

The shrinkage estimator $(n\bar{R}_n + 10)/(n + 20)$ of Section 7.7, which pulled the estimate toward $\frac{1}{2}$ 'as if 20 extra coin flips had come up half heads', is the posterior mean under the Beta$(10, 10)$ prior: $90/144 = 0.625$ for the kiss data. The imaginary flips are the prior's $a + b = 20$ couples. Chapter 7 found that it has lower risk than $\bar{R}_n$ when p is near $\frac{1}{2}$ and higher risk far from it, which in Bayesian terms says that it does best where its prior puts the weight.

From undergraduate statistics: the plus-four estimate

The plus-four interval of Section 8.5, taught in some introductory courses, adds two successes and two failures and is centered at $\tilde{p} = (S+2)/(n+4)$ in place of $\hat{p}$. The estimate $\tilde{p}$ is the posterior mean under the Beta$(2,2)$ prior. For the kiss data it is $82/128 = 0.641$.

Definition 18.4 (Conjugate prior)**.** A family of priors is **conjugate** for a statistical model if, for every prior in the family and every sample, the posterior is again in the family.

In words. With a conjugate prior, updating changes the parameters of the prior and nothing else, so the posterior is a formula instead of an integral to compute.

For the kiss data. The Beta family is conjugate for Bernoulli data (Theorem 18.3): updating adds the number of right turns to a and the number of left turns to b.

Does this make sense? Where must the posterior mean land?

Between the prior mean and the sample proportion, by the weighted average above, and closer to whichever stands for more couples. A posterior mean outside $[0.5, 0.645]$ for a prior with mean $\frac{1}{2}$ and the kiss data signals an arithmetic error. The skeptic's 0.595 passes: $0.347 \times 0.5 + 0.653 \times 0.645 = 0.595$.

18.4 Rates and means: two more conjugate pairs

A rate can be any positive number and a mean any real number, so neither fits a prior on $(0,1)$. For a rate the conjugate family is the Gamma.

Definition 18.5 (The Gamma distribution)**.** For $\alpha, \beta > 0$, the **Gamma distribution** Gamma(α, β), with shape α and rate β, has density

$$f(\lambda) = \frac{\beta^{\alpha} \lambda^{\alpha-1} e^{-\beta\lambda}}{\Gamma(\alpha)}, \qquad \lambda > 0,$$

with mean α/β and variance α/β^2. Gamma$(1, \beta)$ is Expo(β).

In words. A flexible family on $(0, \infty)$, the natural home for a rate. Its mean α/β reads as α events in a time β.

For the kiss data. The kiss proportion lives in $(0,1)$, so its prior is a Beta, not a Gamma; the Gamma is for the subway rate λ, trains per minute, which can be any positive number.

Proposition 18.6 (Gamma prior, exponential data)**.** *Let* $\lambda \sim$ Gamma(α, β) *and, given* λ*, let* $T_1, \ldots, T_n$ *be i.i.d. Expo*(λ)*. The posterior is* Gamma$\big(\alpha + n, \beta + \sum_i T_i\big)$.

Proof strategy (for completeness; skip on a first reading). The same recognition as for the Beta: the likelihood $\lambda^n e^{-\lambda \sum T_i}$ has the shape of a Gamma density in λ.

Proof. $\pi(\lambda \mid \mathbf{t}) \propto \lambda^n e^{-\lambda \sum_i t_i} \cdot \lambda^{\alpha-1} e^{-\beta\lambda} = \lambda^{\alpha+n-1} e^{-(\beta + \sum_i t_i)\lambda}$ for $\lambda > 0$, the shape of the Gamma$(\alpha + n, \beta + \sum_i t_i)$ density. □

In the posterior's parameters the prior again acts as imaginary data, here α waits lasting β minutes in total. For the subway, suppose the timetable promises a train every 5 minutes, and take a prior with mean 0.2 trains per minute that counts as 30 waits: Gamma$(30, 150)$. The 64 observed waits total $64 \times 6.23 = 398.7$ minutes, so the posterior is Gamma$(94, 548.7)$, with mean $94/548.7 = 0.171$ and standard deviation $\sqrt{94}/548.7 = 0.018$. The maximum likelihood estimate is $1/6.23 = 0.161$. The reciprocal of the posterior mean, $548.7/94 = 5.84$ minutes per train, is the average of the timetable's 5 and the observed 6.23, weighted by 30 imaginary waits and 64 real ones.

Proposition 18.7 (Gaussian prior, Gaussian data with known variance)**.** *Let $\mu \sim$ Gauss(μ_0, σ_0^2) and, given μ, let $X_1, \dots, X_n$ be i.i.d. Gauss(μ, σ^2) with σ^2 known. The posterior is Gauss(μ_n, σ_n^2) with*

$$\frac{1}{\sigma_n^2} = \frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}, \qquad \mu_n = \sigma_n^2\Big(\frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}_n}{\sigma^2}\Big).$$

Proof strategy (for completeness; skip on a first reading). Multiply the two Gaussian shapes; the exponent is a quadratic in μ, and completing the square names its center and spread.

Proof. Using $\sum_i (x_i - \mu)^2 = \sum_i (x_i - \bar{x}_n)^2 + n(\bar{x}_n - \mu)^2$ and dropping factors free of μ,

$$\pi(\mu \mid \mathbf{x}) \propto \exp\Big(-\frac{n(\mu - \bar{x}_n)^2}{2\sigma^2} - \frac{(\mu - \mu_0)^2}{2\sigma_0^2}\Big) \propto \exp\Big(-\frac{\mu^2}{2}\Big(\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}\Big) + \mu\Big(\frac{n\bar{x}_n}{\sigma^2} + \frac{\mu_0}{\sigma_0^2}\Big)\Big).$$

Since $-\frac{\mu^2}{2\sigma_n^2} + \frac{\mu\mu_n}{\sigma_n^2} = -\frac{(\mu-\mu_n)^2}{2\sigma_n^2} + \frac{\mu_n^2}{2\sigma_n^2}$, the last expression is proportional to $\exp\big(-(\mu - \mu_n)^2/(2\sigma_n^2)\big)$, the shape of the Gauss$(\mu_n, \sigma_n^2)$ density. □

The reciprocal of a variance is a **precision**, and the proposition says that precisions add: the prior contributes $1/\sigma_0^2$, each observation $1/\sigma^2$, and the posterior mean is the average of μ_0 and $\bar{x}_n$ weighted by their precisions. The prior counts as σ^2/σ_0^2 observations. The same calculation works when the observations have different known variances σ_i^2: each x_i is weighted by $1/\sigma_i^2$, and the prior mean enters as one more observation with variance σ_0^2.

Chapter 14 asked whether the mean wind speed μ at a home is above 10 miles per hour, from 12 monthly measurements with mean 14.3 and standard deviation 4.7. Take $\sigma = 4.7$ as known, and a prior centered at 11, a regional average, with standard deviation 2. The prior counts as $4.7^2/2^2 = 5.5$ months of data, so the posterior mean is $(5.5 \times 11 + 12 \times 14.3)/17.5 = 13.26$ and the posterior standard deviation is $4.7/\sqrt{17.5} = 1.12$. Where the t-test of Chapter 14 rejected $\mu \le 10$, the posterior gives a probability: $\mathbb{P}(\mu > 10 \mid \text{data}) = 1 - \Phi\big((10 - 13.26)/1.12\big) = 0.998$. (Treating σ as unknown needs a prior for σ too. With σ independent of μ and a prior density proportional to $1/\sigma$, a common choice that Chapter 19 explains, the probability is 0.991. That density has an infinite integral, so it is not a probability distribution and Definition 18.1 does not cover it; Chapter 19 says why the posterior is still a proper distribution.) Figure 18.3 draws both examples, and in both the posterior lies between the prior and the likelihood and is narrower than either; on the right, the area under the posterior to the right of the gray line at 10 is the 0.998 above.

model for the data	conjugate prior	posterior
Bern(p)	$p \sim$ Beta(a, b)	Beta$\big(a + \sum X_i, b + n - \sum X_i\big)$
Pois(λ)	$\lambda \sim$ Gamma(α, β)	Gamma$\big(\alpha + \sum X_i, \beta + n\big)$ (Exercise 18.4)
Expo(λ)	$\lambda \sim$ Gamma(α, β)	Gamma$\big(\alpha + n, \beta + \sum X_i\big)$
Gauss(μ, σ^2), σ^2 known	$\mu \sim$ Gauss(μ_0, σ_0^2)	Gauss(μ_n, σ_n^2) of Proposition 18.7
Gauss$(0, v)$	$v \sim$ Inverse Gamma(a, b)	Inverse Gamma$\big(a + \frac{n}{2}, b + \frac{1}{2}\sum X_i^2\big)$ (Exercise 18.6)

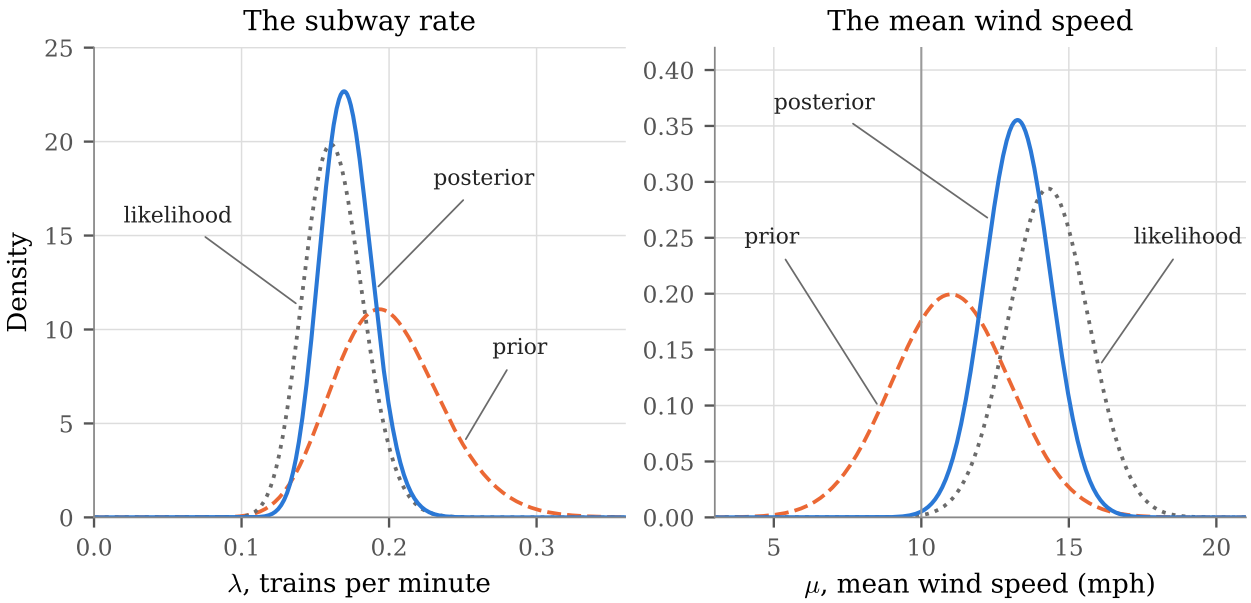


Figure 18.3: Prior (orange, dashed), likelihood scaled to a density (gray, dotted) and posterior (blue). Left: the subway rate, prior Gamma$(30, 150)$, posterior Gamma$(94, 548.7)$. Right: the mean wind speed, prior Gauss$(11, 2^2)$, posterior Gauss$(13.26, 1.12^2)$; gray line at 10.

(The Inverse Gamma is the distribution of $1/V$ for $V \sim \text{Gamma}(a, b)$.) Without a conjugate prior the posterior is still defined by the same formula, and only the computation changes. With one or two parameters a grid like Figure 18.1 with a fine mesh is enough. With many parameters the posterior is explored by simulating draws from it, with methods called Markov chain Monte Carlo, which are beyond this book.

18.5 One observation at a time

Does it matter whether the data arrive all at once or one couple at a time? It does not.

Proposition 18.8 (Updating in stages)**.** *For data i.i.d. given θ and any $m < n$, the posterior given $x_1, \ldots, x_n$ is the posterior obtained by using $\pi(\theta \mid x_1, \ldots, x_m)$ as the prior and $x_{m+1}, \ldots, x_n$ as the data.*

Proof strategy (for completeness; skip on a first reading). The likelihood of all the data is the product of the likelihoods of the two batches.

Proof. $L_n(x_1, \ldots, x_n; \theta) = L_m(x_1, \ldots, x_m; \theta)\, L_{n-m}(x_{m+1}, \ldots, x_n; \theta)$ (Chapter 11), so

$$\begin{aligned} \pi(\theta \mid x_1, \ldots, x_n) &\propto L_{n-m}(x_{m+1}, \ldots, x_n; \theta)\, \big[L_m(x_1, \ldots, x_m; \theta)\, \pi(\theta)\big] \\ &\propto L_{n-m}(x_{m+1}, \ldots, x_n; \theta)\, \pi(\theta \mid x_1, \ldots, x_m). \end{aligned}$$

□

Yesterday's posterior is today's prior. Applying the proposition repeatedly feeds the data in one observation at a time, and the order does not matter, since the likelihood is a product of one factor per observation. Figure 18.4 does it for the kiss data with the flat prior, with the 124 outcomes in a random order, since the study does not report the order. The first couple turned left, and the

posterior Beta$(1, 2)$ already has mean $\frac{1}{3}$; after 10 couples, 5 of them right, the posterior mean is $\frac{1}{2}$ and the middle 95% of the posterior runs from 0.23 to 0.77; after 40 couples it runs from 0.42 to 0.72, and after all 124 from 0.558 to 0.724, which is the solve interval of Chapter 8 to three decimals (Section 18.8 explains why the two nearly agree). Early on, each couple moves the posterior visibly; by the end, a single couple moves the mean by 0.003 if it turns right and by 0.005 if it turns left.

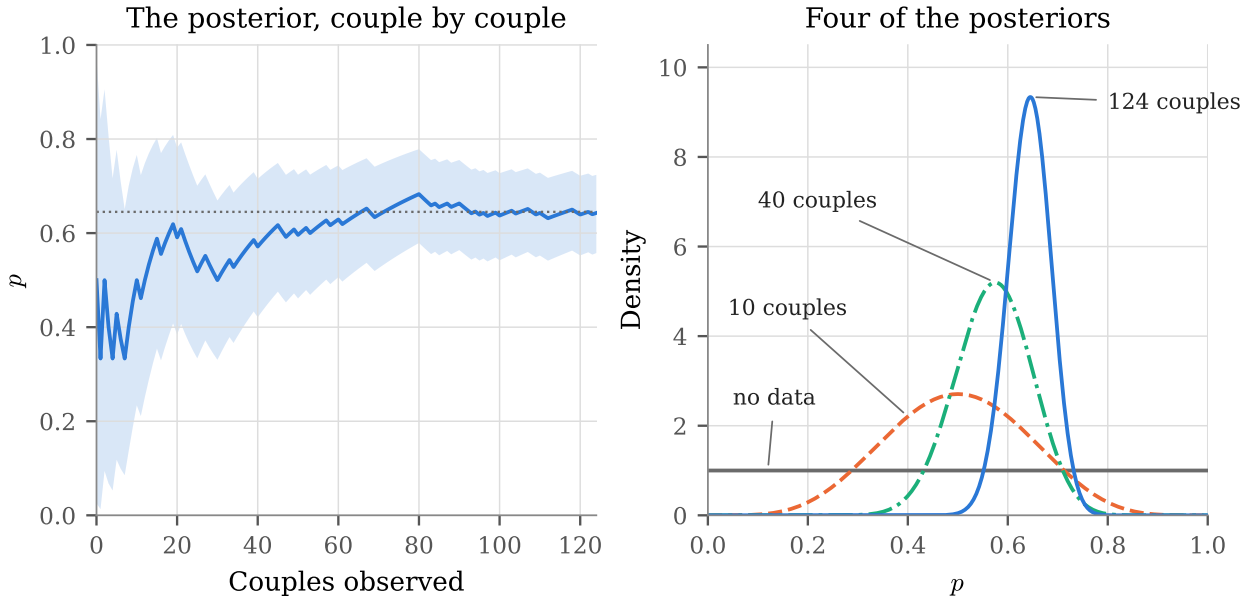


Figure 18.4: Kiss data, one couple at a time, flat prior. Left: posterior mean (blue), middle 95% (shaded), 0.645 (dotted). Right: posteriors after 0 (gray), 10 (orange, dashed), 40 (aqua, dash-dot) and 124 (blue) couples.

18.6 Summarizing the posterior

The posterior is the whole answer, and a report still needs numbers. Three of them describe where it sits.

Definition 18.9 (Posterior mean, median and mode)**.** The **posterior mean** $\mathbb{E}[\theta \mid \mathbf{x}] = \int_\Theta \theta\, \pi(\theta \mid \mathbf{x})\, d\theta$ is also called the **Bayes estimator** for squared error. The **posterior median** is a median of the posterior distribution. The **maximum a posteriori** (MAP) estimate is a value of θ that maximizes $\pi(\theta \mid \mathbf{x})$, the mode of the posterior.

> **In words.** The center of mass, the halfway point and the peak of the posterior. For a symmetric posterior with one peak they coincide; for a skewed one they spread out, usually in the order mode, median, mean, toward the long tail.
>
> **For the kiss data.** With the flat prior the posterior is Beta$(81, 45)$: mean $81/126 = 0.643$, median 0.644, mode $80/124 = 0.645$, which is $\hat{p}$.

Why the mean? Because it is the best single number under squared error, by the same identity as the bias–variance decomposition of Chapter 7.

Proposition 18.10 (The posterior mean minimizes squared error)**.** *For a posterior with finite variance and every number c,*

$$\mathbb{E}\left[(\theta - c)^2 \mid \mathbf{x}\right] = \mathbb{V}ar(\theta \mid \mathbf{x}) + \big(\mathbb{E}[\theta \mid \mathbf{x}] - c\big)^2.$$

So the posterior mean minimizes the posterior expected squared error, and the minimum is the posterior variance.

Proof strategy (for completeness; skip on a first reading). The bias–variance identity of Theorem 7.11, with the posterior in place of the sampling distribution and c in place of the true value: the cross term averages to zero.

Proof. Write $\bar{\theta} = \mathbb{E}[\theta \mid \mathbf{x}]$. Then $(\theta - c)^2 = (\theta - \bar{\theta})^2 + 2(\theta - \bar{\theta})(\bar{\theta} - c) + (\bar{\theta} - c)^2$. Take the posterior expectation of each term: the first gives $\mathbb{V}\mathrm{ar}(\theta \mid \mathbf{x})$; in the second, $\bar{\theta} - c$ is a number and $\mathbb{E}[\theta - \bar{\theta} \mid \mathbf{x}] = 0$; the third is a number. The first term does not involve c, and the last is smallest, 0, at $c = \bar{\theta}$. □

The same reasoning with absolute error picks the posterior median: Proposition 13.5, applied to the posterior distribution, says the median minimizes $\mathbb{E}[|\theta - c| \mid \mathbf{x}]$ (its hypotheses, a continuous density with a finite mean, hold for the Beta posteriors with both parameters above 1 and for the Gamma and Gaussian posteriors here). The mode answers a different question. Taking logarithms,

$$\log \pi(\theta \mid \mathbf{x}) = \ell_n(\theta) + \log \pi(\theta) - \log m(\mathbf{x}),$$

so the MAP estimate maximizes the log-likelihood plus $\log \pi(\theta)$, which is very negative where the prior is small and so works as a penalty. With a flat prior the penalty is constant and the MAP estimate is the maximum likelihood estimator; with a Gaussian prior $\mathrm{Gauss}(\mu_0, \sigma_0^2)$ it is $-(\theta - \mu_0)^2/(2\sigma_0^2)$ plus a constant, a quadratic penalty that pulls toward μ_0 (in regression the same penalty is called ridge).

Pitfall: the MAP estimate depends on the parameterization

The maximum likelihood estimator follows any one-to-one change of parameter (Proposition 11.6); the MAP estimate does not: a density changes shape when its variable is transformed. If $\psi = g(p)$ with g increasing, then $\mathbb{P}(\psi \le t) = \mathbb{P}(p \le g^{-1}(t))$, and differentiating in t shows that the density of ψ is the density of p times $\mathrm{d}p/\mathrm{d}\psi$. With the flat prior, the MAP estimate of p for the kiss data is 0.645. The log-odds $\psi = \log\big(p/(1-p)\big)$ has posterior density $\pi(p \mid \mathbf{x})\, p(1-p) \propto p^{81}(1-p)^{45}$, written in terms of p, which peaks at $p = 81/126 = 0.643$ (Exercise 18.8). The two MAP estimates describe different values of p. The posterior median follows every increasing change of parameter, since it is defined by probabilities.

The posterior also predicts. The probability that the next couple turns right, given the data, averages p over the posterior:

$$\mathbb{P}(X_{n+1} = 1 \mid \mathbf{x}) = \int_0^1 p\, \pi(p \mid \mathbf{x})\, \mathrm{d}p = \mathbb{E}[p \mid \mathbf{x}] = \frac{a + S}{a + b + n}.$$

With the flat prior this is $(S+1)/(n+2)$, Laplace's rule of succession: $81/126 = 0.643$ for the kiss data, close to the plug-in $\hat{p} = 0.645$. The two differ when data are few: after 3 right turns in 3 couples the plug-in answer is 1, a certainty that a fourth will turn right, and Laplace's rule gives 0.8. Laplace's 0.8 is the plug-in 1 pulled toward the prior mean $\frac{1}{2}$ by the two imaginary couples: $0.8 = \frac{2}{5} \times \frac{1}{2} + \frac{3}{5} \times 1$.

18.7 Probabilities of intervals and of hypotheses

Chapter 8 called '$\mathbb{P}(p \in [0.56, 0.73]) = 0.95$' a Bayesian statement and said this chapter would say what it costs. The posterior gives such probabilities directly. For the kiss data, $\mathbb{P}(0.56 \le p \le 0.73 \mid \text{data})$ is 0.955 with the flat prior, 0.941 with Beta(10, 10) and 0.835 with the skeptic's prior. The cost is the prior: the probability belongs to the prior and the data together, and a reader with a different prior gets a different number. With the flat prior it comes out near the confidence level 0.95 (0.952 for the plug-in interval $[0.561, 0.729]$ itself; rounding it out to $[0.56, 0.73]$ gives 0.955), and Section 18.8 explains why. The question Chapter 11 could not answer, the probability that p is between 0.6 and 0.7, now has an answer: 0.75 with the flat prior.

A one-sided hypothesis gets a probability the same way. With the flat prior, $\mathbb{P}(p \le \frac{1}{2} \mid \text{data}) = 0.0006$, the same number as the one-sided p-value of Chapter 9 by the Gaussian approximation (the exact Binomial p-value is 0.0008; Exercise 18.7 explains the near agreement). The numbers agree; the questions do not. The p-value is the probability of data this extreme if $p = \frac{1}{2}$; the posterior probability is the probability that $p \le \frac{1}{2}$ given the data.

A point hypothesis such as $H_0 : p = \frac{1}{2}$ is different, and Chapter 9 promised a different number for it. Under a continuous prior a single value has probability 0, before and after the data, so a prior that takes H_0 seriously must put a lump of probability on $p = \frac{1}{2}$. Take probability $\frac{1}{2}$ on $p = \frac{1}{2}$ and spread the other $\frac{1}{2}$ uniformly over $(0, 1)$. This prior has neither a density nor a PMF, so Definition 18.1 does not cover it, but Bayes' rule for the two events $H_0 : p = \frac{1}{2}$ and $H_1 : p \ne \frac{1}{2}$ does. Write m_0 and m_1 for the probability of the data under H_0 and under H_1; then $m(\mathbf{x}) = m_0\mathbb{P}(H_0) + m_1\mathbb{P}(H_1)$, $\mathbb{P}(H_j \mid \mathbf{x}) = m_j\mathbb{P}(H_j)/m(\mathbf{x})$ for $j = 0, 1$, and dividing the two gives the odds form,

$$\frac{\mathbb{P}(H_0 \mid \mathbf{x})}{\mathbb{P}(H_1 \mid \mathbf{x})} = \frac{m_0}{m_1} \cdot \frac{\mathbb{P}(H_0)}{\mathbb{P}(H_1)},$$

where $m_0 = 2^{-124} = 4.7 \times 10^{-38}$ and $m_1 = \int_0^1 p^{80}(1-p)^{44}\,\mathrm{d}p = 1.0 \times 10^{-36}$, the probability of the data averaged over the uniform prior: the area under the sequence likelihood that Chapter 11 said meant nothing as an area, and that means something here. The ratio $B_{01} = m_0/m_1$ is the **Bayes factor**; here it is 0.0465, the prior odds are 1, and so $\mathbb{P}(H_0 \mid \mathbf{x}) = 0.0465/1.0465 = 0.044$. Among 1,000 studies that saw 80 right turns, in a world where no preference and a uniformly spread preference were equally likely beforehand, about 44 had no preference. The two-sided p-value was 0.0012, about 36 times smaller. The data favor a preference by a factor of $1/B_{01} = 21$; the likelihood ratio of Chapter 15, $e^{10.60/2} = 200$, is about nine times larger, since it compares H_0 with the single best value $\hat{p}$, while the Bayes factor compares it with the average over every p, including values the data rule out.

The gap grows with n. Keep the two-sided p-value at 0.0012, so that the standardized statistic stays 3.23, and let the number of couples grow, with the same prior:

couples n	124	1,240	12,400	124,000	1,240,000
right turns	80	677	6,380	62,569	621,800
$\mathbb{P}(H_0 \mid \text{data})$	0.04	0.13	0.32	0.60	0.83

At a fixed p-value, a larger sample is stronger evidence for the point null, since a true preference would have produced a larger standardized statistic in a larger sample. Lindley (*Biometrika*, 1957) pointed out this divergence, now called Lindley's paradox; Berger and Sellke (*Journal of the American Statistical Association*, 1987) showed that, for a Gaussian mean, at a p-value of 0.05 the

posterior probability of a point null with prior probability $\frac{1}{2}$ is at least 0.29 for every prior on the alternative that is symmetric about the null value with a single peak there, and at least 0.128 for every prior whatsoever.

18.8 When the data outweigh the prior

The priors of Figure 18.2 disagree, and their posteriors disagree less. With more data they would disagree less still.

Recall. The Law of Large Numbers (Theorem 3.5): for i.i.d. random variables with mean μ and finite variance, $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$. Chebyshev's inequality (toolkit A.7): $\mathbb{P}(|Y - \mathbb{E}[Y]| \geq t) \leq \mathbb{V}\mathrm{ar}(Y)/t^2$ for any random variable Y with finite variance and any $t > 0$.

Theorem 18.11 (The prior washes out)**.** *Let $a, b > 0$, and let $X_1, \ldots, X_n$ be i.i.d. Bern(p^*), where p^* is the fixed true value. For the posterior from the prior* Beta(a, b)*, which depends on the data and is therefore random:*

1. *the posterior mean converges to p^* in probability;*
2. *for every $\varepsilon > 0$, the posterior probability $\mathbb{P}(|p - p^*| < \varepsilon \mid X_1, \ldots, X_n)$ converges to 1 in probability.*

Proof strategy (core). The weight on the prior mean is $(a+b)/(a+b+n)$, which goes to 0, and the sample proportion goes to p^*; the posterior variance is at most $1/(4n)$, so the posterior piles up at its mean.

Proof. (1) Fix $\varepsilon > 0$, let $S = \sum_i X_i$, and write $\bar{p}_n$ for the posterior mean. By the weighted average of Section 18.3, $\bar{p}_n - S/n = w_n\big(a/(a+b) - S/n\big)$ with $w_n = (a+b)/(a+b+n)$, which is at most w_n in absolute value since both $a/(a+b)$ and S/n lie in $[0,1]$. So $|\bar{p}_n - p^*| \leq w_n + |S/n - p^*|$. Once $w_n < \varepsilon/2$, the event $|\bar{p}_n - p^*| \geq \varepsilon$ forces $|S/n - p^*| \geq \varepsilon/2$, whose probability tends to 0 by the Law of Large Numbers.

(2) Write $a_n = a + S$ and $b_n = b + n - S$. The posterior variance is $a_n b_n / \big((a_n + b_n)^2(a_n + b_n + 1)\big)$, and $a_n b_n \leq (a_n + b_n)^2/4$, so it is at most $1/\big(4(a+b+n+1)\big)$. By Chebyshev's inequality applied to the posterior distribution, the posterior probability that p is at least $\varepsilon/2$ from $\bar{p}_n$ is at most $c_n = 1/\big(\varepsilon^2(a+b+n+1)\big)$. If p is within $\varepsilon/2$ of $\bar{p}_n$ and $\bar{p}_n$ is within $\varepsilon/2$ of p^*, then $|p - p^*| < \varepsilon$. So on the event E_n that $\bar{p}_n$ is within $\varepsilon/2$ of p^*, $Y_n = \mathbb{P}(|p - p^*| < \varepsilon \mid X_1, \ldots, X_n) \geq 1 - c_n$. Given $\eta > 0$, once $c_n < \eta$ the event $Y_n \leq 1 - \eta$ lies outside E_n, so $\mathbb{P}(|Y_n - 1| \geq \eta) \leq \mathbb{P}(E_n^c) \to 0$ by (1). □

Figure 18.5 draws it. At 124 couples the three posterior means of Figure 18.2 spread over 0.048, about one posterior standard deviation; with ten times as many couples in the same proportion, 800 of 1,240, they spread over 0.007, half a standard deviation of 0.014, and the three posteriors nearly coincide with a single Gaussian centered at $\hat{p}$ with the standard error $\sqrt{\hat{p}(1-\hat{p})/n}$ of Chapter 8. That Gaussian is the general limit.

Recall. Asymptotic normality of the maximum likelihood estimator (Theorem 12.5): under the regularity conditions of Section 12.6, with θ^* in the interior of Θ and $I(\theta^*) > 0$, $\sqrt{n}(\hat{\theta}_n - \theta^*) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}\big(0, 1/I(\theta^*)\big)$.

Theorem 18.12 (Bernstein–von Mises)**.** *Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathbb{P}_{\theta^*}$ in a model satisfying the regularity conditions of Section 12.6, with θ^* in the interior of Θ and $I(\theta^*) > 0$, let $\hat{\theta}_n$ be the maximum likelihood estimator, and suppose the data separate θ^* from distant values: for every $\varepsilon > 0$ there is a*

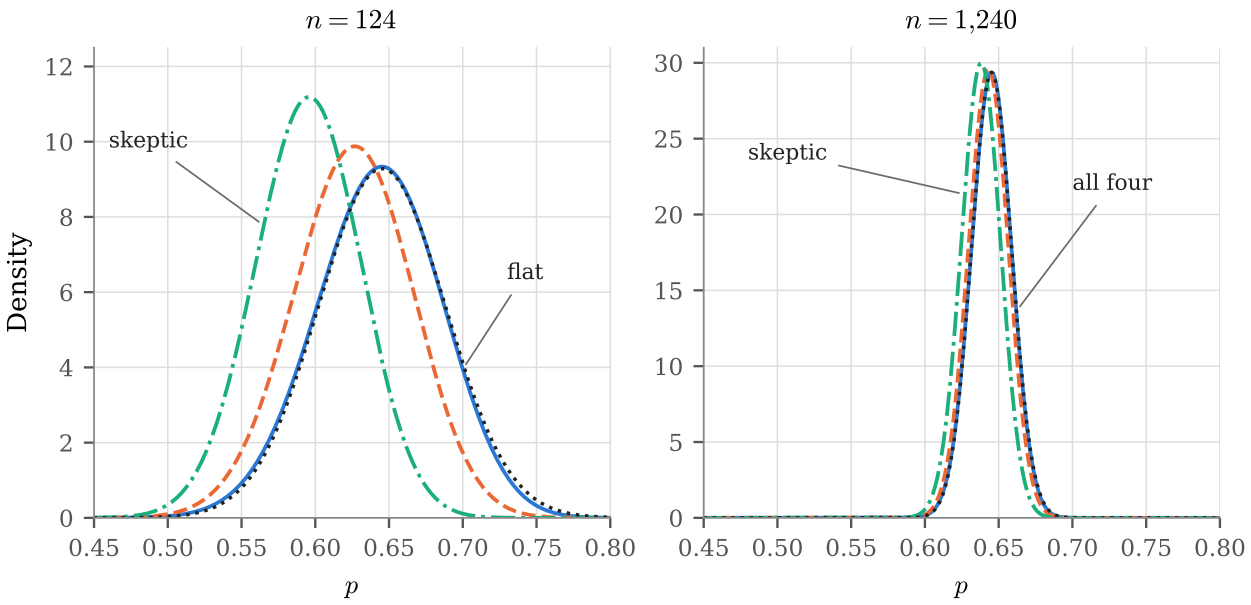


Figure 18.5: Posteriors from the flat prior (blue), Beta(10, 10) (orange, dashed) and the skeptic's Beta(33, 33) (aqua, dash-dot), with Gauss$\big(\hat{p}, \hat{p}(1-\hat{p})/n\big)$ (black, dotted). Left: 80 right turns in 124 couples. Right: 800 in 1,240.

$\delta > 0$ such that, with probability tending to 1, $\frac{1}{n}\big(\ell_n(\theta) - \ell_n(\theta^*)\big) \le -\delta$ *for every* θ *with* $|\theta - \theta^*| \ge \varepsilon$ *(this holds for the models of this chapter). Let the prior have a density that is continuous and positive at* θ^**. Then the total variation distance (Chapter 10) between the posterior distribution of* $\sqrt{n}(\theta - \hat{\theta}_n)$*, with* θ *random and* $\hat{\theta}_n$ *the number computed from the data, and Gauss*$\big(0, 1/I(\theta^*)\big)$ *converges to* 0 *in probability under* $\mathbb{P}_{\theta^*}$*. Informally, for large* n *the posterior is approximately Gauss*$\big(\hat{\theta}_n, 1/(nI(\hat{\theta}_n))\big)$*, for every such prior.*

This theorem is stated without proof; its proof needs more than this book develops. It is named for Bernstein (1917) and von Mises (1931), though Laplace had already found it for a proportion with a flat prior; its modern form is due to Le Cam (1953). The reason it holds is visible in the log-posterior, $\ell_n(\theta) + \log \pi(\theta)$ up to a constant. The log-likelihood grows with n, and near its peak it is a parabola whose curvature is about $nI(\theta^*)$ (Chapter 12); the log-prior stays the same size, and over the width of about $1/\sqrt{nI(\theta^*)}$ where the likelihood matters it is nearly constant, which is why the prior must be continuous and positive at θ^*. So for large n the log-posterior is that parabola, and a density whose logarithm is a parabola is Gaussian. Theorem 12.5 and Theorem 18.12 read the same parabola from two sides: the first describes how the estimate $\hat{\theta}_n$ scatters around the fixed θ^*, the second how the posterior spreads around the fixed $\hat{\theta}_n$, with the same variance $1/(nI)$. That is why the flat-prior probability of the 95% confidence interval came out near 0.95, and why Bayesian and frequentist intervals nearly coincide in large samples.

18.9 What comes next

Every posterior in this chapter came from a prior someone chose. Chapter 19 asks what to use when there is nothing to put in a prior: flat priors and their dependence on the parameterization, improper priors, whose density has an infinite integral and so cannot be rescaled to integrate to 1, and Jeffreys'

prior, built from the Fisher information so that it does not depend on the parameterization. It then builds Bayesian confidence regions, intervals with a stated posterior probability, and compares them with the intervals of Chapter 8, using the Bernstein–von Mises theorem.

18.10 Summary

- Bayesian model: a prior π on Θ, and the data i.i.d. $\mathbb{P}_\theta$ given θ. Posterior: $\pi(\theta \mid \mathbf{x}) \propto L_n(\mathbf{x};\theta)\pi(\theta)$, rescaled by the marginal likelihood $m(\mathbf{x})$. A parameter value with prior probability 0 keeps posterior probability 0.
- Grid: nine candidate values of the kiss proportion; the posterior puts 0.58 on 0.6 and 0.42 on 0.7.
- Conjugate pairs: Beta prior and Bernoulli data give Beta$(a+S, b+n-S)$; Gamma and exponential give Gamma$(\alpha+n, \beta+\sum T_i)$; Gaussian and Gaussian give precisions that add. The prior acts as imaginary data: $a+b$ couples, α waits, σ^2/σ_0^2 observations. Kiss posterior means: 0.643 (flat), 0.625 (Beta$(10,10)$, the shrinkage estimator), 0.595 (skeptic).
- Updating in stages, or one observation at a time in any order, gives the same posterior.
- Posterior mean: best under squared error, with the posterior variance as its error. Median: best under absolute error. MAP: maximizes the log-likelihood plus $\log \pi$, depends on the parameterization. The predictive probability of the next success is the posterior mean, Laplace's $(S+1)/(n+2)$ with a flat prior.
- Interval and one-sided probabilities depend on the prior (0.955, 0.941, 0.835 for $[0.56, 0.73]$). A point null needs prior mass; for the kiss data its posterior probability is 0.044 against a p-value of 0.0012, and at a fixed p-value it grows with n (Lindley).
- The prior washes out: with Beta priors the posterior concentrates at p^*; in general (Bernstein–von Mises) the posterior is approximately Gauss$\big(\hat\theta_n, 1/(nI(\hat\theta_n))\big)$.

18.11 Exercises

Exercise 18.1. Let $P \sim \text{Beta}(a,b)$. (a) Using $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ and $\Gamma(x+1) = x\Gamma(x)$, show that $\mathbb{E}[P] = a/(a+b)$ and $\mathbb{E}[P^2] = a(a+1)/\big((a+b)(a+b+1)\big)$, and deduce the variance. (b) Show that for $a, b > 1$ the density is largest at $(a-1)/(a+b-2)$. *(Solution on page 431.)*

Exercise 18.2. (a) With the flat prior on p, show that the number of successes S in n Bernoulli trials has $\mathbb{P}(S=s) = 1/(n+1)$ for every $s = 0, 1, \ldots, n$, using $B(s+1, n-s+1) = s!\,(n-s)!/(n+1)!$. (b) For the kiss data, compute $m_1 = \int_0^1 p^{80}(1-p)^{44}\,\mathrm{d}p$ in terms of a binomial coefficient, and check the Bayes factor $B_{01} = 0.0465$ of Section 18.7, given that $\binom{124}{80}2^{-124} = 0.000372$. *(Solution on page 432.)*

Exercise 18.3. In the drug trial of Chapter 1, earlier studies showed the drug to be about 80% effective. (a) Express this as a Beta$(40,10)$ prior: what is its mean, and how many patients does it count as? (b) After 78 cures in 100 patients, find the posterior, its mean and its standard deviation, and the weight on the prior mean. (c) Compute $\mathbb{P}(p > 0.7 \mid \text{data})$ approximately with a Gaussian approximation to the posterior. *(Solution on page 432.)*

Exercise 18.4. (a) Show that the Gamma family is conjugate for Poisson data: if $\lambda \sim \text{Gamma}(\alpha, \beta)$ and, given λ, $X_1, \ldots, X_n \overset{iid}{\sim} \text{Pois}(\lambda)$, the posterior is Gamma$(\alpha + \sum_i X_i, \beta + n)$. (b) In the siblings

data of Chapter 2, the 24 values of $X-1$ add up to 24. With the prior Gamma$(2,2)$, find the posterior mean and standard deviation of λ and compare with the maximum likelihood estimate. *(Solution on page 432.)*

Exercise 18.5. Let $X_1, \ldots, X_n$ be i.i.d. geometric given p, with $\mathbb{P}(X = x) = (1-p)^{x-1}p$ for $x = 1, 2, \ldots$ (the trial on which the first success occurs), and let $p \sim \text{Beta}(a, b)$. Find the posterior and the posterior mean. *(Solution on page 432.)*

Exercise 18.6. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(0, v)$ given v, and give v the Inverse Gamma prior with density $\pi(v) = \frac{b^a}{\Gamma(a)} v^{-a-1} e^{-b/v}$, $v > 0$, whose mean is $b/(a-1)$ for $a > 1$. (a) Show that the posterior is Inverse Gamma with parameters $a + n/2$ and $b + \frac{1}{2}\sum_i X_i^2$. (b) Find the posterior mean and its limit as a and b tend to 0 (for $n \geq 3$). Is it the maximum likelihood estimate $\frac{1}{n}\sum_i X_i^2$? *(Solution on page 432.)*

Exercise 18.7. (a) Show that if S of n trials are successes and the prior is flat, then $\mathbb{P}(p \leq x \mid \text{data}) = \mathbb{P}\big(\text{Bin}(n+1, x) \geq S+1\big)$ for $0 < x < 1$. (Hint: the $(S+1)$th smallest of $n+1$ independent Unif$(0,1)$ variables has the Beta$(S+1, n-S+1)$ distribution.) (b) For the kiss data, compare $\mathbb{P}(p \leq \frac{1}{2} \mid \text{data}) = \mathbb{P}(\text{Bin}(125, \frac{1}{2}) \geq 81)$ with the exact one-sided p-value $\mathbb{P}(\text{Bin}(124, \frac{1}{2}) \geq 80)$, which are 0.00060 and 0.00078. *(Solution on page 432.)*

Exercise 18.8. With the flat prior on p and the kiss data, (a) find the posterior density of the log-odds $\psi = \log\big(p/(1-p)\big)$, using $\mathrm{d}p/\mathrm{d}\psi = p(1-p)$, and show that it peaks where $p = 81/126$; (b) explain why the MAP estimate of ψ, transformed back, is not the MAP estimate of p, while the posterior median of ψ, transformed back, is the posterior median of p. *(Solution on page 432.)*

Exercise 18.9. With the flat prior and the kiss data, find the probability that the next 10 couples all turn right, $\mathbb{E}[p^{10} \mid \text{data}]$, as a product of 10 fractions, and compare it with the plug-in answer $\hat{p}^{10}$. *(Solution on page 432.)*

Chapter 19

Jeffreys Priors and Bayesian Intervals

In this chapter

- What a prior for 'knowing nothing' can be, and why the flat prior gives a different answer on each scale a parameter can be written on: the proportion p against its log-odds, a rate against a mean wait.
- Improper priors, whose densities have an infinite integral: when they still give a proper posterior, when they give none, and why they cannot be the prior for the parameter that a point null fixes.
- Jeffreys' prior, built from the Fisher information so that changing the parameterization changes nothing: Beta$(\frac{1}{2}, \frac{1}{2})$ for the kiss data, $1/\lambda$ for the subway rate, $1/\sigma$ for a scale. Why it puts equal mass on what the data can tell apart.
- Bayesian confidence regions, equal-tailed and highest-density, and what their 95% means. When they coincide with the intervals of Chapters 8 and 14 exactly (the subway's exact interval, the t-interval) and when only for large n (Bernstein–von Mises).

Chapter 18 turned a prior and the likelihood into a posterior. This chapter asks which prior to use when there is nothing to put in one, and what the posterior's intervals mean.

19.1 A prior for knowing nothing

Every posterior in Chapter 18 started from a prior somebody chose: a timetable's promise for the subway, a regional average for the wind turbine, a skeptic's 90% belief for the kiss data. Often there is nothing of the sort to write down, or the point of the analysis is to report what the data say on their own. The oldest answer is to give every value of the parameter the same density. Bayes (1763) did this for a proportion, and Laplace made it a rule, later called the principle of insufficient reason: with no reason to prefer one value, prefer none. For the kiss data the flat prior gives the posterior Beta$(81, 45)$ of Theorem 18.3, with mean 0.643 and middle 95% from 0.558 to 0.724.

The trouble is that 'the same density for every value' depends on how the values are written. The kiss proportion p can just as well be described by its log-odds, $\psi = \log(p/(1-p))$, the scale on which logistic regression works (Chapter 23), and someone who knows nothing about p knows nothing about ψ either. But a flat density for p is not flat for ψ. To find the density of ψ, write its CDF and differentiate (toolkit A.6): for a strictly increasing smooth map $\psi = g(p)$, $\mathbb{P}(\psi \le u) = \mathbb{P}(p \le g^{-1}(u))$,

and the derivative in u is the density of p at $g^{-1}(u)$ times the derivative of g^{-1}. In general, for a smooth one-to-one map $\varphi = g(\theta)$,

$$\pi_\varphi(\varphi) = \pi_\theta(\theta) \left| \frac{\mathrm{d}\theta}{\mathrm{d}\varphi} \right|, \qquad \theta = g^{-1}(\varphi), \tag{19.1}$$

where the absolute value covers decreasing maps. For the log-odds, $p = e^\psi/(1+e^\psi)$ and $\mathrm{d}p/\mathrm{d}\psi = p(1-p)$, so the flat prior on p becomes the density $p(1-p) = e^\psi/(1+e^\psi)^2$ on ψ: the logistic density, peaked at $\psi = 0$, which is $p = \frac{1}{2}$. It is far from ignorant about ψ: it puts probability 0.90 on $|\psi| \le 2.94$, which is the event $0.05 \le p \le 0.95$. Run the other way, a flat prior on ψ becomes the density $1/(p(1-p))$ on p, which piles up at 0 and at 1.

Figure 19.1 draws this, together with the prior this chapter ends up recommending. On the left, on the scale of p, the flat prior is the blue line. On the right, on the scale of ψ, the same prior is the blue bell, while the flat rule applied to ψ directly gives the gray line: two people who both claim to know nothing about the kiss data, one thinking in p and one in ψ, start from different priors and end with different posteriors. The orange curves are Jeffreys' prior of Section 19.3, a U on the left and a wider bell on the right. The orange dots on the right were computed on ψ directly, by Jeffreys' rule, and they land on the orange curve: that rule gives one answer on both scales.

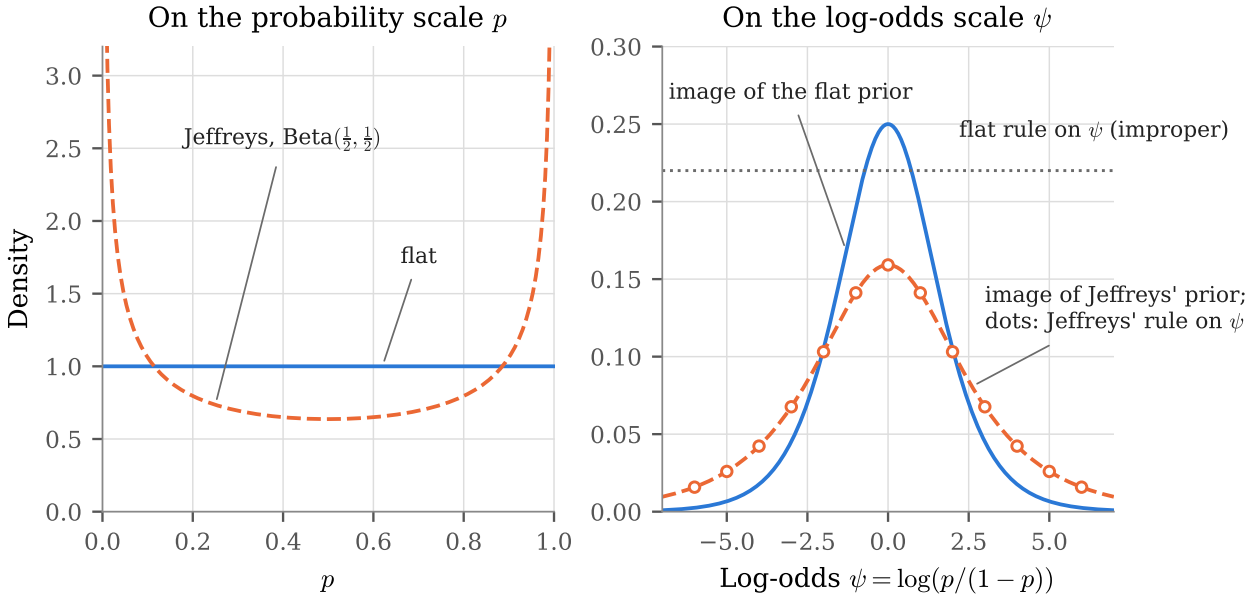


Figure 19.1: Two priors for the kiss proportion. Left, on p: flat (blue) and Jeffreys' Beta($\frac{1}{2}, \frac{1}{2}$) (orange, dashed). Right, on the log-odds ψ: their images (same colors), the flat rule on ψ (gray dotted, height arbitrary) and Jeffreys' rule on ψ (orange dots).

The subway gives a second example. Its rate λ, in trains per minute, and its mean wait $\beta = 1/\lambda$, in minutes per train, are two descriptions of one parameter. A flat prior on λ is the density 1 on $(0, \infty)$; a flat prior on β becomes, by Equation (19.1) with $|\mathrm{d}\beta/\mathrm{d}\lambda| = 1/\lambda^2$, the density $1/\lambda^2$ on λ. The 64 waits total 398.7 minutes, so the likelihood is $\lambda^{64}e^{-398.7\lambda}$ (Chapter 11), and the two posteriors are Gamma$(65, 398.7)$ and Gamma$(63, 398.7)$, with means 0.163 and 0.158 trains per minute and middle 95% intervals $[0.126, 0.205]$ and $[0.121, 0.199]$. Neither prior has a finite integral, which Section 19.2 takes up, yet both posteriors are ordinary Gamma distributions. At $n = 64$ the two posterior means differ by a quarter of a posterior standard deviation (0.005 against 0.020), a difference produced by nothing but the choice between 'rate' and 'wait'.

Pitfall: 'uninformative' is not a property of a prior alone

A prior that is flat for p is informative about ψ, about p^2 and about every other nonlinear function of p, and no density is flat on every scale at once. 'I used a flat prior' says nothing until it says flat *in what*. Chapter 18 met the same problem from the other side: the posterior mode moved from 0.645 to 0.643 when the kiss data were described by their log-odds (Exercise 18.8) because a mode is where a density peaks, and densities change under a change of variables.

19.2 Improper priors

The flat prior on the subway rate gives every $\lambda > 0$ the same density, and $\int_0^\infty 1 \, d\lambda = \infty$, so no rescaling makes it a probability distribution and Definition 18.1, which asks for one, does not cover it. Yet the posterior it produced, Gamma$(65, 398.7)$, is an ordinary distribution. Bayes' formula uses the prior only through the product $L_n\pi$ and its integral, and those can be finite when the integral of π is not.

Definition 19.1 (Improper prior)**.** An **improper prior** on Θ is a nonnegative function π on Θ with $\int_\Theta \pi(\theta) \, d\theta = \infty$. For data $\mathbf{x}$ with

$$0 < m(\mathbf{x}) = \int_\Theta L_n(\mathbf{x}; t) \, \pi(t) \, dt < \infty,$$

its **posterior** is $\pi(\theta \mid \mathbf{x}) = L_n(\mathbf{x}; \theta) \, \pi(\theta)/m(\mathbf{x})$, a probability density on Θ. A prior with a finite integral, rescaled to integrate to 1, is called **proper**.

> **In words.** An improper prior is a weighting of the parameter values whose total weight is infinite, so it is not a distribution. It is used only through Bayes' formula: when likelihood times prior has a finite integral, the posterior is a genuine distribution, and when it does not, there is no posterior. Multiplying an improper prior by a constant changes nothing, since the constant cancels.
>
> **For the kiss data.** The flat prior on the log-odds, $\pi(p) = 1/(p(1-p))$ on $(0, 1)$, is improper: its integral diverges at both ends. Times the likelihood $p^{80}(1-p)^{44}$ it gives $p^{79}(1-p)^{43}$, which has a finite integral, so the posterior is Beta$(80, 44)$, with mean $80/124 = 0.645$, the sample proportion.

This prior is known as Haldane's prior, after J. B. S. Haldane (*Proceedings of the Cambridge Philosophical Society*, 1932). Where improper priors come from is easiest to see for a Gaussian mean with σ known. There Proposition 18.7 turned the prior Gauss(μ_0, σ_0^2) into a posterior with precision $1/\sigma_0^2 + n/\sigma^2$, whose mean is the precision-weighted average of μ_0 and $\bar{x}$.

Intuition: an improper prior is the limit of wider and wider proper ones

Let the prior's standard deviation σ_0 grow. The prior flattens toward the constant 0, which is no density at all, but the posterior settles down: its precision tends to n/σ^2 and its mean to $\bar{x}$, so the posterior tends to Gauss$(\bar{x}, \sigma^2/n)$. That limit is what the improper flat prior $\pi(\mu) = 1$ gives in one step, since the likelihood is $\exp(-n(\mu - \bar{x})^2/(2\sigma^2))$ times a factor free of μ, already a Gaussian shape in μ. For the wind turbine with $\sigma = 4.7$ treated as known, the prior centered at 11 with $\sigma_0 = 2$ gave posterior mean 13.26; with $\sigma_0 = 10$ the mean is 14.24, and in the limit it is $\bar{x} = 14.3$, with standard deviation $4.7/\sqrt{12} = 1.36$. An improper prior is shorthand for 'a prior so wide that its exact width does not matter'.

The condition $m(\mathbf{x}) < \infty$ in Definition 19.1 is the price of that shorthand, and it can fail.

Pitfall: an improper prior can give no posterior

Under Haldane's prior, suppose all n couples had turned right. Then $L_n\pi = p^{n-1}(1-p)^{-1}$, whose integral diverges at $p = 1$, and there is no posterior at all. The formula still produces a function, and normalizing it numerically on a grid even produces a picture, but the picture depends on where the grid stops. With a proper prior and discrete data this cannot happen, since then $m(\mathbf{x}) \le 1$. The results of Chapter 18 that assume a prior distribution, Theorem 18.12 among them, do not apply as stated to an improper prior; for the improper priors of this chapter the Gaussian limit can be checked directly, since their posteriors are Gamma, Beta, Gaussian and t distributions.

Estimation and intervals tolerate improper priors because the arbitrary constant in front of the prior cancels between numerator and denominator. One calculation of Chapter 18 has no denominator to cancel it.

Pitfall: no improper prior for the parameter a point null fixes

Section 18.7 compared $H_0 : p = \frac{1}{2}$ with $H_1 : p \neq \frac{1}{2}$ through the Bayes factor $B_{01} = m_0/m_1$, where m_1 averages the likelihood over the prior on the alternative. If that prior is improper, it is defined only up to a constant c, and c multiplies m_1 but not m_0, which is computed at the single value $p = \frac{1}{2}$: the Bayes factor, and $\mathbb{P}(H_0 \mid \mathbf{x})$ with it, can be made anything. With Haldane's prior on the alternative, for instance, B_{01} is 0.011 for $c = 1$ and 0.000011 for $c = 1{,}000$ (Exercise 19.7). For testing, Jeffreys himself used proper priors for the parameter under test, such as a Cauchy prior for a Gaussian mean, and kept the improper $1/\sigma$ only for the σ that both hypotheses share, whose constant cancels between m_0 and m_1.

19.3 Jeffreys' prior

The flat prior failed because 'constant density' is a rule that produces a different prior on each scale it is applied to. Harold Jeffreys (*Proceedings of the Royal Society A*, 1946) asked for a rule that does not: apply it on the scale of θ and carry the result to $\varphi = g(\theta)$ by Equation (19.1), or apply it on the scale of φ directly, and get the same prior either way. A fixed formula in θ cannot do this, since it ignores the factor $|\mathrm{d}\theta/\mathrm{d}\varphi|$ that every change of variables puts on a density. The rule has to be built from something attached to the model that changes with the parameterization in step with densities, and the Fisher information is such a thing.

Recall. The Fisher information of one observation (Definition 12.3, Theorem 12.4): for a model satisfying the regularity conditions of Section 12.6, $I(\theta) = \mathbb{E}_\theta[\ell'(\theta)^2] = -\mathbb{E}_\theta[\ell''(\theta)]$, where ℓ is the log-likelihood of one observation. A large $I(\theta)$ means one observation pins θ down well: the maximum likelihood estimator has asymptotic variance $1/I(\theta)$, so its standard error is about $1/\sqrt{nI(\theta)}$ (Theorem 12.5).

Lemma 19.2 (Fisher information under a change of parameter)**.** *Let $\varphi = g(\theta)$ with g smooth and one-to-one and $g' \neq 0$, and write I_θ and I_φ for the Fisher information of one observation in the two parameterizations. Then*

$$I_\varphi(\varphi) = I_\theta(\theta)\left(\frac{\mathrm{d}\theta}{\mathrm{d}\varphi}\right)^2, \qquad \theta = g^{-1}(\varphi).$$

Proof strategy (for completeness; skip on a first reading). By the chain rule the score in φ is the score in θ times the number $\mathrm{d}\theta/\mathrm{d}\varphi$, and a constant factor comes out of a variance squared.

Proof. The model is the same, only the labels of its members change: the observation's density at the member labeled φ is $p_{g^{-1}(\varphi)}(x)$. So the log-likelihood in φ is $\tilde{\ell}(\varphi) = \ell(g^{-1}(\varphi))$, and by the chain rule $\tilde{\ell}'(\varphi) = \ell'(\theta)\,\mathrm{d}\theta/\mathrm{d}\varphi$ with $\theta = g^{-1}(\varphi)$. The factor $\mathrm{d}\theta/\mathrm{d}\varphi$ is a number, not a random variable, so $\mathbb{V}\mathrm{ar}(\tilde{\ell}'(\varphi)) = (\mathrm{d}\theta/\mathrm{d}\varphi)^2\,\mathbb{V}\mathrm{ar}(\ell'(\theta))$, which is the claim. □

Exercise 12.2 checked the lemma for the subway: $I = 1/\lambda^2$ for the rate and $1/\beta^2$ for the mean wait. Which function of the information should the prior be? Try a power, $\pi(\theta) \propto I_\theta(\theta)^a$. Applied on the scale of θ and carried to φ by Equation (19.1), it gives $I_\theta(\theta)^a|\mathrm{d}\theta/\mathrm{d}\varphi|$. Applied on the scale of φ directly, it gives, by Lemma 19.2, $I_\varphi(\varphi)^a = I_\theta(\theta)^a|\mathrm{d}\theta/\mathrm{d}\varphi|^{2a}$. The two are proportional for every change of variables only if $|\mathrm{d}\theta/\mathrm{d}\varphi|^{2a-1}$ is constant whatever g is, and that forces $2a = 1$.

Definition 19.3 (Jeffreys' prior)**.** For a one-parameter model with Fisher information $I(\theta)$, **Jeffreys' prior** is

$$\pi_J(\theta) \propto \sqrt{I(\theta)},$$

proper or improper according to whether $\int_\Theta \sqrt{I(\theta)}\,\mathrm{d}\theta$ is finite. For $\theta \in \mathbb{R}^d$ with Fisher information matrix $I(\theta)$, it is $\pi_J(\theta) \propto \sqrt{\det I(\theta)}$.

> **In words.** Take the information one observation carries about θ, take its square root, and use that as the prior density, up to a constant. The prior depends on the model, through I, and on nothing else: not on the data, not on anyone's beliefs. Using the information of all n observations, $nI(\theta)$, gives the same prior, since the factor $\sqrt{n}$ is a constant.
>
> **For the kiss data.** $I(p) = 1/(p(1-p))$ (Section 12.3.1), so $\pi_J(p) \propto p^{-1/2}(1-p)^{-1/2}$: the $\mathrm{Beta}(\frac{1}{2}, \frac{1}{2})$ density, which is proper, with constant $1/\pi$. The posterior is $\mathrm{Beta}(80.5, 44.5)$, with mean $80.5/125 = 0.644$ and middle 95% from 0.558 to 0.725, where the flat prior gave 0.643 and $[0.558, 0.724]$.

Theorem 19.4 (Jeffreys' prior does not depend on the parameterization)**.** *Let $\varphi = g(\theta)$ with g smooth and one-to-one and $g' \neq 0$. If θ has the prior $\pi_J^\theta \propto \sqrt{I_\theta}$, then the prior that Equation (19.1) gives φ is proportional to $\sqrt{I_\varphi(\varphi)}$, which is Jeffreys' prior for the model written in terms of φ. The same holds in $\mathbb{R}^d$, with $\sqrt{\det I}$ and the determinant of the matrix of partial derivatives.*

Proof strategy (core). The factor that a change of variables puts on a density is exactly the factor by which the square root of the information changes.

Proof. By Equation (19.1) and Lemma 19.2,

$$\pi_J^\theta(\theta)\left|\frac{\mathrm{d}\theta}{\mathrm{d}\varphi}\right| \propto \sqrt{I_\theta(\theta)}\left|\frac{\mathrm{d}\theta}{\mathrm{d}\varphi}\right| = \sqrt{I_\theta(\theta)\Big(\frac{\mathrm{d}\theta}{\mathrm{d}\varphi}\Big)^2} = \sqrt{I_\varphi(\varphi)}.$$

In $\mathbb{R}^d$, write J for the $d \times d$ matrix of partial derivatives $\partial\theta_i/\partial\varphi_j$. The chain rule makes the score in φ equal to $J^\top$ times the score in θ, so $I_\varphi = J^\top I_\theta J$ and $\det I_\varphi = (\det J)^2 \det I_\theta$. The density of φ is the density of θ times $|\det J|$ (the substitution rule of multivariable calculus, whose linear case is the change of variables formula of (toolkit A.13): near each point the map from φ to θ is, up to a shift and a negligible error, the linear map J, which multiplies small volumes by $|\det J|$), and $\sqrt{\det I_\theta}\,|\det J| = \sqrt{\det I_\varphi}$. □

Worked example: the kiss data on the log-odds scale

On $\psi = \log(p/(1-p))$, $\mathrm{d}p/\mathrm{d}\psi = p(1-p)$, so Lemma 19.2 gives $I_\psi(\psi) = \frac{1}{p(1-p)}\, p^2(1-p)^2 = p(1-p)$, and Jeffreys' rule applied on ψ gives $\pi_J(\psi) \propto \sqrt{p(1-p)}$, with $p = e^\psi/(1+e^\psi)$. Carrying $\mathrm{Beta}(\frac{1}{2}, \frac{1}{2})$ over by Equation (19.1) instead gives $\frac{1}{\pi\sqrt{p(1-p)}}\, p(1-p) = \sqrt{p(1-p)}/\pi$: the same function, now with its constant. In terms of ψ it is $1/(2\pi\cosh(\psi/2))$, the orange curve on the right of Figure 19.1, and the orange dots there are $\sqrt{I_\psi}/\pi$ at whole numbers ψ. The flat rule has no such property: applied on p and carried over it gives the logistic density $p(1-p)$, and applied on ψ it gives a constant.

Why the square root of the information, beyond the algebra? The information measures how finely the data can tell neighboring values of θ apart, and Jeffreys' prior turns out to give every distinguishable stretch of the parameter the same share.

Intuition: equal mass for what the data can tell apart

Near a value θ, the 95% interval of Chapter 8 has half-width about $1.96/\sqrt{nI(\theta)}$: values closer than that are hard to tell apart with n observations, and values much farther apart are easy. Where $I(\theta)$ is large these intervals are narrow and many of them fit into a unit of θ; where it is small they are wide. Jeffreys' prior gives each of them the same mass: its density $\sqrt{I(\theta)}$ times the width $3.92/\sqrt{nI(\theta)}$ is $3.92/\sqrt{n}$, whatever θ is. For the kiss model the binomial pins p down tightly near 0 and 1, where $p(1-p)$ is small, and loosely near $\frac{1}{2}$, so the U of $\mathrm{Beta}(\frac{1}{2}, \frac{1}{2})$ is the shape that gives every distinguishable interval the same share. Figure 19.2 shows five such intervals at $n = 124$: narrow under the tall ends of the U, wide under its low middle, and each with Jeffreys mass between 0.113 and 0.116 (the approximation $3.92/(\pi\sqrt{124})$, with the constant $1/\pi$, gives 0.112).

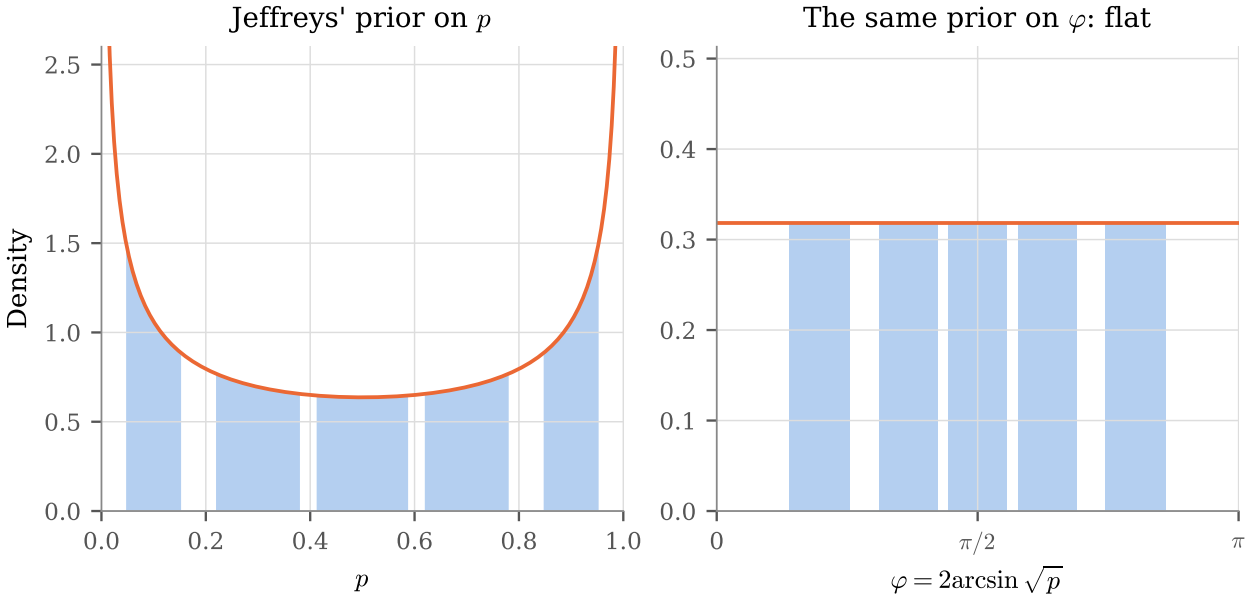


Figure 19.2: Jeffreys' prior for the kiss proportion (orange) with five bands $p_0 \pm 1.96\sqrt{p_0(1-p_0)/124}$, for $p_0 = 0.1, 0.3, 0.5, 0.7, 0.9$ (blue). Left: on p. Right: on $\varphi = 2\arcsin\sqrt{p}$, where the prior is flat.

On the right of Figure 19.2 the same bands are drawn on the scale $\varphi = 2\arcsin\sqrt{p}$, where they all

have nearly the same width and the prior is flat. That is not a coincidence: every one-parameter model has a scale on which its Jeffreys prior is flat.

Proposition 19.5 (Jeffreys' prior is flat where the information is constant)**.** *In a one-parameter model satisfying the regularity conditions of Section 12.6, with Θ an interval and $I(\theta)$ continuous and positive on it, let $\varphi(\theta) = \int_{\theta_0}^{\theta} \sqrt{I(t)}\,\mathrm{d}t$ for a fixed $\theta_0 \in \Theta$. Then $I_\varphi \equiv 1$, Jeffreys' prior is flat in φ, and the maximum likelihood estimator satisfies $\sqrt{n}\big(\varphi(\hat\theta_n) - \varphi(\theta)\big) \xrightarrow[n\to\infty]{(d)} \mathrm{Gauss}(0,1)$ under $\mathbb{P}_\theta$ for every θ in the interior of Θ: φ is a variance-stabilizing transformation.*

Recall. The delta method (Theorem 5.4): if $\sqrt{n}(Z_n - \theta) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0,\tau^2)$ and g is continuously differentiable at θ, then $\sqrt{n}\big(g(Z_n) - g(\theta)\big) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}\big(0, g'(\theta)^2\tau^2\big)$.

Proof strategy (for completeness; skip on a first reading). Lemma 19.2 with $\mathrm{d}\varphi/\mathrm{d}\theta = \sqrt{I(\theta)}$, then the delta method.

Proof. By the fundamental theorem of calculus, $\mathrm{d}\varphi/\mathrm{d}\theta = \sqrt{I(\theta)} > 0$, so φ is strictly increasing, and Lemma 19.2 gives $I_\varphi = I_\theta(\mathrm{d}\theta/\mathrm{d}\varphi)^2 = I_\theta/I_\theta = 1$. Jeffreys' prior in φ is then proportional to $\sqrt{1}$, a constant. By Theorem 12.5, $\sqrt{n}(\hat\theta_n - \theta) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, 1/I(\theta))$, and the delta method with $g = \varphi$, whose derivative $\sqrt{I}$ is continuous, gives the asymptotic variance $I(\theta)\cdot 1/I(\theta) = 1$. □

For the Bernoulli, $\int \mathrm{d}p/\sqrt{p(1-p)} = 2\arcsin\sqrt{p}$, which is why $\mathrm{Beta}(\frac12,\frac12)$ is also called the arcsine distribution. For the Poisson, $I(\lambda) = 1/\lambda$ and $\varphi = 2\sqrt{\lambda}$. For the exponential rate, $I(\lambda) = 1/\lambda^2$ and $\varphi = \log\lambda$: Jeffreys' prior $1/\lambda$ is flat in $\log\lambda$, and since $\log\beta = -\log\lambda$ it is flat in the log of the mean wait too. So for the subway, where the flat prior on the rate and the flat prior on the wait disagreed (Section 19.1), Jeffreys' rule gives one answer: the prior $1/\lambda$ for the rate is the prior $1/\beta$ for the wait, and the posterior is $\mathrm{Gamma}(64, 398.7)$, between the two flat answers, with posterior mean $64/398.7 = 0.1605$, the maximum likelihood estimate.

From undergraduate statistics: the arcsine and square-root transformations

Applied courses in experimental design traditionally transform proportions by $\arcsin\sqrt{\hat p}$ and counts by $\sqrt{x}$ before comparing groups, so that every group has about the same variance whatever its mean. These are the transformations of Proposition 19.5 for the Bernoulli and Poisson models, up to the factor 2. On those scales the Fisher information is constant, the standard error of the transformed estimate is $1/\sqrt{n}$ whatever the parameter, and Jeffreys' prior is flat. For the kiss data, $2\arcsin\sqrt{0.645} = 1.865$ with standard error $1/\sqrt{124} = 0.090$, and the interval $1.865 \pm 1.96 \times 0.090$, mapped back to p, is $[0.559, 0.727]$, close to the Jeffreys posterior interval $[0.558, 0.725]$.

Table 19.1 collects Jeffreys' prior for the book's one-parameter families, with the posterior after n observations whose sum is S. Only the first prior is proper, and each posterior is proper: for the Bernoulli always, for the Poisson, the exponential and the Gaussian mean once $n \geq 1$, and for the Gaussian scale once $n \geq 1$ and not every x_i equals μ.

model	I	Jeffreys' prior	posterior
Bern(p)	$1/(p(1-p))$	Beta($\frac{1}{2},\frac{1}{2}$)	Beta($S+\frac{1}{2}, n-S+\frac{1}{2}$)
Pois(λ)	$1/\lambda$	$\lambda^{-1/2}$, improper	Gamma($S+\frac{1}{2}, n$)
Expo(λ)	$1/\lambda^2$	$1/\lambda$, improper	Gamma(n, S)
Gauss(μ,σ^2), σ known	$1/\sigma^2$	1, improper	Gauss($\bar{x}, \sigma^2/n$)
Gauss(μ,σ^2), μ known, in σ	$2/\sigma^2$	$1/\sigma$, improper	$\sum_i (x_i-\mu)^2/\sigma^2 \sim \chi^2_n$

Table 19.1: Jeffreys' prior and the resulting posterior for the book's one-parameter families, with S the sum of the n observations.

19.4 Several parameters: a Gaussian with unknown variance

Chapter 18 needed a prior for the wind turbine's σ when it was unknown, used a density proportional to $1/\sigma$, independent of μ, and promised this chapter would explain it. The last row of the table is the explanation: $1/\sigma$ is Jeffreys' prior for σ when μ is known, flat in $\log\sigma$. With both parameters unknown, Definition 19.3 asks for the information matrix. Section 12.3.1 computed it for (μ,σ^2), and the matrix rule $I_\varphi = J^\top I_\theta J$ from the proof of Theorem 19.4, with J diagonal and $\mathrm{d}\sigma^2/\mathrm{d}\sigma = 2\sigma$, turns its entry $1/(2\sigma^4)$ into $(2\sigma)^2/(2\sigma^4) = 2/\sigma^2$:

$$I(\mu,\sigma) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 2/\sigma^2 \end{pmatrix}, \qquad \sqrt{\det I(\mu,\sigma)} = \frac{\sqrt{2}}{\sigma^2}.$$

So Jeffreys' rule applied to (μ,σ) together gives the prior $1/\sigma^2$, not $1/\sigma$. Jeffreys himself noticed the difference and recommended treating a location and a scale separately, each with its own one-parameter rule: flat in μ, times $1/\sigma$. The next result shows what the extra factor of $1/\sigma$ does.

Proposition 19.6 (The posterior of a Gaussian mean with unknown variance)**.** *Given (μ,σ), let $X_1,\dots,X_n$ be i.i.d. Gauss(μ,σ^2), with $n \geq 2$, and let $\bar{x}$ and $s>0$ be the observed sample mean and sample standard deviation. Take the improper prior $\pi(\mu,\sigma) = \sigma^{-k}$ on $\mathbb{R}\times(0,\infty)$, with $k=1$ or $k=2$. Then the posterior is proper, and the posterior distribution of μ, with $\bar{x}$ and s fixed, is given by*

$$\frac{\mu-\bar{x}}{s/\sqrt{n}} \sim t_{n-1} \quad (k=1), \qquad \frac{\mu-\bar{x}}{s\sqrt{(n-1)/n}/\sqrt{n}} \sim t_n \quad (k=2).$$

Proof strategy (core). Write the likelihood with $\sum_i (x_i-\mu)^2 = (n-1)s^2 + n(\bar{x}-\mu)^2$, integrate σ out with a Gamma integral, and recognize Student's t density in what is left.

Proof. Adding and subtracting $\bar{x}$ inside the square, the cross term vanishes, and $A(\mu) = \sum_i (x_i-\mu)^2 = (n-1)s^2 + n(\bar{x}-\mu)^2$. The likelihood is $(2\pi)^{-n/2}\sigma^{-n}\exp(-A(\mu)/(2\sigma^2))$, so the posterior is proportional to $\sigma^{-(n+k)}\exp(-A(\mu)/(2\sigma^2))$. For fixed μ, substitute $u = A(\mu)/(2\sigma^2)$, so that $\sigma = (A(\mu)/(2u))^{1/2}$ and $\mathrm{d}\sigma = -\frac{1}{2}(A(\mu)/2)^{1/2}u^{-3/2}\,\mathrm{d}u$:

$$\begin{aligned}\int_0^\infty \sigma^{-(n+k)} e^{-A(\mu)/(2\sigma^2)}\,\mathrm{d}\sigma &= \frac{1}{2}\Big(\frac{A(\mu)}{2}\Big)^{-(n+k-1)/2}\int_0^\infty u^{(n+k-3)/2}e^{-u}\,\mathrm{d}u \\ &= \frac{1}{2}\Big(\frac{A(\mu)}{2}\Big)^{-(n+k-1)/2}\Gamma\Big(\frac{n+k-1}{2}\Big),\end{aligned}$$

finite since $n+k-1 \geq 2$. So the posterior density of μ is proportional to $A(\mu)^{-(n+k-1)/2}$ and, after dividing $A(\mu)$ by the constant $(n-1)s^2$, to

$$\left(1+\frac{t^2}{n-1}\right)^{-(n+k-1)/2}, \qquad t=\frac{\mu-\bar{x}}{s/\sqrt{n}}.$$

For $k=1$ the exponent is $-((n-1)+1)/2$, and this is the t_{n-1} density of Proposition 14.6 in the variable t, without its constant. For $k=2$ the exponent is $-(n+1)/2$; with $t'=t\sqrt{n/(n-1)}$ the ratio $t^2/(n-1)$ becomes t'^2/n, and this is the t_n density in t'. Both are integrable in μ, so the posterior is proper. □

For the wind turbine, $n=12$, $\bar{x}=14.3$ and $s=4.7$. With the prior $1/\sigma$ the posterior of μ is $14.3+(4.7/\sqrt{12})\,t_{11}$. Its middle 95% is $14.3 \pm 2.20 \times 1.357 = [11.3, 17.3]$, the t-interval of Chapter 14 to the digit, and $\mathbb{P}(\mu > 10 \mid \text{data}) = \mathbb{P}(t_{11} > -3.17) = 0.9955$, which is one minus the t-test's one-sided p-value, 0.0045. With the prior $1/\sigma^2$ the posterior is narrower, $[11.5, 17.1]$, with $\mathbb{P}(\mu > 10 \mid \text{data}) = 0.9969$. Read as a frequentist procedure, that narrower interval covers the true μ only 93.9% of the time at $n=12$ (Exercise 19.6), since it pairs t_n quantiles with a scale that has t_{n-1} behind it. The product of the one-parameter rules reproduces the t pivot exactly, the joint rule does not, and that is the usual reason for preferring $1/\sigma$; it is also the prior Chapter 18 used. The same Gamma integral shows why the posterior there was proper, although that chapter kept the prior Gauss$(11, 2^2)$ for μ: integrating σ out leaves that Gaussian density times $A(\mu)^{-n/2}$, which is at most $((n-1)s^2)^{-n/2}$ since $A(\mu) \geq (n-1)s^2 > 0$, so the product is integrable; the 0.991 found there is $\mathbb{P}(\mu > 10)$ under that posterior.

Figure 19.3 draws the coincidence. On the left, before the data and supposing $\mu = 10$, the standardized mean $\sqrt{12}(\bar{X}_{12}-10)/S$ has the t_{11} distribution, and the area beyond the observed 3.17 is the p-value 0.0045. On the right, after the data, the posterior of μ is the same t_{11} shape, centered at 14.3 and scaled by 1.357, and the area below 10 is the same 0.0045, now a probability for μ. The blue area is the 95% t-interval, which here is also a region with posterior probability 0.95. One distribution answers both questions because the quantity $\sqrt{n}(\bar{x}-\mu)/s$ has the t_{n-1} distribution in both readings: over repeated samples with μ fixed, and over μ with the sample fixed.

From undergraduate statistics: the t-interval is also a Bayesian interval

An introductory course computes $\bar{x} \pm t^* s/\sqrt{n}$ and warns that 'there is a 95% probability that μ is in the interval' is the wrong sentence, since μ is fixed and the interval is random. Under the prior flat in μ times $1/\sigma$, Proposition 19.6 makes that sentence true, with the same numbers: the probability is a posterior probability, conditional on the data, and it holds under this prior, not under every prior. The z-interval with σ known has the same double reading under the flat prior on μ, whose posterior is Gauss$(\bar{x}, \sigma^2/n)$.

19.5 Bayesian confidence regions

A posterior answers 'where is θ?' with a whole distribution. A region that holds most of that distribution is the Bayesian counterpart of a confidence interval, and there is more than one way to choose it.

Definition 19.7 (Bayesian confidence region)**.** Let $\alpha \in (0,1)$, and let the posterior have a density. A **Bayesian confidence region of level** $1-\alpha$ is a set $R(\mathbf{x}) \subseteq \Theta$, computed from the data, with

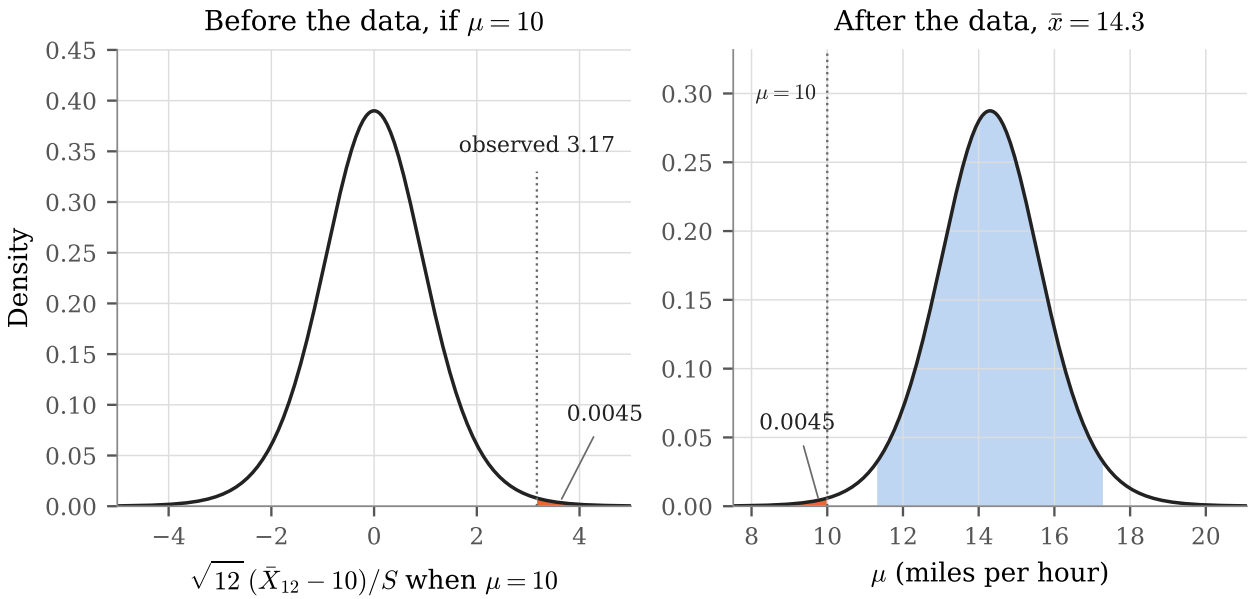


Figure 19.3: One t_{11} density read two ways, for the wind turbine. Left: the standardized mean when $\mu = 10$; orange, the area above the observed 3.17. Right: the posterior of μ under the prior $1/\sigma$; orange, the area below 10; blue, the 95% t-interval.

posterior probability $\mathbb{P}(\theta \in R(\mathbf{x}) \mid \mathbf{x}) = 1 - \alpha$. For a real parameter, the **equal-tailed** region is the interval from the point with posterior probability $\alpha/2$ below it to the point with posterior probability $\alpha/2$ above it. The **highest posterior density** region is $\{\theta : \pi(\theta \mid \mathbf{x}) \geq c\}$, with c chosen so that its posterior probability is $1 - \alpha$.

> **In words.** A region that holds 95% of the posterior. The equal-tailed one cuts 2.5% off each end; the highest-density one keeps the most probable values, those where the posterior density is above some height. Most texts call these credible regions, or credible intervals. This book calls $1 - \alpha$ the level, as for the confidence intervals of Chapter 8, although some texts call α the level of a Bayesian region.
>
> **For the kiss data.** Under Jeffreys' prior the posterior is Beta$(80.5, 44.5)$. Its equal-tailed 95% region is $[0.558, 0.725]$ and its highest-density region is $[0.560, 0.727]$, slightly to the right, toward the posterior's mode 0.646: the posterior has a longer tail to the left.

The two kinds of region promise different things. The highest-density region is the shortest region with the stated probability; the equal-tailed region follows the parameter through any monotone change of scale.

Proposition 19.8 (The highest-density region is the shortest)**.** *Let the posterior have a density f on $\mathbb{R}$, and let $H = \{\theta : f(\theta) \geq c\}$, with $c > 0$, have posterior probability $1 - \alpha$. Every set R with posterior probability at least $1 - \alpha$ is at least as long as H.*

Proof strategy (for completeness; skip on a first reading). Compare the parts where R and H differ: the part only in H has density at least c and the part only in R at most c, so R needs at least as much length to hold at least as much probability.

Proof. Write $|A|$ for the length of a set A and $P(A)$ for its posterior probability. Since $P(R) \geq P(H)$ and the two share $R \cap H$, $P(R \setminus H) \geq P(H \setminus R)$. On $R \setminus H$ the density is below c, so $P(R \setminus H) \leq c|R \setminus H|$;

on $H \setminus R$ it is at least c, so $P(H \setminus R) \geq c|H \setminus R|$. Hence $c|R \setminus H| \geq c|H \setminus R|$, and dividing by c and adding $|R \cap H|$ to both sides gives $|R| \geq |H|$. □

The equal-tailed region has the other property. If $\varphi = g(\theta)$ with g increasing, the point with posterior probability $\alpha/2$ below it for φ is g of the corresponding point for θ, so the equal-tailed region for φ is the image of the one for θ. The highest-density region has no such guarantee, since the posterior density of φ carries the factor $|\mathrm{d}\theta/\mathrm{d}\varphi|$ of Equation (19.1), which changes which values count as most probable.

The wind turbine's σ shows the difference. Under the prior flat in μ times $1/\sigma$, integrating μ out of $\sigma^{-(n+1)} \exp(-A(\mu)/(2\sigma^2))$ leaves the posterior $\sigma^{-n} \exp(-(n-1)s^2/(2\sigma^2))$ for σ, since $A(\mu) = (n-1)s^2 + n(\mu - \bar{x})^2$ and the Gaussian integral $\int \exp(-n(\mu-\bar{x})^2/(2\sigma^2))\,\mathrm{d}\mu = \sigma\sqrt{2\pi/n}$ contributes one factor of σ. The change of variables $v = (n-1)s^2/\sigma^2$, with $\sigma = s\sqrt{(n-1)/v}$ and $|\mathrm{d}\sigma/\mathrm{d}v| \propto v^{-3/2}$, turns this into a density proportional to $v^{n/2}e^{-v/2}v^{-3/2} = v^{(n-1)/2-1}e^{-v/2}$, the χ^2_{n-1} density of Proposition 14.2: $(n-1)s^2/\sigma^2$, with σ random and s fixed, has the χ^2_{n-1} distribution. That is the distribution Cochran's theorem (Theorem 14.3) gives $(n-1)S^2/\sigma^2$ over repeated samples, so here too the equal-tailed region coincides with the frequentist interval built from that pivot: $[3.33, 7.98]$ for σ and, squared, $[11.1, 63.7]$ for σ^2. The highest-density regions are $[3.08, 7.45]$ for σ and $[8.5, 53.9]$ for σ^2, each shorter than the equal-tailed region on its own scale. But the square of the first is $[9.5, 55.5]$, not the second: shortest on the scale of σ and shortest on the scale of σ^2 are different requirements. Figure 19.4 shows both scales.

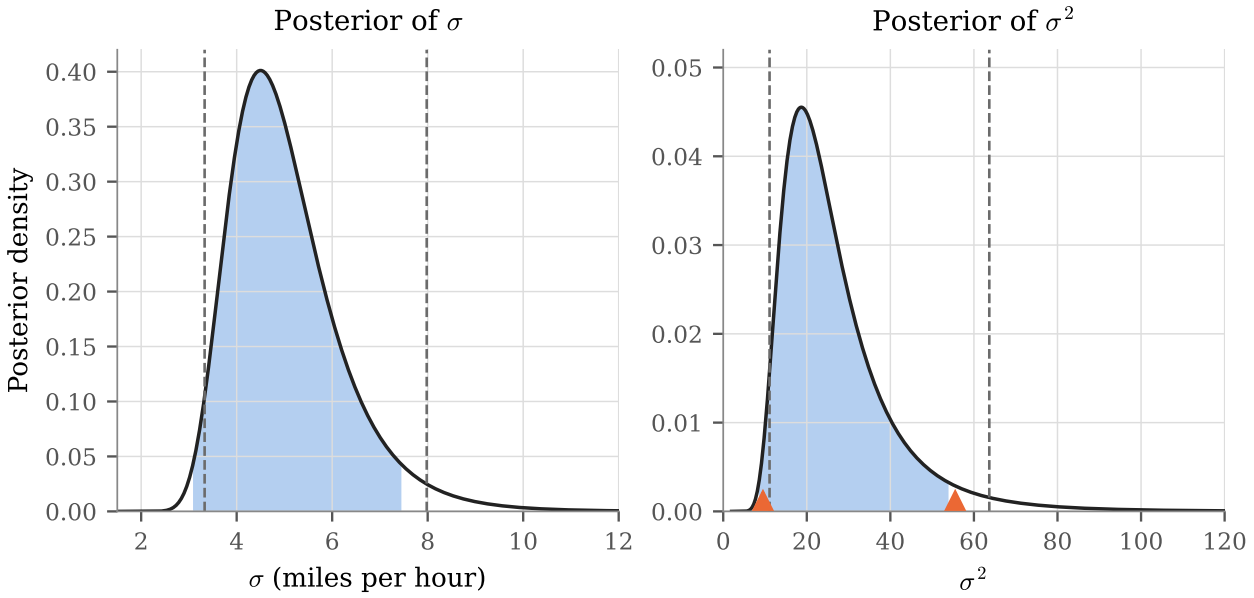


Figure 19.4: Posterior of the wind turbine's σ (left) and σ^2 (right) under the prior $1/\sigma$. Blue: the 95% highest-density regions. Gray dashed: the equal-tailed regions. Orange triangles: the squares of the endpoints of the left highest-density region.

Pitfall: the shortest region depends on the scale

'The 95% highest-density interval' is not a property of the posterior alone: it also depends on whether length is measured in σ or in σ^2, in p or in ψ. When the parameterization is a matter of convention, report the equal-tailed region, which follows the parameter through every monotone

change of scale, or say on which scale the highest-density region was computed.

The subway shows the same exact coincidence for a rate. Under Jeffreys' prior its posterior is Gamma$(64, 398.7)$, whose 2.5% and 97.5% points are $49.29/398.7 = 0.124$ and $80.60/398.7 = 0.202$, where 49.29 and 80.60 are the corresponding points of Gamma$(64, 1)$. That is exactly the interval of Exercise 8.7, built from the pivot $\lambda \sum_i T_i \sim$ Gamma$(64, 1)$. Again one distribution is read twice: over repeated samples with λ fixed, $\lambda \sum_i T_i$ is Gamma$(64, 1)$; over λ with the sample fixed, the posterior Gamma$(64, 398.7)$ makes 398.7λ a Gamma$(64, 1)$ variable. The two sentences come apart only in what they are about. 'Of 1,000 subway studies of 64 waits each, the interval computed from each covers the true rate in about 950' is a statement about the procedure. 'Given these 64 waits and Jeffreys' prior, the rate lies in $[0.124, 0.202]$ with probability 0.95' is a statement about λ: the first of the three readings that Section 8.8 ruled out for a confidence interval, true here under a prior.

Does this make sense? Should the posterior mean wait be $1/0.1605$?

Under Jeffreys' prior the posterior mean of the rate is $64/398.7 = 0.1605$, the maximum likelihood estimate, so it is tempting to call $1/0.1605 = 6.23$ minutes the posterior mean of the wait. It is not: the wait $\beta = 1/\lambda$ has posterior mean $\mathbb{E}[1/\lambda \mid \text{data}] = 398.7/63 = 6.33$ minutes, larger, as Jensen's inequality says it must be, since $1/\lambda$ is convex (toolkit A.8). Means do not follow a nonlinear change of scale, but quantiles do: the posterior median of the rate is 0.1597, and the posterior median of the wait is $1/0.1597 = 6.26$ minutes.

19.6 When the two kinds of interval agree

The exact coincidences of the last two sections, the z- and t-intervals, the chi-squared interval for σ and the subway's Gamma interval, all come from location and scale families with the matching improper prior, flat for a location and $1/\text{scale}$ for a scale: there the posterior distribution of the pivot equals its sampling distribution. In other models the two kinds of interval agree only approximately, and the Bernstein–von Mises theorem says how.

Recall. The Bernstein–von Mises theorem (Theorem 18.12): for i.i.d. data from a model satisfying the regularity conditions of Section 12.6, with θ^* in the interior of Θ, $I(\theta^*) > 0$, data that separate θ^* from distant values, and a prior density continuous and positive at θ^*, the posterior distribution of $\sqrt{n}(\theta - \hat{\theta}_n)$ approaches Gauss$(0, 1/I(\theta^*))$ in total variation, in probability. Slutsky's theorem (Theorem 5.1): if $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$ and $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} c$ for a constant c, then $X_n + Y_n \xrightarrow[n\to\infty]{\text{(d)}} X + c$.

Closeness in total variation makes every posterior probability, and so every posterior quantile of $\sqrt{n}(\theta - \hat{\theta}_n)$, close to the Gaussian one. So the equal-tailed region of level $1 - \alpha$ is $\hat{\theta}_n \pm q_{\alpha/2}/\sqrt{nI(\theta^*)}$ up to an error that is negligible next to $1/\sqrt{n}$. It covers θ^* exactly when $\sqrt{n}(\hat{\theta}_n - \theta^*)$ lies between $-q_{\alpha/2}/\sqrt{I(\theta^*)}$ and $q_{\alpha/2}/\sqrt{I(\theta^*)}$, each moved by a term that converges to 0 in probability, so by Theorem 12.5 and Slutsky's theorem, applied at each end, the Bayesian region has asymptotic coverage $1 - \alpha$, the same as the plug-in interval $\hat{\theta}_n \pm q_{\alpha/2}/\sqrt{nI(\hat{\theta}_n)}$ of Chapter 12, whatever the prior, as long as its density is continuous and positive at θ^*. With an improper prior the same can be checked in each example of this chapter, whose posteriors are explicit. For the kiss data

the priors of this chapter give 95% regions within 0.003 of each other and of the solve interval of Chapter 8, $[0.558, 0.724]$: flat $[0.558, 0.724]$, Jeffreys $[0.558, 0.725]$, Haldane $[0.559, 0.727]$.

How fast the agreement comes depends on the prior, and for one-sided bounds Jeffreys' prior is special. For a one-parameter model with continuous data, Welch and Peers (*Journal of the Royal Statistical Society B*, 1963) showed that the posterior's one-sided bounds have frequentist coverage $1-\alpha$ up to an error of order $1/n$ when the prior is Jeffreys', and of order only $1/\sqrt{n}$ for other smooth priors. In an equal-tailed region the two errors of order $1/\sqrt{n}$ cancel, so its coverage error is of order $1/n$ under any smooth prior. These results are stated without proof; their proof needs more than this book develops, and nothing later depends on them. For a proportion no interval has coverage exactly $1-\alpha$ at every p, since the data are discrete, but the Jeffreys interval, the equal-tailed region of Beta$(S+\frac{1}{2}, n-S+\frac{1}{2})$, comes close. At $p=0.645$ and $n=124$ its coverage is 0.952, the same as the solve interval's, where the plug-in interval has 0.939. At $n=20$, averaged over p from 0.01 to 0.99, the coverage is 0.951 for the Jeffreys interval, 0.954 for the solve interval and 0.861 for the plug-in interval. Figure 19.5 draws the solve and Jeffreys intervals at $n=20$: both oscillate around 0.95, the solve interval dropping lowest at the very ends (0.84 just below $p=0.009$) and the Jeffreys interval at $p=0.12$ and 0.88 (0.89). Of 1,000 studies of 20 couples, the Jeffreys interval covers the true p in about 951 on average over p, the plug-in interval in about 861. Brown, Cai and DasGupta (*Statistical Science*, 2001) recommend the solve (Wilson) interval or the Jeffreys interval for small n, the latter modified when S is 0, 1, $n-1$ or n: as defined here, the interval for $S=0$ starts above 0 (at 0.000024 when $n=20$), so its coverage is 0 for p closer to 0 than that.

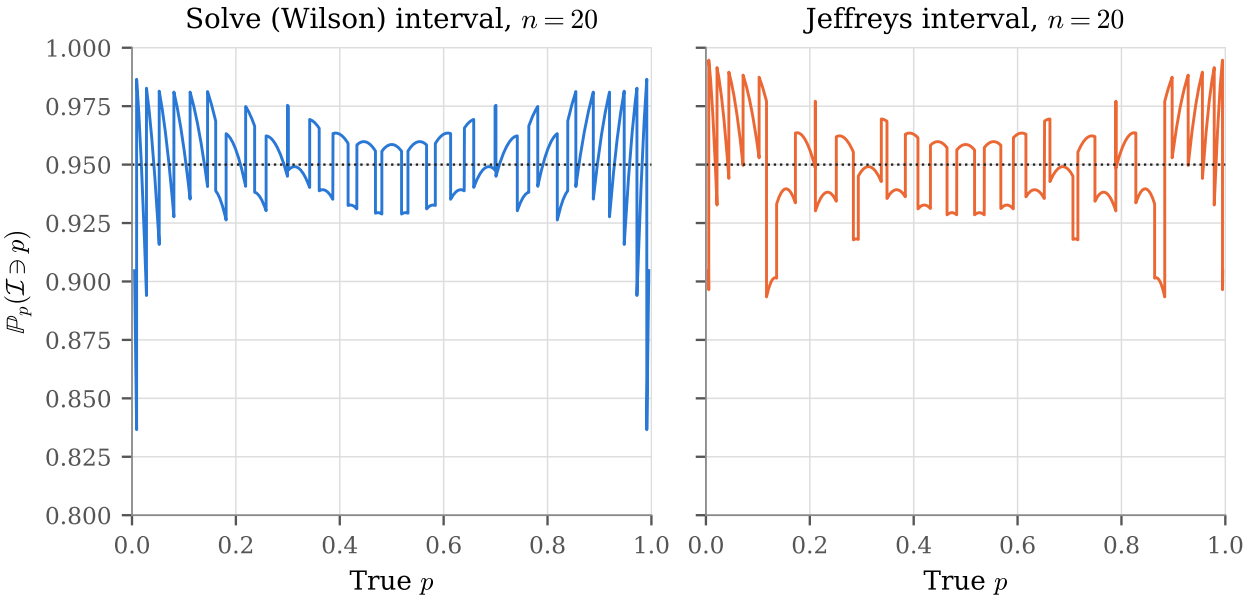


Figure 19.5: Exact coverage $\mathbb{P}_p(\mathcal{I} \ni p)$ of two 95% intervals for a proportion at $n=20$, against the true p: left, the solve (Wilson) interval (blue); right, the equal-tailed Jeffreys interval (orange). Dotted: the nominal 0.95.

The agreement has limits, and Chapter 18 has already shown two. With an informative prior and moderate n the regions differ: the skeptic's prior gave the interval $[0.56, 0.73]$ posterior probability 0.835, not 0.95. For a point null the answers differ at every n: a prior with a lump of probability at $p=\frac{1}{2}$ has no density, so the Bernstein–von Mises theorem does not apply, and $\mathbb{P}(H_0 \mid \mathbf{x}) = 0.044$ stood against a p-value of 0.0012.

19.7 What comes next

This chapter ends Part VII. Part VIII returns to the frequentist methods of the earlier parts and applies them to a model with a predictor, the linear regression model of Chapter 20, where least squares, the method that led Gauss to the distribution named after him (Chapter 6), is the estimator. The Bayesian reading carries over: with Gaussian errors, the least squares estimate is the posterior mode under a flat prior on the coefficients, and with a Gaussian prior centered at zero the posterior mode is the ridge estimate that Chapter 18 mentioned.

19.8 Summary

- A flat prior is flat on one scale only. By the change of variables formula $\pi_\varphi = \pi_\theta |\mathrm{d}\theta/\mathrm{d}\varphi|$, flat in p is the logistic density in the log-odds, and flat in the subway's mean wait is $1/\lambda^2$ in terms of the rate; at $n = 64$ the two flat answers for the subway differ by a quarter of a posterior standard deviation.
- An improper prior has an infinite integral. It gives a posterior when $0 < \int L_n \pi < \infty$, and none otherwise (Haldane's prior when every couple turns right). It can be read as the limit of wider and wider proper priors, and its arbitrary constant cancels in estimation but makes the Bayes factor arbitrary when it is the prior for the parameter that a point null fixes.
- Fisher information changes with the parameterization as $I_\varphi = I_\theta (\mathrm{d}\theta/\mathrm{d}\varphi)^2$, so Jeffreys' prior $\pi_J \propto \sqrt{I}$, or $\sqrt{\det I}$ in several dimensions, is the same prior whichever parameterization it is computed in. It gives equal mass to intervals the data can just tell apart, and it is flat on the variance-stabilizing scale $\int \sqrt{I}$: $2 \arcsin \sqrt{p}$, $2\sqrt{\lambda}$, $\log \lambda$.
- Jeffreys' priors: Beta$(\frac{1}{2}, \frac{1}{2})$ for a proportion (kiss posterior mean 0.644), $\lambda^{-1/2}$ for a Poisson mean, $1/\lambda$ for a rate (subway posterior mean 0.1605, the maximum likelihood estimate), flat for a Gaussian mean, $1/\sigma$ for a scale. For a Gaussian with both parameters unknown the joint rule gives $1/\sigma^2$, and the product of the separate rules, $1/\sigma$, is the one that reproduces the t-interval.
- A Bayesian confidence region of level $1 - \alpha$ has posterior probability $1 - \alpha$. The highest-density region is the shortest; the equal-tailed region follows the parameter through any monotone change of scale, and the highest-density region does not.
- With the matching improper prior, Bayesian regions equal frequentist intervals exactly in location and scale families: the z- and t-intervals, the chi-squared interval for σ, the subway's exact $[0.124, 0.202]$. Elsewhere they agree as n grows (Bernstein–von Mises); for one-sided bounds the agreement is fastest under Jeffreys' prior (Welch and Peers), and the Jeffreys interval for a proportion covers about 95% of the time on average over p even at $n = 20$.

19.9 Exercises

Exercise 19.1. (a) Show that Jeffreys' prior for the Poisson mean is $\lambda^{-1/2}$, that it is improper, and that after $n \geq 1$ observations with sum S the posterior is Gamma$(S + \frac{1}{2}, n)$. (b) The siblings data of Chapter 2 have 24 values of $X - 1$ adding up to 24. Find the posterior mean of λ under Jeffreys' prior and under the flat prior $\pi(\lambda) = 1$, and compare with the maximum likelihood estimate. *(Solution on page 433.)*

Exercise 19.2. A geometric observation counts the trials up to and including the first success, $\mathbb{P}_p(X = x) = p(1-p)^{x-1}$ for $x = 1, 2, \ldots$, with $\mathbb{E}_p[X] = 1/p$. (a) Show that $I(p) = 1/(p^2(1-p))$, so Jeffreys' prior is $p^{-1}(1-p)^{-1/2}$, and that it is improper. (b) Suppose an observer watched kissing couples until the first right turn, five times over, needing 1, 2, 1, 1 and 3 couples. Show that Jeffreys' posterior is Beta$(5, 3.5)$ and find its mean. (c) The same eight couples, read as eight Bernoulli trials with five right turns, have a likelihood proportional to the same function of p. What is Jeffreys' posterior mean for them, and what does the difference say about Jeffreys' prior? *(Solution on page 433.)*

Exercise 19.3. Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Gauss}(\mu, v)$ with μ known, parameterized by the variance $v = \sigma^2$. (a) Show that Jeffreys' prior for v is proportional to $1/v$. (b) Show that carrying the prior $1/\sigma$ on σ to v by Equation (19.1) gives $1/v$ up to a constant, as Theorem 19.4 says it must. (c) Show that the prior $1/\sigma$ is the flat prior on $\log \sigma$, and that rescaling, $\sigma' = c\sigma$ for a constant $c > 0$, such as a change from miles to kilometers per hour, leaves it in the same form. *(Solution on page 433.)*

Exercise 19.4. (a) Show that under Haldane's prior $\pi(p) = 1/(p(1-p))$ the posterior after S successes in n trials is Beta$(S, n-S)$ when $0 < S < n$, and that its mean is the sample proportion. (b) Show that there is no posterior when $S = 0$, and that under Jeffreys' prior there is one for every S. (c) Give the posterior means of p for the kiss data under the flat, Jeffreys and Haldane priors, and say which way each prior pulls the estimate. *(Solution on page 433.)*

Exercise 19.5. For the subway, under each of the three priors of this chapter for the rate (flat in λ, Jeffreys' $1/\lambda$, flat in the wait β), the posterior of λ is Gamma$(a, 398.7)$ with $a = 65$, 64 and 63. (a) Show that the posterior mean of the wait $\beta = 1/\lambda$ is $398.7/(a-1)$, and compute it for the three priors. (b) The 95% equal-tailed regions for λ are $[0.1258, 0.2050]$, $[0.1236, 0.2022]$ and $[0.1214, 0.1994]$. Give the equal-tailed regions for β without further computation. (c) Which prior makes the posterior mean of β equal to the maximum likelihood estimate 6.23, and which makes the posterior mean of λ equal to $1/6.23$? *(Solution on page 433.)*

Exercise 19.6. Under the prior $1/\sigma^2$ of Proposition 19.6, the 95% equal-tailed region for μ is $\bar{x} \pm q_{0.025}(t_n)\sqrt{(n-1)/n}\, s/\sqrt{n}$. (a) Show that, read as a confidence interval, its coverage is $\mathbb{P}(|T| \le q_{0.025}(t_n)\sqrt{(n-1)/n})$ with $T \sim t_{n-1}$, whatever μ and σ are. (b) Evaluate it at $n = 5$, given $q_{0.025}(t_5) = 2.5706$ and $\mathbb{P}(|t_4| \le 2.299) = 0.917$. (c) What is the coverage of the region under the prior $1/\sigma$? *(Solution on page 433.)*

Exercise 19.7. Test $H_0 : p = \frac{1}{2}$ against $H_1 : p \neq \frac{1}{2}$ for the kiss data, with prior probability $\frac{1}{2}$ on each, as in Section 18.7, but with Haldane's prior $c/(p(1-p))$ on H_1. (a) Show that $m_1 = c\,B(80, 44)$, where $B(a, b) = \int_0^1 p^{a-1}(1-p)^{b-1}\,\mathrm{d}p$. (b) Given $B(80, 44) = 4.45 \times 10^{-36}$ and $m_0 = 2^{-124} = 4.70 \times 10^{-38}$, compute B_{01} and $\mathbb{P}(H_0 \mid \mathbf{x})$ for $c = 1$ and for $c = 1{,}000$. (c) Why does the constant not matter when the same prior is used to estimate p? *(Solution on page 433.)*

Exercise 19.8. (a) Show that if a posterior density on $\mathbb{R}$ is symmetric about a point m and decreases as $|\theta - m|$ grows, its highest-density and equal-tailed regions of every level coincide. (b) Which posteriors of this chapter are of that kind, and which are not? *(Solution on page 434.)*

Exercise 19.9. Let X have density $f(x - \theta)$, with f a known density on $\mathbb{R}$ (a location family), and take the improper prior $\pi(\theta) = 1$. (a) Show that the posterior density of θ given $X = x$ is $f(x - \theta)$. (b) Let $u(x)$ be the posterior's upper bound of level $1 - \alpha$: $\mathbb{P}(\theta \le u(x) \mid x) = 1 - \alpha$. Show that $\mathbb{P}_\theta(\theta \le u(X)) = 1 - \alpha$ for every θ: the Bayesian bound has exact frequentist coverage. *(Solution on page 434.)*

Exercise 19.10. In the drug trial of Chapter 1, 78 of 100 patients were cured. (a) Give Jeffreys' posterior, its mean and its 95% equal-tailed region, given that the 2.5% and 97.5% points of the posterior are 0.692 and 0.852. (b) Compute the arcsine interval $2 \arcsin \sqrt{\hat{p}} \pm 1.96/\sqrt{n}$, mapped back to p, and compare with (a), with the solve interval $[0.689, 0.850]$ of Equation (8.2) and with the plug-in interval $[0.699, 0.861]$ of Chapter 1. *(Solution on page 434.)*

Part VIII

Linear Regression

Least squares as a statistical model. One predictor first, with the exact t-test for its slope when the errors are Gaussian and large-sample inference when they are not (Chapter 20); then the matrix form with several predictors, the F-test for groups of coefficients, and the identifiability of Chapter 2 returning as the rank of the design matrix (Chapter 21).

Chapter 20

Univariate Linear Regression

In this chapter

- Predicting one variable from another: the regression function $x \mapsto \mathbb{E}[Y \mid X = x]$, and why it is the best prediction in mean squared error. The example is the Old Faithful geyser, whose next wait can be predicted from the length of the eruption just ended.
- The best line: the theoretical regression line, with slope $\mathbb{C}\text{ov}(X, Y)/\mathbb{V}\text{ar}(X)$, and the noise around it, which has mean zero and is uncorrelated with X whether or not the regression function is a line.
- Least squares as the best line of the data; Gauss's reason for it (maximum likelihood with Gaussian errors); its Bayesian reading, with ridge as the posterior mode under a Gaussian prior on the slope.
- How accurate the estimates are: unbiasedness, the variance σ^2/S_{xx}, the sum of squared residuals divided by $n - 2$, and the exact t distribution when the errors are Gaussian. Tests and intervals for the slope, confidence and prediction bands, and what survives without Gaussian errors.
- Correlation, R^2 and regression toward the mean.

Chapter 11 observed that maximizing a Gaussian likelihood is minimizing a sum of squares; Section 20.4 takes that observation up with a second variable in the picture. The chapter returns from the Bayesian chapters to the frequentist methods of the earlier parts.

20.1 Predicting one variable from another

Old Faithful, the geyser in Yellowstone National Park whose waits Chapter 11 fit with a mixture of two Gaussians, does not erupt on a timetable: in the 272 eruptions of that chapter the wait before the next eruption ranged from 43 to 96 minutes. But the wait is not unpredictable. The data record, for each eruption, how long it lasted and how long the geyser then waited before erupting again, and the left of Figure 20.1 shows that short eruptions are followed by short waits and long eruptions by long ones. The usual physical explanation: a short eruption expels less water, and the geyser's plumbing refills and reheats sooner (O'Hara and Esawi, *GSA Today*, 2013). So someone who times the eruption just ended can predict the next one better than someone who knows only that the average wait is 70.9 minutes.

This is the regression problem: from pairs of a variable that is easy to measure and one that is hard to measure or not yet available, learn to predict the second from the first. The data are pairs $(X_1, Y_1), \ldots, (X_n, Y_n)$, modeled as i.i.d. copies of a random pair (X, Y) with an unknown joint distribution; here X is the duration of an eruption and Y the wait that follows it, both in minutes. Everything there is to know about how Y depends on X is in the conditional distribution of Y given $X = x$, which has density $h(y \mid x) = h(x, y)/h(x)$ when the pair has a joint density $h(x, y)$ with marginal density $h(x) = \int h(x, y)\, \mathrm{d}y$ (toolkit A.15). The right of Figure 20.1 estimates a few of these distributions by grouping the eruptions in half-minute bins of duration: each box spans the middle half of its bin's waits, with the median marked, and the whiskers reach the shortest and longest wait. The distribution moves up with x, fastest between about 2.5 and 3.5 minutes, where only 12 eruptions fall.

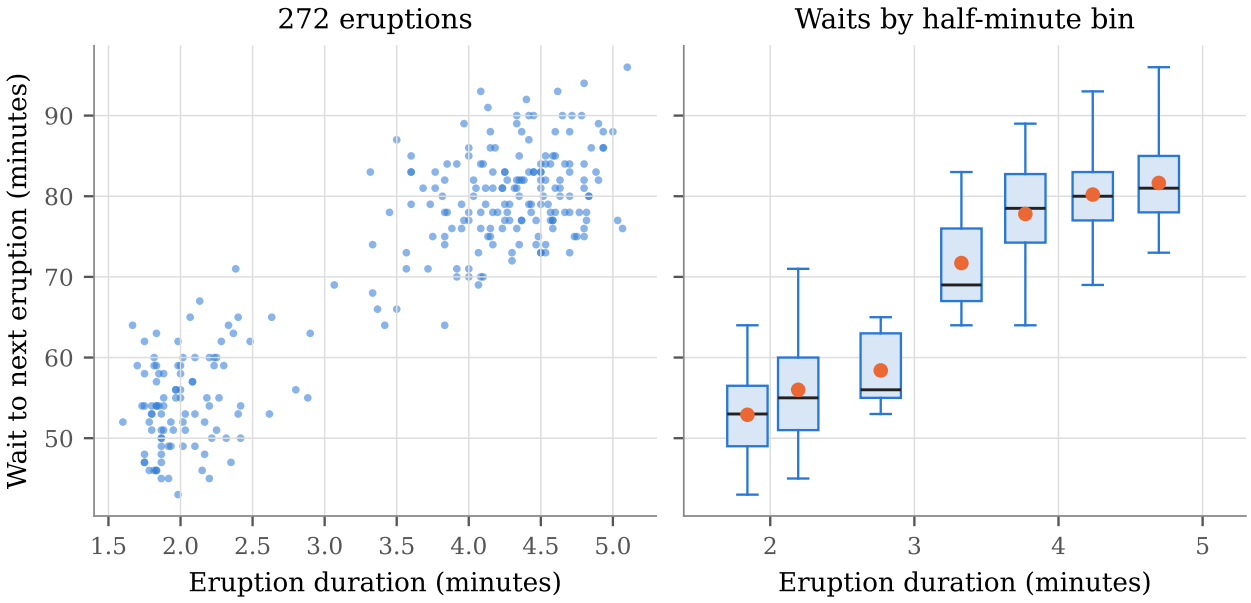


Figure 20.1: Old Faithful, 272 eruptions. Left: the wait until the next eruption against the eruption's duration. Right: the waits in half-minute bins of duration (the last from 4.5 to 5.5), one box per bin, with the bin's mean wait in orange.

A whole conditional distribution for every x is more than most questions need, and more than 272 pairs can estimate well. The usual summary is its mean.

Definition 20.1 (Regression function)**.** For a random pair (X, Y) with $\mathbb{E}|Y| < \infty$, the **regression function** of Y on X is

$$f(x) = \mathbb{E}[Y \mid X = x],$$

the mean of the conditional distribution of Y given $X = x$. X is called the **explanatory variable** (or covariate, or predictor) and Y the **response**.

> The regression function answers the question 'among the cases with $X = x$, how large is Y on average?', one answer for each value of the explanatory variable. Other summaries of the same conditional distributions, such as the conditional median or quartiles, answer other questions; this chapter and the next are about the mean.
> **For the kiss data.** The study recorded only the direction R of each couple's turn, so there is no explanatory variable. For any variable X recorded independently of how couples turn, the day of

> the week for instance, the conditional distribution of R given $X = x$ is Bern(p) whatever x is, and $f(x) = p$: a flat regression function, which is what independence looks like.

The orange dots in the right of Figure 20.1 are the bins' mean waits, estimates of f at each bin's average duration: 52.9 minutes for the 51 eruptions shorter than 2 minutes, rising to 81.6 for the 65 that lasted 4.5 minutes or more. The mean is the summary of choice because of what it does for prediction.

Theorem 20.2 (The regression function is the best predictor)**.** *Let $\mathbb{E}[Y^2] < \infty$ and $f(x) = \mathbb{E}[Y \mid X = x]$. For every function g with $\mathbb{E}[g(X)^2] < \infty$,*

$$\mathbb{E}\left[(Y - g(X))^2\right] = \mathbb{E}\left[(Y - f(X))^2\right] + \mathbb{E}\left[(f(X) - g(X))^2\right].$$

So $f(X)$ has the smallest mean squared prediction error among all predictions computed from X, and a function g that does as well equals f with probability one.

Proof strategy (core). Write $Y - g(X)$ as $(Y - f(X)) + (f(X) - g(X))$ and expand the square. The cross term vanishes: given X, the first factor has mean zero and the second is a fixed number.

Proof. Expanding the square,

$$(Y - g(X))^2 = (Y - f(X))^2 + 2(Y - f(X))(f(X) - g(X)) + (f(X) - g(X))^2.$$

All three terms have finite expectations: $\mathbb{E}[f(X)^2] \le \mathbb{E}[Y^2]$ (toolkit A.15), so the first and last are finite, and the middle one is bounded by their sum, since $|2uv| \le u^2 + v^2$. For the middle term, condition on X: by taking out what is known and then the tower property (toolkit A.15),

$$\begin{aligned}\mathbb{E}\left[(Y - f(X))(f(X) - g(X))\right] &= \mathbb{E}\left[(f(X) - g(X))\ \mathbb{E}[Y - f(X) \mid X]\right] \\ &= \mathbb{E}\left[(f(X) - g(X))(f(X) - f(X))\right] = 0.\end{aligned}$$

Taking expectations in the expansion gives the identity. Its last term is non-negative, and zero only if $g(X) = f(X)$ with probability one. □

The identity says more than that f wins. The error of any prediction rule splits into a part that no rule computed from X can remove, $\mathbb{E}[(Y - f(X))^2]$, which is the average spread of the waits inside each box of Figure 20.1, and the rule's distance from f.

20.2 The best line

The regression function can be any function, and estimating an arbitrary function means borrowing information only from nearby points: the bin means of Figure 20.1 rest on as few as 5 eruptions. The oldest and most useful restriction borrows from all the points at once: predict with a line.

Definition 20.3 (Theoretical regression line)**.** Let X and Y have finite second moments, with $\mathbb{V}\mathrm{ar}(X) > 0$. The **theoretical regression line** of Y on X is the line $x \mapsto a^* + b^* x$ whose coefficients minimize the mean squared error of a linear prediction,

$$(a^*, b^*) = \operatorname*{argmin}_{(a,b)\in\mathbb{R}^2} \mathbb{E}\left[(Y - a - bX)^2\right].$$

Its slope is b^*, and its intercept a^* is its height at $x = 0$. The **noise** is $\epsilon = Y - a^* - b^* X$, so that $Y = a^* + b^* X + \epsilon$.

Among all lines, the theoretical regression line is the one whose predictions miss Y by the least, on average and in squared distance, over the joint distribution of (X, Y). It is a property of the distribution, with no data in sight, and it is what least squares will estimate. The noise is the part of Y the line does not predict.

For the kiss data. For R and any X recorded independently of it, $\mathbb{C}\text{ov}(X, R) = 0$, so by the theorem below the best line is flat at height p: $b^* = 0$ and $a^* = p$. The noise $\epsilon = R - p$ is $1 - p$ for a right turn and $-p$ for a left turn, with mean 0 and variance $p(1 - p)$.

Theorem 20.4 (The theoretical regression line)**.** *Under the conditions of Definition 20.3 the minimizer is unique, with*

$$b^* = \frac{\mathbb{C}ov(X, Y)}{\mathbb{V}ar(X)}, \qquad a^* = \mathbb{E}[Y] - b^* \, \mathbb{E}[X],$$

and the noise satisfies $\mathbb{E}[\epsilon] = 0$ *and* $\mathbb{C}ov(X, \epsilon) = 0$.

Proof strategy (core). Fix b and choose a: the best constant prediction of $Y - bX$ is its mean. What remains is $\mathbb{V}\text{ar}(Y - bX)$, a parabola in b, and its lowest point is the slope.

Proof. For any random variable W with finite variance and any number a, $\mathbb{E}[(W - a)^2] = \mathbb{V}\text{ar}(W) + (\mathbb{E}[W] - a)^2$; this is Theorem 7.11 with W in the role of the estimator. With $W = Y - bX$, for each fixed b the mean squared error $\mathbb{E}[(Y - a - bX)^2]$ is smallest at $a = \mathbb{E}[Y] - b\,\mathbb{E}[X]$, where it equals

$$\mathbb{V}\text{ar}(Y - bX) = \mathbb{V}\text{ar}(Y) - 2b\,\mathbb{C}\text{ov}(X, Y) + b^2\,\mathbb{V}\text{ar}(X).$$

Since $\mathbb{V}\text{ar}(X) > 0$, this is a parabola in b that opens upward, and its only minimum is at $b = \mathbb{C}\text{ov}(X, Y)/\,\mathbb{V}\text{ar}(X)$. So the minimizer is (a^*, b^*) as stated, and it is unique. For the noise, $\mathbb{E}[\epsilon] = \mathbb{E}[Y] - a^* - b^*\,\mathbb{E}[X] = 0$ by the formula for a^*, and $\mathbb{C}\text{ov}(X, \epsilon) = \mathbb{C}\text{ov}(X, Y) - b^*\,\mathbb{V}\text{ar}(X) = 0$ by the formula for b^*. □

The covariance in the numerator gives the slope its sign; the variance in the denominator gives it its units. Measure X in units three times smaller, so that every value of X is multiplied by 3: the covariance is multiplied by 3 and the variance by 9, and the slope is divided by 3, as a rate per unit of X must be.

The two properties of the noise are all that the theoretical line guarantees, and they are weaker than they look.

Pitfall: the best line need not be the regression function

$\mathbb{C}\text{ov}(X, \epsilon) = 0$ says that the noise is uncorrelated with X, not that it averages to zero at each x. When the regression function f is a line, the best line is f itself: by Theorem 20.2 no prediction beats f, so no line does. When f is not a line, take $g(x) = a + bx$ in Theorem 20.2:

$$\mathbb{E}\left[(Y - a - bX)^2\right] = \mathbb{E}\left[(Y - f(X))^2\right] + \mathbb{E}\left[(f(X) - a - bX)^2\right],$$

so (a^*, b^*) is the line closest to f in mean square, with the values of x weighted by the distribution of X. In Figure 20.2, $X \sim \text{Unif}(0, 1)$ and $f(x) = x^2$: the best line is $-\frac{1}{6} + x$ (Exercise 20.2), and the noise has conditional mean $x^2 - x + \frac{1}{6}$, positive at both ends and negative in the middle, although its overall mean is 0 and it is uncorrelated with X.

Least squares estimates (a^*, b^*) in either case. Whether the line is also the regression function is a separate question, and the residual plot of the next section is the first way to ask it.

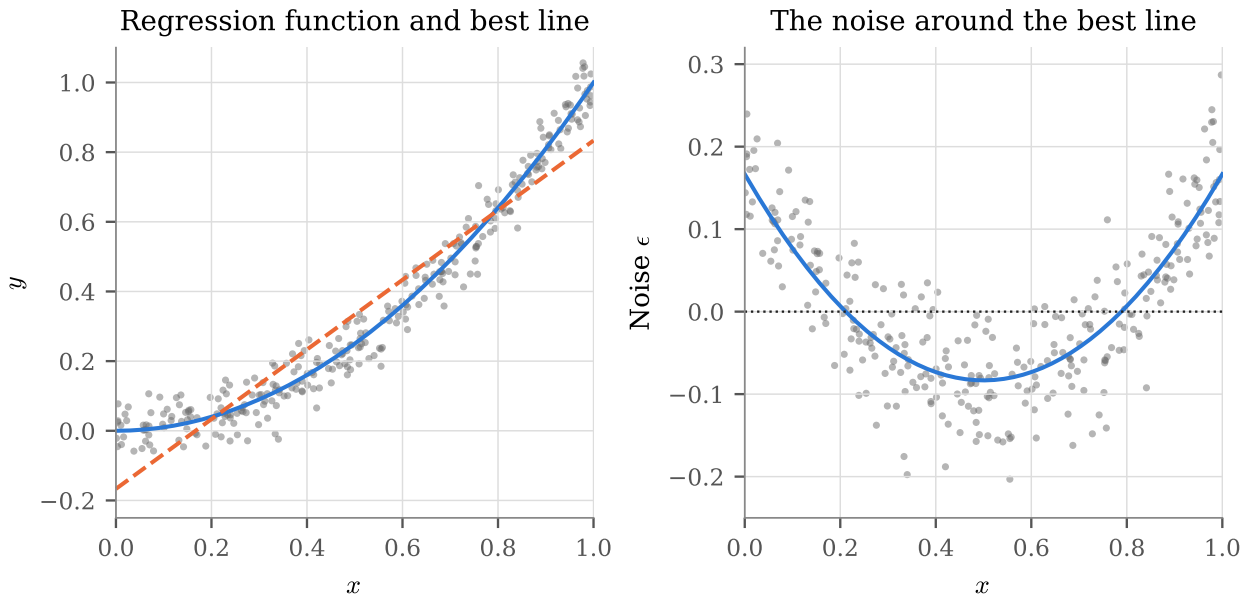


Figure 20.2: 300 draws of $X \sim \text{Unif}(0,1)$ and $Y = X^2$ plus independent $\text{Gauss}(0, 0.05^2)$ noise. Left: the regression function x^2 (blue) and the best line $-\frac{1}{6} + x$ (orange, dashed). Right: the noise around the best line, with its conditional mean (blue).

20.3 Least squares

The joint distribution is unknown, so (a^*, b^*) must be estimated from the pairs. The recipe of the earlier chapters applies: replace the expectation by an average.

Definition 20.5 (Least squares, fitted values, residuals)**.** Given pairs $(X_1, Y_1), \ldots, (X_n, Y_n)$, the **least squares estimator** $(\hat{a}, \hat{b})$ minimizes the sum of squared errors

$$\sum_{i=1}^{n} (Y_i - a - bX_i)^2$$

over $(a, b) \in \mathbb{R}^2$. The **fitted values** are $\hat{Y}_i = \hat{a} + \hat{b}X_i$ and the **residuals** are $e_i = Y_i - \hat{Y}_i$.

> Least squares draws the line that makes the sum of the squared vertical distances from the points to the line as small as possible. The residuals are those vertical distances, with a sign: positive for a point above the line.
> **For the kiss data.** With no explanatory variable, the line has only a height a, and $\sum_i (R_i - a)^2$ is smallest at $a = \bar{R}_n = \hat{p} = 0.645$ (the squared loss of Chapter 13): the sample proportion is the least squares fit of a constant. The residuals are 0.355 for each of the 80 right turns and -0.645 for each of the 44 left turns. A slope would need an explanatory variable that varies: if every X_i had the same value c, every slope b would fit equally well with $a = \bar{R}_n - bc$, which is why the theorem below asks for the X_i not to be all equal.

Write $\bar{X}$ and $\bar{Y}$ for the sample means, and

$$S_{xx} = \sum_{i=1}^{n} (X_i - \bar{X})^2, \qquad S_{xy} = \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y}), \qquad S_{yy} = \sum_{i=1}^{n} (Y_i - \bar{Y})^2.$$

Theorem 20.6 (The least squares line)**.** *If the X_i are not all equal, the least squares estimator is unique, with*

$$\hat{b} = \frac{S_{xy}}{S_{xx}}, \qquad \hat{a} = \bar{Y} - \hat{b}\bar{X},$$

and the residuals satisfy $\sum_i e_i = 0$ and $\sum_i X_i e_i = 0$.

Proof strategy (for completeness; skip on a first reading). The least squares line is the theoretical regression line of the data's empirical distribution, which puts probability $1/n$ on each observed pair. Apply Theorem 20.4 to it.

Proof. Let (X', Y') be a random pair equal to (X_i, Y_i) with probability $1/n$ for each i. Then $\mathbb{E}[(Y' - a - bX')^2] = \frac{1}{n}\sum_i (Y_i - a - bX_i)^2$, so the least squares estimator is the theoretical regression line of (X', Y'). For this pair $\mathbb{E}[X'] = \bar{X}$, $\mathbb{E}[Y'] = \bar{Y}$, $\mathbb{V}\mathrm{ar}(X') = S_{xx}/n$, which is positive when the X_i are not all equal, and $\mathbb{C}\mathrm{ov}(X', Y') = S_{xy}/n$, so Theorem 20.4 gives the formulas and the uniqueness. Its two properties of the noise, for this pair, read $\frac{1}{n}\sum_i e_i = 0$ and $\frac{1}{n}\sum_i (X_i - \bar{X})e_i = 0$, and the second together with the first gives $\sum_i X_i e_i = 0$. □

The two equations for the residuals are called the **normal equations**. They say of the sample what Theorem 20.4 says of the population: the residuals average to zero and are uncorrelated with the explanatory variable. They hold for every data set, so they cannot be used to check the line.

Worked example: predicting Old Faithful

For the 272 eruptions, $\bar{X} = 3.488$ minutes, $\bar{Y} = 70.90$ minutes, $S_{xx} = 353.0$ and $S_{xy} = 3{,}788.0$, so

$$\hat{b} = \frac{3{,}788.0}{353.0} = 10.73, \qquad \hat{a} = 70.90 - 10.73 \times 3.488 = 33.47.$$

Each extra minute of eruption adds about 10.7 minutes to the predicted wait. After a 2-minute eruption the line predicts $33.47 + 10.73 \times 2 = 54.9$ minutes; after a 4.5-minute eruption, 81.8 minutes. The height 33.47 at a duration of 0 is outside the data and is only where the line happens to cross the axis: no eruption lasts 0 minutes.

The left of Figure 20.3 draws the line with each point's residual as a vertical segment. The line passes through $(\bar{X}, \bar{Y}) = (3.49, 70.9)$, the black dot, as the formula $\hat{a} = \bar{Y} - \hat{b}\bar{X}$ forces, although few eruptions lasted about 3.5 minutes: the line borrows from both clusters at once. The right of the figure plots the residuals against the duration, with their means over the half-minute bins in orange. If the line were the regression function, these means would hover near 0; they are -0.3 and -1.0 in the two bins below 2.5 minutes, dip to -4.8 in the sparse bin near 2.8 minutes, reach $+3.9$ near 3.8 minutes and fall to -2.2 in the last bin. The bins near 2.8 and 3.3 minutes hold only 5 and 7 eruptions, but the drop at the right rests on 65: for the longest eruptions the waits rise more slowly than the line. The line is still the estimate of the best line, and Theorem 20.12 below says that $\hat{b}$ converges to b^* whether or not f is a line; it is not an estimate of $f(x)$ at each x.

Does this make sense? Would two group averages give the same slope?

Split the eruptions at 3 minutes. The 97 short ones average 2.04 minutes and are followed by waits averaging 54.5 minutes; the 175 long ones average 4.29 minutes, with waits averaging 80.0. The slope between the two groups' centers is $(80.0 - 54.5)/(4.29 - 2.04) = 11.3$ minutes per minute, close to 10.7: with two clusters, least squares is mostly a line through their centers, adjusted for the tilt inside each.

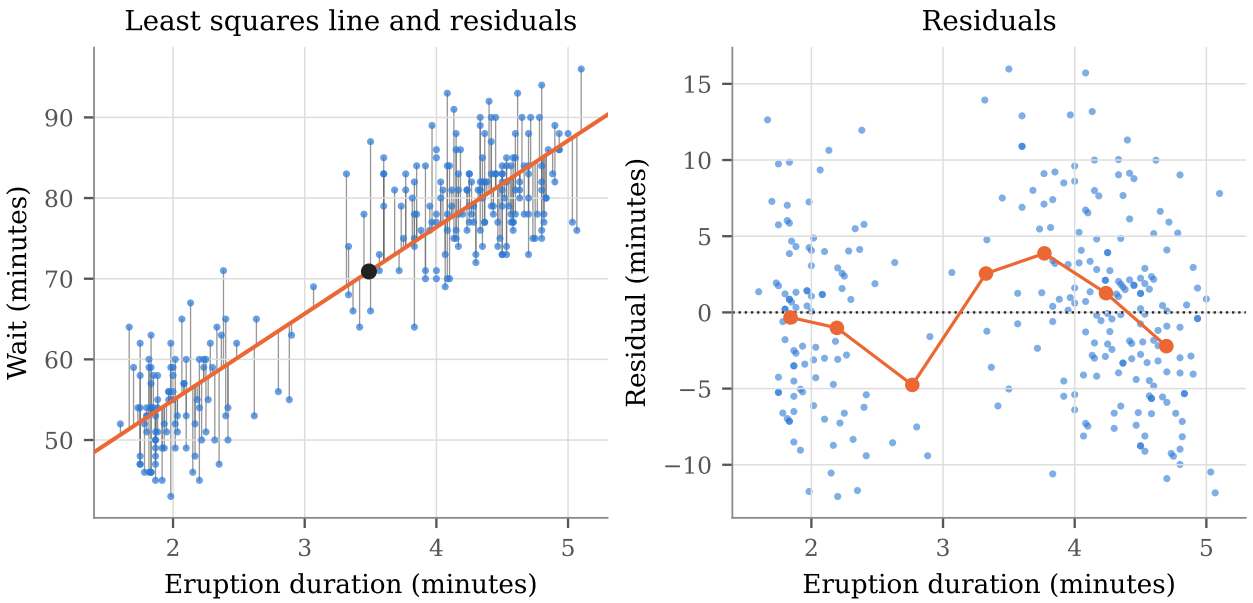


Figure 20.3: Old Faithful. Left: the eruptions (blue), the least squares line $33.47 + 10.73x$ (orange), each residual as a gray segment, and the point $(\bar{X}, \bar{Y})$ (black). Right: the residuals against the duration, with their half-minute bin means (orange).

The units check too: S_{xy} is in minutes of wait times minutes of eruption and S_{xx} in squared minutes of eruption, so $\hat{b}$ is in minutes of wait per minute of eruption. The formula may look different from the one you learned first.

From undergraduate statistics: the regression line you already know

An introductory course writes the least squares line as $\hat{y} = b_0 + b_1 x$, with $b_1 = r\, s_y/s_x$ and $b_0 = \bar{y} - b_1 \bar{x}$, where s_x, s_y are the sample standard deviations and r the correlation of Section 20.7. Since $r = S_{xy}/\sqrt{S_{xx} S_{yy}}$ and $s_y/s_x = \sqrt{S_{yy}/S_{xx}}$, $r\, s_y/s_x = S_{xy}/S_{xx}$: it is the same line. What this chapter adds is what the line estimates, the theoretical line of a distribution, and how good the estimate is, which is the subject of the next three sections.

20.4 Gauss's reason: least squares as maximum likelihood

Least squares was first published by Adrien-Marie Legendre in 1805, in a book on the orbits of comets, as a convenient rule for reconciling more observations than unknowns. Carl Friedrich Gauss published it in 1809, in a book on the orbits of planets and comets, said he had used it since 1795, and gave it a probabilistic reason. He asked for the law of errors under which the arithmetic mean of repeated measurements is the most probable value of the quantity measured, found that the error density must be proportional to $e^{-h^2\Delta^2}$ for an error Δ, and showed that under this law the most probable values of the unknowns are the least squares values (Stigler, *The History of Statistics*, 1986). His most probable values are what Chapter 18 calls the posterior mode under a flat prior, which is the maximum likelihood estimate: with Gaussian errors, least squares is maximum likelihood.

To state it, the explanatory values are treated as fixed numbers $x_1, \ldots, x_n$, as they are when an experimenter sets them, and the randomness is in the errors; Section 20.5 explains when the same

statements hold for observational data like Old Faithful's.

Proposition 20.7 (Least squares is maximum likelihood)**.** *Let $n \geq 3$ and $Y_i = a + bx_i + \epsilon_i$ for $i = 1, \ldots, n$, with $x_1, \ldots, x_n$ fixed numbers not all equal and $\epsilon_1, \ldots, \epsilon_n \overset{iid}{\sim} \text{Gauss}(0, \sigma^2)$. For every $\sigma^2 > 0$ the likelihood is maximized over (a, b) at the least squares estimator, and the maximum likelihood estimator of σ^2 is $\frac{1}{n}\sum_i e_i^2$.*

Proof strategy (for completeness; skip on a first reading). The log-likelihood is a constant minus the sum of squares over $2\sigma^2$; then maximize over σ^2 as for the Gaussian variance in Chapter 11.

Proof. The Y_i are independent with $Y_i \sim \text{Gauss}(a + bx_i, \sigma^2)$, so the log-likelihood is

$$\ell(a, b, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(Y_i - a - bx_i)^2.$$

For each σ^2 it is largest where the sum of squares is smallest, at $(\hat{a}, \hat{b})$. There it equals $-\frac{n}{2}\log(2\pi\sigma^2) - \text{RSS}/(2\sigma^2)$, with $\text{RSS} = \sum_i e_i^2$, whose derivative in σ^2, $-n/(2\sigma^2) + \text{RSS}/(2\sigma^4)$, is positive for $\sigma^2 < \text{RSS}/n$ and negative above, so the maximum is at $\sigma^2 = \text{RSS}/n$ ($\text{RSS} > 0$ with probability one, since $n \geq 3$ points with Gaussian errors lie on one line with probability zero). □

For Old Faithful the residual sum of squares is RSS = 9,443 squared minutes, and the maximum likelihood estimate of the errors' standard deviation is $\sqrt{9{,}443/272} = 5.89$ minutes. Section 20.5 divides by $n - 2$ instead.

The Bayesian reading of Chapter 18 carries over. With σ^2 known, a flat prior on (a, b) makes the posterior density proportional to the likelihood, so the posterior mode is the least squares line. A prior that doubts steep slopes changes the answer. Write the line as $\alpha + b(x - \bar{x})$, so that $\alpha = a + b\bar{x}$ is its height at the mean of the x_i, keep a flat prior on α, and give the slope the prior $\text{Gauss}(0, \tau^2)$. Up to a constant, the log posterior density is

$$-\frac{1}{2\sigma^2}\sum_{i=1}^{n}\big(Y_i - \alpha - b(x_i - \bar{x})\big)^2 - \frac{b^2}{2\tau^2}.$$

Since $\sum_i (x_i - \bar{x}) = 0$, its derivative in α is zero at $\alpha = \bar{Y}$ whatever b is, and its derivative in b is $(S_{xy} - bS_{xx})/\sigma^2 - b/\tau^2$, zero at

$$\tilde{b} = \frac{S_{xy}}{S_{xx} + \lambda} = \hat{b}\,\frac{S_{xx}}{S_{xx} + \lambda}, \qquad \lambda = \frac{\sigma^2}{\tau^2}.$$

This is the **ridge** estimate: it also minimizes the sum of squares plus the penalty λb^2, the quadratic penalty of the MAP discussion in Chapter 18. It shrinks the least squares slope toward 0 by a factor that approaches 1 as S_{xx} grows. For Old Faithful with $\sigma = 5.91$, a skeptical prior with $\tau = 1$, which puts the slope within ± 1.96 minutes per minute with probability 0.95, gives $\lambda = 35.0$ and the factor $353.0/388.0 = 0.91$, so $\tilde{b} = 9.76$ against $\hat{b} = 10.73$; with $\tau = 5$, $\lambda = 1.40$ and $\tilde{b} = 10.69$. Exercise 20.9 shows that the shrunken slope can have a smaller risk than $\hat{b}$, the trade between bias and variance of Chapter 7.

20.5 The linear model and the accuracy of least squares

To say how far $\hat{b}$ falls from the slope it estimates, you need a model for how the Y_i scatter around the line. The standard one treats the x_i as fixed. In an experiment that sets the doses or temperatures, they are; in observational data like Old Faithful's, where the durations are random too, the statements below are about the distribution of the estimates given the observed $x_1, \dots, x_n$. They apply when, given the durations, the waits follow the model; for i.i.d. pairs that means $\mathbb{E}[Y \mid X = x] = a + bx$ with scatter of the same size at every x (and, for the exact results, Gaussian). Then a statement that holds for every set of x's holds without the conditioning as well: an interval that covers with probability 0.95 given every possible set of durations covers with probability 0.95 overall, by the tower property (toolkit A.15). Old Faithful meets neither condition exactly: its waits bend away from the line (Figure 20.3) and consecutive eruptions are dependent; Section 20.6 says what survives.

Definition 20.8 (Linear regression model)**.** Let $x_1, \dots, x_n$ be fixed numbers, not all equal. The **linear regression model** says

$$Y_i = a + bx_i + \epsilon_i, \qquad i = 1, \dots, n,$$

where the errors have mean 0, a common variance and no correlation: $\mathbb{E}[\epsilon_i] = 0$, $\mathbb{V}\mathrm{ar}(\epsilon_i) = \sigma^2$, and $\mathbb{C}\mathrm{ov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$. Its parameter is $(a, b, \sigma^2) \in \mathbb{R}^2 \times (0, \infty)$. The **Gaussian linear model** assumes moreover that $\epsilon_1, \dots, \epsilon_n \overset{iid}{\sim} \mathrm{Gauss}(0, \sigma^2)$.

> The model says three things: the mean of Y at x lies on a line (the regression function is the line $a + bx$, so $a = a^*$ and $b = b^*$); the scatter around the line has the same size at every x; and one observation's error says nothing linear about another's. The Gaussian model adds the shape of the scatter. These are the L, E, I and N of the regression conditions an introductory course checks (Chapter 2); without N the model is semiparametric.
> **For the kiss data.** With no explanatory variable the model is $R_i = a + \epsilon_i$, and it holds with $a = p$ and errors $R_i - p$ of mean 0 and variance $\sigma^2 = p(1-p)$, uncorrelated since the couples are independent. The Gaussian version does not hold: each error takes only the two values $1 - p$ and $-p$.

Proposition 20.9 (Mean and variance of the least squares estimates)**.** *In the linear regression model,*

$$\mathbb{E}[\hat{b}] = b, \quad \mathbb{E}[\hat{a}] = a, \quad \mathbb{V}ar(\hat{b}) = \frac{\sigma^2}{S_{xx}}, \quad \mathbb{V}ar(\hat{a}) = \sigma^2\Big(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\Big), \quad \mathbb{C}ov(\hat{a}, \hat{b}) = -\frac{\sigma^2 \bar{x}}{S_{xx}}.$$

Proof strategy (core). The slope estimate is a weighted sum of the Y_i with weights that sum to zero; the true line passes through the weights unchanged, and what remains is a weighted sum of the errors.

Proof. Since $\sum_i (x_i - \bar{x}) = 0$, $S_{xy} = \sum_i (x_i - \bar{x}) Y_i$, so $\hat{b} = \sum_i w_i Y_i$ with $w_i = (x_i - \bar{x})/S_{xx}$. The weights satisfy $\sum_i w_i = 0$, $\sum_i w_i x_i = \sum_i w_i (x_i - \bar{x}) = S_{xx}/S_{xx} = 1$ and $\sum_i w_i^2 = 1/S_{xx}$. Substituting $Y_i = a + bx_i + \epsilon_i$,

$$\hat{b} = a \sum_i w_i + b \sum_i w_i x_i + \sum_i w_i \epsilon_i = b + \sum_i w_i \epsilon_i.$$

So $\mathbb{E}[\hat{b}] = b$, and since the errors are uncorrelated with variance σ^2, $\mathbb{V}\mathrm{ar}(\hat{b}) = \sigma^2 \sum_i w_i^2 = \sigma^2/S_{xx}$. Next, $\bar{Y} = a + b\bar{x} + \bar{\epsilon}$, so $\hat{a} = \bar{Y} - \hat{b}\bar{x} = a + \bar{\epsilon} - (\hat{b} - b)\bar{x}$, and $\mathbb{E}[\hat{a}] = a$. The covariance of $\bar{\epsilon} = \frac{1}{n}\sum_i \epsilon_i$ and $\hat{b} - b = \sum_i w_i \epsilon_i$ is $\frac{\sigma^2}{n} \sum_i w_i = 0$, so $\mathbb{V}\mathrm{ar}(\hat{a}) = \sigma^2/n + \bar{x}^2 \sigma^2 / S_{xx}$ and $\mathbb{C}\mathrm{ov}(\hat{a}, \hat{b}) = \mathbb{C}\mathrm{ov}(\bar{\epsilon} - (\hat{b} - b)\bar{x}, \hat{b} - b) = -\bar{x}\sigma^2/S_{xx}$. □

Unbiasedness used only $\mathbb{E}[\epsilon_i] = 0$ at every x_i, that is, that the regression function is the line; the errors can have any distribution.

Intuition: spread in x buys precision

Write $S_{xx} = (n-1)s_x^2$, with s_x the sample standard deviation of the x_i: then $\mathbb{V}\mathrm{ar}(\hat{b}) = \sigma^2/((n-1)s_x^2)$. The slope is learned from how Y changes across the x's, and the farther apart the x's are, the less a given amount of noise can tilt the line. Figure 20.4 fits lines to 30 simulated samples of 20 points, all with the same true line and noise standard deviation $\sigma = 6$. With x spread evenly over $[3, 4]$ the slope's standard deviation $\sigma/\sqrt{S_{xx}}$ is 4.42; over $[1.5, 5]$ it is 1.26. Spreading the design 3.5 times as wide does what 12 times as many observations would. An experimenter who can choose the x's should spread them, although putting all of them at the two ends of the range leaves no way to see whether the regression function is a line.

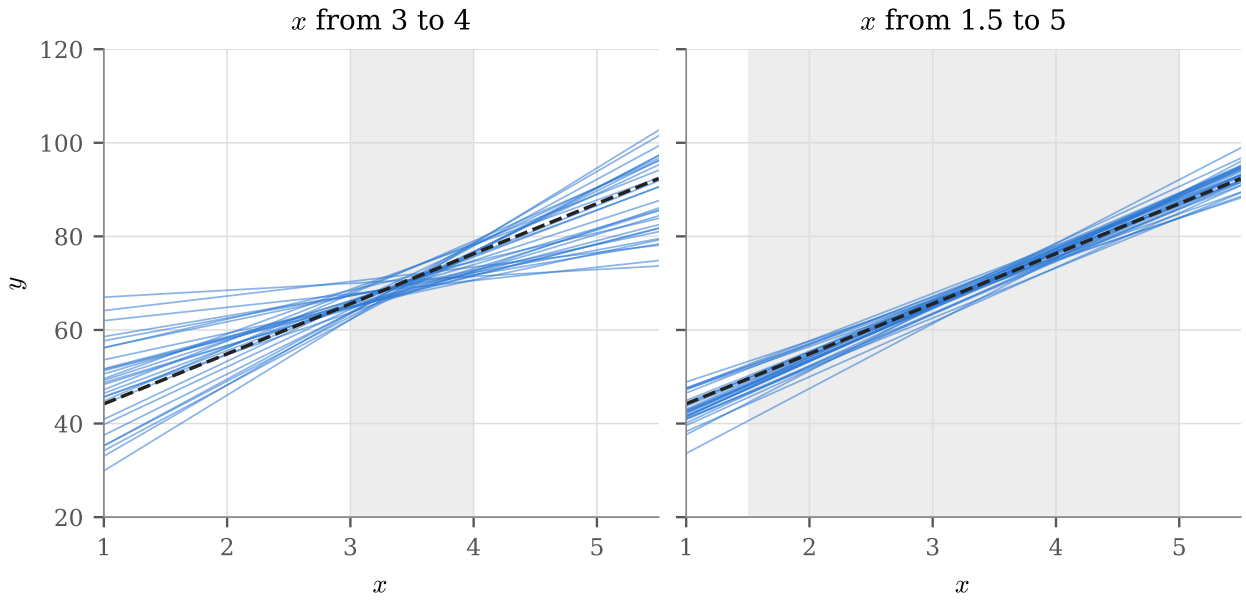


Figure 20.4: Least squares lines (blue) from 30 simulated samples of 20 points equally spaced over the shaded range, $[3, 4]$ (left) or $[1.5, 5]$ (right), with $Y = 33.5 + 10.7x$ plus Gauss$(0, 6^2)$ errors; dashed: the true line.

The slope estimate is not only unbiased; among a natural class of estimators it has the smallest variance.

Proposition 20.10 (Gauss–Markov for the slope)**.** *In the linear regression model, among estimators of b of the form $\sum_i c_i Y_i$, with numbers c_i, that are unbiased for every (a, b), the least squares slope $\hat{b}$ has the smallest variance.*

Proof strategy (for completeness; skip on a first reading). Unbiasedness for every line imposes two linear conditions on the weights. Any weights that meet them are the least squares weights plus a vector perpendicular to them, and the extra part only adds variance.

Proof. $\mathbb{E}[\sum_i c_i Y_i] = a\sum_i c_i + b\sum_i c_i x_i$ equals b for every (a, b) exactly when $\sum_i c_i = 0$ and $\sum_i c_i x_i = 1$. The least squares weights w_i meet both, so $d_i = c_i - w_i$ satisfies $\sum_i d_i = 0$ and

$\sum_i d_i x_i = 0$, and hence $\sum_i w_i d_i = \sum_i (x_i - \bar{x}) d_i / S_{xx} = 0$. Then

$$\mathbb{V}\mathrm{ar}\Big(\sum_i c_i Y_i\Big) = \sigma^2 \sum_i c_i^2 = \sigma^2 \Big(\sum_i w_i^2 + 2\sum_i w_i d_i + \sum_i d_i^2\Big) = \frac{\sigma^2}{S_{xx}} + \sigma^2 \sum_i d_i^2 \geq \mathbb{V}\mathrm{ar}(\hat{b}),$$

with equality only when every $d_i = 0$. □

The variance formulas contain σ^2, which must be estimated. The residuals stand in for the errors, but their sum of squares is divided by $n-2$, not n.

Theorem 20.11 (The residual variance, and exact distributions with Gaussian errors)**.** *In the linear regression model with $n \geq 3$, $\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^n e_i^2$ is an unbiased estimator of σ^2. In the Gaussian linear model, moreover:*

1. *$\hat{b} \sim \mathrm{Gauss}(b, \sigma^2/S_{xx})$ and $\hat{a} \sim \mathrm{Gauss}\big(a, \sigma^2(\frac{1}{n} + \bar{x}^2/S_{xx})\big)$;*
2. *$(n-2)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-2}$;*
3. *$(\hat{a}, \hat{b})$ and $\hat{\sigma}^2$ are independent;*
4. *$\dfrac{\hat{b} - b}{\hat{\sigma}/\sqrt{S_{xx}}} \sim t_{n-2}$, and likewise $\dfrac{\hat{a} - a}{\hat{\sigma}\sqrt{1/n + \bar{x}^2/S_{xx}}} \sim t_{n-2}$.*

Proof strategy (core). Rotate the errors as in Cochran's theorem, with the first two axes along $(1, \ldots, 1)$ and along the centered x's. The fit uses up those two coordinates, and the residuals are the other $n-2$, each with mean 0 and variance σ^2; with Gaussian errors they are independent Gaussians unconnected to the fit.

Recall. The chi-squared distribution (Definition 14.1): a sum of k squared independent standard Gaussians is χ^2_k. Student's distribution (Definition 14.5): if $Z \sim \mathrm{Gauss}(0,1)$ and $V \sim \chi^2_k$ are independent, $Z/\sqrt{V/k} \sim t_k$.

Proof. Let $u_1 = (1, \ldots, 1)/\sqrt{n}$ and $u_2 = (x_1 - \bar{x}, \ldots, x_n - \bar{x})/\sqrt{S_{xx}}$. Both are unit vectors, and they are perpendicular since $\sum_i (x_i - \bar{x}) = 0$. Extend them to an orthonormal basis $u_1, \ldots, u_n$ of $\mathbb{R}^n$ by Gram–Schmidt (toolkit A.13), let Q be the orthogonal matrix with rows $u_1^\top, \ldots, u_n^\top$, and set $Z = Q\epsilon$ for the error vector $\epsilon = (\epsilon_1, \ldots, \epsilon_n)$, so that $Z_k = u_k^\top \epsilon$ and $\epsilon = \sum_k Z_k u_k$. Each Z_k has mean 0, and since the errors are uncorrelated with variance σ^2, $\mathbb{E}[Z_j Z_k] = \sigma^2 \sum_i u_{ji} u_{ki}$, which is σ^2 for $j = k$ and 0 otherwise: the Z_k are uncorrelated with variance σ^2.

Now locate the fit in these coordinates. From the proof of Proposition 20.9, $\hat{b} - b = \sum_i w_i \epsilon_i = Z_2/\sqrt{S_{xx}}$ and $\hat{a} - a + (\hat{b} - b)\bar{x} = \bar{\epsilon} = Z_1/\sqrt{n}$. The residuals are

$$e_i = Y_i - \hat{a} - \hat{b}x_i = \epsilon_i - (\hat{a} - a) - (\hat{b} - b)x_i = \epsilon_i - \bar{\epsilon} - (\hat{b} - b)(x_i - \bar{x}),$$

so the residual vector is $e = \epsilon - Z_1 u_1 - Z_2 u_2 = \sum_{k=3}^n Z_k u_k$. Since the u_k are orthonormal, $\sum_i e_i^2 = \|e\|^2 = Z_3^2 + \cdots + Z_n^2$, whose mean is $(n-2)\sigma^2$: this is the unbiasedness of $\hat{\sigma}^2$.

With Gaussian errors, $Z/\sigma = Q(\epsilon/\sigma)$ has i.i.d. $\mathrm{Gauss}(0,1)$ coordinates by rotation invariance (toolkit A.13). Then $\hat{b} = b + Z_2/\sqrt{S_{xx}}$, and $\hat{a} = a + Z_1/\sqrt{n} - \bar{x}Z_2/\sqrt{S_{xx}}$ is a linear combination of independent Gaussians, hence Gaussian (Chapter 6), with the variance of Proposition 20.9: this is (1). Next, $(n-2)\hat{\sigma}^2/\sigma^2 = (Z_3/\sigma)^2 + \cdots + (Z_n/\sigma)^2$ is a sum of $n-2$ squared independent standard Gaussians, which is (2). The pair $(\hat{a}, \hat{b})$ is a function of (Z_1, Z_2) and $\hat{\sigma}^2$ a function of $(Z_3, \ldots, Z_n)$, and these are independent, which is (3). Finally,

$$\frac{\hat{b} - b}{\hat{\sigma}/\sqrt{S_{xx}}} = \frac{(\hat{b} - b)\sqrt{S_{xx}}/\sigma}{\sqrt{\hat{\sigma}^2/\sigma^2}}$$

is a standard Gaussian divided by the square root of an independent χ^2_{n-2} variable over its degrees of freedom, which is t_{n-2}; the same argument applies to $\hat{a}$. □

For Old Faithful, $\hat{\sigma} = \sqrt{9{,}443/270} = 5.91$ minutes, so the estimated standard errors are $\hat{\sigma}/\sqrt{S_{xx}} = 5.91/\sqrt{353.0} = 0.315$ for the slope and $\hat{\sigma}\sqrt{1/272 + 3.488^2/353.0} = 1.15$ for the intercept.

From undergraduate statistics: the residual standard error and its $n-2$

Regression output reports $\hat{\sigma}$ as the residual standard error 'on $n-2$ degrees of freedom', and the ANOVA table of a regression reports $\hat{\sigma}^2$ as its MSE (Chapter 7). The proof gives $n-2$ the reading that the sample variance's $n-1$ had in Chapter 14: of the n directions the errors can point in, the fit absorbs two, one for the height of the line and one for its slope, and the residuals live in the remaining $n-2$. Dividing by n, as maximum likelihood does, underestimates σ^2 on average by the factor $(n-2)/n$. A fit with a constant and no slope absorbs one direction, and its divisor is $n-1$: that is the sample variance, 0.231 for the kiss data (Exercise 20.5).

20.6 Tests and intervals

With Theorem 20.11, the t methods of Chapter 14 carry over with $n-2$ in place of $n-1$. In the Gaussian linear model, the test of $H_0 : b = b_0$ against $H_1 : b \neq b_0$ that rejects when

$$|T| > q_{\alpha/2}(t_{n-2}), \qquad T = \frac{\hat{b} - b_0}{\hat{\sigma}/\sqrt{S_{xx}}},$$

has level exactly α, and $\hat{b} \pm q_{\alpha/2}(t_{n-2})\,\hat{\sigma}/\sqrt{S_{xx}}$ is a confidence interval of level exactly $1-\alpha$, by the argument of Proposition 14.9. The most common null hypothesis is $b_0 = 0$: x is of no use in predicting Y with a line.

Worked example: the Old Faithful slope

$T = 10.73/0.315 = 34.1$ for $H_0 : b = 0$, against the threshold $q_{0.025}(t_{270}) = 1.969$, with p-value about 10^{-99}. The 95% confidence interval for the slope is $10.73 \pm 1.969 \times 0.315 = [10.11, 11.35]$ minutes of wait per minute of eruption. At $n = 272$ the t and Gaussian quantiles differ in the third digit, 1.969 against 1.960.

Are Old Faithful's errors Gaussian? Hardly: the waits are whole minutes, and the residuals of Figure 20.3 do not even average zero at every duration. What survives without Gaussian errors, and without the line being the regression function, is the large-sample theory for i.i.d. pairs. Old Faithful's pairs are not: the eruptions are consecutive, and a short eruption is almost always followed by a long one (only 6 of the 97 short eruptions were followed by another short one, where independence would give about 35). Methods for dependent data, which this book does not develop, put the slope's standard error between 0.33 and 0.36 instead of 0.315, so T is between 30 and 33 instead of 34.1.

Theorem 20.12 (Least squares without Gaussian errors)**.** *Let $(X_1, Y_1), (X_2, Y_2), \ldots$ be i.i.d. copies of (X, Y) with $\mathbb{E}[X^4]$ and $\mathbb{E}[Y^4]$ finite and $\mathbb{V}ar(X) > 0$, let a^*, b^* and ϵ be the theoretical regression line and its noise, and suppose $v = \mathbb{E}[(X - \mathbb{E}[X])^2\epsilon^2] > 0$. Then $\hat{a} \xrightarrow[n\to\infty]{\mathbb{P}} a^*$, $\hat{b} \xrightarrow[n\to\infty]{\mathbb{P}} b^*$, and*

$$\sqrt{n}(\hat{b} - b^*) \xrightarrow[n\to\infty]{(d)} \mathit{Gauss}\Big(0, \frac{v}{\mathbb{V}ar(X)^2}\Big).$$

If moreover $\mathbb{E}[\epsilon^2 \mid X] = \sigma^2$ *does not depend on* X*, then* $v/\mathbb{V}ar(X)^2 = \sigma^2/\mathbb{V}ar(X)$*,* $\hat{\sigma}^2 \xrightarrow[n\to\infty]{\mathbb{P}} \sigma^2$*, and* $T = (\hat{b} - b^*)/(\hat{\sigma}/\sqrt{S_{xx}}) \xrightarrow[n\to\infty]{(d)} Gauss(0,1)$*.*

Proof strategy (for completeness; skip on a first reading). $\hat{b} - b^*$ is an average of $(X_i - \bar{X})\epsilon_i$ divided by an average of $(X_i - \bar{X})^2$: the Central Limit Theorem for the first, the Law of Large Numbers for the second, Slutsky's theorem for the ratio.

Recall. The Law of Large Numbers (Theorem 3.5): for i.i.d. random variables with mean μ and finite variance, $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mu$. The Central Limit Theorem (Theorem 4.8): if the variance σ^2 is also positive, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, \sigma^2)$. The continuous mapping theorem (Theorem 5.2): if $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$ and g is continuous, $g(X_n) \xrightarrow[n\to\infty]{\mathbb{P}} g(X)$. Slutsky's theorem (Theorem 5.1): if $X_n \xrightarrow[n\to\infty]{(\mathrm{d})} X$ and $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$, a constant, then $X_n + Y_n \xrightarrow[n\to\infty]{(\mathrm{d})} X + y$, $X_n Y_n \xrightarrow[n\to\infty]{(\mathrm{d})} yX$, and $X_n/Y_n \xrightarrow[n\to\infty]{(\mathrm{d})} X/y$ if $y \neq 0$. And a limit in distribution that is a constant is a limit in probability (Proposition 4.7).

Proof. Write $\mu = \mathbb{E}[X]$ and $\epsilon_i = Y_i - a^* - b^* X_i$. Then $Y_i - \bar{Y} = b^*(X_i - \bar{X}) + \epsilon_i - \bar{\epsilon}$, so $S_{xy} = b^* S_{xx} + \sum_i (X_i - \bar{X})\epsilon_i$ (the $\bar{\epsilon}$ term drops out since $\sum_i (X_i - \bar{X}) = 0$), and

$$\hat{b} - b^* = \frac{\frac{1}{n}\sum_i (X_i - \bar{X})\epsilon_i}{S_{xx}/n}.$$

The denominator is $S_{xx}/n = \frac{1}{n}\sum_i X_i^2 - \bar{X}^2 \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[X^2] - \mu^2 = \mathbb{V}\mathrm{ar}(X)$, by the Law of Large Numbers for the X_i^2 and the X_i (both have finite variance since $\mathbb{E}[X^4] < \infty$) and the continuous mapping theorem. The numerator is $\frac{1}{n}\sum_i W_i - (\bar{X} - \mu)\bar{\epsilon}$, with $W_i = (X_i - \mu)\epsilon_i$. The W_i are i.i.d. with mean $\mathbb{E}[(X - \mu)\epsilon] = \mathbb{C}\mathrm{ov}(X, \epsilon) = 0$ (Theorem 20.4) and variance v, which is finite: ϵ, a linear combination of 1, X and Y, has a finite fourth moment, and $2st \leq s^2 + t^2$ bounds $(X - \mu)^2\epsilon^2$ by $\frac{1}{2}((X - \mu)^4 + \epsilon^4)$. By the Law of Large Numbers the numerator converges in probability to $0 - 0 \cdot 0 = 0$, so $\hat{b} \xrightarrow[n\to\infty]{\mathbb{P}} b^*$; and $\hat{a} = \bar{Y} - \hat{b}\bar{X} \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[Y] - b^*\mu = a^*$.

For the limit distribution, $\sqrt{n}\,\frac{1}{n}\sum_i W_i \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, v)$ by the Central Limit Theorem, while $\sqrt{n}(\bar{X} - \mu)\bar{\epsilon} \xrightarrow[n\to\infty]{(\mathrm{d})} 0$ by Slutsky's theorem, since $\sqrt{n}(\bar{X} - \mu)$ converges in distribution and $\bar{\epsilon} \xrightarrow[n\to\infty]{\mathbb{P}} 0$; so this term converges to 0 in probability, and $\sqrt{n}$ times the numerator converges in distribution to $\mathrm{Gauss}(0, v)$ by Slutsky's theorem. Dividing by $S_{xx}/n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{V}\mathrm{ar}(X)$ gives $\sqrt{n}(\hat{b} - b^*) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, v/\mathbb{V}\mathrm{ar}(X)^2)$, by Slutsky's theorem once more.

If $\mathbb{E}[\epsilon^2 \mid X] = \sigma^2$, taking out what is known and the tower property give $v = \mathbb{E}\left[(X - \mu)^2\,\mathbb{E}[\epsilon^2 \mid X]\right] = \sigma^2\,\mathbb{V}\mathrm{ar}(X)$ (toolkit A.15). For $\hat{\sigma}^2$, write $e_i = \epsilon_i - \bar{\epsilon} - (\hat{b} - b^*)(X_i - \bar{X})$, as in the proof of Theorem 20.11; since $\sum_i (\epsilon_i - \bar{\epsilon})(X_i - \bar{X}) = (\hat{b} - b^*)S_{xx}$, expanding the square gives

$$\frac{1}{n}\sum_i e_i^2 = \frac{1}{n}\sum_i (\epsilon_i - \bar{\epsilon})^2 - (\hat{b} - b^*)^2 \frac{S_{xx}}{n} \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[\epsilon^2] - 0 = \sigma^2,$$

and $\hat{\sigma}^2$ differs from this by the factor $n/(n-2) \to 1$. Finally $T = \sqrt{n}(\hat{b} - b^*)/\left(\hat{\sigma}/\sqrt{S_{xx}/n}\right)$, where the numerator converges in distribution to $\mathrm{Gauss}(0, \sigma^2/\mathbb{V}\mathrm{ar}(X))$ and the denominator in probability to $\sigma/\sqrt{\mathbb{V}\mathrm{ar}(X)}$, so $T \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0,1)$ by Slutsky's theorem. □

Three things follow. First, when the scatter around the best line, $\mathbb{E}[\epsilon^2 \mid X = x]$, has the same size at every x, the t-test and t-interval for the slope have asymptotic level α and coverage $1-\alpha$ whatever the errors' distribution. When the regression function bends, that scatter includes the squared distance from $f(x)$ to the line, so it changes with x even when the spread of Y around $f(x)$ does not; the robust standard error below checks whether that matters. Second, when the scatter changes with x, $\hat{\sigma}/\sqrt{S_{xx}}$ estimates the wrong variance; plugging the residuals into v gives the robust, or sandwich, standard error $\sqrt{\sum_i (X_i - \bar{X})^2 e_i^2}\,/S_{xx}$, the sandwich of Chapter 13 for the squared loss. Under the hypotheses of Theorem 20.12, $\frac{1}{n}\sum_i (X_i - \bar{X})^2 e_i^2 \xrightarrow[n\to\infty]{\mathbb{P}} v$, so n times the squared robust standard error converges in probability to $v/\,\mathbb{V}\mathrm{ar}(X)^2$; this follows from the Law of Large Numbers in the form that needs only a finite mean (Theorem B.3) with the bookkeeping of the proof above, and the details are not given here. For Old Faithful the robust standard error is 0.300 against 0.315: the scatter is nearly constant, with residual standard deviations of 5.7 minutes after eruptions shorter than 3 minutes and 6.0 after longer ones. Third, the theorem never assumed that the regression function is a line: $\hat{b}$ estimates b^*, the slope of the best line, which is the sense in which the slope 10.73 means something despite the bend in Figure 20.3.

The other question every regression is asked is how good its predictions are, and at a given x_0 it is two questions. Where is the line at x_0, the mean wait after all 4.5-minute eruptions? And where will the next Y at x_0 fall, the wait after the next 4.5-minute eruption?

Proposition 20.13 (Confidence and prediction intervals)**.** *In the Gaussian linear model, let* $\hat{y}_0 = \hat{a} + \hat{b}x_0$*, and let* $Y_0 = a + bx_0 + \epsilon_0$ *be a new observation at* x_0*, with* $\epsilon_0 \sim$ *Gauss*$(0, \sigma^2)$ *independent of* $\epsilon_1, \ldots, \epsilon_n$*. With* $q = q_{\alpha/2}(t_{n-2})$*, the interval*

$$\hat{y}_0 \pm q\,\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

contains $a + bx_0$ *with probability* $1-\alpha$*, and the interval*

$$\hat{y}_0 \pm q\,\hat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

contains Y_0 *with probability* $1-\alpha$*.*

Proof strategy (for completeness; skip on a first reading). $\hat{y}_0 = \bar{Y} + \hat{b}(x_0 - \bar{x})$ is Gaussian with variance $\sigma^2(1/n + (x_0 - \bar{x})^2/S_{xx})$, and the new error adds σ^2. Standardizing by $\hat{\sigma}$, which is independent of both, gives t_{n-2} as in Theorem 20.11.

Proof. In the coordinates of the proof of Theorem 20.11, $\hat{y}_0 - (a + bx_0) = (\hat{a} - a) + (\hat{b} - b)x_0 = Z_1/\sqrt{n} + (x_0 - \bar{x})Z_2/\sqrt{S_{xx}}$, a Gaussian with mean 0 and variance $\sigma^2(1/n + (x_0 - \bar{x})^2/S_{xx})$, independent of $\hat{\sigma}^2$. Dividing it by $\hat{\sigma}\sqrt{1/n + (x_0 - \bar{x})^2/S_{xx}}$ gives a standard Gaussian divided by the square root of an independent χ^2_{n-2} over $n-2$, which is t_{n-2}; this is the first interval. For the second, $Y_0 - \hat{y}_0 = \epsilon_0 - (\hat{y}_0 - a - bx_0)$, and ϵ_0 is independent of $Z_1, \ldots, Z_n$, so $Y_0 - \hat{y}_0$ is Gaussian with mean 0 and variance $\sigma^2(1 + 1/n + (x_0 - \bar{x})^2/S_{xx})$, and is independent of $\hat{\sigma}^2$; the same standardization gives t_{n-2}. □

After a 4.5-minute eruption the line predicts 81.8 minutes. The 95% confidence interval for the mean wait after such eruptions is $81.8 \pm 0.9 = [80.8, 82.7]$; the 95% prediction interval for the next wait

after one is $81.8 \pm 11.7 = [70.1, 93.4]$. In natural frequencies: of 100 eruptions lasting 4.5 minutes, the model says about 95 are followed by a wait between 70 and 93 minutes. Figure 20.5 draws both intervals for every duration. The confidence band is narrowest at $\bar{x} = 3.49$, where $(x_0 - \bar{x})^2 = 0$, and widens toward the ends, since an error in the slope moves the line more the farther you are from the point it pivots on; the prediction band is nearly parallel, since its 1 dominates. In the data, 260 of the 272 waits, 95.6%, fall inside the prediction band.

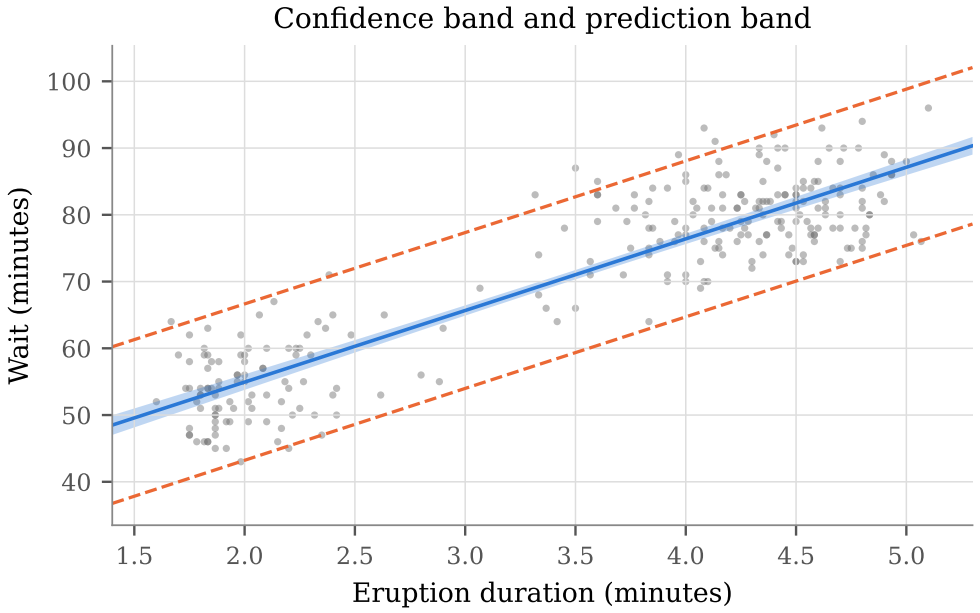


Figure 20.5: Old Faithful's 272 eruptions (gray), the least squares line (blue), the 95% confidence band for the regression line (shaded) and the 95% prediction band for a new wait (orange, dashed).

Pitfall: a prediction interval does not shrink with n

The confidence band's width is of order $1/\sqrt{n}$ and vanishes as the data accumulate. The prediction band's width tends to $2q\sigma$, the spread of a single new wait, which no amount of data removes. And the prediction interval is about one new error ϵ_0, so no limit theorem rescues it: Theorem 20.12 makes $\hat{a}$, $\hat{b}$ and $\hat{\sigma}$ accurate, but the probability that ϵ_0 lands within $\pm q\sigma$ depends on the errors' distribution at every n. Neither interval says anything about durations outside the data's range, 1.6 to 5.1 minutes, where nothing shows that the line still holds.

20.7 Correlation, R^2 and regression toward the mean

The slope depends on the units of both variables. A measure of linear association that does not is the correlation.

Definition 20.14 (Correlation)**.** The **correlation** of random variables X and Y with positive finite variances is

$$\rho = \frac{\mathbb{C}\mathrm{ov}(X, Y)}{\sqrt{\mathbb{V}\mathrm{ar}(X)\,\mathbb{V}\mathrm{ar}(Y)}}.$$

The **sample correlation** of pairs (X_i, Y_i) is $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$.

The correlation is the slope of the best line when both variables are measured in standard deviations: by Theorem 20.4, $b^* = \rho\sqrt{\mathbb{V}\mathrm{ar}(Y)/\mathbb{V}\mathrm{ar}(X)}$, so an increase of one standard deviation in X predicts a change of ρ standard deviations in Y. It lies between -1 and 1, and is ± 1 exactly when Y is a linear function of X (Proposition 20.15 below). The sample correlation is the same quantity for the empirical distribution of the pairs.

For the kiss data. A correlation needs two variables, and the kiss data have one. The correlation of R with itself is 1, a perfect line; with any variable recorded independently of the turns it is 0, and the best line is flat.

Proposition 20.15 (What the line explains)**.** *For the least squares fit,*

$$\sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^{n} e_i^2,$$

and the ***coefficient of determination*** $R^2 = 1 - \sum_i e_i^2/S_{yy}$ *equals* r^2*. In the population,* $\mathbb{V}ar(Y) = b^{*2}\,\mathbb{V}ar(X) + \mathbb{V}ar(\epsilon)$ *and* $\mathbb{V}ar(\epsilon) = (1-\rho^2)\,\mathbb{V}ar(Y)$*, so* $|\rho| \le 1$*, with equality exactly when* $\epsilon = 0$ *with probability one.*

Proof strategy (for completeness; skip on a first reading). $\hat{Y}_i - \bar{Y} = \hat{b}(X_i - \bar{X})$, and the residuals are uncorrelated with the X_i, so the cross term in the square of $(\hat{Y}_i - \bar{Y}) + e_i$ vanishes. The population version is the same computation with the noise.

Proof. $Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + e_i$, with $\hat{Y}_i - \bar{Y} = \hat{b}(X_i - \bar{X})$ since $\hat{a} = \bar{Y} - \hat{b}\bar{X}$. The cross term $2\hat{b}\sum_i (X_i - \bar{X})e_i$ is 0 by the normal equations of Theorem 20.6, which gives the decomposition. Its first term is $\hat{b}^2 S_{xx} = S_{xy}^2/S_{xx} = r^2 S_{yy}$, so $R^2 = r^2$. In the population, $Y = a^* + b^* X + \epsilon$ with $\mathbb{C}\mathrm{ov}(X, \epsilon) = 0$, so $\mathbb{V}\mathrm{ar}(Y) = b^{*2}\,\mathbb{V}\mathrm{ar}(X) + \mathbb{V}\mathrm{ar}(\epsilon)$, and $b^{*2}\,\mathbb{V}\mathrm{ar}(X) = \mathbb{C}\mathrm{ov}(X,Y)^2/\mathbb{V}\mathrm{ar}(X) = \rho^2\,\mathbb{V}\mathrm{ar}(Y)$. Since $\mathbb{V}\mathrm{ar}(\epsilon) \ge 0$, $\rho^2 \le 1$, with equality exactly when $\mathbb{V}\mathrm{ar}(\epsilon) = 0$, that is, when ϵ equals its mean 0 with probability one. □

For Old Faithful, $r = 0.901$ and $R^2 = 0.811$: the line accounts for 81% of the variance of the waits. In standard deviations, the waits vary by 13.6 minutes around their mean and by 5.9 around the line.

Written as $\hat{b} = r\,s_y/s_x$, the least squares line has a consequence that gave regression its name. In standard deviations, the predicted y is r times the value of x, so whenever $|r| < 1$ the prediction is closer to the mean than the explanatory value is. Francis Galton found this in the heights of 928 adult children and their parents: 'the height-deviate of the offspring is, on the average, two-thirds of the height-deviate of its mid-parentage', where the mid-parent's height is the average of the father's and of 1.08 times the mother's (Galton, *Journal of the Anthropological Institute*, 1886). His two-thirds is a slope, inches of child per inch of mid-parent, not a correlation: a mid-parent's height, an average of two heights, varies less than a child's (standard deviations of 1.8 and 2.5 inches in his table), so the correlation, about 0.46, is smaller than the slope, about 0.65. He called the pull toward the mean regression towards mediocrity. It is not a force of heredity: it holds for any two variables that are less than perfectly correlated, and it runs both ways, since tall children have parents who are, on average, less tall than they are. Figure 20.6 draws a bivariate Gaussian with correlation 0.5 in standard units. The ellipses of equal density are tilted along the line $y = x$, but the conditional means of y, the blue dots, lie on the line $y = 0.5x$, which passes through the points where the ellipses have vertical tangents; the best line for predicting x from y is $x = 0.5y$, the orange line, which is steeper than $y = x$.

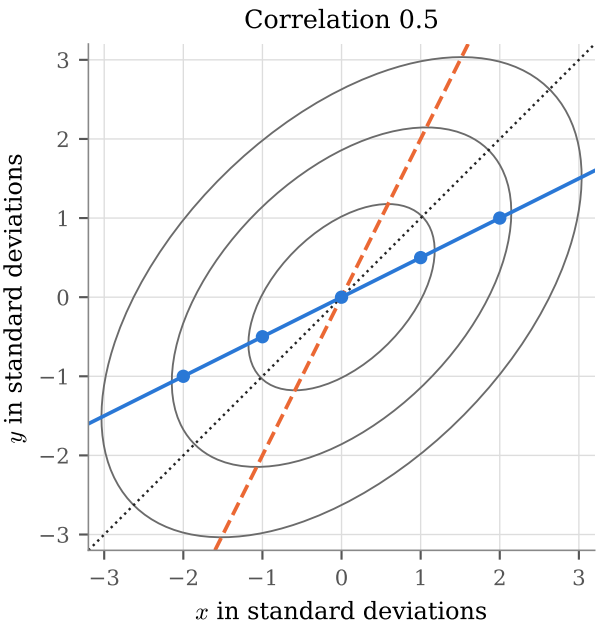


Figure 20.6: A bivariate Gaussian with standard marginals and correlation 0.5: ellipses holding 50%, 90% and 99% of the probability (gray), the line $y = x$ (dotted), the regression of y on x (blue, with dots at $x = -2, \ldots, 2$) and of x on y (orange, dashed).

Does this make sense? Is the line for predicting x from y the same line?

No. Regressing the duration on the wait gives the slope $S_{xy}/S_{yy} = 3{,}788/50{,}087 = 0.0756$ minutes of eruption per minute of wait; turned around, that is 13.2 minutes of wait per minute of eruption, not 10.7. The product of the two slopes is $S_{xy}^2/(S_{xx}S_{yy}) = r^2 = 0.811$, so the two lines coincide only when $r = \pm 1$. Least squares measures misses vertically, in the variable being predicted, and the two predictions answer different questions (Exercise 20.8).

Regression toward the mean is a frequent source of false conclusions.

Pitfall: regression toward the mean is not an effect

Patients chosen for an extreme blood pressure reading, students chosen for a low test score, a team after its best season: measured again, they are on average less extreme, with no treatment and no cause, whenever the two measurements are less than perfectly correlated. An improvement measured only in a group selected for being extreme is not evidence that anything worked; a comparison group selected the same way is. More generally, a slope says how Y and x move together in the data, not what changing x would do to Y: the Old Faithful slope has a physical explanation (Section 20.1), but the line alone would not show that.

20.8 What comes next

Most responses depend on more than one variable: the next wait might depend on the eruption just ended and on the wait before it. Chapter 21 lets x be a vector, writes the n equations as one with a design matrix, and finds that least squares projects the response vector onto the subspace

spanned by the design's columns, which is the rotation in the proof of Theorem 20.11 with more axes absorbed by the fit. The t-test of this chapter tests one coefficient at a time and the F-test tests groups of them, exactly with Gaussian errors and approximately without; full rank of the design is the identifiability of Chapter 2; ridge returns in matrix form. Responses that are proportions or counts, such as the kiss data's 0s and 1s, call for a different link between the mean and the line, the generalized linear models of Chapters 22 and 23.

20.9 Summary

- The regression function $f(x) = \mathbb{E}[Y \mid X = x]$ is the best predictor of Y from X in mean squared error: $\mathbb{E}[(Y - g(X))^2] = \mathbb{E}[(Y - f(X))^2] + \mathbb{E}[(f(X) - g(X))^2]$.
- The theoretical regression line minimizes $\mathbb{E}[(Y - a - bX)^2]$: $b^* = \mathbb{C}\mathrm{ov}(X, Y)/\mathbb{V}\mathrm{ar}(X)$, $a^* = \mathbb{E}[Y] - b^* \mathbb{E}[X]$. The noise $\epsilon = Y - a^* - b^* X$ has mean 0 and is uncorrelated with X; it has conditional mean 0 at every x only when f is a line, and otherwise the best line is the line closest to f.
- Least squares: $\hat{b} = S_{xy}/S_{xx}$, $\hat{a} = \bar{Y} - \hat{b}\bar{X}$, the theoretical line of the empirical distribution; the residuals sum to 0 and are uncorrelated with the X_i. Old Faithful: wait $= 33.47 + 10.73\times$ duration.
- With Gaussian errors least squares is maximum likelihood (Gauss), with $\hat{\sigma}^2_{\mathrm{MLE}} = \mathrm{RSS}/n$, and the posterior mode under a flat prior. A Gauss$(0, \tau^2)$ prior on the slope gives ridge, $S_{xy}/(S_{xx} + \sigma^2/\tau^2)$: 9.76 for Old Faithful with $\tau = 1$.
- In the linear model (fixed x's; uncorrelated errors with mean 0 and variance σ^2): $\hat{a}$ and $\hat{b}$ are unbiased, $\mathbb{V}\mathrm{ar}(\hat{b}) = \sigma^2/S_{xx} = \sigma^2/((n-1)s_x^2)$, smallest among linear unbiased estimators (Gauss–Markov), and $\hat{\sigma}^2 = \mathrm{RSS}/(n-2)$ is unbiased. With Gaussian errors, $(n-2)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-2}$, independent of $(\hat{a}, \hat{b})$, and $(\hat{b} - b)/(\hat{\sigma}/\sqrt{S_{xx}}) \sim t_{n-2}$ exactly.
- Old Faithful: $\hat{\sigma} = 5.91$, standard error 0.315, $T = 34.1$, 95% interval $[10.11, 11.35]$. Without Gaussian errors, for i.i.d. pairs, $\hat{b} \xrightarrow[n\to\infty]{\mathbb{P}} b^*$ and $\sqrt{n}(\hat{b} - b^*)$ is asymptotically Gaussian with variance $\mathbb{E}[(X - \mathbb{E}\,X)^2\epsilon^2]/\mathbb{V}\mathrm{ar}(X)^2$, which is $\sigma^2/\mathbb{V}\mathrm{ar}(X)$ when the scatter is constant; the robust standard error is 0.300, and 0.33 to 0.36 allowing for the dependence between consecutive eruptions.
- At x_0: the confidence interval for $a + bx_0$ is $\hat{y}_0 \pm q\hat{\sigma}\sqrt{1/n + (x_0 - \bar{x})^2/S_{xx}}$, and the prediction interval for a new Y_0 adds 1 under the root; it does not shrink with n and depends on the errors' distribution. After a 4.5-minute eruption: $[80.8, 82.7]$ and $[70.1, 93.4]$.
- $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$, $\hat{b} = r\, s_y/s_x$, $R^2 = r^2$ (0.811 for Old Faithful). In standard deviations a prediction is r times the explanatory value: regression toward the mean (Galton's two-thirds). Predicting x from y is a different line.

20.10 Exercises

Exercise 20.1. A pair (X, Y) takes the values $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$, $(2,1)$, $(2,2)$ with probabilities 0.2, 0.1, 0.1, 0.2, 0.1, 0.3. (a) Find the regression function at $x = 0, 1, 2$. (b) Find the theoretical regression line. (c) Is the regression function a line? Check that the noise has mean 0 and is uncorrelated with X. *(Solution on page 434.)*

Exercise 20.2. Let $Y = X^2 + \eta$, with η independent of X, of mean 0 and finite variance. (a) For $X \sim \text{Unif}(0,1)$, show that the theoretical regression line is $-\frac{1}{6} + x$, and find the conditional mean of the noise. (b) For $X \sim \text{Unif}(-1,1)$, show that the best line is flat, although Y is nearly a function of X when η is small. What is the correlation of X and Y? *(Solution on page 434.)*

Exercise 20.3. Express Old Faithful's durations in seconds and its waits in hours. What happens to $\hat{b}$, $\hat{a}$, r, R^2 and the t statistic for $b = 0$? *(Solution on page 434.)*

Exercise 20.4. Regression through the origin. In the model $Y_i = bx_i + \epsilon_i$, with errors as in Definition 20.8 and not all x_i zero, (a) find the least squares estimator of b; (b) show that it is unbiased with variance $\sigma^2 / \sum_i x_i^2$; (c) find its bias if in fact $Y_i = a + bx_i + \epsilon_i$ with $a \neq 0$. *(Solution on page 434.)*

Exercise 20.5. (a) Fit the constant model $R_i = a + \epsilon_i$ to the kiss data by least squares, and compute the residual sum of squares and $\hat{\sigma}^2$ with divisor $n - 1$. (b) Show that the t statistic for $H_0 : a = \frac{1}{2}$ is the one-sample t statistic of Chapter 14, and compute it. *(Solution on page 435.)*

Exercise 20.6. A 0/1 explanatory variable. (a) Show that if each x_i is 0 or 1, with n_0 zeros and n_1 ones, then $S_{xx} = n_0 n_1 / n$, $\hat{a} = \bar{Y}_0$ and $\hat{b} = \bar{Y}_1 - \bar{Y}_0$, where $\bar{Y}_0, \bar{Y}_1$ are the means of the two groups. (b) In the Physicians' Health Study of Chapter 15, let $x_i = 1$ for aspirin and $Y_i = 1$ for a heart attack: 189 of 11,034 on placebo and 104 of 11,037 on aspirin. Compute $\hat{b}$, $\hat{\sigma}^2$ and the t statistic for $b = 0$, and compare its square with the Wald statistic 25.04 of that chapter. (c) Which assumptions of the Gaussian linear model fail here, and which result supports the test anyway? *(Solution on page 435.)*

Exercise 20.7. For Old Faithful, compute the 95% confidence interval for the mean wait after 2-minute eruptions and the 95% prediction interval for the next wait after one, using $\hat{a} = 33.47$, $\hat{b} = 10.73$, $\bar{x} = 3.488$, $S_{xx} = 353.0$, $\hat{\sigma} = 5.914$, $n = 272$ and $q_{0.025}(t_{270}) = 1.969$. Why is the confidence interval wider than at 4.5 minutes, while the prediction intervals have almost the same width? *(Solution on page 435.)*

Exercise 20.8. (a) Show that the least squares line for predicting X from Y has slope S_{xy}/S_{yy} and passes through $(\bar{X}, \bar{Y})$, and that the product of its slope and $\hat{b}$ is r^2. (b) For Old Faithful, with $S_{yy} = 50{,}087$, predict the duration of an eruption followed by a 90-minute wait, and compare it with the duration at which the line for predicting the wait reaches 90. *(Solution on page 435.)*

Exercise 20.9. Ridge's risk. In the linear model, the ridge slope of Section 20.4 is $\tilde{b}_\lambda = S_{xy}/(S_{xx}+\lambda)$. (a) Find its mean and variance. (b) Show that its mean squared error is $(\sigma^2 S_{xx} + \lambda^2 b^2)/(S_{xx} + \lambda)^2$, and that it is smaller than $\mathbb{V}\text{ar}(\hat{b}) = \sigma^2/S_{xx}$ for every small enough $\lambda > 0$, whatever b is. (c) Which λ minimizes it, and why can't that value be used directly? *(Solution on page 435.)*

Exercise 20.10. A two-point slope. In the linear model with $x_i = i$ for $i = 1, \ldots, 20$, compare $\tilde{b} = (Y_{20} - Y_1)/19$ with $\hat{b}$. (a) Show that $\tilde{b}$ is linear and unbiased, with variance $2\sigma^2/361$. (b) Compute S_{xx} and the ratio $\mathbb{V}\text{ar}(\tilde{b})/\mathbb{V}\text{ar}(\hat{b})$. Which result says that the ratio cannot be below 1? *(Solution on page 436.)*

Chapter 21

Multivariate Linear Regression and Inference

In this chapter

- Several predictors at once: the model $Y = X\beta + \epsilon$ with a design matrix X, which also covers polynomials, 0/1 variables and the previous eruption's wait and duration for Old Faithful.
- Least squares in matrix form, $\hat{\beta} = (X^\top X)^{-1} X^\top Y$, and what it is: the projection of the response vector onto the space spanned by the columns of X. Full rank as identifiability.
- With Gaussian errors: the distribution of $\hat{\beta}$, the residual variance on $n - p$ degrees of freedom by the general form of Cochran's theorem, the exact t-test for one coefficient and the F-test for several, which is the likelihood ratio test. Bonferroni as the alternative.
- Why a coefficient depends on which other predictors are in the model, and why two predictors can matter together when neither does alone.
- Ridge in matrix form, large samples without Gaussian errors, and what a fitted model does and does not show.

Chapter 20 fit one line; this chapter fits a plane, or a hyperplane, with the same arithmetic written in matrices, and the exact t of that chapter becomes a t and an F.

21.1 Several predictors at once

Chapter 20 ended with the remark that Old Faithful's next wait might depend not only on the eruption just ended but on the eruption before it. Most responses are like that. The example that runs through this chapter is older and smaller. A plant oxidizes ammonia to nitric acid, and some of the ammonia escapes up the stack unconverted. Brownlee (*Statistical Theory and Methodology in Science and Engineering*, 1965) reported, for 21 days of operation, the stack loss, ten times the percentage of ammonia lost, together with three things the operators could watch: the air flow into the plant, the temperature of the cooling water, and the concentration of the circulating acid (coded, so that 89 means 58.9%). The data are in the table, and Figure 21.1 plots the loss against each of the three; the question is how the loss depends on the three together.

day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
air flow	80	80	75	62	62	62	62	62	58	58	58	58	58	58	50	50	50	50	50	56	70
water temperature	27	27	25	24	22	23	24	24	23	18	18	17	18	19	18	18	19	19	20	20	20
acid concentration	89	88	90	87	87	87	93	93	87	80	89	88	82	93	89	86	72	79	80	82	91
stack loss	42	37	37	28	18	18	19	20	15	14	14	13	11	12	8	7	8	8	9	15	15

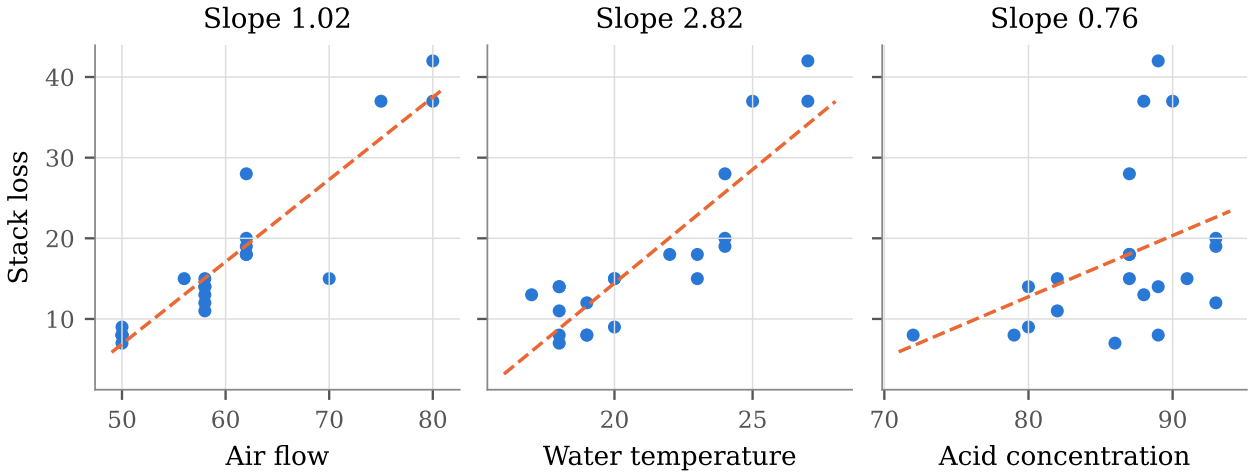


Figure 21.1: The stack loss data, 21 days: the loss against the air flow, the cooling water temperature and the acid concentration, each with the least squares line of the loss on that predictor alone (orange, dashed).

With one predictor the model was $Y_i = a+bx_i+\epsilon_i$. With three it is $Y_i = \beta_1+\beta_2 x_{i2}+\beta_3 x_{i3}+\beta_4 x_{i4}+\epsilon_i$, and the natural way to write n such equations is as one equation between vectors.

Definition 21.1 (Linear regression model with several predictors)**.** Let X be an $n \times p$ matrix of fixed numbers, the **design matrix**, whose row i is $x_i^\top = (x_{i1}, \ldots, x_{ip})$, the p predictor values of observation i; usually $x_{i1} = 1$ for every i, so that β_1 is the intercept. The **linear regression model** says

$$Y = X\beta + \epsilon, \qquad \text{that is,} \qquad Y_i = x_i^\top \beta + \epsilon_i, \quad i = 1, \ldots, n,$$

for an unknown $\beta \in \mathbb{R}^p$, where $Y = (Y_1, \ldots, Y_n)^\top$ and the errors $\epsilon = (\epsilon_1, \ldots, \epsilon_n)^\top$ have mean 0, a common variance σ^2 and no correlation: $\mathbb{E}[\epsilon] = 0$ and $\mathbb{C}\text{ov}(\epsilon) = \sigma^2 I_n$. The **Gaussian linear model** assumes moreover $\epsilon \sim \text{Gauss}_n(0, \sigma^2 I_n)$, that is, $\epsilon_1, \ldots, \epsilon_n \overset{iid}{\sim} \text{Gauss}(0, \sigma^2)$.

Each response is a weighted sum of its predictor values, with the same weights β for every observation, plus an error that behaves like every other error. The design matrix holds the data on the predictors, one row per observation and one column per predictor, and $X\beta$ is the vector of the n means $\mathbb{E}[Y_i]$. The model of Chapter 20 is the case $p = 2$, with columns $(1, \ldots, 1)^\top$ and $(x_1, \ldots, x_n)^\top$ and $\beta = (a, b)^\top$.

For the kiss data. With no predictor the design matrix is the single column $(1, \ldots, 1)^\top$, $p = 1$, and $Y = X\beta + \epsilon$ says $R_i = p + \epsilon_i$: every couple has the same mean. A design with two columns, a column of ones and a column of 0s and 1s marking the couples observed in one of two cities, say, would let the mean differ between the cities, as the aspirin design of Exercise 20.6 let the heart-attack rate differ between the groups.

For the stack loss data, $n = 21$, $p = 4$, and row 1 of X is $(1, 80, 27, 89)$. The model is linear in β, not necessarily in the measured variables, which is what makes it so general. For Old Faithful, a design with columns 1, x_i and x_i^2 fits a parabola to the waits, and x_i^2 is simply a third column of numbers; a design with columns 1, the duration and the previous wait lets the previous wait in; and a column of 0s and 1s, like the aspirin column of Exercise 20.6, lets two groups have different means. Treating the x_i as fixed, and reading observational data through the conditional distribution of Y given X, is as in Section 20.5.

Remark (The random-design version). Chapter 2 wrote the regression model with rows x_i drawn from $\text{Gauss}_d(0, I_d)$ and errors of variance 1, as an example of a parametric model with $\Theta = \mathbb{R}^d$. Condition on the drawn rows and that model is the Gaussian linear model of this chapter with $\sigma^2 = 1$ known; this chapter's $\Theta = \mathbb{R}^p \times (0, \infty)$ adds the unknown variance. Every level and coverage statement below holds given the rows, and so, by the tower-property argument of Section 20.5, unconditionally as well; the distributions and moments themselves, such as $\mathbb{C}\text{ov}(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$, are statements given the rows.

21.2 Least squares in matrix form

With one predictor, least squares minimized a sum of squared vertical distances. In vector form that sum is the squared length of $Y - X\beta$, and minimizing a length is a problem in geometry.

Definition 21.2 (Least squares estimator). The **least squares estimator** $\hat{\beta}$ minimizes the sum of squared errors

$$\|Y - X\beta\|^2 = \sum_{i=1}^{n} (Y_i - x_i^\top \beta)^2$$

over $\beta \in \mathbb{R}^p$; the **fitted values** are $\hat{Y} = X\hat{\beta}$ and the **residuals** are $e = Y - \hat{Y}$.

> Least squares picks, among all vectors of the form $X\beta$ (the column space of X (toolkit A.16)), the one closest to the response vector Y in ordinary Euclidean distance. The residuals are what is left over, observation by observation.
> **For the kiss data.** With the single column of ones, $\|Y - X\beta\|^2 = \sum_i (R_i - \beta)^2$ is smallest at $\beta = \bar{R}_n = 0.645$; the fitted vector is 0.645 in every coordinate and the residuals are 0.355 and -0.645 as in Chapter 20.

The one-predictor theorem of Chapter 20 needed the x_i not to be all equal. The general condition is that the columns of X be linearly independent: no column is a combination of the others, so that $X\beta = X\beta'$ only when $\beta = \beta'$, which is the statement that X has rank p (toolkit A.16).

Theorem 21.3 (The least squares estimator and the projection). *If X has rank p, then $X^\top X$ is invertible and the least squares estimator is unique,*

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

The fitted vector is $\hat{Y} = PY$, with $P = X(X^\top X)^{-1}X^\top$ the matrix of the orthogonal projection onto the column space of X, and the residual vector $e = (I - P)Y$ is perpendicular to every column of X: $X^\top e = 0$.

Proof strategy (core). The closest point to Y in a subspace is the foot of the perpendicular, and the residual is perpendicular to the subspace. Perpendicular to every column means $X^\top(Y - X\hat{\beta}) = 0$, which are p linear equations in $\hat{\beta}$ with the invertible matrix $X^\top X$.

Proof. If $X^\top Xa = 0$ then $a^\top X^\top Xa = \|Xa\|^2 = 0$, so $Xa = 0$ and, the columns being independent, $a = 0$: $X^\top X$ is invertible. Let $\hat{\beta} = (X^\top X)^{-1}X^\top Y$ and $e = Y - X\hat{\beta}$. Then $X^\top e = X^\top Y - X^\top X(X^\top X)^{-1}X^\top Y = 0$: the residual is perpendicular to the columns of X, hence to every vector Xb in the column space. For any β, write $Y - X\beta = e + X(\hat{\beta} - \beta)$, a sum of two perpendicular vectors, so by Pythagoras

$$\|Y - X\beta\|^2 = \|e\|^2 + \|X(\hat{\beta} - \beta)\|^2 \geq \|e\|^2,$$

with equality only when $X(\hat{\beta} - \beta) = 0$, that is, $\beta = \hat{\beta}$. So $\hat{\beta}$ is the unique minimizer. The matrix P satisfies $P^\top = P$ and $P^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = P$, it fixes every vector in the column space ($PXb = Xb$) and sends every vector perpendicular to the columns to 0, which is what the orthogonal projection does (toolkit A.16); and $X\hat{\beta} = PY$ by the formula for $\hat{\beta}$. □

Figure 21.2 draws the theorem: the column space of X as a plane through the origin, the response Y as a vector above it, the fitted vector as the foot of the perpendicular from Y to the plane, and the residual as the perpendicular itself. Least squares is nothing but this picture, in n dimensions. The p equations $X^\top e = 0$ are the **normal equations**; for the design with a column of ones and one predictor they are the two of Theorem 20.6, $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$.

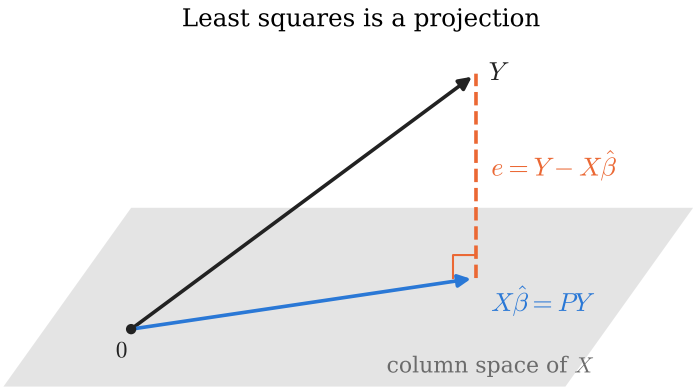


Figure 21.2: Least squares as a projection: the column space of X (gray), the response vector Y (black), the fitted vector $X\hat{\beta}$ (blue), the foot of the perpendicular from Y, and the residual e (orange, dashed).

Worked example: the stack loss fit

For the 21 days, solving the normal equations gives

$$\hat{\beta} = (-39.92,\ 0.716,\ 1.295,\ -0.152)^\top :$$

in words, with the air flow, the water temperature and the acid concentration,

$$\text{predicted loss} = -39.92 + 0.716\,(\text{air}) + 1.295\,(\text{water}) - 0.152\,(\text{acid}).$$

Each unit of air flow adds 0.72 to the loss when the other two are held fixed, each degree of water temperature adds 1.3, and each unit of acid concentration subtracts 0.15. The residual

sum of squares is $\|e\|^2 = 178.8$, against $\sum_i (Y_i - \bar{Y})^2 = 2{,}069.2$ for the constant fit, so the three predictors account for $1 - 178.8/2{,}069.2 = 91\%$ of the variation in the loss. Figure 21.3 plots the observed losses against the fitted ones and the residuals against the fitted values; day 21, with air flow 70 and a loss of only 15, is the worst-fitted day, with residual -7.2.

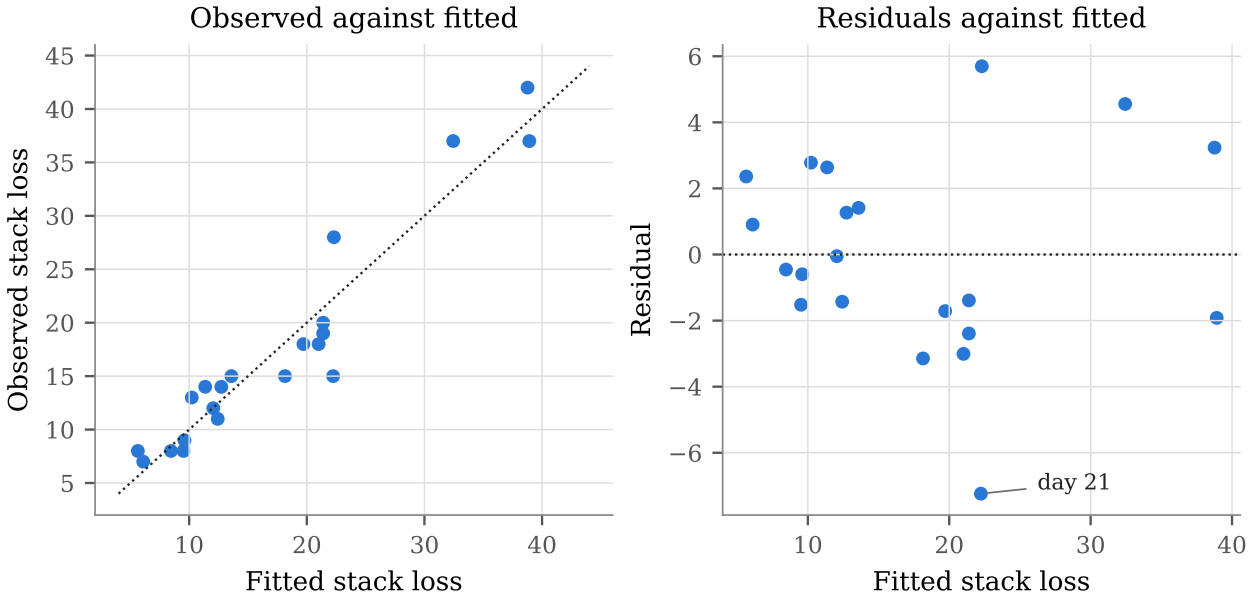


Figure 21.3: The stack loss fit. Left: observed stack loss against the fitted values of the three-predictor model, with the line $y = x$ (dotted). Right: the residuals against the fitted values; day 21 is the lowest point.

The coefficient 0.716 is not the slope of the loss on the air flow alone, which Figure 21.1 drew, and the difference is the first thing to understand about a multiple regression.

Does this make sense? A coefficient on its own, and in company

Regress the loss on the air flow alone and the slope is 1.02, not 0.72. Both are right: 1.02 is the change in loss per unit of air flow when the water temperature and the acid are left to vary as they did, and on these days warmer water went with higher air flow (correlation 0.78), so the air flow alone gets credit for the water's effect as well. The multiple regression coefficient 0.72 is the change per unit of air flow with the other two held fixed.

When the columns of X are dependent, the least squares problem still has a solution, the projection PY onto the column space, but infinitely many β produce it.

Pitfall: collinearity is non-identifiability

If one column is a combination of the others, say $x_{i3} = 2x_{i2}$ for every i, then $X\beta = X\beta'$ whenever β' moves weight from β_2 to β_3 in the ratio $2:1$. The distribution of Y depends on β only through $X\beta$, so two different parameters give the same distribution: the parameter is not identifiable, in the sense of Chapter 2, and no data can tell β from β'. Software reports it by dropping a column. A milder form appears when the columns are nearly dependent (Section 21.4): $\hat{\beta}$ exists but has a large variance in the direction where the columns almost cancel.

21.3 The Gaussian linear model

Everything Chapter 20 proved for one predictor holds for p, with $(X^\top X)^{-1}$ where $1/S_{xx}$ was. The tool is the general form of Cochran's theorem, promised in Chapter 14.

Proposition 21.4 (Cochran's theorem, general form)**.** *Let $\epsilon \sim Gauss_n(0, \sigma^2 I_n)$, and let $\mathbb{R}^n$ be split into mutually perpendicular subspaces $S_1, \dots, S_m$ of dimensions $d_1 + \cdots + d_m = n$, with orthogonal projections $P_1, \dots, P_m$. Then $\|P_1\epsilon\|^2/\sigma^2, \dots, \|P_m\epsilon\|^2/\sigma^2$ are independent, with $\|P_j\epsilon\|^2/\sigma^2 \sim \chi^2_{d_j}$.*

Proof strategy (core). Rotate to an orthonormal basis adapted to the subspaces, d_1 vectors spanning S_1, then d_2 spanning S_2, and so on. The rotated coordinates are independent $\text{Gauss}(0, \sigma^2)$, and each $\|P_j\epsilon\|^2$ is the sum of the squares of its own block of them.

Proof. Choose an orthonormal basis of each S_j and put them together: since the subspaces are perpendicular and their dimensions add to n, these are n orthonormal vectors, and orthonormal vectors are linearly independent (dotting $\sum_k c_k u_k = 0$ with u_j gives $c_j = 0$), so they are an orthonormal basis $u_1, \dots, u_n$ of $\mathbb{R}^n$. Let Q be the orthogonal matrix with rows $u_k^\top$ and $Z = Q\epsilon$, so $Z_k = u_k^\top \epsilon$ and $\epsilon = \sum_k Z_k u_k$; by rotation invariance (toolkit A.13), $Z_1, \dots, Z_n$ are i.i.d. $\text{Gauss}(0, \sigma^2)$. The projection P_j keeps the terms of $\sum_k Z_k u_k$ whose u_k lie in S_j and kills the rest, so $\|P_j\epsilon\|^2$ is the sum of Z_k^2 over the d_j indices of that block. The blocks are disjoint, so the m sums are independent, and each divided by σ^2 is a sum of d_j squared independent standard Gaussians. □

Theorem 14.3 is the case $m = 2$ with S_1 the line through $(1, \dots, 1)$; the proofs below use $m = 2$ and $m = 3$ with the column space of X in place of the line.

Theorem 21.5 (Least squares with p predictors)**.** *In the linear regression model with X of rank p and $n > p$:*

1. *$\mathbb{E}[\hat{\beta}] = \beta$ and $\mathbb{C}ov(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$, so the quadratic risk is $\mathbb{E}\|\hat{\beta} - \beta\|^2 = \sigma^2 \operatorname{tr}\big((X^\top X)^{-1}\big)$;*
2. *$\mathbb{E}\|e\|^2 = (n-p)\sigma^2$, so $\hat{\sigma}^2 = \|e\|^2/(n-p)$ is an unbiased estimator of σ^2.*

In the Gaussian linear model, moreover:

3. *$\hat{\beta} \sim Gauss_p\big(\beta, \sigma^2(X^\top X)^{-1}\big)$, and the maximum likelihood estimators are $\hat{\beta}$ and $\|e\|^2/n$;*
4. *$(n-p)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-p}$, independent of $\hat{\beta}$;*
5. *for each j, with γ_j the jth diagonal entry of $(X^\top X)^{-1}$,*

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{\gamma_j}} \sim t_{n-p}.$$

Proof strategy (core). $\hat\beta-\beta$ is a fixed matrix times the error vector, which gives the mean and covariance at once. For the residuals, rotate the errors so that the first p axes span the column space: the fit absorbs those p coordinates and the residual is the other $n-p$, which is Proposition 21.4 with the column space and its orthogonal complement.

Recall. The chi-squared distribution (Definition 14.1): a sum of k squared independent standard Gaussians is χ^2_k. Student's distribution (Definition 14.5): if $Z\sim\text{Gauss}(0,1)$ and $V\sim\chi^2_k$ are independent, $Z/\sqrt{V/k}\sim t_k$. A Gaussian vector (Appendix A.14): if $W\sim\text{Gauss}_n(0,\sigma^2 I_n)$ and B is a $p\times n$ matrix of rank p, then $BW\sim\text{Gauss}_p(0,\sigma^2BB^\top)$.

Proof. Substituting $Y=X\beta+\epsilon$ into the formula for $\hat\beta$,

$$\hat\beta=(X^\top X)^{-1}X^\top(X\beta+\epsilon)=\beta+B\epsilon,\qquad B=(X^\top X)^{-1}X^\top,$$

so $\mathbb{E}[\hat\beta]=\beta$ and $\mathbb{C}\text{ov}(\hat\beta)=B\,\mathbb{C}\text{ov}(\epsilon)\,B^\top=\sigma^2(X^\top X)^{-1}X^\top X(X^\top X)^{-1}=\sigma^2(X^\top X)^{-1}$ (toolkit A.14). The risk is $\mathbb{E}\|\hat\beta-\beta\|^2=\sum_j\mathbb{V}\text{ar}(\hat\beta_j)$, the sum of the diagonal entries of the covariance matrix, which is (1).

For (2), rotate. By Gram–Schmidt (toolkit A.13) there is an orthonormal basis $u_1,\dots,u_n$ of $\mathbb{R}^n$ whose first p vectors span the column space of X; let Q be the orthogonal matrix with rows $u_k^\top$ and set $Z=Q\epsilon$, so $Z_k=u_k^\top\epsilon$ and $\epsilon=\sum_k Z_ku_k$. The Z_k have mean 0, variance σ^2 and are uncorrelated, since $\mathbb{E}[Z_jZ_k]=u_j^\top\,\mathbb{C}\text{ov}(\epsilon)u_k=\sigma^2u_j^\top u_k$. The projection P fixes $u_1,\dots,u_p$ and kills $u_{p+1},\dots,u_n$, so

$$e=(I-P)Y=(I-P)(X\beta+\epsilon)=(I-P)\epsilon=\sum_{k=p+1}^{n}Z_ku_k,\qquad\|e\|^2=Z_{p+1}^2+\cdots+Z_n^2,$$

whose mean is $(n-p)\sigma^2$. Likewise $\hat\beta-\beta=B\epsilon=B\sum_{k\le p}Z_ku_k$, since $Bu_k=(X^\top X)^{-1}X^\top u_k=0$ for $k>p$ (u_k is perpendicular to the columns): $\hat\beta$ is a function of $Z_1,\dots,Z_p$ alone.

With Gaussian errors, $Z/\sigma=Q(\epsilon/\sigma)$ has i.i.d. Gauss$(0,1)$ coordinates by rotation invariance (toolkit A.13). Then $\hat\beta=\beta+B\epsilon$ is Gaussian with the covariance of (1), since B has rank p (its p rows are independent because $BX=I_p$); the log-likelihood is $-\frac{n}{2}\log(2\pi\sigma^2)-\|Y-X\beta\|^2/(2\sigma^2)$, largest in β where the sum of squares is smallest, so the maximum likelihood estimator of β is $\hat\beta$, and then, as in Proposition 20.7, its derivative in σ^2 at $\hat\beta$ is $-n/(2\sigma^2)+\|e\|^2/(2\sigma^4)$, zero at $\sigma^2=\|e\|^2/n$: this is (3). $(n-p)\hat\sigma^2/\sigma^2=\sum_{k>p}(Z_k/\sigma)^2$ is χ^2_{n-p} by Proposition 21.4 with S_1 the column space and S_2 its orthogonal complement, and it is a function of $Z_{p+1},\dots,Z_n$ while $\hat\beta$ is a function of $Z_1,\dots,Z_p$, so the two are independent: this is (4). Finally $(\hat\beta_j-\beta_j)/(\sigma\sqrt{\gamma_j})$ is a standard Gaussian, independent of the χ^2_{n-p} variable of (4); dividing by $\sqrt{\hat\sigma^2/\sigma^2}$ gives t_{n-p}, which is (5). □

The fit takes p dimensions and the residual the other $n-p$, which is why the divisor is $n-p$: $n-1$ for a mean, $n-2$ for a line, 17 for the stack loss fit.

Worked example: standard errors for the stack loss coefficients

$\hat\sigma=\sqrt{178.8/17}=3.24$, and the diagonal of $(X^\top X)^{-1}$ is $(13.45, 0.00173, 0.0129, 0.00232)$, so the standard errors $\hat\sigma\sqrt{\gamma_j}$ are 11.9, 0.135, 0.368 and 0.156. The 95% confidence interval for the air flow coefficient is $0.716\pm2.110\times0.135=[0.43,1.00]$, with $q_{0.025}(t_{17})=2.110$; for the water temperature, $[0.52,2.07]$; for the acid, $[-0.48,0.18]$, which contains 0. The quadratic risk of $\hat\beta$ is $\sigma^2\times13.47$, almost all of it from the intercept, whose variance is large because the predictors are

far from 0: air flow averages 60, so the height of the fitted plane at air flow 0 is an extrapolation.

21.4 One coefficient at a time

Part (5) of Theorem 21.5 gives the test of Chapter 20 for every coefficient. In the Gaussian linear model, the test of $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$ that rejects when

$$|T_j| > q_{\alpha/2}(t_{n-p}), \qquad T_j = \frac{\hat{\beta}_j}{\hat{\sigma}\sqrt{\gamma_j}},$$

has level exactly α. Its null hypothesis says that predictor j adds nothing to a model that already contains the other $p - 1$.

Worked example: which predictors matter for the stack loss

$T = 5.31$ for the air flow and 3.52 for the water temperature, both beyond 2.110, with p-values 0.0001 and 0.003; $T = -0.97$ for the acid concentration, p-value 0.34. Given the air flow and the water temperature, the acid concentration adds nothing that 21 days can detect. Dropping it gives the fit $-50.36 + 0.671\,(\text{air flow}) + 1.295\,(\text{water temperature})$, with residual sum of squares 188.8 against 178.8.

The sanity check of Section 21.2 already showed that a coefficient depends on the company it keeps. Old Faithful shows something stronger: two predictors can each look useless on its own and be useful together.

Worked example: the previous eruption

Fit, for the 271 eruptions that have a predecessor, the wait on the duration and the previous wait: the previous wait's coefficient is -0.058 with $T = -1.82$, p-value 0.07. Fit the wait on the duration and the previous duration instead: $T = 0.62$. Each, added alone to the line of Chapter 20, falls short of $q_{0.025}(t_{268}) = 1.969$. Now add both: the coefficients are -0.283 for the previous wait ($T = -4.75$) and 3.12 for the previous duration ($T = 4.40$), both far past the threshold, and $\hat{\sigma}$ drops from 5.91 to 5.69. The reason is that the previous wait and the previous duration have correlation 0.90 (long eruptions are followed by long waits), so each stands in for the other when it is alone, and what the pair uses is the previous wait's excess over what the previous duration predicts for it: $-0.283\,(\text{previous wait}) + 3.12\,(\text{previous duration}) = -0.283\,(\text{previous wait} - 11.0 \times \text{previous duration})$, so given the previous duration, a longer previous wait means a shorter wait now.

Figure 21.4 draws the joint 95% confidence region for the two coefficients in the full model, the ellipse $\{b : (\hat{b} - b)^\top C^{-1}(\hat{b} - b) \leq 2q_{0.05}(F_{2,267})\}$ with C the 2×2 block of $\hat{\sigma}^2(X^\top X)^{-1}$ (Exercise 21.6), together with the two separate intervals. The ellipse is long and tilted, from upper left to lower right. Raise the previous wait's coefficient by t, lower the previous duration's by $11t$ and the intercept by $33t$, and the fitted values barely move, since the previous wait is close to $33 + 11\times$ the previous duration, so the three columns nearly cancel in that combination; the data leave it loose, and it is the long axis of the ellipse. The combination along the short axis, 11 times the previous wait's coefficient plus the previous duration's, is pinned down far better than either coefficient alone: its standard error is 0.37, against 0.71 for the previous duration's coefficient. It is a mild form of

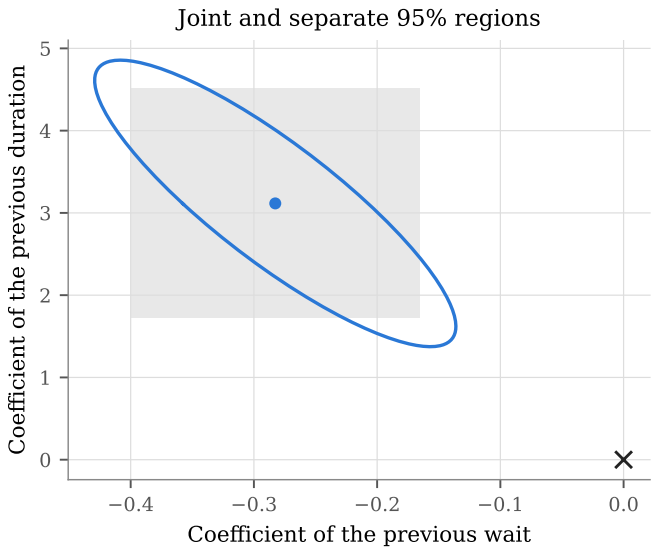


Figure 21.4: Old Faithful, the model with the duration, the previous wait and the previous duration: the joint 95% confidence ellipse for the last two coefficients (blue), the two separate 95% intervals (gray rectangle), the estimates (dot) and the origin (cross).

the collinearity pitfall. The origin lies far outside the ellipse, so the pair is clearly needed, and Section 21.5 gives the test that says so in one number.

Pitfall: what a coefficient does not say

β_j is the change in $\mathbb{E}[Y]$ per unit of predictor j with the others held fixed, in a model containing exactly those others. It is not the effect of changing predictor j: the stack loss plant was observed, not manipulated, and the days with high air flow may have differed in ways the three columns do not record. A regression shows how the response moves with the predictors in the data; an experiment that sets the predictors, as the Physicians' Health Study of Exercise 20.6 assigned aspirin by randomization, is what shows what changing them does.

21.5 Several coefficients at once: the F-test

'Does the acid concentration matter?' is one coefficient; 'do the water temperature and the acid concentration matter, given the air flow?' is two. Testing them one at a time at level α each does not give a test of level α for the pair, as Chapter 14 explained for many tests at once, and the previous worked example shows the stakes: there, both single tests failed and the pair was decisive. The right test compares the fit with the predictors in question to the fit without them.

Let X_0 consist of the first $p-k$ columns of X, the predictors not under test, and let $\|e_0\|^2$ be the residual sum of squares of the least squares fit with X_0 alone; it is at least $\|e\|^2$, since the smaller model is a special case of the larger.

Theorem 21.6 (The F-test)**.** *In the Gaussian linear model with X of rank p, test $H_0 : \beta_{p-k+1} = \cdots = \beta_p = 0$ against the alternative that at least one of them is not 0. Under H_0,*

$$F = \frac{(\|e_0\|^2 - \|e\|^2)/k}{\|e\|^2/(n-p)} \sim F_{k,n-p},$$

the F distribution of Chapter 14, so the test that rejects when $F > q_\alpha(F_{k,n-p})$ has level exactly α. It is the likelihood ratio test of Chapter 15: $T_n^{\mathrm{LR}} = n\log(\|e_0\|^2/\|e\|^2)$, an increasing function of F. For $k = 1$, $F = T_p^2$, and the F-test is the two-sided t-test.

Proof strategy (core). Split the error vector along three orthogonal subspaces: the column space of X_0, the part of the column space of X perpendicular to it (k dimensions), and the residual space ($n - p$ dimensions). Under H_0 the numerator is the squared length of the middle piece and the denominator that of the last; Cochran makes them independent chi-squareds.

Proof. Let P_0 be the projection onto the column space of X_0 and P onto that of X; the first space is inside the second, so $PP_0 = P_0$ and $P - P_0$ is the projection onto the part of the column space of X perpendicular to the column space of X_0, a subspace of dimension $p - (p - k) = k$ (toolkit A.16). Then $e_0 = (I - P_0)Y$, $e = (I - P)Y$, and since $(I - P_0) = (P - P_0) + (I - P)$ with the two pieces perpendicular, Pythagoras gives

$$\|e_0\|^2 - \|e\|^2 = \|(P - P_0)Y\|^2.$$

Under H_0, $X\beta = X_0\beta_0$ lies in the column space of X_0, so $(P-P_0)X\beta = 0$ and $(I-P)X\beta = 0$: both pieces are the same projections of ϵ alone. Choose the orthonormal basis of the proof of Theorem 21.5 so that $u_1, \dots, u_{p-k}$ span the column space of X_0 and $u_1, \dots, u_p$ that of X (Gram–Schmidt applied to the columns of X_0 first, then the rest). With $Z = Q\epsilon$ as before, $\|(P-P_0)\epsilon\|^2 = Z_{p-k+1}^2 + \cdots + Z_p^2$ and $\|(I - P)\epsilon\|^2 = Z_{p+1}^2 + \cdots + Z_n^2$: by Proposition 21.4 with the three subspaces, these are σ^2 times independent χ_k^2 and χ_{n-p}^2 variables, and the ratio of each to its degrees of freedom is $F_{k,n-p}$ by Chapter 14's definition.

For the likelihood ratio, the maximized log-likelihood of a Gaussian linear model is, by (3) of Theorem 21.5, $-\frac{n}{2}\log(2\pi\hat\sigma^2_{\mathrm{MLE}}) - \frac{n}{2}$ with $\hat\sigma^2_{\mathrm{MLE}} = \|e\|^2/n$; the difference of the two models' maxima, times 2, is $n\log(\|e_0\|^2/\|e\|^2) = n\log(1 + kF/(n-p))$, increasing in F. For $k = 1$ the k-dimensional piece is one coordinate, $Z_p = u_p^\top\epsilon$, and under H_0, $\hat\beta_p = \sqrt{\gamma_p}\, Z_p$ (Exercise 21.4), so $F = Z_p^2/\hat\sigma^2 = \hat\beta_p^2/(\gamma_p\hat\sigma^2) = T_p^2$; Exercise 21.4 shows that $F = T_p^2$ holds whether or not H_0 is true. □

Worked example: two F-tests for the stack loss

Do the three predictors together do anything? With X_0 the column of ones, $\|e_0\|^2 = 2{,}069.2$, so

$$F = \frac{(2{,}069.2 - 178.8)/3}{178.8/17} = \frac{630.1}{10.52} = 59.9$$

on $(3, 17)$ degrees of freedom, against $q_{0.05}(F_{3,17}) = 3.20$: p-value 3×10^{-9}. Do the water temperature and the acid add anything to the air flow? The fit on the air flow alone has $\|e_0\|^2 = 319.1$, so $F = (319.1 - 178.8)/2 \div 10.52 = 70.2/10.52 = 6.67$ on $(2, 17)$ against 3.59: p-value 0.007, so of 1,000 plants on which the water and the acid did nothing, about 7 would give an F this large. For Old Faithful, adding the previous wait and the previous duration to the duration gives $F = 11.5$ on $(2, 267)$, p-value 2×10^{-5}, where the two single t-tests found nothing.

Any k independent linear constraints on β, such as $\beta_2 = \beta_3$, take this form after a change of columns (with columns $x_2 + x_3$ and x_3 in place of x_2 and x_3, the coefficients are β_2 and $\beta_3 - \beta_2$), so the F-test covers the implicit hypotheses of Chapter 15 as well. Wilks' theorem (Chapter 15) would

make T_n^{LR} approximately χ^2_k; here Cochran's theorem makes F exactly $F_{k,n-p}$, and as $n \to \infty$ with k fixed, $kF \xrightarrow[n\to\infty]{\text{(d)}} \chi^2_k$ (Exercise 21.5), so the two agree in the limit.

From undergraduate statistics: the ANOVA table and R^2

Regression output's analysis-of-variance table is the first F-test above: the regression sum of squares $\|e_0\|^2 - \|e\|^2 = \sum_i (\hat{Y}_i - \bar{Y})^2$ on $p-1$ degrees of freedom, the residual sum of squares on $n-p$, their mean squares, and their ratio F. Its $R^2 = 1 - \|e\|^2/\|e_0\|^2$ is the share of $\sum_i (Y_i - \bar{Y})^2$ that the fit accounts for (0.914 for the stack loss), and $F = \frac{R^2/(p-1)}{(1-R^2)/(n-p)}$ is the same number read off R^2. Adding a column never lowers R^2, since the larger model contains the smaller, which is why a high R^2 with many predictors proves little and the F-test, which charges each added column its degree of freedom, is the honest comparison.

Remark (Bonferroni's test). An alternative to the F-test rejects H_0 when any one of the k single tests rejects at level α/k: by the union bound of Proposition 14.13, its level is at most α. For the water temperature and the acid, $\alpha/2 = 0.025$ gives the threshold $q_{0.0125}(t_{17}) = 2.458$, and $T = 3.52$ for the water rejects. Bonferroni does not need the k tests to come from one model or one data set, and when it rejects it says which hypothesis failed, which the F-test does not; its price is the conservative level, and a case like Old Faithful's two predictors, where each single T is small in its own smaller model, is exactly where it can miss what the F-test sees.

21.6 Ridge in matrix form

Chapter 20 shrank one slope toward 0 by adding a Gaussian prior; with p coefficients the same prior on each gives the matrix form. Under the Gaussian linear model with σ^2 known and the prior $\beta \sim \mathrm{Gauss}_p(0, \tau^2 I_p)$, the log posterior density is, up to a constant,

$$-\frac{1}{2\sigma^2}\|Y - X\beta\|^2 - \frac{1}{2\tau^2}\|\beta\|^2,$$

whose gradient in β (toolkit A.12) is $\sigma^{-2}(X^\top Y - X^\top X\beta) - \tau^{-2}\beta$, zero at the **ridge estimator**

$$\tilde{\beta}_\lambda = (X^\top X + \lambda I_p)^{-1} X^\top Y, \qquad \lambda = \frac{\sigma^2}{\tau^2},$$

which minimizes $\|Y - X\beta\|^2 + \lambda\|\beta\|^2$.

Since $a^\top (X^\top X + \lambda I)a = \|Xa\|^2 + \lambda\|a\|^2 > 0$ for $a \neq 0$ and $\lambda > 0$, the matrix is invertible whatever the rank of X: ridge has a unique answer even for collinear columns, where least squares has infinitely many, and a steady one when they are nearly collinear. In practice the intercept is left out of the penalty and the predictors are standardized first, so that one λ treats them alike. Figure 21.5 does this for the stack loss data: as λ grows the three standardized coefficients leave their least squares values $(6.56, 4.09, -0.82)$, the air flow's and the water temperature's shrinking toward 0 and the acid's, whose sign is not even determined by the data, crossing 0 near $\lambda = 5$ before it too heads to 0. Choosing λ is choosing τ, the prior spread of the coefficients, and Exercise 20.9 showed why a small positive λ lowers the risk whatever β is.

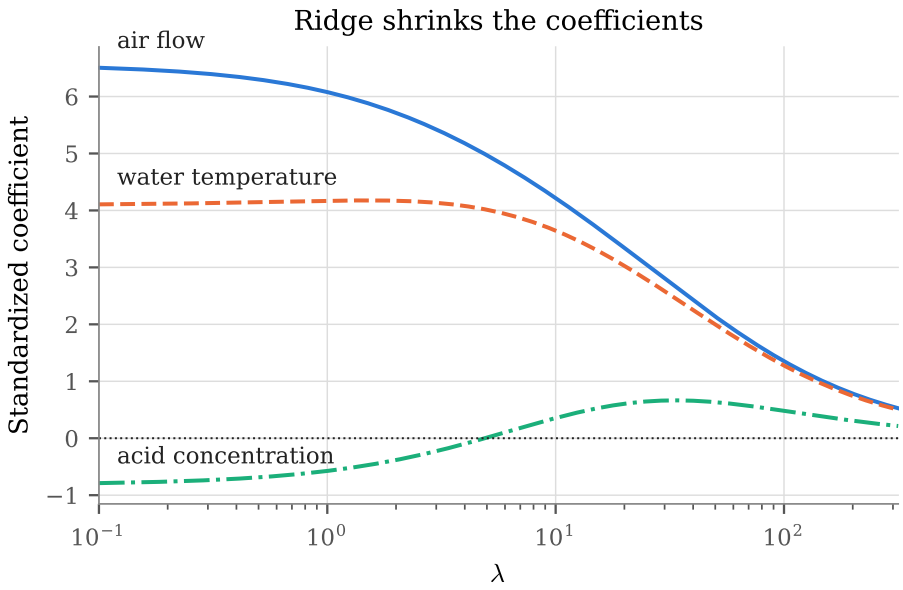


Figure 21.5: Ridge on the stack loss data with standardized predictors: the coefficients of the air flow (blue), the water temperature (orange, dashed) and the acid concentration (aqua, dash-dot) against λ on a logarithmic axis.

21.7 Without Gaussian errors, and checking the model

Nothing in parts (1) and (2) of Theorem 21.5 needed Gaussian errors, and the large-sample theory of Theorem 20.12 extends to p predictors by the same route, with vectors in place of numbers.

Theorem 21.7 (Least squares with p predictors, without Gaussian errors)**.** *Let $(x_1, Y_1), (x_2, Y_2), \ldots$ be i.i.d. copies of $(x, Y) \in \mathbb{R}^p \times \mathbb{R}$ with finite fourth moments, let $M = \mathbb{E}[xx^\top]$ be positive definite, let β^* minimize $\mathbb{E}[(Y - x^\top\beta)^2]$ and $\epsilon = Y - x^\top\beta^*$, and let $V = \mathbb{E}[\epsilon^2 xx^\top]$ be positive definite. Then $\hat{\beta} \xrightarrow[n\to\infty]{\mathbb{P}} \beta^*$ and*

$$\sqrt{n}(\hat{\beta} - \beta^*) \xrightarrow[n\to\infty]{(d)} \mathit{Gauss}_p(0, M^{-1}VM^{-1}).$$

If $\mathbb{E}[\epsilon^2 \mid x] = \sigma^2$, then $M^{-1}VM^{-1} = \sigma^2 M^{-1}$, $\hat{\sigma}^2 \xrightarrow[n\to\infty]{\mathbb{P}} \sigma^2$, and each T_j of Section 21.4 converges in distribution to $\mathit{Gauss}(0,1)$ under $\beta_j^* = 0$.

Proof strategy (for completeness; skip on a first reading). The one-predictor proof with matrices: $\hat{\beta} - \beta^* = (\frac{1}{n}X^\top X)^{-1}\frac{1}{n}X^\top\epsilon$, the Law of Large Numbers for the first factor and the vector Central Limit Theorem for the second.

Recall. The Law of Large Numbers (Theorem 3.5) for each entry of an average of i.i.d. matrices with finite variances; the continuous mapping theorem (Theorem 5.2) for the inverse, continuous at an invertible matrix; the Central Limit Theorem for vectors (the vector-versions remark of Chapter 5): for i.i.d. vectors W_i with mean 0 and positive definite covariance V, $\sqrt{n}\,\bar{W}_n \xrightarrow[n\to\infty]{(d)} \mathrm{Gauss}_p(0, V)$; Slutsky's theorem (Theorem 5.1), as Chapter 15 uses it for vectors: if $A_n \xrightarrow[n\to\infty]{\mathbb{P}} A$, a fixed matrix, and $W_n \xrightarrow[n\to\infty]{(d)} W$, then $(A_n - A)W_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$ entry by entry and $A_nW_n \xrightarrow[n\to\infty]{(d)} AW$.

Proof. The minimizer β^* solves $\mathbb{E}[x(Y - x^\top\beta)] = 0$, since the gradient of $\mathbb{E}[(Y - x^\top\beta)^2]$ is $-2\,\mathbb{E}[x(Y - x^\top\beta)] = -2(\mathbb{E}[xY] - M\beta)$, so $\beta^* = M^{-1}\,\mathbb{E}[xY]$ and $\mathbb{E}[x\epsilon] = 0$. Writing $Y_i = x_i^\top\beta^* + \epsilon_i$ in the formula for $\hat\beta$,

$$\hat\beta - \beta^* = \Big(\frac{1}{n}\sum_i x_i x_i^\top\Big)^{-1}\frac{1}{n}\sum_i x_i\epsilon_i.$$

By the Law of Large Numbers, entry by entry, $\frac{1}{n}\sum_i x_i x_i^\top \xrightarrow[n\to\infty]{\mathbb{P}} M$ (finite fourth moments make the variances finite), so $X^\top X$ is invertible, and $\hat\beta$ defined, with probability tending to 1; the inverse is continuous at M, so the first factor converges in probability to M^{-1}. The second is an average of i.i.d. vectors $W_i = x_i\epsilon_i$ with mean 0 and covariance V, which is finite since ϵ, a linear combination of Y and the entries of x, has a finite fourth moment and $\epsilon^2|x_j x_k| \le \frac{1}{2}(\epsilon^4 + x_j^2 x_k^2)$; so it converges in probability to 0, and $\sqrt{n}$ times it converges in distribution to $\text{Gauss}_p(0, V)$. Slutsky's theorem gives $\hat\beta \xrightarrow[n\to\infty]{\mathbb{P}} \beta^*$ and $\sqrt{n}(\hat\beta - \beta^*) \xrightarrow[n\to\infty]{\text{(d)}} M^{-1}\text{Gauss}_p(0, V) = \text{Gauss}_p(0, M^{-1}VM^{-1})$, by the linear-map rule for Gaussian vectors (toolkit A.14). If $\mathbb{E}[\epsilon^2 \mid x] = \sigma^2$, then $V = \mathbb{E}[xx^\top\,\mathbb{E}[\epsilon^2 \mid x]] = \sigma^2 M$ by taking out what is known and the tower property (toolkit A.15), and $M^{-1}VM^{-1} = \sigma^2 M^{-1}$. For $\hat\sigma^2$, $e = \epsilon - X(\hat\beta - \beta^*)$ and $X^\top e = 0$ give $\|e\|^2 = \|\epsilon\|^2 - \|X(\hat\beta - \beta^*)\|^2$, and $\frac{1}{n}\|X(\hat\beta - \beta^*)\|^2 = (\hat\beta - \beta^*)^\top\frac{1}{n}X^\top X(\hat\beta - \beta^*) \xrightarrow[n\to\infty]{\mathbb{P}} 0$, so $\hat\sigma^2 = \frac{n}{n-p}\cdot\frac{1}{n}\|e\|^2 \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[\epsilon^2] = \sigma^2$, by the Law of Large Numbers for the ϵ_i^2, whose variance is finite by the fourth moments. Then $T_j = \sqrt{n}\hat\beta_j/\big(\hat\sigma\sqrt{n\gamma_j}\big)$, where $n\gamma_j$, the jth diagonal entry of $(\frac{1}{n}X^\top X)^{-1}$, converges in probability to that of M^{-1}, and Slutsky's theorem finishes it. □

So the t-tests and intervals keep their level for large n whatever the errors' distribution, as long as the scatter is the same everywhere; when it is not, the sandwich $M^{-1}VM^{-1}$, estimated by plugging in $\frac{1}{n}\sum_i x_i x_i^\top$ and $\frac{1}{n}\sum_i e_i^2 x_i x_i^\top$, gives the robust standard errors of Chapter 13's bridge, the p-predictor form of the one in Chapter 20; as there, the consistency of the plug-in estimate is stated and not proved. The F-test survives the same way: $kF \xrightarrow[n\to\infty]{\text{(d)}} \chi^2_k$ under H_0 for constant scatter (Exercise 21.5).

What the theory cannot do is tell you whether the model is any good, and the residuals are the place to look. In the Gaussian linear model the residuals are the projection $(I - P)\epsilon$ of the errors, so they should look like mean-zero noise with no pattern against the fitted values or against any predictor: a curve in such a plot says that the mean is not linear, a fan says that the scatter changes with the mean. The right of Figure 21.3 shows neither for the stack loss, apart from day 21. Whether the errors are Gaussian can be tested with the residuals in place of the errors: the Lilliefors test of Chapter 17 on the 21 stack loss residuals gives $D = 0.107$ against a 5% threshold of 0.19, no evidence against normality, with two caveats: the threshold is approximate, since the residuals are correlated and their variances differ (Exercise 21.3), and $n = 21$ has little power (Section 17.6). For the exactness of the t- and F-tests at $n = 21$ that is what matters. At $n = 271$ Old Faithful's whole-minute waits are covered by Theorem 21.7, but its rows are not independent, and in the model with the previous wait as a predictor they cannot be: the previous wait is the response of the row before. The t and F statistics for that model are the ones the Gaussian linear model prescribes, and large-sample theory for such lagged designs, which this book does not develop, supports them approximately, as the remarks of Section 20.6 did for consecutive eruptions.

21.8 What comes next

This chapter closes Part VIII. The response has been a number whose mean is linear in the predictors and whose scatter is the same everywhere. For the kiss data's 0s and 1s, or for counts, neither holds: a mean that is a probability cannot be a line in the predictors without leaving $[0, 1]$, and the variance $p(1-p)$ moves with the mean. Part IX keeps the linear predictor $x^\top\beta$ and changes what it is linked to: Chapter 22 describes the distributions for which this works, the exponential families, and Chapter 23 links their means to $x^\top\beta$ through a function such as the log-odds of Chapter 19, with maximum likelihood, now without a closed form, doing the fitting.

21.9 Summary

- $Y = X\beta + \epsilon$ with an $n \times p$ design matrix, $\mathbb{E}[\epsilon] = 0$, $\mathbb{C}\text{ov}(\epsilon) = \sigma^2 I_n$; polynomials, 0/1 variables and lagged values are columns like any other. Full rank of X is identifiability; collinearity is its failure.
- $\hat\beta = (X^\top X)^{-1}X^\top Y$ is unique when X has rank p; $X\hat\beta = PY$ is the projection of Y onto the column space, $e \perp$ every column (the normal equations $X^\top e = 0$). Stack loss: $-39.92 + 0.716\,\text{air} + 1.295\,\text{water} - 0.152\,\text{acid}$, $R^2 = 0.914$.
- $\mathbb{E}[\hat\beta] = \beta$, $\mathbb{C}\text{ov}(\hat\beta) = \sigma^2(X^\top X)^{-1}$, risk $\sigma^2\,\text{tr}((X^\top X)^{-1})$; $\hat\sigma^2 = \|e\|^2/(n-p)$ is unbiased. Gaussian errors: $\hat\beta \sim \text{Gauss}_p(\beta, \sigma^2(X^\top X)^{-1})$ is the MLE, $(n-p)\hat\sigma^2/\sigma^2 \sim \chi^2_{n-p}$ independent of $\hat\beta$ (Cochran, general form), $(\hat\beta_j - \beta_j)/(\hat\sigma\sqrt{\gamma_j}) \sim t_{n-p}$.
- One coefficient: $T_j = \hat\beta_j/(\hat\sigma\sqrt{\gamma_j})$ against $q_{\alpha/2}(t_{n-p})$; stack loss $T = 5.31, 3.52, -0.97$. A coefficient is a statement about a model: the air flow's 1.02 alone, 0.72 with the others; Old Faithful's previous wait and previous duration, useless alone and decisive together.
- Several coefficients: $F = \frac{(\|e_0\|^2 - \|e\|^2)/k}{\|e\|^2/(n-p)} \sim F_{k,n-p}$ under H_0, the likelihood ratio test made exact; $F = T^2$ for $k = 1$; stack loss 59.9 on $(3, 17)$ and 6.67 on $(2, 17)$; Old Faithful 11.5 on $(2, 267)$. Bonferroni, α/k per test, has level at most α and says which hypothesis failed.
- Ridge: $(X^\top X + \lambda I)^{-1}X^\top Y$, the posterior mode under $\beta \sim \text{Gauss}_p(0, \tau^2 I)$, $\lambda = \sigma^2/\tau^2$; defined for every X.
- Without Gaussian errors, for i.i.d. rows: $\sqrt{n}(\hat\beta - \beta^*) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_p(0, M^{-1}VM^{-1})$, which is $\sigma^2 M^{-1}$ for constant scatter; the sandwich otherwise. Check the residuals against the fitted values, and their normality with Lilliefors.

21.10 Exercises

Exercise 21.1. Take X with columns $(1, \ldots, 1)^\top$ and $(x_1, \ldots, x_n)^\top$. (a) Compute $X^\top X$, $X^\top Y$ and $(X^\top X)^{-1}$, and show that $\hat\beta = (X^\top X)^{-1}X^\top Y$ is the pair $(\hat a, \hat b)$ of Theorem 20.6. (b) Show that $\sigma^2(X^\top X)^{-1}$ is the covariance matrix of Proposition 20.9. *(Solution on page 436.)*

Exercise 21.2. Two groups. Let X have a column of ones and a column z of 0s and 1s, with n_1 ones and $n_0 = n - n_1$ zeros. (a) Compute $(X^\top X)^{-1}$ and show that $\hat\beta = (\bar Y_0, \bar Y_1 - \bar Y_0)^\top$. (b) Show that $\gamma_2 = 1/n_0 + 1/n_1$, so that T_2 is the pooled two-sample t statistic of Theorem 14.10 and the t-test for $\beta_2 = 0$ is the pooled t-test. (c) In the aspirin study, $n_0 = 11{,}034$ and $n_1 = 11{,}037$; verify γ_2 and the standard error 0.00154 of Exercise 20.6. *(Solution on page 436.)*

Exercise 21.3. Let $P = X(X^\top X)^{-1}X^\top$ with X of rank p. (a) Show that $\operatorname{tr}(P) = p$ and $\operatorname{tr}(I-P) = n-p$, using $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. (b) Deduce $\mathbb{E}\|e\|^2 = (n-p)\sigma^2$ directly from $e = (I-P)\epsilon$ and $\mathbb{E}[\epsilon\epsilon^\top] = \sigma^2 I$, without the rotation. (c) The diagonal entry P_{ii} is called the leverage of observation i; show that $\mathbb{V}\mathrm{ar}(e_i) = \sigma^2(1-P_{ii})$, so a high-leverage observation has a small residual however badly it fits. For the stack loss data the largest leverage is 0.41, on day 17, and $\operatorname{tr}(P) = 4$. *(Solution on page 436.)*

Exercise 21.4. For $k = 1$ in Theorem 21.6, show that $\|e_0\|^2 - \|e\|^2 = \hat\beta_p^2/\gamma_p$, so that $F = T_p^2$. (Write u_p, the last basis vector of the proof, as $(I-P_0)x_p/\|(I-P_0)x_p\|$ with x_p the last column of X, and use $\hat\beta_p = u_p^\top Y/\|(I-P_0)x_p\|$ and $\gamma_p = 1/\|(I-P_0)x_p\|^2$; the last two follow from writing the fit of the full model as the fit of X_0 plus the fit of the residual of x_p on X_0.) *(Solution on page 436.)*

Exercise 21.5. (a) Show that if $V \sim \chi^2_k$ and $W_n \sim \chi^2_{n-p}$ are independent, then $k\,\frac{V/k}{W_n/(n-p)} \xrightarrow[n\to\infty]{\text{(d)}} \chi^2_k$ as $n \to \infty$ with k and p fixed. (b) Explain why, under the hypotheses of Theorem 21.7 with constant scatter and H_0 true, $kF \xrightarrow[n\to\infty]{\text{(d)}} \chi^2_k$ even without Gaussian errors. (Write $\|e_0\|^2 - \|e\|^2$ as a quadratic form in $\sqrt{n}(\hat\beta - \beta^*)$'s last k coordinates.) *(Solution on page 436.)*

Exercise 21.6. Joint confidence region. In the Gaussian linear model let $\hat\beta_{(k)}$ and $C_{(k)}$ be as in the previous exercise, and let $\hat C_{(k)} = \hat\sigma^2 C_{(k)}$. Show that $(\hat\beta_{(k)} - \beta_{(k)})^\top \hat C_{(k)}^{-1}(\hat\beta_{(k)} - \beta_{(k)})/k \sim F_{k,n-p}$, so that the ellipsoid $\{b : (\hat\beta_{(k)} - b)^\top \hat C_{(k)}^{-1}(\hat\beta_{(k)} - b) \le k\, q_\alpha(F_{k,n-p})\}$ is a confidence region of level $1-\alpha$ for $\beta_{(k)}$. This is the ellipse of Figure 21.4, with $k = 2$ and $2q_{0.05}(F_{2,267}) = 6.06$. *(Solution on page 437.)*

Exercise 21.7. Polynomial regression. For Old Faithful, fitting the waits on 1, x and x^2 gives $\hat\beta = (17.20, 22.21, -1.766)^\top$ with $T = -3.74$ for the squared term and residual sum of squares 8,976.7, against 9,443.4 for the line. (a) Compute the F statistic for the squared term and check that it equals T^2. (b) The fitted parabola's slope at x is $22.21 - 3.53x$; evaluate it at 2 and at 4.5 minutes and relate the numbers to the residual plot of Figure 20.3. (c) Why is this still a linear regression? *(Solution on page 437.)*

Exercise 21.8. Gauss–Markov in general. In the linear regression model with X of rank p, let $\tilde\beta = AY$ be any estimator that is linear in Y and unbiased for every β. (a) Show that $AX = I_p$. (b) Write $A = (X^\top X)^{-1}X^\top + D$ and show that $DX = 0$. (c) Show that $\mathbb{C}\mathrm{ov}(\tilde\beta) = \sigma^2(X^\top X)^{-1} + \sigma^2 DD^\top$, so that every coordinate of $\hat\beta$ has the smallest variance among linear unbiased estimators. *(Solution on page 437.)*

Exercise 21.9. For the ridge estimator with X of rank p, (a) show that $\mathbb{E}[\tilde\beta_\lambda] = (X^\top X + \lambda I)^{-1}X^\top X\beta$, so it is biased for $\lambda > 0$, and that $\mathbb{C}\mathrm{ov}(\tilde\beta_\lambda) = \sigma^2(X^\top X + \lambda I)^{-1}X^\top X(X^\top X + \lambda I)^{-1}$; (b) when $X^\top X = cI_p$ (orthogonal columns of equal length), show that $\tilde\beta_\lambda = \frac{c}{c+\lambda}\hat\beta$, the one-predictor shrinkage of Section 20.4 applied to every coordinate. *(Solution on page 437.)*

Exercise 21.10. The kiss data as a regression. (a) With X the column of ones, compute $(X^\top X)^{-1}$, $\hat\beta$, P and $\operatorname{tr}(P)$. (b) Suppose the 124 couples had been observed in two cities, 60 in the first with 36 right turns and 64 in the second with 44, and add the city as a 0/1 column. Compute $\hat\beta$, $\|e\|^2$, $\hat\sigma^2$ and T_2, and say what the test of $\beta_2 = 0$ asks. (c) Which assumption of the Gaussian linear model fails, and what makes the test usable anyway? *(Solution on page 437.)*

Part IX

Generalized Linear Models

Regression for responses that are not Gaussian. Exponential families (Chapter 22) are the distributions the method works for; link functions and the fitting algorithm (Chapter 23) connect them to a linear predictor, with maximum likelihood doing the estimation as in Part IV.

Chapter 22

Exponential Families

In this chapter

- One form that the Bernoulli, Poisson, exponential, Gaussian and Gamma families all take, $h(y)\exp(\eta(\theta)^\top T(y) - B(\theta))$, and what it means: a base weight tilted by an exponential. Which families do not take it.
- The canonical one-parameter form, $\exp\big((y\theta - b(\theta))/\phi + c(y,\phi)\big)$, whose one function b carries the mean, the variance and the Fisher information: $\mathbb{E}[Y] = b'(\theta)$ and $\mathbb{V}\mathrm{ar}(Y) = \phi\, b''(\theta)$.
- Why the maximum likelihood estimator of the mean is the sample mean in every one of these families, why the log-likelihood is concave, and why the method of moments agreed with maximum likelihood so often.
- Sufficiency, promised in Chapter 7: the factorization theorem, and the sum $\sum_i T(Y_i)$ that an exponential family makes visible. Why the conjugate priors of Chapter 18 exist.

Chapter 21 ended with a response that is a count or a proportion, whose mean cannot be a line and whose variance moves with its mean. This chapter describes the distributions for which the regression of Chapter 23 will work, and finds that they are the book's old families seen from one angle.

22.1 One form for many families

Chapter 13 noticed that for the Bernoulli, Poisson, exponential and Gaussian families the method of moments and maximum likelihood give the same estimator, and promised a reason. Chapter 11 found that for the first three the likelihood depends on the data only through their sum, and Chapter 7 that the sum is sufficient for the kiss data. Chapter 18 found a conjugate prior for each. These are one fact seen four times. Write the four PMFs and densities with every piece that depends

on the parameter moved into an exponent:

$$\begin{aligned}
\text{Bern}(p): &\quad p^y(1-p)^{1-y} = \exp\Big(y\log\frac{p}{1-p} + \log(1-p)\Big), && y\in\{0,1\};\\
\text{Pois}(\lambda): &\quad \frac{\lambda^y e^{-\lambda}}{y!} = \frac{1}{y!}\exp\big(y\log\lambda - \lambda\big), && y = 0,1,2,\ldots;\\
\text{Expo}(\lambda): &\quad \lambda e^{-\lambda y} = \exp\big(-\lambda y + \log\lambda\big), && y > 0;\\
\text{Gauss}(\mu,\sigma^2): &\quad \frac{e^{-(y-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi}}\exp\Big(\frac{\mu}{\sigma^2}y - \frac{1}{2\sigma^2}y^2 - \frac{\mu^2}{2\sigma^2} - \log\sigma\Big), && y\in\mathbb{R}.
\end{aligned}$$

In each, the exponent is a sum of products of a function of the parameter and a function of y, minus a function of the parameter alone, and what is left outside depends on y alone.

Definition 22.1 (Exponential family)**.** A family of distributions $(\mathbb{P}_\theta)_{\theta\in\Theta}$ on $E\subset\mathbb{R}^q$, with $\Theta\subset\mathbb{R}^k$, is a **$k$-parameter exponential family** if its PMFs or densities can be written as

$$f_\theta(y) = h(y)\exp\Big(\sum_{j=1}^{k}\eta_j(\theta)T_j(y) - B(\theta)\Big) = h(y)\exp\big(\eta(\theta)^\top T(y) - B(\theta)\big)$$

for some functions $\eta_1,\ldots,\eta_k$ and B of θ and $T_1,\ldots,T_k$ and $h\geq 0$ of y. The $\eta_j(\theta)$ are the **natural parameters** and the $T_j(y)$ the **sufficient statistics** of the family (the name is justified in Section 22.4).

Every member of the family is the same base weight $h(y)$ multiplied by an exponential of a linear function of $T(y)$, and $B(\theta)$ is whatever makes the result integrate or sum to 1: $e^{B(\theta)} = \int h(y)e^{\eta(\theta)^\top T(y)}\,\mathrm{d}y$. The parameter and the data meet only in the products $\eta_j(\theta)T_j(y)$, which is the property everything in this chapter rests on. The support, where $f_\theta > 0$, is where $h > 0$, the same for every θ.

For the kiss data. The Bernoulli family is a one-parameter exponential family on $\{0,1\}$ with $h(y) = 1$, $T(y) = y$, $\eta(p) = \log\big(p/(1-p)\big)$, the log-odds of Chapter 19, and $B(p) = -\log(1-p) = \log(1+e^\eta)$. For 124 couples the natural parameter multiplies $\sum_i R_i = 80$.

Reading the four displays against the definition: the Poisson has $h(y) = 1/y!$, $T(y) = y$, $\eta = \log\lambda$, $B = \lambda = e^\eta$; the exponential has $h = 1$, $T(y) = y$, $\eta = -\lambda$, $B = -\log\lambda = -\log(-\eta)$; the Gaussian is a two-parameter family with $T(y) = (y, y^2)$, $\eta = (\mu/\sigma^2, -1/(2\sigma^2))$, $B = \mu^2/(2\sigma^2) + \log\sigma$ and $h = 1/\sqrt{2\pi}$, and with σ^2 known it is a one-parameter family in μ with $h(y) = e^{-y^2/(2\sigma^2)}/(\sigma\sqrt{2\pi})$. The Gamma family Gamma$(\alpha,\beta)$ of Chapter 18, with density $\beta^\alpha y^{\alpha-1}e^{-\beta y}/\Gamma(\alpha)$, is a two-parameter family with $T(y) = (y, \log y)$ and $\eta = (-\beta, \alpha-1)$; the Beta family is another (Exercise 22.3); the binomial with a known number of trials, the geometric and the negative binomial with known size are one-parameter families (Exercise 22.2).

The form is more restrictive than it looks, and the restriction is where the parameter can go.

Pitfall: families that are not exponential families

Unif$([0,\theta])$ is not: its density is $\frac{1}{\theta}\mathbb{1}(0\leq y\leq\theta)$, and the indicator cannot be written as $h(y)e^{\eta(\theta)T(y)}$ because the support moves with θ, while in an exponential family it is fixed by h. The Laplace family with unknown center, density $\frac{1}{2}e^{-|y-\mu|}$, is not: $|y-\mu|$ is not a sum of products of a function of μ and a function of y (Exercise 22.4 proves this for one product;

the same kind of argument handles more, and is not given here). The Student t and Cauchy families are not either. The regular models of Chapter 12 are a wider class than the exponential families, and the uniform, which failed the regularity conditions, fails here for the same reason, its moving support.

There is a picture behind the definition.

Intuition: one base weight, tilted

Start from the base weight $h(y)$ and multiply it by $e^{\theta y}$, for a one-parameter family with $T(y) = y$ and natural parameter θ: this **tilts** the weight, pushing mass toward large y when $\theta > 0$ and toward small y when $\theta < 0$, and dividing by e^B renormalizes it. Figure 22.1 draws two cases. On the left the base is the standard Gaussian density, and tilting by $e^{\theta y}$ moves the bell to the center θ without changing its shape: that is the Gauss$(\mu, 1)$ family, with $\theta = \mu$. On the right the base is the weight $1/y!$ on $0, 1, 2, \ldots$, and tilting by $e^{\theta y}$ gives the Poisson PMF with mean e^θ. An exponential family is the set of all exponential tilts of one base weight, and the parameter says how hard to tilt.

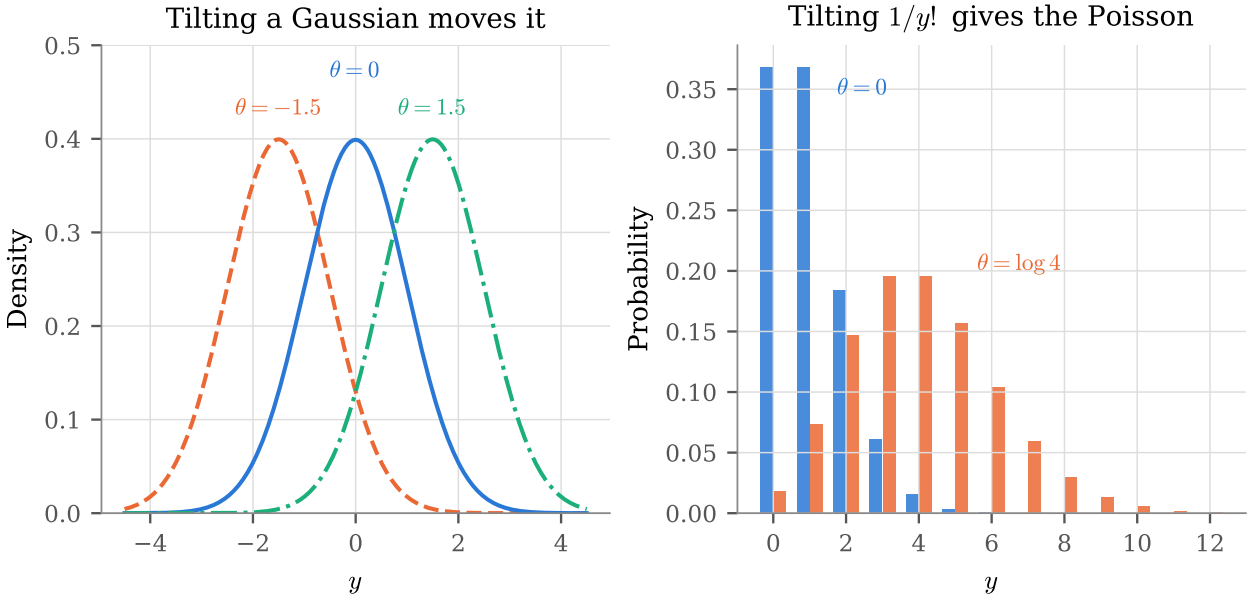


Figure 22.1: Exponential tilting. Left: the standard Gaussian density multiplied by $e^{\theta y}$ and renormalized, for $\theta = -1.5$ (orange, dashed), 0 (blue) and 1.5 (aqua, dash-dot). Right: the weights $1/y!$ tilted and renormalized, for $\theta = 0$ (blue, mean 1) and $\theta = \log 4$ (orange, mean 4).

22.2 The canonical form

In the Bernoulli display of Section 22.1 the parameter entered through one product, $y \log\big(p/(1-p)\big)$; write θ for that log-odds and the PMF is $\exp\big(y\theta - \log(1 + e^\theta)\big)$, one function of θ multiplying y and one function of θ alone. For one parameter and $T(y) = y$, the natural parameter itself is the best coordinate to work in, and the standard way to write the family separates out a scale.

Definition 22.2 (Canonical exponential family)**.** A one-parameter family on $E \subset \mathbb{R}$ is in **canonical form** if its PMFs or densities are

$$f_\theta(y) = \exp\Big(\frac{y\theta - b(\theta)}{\phi} + c(y,\phi)\Big), \qquad \theta \in \Theta,$$

for known functions b and c and a known number $\phi > 0$, the **dispersion parameter**, with Θ an open interval on which b is twice differentiable. θ is the **canonical parameter**.

> This is Definition 22.1 with $k = 1$, $T(y) = y$, $\eta(\theta) = \theta/\phi$, $B(\theta) = b(\theta)/\phi$ and $h(y) = e^{c(y,\phi)}$: the parameter enters only through the product $y\theta$, and one function b does the normalizing. The dispersion ϕ is a known scale, 1 for counts and 0/1 data and σ^2 for a Gaussian with known variance; the family is a one-parameter family in θ with ϕ fixed.
> **For the kiss data.** Bern(p) is canonical with $\theta = \log\big(p/(1-p)\big)$, $b(\theta) = \log(1+e^\theta)$, $\phi = 1$ and $c = 0$: the PMF is $\exp(y\theta - \log(1+e^\theta))$, which at $y = 1$ is $e^\theta/(1+e^\theta) = p$. The kiss data's canonical parameter is the log-odds of turning right, estimated by $\log(80/44) = 0.598$.

The table gives the canonical forms of the book's families; each is checked by writing the PMF or density with $y\theta$ separated out, as in Section 22.1.

family	θ	$b(\theta)$	ϕ	$c(y,\phi)$
Bern(p)	$\log\dfrac{p}{1-p}$	$\log(1+e^\theta)$	1	0
Pois(λ)	$\log\lambda$	e^θ	1	$-\log y!$
Expo(λ)	$-\lambda$	$-\log(-\theta)$	1	0
Gauss(μ,σ^2), σ^2 known	μ	$\theta^2/2$	σ^2	$-\dfrac{1}{2}\Big(\dfrac{y^2}{\phi} + \log(2\pi\phi)\Big)$
Gamma(α,β), α known	$-\beta/\alpha$	$-\log(-\theta)$	$1/\alpha$	$\alpha\log(\alpha y) - \log y - \log\Gamma(\alpha)$

For the Gaussian, $\frac{1}{\sigma\sqrt{2\pi}}e^{-(y-\mu)^2/(2\sigma^2)} = \exp\big(\frac{y\mu-\mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)\big)$, which is the table's row with $\theta = \mu$; the Gamma row is Exercise 22.1. The canonical parameters are the scales on which this book has already found the families easiest to work with: the log-odds for a proportion, the log of a Poisson mean, the mean of a Gaussian, and, for a rate, $-\lambda$ or a multiple of it.

The payoff of the form is that one function determines the moments.

Theorem 22.3 (Mean and variance from b)**.** *In a canonical exponential family,*

$$\mathop{\mathbb{E}}_\theta[Y] = b'(\theta), \qquad \mathop{\mathbb{V}ar}_\theta(Y) = \phi\, b''(\theta).$$

So b is convex, strictly convex wherever $\mathbb{V}ar_\theta(Y) > 0$, and b' is then strictly increasing: the mean $\mu = b'(\theta)$ and the canonical parameter determine each other.

Proof strategy (core). The density integrates to 1 for every θ. Differentiate that identity in θ once and the mean of $Y - b'(\theta)$ comes out as 0; differentiate it twice and the variance comes out as $\phi b''(\theta)$. The same two differentiations are the score identities of Chapter 12.

Recall. The score of one observation has mean zero (Proposition 12.2), and the Fisher information is both the variance of the score and minus the expected second derivative of the log-likelihood (Definition 12.3, Theorem 12.4), under the regularity conditions of Section 12.6, which Proposition 22.6 below verifies for canonical families.

Proof. Write the identity $\int \exp\big((y\theta - b(\theta))/\phi + c(y,\phi)\big)\,\mathrm{d}y = 1$ (a sum for a PMF) and differentiate in θ under the integral sign, which is allowed on the open interval Θ: the integrand is a smooth function of θ and its derivatives are integrable on a neighborhood of each point (Proposition 22.6). The derivative of the integrand is $\frac{y-b'(\theta)}{\phi} f_\theta(y)$, so

$$0 = \int \frac{y - b'(\theta)}{\phi} f_\theta(y)\,\mathrm{d}y = \frac{\mathbb{E}_\theta[Y] - b'(\theta)}{\phi},$$

which is the mean. Differentiating once more,

$$0 = \int \Big(\frac{(y - b'(\theta))^2}{\phi^2} - \frac{b''(\theta)}{\phi}\Big) f_\theta(y)\,\mathrm{d}y = \frac{\mathbb{V}\mathrm{ar}_\theta(Y)}{\phi^2} - \frac{b''(\theta)}{\phi},$$

since $y - b'(\theta) = y - \mathbb{E}_\theta[Y]$, which is the variance. A variance is non-negative, so $b'' \geq 0$ and b is convex; where the variance is positive $b'' > 0$, and b' is strictly increasing there. In the language of Chapter 12: the log-likelihood of one observation is $\ell(\theta) = (Y\theta - b(\theta))/\phi + c(Y,\phi)$, with score $\ell'(\theta) = (Y - b'(\theta))/\phi$ and $\ell''(\theta) = -b''(\theta)/\phi$, so the first identity is $\mathbb{E}_\theta[\ell'(\theta)] = 0$ and the second is $\mathbb{V}\mathrm{ar}_\theta(\ell'(\theta)) = -\,\mathbb{E}_\theta[\ell''(\theta)]$. □

Figure 22.2 draws b for the Bernoulli family. Its slope at θ is the mean $p = e^\theta/(1+e^\theta)$, which rises from 0 to 1 as θ runs over the real line, and its curvature is the variance $p(1-p)$, largest at $\theta = 0$ and vanishing at both ends: a straight line far out, a bend in the middle. At the kiss data's $\hat\theta = 0.598$ the slope is 0.645 and the curvature 0.229.

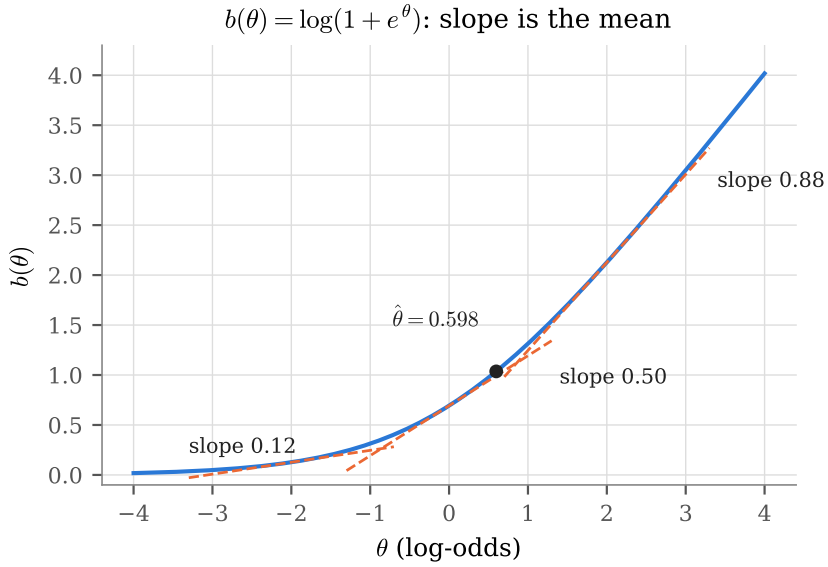


Figure 22.2: The Bernoulli family's $b(\theta) = \log(1 + e^\theta)$ (blue), with tangent lines at $\theta = -2, 0, 2$ (orange, dashed) whose slopes are the means 0.12, 0.50 and 0.88; the dot marks the kiss data's $\hat\theta = 0.598$.

The theorem says how the variance follows the mean in each family, which is the fact the regression of Chapter 23 is built around.

Definition 22.4 (Variance function)**.** For a canonical family with means $\mu = b'(\theta)$ ranging over an interval M, the **variance function** is $V(\mu) = b''\big((b')^{-1}(\mu)\big)$, so that $\mathbb{V}\mathrm{ar}(Y) = \phi\, V(\mu)$ when $\mathbb{E}[Y] = \mu$.

Fix the mean and the family fixes the variance, up to the dispersion. The variance function says which it is: constant for a Gaussian, equal to the mean for a Poisson, the mean squared for an exponential.
For the kiss data. For the Bernoulli, $b'(\theta) = e^\theta/(1+e^\theta) = \mu$ and $b''(\theta) = \mu(1-\mu)$, so $V(\mu) = \mu(1-\mu)$: the variance $p(1-p)$ that every interval for the kiss proportion has used, 0.229 at $p = 0.645$, is the variance function at the mean.

Figure 22.3 draws the four variance functions. They are the reason a Gaussian linear model cannot serve for counts or proportions: it assumes V constant, and for those responses V moves with μ.

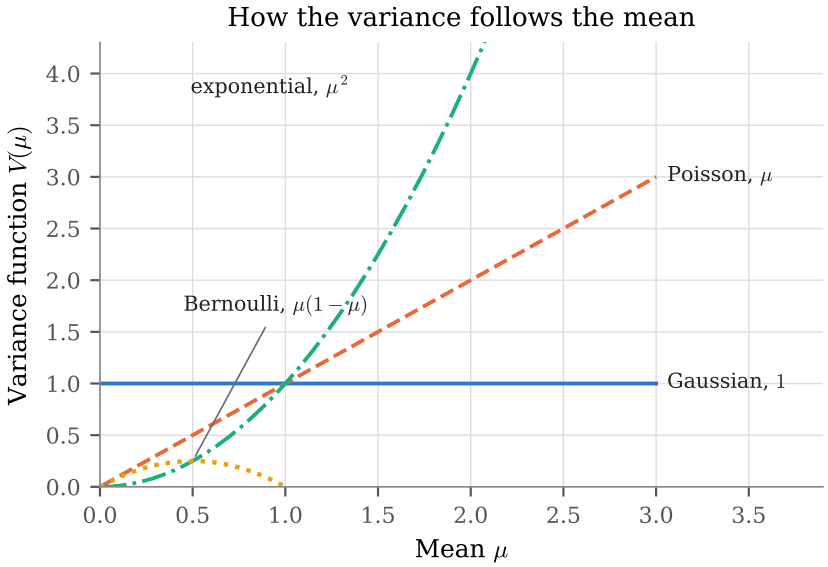


Figure 22.3: The variance function $V(\mu)$ of four families: Gaussian with $\sigma = 1$ (blue), Poisson (orange, dashed), exponential (aqua, dash-dot) and Bernoulli on $(0, 1)$ (yellow, dotted).

Since b' is strictly increasing, it has an inverse, and the inverse is the map from the mean back to the canonical parameter.

Definition 22.5 (Canonical link)**.** The **canonical link** of a canonical exponential family is $g_c = (b')^{-1}$, the function with $g_c(\mu) = \theta$: it sends the mean to the canonical parameter.

Each family has a scale on which its parameter enters the likelihood linearly, and the canonical link is the change of coordinates from the mean to that scale: the identity for the Gaussian, the logarithm for the Poisson, $-1/\mu$ for the exponential.
For the kiss data. $b'(\theta) = e^\theta/(1+e^\theta) = \mu$ solves to $\theta = \log\big(\mu/(1-\mu)\big)$: the canonical link of the Bernoulli family is the log-odds, which Chapter 23 will set equal to $x^\top\beta$ to get logistic regression.

22.3 Likelihood in a canonical family

The regularity conditions of Chapter 12 were listed and checked family by family. In a canonical family they come for free.

Proposition 22.6 (Canonical families are regular)**.** *A canonical exponential family with $b''(\theta) > 0$ on Θ satisfies the regularity conditions of Section 12.6 at every $\theta^* \in \Theta$: the parameter is identifiable, the support does not depend on θ, Θ is open, b is infinitely differentiable, $f_\theta(y)$ is infinitely differentiable in θ with the derivatives movable inside the integral, and $I(\theta) = b''(\theta)/\phi > 0$.*

Proof strategy (for completeness; skip on a first reading). Each condition is read off the form. The one with content is moving derivatives inside the integral, which holds because $\int e^{y\theta/\phi}h(y)\,\mathrm{d}y$ is finite on an open interval, and such an integral can be differentiated under the sign as often as one likes.

Proof. Identifiability: if $f_\theta = f_{\theta'}$ then $(\theta - \theta')y = b(\theta) - b(\theta')$ for every y in the support, and the support has at least two points (one point would make $\mathbb{V}\mathrm{ar}_\theta(Y) = 0$, against $b'' > 0$), so subtracting the equations at two of them gives $\theta = \theta'$. The support is E, where $c(y, \phi)$ is finite, and does not involve θ; Θ is open by definition, so θ^* is interior. The integrand $\exp\big((y\theta - b(\theta))/\phi + c(y,\phi)\big)$ is infinitely differentiable in θ. For the differentiation under the integral sign, write $\int e^{y\theta/\phi}e^{c(y,\phi)}\,\mathrm{d}y = e^{b(\theta)/\phi}$, finite for every $\theta \in \Theta$. The kth derivative of the integrand in θ is $(y/\phi)^k e^{y\theta/\phi}e^{c(y,\phi)}$. Fix θ^* and $\delta > 0$ with $[\theta^* - 2\delta, \theta^* + 2\delta] \subset \Theta$; for θ within δ of θ^*, $|y/\phi|^k e^{y\theta/\phi} \le K_k\big(e^{y(\theta^*+2\delta)/\phi} + e^{y(\theta^*-2\delta)/\phi}\big)$ for a constant K_k depending on k and δ, since a power of $|y|$ is eventually smaller than $e^{\delta|y|/\phi}$. The bound does not depend on θ and is integrable against $e^{c(y,\phi)}$, so the derivative passes through the integral: by the mean value theorem the difference quotient of the $(k-1)$st derivative is a kth derivative at a nearby θ, the bound applies to it, and the dominated convergence theorem (toolkit A.1) lets the limit inside. Applied for $k = 1, 2, \ldots$ this makes $e^{b(\theta)/\phi}$, and so b, infinitely differentiable, and the derivatives of $f_\theta = e^{-b(\theta)/\phi}e^{y\theta/\phi}e^{c(y,\phi)}$ pass through the integral by the product rule. In particular $\ell'''(\theta) = -b'''(\theta)/\phi$ does not involve Y and is continuous in θ, so near θ^* it is bounded by a constant, which is the third-derivative condition, and $\ell''(\theta^*) = -b''(\theta^*)/\phi$ is a constant, with variance 0. Finally $I(\theta) = \mathbb{V}\mathrm{ar}_\theta(\ell'(\theta)) = \mathbb{V}\mathrm{ar}_\theta(Y)/\phi^2 = b''(\theta)/\phi > 0$. □

Now take n i.i.d. observations. The log-likelihood is

$$\ell_n(\theta) = \frac{\theta\sum_{i=1}^n Y_i - n\,b(\theta)}{\phi} + \sum_{i=1}^n c(Y_i, \phi),$$

and the sample enters only through $\sum_i Y_i$, apart from the last term, which does not involve θ.

Theorem 22.7 (Maximum likelihood in a canonical family)**.** *Let $Y_1, \ldots, Y_n$ be i.i.d. from a canonical family with $b'' > 0$ on Θ, and suppose $\bar{Y}_n$ lies in the range of b'. Then:*

1. *ℓ_n is strictly concave, and the maximum likelihood estimator is the unique solution of $b'(\hat{\theta}_n) = \bar{Y}_n$; the maximum likelihood estimator of the mean $\mu = b'(\theta)$ is $\hat{\mu}_n = \bar{Y}_n$;*
2. *$\bar{Y}_n$ is unbiased for μ with variance $\phi\,b''(\theta)/n = 1/(nI_\mu)$, where $I_\mu = 1/(\phi V(\mu))$ is the Fisher information in the mean parameterization: it attains the Cramér–Rao bound;*
3. *$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow[n\to\infty]{(d)} \mathrm{Gauss}\big(0, \phi/b''(\theta)\big)$ and $\sqrt{n}(\bar{Y}_n - \mu) \xrightarrow[n\to\infty]{(d)} \mathrm{Gauss}\big(0, \phi\,b''(\theta)\big)$.*

Proof strategy (core). The score is $(\sum_i Y_i - nb'(\theta))/\phi$, zero exactly where $b'(\theta)$ equals the sample mean, and $\ell_n'' = -nb''/\phi < 0$ everywhere. The rest is the Central Limit Theorem for $\bar{Y}_n$ and Chapter 12's theorem, whose conditions Proposition 22.6 checked.

Recall. The Central Limit Theorem (Theorem 4.8): for i.i.d. Y_i with mean μ and variance $0 < \sigma^2 < \infty$, $\sqrt{n}(\bar{Y}_n - \mu) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, \sigma^2)$. Asymptotic normality of the MLE (Theorem 12.5): in a regular model with θ^* interior and $I(\theta^*) > 0$, $\sqrt{n}(\hat{\theta}_n - \theta^*) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, 1/I(\theta^*))$. The Cramér–Rao bound (Theorem 12.6):

an unbiased estimator in a regular model has variance at least $1/(nI)$. Fisher information under a change of parameter (Lemma 19.2): $I_\mu = I_\theta(\mathrm{d}\theta/\mathrm{d}\mu)^2$. The delta method (Theorem 5.4): if $\sqrt{n}(Z_n - \mu) \xrightarrow[n\to\infty]{\text{(d)}}$ Gauss$(0, \sigma^2)$ and g is continuously differentiable at μ, then $\sqrt{n}(g(Z_n) - g(\mu)) \xrightarrow[n\to\infty]{\text{(d)}}$ Gauss$(0, g'(\mu)^2\sigma^2)$.

Proof. $\ell_n'(\theta) = \big(\sum_i Y_i - nb'(\theta)\big)/\phi$ and $\ell_n''(\theta) = -nb''(\theta)/\phi < 0$, so ℓ_n is strictly concave and has at most one stationary point, which is its maximum (Proposition 11.5); the stationary point is where $b'(\theta) = \bar{Y}_n$, which has a solution since $\bar{Y}_n$ is in the range of the continuous, strictly increasing b', and only one. By the invariance of maximum likelihood (Proposition 11.6), the MLE of $\mu = b'(\theta)$ is $b'(\hat{\theta}_n) = \bar{Y}_n$: this is (1). For (2), $\mathbb{E}[\bar{Y}_n] = b'(\theta) = \mu$ and $\mathbb{V}\mathrm{ar}(\bar{Y}_n) = \phi b''(\theta)/n$ by Theorem 22.3. The information in μ is $I_\mu = I_\theta(\mathrm{d}\theta/\mathrm{d}\mu)^2 = \frac{b''(\theta)}{\phi} \cdot \frac{1}{b''(\theta)^2} = \frac{1}{\phi b''(\theta)} = \frac{1}{\phi V(\mu)}$, since $\mathrm{d}\mu/\mathrm{d}\theta = b''(\theta)$, so $\mathbb{V}\mathrm{ar}(\bar{Y}_n) = 1/(nI_\mu)$, which is the Cramér–Rao bound for μ. For (3), the second limit is the Central Limit Theorem with $\sigma^2 = \phi b''(\theta)$. The first follows from the second by the delta method, since $\hat{\theta}_n = (b')^{-1}(\bar{Y}_n)$ and the derivative of $(b')^{-1}$ at μ is $1/b''(\theta)$, continuous by Proposition 22.6, so the asymptotic variance is $\phi b''(\theta)/b''(\theta)^2 = \phi/b''(\theta)$; it agrees with Theorem 12.5 with $I(\theta) = b''(\theta)/\phi$, whose hypotheses hold by Proposition 22.6. □

The condition that $\bar{Y}_n$ lie in the range of b' is not empty: for the Bernoulli the range is $(0, 1)$, and a sample of all 1s has no maximum likelihood estimate on $\Theta = \mathbb{R}$, the log-likelihood $n\theta - n\log(1 + e^\theta)$ increasing forever. This is the problem that will return, with predictors, in Chapter 23.

This is the explanation Chapter 13 promised. In a canonical family the likelihood equation is $b'(\theta) = \bar{Y}_n$, that is, 'set the model's mean equal to the sample mean', which is exactly the method of moments; and for the uniform and Laplace families, which are not exponential families, the two methods part. With k natural parameters the same differentiation gives the likelihood equations $\mathbb{E}_\eta[T_j(Y)] = \frac{1}{n}\sum_i T_j(Y_i)$ for $j = 1, \ldots, k$ (in the natural parameterization, with B a function of η: differentiating $\int h(y)e^{\eta^\top T(y) - B(\eta)}\,\mathrm{d}y = 1$ in η_j gives $\partial B/\partial\eta_j = \mathbb{E}_\eta[T_j(Y)]$ as in Theorem 22.3, and $\partial\ell_n/\partial\eta_j = \sum_i T_j(Y_i) - n\,\partial B/\partial\eta_j$), which match the moments of the sufficient statistics; for the Gaussian with both parameters unknown, $T = (Y, Y^2)$, and the equations match the first two sample moments, which is how Chapter 13 found $\hat{\mu}$ and $\hat{\sigma}^2$.

From undergraduate statistics: the sample mean is the best unbiased estimator

An introductory course states, without proof, that the sample mean is the best unbiased estimator of a population mean. Part (2) of Theorem 22.7 is the statement with its conditions: for a sample from a canonical exponential family, $\bar{Y}_n$ is unbiased for the mean and its variance $\phi b''(\theta)/n$ is the Cramér–Rao bound, which no unbiased estimator can beat. Outside these families the claim can fail: for the Laplace family of Chapter 13 the sample median has asymptotic variance 1 against the sample mean's 2, and for Unif$([0, \theta])$ the largest observation, rescaled by $(n + 1)/n$ to remove its bias, has error of order $1/n$ against the $1/\sqrt{n}$ of the estimator $2\bar{Y}_n$ built from the mean (Chapter 12).

Here are the three running data sets on their canonical scales.

Worked example: three estimates on the canonical scale

For the kiss data, $b'(\theta) = e^\theta/(1 + e^\theta) = 0.645$ gives $\hat{\theta} = \log(80/44) = 0.598$, the log-odds, with $I(\hat{\theta}) = b''(\hat{\theta}) = 0.645 \times 0.355 = 0.229$ and standard error $1/\sqrt{124 \times 0.229} = 0.188$, the number the odds-ratio bridge of Chapter 5 reached by the delta method; $0.598 \pm 1.96 \times 0.188$,

exponentiated, is the odds interval $[1.26, 2.63]$ found there. For the siblings data of Chapter 2, 24 values of $X_i - 1$ summing to 24, the Poisson mean is $\hat{\lambda} = 1$, so $\hat{\theta} = \log 1 = 0$ with $I = b''(0) = e^0 = 1$ and standard error $1/\sqrt{24} = 0.204$. For the subway's 64 waits averaging 6.23 minutes, the exponential's $b'(\theta) = -1/\theta = 6.23$ gives $\hat{\theta} = -0.1605$ per minute, which is $-\hat{\lambda}$, with $b''(\hat{\theta}) = 1/\hat{\theta}^2 = 38.8$ and standard error $\sqrt{1/(64 \times 38.8)} = 0.020$.

Figure 22.4 shows what the canonical scale buys. The kiss log-likelihood is concave in p and in θ, but in θ it is close to the parabola $-\frac{1}{2}nb''(\hat{\theta})(\theta - \hat{\theta})^2$ over the range drawn, which is what makes the Gaussian approximation to $\hat{\theta}$ accurate; in p the curve is lopsided against its parabola, steep toward 1 and gentle toward 0. Chapter 15, which found that Wald's test depends on the parameterization, was looking at the same fact.

Does this make sense? Two standard errors, two ways

The kiss standard error 0.188 on the log-odds scale came out of $1/\sqrt{nb''(\hat{\theta})} = 1/\sqrt{124 \times 0.229}$ here and out of $\sqrt{1/80 + 1/44}$ in Chapter 5. They are the same number because $1/(np(1-p)) = 1/(np) + 1/(n(1-p))$, and $np = 80$, $n(1-p) = 44$. For the subway, the standard error 0.020 of $\hat{\theta} = -\hat{\lambda}$ is $|\hat{\theta}|/\sqrt{n} = 0.1605/8$, the standard error $\hat{\lambda}/\sqrt{n}$ that the delta method gave for $\hat{\lambda} = 1/\bar{T}_n$ in Chapter 5; changing the sign of a parameter changes nothing.

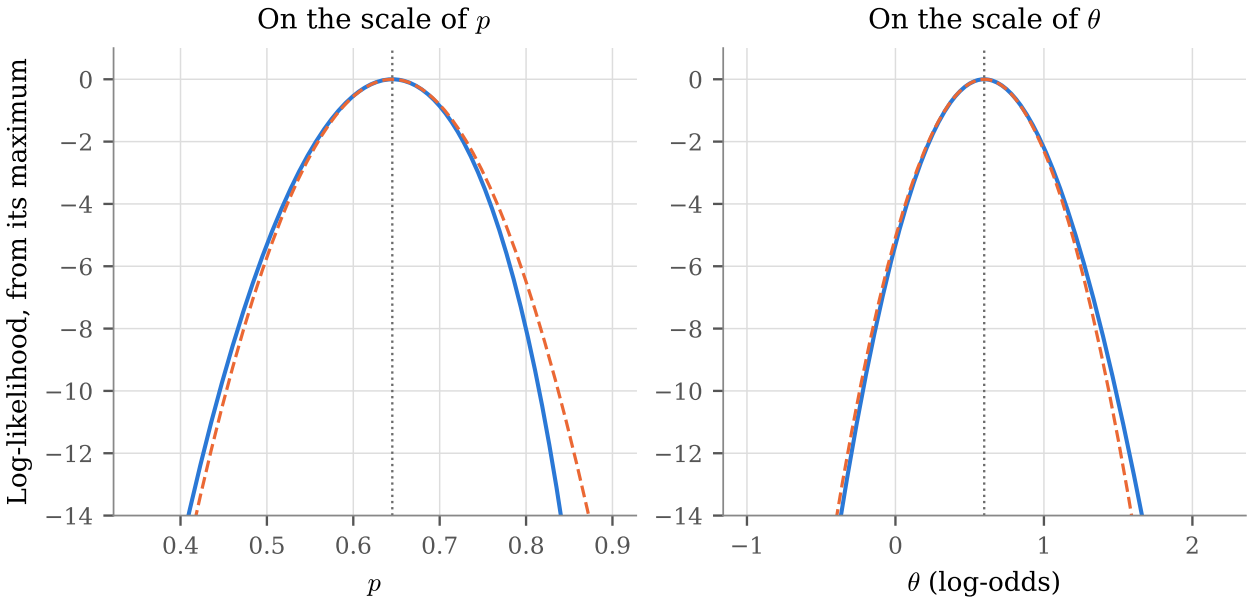


Figure 22.4: The kiss data's log-likelihood from its maximum, with the parabola matching its curvature there (orange, dashed). Left: against p, maximum at 0.645 (dotted), curvature -542. Right: against the canonical parameter θ, maximum at 0.598, curvature -124×0.229.

22.4 Sufficiency

Chapter 7 defined a sufficient statistic, a function T of the sample such that the conditional distribution of the sample given T is free of θ, showed that the count of right turns is sufficient for

the kiss data, and promised the theory would reappear here. Checking the definition directly, as that chapter did, means computing a conditional distribution. A criterion that reads sufficiency off the likelihood is easier.

Theorem 22.8 (Factorization theorem)**.** *Let the sample* $\mathbf{Y} = (Y_1, \dots, Y_n)$ *have joint PMF or density* $f_\theta(\mathbf{y})$*. A statistic* $T(\mathbf{Y})$ *is sufficient for* θ *if and only if*

$$f_\theta(\mathbf{y}) = g_\theta\big(T(\mathbf{y})\big)\, k(\mathbf{y})$$

for some functions g_θ *of the statistic's value and* k *of the sample alone.*

Proof strategy (core). For a discrete sample, the conditional probability of $\mathbf{y}$ given $T = t$ is $f_\theta(\mathbf{y})$ divided by the sum of f_θ over the samples with $T = t$; if f_θ factors, the $g_\theta(t)$ cancels top and bottom. Conversely, if the conditional probability is free of θ, it is the k.

Proof. Take a discrete observation space (the continuous case replaces sums by integrals, and its rigorous version needs more measure theory than this book develops). Fix t and a sample $\mathbf{y}$ with $T(\mathbf{y}) = t$. Then

$$\mathbb{P}_\theta(\mathbf{Y} = \mathbf{y} \mid T = t) = \frac{\mathbb{P}_\theta(\mathbf{Y} = \mathbf{y})}{\mathbb{P}_\theta(T = t)} = \frac{f_\theta(\mathbf{y})}{\sum_{\mathbf{y}' : T(\mathbf{y}') = t} f_\theta(\mathbf{y}')},$$

since $\{\mathbf{Y} = \mathbf{y}\}$ is contained in $\{T = t\}$. If $f_\theta = g_\theta(T)\, k$, the right side is $g_\theta(t)k(\mathbf{y}) \big/ \big(g_\theta(t) \sum_{\mathbf{y}'} k(\mathbf{y}')\big) = k(\mathbf{y}) / \sum_{\mathbf{y}' : T(\mathbf{y}') = t} k(\mathbf{y}')$, free of θ: T is sufficient. Conversely, if T is sufficient, the conditional probability is a function $k(\mathbf{y})$ free of θ, and $f_\theta(\mathbf{y}) = \mathbb{P}_\theta(T = t)\, k(\mathbf{y})$ with $g_\theta(t) = \mathbb{P}_\theta(T = t)$. □

Worked example: four sufficient statistics by factorization

Kiss data: $f_p(\mathbf{r}) = p^s(1-p)^{n-s}$ with $s = \sum_i r_i$, which is $g_p(s)$ times $k = 1$, so S is sufficient, as Chapter 7 found by computing the conditional distribution. Gaussian with both parameters unknown: the joint density is $(2\pi\sigma^2)^{-n/2} \exp\big(-\frac{1}{2\sigma^2}\sum_i y_i^2 + \frac{\mu}{\sigma^2}\sum_i y_i - \frac{n\mu^2}{2\sigma^2}\big)$, a function of $(\sum_i y_i, \sum_i y_i^2)$ and the parameters, so the pair $(\sum_i Y_i, \sum_i Y_i^2)$, equivalently $(\bar{Y}_n, S_n^2)$, is sufficient for (μ, σ^2): the t-test of Chapter 14 used nothing else. Uniform $[0, \theta]$: the joint density is $\theta^{-n}\mathbb{1}(\max_i y_i \le \theta)\mathbb{1}(\min_i y_i \ge 0)$, a function of $\max_i y_i$ and θ times a function of the sample, so the largest observation is sufficient; the MLE of Chapter 11 is a function of it, as the theorem says every MLE must be, since maximizing $g_\theta(T(\mathbf{y}))\, k(\mathbf{y})$ in θ uses the sample only through T. Laplace with unknown center: $2^{-n}\exp(-\sum_i |y_i - \mu|)$ does not change when the observations are reordered, so the sorted sample $(Y_{(1)}, \dots, Y_{(n)})$ is sufficient, and the exponent does not split into a function of μ times a function of one statistic (Exercise 22.4).

In an exponential family the factorization is on the surface.

Corollary 22.9 (Sufficiency in an exponential family)**.** *For* $Y_1, \dots, Y_n$ *i.i.d. from a* k*-parameter exponential family, the vector of sums* $\sum_{i=1}^n T(Y_i) = \big(\sum_i T_1(Y_i), \dots, \sum_i T_k(Y_i)\big)$ *is sufficient for* θ*. In a canonical family,* $\sum_i Y_i$ *is sufficient.*

Proof strategy (for completeness; skip on a first reading). Multiply the n factors: the exponents add, and the parameter meets the data only through the sums.

Proof. $\prod_i f_\theta(y_i) = \big(\prod_i h(y_i)\big) \exp\big(\eta(\theta)^\top \sum_i T(y_i) - nB(\theta)\big)$, which is $k(\mathbf{y}) = \prod_i h(y_i)$ times a function of θ and $\sum_i T(y_i)$. □

A k-parameter exponential family thus has a sufficient statistic of fixed dimension k however large n is: 124 couples reduce to one count, 64 waits to one sum, a Gaussian sample of any size to a sum and a sum of squares. The converse is a classical theorem, due to Darmois, Koopman and Pitman (1935 to 1936): among families with a fixed support and smooth densities, the ones whose i.i.d. samples admit a sufficient statistic of bounded dimension are exactly the exponential families. It is stated without proof; nothing later depends on it. The sufficiency principle of Chapter 7 then says what it has said all along: every method for the kiss data is a function of 80 and 124, and every method for a Gaussian sample is a function of $\bar{Y}_n$ and S_n^2, not because those were convenient choices but because nothing else in the sample speaks about the parameter.

Remark (Rao, 1945, and Blackwell, 1947). Sufficiency improves estimators as well as summarizing data. If T is sufficient and W is any unbiased estimator of θ, then $\mathbb{E}[W \mid T]$ is a function of T alone (the conditional distribution given T does not involve θ), still unbiased by the tower property, and with variance no larger than W's: both have mean θ, and $\mathbb{E}\left[\mathbb{E}[W \mid T]^2\right] \le \mathbb{E}[W^2]$ (toolkit A.15), so $\mathbb{V}\mathrm{ar}(\mathbb{E}[W \mid T]) \le \mathbb{V}\mathrm{ar}(W)$. For the kiss data, $W = R_1$, the first couple's turn, is unbiased for p with variance $p(1-p)$, and $\mathbb{E}[R_1 \mid S] = S/n$, the sample proportion, with variance $p(1-p)/n$: conditioning on the count recovers $\hat{p}$ from a single observation.

22.5 Conjugate priors, explained

Chapter 18 found a conjugate family of priors for each of its models and defined conjugacy as the posterior staying in the prior's family. Written on the canonical scale, every one of those families has the same shape as the likelihood.

Proposition 22.10 (Conjugate priors for a canonical family)**.** *For a canonical family with dispersion ϕ, let the prior density of θ be*

$$\pi(\theta) \propto \exp\Big(\frac{m_0\theta - n_0\, b(\theta)}{\phi}\Big), \qquad \theta \in \Theta,$$

for numbers m_0 and $n_0 > 0$ that make it integrable. Then after n i.i.d. observations with sum $\sum_i Y_i$, the posterior is of the same form with (m_0, n_0) replaced by $(m_0 + \sum_i Y_i, n_0 + n)$.

Proof strategy (for completeness; skip on a first reading). Prior times likelihood: both are exponentials of a linear function of θ minus a multiple of $b(\theta)$, and exponents add.

Proof. $\pi(\theta)L_n(\mathbf{y};\theta) \propto \exp\big((m_0\theta - n_0 b(\theta))/\phi\big)\exp\big((\theta\sum_i y_i - n b(\theta))/\phi\big) = \exp\big(((m_0 + \sum_i y_i)\theta - (n_0 + n)b(\theta))/\phi\big)$, the prior's form with the new numbers. □

The prior behaves like n_0 earlier observations whose sum was m_0, which is the 'imaginary couples' reading of Chapter 18 made general. For the Bernoulli, with θ the log-odds and $b(\theta) = \log(1 + e^\theta)$, the prior $\exp(m_0\theta - n_0\log(1+e^\theta)) = p^{m_0}(1-p)^{n_0 - m_0}$ on the scale of θ is, after the change of variables $\mathrm{d}\theta = \mathrm{d}p/(p(1-p))$ of Chapter 19, the density $p^{m_0 - 1}(1-p)^{n_0 - m_0 - 1}$ on the scale of p: the $\mathrm{Beta}(m_0, n_0 - m_0)$ prior, updated to $\mathrm{Beta}(m_0 + S, n_0 - m_0 + n - S)$, which is Theorem 18.3 with $a = m_0$ and $b = n_0 - m_0$. The Gamma prior for a Poisson mean and for an exponential rate, and the Gaussian prior for a Gaussian mean, are the same proposition in the other rows of the table (Exercise 22.8).

22.6 What comes next

A canonical family has one parameter and n observations that share it. Chapter 23 lets each observation have its own θ_i, tied to that observation's predictors by $g(\mu_i) = x_i^\top \beta$ for a link g, which is the generalized linear model: the Gaussian family with the identity link gives back Chapter 21, the Bernoulli family with its canonical link, the log-odds, gives logistic regression, and the Poisson family with the log link gives regression for counts. The log-likelihood in β stays concave under the canonical link, the maximum likelihood estimator has no closed form but can be found by Newton's method, and the large-sample theory of Chapter 12, verified here for canonical families, carries it through to tests and intervals.

22.7 Summary

- A k-parameter exponential family has densities $h(y)\exp(\eta(\theta)^\top T(y) - B(\theta))$: the base weight h tilted by an exponential, with a fixed support. Bernoulli, Poisson, exponential, Gaussian, Gamma and Beta are; the uniform (moving support), Laplace with unknown center, t and Cauchy are not.
- Canonical form: $\exp\big((y\theta - b(\theta))/\phi + c(y,\phi)\big)$, with θ the canonical parameter and ϕ the known dispersion. Kiss: θ is the log-odds and $b = \log(1+e^\theta)$; Poisson: $\log\lambda$, e^θ; exponential: $-\lambda$, $-\log(-\theta)$; Gaussian with σ^2 known: μ, $\theta^2/2$, $\phi = \sigma^2$.
- $\mathbb{E}[Y] = b'(\theta)$, $\mathbb{V}\mathrm{ar}(Y) = \phi\, b''(\theta)$, so b is convex and b' strictly increasing; the variance function $V(\mu) = b''((b')^{-1}(\mu))$ is 1, μ, μ^2, $\mu(1-\mu)$ for the four families; the canonical link is $g_c = (b')^{-1}$, the log-odds for the Bernoulli.
- Canonical families are regular. For n observations ℓ_n is strictly concave, the MLE solves $b'(\hat\theta_n) = \bar Y_n$ (so it is the method of moments), $\bar Y_n$ attains the Cramér–Rao bound for the mean, $I(\theta) = b''(\theta)/\phi$, and $\sqrt{n}(\hat\theta_n - \theta) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, \phi/b''(\theta))$. Kiss: $\hat\theta = 0.598$, standard error 0.188, odds interval $[1.26, 2.63]$.
- Factorization theorem: T is sufficient exactly when $f_\theta(\mathbf{y}) = g_\theta(T(\mathbf{y}))k(\mathbf{y})$. In an exponential family $\sum_i T(Y_i)$ is sufficient, of dimension k for every n; Darmois, Koopman and Pitman: only exponential families have that (stated). Rao and Blackwell: conditioning an unbiased estimator on a sufficient statistic never raises its variance.
- The prior $\exp((m_0\theta - n_0 b(\theta))/\phi)$ is conjugate for every canonical family, updating to $(m_0 + \sum_i Y_i, n_0 + n)$: the Beta, Gamma and Gaussian conjugates of Chapter 18 are its rows.

22.8 Exercises

Exercise 22.1. (a) Write the Gamma(α, β) density with α known in canonical form and check the table's row: $\theta = -\beta/\alpha$, $b(\theta) = -\log(-\theta)$, $\phi = 1/\alpha$. (b) Use Theorem 22.3 to find the mean and variance, and the variance function. (c) Which row of the table is the case $\alpha = 1$? *(Solution on page 437.)*

Exercise 22.2. (a) Show that Bin(m, p) with m known is a canonical family with θ the log-odds, $b(\theta) = m\log(1+e^\theta)$ and $c(y) = \log\binom{m}{y}$, and recover its mean and variance from b. (b) Show that the geometric family $\mathbb{P}(Y = y) = (1-p)^y p$, $y = 0, 1, 2, \ldots$, is canonical with $\theta = \log(1-p)$ and $b(\theta) = -\log(1-e^\theta)$, and compute $\mathbb{E}[Y]$ from b'. (c) Show that the negative binomial with known

size r, $\mathbb{P}(Y=y)=\binom{y+r-1}{y}(1-p)^y p^r$, is canonical with the same θ and $b(\theta)=-r\log(1-e^\theta)$. *(Solution on page 438.)*

Exercise 22.3. Show that the Beta family Beta(a,b), with density $p^{a-1}(1-p)^{b-1}/B(a,b)$ on $(0,1)$, is a two-parameter exponential family, and name η, T, B and h. *(Solution on page 438.)*

Exercise 22.4. Show that the Laplace family with densities $\frac{1}{2}e^{-|y-\mu|}$, $\mu\in\mathbb{R}$, is not a one-parameter exponential family. (Suppose $|y-\mu|=\eta(\mu)T(y)+B(\mu)+a(y)$ for all y and μ, and look at what the equation says for y far to the right of two fixed centers.) *(Solution on page 438.)*

Exercise 22.5. The variance function determines b, and with it the mean, the variance and the canonical link. (a) Show that if a canonical family with $\phi=1$ has $V(\mu)=\mu$ on $(0,\infty)$, then $b''=b'$, so $b'(\theta)=Ce^\theta$, and with the normalization $C=1$ this is the Poisson row's b. (b) Which rows of the table have $V(\mu)=\mu(1-\mu)$ and $V(\mu)=\mu^2$? *(Solution on page 438.)*

Exercise 22.6. For the kiss data, the sample proportion attains the Cramér–Rao bound for p (Chapter 12). (a) Show from Theorem 22.7 that the same holds for the Poisson mean, the exponential mean and the Gaussian mean with known variance. (b) The MLE of the exponential rate, $\hat\lambda=1/\bar T_n$, has expectation $\frac{n}{n-1}\lambda$ (Chapter 7). Explain why neither it nor its unbiased correction $(n-1)/\sum_i T_i$ attains the bound for λ at any finite n, and why this does not contradict (a). (The bound is attained by an unbiased W exactly when the score is $a(\theta)(W-\tau(\theta))$ for some a, the equality case of the Cauchy–Schwarz step in its proof.) *(Solution on page 438.)*

Exercise 22.7. By the factorization theorem, find a sufficient statistic for (a) θ in Unif$([\theta,\theta+1])$; (b) (α,β) in the Gamma family; (c) λ in the Poisson family, and show that $\bar Y_n$ and $\sum_i Y_i$ are each sufficient while Y_1 alone is not. *(Solution on page 438.)*

Exercise 22.8. Apply Proposition 22.10 to (a) the Poisson family, showing that the conjugate prior on the scale of $\lambda=e^\theta$ is Gamma(m_0,n_0) and the posterior Gamma$(m_0+\sum_i Y_i, n_0+n)$; (b) the Gaussian family with σ^2 known, showing that the prior is Gauss$(m_0/n_0,\sigma^2/n_0)$ and the posterior mean is $(m_0+\sum_i Y_i)/(n_0+n)$, the precision-weighted average of Chapter 18; (c) the Expo(λ) family, showing that the conjugate prior on the scale of $\lambda=-\theta$ is Gamma(n_0+1,m_0) and saying what α and β in Chapter 18's Gamma(α,β) prior for the subway rate count as. *(Solution on page 439.)*

Exercise 22.9. For the siblings data modeled as $Y_i=X_i-1\sim\text{Pois}(\lambda)$, the statistic $W=\mathbb{1}(Y_1=0)$ is unbiased for $e^{-\lambda}$, the probability of having no siblings. (a) Show that $\mathbb{E}[W\mid S]=(1-1/n)^S$ for $S=\sum_i Y_i$. (Given $S=s$, $Y_1\sim\text{Bin}(s,1/n)$.) (b) Compute both estimates for the 24 values with $S=24$, and compare with the plug-in $e^{-\hat\lambda}$. *(Solution on page 439.)*

Exercise 22.10. Figure 22.4 drew the kiss log-likelihood on two scales. (a) Compute the third derivative of ℓ_n in θ and in p at the maximum, for $n=124$ and $\hat p=0.645$, and explain which scale the parabola fits better. (b) Wald's test of Chapter 15 in θ uses $\hat\theta^2/\widehat{\mathbb{V}\text{ar}}(\hat\theta)$; compute it for $H_0:\theta=0$ and compare with Wald's statistic 11.41 in p. *(Solution on page 439.)*

Chapter 23

Generalized Linear Models: Link Functions and Fitting

In this chapter

- The generalized linear model: each response Y_i drawn from a canonical exponential family whose mean μ_i is tied to the predictors by a link, $g(\mu_i) = x_i^\top \beta$. Why a straight line cannot serve for a 0/1 or a count response.
- Link functions: the logit and probit for a proportion, the logarithm for a count, and the canonical link of Chapter 22, under which the log-likelihood is strictly concave in β and the maximum likelihood estimator is unique whenever it exists.
- Fitting by Newton's method, which has a second reading as repeated weighted least squares; when the estimator does not exist (separation).
- Inference from the asymptotic normality of the MLE: standard errors, Wald and likelihood ratio tests. Logistic regression on the kyphosis data, and the aspirin trial as a logistic regression with one 0/1 predictor.

Chapter 22 described the families and found, in each, a scale on which the parameter enters the likelihood linearly. This chapter puts the linear predictor $x^\top \beta$ of Chapter 21 on that scale.

23.1 A response that is not Gaussian

A data set published in Chambers and Hastie's *Statistical Models in S* (1992) records 81 children who had corrective spinal surgery. For each child the record says whether **kyphosis**, a forward curvature of the spine, was present after the operation (17 of the 81), the child's age in months, the number of vertebrae involved, and the **start**, the number of the topmost vertebra operated on (counted from the top of the spine, so a larger start means an operation lower down). The question is how the chance of kyphosis depends on the three.

The response is 0 or 1, so there is no choice of family: given the predictors, $Y_i \sim \text{Bern}(\mu(x_i))$ with $\mu(x) = \mathbb{E}[Y \mid X = x] = \mathbb{P}(Y = 1 \mid X = x)$. What Chapter 21 would do next, $\mu(x) = x^\top \beta$, does not work here. Figure 23.1 fits a line to kyphosis against the start: it reaches 0.64 at start 0 and goes below 0 past start 17, where one child's start is 18. A probability cannot be a line in x over any range of x wide enough to matter, and a count's mean cannot be either, since it must stay positive.

The second trouble is the variance: $\mathbb{V}\mathrm{ar}(Y_i \mid x_i) = \mu(x_i)(1-\mu(x_i))$ moves with the mean, which the Gaussian linear model's constant σ^2 cannot represent. Chapter 22 handled both at once: in a canonical family the variance follows the mean through the variance function, and the mean is a strictly increasing function b' of the canonical parameter, which ranges over an open interval, the whole line for the Bernoulli, Poisson and Gaussian families.

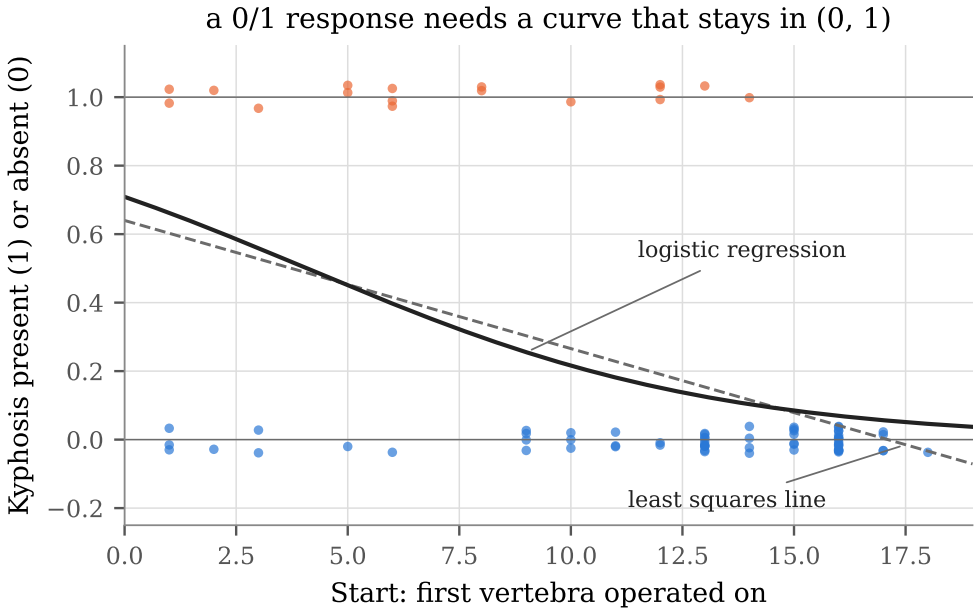


Figure 23.1: Kyphosis (present = 1) against the start vertebra for 81 children, jittered vertically; the least squares line (gray, dashed) and the logistic regression curve $1/(1+e^{-(0.890-0.218x)})$ (black).

Definition 23.1 (Generalized linear model)**.** Let $(x_1, Y_1), \ldots, (x_n, Y_n)$ be independent pairs with $x_i \in \mathbb{R}^p$. A **generalized linear model** has two parts:

1. the **random component**: given x_i, Y_i has a density or PMF in a canonical exponential family, $f_{\theta_i}(y) = \exp\big((y\theta_i - b(\theta_i))/\phi + c(y,\phi)\big)$, the same b and ϕ for every i, with mean $\mu_i = b'(\theta_i)$;
2. the **link**: $g(\mu_i) = x_i^\top \beta$ for a known function g, the **link function**, one-to-one and differentiable on the range of the means, and an unknown $\beta \in \mathbb{R}^p$.

$x_i^\top \beta$ is the **linear predictor**.

The predictors still enter through one linear combination $x^\top\beta$, as in Chapter 21; what changes is that the combination is not the mean but a transformed mean, $g(\mu)$, chosen so that the transformed mean can range over an interval, usually the whole line. The family fixes how the variance follows the mean, and the link fixes how the mean follows the predictors. The Gaussian family with the identity link $g(\mu) = \mu$ is Chapter 21's model exactly.

For the kiss data. With no predictors at all, $x_i = 1$ for every couple and β is one number: the model says $g(p) = \beta$. With the Bernoulli family and g the log-odds, β is the canonical parameter $\theta = \log\big(p/(1-p)\big)$ of Chapter 22, and the fit will be $\hat\beta = \log(80/44) = 0.598$: logistic regression with an intercept alone is the kiss model on the log-odds scale.

The linear model of Chapter 21 treated the predictors as fixed numbers. The definition does the same, writing x_i in lower case and conditioning on it; when the pairs are random draws, as the

kyphosis children are, the conditional model is what the likelihood sees, since the distribution of x does not involve β (Section 23.5).

23.2 Link functions

A link's usual job is to carry the range of the means onto the whole line, so that no value of $x^\top\beta$ is forbidden. For a Gaussian mean the range is already $\mathbb{R}$ and the identity link is natural. For a Poisson or exponential mean the range is $(0,\infty)$, and the logarithm is the obvious choice: $\log\mu = x^\top\beta$, that is, $\mu = e^{x^\top\beta}$, under which each predictor acts multiplicatively on the mean, a unit increase in x_j multiplying it by e^{β_j}. For a Bernoulli mean the range is $(0,1)$, and two links are in common use (Figure 23.2):

- the **logit**, $g(\mu) = \log\big(\mu/(1-\mu)\big)$, the log-odds, with inverse the **logistic function** $\mu = e^t/(1+e^t) = 1/(1+e^{-t})$;
- the **probit**, $g(\mu) = \Phi^{-1}(\mu)$, with Φ the standard Gaussian CDF of Chapter 6, so that the inverse link is $\mu = \Phi(t)$.

The two inverses have the same shape, a sigmoid from 0 to 1, and differ mainly in scale: $\Phi(t/1.7)$ and the logistic function are within 0.01 of each other everywhere, so the two models fit almost the same curve with coefficients that differ by about the factor 1.7. The logit has the advantage that its coefficients are log odds ratios, which Chapter 5 has already made familiar; the probit has a reading in terms of a hidden Gaussian variable (Exercise 23.4).

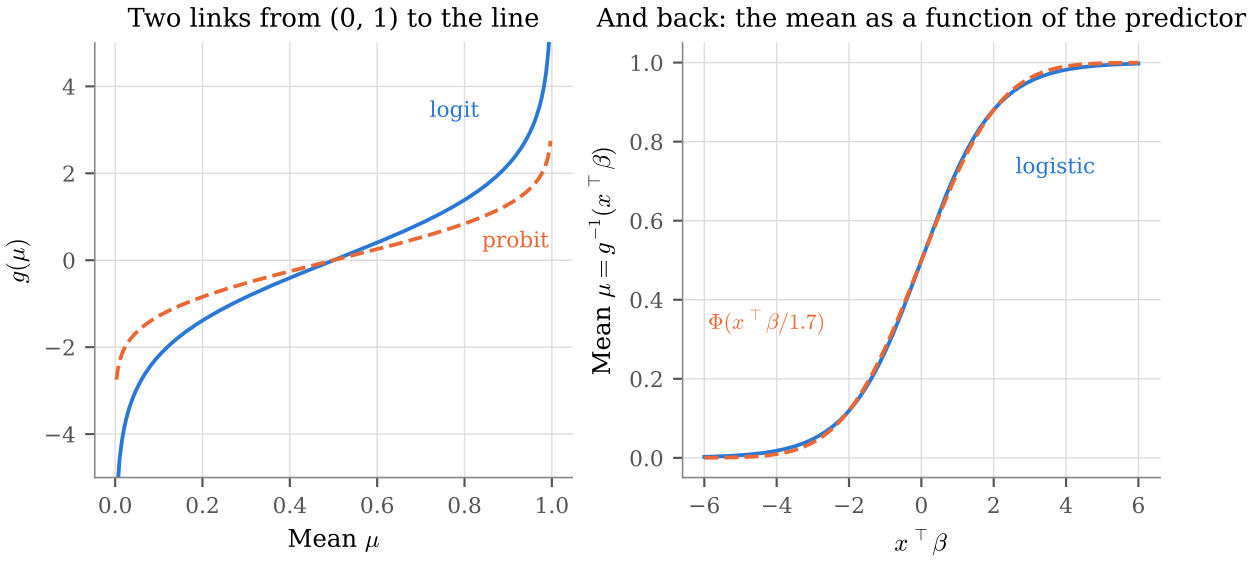


Figure 23.2: Left: the logit (blue) and probit (orange, dashed) links on $(0,1)$. Right: their inverses, the logistic function and the Gaussian CDF with its scale matched, $\Phi(t/1.7)$, as functions of the linear predictor.

Among all links one is singled out by the family.

Definition 23.2 (Canonical link, with predictors)**.** The **canonical link** of a generalized linear model is $g_c = (b')^{-1}$ (Definition 22.5), under which the linear predictor is the canonical parameter itself: $\theta_i = x_i^\top\beta$. Under any other link, $\theta_i = h(x_i^\top\beta)$ with $h = (g\circ b')^{-1}$.

Every link ties the mean to the predictors; the canonical link ties the predictors to the scale on which the family's likelihood is linear in the parameter, so the data meet β only through $Y_i\, x_i^\top \beta$. That is what makes the next section's theorem work. The identity is canonical for the Gaussian, the logarithm for the Poisson, $-1/\mu$ for the exponential and Gamma, and the logit for the Bernoulli: the logarithm and the logit are both the natural choice and the canonical one. For the exponential and Gamma the canonical link reaches only the negative half-line, so $x^\top\beta$ must stay negative, and the logarithm is the usual choice there instead.

For the kiss data. Bernoulli with the logit link is **logistic regression**, and with $x_i = 1$ it is the kiss model with $\theta = \beta$. With a predictor, say $x_i = (1, \text{age of the couple})$, it says the log-odds of turning right is a line in age, $\theta_i = \beta_1 + \beta_2\,\text{age}_i$.

23.3 The likelihood and its shape

Since the pairs are independent, the log-likelihood is a sum over observations, each in Chapter 22's form:

$$\ell_n(\beta) = \sum_{i=1}^{n} \frac{Y_i\theta_i - b(\theta_i)}{\phi} + \sum_{i=1}^{n} c(Y_i, \phi), \qquad \theta_i = h(x_i^\top\beta).$$

Under the canonical link $\theta_i = x_i^\top\beta$ and, dropping the last sum, which does not involve β,

$$\ell_n(\beta) = \frac{1}{\phi}\sum_{i=1}^{n}\big(Y_i\, x_i^\top\beta - b(x_i^\top\beta)\big) = \frac{1}{\phi}\Big(Y^\top X\beta - \sum_{i=1}^{n} b(x_i^\top\beta)\Big),$$

with X the $n \times p$ design matrix of Chapter 21 and $Y = (Y_1, \dots, Y_n)^\top$: linear in β in the first term, with the data entering only through $X^\top Y$, and concave in the second.

Theorem 23.3 (Concavity under the canonical link)**.** *In a generalized linear model with the canonical link, $b'' > 0$, and X of full rank p:*

1. $\nabla\ell_n(\beta) = \frac{1}{\phi}X^\top\big(Y - \mu(\beta)\big)$*, where $\mu(\beta)$ is the vector with entries $\mu_i = b'(x_i^\top\beta)$;*
2. $\nabla^2\ell_n(\beta) = -\frac{1}{\phi}X^\top W(\beta)X$*, where $W(\beta)$ is the diagonal matrix with entries $w_i = b''(x_i^\top\beta) = V(\mu_i)$, and this matrix is negative definite for every β;*
3. *ℓ_n is strictly concave on the convex open set of β with every $x_i^\top\beta \in \Theta$ (all of $\mathbb{R}^p$ for the Bernoulli, Poisson and Gaussian families), so it has at most one maximizer: the maximum likelihood estimator $\hat\beta$ is unique whenever it exists, and it is the unique solution of the* ***likelihood equations*** *$X^\top\big(Y - \mu(\beta)\big) = 0$.*

Proof strategy (core). Differentiate the two terms: the linear term's gradient is $X^\top Y/\phi$, the sum of b's has gradient $\sum_i b'(x_i^\top\beta)x_i/\phi$ and Hessian $\sum_i b''(x_i^\top\beta)x_ix_i^\top/\phi$. The Hessian is a weighted $X^\top X$ with positive weights, negative definite because X has full rank, and a function with a negative definite Hessian everywhere is strictly concave.

Proof. For (1), $\partial(Y_ix_i^\top\beta)/\partial\beta = Y_ix_i$ and $\partial b(x_i^\top\beta)/\partial\beta = b'(x_i^\top\beta)x_i$ by the chain rule (toolkit A.12), so $\nabla\ell_n(\beta) = \frac{1}{\phi}\sum_i(Y_i - b'(x_i^\top\beta))x_i = \frac{1}{\phi}X^\top(Y - \mu(\beta))$. Differentiating once more,

$$\nabla^2\ell_n(\beta) = -\frac{1}{\phi}\sum_i b''(x_i^\top\beta)x_ix_i^\top = -\frac{1}{\phi}X^\top WX,$$

which is (2)'s formula. For $a \in \mathbb{R}^p$, $a^\top X^\top W X a = \sum_i w_i (x_i^\top a)^2 \geq 0$, with equality only if $x_i^\top a = 0$ for every i, that is, $Xa = 0$, which full rank rules out for $a \neq 0$ (toolkit A.16); so $X^\top W X$ is positive definite and the Hessian negative definite. For (3), take $\beta_1 \neq \beta_2$ and $r(t) = \ell_n(\beta_1 + t(\beta_2 - \beta_1))$ on $[0,1]$: $r''(t) = (\beta_2 - \beta_1)^\top \nabla^2 \ell_n(\cdot)(\beta_2 - \beta_1) < 0$, so r is strictly concave in one variable, which is strict concavity of ℓ_n along every segment (toolkit A.12). A strictly concave function cannot have two maximizers (the value at their midpoint would exceed their common value, which is the maximum), and a maximizer on that open set is a stationary point, so it solves $\nabla \ell_n = 0$; conversely a stationary point of a concave function is a global maximizer (Proposition 11.5 applied along the segment from the stationary point) (toolkit A.12), so the solution of the likelihood equations, if there is one, is $\hat{\beta}$. □

The likelihood equations say the fitted means balance the data along every predictor: $\sum_i x_{ij} Y_i = \sum_i x_{ij} \hat{\mu}_i$ for each column j. With an intercept column of ones, the fitted means sum to the observed total, $\sum_i \hat{\mu}_i = \sum_i Y_i$: a logistic regression's fitted probabilities add up to the number of ones, and a Poisson regression's fitted means to the total count. For the Gaussian family with the identity link, $b(\theta) = \theta^2/2$, $\mu(\beta) = X\beta$ and the equations are the normal equations $X^\top(Y - X\beta) = 0$ of Theorem 21.3; the theorem contains least squares as the one case where the equations are linear and solve in closed form.

Existence is the part the theorem leaves open, and it can fail.

Pitfall: separation: a maximum that is not there

Figure 23.3 shows six observations with $y = 0$ at $x = 1, 2, 3$ and $y = 1$ at $x = 4, 5, 6$, and the logistic curves $1/(1 + e^{-b(x-3.5)})$ for several slopes. Every curve classifies the data perfectly, and the steeper the curve the higher the likelihood: the log-likelihood rises toward its supremum 0, the value of a perfect fit, and never reaches it. The MLE does not exist, and Newton's method runs off toward infinite coefficients. This is Chapter 22's all-1s sample with predictors: whenever some hyperplane $x^\top a = c$ puts every $y = 1$ on one side and every $y = 0$ on the other (**complete separation**), the same happens. It is common in small samples with many predictors, and a fit with huge coefficients and huge standard errors is the sign of it; the remedy is fewer predictors, more data, or a prior on β (the ridge of Chapter 21 on the log-odds scale).

23.4 Fitting by Newton's method

Outside the Gaussian case the likelihood equations $X^\top(Y - \mu(\beta)) = 0$ are nonlinear in β, since $\mu_i = b'(x_i^\top \beta)$, and no formula solves them. Newton's method (toolkit A.17) does: from a current β_k, replace ℓ_n by the quadratic with the same gradient and Hessian at β_k and move to its top,

$$\beta_{k+1} = \beta_k - \big(\nabla^2 \ell_n(\beta_k)\big)^{-1} \nabla \ell_n(\beta_k) = \beta_k + \big(X^\top W_k X\big)^{-1} X^\top \big(Y - \mu_k\big),$$

with $W_k = W(\beta_k)$ and $\mu_k = \mu(\beta_k)$; the dispersion ϕ cancels. Because the Hessian is negative definite everywhere, each quadratic has a top and each step is defined, and once the iterate is near $\hat{\beta}$ the error is squared at every step. Figure 23.4 draws the method on the kiss data in one dimension: from $\theta_0 = 0$ the fitted parabola peaks at $\theta_1 = 0 + (80 - 62)/31 = 0.581$, the next at 0.598, and the third step changes nothing in the fourth decimal.

The step has a second reading. Write $z_k = X\beta_k + W_k^{-1}(Y - \mu_k)$, the **working response**: the current linear predictor, corrected by the residual $Y_i - \mu_{k,i}$ divided by the slope $w_{k,i} = \mathrm{d}\mu_i/\mathrm{d}\theta_i$ of

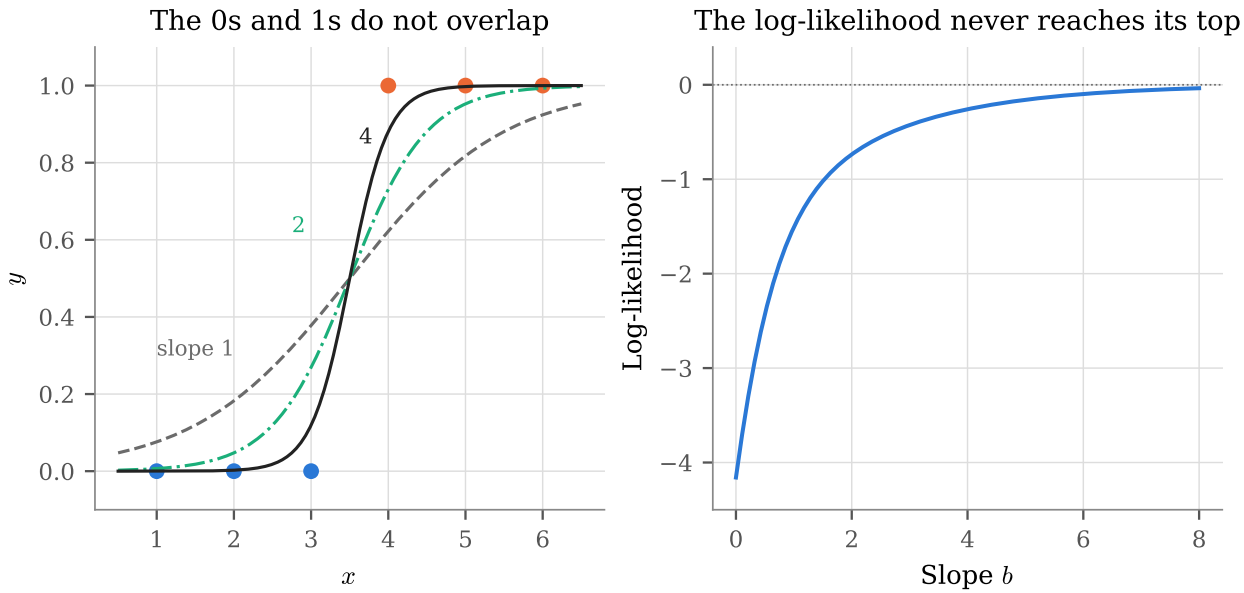


Figure 23.3: Complete separation. Left: six observations and the logistic curves with slopes 1 (gray, dashed), 2 (aqua, dash-dot) and 4 (black). Right: the log-likelihood of the slope, with the intercept fixed at $-3.5b$, climbing toward 0 without a maximum.

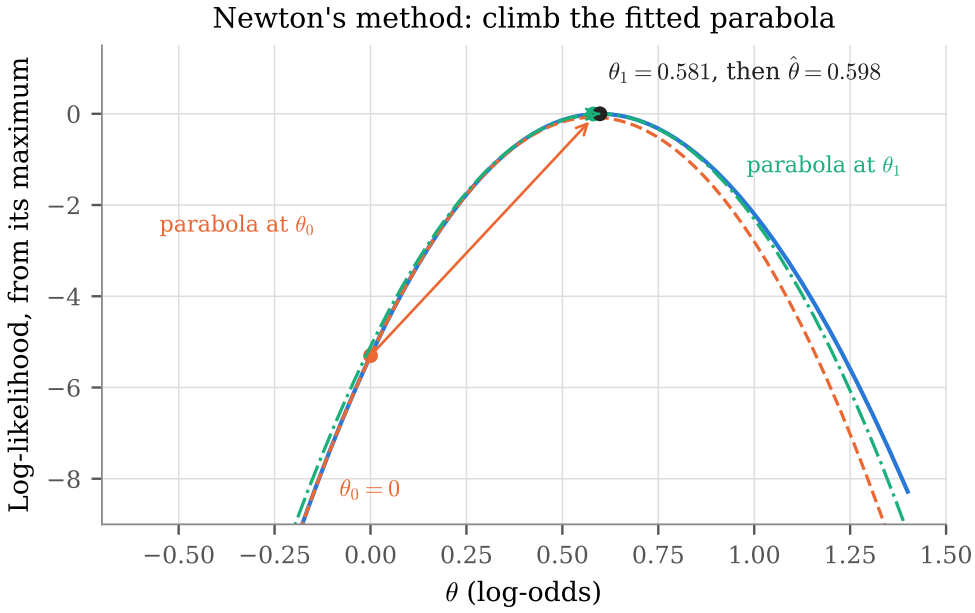


Figure 23.4: Newton's method on the kiss log-likelihood in the log-odds, from $\theta_0 = 0$: the parabola fitted at θ_0 (orange, dashed) peaks at $\theta_1 = 0.581$, the one fitted there (aqua, dash-dot) at 0.598.

the mean, that is, the residual carried from the scale of Y to the scale of the linear predictor. Then

$$\beta_{k+1} = \left(X^\top W_k X\right)^{-1} X^\top W_k z_k,$$

the weighted least squares fit of z_k on X with weights $w_{k,i}$ (Exercise 23.7): Newton's method for a generalized linear model with the canonical link is **iteratively reweighted least squares**. Each round fits a linear regression to a response that stands in for Y on the predictor's scale, weighting

the observations by the family's variance function at the current fit, and refits until the weights stop changing. For the Gaussian with the identity link, $w_i = 1$, $z_k = Y$, and the first step from any start lands on the least squares estimate.

Worked example: fitting the kyphosis data

With $x_i = (1, \text{age}_i, \text{number}_i, \text{start}_i)$, the Bernoulli family and the logit link, Newton's method from $\beta_0 = 0$ (every fitted probability $\frac{1}{2}$) gives

step	ℓ_n	intercept	age	number	start
0	-56.14	0	0	0	0
1	-33.21	-0.955	0.0043	0.210	-0.123
2	-30.93	-1.583	0.0082	0.334	-0.180
3	-30.69	-1.960	0.0105	0.398	-0.203
4	-30.69	-2.035	0.0109	0.410	-0.206
5	-30.69	-2.037	0.0109	0.411	-0.207

and a sixth step changes nothing in the fourth decimal: $\hat\beta = (-2.037, 0.0109, 0.411, -0.207)$ with $\ell_n(\hat\beta) = -30.69$, against -41.62 for the intercept-only fit $\hat\beta_1 = \log(17/64) = -1.326$. The fitted probabilities sum to 17. Read on the odds scale: each month of age multiplies the odds of kyphosis by $e^{0.0109} = 1.011$ (by 1.12 per ten months), each additional vertebra by $e^{0.411} = 1.51$, and each vertebra further down the spine for the start by $e^{-0.207} = 0.81$. For a child of 100 months with 4 vertebrae involved, the fitted probability is 0.42 when the operation starts at vertebra 5 and 0.08 when it starts at 15.

23.5 Inference

The MLE has no formula, but its large-sample distribution does, and it is Chapter 12's. To apply that chapter the observations must be i.i.d., so take the pairs (x_i, Y_i) as i.i.d. draws of (x, Y) with the conditional law of Y given x the generalized linear model's and the law of x not involving β; the joint density of a pair is $f_{h(x^\top\beta)}(y)\, q(x)$, and $\log q(x)$ drops out of every derivative in β, so the log-likelihood of the pairs is the $\ell_n(\beta)$ above plus a constant.

Theorem 23.4 (Asymptotic normality in a generalized linear model)**.** *Let (x_i, Y_i) be i.i.d. pairs following a generalized linear model with the canonical link, $b'' > 0$, $\|x\| \le C$ with probability one for some constant C, $x^\top\beta^*$ in a closed bounded subinterval of Θ with probability one, and $M = \mathbb{E}[b''(x^\top\beta^*)\, xx^\top]$ positive definite at the true β^*. Then the Fisher information of one pair is $I(\beta^*) = M/\phi$, and the MLE satisfies*

$$\sqrt{n}\,(\hat\beta - \beta^*) \xrightarrow[n\to\infty]{(d)} \mathit{Gauss}_p\big(0, \phi M^{-1}\big).$$

In practice: $\hat\beta$ is approximately $\mathit{Gauss}_p\big(\beta^*, \phi(X^\top\hat W X)^{-1}\big)$ with $\hat W = W(\hat\beta)$, and the standard error of $\hat\beta_j$ is the square root of the jth diagonal entry of $\phi(X^\top\hat W X)^{-1}$.

Proof strategy (for completeness; skip on a first reading). The model in β is regular: the density is a canonical family's at $\theta = x^\top\beta$, so Chapter 22's verification of the conditions goes through with x along for the ride, and the information is minus the expected Hessian, $\mathbb{E}[b''(x^\top\beta)xx^\top]/\phi$, which does not even need averaging over Y. Theorem 12.5 in p dimensions does the rest.

Recall. Asymptotic normality of the MLE (Theorem 12.5), d-dimensional form: for i.i.d. observations from a regular model with θ^* interior and $I(\theta^*)$ invertible, $\sqrt{n}(\hat\theta_n - \theta^*) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_d(0, I(\theta^*)^{-1})$. The information as minus the expected Hessian (Theorem 12.4). The Law of Large Numbers (Theorem 3.5): for i.i.d. Z_i with $\mathbb{V}\text{ar}(Z_i) < \infty$, $\frac{1}{n}\sum_i Z_i \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[Z_i]$, applied to each entry of $\frac{1}{n}X^\top\hat W X$. The continuous mapping theorem (Theorem 5.2): if $Z_n \xrightarrow[n\to\infty]{\mathbb{P}} z$ and f is continuous, then $f(Z_n) \xrightarrow[n\to\infty]{\mathbb{P}} f(z)$, applied to matrix inversion, which is continuous on the invertible matrices, an open set containing M.

Proof. The log-likelihood of one pair is $\ell(\beta) = (Yx^\top\beta - b(x^\top\beta))/\phi + c(Y,\phi) + \log q(x)$, with gradient $(Y - b'(x^\top\beta))x/\phi$ and Hessian $-b''(x^\top\beta)xx^\top/\phi$, free of Y. The regularity conditions of Section 12.6 hold: identifiability, since equal conditional densities force $x^\top\beta_1 = x^\top\beta_2$ with probability one, hence $(\beta_1 - \beta_2)^\top \mathbb{E}[xx^\top](\beta_1 - \beta_2) = 0$, and $\mathbb{E}[xx^\top]$ is positive definite (if $a^\top \mathbb{E}[xx^\top]a = 0$ then $x^\top a = 0$ almost surely and $a^\top M a = 0$, against the assumption on M); fixed support and an open parameter set, $\mathbb{R}^p$ when $\Theta = \mathbb{R}$ and otherwise the open set of β with $x^\top\beta \in \Theta$ with probability one; derivatives under the integral by Proposition 22.6 applied conditionally on x; and the moment conditions, since x is bounded, so for β near β^* the values $x^\top\beta$ stay in a bounded interval on which b'' and b''' are bounded: $\nabla^2\ell(\beta^*) = -b''(x^\top\beta^*)xx^\top/\phi$ has finite variance and $\nabla^3\ell$ is bounded by a constant, which serves as the K the conditions ask for; the same bounds give the finite variances the Law of Large Numbers needs below; and $I(\beta^*) = -\mathbb{E}[\nabla^2\ell(\beta^*)] = \mathbb{E}[b''(x^\top\beta^*)xx^\top]/\phi = M/\phi$, invertible. Theorem 12.5 gives $\sqrt{n}(\hat\beta - \beta^*) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_p(0, I(\beta^*)^{-1}) = \text{Gauss}_p(0, \phi M^{-1})$. For the plug-in, $\frac{1}{n}X^\top\hat W X = \frac{1}{n}\sum_i b''(x_i^\top\hat\beta)x_i x_i^\top \xrightarrow[n\to\infty]{\mathbb{P}} M$ by the Law of Large Numbers at β^*, the consistency of $\hat\beta$ and the continuity of b'' (the uniform-law step of Theorem 12.5's proof, for the Hessian at a nearby point), and inversion is continuous at M, so $n\,\phi(X^\top\hat W X)^{-1} \xrightarrow[n\to\infty]{\mathbb{P}} \phi M^{-1}$: the covariance $\phi(X^\top\hat W X)^{-1}$ is the limit's, divided by n and estimated. □

The matrix $X^\top\hat W X/\phi$ is minus the Hessian at the fit, the curvature of the log-likelihood at its top, as in Chapter 12; it is the last matrix Newton's method inverted, so the standard errors come free with the fit. With the covariance in hand, the tests of Chapter 15 apply as they stand.

- **Wald.** For one coefficient, $Z_j = \hat\beta_j/\text{SE}_j$ is approximately standard Gaussian under $\beta_j^* = 0$, and $\hat\beta_j \pm 1.96\,\text{SE}_j$ is a 95% interval; exponentiated, it is an interval for the odds ratio or the rate ratio per unit of x_j. For k coefficients at once, the quadratic form of Definition 15.11 against χ^2_k.
- **Likelihood ratio.** For nested models, twice the gap between the maximized log-likelihoods is approximately χ^2_k under the smaller model, k the number of coefficients set to zero (Theorem 15.12). Twice the gap between a model's maximized log-likelihood and that of the **saturated** model, which gives each observation its own mean, is called the model's **deviance**; the likelihood ratio statistic for nested models is the difference of their deviances (when $\phi \neq 1$ the deviance is usually defined as ϕ times this).

The two agree in large samples and can differ in small ones, as Chapter 15 found for a proportion, and the likelihood ratio test is the one to prefer when both fits are available. The t and F distributions

of Chapter 21 do not return: they came from a Gaussian error with an estimated σ^2, and here the dispersion is known ($\phi = 1$ for a proportion or a count) and the exactness is gone.

Worked example: what the kyphosis fit says

Standard errors from $(X^\top \hat{W} X)^{-1}$: 1.450, 0.0064, 0.225, 0.068, so $Z = -1.41$, 1.70, 1.83, -3.05 for the intercept, age, number and start. The start is the clear predictor, p-value 0.002; age and number are each short of 1.96. The Wald interval for the start coefficient is $-0.207 \pm 1.96 \times 0.068 = [-0.339, -0.074]$, an odds ratio per vertebra between 0.71 and 0.93. The likelihood ratio test of all three predictors against the intercept alone is $2(-30.69 + 41.62) = 21.85$ on 3 degrees of freedom, p-value 7×10^{-5}; of the start alone, refitting without it, $2(-30.69 + 35.81) = 10.25$ on 1 degree of freedom, against Wald's $Z^2 = 9.30$: same verdict, different number, as expected in a sample of 81 with 17 ones. Dropping age and number together costs $2(-30.69 + 34.04) = 6.69$ on 2 degrees of freedom, p-value 0.035: the pair matters at the 5% level even though neither does on its own, the situation of Chapter 21's Old Faithful fit.

A logistic regression with one 0/1 predictor is a case the book has met before, under another name.

From undergraduate statistics: the two-proportion test as a regression

The aspirin trial of Chapter 15 is a logistic regression with one 0/1 predictor: $x_i = (1, a_i)$ with $a_i = 1$ for aspirin, so the log-odds of a heart attack is β_1 on placebo and $\beta_1 + \beta_2$ on aspirin, and β_2 is the log odds ratio. With two groups and two parameters the model is saturated, the likelihood equations say each group's fitted probability is its observed proportion, and the fit is exact. On placebo 189 had a heart attack and 10,845 did not; on aspirin 104 and 10,933. So $\hat{\beta}_1 = \log(189/10{,}845) = -4.050$, $\hat{\beta}_2 = \log\left(\frac{104/10{,}933}{189/10{,}845}\right) = -0.605$, an odds ratio of 0.55. The standard error of $\hat{\beta}_2$ from $(X^\top \hat{W} X)^{-1}$ is $\sqrt{1/104 + 1/10{,}933 + 1/189 + 1/10{,}845} = 0.123$, the four-reciprocals formula of Chapter 5, so $Z = -4.93$, $Z^2 = 24.3$, and the 95% interval for the odds ratio is $[0.43, 0.69]$. The likelihood ratio statistic against $\beta_2 = 0$ is 25.37, the same T^{LR} as in Chapter 15, since the two models compared are the same two. Every two-group comparison of proportions is a logistic regression on a single indicator, and every 2×2 table of Chapter 16 is one as well.

23.6 Counts: Poisson regression

For a count response the Poisson family with its canonical link, the logarithm, is the standard model: $Y_i \sim \mathrm{Pois}(\mu_i)$ with $\log \mu_i = x_i^\top \beta$. Everything above applies with $b(\theta) = e^\theta$, so $w_i = \mu_i$: the Hessian is $-\sum_i \mu_i x_i x_i^\top$, the working weights are the fitted means, and the likelihood equations $X^\top (Y - \mu) = 0$ make the fitted counts add up to the observed ones along every predictor. The coefficients are log rate ratios: a unit increase in x_j multiplies the expected count by e^{β_j}. When the counts are collected over exposures of different lengths (a count of accidents per plant over different numbers of operating hours, say), the exposure E_i enters as $\log \mu_i = \log E_i + x_i^\top \beta$, a predictor with its coefficient fixed at 1, an **offset**, so that β describes the rate per unit of exposure. A warning the Poisson family carries: it fixes $\mathbb{V}\mathrm{ar}(Y_i) = \mu_i$, and real counts are often more variable than that (**overdispersion**); the coefficient estimates are then still sensible, since the likelihood equations are the method of moments for the means, but the standard errors from $(X^\top \hat{W} X)^{-1}$ are too small, and the robust standard errors of Chapter 21 or a model with a dispersion parameter are the repair.

The siblings data of Chapter 2 are the intercept-only case: $\log\lambda = \beta_1$, $\hat\beta_1 = \log 1 = 0$, with standard error $1/\sqrt{\sum_i \hat\mu_i} = 1/\sqrt{24} = 0.204$, Chapter 22's numbers. Exercise 23.5 fits a Poisson model with a link that is not canonical, and sees what the theorem no longer guarantees.

23.7 What comes next

This chapter closes the book's account of regression: a response from any canonical family, a mean linked to a linear predictor, maximum likelihood by iteratively reweighted least squares, and inference from the information matrix. Chapter 24 turns from predicting one variable from others to describing many variables at once: the directions in which multivariate data vary most, found from the eigenvectors of the sample covariance matrix, with the Law of Large Numbers saying why the sample directions approach the population's.

23.8 Summary

- A generalized linear model: independent Y_i from one canonical family with means μ_i, and a one-to-one differentiable link g with $g(\mu_i) = x_i^\top\beta$. The family fixes the variance function, the link carries the range of the means onto an interval, the whole line for the links in common use. Gaussian with the identity link is Chapter 21.
- Links: identity, logarithm for counts, logit (log-odds) and probit for proportions; $\Phi(t/1.7)$ and the logistic function nearly coincide. The canonical link $g_c = (b')^{-1}$ makes $\theta_i = x_i^\top\beta$: logit for the Bernoulli (logistic regression), log for the Poisson.
- Under the canonical link, $\nabla\ell_n = X^\top(Y-\mu)/\phi$, $\nabla^2\ell_n = -X^\top WX/\phi$ with $w_i = b''(x_i^\top\beta)$, negative definite when X has full rank: ℓ_n is strictly concave, the MLE is unique if it exists, and it solves $X^\top(Y-\mu(\beta)) = 0$ (fitted means balance the data along every predictor). Existence fails under complete separation.
- Newton's method: $\beta_{k+1} = \beta_k + (X^\top W_k X)^{-1}X^\top(Y-\mu_k)$, which is weighted least squares of the working response $z_k = X\beta_k + W_k^{-1}(Y-\mu_k)$ with weights $w_{k,i}$, repeated. Kyphosis: five steps from 0; kiss: $0 \to 0.581 \to 0.598$.
- Asymptotic normality: $\sqrt{n}(\hat\beta-\beta^*) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_p(0, \phi M^{-1})$, $M = \mathbb{E}[b''(x^\top\beta^*)xx^\top]$; in practice $\hat\beta \approx \text{Gauss}_p(\beta^*, \phi(X^\top\hat WX)^{-1})$, the inverse curvature at the fit. Wald $Z_j = \hat\beta_j/\text{SE}_j$, likelihood ratio for nested models (differences of deviances). Kyphosis: start $Z = -3.05$, odds ratio per vertebra $[0.71, 0.93]$; all three predictors 21.85 on 3 df. Aspirin: $\hat\beta_2 = -0.605$, SE 0.123, the four-reciprocals formula, $T^{\text{LR}} = 25.37$ as in Chapter 15.
- Poisson regression: $\log\mu_i = x_i^\top\beta$, coefficients as log rate ratios, offsets for exposure, overdispersion inflating the true standard errors above the model's.

23.9 Exercises

Exercise 23.1. Show that the Gaussian family with known σ^2 and the identity link is the linear model of Chapter 21: write $\ell_n(\beta)$, check that $W = I$, that the likelihood equations are the normal equations, that Newton's method from any β_0 reaches $\hat\beta$ in one step, and that Theorem 23.4's covariance $\phi(X^\top\hat WX)^{-1}$ is $\sigma^2(X^\top X)^{-1}$, the exact covariance of Theorem 21.5. *(Solution on page 439.)*

Exercise 23.2. For Poisson regression with the log link, write $\ell_n(\beta)$, its gradient and Hessian, and check Theorem 23.3 directly. If x_j is a 0/1 indicator, show that e^{β_j} is the ratio of the expected counts with and without the attribute, other predictors held fixed. *(Solution on page 440.)*

Exercise 23.3. Verify the aspirin bridge: (a) solve the likelihood equations with $x_i = (1, a_i)$ and show that the fitted probabilities are the two group proportions; (b) with $n_1, \hat{p}_1$ for placebo and $n_2, \hat{p}_2$ for aspirin as in Chapter 15, compute $(X^\top \hat{W} X)^{-1}$ and show that its $(2,2)$ entry is $1/(n_1\hat{p}_1(1-\hat{p}_1)) + 1/(n_2\hat{p}_2(1-\hat{p}_2))$, which equals $1/189 + 1/10{,}845 + 1/104 + 1/10{,}933$; (c) explain why the likelihood ratio statistic equals Chapter 15's 25.37. *(Solution on page 440.)*

Exercise 23.4. A hidden-variable reading of the two Bernoulli links. Suppose $Y = \mathbb{1}(x^\top\beta + \epsilon > 0)$ for an unobserved ϵ with a continuous, symmetric distribution and CDF F. (a) Show that $\mathbb{P}(Y = 1 \mid x) = F(x^\top\beta)$, so that $\epsilon \sim \text{Gauss}(0,1)$ gives the probit model and ϵ with the logistic CDF $F(t) = 1/(1+e^{-t})$ gives the logit model. (b) The logistic distribution has variance $\pi^2/3$; use this to explain the factor 1.7 in Figure 23.2. (c) Why can the scale of ϵ not be estimated from 0/1 data? *(Solution on page 440.)*

Exercise 23.5. A predator and prey model: the number Y of prey a hawk catches in a day is $\text{Pois}(\mu(x))$, where x is the number of prey in its territory and $\mu(x) = mx/(h+x)$ for unknown $m, h > 0$ (m the most it can catch, h the prey count at which it catches half of that). (a) Show that $1/\mu(x) = (1/m) + (h/m)(1/x)$, so this is a generalized linear model with the reciprocal link $g(\mu) = 1/\mu$ and predictor vector $(1, 1/x)$. (b) Is the reciprocal link canonical for the Poisson? Write $\theta_i = h(x_i^\top\beta)$ and show that the Hessian of ℓ_n now contains a term with the factor $Y_i - \mu_i$, so Theorem 23.3 does not apply. (c) What does Newton's method lose? *(Solution on page 440.)*

Exercise 23.6. For the separated data of Figure 23.3, with the model $\mathbb{P}(Y = 1 \mid x) = 1/(1 + e^{-b(x-3.5)})$: (a) show that $\ell_n(b) < 0$ for every finite b and $\ell_n(b) \to 0$ as $b \to \infty$; (b) show that for the full two-parameter logistic model $\mathbb{P}(Y = 1 \mid x) = 1/(1+e^{-(a+bx)})$ the supremum of the log-likelihood is 0 and is not attained; (c) explain why the Newton iterates cannot converge. *(Solution on page 440.)*

Exercise 23.7. Show that Newton's step equals the weighted least squares fit of the working response: with $z_k = X\beta_k + W_k^{-1}(Y - \mu_k)$, verify that $(X^\top W_k X)^{-1} X^\top W_k z_k = \beta_k + (X^\top W_k X)^{-1} X^\top (Y - \mu_k)$, and that the right side minimizes $\sum_i w_{k,i}(z_{k,i} - x_i^\top\beta)^2$ over β. *(Solution on page 440.)*

Exercise 23.8. The start alone. Logistic regression of kyphosis on the start gives $\hat{\beta} = (0.890, -0.218)$ with standard errors $(0.630, 0.060)$ and $\ell_n = -34.04$. (a) Compute the fitted probability of kyphosis for starts 5, 10 and 15. (b) Test the start by Wald and by likelihood ratio against the intercept-only fit ($\ell_n = -41.62$). (c) The full model's start coefficient is -0.207 with standard error 0.068: why are the two coefficients so close here, when Chapter 21 found a coefficient can change a great deal when company is added? *(Solution on page 441.)*

Exercise 23.9. Logistic regression with an intercept alone on the kiss data. (a) Show that the likelihood equation is $\sum_i R_i = n\,e^\beta/(1+e^\beta)$ and solve it. (b) Carry out Newton's method from $\beta_0 = 0$ by hand for two steps, using $\nabla\ell_n = S - np$ and $\nabla^2\ell_n = -np(1-p)$ with $p = e^\beta/(1+e^\beta)$, and compare with Figure 23.4. (c) Compute the standard error from $(X^\top\hat{W}X)^{-1}$ and match it with Chapter 22. *(Solution on page 441.)*

Exercise 23.10. Two plants report $Y_1 = 12$ accidents in $E_1 = 30{,}000$ operating hours and $Y_2 = 9$ in $E_2 = 10{,}000$. Fit the Poisson model $\log\mu_i = \log E_i + \beta_1 + \beta_2 a_i$ with $a_i = \mathbb{1}(i = 2)$: find $\hat{\beta}$, interpret $e^{\hat{\beta}_2}$, and give a Wald interval for it. *(Solution on page 441.)*

Part X

Principal Component Analysis

The book's one chapter of unsupervised statistics: finding the directions in which multivariate data vary most, through the eigenvectors of the sample covariance matrix, and what the Law of Large Numbers says about them (Chapter 24).

Chapter 24

Principal Component Analysis

In this chapter

- A sample of vectors as a cloud of points in $\mathbb{R}^d$: its mean vector and covariance matrix, and the sample variance of the cloud in a direction u, which is $u^\top S u$.
- The direction of greatest spread is the first eigenvector of S, the next the second, and so on: the principal directions, proved from the spectral theorem.
- Principal component analysis as the best flat picture of the cloud: the projection onto the first k principal directions loses less than any other k-dimensional projection, and the eigenvalues say how much.
- What the sample covariance estimates: $S \xrightarrow[n\to\infty]{\mathbb{P}} \Sigma$ entry by entry by the Law of Large Numbers, and the sample directions converge to the population's by the continuous mapping theorem. The units pitfall, and what happens when d is not small against n.

Every chapter so far had one variable to explain or one parameter to estimate. This last chapter looks at many variables at once with no one of them singled out, and asks only in which directions they vary together.

24.1 A cloud of points

The Old Faithful record of Chapter 20 has two numbers per eruption, its duration and the wait to the next one; the stack loss record of Chapter 21 has four per day. Regression picked one of them as the response. Drop that choice: each observation is a point $X_i \in \mathbb{R}^d$, the sample is a cloud of n points, and the question is the cloud's shape.

Definition 24.1 (Mean vector and covariance matrix)**.** Let $X = (X^{(1)}, \dots, X^{(d)})^\top$ be a random vector with $\mathbb{E}[\|X\|^2] < \infty$. Its **mean vector** is $\mathbb{E}[X] = (\mathbb{E}[X^{(1)}], \dots, \mathbb{E}[X^{(d)}])^\top$ and its **covariance matrix** is the $d \times d$ matrix

$$\Sigma = \mathbb{E}\left[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top\right], \qquad \Sigma_{jk} = \mathbb{C}\mathrm{ov}(X^{(j)}, X^{(k)}),$$

with the variances on the diagonal. For a sample $X_1, \dots, X_n$ the **sample mean vector** is $\bar{X} = \frac{1}{n}\sum_i X_i$ and the **sample covariance matrix** is

$$S = \frac{1}{n}\sum_{i=1}^{n} (X_i - \bar{X})(X_i - \bar{X})^\top,$$

whose (j,k) entry is the sample covariance of the jth and kth coordinates, with divisor n.

The mean vector is the center of the cloud and the covariance matrix its shape: each diagonal entry is the spread along one axis, each off-diagonal entry says whether two coordinates move together. The sample versions are the same averages over the points. With $\mathbf{X}$ the $n \times d$ matrix whose rows are the $X_i^\top$ (the design matrix of Chapter 21, in bold here because X is the random vector), $\bar{X} = \frac{1}{n}\mathbf{X}^\top \mathbf{1}$ and $S = \frac{1}{n}\mathbf{X}^\top H \mathbf{X}$ with $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ the projection that removes the column means.

For the kiss data. With $d = 1$ the vector is the single 0/1 variable R_i, the mean vector is $\hat{p} = 0.645$ and the covariance matrix is the one number $\hat{p}(1-\hat{p}) = 0.229$, the sample variance with divisor n (Chapter 7). For Old Faithful, $d = 2$: the mean vector is $(3.49, 70.9)$ and

$$S = \begin{pmatrix} 1.30 & 13.9 \\ 13.9 & 184.1 \end{pmatrix},$$

a duration variance of 1.30 minutes squared, a waiting variance of 184, and a covariance of 13.9, which is the correlation $r = 0.90$ of Chapter 20 times the two standard deviations.

The one fact about S the chapter runs on is what it says about a direction.

Proposition 24.2 (Variance in a direction)**.** *For any $u \in \mathbb{R}^d$, $u^\top \Sigma u = \mathbb{V}ar(u^\top X)$, and $u^\top S u$ is the sample variance, with divisor n, of the n numbers $u^\top X_1, \dots, u^\top X_n$. Both matrices are symmetric and positive semidefinite. If $u^\top S u = 0$ then every X_i lies in the hyperplane $\{x : u^\top x = u^\top \bar{X}\}$.*

Proof strategy (for completeness; skip on a first reading). Pull the fixed vector u out of the expectation or the average: $u^\top (X - \mathbb{E}\,X)(X - \mathbb{E}\,X)^\top u = (u^\top (X - \mathbb{E}\,X))^2$.

Proof. $u^\top \Sigma u = \mathbb{E}[u^\top (X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top u] = \mathbb{E}[(u^\top X - u^\top \mathbb{E}[X])^2] = \mathbb{V}\mathrm{ar}(u^\top X)$, since $u^\top \mathbb{E}[X] = \mathbb{E}[u^\top X]$; likewise $u^\top S u = \frac{1}{n}\sum_i (u^\top X_i - u^\top \bar{X})^2$, and $u^\top \bar{X}$ is the mean of the $u^\top X_i$. A variance is non-negative, so both matrices are positive semidefinite, and they are symmetric because $\mathbb{C}\mathrm{ov}(X^{(j)}, X^{(k)}) = \mathbb{C}\mathrm{ov}(X^{(k)}, X^{(j)})$. If the sample variance of the $u^\top X_i$ is 0 they all equal their mean $u^\top \bar{X}$. □

For a unit vector u, the $u^\top X_i$ are the coordinates of the points along the axis through the origin in direction u, so $u^\top S u$ is how far the cloud spreads along that axis, and the proposition turns a question about the cloud into a question about one matrix. Figure 24.1 plots $u^\top S u$ against the angle of u for the Old Faithful cloud after each variable has been standardized (write Z_i for the standardized X_i, with coordinates z_{i1}, z_{i2}: each coordinate is centered and divided by its standard deviation, so that S becomes the correlation matrix, with 1s on the diagonal and $r = 0.90$ off it): the spread is $1 + 0.90 \sin 2t$ at angle t, largest along the diagonal at 45°, where it is 1.90, and smallest at 135°, where it is 0.10, with the two extremes at right angles. Which directions those are, and why they are perpendicular, is the subject of the next section.

24.2 The principal directions

S is symmetric, so the spectral theorem (toolkit A.18) writes it as

$$S = PDP^\top = \sum_{j=1}^{d} \lambda_j v_j v_j^\top, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0,$$

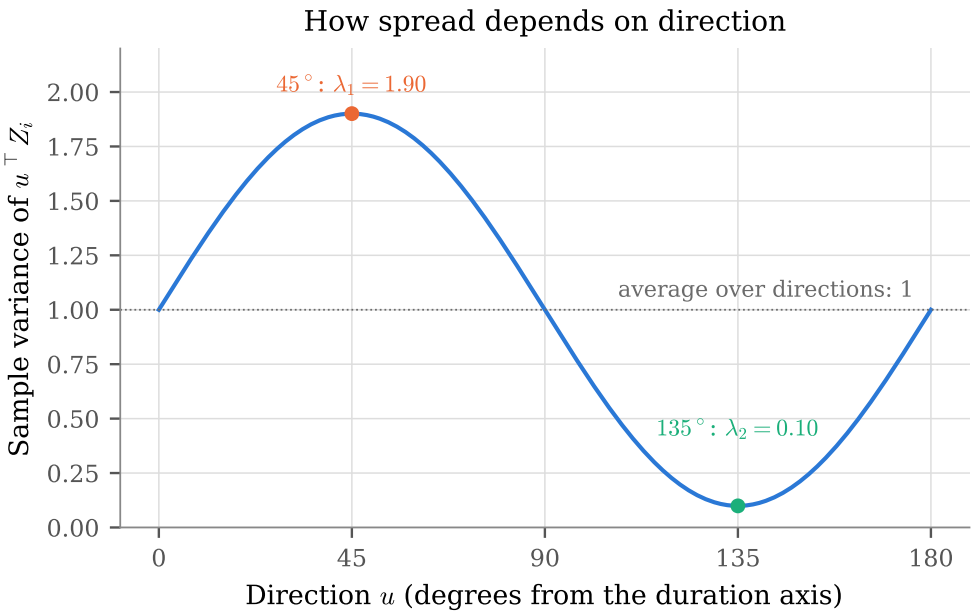


Figure 24.1: The sample variance of $u^\top Z_i$ for the standardized Old Faithful data against the angle of the unit vector u: $1 + 0.90 \sin 2t$, with its maximum 1.90 at 45° and its minimum 0.10 at 135°.

with $P = (v_1, \dots, v_d)$ orthogonal, so the v_j are orthonormal, and the eigenvalues non-negative because S is positive semidefinite. The theorem of this chapter says the eigenvectors are the directions the cloud spreads along and the eigenvalues are the spreads.

Theorem 24.3 (Principal directions)**.** *Let $S = PDP^\top$ as above. Then:*

1. *v_1 maximizes $u^\top S u$ over unit vectors u, and the maximum is λ_1;*
2. *for $j = 2, \dots, d$, v_j maximizes $u^\top S u$ over unit vectors perpendicular to $v_1, \dots, v_{j-1}$, and the maximum is λ_j;*
3. *the sample covariance matrix of the rotated points $P^\top X_i$ is D: the coordinates along the v_j have sample variances λ_j and sample covariances 0; and $\sum_j \lambda_j = \operatorname{tr}(S)$, the sum of the d sample variances.*

Proof strategy (core). In the eigenvector coordinates $c = P^\top u$, the quadratic form is $\sum_j \lambda_j c_j^2$ with $\sum_j c_j^2 = 1$: a weighted average of the eigenvalues, largest when all the weight sits on the largest one. Perpendicularity to the first $j-1$ eigenvectors sets their coordinates to zero, and the same argument runs on the rest.

Proof. For a unit vector u put $c = P^\top u$, so $u = Pc$ and $\|c\|^2 = c^\top P^\top P c = \|u\|^2 = 1$. Then $u^\top S u = c^\top P^\top P D P^\top P c = c^\top D c = \sum_j \lambda_j c_j^2 \le \lambda_1 \sum_j c_j^2 = \lambda_1$, with equality when $c = e_1$, that is, $u = v_1$: this is (1). If u is perpendicular to $v_1, \dots, v_{j-1}$ then $c_1 = \cdots = c_{j-1} = 0$, since $c_k = v_k^\top u$, so $u^\top S u = \sum_{k \ge j} \lambda_k c_k^2 \le \lambda_j$, with equality at $u = v_j$: this is (2). For (3), the rotated points have mean $P^\top \bar{X}$ and sample covariance $\frac{1}{n} \sum_i P^\top (X_i - \bar{X})(X_i - \bar{X})^\top P = P^\top S P = D$; and $\operatorname{tr}(S) = \operatorname{tr}(PDP^\top) = \operatorname{tr}(DP^\top P) = \operatorname{tr}(D)$. □

Definition 24.4 (Principal directions and components)**.** The eigenvectors $v_1, \dots, v_d$ of S, ordered by decreasing eigenvalue, are the sample's **principal directions**; the numbers $v_j^\top (X_i - \bar{X})$, $i = 1, \dots, n$, are its jth **principal component** (or **scores**); and $\lambda_j / \operatorname{tr}(S)$ is the **proportion of the variance explained** by the jth component.

Rotate the axes so that the first points along the cloud's longest extent, the second along the longest extent perpendicular to it, and so on. In the new axes the coordinates are uncorrelated, their variances are the eigenvalues, and the total variance is the same as before, since a rotation does not change distances. The scores are the coordinates of each point in the new axes, measured from the center.

For the kiss data. With one variable there is one direction and nothing to rotate. For the standardized Old Faithful data (Figure 24.2), $v_1 = (0.71, 0.71)$ and $v_2 = (-0.71, 0.71)$ with $\lambda_1 = 1.90$ and $\lambda_2 = 0.10$: the first component is $(z_{i1} + z_{i2})/\sqrt{2}$, an eruption's size on the two measures combined, and carries $1.90/2 = 95\%$ of the variance; the second is $(z_{i2} - z_{i1})/\sqrt{2}$, how far the standardized wait sits above the standardized duration, and carries 5%.

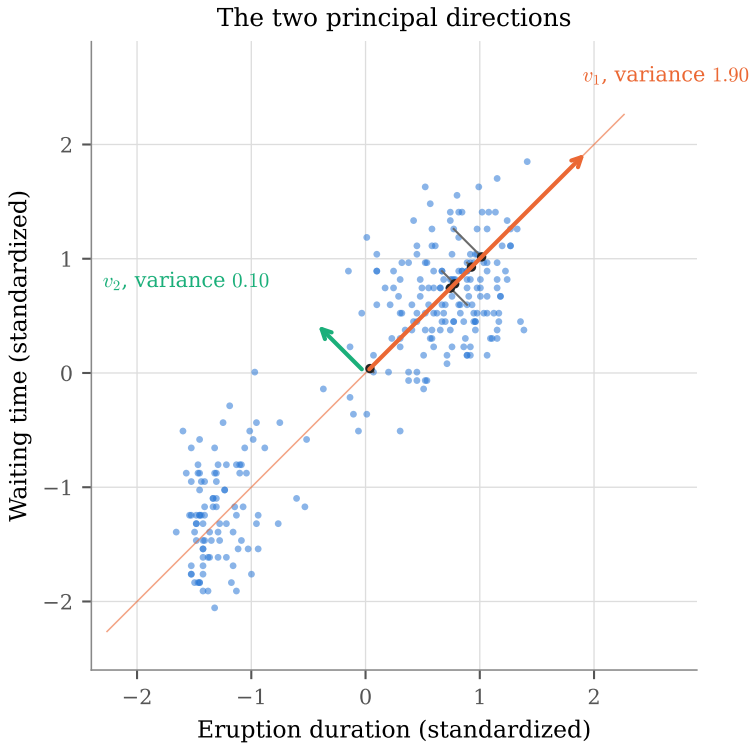


Figure 24.2: The standardized Old Faithful cloud with the line along its first principal direction, the two principal directions from the mean, each drawn with length $2\sqrt{\lambda_j}$, and five points of the upper cluster projected onto the first (gray segments, black dots).

A correlation matrix with 1s on the diagonal and r off it always has eigenvectors $(1, 1)/\sqrt{2}$ and $(-1, 1)/\sqrt{2}$ with eigenvalues $1 + r$ and $1 - r$ (Exercise 24.1), so for two standardized variables the principal directions are the two diagonals whatever the data, the rising one first when $r > 0$, and only the eigenvalues carry information. With more variables the directions are no longer fixed in advance, and they are what the method is for. The method is Pearson's (1901), who found the line and the plane of closest fit to a cloud of points, which is Theorem 24.5 below; Hotelling (1933) named the principal components and tied them to the eigenvectors of the covariance matrix.

Pitfall: units decide the directions

$u^\top Su$ is a variance, in the units of $u^\top X$, and changing the unit of one coordinate changes the eigenvectors, not just the eigenvalues. Figure 24.3 shows Old Faithful in its raw units, duration

in minutes with variance 1.30 and waiting in minutes with variance 184: the first principal direction is tilted 86° toward the waiting axis and explains 99.9% of the variance, which says only that 184 is much larger than 1.30. Measure the duration in seconds instead and the picture tilts the other way. When the coordinates are measured in different units, or in the same unit on very different scales, standardize them first; the analysis is then of the correlation matrix, and the first direction is the one along which the standardized variables move together most. When the coordinates share a unit and their scales mean something, as with d measurements of the same kind, the raw covariance matrix is the right object.

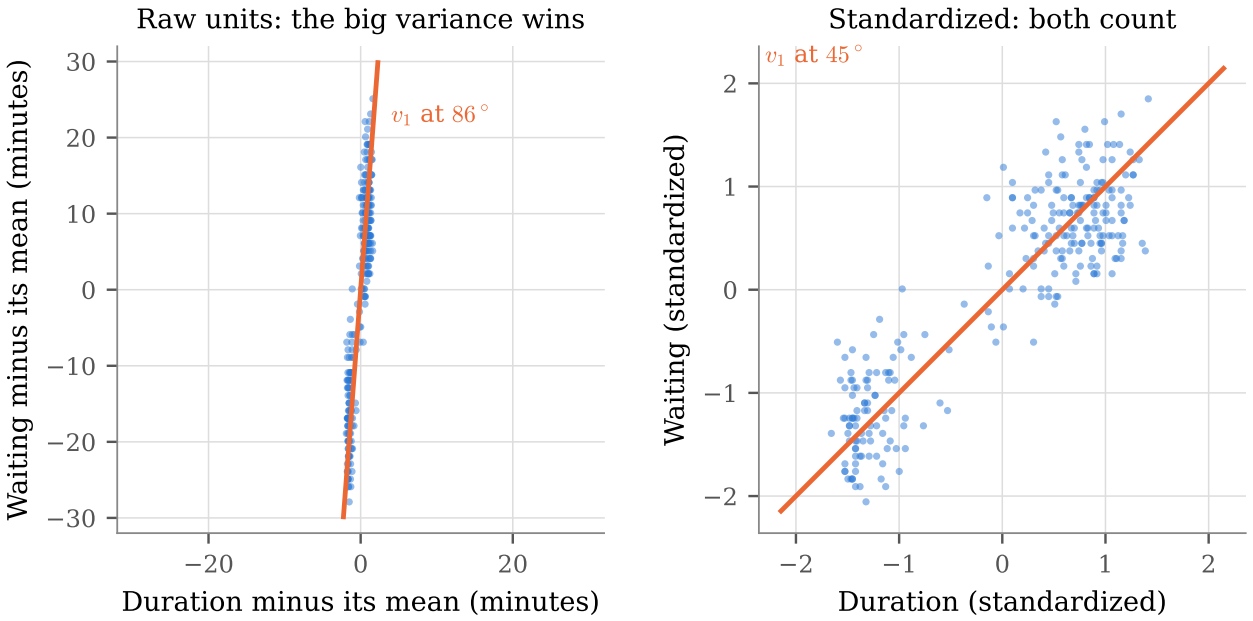


Figure 24.3: Old Faithful with both axes in minutes (left, equal scales) and standardized (right). The first principal direction (orange) follows the variable with the larger variance on the left and the diagonal on the right.

24.3 The best flat picture

A cloud in $\mathbb{R}^d$ with $d > 3$ cannot be drawn. Principal component analysis draws it anyway, by projecting it onto the plane (or the line, or a k-dimensional subspace) along which it is most spread out, and the point of the next theorem is that the plane which keeps the most variance is also the plane which loses the least distance.

Theorem 24.5 (Projection onto the first k principal directions)**.** *Let $S = PDP^\top$ as above, let $1 \le k < d$, and for an orthonormal $u_1, \dots, u_k$ let Q be the projection onto their span, $Qx = \sum_{j \le k} (u_j^\top x) u_j$. Over all such choices, the sum of squared distances from the centered points to their projections,*

$$\sum_{i=1}^{n} \|(X_i - \bar{X}) - Q(X_i - \bar{X})\|^2 = n\Big(\operatorname{tr}(S) - \sum_{j=1}^{k} u_j^\top S u_j\Big),$$

is smallest when $u_1, \dots, u_k$ span the same subspace as $v_1, \dots, v_k$, and its minimum is $n(\lambda_{k+1} + \cdots +$

λ_d). *The projected cloud then has total variance (the sum of its k sample variances) $\lambda_1 + \cdots + \lambda_k$, a proportion $(\lambda_1 + \cdots + \lambda_k)/\mathrm{tr}(S)$ of $\mathrm{tr}(S)$, the cloud's total.*

Proof strategy (core). Pythagoras splits each centered point's squared length into the part along the subspace and the part lost; summing, the lost part is the total variance minus the variance kept, $\sum_{j\le k} u_j^\top S u_j$. Keeping the most is choosing k orthonormal directions whose quadratic forms add up to the most, and in eigenvector coordinates that sum is $\sum_j \lambda_j w_j$ with weights $w_j \in [0,1]$ adding up to k, which is largest with the weight on the k largest eigenvalues.

Proof. Write $Y_i = X_i - \bar{X}$. Since Q is an orthogonal projection, $\|Y_i\|^2 = \|QY_i\|^2 + \|Y_i - QY_i\|^2$ (toolkit A.16), and $\|QY_i\|^2 = \sum_{j\le k}(u_j^\top Y_i)^2$ by orthonormality. Summing over i and dividing by n, $\frac{1}{n}\sum_i \|Y_i\|^2 = \mathrm{tr}(S)$ (the sum of the d sample variances) and $\frac{1}{n}\sum_i (u_j^\top Y_i)^2 = u_j^\top S u_j$ (Proposition 24.2, the $u_j^\top Y_i$ having mean 0), which gives the displayed identity. It remains to maximize $\sum_{j\le k} u_j^\top S u_j$. Put $c_j = P^\top u_j$, orthonormal since P is orthogonal, so that $u_j^\top S u_j = \sum_m \lambda_m c_{jm}^2$ and

$$\sum_{j=1}^{k} u_j^\top S u_j = \sum_{m=1}^{d} \lambda_m w_m, \qquad w_m = \sum_{j=1}^{k} c_{jm}^2.$$

Each $w_m \le 1$, since $c_{1m}, \ldots, c_{km}$ are the mth coordinates of k orthonormal vectors, which extend to an orthonormal basis of $\mathbb{R}^d$, whose mth coordinates have squares summing to 1 (toolkit A.13); and $\sum_m w_m = \sum_j \|c_j\|^2 = k$. A sum $\sum_m \lambda_m w_m$ with $\lambda_1 \ge \cdots \ge \lambda_d$, $0 \le w_m \le 1$ and $\sum_m w_m = k$ is at most $\lambda_1 + \cdots + \lambda_k$: moving weight from any $m > k$ to any $m' \le k$ with $w_{m'} < 1$ cannot decrease it, and when no such move is left the weights are 1 on $m \le k$ and 0 after. The bound is attained by $u_j = v_j$, where $c_j = e_j$; and any orthonormal $u_1, \ldots, u_k$ spanning the same subspace give the same projection Q, hence the same loss. □

Among all flat pictures of dimension k, the one drawn on the first k principal directions is the one in which the points move least from where they were, and what it loses is exactly the sum of the eigenvalues it leaves out. The proportion of the variance explained says how much of the cloud's variance the picture keeps.

For the kiss data. With $d = 1$ there is nothing to project. The standardized Old Faithful cloud projected onto v_1 (Figure 24.2) keeps 95% of its variance, and the squared distances the points move add up to $n\lambda_2 = 272 \times 0.10 = 27$.

In practice the analysis runs in three steps: center the data (and standardize them unless the units agree), compute the eigenvectors and eigenvalues of S, and choose k. The choice of k is a judgment with two common guides. The **scree plot** draws λ_j against j and stops where the fall flattens, the eigenvalues past that point being read as noise; the **proportion explained** chooses the smallest k with $(\lambda_1 + \cdots + \lambda_k)/\mathrm{tr}(S)$ above a threshold such as 0.9. For a picture, k is 2 or 3 whatever the eigenvalues say.

Worked example: the stack loss data in four dimensions

The 21 days of Chapter 21 have four measurements each, air flow, water temperature, acid concentration and stack loss, in different units, so the analysis is of the correlation matrix, whose off-diagonal entries are 0.78 (air, water), 0.50 (air, acid), 0.92 (air, loss), 0.39 (water, acid), 0.88 (water, loss) and 0.40 (acid, loss). Its eigenvalues are 3.00, 0.73, 0.22 and 0.05, which explain 75%, 18%, 5% and 1% of the variance (Figure 24.4): two components keep 93%. The first direction is $v_1 = (0.55, 0.52, 0.35, 0.55)$, close to an equal-weight average of the four standardized

variables, an overall 'how hard the plant was running' score, on which days 1 to 3 sit far to the right and days 17 to 19 far to the left. The second, $v_2 = (-0.08, -0.26, 0.93, -0.26)$, is the acid concentration against everything else: acid is the variable least tied to the other three (its correlations are 0.39 to 0.50), and the second component separates days by it, day 14 at the top and day 17 at the bottom. The regression of Chapter 21 explained the loss by the other three; the first component says the three and the loss mostly rise and fall together.

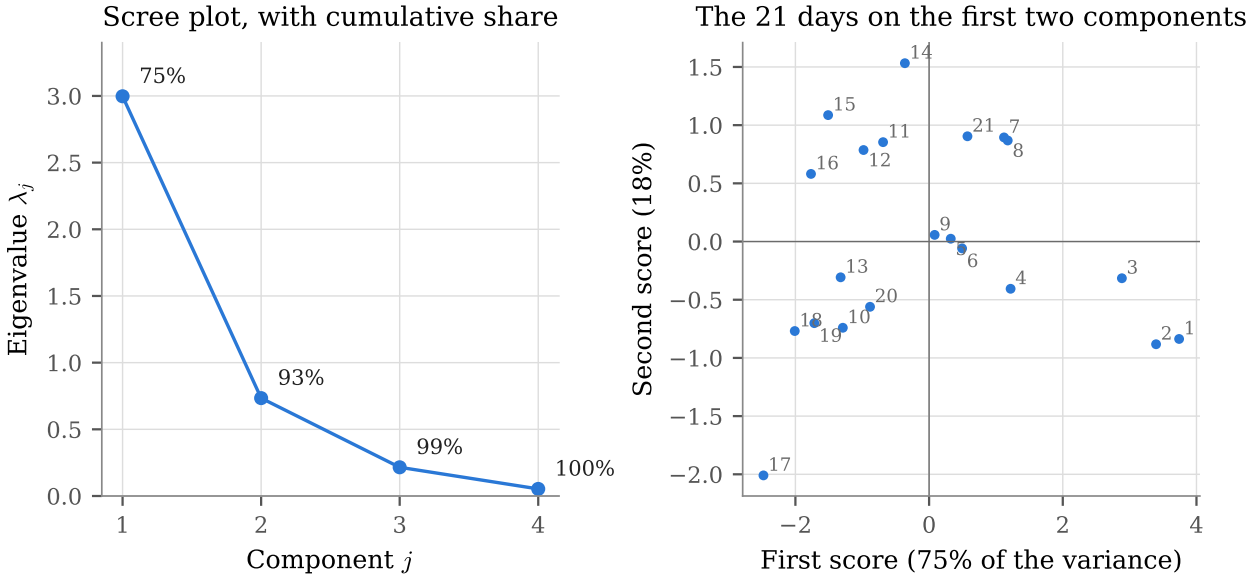


Figure 24.4: Stack loss, four standardized variables. Left: the scree plot of the eigenvalues 3.00, 0.73, 0.22, 0.05 with the cumulative share explained. Right: the 21 days on their first two principal components, numbered.

24.4 What the sample directions estimate

Everything so far was a description of n points, with no model behind it, and principal component analysis is used in that spirit on clouds that are not random at all. When the X_i are i.i.d. copies of a random vector X, the sample quantities estimate population ones, and the book's limit theorems say in what sense.

Theorem 24.6 (Consistency of the sample covariance and its eigenvectors)**.** *Let $X_1, X_2, \dots$ be i.i.d. copies of $X \in \mathbb{R}^d$ with $\mathbb{E}[\|X\|^4] < \infty$, mean m and covariance matrix Σ, with eigenvalues $\lambda_1^* \geq \cdots \geq \lambda_d^*$ and orthonormal eigenvectors $v_1^*, \dots, v_d^*$. Then:*

1. *$\bar{X} \xrightarrow[n\to\infty]{\mathbb{P}} m$ and $S \xrightarrow[n\to\infty]{\mathbb{P}} \Sigma$, entry by entry;*
2. *each sample eigenvalue converges: $\lambda_j \xrightarrow[n\to\infty]{\mathbb{P}} \lambda_j^*$ for $j = 1, \dots, d$;*
3. *if λ_j^* differs from both its neighbors, $\lambda_{j-1}^* > \lambda_j^* > \lambda_{j+1}^*$ (only the second inequality when $j = 1$, only the first when $j = d$), the sample direction converges to the population direction: with the signs chosen so that $v_j^\top v_j^* \geq 0$, $v_j \xrightarrow[n\to\infty]{\mathbb{P}} v_j^*$.*

Proof strategy (core). Each entry of S is an average of products of coordinates minus a product of averages, so the Law of Large Numbers and the continuous mapping theorem give (1); the eigenvalues, and the eigenvector lines where the eigenvalue is isolated, are continuous functions of the matrix, so the continuous mapping theorem gives (2) and (3).

Recall. The Law of Large Numbers (Theorem 3.5): for i.i.d. Z_i with $\mathbb{V}\mathrm{ar}(Z_i) < \infty$, $\frac{1}{n}\sum_i Z_i \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[Z_i]$. The continuous mapping theorem (Theorem 5.2): if $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$ and f is continuous, then $f(Y_n) \xrightarrow[n\to\infty]{\mathbb{P}} f(y)$; it holds for a vector $Y_n \xrightarrow[n\to\infty]{\mathbb{P}} y$ and a continuous f of several variables, since continuity of f at y gives a δ with $\|f(z) - f(y)\| \le \epsilon$ whenever $\|z - y\| \le \delta$, so $\mathbb{P}(\|f(Y_n) - f(y)\| > \epsilon) \le \mathbb{P}(\|Y_n - y\| > \delta) \to 0$; and a vector converges in probability as soon as each of its q coordinates does, since $\mathbb{P}(\|Y_n - y\| > \epsilon) \le \sum_{j=1}^{q} \mathbb{P}(|Y_n^{(j)} - y_j| > \epsilon/\sqrt{q})$.

Proof. Write $S = \frac{1}{n}\sum_i X_i X_i^\top - \bar{X}\bar{X}^\top$ (expand the square in Definition 24.1). The (j,k) entry of the first term is the average of the i.i.d. products $X_i^{(j)} X_i^{(k)}$, whose variance is finite by the fourth-moment assumption, so it converges in probability to $\mathbb{E}[X^{(j)}X^{(k)}]$; and $\bar{X}^{(j)} \xrightarrow[n\to\infty]{\mathbb{P}} m_j$ by the Law of Large Numbers with the second moments finite. Since the vector of all $d + d^2$ quantities then converges in probability (each coordinate does), and $(a, b) \mapsto a - bb^\top$ is continuous, the continuous mapping theorem gives $S \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[XX^\top] - mm^\top = \Sigma$ entry by entry, which is (1). The map from a symmetric matrix to its jth largest eigenvalue is continuous (toolkit A.18), so (2) follows from (1) by the continuous mapping theorem; and where λ_j^* is isolated from its neighbors, the map to the unit eigenvector with the sign fixed by $v_j^\top v_j^* \ge 0$ is continuous at Σ, which gives (3) (the map need only be continuous where the limit lives, as Chapter 5 notes). □

The condition in (3) is not decoration: when two population eigenvalues are equal, any direction in the plane they span is an eigenvector, and the sample direction wanders in that plane however large n is. For the standardized Old Faithful data, the gap 1.90 against 0.10 is wide, and the sample directions are the population's diagonals in any case. There is a Central Limit Theorem to go with the Law: for Gaussian data, $\sqrt{n}(\lambda_j - \lambda_j^*) \xrightarrow[n\to\infty]{(\mathrm{d})} \mathrm{Gauss}(0, 2\lambda_j^{*2})$ when λ_j^* is isolated (Anderson, 1963), so a sample eigenvalue has a standard error of about $\lambda_j\sqrt{2/n}$; it is stated here and not proved, and nothing depends on it.

Pitfall: many variables, few observations

The theorem lets n grow with d fixed. When d is comparable to n, or larger, S is a poor estimate of Σ even though each of its entries is consistent: with $d > n$ the cloud spans at most $n - 1$ dimensions, S has rank at most $n - 1$, and at least $d - n + 1$ of its eigenvalues are exactly 0 whatever Σ is. The largest sample eigenvalue is then biased upward and the smallest downward, and the first sample direction can be far from v_1^* even when the gap below λ_1^* is wide. Gene expression data, with thousands of genes measured on hundreds of people, are the standard case. The remedies assume something about Σ: that it has low rank, so that only the first k sample components are kept and the rest treated as noise, or that v_1^* has few nonzero coordinates (sparse principal components), both outside this book.

24.5 Where the book ends

The book began with 124 couples and one proportion, and it ends with 21 days and a four-by-four matrix, but the tools did not change. A model says what the data could have been; an estimator is a function of the data; the Law of Large Numbers says it settles down, the Central Limit Theorem says how, and Slutsky, the continuous mapping theorem and the delta method carry those two facts to every estimator built from averages, which is nearly all of them. Maximum likelihood turned those tools into a method, with the Fisher information as its yardstick; tests and intervals were the same calculation read two ways; the Bayesian chapters put a prior on the parameter and found, in large samples, the same answers; regression and its generalizations put a linear predictor behind the mean; and this chapter dropped the response and kept the covariance. What comes after, in the next book of the series, is the same tools, the theorems of Chapters 3 to 5, applied to harder models.

24.6 Summary

- A sample of vectors is a cloud in $\mathbb{R}^d$ with mean vector $\bar{X}$ and sample covariance matrix $S = \frac{1}{n}\sum_i (X_i - \bar{X})(X_i - \bar{X})^\top$; $u^\top S u$ is the cloud's variance along u. Old Faithful: variances 1.30 and 184, covariance 13.9, correlation 0.90.
- Spectral theorem: $S = PDP^\top$ with orthonormal eigenvectors and eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$. The eigenvectors are the principal directions: v_1 maximizes $u^\top S u$ with value λ_1, v_j does so among directions perpendicular to the earlier ones, the rotated coordinates are uncorrelated with variances λ_j, and $\sum_j \lambda_j = \operatorname{tr}(S)$.
- Projecting onto the first k principal directions loses $n(\lambda_{k+1} + \cdots + \lambda_d)$ in squared distance, less than any other k-dimensional projection, and keeps the proportion $(\lambda_1 + \cdots + \lambda_k)/\operatorname{tr}(S)$ of the variance. Choose k by the scree plot or a proportion threshold. Stack loss: 75%, 93%, 99%, with an overall-activity first direction and an acid-against-the-rest second.
- Standardize when the units differ: in raw minutes Old Faithful's first direction is at 86° and explains 99.9%, which only reports the ratio of two variances.
- With i.i.d. data and fourth moments, $S \xrightarrow[n\to\infty]{\mathbb{P}} \Sigma$ entry by entry (LLN and CMT), the eigenvalues converge, and each eigenvector converges where its eigenvalue is isolated. For Gaussian data $\sqrt{n}(\lambda_j - \lambda_j^*) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, 2\lambda_j^{*2})$ (stated). With d near or above n the sample eigenvalues are badly biased and extra assumptions are needed.

24.7 Exercises

Exercise 24.1. (a) Show that $\begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ has eigenvectors $(1,1)/\sqrt{2}$ and $(-1,1)/\sqrt{2}$ with eigenvalues $1+r$ and $1-r$, and check the Old Faithful values 1.90 and 0.10. (b) Show that for the unit vector $u = (\cos t, \sin t)$, $u^\top S u = 1 + r\sin 2t$, the curve of Figure 24.1, and find its average over t. (c) For a general 2×2 covariance matrix $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$, show that the first principal direction makes the angle $\frac{1}{2}\arctan\big(2c/(a-b)\big)$ with the first axis (when $a > b$; when $a < b$ the maximum is at the other root of $\tan 2t = 2c/(a-b)$, 90° away), and compute it for Old Faithful's raw S. *(Solution on page 441.)*

Exercise 24.2. Let $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ with $\mathbf{1} \in \mathbb{R}^n$ the vector of ones. Show that H is symmetric with $H^2 = H$, say onto which subspace it projects, and verify $S = \frac{1}{n}\mathbf{X}^\top H \mathbf{X}$ for the $n \times d$ data matrix $\mathbf{X}$. *(Solution on page 441.)*

Exercise 24.3. Show that the principal components are uncorrelated: for $j \neq k$ the sample covariance of the scores $v_j^\top(X_i - \bar{X})$ and $v_k^\top(X_i - \bar{X})$ is $v_j^\top S v_k = 0$. Then show that the $n \times d$ matrix of scores is $H\mathbf{X}P$ and that its columns have squared lengths $n\lambda_j$. *(Solution on page 441.)*

Exercise 24.4. With $P_k = (v_1, \dots, v_k)$, the reconstruction of X_i from its first k scores is $\hat{X}_i = \bar{X} + P_k P_k^\top (X_i - \bar{X})$. (a) Show that $\frac{1}{n}\sum_i \|X_i - \hat{X}_i\|^2 = \lambda_{k+1} + \cdots + \lambda_d$. (b) For the stack loss data, what fraction of the standardized cloud's squared size does a two-component reconstruction lose, and how many days' worth of squared distance is that? *(Solution on page 441.)*

Exercise 24.5. Let $X \sim \text{Gauss}_d(m, \Sigma)$ with $\Sigma = PDP^\top$ positive definite. (a) Show that the population components $v_j^{*\top}(X - m)$ are independent Gauss$(0, \lambda_j^*)$. (b) Show that the contours of the density of X are the ellipsoids $(x - m)^\top \Sigma^{-1}(x - m) = c$, whose axes are the eigenvectors with half-lengths proportional to $\sqrt{\lambda_j^*}$: the principal directions are the axes of the Gaussian's ellipse, as in the confidence ellipse of Chapter 21. *(Solution on page 442.)*

Exercise 24.6. In Theorem 24.6, where exactly is the fourth-moment assumption used, and what would fail without it? Show by an example with $d = 1$ that S can be consistent for Σ when only $\mathbb{E}[X^2] < \infty$ holds, so that the assumption is a convenience of the proof rather than a necessity. *(Solution on page 442.)*

Exercise 24.7. Let $\Sigma = I_2$, so that every direction is a population eigenvector with eigenvalue 1. Explain why the first sample principal direction has no limit as $n \to \infty$, even though $S \xrightarrow[n\to\infty]{\mathbb{P}} I_2$, and why this is consistent with Theorem 24.6. What does a scree plot look like in this case? *(Solution on page 442.)*

Exercise 24.8. Principal components and regression. (a) Show that regressing the first score $v_1^\top(X_i - \bar{X})$ on the second by least squares through the origin gives slope 0. (b) For the standardized Old Faithful data, the regression of waiting on duration has slope $r = 0.90$ (Chapter 20), while the first principal direction is the 45° line with slope 1. Explain why the two lines differ, and which one minimizes which distance. *(Solution on page 442.)*

Appendices

Appendix A

Mathematical Toolkit

This appendix collects the mathematical habits the book uses: not statistics, but the handful of ideas from calculus, sets and probability that the proofs lean on. Each entry says what the idea is in plain words, draws it, and names the first place in the book that needs it. When a chapter marks a step with '(toolkit A.9)', this is where to come. Nothing here is needed to state a result; everything here is needed to follow a proof. The later entries (Gaussian vectors, matrices and projections, Newton's method, eigenvalues) are the ones the regression, GLM and PCA chapters need.

A.1 Limits of sequences of numbers

A sequence of numbers $x_1, x_2, x_3, \ldots$ **converges** to x if the terms eventually stay as close to x as you like. The definition makes 'as close as you like' and 'eventually' precise:

$$\text{for every } \epsilon > 0 \text{ there is an index } N \text{ such that } |x_n - x| < \epsilon \text{ for all } n \geq N.$$

Read it as a game with an opponent. The opponent names a tolerance ϵ, any positive number, however small. You must answer with an index N past which every term of the sequence is within ϵ of x. If you can always answer, the sequence converges. The order of the two moves is the whole content: ϵ is chosen first and N is allowed to depend on it, which is why a smaller ϵ is usually met with a larger N (Figure A.1). Writing the moves in the other order ('there is an N that works for every ϵ') would say the sequence is eventually *equal* to x, a much stronger and usually false statement. The left of Figure A.1 plays the game for $x_n = 2 + (-1)^n/\sqrt{n}$ with two tolerances: for $\epsilon = 0.3$ every term past $N = 12$ is inside the band, and for $\epsilon = 0.1$ you need $N = 101$ (at $n = 100$ the term sits exactly on the band's edge, and the definition asks for strictly inside). The right draws the index that works as a function of the tolerance; it climbs as ϵ shrinks, and convergence means the staircase is defined for every $\epsilon > 0$, however small.

Where the book uses it. Section 3.1 starts from this definition and explains why it cannot be applied directly to random variables; Definition 3.1 keeps its shape but applies it to the *probabilities* $\mathbb{P}(|X_n - X| \geq \epsilon)$, which are ordinary numbers. Every '$\to 0$' in a proof of convergence in probability is this definition applied to a sequence of numbers.

Limits inside an expectation. One further fact is used in the uniform law of large numbers of Appendix C.3 and, for an integral in place of an expectation, in Chapter 22: if random variables W_r converge to 0 as $r \to 0$ for every outcome, and all of them are bounded by one fixed random

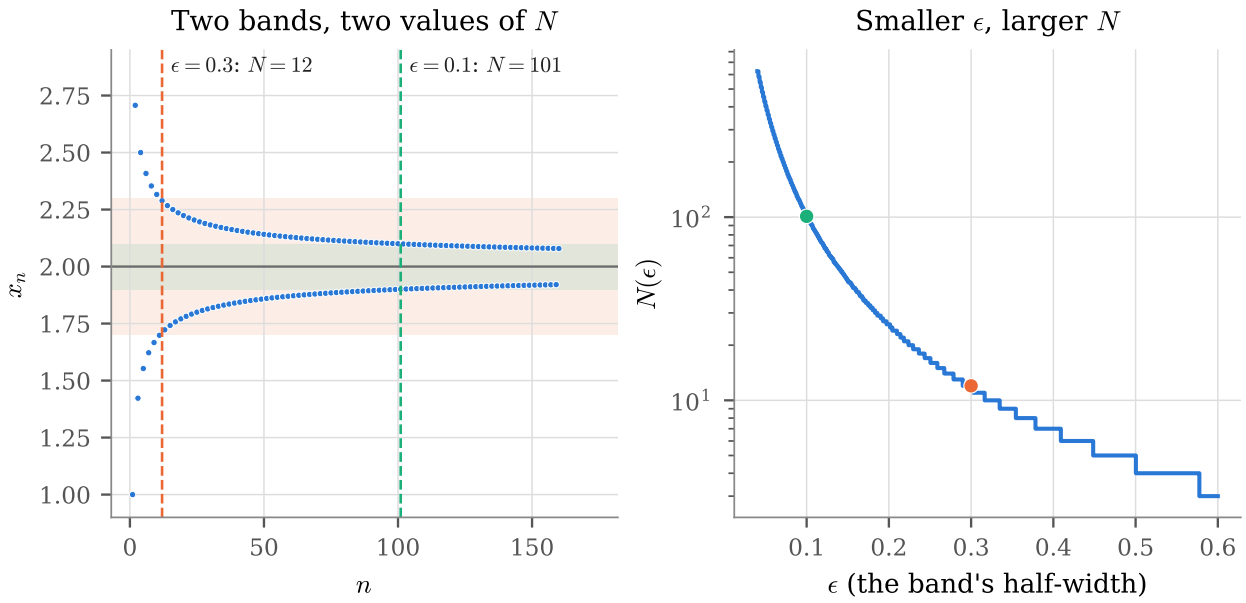


Figure A.1: The ϵ-N definition as a game. Left: the sequence $x_n = 2 + (-1)^n/\sqrt{n}$ with the bands 2 ± 0.3 and 2 ± 0.1. Right: the smallest index N that works, as a function of the tolerance ϵ.

variable G with $\mathbb{E}[G] < \infty$, then $\mathbb{E}[W_r] \to 0$ as well. This is the dominated convergence theorem. The bound is what makes it true: without it, a W_r that is huge on a set of tiny probability can have an expectation that refuses to go to 0 even though W_r itself goes to 0 at every outcome.

A.2 Supremum and infimum

The **maximum** of a set of numbers is its largest member. Some sets have none: $\{1 - \frac{1}{n} : n = 1, 2, \ldots\}$ climbs toward 1 and never reaches it. The **supremum**, written sup, is the least upper bound: the smallest number that no member exceeds. For that set the supremum is 1, which is not a member. When a maximum exists it equals the supremum; the supremum exists more often, which is why proofs prefer it. The **infimum**, inf, is the greatest lower bound, and the same remarks apply with 'minimum'. Figure A.2 shows both cases. On the left, the set $\{1 - 1/n\}$ has no largest element, but 1 is its least upper bound. On the right, xe^{-x} on $[0, 4]$ attains its supremum at $x = 1$ and its infimum at $x = 0$, so there sup = max and inf = min; a continuous function on a closed, bounded interval always attains both. The same holds on any closed, bounded set in $\mathbb{R}^d$, such as a closed ball (the extreme value theorem).

Where the book uses it. The conservative confidence interval of Section 8.4 replaces $\sqrt{p(1-p)}$ by its supremum over $p \in (0, 1)$, which is $\frac{1}{2}$ and is attained; the subway version has $\sup_\lambda \lambda = \infty$, which is why no conservative interval exists there. Chapter 7 mentions that a statistic may be defined through a sup or inf; the maximum of a sample, $\max_i X_i$, is a maximum of finitely many numbers and causes no trouble. The proofs of consistency in Theorems 12.1 and 16.9 use the extreme value theorem on a closed, bounded set of parameters.

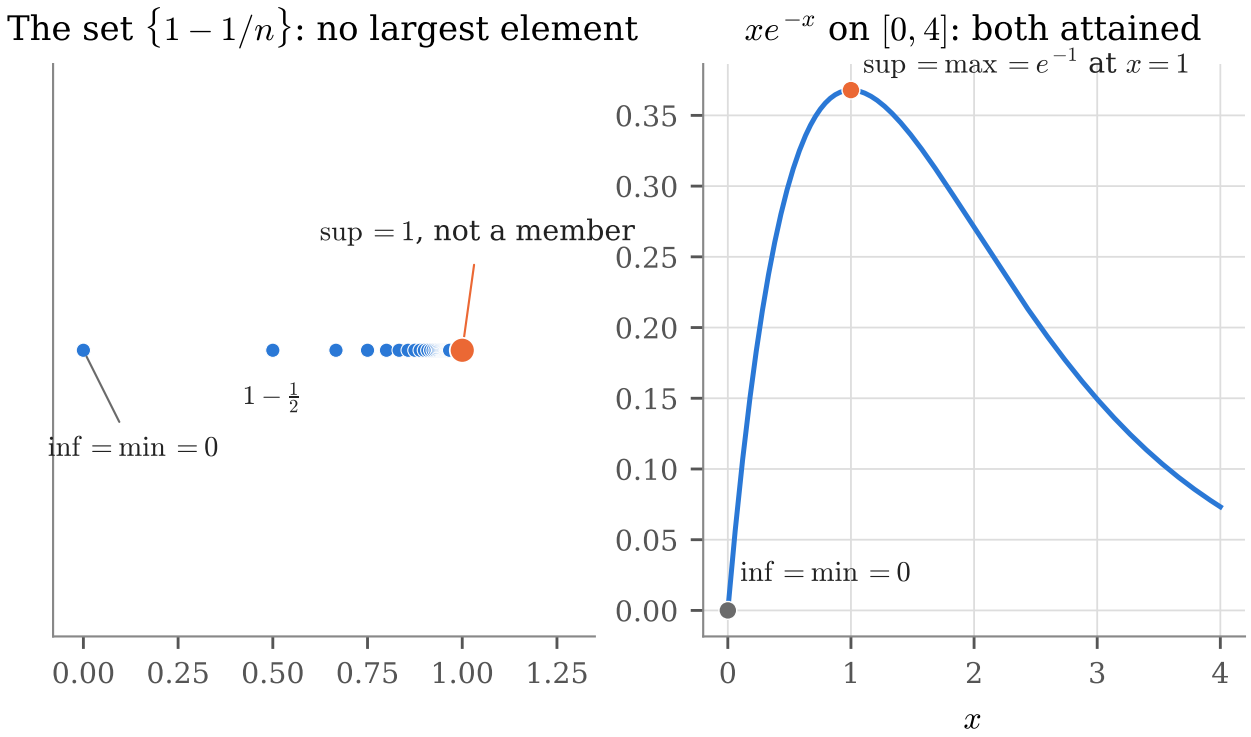


Figure A.2: Left: the set $\{1 - 1/n : n = 1, 2, \ldots\}$ and its supremum 1. Right: the function xe^{-x} on $[0, 4]$.

A.3 Sets, maps and one-to-one

A **map** (or function) f from a set A to a set B assigns to each element of A exactly one element of B. Its **image** is the set of elements of B that are actually hit. The map is **injective** (one-to-one) if different inputs always give different outputs: $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$, or, turned around, $f(a_1) = f(a_2)$ implies $a_1 = a_2$. An injective map can be undone: from the output you can recover the input. For a function of one real variable the test is visual, the horizontal-line test of Figure A.3: if some horizontal line crosses the graph twice, two inputs share an output and the map is not injective. On the left of the figure, $\lambda \mapsto e^{-5\lambda}$ is strictly decreasing, so every height is hit once and λ can be read off the output; this is the map of the censored-exponential example of Example 2.5. On the right, $x \mapsto x^2$ on $[-2, 2]$ hits the height 1 twice, so the input cannot be recovered from the output. A strictly increasing or strictly decreasing function always passes the test.

A **partition** of a set is a way of cutting it into pieces that do not overlap and together cover everything; any map partitions its domain into the sets of inputs that share an output.

Where the book uses it. Identifiability (Definition 2.3) is exactly the statement that the map $\theta \mapsto \mathbb{P}_\theta$ is injective; every identifiability argument in Chapter 2 is a horizontal-line test on that map. A statistic partitions the sample space (Section 7.2).

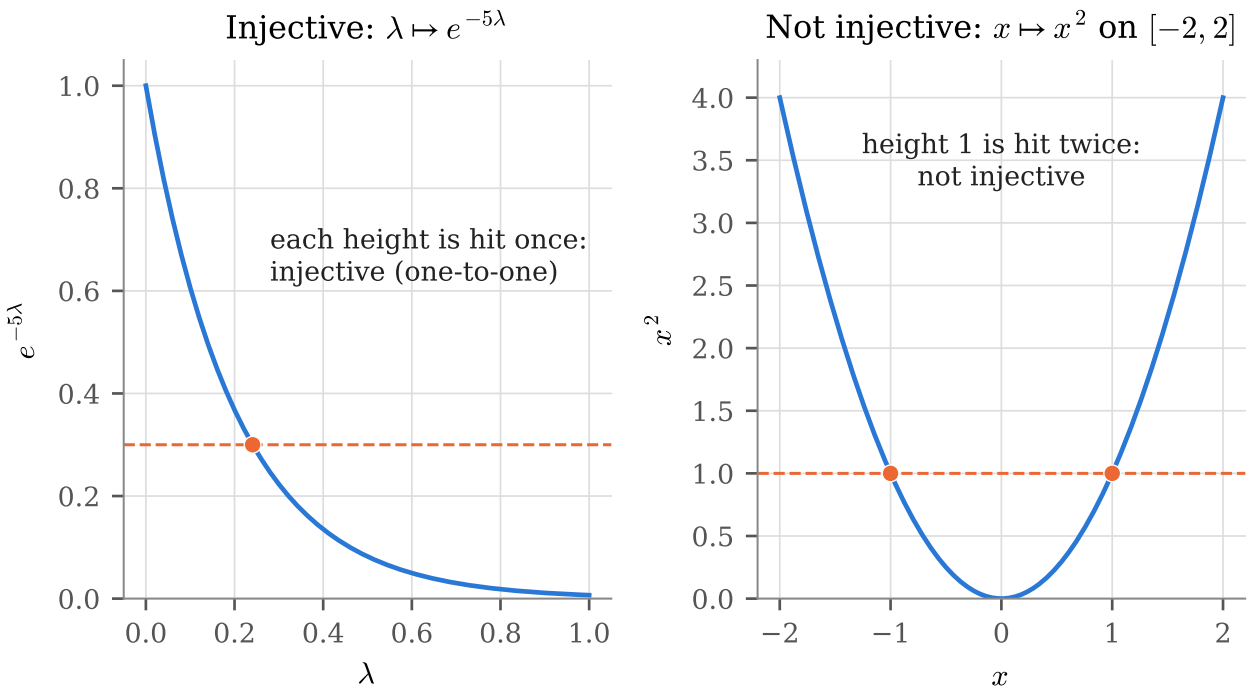


Figure A.3: The horizontal-line test. Left: $\lambda \mapsto e^{-5\lambda}$, the map of the censored-exponential example of Example 2.5. Right: $x \mapsto x^2$ on $[-2, 2]$.

A.4 Indicator functions

The **indicator** of an event or a condition is the function that is 1 when the condition holds and 0 when it does not: $\mathbb{1}(x \geq 5)$ is 0 for $x < 5$ and 1 from 5 on. It is the simplest random variable there is, a coin that comes up 1 with probability $\mathbb{P}(\text{condition})$, and its expectation is that probability:

$$\mathbb{E}[\mathbb{1}(X \geq 5)] = 1 \cdot \mathbb{P}(X \geq 5) + 0 \cdot \mathbb{P}(X < 5) = \mathbb{P}(X \geq 5).$$

That one line, *the expectation of an indicator is a probability*, is how probabilities enter computations that are written with expectations, and how a sum of indicators becomes a count (the number of couples turning right is $\sum_i \mathbb{1}(R_i = 1)$). Figure A.4 draws the two sides of the line: on the left the indicator $\mathbb{1}(x \geq 5)$, a step from 0 to 1; on the right, for $X \sim \text{Expo}(0.16)$, its expectation as the shaded probability $\mathbb{P}(X \geq 5) = e^{-5 \times 0.16} = 0.45$.

Where the book uses it. The CDF of a constant is a step function (Example 4.2); the censored observations of Chapter 2 are indicators of the unobserved variable; the Strong Law in Appendix B counts failures with a sum of indicators; Markov's inequality below is proved with one.

A.5 Random variables as functions, and 'almost surely'

A **random variable** is a function from the sample space Ω (the set of possible outcomes of an experiment, written ω) to the real line. Figure A.5 draws the simplest case: the 36 outcomes of two dice, and the function $X =$ 'the sum'. The probability $\mathbb{P}(X = 7)$ is the probability of the set of outcomes that X sends to 7: six of the 36, so $\mathbb{P}(X = 7) = 6/36$ on the right of the figure.

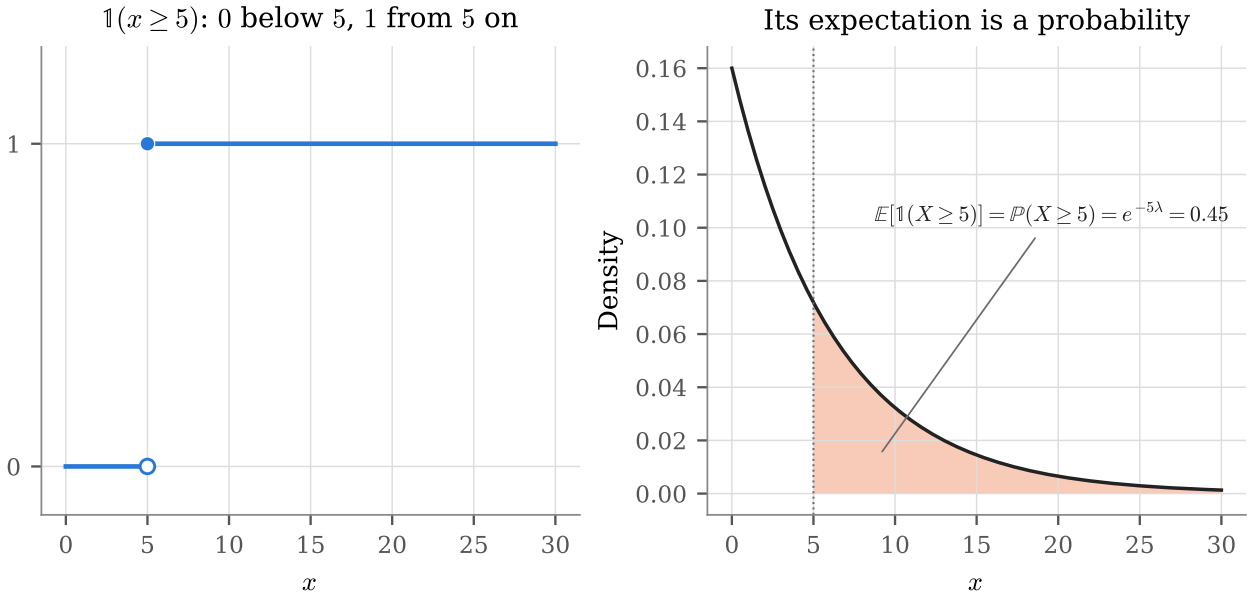


Figure A.4: Left: the indicator $\mathbb{1}(x \geq 5)$ as a function of x. Right: the density of $X \sim \text{Expo}(0.16)$, with the probability $\mathbb{P}(X \geq 5)$ shaded.

Two random variables on the same sample space, say the sum X and the larger die Y, are two functions of the *same* outcome, which is what gives meaning to $X - Y$ and to 'X and Y are close': you compare them outcome by outcome. This is the viewpoint Section 3.1 needs for a sequence of random variables, which is a sequence of functions on one sample space.

Three phrases from probability theory appear in the book and mean less than they sound.

- **Almost surely** means 'with probability one'. A continuous random variable equals any particular number with probability zero, so '$X \neq 5$ almost surely' is true even though the outcome $X = 5$ is not impossible; the set of outcomes where a statement fails is allowed to be nonempty as long as it has probability zero. The book uses it in Hoeffding's inequality ('bounded almost surely': the probability of landing outside $[a, b]$ is zero) and in Appendix B.
- **Measurable** (a measurable function, a measurable sample space) is the technical condition that makes probabilities of the relevant events well defined. Every function you can write down with formulas, limits, sums and integrals is measurable; the usual rule of thumb, that anything you can compute from the data is a statistic, is correct in practice. The book never needs more.
- A **probability measure** (or distribution) assigns a probability to every event. A PMF or PDF is a recipe for one: sum or integrate it over the event. The two determine each other, and the book writes down families of distributions through their PMFs and PDFs (Definition 2.1).

A.6 Densities, CDFs and quantiles

For a continuous random variable, **probability is area**: $\mathbb{P}(a \leq X \leq b)$ is the area under the density f between a and b. The **CDF** $F(x) = \mathbb{P}(X \leq x)$ is the area accumulated up to x, so $\mathbb{P}(a \leq X \leq b) = F(b) - F(a)$, and F is the antiderivative of f: $f = F'$. The **quantile** reads the CDF backwards: given a probability u, the quantile is the x with $F(x) = u$; the book's q_α is the

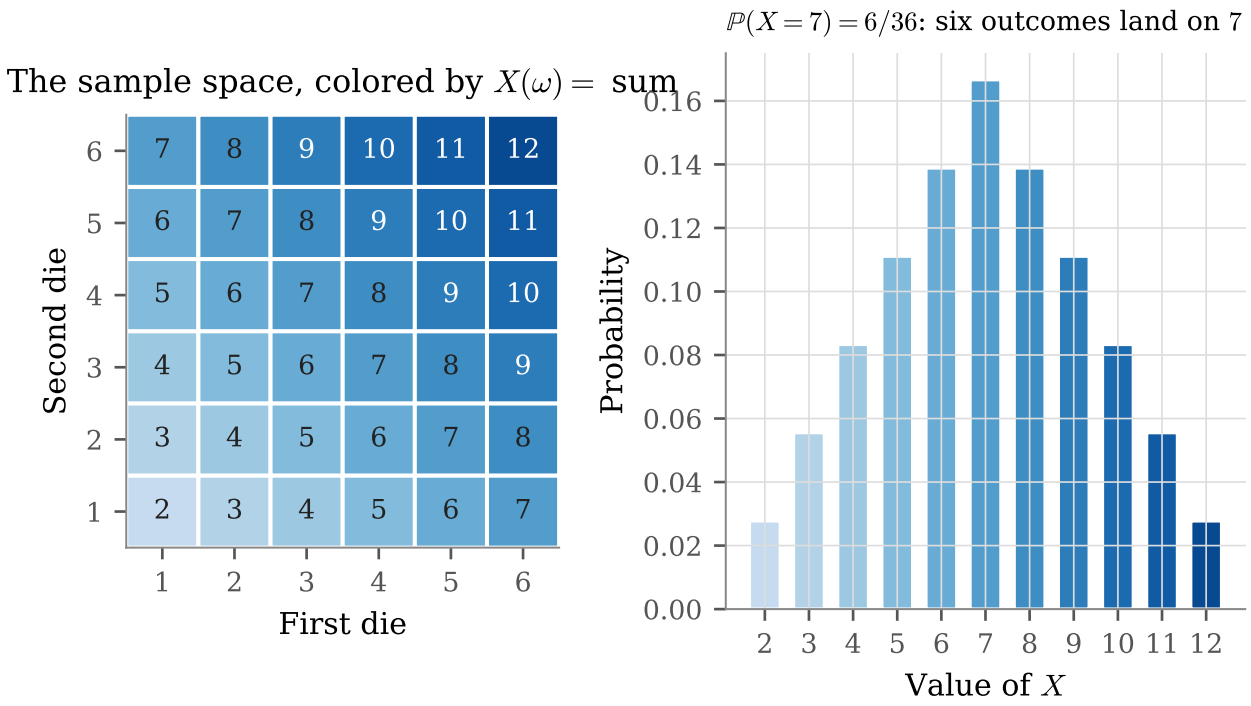


Figure A.5: A random variable as a function on the sample space. Left: the 36 outcomes of two dice, each colored by the value the function $X =$ 'sum' assigns it. Right: the distribution of X on the real line.

case $u = 1 - \alpha$, indexed by the probability *above* the cutoff. Figure A.6 draws the three on one example, the standard Gaussian. At the top, the shaded area under the density between $a = -1$ and $b = 1$ is $\mathbb{P}(a \le X \le b) = 0.683$. At the bottom, the same probability is the rise of the CDF from $F(a)$ to $F(b)$, and reading the CDF sideways at 0.95 gives the quantile $q_{0.05} = 1.645$ of Section 6.5. The same three ideas work for a discrete random variable with sums in place of integrals: the PMF is the list of probabilities, the CDF the running total, and the CDF jumps at each possible value instead of rising smoothly.

Where the book uses it. Convergence in distribution (Definition 4.1) compares CDFs, and the jumps of a discrete CDF are why only continuity points count. Every probability in Chapter 6 is a CDF lookup. The density of a transformed variable, such as $aX + b$ or X^2, is found by writing its CDF and differentiating (Section 6.3, Chapter 2).

A.7 Three inequalities

The triangle inequality: $|x + y| \le |x| + |y|$, and so $|x - z| \le |x - y| + |y - z|$: a detour through y is never shorter than the direct distance. Proofs use it to split an error into two pieces, for instance $|T_n - \mu| \le |T_n - \mathbb{E}[T_n]| + |\mathbb{E}[T_n] - \mu|$ in Proposition 3.3: if the total is at least ϵ, one of the two pieces is at least $\epsilon/2$.

Markov's inequality: if $Y \ge 0$ then $\mathbb{P}(Y \ge a) \le \mathbb{E}[Y]/a$ for every $a > 0$. The picture (Figure A.7, left) is the proof: the step function $a \cdot \mathbb{1}(Y \ge a)$ never exceeds Y, so its expectation, $a\mathbb{P}(Y \ge a)$, is at most $\mathbb{E}[Y]$. A non-negative variable with a small mean is rarely large.

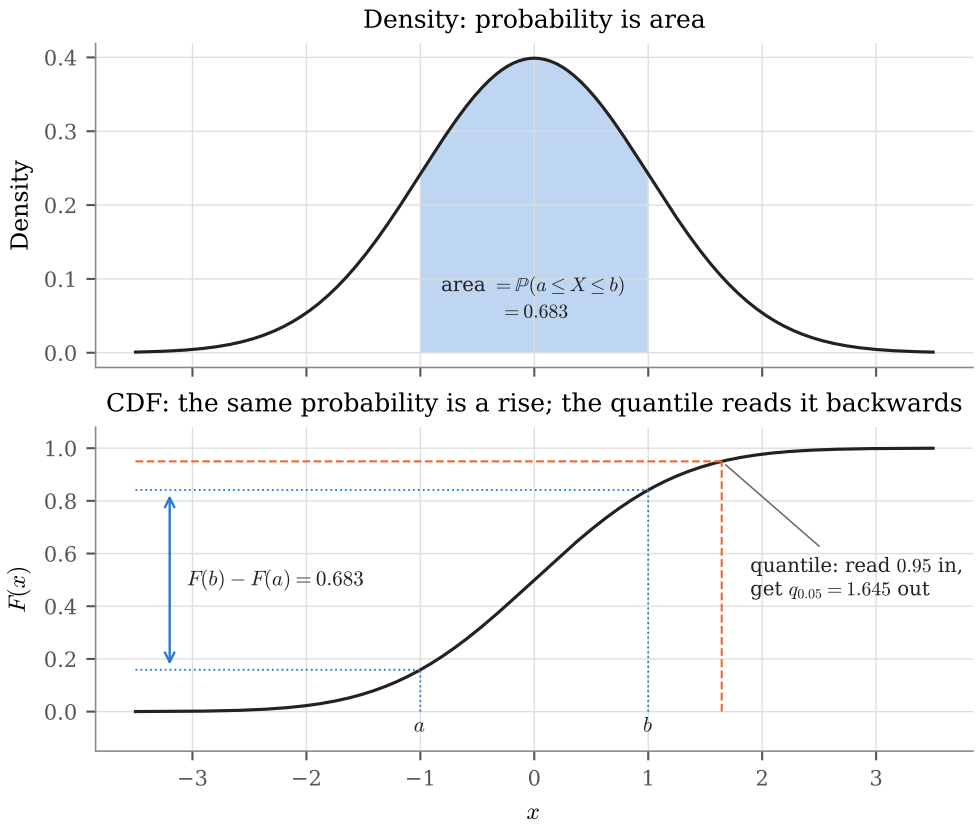


Figure A.6: Density, CDF and quantile for Gauss$(0, 1)$. Top: the density, with the area between $a = -1$ and $b = 1$ shaded. Bottom: the CDF, with the rise from $F(a)$ to $F(b)$ and the quantile $q_{0.05}$ marked.

Chebyshev's inequality: for any random variable with finite variance, $\mathbb{P}(|T - \mathbb{E}[T]| \ge \epsilon) \le \mathbb{V}\mathrm{ar}(T)/\epsilon^2$. It is Markov's inequality applied to $Y = (T - \mathbb{E}[T])^2$ with $a = \epsilon^2$. In standard-deviation units it says $\mathbb{P}(|T - \mathbb{E}[T]| \ge k\sigma) \le 1/k^2$ for every distribution whatsoever, which is both its strength (universal) and its weakness (loose: for a Gaussian the true two-standard-deviation tail is 0.046, the bound says 0.25). The right of Figure A.7 draws the bound $1/k^2$ against the exact tail probabilities of a Gaussian and of an Expo(1) variable: the bound holds for every distribution and is far from tight for these two.

Where the book uses it. The Law of Large Numbers (Theorem 3.5) is Chebyshev's inequality with $n \to \infty$; the shrinking-variance criterion is the same with the triangle inequality in front; convergence in mean implies convergence in probability by Markov (Appendix B); Hoeffding's inequality (Theorem 4.9) is the sharper cousin for bounded variables.

A.8 Convex functions

A function is **convex** if the chord joining any two points of its graph lies on or above the graph, and **concave** if every chord lies on or below (Figure A.8). For a function with a second derivative, convex means $g'' \ge 0$ (the slope keeps increasing) and concave means $g'' \le 0$. x^2, e^x and $1/x$ on $(0, \infty)$ are convex; $\log x$ and $\sqrt{x}$ are concave; $\sin x$ is neither. Averaging interacts with convexity

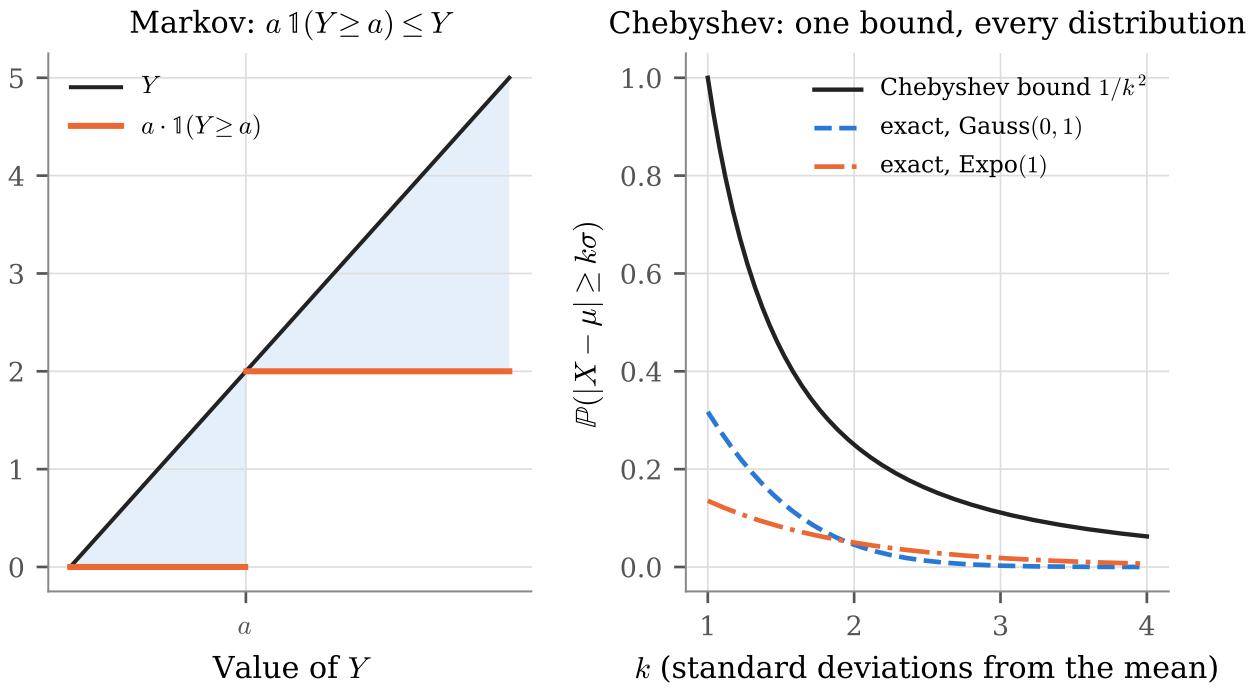


Figure A.7: Left: Y and the step function $a \cdot \mathbb{1}(Y \geq a)$. Right: Chebyshev's bound $1/k^2$ against the exact probabilities $\mathbb{P}(|T - \mathbb{E}[T]| \geq k\sigma)$ for a Gaussian T and for $T \sim \text{Expo}(1)$.

in one direction only: for convex g, the average of g over two points is at least g of their average, which is the chord sitting above the curve at its midpoint.

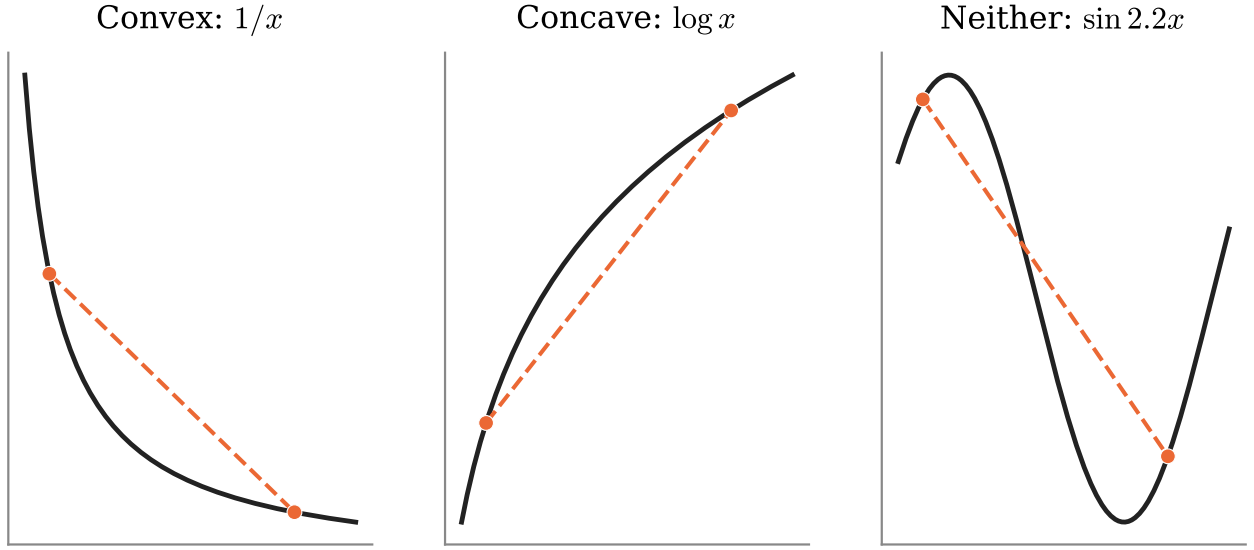


Figure A.8: Convex: chords above the graph. Concave: chords below. Neither: a chord can cross the graph.

Where the book uses it. Jensen's inequality (Theorem 7.8) is the chord statement with the two-point average replaced by an expectation; it explains why $1/\bar{T}_n$ overestimates λ and why $\sqrt{\cdot}$ of an unbiased estimator is biased (Chapter 7). The quadratic in the solve interval of Chapter 8 is

convex, which is why the inequality holds between its roots. Likelihoods in Part IV are maximized by finding where a concave function's slope is zero.

A.9 Tangent lines, the mean value theorem and Taylor's expansion

The derivative $g'(\theta)$ is the slope of the tangent line to the graph of g at θ, and the tangent line $g(\theta) + g'(\theta)(x - \theta)$ is the best straight-line stand-in for g near θ. Two facts make this precise.

The mean value theorem. For a smooth g and any $a < b$ there is a point c between them where the tangent is parallel to the chord:

$$g(b) - g(a) = g'(c)\,(b - a) \quad \text{for some } c \in (a, b).$$

The change in g over an interval equals the slope somewhere inside the interval times the interval's length: on the left of Figure A.9, the tangent at c is parallel to the chord from a to b. You never learn where c is, and you never need to: in the delta method it is squeezed between two points that converge to θ, so $g'(c)$ converges to $g'(\theta)$.

The first-order Taylor expansion. Near θ,

$$g(x) = g(\theta) + g'(\theta)(x - \theta) + (\text{an error of order } (x - \theta)^2),$$

so on a window of width h around θ the tangent line is wrong by about h^2, which is negligible compared with the h-sized changes along the line. The right of Figure A.9 draws this for $g(x) = 1/x$: the shaded error is tiny on the window $\theta \pm \sigma/\sqrt{n}$ where an estimator lives, which is why the delta method can treat g as linear there. Adding the next term, $\frac{1}{2}g''(\theta)(x - \theta)^2$, gives the second-order expansion, and it too has an exact form with an unknown in-between point, **Taylor's theorem with the Lagrange remainder**: for a twice differentiable g,

$$g(b) = g(a) + g'(a)(b - a) + \tfrac{1}{2}g''(c)(b - a)^2 \quad \text{for some } c \text{ between } a \text{ and } b.$$

At a maximum, where $g'(a) = 0$, it says that the drop from the top to b is exactly half a squared distance times a curvature taken somewhere in between, which is the step behind the likelihood ratio test. The pattern continues: for a g with $k + 1$ derivatives, $g(x) = \sum_{i=0}^{k} \frac{1}{i!} g^{(i)}(\theta)(x - \theta)^i +$ (an error of order $(x - \theta)^{k+1}$), where $g^{(i)}$ is the ith derivative and $g^{(0)} = g$.

Where the book uses it. The delta method (Theorem 5.4) is the mean value theorem followed by Slutsky's theorem; its intuition box is the tangent-line picture. The second-order expansion is behind the remark on $g'(\theta) = 0$ in Section 5.4, and Taylor's theorem with the Lagrange remainder proves Wilks' theorem in Chapter 15. Exercise 16.7 uses the terms of order three and four to compare G^2 with Pearson's statistic.

A.10 Exponentials and logarithms

Two facts about e recur. First, $(1 - x/n)^n \to e^{-x}$ as $n \to \infty$; the left of Figure A.10 shows the curves for $n = 1, 2, 5, 20$ climbing toward e^{-x}, each drawn up to $x = n$, where it reaches 0. More generally $(1 + a/n)^n \to e^a$. It is the step that turns the finite-n formula in the maximum-of-uniforms proof (Example 4.6) into an exponential limit, and it reappears whenever a product of n nearly-equal factors is pushed to the limit. Second, the logarithm turns multiplication into addition,

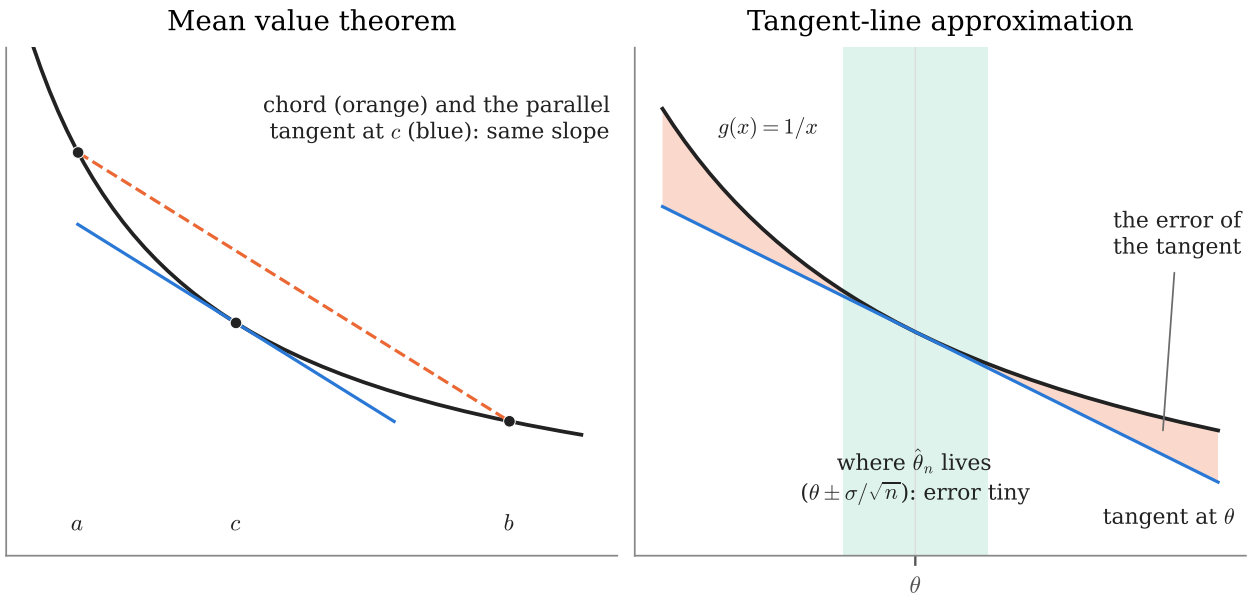


Figure A.9: Left: the mean value theorem, with the chord from a to b (orange) and the parallel tangent at c (blue). Right: $g(x) = 1/x$ and its tangent line at θ, with the error between them shaded over the window $\theta \pm \sigma/\sqrt{n}$.

$\log(ab) = \log a + \log b$: on the right of Figure A.10, equal ratios along the axis become equal steps up the curve, so a product becomes a sum. A likelihood is a product of n densities, a number so small it underflows a computer for moderate n; its logarithm is a sum of n ordinary terms, and a sum of n terms is what the Law of Large Numbers and the Central Limit Theorem know how to handle. That is the entire reason Part IV works with log-likelihoods. Because log is strictly increasing, maximizing the likelihood and maximizing the log-likelihood give the same answer.

A.11 Orders of magnitude: how fast is 'fast'

The book keeps saying that an error is 'of order $1/\sqrt{n}$', that a bound 'goes to zero like $1/n$', or 'exponentially fast'. These are statements about how a quantity scales with n, ignoring constants. Figure A.11 draws the three rates that matter here on ordinary axes and on log-log axes, where a power law n^{-c} is a straight line of slope $-c$ and an exponential bends away below every power law. On ordinary axes the three look alike at first; on log-log axes $1/\sqrt{n}$ and $1/n$ are straight lines whose slopes are the exponents, and the exponential drops away below both. Practical translation of each rate:

- $1/\sqrt{n}$: four times the data halves the error; a hundred times the data buys one decimal place. This is the rate of every estimator built from an average and of every confidence interval in this book.
- $1/n$: twice the data halves the error. Chebyshev's bound and the maximum-of-uniforms example have this rate; so does the bias of $1/\bar{T}_n$ in Chapter 7, which is why the bias is swamped by the $1/\sqrt{n}$ standard error.
- e^{-cn}: each fixed addition to n multiplies the error by the same factor below 1. Hoeffding's bound has this rate, which is why it overtakes Chebyshev's for large n (Figure 4.5).

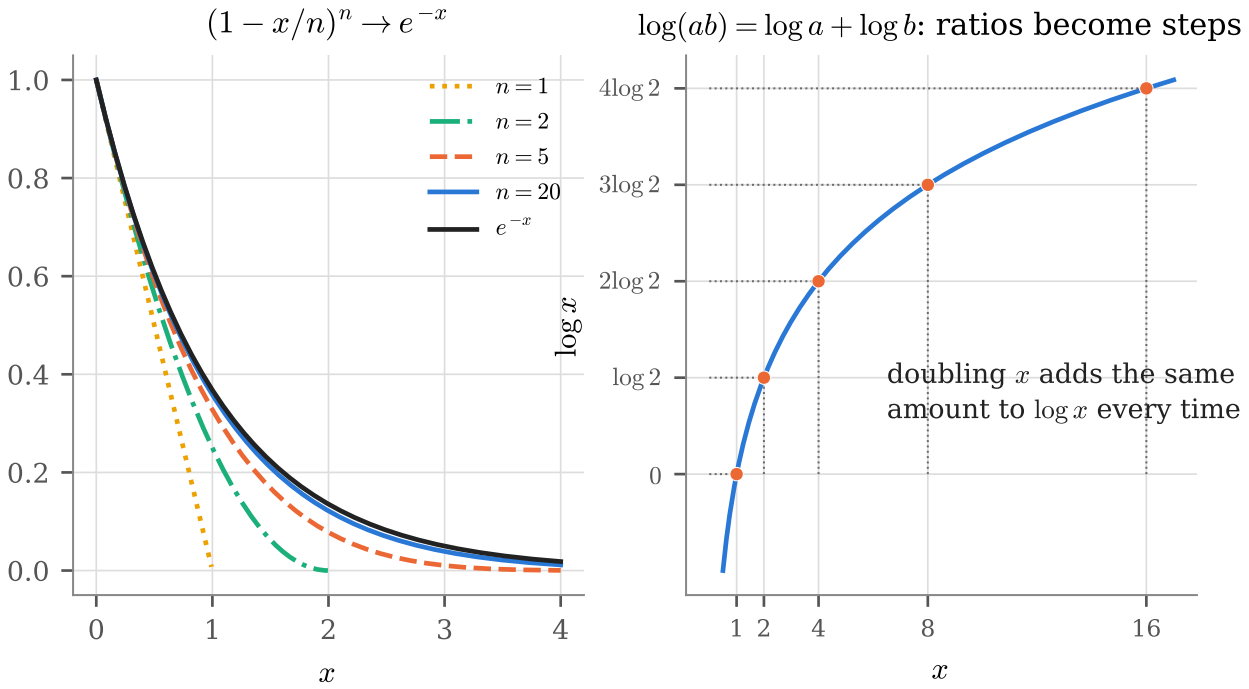


Figure A.10: Left: $(1 - x/n)^n$ for $n = 1, 2, 5, 20$, each drawn up to $x = n$, and the limit e^{-x}. Right: the graph of the logarithm.

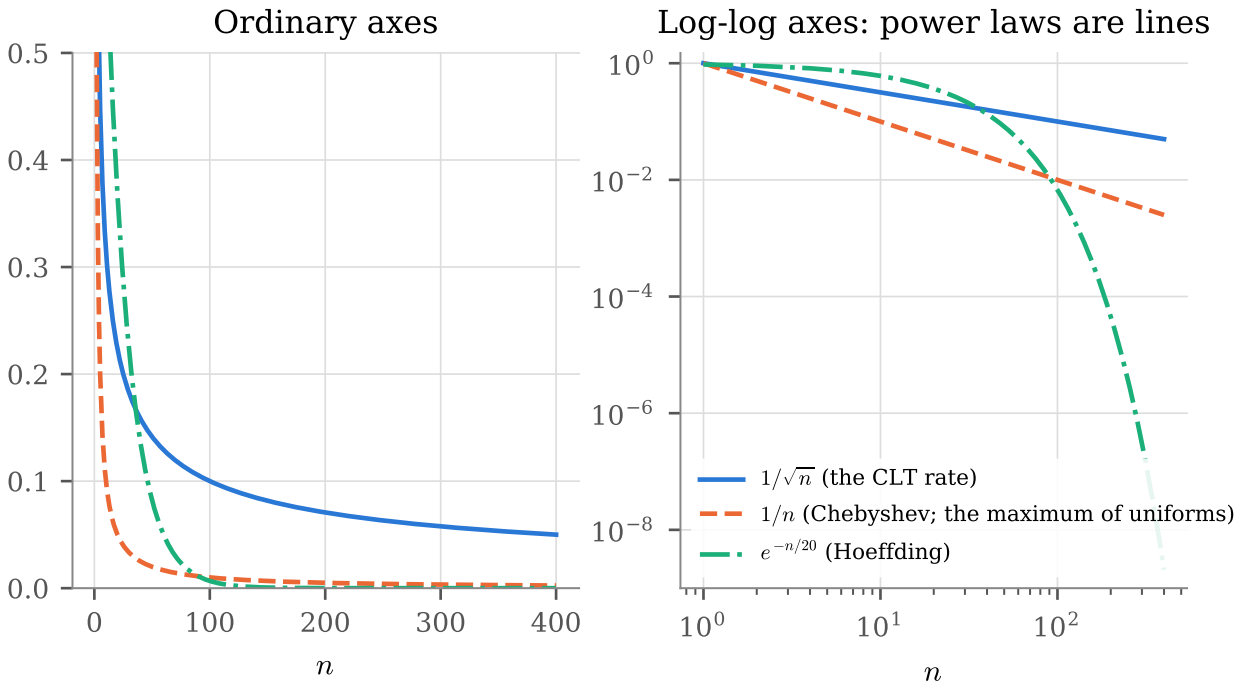


Figure A.11: Three rates of decay, $1/\sqrt{n}$, $1/n$ and an exponential. Left: on ordinary axes. Right: on log-log axes.

A.12 Functions of several variables: gradient and Hessian

A parameter with several coordinates, like (μ, σ^2) for a Gaussian, turns a log-likelihood into a function of several variables, and maximizing it needs the several-variable versions of the derivative and the second derivative.

Partial derivatives and the gradient. For $h(a, b)$, the partial derivative $\partial h/\partial a$ is the ordinary derivative in a with b held fixed: the slope of the surface walking in the a direction. The **gradient** collects them in a vector, $\nabla h = (\partial h/\partial a, \partial h/\partial b)$. It points in the direction in which h climbs fastest, it is perpendicular to the contour line through its point, and its length is the steepness of the climb. On the left of Figure A.12, for the concave function $-(a^2 + ab + b^2)$, each arrow points uphill, perpendicular to the contour through its base, and the arrows shrink to nothing at the top, the one point where the gradient is 0. At the top of a hill the surface is flat in every direction, so the gradient is the zero vector: $\nabla h = 0$ is the first-order condition, one equation per coordinate.

The Hessian and the second-derivative test. The **Hessian** is the matrix of second partial derivatives,

$$\nabla^2 h = \begin{pmatrix} \partial^2 h/\partial a^2 & \partial^2 h/\partial a\,\partial b \\ \partial^2 h/\partial b\,\partial a & \partial^2 h/\partial b^2 \end{pmatrix},$$

which is symmetric for the smooth functions of this book. A zero gradient alone does not certify a maximum: the saddle $h(a, b) = a^2 - b^2$ has $\nabla h = 0$ at the origin, which is a minimum along a and a maximum along b (Figure A.12, right). The second-derivative test settles it. A symmetric matrix H is **negative definite** if $u^\top H u < 0$ for every nonzero vector u: the function curves down in every direction. If at a point where $\nabla h = 0$ the Hessian is negative definite, the point is a local maximum. For a 2×2 Hessian, negative definite means the top-left entry is negative and the determinant is positive. For the saddle the Hessian is $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$, whose determinant is -4: not negative definite. For $h(a, b) = -(a^2 + ab + b^2)$ it is $\begin{pmatrix} -2 & -1 \\ -1 & -2 \end{pmatrix}$, top-left $-2 < 0$, determinant $3 > 0$: negative definite, and the origin is the maximum.

Concave in several variables. Along the line $\theta + tu$, the function $t \mapsto h(\theta + tu)$ has first derivative $\nabla h(\theta + tu)^\top u$ and second derivative $u^\top \nabla^2 h(\theta + tu)\, u$, by the chain rule. Suppose the Hessian is negative definite at every point of a domain that contains the segment between any two of its points. Then the second derivative along every line is negative, so every slice through the function, in every direction, is strictly concave in one variable. If $\nabla h(\hat{\theta}) = 0$, the slice from $\hat{\theta}$ toward any other point θ has zero slope at $\hat{\theta}$, so by the one-variable result (Proposition 11.5) it is highest there: $h(\theta) < h(\hat{\theta})$. A point with zero gradient, if there is one, is the unique maximizer, exactly as in one variable.

Where the book uses it. The Gaussian maximum likelihood estimator (Section 11.6) is found by a two-step argument, and its Hessian at the maximum is negative definite; the EM algorithm (Section 11.8) maximizes in five coordinates; Chapter 12 turns the Hessian of the log-likelihood into the Fisher information matrix.

A.13 Vectors of independent Gaussians and rotations

Cochran's theorem (Theorem 14.3) rests on one fact: a vector of independent standard Gaussians looks the same after any rotation of the axes. Three pieces of calculus and linear algebra make it precise.

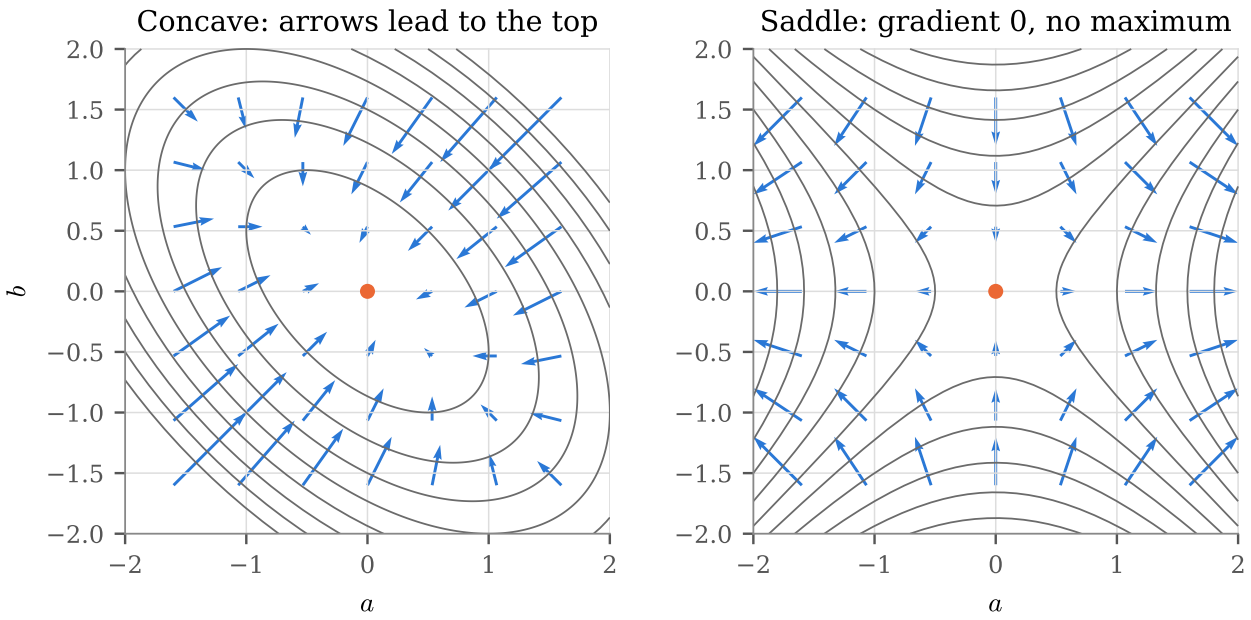


Figure A.12: Left: contours of $h(a,b) = -(a^2 + ab + b^2)$, with the gradient drawn as an arrow at a grid of points. Right: the saddle $h(a,b) = a^2 - b^2$.

Joint densities. If $Z_1, \dots, Z_n$ are independent continuous random variables with densities $f_1, \dots, f_n$, the vector $Z = (Z_1, \dots, Z_n)$ has the joint density $f(z) = f_1(z_1) \cdots f_n(z_n)$ on $\mathbb{R}^n$: the probability that Z lands in a region is the integral of f over it, and the product form is what independence means for densities. Conversely, if a joint density factors as a product of a function of z_1 alone, a function of z_2 alone and so on, the coordinates are independent, each with the density its factor gives after normalizing. For i.i.d. standard Gaussians,

$$f(z) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-z_i^2/2} = (2\pi)^{-n/2} \exp\Big(-\frac{\|z\|^2}{2}\Big), \qquad \|z\|^2 = z_1^2 + \cdots + z_n^2.$$

The density depends on z only through its length $\|z\|$, so it is the same at every point of a sphere around the origin: the cloud of draws is round (Figure 14.3).

Orthogonal matrices. An $n \times n$ matrix Q is **orthogonal** if $Q^\top Q = I$, which says that its columns are unit vectors perpendicular to each other: an orthonormal basis of $\mathbb{R}^n$ written as a matrix. Since Q is square, $Q^\top Q = I$ makes $Q^\top$ the inverse of Q, so $QQ^\top = I$ as well: the rows are orthonormal too, and $Q^\top$ is itself orthogonal. Multiplying by Q rotates (and possibly reflects) the axes. Two consequences: Q preserves lengths, $\|Qz\|^2 = (Qz)^\top(Qz) = z^\top Q^\top Q z = \|z\|^2$, and $\det Q = \pm 1$, since $\det(Q^\top Q) = (\det Q)^2 = 1$. Any unit vector u_1 can be made the first row of an orthogonal matrix: extend it to a basis of $\mathbb{R}^n$ and run the Gram–Schmidt process, which subtracts from each new vector its components along the previous ones and normalizes, producing an orthonormal basis $u_1, \dots, u_n$ that keeps u_1 as it is; the same process keeps any orthonormal $u_1, \dots, u_k$ as the first k rows. The proof of Theorem 14.3 uses $u_1 = (1, \dots, 1)/\sqrt{n}$.

Change of variables for a linear map. If Z has density f on $\mathbb{R}^n$ and A is an invertible $n \times n$ matrix, then $Y = AZ$ has density

$$f_Y(y) = \frac{f(A^{-1}y)}{|\det A|}.$$

This is the substitution rule of multivariable calculus: a linear map multiplies every volume by $|\det A|$, so the probability $f(z)\,\mathrm{d}z$ of a small box around z is spread over a box of volume $|\det A|\,\mathrm{d}z$ around $y = Az$. For $n = 1$ it is the familiar $f_Y(y) = f(y/a)/|a|$ for $Y = aZ$, the affine property of Chapter 6.

Rotation invariance. Put the three together. Let Z have i.i.d. Gauss$(0,1)$ coordinates and Q be orthogonal, and set $Y = QZ$. Then $A^{-1} = Q^{-1} = Q^\top$, $|\det Q| = 1$, and $\|Q^\top y\| = \|y\|$, so

$$f_Y(y) = (2\pi)^{-n/2} \exp\Big(-\frac{\|Q^\top y\|^2}{2}\Big) = (2\pi)^{-n/2} \exp\Big(-\frac{\|y\|^2}{2}\Big) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-y_i^2/2}.$$

The density of Y is the same product, so $Y_1, \dots, Y_n$ are again i.i.d. Gauss$(0,1)$, and $\|Y\| = \|Z\|$. In words: rotate a round Gaussian cloud and you get the same round Gaussian cloud, with the new coordinates as independent as the old ones. Exercise 14.2 does the computation by hand for $n = 2$ and the 45° rotation.

Where the book uses it. Theorem 14.3 rotates the standardized data so that one coordinate is $\sqrt{n}\bar{Z}_n$ and the others carry the deviations; Theorem 16.5 rotates the standardized deviations of the counts with an orthogonal matrix whose first row is $(\sqrt{p_1^0}, \dots, \sqrt{p_K^0})$, the direction in which those deviations cannot move; Theorem 20.11 rotates the errors of a regression with the first two axes along $(1, \dots, 1)$ and along the centered explanatory values; Chapter 21 uses the same rotation with the regressors' subspace in place of the line through $(1, \dots, 1)$.

A.14 Gaussian vectors with a covariance matrix

The limits in Chapters 12, 13 and 15 are Gaussian vectors whose coordinates have different variances and are correlated. This entry says what such a vector is, building on the round Gaussian cloud of the previous entry, and proves the facts the book uses about it.

Covariance matrices. For a random vector $Z = (Z_1, \dots, Z_d)$ with finite variances, the covariance matrix $\mathbb{C}\mathrm{ov}(Z)$ has entries $\mathbb{C}\mathrm{ov}(Z_j, Z_k)$, with the variances on the diagonal. For a matrix B, $\mathbb{C}\mathrm{ov}(BZ) = B\,\mathbb{C}\mathrm{ov}(Z)\,B^\top$: each entry is the covariance of two linear combinations of the Z_j, expanded term by term. A covariance matrix Σ is symmetric, and $a^\top \Sigma a = \mathbb{V}\mathrm{ar}(a^\top Z) \geq 0$ for every vector a. It is **positive definite** when $a^\top \Sigma a > 0$ for every $a \neq 0$, that is, when no nonzero linear combination of the coordinates is constant; a positive definite matrix is invertible, since $\Sigma a = 0$ would give $a^\top \Sigma a = 0$. Conversely, an invertible covariance matrix is positive definite. If $a^\top \Sigma a = \mathbb{V}\mathrm{ar}(a^\top Z) = 0$, then $a^\top Z$ is constant, so $(\Sigma a)_j = \mathbb{C}\mathrm{ov}(Z_j, a^\top Z) = 0$ for every j; that is, $\Sigma a = 0$, and invertibility forces $a = 0$. The inverse of a positive definite matrix is positive definite too: for $a \neq 0$, $b = \Sigma^{-1} a$ is not 0, and $a^\top \Sigma^{-1} a = b^\top \Sigma b > 0$. The Fisher information matrix is the covariance matrix of the score (Definition 12.3), so when it is invertible it is positive definite, and so is its inverse.

Definition. Let $m \in \mathbb{R}^d$ and let Σ be a symmetric positive definite $d \times d$ matrix. A random vector Z has the **Gaussian distribution** $\mathrm{Gauss}_d(m, \Sigma)$ if it has the distribution of $m + AW$, where W has d independent Gauss$(0,1)$ coordinates and A is an invertible $d \times d$ matrix with $AA^\top = \Sigma$. The definition concerns only the distribution of Z, which is all that a limit in distribution, such as the one in the Central Limit Theorem, delivers. Such an A exists for every positive definite Σ; one choice is the lower-triangular matrix of the Cholesky factorization, a standard result of linear algebra, and for $d = 2$

it is explicit: for $\Sigma = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, $A = \begin{pmatrix} \sqrt{a} & 0 \\ b/\sqrt{a} & \sqrt{c - b^2/a} \end{pmatrix}$. $\mathbb{C}\mathrm{ov}(Z) = A\,\mathbb{C}\mathrm{ov}(W)A^\top = AA^\top = \Sigma$, and every nonzero linear combination $a^\top Z$ has the distribution of $a^\top m + (A^\top a)^\top W$, a sum of independent Gaussians plus a constant, so it is $\mathrm{Gauss}(a^\top m, a^\top \Sigma a)$ by the sum rule and the affine property of Chapter 6. That is the description of Gauss_d used in the recall lines of Chapter 13. (Conversely, a vector all of whose linear combinations are Gaussian, with positive definite covariance matrix, has this form; the proof uses characteristic functions, which this book does not develop, and nothing in the book needs it.)

The distribution depends on A only through Σ. By the change of variables formula of the previous entry, with f the density of W,

$$f_Z(z) = \frac{f\big(A^{-1}(z-m)\big)}{|\det A|} = \frac{1}{(2\pi)^{d/2}|\det A|} \exp\Big(-\tfrac{1}{2}\|A^{-1}(z-m)\|^2\Big).$$

Now $\|A^{-1}(z-m)\|^2 = (z-m)^\top (A^{-1})^\top A^{-1}(z-m) = (z-m)^\top (AA^\top)^{-1}(z-m)$, and $|\det A| = \sqrt{\det(AA^\top)}$, so

$$f_Z(z) = \frac{1}{(2\pi)^{d/2}\sqrt{\det \Sigma}} \exp\Big(-\tfrac{1}{2}(z-m)^\top \Sigma^{-1}(z-m)\Big),$$

a formula with only m and Σ in it: two choices of A with the same $AA^\top$ give the same distribution. For $d = 1$ it is the Gaussian density of Chapter 6. The contours of the density are the ellipses $(z-m)^\top \Sigma^{-1}(z-m) = \text{constant}$, drawn in Figure 15.2.

Linear maps. If $Z \sim \mathrm{Gauss}_d(m, \Sigma)$ and B is a $k \times d$ matrix of rank k (its rows are linearly independent), then $BZ \sim \mathrm{Gauss}_k(Bm, B\Sigma B^\top)$. To see it, note that BZ has the distribution of $Bm + CW$ with $C = BA$, a $k \times d$ matrix of rank k, since A is invertible. Run the Gram–Schmidt process on the rows of C to get orthonormal vectors $q_1, \ldots, q_k$ spanning the same space, and let Q_1 be the $k \times d$ matrix with these rows. Each row of C is a combination of $q_1, \ldots, q_k$, so $C = LQ_1$ with $L = CQ_1^\top$, a $k \times k$ matrix; it is invertible, since $k = \operatorname{rank} C = \operatorname{rank}(LQ_1) \le \operatorname{rank} L$. Extend $q_1, \ldots, q_k$ to a basis of $\mathbb{R}^d$ and continue the Gram–Schmidt process, as in the previous entry, to get an orthogonal matrix Q whose first k rows are $q_1, \ldots, q_k$. By rotation invariance QW has independent standard Gaussian coordinates, and $W' = Q_1 W$ is its first k of them. So BZ has the distribution of $Bm + LW'$, with L invertible and $LL^\top = LQ_1Q_1^\top L^\top = CC^\top = B\Sigma B^\top$, which is the definition.

The quadratic form. If $Z \sim \mathrm{Gauss}_d(m, \Sigma)$, then $(Z-m)^\top \Sigma^{-1}(Z-m)$ has the distribution of $W^\top A^\top (AA^\top)^{-1} AW = W^\top W$, which is χ^2_d (Lemma 15.1): the squared distance from the center in units of the noise, which Figure 15.2 draws.

Where the book uses it. The multivariate CLT and the asymptotic normality of estimators with several coordinates (Chapters 5, 12 and 13) have Gauss_d limits when their covariance matrices are positive definite. The quadratic form gives the chi-squared limits of Wald's, the likelihood ratio and the score tests (Chapter 15), and the linear-map rule gives the common limit of all three with nuisance parameters (Theorem 15.12). The vector of counts in Chapter 16, whose coordinates add up to n, has a covariance matrix that is not positive definite; that chapter rotates the standardized deviations of the counts so that the direction in which they cannot move is one coordinate and sets it aside (Theorem 16.5), and for the three tests drops one category (Section 16.3); either way the covariance matrix that remains is positive definite.

A.15 Conditional distributions and conditional expectation

For a pair (X, Y) with a joint density $h(x, y)$, the **marginal density** of X is $h(x) = \int h(x, y)\,\mathrm{d}y$, and wherever $h(x) > 0$ the **conditional density** of Y given $X = x$ is

$$h(y \mid x) = \frac{h(x, y)}{h(x)} :$$

the slice of the joint density along the vertical line at x, rescaled so that it integrates to 1. For discrete pairs, PMFs replace densities and sums replace integrals: $\mathbb{P}(Y = y \mid X = x) = \mathbb{P}(X = x, Y = y)/\mathbb{P}(X = x)$. The **conditional expectation** $\mathbb{E}[Y \mid X = x] = \int y\, h(y \mid x)\,\mathrm{d}y$ is the mean of that slice, one number for each x. Figure A.13 draws the simplest case: two dice, X the first and Y the sum. Given $X = x$, the sum is $x + d$ with $d = 1, \ldots, 6$ equally likely, so $\mathbb{E}[Y \mid X = x] = x + 3.5$.

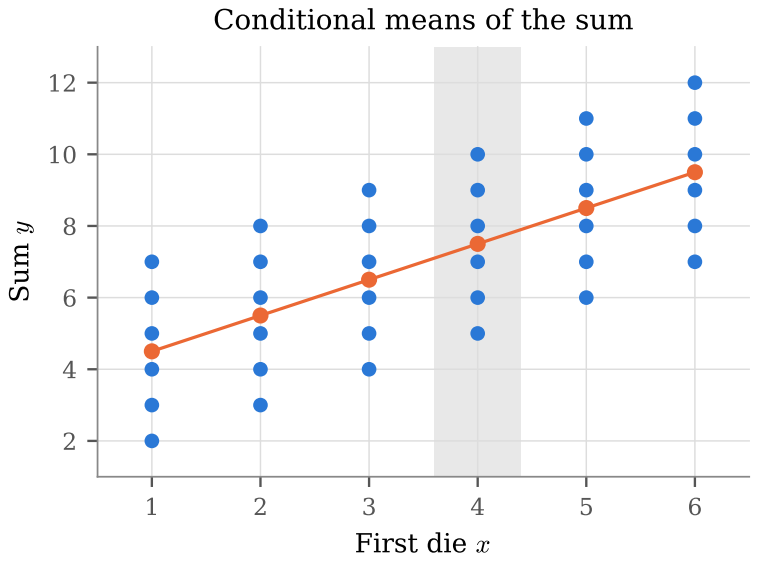


Figure A.13: Two dice: the 36 equally likely outcomes as points (first die, sum), and the conditional means of the sum given the first die (orange); the shaded column holds the six outcomes with first die 4.

Writing $f(x) = \mathbb{E}[Y \mid X = x]$, the random variable $f(X)$, the conditional mean evaluated at the observed X, is written $\mathbb{E}[Y \mid X]$. Three facts handle it.

- **The tower property**: $\mathbb{E}\big[\mathbb{E}[Y \mid X]\big] = \mathbb{E}[Y]$. For densities, $\mathbb{E}[f(X)] = \int f(x)h(x)\,\mathrm{d}x = \iint y\, h(y \mid x)h(x)\,\mathrm{d}y\,\mathrm{d}x = \iint y\, h(x, y)\,\mathrm{d}y\,\mathrm{d}x = \mathbb{E}[Y]$. For the dice, the average of $x + 3.5$ over $x = 1, \ldots, 6$ is 7, the mean of the sum.
- **Taking out what is known**: $\mathbb{E}[g(X)Y \mid X] = g(X)\,\mathbb{E}[Y \mid X]$ for any function g with $\mathbb{E}|g(X)Y| < \infty$: once $X = x$ is fixed, $g(x)$ is a constant and comes out of the mean of the slice. With the tower property, $\mathbb{E}[g(X)Y] = \mathbb{E}\big[g(X)\,\mathbb{E}[Y \mid X]\big]$.
- **Conditional variance**: $\mathbb{V}\mathrm{ar}(Y \mid X = x) = \mathbb{E}[Y^2 \mid X = x] - \big(\mathbb{E}[Y \mid X = x]\big)^2$ is the variance of the slice, so it is non-negative; hence $\mathbb{E}[Y \mid X]^2 \le \mathbb{E}[Y^2 \mid X]$, and by the tower property $\mathbb{E}\big[\mathbb{E}[Y \mid X]^2\big] \le \mathbb{E}[Y^2]$.

Where the book uses it. The posterior of Chapter 18 is a conditional density of the parameter given the data; the regression function of Chapter 20 is a conditional expectation, and the proof that it is the best predictor (Theorem 20.2) uses all three facts.

A.16 Matrices, rank and projections

Chapter 21 writes n regression equations as one, $Y = X\beta + \epsilon$, and solves them by geometry. This entry collects the linear algebra that takes.

Vectors and matrices. A vector $v \in \mathbb{R}^n$ is a column of n numbers; $v^\top$ is the same numbers as a row, so $u^\top v = \sum_i u_i v_i$ is a number, the inner product, and $\|v\|^2 = v^\top v$ is the squared length. Two vectors are **perpendicular** (orthogonal) when $u^\top v = 0$, and then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$ (Pythagoras: the cross term $2u^\top v$ vanishes). An $n \times p$ matrix X sends $\beta \in \mathbb{R}^p$ to $X\beta \in \mathbb{R}^n$, which is the combination $\beta_1 x_{(1)} + \cdots + \beta_p x_{(p)}$ of the columns of X with the entries of β as weights. The transpose rule $(AB)^\top = B^\top A^\top$ and the inverse rule $(AB)^{-1} = B^{-1}A^{-1}$ hold whenever the products and inverses exist; the **trace** $\operatorname{tr}(A)$ of a square matrix is the sum of its diagonal entries, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ whenever both products are square, and tr is linear, so it passes through expectations.

Column space and rank. The **column space** of X is the set of all vectors $X\beta$, a subspace of $\mathbb{R}^n$ (closed under sums and multiples). The columns are **linearly independent** when $X\beta = 0$ only for $\beta = 0$, that is, when no column is a combination of the others; then every vector in the column space is $X\beta$ for exactly one β, and the column space has dimension p. The **rank** of X is the dimension of its column space, so 'rank p' with $p \le n$ means the columns are independent. Independent columns make $X^\top X$ invertible: $X^\top X a = 0$ gives $a^\top X^\top X a = \|Xa\|^2 = 0$, so $Xa = 0$ and $a = 0$.

Orthogonal projection. Let S be a subspace of $\mathbb{R}^n$ of dimension d. The Gram–Schmidt process of Appendix A.13, fed a basis of S and then vectors completing it to a basis of $\mathbb{R}^n$, gives an orthonormal basis $u_1, \ldots, u_n$ of $\mathbb{R}^n$ whose first d vectors span S; fed a basis of a smaller subspace $S_0 \subset S$ first, it gives one adapted to the chain $S_0 \subset S \subset \mathbb{R}^n$. Every y then splits as $y = s + r$ with $s = \sum_{k \le d} (u_k^\top y) u_k \in S$ and $r = \sum_{k > d} (u_k^\top y) u_k$ perpendicular to every vector of S, and uniquely, since two such splits would differ by a vector of S perpendicular to S, which is 0; s is the **orthogonal projection** of y onto S, and it is the point of S closest to y, since for any $s' \in S$, $\|y - s'\|^2 = \|r\|^2 + \|s - s'\|^2 \ge \|r\|^2$ by Pythagoras. Figure A.14 draws the case of a line. The map $y \mapsto s$ is linear, so it is a matrix P, with $P^2 = P$ (projecting twice changes nothing) and $P^\top = P$; for S the column space of a rank-p matrix X, the condition that $y - Py$ be perpendicular to every column reads $X^\top(y - Xb) = 0$, whose solution $b = (X^\top X)^{-1} X^\top y$ gives

$$P = X(X^\top X)^{-1} X^\top, \qquad \operatorname{tr}(P) = \operatorname{tr}\big((X^\top X)^{-1} X^\top X\big) = p.$$

$I - P$ is the projection onto the **orthogonal complement** of S, the vectors perpendicular to S, a subspace of dimension $n - p$. In the adapted basis, $P = \sum_{k \le d} u_k u_k^\top$, so if $S_0 \subset S$ are nested subspaces of dimensions $d_0 \le d$ with projections $P_0 = \sum_{k \le d_0} u_k u_k^\top$ and P, then $PP_0 = P_0$, and $P - P_0 = \sum_{d_0 < k \le d} u_k u_k^\top$ is the projection onto the span of $u_{d_0+1}, \ldots, u_d$, the part of S perpendicular to S_0, of dimension $d - d_0$.

Where the book uses it. Least squares is the projection of the response onto the column space of the design matrix (Theorem 21.3); the residual sum of squares has $n - p$ degrees of freedom because $I - P$ projects onto an $(n-p)$-dimensional space (Theorem 21.5); the F-test splits the column space into nested pieces (Theorem 21.6); and full rank is the identifiability of the regression model (Chapter 2).

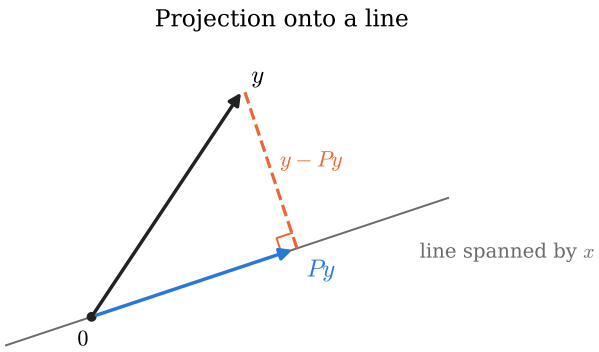


Figure A.14: Orthogonal projection onto a line: the vector y (black), the line spanned by x (gray), the projection Py (blue) and the residual $y - Py$ (orange, dashed), perpendicular to the line.

A.17 Newton's method

Chapter 23 maximizes a log-likelihood that has no closed-form maximizer, and does it by a method that goes back to Newton (1669) and Raphson (1690); the form with derivatives used here was given by Simpson (1740).

The idea. To find where a smooth function ℓ of one variable is largest, start at a guess t_0, replace ℓ near t_0 by the parabola with the same value, slope and curvature there,

$$q(t) = \ell(t_0) + \ell'(t_0)(t - t_0) + \tfrac{1}{2}\ell''(t_0)(t - t_0)^2,$$

and move to the top of the parabola: $q'(t) = 0$ at $t_1 = t_0 - \ell'(t_0)/\ell''(t_0)$. Repeat from t_1. Equivalently, Newton's method solves the equation $\ell'(t) = 0$ by following the tangent line of ℓ' at the current point down to the axis. In d variables the parabola is a quadratic with the gradient and the Hessian, and the step is

$$t_{k+1} = t_k - \big(\nabla^2 \ell(t_k)\big)^{-1} \nabla \ell(t_k),$$

one linear system per step (toolkit A.12).

When it works. If ℓ is strictly concave with an invertible Hessian, the parabola at each step is concave too, so every step goes somewhere definite, and once the iterate is close to the maximizer the error is squared at each step: for a one-variable ℓ with ℓ''' bounded near the maximizer $\hat{t}$, a Taylor expansion of ℓ' gives $t_{k+1} - \hat{t} = \frac{\ell'''(\xi)}{2\ell''(t_k)}(t_k - \hat{t})^2$ for a ξ between t_k and $\hat{t}$. An error of 10^{-2} becomes about 10^{-4}, then 10^{-8}: three or four steps from a reasonable start usually finish the job, which Chapter 23 sees in the kyphosis fit. Far from the maximizer the quadratic model can be poor and a full step can overshoot; the remedy is to shorten the step until the function goes up (a line search), which strict concavity guarantees is possible. If the function has no maximum, as a separated logistic regression does not, the iterates run away and the coefficients grow without bound, which is how the problem shows itself in a fit.

Where the book uses it. Fitting a generalized linear model (Section 23.4), where the step has a second reading as a weighted least squares fit.

A.18 Eigenvalues, eigenvectors and the spectral theorem

Chapter 24 asks in which direction a cloud of points is most spread out, and the answer is an eigenvector of its covariance matrix. This entry says what that is and proves the one theorem the chapter needs.

Eigenvalues and eigenvectors. For a $d \times d$ matrix A, a nonzero vector v with $Av = \lambda v$ for a number λ is an **eigenvector** of A with **eigenvalue** λ: A acts on v by stretching it, without turning it. A matrix A is **symmetric** if $A^\top = A$, and then $u^\top Av = (Au)^\top v$ for all u, v. A symmetric matrix is **positive semidefinite** if $u^\top Au \geq 0$ for every u, and **positive definite** if $u^\top Au > 0$ for every $u \neq 0$; every covariance matrix is positive semidefinite, since $u^\top \Sigma u = \mathbb{V}\mathrm{ar}(u^\top X) \geq 0$. The **trace** of A is the sum of its diagonal entries, and $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ whenever both products are square.

The spectral theorem. Every symmetric $d \times d$ matrix A has d orthonormal eigenvectors $v_1, \ldots, v_d$ with real eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$:

$$A = PDP^\top = \sum_{j=1}^{d} \lambda_j v_j v_j^\top, \qquad P = (v_1, \ldots, v_d) \text{ orthogonal}, \quad D = \mathrm{diag}(\lambda_1, \ldots, \lambda_d).$$

Proof. The function $u \mapsto u^\top Au$ is continuous on the unit sphere $\{u : \|u\| = 1\}$, which is closed and bounded, so it attains its maximum, at some v_1 say, with value λ_1 (toolkit A.2). For any unit w with $w^\top v_1 = 0$, the curve $u(t) = \cos t\, v_1 + \sin t\, w$ stays on the sphere and passes through v_1 at $t = 0$, so $t \mapsto u(t)^\top Au(t)$ has a maximum at $t = 0$ and its derivative there is zero; by the product rule and symmetry that derivative is $2u'(0)^\top Au(0) = 2w^\top Av_1$. So Av_1 is perpendicular to every w perpendicular to v_1: writing $Av_1 = \alpha v_1 + w'$ with $w' \perp v_1$ gives $w'^\top Av_1 = \|w'\|^2 = 0$, hence $Av_1 = \alpha v_1$, and $\alpha = v_1^\top Av_1 = \lambda_1$. Now restrict A to the vectors perpendicular to v_1: for such u, $v_1^\top Au = (Av_1)^\top u = \lambda_1 v_1^\top u = 0$, so Au stays perpendicular to v_1, and the same argument on the unit sphere of that $(d-1)$-dimensional subspace gives v_2 and $\lambda_2 \leq \lambda_1$; after d rounds the eigenvectors are orthonormal by construction. Then $AP = PD$, and $P^\top P = I$ gives $A = PDP^\top$.

Consequences. Writing $u = Pc$, that is, $c = P^\top u$ with $\|c\| = \|u\|$, gives $u^\top Au = \sum_j \lambda_j c_j^2$: a weighted sum of squares with the eigenvalues as weights. So A is positive semidefinite exactly when every $\lambda_j \geq 0$, and positive definite exactly when every $\lambda_j > 0$; $\lambda_d \leq u^\top Au \leq \lambda_1$ for every unit u, with the ends attained at v_d and v_1; $\mathrm{tr}(A) = \mathrm{tr}(PDP^\top) = \mathrm{tr}(DP^\top P) = \sum_j \lambda_j$; and $\det A = \prod_j \lambda_j$. The eigenvalues of a symmetric matrix depend continuously on its entries, and when λ_j differs from its neighbors, $\lambda_{j-1} > \lambda_j > \lambda_{j+1}$ (only the inequality that exists when $j = 1$ or $j = d$), so does the line through v_j (the vector itself is determined only up to sign); Chapter 24 uses both with the continuous mapping theorem, and both follow from the theorem. For symmetric A and B write $\|A - B\|$ for the square root of the sum of the squared entries of $A - B$. For a unit vector u, $|u^\top Au - u^\top Bu| = |\sum_{j,k}(A-B)_{jk} u_j u_k| \leq \|A - B\|$ by the Cauchy–Schwarz inequality, since the squares of the $u_j u_k$ add up to $\|u\|^4 = 1$. The proof of Theorem 24.5 uses only the decomposition $A = PDP^\top$, not the sign of the eigenvalues, and shows that $\lambda_1 + \cdots + \lambda_k$ is the largest value of $\sum_{j \leq k} u_j^\top Au_j$ over orthonormal $u_1, \ldots, u_k$ (for $k = d$ every orthonormal basis gives the same sum, $\mathrm{tr}(A) = \lambda_1 + \cdots + \lambda_d$). Replacing A by B changes each such sum by at most $k\|A - B\|$, so it changes the largest value by at most that, and changes $\lambda_k = (\lambda_1 + \cdots + \lambda_k) - (\lambda_1 + \cdots + \lambda_{k-1})$ by at most $(2k-1)\|A - B\|$: each eigenvalue is a continuous function of the matrix. For the eigenvector, let $A_n \to A$ with $\lambda_{j-1} > \lambda_j > \lambda_{j+1}$ for A, so that the eigenvectors of A for λ_j are the multiples of v_j (if $Aw = \lambda_j w$ and $w = \sum_i c_i v_i$, then $c_i(\lambda_i - \lambda_j) = 0$ for every i, so $c_i = 0$ for $i \neq j$). Let u_n be a unit eigenvector of A_n for its jth eigenvalue with $u_n^\top v_j \geq 0$. Every subsequence of the u_n has a further

subsequence converging to some unit vector w, since the sphere is closed and bounded (toolkit A.2), and passing to the limit in $A_n u_n = \lambda_j(A_n) u_n$, with $\lambda_j(A_n) \to \lambda_j$ by the bound just proved, gives $Aw = \lambda_j w$, so $w = \pm v_j$, and $w^\top v_j = \lim_n u_n^\top v_j \geq 0$ makes it v_j. A sequence on the sphere whose convergent subsequences all have the same limit converges to that limit, so $u_n \to v_j$.

Where the book uses it. The principal directions of a sample are the eigenvectors of its covariance matrix (Theorem 24.3), and the best k-dimensional picture of the cloud is the projection onto the first k of them (Theorem 24.5).

Appendix B

The Modes of Convergence in Full

Chapters 3 and 4 used two modes of convergence, in probability and in distribution: the Law of Large Numbers lives in the first and the Central Limit Theorem in the second. There is a third, almost sure convergence, and a fourth, convergence in mean; all four can be ranked by strength. Nothing later in the book is proved with the two extra modes, which is why they're here rather than in the main sequence; where a later chapter mentions almost sure convergence, as Chapters 5, 7, 17 and 24 do, the in-probability version is the one in use. This appendix is for orientation, and for the reader who wants to know what the Strong Law of Large Numbers actually says.

B.1 Intuition for the three main modes

1. **Convergence in probability:** Imagine you have a sequence of r.v.s (like a sequence of coin flips or dice rolls). If this sequence converges in probability to a specific value, it means that as you progress through the sequence, the probability of the r.v.s being 'close' to that specific value tends to 1. In other words, with enough trials you can make it as likely as you please that the result is close to the value the sequence is converging to.

2. **Almost sure convergence (convergence almost surely):** Almost sure convergence is a stronger notion than convergence in probability. If a sequence of r.v.s converges almost surely to a value, it means that the probability of the sequence eventually staying close to that value is 1. More concretely, if you were to run the experiment and watch the whole infinite sequence unfold, the sequence you see converges, in the plain calculus sense, with probability 1. Almost sure convergence implies convergence in probability, but not conversely. Given that a sequence of r.v.s converging almost surely means that it converges to the value with probability 1, this mode of convergence is the closest analog to convergence of sequences of numbers as studied in calculus; it's a deterministic understanding of convergence applied one outcome at a time.

3. **Convergence in distribution:** Convergence in distribution is a weaker notion compared to the previous two. It doesn't focus on the actual values of the random variables but rather on their distribution. If a sequence of r.v.s converges in distribution to an r.v., it means that the shape of the distribution of the sequence becomes increasingly similar to the distribution of the limiting r.v. In simpler terms, the probabilities of the sequence falling within certain ranges will approximate the probabilities of the limiting variable falling within the same ranges.

The modes come in the order the book uses them. Convergence in probability first since the Law of Large Numbers lives there and it's where estimators are shown to work; convergence in distribution second since the Central Limit Theorem lives there and it's where confidence intervals and tests come from; and almost sure convergence last since the Strong Law is worth stating on its own account and is never needed again.

B.2 Convergence almost surely

Like convergence in probability, convergence almost surely involves a limiting process between random variables, say X_n and X, that live on the same sample space. Unlike convergence in probability, though, the limiting process for almost sure convergence is located *within* the probability measure. Compare the two definitions side by side:

$$\begin{aligned} &\text{in probability:} && \lim_{n\to\infty} \mathbb{P}\left(|X_n - X| \geq \epsilon\right) = 0 \quad \text{for every } \epsilon > 0, \\ &\text{almost surely:} && \mathbb{P}\left(\lim_{n\to\infty} X_n = X\right) = 1. \end{aligned}$$

In the first, you compute a probability for each n and then take a limit of numbers. In the second, you fix an outcome ω of the experiment, which fixes the entire sequence of numbers $X_1(\omega), X_2(\omega), \ldots$, ask whether that sequence of numbers converges to the number $X(\omega)$ in the calculus sense, and then ask for the probability of the set of outcomes for which the answer is yes.

Definition B.1 (Convergence almost surely)**.** Let $(X_n)_{n\geq 1}$ be a sequence of r.v.s defined on a sample space Ω. The sequence **converges almost surely** to an r.v. X if:

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 1.$$

If $(X_n)_{n\geq 1}$ converges almost surely to the r.v. X, this is denoted by $X_n \xrightarrow[n\to\infty]{\text{a.s.}} X$.

> **In words.** Run the experiment once and watch the whole infinite sequence of numbers it produces; with probability one, that sequence converges in the plain calculus sense (toolkit A.1). The set of runs where it fails is allowed to be nonempty, as long as it has probability zero (toolkit A.5).
>
> **For the kiss data.** Keep observing couples forever and keep updating the running proportion; the Strong Law says the running proportion settles on p and stays there, for all but a probability-zero set of infinite runs.

'Almost surely' is probability's phrase for 'with probability 1'. The set of outcomes on which the sequence fails to converge is allowed to be nonempty, as long as it has probability 0.

Example B.2. Let $U \sim \text{Unif}([0,1])$ and define $X_n = U + U^n$. For every outcome with $U \in [0,1)$, $U^n \to 0$ as $n \to \infty$, so $X_n \to U$. For the outcome $U = 1$ the sequence is constant at 2 and does not converge to $U = 1$. But $\mathbb{P}(U = 1) = 0$, so $\mathbb{P}(X_n \to U) = \mathbb{P}(U \in [0,1)) = 1$ and $X_n \xrightarrow[n\to\infty]{\text{a.s.}} U$. This is the sense in which 'almost' is used: convergence can fail on a set of outcomes, provided that set has probability zero.

B.2.1 The Strong Law of Large Numbers (SLLN)

Theorem B.3 (Strong Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be i.i.d. r.v.s with $\mathbb{E}|X_i| < \infty$ and $\mathbb{E}[X_i] = \mu$. Then:*

$$\bar{X}_n \xrightarrow[n\to\infty]{a.s.} \mu.$$

Two things are notable about the statement. First, it requires only a finite mean, not a finite variance; the Weak Law was proved in Chapter 3 using Chebyshev's inequality, which needs the variance, but the conclusion of both laws holds under the weaker assumption. Second, the conclusion is about sample paths: run the experiment once, forever, and the sequence of running averages you record converges to μ, with probability one. Figure B.1 shows six such paths for a fair coin, on the left, with the band $\mu \pm 0.05$ around the mean. Each path wanders early, enters the band and, as the Strong Law says, stays there; with probability one every path eventually does this. The two laws are often written together as

$$\bar{X}_n \xrightarrow[n\to\infty]{\text{a.s./}\mathbb{P}} \mu,$$

allowing whichever one is convenient to be used; the Weak Law of Chapter 3 is enough for almost everything in this book, and Chapters 5, 20 and 24 use the conclusion for averages whose terms have a finite mean but perhaps no finite variance.

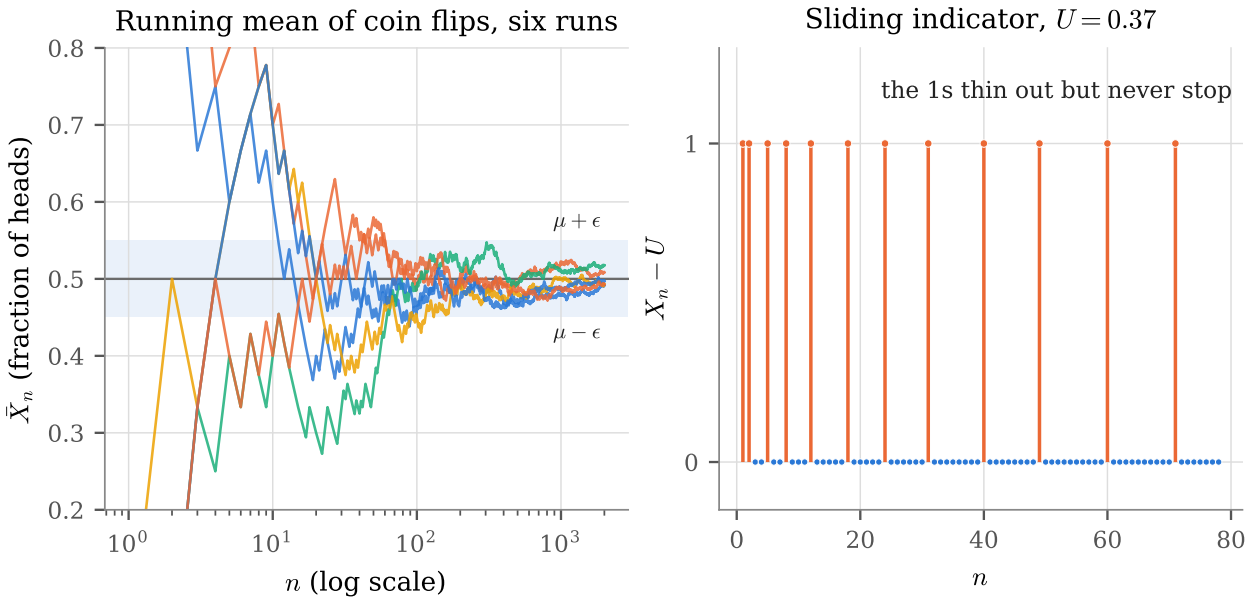


Figure B.1: Left: six realizations of the running proportion of heads in a fair coin, on a logarithmic n axis, with the band $\mu \pm 0.05$. Right: one outcome, $U = 0.37$, of the sliding-indicator sequence of Example B.4.

B.2.2 Convergence almost surely versus convergence in probability

While the Weak Law of Large Numbers illustrates convergence in probability and the Strong Law of Large Numbers shows convergence almost surely, what's the difference between the two? The following provides some intuition on the distinction between these two modes of convergence.

Suppose you have some device that improves over time. So every time you use the device the probability of it failing is less than before.

Convergence in probability says that the chance of failure goes to zero as the number of usages goes to infinity. So after using the device a large number of times, you can be very confident of it working correctly; it still might fail, but that's very unlikely.

On the other hand, convergence almost surely is a bit stronger. It says that the total number of failures is *finite*. That is, if you count the number of failures as the number of usages goes to infinity, you will get a finite number. The impact of this is as follows: As you use the device more and more, you will, after some finite number of usages, exhaust all failures. From then on the device will work perfectly. You don't actually know when you have exhausted all failures, so from a purely practical point of view, there isn't much difference between the two modes of convergence.

It's a good thing that the Strong Law of Large Numbers exists, as opposed to just the Weak Law. For example, a scientific experiment is justified in taking averages: after obtaining enough data, you can get arbitrarily close to the true average, and there will be only finitely many failures (however improbable) in the averaging process, after which the average stays close.

Here is what 'failures (however improbable) in the averaging process' means. Choose some $\delta > 0$ arbitrarily small. You obtain n estimates $X_1, \ldots, X_n$ of some other quantity that has some 'true' value, say μ. You compute the average:

$$S_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

As you obtain more data (n increases) you can compute S_n for $n = 1, 2, \ldots$. The Weak Law says (under some assumptions about the X_n) that the probability $\mathbb{P}(|S_n - \mu| > \delta) \to 0$, as n goes to ∞. The Strong Law says that the number of times that $|S_n - \mu|$ is larger than δ is finite (with probability 1). That is, if we define the indicator function $\mathbb{1}(|S_n - \mu| > \delta)$ that returns one when $|S_n - \mu| > \delta$ and zero otherwise, then $\sum_{n=1}^{\infty} \mathbb{1}(|S_n - \mu| > \delta)$ is finite. This gives you considerable confidence in the value of S_n because it guarantees, with probability 1, the existence of some finite n_0 such that $|S_n - \mu| \leq \delta$ for all $n > n_0$. That is, the average never fails for $n > n_0$. Note that the Weak Law gives no such guarantee: it says failures become rare, not that they stop.

The following example, from Casella and Berger, is the standard illustration of a sequence for which failures become rare but never stop.

Example B.4 (The sliding indicator)**.** Let $U \sim \text{Unif}([0, 1])$ and define

$$\begin{aligned}
&X_1 = U + \mathbb{1}(U \in [0, 1]), \\
&X_2 = U + \mathbb{1}(U \in [0, \tfrac{1}{2}]), \qquad X_3 = U + \mathbb{1}(U \in [\tfrac{1}{2}, 1]), \\
&X_4 = U + \mathbb{1}(U \in [0, \tfrac{1}{3}]), \qquad X_5 = U + \mathbb{1}(U \in [\tfrac{1}{3}, \tfrac{2}{3}]), \qquad X_6 = U + \mathbb{1}(U \in [\tfrac{2}{3}, 1]), \\
&\quad \vdots
\end{aligned}$$

so the indicator's interval slides across $[0, 1]$ in blocks, and the intervals in the m-th block have length $1/m$. Each X_n equals either U or $U + 1$. *Convergence in probability holds:* for any $0 < \epsilon < 1$, $\mathbb{P}(|X_n - U| \geq \epsilon) = \mathbb{P}(U \in [a_n, b_n]) = b_n - a_n$, the length of the n-th interval, which goes to zero. So $X_n \xrightarrow[n\to\infty]{\mathbb{P}} U$. *Almost sure convergence fails:* fix any outcome $U = u$. In every block, exactly

one interval (or two, at a shared endpoint) contains u, so $X_n(u) - u$ equals 1 at least once in every block, forever. The sequence of numbers $X_n(u)$ therefore takes the value $u + 1$ infinitely often and cannot converge to u. This happens for every outcome, so $\mathbb{P}(X_n \to U) = 0$, the opposite of what almost sure convergence requires. The right of Figure B.1 shows one outcome, $U = 0.37$: the value 1 keeps recurring, at ever longer intervals, forever, so the sequence converges in probability but not almost surely.

In the device analogy, the sliding indicator is a device whose failures get rarer and rarer but never run out.

B.3 Convergence in mean

There's a fourth mode, which this book needs only in passing but which you've already used: the shrinking-variance criterion of Proposition 3.3 is a statement about it.

Definition B.5 (Convergence in r-th mean)**.** For $r \geq 1$, a sequence $(X_n)_{n\geq 1}$ **converges in r-th mean** (or in L^r) to X, written $X_n \xrightarrow[n\to\infty]{L^r} X$, if

$$\lim_{n\to\infty} \mathbb{E}\left[|X_n - X|^r\right] = 0.$$

The case $r = 2$ is called convergence in mean square (or quadratic mean).

> **In words.** The average size of the error, measured by its r-th power, goes to zero. For $r = 2$ this is 'the quadratic risk goes to zero'.
>
> **For the kiss data.** $\mathbb{E}[(\bar{R}_n - p)^2] = p(1-p)/n \to 0$, so $\bar{R}_n \to p$ in mean square; by Markov's inequality (toolkit A.7) that implies convergence in probability, which is the shrinking-variance criterion of Chapter 3 again.

Convergence in mean square to a constant μ is exactly the condition $\mathbb{E}[(X_n - \mu)^2] \to 0$, and since $\mathbb{E}[(X_n - \mu)^2] = \mathbb{V}\mathrm{ar}(X_n) + (\mathbb{E}[X_n] - \mu)^2$ (variance plus squared bias, the identity behind the quadratic risk of Chapter 7), it holds if and only if $\mathbb{E}[X_n] \to \mu$ and $\mathbb{V}\mathrm{ar}(X_n) \to 0$. Convergence in r-th mean implies convergence in probability, by Markov's inequality applied to $|X_n - X|^r$:

$$\mathbb{P}(|X_n - X| \geq \epsilon) = \mathbb{P}(|X_n - X|^r \geq \epsilon^r) \leq \frac{\mathbb{E}|X_n - X|^r}{\epsilon^r} \to 0.$$

Neither convergence in mean nor almost sure convergence implies the other.

B.4 The hierarchy of convergence types

A sequence of r.v.s might converge for one type of convergence, but not for another. Some convergence types are 'stronger' than others by which it's meant that if Type A convergence is stronger than Type B convergence, then Type A convergence implies Type B convergence. In Figure B.2, the stronger types of convergence are given at the top and as you move to the bottom the type of convergence becomes weaker. By way of example, if a sequence of r.v.s converges in probability, then it converges in distribution as well. Each solid arrow is an implication that always holds, and

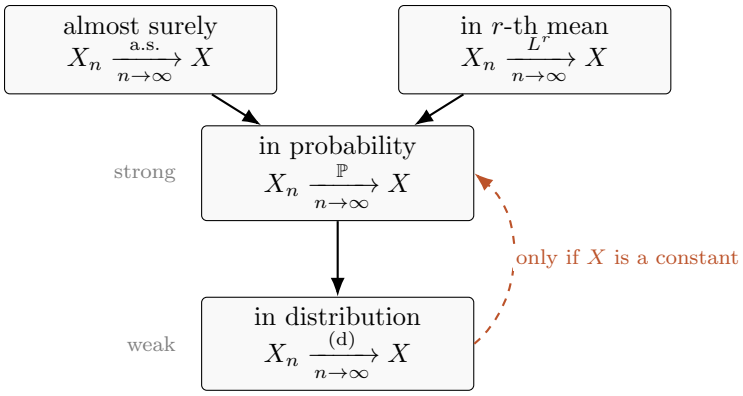


Figure B.2: The four modes of convergence, strongest at the top. Solid arrows are implications; the dashed arrow is the partial converse of Proposition 4.7.

no solid arrow reverses. Almost sure convergence and convergence in mean, side by side at the top, don't imply each other.

Why does this hierarchy exist? Let's focus on the 3 main modes of convergence. Consider the following points:

- **Almost sure convergence:** If a sequence of r.v.s converges almost surely to an r.v., it means that the probability of the sequence eventually staying close to the limiting r.v. is 1. This is the strongest form of convergence because it tells us that the individual values of the r.v.s in the sequence will eventually be close to the limiting r.v. with probability 1. Since the sequence eventually stays within ϵ of the limit, the probability of being more than ϵ away at time n must go to zero, which is convergence in probability.

- **Convergence in probability:** If a sequence of r.v.s converges in probability to an r.v., it means that for any given $\epsilon > 0$, the probability of the difference between the random variables in the sequence and the limiting r.v. being greater than ϵ goes to 0 as the sequence progresses. This is weaker than almost sure convergence because it only tells us about the probability of the r.v.s being close to the limiting r.v. at each n separately, not that they will eventually be close with probability 1. The sliding indicator of Example B.4 is the example of the gap.

- **Convergence in distribution:** If a sequence of r.v.s converges in distribution to an r.v., it means that the cumulative distribution functions (CDFs) of the r.v.s in the sequence converge pointwise to the CDF of the limiting r.v. This is the weakest form of convergence because it only tells us about the behavior of the distribution of the r.v.s, not about the individual values or their probabilities of being close to the limiting r.v. If X_n is close to X with high probability then their CDFs must be close, which is why convergence in probability implies convergence in distribution; the $X_n = -Z$ example in the pitfall box of Section 4.1 shows the converse fails.

The dashed arrow in Figure B.2, the one partial converse that matters, is Proposition 4.7: convergence in distribution to a *constant* is the same as convergence in probability to it.

B.5 Summary

Mode	Definition	Where it's used in this book
almost surely, $X_n \xrightarrow[n\to\infty]{\text{a.s.}} X$	$\mathbb{P}(X_n \to X) = 1$	Strong Law of Large Numbers
in probability, $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$	$\mathbb{P}(\lvert X_n - X\rvert \geq \epsilon) \to 0$ for every $\epsilon > 0$	Weak LLN; consistency of estimators; the constant in Slutsky's theorem
in distribution, $X_n \xrightarrow[n\to\infty]{\text{(d)}} X$	$F_{X_n}(x) \to F_X(x)$ at every continuity point of F_X	CLT; confidence intervals; test statistics
in r-th mean, $X_n \xrightarrow[n\to\infty]{L^r} X$	$\mathbb{E}\lvert X_n - X\rvert^r \to 0$	shrinking-variance criterion ($r = 2$); quadratic risk

B.6 Exercise

Exercise B.1. Is the following statement correct? 'If $(T_n)_{n\geq 1}$ converges in probability, then it also converges almost surely.' *(Solution on page 442.)*

Appendix C

Supplementary Proofs

Some results are stated in a chapter because they explain something there, but nothing later in the book depends on them. Their proofs are collected here so the chapters can keep moving. Each section says where the result is stated and confirms that no later argument uses it.

C.1 Pinsker's inequality

Theorem 10.8 in Chapter 10 states that for any distributions $\mathbb{P}$ and $\mathbb{Q}$ on the same observation space,

$$\mathrm{TV}(\mathbb{P}, \mathbb{Q}) \leq \sqrt{\tfrac{1}{2}\mathrm{KL}(\mathbb{P}, \mathbb{Q})}.$$

It is there to say that a fit that is good in KL is good for every probability computed from it. **Nothing later in the book depends on it**: no theorem, proof or exercise after Chapter 10 uses Pinsker's inequality, and you can skip this section without losing anything.

Proof strategy (for completeness; skip on a first reading). Two steps. First prove the inequality for two Bernoulli distributions, by calculus in one variable. Then show that collapsing $\mathbb{P}$ and $\mathbb{Q}$ to 'in the event $A^\star$ or not' turns them into two Bernoullis with the same total variation and no larger KL divergence. The second step uses the same tangent-line bound $-\log y \geq 1 - y$ as the proof that KL is nonnegative (Proposition 10.6).

Step 1: two Bernoulli distributions

Lemma C.1. *For $p, q \in (0, 1)$,*

$$\mathrm{KL}(\mathit{Bern}(p), \mathit{Bern}(q)) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q} \geq 2(p - q)^2.$$

The inequality also holds for $p \in [0, 1]$ and $q \in (0, 1)$, with the convention $0 \log 0 = 0$.

Proof strategy (for completeness; skip on a first reading). Fix p, treat the gap between the two sides as a function of q, and show by its derivative that it is smallest, and equal to 0, at $q = p$.

Proof. Fix p and look at the gap as a function of q:

$$g(q) = p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q} - 2(p - q)^2.$$

At $q = p$ both terms are 0, so $g(p) = 0$. Differentiate in q:

$$g'(q) = -\frac{p}{q} + \frac{1-p}{1-q} + 4(p-q).$$

The first two terms combine over the common denominator $q(1-q)$: $-p(1-q) + (1-p)q = q - p$. So

$$g'(q) = \frac{q-p}{q(1-q)} - 4(q-p) = (q-p)\left(\frac{1}{q(1-q)} - 4\right).$$

The product $q(1-q)$ is at most $\frac{1}{4}$, since $\frac{1}{4} - q(1-q) = (q - \frac{1}{2})^2 \geq 0$ (it is the variance of one Bernoulli observation, largest for a fair coin). So $1/(q(1-q)) \geq 4$ and the bracket is never negative. The sign of $g'(q)$ is therefore the sign of $q - p$: g decreases until $q = p$ and increases after. Its smallest value is $g(p) = 0$, so $g(q) \geq 0$ for every q. □

Figure C.1 draws the lemma, for $p = 0.645$ on the left. The divergence curve sits above the parabola $2(p-q)^2$ and touches it only at $q = p$. Near $q = p$ the two are close: the divergence is approximately $(p-q)^2/(2p(1-p))$ (Section 10.5), and $1/(2p(1-p)) \geq 2$, with equality only for $p = \frac{1}{2}$. The parabola $2(p-q)^2$ is the divergence's own parabola for the hardest case, the fair coin, and every other Bernoulli curves up faster. The right of the figure draws the gap $g(q)$ for $p = 0.2$, 0.5 and 0.645: each is 0 at $q = p$ and positive everywhere else.

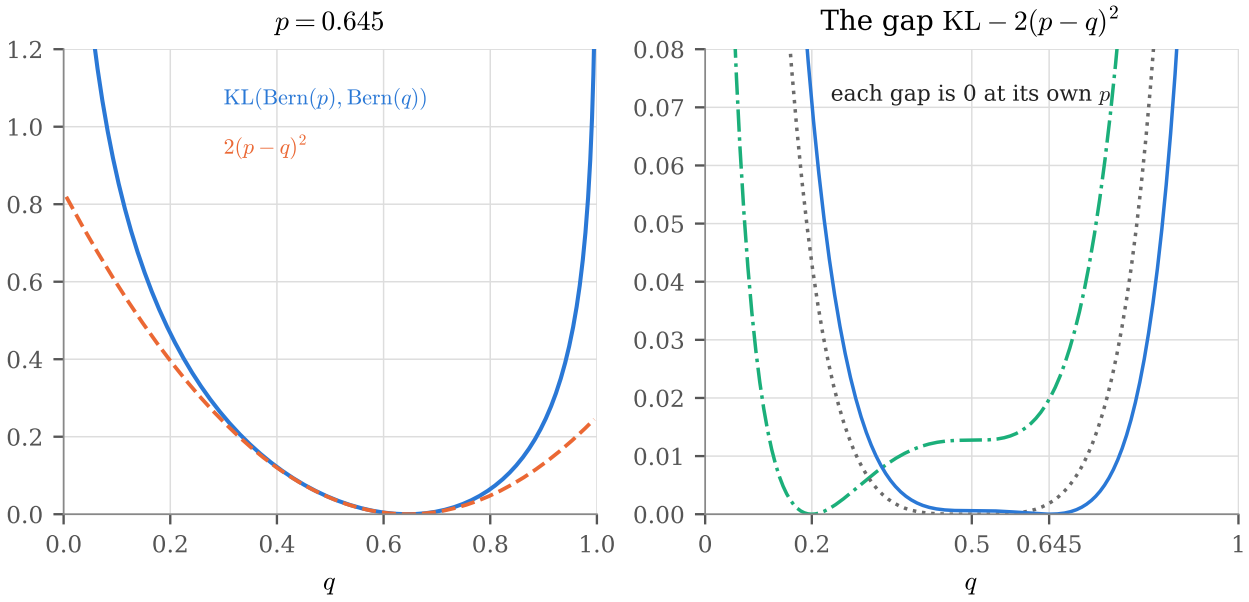


Figure C.1: The Bernoulli case of Pinsker's inequality. Left: $\mathrm{KL}(\mathrm{Bern}(p), \mathrm{Bern}(q))$ (blue) and $2(p-q)^2$ (orange, dashed) for $p = 0.645$. Right: the gap between the two, as a function of q, for $p = 0.2$ (aqua, dash-dot), 0.5 (gray, dotted) and 0.645 (blue, solid).

For the kiss data, $p = 80/124$ and $q = \frac{1}{2}$: KL $= 0.0428$ and $2(p-q)^2 = 2 \times 0.1452^2 = 0.0422$. Taking square roots, $|p-q| = 0.1452 \leq \sqrt{0.0428/2} = 0.1462$. Since $\mathrm{TV}(\mathrm{Bern}(p), \mathrm{Bern}(q)) = |p-q|$, the lemma is Pinsker's inequality for Bernoulli distributions.

Step 2: reducing any pair of distributions to two Bernoullis

Lemma C.2 (The log-sum inequality)**.** *For nonnegative numbers $a_1, \dots, a_k$ and nonnegative numbers $b_1, \dots, b_k$ with $b_i > 0$ whenever $a_i > 0$, and with $a = \sum_i a_i > 0$ and $b = \sum_i b_i$,*

$$\sum_{i=1}^{k} a_i \log \frac{a_i}{b_i} \geq a \log \frac{a}{b},$$

with the convention $0 \log 0 = 0$. The same holds with the sums replaced by integrals over a region.

Proof strategy (for completeness; skip on a first reading). Rewrite the difference of the two sides as a weighted average of $-\log$ of ratios and apply the tangent-line bound $-\log y \geq 1 - y$ to each.

Proof. Only the terms with $a_i > 0$ contribute on the left. For those, write

$$\sum_{i:a_i>0} a_i \log \frac{a_i}{b_i} - a \log \frac{a}{b} = \sum_{i:a_i>0} a_i \log \left(\frac{a_i}{b_i} \cdot \frac{b}{a} \right) = \sum_{i:a_i>0} a_i \left[-\log \left(\frac{b_i}{a_i} \cdot \frac{a}{b} \right) \right],$$

using $\sum_{i:a_i>0} a_i = a$ to bring $a \log(a/b)$ inside the sum. Apply $-\log y \geq 1 - y$ to each bracket:

$$\geq \sum_{i:a_i>0} a_i \left(1 - \frac{b_i}{a_i} \cdot \frac{a}{b} \right) = a - \frac{a}{b} \sum_{i:a_i>0} b_i \geq a - \frac{a}{b} \cdot b = 0.$$

The last step uses that the b_i with $a_i > 0$ add up to at most b. For integrals the same lines go through with $\int$ in place of $\sum$. □

In words: lumping several outcomes into one can only lower the divergence. If you record only whether an outcome fell in a set, and not which outcome it was, the evidence it carries cannot go up.

Proof strategy (for completeness; skip on a first reading). Collapse both distributions to the indicator of $A^\star$: total variation is unchanged, KL can only fall (Lemma C.2), and for the two resulting Bernoullis Lemma C.1 applies.

Proof of Theorem 10.8. If $\mathrm{KL}(\mathbb{P}, \mathbb{Q}) = +\infty$ there is nothing to prove, so assume it is finite; then $q(x) > 0$ wherever $p(x) > 0$. Let $A^\star = \{x : p(x) > q(x)\}$, and write $a = \mathbb{P}(A^\star)$, $b = \mathbb{Q}(A^\star)$. By Proposition 10.2, $\mathrm{TV}(\mathbb{P}, \mathbb{Q}) = a - b$. If $a = b$, TV $= 0$ and there is nothing to prove, so assume $a > b$; then $0 < a$ and $b < 1$.

Split the sum (or integral) defining $\mathrm{KL}(\mathbb{P}, \mathbb{Q})$ into the part over $A^\star$ and the part over its complement, and apply Lemma C.2 to each part, with $a_i = p(x)$ and $b_i = q(x)$:

$$\mathrm{KL}(\mathbb{P}, \mathbb{Q}) = \sum_{x \in A^\star} p(x) \log \frac{p(x)}{q(x)} + \sum_{x \notin A^\star} p(x) \log \frac{p(x)}{q(x)} \geq a \log \frac{a}{b} + (1 - a) \log \frac{1 - a}{1 - b}.$$

(If a part has total $\mathbb{P}$-probability 0, it contributes 0 on both sides.) The right-hand side is $\mathrm{KL}(\mathrm{Bern}(a), \mathrm{Bern}(b))$, the divergence between the two distributions of the indicator $\mathbb{1}(X \in A^\star)$. By Lemma C.1 (in the extended form for $a \in [0, 1]$; if $b = 0$ the divergence on the right is $+\infty$, which contradicts the finite left side, so $b \in (0, 1)$),

$$\mathrm{KL}(\mathbb{P}, \mathbb{Q}) \geq \mathrm{KL}(\mathrm{Bern}(a), \mathrm{Bern}(b)) \geq 2(a - b)^2 = 2\,\mathrm{TV}(\mathbb{P}, \mathbb{Q})^2.$$

Divide by 2 and take square roots. □

Intuition: why the proof works

Total variation is decided by one event, $A^\star$, where $\mathbb{P}$ beats $\mathbb{Q}$. Throw away everything about an outcome except whether it fell in $A^\star$. That leaves a coin, with probability a under $\mathbb{P}$ and b under $\mathbb{Q}$, and the same total variation $a - b$. Lumping can only lower KL, so the coin's KL is at most the original one. And for coins, the lemma says KL is at least twice the squared total variation.

C.2 Each step of the EM algorithm raises the likelihood

Section 11.8 in Chapter 11 states that one E step followed by one M step never lowers the log-likelihood of a mixture. It explains the climbing curves of Figure 11.7. **Nothing later in the book depends on it.**

Proof strategy (for completeness; skip on a first reading). Split the log-likelihood of the observed data into two pieces: the quantity the M step maximizes, minus a term that measures how well the parameter predicts the hidden labels. The M step raises the first piece. The second piece can only fall away from its value at the current parameter, by an amount that is a KL divergence and so is never negative.

Proof. Write θ for all five parameters, θ_t for the current values and θ_{t+1} for the values after one E step and one M step. For observation i, with hidden label $z \in \{0, 1\}$, let $p_\theta(x_i, z)$ be the joint probability (density in x_i) of the observation and the label, $p_\theta(x_i) = \sum_z p_\theta(x_i, z)$ the mixture density, and $p_\theta(z \mid x_i) = p_\theta(x_i, z)/p_\theta(x_i)$ the conditional probability of the label. Write $q_i(z) = p_{\theta_t}(z \mid x_i)$ for the current weights: $q_i(1) = w_i$ and $q_i(0) = 1 - w_i$.

For any θ and each z, $\log p_\theta(x_i) = \log p_\theta(x_i, z) - \log p_\theta(z \mid x_i)$. Average this over z with the weights $q_i(z)$, which add up to 1, and add over i:

$$\ell_n(\theta) = \underbrace{\sum_{i=1}^{n} \sum_{z} q_i(z) \log p_\theta(x_i, z)}_{Q(\theta)} \; - \; \underbrace{\sum_{i=1}^{n} \sum_{z} q_i(z) \log p_\theta(z \mid x_i)}_{H(\theta)} .$$

$Q(\theta)$ is the complete-data log-likelihood with each label replaced by its weight, which is exactly what the M step maximizes over all five parameters (Exercise 11.9), so $Q(\theta_{t+1}) \geq Q(\theta_t)$. This holds for the sum over i, not observation by observation: a step can lower the density of some observations while raising the total. For the second term,

$$H(\theta_t) - H(\theta) = \sum_{i=1}^{n} \sum_{z} q_i(z) \log \frac{p_{\theta_t}(z \mid x_i)}{p_\theta(z \mid x_i)} = \sum_{i=1}^{n} \mathrm{KL}\big(p_{\theta_t}(\cdot \mid x_i),\, p_\theta(\cdot \mid x_i)\big) \geq 0$$

by Proposition 10.6, term by term, since $q_i = p_{\theta_t}(\cdot \mid x_i)$. So $H(\theta_{t+1}) \leq H(\theta_t)$, and

$$\ell_n(\theta_{t+1}) - \ell_n(\theta_t) = \big(Q(\theta_{t+1}) - Q(\theta_t)\big) + \big(H(\theta_t) - H(\theta_{t+1})\big) \geq 0.$$

□

Ascent says nothing about where the climb ends: EM stops where the gradient of ℓ_n is 0, which can be a local maximum or a saddle point.

C.3 The uniform law of large numbers

The consistency proof of Theorem 12.1 in Chapter 12 needs the Law of Large Numbers to hold for all candidates θ at once, not one at a time. That is the one step of that chapter not proved there. **Nothing later in the book depends on this section** beyond Theorem 12.1 itself; the asymptotic normality theorem uses consistency, not the uniform law directly.

Recall. The Law of Large Numbers (Theorem 3.5): for i.i.d. $X_1, X_2, \dots$ and a function h with $\mathbb{V}\mathrm{ar}(h(X)) < \infty$, $\frac{1}{n}\sum_{i=1}^{n} h(X_i) \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[h(X)]$.

Theorem C.3 (Uniform law of large numbers)**.** *Let $X_1, \dots, X_n$ be i.i.d. and let $g(x, \theta)$ be a function of an observation and a parameter $\theta \in \Theta$, with $\Theta \subset \mathbb{R}^d$ bounded and closed. Suppose that $g(x, \theta)$ is continuous in θ for every x, and that there is an envelope $G(x) \geq \sup_\theta |g(x, \theta)|$ with $\mathbb{E}[G(X)^2] < \infty$. Then, with $M(\theta) = \mathbb{E}[g(X, \theta)]$,*

$$\sup_{\theta\in\Theta} \left| \frac{1}{n}\sum_{i=1}^{n} g(X_i, \theta) - M(\theta) \right| \xrightarrow[n\to\infty]{\mathbb{P}} 0.$$

Proof strategy (for completeness; skip on a first reading). Cover Θ by finitely many small balls. On each ball the function g varies little, so the average over the ball's center controls the average everywhere in the ball; the ordinary LLN handles the finitely many centers at once, and continuity plus the envelope make the within-ball variation as small as you like.

Proof. Fix $\epsilon > 0$. For $\theta \in \Theta$ and $r > 0$ let $B(\theta, r)$ be the ball of radius r around θ, and define the local oscillation

$$w_r(x, \theta) = \sup_{\theta' \in B(\theta, r)\cap\Theta} |g(x, \theta') - g(x, \theta)|.$$

For each fixed x and θ, continuity of g in θ gives $w_r(x, \theta) \to 0$ as $r \to 0$, and $0 \leq w_r(x, \theta) \leq 2G(x)$. So by dominated convergence (toolkit A.1) (the dominating function is $2G$, integrable), $\mathbb{E}[w_r(X, \theta)] \to 0$ as $r \to 0$, for each θ.

A bounded closed set has the following property: for any way of attaching a ball $B(\theta, r_\theta)$ to each of its points, finitely many of those balls already cover it. Choose r_θ for each θ so small that $\mathbb{E}[w_{r_\theta}(X, \theta)] < \epsilon$, and pick a finite cover $B(\theta_1, r_1), \dots, B(\theta_K, r_K)$.

Now take any $\theta \in \Theta$, say in the ball around θ_k. Split the error at θ into three pieces:

$$\left|\frac{1}{n}\sum_i g(X_i, \theta) - M(\theta)\right| \leq \underbrace{\frac{1}{n}\sum_i |g(X_i, \theta) - g(X_i, \theta_k)|}_{\leq \frac{1}{n}\sum_i w_{r_k}(X_i, \theta_k)} + \underbrace{\left|\frac{1}{n}\sum_i g(X_i, \theta_k) - M(\theta_k)\right|}_{\text{LLN at } \theta_k} + \underbrace{|M(\theta_k) - M(\theta)|}_{\leq \mathbb{E}[w_{r_k}(X, \theta_k)] < \epsilon}.$$

Taking the supremum over θ amounts to taking the maximum over the finitely many k, so

$$\sup_\theta \left|\frac{1}{n}\sum_i g(X_i, \theta) - M(\theta)\right| \leq \max_{k\leq K} \frac{1}{n}\sum_i w_{r_k}(X_i, \theta_k) + \max_{k\leq K} \left|\frac{1}{n}\sum_i g(X_i, \theta_k) - M(\theta_k)\right| + \epsilon.$$

Each of the $2K$ averages on the right is an ordinary average of i.i.d. terms with finite variance (bounded by $2G$ or G, with $\mathbb{E}[G^2] < \infty$), so by the LLN the first maximum converges in probability

to $\max_k \mathbb{E}[w_{r_k}(X,\theta_k)] < \epsilon$ and the second to 0. A maximum of finitely many sequences converging in probability converges in probability to the maximum of the limits. Hence with probability tending to 1 the right side is below 3ϵ, and since ϵ was arbitrary, the left side converges in probability to 0. □

The theorem is applied in Theorem 12.1 with $g(x,\theta) = \log p_\theta(x)$, where the envelope $G = \sup_\theta |\log p_\theta(X)|$ is the hypothesis stated there, with the finite second moment the book's version of the Law of Large Numbers needs.

C.4 The densities of the chi-squared and Student distributions

Proposition 14.2(3) in Chapter 14 states the density of χ^2_k, and Proposition 14.6(2) the density of t_k. The chapter proves the $k = 1$ case of the first, and afterwards uses the chi-squared density only in Exercise 14.4, for the variance of t_k: its tests and intervals need only the definitions of the two distributions as a sum of squares and as a ratio, and their quantiles. **Nothing later in the book depends on this section.**

Proposition C.4. *For $k \geq 1$, the density of χ^2_k is $f_k(v) = \dfrac{1}{2^{k/2}\Gamma(k/2)} v^{k/2-1} e^{-v/2}$ for $v > 0$.*

Proof strategy (for completeness; skip on a first reading). Induction on k: adding one more squared Gaussian convolves f_k with f_1, and the convolution integral is a Beta integral.

Proof. The case $k = 1$ is in the chapter. Suppose the formula holds for k, and let $V \sim \chi^2_k$ and $Z^2 \sim \chi^2_1$ be independent, so $V + Z^2 \sim \chi^2_{k+1}$. The density of a sum of independent variables is the convolution of the densities: for $x > 0$,

$$f_{k+1}(x) = \int_0^x f_k(u)\, f_1(x-u)\, \mathrm{d}u = \frac{1}{2^{k/2}\Gamma(k/2)} \cdot \frac{1}{2^{1/2}\Gamma(1/2)}\, e^{-x/2} \int_0^x u^{k/2-1} (x-u)^{-1/2}\, \mathrm{d}u,$$

since $e^{-u/2} e^{-(x-u)/2} = e^{-x/2}$ comes out of the integral. Substituting $u = xs$,

$$\int_0^x u^{k/2-1}(x-u)^{-1/2}\,\mathrm{d}u = x^{(k+1)/2-1} \int_0^1 s^{k/2-1}(1-s)^{-1/2}\,\mathrm{d}s = x^{(k+1)/2-1}\, \frac{\Gamma(k/2)\,\Gamma(1/2)}{\Gamma((k+1)/2)},$$

where the last step is the Beta integral $\int_0^1 s^{a-1}(1-s)^{b-1}\,\mathrm{d}s = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, a standard identity of calculus. The Gamma factors cancel against those in front, leaving

$$f_{k+1}(x) = \frac{1}{2^{(k+1)/2}\,\Gamma\!\left(\frac{k+1}{2}\right)}\, x^{(k+1)/2-1} e^{-x/2},$$

the formula for $k+1$. □

Proposition C.5. *For $k \geq 1$, the density of t_k is $g_k(t) = \dfrac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi}\,\Gamma\left(\frac{k}{2}\right)} \left(1 + \dfrac{t^2}{k}\right)^{-(k+1)/2}$. For $k = 1$ it is the Cauchy density $1/(\pi(1+t^2))$.*

Proof strategy (for completeness; skip on a first reading). Condition on the denominator: given $V = v$, the ratio is a Gaussian scaled by $\sqrt{k/v}$. Integrate the resulting Gaussian density against the density of V; the integral is a Gamma integral.

Proof. Let $T = Z/\sqrt{V/k}$ with $Z \sim \text{Gauss}(0,1)$ and $V \sim \chi^2_k$ independent, and let φ and Φ be the standard Gaussian density and CDF. Since $V > 0$, $\{T \le t\} = \{Z \le t\sqrt{V/k}\}$, and by independence, conditioning on V,

$$\mathbb{P}(T \le t) = \int_0^\infty \Phi\big(t\sqrt{v/k}\big)\, f_k(v)\, \mathrm{d}v.$$

Differentiating in t under the integral (the integrand's derivative, $\varphi(t\sqrt{v/k})\sqrt{v/k}\, f_k(v)$, is bounded by $\sqrt{v/k} f_k(v)$, which is integrable),

$$\begin{aligned} g_k(t) &= \int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2 v/(2k)} \sqrt{\frac{v}{k}} \cdot \frac{v^{k/2-1} e^{-v/2}}{2^{k/2}\Gamma(k/2)}\, \mathrm{d}v \\ &= \frac{1}{\sqrt{2\pi k}\, 2^{k/2}\Gamma(k/2)} \int_0^\infty v^{\frac{k+1}{2}-1} \exp\Big(-\frac{v}{2}\Big(1 + \frac{t^2}{k}\Big)\Big)\mathrm{d}v. \end{aligned}$$

The integral is $\int_0^\infty v^{a-1}e^{-bv}\,\mathrm{d}v = \Gamma(a)/b^a$ with $a = (k+1)/2$ and $b = \frac{1}{2}(1 + t^2/k)$, that is, $\Gamma\big(\frac{k+1}{2}\big)\, 2^{(k+1)/2}(1 + t^2/k)^{-(k+1)/2}$. Collecting the powers of 2, $2^{-1/2} \cdot 2^{-k/2} \cdot 2^{(k+1)/2} = 1$, gives the formula. For $k = 1$, $\Gamma(1) = 1$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, so $g_1(t) = 1/(\pi(1 + t^2))$. □

C.5 The Benjamini–Hochberg procedure controls the false discovery rate

Theorem 14.14 in Chapter 14 states that the Benjamini–Hochberg procedure has FDR $= (N_0/N)\alpha$ when the p-values of the true nulls are uniform on $[0,1]$ and independent of each other and of the remaining p-values. **Nothing later in the book depends on this section.**

Recall the procedure: with $P_{(1)} \le \cdots \le P_{(N)}$ the sorted p-values, $i_{\max} = \max\{i : P_{(i)} \le i\alpha/N\}$ (0 if the set is empty), and the tests with the $i_{\max}$ smallest p-values are rejected. Write $R = i_{\max}$ for the number of rejections and V for the number of rejected true nulls. Two observations about the procedure are the heart of the proof.

Lemma C.6. *(a) Lowering any one p-value cannot decrease R. (b) Fix a test i and let $R^{(i)}$ be the number of rejections when P_i is replaced by 0 and the others are unchanged. Then test i is rejected if and only if $P_i \le \alpha R^{(i)}/N$, and on that event $R = R^{(i)}$.*

Proof strategy (for completeness; skip on a first reading). (a) Sorted values move down when any value does. (b) Zeroing a rejected p-value shuffles only the ranks below it, so the count of rejections is unchanged; conversely, if P_i is under the line set by the zeroed vector, enough values are under it in the original vector too.

Proof. (a) If one p-value is lowered, every sorted value $P_{(j)}$ can only go down, so every index j with $P_{(j)} \le j\alpha/N$ before still satisfies it after, and the largest such index cannot fall.

(b) Write P' for the modified vector (with $P'_i = 0$) and $R' = R^{(i)}$. By (a), $R' \ge R$. Suppose first that test i is rejected, so its rank is at most R and $P_i \le P_{(R)} \le R\alpha/N$. Moving P_i to 0 changes the sorted vector only at ranks up to the rank of P_i, which is at most R; for $j > R$, $P'_{(j)} = P_{(j)} > j\alpha/N$ by the

maximality of R, so $R' \le R$. Hence $R' = R$ and $P_i \le \alpha R'/N$. Conversely, suppose $P_i \le \alpha R'/N$. Under P' the R' smallest values are rejected and $P'_i = 0$ is among them; the other $R' - 1$ rejected values are unchanged original p-values, each at most $P'_{(R')} \le R'\alpha/N$. Together with P_i itself, at least R' of the original p-values are at most $R'\alpha/N$, so $P_{(R')} \le R'\alpha/N$ and $R \ge R'$; with (a), $R = R'$. Finally, test i has rank at most R: otherwise $P_i \ge P_{(R+1)} > (R+1)\alpha/N > R\alpha/N = R'\alpha/N$, contradicting the assumption. So test i is rejected. □

Proof strategy (for completeness; skip on a first reading). Write the false discovery proportion as a sum over the true nulls, replace R by $R^{(i)}$ using the lemma, and condition on the other p-values: the uniform P_i falls under its threshold with probability exactly $\alpha R^{(i)}/N$, and the $R^{(i)}$ cancels.

Proof of Theorem 14.14. On the event that nothing is rejected, $V/\max(R,1) = 0$; otherwise $R \ge 1$ and V/R counts the rejected true nulls. So, summing over the true nulls i and using Lemma C.6(b) to rewrite each indicator and to replace R by $R^{(i)}$ on the event,

$$\mathrm{FDR} = \mathbb{E}\left[\frac{V}{\max(R,1)}\right] = \sum_{i \text{ true null}} \mathbb{E}\left[\frac{\mathbb{1}(P_i \le \alpha R^{(i)}/N)}{R^{(i)}}\right].$$

Note that $R^{(i)} \ge 1$ always, since the modified vector has a p-value equal to $0 \le \alpha/N$. The number $R^{(i)}$ is a function of the other p-values only (with P_i replaced by the constant 0), so by independence, conditioning on them and using that P_i is uniform,

$$\mathbb{E}\left[\frac{\mathbb{1}(P_i \le \alpha R^{(i)}/N)}{R^{(i)}}\right] = \mathbb{E}\left[\frac{\mathbb{P}(P_i \le \alpha R^{(i)}/N \mid \text{others})}{R^{(i)}}\right] = \mathbb{E}\left[\frac{\alpha R^{(i)}/N}{R^{(i)}}\right] = \frac{\alpha}{N},$$

where $\alpha R^{(i)}/N \le \alpha \le 1$ keeps the uniform probability equal to its argument. Summing over the N_0 true nulls gives $\mathrm{FDR} = N_0\alpha/N$. □

Without independence the identity can fail; Benjamini and Yekutieli (2001) proved, by an argument not reproduced here, the general bound $\mathrm{FDR} \le (N_0/N)\,\alpha\,(1 + \frac{1}{2} + \cdots + \frac{1}{N})$, which is why the procedure is sometimes run at the level $\alpha/(1+\frac{1}{2}+\cdots+\frac{1}{N})$ when the tests are arbitrarily dependent.

Solutions to the Exercises

Chapter 1. The Kiss Experiment and What Statistics Is For

Exercise 1.1 *(page 10).* Call 7: six of the 36 equally likely pairs sum to 7, more than any other total, so the probability is 1/6 and the expected winning is $100/6 \approx \$16.67$. The mechanism (fair, independent dice) is known in full, so there is nothing to learn from data; the answer is deduced.

Exercise 1.2 *(page 10).* The count is $\text{Bin}(100, 0.8)$ with mean $np = 80$ and standard deviation $\sqrt{np(1-p)} = \sqrt{16} = 4$. The value 65 is $15/4 = 3.75$ standard deviations below the mean; a Gaussian approximation gives $\mathbb{P}(\text{at least } 65) \approx \Phi(3.75) \approx 0.9999$, and the exact Binomial computation gives 0.99985.

Exercise 1.3 *(page 10).* $\mathbb{P}(\text{at least } 2) = 3p^2(1-p) + p^3$, which is 0.282 at $p = 0.35$ and 0.718 at $p = 0.65$. The data are only about 2.5 times more probable under the larger p, so three couples cannot distinguish the two truths; with 124 couples the corresponding ratio is astronomically large (Figure 1.2).

Exercise 1.4 *(page 10).* (a) Independence: the 124 records are really 62 independent observations, so $\hat{p}$ is as variable as a sample of 62, not 124, though its expectation is still p. (b) Identical distribution and independence: those couples share a p_i near 1 that is not the population's p, and they all followed the same instruction, so their outcomes are not independent; $\hat{p}$ is pushed upward. (c) The sample is not drawn from the population of interest at all: if being visible is related to the direction of the turn (it plausibly is, since from one side the observer sees the faces of couples who turn one way and the backs of the heads of couples who turn the other), the R_i are Bernoulli with a parameter that differs from p, and no amount of data fixes it. The third is the most dangerous because it is invisible in the data.

Chapter 2. Statistical Models and Identifiability

Exercise 2.1 *(page 28).* (2). The first has an observation space that excludes the value 0, which every $\text{Bern}(p)$ puts mass on. The second is a legitimate model with a restricted parameter space; whether it's well-specified is a separate question.

Exercise 2.2 *(page 28).* The intended answer is (2), and (1) is the one that is definitely wrong: the observation space can't depend on the unknown a. Both $\mathbb{R}_+$ and $\mathbb{R}$ contain every observation for every a, so by Definition 2.1 (3) is a legitimate model too; the convention is to take $E = \mathbb{R}_+$, the smallest natural set that works, and that convention is what the question is testing.

Exercise 2.3 *(page 28).* B, C, D, E, F. In (A) a member of the family needs infinitely many numbers (a probability for every non-negative integer). (B), (C), (F) need one number, (E) two, (D) three (the fourth p_i is determined by the other three). In (G) a member is a continuous function on $[0, 1]$, which no finite list of numbers specifies.

Exercise 2.4 *(page 28).* $\mathbb{P}(Y_i \geq c) = 1 - \Phi(c - \mu) = \Phi(\mu - c)$, strictly increasing in μ, so μ is identifiable: $\mu = c + \Phi^{-1}(p)$. With $\mu = 0$ known and σ unknown, $\mathbb{P}(Y_i \geq c) = \Phi(-c/\sigma)$: for $c \neq 0$ this is strictly monotone in σ and σ is identifiable; for $c = 0$ it equals $1/2$ for every σ, and σ is not. The threshold matters: thresholding at the known mean throws away all information about the spread, as in the uniform example.

Exercise 2.5 *(page 28).* Ordering the means removes the label-switching ambiguity, and it's a theorem (not proved here) that finite Gaussian mixtures are then identifiable. The endpoint $\pi = 1$ is excluded from Θ for a good reason: at $\pi = 1$ the density is f_1 whatever (μ_2, σ_2^2) are, so every value of the second component's parameters gives the same distribution. A component with zero weight is invisible, which is why the weight is required to be in the open interval $(0, 1)$.

Chapter 3. The Law of Large Numbers

Exercise 3.1 *(page 44).* (a) $\mathbb{P}(|X_n| > \epsilon) = \mathbb{P}(X_n = 1) = 1/n$. (b) Yes, to 0, since $1/n \to 0$. (c) Yes, to 0: convergence in probability implies convergence in distribution (Proposition 4.4 in the next chapter), or directly, the CDF of X_n is $1 - 1/n$ on $[0, 1)$ and 1 from 1 on, which converges to the CDF of the constant 0 at every x (at $x = 0$ too, here). (d) $\mathbb{E}[\cos X_n] = (1 - \frac{1}{n}) \cos 0 + \frac{1}{n} \cos 1 \to 1$; the next chapter's Proposition 4.3 says this is no accident, since cos is continuous and bounded.

Exercise 3.2 *(page 45).* $\mathbb{E}[\bar{X}_n] = 1/\lambda$ for every n and $\mathbb{V}\mathrm{ar}(\bar{X}_n) = 1/(n\lambda^2) \to 0$, so Proposition 3.3 applies. Chebyshev with $\epsilon = 0.1/\lambda$ gives $\mathbb{P}(|\bar{X}_n - 1/\lambda| \geq 0.1/\lambda) \leq \frac{1/(n\lambda^2)}{0.01/\lambda^2} = \frac{100}{n}$, which is at most 0.05 once $n \geq 2000$. The CLT of the next chapter, used as in its exercise 4, gives $n \approx 1.96^2/0.01 \approx 385$, which is the usual gap between a bound and the truth.

Exercise 3.3 *(page 45).* $\mathbb{P}(|\bar{R}_n - p| \geq 0.02) \leq p(1-p)/(n \times 0.0004) \leq 0.05$ requires $n \geq p(1 - p)/(0.00002) = 0.229/0.00002 \approx 11{,}450$. With the worst case $p(1 - p) = 1/4$ the requirement is 12,500; the dependence on p is mild because $p(1 - p)$ is flat near $1/2$.

Chapter 4. The Central Limit Theorem

Exercise 4.1 *(page 59).* (a) $|T_n - 5| \leq \frac{1}{2n}$ always, so the probability is 0 as soon as $\frac{1}{2n} \leq \delta$, i.e., for $n \geq \frac{1}{2\delta}$; any $N \geq \frac{1}{2\delta}$ works. (b) $T_n \xrightarrow[n\to\infty]{\mathbb{P}} 5$ (and so also in distribution). (c) For all $t \neq 5$: $F_n(5) = \frac{1}{2}$ for every n while $F(5) = 1$. See Example 4.2 and Figure 4.1.

Exercise 4.2 *(page 59).* (a) Since $M_n \leq 1$, $\mathbb{P}(|M_n - 1| > \delta) = \mathbb{P}(M_n < 1 - \delta) = \mathbb{P}(Y_1 < 1 - \delta)^n = (1 - \delta)^n$. (b) Yes: $(1 - \delta)^n \to 0$, so $M_n \xrightarrow[n\to\infty]{\mathbb{P}} 1$. (c) $F_{M_n}(x) = \mathbb{P}(Y_1 \leq x, \ldots, Y_n \leq x) = x^n$. As $n \to \infty$ this tends to 0 for $x < 1$ and equals 1 at $x = 1$: the CDF of the constant 1, at its continuity points.

Exercise 4.3 *(page 59).* Chebyshev with $\epsilon = c\sigma/\sqrt{n}$ gives $\frac{\sigma^2/n}{c^2\sigma^2/n} = 1/c^2$: 0.25 for $c = 2$ and 0.028 for $c = 6$. Hoeffding with $(b-a)^2 = b^2$ gives $2\exp(-2n\epsilon^2/b^2) = 2\exp(-2c^2\sigma^2/b^2) = 2e^{-c^2/6}$: $2e^{-2/3} = 1.03$ (useless, since a probability is at most 1) for $c = 2$, and $2e^{-6} = 0.005$ for $c = 6$. Chebyshev wins for small c; Hoeffding's bound decays like $e^{-c^2/6}$ against Chebyshev's $1/c^2$, so Hoeffding wins once c is large, exactly as in Figure 4.5 with n in place of c.

Exercise 4.4 *(page 59).* The CLT gives $\mathbb{P}(|\bar{R}_n - p| \geq \epsilon) \approx 2(1 - \Phi(\epsilon\sqrt{n}/\sqrt{p(1-p)}))$, which equals 0.05 when $\epsilon\sqrt{n}/\sqrt{p(1-p)} = 1.96$, i.e., $n = 1.96^2 \times 0.229/0.02^2 \approx 2{,}200$. Chebyshev demanded about 11,450, five times as many, for the same guarantee.

Chapter 5. Working with Limits

Exercise 5.1 *(page 70).* With $g(p) = p(1-p)$, $g'(p) = 1 - 2p$, and the CLT gives asymptotic variance $p(1-p)$ for $\bar{X}_n$, so the limit is $\text{Gauss}(0, (1-2p)^2 p(1-p))$. At $p = \frac{1}{2}$ the derivative is 0 and the limit is degenerate: the sample variance of a fair coin is estimated with error of order $1/n$ rather than $1/\sqrt{n}$, and the second-order remark of Section 5.4 applies.

Exercise 5.2 *(page 70).* The CLT gives $\sqrt{n}(\bar{X}_n - 1/\lambda) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, 1/\lambda^2)$. With $g(x) = \log x$, $g'(1/\lambda) = \lambda$, so the asymptotic variance of $\log \bar{X}_n$ is $\lambda^2 \cdot \lambda^{-2} = 1$, free of λ. A limit whose variance doesn't depend on the unknown parameter needs no plug-in: the interval $\log \bar{X}_n \pm 1.96/\sqrt{n}$ for $\log(1/\lambda)$ has asymptotic level 95% with no further work, and can be exponentiated afterwards. This is called a variance-stabilizing transformation.

Exercise 5.3 *(page 70).* By the CLT, $\sqrt{n}(\bar{X}_n - p)/\sqrt{p(1-p)} \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1)$. By the LLN, $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} p$, and $x \mapsto \sqrt{p(1-p)/(x(1-x))}$ is continuous at p, so by the continuous mapping theorem $\sqrt{p(1-p)/(\bar{X}_n(1-\bar{X}_n))} \xrightarrow[n\to\infty]{\mathbb{P}} 1$. Slutsky's multiplicative property with $y = 1$ gives the product's limit, $1 \cdot \text{Gauss}(0,1)$.

Chapter 6. The Gaussian Distribution and Quantiles

Exercise 6.1 *(page 81).* From the table, $\Phi(1) = 0.8413$. By symmetry $\mathbb{P}(Z \geq -1) = \mathbb{P}(Z \leq 1) = 0.8413$, and $\mathbb{P}(|Z| > 1) = 2(1 - \Phi(1)) = 0.3173$. So about a third of standard Gaussian draws fall more than one standard deviation from the mean, the complement of the 68% in Figure 6.1.

Exercise 6.2 *(page 81).* By the affine property $Y \sim \text{Gauss}(2\mu, 4\sigma^2)$, so $f_Y(y) = \frac{1}{2\sigma\sqrt{2\pi}} \exp\left(-\frac{(y-2\mu)^2}{2\cdot 4\sigma^2}\right)$. By change of variables with $g(x) = 2x$, $g^{-1}(y) = y/2$ and $g' = 2$: $f_Y(y) = \frac{1}{2} f_X(y/2) = \frac{1}{2\sigma\sqrt{2\pi}} \exp\left(-\frac{(y/2-\mu)^2}{2\sigma^2}\right)$, and $(y/2-\mu)^2/(2\sigma^2) = (y-2\mu)^2/(8\sigma^2)$, the same thing.

Exercise 6.3 *(page 81).* Standardize: $(X-2)/\sqrt{2} \sim \text{Gauss}(0,1)$, so $\mathbb{P}(X \geq -x) = \mathbb{P}\left(Z \geq \frac{-x-2}{\sqrt{2}}\right) = 1 - \Phi\left(-\frac{x+2}{\sqrt{2}}\right) = \Phi\left(\frac{x+2}{\sqrt{2}}\right)$ by symmetry. For $x = 1$ this is $\Phi(2.12) = 0.983$.

Exercise 6.4 *(page 81).* $2\Phi(1)-1 = 0.6827$, $2\Phi(2)-1 = 0.9545$, $2\Phi(3)-1 = 0.9973$. The complement of the second is $\mathbb{P}(|X-\mu| > 2\sigma) = \mathbb{P}(|Z| > 2) = 0.0455$. The event $|X-\mu| > 2\sigma$ is the event $|Z| > 2$ for the standardized variable, whose distribution is the same whatever μ and σ are.

Exercise 6.5 *(page 81).* $c = q_{0.10} = 1.28$ and $c' = q_{0.05} = 1.645$. A confidence interval is a two-sided statement (the truth may be on either side of the estimate), so it needs $c' = 1.645$: $\mathbb{P}(|\bar{X}_n - \mu| \le 1.645\,\sigma/\sqrt{n}) \approx 0.90$. Using 1.28 would give an 80% interval.

Chapter 7. Estimators: Bias, Variance and Risk

Exercise 7.1 *(page 97).* $\mathbb{E}[\bar{X}_n] = a + \frac{1}{2}$, bias $\frac{1}{2}$. $\mathbb{V}\mathrm{ar}(X_1) = \int_a^{a+1} x^2\,\mathrm{d}x - (a+\frac{1}{2})^2 = a^2 + a + \frac{1}{3} - a^2 - a - \frac{1}{4} = \frac{1}{12}$, so $\mathbb{V}\mathrm{ar}(\bar{X}_n) = \frac{1}{12n}$. The estimator $\bar{X}_n - \frac{1}{2}$ is unbiased with the same variance, so its risk is $\frac{1}{12n}$, which tends to 0.

Exercise 7.2 *(page 97).* $\hat{p} \sim \mathrm{Bern}(p^2)$, so bias $= p^2 - p$, variance $= p^2(1-p^2)$, and risk $= p^2(1-p^2) + (p^2-p)^2 = p^2(1-p^2) + p^2(1-p)^2$. None of these depend on n, so the risk does not go to 0. The estimator is not consistent either, for the direct reason that its distribution never changes with n: $\mathbb{P}(|\hat{p} - p| > \epsilon)$ is the same positive number for every n. It never uses $X_3, \ldots, X_n$.

Exercise 7.3 *(page 97).* $\mathbb{E}[\tilde{p}_n] = (np+10)/(n+20)$ and $\mathbb{V}\mathrm{ar}(\tilde{p}_n) = np(1-p)/(n+20)^2$. At $p = 0.645$: mean 0.6249, bias -0.0201, variance 0.001369, risk 0.001775, against $\bar{R}_n$'s risk 0.001847. At $p = 0.9$: mean 0.8444, bias -0.0556, variance 0.000538, risk 0.003625, against $\bar{R}_n$'s 0.000726. Shrinking toward $\frac{1}{2}$ pays when p is near $\frac{1}{2}$ and costs five times over when it isn't; Figure 7.5 shows the crossover at $p \approx 0.34$ and 0.66.

Exercise 7.4 *(page 97).* $g(x) = 1/x$ is convex on $(0,\infty)$, so $\mathbb{E}[1/\bar{X}_n] \ge 1/\mathbb{E}[\bar{X}_n] = \lambda$: the bias is non-negative. Exactly, the bias is $\frac{n}{n-1}\lambda - \lambda = \frac{\lambda}{n-1}$, positive and of order $1/n$. The estimator $\frac{n-1}{n} \cdot \frac{1}{\bar{X}_n} = (n-1)/\sum X_i$ is unbiased.

Exercise 7.5 *(page 97).* Bias and variance tending to 0 means $R(\hat{\theta}_n) = \mathbb{E}[(\hat{\theta}_n - \theta)^2] \to 0$, which implies convergence in probability by Chebyshev's inequality (Proposition 3.3; in the language of Appendix B, convergence in quadratic mean implies convergence in probability). The converse fails: convergence in probability does not require any moments to exist. For an example, take $\hat{\theta}_n = \theta$ with probability $1 - 1/n$ and $\hat{\theta}_n = n$ with probability $1/n$; then $\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta$, but $\mathbb{E}[\hat{\theta}_n] = \theta(1-1/n) + 1 \to \theta + 1$, so the bias tends to 1, not 0.

Chapter 8. Confidence Intervals

Exercise 8.1 *(page 112).* $q_{0.01} = 2.326$. Conservative: the half-width is $2.326/(2\sqrt{124}) = 0.1045$, so $0.6452 \pm 0.1045 = [0.541, 0.750]$. Plug-in: the half-width is $2.326 \times 0.0430 = 0.1000$, so $0.6452 \pm 0.1000 = [0.545, 0.745]$. Both contain their 95% versions, as they must for intervals of the same type. A 95% interval is a 90% interval (level is a floor) but is not in general a 98% interval, so the answers are No to the 98% question and Yes to the 90% one.

Exercise 8.2 *(page 112).* (1) is the realization of $[0, \bar{R}_n]$, and $\mathbb{P}_p([0, \bar{R}_n] \ni p) = \mathbb{P}_p(\bar{R}_n \geq p) \to \mathbb{P}(Z \geq 0) = \frac{1}{2}$ for every p, by the CLT: a 50% asymptotic interval, a strange one. (2) is $[\bar{R}_n, 1]$, the same argument. (3) is the realization at $n = 150$ of the rule $\bar{R}_n \pm c/\sqrt{n}$ with $c = 0.0335\sqrt{150} = 0.41$ (a fixed half-width would have limiting coverage 1, so the rule has to shrink with n), and the conservative bound gives $\lim_{n\to\infty} \mathbb{P}_p(|\bar{R}_n - p| \leq c/\sqrt{n}) \geq \mathbb{P}(|Z| \leq 2c) = \mathbb{P}(|Z| \leq 0.82) = 0.59 \geq 0.5$ for every p: yes. (4) is the rule $\bar{R}_n \pm c/\sqrt{n}$ with $c = 0.02\sqrt{150} = 0.245$, and at $p = \frac{1}{2}$ its limiting coverage is $\mathbb{P}(|Z| \leq 2c) = \mathbb{P}(|Z| \leq 0.49) = 0.38 < 0.5$. The definition asks for coverage at every p, and $p = \frac{1}{2}$ is a p, so (4) is not a 50% interval even though it is one for p far from $\frac{1}{2}$. The realizations of 50% asymptotic intervals are therefore (1), (2) and (3).

Exercise 8.3 *(page 112).* $\hat{p}(1-\hat{p}) \leq \frac{1}{4}$ for every $\hat{p} \in [0, 1]$, so the plug-in half-width $q_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$ is at most the conservative one $q_{\alpha/2}/(2\sqrt{n})$, with equality exactly when $\hat{p} = \frac{1}{2}$. From Equation (8.2), the center of $\mathcal{I}_{\text{solve}}$ is $(\bar{R}_n + q^2_{\alpha/2}/(2n))/(1 + q^2_{\alpha/2}/n)$, which equals $\bar{R}_n$, the center of $\mathcal{I}_{\text{plug-in}}$, if and only if $\bar{R}_n + q^2/(2n) = \bar{R}_n + \bar{R}_n q^2/n$, that is, $\bar{R}_n = \frac{1}{2}$. Otherwise the solve interval's center lies strictly between $\bar{R}_n$ and $\frac{1}{2}$.

Exercise 8.4 *(page 112).* (a) $\bar{X}_n \sim \text{Gauss}(\theta, \theta/n)$ exactly, so $\sqrt{n/\theta}\,(\bar{X}_n - \theta) \sim \text{Gauss}(0, 1)$ for every n and θ. (b) With $t = q_{0.05} = 1.645$: $\mathcal{I}_\theta = [\bar{X}_n - t\sqrt{\theta/n},\ \bar{X}_n + t\sqrt{\theta/n}]$, exact coverage 0.90, not computable. (c) $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta$, so Slutsky gives $\sqrt{n/\bar{X}_n}\,(\bar{X}_n - \theta) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, 1)$ and $\mathcal{I}_{\text{plug-in}} = \bar{X}_n \pm t\sqrt{\bar{X}_n/n}$, asymptotic level 90%. (d) $\frac{n}{\theta}(\bar{X}_n - \theta)^2 \leq t^2$ is $\theta^2 - (2\bar{X}_n + t^2/n)\theta + \bar{X}_n^2 \leq 0$, so

$$\mathcal{I}_{\text{solve}} = \left[\bar{X}_n + \frac{t^2}{2n} - \sqrt{\Delta},\ \bar{X}_n + \frac{t^2}{2n} + \sqrt{\Delta}\right], \qquad \Delta = \frac{t^4}{4n^2} + \frac{t^2\bar{X}_n}{n}.$$

Because the pivot in (a) is exact, so is this interval's level: it is a confidence interval of level 90% for every n, not only asymptotically. Solving adds no approximation beyond the pivot's own; plugging in always adds one.

Exercise 8.5 *(page 112).* (a) $\mathbb{E}[X_1] = a + 1$ and $\mathbb{V}\text{ar}(X_1) = 1$ (a shift doesn't change the variance of Expo(1)), so $\hat{a}_1 \xrightarrow[n\to\infty]{\mathbb{P}} a$ by the LLN and $\sqrt{n}(\hat{a}_1 - a) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0, 1)$ by the CLT. The asymptotic variance is known, so there is nothing to plug in: $\mathcal{I}_1 = \hat{a}_1 \pm q_{\alpha/2}/\sqrt{n}$. (b) For $t \geq a$, $\mathbb{P}(\hat{a}_2 > t) = \mathbb{P}(X_i > t\ \forall i) = (e^{-(t-a)})^n = e^{-n(t-a)}$, so $\mathbb{P}(n(\hat{a}_2 - a) > s) = e^{-s}$: $n(\hat{a}_2 - a) \sim \text{Expo}(1)$, for every n. Since $\hat{a}_2 \geq a$ always, the natural interval is $[\hat{a}_2 - s, \hat{a}_2]$, and $\mathbb{P}(a \in [\hat{a}_2 - s, \hat{a}_2]) = \mathbb{P}(n(\hat{a}_2 - a) \leq ns) = 1 - e^{-ns}$. Setting this to $1 - \alpha$ gives $s = \log(1/\alpha)/n$ and $\mathcal{I}_2 = [\hat{a}_2 - \log(1/\alpha)/n,\ \hat{a}_2]$, exact level $1 - \alpha$. (c) $\mathcal{I}_1$ has width $2q_{\alpha/2}/\sqrt{n}$ and $\mathcal{I}_2$ has width $\log(1/\alpha)/n$: at $n = 100$ and $\alpha = 0.05$, 0.39 against 0.03. The minimum is tailored to the model, uses its exact distribution, and needs no asymptotics. It is also fragile: one observation below a, impossible under the model but possible under a slightly wrong one, breaks it completely, while the sample mean barely notices. The generic $1/\sqrt{n}$ interval is the price of robustness.

Exercise 8.6 *(page 112).* One-sided, so $q_{0.05} = 1.645$. Plug-in: $[0.645 - 1.645 \times 0.0430, 1] = [0.574, 1]$. Conservative: $[0.645 - 1.645/(2\sqrt{124}), 1] = [0.571, 1]$. With 95% confidence, at least 57% of couples turn right. The conservative lower bound exceeds $\frac{1}{2}$ when $\bar{R}_n - 1.645/(2\sqrt{n}) > \frac{1}{2}$, that is, $\bar{R}_n > 0.5 + 0.0739 = 0.5739$, that is, $n\bar{R}_n > 71.2$, that is, $n\bar{R}_n \geq 72$. A confidence interval excluding the value $\frac{1}{2}$ and a test rejecting $p = \frac{1}{2}$ are the same decision looked at from two sides, and the conservative scale $\frac{1}{2}$ is exactly the scale the test uses under its null hypothesis; Chapter 9 makes this duality general.

Exercise 8.7 *(page 112).* The pivot is $G = \lambda n\bar{T}_n$, whose distribution doesn't involve λ. Trap it: $\mathbb{P}(49.29 \le \lambda n\bar{T}_n \le 80.60) = 0.95$ for every λ and this n. Solve for λ: $\lambda \in [49.29/(n\bar{T}_n),\ 80.60/(n\bar{T}_n)]$, an interval of exact level 95% with computable endpoints. With $n\bar{T}_n = 398.7$: $[0.124, 0.202]$, between the solve interval $[0.129, 0.213]$ and the plug-in interval $[0.121, 0.200]$ of Section 8.6, and like the solve interval not centered at $\hat{\lambda} = 0.1605$. This is the exact interval Section 4.4 mentioned; it exists because the parent distribution is known in full, which is exactly what the CLT lets you do without.

Chapter 9. Hypothesis Tests

Exercise 9.1 *(page 129).* $H_0 : p = \frac{1}{2}$ against $H_1 : p \neq \frac{1}{2}$ in the model $(\{0,1\}, (\text{Bern}(p))_{p\in(0,1)})$. $T_n = 2\sqrt{124}(0.6452 - 0.5) = 3.23$. CLT p-value: $2(1 - \Phi(3.23)) = 0.0012$. Exact: $\mathbb{P}_{1/2}(n\bar{R}_n \ge 80) = 0.0008$, doubled for two sides, 0.0016. Both are far below 0.05 (and below 0.01): reject H_0 at any conventional level; the data are not compatible with equal appeal. One-sided, $H_1 : p > \frac{1}{2}$: p-values $1 - \Phi(3.23) = 0.0006$ (CLT) and 0.0008 (exact), half the two-sided ones.

Exercise 9.2 *(page 129).* Let μ be the mean number of chips. The claim is the null, $H_0 : \mu \ge 20$, and the student's doubt is the alternative, $H_1 : \mu < 20$: a one-sided test with the threshold computed at the boundary $\mu_0 = 20$. With σ replaced by the sample standard deviation, $T_n = \sqrt{32}\,(14.77 - 20)/4.37 = -6.77$. By the CLT and Slutsky the p-value is approximately $\Phi(-6.77) \approx 6 \times 10^{-12}$: reject at every conventional level; the difference is not chance. The approximation is the Gaussian limit at $n = 32$ with an estimated σ; Chapter 14 replaces it by Student's t with 31 degrees of freedom, for which -6.77 is just as decisive.

Exercise 9.3 *(page 129).* The claim needs evidence, so it is the alternative: $H_0 : p \ge 0.95$ against $H_1 : p < 0.95$, one-sided, threshold at the boundary $p_0 = 0.95$. $\hat{p} = 206/297 = 0.6936$ and $T_n = \sqrt{297}\,(0.6936 - 0.95)/\sqrt{0.95 \times 0.05} = -20.3$. The p-value $\Phi(-20.3)$ is zero to every decimal place anyone prints: reject H_0; the procedure's false positive rate is below 95%. (Whether it is below by enough to matter is the confidence interval's question: $0.694 \pm 1.96\sqrt{0.694 \times 0.306/297} = [0.64, 0.75]$.)

Exercise 9.4 *(page 129).* $T_n = -0.73$, CLT p-value $2\Phi(-0.73) = 0.47$. Exactly, $\mathbb{P}_{1/2}(\text{Bin}(30, \frac{1}{2}) \le 13) = 0.292$, doubled 0.58. Neither is close to rejecting. The gap between 0.47 and 0.58 is the discreteness of Figure 9.4 at $n = 30$: the Gaussian tail beyond 13 undercounts the mass of the bar at 13 itself, which the continuity correction (evaluating at 13.5) largely repairs: $2\Phi(\sqrt{30}(13.5/30 - 0.5)/0.5) = 2\Phi(-0.55) = 0.58$.

Exercise 9.5 *(page 129).* $\Theta_0 = [1, 3]$. Reject when $\hat{\lambda}$ is far outside it on either side, with a margin that shrinks like $1/\sqrt{n}$ and uses the standard deviation $\sqrt{\lambda}$ at its own endpoint: $\psi = \mathbb{1}(\hat{\lambda} > 3 + s\sqrt{3/n}) + \mathbb{1}(\hat{\lambda} < 1 - s\sqrt{1/n})$ for a number s to be chosen (the two events are disjoint, since $\hat{\lambda}$ cannot be both above 3 and below 1). The type 1 error at $\lambda \in [1, 3]$ is largest at the endpoints: at $\lambda = 3$ the first indicator is the event $\sqrt{n}(\hat{\lambda} - 3)/\sqrt{3} > s$, whose probability tends to $\mathbb{P}(Z > s)$ by the CLT, and the second indicator's probability tends to 0 by the LLN; at $\lambda = 1$ the reverse, with limit $\mathbb{P}(Z < -s)$. At any interior λ both probabilities tend to 0. Choosing $s = 1.645$ makes each endpoint's limiting type 1 error 0.05, so the test has asymptotic level 5% (not 2.5% per side: at any one λ only one side can fire in the limit). Each side uses the standard deviation $\sqrt{\lambda}$ at its own endpoint, which is the 'parameter known under the null' idea applied endpoint by endpoint.

Exercise 9.6 *(page 129).* For $u \in (0,1)$, $\mathbb{P}_{H_0}(P \le u) = \mathbb{P}_{H_0}(1 - F_0(T) \le u) = \mathbb{P}_{H_0}(F_0(T) \ge 1-u) = \mathbb{P}_{H_0}(T \ge F_0^{-1}(1-u)) = 1 - F_0(F_0^{-1}(1-u)) = u$, using that F_0 is continuous and strictly increasing so that F_0^{-1} exists and $\mathbb{P}(T = t) = 0$ (toolkit A.6). A CDF equal to u on $(0,1)$ is the uniform CDF. Then $\mathbb{P}_{H_0}(P \le \alpha) = \alpha$, which is the golden rule's 'reject when $P \le \alpha$' having type 1 error exactly α. For a discrete T the equality becomes $\le \alpha$, the staircase of Section 9.6.

Exercise 9.7 *(page 129).* Not rejecting p_0 means $|\bar{R}_n - p_0| \le q_{\alpha/2}\sqrt{p_0(1-p_0)/n}$, that is, $(\bar{R}_n - p_0)^2 \le q_{\alpha/2}^2\, p_0(1-p_0)/n$, the quadratic inequality in p_0 whose solution set is the solve interval (Equation (8.2)). With $\sqrt{\bar{R}_n(1-\bar{R}_n)/n}$ in place of $\sqrt{p_0(1-p_0)/n}$ the inequality is $|\bar{R}_n - p_0| \le q_{\alpha/2}\sqrt{\bar{R}_n(1-\bar{R}_n)/n}$, whose solution set is $\bar{R}_n \pm q_{\alpha/2}\sqrt{\bar{R}_n(1-\bar{R}_n)/n}$, the plug-in interval. The conservative interval inverts the test that uses the worst-case standard error $1/(2\sqrt{n})$ for every p_0.

Chapter 10. Distances Between Distributions

Exercise 10.1 *(page 150).* On $E = \{0,1\}$ the events are $\emptyset$, $\{0\}$, $\{1\}$ and E, and the first and last have the same probability under any two distributions. For Bern($\frac{1}{2}$) against Bern(q), the event $\{1\}$ has probabilities $\frac{1}{2}$ and q, and $\{0\}$ has $\frac{1}{2}$ and $1-q$; both differences are $|\frac{1}{2} - q|$. So the three distances are 0, $\frac{1}{6} = 0.167$ and 0.4, matching $|p - q|$. The first is 0 whether or not two Bernoulli variables with these distributions are independent: total variation compares distributions, not random variables, and their joint behavior doesn't enter.

Exercise 10.2 *(page 150).* The computation is the third item of the worked example in Section 10.2. The event on which Expo(1) puts more mass is $A^\star = \{x > 1\}$, where the uniform density is 0; its probabilities are e^{-1} and 0, so $|\mathbb{P}(A^\star) - \mathbb{Q}(A^\star)| = e^{-1} = 0.368$, the same answer as the integral. The complementary event $[0,1]$ favors the uniform by the same amount, $1 - (1 - e^{-1}) = e^{-1}$.

Exercise 10.3 *(page 150).* X takes values in $\{0,1\}$ and $X + a$ in $\{a, 1+a\}$, disjoint from it since $0 < a < 1$. The event $A = \{0,1\}$ has $\mathbb{P}(X \in A) = 1$ and $\mathbb{P}(X + a \in A) = 0$, so TV $= 1$, the largest possible value. For the second, $2\sqrt{n}(\bar{X}_n - \frac{1}{2})$ takes values in the finite set $S_n = \{2\sqrt{n}(k/n - \frac{1}{2}) : k = 0, \dots, n\}$, so $\mathbb{P}(2\sqrt{n}(\bar{X}_n - \frac{1}{2}) \in S_n) = 1$, while $\mathbb{P}(Z \in S_n) = 0$ because a continuous random variable has probability 0 of landing in any finite set. Hence TV $= 1$. The CLT says $\mathbb{P}(2\sqrt{n}(\bar{X}_n - \frac{1}{2}) \le t) \to \Phi(t)$ for every t: convergence of the probabilities of the intervals $(-\infty, t]$. Total variation asks for agreement on all events at once, including the finite set S_n, which changes with n. Convergence in distribution is the weaker requirement.

Exercise 10.4 *(page 150).* $\alpha = \mathbb{P}_{p_0}(R)$, $\beta = \mathbb{P}_{p_1}(R^c)$. $R = \emptyset$: $\alpha = 0$, $\beta = 1$, sum 1. $R = \{0,1\}$: $\alpha = 1$, $\beta = 0$, sum 1. $R = \{1\}$: $\alpha = p_0$, $\beta = 1 - p_1$, sum $1 - (p_1 - p_0)$. $R = \{0\}$: $\alpha = 1 - p_0$, $\beta = p_1$, sum $1 + (p_1 - p_0)$. The smallest is $1 - (p_1 - p_0) = 1 - \text{TV}$, from the test that rejects when $X = 1$, the outcome Bern(p_1) favors. For the kiss values, 0.855: one couple barely distinguishes the two hypotheses.

Exercise 10.5 *(page 150).* The binomial coefficients cancel in the ratio: $\log\left(p_{\text{Bin}(n,p)}(k)/p_{\text{Bin}(n,q)}(k)\right) = k\log(p/q) + (n-k)\log((1-p)/(1-q))$. This is linear in k, and under Bin(n,p), $\mathbb{E}[K] = np$, so the expectation is $np\log(p/q) + n(1-p)\log((1-p)/(1-q)) = n\,\text{KL}(\text{Bern}(p), \text{Bern}(q))$. For the kiss data, $124 \times 0.0428 = 5.30$. The count of right turns carries the same evidence as the full sequence of turns, which is why the distributions of the sequence and of the count give the same divergence.

Exercise 10.6 *(page 150).* The log-ratio of the densities is

$$\log \frac{p(x)}{q(x)} = \log \frac{\tau}{\sigma} - \frac{(x-a)^2}{2\sigma^2} + \frac{(x-b)^2}{2\tau^2}.$$

Under $X \sim \text{Gauss}(a, \sigma^2)$, $\mathbb{E}[(X-a)^2] = \sigma^2$ and $\mathbb{E}[(X-b)^2] = \sigma^2 + (a-b)^2$, so the expectation is $\log(\tau/\sigma) - \frac{1}{2} + (\sigma^2 + (a-b)^2)/(2\tau^2)$. With $\tau = \sigma$ the first term is 0 and the rest is $\frac{1}{2} + (a-b)^2/(2\sigma^2) - \frac{1}{2}$. For $\sigma = 0.01$, $\tau = 0.02$, $a = b = 0$: $\log 2 + \frac{1}{8} - \frac{1}{2} = 0.318$. The other way round, $\sigma = 0.02$, $\tau = 0.01$: $\log \frac{1}{2} + 2 - \frac{1}{2} = 0.807$. Believing the narrow distribution when the wide one is true is costlier: the wide one regularly produces values the narrow one calls very unlikely.

Exercise 10.7 *(page 150).* The derivation is the third item of the worked example in Section 10.5. With $t = \mu/\lambda > 0$ the divergence is $-\log t + t - 1 = g(t)$. Then $g(1) = 0$, and $g'(t) = 1 - 1/t$ is negative for $t < 1$ and positive for $t > 1$, so g has its minimum 0 at $t = 1$ and is positive elsewhere. This is Proposition 10.6 for this family, by direct calculation instead of Jensen.

Exercise 10.8 *(page 150).* $\mathbb{P}_n$ puts mass $e^{-1/n}$ at 0 and $1 - e^{-1/n}$ on $\{1, 2, \ldots\}$. By Proposition 10.2, $\text{TV} = \frac{1}{2}\big(|e^{-1/n} - 1| + (1 - e^{-1/n})\big) = 1 - e^{-1/n} \to 0$ (0.095 at $n = 10$). $\text{KL}(\mathbb{P}_n, \delta_0) = +\infty$ for every n, since $\mathbb{P}_n$ gives positive probability to 1, which δ_0 calls impossible. $\text{KL}(\delta_0, \mathbb{P}_n)$ has a single term, at $x = 0$: $1 \cdot \log\big(1/e^{-1/n}\big) = 1/n \to 0$. So $\mathbb{P}_n$ approaches δ_0 in total variation and in one direction of KL, and stays infinitely far in the other.

Exercise 10.9 *(page 150).* $\log p_\lambda(x) = -\lambda + x \log \lambda - \log(x!)$, so $\widehat{\text{KL}} = \text{constant} - \frac{1}{n}\sum_i \log p_\lambda(X_i) = \text{constant}' + \lambda - \bar{X}_n \log \lambda$, where $\frac{1}{n}\sum_i \log(X_i!)$ has been absorbed into the constant since it doesn't depend on λ. The derivative $1 - \bar{X}_n/\lambda$ is zero at $\lambda = \bar{X}_n$, negative before and positive after. The minimizer, which is the maximum likelihood estimator, is the sample mean. It is the sample version of Proposition 10.10(2), with the true mean replaced by the sample mean.

Exercise 10.10 *(page 150).* $-\mathbb{E}[\log p_\theta(X)] = -\mathbb{E}[\log \binom{2}{X}] - \mathbb{E}[X] \log \theta - (2 - \mathbb{E}[X]) \log(1-\theta)$, and only the last two terms depend on θ. Setting the derivative $-\mathbb{E}[X]/\theta + (2 - \mathbb{E}[X])/(1-\theta)$ to zero gives $\theta^\circ = \mathbb{E}[X]/2 = (0.2 + 0.6)/2 = 0.4$: the Binomial with the same mean. $\text{Bin}(2, 0.4)$ has PMF $(0.36, 0.48, 0.16)$, and $\text{KL} = 0.5 \log(0.5/0.36) + 0.2 \log(0.2/0.48) + 0.3 \log(0.3/0.16) = 0.164 - 0.175 + 0.189 = 0.178$. The model is not well-specified: every $\text{Bin}(2, \theta)$ satisfies $\mathbb{P}(X = 1)^2 = 4\mathbb{P}(X = 0)\mathbb{P}(X = 2)$, and here $0.04 \neq 0.6$. The fitted model matches the mean but puts far too much mass on 1 and too little on 0 and 2.

Chapter 11. The Likelihood and Maximum Likelihood

Exercise 11.1 *(page 167).* $L_2(1, 2; \lambda) = e^{-\lambda}\frac{\lambda^1}{1!} \cdot e^{-\lambda}\frac{\lambda^2}{2!} = e^{-2\lambda}\frac{\lambda^3}{2}$. From the definition, $L_3(1, 2, 3; \lambda) = e^{-3\lambda}\frac{\lambda^{1+2+3}}{1!\,2!\,3!} = e^{-3\lambda}\frac{\lambda^6}{12}$. By updating, $L_3 = L_2 \cdot e^{-\lambda}\frac{\lambda^3}{3!} = e^{-2\lambda}\frac{\lambda^3}{2} \cdot e^{-\lambda}\frac{\lambda^3}{6} = e^{-3\lambda}\frac{\lambda^6}{12}$. $L_3(2, 3, 1; \lambda)$ is the same product with the factors in a different order, so it is also $e^{-3\lambda}\lambda^6/12$: the likelihood of i.i.d. data does not depend on the order of the observations. The MLE from the three observations is $\bar{X}_3 = 2$.

Exercise 11.2 *(page 167).* True. $L_n(\theta) = \prod_{i=1}^n p_\theta(x_i)$ is a product of n continuous functions of θ, and a product of continuous functions is continuous. The Bernoulli likelihood $p^S(1-p)^{n-S}$ is a polynomial in p, as smooth as a function can be, although the data take only the values 0 and 1. The Poisson likelihood $e^{-n\lambda}\lambda^{\sum x_i}/\prod x_i!$ is another.

Exercise 11.3 *(page 167).* $\ell_n(\sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i X_i^2$. With $v = \sigma^2$ and $Q = \sum_i X_i^2$, $\ell_n'(v) = -\frac{n}{2v} + \frac{Q}{2v^2} = \frac{Q-nv}{2v^2}$, positive for $v < Q/n$ and negative for $v > Q/n$, so the unique maximizer is $\widehat{\sigma^2} = \frac{1}{n}\sum_i X_i^2$. (The log-likelihood is not concave in v for all v, which is why the sign of the derivative is used instead of the second derivative.) It is unbiased here, since the mean is known: $\mathbb{E}[X_i^2] = \sigma^2$. By invariance, $\hat\sigma = \sqrt{\frac{1}{n}\sum_i X_i^2}$.

Exercise 11.4 *(page 167).* $M \le m$ exactly when every $X_i \le m$, and by independence $\mathbb{P}(M \le m) = \prod_i \mathbb{P}(X_i \le m) = (m/\theta)^n$. Differentiating, M has density nm^{n-1}/θ^n on $[0,\theta]$. Then $\mathbb{E}[M] = \int_0^\theta m \cdot \frac{nm^{n-1}}{\theta^n}\,\mathrm{d}m = \frac{n}{\theta^n}\cdot\frac{\theta^{n+1}}{n+1} = \frac{n}{n+1}\theta$. So $\frac{n+1}{n}M$ is unbiased. For the five observations of Figure 11.6, $\frac{6}{5}\times 7.6 = 9.12$. The bias of M is $-\theta/(n+1)$, of order $1/n$, and its standard deviation is also of order $1/n$, not $1/\sqrt{n}$: this model is not one of the regular models whose large-sample theory Chapter 12 develops.

Exercise 11.5 *(page 167).* Each density is $\mathbb{1}(\theta \le x_i \le \theta+1)$, so $L(\theta) = \prod_i \mathbb{1}(\theta \le x_i)\mathbb{1}(x_i \le \theta+1) = \mathbb{1}(\theta \le \min_i x_i)\,\mathbb{1}(\theta \ge \max_i x_i - 1)$. It is 1 on $[\max_i x_i - 1, \min_i x_i]$ and 0 elsewhere, so every point of that interval maximizes it and the MLE is not unique. The interval is nonempty because all the data lie in an interval of length 1, $[\theta^*, \theta^*+1]$, so $\max_i x_i - \min_i x_i \le 1$. Its midpoint is a sensible choice of estimate, but the likelihood alone doesn't pick it.

Exercise 11.6 *(page 167).* $\ell_n(p) = \sum_i \big[(X_i - 1)\log(1-p) + \log p\big] = (S-n)\log(1-p) + n\log p$ with $S = \sum_i X_i$. $\ell_n'(p) = -\frac{S-n}{1-p} + \frac{n}{p} = 0$ gives $n(1-p) = (S-n)p$, so $p = n/S$: $\hat p = 1/\bar X_n$. $\ell_n''(p) = -\frac{S-n}{(1-p)^2} - \frac{n}{p^2} < 0$, so this is the unique maximum. For the data, $S = 12$ and $n = 5$, so $\hat p = 5/12 = 0.417$: five makes in twelve shots.

Exercise 11.7 *(page 167).* $L(a) = \prod_i e^{-(x_i - a)}\mathbb{1}(x_i \ge a) = e^{na - \sum_i x_i}\,\mathbb{1}(a \le \min_i x_i)$. For $a \le \min_i x_i$ this is increasing in a; beyond, it is 0. So $\hat a = \min_i X_i$, found at the jump, as for the uniform. Again no derivative vanishes where $L > 0$, and again the estimator is biased: $\min_i X_i - a \sim \text{Expo}(n)$, so $\mathbb{E}[\hat a] = a + 1/n$.

Exercise 11.8 *(page 167).* $\ell_n(m) = -n\log 2 - \sum_i |X_i - m|$, so maximizing it is minimizing $g(m) = \sum_i |X_i - m|$. Away from the data points, $g'(m) = (\text{number of } X_i < m) - (\text{number of } X_i > m)$, which is negative when fewer than half the observations lie below m and positive when more than half do. With n odd, g decreases until the middle observation and increases after it, so the minimizer is the sample median. The Gaussian's squared deviations give the mean; the Laplace's absolute deviations give the median. Chapter 13 takes this up as M-estimation.

Exercise 11.9 *(page 167).* The terms involving π are $\big(\sum_i w_i\big)\log\pi + \big(\sum_i (1-w_i)\big)\log(1-\pi)$: the Bernoulli log-likelihood with $\sum_i w_i$ 'successes' out of n, maximized at $\hat\pi = \frac{1}{n}\sum_i w_i$. The terms involving μ_1 are $-\sum_i w_i(X_i - \mu_1)^2/(2\sigma_1^2)$, a concave quadratic in μ_1 with derivative $\sum_i w_i(X_i - \mu_1)/\sigma_1^2$, zero at $\hat\mu_1 = \sum_i w_i X_i / \sum_i w_i$. For the variance, write $W = \sum_i w_i$ and $D_1 = \sum_i w_i(X_i - \hat\mu_1)^2$. The terms involving (μ_1, σ_1^2) are $-\frac{W}{2}\log(2\pi\sigma_1^2) - \sum_i w_i(X_i - \mu_1)^2/(2\sigma_1^2)$. For each σ_1^2 this is largest at $\mu_1 = \hat\mu_1$, as just shown; with $\mu_1 = \hat\mu_1$ it is $-\frac{W}{2}\log\sigma_1^2 - D_1/(2\sigma_1^2)$ plus a constant, whose derivative $-\frac{W}{2\sigma_1^2} + \frac{D_1}{2\sigma_1^4}$ is positive below D_1/W and negative above. Chaining the two steps as in the unweighted Gaussian case gives the joint maximum $\hat\sigma_1^2 = D_1/W$. The terms for component 2 are maximized separately in the same way, with weights $1 - w_i$.

Exercise 11.10 *(page 167).* With $S = 0$, $L(p) = (1-p)^n$, strictly decreasing on $(0,1)$, so for every p in Θ there is a smaller p with a larger likelihood: no maximum is attained. Allowing $p \in [0,1]$ gives $\hat{p} = 0$. The event has probability $0.9^5 = 0.59$ for $n = 5$ and $0.9^{50} = 0.005$ for $n = 50$.

Chapter 12. Properties of the MLE

Exercise 12.1 *(page 182).* $\ell(\lambda) = -\lambda + X\log\lambda - \log X!$, $\ell'(\lambda) = -1 + X/\lambda$, $\ell''(\lambda) = -X/\lambda^2$. $\mathbb{V}\mathrm{ar}(\ell') = \mathbb{V}\mathrm{ar}(X)/\lambda^2 = \lambda/\lambda^2 = 1/\lambda$; $-\mathbb{E}[\ell''] = \mathbb{E}[X]/\lambda^2 = 1/\lambda$. So $\sqrt{n}(\bar{X}_n - \lambda) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,\lambda)$. The CLT directly: $\bar{X}_n$ has mean λ and variance λ/n, so the same. For the siblings, standard error $\sqrt{1.0/24} = 0.20$.

Exercise 12.2 *(page 182).* In β the density is $\beta^{-1}e^{-x/\beta}$, so $\ell(\beta) = -\log\beta - X/\beta$, $\ell'(\beta) = -1/\beta + X/\beta^2$, $\ell''(\beta) = 1/\beta^2 - 2X/\beta^3$, and with $\mathbb{E}[X] = \beta$, $I(\beta) = -\mathbb{E}[\ell''] = -1/\beta^2 + 2/\beta^2 = 1/\beta^2$. Check: $I(\lambda) = 1/\lambda^2 = \beta^2$ and $\mathrm{d}\lambda/\mathrm{d}\beta = -1/\beta^2$, so $I(\lambda)(\mathrm{d}\lambda/\mathrm{d}\beta)^2 = \beta^2/\beta^4 = 1/\beta^2$. The theorem gives $\sqrt{n}(\bar{X}_n - \beta) \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,\beta^2)$, and the CLT gives the same since $\mathbb{V}\mathrm{ar}(X) = \beta^2$. For the subway, $\hat{\beta} = 6.23$ with standard error $6.23/8 = 0.78$ minutes.

Exercise 12.3 *(page 182).* $\ell(\mu) = -\frac{1}{2}\log(2\pi\sigma^2) - (X-\mu)^2/(2\sigma^2)$, $\ell''(\mu) = -1/\sigma^2$, so $I(\mu) = 1/\sigma^2$ and the bound is σ^2/n. $\mathbb{V}\mathrm{ar}(\bar{X}_n) = \sigma^2/n$ exactly, and $\bar{X}_n$ is unbiased, so it is efficient. (The Gaussian and Bernoulli sample means are efficient at every n; most MLEs reach the bound only in the limit.)

Exercise 12.4 *(page 182).* $\ell_n'(p) = 80/p - 44/(1-p)$: at $\frac{1}{2}$, $160 - 88 = 72$; at 0.6, $133.3 - 110 = 23.3$; at 0.645, 0; at 0.7, $114.3 - 146.7 = -32.4$. The score is positive below $\hat{p}$ (the data push the estimate up), zero at $\hat{p}$, negative above. At $p_0 = \frac{1}{2}$, $nI(p_0) = 124 \times 4 = 496$ and $72/\sqrt{496} = 3.23$, which is the $T_n = 2\sqrt{124}(0.645 - 0.5) = 3.23$ of Chapter 9: the standardized score at the null value is the score test statistic, and under H_0 it is approximately Gauss$(0,1)$ by the CLT for scores. Chapter 15 develops this.

Exercise 12.5 *(page 182).* From Exercise 11.4, $\mathbb{P}(M \le m) = (m/\theta)^n$. For $t \ge 0$, $\mathbb{P}\big(n(\theta - M)/\theta > t\big) = \mathbb{P}\big(M < \theta(1 - t/n)\big) = (1 - t/n)^n \to e^{-t}$ (toolkit A.10). So the CDF of $n(\theta - M)/\theta$ converges to $1 - e^{-t}$, the Expo(1) CDF. The error $\theta - M$ is therefore about θ/n times an exponential variable: order $1/n$, always positive, with a skewed limit. Scaling by $\sqrt{n}$ instead would send $\sqrt{n}(\theta - M) \xrightarrow[n\to\infty]{\mathbb{P}} 0$, a degenerate limit, which is what asymptotic normality at rate $1/\sqrt{n}$ would look like if it held with asymptotic variance 0; the model is not regular and the theorem does not apply.

Exercise 12.6 *(page 182).* $\ell(p) = (X-1)\log(1-p) + \log p$, $\ell''(p) = -(X-1)/(1-p)^2 - 1/p^2$, and $\mathbb{E}[X] = 1/p$ gives $\mathbb{E}[X-1] = (1-p)/p$, so $I(p) = \frac{1}{p(1-p)} + \frac{1}{p^2} = \frac{p+(1-p)}{p^2(1-p)} = \frac{1}{p^2(1-p)}$. At $\hat{p} = 5/12$ with $n = 5$: $nI = 5/((5/12)^2 \times 7/12) = 49.4$ and the standard error is $1/\sqrt{49.4} = 0.14$. Five rounds say p is somewhere around 0.42 ± 0.28.

Exercise 12.7 *(page 182).* $\ell_n'(\theta) = \sum_{i=1}^n \ell_i'(\theta)$ is a sum of n independent random variables each with variance $I(\theta)$, so its variance is $nI(\theta)$. And $-\mathbb{E}[\ell_n''(\theta)] = -\sum_i \mathbb{E}[\ell_i''(\theta)] = nI(\theta)$ by Theorem 12.4 applied to each term. Information adds over independent observations, the same way KL divergence did in Proposition 10.7, and for the same reason: logs turn products into sums.

Exercise 12.8 *(page 182).* The theorem (misspecified case) says $\hat{\theta}_n \xrightarrow[n\to\infty]{\mathbb{P}} \theta^\circ = (1,1)$. Directly: $\bar{X}_n \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[X] = 1$ by the LLN, and $\frac{1}{n}\sum(X_i - \bar{X}_n)^2 = \frac{1}{n}\sum X_i^2 - \bar{X}_n^2 \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[X^2] - 1 = 2 - 1 = 1$ by the LLN and the continuous mapping theorem (Theorem 5.2, since $x \mapsto x^2$ is continuous). The two agree. The fitted Gaussian is the closest one to the truth in KL, at divergence 0.419, and no amount of data moves it closer.

Chapter 13. Method of Moments and M-Estimation

Exercise 13.1 *(page 195).* $\hat{m}_1 = (0.5 + 1.8 - 2.3 + 0.9)/4 = 0.225$ and $\hat{m}_2 = (0.25 + 3.24 + 5.29 + 0.81)/4 = 2.3975$. So $\hat{\mu} = 0.225$ and $\hat{\sigma} = \sqrt{2.3975 - 0.225^2} = \sqrt{2.347} = 1.532$. These are the MLEs as well; the divide-by-$(n-1)$ sample standard deviation would be $1.532\sqrt{4/3} = 1.769$.

Exercise 13.2 *(page 195).* $m_1 = \theta/2$, so $\hat{\theta}^{\mathrm{MM}} = 2\bar{X}_n$. With $M^{-1}(m) = 2m$ and $\mathbb{V}\mathrm{ar}(X) = \theta^2/12$, the asymptotic variance is $4\theta^2/12 = \theta^2/3$, and the standard deviation at $n = 100$ is $\sqrt{100/3}/10 = 0.577$. The MLE's error is of order $\theta/n = 0.1$, six times smaller, and shrinking like $1/n$ rather than $1/\sqrt{n}$. The method of moments estimator is unbiased and asymptotically normal, and the MLE beats it anyway: the model is not regular, so the Cramér–Rao floor that would protect a regular MLE's variance does not exist here, and the MLE goes below it.

Exercise 13.3 *(page 195).* From $m_2 - m_1^2 = a/b^2$ (the variance) and $m_1 = a/b$: dividing, $m_1/(m_2 - m_1^2) = b$, and then $a = m_1 b = m_1^2/(m_2 - m_1^2)$. So $\hat{b} = \hat{m}_1/(\hat{m}_2 - \hat{m}_1^2) = \bar{X}_n/\widehat{\sigma^2}$ and $\hat{a} = \bar{X}_n^2/\widehat{\sigma^2}$, with $\widehat{\sigma^2}$ the divide-by-n sample variance. Two moments, two unknowns, solved in two lines; the likelihood equations involve the derivative of $\log\Gamma(a)$ and need a numerical solver.

Exercise 13.4 *(page 195).* $Q(\mu) = \alpha\int_\mu^\infty (x-\mu)f(x)\,\mathrm{d}x + (1-\alpha)\int_{-\infty}^\mu (\mu - x)f(x)\,\mathrm{d}x$. Differentiating as in Proposition 13.5, $Q'(\mu) = -\alpha\mathbb{P}(X > \mu) + (1-\alpha)\mathbb{P}(X < \mu) = -\alpha(1 - F(\mu)) + (1-\alpha)F(\mu) = F(\mu) - \alpha$. This is zero exactly when $F(\mu) = \alpha$, negative below and positive above, so the minimizer is the α-quantile. With $\alpha = \frac{1}{2}$, $Q' = F - \frac{1}{2}$, half the median's $2F - 1$, and the same minimizer. (The quantile here is the point with probability α *below* it; the book's q_α for Gaussian tails has probability α above, as Chapter 6 explains.)

Exercise 13.5 *(page 195).* $f(m) = 1/(\sigma\sqrt{2\pi})$, so $1/(4f(m)^2) = 2\pi\sigma^2/4 = (\pi/2)\sigma^2 = 1.571\sigma^2$, against σ^2 for the mean. The median needs $\pi/2 - 1 = 57\%$ more observations. In the simulation of Figure 13.3, $0.124/0.100 = 1.24 \approx \sqrt{\pi/2} = 1.253$.

Exercise 13.6 *(page 195).* $K/J^2 = 0.710/0.674 = 1.053$. The mean has asymptotic variance 1, so Huber's estimator is 5.3% noisier on clean Gaussian data, which is the price of capping every observation's influence at δ. For $\delta = 1$ the figures are $J = 0.683$, $K = 0.516$ and variance 1.107; for $\delta = 2$, variance 1.010. The larger δ, the closer to the mean in both efficiency and fragility.

Exercise 13.7 *(page 195).* Both theorems assume a finite mean (the LLN as stated in this book assumes a finite variance). The Cauchy has neither: $\int |x| f(x)\,\mathrm{d}x = \infty$. So the theorems say nothing, and the fact is consistent with them. $\bar{X}_{1000}$ is exactly as spread out as a single observation: its probability of landing within 1 of the center is $\mathbb{P}(|X_1 - m| \le 1) = \frac{1}{2}$, the same as for $n = 1$. A thousand observations averaged are worth one. The sample median of the same thousand has standard deviation about $\sqrt{\pi^2/4}/\sqrt{1000} = 0.050$.

Exercise 13.8 *(page 195).* $\nabla_\theta \rho = -\nabla_\theta \log p_\theta(X)$, the negative score, whose covariance is the covariance of the score, $I(\theta^*)$ by Definition 12.3 (a sign flip does not change a covariance). $\nabla^2_\theta \rho = -\nabla^2_\theta \log p_\theta(X)$, whose expectation is $-\mathbb{E}[\nabla^2 \ell] = I(\theta^*)$ by Theorem 12.4. So $J = K = I$ and $J^{-1}KJ^{-1} = I^{-1}II^{-1} = I^{-1}$.

Chapter 14. The t-Test

Exercise 14.1 *(page 216).* (a) $(X_i - 3)/\sigma \overset{iid}{\sim} \text{Gauss}(0,1)$, so $a = 1/\sigma^2$ and $k = n$. (b) $\tilde{\sigma}^2 = \sigma^2 V/n$ with $V \sim \chi^2_n$: mean σ^2, variance $2\sigma^4/n$, against $2\sigma^4/(n-1)$ for S^2_n. Knowing the mean buys one degree of freedom. (c) $\mathbb{P}(Z_1^2 + Z_2^2 \le 4) = \mathbb{P}(\chi^2_2 \le 4) = 1 - e^{-2} = 0.865$.

Exercise 14.2 *(page 216).* The joint density of (X_1, X_2) is $\frac{1}{2\pi} e^{-(x_1^2+x_2^2)/2}$. The map $(x_1, x_2) \mapsto (u, w) = ((x_1 + x_2)/\sqrt{2}, (x_1 - x_2)/\sqrt{2})$ is linear with matrix $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, which is orthogonal ($Q^\top Q = I$, $\det Q = -1$), and $u^2 + w^2 = x_1^2 + x_2^2$. By the change of variables formula (toolkit A.13), the density of (U, W) is $\frac{1}{2\pi} e^{-(u^2+w^2)/2} / |\det Q| = \frac{1}{2\pi} e^{-u^2/2} e^{-w^2/2}$, a product of two standard Gaussian densities, so U and W are independent standard Gaussians. Then $\bar{X}_2 = U/\sqrt{2}$ and $S_2^2 = (X_1 - X_2)^2/2 = W^2 \sim \chi^2_1$, independent of $\bar{X}_2$.

Exercise 14.3 *(page 216).* $\sum_i (X_i - \bar{X}_n)^2 = \sigma^2 V$ with $V \sim \chi^2_{n-1}$, so $\mathbb{E}[\hat{\sigma}^2_c] = c(n-1)\sigma^2$, bias $(c(n-1) - 1)\sigma^2$, variance $2c^2(n-1)\sigma^4$, and risk $\sigma^4\big[2c^2(n-1) + (c(n-1) - 1)^2\big]$. The derivative in c is $\sigma^4[4c(n-1) + 2(n-1)(c(n-1) - 1)]$, zero at $c(2 + n - 1) = 1$, that is, $c = 1/(n+1)$, a minimum since the risk is a convex parabola in c. The risks are $2\sigma^4/(n-1)$ for $c = 1/(n-1)$, $(2n-1)\sigma^4/n^2$ for $c = 1/n$ and $2\sigma^4/(n+1)$ for $c = 1/(n+1)$: at $n = 12$, 0.182, 0.160 and 0.154. This is the claim of the bridge in Chapter 7.

Exercise 14.4 *(page 216).* (a) $\mathbb{E}[1/V] = \frac{1}{2^{k/2}\Gamma(k/2)} \int_0^\infty v^{k/2-2} e^{-v/2} \, \mathrm{d}v$. Substituting $v = 2s$, the integral is $2^{k/2-1} \int_0^\infty s^{k/2-2} e^{-s} \, \mathrm{d}s$, which is $2^{k/2-1}\Gamma(k/2 - 1)$, finite when $k/2 - 1 > 0$. So $\mathbb{E}[1/V] = \frac{1}{2}\Gamma(k/2-1)/\Gamma(k/2) = \frac{1}{2} \cdot \frac{1}{k/2-1} = 1/(k-2)$. (b) By independence, $\mathbb{E}[T^2] = \mathbb{E}[Z^2] \, \mathbb{E}[k/V] = k/(k-2)$, and $\mathbb{E}[T] = \mathbb{E}[Z] \, \mathbb{E}[\sqrt{k/V}] = 0$ provided $\mathbb{E}[1/\sqrt{V}] < \infty$, which needs $\int_0 v^{k/2-3/2} \, \mathrm{d}v$ to converge at 0, that is, $k \ge 2$; for $k = 1$ the mean is the Cauchy's, undefined. (c) 3, 1.67 and 1.07: t_3 has standard deviation 1.73, so its quantiles are far outside the Gaussian's; by $k = 30$ the standard deviation is 1.04 and the quantiles are within a few percent.

Exercise 14.5 *(page 216).* (a) $\hat{\sigma}^2 = \frac{n-1}{n} S_n^2$, so $\sqrt{n-1}(\hat{\mu} - \mu)/\hat{\sigma} = \sqrt{n-1}(\bar{X}_n - \mu)/\big(\sqrt{(n-1)/n}\, S_n\big) = \sqrt{n}(\bar{X}_n - \mu)/S_n \sim t_{n-1}$ by Theorem 14.7. (b) Reject when $T = \sqrt{n-1}\,\hat{\mu}/\hat{\sigma} < -q_\alpha(t_{n-1})$. At $\mu = 0$, T is the pivot and the probability is α by symmetry of t_{n-1}. For $\mu > 0$, $T = T^* + \sqrt{n-1}\,\mu/\hat{\sigma}$ with $T^* \sim t_{n-1}$ and the added term positive, so $T > T^*$ and $\mathbb{P}_\mu(T < -q) \le \mathbb{P}(T^* < -q) = \alpha$.

Exercise 14.6 *(page 216).* (a) The interval contains μ exactly when $|\sqrt{n}(\bar{X}_n - \mu)/S_n| \le 1.96$, so the coverage is $\mathbb{P}(|t_{n-1}| \le 1.96) = 1 - \mathbb{P}(|t_{n-1}| > 1.96)$: from the figure's values 0.189, 0.122, 0.076, 0.060, the coverages are 81%, 88%, 92% and 94%. (b) $\mathbb{P}(|t_{n-1}| > 1.96) \le 0.06$ first holds at $n = 30$ (0.0597); at $n = 29$ it is 0.06002, just above.

Exercise 14.7 *(page 216).* $T = (156.4 - 132.7)/\sqrt{5198.4/70 + 3867.0/50} = 23.7/12.31 = 1.925$. $\hat{\nu} = (74.26 + 77.34)^2/\big(74.26^2/69 + 77.34^2/49\big) = 22{,}983/(79.9 + 122.1) = 113.8$, rounded down to 113, between 49 and 118. The threshold is $q_{0.05}(t_{113}) = 1.66$, so the test rejects, with one-sided p-value $\mathbb{P}(t_{113} > 1.925) = 0.028$; the Gaussian p-value is 0.027, since at these sizes the two agree. The two-sided p-value is twice that, 0.057, which would wrongly fail to reject at 5%: the hypothesis is one-sided, and the p-value must match it.

Exercise 14.8 *(page 216).* (a) By Theorem 14.3, $V = (n-1)S_1^2/\sigma_1^2 \sim \chi^2_{n-1}$ and $W = (m-1)S_2^2/\sigma_2^2 \sim \chi^2_{m-1}$, independent since the samples are, and $(S_1^2/\sigma_1^2)/(S_2^2/\sigma_2^2) = (V/(n-1))/(W/(m-1))$ is $F_{n-1,m-1}$ by definition. (b) $2.1^2/1.3^2 = 2.61$. If the variances were equal this ratio would be an $F_{71,55}$ draw, and a value of 2.61 or more happens less than once in 1,000 such draws: the variances differ, and the pooled pivot's hypothesis fails. Welch's statistic is the one with a justification; here both are 27 or more standard errors from zero, so the decision is the same.

Exercise 14.9 *(page 217).* (a) $8064 \times 0.001 = 8.06$; $1 - 0.999^{8064} = 0.9997$; $0.001/8064 = 1.24 \times 10^{-7}$. (b) $8000 \times 0.001 = 8$; with Bonferroni each test has level $0.05/8064 = 6.2 \times 10^{-6}$, so $8000 \times 6.2 \times 10^{-6} = 0.05$ false rejections are expected, and the FWER is at most 0.05. (c) About 8 of 50, or 16%: the uncorrected procedure's realized false discovery proportion in this batch.

Exercise 14.10 *(page 217).* (a) The threshold is 0.005: only the first. (b) The thresholds $i\alpha/N$ are 0.005, 0.010, 0.015, 0.020, 0.025 and so on. The first three p-values are under their thresholds ($0.001 \le 0.005$, $0.008 \le 0.010$, $0.012 \le 0.015$), the fourth is not ($0.03 > 0.020$), the fifth is not ($0.06 > 0.025$), and no later one is, so $i_{\max} = 3$ and the first three are rejected. (c) Now $P_{(4)} = 0.022 > 0.020$ still fails its own threshold, but $P_{(5)} = 0.024 \le 0.025$ passes, so $i_{\max} = 5$ and the first five are rejected, the fourth included. The procedure takes the largest i whose p-value is under the line, not the first one that fails, so a small enough fifth p-value pulls the fourth in with it: the small p-values vouch for each other.

Chapter 15. Wald, Likelihood Ratio and Score Tests

Exercise 15.1 *(page 235).* (a) χ^2_2 is Expo($\frac{1}{2}$) (Proposition 14.2), so $\mathbb{P}(\chi^2_2 > q) = e^{-q/2}$; setting this to 0.05 gives $q = -2\log 0.05 = 5.99$. (b) $\mathbb{P}(Z^2 > c) = \mathbb{P}(|Z| > \sqrt{c})$, which is α when $\sqrt{c} = q_{\alpha/2}$; so $q_\alpha(\chi^2_1) = q^2_{\alpha/2}$, and $3.84 = 1.96^2$. (c) $\det\Sigma = 2.25 - 1.44 = 0.81$ and $\Sigma^{-1} = \frac{1}{0.81}\begin{pmatrix} 2.25 & -1.2 \\ -1.2 & 1 \end{pmatrix}$. For $(1,1)$: $(2.25 - 2.4 + 1)/0.81 = 1.05$, inside. For $(1,-1)$: $(2.25 + 2.4 + 1)/0.81 = 6.98 > 5.99$, outside. Both are at ordinary distance $\sqrt{2}$ from the center; $(1,1)$ points along the cloud, where the noise is large, and $(1,-1)$ across it, where the noise is small.

Exercise 15.2 *(page 235).* $\hat{p} = 0.78$. Wald: $100(0.02)^2/(0.78 \times 0.22) = 0.233$. Score: $100(0.02)^2/(0.8 \times 0.2) = 0.250$. Likelihood ratio: $2\big(78\log\frac{0.78}{0.8} + 22\log\frac{0.22}{0.2}\big) = 2(-1.975 + 2.097) = 0.244$. The χ^2_1 p-values are 0.63, 0.62 and 0.62: the trial is entirely consistent with 80% effectiveness, and the three tests agree because the null is far inside every reasonable interval and the log-likelihood is close to a parabola there.

Exercise 15.3 *(page 235).* $\ell_n(\lambda) = n\bar{X}_n \log\lambda - n\lambda + \text{const}$, $\ell_n'(\lambda) = n\bar{X}_n/\lambda - n$, $I(\lambda) = 1/\lambda$ (Chapter 12). Wald: $nI(\hat{\lambda})(\hat{\lambda} - \lambda_0)^2 = n(\hat{\lambda} - \lambda_0)^2/\hat{\lambda}$. Score: $\ell_n'(\lambda_0)^2/(nI(\lambda_0)) = n^2(\hat{\lambda}/\lambda_0 - 1)^2\lambda_0/n =$

$n(\hat\lambda-\lambda_0)^2/\lambda_0$. Likelihood ratio: $2(\ell_n(\hat\lambda)-\ell_n(\lambda_0)) = 2n(\hat\lambda\log(\hat\lambda/\lambda_0)-\hat\lambda+\lambda_0)$. With $n=24$, $\hat\lambda=1$, $\lambda_0=1.5$: $T^{\mathrm{W}}=6.0$, $T^{\mathrm{S}}=4.0$, $T^{\mathrm{LR}}=48(\log\frac{2}{3}+0.5)=4.54$, with p-values 0.014, 0.046 and 0.033. All three reject at 5%, the score test only just: with 24 observations the three can still differ by a factor of 1.5.

Exercise 15.4 *(page 236).* (a) $L_n(0.75)/L_n(0.25) = (0.75/0.25)^S(0.25/0.75)^{n-S} = 3^{2S-n}$, which exceeds 1 exactly when $S > n/2$. (b) For $n=16$ the test rejects when $S\ge 9$: level $\mathbb{P}_{0.25}(S\ge 9) = 0.0075$, power $\mathbb{P}_{0.75}(S\ge 9)=0.973$. With 6 heads the ratio is $3^{-4}=0.012$: the null explains the data 81 times better, and the test does not reject. (c) No test of 16 flips with a type 1 error no larger than this test's has power above 0.973 against $p=0.75$.

Exercise 15.5 *(page 236).* $\ell_n(\mu) = -\frac{1}{2\sigma^2}\sum_i(X_i-\mu)^2+\text{const} = -\frac{n}{2\sigma^2}(\bar X_n-\mu)^2+\text{const}'$, an exact parabola in μ with curvature $n/\sigma^2 = nI$ for every μ. Wald: $nI(\bar X_n-\mu_0)^2 = n(\bar X_n-\mu_0)^2/\sigma^2$. Likelihood ratio: twice the drop of the parabola, the same. Score: $\ell_n'(\mu_0) = n(\bar X_n-\mu_0)/\sigma^2$, squared and divided by n/σ^2, the same. Under H_0, $\sqrt{n}(\bar X_n-\mu_0)/\sigma \sim \text{Gauss}(0,1)$ exactly (Chapter 6), so its square is exactly χ^2_1.

Exercise 15.6 *(page 236).* (a) For fixed μ the log-likelihood is maximized at $\sigma^2(\mu) = \frac{1}{n}\sum_i(X_i-\mu)^2 = \hat\sigma^2+(\bar X_n-\mu)^2$, where it equals $-\frac{n}{2}\log(2\pi\sigma^2(\mu))-\frac{n}{2}$. So $T_n^{\mathrm{LR}} = n\log\big(\sigma^2(\mu_0)/\hat\sigma^2\big) = n\log\big(1+(\bar X_n-\mu_0)^2/\hat\sigma^2\big)$, and since $\hat\sigma^2=\frac{n-1}{n}S_n^2$, $(\bar X_n-\mu_0)^2/\hat\sigma^2 = T^2/(n-1)$. (b) $x\mapsto n\log(1+x^2/(n-1))$ is increasing in $|x|$. (c) $T_n^{\mathrm{LR}}>3.84$ when $|T|>\sqrt{11(e^{3.84/12}-1)}=2.04$, while the exact 5% threshold is $q_{0.025}(t_{11})=2.20$; the chi-squared version has exact level $\mathbb{P}(|t_{11}|>2.04)=0.066$.

Exercise 15.7 *(page 236).* (a) $\ell_n(\theta) = n\log\theta-(\theta+1)\sum_i\log X_i$, so $\hat\theta = n/\sum_i\log X_i$ and $\ell''(\theta) = -1/\theta^2$, $I(\theta)=1/\theta^2$. (b) $\sum_i\log X_i = n/\hat\theta$, so $T^{\mathrm{LR}} = 2n\big(\log(\hat\theta/\theta_0)-(\hat\theta-\theta_0)/\hat\theta\big) = 200(0.2029-0.1837) = 3.85 > 3.84$: reject, just ($p=0.0496$). $T^{\mathrm{W}} = n(\hat\theta-\theta_0)^2/\hat\theta^2 = 100(0.45/2.45)^2 = 3.37$: do not reject ($p=0.066$). (c) $\mathbb{P}(\log X>t) = \mathbb{P}(X>e^t) = e^{-\theta t}$, the Expo($\theta$) tail. The model is the exponential rate model in disguise, and the two tests sit on either side of the threshold for the same reason as in Figure 15.3; here they disagree in the other direction, since $\hat\theta>\theta_0$ puts the null on the other side of the asymmetric log-likelihood.

Exercise 15.8 *(page 236).* $\hat\mu = -0.8$ and $\hat\sigma^2 = 3.86/5 = 0.772$. Wald at the estimate: $5\big(0.64/0.772+0.228^2/(2\times 0.772^2)\big) = 4.36$. Wald with $I(0,1)=\text{diag}(1,\frac12)$: $5(0.64+0.228^2/2) = 3.33$. Likelihood ratio: $2\big(-\frac52\log 0.772-\frac52+\frac12\sum_i X_i^2\big)$ with $\sum_i X_i^2 = 7.06$, which is $2(0.647-2.5+3.53) = 3.35$. Score: the gradient at $(0,1)$ is $(\sum_i X_i, -\frac52+\frac12\sum_i X_i^2) = (-4, 1.03)$, and $T^{\mathrm{S}} = \frac15\big(16/1+1.03^2/\frac12\big) = 3.62$. All are below $q_{0.05}(\chi^2_2)=5.99$: five observations are consistent with Gauss$(0,1)$.

Exercise 15.9 *(page 236).* (a) $\ell_n(\mu) = -\frac{n}{2}(\bar X_n-\mu)^2+\text{const}$ is maximized over $[0,\infty)$ at $\bar X_n$ if $\bar X_n>0$ and at 0 otherwise; then $T_n^{\mathrm{LR}} = n\bar X_n^2$ or 0. (b) Under H_0, $Z=\sqrt{n}\bar X_n\sim\text{Gauss}(0,1)$ and for $c>0$, $\mathbb{P}(T_n^{\mathrm{LR}}>c) = \mathbb{P}(Z>\sqrt{c}) = \frac12\mathbb{P}(Z^2>c)$; at $c=3.84$ this is 0.025. (c) $\mathbb{P}(Z>\sqrt{c})=0.05$ when $\sqrt{c}=1.645$, so $c=2.71$: the one-sided test of Chapter 9, written in squares.

Exercise 15.10 *(page 236).* (a) ℓ_n as a function of η is $\ell_n(h(\eta))$; its maximum is the same number, attained at $\hat\eta$ with $h(\hat\eta)=\hat\theta$ (Proposition 11.6), and its value at η_0 is $\ell_n(\theta_0)$. (b) The score

statistic in η is $\big(\ell_n'(\theta_0)h'(\eta_0)\big)^2/\big(nI_\theta(\theta_0)h'(\eta_0)^2\big) = \ell_n'(\theta_0)^2/(nI_\theta(\theta_0))$. (c) $\ell_n(\mu) = -n\log\mu - n\bar{T}_n/\mu$, $\ell_n'(\mu_0) = n(\bar{T}_n - \mu_0)/\mu_0^2$ and $I(\mu) = 1/\mu^2$, so $T^{\mathrm{S}} = n(\bar{T}_n - \mu_0)^2/\mu_0^2 = 64(1.23/5)^2 = 3.87$, the value found in λ. Wald's statistic changed from 3.87 to 2.49 because it evaluates the information at $\hat{\theta}$ and measures the distance $\hat{\theta} - \theta_0$ in the chosen coordinate, and neither is preserved by a nonlinear h.

Chapter 16. Goodness of Fit for Discrete Distributions

Exercise 16.1 *(page 255).* (a) $\mathbb{E}[(N_j - np_j^0)^2] = \mathbb{V}\mathrm{ar}(N_j) = np_j^0(1-p_j^0)$ (Proposition 16.3), so $\mathbb{E}[(N_j - np_j^0)^2/(np_j^0)] = 1 - p_j^0$, and $\mathbb{E}[T_n^{\mathrm{S}}] = \sum_j(1-p_j^0) = K - (p_1^0 + \cdots + p_K^0) = K - 1$. (b) $\mathbb{P}(N_j = 0) = (1-p_j^0)^n > 0$, and on that event $T_n^{\mathrm{W}} = +\infty$, so its mean is $+\infty$. Convergence in distribution is a statement about probabilities, not means: the event has probability tending to 0, so it does not affect the limit of $\mathbb{P}(T_n^{\mathrm{W}} \le t)$.

Exercise 16.2 *(page 255).* (a) $T^{\mathrm{S}} = 2 \times 18^2/62 = 10.45$; $G^2 = 2(80\log\frac{80}{62} + 44\log\frac{44}{62}) = 10.60$; $T^{\mathrm{W}} = 18^2/80 + 18^2/44 = 4.05 + 7.36 = 11.41$. These are the score, likelihood ratio and Wald statistics of Chapter 15, as Proposition 16.6 says they must be. (b) With $N_2 - np_2^0 = -(N_1 - np_1^0)$, $T_n^{\mathrm{W}} = (N_1 - np_1^0)^2(1/N_1 + 1/N_2) = (N_1 - np_1^0)^2\, n/(N_1N_2)$, and with $N_1 = n\hat{p}$, $N_2 = n(1-\hat{p})$ and $p_0 = p_1^0$ this is $n^2(\hat{p}-p_0)^2\, n/(n^2\hat{p}(1-\hat{p})) = n(\hat{p}-p_0)^2/(\hat{p}(1-\hat{p}))$.

Exercise 16.3 *(page 255).* Expected: $7{,}324 \times \frac{3}{4} = 5{,}493$ round and 1,831 wrinkled; the deviations are -19 and $+19$. $T^{\mathrm{S}} = 19^2/5{,}493 + 19^2/1{,}831 = 0.066 + 0.197 = 0.263$, with $K - 1 = 1$ degree of freedom and p-value $\mathbb{P}(\chi_1^2 > 0.263) = 0.61$; $G^2 = 0.262$. The observed ratio is $2.96 : 1$, and the test finds nothing against $3 : 1$.

Exercise 16.4 *(page 255).* (a) Each face expects 2.5: $T^{\mathrm{S}} = (0.5^2 + 1.5^2 + 1.5^2 + 0.5^2 + 0.5^2 + 3.5^2)/2.5 = 17.5/2.5 = 7.0$, below $q_{0.05}(\chi_5^2) = 11.07$, with p-value $\mathbb{P}(\chi_5^2 > 7.0) = 0.22$: no evidence that the die is unfair, despite six 6s in 15 rolls. (b) Every deviation and every expected count is four times larger, so $T^{\mathrm{S}} = 4 \times 7.0 = 28.0$, with p-value 0.00004: reject. The same proportions are strong evidence in 60 rolls and weak evidence in 15. (c) All six expected counts are 2.5, below 5. Draw, say, 100,000 tables of 15 rolls of a fair die, that is, from $\mathrm{Mult}(15, (\frac{1}{6}, \ldots, \frac{1}{6}))$, compute T^{S} for each, and report the fraction that are at least 7.0; it estimates the exact 0.256, a little larger than the chi-squared 0.22.

Exercise 16.5 *(page 255).* (a) Each sign expects $256/12 = 21.33$. Since $\sum_j (N_j - E)^2/E = \sum_j N_j^2/E - n$, $T^{\mathrm{S}} = 5{,}570 \times 12/256 - 256 = 5.09$, with $K - 1 = 11$ degrees of freedom and p-value $\mathbb{P}(\chi_{11}^2 > 5.09) = 0.93$: no evidence that the signs are unequally represented. Pisces contributes $(29 - 21.33)^2/21.33 = 2.76$, more than half of the total. (b) On its own, the Pisces count is $\mathrm{Bin}(256, \frac{1}{12})$, with standard deviation $\sqrt{256 \times \frac{1}{12} \times \frac{11}{12}} = 4.42$, so 29 is 1.73 standard deviations above its mean, a two-sided p-value of about 0.08 by the Gaussian approximation. (c) Pisces was singled out after the data were seen because its count was the largest. The relevant chance is that the largest of twelve counts is at least 29, which is 56%: when all the signs are equally likely, some sign reaches 29 more often than not. Testing the most extreme of many categories as if it had been named in advance is the multiple-testing error of Section 14.7; Pearson's statistic, which looks at all twelve signs at once, does not make it.

Exercise 16.6 *(page 255).* (a) The expected counts are $275 \times 0.72 = 198$, 19.25, 33 and 24.75. $T^{\mathrm{S}} = 7^2/198 + 6.75^2/19.25 + 8^2/33 + 5.75^2/24.75 = 0.247 + 2.367 + 1.939 + 1.336 = 5.89$, with 3 degrees of freedom and p-value $\mathbb{P}(\chi^2_3 > 5.89) = 0.12$: not significant at 5%. Black jurors are 9.5% of the sample against 7% of the voters, Hispanic jurors 9.1% against 12%. (b) With the proportions fixed, $T^{\mathrm{S}} = n\sum_j (\hat{p}_j - p^0_j)^2/p^0_j = 0.0214\, n$, which exceeds the threshold 7.81 from about $n = 365$ (Section 16.6).

Exercise 16.7 *(page 255).* (a) $f(m) = 0$; $f'(x) = \log(x/m) + 1$, so $f'(m) = 1$; $f''(x) = 1/x$, so $f''(m) = 1/m$. Then $f(x) = (x-m) + (x-m)^2/(2m) +$ (terms of order three and higher) (toolkit A.9). Summing over the categories with $x = N_j$ and $m = np^0_j$, the first-order terms add up to $\sum_j (N_j - np^0_j) = 0$, so $G^2 = 2\sum_j f(N_j) = \sum_j (N_j - np^0_j)^2/(np^0_j) + \cdots = T^{\mathrm{S}}_n + \cdots$: Pearson's statistic is the parabola approximation of G^2, as in Chapter 10. (b) $f'''(x) = -1/x^2$ and $f''''(x) = 2/x^3$, so the next terms are $-(x-m)^3/(6m^2)$ and $(x-m)^4/(12m^3)$. With deviations +18 and −18 from the same expected count 62, the cubic terms cancel, and the quartic terms add $2 \times 2 \times 18^4/(12 \times 62^3) = 0.147$ to G^2, nearly all of the difference 0.152.

Exercise 16.8 *(page 255).* The log-likelihood is $\sum_j N_j \log\left(\binom{3}{j}\theta^j(1-\theta)^{3-j}\right) = S\log\theta + (3n - S)\log(1-\theta) + \text{const}$, with $S = \sum_j jN_j = 455 + 360 + 78 = 893$, so $\hat{\theta} = S/(3n) = 893/3{,}000 = 0.2977$. Expected counts: $1{,}000(1-\hat{\theta})^3 = 346.44$, $3{,}000\hat{\theta}(1-\hat{\theta})^2 = 440.49$, $3{,}000\hat{\theta}^2(1-\hat{\theta}) = 186.69$ and $1{,}000\hat{\theta}^3 = 26.37$. $T^{\mathrm{S}} = 7.44^2/346.44 + 14.51^2/440.49 + 6.69^2/186.69 + 0.37^2/26.37 = 0.160 + 0.478 + 0.240 + 0.005 = 0.88$, with $K - 1 - d = 4 - 1 - 1 = 2$ degrees of freedom and p-value $\mathbb{P}(\chi^2_2 > 0.88) = 0.64$: nothing against the binomial family.

Exercise 16.9 *(page 256).* (a) $n = 2{,}201$, with 711 survivors and 1,490 deaths, and class totals 325, 285, 706 and 885. The expected numbers of survivors are $325 \times 711/2{,}201 = 104.99$, then 92.06, 228.06 and 285.89; of deaths, 220.01, 192.94, 477.94 and 599.11. (b) The eight terms are 91.50 and 43.66 (first class, survivors and deaths), 7.31 and 3.49 (second), 10.99 and 5.24 (third) and 19.10 and 9.11 (crew), so $T^{\mathrm{S}} = 190.4$, with $(4-1)(2-1) = 3$ degrees of freedom; the p-value is about 10^{-40}. (c) First-class survivors: $(203 - 104.99)^2/104.99 = 91.5$, almost half the total. Nearly twice as many first-class passengers survived as independence predicts: 62% of the first class survived, against 32% of everyone aboard.

Exercise 16.10 *(page 256).* (a) $\sum_k E_{jk} = N_{j\cdot}\sum_k N_{\cdot k}/n = N_{j\cdot}$, and likewise $\sum_j E_{jk} = N_{\cdot k}$. (b) Subtracting, $\sum_k (N_{jk} - E_{jk}) = 0$ for every j and $\sum_j (N_{jk} - E_{jk}) = 0$ for every k. Given the deviations in the first $r-1$ rows and $c-1$ columns, the last entry of each of those rows is minus the sum of the others, and then each entry of the last row is minus the sum of the rest of its column. (c) With δ the deviation in cell $(1,1)$, the row condition gives $-\delta$ in $(1,2)$, the column conditions give $-\delta$ in $(2,1)$, and then $+\delta$ in $(2,2)$; the aspirin table has $\delta = 42.52$.

Exercise 16.11 *(page 256).* (a) $p^0_1(1-p^0_1) = p^0_2(1-p^0_2)$ and $(N_2 - np^0_2)^2 = (N_1 - np^0_1)^2$, so the two terms of V_n are equal, each equal to $(N_1 - np^0_1)^2/(np^0_1(1-p^0_1))$, which is T^{S}_n by the sanity check of Section 16.2. (b) Under the null $T^{\mathrm{S}}_n \xrightarrow[n\to\infty]{\text{(d)}} \chi^2_1$, so $V_n \xrightarrow[n\to\infty]{\text{(d)}} 2\chi^2_1$ by the continuous mapping theorem (Theorem 5.2: if $T_n \xrightarrow[n\to\infty]{\text{(d)}} T$ and g is continuous, $g(T_n) \xrightarrow[n\to\infty]{\text{(d)}} g(T)$; here $g(t) = 2t$), and $\mathbb{P}(2\chi^2_1 > 5.99) = \mathbb{P}(\chi^2_1 > 3.00) = 0.083$: a nominal 5% test with type 1 error 8.3%.

Exercise 16.12 *(page 256).* (a) In decimals the alternative is $(0.5714, 0.1905, 0.1905, 0.0476)$ and the null is $(0.5625, 0.1875, 0.1875, 0.0625)$, so $c = 0.0089^2/0.5625 + 2 \times 0.0030^2/0.1875 + 0.0149^2/0.0625 = 0.00378$, most of it from the last category. (b) $\mathbb{E}[(N_j - np_j^0)^2] = \mathbb{V}\mathrm{ar}(N_j) + (\mathbb{E}[N_j] - np_j^0)^2 = np_j(1-p_j) + n^2(p_j - p_j^0)^2$; divide by np_j^0 and add over j. (c) The second term is 2.806, close to $K - 1 = 3$. At $n = 556$ the mean is $2.101 + 2.806 = 4.91$, below the threshold 7.81, and the test rejects only 18% of the time; at $n = 2{,}000$ it is $7.559 + 2.806 = 10.36$, above the threshold, and the test rejects 64% of the time. The mean crosses the threshold $q_{0.05}(\chi^2_3) = 7.815$ at about $n = (7.815 - 2.806)/0.00378 = 1{,}325$.

Chapter 17. Kolmogorov–Smirnov, Lilliefors and QQ Plots

Exercise 17.1 *(page 273).* (a) For x with $0 < F(x) < 1$, $F^{-1}(U) \le x$ exactly when $U \le F(x)$, since F and F^{-1} are increasing; so $\mathbb{P}(F^{-1}(U) \le x) = \mathbb{P}(U \le F(x)) = F(x)$. If $F(x) = 0$, then $F^{-1}(U)$, which lies where $0 < F < 1$, is above x, and both sides are 0; if $F(x) = 1$, it is below x, and both sides are 1. (b) $F(x) = 1 - e^{-x/5}$ for $x \ge 0$, so $F^{-1}(u) = -5\log(1-u)$: 0.42 and 3.57 minutes, the first and fourth points of the right of Figure 17.3. (c) $F(0) = \frac{1}{2}$ and $F(1) = 1$, so $F(X)$ equals $\frac{1}{2}$ or 1, each with probability $\frac{1}{2}$: not uniform.

Exercise 17.2 *(page 273).* Sorted: $0.2, 0.4, 0.7, 0.7, 0.8$, so F_5 is 0.2 from 0.2, 0.4 from 0.4, 0.8 from 0.7 (two values) and 1 from 0.8. The gaps just below and at each step are 0.2 and 0 at 0.2; 0.2 and 0 at 0.4; $0.7 - 0.4 = 0.3$ and $0.8 - 0.7 = 0.1$ at 0.7; 0 and 0.2 at 0.8. So $D_5 = 0.3$, attained just below 0.7. It is below the exact threshold 0.563 and below $1.358/\sqrt{5} = 0.607$ (equivalently, $\sqrt{5}D_5 = 0.67 < 1.358$): do not reject. The value 0.7 appears twice, which has probability 0 under a continuous distribution; the data were rounded or made up. (The formula of Proposition 17.6, applied to the five sorted values, gives the same 0.3.)

Exercise 17.3 *(page 273).* (a) Proposition 17.2 with $F(u) = u$ gives $\mathbb{C}\mathrm{ov}(G_n(u), G_n(v)) = u(1-v)/n$; multiply by n. (b) The pair is $\sqrt{n}(\bar{Y}_n - m)$ with $Y_i = (\mathbb{1}(U_i \le u), \mathbb{1}(U_i \le v))$, $m = (u, v)$ and covariance matrix with diagonal $u(1-u)$, $v(1-v)$ and off-diagonal $u(1-v)$. The diagonal is positive and the determinant is $u(1-u)v(1-v) - u^2(1-v)^2 = u(1-v)(v-u) > 0$, so the matrix is positive definite, and the CLT for vectors gives the Gaussian limit. (c) The variance $u(1-u)$ is largest at $u = \frac{1}{2}$, where it is $\frac{1}{4}$. At $\frac{1}{4}$ and $\frac{3}{4}$ the covariance is $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$ and each variance is $\frac{3}{16}$, so the correlation is $\frac{1}{3}$.

Exercise 17.4 *(page 273).* (a) $F_n - F^0$ is 0 for $t < 0$ and for $t \ge 1$, and $(1 - \hat{p}) - \frac{1}{2} = \frac{1}{2} - \hat{p}$ for $0 \le t < 1$, so $D_n = |\hat{p} - \frac{1}{2}|$, and $\sqrt{n}D_n = \sqrt{n}|\hat{p} - \frac{1}{2}| = |T_n|/2$. (b) $T_n \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}(0,1)$ by the CLT (Theorem 4.8), so by the continuous mapping theorem (Theorem 5.2: if $T_n \xrightarrow[n\to\infty]{\text{(d)}} T$ and g is continuous, $g(T_n) \xrightarrow[n\to\infty]{\text{(d)}} g(T)$; here $g(t) = |t|/2$), $\sqrt{n}D_n \xrightarrow[n\to\infty]{\text{(d)}} |Z|/2$ with $Z \sim \text{Gauss}(0,1)$, and $\mathbb{P}(|Z|/2 > 1.358) = \mathbb{P}(|Z| > 2.716) = 0.0066$. (At $n = 124$ the threshold rejects for 46 or fewer or 78 or more right turns, probability 0.0052.) (c) Lemma 17.7: $F^0(X)$ takes only the values $\frac{1}{2}$ and 1, so the transformed data are not uniform.

Exercise 17.5 *(page 273).* (a) $0.305 \pm 1.96\sqrt{0.305 \times 0.695/272} = 0.305 \pm 0.055 = [0.250, 0.360]$, and $0.305 \pm 1.358/\sqrt{272} = 0.305 \pm 0.082 = [0.223, 0.387]$. (b) The interval covers $F(60)$ alone with

probability about 95%; the band must cover $F(t)$ at every t at once, and the largest of $\sqrt{n}|F_n(t)-F(t)|$ over all t is larger than its value at one point: 1.358 against $1.96\sqrt{0.305\times 0.695}=0.90$ at $t=60$. (c) Every Gaussian CDF is somewhere at least $0.102>0.082$ from the staircase, so every Gaussian leaves the band. If the waits were Gaussian, the band with half-width $1.3581/\sqrt{272}$, also 0.082, would contain their CDF with probability at least 95%, so this rejects the whole Gaussian family at level 5%. (Rounding the Gaussian to the minute, as the waits are, still leaves it 0.092 away.)

Exercise 17.6 *(page 273).* (a) Let $\mathbb{P}$ and $\mathbb{Q}$ have CDFs F and G. For each t, $|F(t)-G(t)| = |\mathbb{P}(A)-\mathbb{Q}(A)|$ with $A=(-\infty,t]$, which is at most the supremum over all events, the total variation distance. (b) At $t=0$ the Gaussian puts $\Phi(-1)=0.159$ below 0 and the exponential puts nothing there. For $t<0$ the gap is $\Phi(t-1)$, smaller still. For $t>0$ the gap first shrinks; the exponential CDF then runs above the Gaussian one between $t=0.26$ and $t=2.25$, by at most 0.133 (at $t=0.92$, where the densities cross), and beyond $t=2.25$ below it, by at most 0.027 (at $t=3.08$, where they cross again). So 0.159 is the largest gap. (c) Total variation compares all events, including the finite set of observed values, which has empirical probability 1 and probability 0 under F. The Kolmogorov distance compares only half-lines, and Theorem 17.3 controls all of them at once.

Exercise 17.7 *(page 273).* (a) Substitute $u=F^0(t)$, $\mathrm{d}u=f^0(t)\,\mathrm{d}t$, and use $F_n(t)=G_n(F^0(t))$ from the proof of Theorem 17.8. (b) $\int_{U_{(i)}}^{U_{(i+1)}}(i/n-u)^2\,\mathrm{d}u=\frac{1}{3}\big[(U_{(i+1)}-\frac{i}{n})^3-(U_{(i)}-\frac{i}{n})^3\big]$. Summing over $i=0,\dots,n$, the terms at $U_{(0)}=0$ and $U_{(n+1)}=1$ vanish, and each $U_{(i)}$ with $1\le i\le n$ contributes $\frac{1}{3}\big[(U_{(i)}-\frac{i-1}{n})^3-(U_{(i)}-\frac{i}{n})^3\big]$. With $a_i=U_{(i)}-\frac{2i-1}{2n}$ this is $\frac{1}{3}\big[(a_i+\frac{1}{2n})^3-(a_i-\frac{1}{2n})^3\big]=\frac{a_i^2}{n}+\frac{1}{12n^3}$. Multiplying the sum by n gives $\sum_i a_i^2+\frac{1}{12n}$. (c) The differences a_i are $0.03, 0.02, 0.13, 0.16, 0.09, 0.06, 0.15, 0.12, 0.09, 0.03$, whose squares add up to 0.1014; with $\frac{1}{120}=0.0083$, $W_{10}^2=0.110$, far below 0.46.

Exercise 17.8 *(page 273).* Write $X_i=Y_i/\lambda$ with $Y_1,\dots,Y_n\overset{iid}{\sim}\mathrm{Expo}(1)$, so $\bar{X}_n=\bar{Y}_n/\lambda$. For $t\ge 0$ put $s=\lambda t$: $X_i\le t$ exactly when $Y_i\le s$, so $F_n(t)=F_n^Y(s)$, and $t/\bar{X}_n=s/\bar{Y}_n$. As t runs over $[0,\infty)$ so does s, so $\hat{D}_n=\sup_{s\ge 0}|F_n^Y(s)-(1-e^{-s/\bar{Y}_n})|$, a function of the Y_i alone.

Exercise 17.9 *(page 274).* (a) $h(-2)=-\log(0.977)=0.023$, $h(0)=\log 2=0.693$ and $h(2)=-\log(0.023)=3.77$. (b) The slopes are $(0.693-0.023)/2=0.335$ and $(3.77-0.693)/2=1.54$, more than four times larger on the right: the long right tail stretches the upper quantiles and the short left tail squeezes the lower ones, so the points bend upward. (c) $h(x)=\Phi(x)$, which flattens toward 0 on the left and toward 1 on the right: the ends of a uniform sample are closer to the middle than Gaussian ends would be, the light-tailed pattern.

Chapter 18. Priors, Posteriors and Conjugacy

Exercise 18.1 *(page 289).* (a) $\mathbb{E}[P]=\int_0^1 p\cdot p^{a-1}(1-p)^{b-1}\,\mathrm{d}p/B(a,b)=B(a+1,b)/B(a,b)=\frac{\Gamma(a+1)\Gamma(a+b)}{\Gamma(a)\Gamma(a+b+1)}=\frac{a}{a+b}$. Likewise $\mathbb{E}[P^2]=B(a+2,b)/B(a,b)=\frac{a(a+1)}{(a+b)(a+b+1)}$. So $\mathbb{V}\mathrm{ar}(P)=\frac{a(a+1)}{(a+b)(a+b+1)}-\frac{a^2}{(a+b)^2}=\frac{a\big[(a+1)(a+b)-a(a+b+1)\big]}{(a+b)^2(a+b+1)}=\frac{ab}{(a+b)^2(a+b+1)}$, since $(a+1)(a+b)-a(a+b+1)=b$. (b) The log-density is $(a-1)\log p+(b-1)\log(1-p)$ plus a constant, with derivative $(a-1)/p-(b-1)/(1-p)$, which is 0 at $p=(a-1)/(a+b-2)$; the second derivative $-(a-1)/p^2-(b-1)/(1-p)^2$ is negative, so this is the maximum.

Exercise 18.2 *(page 289).* (a) $\mathbb{P}(S = s) = \int_0^1 \binom{n}{s} p^s (1-p)^{n-s}\,\mathrm{d}p = \binom{n}{s} B(s+1, n-s+1) = \frac{n!}{s!\,(n-s)!} \cdot \frac{s!\,(n-s)!}{(n+1)!} = \frac{1}{n+1}$. Before any data, every count is equally likely, which is how Bayes himself chose the flat prior in 1763. (b) $m_1 = B(81, 45) = 80!\,44!/125! = 1/\bigl(125\binom{124}{80}\bigr)$, so $B_{01} = 2^{-124}/m_1 = 125\binom{124}{80}2^{-124} = 125 \times 0.000372 = 0.0465$.

Exercise 18.3 *(page 289).* (a) Mean $40/50 = 0.8$; it counts as 50 patients, 40 cured. (b) Beta$(118, 32)$: mean $118/150 = 0.787$, standard deviation $\sqrt{118 \times 32/(150^2 \times 151)} = 0.033$; the weight on the prior mean is $50/150 = \frac{1}{3}$, and $0.787 = \frac{1}{3} \times 0.8 + \frac{2}{3} \times 0.78$. (c) $\mathbb{P}(p > 0.7) \approx 1 - \Phi\bigl((0.7 - 0.787)/0.033\bigr) = 1 - \Phi(-2.6) = 0.995$ (the exact Beta value is 0.992).

Exercise 18.4 *(page 290).* (a) $\pi(\lambda \mid \mathbf{x}) \propto \prod_i e^{-\lambda}\lambda^{x_i} \cdot \lambda^{\alpha-1}e^{-\beta\lambda} = \lambda^{\alpha+\sum x_i - 1}e^{-(\beta+n)\lambda}$, the shape of Gamma$(\alpha + \sum x_i, \beta + n)$. (b) Gamma$(26, 26)$: mean 1.0, standard deviation $\sqrt{26}/26 = 0.196$. The maximum likelihood estimate is the sample mean, 1.0; the prior, with mean 1 and worth 2 observations, agrees with the data and only narrows the posterior.

Exercise 18.5 *(page 290).* $\pi(p \mid \mathbf{x}) \propto \prod_i (1-p)^{x_i - 1}p \cdot p^{a-1}(1-p)^{b-1} = p^{a+n-1}(1-p)^{b+\sum x_i - n - 1}$, so the posterior is Beta$\bigl(a + n, b + \sum_i x_i - n\bigr)$, with mean $(a+n)/(a+b+\sum_i x_i)$. The prior again counts as data: a successes and b failures.

Exercise 18.6 *(page 290).* (a) $\pi(v \mid \mathbf{x}) \propto \prod_i v^{-1/2}e^{-x_i^2/(2v)} \cdot v^{-a-1}e^{-b/v} = v^{-(a+n/2)-1}e^{-(b+\sum x_i^2/2)/v}$, the shape of the Inverse Gamma with parameters $a + n/2$ and $b + \frac{1}{2}\sum_i x_i^2$. (b) The posterior mean is $\bigl(b + \frac{1}{2}\sum_i x_i^2\bigr)/(a + n/2 - 1)$, which tends to $\sum_i x_i^2/(n-2)$, not the maximum likelihood estimate $\frac{1}{n}\sum_i x_i^2$. (The posterior mode, $\bigl(b + \frac{1}{2}\sum_i x_i^2\bigr)/(a + n/2 + 1)$, tends to $\sum_i x_i^2/(n+2)$; neither summary of this posterior is the maximum likelihood estimate.)

Exercise 18.7 *(page 290).* (a) The posterior is Beta$(S+1, n-S+1)$, the distribution of the $(S+1)$th smallest of $n+1$ independent uniforms, whose density is $\frac{(n+1)!}{S!\,(n-S)!}u^S(1-u)^{n-S}$. That order statistic is at most x exactly when at least $S+1$ of the $n+1$ uniforms are at most x, and the number of uniforms at most x is Bin$(n+1, x)$. (b) The posterior probability uses one more trial and one more success than the p-value: $\mathbb{P}(\mathrm{Bin}(125, \frac{1}{2}) \geq 81)$ against $\mathbb{P}(\mathrm{Bin}(124, \frac{1}{2}) \geq 80)$. The two tail probabilities are of the same size, 0.00060 and 0.00078, which is why the posterior probability and the one-sided p-value nearly agree, though they answer different questions.

Exercise 18.8 *(page 290).* (a) A density of p becomes a density of ψ when multiplied by $\mathrm{d}p/\mathrm{d}\psi = p(1-p)$, so the posterior of ψ is proportional to $p^{80}(1-p)^{44} \cdot p(1-p) = p^{81}(1-p)^{45}$, written in terms of p. As p increases with ψ, this is largest where $p^{81}(1-p)^{45}$ is, at $p = 81/126 = 0.643$, that is $\psi = \log(81/45) = 0.588$. (b) The factor $p(1-p)$ changes where the density peaks, so the modes of the two densities correspond to different values of p: 0.645 and 0.643. The median is defined by probabilities, $\mathbb{P}(p \leq m) = \frac{1}{2}$, and an increasing map preserves them: $\mathbb{P}(\psi \leq \log(m/(1-m))) = \mathbb{P}(p \leq m) = \frac{1}{2}$.

Exercise 18.9 *(page 290).* Given p the probability is p^{10}, so the predictive probability is $\mathbb{E}[p^{10} \mid \text{data}] = B(91, 45)/B(81, 45) = \prod_{j=0}^{9} \frac{81+j}{126+j} = 0.0145$, against $0.645^{10} = 0.0125$. By Jensen's inequality (toolkit A.8), $\mathbb{E}[p^{10} \mid \text{data}] \geq (\mathbb{E}[p \mid \text{data}])^{10} = 0.643^{10} = 0.0121$: values of p above the mean raise p^{10} more than values below it lower it. The bound alone is below $\hat{p}^{10}$, since the posterior mean is below $\hat{p}$; the exact 0.0145 shows that the spread of the posterior adds enough to pass $\hat{p}^{10}$.

Chapter 19. Jeffreys Priors and Bayesian Intervals

Exercise 19.1 *(page 304).* (a) $I(\lambda) = 1/\lambda$ (Section 12.3.1), so $\pi_J(\lambda) \propto \lambda^{-1/2}$, and $\int_0^M \lambda^{-1/2}\,\mathrm{d}\lambda = 2\sqrt{M} \to \infty$: improper. The likelihood is proportional to $\lambda^S e^{-n\lambda}$, so the posterior is proportional to $\lambda^{S-1/2}e^{-n\lambda}$, the Gamma$(S+\frac{1}{2}, n)$ density, whose integral is finite for every $S \geq 0$ once $n \geq 1$. (b) Jeffreys: Gamma$(24.5, 24)$, mean $24.5/24 = 1.021$. Flat: Gamma$(25, 24)$, mean $25/24 = 1.042$. The maximum likelihood estimate is $24/24 = 1$. Jeffreys' prior adds half a count to S and the flat prior a whole count.

Exercise 19.2 *(page 305).* (a) $\ell(p) = \log p + (X-1)\log(1-p)$, so $\ell''(p) = -1/p^2 - (X-1)/(1-p)^2$, and with $\mathbb{E}_p[X-1] = (1-p)/p$, $I(p) = 1/p^2 + 1/(p(1-p)) = (1-p+p)/(p^2(1-p)) = 1/(p^2(1-p))$. So $\pi_J(p) \propto p^{-1}(1-p)^{-1/2}$, whose integral diverges at $p = 0$ like $\int \mathrm{d}p/p$. (b) The likelihood of $n = 5$ observations with sum 8 is $p^5(1-p)^{8-5}$, and times the prior it is $p^4(1-p)^{2.5}$: the Beta$(5, 3.5)$ density, with mean $5/8.5 = 0.588$. (c) Jeffreys' prior for Bernoulli trials is Beta$(\frac{1}{2}, \frac{1}{2})$, and the posterior is Beta$(5.5, 3.5)$, with mean $5.5/9 = 0.611$. The likelihoods are the same, but the priors differ because the Fisher information depends on how the data were collected, not only on the likelihood they produced: Jeffreys' prior depends on the sampling plan.

Exercise 19.3 *(page 305).* (a) By Section 12.3.1, the information for the variance is $1/(2v^2)$, and its square root is $1/(\sqrt{2}\,v) \propto 1/v$. (b) $\sigma = \sqrt{v}$ and $\mathrm{d}\sigma/\mathrm{d}v = 1/(2\sqrt{v})$, so the prior on v is $\frac{1}{\sqrt{v}} \cdot \frac{1}{2\sqrt{v}} = \frac{1}{2v} \propto \frac{1}{v}$. (c) With $\eta = \log\sigma$, $\mathrm{d}\eta/\mathrm{d}\sigma = 1/\sigma$, so a constant density in η is $1/\sigma$ in σ by Equation (19.1). Under $\sigma' = c\sigma$, $\mathrm{d}\sigma/\mathrm{d}\sigma' = 1/c$ and $\pi(\sigma') = \frac{1}{\sigma'/c} \cdot \frac{1}{c} = \frac{1}{\sigma'}$, the same form.

Exercise 19.4 *(page 305).* (a) $L_n\pi = p^S(1-p)^{n-S}/(p(1-p)) = p^{S-1}(1-p)^{n-S-1}$, integrable when $S \geq 1$ and $n - S \geq 1$; it is the Beta$(S, n-S)$ density, with mean S/n. (b) With $S = 0$, $L_n\pi = p^{-1}(1-p)^{n-1}$, whose integral diverges at $p = 0$ like $\int \mathrm{d}p/p$, so $m(\mathbf{x}) = \infty$. Under Jeffreys' prior the posterior is Beta$(S+\frac{1}{2}, n-S+\frac{1}{2})$, both of whose parameters are at least $\frac{1}{2}$ for every S. (c) Flat: $81/126 = 0.643$; Jeffreys: $80.5/125 = 0.644$; Haldane: $80/124 = 0.645$. The flat prior adds one imaginary success and one failure and pulls toward $\frac{1}{2}$ the most, Jeffreys' prior adds half of each, and Haldane's adds none.

Exercise 19.5 *(page 305).* (a) $\mathbb{E}[1/\lambda] = \int_0^\infty \frac{1}{\lambda}\,\frac{b^a}{\Gamma(a)}\lambda^{a-1}e^{-b\lambda}\,\mathrm{d}\lambda = \frac{b^a}{\Gamma(a)} \cdot \frac{\Gamma(a-1)}{b^{a-1}} = \frac{b}{a-1}$, using $\Gamma(a) = (a-1)\Gamma(a-1)$. With $b = 398.7$: 6.23, 6.33 and 6.43 minutes. (b) The map $\lambda \mapsto 1/\lambda$ is decreasing, so the equal-tailed region for β is $[1/U, 1/L]$: $[4.88, 7.95]$, $[4.95, 8.09]$ and $[5.02, 8.24]$ minutes. (c) The flat prior on λ ($a = 65$) gives posterior mean wait $398.7/64 = 6.23$; Jeffreys' prior ($a = 64$) gives posterior mean rate $64/398.7 = 0.1605 = 1/6.23$. No prior makes both means equal to the maximum likelihood estimates, since $\mathbb{E}[1/\lambda] > 1/\mathbb{E}[\lambda]$.

Exercise 19.6 *(page 305).* (a) The interval contains μ exactly when $|\sqrt{n}(\bar{X}_n - \mu)/S| \leq q_{0.025}(t_n)\sqrt{(n-1)/n}$, and by Theorem 14.7 the left side is $|T|$ with $T \sim t_{n-1}$ for every μ and σ. (b) $2.5706 \times \sqrt{4/5} = 2.299$, so the coverage is 0.917, against the nominal 0.95; at $n = 12$ it is 0.939. (c) Exactly 0.95: under $1/\sigma$ the region is $\bar{x} \pm q_{0.025}(t_{n-1})s/\sqrt{n}$, the t-interval.

Exercise 19.7 *(page 305).* (a) $m_1 = \int_0^1 p^{80}(1-p)^{44}\,\frac{c}{p(1-p)}\,\mathrm{d}p = c\int_0^1 p^{79}(1-p)^{43}\,\mathrm{d}p = c\,B(80, 44)$. (b) $c = 1$: $B_{01} = 4.70 \times 10^{-38}/(4.45 \times 10^{-36}) = 0.0106$ and $\mathbb{P}(H_0 \mid \mathbf{x}) = 0.0106/1.0106 = 0.0105$. $c = 1{,}000$: $B_{01} = 1.06 \times 10^{-5}$ and $\mathbb{P}(H_0 \mid \mathbf{x}) = 1.06 \times 10^{-5}$. Nothing in the data or the model picks c. (c) The posterior under H_1 is $c\,p^{79}(1-p)^{43}/(c\,B(80, 44))$, and c cancels.

Exercise 19.8 *(page 305).* (a) The set where the density is at least c is an interval $[m-h, m+h]$, symmetric about m, and by symmetry it leaves the same posterior probability $\alpha/2$ outside it on each side. So the highest-density region with probability $1-\alpha$ is the equal-tailed one. (b) The posterior Gauss$(\bar{x}, \sigma^2/n)$ of a Gaussian mean and the t posteriors of Proposition 19.6 are of that kind. The Beta posteriors of the kiss data, the Gamma posteriors of the subway and the posteriors of σ and σ^2 are skewed, and there the two regions differ.

Exercise 19.9 *(page 305).* (a) $L(x;\theta)\pi(\theta) = f(x-\theta)$, and $\int f(x-\theta)\,\mathrm{d}\theta = \int f(w)\,\mathrm{d}w = 1$, so $m(x) = 1$ and the posterior is $f(x-\theta)$. (b) Given x, $\theta = x - W$ with W having density f, so $\theta \le u(x)$ exactly when $W \ge x - u(x)$. The posterior condition says $\mathbb{P}(W \ge x - u(x)) = 1-\alpha$, so $x - u(x) = w$, the point with probability α below it under f, the same for every x, and $u(x) = x - w$. Under $\mathbb{P}_\theta$, $X = \theta + W'$ with W' having density f, and $\theta \le X - w$ exactly when $W' \ge w$, which has probability $1-\alpha$.

Exercise 19.10 *(page 306).* (a) Beta$(78.5, 22.5)$, with mean $78.5/101 = 0.777$ and 95% region $[0.692, 0.852]$. (b) $2\arcsin\sqrt{0.78} = 2.165$ and $1.96/\sqrt{100} = 0.196$, so $\varphi \in [1.969, 2.361]$, and $p = \sin^2(\varphi/2)$ gives $[0.694, 0.855]$. The Jeffreys, arcsine and solve intervals agree to within about 0.005 and all sit to the left of the plug-in interval, which is symmetric about 0.78 although the binomial is skewed this close to 1.

Chapter 20. Univariate Linear Regression

Exercise 20.1 *(page 325).* (a) $\mathbb{P}(X=0) = 0.3$ and $f(0) = (0\times0.2+1\times0.1)/0.3 = \frac{1}{3}$; $\mathbb{P}(X=1) = 0.3$ and $f(1) = 0.2/0.3 = \frac{2}{3}$; $\mathbb{P}(X=2) = 0.4$ and $f(2) = (0.1 + 2\times0.3)/0.4 = 1.75$. (b) $\mathbb{E}[X] = 1.1$, $\mathbb{E}[X^2] = 1.9$, $\mathbb{V}\mathrm{ar}(X) = 0.69$, $\mathbb{E}[Y] = 1$ and $\mathbb{E}[XY] = 0.2 + 0.2 + 1.2 = 1.6$, so $\mathbb{C}\mathrm{ov}(X,Y) = 0.5$, $b^* = 0.5/0.69 = 50/69 = 0.725$ and $a^* = 1 - 1.1\times50/69 = 14/69 = 0.203$. (c) The line gives 0.203, 0.928, 1.652 at $x = 0, 1, 2$ against $f = 0.333, 0.667, 1.75$: f rises by $\frac{1}{3}$ and then by $\frac{13}{12}$, so it is not a line. The noise's conditional means are $f(x) - a^* - b^*x = 9/69, -18/69, 27/276$, so $\mathbb{E}[\epsilon] = 0.3\cdot\frac{9}{69} - 0.3\cdot\frac{18}{69} + 0.4\cdot\frac{27}{276} = \frac{10.8-21.6+10.8}{276} = 0$ and $\mathbb{E}[X\epsilon] = -0.3\cdot\frac{18}{69} + 0.8\cdot\frac{27}{276} = \frac{-21.6+21.6}{276} = 0$, so $\mathbb{C}\mathrm{ov}(X,\epsilon) = 0$.

Exercise 20.2 *(page 326).* (a) $\mathbb{E}[X] = \frac{1}{2}$, $\mathbb{E}[X^2] = \frac{1}{3}$, $\mathbb{E}[X^3] = \frac{1}{4}$, so $\mathbb{C}\mathrm{ov}(X,Y) = \mathbb{C}\mathrm{ov}(X,X^2) = \frac{1}{4} - \frac{1}{6} = \frac{1}{12} = \mathbb{V}\mathrm{ar}(X)$: $b^* = 1$ and $a^* = \mathbb{E}[Y] - \mathbb{E}[X] = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$. The noise has conditional mean $f(x) - a^* - b^*x = x^2 - x + \frac{1}{6}$. (b) $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^3] = 0$, so $\mathbb{C}\mathrm{ov}(X,Y) = \mathbb{C}\mathrm{ov}(X,X^2) = 0$: $b^* = 0$, $a^* = \mathbb{E}[X^2] = \frac{1}{3}$, and the correlation is 0.

Exercise 20.3 *(page 326).* With $x' = 60x$ and $y' = y/60$: $S_{x'y'} = S_{xy}$ and $S_{x'x'} = 3{,}600\,S_{xx}$, so $\hat{b}' = \hat{b}/3{,}600 = 0.00298$ hours per second, and $\hat{a}' = \bar{Y}' - \hat{b}'\bar{X}' = (\bar{Y} - \hat{b}\bar{X})/60 = 0.558$ hours. The correlation and R^2 do not change, since rescaling a variable rescales its standard deviation by the same factor. $\hat{\sigma}' = \hat{\sigma}/60$ and $\sqrt{S_{x'x'}} = 60\sqrt{S_{xx}}$, so the standard error is divided by 3,600, as $\hat{b}$ is: $T = 34.1$ as before.

Exercise 20.4 *(page 326).* (a) The derivative of $\sum_i (Y_i - bx_i)^2$ in b is $-2\sum_i x_i(Y_i - bx_i)$, zero at $\tilde{b} = \sum_i x_iY_i / \sum_i x_i^2$, a minimum since the sum is a parabola in b that opens upward. (b) $\tilde{b} = b + \sum_i x_i\epsilon_i / \sum_i x_i^2$, with mean b and variance $\sigma^2\sum_i x_i^2/(\sum_i x_i^2)^2 = \sigma^2/\sum_i x_i^2$. Since $\sum_i x_i^2 = S_{xx} + n\bar{x}^2 \ge S_{xx}$, it is more precise than $\hat{b}$ when the model holds. (c) $\mathbb{E}[\tilde{b}] = (a\sum_i x_i + b\sum_i x_i^2)/\sum_i x_i^2$, so the bias is $a\sum_i x_i / \sum_i x_i^2$, zero only when the x_i average 0.

Exercise 20.5 *(page 326).* (a) $\hat{a} = \bar{R}_n = 0.645$. The residuals are $44/124 = 0.355$ for the 80 right turns and $-80/124 = -0.645$ for the 44 left turns, so $\text{RSS} = 80 \times 0.355^2 + 44 \times 0.645^2 = 28.39$ and $\hat{\sigma}^2 = 28.39/123 = 0.231$, the sample variance S_n^2 of Chapter 14: one coefficient fitted, one direction absorbed, divisor $n-1$. (b) $\mathbb{V}\text{ar}(\hat{a}) = \sigma^2/n$, so $T = (\hat{a} - \frac{1}{2})/(\hat{\sigma}/\sqrt{n}) = \sqrt{124}\,(0.645 - 0.5)/0.480 = 3.36$, the t statistic of Section 14.5.

Exercise 20.6 *(page 326).* (a) $\bar{x} = n_1/n$ and $S_{xx} = \sum_i x_i^2 - n\bar{x}^2 = n_1 - n_1^2/n = n_0 n_1/n$. $S_{xy} = \sum_i x_i Y_i - n\bar{x}\bar{Y} = n_1\bar{Y}_1 - \frac{n_1}{n}(n_0\bar{Y}_0 + n_1\bar{Y}_1) = \frac{n_0 n_1}{n}(\bar{Y}_1 - \bar{Y}_0)$, so $\hat{b} = \bar{Y}_1 - \bar{Y}_0$ and $\hat{a} = \bar{Y} - \hat{b}\bar{x} = \bar{Y}_0$. (b) $\bar{Y}_0 = 0.01713$ and $\bar{Y}_1 = 0.00942$, so $\hat{b} = -0.00771$: 7.7 fewer heart attacks per 1,000 physicians. The residual sum of squares is the sum of squared deviations within the groups, $189 \times (1 - 0.01713) + 104 \times (1 - 0.00942) = 288.8$, so $\hat{\sigma}^2 = 288.8/22{,}069 = 0.01309$; $S_{xx} = 5{,}517.8$, the standard error is $\sqrt{0.01309/5{,}517.8} = 0.00154$, and $T = -5.00$, with $T^2 = 25.04$, the Wald statistic to two decimals (with groups of almost equal size, $\hat{\sigma}^2(1/n_0 + 1/n_1)$ is almost the unpooled variance $\hat{p}_0(1-\hat{p}_0)/n_0 + \hat{p}_1(1-\hat{p}_1)/n_1$). (c) The errors take two values in each group, so they are not Gaussian, and their variance $p(1-p)$ differs between the groups. If the physicians are modeled as i.i.d. draws, each assigned to aspirin by a fair coin, the pairs are i.i.d. and Theorem 20.12 applies without its constant-scatter condition: the robust standard error, which here is the unpooled one, makes the test asymptotically valid, and with 22,071 physicians in two groups of nearly equal size the two standard errors agree.

Exercise 20.7 *(page 326).* $\hat{y}_0 = 33.47 + 10.73 \times 2 = 54.93$. The standard error for the mean is

$$5.914\sqrt{\frac{1}{272} + \frac{(2 - 3.488)^2}{353.0}} = 5.914 \times 0.0997 = 0.590,$$

so the confidence interval is $54.93 \pm 1.16 = [53.8, 56.1]$. For a new wait it is $5.914\sqrt{1 + 0.00995} = 5.943$, so the prediction interval is $54.93 \pm 11.70 = [43.2, 66.6]$. Two minutes is farther from $\bar{x}$ than 4.5 minutes is (1.49 against 1.01), and the error in the slope moves the line more the farther you are from $\bar{x}$; in the prediction interval the 1 for the new wait dominates both.

Exercise 20.8 *(page 326).* (a) Exchange the roles of the variables in Theorem 20.6: the slope is S_{xy}/S_{yy} and the line is $\hat{x} = \bar{X} + (S_{xy}/S_{yy})(y - \bar{Y})$, which passes through $(\bar{X}, \bar{Y})$. The product of the slopes is $S_{xy}^2/(S_{xx}S_{yy}) = r^2$. (b) The slope is $3{,}788/50{,}087 = 0.0756$, and $\hat{x} = 3.488 + 0.0756 \times (90 - 70.90) = 4.93$ minutes. The line for the wait reaches 90 at $(90 - 33.47)/10.73 = 5.27$ minutes. The two answers differ because they answer different questions: the first is the best line's prediction of the duration from the wait, pulled back toward the average duration, 3.49 minutes, since 90 minutes is a long wait; the second, 5.27 minutes, lies beyond the longest eruption in the data, 5.1 minutes. Neither is the mean duration after a 90-minute wait: the 17 eruptions followed by waits of 88 to 92 minutes lasted 4.5 minutes on average, since the durations level off at long waits as the waits do at long durations (Figure 20.3).

Exercise 20.9 *(page 326).* (a) $S_{xy} = \sum_i (x_i - \bar{x})Y_i = bS_{xx} + \sum_i (x_i - \bar{x})\epsilon_i$, so $\mathbb{E}[\tilde{b}_\lambda] = bS_{xx}/(S_{xx} + \lambda)$, a bias of $-\lambda b/(S_{xx} + \lambda)$, and $\mathbb{V}\text{ar}(\tilde{b}_\lambda) = \sigma^2 S_{xx}/(S_{xx} + \lambda)^2$. (b) Variance plus squared bias gives the formula, which equals σ^2/S_{xx} at $\lambda = 0$. Its derivative in λ is $2S_{xx}(\lambda b^2 - \sigma^2)/(S_{xx} + \lambda)^3$, negative for $0 \le \lambda < \sigma^2/b^2$ (for every λ if $b = 0$), so the mean squared error decreases from σ^2/S_{xx} as λ increases from 0. (c) The derivative is zero at $\lambda = \sigma^2/b^2$, the minimum, which depends on the unknown b: the same trade-off as the shrinkage estimators of Chapter 7. In the Bayesian reading, $\lambda = \sigma^2/\tau^2$ with τ^2 a prior guess at the size of b^2.

Exercise 20.10 *(page 326).* (a) $\mathbb{E}[\tilde{b}] = (a + 20b - a - b)/19 = b$ and $\mathbb{V}\mathrm{ar}(\tilde{b}) = 2\sigma^2/19^2 = 2\sigma^2/361$. (b) $S_{xx} = \sum_{i=1}^{20}(i - 10.5)^2 = 665$, so the ratio is $(2/361)/(1/665) = 3.68$: the two-point slope needs about 3.7 times as much data to be as precise. Proposition 20.10 says the ratio is at least 1 for every linear unbiased estimator.

Chapter 21. Multivariate Linear Regression and Inference

Exercise 21.1 *(page 340).* (a) $X^\top X = \begin{pmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{pmatrix}$, $X^\top Y = (\sum_i Y_i, \sum_i x_i Y_i)^\top$, and $\det(X^\top X) = n\sum_i x_i^2 - (\sum_i x_i)^2 = nS_{xx}$, so $(X^\top X)^{-1} = \frac{1}{nS_{xx}}\begin{pmatrix} \sum_i x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix}$. The second row gives

$$\hat{b} = \frac{n\sum_i x_i Y_i - n\bar{x}\sum_i Y_i}{nS_{xx}} = \frac{S_{xy}}{S_{xx}}, \qquad \hat{a} = \frac{\sum_i x_i^2 \sum_i Y_i - n\bar{x}\sum_i x_i Y_i}{nS_{xx}} = \bar{Y} - \hat{b}\bar{x},$$

the second after writing $\sum_i x_i^2 = S_{xx} + n\bar{x}^2$. (b) σ^2 times the entries: $\mathbb{V}\mathrm{ar}(\hat{b}) = \sigma^2/S_{xx}$, $\mathbb{V}\mathrm{ar}(\hat{a}) = \sigma^2\sum_i x_i^2/(nS_{xx}) = \sigma^2(1/n + \bar{x}^2/S_{xx})$, $\mathbb{C}\mathrm{ov}(\hat{a}, \hat{b}) = -\sigma^2\bar{x}/S_{xx}$.

Exercise 21.2 *(page 340).* (a) $X^\top X = \begin{pmatrix} n & n_1 \\ n_1 & n_1 \end{pmatrix}$, with inverse $\frac{1}{nn_1 - n_1^2}\begin{pmatrix} n_1 & -n_1 \\ -n_1 & n \end{pmatrix} = \frac{1}{n_0}\begin{pmatrix} 1 & -1 \\ -1 & n/n_1 \end{pmatrix}$, and $X^\top Y = (n_0\bar{Y}_0 + n_1\bar{Y}_1, n_1\bar{Y}_1)^\top$, giving $\hat{\beta}_1 = \bar{Y}_0$ and $\hat{\beta}_2 = \bar{Y}_1 - \bar{Y}_0$. (b) $\gamma_2 = n/(n_0 n_1) = 1/n_0 + 1/n_1$, and $\hat{\sigma}^2$ is the pooled variance (the residuals are the deviations within each group, on $n - 2$ degrees of freedom), so $T_2 = (\bar{Y}_1 - \bar{Y}_0)/(\hat{\sigma}\sqrt{1/n_0 + 1/n_1})$. (c) $\gamma_2 = 1/11{,}034 + 1/11{,}037 = 1.8124 \times 10^{-4}$, and with $\hat{\sigma}^2 = 0.01309$ the standard error is $\sqrt{0.01309 \times 1.8124 \times 10^{-4}} = 0.00154$.

Exercise 21.3 *(page 341).* (a) $\mathrm{tr}(P) = \mathrm{tr}\big((X^\top X)^{-1}X^\top X\big) = \mathrm{tr}(I_p) = p$, and $\mathrm{tr}(I_n - P) = n - p$. (b) $\|e\|^2 = \epsilon^\top(I-P)^\top(I-P)\epsilon = \epsilon^\top(I-P)\epsilon = \mathrm{tr}\big((I-P)\epsilon\epsilon^\top\big)$, whose expectation is $\mathrm{tr}\big((I-P)\sigma^2 I\big) = (n-p)\sigma^2$. (c) $\mathbb{C}\mathrm{ov}(e) = (I - P)\sigma^2 I(I - P)^\top = \sigma^2(I - P)$, so $\mathbb{V}\mathrm{ar}(e_i) = \sigma^2(1 - P_{ii})$; as $P_{ii} \to 1$ the residual's variance vanishes, so the fit passes near the observation whatever its value.

Exercise 21.4 *(page 341).* Let $w = (I - P_0)x_p$, the part of the last column perpendicular to the column space of X_0; $w \neq 0$ since X has full rank, and the column space of X is spanned by the columns of X_0 together with w, with w perpendicular to the first. So $P = P_0 + ww^\top/\|w\|^2$ and $(P - P_0)Y = (w^\top Y/\|w\|^2)w$, whose squared length is $(w^\top Y)^2/\|w\|^2$. The coefficient of x_p in the full fit is the coefficient of w in PY, since replacing x_p by w only reparameterizes the X_0 coefficients: $\hat{\beta}_p = w^\top Y/\|w\|^2$, and its variance $\sigma^2\gamma_p$ is $\sigma^2\|w\|^2/\|w\|^4 = \sigma^2/\|w\|^2$. Hence $\|e_0\|^2 - \|e\|^2 = \hat{\beta}_p^2\|w\|^2 = \hat{\beta}_p^2/\gamma_p$ and $F = \hat{\beta}_p^2/(\hat{\sigma}^2\gamma_p) = T_p^2$. For the stack loss acid coefficient, $(188.8 - 178.8)/10.52 = 0.95 = (-0.973)^2$.

Exercise 21.5 *(page 341).* (a) $W_n/(n-p)$ is the mean of $n - p$ i.i.d. χ_1^2 variables, which converges in probability to 1 by the Law of Large Numbers, so $kF = V/(W_n/(n-p)) \xrightarrow[n\to\infty]{\text{(d)}} V$ by Slutsky's theorem. (b) Let X_1 be the last k columns of X, $\hat{\beta}_{(k)}$ their coefficients and $C_{(k)}$ the matching block of $(X^\top X)^{-1}$, and let $W = (I - P_0)X_1$, the parts of those columns perpendicular to the

column space of X_0; W has rank k since X has rank p, and $X_1 = X_0A + W$ for some matrix A. As in the previous exercise the column space of X is that of X_0 plus that of W, perpendicular to each other, so $P = P_0 + W(W^\top W)^{-1}W^\top$; and $X\beta = X_0(\beta_0 + A\beta_{(k)}) + W\beta_{(k)}$ shows that $\hat\beta_{(k)} = (W^\top W)^{-1}W^\top Y$, with covariance $\sigma^2(W^\top W)^{-1}$, so $C_{(k)} = (W^\top W)^{-1}$. Hence $\|e_0\|^2 - \|e\|^2 = \|(P - P_0)Y\|^2 = Y^\top W(W^\top W)^{-1}W^\top Y = \hat\beta_{(k)}^\top C_{(k)}^{-1}\hat\beta_{(k)}$. Under H_0, $\sqrt{n}\hat\beta_{(k)} \xrightarrow[n\to\infty]{\text{(d)}} \text{Gauss}_k(0, \sigma^2 M_{(k)}^{-1})$ and $nC_{(k)} \xrightarrow[n\to\infty]{\mathbb{P}} M_{(k)}^{-1}$ by Theorem 21.7, so the quadratic form converges in distribution to $\sigma^2\chi_k^2$ by Lemma 15.1 and the continuous mapping theorem; dividing by $\hat\sigma^2 \xrightarrow[n\to\infty]{\mathbb{P}} \sigma^2$ gives $kF \xrightarrow[n\to\infty]{\text{(d)}} \chi_k^2$.

Exercise 21.6 *(page 341).* $\hat\beta_{(k)} \sim \text{Gauss}_k(\beta_{(k)}, \sigma^2 C_{(k)})$ by the linear-map rule, so by Lemma 15.1 the quadratic form $(\hat\beta_{(k)} - \beta_{(k)})^\top(\sigma^2 C_{(k)})^{-1}(\hat\beta_{(k)} - \beta_{(k)})$ is χ_k^2, independent of $(n-p)\hat\sigma^2/\sigma^2 \sim \chi_{n-p}^2$ by Theorem 21.5. Replacing σ^2 by $\hat\sigma^2$ multiplies the quadratic form by $\sigma^2/\hat\sigma^2$, so the quantity in the exercise is $(\chi_k^2/k)/(\chi_{n-p}^2/(n-p))$ with independent numerator and denominator: $F_{k,n-p}$. The ellipsoid contains $\beta_{(k)}$ exactly when this statistic is at most $q_\alpha(F_{k,n-p})$, which has probability $1-\alpha$.

Exercise 21.7 *(page 341).* (a) $F = (9{,}443.4 - 8{,}976.7)/1 \div 8{,}976.7/269 = 466.7/33.37 = 13.99 = (-3.74)^2$. (b) 15.1 minutes per minute at 2 minutes and 6.3 at 4.5: the waits rise faster than the line for short eruptions and more slowly for long ones, which is the bend in the bin means of Figure 20.3. (c) The model is linear in the coefficients: x^2 is a column of numbers computed from the durations, and $Y = X\beta + \epsilon$ with that column is Definition 21.1.

Exercise 21.8 *(page 341).* (a) $\mathbb{E}[AY] = AX\beta$ equals β for every β only if $AX = I_p$. (b) $DX = AX - (X^\top X)^{-1}X^\top X = I_p - I_p = 0$. (c) $\mathbb{C}\text{ov}(AY) = \sigma^2 AA^\top$, and $AA^\top = (X^\top X)^{-1}X^\top X(X^\top X)^{-1} + (X^\top X)^{-1}X^\top D^\top + DX(X^\top X)^{-1} + DD^\top$; the two middle terms vanish since $DX = 0$ and $(DX)^\top = X^\top D^\top = 0$. The diagonal of $DD^\top$ is $\sum_k D_{jk}^2 \geq 0$, so $\mathbb{V}\text{ar}(\tilde\beta_j) \geq \mathbb{V}\text{ar}(\hat\beta_j)$ for each j; this is Proposition 20.10 for every coordinate at once.

Exercise 21.9 *(page 341).* (a) $\tilde\beta_\lambda = (X^\top X + \lambda I)^{-1}X^\top(X\beta + \epsilon)$; the expectation is the first term and the covariance is $(X^\top X + \lambda I)^{-1}X^\top \sigma^2 I\, X(X^\top X + \lambda I)^{-1}$. (b) $(cI + \lambda I)^{-1}X^\top Y = \frac{1}{c+\lambda}X^\top Y$ and $\hat\beta = \frac{1}{c}X^\top Y$, so $\tilde\beta_\lambda = \frac{c}{c+\lambda}\hat\beta$, which is the factor $S_{xx}/(S_{xx} + \lambda)$ with c in place of S_{xx}.

Exercise 21.10 *(page 341).* (a) $X^\top X = 124$, $(X^\top X)^{-1} = 1/124$, $\hat\beta = \bar R_n = 0.645$, P is the matrix with every entry $1/124$, whose trace is 1. (b) By Exercise 21.2, $\hat\beta = (36/60, 44/64 - 36/60)^\top = (0.600, 0.0875)^\top$; $\|e\|^2 = 36(0.4)^2 + 24(0.6)^2 + 44(0.3125)^2 + 20(0.6875)^2 = 5.76 + 8.64 + 4.30 + 9.45 = 28.15$, $\hat\sigma^2 = 28.15/122 = 0.2307$, $\gamma_2 = 1/60 + 1/64 = 0.03229$, and $T_2 = 0.0875/\sqrt{0.2307 \times 0.03229} = 1.01$: no evidence that the proportion turning right differs between the cities. (c) The errors take two values, so they are not Gaussian, and their variance $p(1-p)$ differs slightly between the cities (0.240 and 0.215). With the couples modeled as i.i.d. draws whose city is recorded, Theorem 21.7 applies without its constant-scatter condition, and the robust standard error, here the unpooled $\sqrt{0.240/60 + 0.215/64} = 0.0858$, makes the test asymptotically valid; it is within 1% of the pooled $\hat\sigma\sqrt{\gamma_2} = 0.0863$, as in Exercise 20.6.

Chapter 22. Exponential Families

Exercise 22.1 *(page 354).* (a) $\frac{\beta^\alpha}{\Gamma(\alpha)}y^{\alpha-1}e^{-\beta y} = \exp\big(-\beta y + \alpha\log\beta + (\alpha-1)\log y - \log\Gamma(\alpha)\big)$. With $\theta = -\beta/\alpha$ and $\phi = 1/\alpha$, $-\beta y = y\theta/\phi$ and $\alpha\log\beta = \alpha\log(-\theta) + \alpha\log\alpha = -b(\theta)/\phi + \alpha\log\alpha$, so the

exponent is $(y\theta - b(\theta))/\phi + \alpha\log\alpha + (\alpha-1)\log y - \log\Gamma(\alpha)$, and $c(y,\phi) = \alpha\log(\alpha y) - \log y - \log\Gamma(\alpha)$. (b) $b'(\theta) = -1/\theta = \alpha/\beta$ and $b''(\theta) = 1/\theta^2 = \alpha^2/\beta^2$, so $\mathbb{E}[Y] = \alpha/\beta$ and $\mathbb{V}\mathrm{ar}(Y) = \phi b'' = \alpha/\beta^2$, the Gamma's mean and variance of Definition 18.5; $V(\mu) = \mu^2$, with dispersion $1/\alpha$. (c) $\alpha = 1$ is Expo(β): $\theta = -\beta$, $\phi = 1$, $c = 0$.

Exercise 22.2 *(page 355).* (a) $\binom{m}{y}p^y(1-p)^{m-y} = \binom{m}{y}\exp\big(y\log\frac{p}{1-p} + m\log(1-p)\big)$, and $m\log(1-p) = -m\log(1+e^\theta)$. Then $b'(\theta) = me^\theta/(1+e^\theta) = mp$ and $b''(\theta) = mp(1-p)$. (b) $(1-p)^y p = \exp\big(y\log(1-p) + \log p\big)$ with $\log p = \log(1-e^\theta)$, so $b(\theta) = -\log(1-e^\theta)$ on $\theta < 0$; $b'(\theta) = e^\theta/(1-e^\theta) = (1-p)/p$, the geometric mean. (c) $\log p^r = -r\log(1-e^\theta)$, so $b(\theta) = -r\log(1-e^\theta)$ with $c(y) = \log\binom{y+r-1}{y}$, and $b'(\theta) = re^\theta/(1-e^\theta) = r(1-p)/p$, r times the geometric mean.

Exercise 22.3 *(page 355).* $p^{a-1}(1-p)^{b-1}/B(a,b) = \exp\big((a-1)\log p + (b-1)\log(1-p) - \log B(a,b)\big)$: $T(p) = (\log p, \log(1-p))$, $\eta = (a-1, b-1)$, the normalizer is $\log B(a,b)$, the log of the Beta function, and $h = 1$ on $(0,1)$.

Exercise 22.4 *(page 355).* Suppose $|y-\mu| = \eta(\mu)T(y) + B(\mu) + a(y)$ for all y, μ. Pick $\mu_1 < \mu_2$ with $\eta(\mu_1) \neq \eta(\mu_2)$ (if η were constant, $\eta T(y)$ could be absorbed into $a(y)$, and then $|y-\mu_1| - |y-\mu_2| = B(\mu_1) - B(\mu_2)$ would be constant in y for any two centers; but it equals $\mu_2 - \mu_1$ to the right of both and $\mu_1 - \mu_2$ to the left). Subtracting the equations for μ_1 and μ_2, $|y-\mu_1| - |y-\mu_2| = (\eta(\mu_1) - \eta(\mu_2))T(y) + \text{constant}$, and the left side equals $\mu_2 - \mu_1$ for every $y \geq \mu_2$; so T is constant on $[\mu_2, \infty)$. Then for every μ, $|y-\mu| - a(y) = \eta(\mu)T(y) + B(\mu)$ is constant in y on $[\mu_2, \infty)$, so for two centers $\mu_2 < \mu < \mu'$ the difference $|y-\mu| - |y-\mu'|$ is constant in y on $[\mu_2, \infty)$; but between μ and μ' it equals $2y - \mu - \mu'$, which is not constant. No such representation exists.

Exercise 22.5 *(page 355).* (a) $V(\mu) = b''(\theta)$ at $\mu = b'(\theta)$, so $b''(\theta) = b'(\theta)$; the solutions are $b'(\theta) = Ce^\theta$, and $C = e^{\theta_0}$ only shifts θ by θ_0, so take $b' = e^\theta$ and $b = e^\theta$ up to a constant, which is absorbed in c: the Poisson row's b. (b) $b'' = b'(1-b')$ gives $b' = e^\theta/(1+e^\theta)$ up to a shift of θ, the Bernoulli row's b; $b'' = (b')^2$ gives $b' = -1/\theta$ up to a shift, the exponential row's, and with dispersion $1/\alpha$ the Gamma row's. (The base weight $e^{c(y)}$ is then forced as well, since $\int e^{y\theta}e^{c(y)}\,\mathrm{d}y = e^{b(\theta)}$ for every θ determines it; that a weight is determined by this transform is not proved in this book.)

Exercise 22.6 *(page 355).* (a) In each family the mean is $b'(\theta)$ and $\bar{Y}_n$ is unbiased with variance $\phi b''(\theta)/n = 1/(nI_\mu)$ by part (2) of the theorem. (b) $\hat{\lambda}$ is biased, so the bound, which is for unbiased estimators, does not apply to it. The score of the exponential sample is $n/\lambda - \sum_i T_i$, an affine function of $\bar{T}_n$, so it is an affine function of $W - \tau$ only for W affine in $\bar{T}_n$; $(n-1)/\sum_i T_i$ is not, so it does not attain the bound (its variance is $\lambda^2/(n-2)$ for $n > 2$, against the bound λ^2/n). No contradiction: (a) is about the mean $1/\lambda$, and in a canonical family the bound is attained for the mean and for no parameter that is not an affine function of it: the score $(\sum_i Y_i - nb'(\theta))/\phi$ is affine in $\bar{Y}_n$, so an estimator W free of θ with $W - \tau(\theta) = c(\theta)\,\ell_n'(\theta)$ for every θ must be $\alpha\bar{Y}_n + \beta$ with constants α, β, which forces $\tau = \alpha b'(\theta) + \beta$.

Exercise 22.7 *(page 355).* (a) The joint density is $\mathbb{1}(\theta \leq \min_i y_i)\,\mathbb{1}(\max_i y_i \leq \theta + 1)$, a function of the pair $(\min_i y_i, \max_i y_i)$ and θ: the two extremes together are sufficient. (b) The joint density is

$$\prod_{i=1}^n \frac{\beta^\alpha}{\Gamma(\alpha)} y_i^{\alpha-1} e^{-\beta y_i} = \frac{\beta^{n\alpha}}{\Gamma(\alpha)^n}\Big(\prod_i y_i\Big)^{\alpha-1} e^{-\beta\sum_i y_i},$$

so $(\sum_i Y_i, \prod_i Y_i)$, or equivalently $(\sum_i Y_i, \sum_i \log Y_i)$, is sufficient, as Corollary 22.9 says with $T = (y, \log y)$. (c) $\prod_i \lambda^{y_i} e^{-\lambda}/y_i! = \lambda^{\sum_i y_i} e^{-n\lambda} / \prod_i y_i!$, so $\sum_i Y_i$ is sufficient, and so is $\bar{Y}_n$, a one-to-one function of it. Y_1 is not: given $Y_1 = y_1$ the remaining $Y_2, \ldots, Y_n$ still have the Pois(λ) distribution, which depends on λ.

Exercise 22.8 *(page 355).* (a) $\pi(\theta) \propto \exp(m_0\theta - n_0 e^\theta)$, and with $\lambda = e^\theta$, $\mathrm{d}\theta = \mathrm{d}\lambda/\lambda$, the density of λ is proportional to $\lambda^{m_0 - 1} e^{-n_0\lambda}$: Gamma($m_0, n_0$), updated to Gamma($m_0 + \sum_i Y_i, n_0 + n$), the Gamma–Poisson pair of Chapter 18. (b) $\pi(\theta) \propto \exp\big((m_0\theta - n_0\theta^2/2)/\sigma^2\big) = \exp\big(-\frac{n_0}{2\sigma^2}(\theta - m_0/n_0)^2\big)$ up to a constant: Gauss($m_0/n_0, \sigma^2/n_0$), a prior worth n_0 observations averaging m_0/n_0. The posterior is Gauss$\big((m_0 + \sum_i Y_i)/(n_0 + n), \sigma^2/(n_0 + n)\big)$, whose mean weights the prior mean by n_0 and the sample mean by n, that is, by the precisions n_0/σ^2 and n/σ^2 of Chapter 18. (c) With $\theta = -\lambda$ and $b(\theta) = -\log(-\theta) = -\log\lambda$, $\pi(\theta) \propto \exp(-m_0\lambda + n_0 \log\lambda) = \lambda^{n_0} e^{-m_0\lambda}$, and $\mathrm{d}\theta = -\mathrm{d}\lambda$ changes nothing but the sign, so the prior on λ is Gamma($n_0 + 1, m_0$): $\alpha = n_0 + 1$ and $\beta = m_0$. Chapter 18 read the prior as α waits lasting β minutes, since the data add n to α and $\sum_i T_i$ to β; on the canonical scale the count is $n_0 = \alpha - 1$, one fewer, because the flat prior on λ, which is Gamma(1, 0), counts as no waits at all. The posterior is Gamma($n_0 + n + 1, m_0 + \sum_i T_i$) = Gamma($\alpha + n, \beta + \sum_i T_i$), the update of Chapter 18.

Exercise 22.9 *(page 355).* (a) A sum of independent Poissons is Poisson with the summed mean (its PMF is the convolution, which collapses by the binomial theorem), so $Y_2 + \cdots + Y_n \sim \text{Pois}((n-1)\lambda)$ and $S \sim \text{Pois}(n\lambda)$, and

$$\begin{aligned}\mathbb{P}(Y_1 = y \mid S = s) &= \frac{\mathbb{P}(Y_1 = y)\mathbb{P}(Y_2 + \cdots + Y_n = s - y)}{\mathbb{P}(S = s)} \\ &= \frac{e^{-\lambda}\lambda^y}{y!} \cdot \frac{e^{-(n-1)\lambda}((n-1)\lambda)^{s-y}}{(s-y)!} \cdot \frac{s!}{e^{-n\lambda}(n\lambda)^s} = \binom{s}{y}\Big(\frac{1}{n}\Big)^y\Big(1 - \frac{1}{n}\Big)^{s-y},\end{aligned}$$

the Bin($s, 1/n$) PMF, free of λ; so $\mathbb{P}(Y_1 = 0 \mid S = s) = (1 - 1/n)^s$. (b) The first student in Chapter 2's list has $X_1 = 2$, so $Y_1 = 1$ and $W = 0$: an unbiased estimate of $e^{-\lambda}$ that uses one person. The Rao–Blackwell estimate is $(1 - 1/24)^{24} = 0.360$, against the plug-in $e^{-1} = 0.368$ and the raw proportion of only children, $8/24 = 0.333$. The Rao–Blackwell estimate is unbiased, the plug-in is the MLE by invariance and is not, and they agree to two decimals here.

Exercise 22.10 *(page 355).* (a) In θ: $\ell_n'''(\theta) = -nb'''(\theta) = -np(1-p)(1-2p)$, which is $-124 \times 0.229 \times (-0.29) = 8.2$ at the maximum; in p: $\ell_n(p) = S\log p + (n - S)\log(1 - p)$ has $\ell_n''' = 2S/p^3 - 2(n-S)/(1-p)^3$, which is $2 \times 80/0.645^3 - 2 \times 44/0.355^3 = 596 - 1{,}967 = -1{,}371$ (with the rounded $\hat{p}$; 80/124 gives $-1{,}374$). The two numbers are in different units, so compare them at one standard error from the maximum, where each parabola has dropped by $\frac{1}{2}$: the cubic term $\frac{1}{6}\ell_n'''\,\mathrm{SE}^3$ is $\frac{1}{6} \times 8.2 \times 0.188^3 = 0.009$ in θ and $\frac{1}{6} \times 1{,}371 \times 0.043^3 = 0.018$ in p, twice as large, so the parabola fits better in θ. (b) $\hat{\theta}^2/\widehat{\mathbb{V}\mathrm{ar}}(\hat{\theta}) = 0.598^2/0.188^2 = 10.1$, against 11.41 in p: the two Wald statistics differ, as Chapter 15 said they would, and the likelihood ratio statistic 10.60, which does not depend on the scale, sits between them.

Chapter 23. Link Functions and Fitting

Exercise 23.1 *(page 365).* $b(\theta) = \theta^2/2$, $\phi = \sigma^2$, so $\ell_n(\beta) = \frac{1}{\sigma^2}\sum_i (Y_i x_i^\top\beta - (x_i^\top\beta)^2/2) + \text{const} = -\frac{1}{2\sigma^2}\|Y - X\beta\|^2 + \text{const}'$. $b'' = 1$ gives $W = I$, $\mu(\beta) = X\beta$, and $X^\top(Y - X\beta) = 0$ are the normal

equations. Newton's step is $\beta_0 + (X^\top X)^{-1}X^\top(Y - X\beta_0) = (X^\top X)^{-1}X^\top Y = \hat\beta$ whatever β_0 is, since the log-likelihood is exactly quadratic. The covariance is $\sigma^2(X^\top X)^{-1}$, and here the Gaussian distribution of $\hat\beta$ is exact, not asymptotic.

Exercise 23.2 *(page 366).* $b(\theta) = e^\theta$, $\phi = 1$, $c(y) = -\log y!$: $\ell_n(\beta) = \sum_i (Y_i x_i^\top \beta - e^{x_i^\top \beta}) - \sum_i \log Y_i!$, $\nabla \ell_n = \sum_i (Y_i - e^{x_i^\top \beta})x_i = X^\top(Y - \mu)$, $\nabla^2 \ell_n = -\sum_i e^{x_i^\top \beta} x_i x_i^\top = -X^\top W X$ with $w_i = \mu_i > 0$: negative definite for X of full rank. With $x_j \in \{0, 1\}$ and the other coordinates fixed, $\mu(x_j = 1)/\mu(x_j = 0) = e^{x^\top\beta + \beta_j}/e^{x^\top\beta} = e^{\beta_j}$.

Exercise 23.3 *(page 366).* (a) The two equations are $\sum_i (Y_i - \mu_i) = 0$ and $\sum_{i:a_i=1}(Y_i - \mu_i) = 0$; μ_i takes one value $\mu^{(1)}$ on aspirin and one $\mu^{(0)}$ on placebo, so the second says $n_2\mu^{(1)} = 104$ and the first then says $n_1\mu^{(0)} = 189$: the proportions. (b) $X^\top \hat W X = \begin{pmatrix} s_0 + s_1 & s_1 \\ s_1 & s_1 \end{pmatrix}$ with $s_0 = n_1\hat p_1(1 - \hat p_1)$ and $s_1 = n_2\hat p_2(1 - \hat p_2)$; its inverse has $(2, 2)$ entry $(s_0 + s_1)/(s_0 s_1) = 1/s_0 + 1/s_1$, and $1/(n\hat p(1 - \hat p)) = 1/(n\hat p) + 1/(n(1 - \hat p))$ gives the four reciprocals. (c) The free model fits two proportions and the null model ($\beta_2 = 0$) one common proportion, which are exactly the two fits Chapter 15's likelihood ratio test compared; same maximized likelihoods, same statistic.

Exercise 23.4 *(page 366).* (a) $\mathbb{P}(x^\top\beta + \epsilon > 0) = \mathbb{P}(\epsilon > -x^\top\beta) = 1 - F(-x^\top\beta) = F(x^\top\beta)$ by symmetry. (b) A logistic ϵ has standard deviation $\pi/\sqrt{3} = 1.81$, so matching standard deviations would give $\Phi(t/1.81)$; matching the two curves so that the largest gap between them is as small as possible gives the slightly smaller 1.7, with the gap below 0.01 everywhere (the two densities have different shapes, the logistic heavier in the tails, so no single scale makes them coincide). (c) Multiplying β and the scale of ϵ by the same constant leaves $\mathbb{P}(Y = 1 \mid x)$ unchanged: only β divided by the scale is identifiable, which is why the scale is fixed at 1.

Exercise 23.5 *(page 366).* (a) $1/\mu = (h + x)/(mx) = 1/m + (h/m)/x$, linear in $1/x$ with $\beta = (1/m, h/m)$. (b) The Poisson's canonical link is log, so no: $\theta_i = \log\mu_i = -\log(x_i^\top\beta)$, $h(t) = -\log t$. Then $\ell_n = \sum_i (Y_i\theta_i - e^{\theta_i})$ with $\partial\theta_i/\partial\beta = h'(x_i^\top\beta)x_i$, so $\nabla\ell_n = \sum_i (Y_i - \mu_i)h'(x_i^\top\beta)x_i$ and the Hessian is $\sum_i \big[-\mu_i h'(x_i^\top\beta)^2 + (Y_i - \mu_i)h''(x_i^\top\beta)\big]x_i x_i^\top$. The first term is negative definite; the second has the sign of $Y_i - \mu_i$ and can make the whole positive somewhere. (c) The guarantee that every quadratic model has a top and that a stationary point is the global maximum: a step can go downhill, since the quadratic model need not have a top, and a stationary point it reaches need not be the global maximum, so the iteration must be started well, or the Hessian replaced by its expectation, $-\sum_i \mu_i h'(x_i^\top\beta)^2 x_i x_i^\top$ (Fisher scoring), which is negative definite.

Exercise 23.6 *(page 366).* (a) Each term is $\log\mathbb{P}(Y_i = y_i)$, strictly negative for finite b since every probability is strictly between 0 and 1; as $b \to \infty$, $\mathbb{P}(Y_i = y_i) \to 1$ for every i since $x_i - 3.5$ has the sign of $2y_i - 1$, so the sum of logarithms tends to 0. (b) The log-likelihood is a sum of logarithms of probabilities, so it is at most 0, and along $a = -3.5b$, $b \to \infty$ it tends to 0 by (a); it is never 0 since no finite (a, b) gives probabilities 0 and 1. (c) If the iterates converged to some $\hat\beta$, continuity would make $\hat\beta$ a stationary point, hence by concavity a maximizer, which does not exist.

Exercise 23.7 *(page 366).* $X^\top W_k z_k = X^\top W_k X\beta_k + X^\top(Y - \mu_k)$, and multiplying by $(X^\top W_k X)^{-1}$ gives the step. The weighted sum of squares is $(z_k - X\beta)^\top W_k(z_k - X\beta)$, whose gradient $-2X^\top W_k(z_k - X\beta)$ vanishes at $\beta = (X^\top W_k X)^{-1}X^\top W_k z_k$, and whose Hessian $2X^\top W_k X$ is positive definite, so that is its minimizer: the normal equations of Chapter 21 with the weights $w_{k,i}$ inserted.

Exercise 23.8 *(page 366).* (a) $1/(1+e^{-(0.890-0.218s)})$: 0.45, 0.22, 0.085. (b) $Z=-0.218/0.060=-3.61$, $Z^2=13.0$; likelihood ratio $2(-34.04+41.62)=15.16$ on 1 degree of freedom, p-value 10^{-4}. (c) A coefficient changes with company when the added predictors are correlated with the one already in and carry their own effect. Here the start is nearly uncorrelated with age (correlation 0.06) and moderately correlated with the number of vertebrae (-0.43), and the number's own effect is modest, so the start's coefficient moves only from -0.218 to -0.207; its standard error grows from 0.060 to 0.068, the price of estimating two more coefficients from columns that overlap with it.

Exercise 23.9 *(page 366).* (a) X is a column of ones, so $X^\top(Y-\mu)=S-n\,e^\beta/(1+e^\beta)=0$ gives $\hat\beta=\log\big(S/(n-S)\big)=\log(80/44)=0.598$. (b) At $\beta_0=0$, $p=\frac12$, gradient $80-62=18$, curvature -31, so $\beta_1=18/31=0.581$; at β_1, $p=0.6412$, gradient $80-79.51=0.49$, curvature -28.53, so $\beta_2=0.581+0.0171=0.598$. (c) $X^\top\hat WX=n\hat p(1-\hat p)=28.39$, standard error $1/\sqrt{28.39}=0.188$, Chapter 22's $1/\sqrt{nb''(\hat\theta)}$.

Exercise 23.10 *(page 366).* The model is saturated, so the fitted means are the counts: $\mu_1=E_1e^{\beta_1}=12$ gives $\hat\beta_1=\log(12/30{,}000)=-7.824$ (a rate of 0.4 per 1,000 hours), and $\mu_2=E_2e^{\beta_1+\beta_2}=9$ gives $\hat\beta_2=\log\big(\frac{9/10{,}000}{12/30{,}000}\big)=\log 2.25=0.811$: the second plant's accident rate is 2.25 times the first's. $X^\top\hat WX$ has the form of the aspirin exercise with $s_0=12$ and $s_1=9$, so $\mathrm{SE}_2=\sqrt{1/12+1/9}=0.441$, and the interval for the rate ratio is $e^{0.811\pm1.96\times0.441}=[0.95,5.34]$: the difference is suggestive and not established on 21 accidents.

Chapter 24. Principal Component Analysis

Exercise 24.1 *(page 376).* (a) $\begin{pmatrix}1&r\\r&1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}=(1+r)\begin{pmatrix}1\\1\end{pmatrix}$ and likewise $(1-r)$ for $(-1,1)$; with $r=0.90$, 1.90 and 0.10. (b) $u^\top Su=\cos^2t+\sin^2t+2r\sin t\cos t=1+r\sin 2t$; its average over a full turn is 1, the average of the eigenvalues, that is, $\mathrm{tr}(S)/d$. (c) $u^\top Su=a\cos^2t+b\sin^2t+2c\sin t\cos t=\frac{a+b}{2}+\frac{a-b}{2}\cos 2t+c\sin 2t$, whose derivative in t vanishes when $\tan 2t=2c/(a-b)$. For Old Faithful $a<b$, and $2\times13.9/(1.30-184.1)=-0.152$ gives $2t=-8.6°$ for the minimum and $2t=180°-8.6°$ for the maximum, so $t=85.7°$, the 86° of Figure 24.3.

Exercise 24.2 *(page 377).* $H^\top=H$ is clear, and $H^2=I-\frac{2}{n}\mathbf{1}\mathbf{1}^\top+\frac{1}{n^2}\mathbf{1}(\mathbf{1}^\top\mathbf{1})\mathbf{1}^\top=I-\frac1n\mathbf{1}\mathbf{1}^\top=H$ since $\mathbf{1}^\top\mathbf{1}=n$. $H\mathbf{1}=0$ and $Hv=v$ for $v\perp\mathbf{1}$, so H projects onto the vectors with coordinates summing to 0, the orthogonal complement of the line through $\mathbf{1}$ (the residual projection of Chapter 21 for the intercept-only model). $H\mathbf{X}$ subtracts each column's mean from the column, so its rows are $(X_i-\bar X)^\top$, and $\frac1n\mathbf{X}^\top H\mathbf{X}=\frac1n(H\mathbf{X})^\top(H\mathbf{X})=\frac1n\sum_i(X_i-\bar X)(X_i-\bar X)^\top=S$, using $H^\top H=H$.

Exercise 24.3 *(page 377).* The scores have mean 0, so their sample covariance is $\frac1n\sum_i v_j^\top(X_i-\bar X)(X_i-\bar X)^\top v_k=v_j^\top Sv_k=v_j^\top\lambda_kv_k=0$ by orthonormality. The score matrix has (i,j) entry $(X_i-\bar X)^\top v_j$, which is the (i,j) entry of $H\mathbf{X}P$; the jth column's squared length is $\sum_i(v_j^\top(X_i-\bar X))^2=nv_j^\top Sv_j=n\lambda_j$.

Exercise 24.4 *(page 377).* (a) $P_kP_k^\top$ is the projection onto the span of $v_1,\dots,v_k$, so by Theorem 24.5 with $u_j=v_j$ the sum of squared distances is $n(\lambda_{k+1}+\cdots+\lambda_d)$. (b) $(0.215+0.053)/4=6.7\%$; the squared distances lost add up to $21\times0.268=5.6$, against a total squared size $21\times4=84$.

Exercise 24.5 *(page 377).* (a) $P^\top(X-m)$ is Gaussian with mean 0 and covariance $P^\top\Sigma P = D$ (toolkit A.13); a Gaussian vector with diagonal covariance has independent coordinates, with variances the diagonal entries. (b) The density is proportional to $\exp(-\frac{1}{2}(x-m)^\top\Sigma^{-1}(x-m))$, and $\Sigma^{-1} = PD^{-1}P^\top$, so with $z = P^\top(x-m)$ the exponent is $-\frac{1}{2}\sum_j z_j^2/\lambda_j^*$: a level set $\sum_j z_j^2/\lambda_j^* = c$ is an ellipsoid in the eigenvector coordinates with semi-axes $\sqrt{c\lambda_j^*}$ along v_j^*.

Exercise 24.6 *(page 377).* It makes $\mathbb{V}\mathrm{ar}(X^{(j)}X^{(k)}) \le \mathbb{E}[(X^{(j)})^2(X^{(k)})^2] \le \mathbb{E}[\|X\|^4] < \infty$, which is the finite-variance hypothesis of Theorem 3.5 for the averages $\frac{1}{n}\sum_i X_i^{(j)}X_i^{(k)}$; without it that theorem does not apply. With $d=1$ and $\mathbb{E}[X^2] < \infty$, $\frac{1}{n}\sum_i X_i^2$ is an average of i.i.d. variables with finite mean, which the Strong Law (Theorem B.3, needing only a finite mean; almost sure convergence implies convergence in probability, Appendix B) covers without a variance, so $\frac{1}{n}\sum_i X_i^2 \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{E}[X^2]$, $\bar{X} \xrightarrow[n\to\infty]{\mathbb{P}} m$, and $S = \frac{1}{n}\sum_i X_i^2 - \bar{X}^2 \xrightarrow[n\to\infty]{\mathbb{P}} \mathbb{V}\mathrm{ar}(X)$ by the continuous mapping theorem; the same argument works entrywise for any d, since $\mathbb{E}|X^{(j)}X^{(k)}| \le \sqrt{\mathbb{E}[(X^{(j)})^2]\,\mathbb{E}[(X^{(k)})^2]} < \infty$ by the Cauchy–Schwarz inequality, so the fourth moment is only what the book's weak law asks for.

Exercise 24.7 *(page 377).* $S - I_2$ is a small symmetric matrix of random sign, and the first eigenvector of S is the first eigenvector of $S - I_2$ (adding I_2 shifts every eigenvalue by 1 and changes no eigenvector), whose direction depends on which way the random perturbation happens to point and does not settle down. The theorem's (3) requires $\lambda_1^* > \lambda_2^*$, which fails here; (1) and (2) hold, and both sample eigenvalues converge to 1. The scree plot is nearly flat, two eigenvalues a little above and below 1, which is the plot's way of saying there is no preferred direction.

Exercise 24.8 *(page 377).* (a) The slope is the sample covariance of the two scores over the variance of the second, and the covariance is 0 by Exercise 24.3. (b) Least squares minimizes the sum of squared vertical distances, from each point to the line along the waiting axis, and gives the regression line of Chapter 20, whose slope $r < 1$ is the regression toward the mean of that chapter; the first principal direction minimizes the sum of squared perpendicular distances (Theorem 24.5 with $k=1$), and treats the two variables alike, which for standardized variables with positive correlation always gives the diagonal. The regression line is the right one for predicting waiting from duration; the principal direction is the right one for describing the cloud.

Appendix B. The Modes of Convergence in Full

Exercise B.1 *(page 405).* No. The sliding indicator of Example B.4 converges in probability to U and almost surely to nothing.